SEMANTIC

NORMALIZATION AND REALIZATION

A Unified General Theory:

Geometry, Chronology, Information, and Resources

A theory of how mathematically defined behavior can be realized. It explores which choices are genuinely possible and which costs are unavoidable. It also brings together independent research programs in combinatorics, arithmetic, topology, and algorithms.

ALP EREN BÜTÜN

THEORY · CONSTRUCTION · PROOFS · GENERALIZATION

github.com/alperenbutun

2023–2026

AUTOMATA THEORY · NORMALIZATION · COMPLETION · STRUCTURE

Research Monograph

Semantic Normalization and Realization

A Unified General Theory: Geometry, Chronology, Information, and Resources

Author

Alp Eren Bütün
Independent Researcher, Türkiye
ORCID: 0009-0001-2479-6037
`github.com/alperenbutun`

Research period

2023–2026

Scope and proof standards

This monograph presents the research conducted from 2023 to 2026 as a coherent mathematical account of semantic normalization and physical realization. The chapters are arranged by the logical connections between results: operational models lead to semantic invariants, completion geometry, physical realization, arithmetic structure, and computational complexity. Definitions and theorem hypotheses remain attached to the models for which they were proved. The book distinguishes unrestricted machines from restricted implementations, formal sources from physical constructions, and exact optimization from representation-dependent bounds.

The work has been subjected to mathematical and editorial audits covering statements, proof structure, symbolic identities, finite arithmetic claims, references, and compatibility between related results. These checks support a unified exposition in which each theorem can be followed through its stated assumptions and deductions. Purely symbolic proofs are given as mathematical arguments; computer-assisted proofs identify the finite algebraic conditions and exact arithmetic on which the conclusions depend. Numerical examples and consistency tests are identified by their own scope, alongside the theorem statements they illustrate.

The relevant mathematical results of the finite computations are recorded in the book through explicit congruences, equations, coefficient identities, certified ranges, and finite-case outcomes. Their presentation permits the claims to be examined and audited again by exact mathematical methods, without requiring the reader to consult implementation-specific programs. Verification outcomes are presented as mathematical statements and exact arithmetic records in the exposition, rather than as dependencies on code files or software packages. Appendix D brings together the validation methods and the interpretation of exact finite certificates, so that the role of each verification can be read consistently with the surrounding proofs.

Completion date

November 2026.

Preface

This book began with a particular deterministic pushdown automaton rather than with a general theory. In 2023 I was studying the binary counting language

$$L_{2,1} = \{w \in \{0,1\}^* : N_1(w) = 2N_0(w)\}.$$

Its center state, two side states, and cancellation-based stack transitions were initially an operational construction. The decisive observation was that a configuration did more than recognize the language: after reading a prefix, it represented a signed residual and a shortest demand for completing that prefix. The stack expressed completion demand, finite control corrected its interpretation, and local transitions traced a directed geometry of shortest continuations. That small machine remains the starting object of the book.

The first chapters make this observation precise. They recover exact residual invariants, reachable stores, prefix quotients, completion lengths, and representation-sensitive resource bounds. Higher-dimensional balance systems lead to cubical completion geometry and periodic geodesic compilers. In this setting, optimal paths have compatibility data of genuinely different orders: lower-order data do not determine the next order, and exact coefficient-bank constructions realize arbitrary labeled simplicial complexes at one canonical distance. A related line develops cancellation statistics, chronological transport, and arithmetic consequences. These are branches of the original completion question, not alternative starting points.

The central construction then returns to the physical stack. For coprime positive p, q, a centered literal-demand machine exposes the WRITE–SEARCH–CANCEL calculus. Its semantic update can be uniquely specified even when its physical realization is not: stack locality, normalization schedules, control sheets, and delayed materialization produce distinct lifts and measurable representation debt. The resulting theory of sections, protected interfaces, causal schedules, bounded recoding, confluence, and minimal cores culminates in a General Theory that begins with a semantic system and asks which exact physical realizations are possible. It does not presuppose a weighted counting language or confuse semantic equivalence with physical interchangeability.

Further chapters examine what that separation makes visible. Optimal completion fibers admit lattice-convex and higher-rank realizations; rational path enumeration, Farey recursion, and Cohn–Markoff arithmetic emerge from shortest-completion data. Chronological carry develops into typed CCR-k transport and its arithmetic refinements, alongside distinct Nörlund–Stirling and primitive-orbit questions. Inverse CCR-k observability also recovers masked sources through nonresetting adaptive pulses, with a sharp advantage over preset schedules under a restricted-input contract. The complementary computational CCR-k study proves a sharp minimum retained forecast message, constructs the causal archive, computes exact finite-ring answer families, and derives lossless source-mask coding laws, with independent algorithmic and event-log measurements. The subsequent independent CCR archive continuation identifies the entire future-response equivalence class of a freely selectable history, sharp all-horizon minimum checkpoint messages, and exceptional finite-field rank recovery. Its five-column mandatory transfer does not claim unrestricted physical machine-memory minimality. Exact state–stack frontiers, bounded replacement, and source-stack exposure show how semantic information imposes physical resource costs. Other chapters establish recognition and algorithmic boundaries, including the distinction between undecidable one-counter decomposition and decidable fixed one-counter comparison problems. Formal integral-source statements are not silently extended to arbitrary p-adic sources or to arbitrary physical machines.

The synthesis is intentionally wider than automata. Unified Semantic Normalization and Realization Theory organizes typed fibers, protected obligations, transport, information loss, and representation cost without identifying them as one invariant. Two independent tagged-Dyck studies classify source-faithful one-stack realizability under linear-modular and periodic height guards. Extrema–Frame Foreground–Background theory applies exact common-realization reasoning to finite scalar fields, with its own tractability, hardness, universality, and uncertainty results. The point of this extension is not to disguise a new domain as a pushdown problem. It is to test which aspects of a realization theory can be carried across domains, and which require new hypotheses. The subsequent thermal-network study makes this test explicitly physical: it fixes a diffusion plant and a shared input budget, proves a sharp three-query impossibility threshold, and determines how correlated terminal responses and optimal physical freedom change under a small conductance perturbation. Its own heat-flow hypotheses and exact certification are not smuggled into the abstract theory.

The distinction between a mathematical description and its realization also guides the book's use of examples. A shortest completion can be uniquely determined by a residual even when several stack schedules realize it, and a finite-state summary can preserve recognition while discarding precisely the information needed for optimal completion. These differences motivate the separate treatment of semantic cost, physical cost, and information retained by a representation. They explain why a lower bound proved for a literal encoding cannot be presented as a lower bound for every deterministic pushdown machine. In the opposite direction, a construction that works for all valid source assignments requires more than the success of a few representative assignments. The appropriate quantifiers are therefore part of the central narrative, not qualifications added after a proof.

Several themes reappear at different scales. The geometry of prefixes becomes the geometry of optimal words and common realizations. Chronology appears both as the order of physical stack actions and as a source of exact arithmetic transport. Protected information gives rise to the resource cost of reading and rewriting stored obligations. Local compatibility can be proved by a finite invariant, while global compatibility may require an additional synchronization or obstruction theorem. Taken together, these connections make the original machine a useful reference point for results whose later formulations are substantially more abstract.

The intended reader may enter the work through several doors. The automata chapters supply concrete transition systems and machine-specific proofs; the geometric chapters describe exact distances, optimal fibers, and associated finite combinatorics; the arithmetic chapters follow integer recurrences and congruence constraints; the general-theory chapters identify the shared organizational principles. These routes meet through stated constructions and reductions, not through informal analogies. The cross-references and appendices are designed to make those routes traceable while keeping the individual proofs at the level of precision their claims require.

Two editorial principles govern the exposition. First, the route of discovery is retained when it helps explain a construction, but later results are placed beside the prerequisites that make them intelligible. The simpler realizations and larger abstractions discovered later do not replace the machine that motivated the work. Second, *scope is part of the mathematics*. Raw residual-faithful optimality, optimality within a literal-demand model, and lower bounds against general deterministic pushdown machines are different claims. A finite computation may establish a precisely reduced finite case; it cannot replace a proof for an unrestricted family. Appendix D gives the validation conventions, while each substantive theorem states its own assumptions and proof dependencies.

The book is therefore both a consolidation of research studies and a single narrative about geometry, chronology, information, and resources. Its final synthesis returns to the original center/left/right machine so that residual meaning, literal completion, SEARCH debt, schedule choice, locality, and resource tradeoffs can be seen together. The purpose of the theory is not to make that machine disappear. It is to explain, in exact mathematical terms, why it carried more structure than was initially visible.

Alp Eren Bütün
November 2026

Abstract

This monograph develops a unified mathematical theory of semantic normalization and physical realization, beginning with a completion-aware deterministic pushdown automaton for $L_{2,1} = \{w : N_1(w) = 2N_0(w)\}$. For coprime positive p, q, the signed residual $\rho_{p,q}(w) = qN_1(w) - pN_0(w)$ determines the prefix quotient and canonical shortest completion demand. Its primitive kernel (q, p) induces a directed completion geometry, exact descriptions of reachable stores, and resource bounds whose validity depends explicitly on the representation class. The construction extends to centered literal-demand machines with WRITE–SEARCH–CANCEL normalization; it separates a well-defined semantic process from the LIFO constraints, scheduling choices, and representation debt of its physical implementations.

Higher-dimensional completion systems and finite-phase periodic actions admit exact cubical geodesic compilers for distances, optimal moves, shortest-prefix membership, and local optimal compatibility. The compatibility hierarchy is strict: data through order $k-1$ can be held fixed while the order-k layer varies freely. On a structural simplex with s vertices, realizing every labeled simplicial complex requires exactly $\binom{s}{\lfloor s/2 \rfloor}$ coefficient rows in advance; common-shell synchronization does not increase that minimum. The corresponding periodic Bellman realization is exact at one canonical distance. Higher-rank optimal fibers, physical holonomy, rational paths, and Farey–Cohn–Markoff structures extend these geometric consequences.

Physical realization results include exact semantic lifts, protected readout ports, bounded bi-recoding, causal schedules, confluence, and typed locality obstructions. State–stack bounds, guarded two-layer alphabet costs, and saturation gaps retain their distinct physical contracts and resource assumptions. Two independent tagged-Dyck studies establish exact one-stack realizability versus non-context-freeness dichotomies for linear-modular and periodic signed-winding height guards. Unified Semantic Normalization and Realization Theory organizes representation fibers, protected obligations, source-faithful transport, retained information, and realization cost. In a distinct domain, Extrema–Frame Foreground–Background theory treats exact common realization of finite scalar-field observations, yielding tractability on restricted graphs, hardness on shallow graph classes, universality of finite common-realization complexes, and parameterized ambiguity and uncertainty results. An additional physical realization chapter brings these relational predicates to controlled finite heat networks under one shared actuation budget. On a rational degree-three tree it proves a strict joint-feasibility threshold $Q_* = 108.32865113364069\ldots$ and a unique critical two-switch realizing control, although every proper subfamily of three queries is feasible at a smaller budget. An active-tie Krylov criterion characterizes optimum-face dimensions; on eight-vertex trees it separates a nonsemialgebraic optimal physical face with an exactly attainable correlated terminal-cap segment from its one-percent conductance perturbation, where the terminal-difference Krylov rank jumps from 2 to 7 and the optimal physical face collapses to a point. The statements concern mathematical heat-network solutions, not physical measurements or unrestricted optimal-control superiority.

The arithmetic branches develop exact chronological CCR-k source transport, fixed-column and valuation laws, Nörlund–Stirling rigidity, and primitive-denominator cancellation spectra. For a restricted-input CCR-10 family over primes $q \geq 41$, active observability recovers masked old sources without resets and proves a sharp two-adaptive-versus-three-

preset intervention gap. Complementarily, typed CCR-k admits an exact five-step forecast archive of $5J-31$ field symbols, attained by a causal construction under stated prime-field conditions, and a $6J-21$ source-response support law. Exact finite-ring profiles and optimal $\Theta(N \log N)$-bit lossless event-equality mask coding accompany measured task-specific speed and data-representation savings. A subsequent independent all-horizon typed CCR archive theorem gives the exact minimum for every admissible prime-field horizon as a direct-read historical coordinate count plus the rank of a five-column mandatory transfer. It classifies rank exceptions at horizons four through six, shows that full transfer rank returns by horizon seven, and proves eventual archive saturation at $5 + J(J-7)/2$ symbols under the separate hypotheses $T \geq J \geq 8$ and prime $\ell > T + H$. The conclusions concern free historical sources and fixed-length checkpoint messages, not total execution memory or one fixed prime over unbounded horizons. Three-terminal rational CCR completion is characterized by explicit integrality conditions and a finite local–global decision principle. Separately, ordinary-integer-source triple-zero impossibility holds for odd orders $5 \leq n \leq 75$ under the specified stationary model; the corresponding unrestricted all-order classification is not claimed. Primitive-orbit results establish rational predecessor noncoverage at every odd order and polynomial noncoverage at every good prime through odd order 71, leaving the general good-prime question open. Other consequences include exact enumerative results for rational Dyck-path contacts, Farey parent synchronization obstructions, and sharp retained-information bounds.

For stationary source polynomials, the results make the contrast between rational completion and integral obstruction especially explicit. A formal system may admit a unique rational choice of terminal sources while no ordinary-integer choice satisfies the same three coefficient equations. The exact terminal criteria turn this contrast into valuation inequalities, finite congruences, and lattice membership tests. Other arithmetic results treat prime-indexed rigidity, local lifting, and reductions of predecessor polynomials. Their statements carefully distinguish rational identities, finite ranges of good primes, and full universal assertions. In the geometric branches, analogous distinctions separate a selected optimal word from its entire tied optimum fiber, and source-aware transfer from rules that are permitted to use only reduced or unordered information.

The algorithmic results distinguish existential decomposition from fixed comparison. Presentation-relative state-primality and fixed-budget factorization for deterministic one-counter automata are undecidable under the stated restrictions; in contrast, the treated deterministic-versus- nondeterministic one-counter-net and zero-testing comparison problems are decidable in their defined models. Tree-walking recognition boundaries and finite resource-existence procedures likewise retain their precise model conditions. The proof standards distinguish symbolic arguments, exact finite certificates, and consistency checks (Appendix D). Throughout, the semantic object and the resources needed for its realization are specified explicitly.

The book distinguishes mathematical construction from exact finite certification, and finite-range statements from unrestricted theorems. Representation-sensitive bounds are never presented as universal bounds; model assumptions and proof dependencies accompany the stated results. The concluding synthesis relates these distinctions to the original completion-aware machine, while maintaining the separate semantics of physical implementation and numerical common realization.

A second, distinct positive-thermal realization experiment adds a temporal resource axis. Under fixed sampled guard conditions on a leaky six-node network, one held input per observation interval cannot realize any word of length at least two at any finite power, whereas two half-interval holds realize every finite word with a causal controller at the exact nominal power Q_0. A cross-coupled twelve-node construction admits a locally unique critical conductance and an explicit 81-action causal controller realizing all independently assigned two-module vector words at every horizon. The labels are invariant quantitative projections of EF–FG terminal records, not the full intrinsic three-color EF–FG language. The one-time conservative and sampled leaky heat models keep distinct initial states, control spaces, and power thresholds.

New NP-Complete Problems, Resolved Open Problems, Impossibility Theorems, Remaining Open Problems, and Main Contributions

The opening catalogue separates new complexity theorems, solutions to externally posed problems, model-specific impossibility results, and questions that remain open. Their precise assumptions are stated below.

New NP-complete problems introduced by the theory

The EF–FG branch establishes three native realization hardness results and one underlying scheduling-kernel hardness result. A separate guarded two-layer assembly model yields a fifth vertex-cover-equivalent family-profile decision problem.

1. **One-Polarity Common-Realization Observer Safety.** Deciding whether one scalar realization can make the one-polarity observer saliency safe is NP-complete even on interval-valued *radius-two trees*, with fixed $\theta = 1/2$ and $H = 0$.
2. **Possible Background Polarity-Core Ownership.** Deciding whether the self-dual observer state can be made background is NP-complete even on interval-valued *radius-two trees*, with fixed $\theta = 1/2$ and $\Delta = 0$.
3. **Complete Categorical Spatial-Frame Realization.** Deciding whether prescribed finite spatial-frame states can all be realized by one common scalar field is NP-complete even on *bipartite graphs of radius at most three*, with fixed seed threshold and uncertainty confined to identical terminal intervals. Its terminal-mask hardness core is already NP-complete on hub-radius-two bipartite graphs.
4. **Underlying kernel: Threshold-Band Owner-Scheduling Completion.** The threshold-band owner-scheduling kernel is NP-complete already for fixed $\theta = 1/2$ and $H = 0$. It isolates the source of the ownership hardness as global order selection under one-use owner resources, rather than evaluation of a fixed order.
5. **Guarded Two-Layer Family-Profile Saturation Gap.** For a rooted two-vertex-connected graph H with M vertices, $U = V(H) \setminus N_H[r]$, and threshold t, decide whether the variable-mask family-profile gap obeys $\Delta_M \leq t$ at the first feasible alphabet budget $g = M$. This is NP-complete since $\Delta_M = \tau(H[U])$, the induced vertex-cover number. Hardness persists at $g = M + 1$ and fixed stack height two; at height one the profile is explicit. The result concerns top-sensitive one-state guarded families, not arbitrary DPDA synthesis (Chapter 114).

Why these NP-complete problems are significant. The first four remain NP-complete on shallow trees or restricted bipartite graphs because their joint-realization, scheduling and ownership constraints are global. The fifth transfers vertex-cover hardness to joint mask-family saturation on rooted two-vertex- connected graphs, even at stack height two; its height-one profile is tractable.

Six externally posed problems resolved in the monograph

The six statements below answer questions posed outside the present research programme. They are listed separately because enumeration, hierarchy strictness, undecidability and language-equivalence decidability require different mathematical certificates.

1. **Dai–Fu–Qiu: run–return enumeration.** Dai, Fu, and Qiu asked for explicit enumeration with prescribed ordinary run values and returns in rational paths [23]. Returns are identified with coarea-barrier contacts, giving a finite determinant for their joint distribution with the run statistic. The formula is direct: it does not leave the requested answer as an unevaluated coefficient extraction.

2. **Dai–Fu–Qiu: ratio-run–return enumeration.** Their companion question replaces ordinary runs by *ratio-runs* while retaining the return statistic [23]. A different sampling width in the same contact determinant yields its explicit joint enumeration. The common barrier language explains the parallel forms, without conflating the two distinct path statistics.

3. **Masopust: deterministic stack-alphabet hierarchy beyond realtime.** For every $m \geq 2$ and $k \geq 1$, binary witness languages separate $\mathcal{D}_{m,k} \subsetneq \mathcal{D}_{m,k+1}$ in the general deterministic pushdown model permitting ε-moves [16]. The witness has the representation-independent lower bound $|Q|\,|\Gamma| \geq mk+1$. This establishes strictness beyond the realtime convention, while keeping the stateless convention and the physical resource accounting distinct.

4. **Totzke: Prime-DOCA decidability.** Presentation-relative state-primality is *undecidable* [99], through deterministic two-counter halting. The negative result already holds for complete realtime one-work-symbol DOCAs with a permanent bottom marker, all states final and counter updates in $\{-1, 0, +1\}$. Moreover, the fixed-factor-budget question is undecidable at $k = 2$ with two half-size factors, and thus at each fixed $k \geq 2$; the restriction to a small budget does not remove undecidability.

5. **Automata Exchange 24.04(B): deterministic–nondeterministic one-counter-net equivalence.** Finite-word equivalence of a deterministic and a nondeterministic one-counter net is *decidable* [109], with final-control-state acceptance, arbitrary integer weights and counter initially zero. One inclusion uses [108]; for the remaining inclusion, exact forward envelopes, backward requirements, return compactness and a well-quasi-order tree give a computable finite witness bound. The result is a decision procedure, not an asserted complexity upper bound.

6. **Automata Exchange 24.04(A): zero-testing automata versus one-counter nets.** Finite-word equivalence of a deterministic zero-testing one-counter automaton and a nondeterministic one-counter net is likewise *decidable* [109]. This includes realtime unit-step transitions and the extension with deterministic ε-moves on the zero-testing side and arbitrary ε-moves on the net side. The hard inclusion follows by counter envelopes, first-return compactness, a finite zero-state analysis and threshold-affine macros; the reverse reduces to reachability in a vector addition system with one zero test. No complexity upper bound is claimed.

Impossibility theorems and precise recognition boundaries

Each obstruction belongs to a specified recognition, arithmetic or resource model. The hypotheses are part of the conclusion: a finite-range or model-relative impossibility is not promoted to an unrestricted theorem.

1. **Right-turn spectra: deterministic recognition obstructions.** On ordered full binary trees with three marked leaves, $\kappa(t)$ counts right-child edges between their relevant marked ancestors. Then $\mathcal{L}_E = \{t : \kappa(t) \in E\}$ is regular exactly when $E \subseteq \mathbb{N}$ is ultimately periodic, with bottom-up state complexity linear in the least preperiod and period. Right-edge dilation and the three-marker separation [156] exclude deterministic tree-walking recognition for fixed upper and lower thresholds and some infinite ultimately periodic spectra. Every regular spectrum is recognizable with one visible pebble; the DTWA versus UTWA question and the pure even spectrum remain outside these exclusions.

2. **Farey price-blind transfer obstruction.** For genuine positive unimodular rank-three parent steps, the complete unordered synchronized discrepancy alphabet and daughter congruence lattice cannot uniformly recover the optimal primitive daughter ray when its numerical price is omitted. The discrepancy module may have unbounded index or rank-one differences even when the daughter optimum requires a further physical letter. This forbids only *price-blind* transfer, not price-aware synchronization (Section 84.5).

3. **Primitive denominator cancellation.** For nonsymmetric nonnegative integral determinant-one matrices and positive primitive vectors, each attainable cancellation factor c satisfies $v_2(c) \leq 1$, and every odd prime dividing c is 1 (mod 4). Even diagonal entries impose a further specified 2-adic restriction. In particular, odd primes 3 (mod 4) cannot cancel in this reconstruction model; the companion spectrum and local-density results retain the same assumptions (Section 86.5).

4. **Nörlund–Stirling triple-root exclusions.** A fixed-centre three-order common root at $X = -3$ modulo p^3 is excluded for good $2n_2 < p \leq 10^7$ by a two-base Wieferich sieve. Separately, let $E \geq 6$ and $2 \leq h \leq 120$ be even, and set $n_j = E + jh - 1$ for $j = 0, 1, 2$. For *every* prime $p > 2n_2$, there is no common root $X \equiv -3 \pmod{p}$ of the three quotients modulo p^2. The second theorem has no prime upper bound but has the stated gap and precision conditions (Section 34.53).

5. **Stationary CCR integer-source triple-zero exclusion.** For every odd $5 \leq n \leq 75$, prime transport parameter $P > 2n$ and *every ordinary-integer* source vector in the formal stationary CCR model, $X^3 \nmid F_n(X)$. All 36 orders are covered through Bernoulli transport sieves and exact all-source finite-ring certificates. The statement is not a theorem for all odd orders, nor for rational or arbitrary local q-adic sources (Section 34.55).

6. **Nörlund–Stirling predecessor noncoverage.** At every odd $n \geq 5$, a rational irreducible target factor is absent from the predecessor product. Thus, at each fixed n, polynomial coverage fails at all but finitely many good primes. More strongly, $\overline{q}_n \nmid \overline{\mathcal{B}}_n$ for every good prime and odd $5 \leq n \leq 71$, and in the separately proved factor families. For nonsquarefree reductions, polynomial noncoverage alone need not give a new geometric root (Section 34.56).

7. **Restricted-input chronological source masking and intervention costs.** For a typed CCR-10 source family, prepared complete coefficient rows may conceal an old rank-four source while later nonresetting interventions recover it. Over every prime $q \geq 41$, the specified adaptive controller requires two pulses in the worst case, whereas preset schedules require three. The lower bounds allow a stated family of higher-rank nuisance sources; they do not concern unrestricted source access (Chapter 35).

8. **Protected one-pass readout lower bound.** For one adaptive choice of a genuine immutable and repeatedly inspectable protected record, a cyclic query fixed before scanning, and all control phases charged to states, the deterministic one-pass complexity satisfies $q_{\mathrm{cyc}}(n, 1; m) = \Theta_m(m^n/n)$ for fixed $m \geq 2$. Hence asymptotically smaller controllers cannot realize all the specified cyclic readouts. This is a protected-model bound, not a bound on unrestricted DPDAs (Section 101.1).

9. **Fixed-Parikh direct-realization LIFO obstruction.** In the specified literal WRITE–SEARCH–CANCEL model for fixed-Parikh balanced words, the zero-obstruction realizations are exactly the unique concatenations of the two primitive direct blocks B_0, B_1. Thus only 2^g words in the (gq, gp) fiber are directly realizable without SEARCH or outer WRITE; the first-obstruction survival law is exact. This excludes *direct* realization of the other words, not their realization with additional physical operations (Theorems 87.6 and 87.7).

10. **Physical holonomy and literal LIFO carrier minimum.** For the specified nonabelian free shortest-completion episode, every persistent nonempty zero-weight literal order carrier has height at least d, and this minimum is attained by an explicit real-time one-top construction. The associated toral comparison problem has its own minimum global correction and Pareto law. These are costs for the stated literal-carrier and comparison contracts, not universal lower bounds on all encodings (Chapter 94).

11. **Protected-role capacity and Pareto down-set exclusion.** When a refined representation contract requires N separately observable protected roles but the available resources provide fewer than N observation cells, that resource down-set contains no valid realization. For a coarse attainable point the exclusion forces strict Pareto deterioration under refinement; the necessary capacity and contract hypotheses are explicit (Theorem 103.6 and Corollary 103.7).

12. **Schedule-confluence obstruction from fatal supports.** For finite downward-closed schedule-activation families whose admissibility is an independence complex, two distinct maximal admissible schedules are non-joinable normal forms. In particular, activation is confluent precisely when every minimal fatal support is a singleton; any non-singleton minimal fatal support forces nonconfluence in this rewriting model (Theorems 63.9 and 63.10).

13. **All-odd-order stationary CCR integer-source obstruction.** For every odd $r \geq 5$ and every ordinary-integer stationary source vector in the normalized CCR model, the first-order Stirling-source invariant satisfies $A_r \equiv 1 \pmod 6$ and thus $A_r \neq 0$. Its exact source image is an arithmetic coset $1 + d_r\mathbb{Z}$; the result is not an exclusion for arbitrary p-adic source choices (Theorems 34.50 and 34.51).

14. **Canonical third-source CCR depth-three obstruction.** For each odd $r \geq 5$, prime $p > 2r$, canonical stationary source $\lambda_3 = 1$, and arbitrary higher p-adic stationary sources, the near-leading coefficient satisfies $0 \leq v_p(C_{r,p}) \leq 2$, sharply. Hence valuation three or higher is impossible under the canonical-source contract. Allowing λ_3 to vary changes the possible valuations and is not covered by this prohibition (Theorem 34.73).

15. **Protected paired-session and uninterrupted-trace no-go.** For the completion-total source-native shortest-fiber model, a concatenation of two or more shortest rooted sessions with nonzero target b has target kb, not b; it is not one original shortest rooted execution. Moreover, every finite source word extends to a zero-return word, so an irreversible prefix-exclusive comparison-mode routing rule cannot coexist with recognition of every zero-return continuation. These restrictions do not forbid the explicitly host-selected resettable paired-session service (Proposition 113.23).

16. **Linear-modular height-guard non-context-freeness.** For two fully tagged independent Dyck stacks subject to the all-prefix linear-modular height guard modulo p, if none of $a \equiv 0$, $b \equiv 0$, $a + b \equiv 0 \pmod{p}$ holds, the guarded empty-to-empty language is non-context-free. Consequently no single-stack PDA, even nondeterministic, recognizes it. Each complementary coefficient regime has an exact deterministic real-time single-stack construction (Theorem 116.2).

17. **Periodic signed-winding non-context-freeness.** For the specified rectangularly periodic tagged two-source height guards, a reachable quotient component of signed-winding rank two, or one of rank one with opposite-signed generator coordinates, forces non-context-freeness and excludes every single-stack PDA. Components meeting the complementary rank-and-sign criterion have canonical source-faithful deterministic real-time realizations (Theorems 117.3 and 117.4).

18. **Joint EF–FG physical heat-realization obstruction and correlated-cap restriction.** For the specified seven-node rational thermal tree, shared measurable nonnegative heater control, one horizon $\tau = 0.01$, and strict observer ownership with two intrinsic terminal-region requests, every proper subfamily is realizable at pointwise power $Q = 100$ but their conjunction is realizable precisely when $Q > Q_* = 108.32865113364069\ldots$. The critical weak optimum is a unique two-switch control. On the separate eight-node symmetric tree, optimal terminal caps satisfy an *exact correlated segment* rather than a product of independent intervals, even though the optimum face is nonsemialgebraic; a 1% conductance perturbation raises terminal-difference Krylov rank from 2 to 7 and collapses the optimum physical face to a singleton. These are exact instance- and control-category-specific obstructions, not a new NP-completeness theorem or a claim for arbitrary thermal networks (Theorems 124.6, 124.14 and 124.16).

19. **Held-actuation resolution and complete sampled-language collapse.** For the declared leaky six-node thermal plant, with five nonnegative heater ports, sampled numeric terminal-record guards, and a common per-held-phase power bound, one full-period hold (H1) admits no observation word of length at least two at *any* finite power; two half-period holds (H2) realize every finite word with an exact sharp all-word power threshold and a causal policy. For the separate cross-linked twelve-node pair at every coupling $g \in [0.0054757560, 0.0054757561]$, H1 likewise admits no two-sample vector word at any finite power, while an explicit 81-action H2 policy realizes all 81^n independent vector words, for every $n \geq 1$, under per-phase total power 421. This is a model- and sampled-numeric-projection-specific obstruction, not an impossibility theorem for native categorical EF–FG records or arbitrary heat networks (Theorems 125.7, 125.10,125.32 and 125.31).

Remaining Open Problems

Eight questions remain unresolved. Each formulation retains the stated model and distinguishes proven restricted cases from the general claim.

1. **Higher-order common-root lifting for Nörlund–Stirling quotients.** For distinct odd $5 \le r < s$, do some prime $p > 2s$ and $X \in \mathbb{Z}/p^4\mathbb{Z}$ satisfy $q_r(X) \equiv q_s(X) \equiv 0 \pmod{p^4}$? Existing prime-indexed rigidity and sporadic classifications are mod-p results; the separate first-Hensel-derivative nonvanishing hypothesis is conjectural, not an assumed proof premise.

2. **Complete two-resource Pareto classification for deterministic pushdown witnesses.** For every $m \ge 2$, $k \ge 1$, determine all Pareto-minimal physical state/stack-alphabet pairs for binary $B_{m,k}$ in the deterministic ε-pushdown model with one-symbol initial stack and final-state plus empty-stack acceptance. The product bound, one-state threshold, exact corner families and $\Theta(\sqrt{mk})$ frontier cardinality leave individual coordinates and threshold plateaux undetermined; ε-free results do not settle the general model.

3. **Original-parent-to-daughter optimal WORD transfer.** For positive unimodular rank-three Farey parent–child steps, can the complete *ordered, original-labelled* optimal daughter WORD, including all ties, be recovered from the two independently optimized complete parent WORDs and source frames, with numerical daughter price, *without new daughter minimization*? The discrepancy formulation (Section 84.5) requires full-fiber, source-aware price information; the Apéry-tie formulation (Section 84.4) retains all labels and ties. Price-blind failure and a family-specific construction settle neither the general transfer nor necessity of both parents.

4. **Exact finite protected-source cyclic readout complexity.** For each $n, m \ge 2$, determine the exact $q_{\mathrm{cyc}}(n, 1; m)$ for a deterministic one-pass controller with one adaptively selected immutable, repeatedly inspectable genuine record, preannounced cyclic query and all finite-control phases charged. The known $\Theta_m(m^n/n)$ bound gives only asymptotic order (Section 101.1).

5. **Uniform predecessor polynomial noncoverage.** Is $\overline{q}_n \nmid \overline{\mathcal{B}}_n$ in $\mathbb{F}_p[X]$ for *every* odd $n \ge 5$ and prime $p > 2n$? Equivalently, can the exact content C_n of the rational predecessor remainder have no prime factor $> 2n$ at every odd order? The theorem covers $n \le 71$, certain additional factor types and the all-order rational analogue, but not this uniform statement. At nonsquarefree reductions, polynomial nondivisibility is weaker than new geometric-root existence (Section 34.56).

6. **Zero-pebble deterministic recognition of the pure even right-turn spectrum.** For ordered full binary trees with three marked leaves, let $\mathcal{L}_{\mathrm{even}}$ consist of those trees for which the designated right-turn count is even. Is this regular language recognizable by an ordinary deterministic tree-walking automaton without pebbles? Its one-pebble deterministic recognition and zero-pebble parity-of-runs acceptance do not settle ordinary zero-pebble recognition (Question 115.24).

7. **Fixed-resource existence for epsilon-enabled deterministic pushdown competitors.** Given the binary witness language $B_{m,k}$ and fixed state and stack-alphabet budgets (q, g), is existence of an equivalent deterministic PDA with ε-moves decidable, in the stated singleton-initial-stack, final-state-and-empty-stack acceptance model with arbitrary finite replacement words? The ε-free version has a terminating decision procedure; its conclusion is not known for general ε-enabled competitors (Theorem 11.28 and the following discussion).

8. **Sharp two-hold power for the cross-linked twelve-node heat plant.** For the specified *local sampled-numeric* two-cell model and fixed $g \in [0.0054757560, 0.0054757561]$, determine the least total per-phase heater budget enabling every independent vector word under causal two-hold control. The 81-action policy gives the sufficient budget $Q = 421$, but does not determine the sharp optimum. Is this causal minimum equal to the full-preview all-word minimum, as proved for the isolated six-node plant (Chapter 125)?

Main Contributions

This monograph develops a unified theory of exact semantic descriptions and faithful physical computations. It distinguishes representation invariants, information retained by a source, and resources required by a realization. Two independent tagged-Dyck studies classify one-stack realizability against non-context-freeness for linear-modular and periodic signed-winding guards, within their specified models. The framework does not assert a universal multistack impossibility or a general assembly-deficit formula.

The theory resolves six externally posed problems and establishes five new NP-complete problem families or kernels, each under explicit structural restrictions. The EF–FG thermal realization chapter contributes a different kind of exact result: a model-specific global shared-power obstruction with a matching two-switch controller, a nonsemialgebraic optimal face whose cap outputs obey a sharp correlated segment, and a one-percent network perturbation exhibiting physical rank rigidity. It neither changes the count of NP-complete problems nor adds a claim of independent resolution of an externally posed problem (Chapter 124). Recognition, equivalence, ownership, and joint-realization problems have distinct quantifiers and proof obligations. Negative results range from bounds on restricted encodings to language-level non-context-freeness and arithmetic source exclusions; none is extended beyond the contract appearing in its theorem.

Deterministic-pushdown normalization is analyzed through stack exposure, input chronology, resets, protected readout, and finite control. Four complementary resource results form an exact progression: $|Q| \geq \max(p, q)$ for general ε-DPDAs recognizing $L_{p,q}$ (Article 18); $|Q| \geq p + q$ with a permanent bottom and one work symbol (Article 24); a sharp bounded-replacement state–push staircase (Article 30); and the representation-independent collapse-retention and source-stack exposure capacity, with sharp witnesses (Article 31). Their resource contracts are not interchangeable (Theorems 97.3, 98.13, 98.19, and 100.1).

Canonical completion fibers, geodesic compilers, confluence, and lattice-geometric constructions link automata to topology and optimization. Farey and Apéry results characterize shortest completions and source-aware optimal transfers. For fixed-bank simultaneous universality of labelled simplicial complexes, the exact number of coefficient rows is the Boolean-lattice width $\binom{s}{\lfloor s/2 \rfloor}$; a common shell has zero extra row defect (Theorem 20.2). This is both a sharp bound and a matching construction.

The arithmetic programme establishes CCR-k chronological transport, primitive-matrix cancellation restrictions, Nörlund–Stirling rigidity, explicit rational completion formulae, and certified local–global decisions. It distinguishes ordinary integers from rational and p-adic source choices. The restricted-input CCR-10 active-observability theorem gives a sharp two-adaptive-versus-three- preset intervention gap, not a bound for unrestricted sources. For the specified uniform-prior CCR–10 diagnostic experiment, the finite Bellman computation additionally gives the *exact minimum expected pulse costs* 199/125 when exceptional three-pulse trajectories are permitted and 44/25 when every trajectory is capped at two pulses. These are expectation values for that diagnostic source and nuisance contract, not universal actuation-power thresholds (Theorem 35.3). A complementary forecast-information theorem determines the minimum five-step fixed-length archive for free-source prime-field CCR inputs, $m(T, J, 5) = 5J - 31$, under its stated accessibility and determinant conditions, with a matching causal encoder. It also gives a sparse $6J - 21$ future-source support count, exact composite-ring response-profile cardinalities and an optimal $\Theta(N \log N)$-bit lossless encoding law for realizable event-equality masks, rather than raw quadratic bits. Task-specific exact computations and event-log studies demonstrate algorithmic and payload savings without reinterpreting these results as universal physical memory bounds (Chapter 36). A separate all-horizon continuation (Article 66) determines the exact free-source field-symbol archive by a direct-read count plus the rank of a five-column boundary-transfer matrix. Under $T \geq J \geq 8$ and prime $\ell > T + H$, the boundary rank has two exceptional pairs at horizon four, a determinant congruence criterion at horizon five, and three exceptional pairs at horizon six; full rank is recovered by horizon seven. The minimum archive eventually stabilizes at $5 + J(J - 7)/2$ symbols for admissible horizon/field pairs (Chapter 37). This is a new information-realization result, not an additional NP-completeness or externally posed problem resolution. All-order rational noncoverage and finite-range good-prime and integer-source exclusions retain their separate quantifiers.

The protected-port theory also proves that the full cube-face dimension vector does not

determine minimum port-state complexity: there are pairs with identical face vectors but different exact port indices in every dimension $d \geq 3$ (Proposition 102.5). The missing-class correction therefore contains information invisible to face counts. Across the book, exact identities and finite arithmetic certificates are embedded in proofs with explicit assumptions; computations do not replace universal reasoning.

The controlled-heat EF–FG branch also makes the distinction between individually feasible semantic records and jointly attainable physical records quantitative. Its Q_* threshold is proved using a global support-function bound with exact complementarity, not merely a local numerical optimum; the cap segment and Krylov dimension formulas identify physical and semantic nonuniqueness separately.

These results organize semantic normalization and physical realization through constructive algorithms, sharp resource frontiers, exact classification, and rigorous obstruction principles. The synthesis explains when abstractly compatible descriptions have a common physical realization, what information must survive normalization, and which limitations cannot be removed within a stated model.

The later sampled EF–FG projection experiment introduces a logically separate actuation-resolution obstruction. At the stated leaky six-node plant, H1 cannot realize a word of length at least two, even at unbounded finite power, while H2 realizes every finite word at an analytically specified sharp nominal common-power threshold, with coincident online/offline optima. For a cross-coupled twelve-node network it certifies an exact local critical conductance, one-hold language extinction after the first sample, and a memoryless 81-action two-hold controller for every vector word at every finite horizon at the *sufficient*, not proved sharp, total per-phase budget $Q = 421$ (Chapter 125). These conclusions concern finite-sample numeric projection records; they neither raise the count of NP-complete problems nor assert that arbitrary native categorical EF–FG outputs are physically realizable.

A recurring contribution is the treatment of *common realization* as a problem with quantifiers different from separate satisfiability. The existence of a witness for each individual description does not supply a single witness for their conjunction. In the categorical EF–FG branch this gap is detected through shared ownership and observer-state constraints; in the guarded stack branch it is measured by simultaneous family-profile saturation under an alphabet budget. In the continuous thermal branch it arises because every requested terminal inequality must be met by the same nonnegative heater inputs at the same observation time. The mathematical witnesses and resource currencies differ, so the common principle is a comparison of realization contracts rather than an identification of unrelated obstruction invariants.

A second contribution concerns the information that a realization must preserve. A language recognizer may correctly decide membership without preserving a source-labelled explanation of the accepted computation; conversely, a normalized shortest-completion representation can retain source chronology, ties and exact residuals that a coarser membership description forgets. The protected-readout and stack-resource results make this distinction operational: access to a genuine immutable record, its permitted scan direction, and the counting of finite-control phases are parts of the resource contract. The two thermal studies expose a further distinction between *simultaneous feasibility* and *causal sequence feasibility*. In the first, a prescribed collection of terminal conditions competes for a shared power envelope at one horizon; exact dual bounds and switching controls establish the threshold under its own heat equation. In the second, an entire sequence of requested sampled numeric labels must be realized online. The resulting H1/H2 separation depends not on a new categorical EF–FG alphabet, but on the allowed actuation holds and on a fixed quantitative projection of terminal records. Its all-horizon certificate combines local input feasibility, a finite history calculation and decay of older thermal states. Finally, the book records which forms of exactness are available in each branch. Some results are all-order algebraic identities or full language classifications; others are sharp within an explicit finite parameter range, a fixed machine model, a prescribed network, or a restricted field of sources. Determinant formulas, dual certificates, matching constructions and finite verification tables establish these statements at different logical levels. Retaining their hypotheses is part of the contribution: it permits one to use a proved impossibility or resource lower bound without extrapolating it to a system in which the relevant interface, alphabet or admissible controller has changed.

Contents

Preface iii

Abstract v

New NP-Complete Problems, Resolved Open Problems, Impossibility Theorems, Remaining Open Problems, and Main Contributions vii

Notation and Conventions xlix

Prologue — The Machine Before the Theory 1

I Foundations: From the 2023 Machine to Representation-Independent Questions 3

1 The 2023 Completion-Aware DPDA 4

1.1 The motivating language . . . 4
1.2 Two formal variants of the same core design . . . 4
1.3 Recognition and completion are different operations . . . 8
1.4 Removing the wrapper from the semantic analysis . . . 9
1.5 The reachable stack is much more restricted than it looks . . . 10
1.6 The state carries part of the arithmetic . . . 11
1.7 End-of-input completion transfers the state correction into the stack 12
1.8 A worked run: the prefix 111 . . . 13
1.9 The first structural surprise: two stack cells are enough . . . 14
1.10 The cancellation transition that will reappear much later . . . 14
1.11 What has been learned without leaving the original machine . . . 15

2 What the Machine Is Really Storing 16

2.1 From the 2:1 imbalance to the general p:q residual . . . 16
2.2 The completion equation . . . 17
2.3 The primitive balanced cycle . . . 18
2.4 Every completion lies on one affine ray . . . 19
2.5 Minimum completion demand . . . 20
2.6 A canonical shortest word . . . 21
2.7 The original DPDA implements this decoder at 2:1 . . . 23
2.8 Physical information versus semantic information . . . 24
2.9 The residual transition system . . . 25
2.10 Completion length as distance . . . 25
2.11 How many shortest completions are there? . . . 26
2.12 Why the minimum pair is a better semantic object than the canonical word . . . 26
2.13 A first exact notion of redundancy . . . 27
2.14 The original machine viewed through the new semantics . . . 28
2.15 The first major conceptual compression . . . 28

3 Minimum Demand and the Completion Ladder **30**
3.1 Geodesic input symbols . . . 30
3.2 The left branch as an exact shortest-path process . . . 32
3.3 The ladder sources are actually reachable . . . 33
3.4 Every position of the required one is shortest . . . 34
3.5 Local commuting diamonds . . . 36
3.6 Parikh synchronization . . . 37
3.7 Completion and cancellation are opposite boundaries of one object . . . 38
3.8 A commutation potential . . . 38
3.9 Why this is more than a counting argument . . . 39
3.10 The canonical completion is a boundary choice . . . 40
3.11 Limits of the ladder description . . . 40

4 Quotients, Geodesics, and What Can Be Seen Locally **41**
4.1 The residual classifies all future behavior . . . 41
4.2 The quotient graph . . . 43
4.3 Completion length is exactly directed distance . . . 43
4.4 The fundamental balanced cycle . . . 44
4.5 Balanced-cycle reduction . . . 45
4.6 Which words are geodesic? . . . 46
4.7 Exact quotient spheres . . . 46
4.8 What the original two-cell observation really reveals . . . 47
4.9 General canonical action visibility . . . 49
4.10 A second bounded quantity: the completion frontier . . . 51
4.11 One extremal word, two different bounds . . . 53
4.12 The next representation idea is now forced . . . 54

5 General p:q Completion Rectangles **55**
5.1 Remaining demand as a grid coordinate . . . 55
5.2 The shortest interval is an actual rectangle . . . 56
5.3 Exact number of shortest completions . . . 58
5.4 Exact number of commuting squares . . . 58
5.5 A worked 3:2 rectangle . . . 59
5.6 Canonical order as a boundary normal form . . . 60
5.7 Exactly when does a geodesic fork occur? . . . 61
5.8 First obstruction at length $p + q$ . . . 62
5.9 Exact geodesic-word growth . . . 63
5.10 Distinct paths versus distinct quotient vertices . . . 65
5.11 The 2:1 ladder is now completely explained . . . 65
5.12 From geometry back to representation . . . 66

6 The First General Physical Lift and Its Defects **67**
6.1 Finite phase plus unary magnitude . . . 67
6.2 The stable four-symbol encoding . . . 68
6.3 Stable rules away from the boundary . . . 69
6.4 Forced boundary normalization . . . 70
6.5 Exact normalized residual conjugacy . . . 70
6.6 The abstract rectangles lift to physical configurations . . . 71
6.7 A concrete 3:2 boundary crossing . . . 72
6.8 Raw residual conjugacy . . . 73
6.9 The defect set is finite and exact . . . 73
6.10 Why the defects cannot be patched locally in the same coordinates . . . 74
6.11 Which shortest rectangles are already raw? . . . 75
6.12 The defect is coordinate-relative . . . 76

7 From Boundary Defects to a Raw Three-Symbol DPDA 77
7.1 Before using two work symbols: the unary baseline . . . 77
7.2 Boundary-compatible coordinates . . . 79
7.3 Positive-tail arithmetic . . . 80
7.4 The repaired sign boundary . . . 80
7.5 Negative-tail arithmetic and the second repaired boundary . . . 81
7.6 Global raw conjugacy . . . 81
7.7 The 3:2 boundary is now literal . . . 83
7.8 Raw rectangles . . . 84
7.9 What has actually been optimized? . . . 84
7.10 The motivating 2:1 ratio survives arbitrary three-symbol encodings . . . 85
7.11 Further exact special ratios . . . 86
7.12 The right endmarker saves one control state in the scoped models . . . 87
7.13 The coordinate-shift improvement . . . 88
7.14 Three different meanings of "better" . . . 88
7.15 Removing the representation shape . . . 89

8 The Complete Raw Residual-Faithful Frontier 90
8.1 Model R . . . 90
8.2 Why bounded physical height is impossible . . . 91
8.3 The last occurrence of a height creates a stable suffix . . . 92
8.4 Positive residue classes synchronize . . . 93
8.5 Why one work symbol is worse . . . 94
8.6 The complete raw frontier . . . 95
8.7 Scope of the theorem . . . 96
8.8 Acceptance without a right endmarker . . . 96
8.9 Can the optimum survive bounded pushes? . . . 96
8.10 A stronger semantic-visibility budget . . . 98
8.11 A superlinear exact family . . . 99
8.12 Direct lossless factorization: counting semantic cells . . . 100
8.13 Exact normalization-rule cost under push-at-most-two . . . 100
8.14 A top-window implementation lemma . . . 101
8.15 Four resource questions, four different answers . . . 102
8.16 What is now genuinely settled . . . 103

9 From Residual Frontiers to a Representation-Independent Hierarchy Theorem 105
9.1 The methodological jump . . . 105
9.2 The external hierarchy question . . . 106
9.3 Why realtime separation does not automatically survive ε-moves . . . 107
9.4 Lineage from the original DPDA . . . 107
9.5 The improved 2:1 raw DPDA as the immediate precursor . . . 107
9.6 Why the witness development did not end with a direct encoding trick 108
9.7 The general DPDA model used in the lower bound . . . 109
9.8 Modes . . . 109
9.9 The lower-bound target . . . 110

10 Completion Slopes and ε-Robust Pumping 111
10.1 The binary realtime witness . . . 111
10.2 Four rule families . . . 112
10.3 Upper-bound theorem . . . 113
10.4 The exact completion-slope slice . . . 113
10.5 The smallest new case . . . 115
10.6 The lower-bound opponent is arbitrary . . . 115
10.7 Continuation determinism . . . 116
10.8 No raw configuration can repeat on a live slope ray . . . 116

10.9 Raw stack height must escape . . . 117
10.10 Suffix-minimum positions . . . 117
10.11 The ε-robust step pump . . . 118
10.12 The remaining proof obligation . . . 119

11 One Mode, One Slope 120
11.1 Unary completion uniqueness . . . 120
11.2 Slope separation . . . 121
11.3 The product lower bound . . . 123
11.4 The hierarchy theorem . . . 124
11.5 What exactly has been resolved . . . 125
11.6 Why this is not a realtime lower bound in disguise . . . 125
11.7 Why the product, rather than states or symbols separately, appears . 126
11.8 Scope and excluded claims . . . 126
11.9 The methodological meaning for the monograph . . . 127
11.10 Exact Pareto Geometry and Resource Thresholds for the Binary Hierarchy . . . 128
11.10.1 Representation-independent lower bound and its arithmetic consequence . . . 128
11.10.2 Constructive resource conversions . . . 129
11.10.3 The exact one-state endpoint . . . 131
11.10.4 Exact resource thresholds and Pareto corners . . . 133
11.10.5 Uniform order of Pareto-frontier width . . . 133
11.10.6 Full state-coordinate support in a square-root wedge . . . 134
11.10.7 Illustrative exact points and scope . . . 136
11.10.8 Scope and remaining resource questions . . . 136
11.11 Epsilon-Free Fixed-Resource Existence and Zero-Drain Compactness 137
11.11.1 Finite-ones language calibration . . . 137
11.11.2 Deletion-only normalization . . . 137
11.11.3 Finite-level decision and compactness . . . 138
11.11.4 A terminating resource-existence decision procedure . . . 139
11.11.5 Finite refutation packets . . . 140
11.11.6 A calibrated zero-drain principle . . . 141
11.11.7 A transparent finite infeasibility certificate . . . 142
11.11.8 Computational Pareto case study for $B_{3,1}$ . . . 142
11.11.9 Discussion and conclusion . . . 151
11.12 Transition to higher-dimensional completion geometry . . . 151

II Completion Geometry, Geodesic Compilation, and Universal Compatibility 153

12 When Minimum Demand Becomes a Fiber 154
12.1 Matrix counting languages . . . 154
12.2 Completion totality . . . 155
12.3 Residuals still classify right quotients . . . 156
12.4 Minimum completion length . . . 157
12.5 The optimal Parikh fiber . . . 157
12.6 Equal-length trades . . . 158
12.7 Affine-semigroup viewpoint . . . 158
12.8 Continuous optimum geometry is not the same as integer multiplicity 159
12.9 The shortest-prefix down-set . . . 159
12.10 Exact geodesic-prefix criterion . . . 160
12.11 First actions and compatible action sets . . . 161
12.12 The conceptual transition . . . 162

13 The Filled Shortest-Prefix Quotient **163**
13.1 Why vertices alone are not enough . . . 163
13.2 Cubical realization of a down-set . . . 163
13.3 Equal-length identifications . . . 164
13.4 The quotient theorem . . . 165
13.5 Recovery of the binary rectangle . . . 166
13.6 When is the quotient literally a box? . . . 166
13.7 Unique optimum versus unique shortest word . . . 168
13.8 Local compatibility and global identification are different . . . 168
13.9 The rank of equal-length trade . . . 169

14 Local and Global Topology **170**
14.1 The local action complex . . . 170
14.2 Squarefree-divisor interpretation . . . 171
14.3 Rank zero: one simplex . . . 172
14.4 Rank one: the optimum fiber is an arithmetic interval . . . 172
14.5 Rank two already permits a local hole . . . 173
14.6 Rank-one global geometry . . . 174
14.7 Overlap complexes . . . 175
14.8 Periodic saturation . . . 175
14.9 The global circle . . . 176
14.10 Local triviality, global nontriviality . . . 178
14.11 What the topology means operationally . . . 178

15 Scalar Rank Hierarchy and Exact Three-Letter Geometry **179**
15.1 Scalar counting systems . . . 179
15.2 The $d = 2, 3, 4+$ hierarchy . . . 179
15.3 Three-letter normal form . . . 181
15.4 Parameterizing the optimum fiber . . . 181
15.5 Exact three-letter quotient vertex count . . . 182
15.6 Exact cubical f-vector . . . 183
15.7 Worked example . . . 183
15.8 What changed relative to the binary rectangle? . . . 184
15.9 Why this hierarchy belongs in the same monograph . . . 185

16 Eventual Geodesic Arithmetic **186**
16.1 The continuous directional slope . . . 186
16.2 Dual certificate and reduced costs . . . 187
16.3 Zero-defect shifts . . . 188
16.4 Stabilized defect profile . . . 188
16.5 Why τ_b already contains the slope denominator . . . 190
16.6 Fundamental defect period . . . 190
16.7 Eventual local action geometry . . . 191
16.8 What is and is not being claimed . . . 193
16.9 The higher-dimensional branch in one picture . . . 193
16.10 From completion geometry to finite compilation . . . 193

17 From Completion Geometry to Finite Geodesic Compilers **195**
17.1 Finite-phase periodic graphs . . . 195
17.2 Gain-cost relations . . . 196
17.3 Bellman guards . . . 197
17.4 Optimal unfolding . . . 197

18 Cubical Geodesic Compilation **199**
18.1 Periodic action systems . . . 199
18.2 Structural cubes . . . 199
18.3 Optimal compatibility . . . 200
18.4 Arithmetic face guards . . . 201
18.5 Signature regions . . . 201
18.6 Affine-semigroup interpretation . . . 201
18.7 Complexity of the Compiled Representation . . . 203
18.7.1 Query parameters . . . 203
18.7.2 Prefix queries . . . 203
18.7.3 Face queries . . . 204
18.7.4 Canonical routes . . . 204
18.7.5 Enumeration of all shortest paths . . . 204
18.7.6 Normalized arithmetic compilers . . . 205

19 Prescribed-Order Separation, Boolean Amplification, and Universal k-Fibers **206**
19.1 Completion and nondegeneracy . . . 206
19.2 Binary-deficit distance . . . 207
19.3 Adjacent equal-distance states . . . 207
19.4 First discrepancy at order k . . . 208
19.5 Exact full complexes . . . 209
19.6 Large optimization-induced minimal nonfaces . . . 210
19.7 Boolean Amplification . . . 211
19.7.1 Product formulas . . . 211
19.7.2 Product family . . . 212
19.7.3 Exponential fibers . . . 212
19.7.4 Conditional discrimination . . . 213
19.8 From Boolean amplification to k-fibers . . . 213
19.9 The universal two-phase construction . . . 214
19.10 Exact metric and structural isolation . . . 215
19.11 Hypergraph-coded witness states . . . 215
19.12 Universal k-fiber realization . . . 216
19.13 A worked universal fiber: $s = 3$, $k = 2$ . . . 218
19.14 Exact next-order capacity . . . 218
19.15 From separation to universality . . . 219
19.16 Relation to realization theories and scope . . . 219
19.17 Synthesis: finite compilation and universal next-order freedom . . . 219

20 Universal Coefficient Banks and Simultaneous Simplicial Representation **221**
20.1 Definitions and Main Results . . . 221
20.2 Universal Coefficient Construction . . . 222
20.2.1 Exact one-row separation . . . 223
20.2.2 Minimal nonfaces and canonical quotas . . . 223
20.2.3 Canonical quota mass . . . 223
20.2.4 Injectivity of subset scores . . . 224

21 Exact Row Complexity and Common-Shell Compression **225**
21.1 Maximum antichains . . . 225
21.2 Two-for-one middle-level compression . . . 226
21.3 Lower-orthant shattering . . . 226
21.4 Full-vertex and one-vertex boundary cases . . . 227

22 Periodic Shortest-Path Lift and Exact Bellman Universality **228**

23 Extension Fibers, Full Metric-Shell Capacity, and Topological Universality 230
23.1 Extension fibers and simultaneous all-rank saturation 230
23.2 Full Metric-Shell Capacity and Entropy 231
23.3 Topological Universality and Effective Consequences 232
23.4 Returning to the original machine from the opposite direction 233

III Cancellation Geometry, Chronology, and the General Real-Time p:q Machine 235

24 Zooming Back into the Original DPDA 236
24.1 The machine facts we reuse, not re-prove 236
24.2 The nonnegative path sector 237
24.3 Exact landing-height correspondence 237
24.4 From one machine event to a d-Dyck statistic 238
24.5 Chronological correction signatures 239
24.6 Path defect 240
24.7 The new inverse problem 240

25 Correction Signatures and Minimum Defect 241
25.1 The threshold function 241
25.2 Local Gap Cost 242
25.3 Terminal Cost 243
25.4 Intrinsic signature defect 243
25.5 Canonical blocks 244
25.6 Defect–Signature Characterization 244
25.7 Canonical does not mean unique 246
25.8 Signature persistence 246
25.9 Maximum number of corrections 246
25.10 Zero defect 247
25.11 Defect as penalized overshoot 247

26 The Catalan Boundary and the Binary Cancellation Skeleton 248
26.1 Two unavoidable unmarked nodes 248
26.2 The d entrance choices 249
26.3 Why correction nodes become binary 249
26.4 Binary Cancellation Skeleton 250
26.5 Tree coordinates 250
26.6 Correction height from the binary address 251
26.7 Inorder chronology 251
26.8 The signature is lossless 252
26.9 Area statistic 253
26.10 What the original transition has become 253

27 Exact Correction and Defect Enumeration 254
27.1 Scalar correction generating function 254
27.2 The auxiliary kernel and A371395 255
27.3 Exact correction triangle 256
27.4 Row sums and boundary columns 256
27.5 Intrinsic signature enumeration 257
27.6 Translation and first-zero decomposition 257
27.7 Defect–Signature Master Equation 258
27.8 Exact defect coefficients 258
27.9 Cumulative realizability 259
27.10 Three arrays, three questions 259

28 The Asymptotic Defect Hierarchy **260**
28.1 Fixed-defect generating functions . . . 260
28.2 Algebraicity and singularity location . . . 261
28.3 Critical scaling . . . 261
28.4 Pairwise leading singularities . . . 262
28.5 Asymptotic Defect Hierarchy . . . 262
28.6 The defect pairs . . . 263
28.7 Geometric interpretation . . . 264
28.8 Cumulative asymptotics . . . 264
28.9 What the cancellation branch has established . . . 265
28.10 Chronological carry as an arithmetic offshoot . . . 266

29 From Ordered Corrections to CCR-3 **267**
29.1 Scientific boundary . . . 267
29.2 Connected supports through order three . . . 268
29.3 The CCR-3 recurrence . . . 268
29.4 Hessenberg realization . . . 269
29.5 Operator form . . . 270
29.6 What this chapter has and has not established . . . 271

30 Fixed Columns and the Universal Stirling Shadow **272**
30.1 Column coefficients . . . 272
30.2 Transport equations . . . 272
30.3 First three fixed columns . . . 273
30.4 Reduction modulo p . . . 274
30.5 Normalized shadow polynomials . . . 274
30.6 Why the third column is the first singular one . . . 275

31 The Third-Column p-Adic Split **276**
31.1 The simple branch . . . 276
31.2 The double shadow branch . . . 277
31.3 Generic residue classes . . . 278
31.4 The scaled third column . . . 278
31.5 Independent consistency checks . . . 279
31.6 Ramified location of the remaining algebraic roots . . . 279

32 Ramified Recovery **280**
32.1 Exact discriminant factorization . . . 280
32.2 Removing the integral root . . . 280
32.3 Half-integral roots . . . 281
32.4 Ramified square class by $p \bmod 8$ . . . 281
32.5 The complete third-order arithmetic picture . . . 282
32.6 Novelty boundary . . . 283

33 CCR-4: Fourth-Order Chronological Carry, Connected Quadruples, and the Stirling Shadow **284**
33.1 The CCR-3 baseline . . . 284
33.2 What is genuinely new at order four . . . 285
33.2.1 Exact reconstruction of a general p-divisible fourth source 286
33.3 The stationary p-divisible fourth-order family . . . 287
33.4 Four-band Hessenberg and operator realization . . . 288
33.5 Fourth-column transport and exact identifiability . . . 289
33.6 The fourth-jet Stirling shadow . . . 291

34 CCR-k General Theory: Higher-Order Chronological Carry, Hessenberg Hierarchies, and Universal Stirling Shadows 292
34.1 Chronological support geometry and scope . . . 292
34.2 General terminal-block CCR-k . . . 293
34.3 Higher-band lower-Hessenberg realization . . . 294
34.4 The subdiagonal light cone . . . 295
34.5 Normalized fixed-column transport and inverse reconstruction . . . 295
34.6 Stationary p-divisible extensions . . . 297
34.7 Operator realization . . . 297
34.8 Polynomial fixed columns . . . 298
34.9 Exact parameter translation and identifiability . . . 299
34.10 Universal Stirling shadows . . . 300
34.11 The all-order closure theorem . . . 301
34.12 The first genuinely new checkpoint beyond CCR-4: order five . . . 302
34.13 Synthesis . . . 302
34.14 Proof-dependency ledger . . . 303
34.15 Scope safeguards and nonclaims . . . 303
34.16 Arithmetic stratification of the normalized CCR columns . . . 304
34.16.1 Fourth-column normalization and the integral numerator . 304
34.17 The complete local factorization classification . . . 304
34.18 The 419-adic critical coordinate . . . 305
34.19 An exact integer discriminant certificate . . . 306
34.20 Prime-axis density: S_3 Chebotarev law . . . 307
34.21 Parameter-axis density and precision law . . . 307
34.22 Prime-power congruence fibers and a resonance moment law . . . 308
34.23 Universal odd-column fold and p-adic source resonance . . . 310
34.23.1 The all-order Stirling generating function . . . 310
34.23.2 An all-odd-order arithmetic fold theorem . . . 310
34.23.3 Universal local prime-power fiber and moment laws . . . 311
34.23.4 The entire fifth column away from $p = 73$ . . . 312
34.24 The exceptional fifth-column prime 73: two coupled folds . . . 313
34.25 Joint source equidistribution and the exact five-root density at 73 . . . 315
34.26 Fixed integer sources: finite resonance support across primes . . . 317
34.27 All odd orders: a Stirling source obstruction and finite prime support 318
34.28 Chronological integer-source observations and their limits . . . 323
34.29 Explicit Galois laws at the seventh and ninth columns . . . 325
34.30 Exact certificates, higher odd columns, and adelic laws . . . 327
34.31 Haar-product independence across different odd CCR columns . . . 327
34.32 Adelic resonance dichotomy: fixed integer sources versus profinite Haar sources . . . 329
34.33 Boundary of arithmetic specialization . . . 332
34.34 Sharp p-adic rigidity for the third near-leading coefficient . . . 332
34.35 The high-jet formula and source translation . . . 333
34.36 A bounded cofactor obstruction . . . 334
34.37 Exhaustive finite exclusion and sharpness . . . 336
34.38 Exact source fibers and resonance engineering . . . 337
34.39 Sheffer structure and classical comparison . . . 337
34.40 Computational reproducibility and scope . . . 338
34.41 Novelty boundary and related arithmetic literature . . . 338
34.42 Technical identities for the coefficient bound . . . 339
34.43 Exact reproduction of the finite exclusion . . . 339
34.44 Prime-indexed Nörlund–Stirling quotient pairs and single-site rigidity 339
34.45 Finite-difference reduction . . . 340
34.46 An effective single-site rigidity theorem . . . 341
34.47 Computational proof certificates . . . 342

34.48 Exact exceptional-prime classification through twelve sites 343
34.49 Exact natural densities and first failure 344
34.50 A sharp local resultant example and scope 345
34.51 Open Problems and Future Directions 345
34.52 Novelty boundary and related arithmetic literature 346
34.53 Fixed-centre and finite-gap Nörlund–Stirling exclusions 346
34.53.1 Notation and exact negative-integer coefficients 346
34.53.2 An exact criterion for three odd orders at -3 348
34.53.3 The two-base Wieferich obstruction 348
34.53.4 General negative integer centres 349
34.53.5 The first shifted Hensel jet at -3 349
34.53.6 All-prime finite-gap exclusion of shifted triples 351
34.53.7 Finite verification, certification and relation to earlier results 352
34.54 Integral terminal-source completion and finite local–global decision . 353
34.54.1 Problem and precise scope 353
34.54.2 CCR response series and high-source filtration 353
34.54.3 Explicit three-source integrality formulas 354
34.54.4 Half-source affine elimination and a lattice criterion 355
34.54.5 Effective finite local–global principle 356
34.54.6 Exact examples and decision precision 357
34.54.7 Relation to prior papers and novelty boundary 358
34.54.8 Exact computational verification 358
34.55 Finite-range stationary CCR triple-zero impossibility 359
34.55.1 Introduction and stationary transport 359
34.55.2 Contributions and novelty 360
34.55.3 Exact generating representation 360
34.55.4 Bernoulli transport condition and auxiliary-prime sieves . 361
34.55.5 Certified order reductions below 57 362
34.55.6 Exceptional orders 57 through 75 364
34.55.7 Completion of the finite-range proof 366
34.56 Primitive-orbit rigidity and predecessor noncoverage 367
34.56.1 Introduction and provenance 368
34.56.2 The stationary model and good-prime normalization . . . 368
34.56.3 Source interpolation and positional rigidity 369
34.56.4 The exact shift-companion identity 371
34.56.5 Primitive long Frobenius factors at every odd order 372
34.56.6 Every odd order has a new rational irreducible factor . . . 374
34.56.7 Chebotarev densities from long irreducible factors 375
34.56.8 Exact content obstructions and an unconditional finite range 376
34.56.9 Bernoulli anchors and primitive integer-coordinate ladders 377
34.56.10 Proof provenance, scope and the uniform open problem . 378

35 Active Observability of CCR-k 380
35.1 Introduction 380
35.2 Contributions 381
35.3 Chronological recurrence and normalized readouts 381
35.4 Passive masking and the need for active interrogation 381
35.4.1 A sharp adaptive–predetermined separation in CCR–10 . 382
35.5 Sharp expected intervention cost 383
35.6 A finite-field criterion and a cross-prime family 384
35.7 An explicit infinite-prime adaptive gap 386
35.8 Full-row robust observability under restricted fresh-source freedom . 388
35.8.1 Exact rational certificate for the circle lemma 389

36 Computational Power of Typed CCR-k **391**
36.1 Introduction . . . 391
36.1.1 Contributions . . . 392
36.1.2 Main achievements in one view . . . 392
36.1.3 Relationship to the author's CCR series . . . 393
36.1.4 What counts as an advantage . . . 393
36.2 CCR recurrence, source interfaces, and query tasks . . . 394
36.2.1 General parameterization and source distinctions . . . 394
36.3 An exact weighted-path interpretation . . . 394
36.4 Exact source diagnosis and observability obstructions . . . 395
36.4.1 The independent-event factorization gate . . . 395
36.4.2 Two terminal coefficients can recover a limited parity load 395
36.4.3 A passive terminal jet can conceal an interaction . . . 395
36.4.4 Information cannot be generated by deterministic features 396
36.5 Selective monitors and source-polynomial compilation . . . 396
36.5.1 Selective monitoring . . . 396
36.5.2 Full source expansion is not always a good compiler . . . 396
36.5.3 Compiling future response at a checkpoint . . . 397
36.5.4 Homogeneous degree-four interaction channels . . . 397
36.6 Forecast-response information and sharp archives . . . 397
36.6.1 One-step read coordinates versus encoded archives . . . 397
36.6.2 General response-profile principle . . . 398
36.6.3 The first four future horizons . . . 398
36.6.4 The sharp five-step theorem . . . 398
36.6.5 An exact future-source support budget . . . 399
36.6.6 A causal realization with the minimum retained message . 400
36.6.7 Composite moduli and exact response-profile cardinalities 400
36.6.8 Why the information optimum does not imply a working-memory optimum . . . 401
36.7 Event-equality signatures: graph geometry and coding . . . 401
36.7.1 Sharp lossless payload bits . . . 401
36.7.2 What the terminal CCR outputs retain . . . 402
36.7.3 A real-log round-trip coding check . . . 403
36.7.4 Relation to exact terminal diagnosis . . . 403
36.8 Algorithm contracts, equality checks, and resources . . . 403
36.8.1 Four exact evaluators for four kinds of work . . . 403
36.8.2 Verifiable equality and modular safety . . . 404
36.8.3 Benchmark taxonomy . . . 404
36.9 Reproducible timing and memory evidence . . . 404
36.9.1 Long-stream and real-prefix checks . . . 405
36.9.2 Precomputed small-mask lookup . . . 405
36.9.3 External prediction: no established added benefit . . . 406
36.9.4 Why repeated forecasting is an accelerator rather than a mere compression trick . . . 406
36.9.5 Executable architecture: minimal message, fast decoder, explicit cost switches . . . 406
36.9.6 Evidence-based method selection . . . 407
36.9.7 Measured favorable results in separate tasks . . . 407
36.9.8 The strength of the measured cases . . . 408
36.9.9 Negative findings that control the deployment claim . . . 409
36.10 Independent event logs: exact features versus predictive utility . . . 409
36.10.1 Primary data identity and validity gates . . . 409
36.10.2 The independent Helpdesk test . . . 410
36.10.3 Measured real computational work and reproducibility boundaries . . . 410

36.11 Interpreting CCR computational strength . . . 410
36.11.1 What follows mathematically . . . 410
36.11.2 What follows computationally . . . 411
36.11.3 What requires further validation before an application claim 411
36.12 From mathematical theorem to reproducible software system . . . 411
36.12.1 A precision-preserving implementation contract . . . 411
36.12.2 Portable benchmark protocol for an independent implementation . . . 411
36.12.3 An application-oriented significance statement . . . 412
36.13 Conclusion . . . 412
36.14 Further derivations and mathematical safeguards . . . 413
36.14.1 The coefficient-wise recurrence . . . 413
36.14.2 Mandatory-response determinant and its hypotheses . . . 413
36.14.3 Composite-ring profiles and finite-field rank are different . 413
36.14.4 Formal and empirical certification are different kinds of proof 413
36.15 Detailed resource measurements and failure cases . . . 414
36.15.1 Cold forecasting at low and high query volumes . . . 414
36.15.2 A chronological performance ledger . . . 414
36.15.3 A fully charged algorithm selection criterion . . . 415
36.16 Coverage of mathematical structures, algorithms, and evidence . . . 415
36.17 Worked certificates and executable design templates . . . 417
36.17.1 A concrete one-step coordinate menu . . . 417
36.17.2 The first interaction at future horizon five . . . 417
36.17.3 An exact interaction-channel test case . . . 417
36.17.4 An explicit finite-ring non-identifiability profile . . . 417
36.17.5 Source-mask counts and terminal coordinate tests . . . 418
36.17.6 Reference pseudocode for a fixed-query compiler . . . 418
36.18 Reproducibility, exact certificates, and empirical scope . . . 419
36.18.1 A self-contained mathematical record . . . 419
36.18.2 Algebraic and finite-ring certificate summary . . . 419
36.18.3 Computational verification and measured applications . . 420

37 CCR: General Future-Archive Complexity 421
37.1 Introduction . . . 421
37.1.1 Contributions . . . 421
37.1.2 Relationship to realization theory and earlier CCR work . 422
37.2 The recurrence and archive contract . . . 422
37.3 Historical accessibility and direct-read information . . . 423
37.4 The five-dimensional mandatory boundary . . . 424
37.5 Complete characteristic-dependent rank classification . . . 425
37.6 Examples and information interpretation . . . 427
37.7 Reconstruction of the algebraic certificates . . . 427
37.8 Discussion . . . 428
37.9 Re-entry to the main DPDA story . . . 428

38 Why Static Two-Visibility Fails 429
38.1 We do not re-prove minimum demand . . . 429
38.2 A two-visible canonical word . . . 430
38.3 The 3:2 obstruction . . . 430
38.4 Static two-visible obstruction . . . 431
38.5 SEARCH . . . 431

39 Debt, Virtual Demand, and Minimal WRITE **432**
39.1 Virtual demands . . . 432
39.2 Virtual-demand projection . . . 432
39.3 The WRITE equation . . . 433
39.4 Minimal kernel carry . . . 434
39.5 Why the historical long writes existed . . . 434
39.6 From arbitrary debt to a finite rail . . . 435

40 The Centered Real-Time p:q DPDA **436**
40.1 The centered rail . . . 436
40.2 Residual invariant . . . 438
40.3 Why the debt rail can stop at h_{ctr} . . . 439
40.4 Boundary guard . . . 441
40.5 Canonical segments . . . 442
40.6 Two-cell qualitative visibility . . . 443
40.7 Exact WRITE normalization . . . 443
40.8 Zero residual . . . 443
40.9 Main construction theorem . . . 444
40.10 Shortest-demand quotient . . . 445

41 What "Optimal" Means in the Deferred-SEARCH Architecture **446**
41.1 Canonical deferred-SEARCH model . . . 446
41.2 Residual distinguishability . . . 447
41.3 One witness containing enough of both symbols . . . 447
41.4 Exhaust the same stack, leave different debts . . . 448
41.5 Exact deferred width . . . 449
41.6 Scope — place this beside the theorem . . . 450
41.7 Why Chapter 8 gives a different optimum . . . 450
41.8 Why this chapter belongs before the 2:1 recovery . . . 451

42 Closing the First Loop: Recovering the Original 2:1 Machine **452**
42.1 Center mismatch directions . . . 453
42.2 Left cancellation rail . . . 453
42.3 Right locality-repair rail . . . 453
42.4 Historical WRITE arithmetic . . . 453
42.5 Specialization and recovery theorem . . . 454
42.6 Lazy versus eager discharge . . . 454
42.7 Canonical materialization map . . . 455
42.8 Eager–lazy semantic equivalence . . . 455
42.9 Historical machine as an eager refinement . . . 456
42.10 Computational audit . . . 456
42.11 The structural consequence of the p:q generalization . . . 457

IV Structural Pushdown Normalization: WSC, Lift Freedom, Causality, and Confluence **458**

43 Semantic Arithmetic Before Representation **459**
43.1 Dominating kernel level . . . 459
43.2 One input symbol: CANCEL plus optional carry . . . 460
43.3 Semantic WRITE count . . . 461
43.4 Demand-length clock . . . 462
43.5 Semantic chronology is unique . . . 462

44 What One-Top LIFO Locality Forbids **463**
44.1 Exact literal-demand lift . . . 463
44.2 Fixed block orientation . . . 464
44.3 Top-rewrite suffix criterion . . . 464
44.4 Complete obstruction classification . . . 464
44.5 Two fundamentally different failures . . . 465

45 Quantity, Order, and Control Debt **467**
45.1 Quantity defect . . . 467
45.2 Universal representation conservation . . . 467
45.3 Order defect . . . 468
45.4 Three-channel canonicality . . . 468
45.5 Kernel economy . . . 469
45.6 Forced unit SEARCH . . . 469
45.7 Carry obstruction with zero quantity debt . . . 470
45.8 Minimum order debt . . . 471

46 The WSC Count Shadow **472**
46.1 Phase coherence . . . 472
46.2 WSC defect lattice . . . 472
46.3 Defect coordinates . . . 473
46.4 Defect quotient . . . 474
46.5 WSC lattice is not the full LIFO fiber . . . 474
46.6 Kernel economy in lattice coordinates . . . 474
46.7 Abstract count-shadow fiber region . . . 475

47 Physical Materialization, Backlog, and WSC Factorization **476**
47.1 Materialization level . . . 476
47.2 Physical length formula . . . 476
47.3 Materialization amount . . . 477
47.4 Backlog law . . . 477
47.5 Canonical carry dichotomy . . . 478
47.6 Unique physical WSC role factorization . . . 478
47.7 Reconstructing the charges . . . 479
47.8 Global materialization budget . . . 480
47.9 Deferral capacity . . . 481

48 Unique Semantic Quotient, Nonunique Physical Presentation **482**
48.1 Semantic fibers . . . 482
48.2 Fiber congruence . . . 482
48.3 Canonical retraction . . . 483
48.4 What is actually unique . . . 484
48.5 Cheap nonuniqueness already exists . . . 484
48.6 Scope of the confluence claim . . . 485

49 The Original 2:1 Machine as WSC **486**
49.1 Historical states as SEARCH sheets . . . 486
49.2 Historical stack-length changes . . . 487
49.3 The global identity $b + 2c = n$ . . . 487
49.4 What Structure Theory has accomplished . . . 488
49.5 The Structure Theory endpoint . . . 488

50 The Semantic Core Belongs to the Language **489**
50.1 Right residuals . . . 489
50.2 Universal semantic quotient . . . 490
50.3 Minimum-demand semantics is a coordinate change . . . 490
50.4 Three levels of description . . . 491

51 Exact Sectioned Lifts and Category-Relative Minimality 492
51.1 Reachable versus ambient configurations . . . 492
51.2 Exact sectioned lift . . . 492
51.3 Section-preserving morphisms . . . 493
51.4 Minimality is category-relative . . . 493

52 When LIFO Locality Forces Fiber Branching 494
52.1 Uniform local rule . . . 494
52.2 Canonical-section closure dichotomy . . . 494
52.3 Essential physical fiber branching . . . 495
52.4 Quotient-protected branching . . . 495

53 Two Reduced 2:1 Lifts 496
53.1 Historical eager lift . . . 496
53.2 Centered lazy lift . . . 496
53.3 Section incomparability . . . 496
53.4 Coordinate reducedness . . . 497
53.5 Morphism-relative nonuniqueness . . . 497

54 The Six 2:1 Normalization Schedules 498
54.1 Sites . . . 498
54.2 Exact schedule classification . . . 498
54.3 Schedule geometry . . . 499
54.4 Six reduced lifts . . . 499
54.5 Terminal normalization coherence . . . 500

55 Why A Causes B 501
55.1 Local carry visibility . . . 501
55.2 Portability . . . 501
55.3 A expands the B-fiber . . . 501
55.4 Direct canonicalization becomes impossible . . . 502
55.5 Forced kernel excess . . . 502

56 Directed Causal Scheduling 504
56.1 First-hit exposure . . . 504
56.2 Exact 2:1 primary dependency . . . 504
56.3 Concrete causal certificate . . . 505
56.4 Coordinate invariance . . . 506
56.5 But semantics does not determine causality . . . 506

57 Which Obstructions Are Really Forced? 507
57.1 Nested representation classes . . . 507
57.2 Obstruction monotonicity . . . 507
57.3 Branching profile . . . 508
57.4 Category-relative status of $A \to B$ . . . 508
57.5 Representation–obstruction separation . . . 508
57.6 Computational validation note . . . 508
57.7 Transition to General Theory . . . 509

58 From the Six 2:1 Schedules to the Arbitrary-Coprime Compiler 510
58.1 Centered literal-demand normalization . . . 510
58.2 The uniform eager-return compiler . . . 511
58.2.1 The mandatory reset envelope . . . 511
58.3 The finite baseline-portable site universe . . . 512

59 Exact Causal-Hypergraph Classification for All Coprime Ratios 514
59.1 The high-minimum regime: every core schedule is safe . . . 514
59.2 Extra-site self-exposure for $\min(p, q) \geq 3$ . . . 515
59.3 Low-minimum regimes and arbitrary-coprime classification . . . 516
59.3.1 The line $\min(p, q) = 2$ . . . 516
59.3.2 The exceptional line $q = 1$ . . . 516
59.4 Arithmetic details for the Eight-Site Theorem . . . 517
59.5 Low-minimum certificates . . . 518
59.5.1 Self-contained odd-p:2 fork certificate . . . 518

60 Downward Closure and Category-Relative Causal Persistence 522
60.1 Downward closure is not universal . . . 522
60.2 Causal persistence depends on the presentation category . . . 523
60.3 Persistence signature and category typing . . . 523
60.4 Monograph synthesis . . . 524

61 Strict Coordinate Quotients and Protected Interfaces 525
61.1 Exact sectioned lifts and used coordinates . . . 525
61.2 Strict coordinate quotients . . . 525
61.3 Protected interfaces and maximal safe quotients . . . 526

62 Bounded Bi-Recoding and Minimal Protected Cores 528
62.1 Bounded bi-recoding and recoding-invariant resource spectra . . . 528
62.2 Typed protected contracts and BBR equivalence . . . 529
62.3 Interface with General Theory . . . 529

63 Confluence Across Exact Pushdown Normalization Layers 530
63.1 Exact lifts and four notions of normalization coherence . . . 530
63.2 Finite schedule-activation rewriting . . . 532
63.3 Rational schedule confluence . . . 532
63.4 The full rational completion fiber . . . 534
63.5 Finite-control sheets and three-channel confluence . . . 535
63.6 Protected confluence and canonical Pareto anchoring . . . 536
63.7 Rational fatal-support data used in Section 63.3 . . . 538
63.7.1 Complete site universe for $\min(p, q) \geq 2$ . . . 539
63.7.2 High-minimum fatal supports . . . 540
63.7.3 Width two . . . 540
63.7.4 The boundary line $q = 1$ . . . 541
63.8 Interface with the General Theory . . . 543

V General Theory of Exact Pushdown Normalization 544

64 Semantic Fibers and Canonical Retraction 545
64.1 Rooted exact semantic lifts and defined domains . . . 545
64.2 Canonical fibers . . . 546
64.3 Canonical absorption . . . 547
64.4 Zero debt and raw conjugacy . . . 547
64.5 Three levels now separated permanently . . . 547

65 Bounded-Local Realizability, Collapse Capacity, and the Three-Stage Obstruction 549
65.1 Collapse interfaces and retained information . . . 549
65.2 Observation filtrations and hidden information . . . 550
65.3 Uniform one-top portability . . . 551
65.4 Finite observation capacity . . . 551
65.5 Bounded-d-top pushdown capacity . . . 551
65.6 The three stages . . . 551

65.7 One-top specialization . . . 552
65.8 Locality–debt–recoordination principle . . . 552

66 Representation Debt and Behavioral Safety 553
66.1 Terminal semantics . . . 553
66.2 Immediate disagreement . . . 553
66.3 Behavioral hull . . . 554
66.4 Forward invariance . . . 555
66.5 Debt birth dichotomy . . . 555
66.6 Two opposite examples . . . 555

67 Semantic Containment Cores 557
67.1 Semantic trap containment . . . 557
67.2 Absorbing-fiber harmlessness . . . 557
67.3 Maximal full-fiber containment core . . . 558
67.4 Reachable-fiber refinement . . . 558
67.5 Root correctness via containment . . . 558

68 Safe Reset and the Fatal Frontier 560
68.1 Wordwise safe-reset domain . . . 560
68.2 Structural unsafe targets . . . 560
68.3 Fatal Frontier . . . 560
68.4 Activation tightness . . . 561
68.5 Site-free diagnostics . . . 561
68.6 Fatal causality . . . 561

69 Conflict Complexes and Weighted-Balance Stress Tests 562
69.1 Minimal conflicts . . . 562
69.2 Robust fatal supports . . . 562
69.3 Reset-ray fork . . . 562
69.4 Imported weighted-balance foundations . . . 563
69.5 Exact 3:2 case . . . 563
69.6 Infinite odd-p:2 family . . . 564

70 Rooted Section Synthesis 567
70.1 Rooted path-outcome fibers . . . 567
70.2 Path-outcome coherence . . . 567
70.3 Finite synthesis algorithm . . . 568
70.4 Short obstruction certificate . . . 568
70.5 Counting sections . . . 568
70.6 Infinite semantic cores require symbolic synthesis . . . 569

71 Symbolic Re-Coordination and Forced-Debt Frontiers 570
71.1 Modular quotient/remainder synthesis . . . 570
71.2 Finite symbolic skeleton . . . 571
71.3 Partial canonical sections . . . 571
71.4 Canonical-hit capacity . . . 571
71.5 Quantitative forced debt . . . 571
71.6 Multi-type Dyck frontier . . . 572

72 Effective Verification Hierarchy 573
72.1 Level I — Structural safe-reset saturation . . . 573
72.2 Level II — Exact immediate-disagreement saturation . . . 573
72.3 Reachable-containment shortcut . . . 574
72.4 Finite semantic containment computation . . . 574
72.5 Level III — DPDA-equivalence fallback . . . 574
72.6 Verification hierarchy . . . 575
72.7 General Theory synthesis . . . 575

VI Enumerative and Arithmetic Consequences of Completion Semantics **577**

73 Primitive Operational Statistics and Schubert Cancellation Geometry **578**
73.1 The centered p:q event process . . . 578
73.2 Primitive balanced words: fixed-stack Hamming geometry . . . 579
73.3 Primitive rational-Dyck paths and the Schubert statistic . . . 580
73.3.1 Narayana windows . . . 581

74 Repeated Kernels, Carry Histories, and Algebraic Closure **582**
74.1 Repeated kernels and operational Bizley factorization . . . 582
74.2 Physical carry hierarchy and WRITE-weight conservation . . . 582
74.3 Minimal multi-carry paths: Raney signatures and Schubert edge weights 584
74.3.1 Stationary bulk kernel and finite wall . . . 585
74.4 Algebraic closure of the minimal-onset family . . . 585
74.5 Carry-onset lower bound and wall-rank lemma . . . 586
74.6 Formal domain of the small-root formulas . . . 587

75 Phase Normality and Exact One-Counter Compression **588**
75.1 Centered phase normal form and arbitrary-scale compression . . . 588
75.1.1 Exact one-counter recurrence . . . 589
75.2 The original 2:1 family at arbitrary scale . . . 590
75.3 Computational validation and structural checks . . . 591
75.4 Operational synthesis: the loss of primitive log-concavity . . . 591
75.5 Phase reconstruction and exact compressed transition table . . . 592
75.6 Operational-to-intrinsic bridge . . . 593

76 From Residual Rank to Explicit Run–Return Enumeration **594**
76.1 Rational Dyck paths and sampled runs . . . 595
76.2 Returns are coarea contacts . . . 595
76.3 Deleting forbidden coarea values . . . 596

77 Weighted Determinants and Explicit Run–Return Enumeration **598**
77.1 A weighted binomial determinant . . . 598
77.2 Explicit run–return enumeration . . . 600
77.2.1 A coefficient-free form . . . 600
77.3 Examples . . . 601
77.4 Computational verification . . . 602
77.5 Interpretation and scope . . . 602

78 Contact-Preserving Barrier Compression **604**
78.0.1 Deleting forbidden values . . . 604
78.0.2 Multivariate contacts . . . 605
78.0.3 Presence, absence, and simultaneous first-missing statistics 606
78.1 Fixed-composition rational Dyck paths . . . 607
78.1.1 Component-column coordinates and returns . . . 607
78.1.2 Forbidden columns and the effective barrier . . . 608

79 Four Composition-Refined Enumeration Problems **610**
79.1 Explicit fixed-composition run–return enumeration . . . 610
79.2 Composition-refined ratio-run and returns . . . 612
79.3 Simultaneous run, ratio-run, and exact returns . . . 613
79.4 Explicit enumeration at fixed ratio-signature . . . 614

80 Arbitrary Boundaries and Labeled Applications 617
80.1 Paths below an arbitrary fixed boundary . . . 617
80.1.1 Partial Dyck paths . . . 617
80.2 Column-increasing rational parking-function labelings . . . 618
80.3 Computational validation . . . 619
80.4 Conclusion . . . 619

81 Canonical Completion Profiles and Farey Unit Lifts 621
81.1 Shortest-completion profiles and canonical strips . . . 621
81.2 Canonical Farey unit-lift dynamics . . . 623

82 Ordered Farey Recursion and the Completion Tree 626
82.1 Local Farey recovery and ordered parent recursion . . . 626
82.1.1 Local encoding and parent recovery . . . 626
82.1.2 Ordered parent-to-mediant recursion . . . 627
82.2 The global Farey completion family . . . 628

83 Universal Completion Spectra and Arithmetic Interfaces 630
83.1 Scalar projection and universal spectra . . . 630
83.1.1 Fundamental spectra . . . 631
83.1.2 Universal shell spectra . . . 632
83.1.3 Reflection symmetry . . . 632
83.2 Classical arithmetic interfaces . . . 633
83.2.1 Mechanical and Christoffel coding . . . 633
83.2.2 Unimodular realization and the distinction from standard factorization . . . 633
83.2.3 Dedekind structure of the one-sided second moment . . . 634
83.3 Computational discovery and validation . . . 635
83.4 Conclusion . . . 636

84 Cohn Word States and Farey Completion Coordinates 637
84.1 The Cohn word-state system . . . 637
84.2 Farey completion coordinates . . . 639
84.3 Relative completion coordinates . . . 640
84.4 Sharp Apéry multiplicities in genuine Farey steps . . . 641
84.4.1 Introduction and relation to previous work . . . 642
84.4.2 Minimum Parikh fibers and the Apéry correspondence . . . 642
84.4.3 The exact two-cost tie count . . . 643
84.4.4 Sharp total multiplicity at fixed embedding dimension . . . 644
84.4.5 An authentic extremal Farey edge at every multiplicity . . . 645
84.4.6 Complete physical daughter word and a restricted parent reblocking . . . 646
84.4.7 Verification and scope of the publication claim . . . 648
84.5 Saturation defects and price-dependent Farey parent synchronization 648
84.5.1 Genuine source and CRT synchronization . . . 649
84.5.2 Compressed raw and difference modules . . . 650
84.5.3 Exact determinant quantization . . . 651
84.5.4 Two authentic infinite obstructions . . . 652
84.5.5 A genuine fixed-index family with a complete price-wall repair . . . 654

85 Quadratic Swap Curvature and Completion Returns 658
85.1 Quadratic chronology and Farey curvature . . . 658
85.2 Kernel lifts and quadratic returns . . . 660

86 Characteristic Parents, Centered Fibers, and Reversal Defects 662
86.1 Characteristic roots and Farey parents . . . 662
86.2 Centered fibers and reversal defects . . . 663
86.3 Relation to the classical Markoff setting . . . 664
86.4 Return to the monograph's completion language . . . 664
86.5 Primitive denominator spectra and quadratic swap returns . . . 664
86.5.1 Introduction . . . 664
86.5.2 Canonical reconstruction and its exact denominator . . . 665
86.5.3 The complete primitive cancellation spectrum . . . 666
86.5.4 Exact local densities and coefficient universality . . . 667
86.5.5 A fixed factor for a family of adjacent-swap defects . . . 668
86.5.6 A structural return theorem for free positive monoids . . . 670
86.5.7 Prescribing a unique return word . . . 670
86.5.8 Scope of the conclusions . . . 671
86.5.9 Finite verification and its scope . . . 672

VII Chronology, Topology, and Higher-Rank Completion Structure 673

87 Fixed-Parikh Random Ordering and Direct Obstruction 674
87.1 Shortest completion demand . . . 674
87.2 Transition classes . . . 675
87.3 The fixed-Parikh model as a critical bridge . . . 676
87.4 Direct realization and the first LIFO obstruction . . . 677
87.4.1 Primitive direct blocks . . . 677
87.4.2 Exact first-obstruction law . . . 678
87.4.3 Primitive balanced SEARCH distribution . . . 679

88 Physical-Lift Excursions and Rayleigh Rematerialization 680
88.1 The block-boundary physical-lift chain . . . 680
88.2 Rayleigh law for Bottom-WRITE rematerialization . . . 681

89 Macroscopic LIFO Cost, Duality, and Finite Kernels 684
89.1 Two regenerative bulk regimes . . . 684
89.2 Finite-to-macroscopic operational transfer . . . 685
89.2.1 Bridge perturbation . . . 686
89.2.2 The central band . . . 687
89.2.3 Poisson decomposition under the bridge . . . 687
89.3 Macroscopic operational limit laws . . . 688
89.4 Physical-lift duality . . . 689
89.5 Near-diagonal universality . . . 690
89.6 Finite physical-lift kernels and explicit computation of $\kappa_{p,q}$ . . . 692
89.6.1 Affine shortest-demand tails . . . 692
89.6.2 Small explicit kernel examples . . . 693

90 Optimal-Fiber Compression and Active Trade Geometry 696
90.1 Minimum-completion geometry . . . 696
90.1.1 The shortest-prefix down-set . . . 697
90.2 Active trades and optimal-fiber lattice geometry . . . 697
90.3 Optimal-fiber nerve compression . . . 698
90.4 Finite presentations and first homology . . . 699
90.4.1 Freiman dimension and first homology . . . 700
90.5 The optimal-fiber Taylor slice . . . 701
90.5.1 Principal truncation and character twists . . . 702

91 Lattice-Ideal Slices and Rank-Two Structural Rigidity 703
91.1 Rank-two homological dimension . . . 703
91.2 Planar lattice geometry and the fundamental group . . . 704
91.2.1 The parallelogram branch . . . 704
91.2.2 The parallelogram-free branch . . . 706
91.3 The parallelogram branch and exact homotopy type . . . 708

92 Complete Rank-Two Classification by Lattice Geometry 711
92.1 Complete active-rank-two classification . . . 711
92.2 Explicit scalar realizations . . . 712
92.2.1 A fixed-system residual bifurcation . . . 712
92.2.2 Unbounded second homology at fixed rank two . . . 713
92.2.3 A free-rank-three example . . . 714
92.3 Computational consistency checks . . . 715

93 From Displacement Kernel to Physical Completion Holonomy 716
93.1 Optimal fibers, displacement, and pure loops . . . 716
93.2 A fixed completion system with a free/toral residual bifurcation . . . 717
93.3 Stable-LIFO comparison transport . . . 718
93.4 The full free shortest-completion episode . . . 718

94 Sharp LIFO Carrier Costs and Toral Correction 720
94.1 Sharp literal-stack carrier cost . . . 720
94.2 Toral curvature and the difference between local flattening and torus descent . . . 721
94.3 The full toral episode and repair/debt tradeoff . . . 723
94.4 The free–toral topology-to-LIFO dichotomy . . . 723
94.5 Scope and representation dependence . . . 724
94.6 Full free-episode connection certificate . . . 724
94.7 Ancestry model and sharp local-memory bounds . . . 725
94.8 Global torus-descent audit . . . 725
94.9 Topology-to-LIFO synthesis . . . 726

95 Lattice-Convex Realization, Higher-Rank Topology, and Marked Holonomy 727
95.1 Intrinsic periodic-fiber model . . . 727
95.2 Every lattice-convex configuration is an optimal fiber . . . 727
95.3 Minimal fibers and free groups in every rank . . . 729
95.4 Products and higher tori . . . 730
95.5 Unequal simplex-strand product rigidity . . . 731
95.5.1 Lattice normalization, full faces, and marked displacement 731
95.6 The corrected full-face lemma . . . 732
95.7 Product rigidity and marked displacement . . . 733
95.8 An exact shortest-completion example with a topological transition . 735
95.9 Toric-fiber interpretation and theorem hypotheses . . . 737
95.10 Two one-rank extensions: apex versus prism . . . 737
95.11 Fixed-rank higher homology and displacement . . . 738
95.11.1 The displacement map contains information beyond π_1 . . 741
95.12 The displacement-marked group in arbitrary rank . . . 742
95.13 Marked product, prism, and apex transfer . . . 743
95.14 Finite physical holonomy and literal carrier costs . . . 744
95.15 Connection memory and repair . . . 746
95.16 An all-rank exact finite-holonomy family . . . 747

VIII Representation-Independent Physical Complexity and Resource Synthesis **749**

96 Protected Phase Pumping and Deferred Exposure **750**
96.1 Model and semantic quotient . . . 750
96.1.1 Pushdown model . . . 750
96.1.2 Residuals . . . 751
96.2 Unary nonrecurrence and height escape . . . 752
96.3 Protected stack pumping and synchronized phases . . . 752
96.3.1 Iteration and protected cuts . . . 753
96.4 Exposure, deferral, and the diagonal obstruction . . . 754
96.4.1 Fixed-continuation regularity . . . 754

97 Exact Unrestricted State Complexity of $L_{p,q}$ **756**
97.1 The unrestricted lower bound . . . 756
97.2 A matching three-symbol construction . . . 757
97.3 Exact state complexity . . . 758

98 Exact Unrestricted Bounded-Replacement State Complexity **759**
98.1 Model and semantic rigidity . . . 760
98.1.1 Bounded-replacement one-work-symbol pushdown recognizers . . . 760
98.1.2 Residual quotient . . . 760
98.2 A matching bounded one-work-symbol construction . . . 761
98.3 Protected unary-stack phases and finite-depth descent . . . 762
98.4 Outward first-hit bands . . . 765
98.5 Synchronization and the exact theorem . . . 766
98.6 The unrestricted state–push staircase . . . 768
98.7 Why the lower bound is genuinely unrestricted . . . 770
98.8 A separate three-symbol bounded-push optimum . . . 770
98.9 Resource-theoretic synthesis . . . 771

99 Filtered Collapse Capacity for Unrestricted Pushdown Stores **772**
99.1 Unrestricted deterministic pushdown exposure . . . 773
99.1.1 Raw model . . . 773
99.2 Optimal exposure spectrum and pointwise sharpness . . . 774
99.3 Universal k-fibers as the semantic witness family . . . 776
99.3.1 From universal freedom to a common collapse . . . 776

100 Universal Retained Information Forces Physical Exposure **778**
100.0.1 The graph case . . . 779

101 Strong Converse and Simultaneous-Query Hardness **780**
101.0.1 Strong converse . . . 780
101.0.2 A direct-sum theorem across all k-face queries . . . 780
101.1 Adaptive protected-source readout and exact resource scaling . . . 782
101.1.1 Scope and prior work . . . 782
101.1.2 Contributions and novelty . . . 782
101.1.3 Machine model and main theorem . . . 782
101.1.4 The indistinguishability lower bound . . . 783
101.1.5 A genuine-owner online encoder . . . 783
101.1.6 Assembly for all cyclic queries . . . 785
101.1.7 A quantitative separation from fixed-owner staging . . . 785
101.1.8 Finite verification and scope . . . 786
101.2 Transition to protected-contract synthesis . . . 786

102 Future-Stable Readout Monitors and Exact Protected-Port Complexity **787**
102.1 Introduction . . . 787
102.1.1 Main contributions . . . 787
102.2 The physical readout-port contract . . . 788
102.3 Extremal protected budgets . . . 788
102.4 Cube-face statistics and the missing-class correction . . . 789
102.5 Exact reduction to contained faces and VC shattering . . . 790
102.6 The protected-set objective as a polymatroid rank . . . 790

103 Contract Refinement and Protected-Information Capacity **792**
103.1 Protected representation contracts and refinement . . . 792
103.2 Strictness from protected-information capacity . . . 793

104 Exact Pareto Frontiers for Rational Balance Lifts **794**
104.1 Rational balance application I: an exact contract state premium . . . 794
104.2 Rational balance application II: exact two-resource frontiers . . . 795
104.2.1 The audited raw-safe nesting . . . 796
104.3 Endpoint resource theorems used in the rational-balance application 797
104.4 Synthesis and scope . . . 797

IX State Decomposition and the Algorithmic Boundary **798**

105 From State Complexity to State-Primality **799**
105.1 Preliminaries . . . 799
105.1.1 Deterministic one-counter automata . . . 799
105.1.2 Deterministic two-counter machines . . . 800

106 Transcript Encoding and the Elementary-Abelian Prime Core **802**
106.1 A transcript lemma for deterministic two-counter halting . . . 802
106.1.1 Branch-labelled transition alphabet . . . 802
106.1.2 The three predicates . . . 802
106.2 The elementary-abelian prime core and failure lattice . . . 803

107 The Target DOCA: Primality, Failure Factors, and Exact Decomposition **805**
107.1 Construction of the target DOCA . . . 805
107.1.1 Coloring and alphabet . . . 805
107.1.2 Core and poison transitions . . . 805
107.1.3 Counter-color transitions . . . 806
107.1.4 Regular-color transitions . . . 806
107.2 Halting implies primality . . . 807
107.3 Explicit half-size factors in the nonhalting case . . . 808
107.3.1 Core behavior and the special coset . . . 809
107.3.2 Non-special zero cosets . . . 809
107.3.3 Special counter factors . . . 809
107.3.4 Special regular factors . . . 809
107.4 Exactness of the factorization . . . 810

108 State-Primality Undecidability and Decomposition **812**
108.1 Undecidability and consequences . . . 812
108.2 Proof anatomy, edge cases, and verification points . . . 813
108.2.1 Why the target does not need two counters . . . 813
108.2.2 Why the state budget is exact . . . 813
108.2.3 Malformed and unfinished inputs . . . 814
108.2.4 Underflow and zero tests . . . 814
108.2.5 Epsilon-enabled adversarial factors . . . 814

108.2.6 One reject point versus many . . . 814
108.3 Related work and positioning . . . 814
108.4 The verification–decomposition boundary . . . 815

109 From Unrestricted Decomposition to Two Factors 816
109.1 Phase coding and alternating sentinels . . . 816
109.1.1 Checker 1 . . . 817
109.1.2 Checker 2 . . . 817
109.1.3 Terminal block . . . 817
109.2 Synchronization lemmas . . . 817

110 Two Half-Size Factors and the Sharp Factor-Budget Boundary 819
110.1 Target and two half-size factors . . . 819
110.1.1 The target T_M . . . 819
110.1.2 The factors F_1, F_2 . . . 820
110.2 The quotient invariant and forward inclusion . . . 820
110.3 Halting exposes a prime section . . . 821
110.4 Nonhalting gives an exact two-factor decomposition . . . 821
110.5 Undecidability and consequences . . . 822
110.6 Interpretation . . . 823
110.7 Construction reference tables . . . 823
110.7.1 Checker 1 . . . 824
110.7.2 Checker 2 . . . 824
110.8 Edge conditions for repeated queries . . . 824

111 Fixed One-Counter Language Equivalence: Problems 24.04(B) and 24.04(A) 826
111.1 Model, acceptance, and integer-weight normalization . . . 826
111.2 Exact residual semantics for the nondeterministic side . . . 827
111.3 Shortest counterexamples and bounded-height compactness . . . 828
111.4 Promise geometry: sources, sinks, and slope budgets . . . 829
111.4.1 Protected exposure and bounded event cores . . . 829
111.4.2 Interior full phases and sign locking . . . 831
111.4.3 Boundary and low-active branches . . . 833
111.4.4 Anchored relative envelopes and the no-core case . . . 833
111.4.5 Slope potentials and SCC budgets . . . 835
111.5 Full-envelope control inside same-level high returns . . . 838
111.5.1 Differential potentials and canonical owners . . . 838
111.5.2 Positive relative escape and lag/catch-up reduction . . . 840
111.6 Context-safe recurrence and exact-entry bounded-return compactness 842
111.6.1 Negative and zero mean . . . 843
111.6.2 Strict and exact source–sink routes . . . 844
111.7 Terminal growth and the exact LOW continuation tree . . . 845
111.7.1 Node-dependent successor enumeration . . . 845
111.7.2 Finite bad-sequence tree . . . 846
111.7.3 Prefix-safe decisive accounting . . . 846
111.8 Equality and the resolved comparison problem . . . 847
111.9 Scope of the net theorem and the zero-test frontier . . . 848
Problem 24.04(A): deterministic zero-testing OCA versus nondeterministic OCN 848
111.10 Main theorems and proof architecture . . . 848
111.11 Machines and acceptance . . . 850
111.11.1 Deterministic one-counter automata . . . 850
111.11.2 Nondeterministic one-counter nets . . . 850
111.11.3 Model scope and normalization . . . 850
111.12 Exact residual semantics of the net . . . 851
111.13 Shortest counterexamples and zero cuts . . . 852

111.14 Positive first-return compactness . . . 852
111.14.1 Replacement principles and anchored shells . . . 853
111.14.2 Finite owner/provenance refinement and macro systems . 854
111.14.3 Exact projective modes and finite-projective compression . 855
111.14.4 Owner guards, recurrent guard rank, and the rank-one base 856
111.14.5 Directional walls and exact finite wall signatures . . . 858
111.14.6 Dense same-signature atoms and carrier colors . . . 858
111.14.7 High walls and translated arches . . . 859
111.14.8 Finite scalar/projective reserve split and the vector shell defect . . . 860
111.14.9 Superlevel activation and the vector-defect/activation split 862
111.14.10 Literal activation annuli and fixed affine cells . . . 863
111.14.11 Pure-scalar recurrence and the virtual threshold shadow . 864
111.14.12 Collapse of the tight shadow to a bounded high contact . 865
111.14.13 From bounded contacts to an actual-trace relative valley . 866
111.14.14 Trace fidelity: actual cycles versus alternative paths . . . 867
111.14.15 Activation gain, threshold contacts, and noncircular return 869
111.14.16 Bounded-context domination and the rank lift . . . 870
111.14.17 Quantitative effectivity and the proof-state order . . . 871
111.15 Effective next-zero successors . . . 872
111.16 The finite zero-state tree . . . 874
111.17 The terminal positive tail . . . 875
111.18 Global small witnesses and decidability . . . 876
111.19 The reverse inclusion . . . 876
111.20 The epsilon-move extension: models and deterministic reset normalization . . . 877
111.20.1 Deterministic epsilon one-counter automata . . . 877
111.20.2 Nondeterministic epsilon one-counter nets . . . 878
111.20.3 Path requirement and effect . . . 878
111.20.4 Stable cuts and visible length . . . 878
111.21 Weak epsilon-OCN envelope calculus . . . 878
111.21.1 Cycle deletion and pumping . . . 879
111.21.2 Forward envelopes and backward requirements . . . 880
111.21.3 Initial and terminal epsilon closure . . . 881
111.22 From unit edges to threshold-affine weak macros . . . 881
111.22.1 Epsilon separator margin . . . 881
111.22.2 Threshold contacts and protected propagation . . . 882
111.22.3 High-zone pump obstruction . . . 883
111.22.4 First amplification and trace fidelity . . . 883
111.23 Guarded-macro robustness of positive first-return compactness . . . 884
111.24 Reset-ending first returns and effective zero successors . . . 887
111.24.1 Consecutive stable zero cuts . . . 887
111.24.2 Reset signatures . . . 887
111.24.3 A genuine reset-drain auxiliary instance . . . 887
111.24.4 Effective next-zero successors . . . 890
111.25 The finite stable-zero tree . . . 890
111.26 Terminal tails and the global hard-inclusion bound . . . 891
111.26.1 A syntactic terminal auxiliary automaton . . . 891
111.27 Reverse inclusion . . . 893
111.27.1 A marked noninclusion reduction . . . 893
111.27.2 Reset decompression . . . 894
111.27.3 The asynchronous marked product . . . 894
111.27.4 Compilation to one zero-test VASS . . . 894
111.28 Conclusion . . . 895
111.29 Real-time effectivity ledger . . . 895

111.30 Real-time canonical dependency graph and excluded proof routes . . 896
111.31 Real-time trace-preservation checklist 898
111.32 Effective weak-envelope construction 898
111.32.1 Effective finite and pump enumeration 898
111.32.2 Exact propagation from an omega source 898
111.32.3 Backward transformer 898
111.33 Macro-transfer and constant-rescaling ledger 899
111.33.1 Per-step observable bounds 899
111.33.2 Threshold events 900
111.33.3 Projective cutoffs 900
111.33.4 Transfer certificate 900
111.33.5 Literal-trace interface 900
111.34 Formal one-zero-test product gadget 900
111.34.1 Semantic product edges 900
111.34.2 Positive-source gadget 901
111.34.3 Unique zero-test gate 901
111.34.4 Reset decompression and asynchronous commutation . . . 901
111.34.5 Exact cleanup target 901

X Unified Realization Theory and External Scalar-Field Generativity 902

112 Exact Realizations, Path Outcomes, and Fixed-Template Debt 903
112.1 Typed architecture and theorem scope 903
112.2 Semantic action systems and exact realizations 904
112.2.1 Sections are optional structure 905
112.3 Path outcomes, canonicalization, and fixed-template debt 905

113 Protected Obligations, Products, and Same-Model Saturation 907
113.1 Protected obligations and compatibility complexes 907
113.1.1 Refinement and resource exclusion 908
113.2 Products, amplification, and asymptotic obligation rate 908
113.3 Transfer calculus and product realization 909
113.3.1 Exact product realization 909
113.4 Same-model transfer saturation 910
113.4.1 Pairwise defect versus assembly defect 911
113.4.2 Shared-interface bounds 911
113.4.3 A source-stack capacity specialization 912
113.5 Source-native shortest fibers and protected paired-session comparison 913
113.5.1 Native source and comparison semantics 913
113.6 An exact facet-and-fiber compiler 913
113.7 Whole-language recognition after lossless scalar folding 914
113.8 A source-faithful protected comparison service 915
113.9 A seven-letter exact certificate 916

114 Guarded Two-Layer Graph Actions: Stack-Alphabet Cost and Saturation Gaps 917
114.1 Introduction 917
114.1.1 Main contributions 917
114.2 Guarded partial actions and physical devices 918
114.3 An exact alphabet-size theorem 918
114.4 Protected saturation profiles 919
114.5 Complexity at the first feasible budget 921
114.6 A height-induced complexity transition 921
114.7 Related work 922
114.8 An illustrative four-cycle and concluding remarks 922

115 Marked Transport, Four-Axis Recovery, and Non-Transfer 923
115.1 Marked topology, transport, and holonomy . . . 923
115.1.1 A typed transfer theorem . . . 924
115.2 Recovery of four source theories . . . 924
115.2.1 Recovery datum . . . 924
115.2.2 Axis I: completion geometry . . . 925
115.2.3 Axis II: exact pushdown normalization . . . 925
115.2.4 Axis III: chronological arithmetic . . . 925
115.2.5 Axis IV: information and resource complexity . . . 926
115.2.6 Recovery/transfer consistency . . . 926
115.2.7 Dependency boundary . . . 927
115.3 Non-transfer principles and scope . . . 927
115.4 Relation to neighboring frameworks . . . 928
115.5 Theorem-scope and bridge ledger . . . 928
115.5.1 Intrinsic/formal results . . . 928
115.5.2 Conditional bridges . . . 929
115.5.3 Category-sensitive conclusions . . . 929
115.6 Four-axis recovery ledger . . . 929
115.6.1 Axis I: completion geometry . . . 929
115.6.2 Axis II: exact pushdown normalization . . . 930
115.6.3 Axis III: chronological arithmetic . . . 930
115.6.4 Axis IV: information and resource complexity . . . 930
115.6.5 Ledger conclusion . . . 930
115.7 Right-turn spectra and deterministic recognition limits . . . 931
115.7.1 Spectrum model and exact regularity boundary . . . 931
115.8 Right-edge dilation and a nonrecognizability criterion . . . 933
115.9 Exactly what the dilation method proves . . . 934
115.10 One visible pebble recognizes every regular spectrum . . . 934
115.11 The pure even spectrum: resolved and unresolved aspects . . . 935
115.12 Discussion, limitations and future work . . . 937
115.13 Finite-control realization details . . . 937
115.13.1 Local virtual-head transitions for dilation . . . 937
115.13.2 Anchored one-pebble probes and the monotone ascent . . . 938
115.13.3 Reproducibility scope . . . 938
115.14 Bridge to the external scalar-field realization . . . 939

116 Linear Modular Height Guards: Exact One-Stack Realizability 940
116.1 Model and main theorem . . . 940
116.2 Canonical positive realizations . . . 941
116.2.1 Axis and vertical-stripe cases . . . 941
116.2.2 Diagonal-stripe case . . . 942
116.3 From bounded repairs to a strict real-time PDA . . . 943
116.4 Two independent impossibility mechanisms . . . 944
116.4.1 Anti-diagonal transfer: three equal blocks . . . 944
116.4.2 Isolated holes: periodic detours . . . 945
116.5 Consequences and action-sensitive scope . . . 945

117 Periodic Height Guards: Signed-Winding Realizability 947
117.1 The guarded two-source model and its finite quotient . . . 947
117.2 Two unconditional negative obstructions . . . 948
117.3 The canonical positive construction . . . 949
117.3.1 Rank-zero lifted islands and short-source modes . . . 950
117.3.2 Axis-parallel rank-one components . . . 950
117.3.3 Same-sign rank-one components: aligned tagged frames . . . 950
117.3.4 Free-axis interfaces, top locality and one-stack packing . . . 951
117.4 An effective signed-winding test and examples . . . 953

117.4.1 A sharp island bound and a physical-information lower bound . . . 953

118 Extrema–Frame Semantics, Observer Records, and Exact Realizations 955

118.1 EF-FG model . . . 955
118.1.1 Filtration conventions and birth atoms . . . 955
118.1.2 Scalar fields, births, and maximin attachment . . . 955
118.1.3 Observer records . . . 956
118.1.4 Observer-core saliency . . . 956
118.1.5 Spatial-frame projection . . . 957
118.1.6 Interval uncertainty and exact semantics . . . 957
118.2 Hierarchy-free observer normal form . . . 957
118.2.1 Exact observer-record death . . . 957
118.2.2 Record chains . . . 958
118.2.3 A single elder-supply profile . . . 958
118.2.4 Self-duality . . . 959
118.3 Exact common-realization perturbation theory . . . 959
118.3.1 Why marginal envelopes are insufficient . . . 959
118.3.2 Ownership relation . . . 959
118.3.3 Realization-wise self-dual conjugacy . . . 959
118.3.4 Local witness versus global synchronization . . . 959
118.4 Proof package: observer records and exact common-realization conjugacy . . . 960
118.4.1 Maximin equality . . . 960
118.4.2 Proof of Theorem 118.2 . . . 960
118.4.3 Adjacent-record chain and supply profile . . . 960
118.4.4 Self-dual conjugacy . . . 961
118.5 Exact spatial-frame construction . . . 961

119 Threshold-Band Scheduling and Ownership Complexity 963

119.1 The τ-band owner-scheduling kernel . . . 963
119.1.1 Hard-band state . . . 963
119.1.2 τ-band coordinates . . . 964
119.1.3 Fixed-order reachability . . . 964
119.2 NP-completeness of τ-band owner scheduling . . . 965
119.2.1 Decision problem . . . 965
119.2.2 Reduction spine . . . 965
119.3 Exact lift to interval-valued scalar trees . . . 965
119.3.1 Affine rectangle conjugacy . . . 965
119.3.2 Radius-two physical realization . . . 966
119.3.3 Physical complexity theorem . . . 966
119.4 Self-dual ownership complexity . . . 967
119.4.1 Dark-anchor neutralization . . . 967
119.4.2 Possible background ownership . . . 967
119.5 The local/global algorithmic boundary . . . 967
119.5.1 Local possible exceedance . . . 967
119.5.2 Complexity separation . . . 968
119.6 Proof package: threshold-band derivation and fixed-order theorem . . . 968
119.7 Complete threshold-band NP-completeness reduction . . . 969
119.7.1 The normalized Boolean interface . . . 969
119.7.2 Exact headroom certification . . . 972
119.7.3 Clause audit . . . 972
119.7.4 Global scale separation and phase-leak repair . . . 973
119.7.5 Correctness and NP membership . . . 975
119.8 Proof package: exact physical lift and physical NP membership . . . 975

119.9 Proof package: dark-anchor lift and ownership NP-completeness . . . 978
119.10 Proof package: local possible-exceedance witness on cactus graphs . . 978

120 All-Orders Common-Realization Obstructions and EF–FG Synthesis 980
120.1 Unbounded common-realization obstruction order 980
120.1.1 Protected record complexes 980
120.1.2 Canonical boundary theorem 980
120.1.3 Raw boundary theorem 981
120.1.4 Consequence 982
120.2 Spatial-frame layer and complete EF-FG output 982
120.3 Finite evaluation and structural consistency checks 982
120.3.1 Fixed-realization evaluation 982
120.3.2 Interval-valued instances 983
120.3.3 Finite consistency checks 983
120.4 Discussion 983
120.4.1 What the theory adds 983
120.4.2 Why radius two matters 983
120.4.3 Scope and limitations 983
120.4.4 Second-stage program 984
120.5 Conclusion 984
120.6 External generativity: interval-valued scalar fields 984

121 EF–FG II: Spatial-Frame Complexity and Finite-Complex Universality 986
121.1 Preliminaries from EF–FG I 986
121.2 Global spatial-frame realization complexity 987
121.2.1 Terminal mixedness 988
121.2.2 Complete categorical spatial-frame realization 989
121.3 Finite-complex realization universality 990
121.3.1 Facet-selector construction 990
121.3.2 Canonical freezing 991

122 EF–FG II: Parameterized Ownership and Correlated Uncertainty 992
122.1 Parameterized complexity of the ownership kernel 992
122.1.1 Elder-order ambiguity 992
122.1.2 Uncertain attachment gates 993
122.2 Correlated rational-polyhedral uncertainty 994

123 EF–FG II: Layered Semantics and Closure of the Open Program 996
123.1 Finite multichannel and spatiotemporal transfer 996
123.2 Synthesis: the closed theorem package 998
123.3 Conclusion 998

124 EF–FG: Joint Physical Realization 1000
124.1 Introduction 1000
124.2 Physical model and source-faithful EF–FG queries 1002
124.2.1 What is retained from the intrinsic spatial semantics . . . 1003
124.3 Shared-control duality and exact joint infeasibility 1003
124.3.1 Exact support equality and complementary slackness . . . 1004
124.4 A sharp minimal resource obstruction on a nonsymmetric tree 1004
124.4.1 From controls to one physical state: the seven-vertex case 1006
124.5 A finite spectral test for the dimension of an optimal physical fiber . 1007
124.5.1 Why active ties, not graph automorphisms alone, govern optimal fibers 1008
124.6 A nonsemialgebraic optimal face with a sharp correlated cap segment 1008
124.6.1 Why the optimal physical face can be nonsemialgebraic . 1010
124.6.2 An exact relational obstruction inside an optimal face . . 1011

124.7 Conductance perturbation and relational rigidity 1012
124.7.1 A controlled comparison of the two physical regimes . . . 1013
124.8 Physical meaning and comparison with existing methods 1014
124.9 Conclusion 1014
124.10 Certificate construction and independent reproduction 1014

125 Actuation Resolution of EF–FG Projections 1017
125.1 Introduction 1017
125.2 Plant, semantics, and languages 1019
125.3 Sampled input operators and the one-piece obstruction 1022
125.4 The sharp two-piece budget 1023
125.5 An exact causal all-word policy 1024
125.6 Strict and parameter-robust language separation 1024
125.7 An unbounded family of physically coupled thermal modules 1026
125.8 Nonidentical modules: a uniform robust language phase 1028
125.9 Cross-type coupling and independently prescribed local words 1030
125.10 A certified first-sample cross-link resolution phase 1033
125.11 An exact local cross-link critical threshold 1035
125.12 Complete all-horizon language phase for the crossed network 1037
125.13 Reproducibility and comparisons 1039
125.14 Conclusion 1040
125.15 Integrated exact certificate proofs and construction data 1040
125.16 Exact modular lift certificate 1041
125.17 Uniform heterogeneous-network certificate 1041
125.18 One-cell exact Farkas duals and causal input ledgers 1042
125.19 Analytic initial 22 action 1044
125.20 Exact common mean and reference shapes 1044
125.21 Exact held-input tables 1046
125.22 Proof-verification protocol 1055
125.23 Arbitrary-type intermodule couplings 1055
125.24 Cross-link first-sample phase and quotient obstruction 1056
125.25 Certified unique local coupling threshold 1058
125.26 Exact all-horizon two-cross-link certificates 1059
125.27 Full finite certificate tables for the crossed network 1061
125.28 Eighty-one exact all-horizon H2 controls 1061
125.29 Eighty-one exact two-sample H1 Farkas duals 1063
125.30 First-sample H1 duals at the right critical endpoint 1068
125.31 First-sample H2 primal controls at the right critical endpoint 1070
125.32 First-sample H1 duals at $g = 7/1000$ 1073
125.33 First-sample H2 controls at $g = 7/1000$ 1075
125.34 First-sample H1 duals at $g = 1/100$ 1077
125.35 First-sample H2 controls at $g = 1/100$ 1080

XI Synthesis and Return to the Original Machine 1083

126 The 2023 DPDA Revisited 1084
126.1 The two origin figures remain distinct 1085
126.2 What the stack was storing 1087
126.3 What the completion rules were doing 1087
126.4 Why depth two appeared 1087
126.5 The cancellation transition 1088
126.6 Why the improved raw DPDA looked so different 1088
126.7 Why the general p:q machine returned to the original architecture . . 1088
126.8 Why the historical table was not literally the centered specialization 1089
126.9 C/L/R after Structure Theory 1089

126.10 C/L/R after Uniqueness Theory . . . 1090
126.11 The original machine after General Theory . . . 1090

127 Forced, Chosen, and General 1093
127.1 Language-semantic necessities . . . 1093
127.2 Necessities of a chosen locality category . . . 1093
127.3 Coordinate artifacts . . . 1094
127.4 Normalization-schedule choices . . . 1094
127.5 Special 2:1 features . . . 1095
127.6 What generalized to all coprime p:q . . . 1095
127.7 What generalized beyond rational binary counting . . . 1095
127.8 The dependency structure in one chain . . . 1096
127.9 Final perspective . . . 1098

A Complete Transition Audit of the Original 2023 DPDA 1101
A.1 Purpose and scope . . . 1101
A.2 Wrapper projection . . . 1102
A.3 Center-state recognition transitions . . . 1103
A.4 Left-state recognition transitions . . . 1104
A.5 Right-state recognition transitions . . . 1105
A.6 Complete recognition-phase audit table . . . 1107
A.7 Reachable-store closure . . . 1108
A.8 Audit of the end-of-input completion rules . . . 1109
A.9 From residual preservation to the canonical shortest completion . . . 1110
A.10 Two representative runs . . . 1111
A.11 Determinism of the completion extension . . . 1112
A.12 Finite transition consistency checks . . . 1112
A.13 Conclusion . . . 1112

B General Construction Transition Schemas 1113
B.1 Purpose and model separation . . . 1113
B.2 Four-symbol normalized paired-sign lift . . . 1113
B.3 One-work-symbol raw conjugacy . . . 1115
B.4 Boundary-compatible three-symbol raw DPDA . . . 1116
B.5 Centered real-time literal-demand construction . . . 1117
B.6 Minimal kernel-carry compiler identity . . . 1119
B.7 The 2:1 specialization and the eager–lazy distinction . . . 1119
B.8 Determinism and implementation conventions . . . 1120
B.9 Summary table . . . 1120

C Scoped Lower Bounds and Finite Certificates 1121
C.1 Why this appendix is necessary . . . 1121
C.2 Lower-bound scope table . . . 1121
C.3 Raw residual-conjugacy frontier . . . 1122
C.4 The 3:2 finite reduction . . . 1123
C.5 Normalization of the minimum configuration . . . 1123
C.6 Legal branching and pruning rules . . . 1124
C.7 Exact survivor vectors . . . 1124
C.8 Computational certificate validation . . . 1125
C.9 Architecture-relative deferred-SEARCH width . . . 1125
C.10 Representation-independent hierarchy product bound . . . 1126
C.11 Final scope rule . . . 1126

D Computational Validation **1128**
D.1 Role of computation . . . 1128
D.2 Validation-status terminology . . . 1128
D.3 Independently rerun audits . . . 1129
D.4 Independent finite-range comparisons . . . 1130
D.5 General-Theory computational support . . . 1131
D.6 Enumerative and arithmetic computational audits . . . 1131
D.7 Active chronological-source recovery certificates . . . 1132
D.8 EF–FG controlled-heat realization: exact-certificate ledger . . . 1132
D.9 Interpretation rule . . . 1132
D.10 Post-theory validation . . . 1133
D.10.1 Fixed-Parikh quantitative LIFO chronology . . . 1133
D.10.2 Optimal-fiber rank-two topology . . . 1133
D.10.3 Unrestricted state complexity . . . 1133
D.11 Geodesic-compiler validation . . . 1133
D.12 Universal k-fiber validation . . . 1134
D.13 Finite-horizon and epsilon-free resource-certificate provenance . . . 1134
D.14 Temporal relational heat control: finite rational certificates . . . 1135

E Lagrange–Bürmann Derivation of the Exact Correction Triangle **1136**
E.1 The marked first-return grammar . . . 1136
E.2 The auxiliary kernel variable . . . 1136
E.3 Coefficient formula for the auxiliary kernel . . . 1137
E.4 Recovering the correction generating function . . . 1138
E.5 Lagrange–Bürmann extraction for a fixed summand . . . 1138
E.6 Exact correction triangle . . . 1139
E.7 Equivalent elimination through $U = A - 1$ . . . 1139
E.8 Row sum . . . 1140
E.9 Why the kernel and correction arrays differ . . . 1140

F General-Theory Stress-Test Catalogues **1141**
F.1 Purpose and scope . . . 1141
F.2 The exact centered 3:2 certificate . . . 1141
F.2.1 Finite reset repertoires . . . 1142
F.2.2 Direct verification of reset closure . . . 1143
F.2.3 Exact schedule classification and fatal witnesses . . . 1143
F.3 The infinite odd-p:2 family . . . 1144
F.3.1 Arithmetic catalogue . . . 1144
F.3.2 Source-geometry catalogue . . . 1145
F.3.3 Reset-target catalogue and closure . . . 1146
F.3.4 Uniform D-induced source-fiber expansion . . . 1146
F.3.5 Uniform reachability of the length-three frontier . . . 1146
F.3.6 Uniform kernel-suffix fatality . . . 1147
F.3.7 Exact odd-p:2 classification . . . 1147
F.4 Interpretive boundary . . . 1148

Bibliography **1149**

Result Index **1166**

Notation Index **1177**

List of Figures

1.1 Formalized redrawing of the original 2:1 DPDA variant with the empty input excluded. Solid arrows are ordinary recognition transitions; dashed callout boxes contain only right-endmarker completion rules. The listed left-state completion rule with top 1 is retained for fidelity but has an unreachable source condition. 5

1.2 Formalized redrawing of the original 2:1 DPDA variant with the empty input included. The center/left/right recognition logic and end-of-input completion rules are identical to Figure 1.1; only the initialization/final-state wrapper changes so that ε is accepted. 7

1.3 Improved 2:1 raw residual-conjugate DPDA, following the two original construction variants. The initial arrow enters R_0 from above. The states R_0, R_1 form the two-state residual control; A is the separate endmarker acceptance wrapper. The right-hand box describes the exact stack encoding. The scoped optimality theorem and proof appear in Chapter 7, Figure 7.1. 8

3.1 Exact configuration ladder for $k = 2$. The upper boundary is canonical completion; the lower boundary postpones the unique required 1 as long as possible. 36

5.1 The 3:2, $r = 2$ shortest-completion rectangle. Horizontal edges read 0 and subtract 3 from the residual; vertical edges read 1 and add 2. The nine residual vertices form four commuting squares, and every monotone path from residual 2 to residual 0 is one of the six shortest completions with Parikh pair $(2, 2)$. 60

7.1 The improved three-symbol raw DPDA for $L_{2,1}$, with initial arrow and residual-code key. The two raw residual-control states are R_0, R_1; the double-circled A is a separate endmarker acceptance wrapper. Labels are input, top/replacement; only residual-orbit cells are shown. The endmarker accepts exactly at (R_0, Z_0). 86

14.1 The local action complex for $\omega = (-11, -2, 1, 10)$ at residual 7. Its maximal faces are $\{2, 3\}$, $\{1, 3\}$, and the filled triangle $\{1, 2, 4\}$. The cycle $1-2-3-1$ remains unfilled, giving $H_1 \cong \mathbb{Z}$. 174

15.1 The complete rank-one quotient for $\omega = (-5, -3, 1)$ and $r = 9$. The down-set below $(2, 0, 1)$ contributes two filled squares; the path below $(0, 3, 0)$ shares the origin, and its far endpoint is identified with $(2, 0, 1)$ by the equal-length kernel translation. The quotient has $(f_0, f_1, f_2, f_3) = (8, 10, 2, 0)$ and is homotopy equivalent to S^1. 184

19.1 Prescribed-order separation for $k = 3$. The adjacent states $r = 89$ and $r' = 90$ have equal shortest distance and identical optimal-compatibility data through order 2. On the distinguished action set $J_\star = \{a_1, a_2, a_3\}$, the restriction of the optimal-compatibility complex changes from a filled triangle to its boundary. Thus the first discrepancy occurs exactly at order 3. 210
19.2 Boolean amplification for three independent factors. The eight product states form a directed state cube. Exact distance and the complete order-$(k-1)$ signature are constant throughout the cube. Moving in coordinate t toggles exactly the distinguished order-k face $J_\star^{(t)}$, producing eight distinct order-k signatures. 213

40.1 Schematic centered debt rail. Labels 1/0 and 0/1 denote input/top mismatches. Interior mismatches pop the visible cell and move debt by one; an outward mismatch at $\pm h_{\mathrm{ctr}}$ triggers WRITE back to the center. . . . 438

54.1 Hasse diagram of the six admissible 2:1 schedules. The schedule bits are ordered by inclusion of eager sites A, B, C. The two omitted cube vertices 110 and 111 are shown separately as forbidden because A and B cannot be simultaneously eager. 499

98.1 Architecture of the new unrestricted bounded-replacement lower bound for $p \geq q$. The bounded rewrite length is used only to create the finite first-hit band; all representation freedom before that crossing is absorbed by residual rigidity and protected finite-depth descent. 766

107.1 Proof architecture. The target never stores both source counters simultaneously. The halting direction extracts a prime finite section; the nonhalting direction is witnessed by explicit quotient factors. 807

124.1 Seven-vertex nonsymmetric rational thermal tree. Shaded nodes are controllable heater ports. Both terminal queries concern the same separator b and final field. 1005
124.2 An eight-node thermal tree with exchangeable terminal arms. Replacing only the x_2t_2 conductance 1 by 1.01 yields the nonsymmetric comparison in Section 124.7. 1009

126.1 The empty-input-excluded original DPDA revisited after the general theory. This is the same formalized machine as Figure 1.1, deliberately repeated so that the final interpretation can be read against the original object. . . . 1085
126.2 The empty-input-included original DPDA revisited. This is the same recognition/completion core as Figure 1.2; the wrapper accepts ε. 1086
126.3 The original $C/L/R$ architecture annotated by the complete theory. The figure is interpretive rather than a replacement transition table: it overlays the residual/state-potential invariant, WSC SEARCH-sheet coordinates, literal-demand magnitude, kernel materialization, end-of-input completion, schedule freedom, and the semantic-lift viewpoint on the original machine architecture. 1091

Notation and Conventions

This page records the notation used across several layers of the monograph. Symbols that are intentionally local to a chapter are defined again where they are used.

Words and weighted balance

$N_a(w)$ — Number of occurrences of the symbol a in the word w.

$L_{p,q}$ — For coprime positive integers p, q,

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}.$$

$\rho_{p,q}(w)$ — Signed residual

$$\rho_{p,q}(w) = qN_1(w) - pN_0(w).$$

Thus reading 0 changes the residual by $-p$, while reading 1 changes it by $+q$.

m — The total weight $m = p + q$ when the p:q family is under discussion.

$w_{\max}, w_{\min}$ — The larger and smaller balance weights, respectively,

$$w_{\max} = \max(p, q), \qquad w_{\min} = \min(p, q).$$

These symbols are used in the unrestricted resource theorems so that those global endpoint formulas do not overload the semantic-state symbol M of the General Theory. Earlier chapter-local uses of M remain local and are defined where they occur.

K — Primitive nonnegative kernel vector

$$K = (q, p),$$

which preserves the weighted completion equation.

$\mu(r)$ — The canonical minimum nonnegative completion-demand vector associated with residual r; when written (x, y), x counts appended zeros and y appended ones.

$\eta(x, y)$ — The canonical literal stack word associated with demand (x, y) in the relevant orientation, typically a fixed two-block representative such as 1^y0^x.

Original and centered machines

C, L, R — Center, left, and right states of the original 2:1 completion-aware DPDA. In the original-machine chapters their state potentials are $\Phi(C) = 0$, $\Phi(L) = 3$, and $\Phi(R) = -3$.

$D(S)$ — Original-machine logical-stack weight

$$D(S) = 2N_0(S) - N_1(S).$$

Together with the state potential it represents the prefix residual.

h_{ctr} — Centered-rail half-width

$$h_{\mathrm{ctr}} = \left\lfloor \frac{p+q-1}{2} \right\rfloor.$$

The subscript distinguishes this scalar from the WSC exchange vector below.

D_d — State at centered-rail coordinate d; $D_0 = C$. In the centered construction the state potential is proportional to dm.

WRITE–SEARCH–CANCEL coordinates

P — Physical count shadow of a literal stack or physical representation.

Θ — Canonical semantic demand/count target represented by a physical configuration. In the abstract General Theory the same letter denotes the semantic projection from physical configurations to semantic states.

Q — Quantity defect, typically

$$Q = \Theta - P.$$

Ω — Order defect: the part of physical noncanonicity that is invisible to Parikh counts but visible to LIFO order.

$\mathbf{h}$ — WSC exchange direction

$$\mathbf{h} = (1, -1).$$

It is typographically distinct from h_{ctr}.

M_t — Kernel-materialization count at time t in a physical WSC factorization.

General Theory

M — Semantic state space of a deterministic semantic system. This use of M is local to the General Theory and should not be confused with the WSC scalar M_t.

Σ — Input alphabet.

$U_a : M \rightharpoonup M$ — Semantic update induced by input symbol a on its defined semantic domain.

C — Physical policy/configuration domain in the abstract General Theory. This is a context-local reuse of the letter C; it is not the center state of the earlier machine chapters.

$\delta_a : C \rightharpoonup C$ — Physical settled-to-settled macro-update under input symbol a, on the policy domain where it is defined. A macro-update may include uniquely forced finite internal normalization before or after consuming a.

$\Theta : C \to M$ — Exact semantic projection satisfying domain reflection and

$$\Theta\delta_a = U_a\Theta$$

where the transitions are defined.

$\sigma : M \to C$ — Chosen canonical section, with $\Theta\sigma = \mathrm{id}_M$.

N — Canonical retraction

$$N = \sigma\Theta.$$

It satisfies $N^2 = N$ on its domain.

$\mathcal{D}_H$ — Immediate-disagreement or bad set associated with a fixed physical policy or schedule H, as defined in the relevant chapter.

$\mathcal{B}_H$ — Behavioral safe hull for policy H, equivalently the greatest forward-invariant subset that avoids eventual observable disagreement under the stated hypotheses.

$\mathrm{Adm}(\mathcal{E})$ — Admissible schedule family for a schedule system or environment $\mathcal{E}$.

Unified realization and EF–FG

$\mathrm{Ob}_h(P_h)$ — Protected-obligation compatibility complex at observation level h; its faces are source families that can share one observation cell while satisfying the declared confinement and protected-future contract.

$\mathrm{Sat}_{\mathsf{Cat}}(A, B)$ — Maximum joint protected-label rank simultaneously compilable inside one realization of the declared physical category, while preserving the two marginal protected families.

$\delta_{\mathrm{pair}}, \delta_{\mathrm{asm}}$ — Pairwise-availability and simultaneous-assembly defects in same-model transfer saturation.

$H_{c_0}(m)$ — Rooted path-outcome fiber: physical endpoints actually produced from the root by semantic words ending at m. It is generally smaller than the full representation fiber $\Theta^{-1}(m)$.

$\kappa_F(p, x)$ — EF–FG maximin attachment of a birth carrier p to observer x in scalar field F.

$\Sigma^{\pm}_{\theta}(x; F)$ — Bright/dark observer-core saliency in EF–FG; the dark quantity is the bright construction applied to $-F$.

$\mathsf{SF}_x, \mathsf{PO}_x$ — Spatial-frame and polarity-core projections. They are evaluated from the same scalar realization but neither determines the other in general.

τ — Local critical birth floor in the EF–FG threshold-band scheduling kernel; the symbol is distinct from the global $p{:}q$ balance ratio.

$\xi = \tau - D$ — EF–FG threshold-band deficit coordinate, paired with headroom $h = M - \tau$; the scalar-field observer coordinate remains x.

$\mathcal{K}_Q$ — Terminal temperature fields reachable on a fixed weighted heat graph with one pointwise heater power budget Q and one specified time horizon.

$\sigma_Q(v)$ — Exact support of $\mathcal{K}_Q$ in direction v, obtained by maximizing the heat-kernel control gains pointwise in backward time.

$\mathcal{C}_d$ — Terminal-difference Krylov matrix $[d, Ld, \dots, L^{n-1}d]$ for a weighted graph Laplacian L; its rank governs active-tie spectral rigidity.

Q_* — Strict common-realization threshold of the specified seven-vertex EF–FG thermal network; this numerical value is not a universal heat-network constant.

Rational-path enumeration

$\mathcal{D}_{m,n}$ — The set of $m \times n$ rational Dyck paths. In the rational-path chapters, m may denote a rectangle width; this local use is distinct from the earlier shorthand $m = p + q$ when that shorthand is explicitly in force.

$\mathcal{U}(P) = (u_1, \dots, u_n)$ — The coarea sequence of a rational Dyck path, where u_i is the starting column of the ith north step.

b_i — The rational-Dyck upper coarea barrier

$$b_i = \left\lfloor \frac{m(i-1)}{n} \right\rfloor.$$

A return is an upper-barrier contact $u_i = b_i$.

run, $\widetilde{\text{run}}$, ret — Ordinary rational run, ratio-run, and return statistics. The sampled notation run_d unifies the two run specializations only where explicitly stated.

$\alpha \vDash n$ — Ordered vertical-run composition; RetComp and ReturnSet record its exact return components and positions.

Farey shortest-completion arithmetic

$\mu_{p,q}(r)$ — The unique shortest nonnegative completion vector

$$\mu_{p,q}(r) = \operatorname*{arg\,min}_{x,y \geq 0} \{x + y : px - qy = r\}.$$

This is the ratio-indexed form of the earlier canonical minimum-demand map $\mu(r)$.

$L_{p,q}(r)$ — The scalar completion length $|\mu_{p,q}(r)|_1$.

$((c,u),(v,d))$ — The oriented Farey parents of a proper coprime pair (p,q), normalized in the Farey-completion chapters by

$$(p,q) = (c,u) + (v,d), \qquad pu - qc = qv - pd = 1.$$

Scope conventions

"Optimal", "minimal", and "unique" are always read relative to the model specified in the theorem. In particular, the monograph distinguishes raw residual-faithful resource frontiers, deferred-SEARCH literal-demand width, and unrestricted DPDA recognition resources. A common terminal normal form is not, by itself, a claim of Church–Rosser confluence.

The symbol $\dashv$ denotes an explicit right endmarker. Completion callouts selected by $\dashv$ are not ordinary competing ε-moves.

Sampled thermal-language notation

H_1, H_2	One versus two piecewise constant held-input segments in a single sampled observation period of the leaky heat model.
Σ	The nine-letter alphabet of paired numerical terminal record bands for a six-node local module; its two-module product has 81 letters.
Q_0	Exact nominal minimum per-phase power for realizing every word with a causal H_2 controller in the specified six-node plant. Distinct from the one-shot Q_* used in the conservative thermal chapter.
g_*	Certified, locally unique cross-link conductance separating first-sample H_1 realization phases in the specified twelve-node crossed network.

Post-theory notation

$\mathcal{F}_{gq,gp}$	Balanced fixed-Parikh fiber used in the random-ordering experiment.
$\kappa_{p,q}$	Physical rematerialization constant of the block-boundary lift chain.
$\mathcal{O}_A(r)$	Optimal minimum-completion Parikh fiber for matrix system A and residual r.
$L_A(r)$	Active trade lattice generated by differences of optimal completions.
$I_A(r)$	Filled shortest-prefix quotient at residual r.
$\mathfrak{M}(P)$	Displacement-marked optimal-fiber group $$\mathfrak{M}(P) = (\pi_1(Q(P)), L, \delta_P).$$
$K_0(P), K_{\mathrm{lin}}(P)$	Full zero-displacement loop sector and its abelian shadow.
$\Pi_\rho(P)$	Realized pure holonomy image $\rho(K_0(P))$ for a chosen finite physical representation.
$g_k^{\mathrm{ord}}(A)$	Minimum zero-semantic literal carrier height whose Parikh vector has at least k distinct orderings.
$\chi(\mathcal{C})$	Deferred-connection terminal collision multiplicity controlling the transient path-state lower bound.
$Q_{\min}(L_{p,q})$	Minimum control-state count under the deterministic-pushdown convention stated in the relevant theorem.

Binary hierarchy and epsilon-free synthesis resource notation

For the separate binary witness $B_{m,k}$ with $m \geq 2, k \geq 1$,

$\mathrm{Pareto}(B_{m,k})$	Minimal attainable state/physical-stack-alphabet pairs in the deterministic ε-PDA model with a single initial stack letter and final-plus-empty acceptance.
$b_p(B_{m,k}), a_g(B_{m,k})$	Least feasible alphabet budget at p controls, and least feasible control budget at g physical stack symbols, respectively. These differ from resource functions for the rational-balance languages $L_{p,q}$.
$\mathsf{SAT}_{\mathrm{rt}}(m, k, q, g)$	The epsilon-free existence predicate at control/stack-alphabet budgets (q, g), with unrestricted but finite transition replacements.
R_H, S_H, K_H	Respectively, the finite-ones regular horizon, its positive accepted slice $B_{m,k} \cap R_H$, and the effective zero-drain bound $\max(1, (mk+1)H)$.

The Pareto model allows deterministic epsilon moves; the synthesis predicate does not. Neither is a reparametrization of the bounded-replacement/immutable-bottom-marker $L_{p,q}$ frontiers proved in later chapters.

Prologue — The Machine Before the Theory

This monograph begins with a machine, not with an abstract theory.

The starting problem is the binary counting language

$$L_{2,1} = \{w \in \{0,1\}^* : N_1(w) = 2N_0(w)\}.$$

At the level of arithmetic, the condition is elementary. One can measure the imbalance between the numbers of zeros and ones by a signed linear expression. At the level of language recognition, this already suggests counter-like memory.

The research program developed in this monograph began somewhere more concrete. In 2023, a deterministic pushdown design was constructed for this language using a center state, two side states, and a stack whose contents were manipulated by cancellation and replacement rules. The machine was not originally designed as an instance of a general theory of semantic fibers, local realizability, normalization debt, or causal schedules. Those notions did not yet exist in the project.

What made the design mathematically productive was a more elementary observation: after a prefix had been processed, the machine appeared to remember not only whether the prefix was currently balanced, but also **what was still needed to make it balanced**.

That observation changed the direction of the work.

The first question was therefore not whether $L_{2,1}$ was deterministic context-free. The more useful questions were:

- Which stack contents can actually occur?
- Why do the side states matter arithmetically?
- What exactly is the cancellation operation doing?
- When the input ends away from balance, does the machine encode a continuation that repairs the deficit?
- Is that continuation shortest?
- Why does looking only a small distance into the stack reveal information about the entire store?
- Can the same structure be built for a general ratio $p{:}q$?
- If several machines realize the same counting semantics, which aspects of their design are forced and which are representational choices?

The answers do not arrive all at once. They are the subject of the book.

The development will repeatedly return to the original machine. First it will be analyzed as a concrete DPDA. Its state and stack behavior will then be reinterpreted arithmetically. The arithmetic will become a geometry of shortest completions. That geometry will lead to more efficient pushdown representations and exact resource frontiers. A resource question will then escape the original representation class and lead to a stack-alphabet hierarchy theorem for general deterministic pushdown automata, followed by quantitative Pareto corner and width bounds for that witness and an epsilon-free resource-existence decision procedure. The completion geometry will branch into higher-dimensional topology and then into finite geodesic compilation with a strict hierarchy of higher-order optimal compatibility, while

one local cancellation transition will generate a separate Fuss–Catalan defect geometry and, later, an arithmetic p-adic offshoot.

After those branches, the book will return to the machine-construction problem. A uniform real-time p:q DPDA will explain the arithmetic hidden inside the original center/left/right architecture. WRITE, SEARCH, and CANCEL will then cease to be merely names for transition patterns and become components of a normalization calculus. Finally, the distinction between a unique semantic process and its nonunique physical pushdown realizations will lead to a general theory of locality, representation debt, causal schedules, synthesis, and verification.

The final chapters return once more to the 2023 design.

The aim is therefore not to replace the original machine by abstraction. It is to understand how much theory was already latent in it.

Part I

Foundations: From the 2023 Machine to Representation-Independent Questions

---

Part guide. **Question.** What semantic quantity is the original completion-aware DPDA really representing, and which resource constraints survive when its particular encoding is discarded? **Main result.** Signed residual and shortest-completion demand give the canonical quotient coordinates and exact raw frontiers, while a representation-independent hierarchy theorem resolves the fixed-state general-DPDA stack-alphabet case left open after Masopust's realtime result and its binary witness subsequently yields exact Pareto corner families, uniform frontier width, and an independent epsilon-free fixed-resource synthesis decision theorem. **Why here.** This Part separates the historical machine from the invariant it realizes and then tests which resource information is visible in the language itself. **Dependency.** It requires only the original construction and supplies the semantic coordinates used throughout the book.

Chapter 1

The 2023 Completion-Aware DPDA

1.1 The motivating language

For a binary word w, let $N_0(w)$ and $N_1(w)$ denote the numbers of zeros and ones in w. The motivating language is

$$L_{2,1} = \{w \in \{0,1\}^* : N_1(w) = 2N_0(w)\}.$$

Equivalently,

$$2N_0(w) - N_1(w) = 0.$$

It is useful to define already, without yet turning it into a general theory, the imbalance

$$\rho_{2,1}(w) = N_1(w) - 2N_0(w).$$

Thus $\rho_{2,1}(w) = 0$ exactly when $w \in L_{2,1}$.

The original 2023 DPDA was built directly at the machine level. Its characteristic structure consists of three semantic core states:

$$C, \qquad L, \qquad R,$$

which we call the **center**, **left**, and **right** states.

The center is the balanced operational hub. The side states are not independent counters. They record a finite correction to the meaning of the physical stack. This interpretation will be proved rather than assumed.

Before doing so, we must distinguish two formal variants of the original design.

1.2 Two formal variants of the same core design

There are two variants of the 2023 machine that we will preserve throughout the monograph.

The first excludes the empty input and recognizes

$$L_{2,1} \setminus \{\varepsilon\}.$$

The second includes the empty input and recognizes the standard language

$$L_{2,1}.$$

Their center/left/right recognition logic is the same. Their difference lies in the initialization and acceptance wrapper.

This distinction is worth preserving for two reasons.

First, it records the actual development of the design rather than silently collapsing two formal choices into one picture. Second, it separates a semantic question—how the $C/L/R$ core represents imbalance—from an acceptance-convention question—whether ε is accepted by the wrapper.

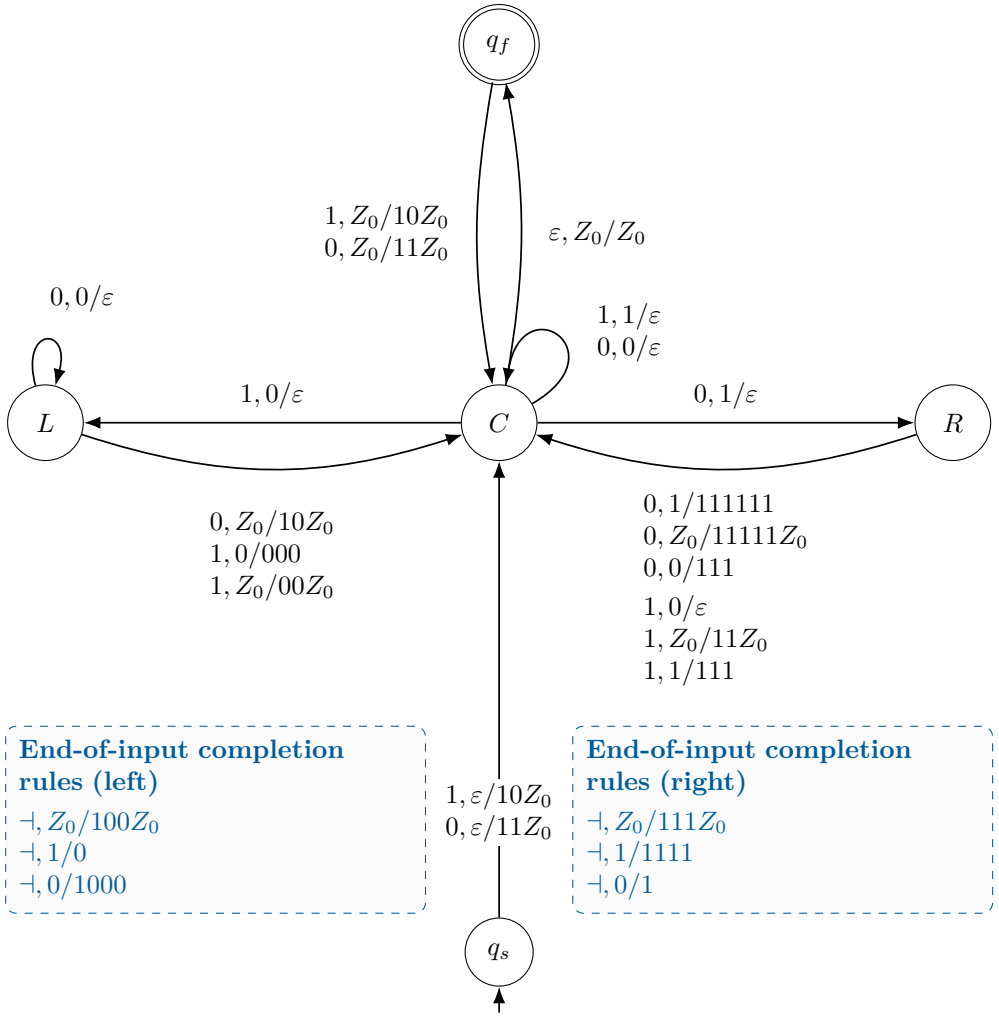


Figure 1.1: Formalized redrawing of the original 2:1 DPDA variant with the empty input excluded. Solid arrows are ordinary recognition transitions; dashed callout boxes contain only right-endmarker completion rules. The listed left-state completion rule with top 1 is retained for fidelity but has an unreachable source condition.

The logical core has states C, L, R. The wrapper contains the source state q_s and accepting state q_f.

The principal recognition transitions are the following.

From C:

$$1, 0/\varepsilon : C \to L,$$

$$0, 1/\varepsilon : C \to R,$$

and the center cancellation loops

$$1, 1/\varepsilon, \qquad 0, 0/\varepsilon.$$

The left state has the characteristic cancellation loop

$$\boxed{0, 0/\varepsilon}$$

and returns toward the center through the replacement rules

$$0, Z_0/10Z_0,$$

$$1, 0/000,$$

$$1, Z_0/00Z_0.$$

The right state returns toward the center through

$$0, 1/111111,$$

$$0, Z_0/11111Z_0,$$

$$0, 0/111,$$

$$1, 0/\varepsilon,$$

$$1, Z_0/11Z_0,$$

$$1, 1/111.$$

The wrapper moves between C and q_f at empty logical stack and initializes nonempty inputs appropriately.

This variant does not accept ε.

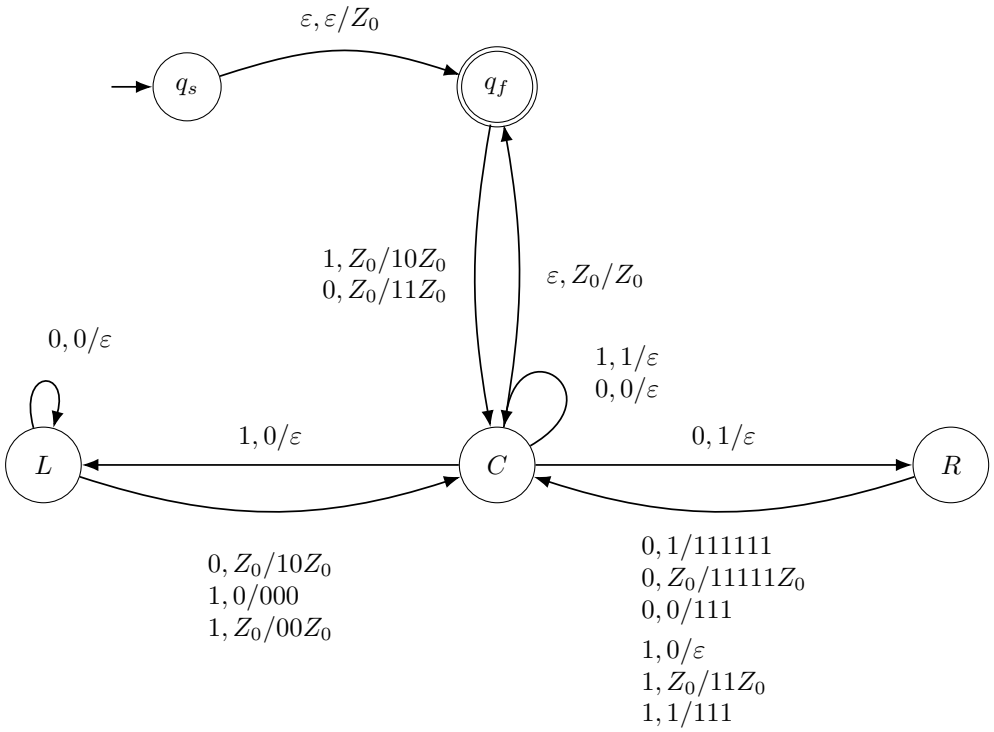


Figure 1.2: Formalized redrawing of the original 2:1 DPDA variant with the empty input included. The center/left/right recognition logic and end-of-input completion rules are identical to Figure 1.1; only the initialization/final-state wrapper changes so that ε is accepted.

The $C/L/R$ transition system is unchanged.

The difference is the initialization/final-state wrapper. In particular, an initial

$$\varepsilon, \varepsilon / Z_0$$

wrapper move allows the physical machine to represent the empty prefix at the accepting bottom-marker configuration.

Consequently this variant recognizes

$$L_{2,1},$$

including ε.

The original figures share the $C/L/R$ recognition core. An improved, representation-faithful realization of the same counting language is shown in Figure 1.3 below. The two-state optimality concerns the R_0, R_1 residual control; the endmarker-acceptance node A is a separate terminal wrapper, not a third residual-control state.

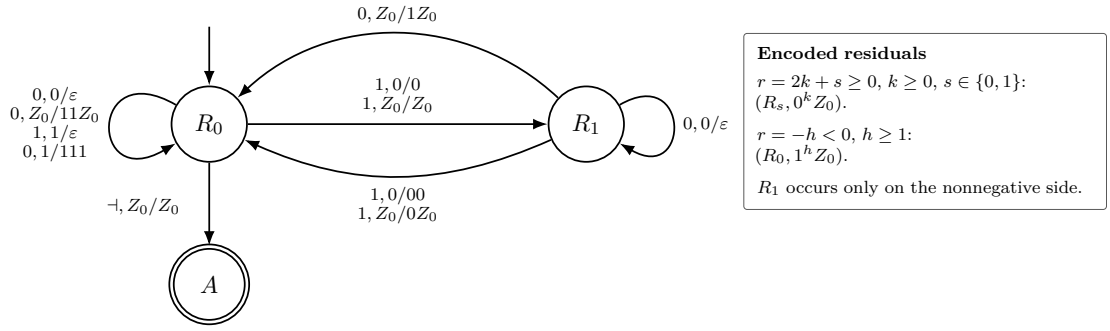


Figure 1.3: Improved 2:1 raw residual-conjugate DPDA, following the two original construction variants. The initial arrow enters R_0 from above. The states R_0, R_1 form the two-state residual control; A is the separate endmarker acceptance wrapper. The right-hand box describes the exact stack encoding. The scoped optimality theorem and proof appear in Chapter 7, Figure 7.1.

1.3 Recognition and completion are different operations

The original design also contains rules whose purpose is not to consume another binary input symbol, but to expose the continuation represented by a nonbalanced final configuration.

These are **completion rules**.

In the formal monograph model, they are enabled only when a distinguished right endmarker

$$\dashv$$

is the next unread symbol.

For the left state, the completion list is

$$\dashv, Z_0/100Z_0,$$

$$\dashv, 1/0,$$

$$\dashv, 0/1000.$$

For the right state, it is

$$\dashv, Z_0/111Z_0,$$

$$\dashv, 1/1111,$$

$$\dashv, 0/1.$$

The rule with left-state top 1 is retained as part of the original completion list, but we will prove shortly that its source condition is unreachable.

The endmarker convention is essential. If these operations were interpreted as ordinary ε-moves, some state/top pairs would also have input-consuming moves, producing a conflict with standard DPDA determinism. With the right endmarker, recognition on 0 or 1 and completion on $\dashv$ are disjoint cases.

Lemma 1.1 — Endmarker determinism.

If every post-input completion rule is interpreted as a $\dashv$-labeled transition to a completion sink with no outgoing transitions, then the completion-extended machine is deterministic in the standard DPDA sense.

Proof.

Fix a state/top pair.

During binary recognition, the next input symbol is 0 or 1. A completion rule is unavailable.

After binary input has been exhausted, the next input symbol is $\dashv$. The binary recognition rules are unavailable, and at most one completion rule applies to the current state/top pair.

Thus every instantaneous configuration has at most one successor. The completion sink introduces no further conflict. □

This formalization will be used consistently in the rest of the monograph.

1.4 Removing the wrapper from the semantic analysis

The physical states q_s and q_f serve initialization and acceptance. They should not be confused with the semantic $C/L/R$ core.

Write a physical stack as

$$SZ_0, \qquad S \in \{0,1\}^*,$$

where S is the **logical stack**.

Define the core projection

$$\pi(C, SZ_0) = (C, S),$$

$$\pi(L, SZ_0) = (L, S),$$

$$\pi(R, SZ_0) = (R, S),$$

together with

$$\pi(q_s, \varepsilon) = (C, \varepsilon),$$

$$\pi(q_f, Z_0) = (C, \varepsilon).$$

Thus the empty prefix and every balanced prefix accepted at q_f receive the same zero-residual semantic core meaning.

From now on, unless the wrapper itself matters, "the configuration after a prefix" means its core-projected configuration.

1.5 The reachable stack is much more restricted than it looks

The transition table contains replacements of several different lengths. One might therefore expect the machine to generate a complicated set of stack words.

It does not.

Let

$$\mathcal{S}_C = 0^* \cup 1^* \cup \{10\},$$

$$\mathcal{S}_L = 0^*,$$

$$\mathcal{S}_R = 1^* \cup \{0\}.$$

Theorem 1.2 — Reachable-store normal form.

For every binary prefix processed by either formal variant, if

$$(q, S)$$

is its core-projected configuration, then

$$S \in \mathcal{S}_q.$$

In particular:

- every reachable left-state stack is unary 0^k;
- every reachable right-state stack is either unary 1^k or the exceptional word 0;
- every reachable center stack is unary or the exceptional mixed word 10.

Proof.

We proceed by induction on the number of consumed input symbols.

The projected empty configuration is

$$(C, \varepsilon),$$

and

$$\varepsilon \in 0^* \cap 1^*.$$

Assume the current projected configuration belongs to the stated family.

The center loops remove matching top symbols. Hence they preserve unary forms.

A center mismatch on top 0 moves to L and removes that top. The resulting left-state stack is therefore still in 0^*.

A center mismatch on top 1 moves to R and removes that top. The resulting right-state stack lies in 1^*, with the special bottom-related construction accounting for the exceptional reachable 0 case.

The left-state loop

$$0, 0/\varepsilon$$

preserves 0^*. Its return transitions create only the allowed center forms 0^*, 1^*, or 10.

The right-state return transitions similarly produce only the allowed center families.

No recognition transition enters L with top 1. Therefore a left-state source condition with logical top 1 is unreachable.

All reachable configurations remain in the stated families. □

The theorem is stronger than a convenient regular-language description. It tells us that the apparently global question

"Does a zero or a one occur somewhere in the stack?"

may have a bounded local answer.

Before proving that, we need to understand why the side states exist.

1.6 The state carries part of the arithmetic

Define the stack weight

$$D(S) = 2N_0(S) - N_1(S).$$

Define the state potential

$$\Phi(C) = 0,$$

$$\Phi(L) = 3,$$

$$\Phi(R) = -3.$$

Theorem 1.3 — Original-machine state-potential invariant.

After every consumed prefix w, let

$$(q, S)$$

be the core-projected configuration. Then

$$\boxed{\rho_{2,1}(w) = D(S) + \Phi(q).}$$

Proof.

For the empty prefix,

$$\rho_{2,1}(\varepsilon) = 0,$$

and the projected configuration is

$$(C, \varepsilon),$$

so

$$D(\varepsilon) + \Phi(C) = 0.$$

It remains to compare the change of the two sides under each recognition transition.

Reading 0 decreases $\rho_{2,1}$ by 2. Reading 1 increases it by 1.

Each stack replacement changes $D(S)$, and every state change changes Φ. The transition table is constructed so that

$$\Delta\big(D(S) + \Phi(q)\big) = \Delta\rho_{2,1}.$$

For the transition that will later generate the cancellation geometry,

$$(L, 0S) \xrightarrow{0} (L, S),$$

the stack weight decreases by 2, while the state potential remains 3. Thus the right-hand side changes by -2, exactly matching the change induced by consuming the relevant symbol under the selected residual sign convention.

The remaining transition types satisfy the same identity by direct finite verification. The complete transition-by-transition audit is recorded in Appendix A. □

Interpretation

The logical stack alone does **not** always equal the current arithmetic deficit.

The side state stores a finite correction:

$$+3 \quad \text{in } L,$$

$$-3 \quad \text{in } R.$$

This explains a configuration such as

$$(L, \varepsilon).$$

Its empty logical stack does not mean zero residual. The missing value is carried by the state.

This is the first place where the machine reveals a distinction that will become much more important later:

$$\boxed{\text{semantic information} = \text{stack information} + \text{finite-control information.}}$$

At this point, however, we keep the observation concrete.

1.7 End-of-input completion transfers the state correction into the stack

Suppose the binary input has been exhausted.

If the projected configuration is already in C, no side-state correction is required.

If it is in L or R, the unique endmarker completion rule converts the side-state potential into a normalized stack representation.

The result has an exceptionally simple form.

Theorem 1.4 — Original-machine shortest-completion extraction.

After end-of-input normalization, the logical stack is

$$\widehat{S}(w) = \begin{cases} 0^m, & \rho_{2,1}(w) = 2m, \quad m \geq 0, \\ 10^{m+1}, & \rho_{2,1}(w) = 2m+1, \quad m \geq 0, \\ 1^m, & \rho_{2,1}(w) = -m, \quad m \geq 1. \end{cases}$$

When interpreted as a continuation word, $\widehat{S}(w)$ is a shortest suffix that completes w to $L_{2,1}$.

Proof.

By Theorems 1.2 and 1.3, every side-state configuration belongs to a one-parameter stack family and differs from its stack weight by exactly the fixed state potential.

The endmarker completion rule removes that finite state correction and returns the representation to the zero-potential center convention.

Consequently, after normalization,

$$D(\widehat{S}(w)) = \rho_{2,1}(w).$$

The reachable normalized center family is

$$0^* \cup 1^* \cup \{10^+\},$$

so the displayed residual-to-stack map is injective.

Now let a completion contain x zeros and y ones. The completed word must satisfy

$$N_1(w) + y = 2\big(N_0(w) + x\big).$$

Since

$$\rho_{2,1}(w) = N_1(w) - 2N_0(w),$$

this is equivalent to

$$\boxed{2x - y = \rho_{2,1}(w).}$$

The three displayed normalized forms give the unique minimum nonnegative Parikh pair for this equation. Hence the continuation selected by the normalized stack has minimum possible length. □

The important discovery is independent of notation:

the machine's end configuration can be decoded as a shortest repair word.

This is the observation from which the completion theory begins.

1.8 A worked run: the prefix 111

Consider the empty-input-excluded presentation and ignore Z_0 in the logical-stack notation.

Starting in the center:

$$(C, 111\ \dashv, \varepsilon) \vdash (C, 11\ \dashv, 10) \vdash (C, 1\ \dashv, 0) \vdash (L, \dashv, \varepsilon).$$

The consumed prefix has

$$\rho_{2,1}(111) = 3.$$

The physical logical stack is empty, but the machine is in L, whose potential carries the missing correction.

The endmarker completion rule

$$\dashv, Z_0/100Z_0$$

materializes the continuation

$$100.$$

Indeed,

$$111100$$

contains four ones and two zeros, so

$$111100 \in L_{2,1}.$$

No shorter continuation can repair the count imbalance.

Thus a side state that initially looks like an implementation detail has a precise role: it stores a finite amount of the completion demand outside the stack until the end-of-input normalizer materializes it.

1.9 The first structural surprise: two stack cells are enough

For a word v, let

$$\mathrm{alph}(v) = \{a \in \{0,1\} : a \text{ occurs in } v\}.$$

Let

$$\mathrm{pref}_d(v)$$

denote the prefix of v of length $\min(d, |v|)$.

Theorem 1.5 — Tight depth-two visibility.

For every completion-normalized stack S,

$$\boxed{\mathrm{alph}(S) = \mathrm{alph}(\mathrm{pref}_2(S)).}$$

Depth two is necessary.

Proof.

By Theorem 1.4, every normalized stack belongs to one of the families

$$0^*, \qquad 1^*, \qquad 10^+.$$

Every symbol that occurs anywhere in one of those words therefore occurs within the first two positions.

Depth one does not suffice. The normalized words

$$1$$

and

$$10$$

have the same top symbol but different global symbol supports.

Hence the exact visibility depth is two. □

This theorem formalizes an intuition present in the original design: a question that appears to require searching the entire stack can, on the reachable normalized family, be answered from only the first two cells.

The reason is not that arbitrary pushdown stacks are locally transparent. It is that the machine's reachable stores occupy a highly restricted language.

That distinction will matter repeatedly later.

1.10 The cancellation transition that will reappear much later

One transition deserves to be marked now even though its deeper significance will not be developed until the later cancellation-geometry branch:

$$\boxed{(L, 0S) \xrightarrow{0} (L, S).}$$

At the machine level it is simply a cancellation operation.

At the arithmetic level it participates in the state-potential invariant.

Much later, when prefixes are represented as nonnegative 2-Dyck paths, the same transition will be characterized by an exact family of landing heights. That observation will generate correction signatures, an intrinsic defect, a Catalan boundary, and a binary cancellation skeleton.

We do not need any of that theory yet.

For now we record only the transition and its exact place in the original machine.

1.11 What has been learned without leaving the original machine

We can now answer several questions that were not visible from the transition diagram alone.

First, the reachable stack language is extremely small.

Second, the side states carry an exact finite arithmetic potential.

Third, the post-input completion rules can be formalized deterministically with a right endmarker.

Fourth, the final state/stack configuration can be normalized into a shortest continuation.

Fifth, global symbol support of the normalized store is exactly visible at depth two.

These facts suggest that the original DPDA is not merely storing an arbitrary history of the input. Its configuration is representing a much smaller mathematical object.

The next chapter identifies that object.

The progression will be:

$$\text{physical state and stack} \longrightarrow \text{residual} \longrightarrow \text{minimum completion demand} \longrightarrow \text{directed completion geometry.}$$

Only after that geometry is understood will we return to the engineering question:

> Can the representation itself be improved?

That question leads to the improved and scoped-optimal DPDA constructions, the exact resource frontiers, and the general $p{:}q$ architectures developed below.

Chapter 2

What the Machine Is Really Storing

Chapter 1 stayed deliberately close to the 2023 transition diagram. We saw that the reachable stores are highly restricted, that the side states carry a finite arithmetic potential, and that an end-of-input normalization converts a nonbalanced configuration into a shortest continuation.

Those observations suggest a simpler object behind the physical machine.

The purpose of this chapter is to identify it.

The central distinction will be:

> physical configuration $\longrightarrow$ arithmetic residual
> $\longrightarrow$ minimum completion demand.

The first object depends on a particular DPDA representation. The second and third do not.

That separation is the first decisive abstraction in the monograph.

2.1 From the 2:1 imbalance to the general p:q residual

The original language satisfies

$$N_1(w) = 2N_0(w).$$

The natural general family is

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\},$$

where

$$p, q \geq 1, \qquad \gcd(p, q) = 1.$$

The coprimality assumption removes a redundant common scale. Nothing about the language changes if p and q are divided by their greatest common divisor.

Define the residual

$$\boxed{\rho_{p,q}(w) = qN_1(w) - pN_0(w).}$$

Thus

$$w \in L_{p,q} \iff \rho_{p,q}(w) = 0.$$

Reading a symbol changes the residual by a fixed amount:

$$\rho_{p,q}(w0) = \rho_{p,q}(w) - p,$$

$$\rho_{p,q}(w1) = \rho_{p,q}(w) + q.$$

For the motivating ratio $(p, q) = (2, 1)$,

$$\rho_{2,1}(w) = N_1(w) - 2N_0(w),$$

exactly the convention used in Chapter 1.

The residual is already a substantial simplification. Instead of retaining the whole Parikh pair

$$(N_0(w), N_1(w)),$$

we retain one integer.

But the stronger observation is that this one integer determines exactly what is needed on the right to reach the language.

2.2 The completion equation

Let w be a prefix with residual

$$r = \rho_{p,q}(w).$$

Suppose a continuation z contains

$$x = N_0(z)$$

zeros and

$$y = N_1(z)$$

ones.

Then

$$wz \in L_{p,q}$$

exactly when

$$q\big(N_1(w) + y\big) = p\big(N_0(w) + x\big).$$

Rearranging gives the basic completion equation.

Lemma 2.1 — Completion equation.

For every prefix w and continuation z with Parikh pair (x, y),

$$\boxed{z \in w^{-1}L_{p,q} \iff px - qy = \rho_{p,q}(w).}$$

Proof.

Expand the defining equality

$$qN_1(wz) = pN_0(wz).$$

Since

$$N_0(wz) = N_0(w) + x, \qquad N_1(wz) = N_1(w) + y,$$

we obtain

$$qN_1(w) + qy = pN_0(w) + px.$$

Therefore

$$px - qy = qN_1(w) - pN_0(w) = \rho_{p,q}(w).$$

The steps are reversible. □
This equation is the semantic heart of the original completion mechanism.
For $p = 2, q = 1$, it becomes

$$2x - y = r.$$

Thus the Chapter 1 example

$$w = 111, \qquad r = 3,$$

requires

$$2x - y = 3.$$

The minimum nonnegative solution is

$$(x, y) = (2, 1).$$

Hence every shortest completion has two zeros and one one.
There are three such words:

$$100, \qquad 010, \qquad 001.$$

The original DPDA selects 100.
This distinction will be important throughout the book:

minimum Parikh demand is unique; a shortest word need not be unique.

2.3 The primitive balanced cycle

The completion equation

$$px - qy = r$$

has an elementary but fundamental symmetry.
If (x, y) is a solution, then

$$(x + q,\ y + p)$$

is another solution, because

$$p(x + q) - q(y + p) = px - qy.$$

The vector

$$\boxed{K = (q, p)}$$

is therefore the primitive nonnegative kernel direction.
Its meaning is transparent:

- adding q zeros contributes pq;
- adding p ones contributes $-pq$;
- the two contributions cancel.

At the word level, any word containing exactly q zeros and p ones has net residual change zero.

Its length is

$$\boxed{p+q.}$$

Later this will become the fundamental directed cycle length of the residual graph.
At this stage we need only its Diophantine role.

2.4 Every completion lies on one affine ray

Because $\gcd(p,q)=1$, the integer solutions of

$$px-qy=r$$

form a one-dimensional affine lattice.

Proposition 2.2 — Affine completion ray.

For every

$$r\in\mathbb{Z},$$

there is a unique nonnegative solution

$$\mu(r)=(x_r,y_r)$$

minimizing

$$x+y$$

subject to

$$px-qy=r.$$

Every nonnegative solution is

$$\boxed{(x_r+qk,\ y_r+pk),\qquad k\in\mathbb{N}_0.}$$

Consequently the possible completion lengths are exactly

$$\boxed{\Delta(r)+(p+q)k,\qquad k\in\mathbb{N}_0,}$$

where

$$\Delta(r)=x_r+y_r.$$

Proof.

Since p and q are coprime, Bézout's identity guarantees an integer solution

$$(x_0,y_0)$$

to

$$px-qy=r.$$

Every integer solution has the form

$$(x_0+qk,\ y_0+pk),\qquad k\in\mathbb{Z}.$$

For sufficiently large k, both coordinates are nonnegative.

Let k_* be the least integer for which this happens, and define

$$(x_r, y_r) = (x_0 + qk_*,\ y_0 + pk_*).$$

Every other nonnegative solution is obtained by replacing k_* with

$$k_* + j, \qquad j \in \mathbb{N}_0.$$

Therefore every other nonnegative solution adds

$$j(q, p).$$

Its length increases by

$$j(p + q).$$

Hence (x_r, y_r) is the unique minimum-length nonnegative Parikh pair. □
This proposition gives the first exact picture of the completion space:

$$\boxed{\mu(r), \quad \mu(r) + K, \quad \mu(r) + 2K, \quad \mu(r) + 3K, \dots}$$

The minimum demand is the first lattice point of an affine ray inside the nonnegative quadrant.

The later geometry of the monograph begins here.

2.5 Minimum completion demand

We now give the central object a name.

Definition 2.3 — Minimum completion demand.

For residual

$$r \in \mathbb{Z},$$

let

$$\boxed{\mu_{p,q}(r) = (x_r, y_r)}$$

be the unique nonnegative solution of

$$px - qy = r$$

minimizing

$$x + y.$$

Define the minimum completion length

$$\boxed{\Delta_{p,q}(r) = x_r + y_r.}$$

When p, q are fixed, we write simply

$$\mu(r), \qquad \Delta(r).$$

The word **demand** is intentional.
The pair

$$(x_r, y_r)$$

does not describe the prefix already read. It describes what the prefix still requires:

- x_r additional zeros;

- y_r additional ones.

The physical DPDA need not store those two integers literally.
The semantic claim is only that a configuration representing residual r implicitly represents the unique pair $\mu(r)$.
For $L_{2,1}$,

$$2x - y = r.$$

The minimum demand has the closed form

$$\mu(2m) = (m, 0), \qquad m \geq 0,$$

$$\mu(2m+1) = (m+1, 1), \qquad m \geq 0,$$

and

$$\mu(-m) = (0, m), \qquad m \geq 1.$$

These are exactly the three families already materialized by the original 2023 DPDA:

$$0^m, \qquad 10^{m+1}, \qquad 1^m.$$

So the stack forms discovered operationally in Chapter 1 are not arbitrary normal forms.
They are physical encodings of $\mu(r)$.

2.6 A canonical shortest word

The pair

$$\mu(r) = (x_r, y_r)$$

is unique, but its letters can usually be ordered in several ways.
For semantic decoding it is convenient to choose one fixed representative.
The minimum pair has a simple modular description.

Theorem 2.4 — Canonical two-block shortest completion.
Let $r > 0$.
There is a unique

$$y_r \in \{0, \ldots, p-1\}$$

such that

$$qy_r \equiv -r \pmod{p}.$$

Set

$$x_r = \frac{r + qy_r}{p}.$$

Then

$$\boxed{C(r) = 1^{y_r} 0^{x_r}}$$

is a shortest completion word.
For $r < 0$, let

$$x_r \in \{0, \ldots, q-1\}$$

be the unique solution of

$$px_r \equiv r \pmod{q}.$$

Set

$$y_r = \frac{px_r - r}{q}.$$

Then

$$\boxed{C(r) = 0^{x_r}1^{y_r}}$$

is a shortest completion.
Finally,

$$C(0) = \varepsilon.$$

Proof.
Suppose first that $r > 0$.
The congruence

$$qy \equiv -r \pmod{p}$$

has a unique solution modulo p because q is invertible modulo p.
Selecting the representative

$$0 \le y_r < p$$

makes

$$x_r = \frac{r + qy_r}{p}$$

a nonnegative integer.
Any other integer solution differs by the kernel vector

$$(q, p).$$

Subtracting one copy of K would make the bounded coordinate y_r negative. Therefore this is the first nonnegative point on the affine completion ray and hence the unique minimum Parikh pair.
The argument for $r < 0$ is symmetric, now bounding the zero coordinate modulo q.
The displayed block word is simply a fixed ordering of the unique minimum pair. □
The canonical word is a **decoder convention**, not a claim that other shortest orderings are invalid.
For example, for

$$p = 3, \qquad q = 2, \qquad r = 4,$$

we solve

$$2y \equiv -4 \pmod{3}.$$

Thus

$$y = 1, \qquad x = \frac{4 + 2}{3} = 2.$$

Hence

$$\mu(4) = (2, 1)$$

and

$$C(4) = 100.$$

The other shortest words with the same Parikh pair are equally valid as completions. The canonical map simply chooses one of them reproducibly.

2.7 The original DPDA implements this decoder at 2:1

We can now restate the Chapter 1 completion theorem in semantic language.

Let

$$c = (q, S)$$

be any reachable core configuration of the original 2:1 DPDA.

Its residual is decoded by

$$\boxed{\Theta_{\text{orig}}(c) = D(S) + \Phi(q).}$$

Thus

$$\Theta_{\text{orig}}(c) = \rho_{2,1}(w)$$

for the prefix w that reached c.

The endmarker normalizer removes the side-state correction and produces

$$\widehat{S}(c).$$

Chapter 1 proved that

$$\widehat{S}(c) = \begin{cases} 0^m, & r = 2m, \\ 10^{m+1}, & r = 2m+1, \\ 1^m, & r = -m, \end{cases}$$

where

$$r = \Theta_{\text{orig}}(c).$$

But these are exactly the canonical two-block words

$$C(r)$$

for

$$p = 2, \qquad q = 1.$$

Therefore the original machine realizes the diagram

$$\boxed{c \overset{\Theta_{\text{orig}}}{\longmapsto} r \overset{\mu}{\longmapsto} (x_r, y_r) \overset{C}{\longmapsto} C(r).}$$

The machine is not being claimed to be an output transducer.

Rather, its configuration is a mathematical certificate from which the shortest canonical continuation can be decoded.

This distinction between **stored information** and **explicit output** will remain important later.

2.8 Physical information versus semantic information

We can now answer the question posed at the end of Chapter 1.

What is the machine really storing?

Not the entire prefix.

Not even the complete Parikh pair of the prefix.

At the semantic level, it stores only the residual

$$r.$$

From r, the minimum completion demand

$$\mu(r)$$

is uniquely determined.

From $\mu(r)$, the canonical shortest continuation

$$C(r)$$

is determined by convention.

This gives three layers.

Layer 1 — Physical representation

A concrete configuration such as

$$(L, 0^k),$$

$$(R, 1^k),$$

or

$$(C, 10)$$

belongs to the chosen 2023 machine.

Layer 2 — Semantic residual

The decoder

$$\Theta_{\text{orig}}$$

maps that configuration to one integer

$$r.$$

Layer 3 — Completion semantics

The residual determines

$$\mu(r), \qquad \Delta(r), \qquad C(r).$$

The crucial fact is that Layer 1 is not unique in principle.

Later chapters construct different physical DPDAs representing the same Layer 2 and Layer 3 objects.

That is the beginning of the distinction between semantics and representation.

The exact-lift, section, and semantic-fiber formalism is developed later in the book.

For now it is enough to observe that a complicated state/stack configuration may be merely a coordinate system for a much simpler arithmetic state.

2.9 The residual transition system

Because the residual changes by fixed amounts,

$$r \xrightarrow{0} r - p,$$

$$r \xrightarrow{1} r + q.$$

Thus every binary prefix traces a path on the integer line.
For the motivating language,

$$r \xrightarrow{0} r - 2,$$

$$r \xrightarrow{1} r + 1.$$

This graph is independent of the original DPDA's stack representation.
The 2023 machine is one physical realization of motion over this arithmetic system.
That observation immediately suggests a new question:

> If the semantic state is only one integer, why does the physical machine need the particular stack words and side states that it uses?

There are two very different possible answers.
One possibility is that the physical complexity is largely accidental and can be reduced by choosing better coordinates.
The other is that LIFO locality forces some of it.
Both possibilities will eventually occur.
Before studying resources, however, we need to understand the shortest paths in the residual graph.

2.10 Completion length as distance

The quantity

$$\Delta(r) = x_r + y_r$$

has an immediate graph interpretation.
It is the length of a shortest directed path from residual r to residual 0.
Indeed:

- appending a zero follows the edge
$$r \to r - p;$$
- appending a one follows
$$r \to r + q;$$
- a continuation reaches the language exactly when the resulting residual is zero.

Therefore

$$\boxed{\Delta(r) = \operatorname{dist}^{\to}(r, 0).}$$

We will prove and exploit this metric statement systematically in later chapters.
Here it already changes the interpretation of the minimum demand.
The pair

$$\mu(r) = (x_r, y_r)$$

does not merely count missing symbols.
It counts the two types of edges used by a shortest path to zero.
The canonical completion

$$C(r)$$

selects one distinguished shortest path among all orderings of those edges.
Thus the completion problem has become a geodesic problem.

2.11 How many shortest completions are there?

Because every shortest completion has the same Parikh pair

$$(x_r, y_r),$$

the number of shortest completion words is

$$\boxed{\binom{x_r + y_r}{x_r}.}$$

This simple count is the first indication that shortest completion has an internal geometry.

If both coordinates are positive, there are several ways to interleave the required zeros and ones.

For the prefix 111,

$$\mu(3) = (2, 1),$$

so there are

$$\binom{3}{1} = 3$$

shortest completions.
The original machine chooses the boundary ordering

$$100.$$

Later we will see that different shortest orderings are connected by elementary commuting moves, and the left branch of the original machine realizes a complete ladder of such shortest schedules at configuration level.

The multiplicity of shortest words is therefore not noise.
It is the first object that turns completion arithmetic into geometry.

2.12 Why the minimum pair is a better semantic object than the canonical word

It would be possible to define the semantics directly by the canonical word

$$C(r).$$

For the structural developments that follow, however, the pair

$$\mu(r) = (x_r, y_r)$$

is more fundamental.
There are several reasons.
First, the completion equation is linear in (x, y).
Second, the kernel direction

$$K = (q, p)$$

acts naturally on Parikh pairs.
Third, shortest-path descent becomes coordinate deletion:

- consuming a geodesic zero removes one unit of the zero demand;
- consuming a geodesic one removes one unit of the one demand.

Fourth, later physical representations will use different word orderings while preserving the same minimum pair.

Finally, in higher-dimensional counting languages the minimum demand will cease to be a single pair and become a fiber of optimal Parikh vectors. That generalization is visible only if we separate the Parikh object from its chosen word ordering now.

Thus our semantic hierarchy is

$$\boxed{r \quad \longrightarrow \quad \mu(r) \quad \longrightarrow \quad C(r),}$$

with the residual and minimum demand carrying the mathematics, and the canonical word supplying a convenient physical or decoding coordinate.

2.13 A first exact notion of redundancy

Suppose two different prefixes u and v have the same residual:

$$\rho_{p,q}(u) = \rho_{p,q}(v) = r.$$

Then both prefixes have exactly the same completion equation

$$px - qy = r.$$

Consequently they have:

- the same minimum demand;
- the same minimum completion length;
- the same set of feasible completion Parikh pairs;
- the same canonical shortest continuation.

Already, therefore, their past histories have become irrelevant to the completion problem.

This foreshadows the quotient theorem proved later:

$$u^{-1}L_{p,q} = v^{-1}L_{p,q} \iff \rho_{p,q}(u) = \rho_{p,q}(v).$$

We postpone the full quotient interpretation to Chapter 4, where it will be connected to directed geometry.

For the present chapter, the important message is simpler:

the residual is a sufficient statistic for all right-completion arithmetic.

2.14 The original machine viewed through the new semantics

We can now revisit several Chapter 1 observations without introducing any new transition rule.

The side states

The state potential means that a side state stores part of the residual outside the logical stack.

For example,

$$(L, \varepsilon)$$

represents

$$r = 3.$$

This is why the stack may be physically empty even though the completion demand is nonzero.

Endmarker materialization

The completion rule does not create new semantic information.

It changes coordinates.

Before the endmarker, the residual may be split between:

$$\text{stack weight} + \text{state potential}.$$

After normalization, the state potential is zero and the entire minimum demand is visible in the canonical stack word.

Tight depth-two visibility

The normalized words

$$0^*, \qquad 1^*, \qquad 10^+$$

are exactly the 2:1 canonical completion words.

Their global support is visible in the first two positions because the canonical decoder has only these three structural forms.

Thus the depth-two theorem is not an isolated stack-language coincidence.

It is a property of the chosen canonical ordering of the minimum demand.

This distinction will later become crucial when we compare alternative physical representations.

2.15 The first major conceptual compression

The entire original configuration can now be summarized by the following commuting picture:

$$\begin{array}{ccc} \text{reachable DPDA configuration} & \xrightarrow{\Theta_{\text{orig}}} & r \in \mathbb{Z} \\ & & \downarrow \mu \\ & & (x_r, y_r) \\ & & \downarrow C \\ & & \text{canonical shortest continuation} \end{array}$$

The physical DPDA is rich.
The semantic state is one-dimensional.
The completion demand is exact.
This is the first compression of the research program.
It also exposes the next structural question.
If

$$\Delta(r)$$

is the shortest distance to balance, what does a single input symbol do to that distance?
When does reading a zero move one step closer to acceptance?
When does reading a one do so?
Can both actions be shortest simultaneously?
And do the original DPDA's cancellation and completion moves physically realize those shortest paths?

Those questions lead directly to the **minimum-demand geodesic descent** and the **completion ladder**.

That is the subject of Chapter 3.

Chapter 3

Minimum Demand and the Completion Ladder

Research branch — Completion Becomes Geometry. *This branch develops the configuration ladder, minimum-demand geometry, and rectangular shortest-completion theory from the exact structural analysis of the original machine and its quotient/completion semantics. The use of shortest-path structure in pushdown graphs has prior literature; the claim here is the exact arithmetic specialization and configuration-level geometry of this family [5, 6].*

Chapter 2 replaced the original machine's physical configuration by a simpler semantic object:

$$r \longmapsto \mu(r) = (x_r, y_r).$$

For the motivating language,

$$L_{2,1} = \{w : N_1(w) = 2N_0(w)\},$$

the residual evolves by

$$r \xrightarrow{0} r - 2, \qquad r \xrightarrow{1} r + 1.$$

The minimum completion demand tells us how many zeros and ones are still required. We now ask what happens to that demand when another input symbol is read.

The answer reveals that the original DPDA is traversing shortest paths.

3.1 Geodesic input symbols

Fix

$$p = 2, \qquad q = 1.$$

For residual r, write

$$\mu(r) = (x_r, y_r),$$

and

$$\Delta(r) = x_r + y_r.$$

Recall:

$$\mu(2m) = (m, 0),$$

$$\mu(2m+1) = (m+1, 1),$$

and

$$\mu(-m) = (0, m).$$

Define the residual updates

$$T_0(r) = r - 2,$$

$$T_1(r) = r + 1.$$

A symbol $a \in \{0, 1\}$ will be called **geodesic at residual** r if reading it decreases the remaining shortest-completion distance by exactly one:

$$\boxed{\Delta(T_a(r)) = \Delta(r) - 1.}$$

This is the semantic form of the question:

> Does this next input symbol perform one unit of the completion work that was already required?

Lemma 3.1 — Minimum-demand geodesic descent.

Let

$$\mu(r) = (x_r, y_r).$$

If

$$x_r > 0,$$

then

$$\boxed{\mu(r-2) = (x_r - 1, y_r)}$$

and

$$\Delta(r-2) = \Delta(r) - 1.$$

If

$$y_r > 0,$$

then

$$\boxed{\mu(r+1) = (x_r, y_r - 1)}$$

and

$$\Delta(r+1) = \Delta(r) - 1.$$

Therefore the set of geodesic first symbols is

$$\boxed{\mathrm{GeoFirst}(r) = \{0 : x_r > 0\} \cup \{1 : y_r > 0\}.}$$

Proof.

Suppose

$$x_r > 0.$$

Since

$$2x_r - y_r = r,$$

we have

$$2(x_r - 1) - y_r = r - 2.$$

Thus

$$(x_r - 1, y_r)$$

is a completion demand for residual $r - 2$.
If $r - 2$ had a completion pair of length strictly less than

$$x_r + y_r - 1,$$

adding one zero to that pair would produce a completion of r shorter than

$$\mu(r),$$

contradicting minimality.
Hence

$$\mu(r - 2) = (x_r - 1, y_r).$$

The argument for $y_r > 0$ is symmetric:

$$2x_r - (y_r - 1) = r + 1.$$

Again, any shorter completion after reading 1 would yield a shorter completion before reading it.

Thus precisely those symbols whose coordinate in $\mu(r)$ is positive decrease the minimum distance by one. □

This lemma has a useful interpretation.
If the minimum demand is

$$(x, y),$$

then every geodesic input symbol simply deletes one required unit:

$$0 : (x, y) \mapsto (x - 1, y),$$

$$1 : (x, y) \mapsto (x, y - 1).$$

At the semantic level, shortest completion is therefore almost trivial.

The interesting question is whether the physical DPDA realizes the same simple coordinate deletion.

3.2 The left branch as an exact shortest-path process

Consider the reachable left configurations

$$c_k = (L, 0^k), \qquad k \geq 0.$$

From the state-potential invariant,

$$\rho(c_k) = 2k + 3.$$

Therefore

$$\boxed{\mu(c_k) = (k+2, 1)}$$

and

$$\boxed{\Delta(c_k) = k+3.}$$

So every such configuration requires:

- $k+2$ zeros;
- one one.

The left cancellation transition

$$(L, 0^k) \xrightarrow{0} (L, 0^{k-1})$$

does exactly what Lemma 3.1 predicts:

$$(k+2, 1) \mapsto (k+1, 1).$$

It decreases the shortest-completion distance by exactly one.
This gives the original cancellation loop a new meaning.
It is not merely deleting matching physical stack symbols.
It is executing one edge of a shortest path in completion space.
The boundary sequence

$$(L, \varepsilon) \xrightarrow{0} (C, 10) \xrightarrow{0} (R, 0) \xrightarrow{1} (C, \varepsilon)$$

continues the same exact descent.
Thus the entire route from a left configuration to balance can be read geodesically.

3.3 The ladder sources are actually reachable

The configurations

$$(L, 0^k)$$

used in the next theorem are not hypothetical legal stack patterns.
They are reachable.
The prefix

$$111$$

reaches

$$(L, \varepsilon).$$

Two additional ones increase the left unary tail by one unit through the machine's center-side transition pattern.
Inductively,

$$\boxed{1^{2k+3} \text{ reaches } (L, 0^k).}$$

This reachability matters.
The geometry we are about to describe is therefore a geometry of actual machine computations, not merely of syntactically possible configurations.

3.4 Every position of the required one is shortest

From

$$(L, 0^k),$$

the minimum demand is

$$(k+2, 1).$$

Hence every shortest continuation contains exactly one 1 and $k+2$ zeros.
There are

$$k+3$$

possible positions for that unique one:

$$1\,0^{k+2},$$

$$0\,1\,0^{k+1},$$

$$0^2\,1\,0^k,$$

$$\ldots,$$

$$0^{k+2}1.$$

The remarkable fact is that the original DPDA realizes **every one of these shortest schedules**, and it realizes them in a rectangular ladder of configurations.

Theorem 3.2 — Exact geodesic ladder.

For every

$$k \geq 0,$$

define the lower row

$$\Psi_k(i,0) = \begin{cases} (L, 0^{k-i}), & 0 \leq i \leq k, \\ (C, 10), & i = k+1, \\ (R, 0), & i = k+2, \end{cases}$$

and the upper row

$$\Psi_k(i,1) = (C, 0^{k+2-i}), \qquad 0 \leq i \leq k+2.$$

Then:

1. every horizontal ladder edge is a solid input-0 transition;
2. every vertical ladder edge is a solid input-1 transition;
3. every monotone path from
$$(L, 0^k)$$
to
$$(C, \varepsilon)$$
consumes the minimum demand
$$(k+2, 1);$$
4. therefore every such path is shortest;

5. the shortest continuation words are exactly

$$\boxed{0^i 1\, 0^{k+2-i}, \qquad 0 \le i \le k+2.}$$

There are exactly

$$\boxed{k+3}$$

shortest continuations.

Proof.

Along the lower row, while a zero remains on the left stack, the horizontal edge is the cancellation loop

$$(L, 0^m) \xrightarrow{0} (L, 0^{m-1}).$$

At the lower boundary,

$$(L, \varepsilon) \xrightarrow{0} (C, 10),$$

followed by

$$(C, 10) \xrightarrow{0} (R, 0).$$

Along the upper row, each horizontal edge is a center cancellation

$$(C, 0^m) \xrightarrow{0} (C, 0^{m-1}).$$

The vertical edges consume the unique required 1.
For an interior left configuration,

$$(L, 0^m) \xrightarrow{1} (C, 0^{m+2}).$$

At the lower boundary,

$$(L, \varepsilon) \xrightarrow{1} (C, 00).$$

At

$$(C, 10),$$

the input 1 removes the visible 1 and reaches

$$(C, 0).$$

At

$$(R, 0),$$

the input 1 reaches

$$(C, \varepsilon).$$

Thus the entire two-row grid is physically realized by the DPDA.
Every source-to-target monotone path contains exactly:

$$k+2$$

horizontal zero edges and one vertical one edge.
Hence every path consumes the unique minimum Parikh demand

$$(k+2, 1).$$

By Chapter 2, no shorter continuation exists.

The position of the vertical edge determines the position of the unique 1 in the continuation, giving exactly

$$0^i 1\, 0^{k+2-i}, \qquad 0 \le i \le k+2.$$

□

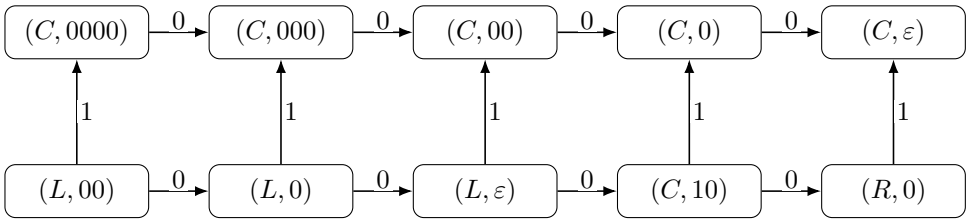


Figure 3.1: Exact configuration ladder for $k = 2$. The upper boundary is canonical completion; the lower boundary postpones the unique required 1 as long as possible.

The source is

$$(L, 00),$$

with residual

$$7$$

and minimum demand

$$(4, 1).$$

Its five shortest completions are

$$10000,$$

$$01000,$$

$$00100,$$

$$00010,$$

$$00001.$$

The upper route materializes the required 1 as early as possible.

The lower route postpones it as long as possible.

Both are shortest.

Everything between them is shortest as well.

3.5 Local commuting diamonds

A ladder square gives two ways to consume one zero and one one:

$$01$$

or

$$10.$$

The machine reaches the same configuration either way.

Lemma 3.3 — Local commuting diamonds.

At every square of the geodesic ladder,

$$\boxed{\delta^*(c, 01) = \delta^*(c, 10).}$$

Proof.

For an interior left configuration

$$(L, 0^m), \qquad m \geq 1,$$

one route is

$$(L, 0^m) \xrightarrow{0} (L, 0^{m-1}) \xrightarrow{1} (C, 0^{m+1}).$$

The other is

$$(L, 0^m) \xrightarrow{1} (C, 0^{m+2}) \xrightarrow{0} (C, 0^{m+1}).$$

Both end at the same configuration.
The two boundary squares are checked directly from the corresponding transition rules:

- at (L, ε), both 01 and 10 reach the same center configuration;
- at $(C, 10)$, both orders reach (C, ε).

These are exactly the ladder squares. □
The point is not that 0 and 1 commute as input symbols globally.
They do not.
The point is much more specific:

Within this shortest-completion interval, adjacent geodesic actions commute physically.

3.6 Parikh synchronization

Because adjacent 01 and 10 factors can be exchanged locally, any two shortest prefixes that have consumed the same numbers of zeros and ones reach the same ladder position.

Corollary 3.4 — Geodesic Parikh synchronization.

Let u and v be prefixes of shortest continuations from

$$(L, 0^k).$$

If

$$N_0(u) = N_0(v)$$

and

$$N_1(u) = N_1(v),$$

then

$$\boxed{\delta^*((L, 0^k), u) = \delta^*((L, 0^k), v).}$$

In particular, every shortest continuation reaches

$$(C, \varepsilon).$$

Proof.

Inside the ladder, repeatedly exchange adjacent factors

$$01 \leftrightarrow 10$$

using Lemma 3.3.

The consumed Parikh pair uniquely determines the grid coordinate. □

This is the first place where the physical machine's computation depends only on the consumed Parikh pair **inside a geodesic interval**, even though the machine is still a stack automaton processing an ordered word.

That phenomenon will later generalize.

3.7 Completion and cancellation are opposite boundaries of one object

The original machine suggested two seemingly different strategies.

One strategy is to expose or materialize the canonical completion as early as possible.

The other is to continue cancelling zeros on the left before finally consuming the required one.

The ladder shows that these are not different semantic processes.

They are boundary schedules of the same shortest-path rectangle.

Corollary 3.5 — Completion–cancellation boundary duality.

For the source

$$(L, 0^k),$$

the canonical completion

$$\boxed{10^{k+2}}$$

and the maximal-cancellation continuation

$$\boxed{0^{k+2}1}$$

are the two extremal boundary paths of the same geodesic ladder.

The word **duality** here is operational.

One boundary performs the unique required 1 first.

The other performs it last.

Both consume the same minimum demand.

This is a major reinterpretation of the original design:

> completion and cancellation are scheduling choices
> inside one geodesic object.

3.8 A commutation potential

For residual r, define

$$\Xi(r) = x_r y_r$$

when

$$\mu(r) = (x_r, y_r).$$

For the left-ladder source,

$$\mu(2k+3) = (k+2, 1),$$

so

$$\Xi(2k+3) = k+2.$$

This number has several simultaneous meanings.
It is:

1. the number of elementary squares in the two-row ladder;
2. the maximum number of geodesic zeros that can be consumed before the unique one becomes unavoidable;
3. the number of adjacent $01 \leftrightarrow 10$ swaps needed to move the unique one from the first position to the last.

Corollary 3.6 — Cancellation–diamond identity.
For every reachable left configuration,

$$\Xi = \text{diamond count} = \text{cancellation horizon} = \text{extremal adjacent-swap distance}.$$

This identity is small, but it is conceptually useful.
A quantity defined purely from minimum-demand arithmetic,

$$x_r y_r,$$

simultaneously measures:

- a piece of configuration-space geometry;
- a cancellation horizon in the DPDA;
- a word-reordering distance.

The different descriptions are beginning to converge.

3.9 Why this is more than a counting argument

At the arithmetic level, it is easy to observe that a demand

$$(k+2, 1)$$

has

$$k+3$$

possible shortest word orderings.
The DPDA theorem says more.
It says that the machine realizes these orderings in a coherent configuration-level structure:

$$\text{shortest words} \longleftrightarrow \text{monotone paths} \longleftrightarrow \text{reachable configurations}.$$

Adjacent word commutations become literal commuting squares of machine transitions.
This is why the completion ladder matters.
It is the first exact bridge between:

- Diophantine completion arithmetic;
- shortest-path geometry;
- pushdown-store evolution.

3.10 The canonical completion is a boundary choice

The original endmarker normalizer selects

$$10^{k+2}$$

from the $k+3$ shortest completions of a left source.

Nothing about the minimum Parikh demand itself forces the 1 to occur first.

The ordering

$$10^{k+2}$$

is a canonical representation choice.

The geodesic ladder makes this distinction visible:

$$\boxed{\text{semantic minimum demand} \neq \text{unique physical schedule.}}$$

This sentence will acquire much deeper meanings later.

Later, the original and centered machines will differ by eager versus lazy normalization chronology.

The Structure Theory will separate semantic carry chronology from physical materialization chronology.

The Uniqueness/Lift Theory will exhibit several reduced physical lifts with the same semantics.

At this early stage, however, the ladder already contains the seed of that distinction.

3.11 Limits of the ladder description

The ladder gives a complete operational picture for the motivating left branch.

It does not yet answer three broader questions.

First, when we look at an arbitrary residual, how much of its shortest-path structure can be inferred from a bounded part of the physical stack?

Second, do equal residuals really represent identical future language behavior, not merely identical completion lengths?

Third, what replaces this two-row ladder for a general ratio

$$p{:}q,$$

where a shortest demand may contain several copies of both symbols?

These questions lead in two directions:

$$\boxed{\text{quotient and visibility}}$$

and

$$\boxed{\text{general } p{:}q \text{ rectangles.}}$$

The next chapter takes the first direction.

Chapter 4

Quotients, Geodesics, and What Can Be Seen Locally

Chapter 2 showed that the residual

$$\rho_{p,q}(w) = qN_1(w) - pN_0(w)$$

determines the entire arithmetic of right completion. Chapter 3 then showed, in the original 2:1 machine, that shortest completions are realized by an actual configuration-level ladder of geodesic schedules.

We now strengthen both observations.

The residual does not merely determine the length or Parikh vector of a shortest repair. It determines the **entire future right language** of the prefix.

At the same time, the canonical shortest-completion word has a bounded visible frontier: one need not inspect its unbounded tail to know which types of geodesic action are available.

These two facts connect three levels:

prefix quotient $\longleftrightarrow$ residual vertex $\longleftrightarrow$ shortest-completion geometry.

4.1 The residual classifies all future behavior

For a language L and prefix u, the left quotient

$$u^{-1}L = \{z : uz \in L\}$$

is the set of all continuations that lead from u into L.

For $L_{p,q}$, Chapter 2 already gave the completion equation

$$z \in u^{-1}L_{p,q} \iff pN_0(z) - qN_1(z) = \rho_{p,q}(u).$$

The right-hand side depends on u only through its residual.

That observation is exact.

Lemma 4.1 — Every integer residual is reachable.

For every

$$r \in \mathbb{Z},$$

there exists a word

$$w \in \{0,1\}^*$$

such that

$$\rho_{p,q}(w) = r.$$

Proof.

Since

$$\gcd(p, q) = 1,$$

Bézout's identity gives integers x_0, y_0 satisfying

$$qy_0 - px_0 = r.$$

Every pair

$$(x_0 + qk,\ y_0 + pk)$$

has the same value of $qy - px$.

For sufficiently large k, both coordinates are nonnegative. Any binary word with those numbers of zeros and ones therefore has residual r. □

Thus the semantic residual graph will genuinely contain every integer, not merely a subset reached by special prefixes.

Theorem 4.2 — Prefix quotient classification.

For all

$$u, v \in \{0, 1\}^*,$$

$$\boxed{u^{-1}L_{p,q} = v^{-1}L_{p,q} \iff \rho_{p,q}(u) = \rho_{p,q}(v).}$$

Proof.

If the residuals agree, the completion equation is identical for u and v. Hence every continuation completes one prefix exactly when it completes the other.

Conversely, suppose

$$\rho_{p,q}(u) = r, \qquad \rho_{p,q}(v) = s, \qquad r \neq s.$$

Choose any nonnegative solution

$$(x, y)$$

of

$$px - qy = r,$$

for example the minimum completion demand $\mu(r)$. Let z be any word with that Parikh pair.

Then

$$z \in u^{-1}L_{p,q}.$$

But the same pair cannot also satisfy

$$px - qy = s.$$

Therefore

$$z \notin v^{-1}L_{p,q}.$$

The quotients differ. □

This theorem gives the residual a stronger interpretation than the one used so far.

It is not merely a useful numeric invariant.

It is an exact coordinate for right-continuation behavior:

$$\boxed{\text{same residual} \iff \text{same future language.}}$$

Left quotients themselves are classical objects. The role of the theorem in this monograph is not to claim a new general quotient theory. Its importance is that the physical DPDA configurations developed from the original machine can later be shown to encode this quotient coordinate exactly.

4.2 The quotient graph

Define the directed labeled graph

$$\mathcal{G}_{p,q}$$

with vertex set

$$\mathbb{Z}$$

and edges

$$r \xrightarrow{0} r - p,$$

$$r \xrightarrow{1} r + q.$$

By Theorem 4.2 and Lemma 4.1, the vertices of this graph may equally well be regarded as the prefix left quotients of $L_{p,q}$.

Thus the arithmetic picture

$$\cdots, -2, -1, 0, 1, 2, \cdots$$

is simultaneously a quotient graph.

A word labels a path by applying its symbols from left to right.

The acceptance quotient is the vertex

$$0.$$

4.3 Completion length is exactly directed distance

Chapter 2 interpreted the minimum completion length

$$\Delta(r)$$

as a shortest path to zero.

The quotient graph allows a slightly stronger translation-invariant form.

Theorem 4.3 — Directed completion metric.

Let

$$d_{\mathcal{G}}(r, s)$$

denote directed shortest-path distance in $\mathcal{G}_{p,q}$. Then

$$\boxed{d_{\mathcal{G}}(r, s) = \Delta(r - s).}$$

Proof.

A path from r to s containing x zeros and y ones has displacement

$$s - r = qy - px.$$

Equivalently,

$$px - qy = r - s.$$

Thus the Parikh vectors of directed r-to-s paths are exactly the nonnegative solutions of the completion equation for residual $r - s$.

Minimizing path length

$$x + y$$

is therefore exactly the minimum-completion problem. □

The geometry is homogeneous:

$$d_{\mathcal{G}}(r, s)$$

depends only on the difference

$$r - s.$$

The language-specific shortest-completion problem is therefore one distinguished family of distances in a translation-invariant directed graph.

4.4 The fundamental balanced cycle

A directed closed walk has zero net residual displacement.

If it contains x zeros and y ones, then

$$qy - px = 0.$$

Coprimality forces a rigid form.

Theorem 4.4 — Fundamental directed cycle.

Every nonempty directed closed walk satisfies

$$N_0 = qk, \qquad N_1 = pk$$

for some integer

$$k \geq 1.$$

Consequently every directed cycle length is a positive multiple of

$$p + q,$$

and the shortest possible directed cycle length is

$$\boxed{g_{p,q} = p + q.}$$

Proof.

A closed walk has

$$qN_1 = pN_0.$$

Since p and q are coprime,

$$N_0 = qk, \qquad N_1 = pk$$

for some nonnegative integer k.

A nonempty walk has $k \geq 1$.

Conversely, any ordering of q zeros and p ones has net residual displacement zero and therefore gives a closed walk of length $p + q$. $\square$

The primitive kernel vector

$$K = (q, p)$$

from Chapter 2 has now become a geometric object:

$$\boxed{K = \text{Parikh vector of a shortest directed balanced cycle.}}$$

This is the global analogue of cancellation.

4.5 Balanced-cycle reduction

Suppose a word z has Parikh pair

$$(x, y).$$

If

$$x \geq q \qquad \text{and} \qquad y \geq p,$$

then one full balanced kernel packet

$$(q, p)$$

can be removed without changing the displacement of the word.

Repeat this as often as possible.

Theorem 4.5 — Balanced-cycle reduction.

Let a word have Parikh pair

$$(x, y).$$

Define

$$k = \min\left(\left\lfloor \frac{x}{q} \right\rfloor, \left\lfloor \frac{y}{p} \right\rfloor\right).$$

Then

$$(x', y') = (x - kq,\ y - kp)$$

has the same residual displacement as (x, y) and is the unique minimum Parikh pair for that displacement.

Proof.

Subtracting

$$(q, p)$$

preserves

$$px - qy.$$

By maximality of k,

$$x' < q \qquad \text{or} \qquad y' < p.$$

Every integer solution with the same displacement differs by an integer multiple of

$$(q, p).$$

No further positive multiple can be removed while remaining nonnegative.

Therefore (x', y') is the first nonnegative point on the affine solution ray and hence the unique minimum pair. □

This theorem gives a global quotient-graph interpretation of the minimum demand:

> minimum demand = Parikh vector after deleting
> every removable balanced cycle.

The original machine's cancellation behavior was local and representation-specific.

Balanced-cycle reduction is semantic and representation-independent.

4.6 Which words are geodesic?

A word is geodesic between its endpoints precisely when no balanced kernel packet can be deleted.

Corollary 4.6 — Exact geodesic domain.

A word z is shortest among all words with the same residual displacement if and only if

$$\boxed{N_0(z) < q \quad \text{or} \quad N_1(z) < p.}$$

Equivalently,

$$\mathrm{Geo}_{p,q} = \{z : N_0(z) < q\} \cup \{z : N_1(z) < p\}.$$

The geodesic-word language is therefore regular.

The criterion is strikingly simple.

A word ceases to be shortest exactly when it contains enough of **both** symbol types to hide one complete balanced cycle in its Parikh vector.

Consequently every binary word of length less than

$$p + q$$

is geodesic.

The first possible failure occurs exactly at the fundamental cycle length.

4.7 Exact quotient spheres

Let

$$S_n = \{r \in \mathbb{Z} : d_{\mathcal{G}}(r, 0) = n\}.$$

A length-n Parikh pair has the form

$$(x, n - x).$$

Its residual displacement is

$$(p + q)x - qn,$$

which is strictly increasing in x. Hence distinct values of x give distinct residual vertices.

Combining this with the geodesic criterion gives an exact growth law.

Proposition 4.7 — Exact quotient spheres.

For every

$$n \geq 0,$$

$$\boxed{|S_n| = \min(n+1, p+q).}$$

Therefore

$$\boxed{\sum_{n \geq 0} |S_n| z^n = \frac{1 - z^{p+q}}{(1-z)^2}.}$$

Proof.
If

$$n < p + q,$$

every length-n Parikh pair is geodesic, so all

$$n + 1$$

possible zero counts give distinct vertices at distance n.
If

$$n \geq p + q,$$

a length-n pair is geodesic precisely when

$$x < q$$

or

$$n - x < p.$$

These two regions are disjoint and contain respectively q and p pairs.
Hence exactly

$$p + q$$

residuals occur at distance n. □

The graph therefore has linear growth only until the first balanced cycle appears. After that, every directed sphere has constant size

$$p + q.$$

This small exact formula will later help distinguish semantic growth from the sometimes much larger physical state/stack resources used by a particular lift.

4.8 What the original two-cell observation really reveals

Chapter 1 proved that the original 2:1 normalized completion word exposes its global symbol support within two stack cells.

Chapter 3 gave a more operational interpretation.

For a reachable configuration c, define

$$\mathrm{GeoFirst}(c)$$

to be the set of symbols that begin some shortest continuation from the residual represented by c.

If

$$\mu(r) = (x_r, y_r),$$

then

$$\boxed{\mathrm{GeoFirst}(r) = \{0 : x_r > 0\} \cup \{1 : y_r > 0\}.}$$

For the original machine, the reachable qualitative roles are:

$$(C, \varepsilon) \mapsto \varnothing,$$

$$(C, 0^m) \mapsto \{0\},$$

$$(C, 1^m) \mapsto \{1\},$$

$$(C, 10) \mapsto \{0, 1\},$$

every reachable left configuration maps to

$$\{0, 1\},$$

and every reachable right configuration maps to

$$\{1\}.$$

Thus the old "bounded stack search" intuition can be stated exactly.

Theorem 4.8 — Exact depth-two geodesic locality of the original machine.
For the motivating 2:1 DPDA:

1. the complete geodesic first-action set is determined by the control state and the first two logical stack symbols;
2. depth one is insufficient;
3. after end-of-input normalization,

$$\boxed{\mathrm{alph}(\widehat{S}(c)) = \mathrm{GeoFirst}(c);}$$

4. no fixed finite observation depth determines the full quantitative demand vector on all reachable configurations.

Proof.
The reachable-store normal form from Chapter 1 leaves only the center forms

$$\varepsilon, \quad 0^m, \quad 1^m, \quad 10,$$

together with the fixed qualitative left and right roles.

At the center, two visible symbols distinguish exactly the four possible geodesic-action sets:

$$\varnothing, \quad \{0\}, \quad \{1\}, \quad \{0, 1\}.$$

Depth one fails because

$$(C, 1)$$

and

$$(C, 10)$$

have the same state and top symbol but different geodesic first-action sets.

End-of-input normalization produces the canonical shortest continuation represented by the residual, and its symbol support is exactly the set of positive coordinates of the minimum demand.

Finally, arbitrarily large reachable unary center stacks

$$(C, 0^m)$$

share every fixed top-d observation once $m \geq d$, yet their demands

$$(m, 0)$$

have different magnitudes.

Thus qualitative geometry is exactly two-local, while quantitative demand is not fixed-depth local. □

This is the precise form of an important distinction:

$$\boxed{\text{bounded local observation can reveal the shape of the shortest-path choice}}$$

without revealing

$$\boxed{\text{the unbounded distance still remaining.}}$$

4.9 General canonical action visibility

For general $p{:}q$, recall the canonical shortest word

$$C(r) = \begin{cases} 1^{y_r}0^{x_r}, & r > 0, \\ 0^{x_r}1^{y_r}, & r < 0, \\ \varepsilon, & r = 0. \end{cases}$$

The minimum pair satisfies the bounded-coordinate property:

$$r > 0 \Longrightarrow 0 \leq y_r < p,$$

$$r < 0 \Longrightarrow 0 \leq x_r < q.$$

Therefore the first change of symbol, if it occurs, cannot be arbitrarily deep.

Definition 4.9 — Canonical visibility depth.

Let $d_{p,q}$ be the least integer d such that for every residual r,

$$\mathrm{alph}(C(r)) = \mathrm{alph}(\mathrm{pref}_d(C(r))).$$

Theorem 4.10 — Exact bounded visibility.

$$\boxed{d_{p,q} = \max(p, q).}$$

Proof.

Suppose first that

$$r > 0.$$

Then

$$C(r) = 1^{y_r}0^{x_r}$$

with

$$y_r < p.$$

If a zero occurs, the first zero appears no later than position

$$p.$$

Thus depth p suffices on the positive side.
For

$$r < 0,$$

the symmetric argument shows that depth q suffices.
Hence

$$d_{p,q} \leq \max(p, q).$$

For tightness, suppose

$$p > q.$$

Take residual

$$r = q.$$

The minimum solution is

$$(x_r, y_r) = (q, p - 1),$$

so

$$C(q) = 1^{p-1}0^q.$$

Depth $p - 1$ sees only ones even though zeros occur later. Therefore

$$d_{p,q} \geq p.$$

If

$$q > p,$$

the symmetric residual

$$r = -p$$

gives

$$C(-p) = 0^{q-1}1^p,$$

forcing

$$d_{p,q} \geq q.$$

For

$$p = q = 1,$$

depth one is plainly necessary and sufficient.
Therefore

$$d_{p,q} = \max(p, q).$$

□

Since

$$\mathrm{alph}(C(r)) = \mathrm{GeoFirst}(r),$$

the same theorem can be read operationally:

> The complete set of locally shortest first actions is visible within $\max(p, q)$ canonical symbols.

For the original machine,

$$(p, q) = (2, 1),$$

so

$$d_{2,1} = 2,$$

exactly reproducing the original tight depth-two theorem.

4.10 A second bounded quantity: the completion frontier

Visibility asks one question:

> How far into a canonical shortest completion must we inspect before every symbol type that occurs anywhere in it has appeared?

There is another bounded quantity that will become important when we redesign the physical DPDA.

Write positive and negative residuals in Euclidean form:

$$r = pm + s, \qquad 0 \le s < p,$$

and

$$r = -(qm + t), \qquad 0 \le t < q.$$

Define the finite residue-dependent bases

$$A_s = C(s),$$

$$B_t = C(-t),$$

with

$$A_0 = B_0 = \varepsilon.$$

Then

$$\boxed{C(pm + s) = A_s 0^m,}$$

and

$$\boxed{C(-(qm + t)) = B_t 1^m.}$$

Thus every canonical completion decomposes into:

bounded residue-dependent frontier + unbounded unary tail.

Definition 4.11 — Completion frontier.

Define

$$K_{p,q} = \max\left(\max_{0\le s<p} |A_s|, \max_{0\le t<q} |B_t|\right).$$

Theorem 4.12 — Exact almost-cycle frontier.

For every coprime pair (p, q),

$$\boxed{K_{p,q} = \begin{cases} 0, & (p,q) = (1,1), \\ p+q-1, & \text{otherwise.} \end{cases}}$$

Proof.

For

$$(p, q) = (1, 1),$$

the only residue bases are

$$A_0 = B_0 = \varepsilon,$$

so

$$K_{1,1} = 0.$$

Now assume the coprime pair is nontrivial. Then

$$p \neq q.$$

The bounded-coordinate formulas for $C(r)$ imply

$$K_{p,q} \le p + q - 1.$$

If

$$p > q,$$

the base residual

$$q$$

has

$$A_q = C(q) = 1^{p-1}0^q,$$

whose length is

$$p + q - 1.$$

If

$$q > p,$$

the symmetric base is

$$B_p = C(-p) = 0^{q-1}1^p,$$

again of length

$$p + q - 1.$$

Hence the upper bound is attained. □
Since the shortest balanced cycle length is

$$g_{p,q} = p + q,$$

we obtain, for every nontrivial coprime pair,

$$\boxed{K_{p,q} = g_{p,q} - 1.}$$

The largest finite frontier is literally one symbol short of a complete balanced cycle.

4.11 One extremal word, two different bounds

The same almost-cycle word makes both visibility and frontier tight.
If

$$p > q,$$

the witness is

$$C(q) = 1^{p-1}0^q.$$

If

$$q > p,$$

the witness is

$$C(-p) = 0^{q-1}1^p.$$

But the two parameters measure different things.

Visibility depth

$$\boxed{d_{p,q} = \max(p, q)}$$

asks:

> How many leading symbols must be inspected before the **qualitative support** of the canonical shortest completion is known?

Completion frontier

$$\boxed{K_{p,q} = p + q - 1}$$

for nontrivial pairs asks:

> How large can the complete **residue-dependent non-unary base** be before only an unbounded unary magnitude remains?

These should not be conflated.
In particular, their numerical values are generally different even though one extremal word witnesses both lower bounds.
This distinction is crucial for later resource arguments.
A finite observation depth is not the same thing as a finite amount of residue information that can be moved into control.

4.12 The next representation idea is now forced

We have reached a useful decomposition:

$$C(r) = \text{finite residue-dependent base} + \text{unbounded unary tail.}$$

The finite base has exact worst-case size

$$K_{p,q}.$$

The unary tail stores the unbounded magnitude.

This immediately suggests a new physical representation:

$$\boxed{\text{store the bounded frontier in finite control}}$$

and

$$\boxed{\text{leave only the unbounded unary magnitude on the pushdown.}}$$

Such a factorization can be lossless because the pair

$$\text{control residue role} + \text{unary tail length}$$

reconstructs the exact residual.

Once the residual is known, Theorem 4.2 gives the exact prefix quotient, Theorem 4.3 gives the exact distance to acceptance, and Chapter 2 gives a canonical shortest continuation.

This will eventually yield a compact physical realization.

But before we redesign the machine, one more piece of geometry is needed.

The 2:1 ladder of Chapter 3 was only two rows high because its minimum demand contained a single 1.

For a general residual,

$$\mu(r) = (x_r, y_r)$$

may have several copies of both symbols.

What is the full shortest-path interval then?

The answer is a rectangle.

That is the subject of Chapter 5.

Chapter 5

General p:q Completion Rectangles

The 2:1 ladder of Chapter 3 had two rows because every left-ladder source had minimum demand

$$(k+2, 1).$$

Only one 1 was required, so the position of that single action was the only scheduling choice.

For a general ratio p:q, the minimum demand

$$\mu(r) = (x_r, y_r)$$

may contain several copies of both symbols.

The ladder then becomes a rectangle.

This chapter identifies that rectangle exactly and shows that three objects coincide:

> minimum-demand coordinates $\longleftrightarrow$ directed shortest-path interval
> $\longleftrightarrow$ commuting schedule geometry.

5.1 Remaining demand as a grid coordinate

Fix coprime positive integers

$$p, q$$

and residual

$$r.$$

Let

$$\mu(r) = (x_r, y_r).$$

Suppose a shortest completion has so far consumed

$$i$$

zeros and

$$j$$

ones.

Its current residual is

$$r - pi + qj.$$

If the prefix is still geodesic, the remaining demand should be

$$(x_r - i,\ y_r - j).$$

Thus the natural coordinate set is

$$[0, x_r] \times [0, y_r],$$

where the point

$$(i, j)$$

records how much of the unique minimum Parikh demand has already been consumed. A horizontal step consumes a zero:

$$(i, j) \to (i + 1, j).$$

A vertical step consumes a one:

$$(i, j) \to (i, j + 1).$$

The source is

$$(0, 0),$$

and the balanced target is

$$(x_r, y_r).$$

The only possible problem is that two different grid coordinates might correspond to the same residual before the target is reached.

The minimum-demand property prevents exactly that.

5.2 The shortest interval is an actual rectangle

Theorem 5.1 — Rectangular shortest-completion geometry.

Let

$$\mu(r) = (x_r, y_r).$$

The map

$$\boxed{\varphi_r(i, j) = r - pi + qj}$$

is injective on

$$[0, x_r] \times [0, y_r].$$

Consequently the directed shortest-path interval from r to 0 in the residual graph is exactly the rectangular grid

$$\boxed{[0, x_r] \times [0, y_r].}$$

Every monotone source-to-target path is a shortest completion, and every shortest completion occurs in this way.

Proof.

Suppose two rectangle coordinates satisfy

$$r - pi + qj = r - pi' + qj'.$$

Then

$$p(i - i') = q(j - j').$$

Since

$$\gcd(p, q) = 1,$$

there is an integer t such that

$$i - i' = qt, \qquad j - j' = pt.$$

If

$$t \neq 0,$$

then the rectangle contains two points separated by at least one full kernel vector

$$(q, p).$$

In particular, the side lengths must satisfy

$$x_r \geq q, \qquad y_r \geq p.$$

But then

$$(x_r - q,\ y_r - p)$$

would be a shorter nonnegative solution of the same completion equation

$$px - qy = r,$$

contradicting the minimality of

$$\mu(r).$$

Hence

$$t = 0,$$

so

$$i = i', \qquad j = j'.$$

Thus φ_r is injective.
Every monotone path from

$$(0, 0)$$

to

$$(x_r, y_r)$$

consumes exactly x_r zeros and y_r ones, hence exactly the minimum demand.

Conversely, every shortest completion has that same unique minimum Parikh vector, so its sequence of consumed zero/one counts traces one monotone path through the rectangle. □

This theorem generalizes the Chapter 3 ladder without changing its meaning.

When

$$y_r = 1,$$

the rectangle has exactly two rows.
When

$$x_r = 0$$

or

$$y_r = 0,$$

it degenerates to a one-dimensional corridor.
When both coordinates are positive, the residual has a genuine scheduling fork.

5.3 Exact number of shortest completions

A monotone path in an

$$x_r \times y_r$$

rectangle consists of:

- x_r horizontal moves;
- y_r vertical moves.

Therefore the number of source-to-target paths is immediate.

Corollary 5.2 — Exact shortest-completion count.
For every residual r,

$$\boxed{|\operatorname{GeoComp}(r)| = \binom{x_r + y_r}{x_r} = \binom{x_r + y_r}{y_r}.}$$

Equivalently,

$$\operatorname{GeoComp}(r) = \{z : N_0(z) = x_r,\ N_1(z) = y_r\}.$$

The statement is stronger than "there exists a shortest completion."
It says that the unique minimum demand completely classifies **all** shortest words.
Their only freedom is order.

5.4 Exact number of commuting squares

Every elementary unit square of the rectangle corresponds to one local choice between

$$01$$

and

$$10.$$

There are

$$x_r y_r$$

such squares.
Define

$$\boxed{\Xi(r) = x_r y_r.}$$

Corollary 5.3 — Exact commutation potential.

The number of elementary commuting squares in the shortest-completion rectangle of residual r is

$$\boxed{\Xi(r) = x_r y_r.}$$

Thus the same arithmetic quantity introduced in the 2:1 ladder has a general meaning.

It measures the number of elementary places at which the two required symbol types can interchange their local chronological order while remaining inside the shortest interval.

For the Chapter 3 sources,

$$\mu(2k+3) = (k+2, 1),$$

so

$$\Xi(2k+3) = k+2,$$

exactly recovering the number of ladder diamonds.

5.5 A worked 3:2 rectangle

Take

$$p = 3, \qquad q = 2, \qquad r = 2.$$

The completion equation is

$$3x - 2y = 2.$$

The minimum nonnegative solution is

$$\boxed{\mu(2) = (2, 2).}$$

Thus

$$\Delta(2) = 4.$$

The shortest interval is

$$[0, 2] \times [0, 2].$$

It contains

$$(2+1)(2+1) = 9$$

residual vertices.

There are

$$\binom{4}{2} = 6$$

shortest completion words:

$$0011, \quad 0101, \quad 0110, \quad 1001, \quad 1010, \quad 1100.$$

The canonical two-block completion is

$$\boxed{C(2) = 1100.}$$

The rectangle contains

$$2 \cdot 2 = 4$$

elementary commuting squares.

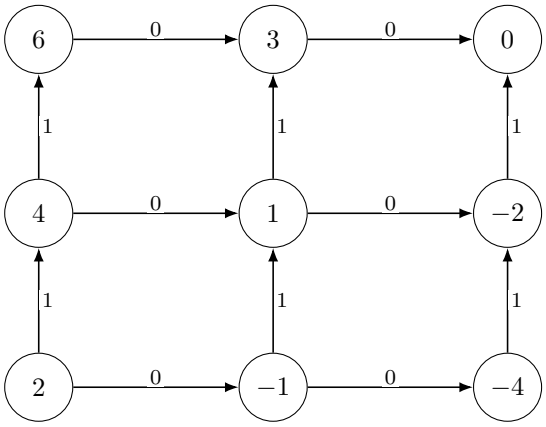


Figure 5.1: The 3:2, $r = 2$ shortest-completion rectangle. Horizontal edges read 0 and subtract 3 from the residual; vertical edges read 1 and add 2. The nine residual vertices form four commuting squares, and every monotone path from residual 2 to residual 0 is one of the six shortest completions with Parikh pair $(2, 2)$.

A useful feature of this example is that neither symbol type occurs only once. The two-row ladder has become a genuinely two-dimensional scheduling space.

5.6 Canonical order as a boundary normal form

For

$$r > 0,$$

our canonical representative is

$$C(r) = 1^{y_r} 0^{x_r}.$$

Among all shortest words with Parikh pair

$$(x_r, y_r),$$

this is one boundary ordering: every required 1 is performed before every required 0. For

$$r < 0,$$

the canonical representative

$$C(r) = 0^{x_r} 1^{y_r}$$

uses the opposite boundary orientation.
The rest of the shortest words lie between these extremal orderings.

For positive residuals, repeatedly apply the adjacent rewrite

$$01 \longrightarrow 10.$$

Each rewrite preserves the Parikh pair and therefore remains inside the same shortest-completion rectangle.

If we count inversions of the form "a 0 occurring before a 1," every rewrite decreases that count by one.

The unique normal form is

$$1^{y_r}0^{x_r} = C(r).$$

The maximum possible inversion count is

$$x_r y_r.$$

Thus:

$$\Xi(r) = \text{rectangle-square count} \\ = \text{maximum adjacent-swap normalization distance.}$$

For negative residuals the symmetric rewrite

$$10 \longrightarrow 01$$

normalizes toward

$$0^{x_r}1^{y_r}.$$

The canonical shortest word is therefore not arbitrary.

It is a boundary normal form for a finite commuting geometry.

5.7 Exactly when does a geodesic fork occur?

A residual has a genuine first-action fork exactly when both minimum-demand coordinates are positive.

For positive residuals, the bounded coordinate satisfies

$$0 \le y_r < p.$$

The completion congruence gives

$$qy_r \equiv -r \pmod{p}.$$

Since q is invertible modulo p,

$$y_r = 0 \iff p \mid r.$$

The negative case is symmetric.

Corollary 5.4 — Exact geodesic-fork classification.

For

$$r > 0,$$

$$\mathrm{GeoFirst}(r) = \begin{cases} \{0\}, & p \mid r, \\ \{0,1\}, & p \nmid r. \end{cases}$$

For

$$r < 0,$$

$$\boxed{\mathrm{GeoFirst}(r) = \begin{cases} \{1\}, & q \mid r, \\ \{0,1\}, & q \nmid r. \end{cases}}$$

At

$$r = 0,$$

the set is empty.
This gives an exact arithmetic classification of local shortest-path shape.
There are only three nonterminal qualitative possibilities:

$$\boxed{\text{zero corridor,} \qquad \text{one corridor,} \qquad \text{commuting fork.}}$$

The original 2:1 machine has forks only on the positive side because

$$q = 1.$$

Every negative residual is divisible by q, so its shortest path is a one-way 1-corridor.
The one-sided appearance of the original transition diagram is therefore arithmetic, not merely graphical.

5.8 First obstruction at length $p + q$

Chapter 4 proved that a word is geodesic precisely when

$$N_0 < q \qquad \text{or} \qquad N_1 < p.$$

This immediately gives the first scale at which a word can fail to be shortest.
If

$$n < p + q,$$

a length-n word cannot simultaneously contain at least q zeros and at least p ones.
Therefore:

$$\boxed{\text{every binary word of length } n < p + q \text{ is geodesic.}}$$

At length

$$p + q,$$

the first failures appear.
They are exactly the words with Parikh vector

$$(q, p),$$

that is, the orderings of one primitive balanced cycle.
There are

$$\boxed{\binom{p+q}{q}}$$

such first non-geodesic words.
This is another appearance of the same threshold:

$p + q$ = kernel packet length = fundamental cycle length
= first possible non-geodesic length.

5.9 Exact geodesic-word growth

The quotient-sphere count of Chapter 4 counted **distinct residual endpoints** at a given directed distance.

We now count something different:

> How many length-n binary words themselves are geodesic between their endpoints?

Let

$$G_{p,q}(n)$$

denote that number.

Theorem 5.5 — Exact geodesic-word growth.

For

$$0 \le n < p + q,$$

$$\boxed{G_{p,q}(n) = 2^n.}$$

For

$$n \ge p + q,$$

$$\boxed{G_{p,q}(n) = \sum_{i=0}^{q-1} \binom{n}{i} + \sum_{j=0}^{p-1} \binom{n}{j}.}$$

The first non-geodesic words occur at length

$$p + q$$

and are exactly the

$$\binom{p+q}{q}$$

orderings of the primitive balanced Parikh vector

$$(q, p).$$

If

$$(p, q) \neq (1, 1)$$

and

$$M = \max(p, q),$$

then

$$\boxed{G_{p,q}(n) \sim \frac{n^{M-1}}{(M-1)!}.}$$

For

$$(p, q) = (1, 1),$$

$$G_{1,1}(0) = 1, \qquad G_{1,1}(n) = 2 \quad (n \geq 1).$$

Proof.
A length-n word with i zeros is geodesic exactly when

$$i < q$$

or

$$n - i < p.$$

For

$$n < p + q,$$

these inequalities cannot both fail, so every word is geodesic.
For

$$n \geq p + q,$$

the two geodesic regions are disjoint:

- words with fewer than q zeros;
- words with fewer than p ones.

Counting them gives the two fixed binomial tails.
For a nontrivial coprime pair,

$$p \neq q.$$

The larger fixed binomial tail has highest index

$$M - 1.$$

Hence the dominant term is

$$\binom{n}{M-1} \sim \frac{n^{M-1}}{(M-1)!}.$$

The balanced case $(1, 1)$ reduces to the two unary words at every positive length. □
This theorem shows a sharp change of growth.
Before the first balanced cycle, geodesic words grow exponentially:

$$2^n.$$

After the threshold, only two bounded-width binomial tails survive, and growth becomes polynomial.

Again, the primitive balanced cycle is the obstruction.

5.10 Distinct paths versus distinct quotient vertices

Two counting functions have now appeared:

$$|S_n| = \min(n+1, p+q)$$

for quotient vertices at directed distance n, and

$$G_{p,q}(n)$$

for geodesic words of length n.

They measure different kinds of multiplicity.

The first forgets word order and remembers only the endpoint residual.

The second counts all geodesic orderings.

Inside a fixed shortest-completion rectangle, the same distinction appears as:

grid vertices versus monotone paths.

The semantic quotient collapses many word histories into one residual coordinate.

The rectangle restores exactly the scheduling information relevant to a shortest interval.

This interplay between quotienting and lifting will become a recurring theme later in the monograph.

5.11 The 2:1 ladder is now completely explained

Return to the Chapter 3 source

$$(L, 0^k).$$

Its residual is

$$2k+3,$$

and its minimum demand is

$$(k+2, 1).$$

The general rectangle is therefore

$$[0, k+2] \times [0, 1].$$

That is precisely a two-row ladder.

Its path count is

$$\binom{k+3}{1} = k+3.$$

Its square count is

$$(k+2) \cdot 1 = k+2.$$

The positive residual is odd, hence not divisible by $p = 2$, so it has a fork:

$$\mathrm{GeoFirst}(2k+3) = \{0, 1\}.$$

Every feature of the original operational ladder is therefore a specialization of the general minimum-demand rectangle.

This is an important methodological point.

We did not begin by defining a rectangle and then force the original machine into it.

The rectangle was discovered by first analyzing a concrete cancellation/completion ladder in the machine and then identifying the arithmetic that generalized it.

That discovery order will be preserved in the monograph.

5.12 From geometry back to representation

We now have two complementary pieces of structure.

Chapter 4 gave the canonical completion decomposition

$$\boxed{C(r) = \text{finite residue frontier} + \text{unbounded unary tail.}}$$

This suggests moving the finite frontier into control.

Chapter 5 gives the full shortest-path rectangle

$$\boxed{[0, x_r] \times [0, y_r].}$$

This gives an exact semantic graph that a physical DPDA representation ought to realize.

These facts create a new engineering objective.

Can we construct a pushdown representation in which:

1. the exact residual is decodable;
2. the exact prefix quotient is therefore known;
3. the unbounded magnitude remains stack-like;
4. the finite residue information is moved into control;
5. ordinary input transitions reproduce the residual graph as faithfully as possible?

The first attempt will work semantically but reveal a boundary defect.

That defect will turn out to be useful.

It will teach us that the problem is not the residual semantics.

The problem is the coordinate system.

The next chapter begins that redesign.

Chapter 6

The First General Physical Lift and Its Defects

Research branch — Better Coordinates and Exact Resource Frontiers. *This branch consolidates the coordinate constructions and exact residual-faithful resource frontiers developed in the preceding structural stages of the monograph. State/stack-symbol tradeoffs and limited pushdown alphabets have a substantial descriptional-complexity literature [10, 9, 16, 20]; the frontier statements below are therefore kept explicitly scoped to their representation classes.*

The previous chapters separated two objects that were entangled in the original 2023 machine:

$$\text{semantic residual} \qquad \text{and} \qquad \text{physical pushdown representation.}$$

The residual graph is now completely explicit:

$$r \xrightarrow{0} r - p, \qquad r \xrightarrow{1} r + q.$$

Its shortest intervals are rectangles, its quotient coordinate is one integer, and its canonical completion has a bounded finite frontier followed by an unbounded unary tail.

This makes a new design problem unavoidable.

> Can the residual graph itself be lifted into a compact DPDA configuration graph?

The first answer is yes—but only after a deterministic normalization step at a finite boundary.

That apparent imperfection will be useful. It will identify exactly what the first coordinates got wrong.

6.1 Finite phase plus unary magnitude

Let

$$m = \max(p, q).$$

The idea is to move the bounded residue information into finite control while leaving only an unbounded magnitude on the stack.

Use stable control states

$$R_0, \ldots, R_{m-1}$$

and stack alphabet

$$\Gamma_4 = \{0, 1, X, Z_0\}.$$

The positive and negative ends are represented by different unary symbols:

- 0 for the positive end;
- 1 for the negative end.

The additional symbol X is a negative-side sentinel.
For

$$0 \le s < p,$$

define

$$a_s = \begin{cases} 0, & s = 0, \\ s - p, & 1 \le s < p. \end{cases}$$

For

$$0 \le t < q,$$

define

$$b_t = \begin{cases} 0, & t = 0, \\ q - t, & 1 \le t < q. \end{cases}$$

These are not arbitrary constants. They choose one finite base residual for each positive and negative residue class.

6.2 The stable four-symbol encoding

Define

$$\eta(0) = (R_0, Z_0).$$

If

$$r > 0,$$

let

$$s = r \bmod p$$

and write uniquely

$$r = pk + a_s, \qquad k \ge 1.$$

Set

$$\boxed{\eta(r) = (R_s, 0^k Z_0).}$$

If

$$r < 0,$$

let

$$t = (-r) \bmod q$$

and write uniquely

$$r = -qk + b_t, \qquad k \geq 1.$$

Set

$$\boxed{\eta(r) = (R_t, 1^k X).}$$

The representation is injective.

The control state records the finite phase. The unary tail records the unbounded quotient magnitude. The sign is visible physically from the top/sentinel convention.

This is the first systematic realization of the Chapter 4 factorization

$$\text{finite boundary role} + \text{unbounded unary magnitude}.$$

6.3 Stable rules away from the boundary

On the positive side, input 0 should subtract p. Deep inside the unary tail, this is simply a pop:

$$\delta(R_s, 0, 0) = (R_s, \varepsilon).$$

Input 1 adds q. Let

$$s' = (s + q) \bmod p$$

and define

$$c_s = \frac{a_s + q - a_{s'}}{p}.$$

Then

$$pk + a_s + q = p(k + c_s) + a_{s'}.$$

Thus the correct stable rule is

$$\delta(R_s, 1, 0) = (R_{s'}, 0^{c_s+1}).$$

The negative side is symmetric.

Let

$$t' = (t + p) \bmod q$$

and

$$d_t = \frac{p - b_t + b_{t'}}{q}.$$

Then

$$\delta(R_t, 1, 1) = (R_t, \varepsilon)$$

and

$$\delta(R_t, 0, 1) = (R_{t'}, 1^{d_t+1}).$$

Away from the finite boundary, the design is already raw: one input symbol produces the exact next stable residual encoding.

The only difficulty appears when a unary tail is completely consumed.

6.4 Forced boundary normalization

Suppose a positive configuration has only one unary 0 left.

Consuming that 0 exposes

$$Z_0.$$

For residue class $s = 0$, this is already the correct zero configuration.

For

$$1 \le s < p,$$

the configuration

$$(R_s, Z_0)$$

is only transient. A unique forced normalization moves it to the stable encoding of

$$a_s = s - p < 0.$$

Likewise, on the negative side, consuming the last 1 exposes

$$X,$$

and a unique forced normalization moves to the stable encoding of the corresponding nonnegative base residual b_t.

This motivates a semantic macro-step.

Definition 6.1 — Normalized input macro-step.

Write

$$c \overset{a}{\Rightarrow} c'$$

when the machine:

1. consumes the binary symbol a;
2. performs the unique forced finite normalization chain, if one is enabled;
3. stops at the next stable encoded configuration.

The normalization consumes no additional binary input and introduces no choice.

6.5 Exact normalized residual conjugacy

Theorem 6.2 — Normalized residual–configuration conjugacy.

For every

$$r \in \mathbb{Z},$$

$$\boxed{\eta(r) \overset{0}{\Rightarrow} \eta(r-p), \qquad \eta(r) \overset{1}{\Rightarrow} \eta(r+q).}$$

Consequently, for every binary word u,

$$\boxed{\widehat{\delta}^*(\eta(r), u) = \eta(r - pN_0(u) + qN_1(u))\,.}$$

Thus η is a label-preserving graph isomorphism between the residual graph and the stable configuration graph **modulo deterministic normalization**.

Proof.

Let

$$r = pk + a_s > 0.$$

On input 0, if $k > 1$, one unary 0 is popped and the result is immediately

$$\eta(r - p).$$

If $k = 1$, the pop reaches the finite boundary. The uniquely forced normalization sends the exposed boundary pair to the stable encoding of $r - p$.

On input 1,

$$pk + a_s + q = p(k + c_s) + a_{s'},$$

so the positive stable rule lands directly in

$$\eta(r + q).$$

The negative case is symmetric. Input 1 pops one unary 1, with the $k = 1$ case followed by its unique sentinel normalization; input 0 uses the identity defining d_t.

At residual zero the two binary moves are defined directly toward the encodings of $-p$ and q.

The one-step identities imply the word formula by induction. □

This theorem is already a major improvement in understanding.

The original $C/L/R$ machine encoded the residual through a more specialized stack/state interaction.

The new construction makes the quotient coordinate explicit:

finite residual phase in control + unbounded quotient magnitude on a unary tail.

6.6 The abstract rectangles lift to physical configurations

Let

$$\mu(r) = (x_r, y_r).$$

Define

$$\widehat{\Psi}_r(i, j) = \eta(r - pi + qj).$$

Corollary 6.3 — Exact normalized configuration rectangle.

The map

$$\widehat{\Psi}_r$$

is a label-preserving isomorphism from

$$[0, x_r] \times [0, y_r]$$

onto the normalized shortest-path interval from

$$\eta(r)$$

to

$$\eta(0).$$

Every elementary rectangle square is a genuine normalized commuting diamond:

$$\boxed{\widehat{\delta}^*\big(\widehat{\Psi}_r(i,j),01\big) = \widehat{\Psi}_r(i+1,j+1) = \widehat{\delta}^*\big(\widehat{\Psi}_r(i,j),10\big).}$$

The shortest-completion rectangle is therefore not merely an arithmetic diagram.
It is realized by actual DPDA configurations.
The qualification is important:

one rectangle edge may be	one input move + forced normalization.

The original 2:1 ladder was stronger: its relevant edges were ordinary binary moves.
The question is now whether the general construction can be made equally literal.

6.7 A concrete 3:2 boundary crossing

Take

$$p = 3, \qquad q = 2, \qquad r = 4.$$

Then

$$\mu(4) = (2,1).$$

The normalized rectangle contains

$$\eta(4) = (R_1, 00Z_0),$$

$$\eta(1) = (R_1, 0Z_0),$$

$$\eta(-2) = (R_0, 1X),$$

$$\eta(6) = (R_0, 00Z_0),$$

$$\eta(3) = (R_0, 0Z_0),$$

$$\eta(0) = (R_0, Z_0).$$

The lower horizontal edge

$$1 \xrightarrow{0} -2$$

is semantically correct, but in these coordinates the binary pop first exposes a transient boundary pair before the forced normalization reaches

$$\eta(-2).$$

The rectangle remains exact.
The edge is simply not yet **raw**.

6.8 Raw residual conjugacy

We now strengthen the representation requirement.

Definition 6.4 — Raw residual–configuration conjugacy.

An injective encoding

$$\zeta : \mathbb{Z} \to \text{DPDA configurations}$$

is a raw residual conjugacy if

$$\boxed{\delta(\zeta(r), 0) = \zeta(r - p), \qquad \delta(\zeta(r), 1) = \zeta(r + q)}$$

for every residual r, using exactly one ordinary binary-input transition and no normalization step.

A raw defect of an encoding is therefore an edge on which this stronger identity fails.

The four-symbol construction is normalized-exact.

We now ask exactly where it fails to be raw.

6.9 The defect set is finite and exact

Proposition 6.5 — Exact four-symbol raw-defect set.

For the four-symbol encoding η,

$$\boxed{\mathcal{D}_{p,q} = \{(s, 0) : 1 \le s < p\} \cup \{(-j, 1) : 1 \le j \le q\}.}$$

Therefore

$$\boxed{|\mathcal{D}_{p,q}| = p + q - 1.}$$

Every residual edge outside this set is already one raw binary transition between stable configurations.

Proof.

On the positive side, input 1 always uses a stable rule.

Input 0 is also raw whenever at least two unary zeros are present. The only problem occurs when the source residual is one of

$$1, 2, \ldots, p - 1.$$

The last zero is popped, leaving the transient state/bottom pair

$$(R_s, Z_0)$$

instead of the required negative encoding.

These are exactly the $p - 1$ positive defects.

On the negative side, input 0 is always raw.

Input 1 is raw while at least two unary ones remain. At the final unary one, the pop exposes X, and the unique normalization is still required.

The affected residuals are

$$-q, -(q - 1), \ldots, -1,$$

giving exactly q negative defects.

The two families are disjoint. □

The count

$$p + q - 1$$

is numerically the same as the completion-frontier scale of Chapter 4 for nontrivial coprime pairs.

The meanings are different.

One counts the largest bounded canonical completion base.

The other counts the raw boundary defects of this particular four-symbol encoding.

The equality is evidence that the same finite arithmetic boundary is being seen from two physical viewpoints—not that the notions are identical.

6.10 Why the defects cannot be patched locally in the same coordinates

It is tempting to keep the encoding η and merely change the boundary transition rules.

That does not work.

Proposition 6.6 — Boundary obstruction for the fixed encoding.

Keeping the four-symbol physical representations $\eta(r)$, no standard top-rewrite PDA transition function can make all edges in

$$\mathcal{D}_{p,q}$$

raw while preserving the already correct behavior on the adjacent deeper unary tails.

Hence every defect of Proposition 6.5 is unavoidable **within these coordinates**.

Proof.

Fix

$$1 \leq s < p.$$

Compare

$$\eta(p+s) = (R_s, 00Z_0)$$

with

$$\eta(s) = (R_s, 0Z_0).$$

The two sources have the same state and visible top symbol.

Correct realization of

$$p + s \xrightarrow{0} s$$

forces the transition on the local pair

$$(R_s, 0)$$

to pop one zero and remain in state R_s.

But the same deterministic rule applied to

$$(R_s, 0Z_0)$$

must produce

$$(R_s, Z_0),$$

whereas the correct target is the negative encoding

$$\eta(s-p).$$

The transition cannot distinguish the two hidden suffixes.

Thus the positive boundary edge cannot be raw without breaking the already correct deeper edge.

The negative case is analogous. Compare

$$(R_t, 11X)$$

and

$$(R_t, 1X).$$

Correct deep behavior forces the visible-1 rule to pop. Applied at the boundary, the same pop leaves

$$(R_t, X),$$

not the required nonnegative target.

The obstruction depends only on the fixed state/top encoding and one-top locality. □

This is the real lesson of the defect theorem.

The problem is not that we failed to invent a clever enough boundary rule.

The local rule is already determined by the deeper tail.

6.11 Which shortest rectangles are already raw?

Corollary 6.7 — Raw rectangle criterion in the four-symbol orientation.

The entire shortest-completion rectangle from residual r is raw under η exactly when

$$\boxed{r = 0 \quad \text{or} \quad (r > 0 \text{ and } p \mid r).}$$

Interpretation. If

$$r > 0$$

and

$$p \mid r,$$

then

$$\mu(r) = (r/p, 0),$$

so the rectangle degenerates to a pure zero corridor and never crosses a defective boundary.

Every positive residual with a genuine two-symbol fork eventually exposes one of the positive boundary defects.

Every negative nonzero rectangle contains the final edge

$$-q \xrightarrow{1} 0,$$

which is defective in this orientation.

The asymmetry is representational.

Zero shares the positive sentinel Z_0, while the negative side uses X.

That observation points directly toward the cure.

6.12 The defect is coordinate-relative

The fixed-coordinate obstruction proves:

> these coordinates cannot be made globally raw.

It does **not** prove:

> four stack symbols are necessary.

Those are very different statements.
The first is a theorem.
The second would be an unjustified extrapolation.
The right design question is therefore not:

> What extra symbol should we add?

It is:

> Can we align the finite residual bases differently so that the sign-changing boundary is itself an ordinary top rewrite?

The answer is yes.
Even better, the new coordinates remove the extra sentinel X.
The next chapter performs that coordinate change.

Chapter 7

From Boundary Defects to a Raw Three-Symbol DPDA

Chapter 6 ended with a precise failure.

The four-symbol representation

$$\eta$$

was semantically exact, but at

$$p + q - 1$$

boundary edges an ordinary binary transition could not land directly in the next stable residual encoding.

The obstruction was not global.

It was local:

the same state/top pair had to serve both a deep unary tail and a sign boundary.

The first coordinates placed the finite residual bases on the wrong side of that boundary.

We now shift those bases.

The result is stronger in three ways:

1. every residual edge becomes one raw input transition;
2. the extra sentinel X disappears;
3. the construction still uses only

$$\max(p, q)$$

control states.

This is the first major example in the monograph where understanding the mathematics of a DPDA leads directly to a better DPDA.

7.1 Before using two work symbols: the unary baseline

It is useful to know what can be done with only one work-stack symbol.

Let

$$\Gamma_2 = \{A, Z_0\},$$

where Z_0 is a permanent bottom marker.

Every encoded stack then has the form

$$A^h Z_0.$$

The state must therefore distinguish every finite role that cannot be carried by stack-symbol type.

Theorem 7.1 — Exact one-work-symbol raw-conjugacy complexity.

For every coprime

$$p, q \geq 1,$$

any ε-free permanent-bottom raw residual conjugacy using only one work symbol requires

$$\boxed{|Q| \geq p + q.}$$

There is a right-endmarker construction attaining

$$\boxed{|Q| = p + q, \qquad |\Gamma| = 2.}$$

Hence $p + q$ is exact in this model.

Proof idea

Input 0 partitions the residual line into p bi-infinite orbits modulo p.

Far out on the positive end, injectivity forces the stack height to escape every bounded region. Since a single top rewrite can decrease work height by at most one, the eventual input-0 dynamics must contain enough negative-drift state phases to distinguish all p positive residue classes.

Thus at least p state roles are required.

Symmetrically, input 1 forces at least q state roles on the negative end.

With only one work symbol, those two tail families cannot safely share the same control roles: otherwise repeated negative drift would send infinitely many distinct residuals into finitely many bottom configurations.

Therefore the roles are disjoint:

$$|Q| \geq p + q.$$

A matching construction uses state families

$$P_0, \ldots, P_{p-1}$$

and

$$N_0, \ldots, N_{q-1},$$

with the single work symbol storing only unary magnitude.

The full transition table is recorded in Appendix B.

The theorem establishes an exact baseline:

$$\boxed{\text{one work symbol} \Longrightarrow p + q \text{ states.}}$$

The next question is whether a second work symbol can let the positive and negative phase roles share control states.

It can.

7.2 Boundary-compatible coordinates

Assume first

$$p \geq q.$$

Use only

$$\boxed{\Gamma_3 = \{0, 1, Z_0\}.}$$

Use control states

$$\boxed{Q_3 = \{R_0, \ldots, R_{p-1}\}.}$$

There are exactly

$$p = \max(p, q)$$

states.
For nonnegative residuals, use ordinary Euclidean division:

$$r = pk + s, \qquad k \geq 0, \qquad 0 \leq s < p.$$

Define

$$\boxed{\zeta(r) = (R_s, 0^k Z_0).}$$

For negative residuals, write uniquely

$$r = u - qh, \qquad h \geq 1, \qquad 0 \leq u < q.$$

Define

$$\boxed{\zeta(r) = (R_u, 1^h Z_0).}$$

This is the decisive shift.
The negative side no longer terminates in a separate sentinel X.
Its last unary 1 sits directly above the ordinary bottom marker.
The states

$$R_0, \ldots, R_{q-1}$$

are deliberately shared between:

- nonnegative boundary roles;
- negative residue phases.

The sign remains visible because:

- a negative configuration has top 1;
- a positive nonboundary configuration has top 0;
- a nonnegative boundary configuration has top Z_0.

Thus state sharing does not destroy injectivity.

7.3 Positive-tail arithmetic

For

$$0 \le s < p,$$

define

$$s^+ = (s + q) \bmod p$$

and

$$c_s = \frac{s + q - s^+}{p}.$$

Since

$$0 < q \le p,$$

we have

$$c_s \in \{0, 1\}.$$

On a positive unary tail:

$$\boxed{\delta(R_s, 0, 0) = (R_s, \varepsilon),}$$

and

$$\boxed{\delta(R_s, 1, 0) = (R_{s^+}, 0^{c_s+1}).}$$

The arithmetic identity is

$$pk + s + q = p(k + c_s) + s^+.$$

So the visible rewrite is exactly the residual update.

7.4 The repaired sign boundary

For the nonnegative boundary, define

$$u_s = (s - p) \bmod q$$

and

$$h_s = \frac{u_s + p - s}{q}.$$

Then

$$s - p = u_s - qh_s.$$

Therefore input 0 at the bottom marker can cross directly into the negative encoding:

$$\boxed{\delta(R_s, 0, Z_0) = (R_{u_s}, 1^{h_s} Z_0).}$$

No transient pair is created.
No normalization follows.
The boundary itself is now raw.
Input 1 at a nonnegative boundary uses

$$\boxed{\delta(R_s, 1, Z_0) = (R_{s^+}, 0^{c_s} Z_0).}$$

Again the target is already the exact stable encoding.

7.5 Negative-tail arithmetic and the second repaired boundary

For

$$0 \le u < q,$$

define

$$u^- = (u - p) \bmod q$$

and

$$d_u = \frac{u^- - u + p}{q}.$$

On the negative tail:

$$\boxed{\delta(R_u, 1, 1) = (R_u, \varepsilon),}$$

and

$$\boxed{\delta(R_u, 0, 1) = (R_{u^-}, 1^{d_u+1}).}$$

Now observe what happens when the negative stack has only one 1:

$$(R_u, 1Z_0) \xrightarrow{1} (R_u, Z_0).$$

But by definition

$$(R_u, Z_0) = \zeta(u).$$

The final negative 1-pop therefore crosses the sign boundary automatically.

The very rule that was forced by the deep negative tail is now also the correct boundary rule.

This is exactly what the four-symbol coordinates failed to achieve.

7.6 Global raw conjugacy

Theorem 7.2 — Three-symbol raw residual–configuration conjugacy.

For every coprime

$$p, q \ge 1,$$

there is an ε-free right-endmarker DPDA with

$$\boxed{|Q| = \max(p, q), \qquad |\Gamma| = 3,}$$

whose stack alphabet is

$$\{0, 1, Z_0\}$$

and whose injective encoding

$$\zeta : \mathbb{Z} \to Q \times \Gamma^*$$

satisfies

$$\boxed{\delta(\zeta(r), 0) = \zeta(r - p), \qquad \delta(\zeta(r), 1) = \zeta(r + q)}$$

for every residual r.

Hence, for every word w,

$$\boxed{\delta^*(\zeta(r), w) = \zeta(r - pN_0(w) + qN_1(w))\,.}$$

The right endmarker is accepted exactly at

$$\zeta(0).$$

Proof.

Assume

$$p \geq q.$$

The nonnegative and negative encodings are injective within their own Euclidean decompositions.

They are mutually disjoint physically because the negative family has top 1, whereas the nonnegative family has top 0 or Z_0.

If

$$r = pk + s$$

with

$$k \geq 1,$$

input 0 pops one zero:

$$pk + s - p = p(k - 1) + s.$$

Input 1 uses

$$pk + s + q = p(k + c_s) + s^+.$$

If

$$k = 0,$$

the same input-1 identity works at Z_0, while the boundary identity

$$s - p = u_s - qh_s$$

makes input 0 land directly in the negative encoding.

Now let

$$r = u - qh < 0.$$

Input 1 pops one unary 1.

If

$$h > 1,$$

the result is still negative.

If

$$h = 1,$$

the pop leaves

$$(R_u, Z_0) = \zeta(u),$$

which is exactly the nonnegative target.

Finally,

$$u - p = u^- - qd_u$$

gives

$$r - p = u^- - q(h + d_u),$$

so the input-0 rewrite produces exactly the required deeper negative encoding.
No case uses an ε-move.
For

$$q > p,$$

exchange the roles of 0 and 1, exchange p and q, and reverse the residual sign. $\square$
This theorem removes both weaknesses of Chapter 6 at once:

$$\boxed{(\max(p, q), 4, \text{normalized macro-edges})}$$

becomes

$$\boxed{(\max(p, q), 3, \text{raw input edges}).}$$

7.7 The 3:2 boundary is now literal

Return to

$$p = 3, \qquad q = 2, \qquad r = 4.$$

The new encodings include

$$\zeta(4) = (R_1, 0Z_0),$$

$$\zeta(1) = (R_1, Z_0),$$

$$\zeta(-2) = (R_0, 1Z_0),$$

$$\zeta(0) = (R_0, Z_0).$$

The old defective edge

$$1 \xrightarrow{0} -2$$

is now exactly

$$\boxed{\delta(R_1, 0, Z_0) = (R_0, 1Z_0).}$$

The old negative normalization edge

$$-2 \xrightarrow{1} 0$$

is now the ordinary pop

$$\boxed{\delta(R_0, 1, 1) = (R_0, \varepsilon),}$$

which exposes Z_0.
The same semantic rectangle now exists literally in the raw configuration graph.

7.8 Raw rectangles

Corollary 7.3 — Three-symbol raw configuration rectangles.

Let

$$\mu(r) = (x_r, y_r).$$

Define

$$\Psi_r^{\mathrm{raw}}(i, j) = \zeta(r - pi + qj).$$

Then

$$\Psi_r^{\mathrm{raw}}$$

is a label-preserving isomorphism from

$$[0, x_r] \times [0, y_r]$$

onto the ordinary DPDA shortest-path interval from

$$\zeta(r)$$

to

$$\zeta(0).$$

Every elementary square is a raw commuting diamond:

$$\boxed{\delta^*\big(\Psi_r^{\mathrm{raw}}(i, j), 01\big) = \Psi_r^{\mathrm{raw}}(i + 1, j + 1) = \delta^*\big(\Psi_r^{\mathrm{raw}}(i, j), 10\big).}$$

The Chapter 5 rectangle has now been fully lifted into one-symbol DPDA dynamics. No quotient by normalization is needed.

7.9 What has actually been optimized?

The construction uses

$$\max(p, q)$$

states and two work symbols $0, 1$ above the bottom marker.

At this point it is tempting to call it simply "the optimal DPDA."

That wording is too broad.

We must specify the representation category.

The Chapter 7 machine is optimal under several increasingly strong statements, each tied to an explicit representation category. Unrestricted recognition complexity is treated separately in the representation-independent theory.

Theorem 7.4 — Exact homogeneous-tail frontier.

In the permanent-bottom, ε-free raw-conjugacy model with asymptotically homogeneous positive and negative tails:

$$\boxed{S_{\mathrm{hom}}(k) = \begin{cases} p + q, & k = 1, \\ \max(p, q), & k \geq 2, \end{cases}}$$

where k is the number of work symbols.

Thus a second work symbol gives the entire possible state reduction in this model.

Additional homogeneous-tail work symbols cannot reduce the state count below

$$\max(p, q).$$

The proof uses preserved deep suffixes and the cyclic action of:

- addition by q on positive classes modulo p;
- subtraction of p on negative classes modulo q.

Coprimality forces one common positive tail symbol and one common negative tail symbol. With one work symbol the two ends must remain state-disjoint; with two symbols they may share control phases.

This is already an exact state/stack tradeoff.

But it still assumes a homogeneous-tail representation shape.

7.10 The motivating 2:1 ratio survives arbitrary three-symbol encodings

The original research program began at

$$(p, q) = (2, 1).$$

Here the stronger lower bound can be proved without assuming homogeneous tails or sign separation.

Theorem 7.5 — Exact three-symbol state complexity for 2:1.

Consider the full permanent-bottom stack alphabet

$$\Gamma_3 = \{A, B, Z_0\}$$

with:

- no ε-moves;
- injective raw residual conjugacy;
- no homogeneous-tail assumption;
- no prescribed sign-separated encoding.

Then

$$\boxed{|Q| \geq 2.}$$

The construction of Theorem 7.2 uses exactly two states, so

$$\boxed{|Q|_{\min} = 2}$$

for three-symbol raw conjugacy of the motivating 2:1 residual graph in this model.

Proof strategy

Assume one state.

Let

$$G : S_r \mapsto S_{r+1}$$

be input 1 and

$$F : S_r \mapsto S_{r-2}$$

be input 0.

On the encoded orbit,

$$\boxed{FG^2 = G^2F = \mathrm{id}.}$$

Choose an encoded residual of minimum work-stack height.

The inverse identity bounds the lengths of all possible top replacements, reducing the problem to finitely many local rewrite patterns.

If the minimum height is positive, a common inert suffix can be stripped. The remaining possibilities for the first few G-iterates are finite, and each is eliminated by one of:

- preserved-suffix impossibility;
- forced rule reuse;
- violation of minimum height;
- collision of two residuals.

If the minimum height is zero, the bottom write is likewise bounded and only finitely many cases remain. Every case again contradicts injectivity or the inverse relation.

The scope and finite-certificate framework for these lower bounds are recorded in Appendix C.

Thus one state is impossible.

This theorem is particularly important for the monograph because it ties the improved machine back to the original ratio:

> The analysis of the 2023 2:1 DPDA leads to an exact two-state, three-symbol raw residual-conjugate realization.

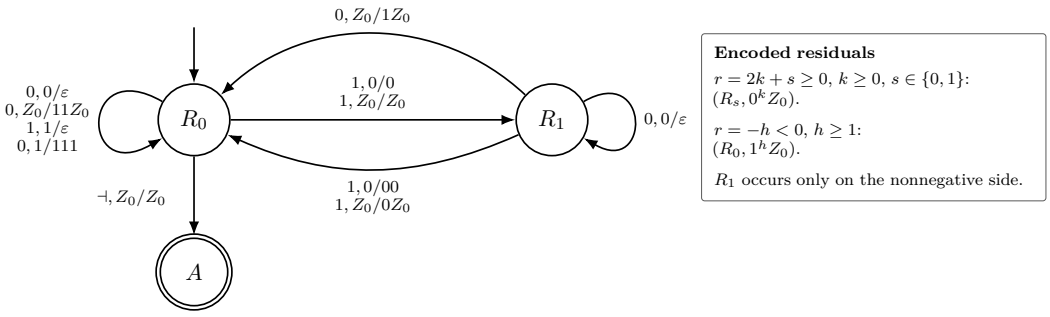


Figure 7.1: The improved three-symbol raw DPDA for $L_{2,1}$, with initial arrow and residual-code key. The two raw residual-control states are R_0, R_1; the double-circled A is a separate endmarker acceptance wrapper. Labels are input, top/replacement; only residual-orbit cells are shown. The endmarker accepts exactly at (R_0, Z_0).

Figure 7.1 is exact in the permanent-bottom, ε-free raw residual-conjugacy model. It is not a claim of unrestricted global optimality among all DPDAs recognizing $L_{2,1}$.

The scope words remain essential.

7.11 Further exact special ratios

The same unrestricted permanent-bottom three-symbol question can be solved at two further small ratios.

Theorem 7.6 — Exact 3:1 three-symbol state complexity.

For

$$(p, q) = (3, 1),$$

every ε-free permanent-bottom three-symbol raw conjugacy requires

$$\boxed{|Q| \geq 3.}$$

The construction of Theorem 7.2 attains three states.
Hence

$$\boxed{|Q|_{\min} = 3.}$$

The symmetric result holds for

$$(1, 3).$$

The proof again reduces an assumed smaller-state conjugacy to finitely many bounded top-rewrite possibilities, but the case structure is substantially larger than at 2:1. Its role and scope within the small-ratio finite certificates are recorded in Appendix C.

Theorem 7.7 — Computer-assisted exact 3:2 result.

For

$$(p, q) = (3, 2),$$

a finite mathematically bounded search eliminates every two-state permanent-bottom three-symbol raw conjugacy.
Therefore

$$\boxed{|Q| \geq 3,}$$

and the three-state construction is exact:

$$\boxed{|Q|_{\min} = 3.}$$

The symmetric result holds for

$$(2, 3).$$

This theorem is explicitly **computer-assisted**.
The computer is used only after the proof reduces the unrestricted-looking problem to a finite bounded search space. The survivor vectors and branch counts belong in the computational-validation appendix; the theorem should never be presented as a purely symbolic proof.
The small-ratio finite eliminations were an intermediate stage only. Chapter 8 closes the arbitrary-encoding theorem for every coprime pair by a representation-shape-free argument, so no ratio-by-ratio extension is required.

7.12 The right endmarker saves one control state in the scoped models

The constructions above use a distinguished right endmarker and accept exactly at the zero residual configuration.
If ordinary final-state acceptance on the binary alphabet is required instead, the zero configuration must use a final state that does not also occur on a nonzero residual.
In the scoped raw-conjugacy models this costs exactly one state.

Corollary 7.8 — Exact no-endmarker cost in the early raw models.

For the one-work-symbol model:

$$\boxed{|Q| = p + q + 1, \qquad |\Gamma| = 2.}$$

For the sign-separated homogeneous-tail three-symbol model:

$$\boxed{|Q| = \max(p, q) + 1, \qquad |\Gamma| = 3.}$$

Both bounds are tight.
This acceptance cost is separate from the stack-alphabet tradeoff.
It changes only how residual zero is represented.

7.13 The coordinate-shift improvement

The conclusion of Chapters 6 and 7 can now be stated cleanly.

Corollary 7.9 — Coordinate-shift improvement.

The $p + q - 1$ raw defects of the first four-symbol encoding are not a lower bound on the number of stack symbols.
They are properties of those coordinates.
A boundary-compatible shift gives

$$\boxed{(\max(p, q),\ 4,\ \text{normalized macro-steps}) \longrightarrow (\max(p, q),\ 3,\ \text{raw binary steps}).}$$

The missing ingredient was not an additional endpoint bit.
The finite bases had been aligned incorrectly with the sign-changing unary pops.
After moving the bases to adjacent residual classes:

- the same state can serve the post-pop boundary target;
- the ordinary bottom marker is sufficient;
- the extra sentinel X disappears;
- every residual edge becomes literal.

This is the first major design lesson of the monograph:

> a locality defect may be a coordinate defect
> rather than a semantic necessity.

7.14 Three different meanings of "better"

The evolution from the original machine to the raw three-symbol construction improves the representation in several non-equivalent ways.

Semantic transparency

The residual is decoded by a simple Euclidean state/tail coordinate.

Transition literalness

Every binary residual edge is exactly one DPDA input transition.

Stack alphabet

The general raw construction needs only

$$\{0, 1, Z_0\}.$$

Control width

It uses

$$\max(p, q)$$

states.

But the exactness claims established so far belong to stated raw-conjugacy categories.

Unrestricted recognition complexity is representation-independent and is treated separately in the later physical-complexity macro-Part.

That distinction is one of the central methodological themes of the book.

7.15 Removing the representation shape

The current lower bounds still come from several different levels of assumption:

- one work symbol;
- homogeneous positive/negative tails;
- special-ratio arbitrary three-symbol encodings.

The obvious next question is:

> Are the resource points
>
> $$p + q \quad \text{and} \quad \max(p, q)$$
>
> accidents of these structured coordinates, or are they forced by raw residual conjugacy itself?

To answer that, we must stop assuming homogeneous tails.

We will keep only the semantic requirement:

Every integer residual has one injective stable physical encoding, and every residual edge is one raw input transition.

From that weak assumption alone, the next chapter derives the complete exact frontier.

That result will turn the improved construction discovered here into a mature resource theorem.

Chapter 8

The Complete Raw Residual-Faithful Frontier

Chapter 7 produced a sharp construction:

$$\boxed{|Q| = \max(p, q), \qquad |\Gamma| = 3,}$$

with one raw binary transition for every residual edge.
It also produced exact lower bounds in several structured settings:

- one work symbol;
- homogeneous positive/negative tails;
- unrestricted three-symbol encodings for several small ratios.

The remaining question is whether the same resource points are forced **without assuming a particular stack shape**.
This chapter removes that assumption.
The result is the mature form of the early resource story:

$$\boxed{S_{\mathrm{raw}}(g) = \begin{cases} \infty, & g = 0, \\ p + q, & g = 1, \\ \max(p, q), & g \geq 2. \end{cases}}$$

The proof is deliberately representation-shape-free.
It does not assume:

- homogeneous tails;
- sign separation;
- a canonical coordinate formula;
- fixed stack depth;
- bounded replacement length.

Only raw residual conjugacy remains fixed.

8.1 Model R

Let

$$\Gamma = \Delta \cup \{Z_0\},$$

where Z_0 is a permanent bottom marker and

$$g = |\Delta|$$

is the number of work symbols.

Definition 8.1 — Model R.

A raw residual conjugacy is an injective map

$$\eta : \mathbb{Z} \to Q \times \Delta^* Z_0$$

such that, for every residual r,

$$\boxed{\delta(\eta(r), 0) = \eta(r - p), \qquad \delta(\eta(r), 1) = \eta(r + q),}$$

with no ε-normalization on encoded residual configurations.

Let

$$S_{\mathrm{raw}}(g)$$

be the minimum number of control states among such realizations, with value ∞ if no finite-state realization exists.

The semantic restriction is exact but narrow:

$$\boxed{\text{one residual} \leftrightarrow \text{one stable physical configuration.}}$$

The physical shape of that configuration is otherwise unrestricted.

This is the right setting for deciding whether the Chapter 7 optimum was merely a feature of its chosen coordinates.

8.2 Why bounded physical height is impossible

Fix a positive residue class

$$s \bmod p.$$

Consider the encoded chain

$$\eta(s), \quad \eta(s + p), \quad \eta(s + 2p), \quad \dots$$

or, if a finite initial segment crosses zero, begin sufficiently far into the positive tail.

Let

$$h(c)$$

denote the number of work symbols above Z_0.

Lemma 8.2 — Tail escape.

For every positive residue class $s \bmod p$,

$$\boxed{h(\eta(s + pk)) \to \infty \qquad (k \to \infty).}$$

The analogous statement holds on every negative residue class modulo q.

Proof.

Suppose infinitely many residuals on the chain were encoded at height at most H.

There are only finitely many physical configurations of bounded height because:

- Q is finite;
- Δ is finite;
- only finitely many stack words have length at most H.

Injectivity of η would then fail.

Therefore the height must escape every finite bound. □

This elementary argument replaces every earlier homogeneous-tail assumption.

No stack-shape hypothesis is available or needed here. The argument forces only unbounded growth along the residual chain.

8.3 The last occurrence of a height creates a stable suffix

Write

$$c_k = \eta(s + pk)$$

and

$$h_k = h(c_k).$$

Because

$$c_{k+1} \xrightarrow{0} c_k,$$

one ordinary PDA transition can remove at most the visible top work symbol.

Hence

$$\boxed{h_{k+1} \le h_k + 1.}$$

This one-sided height bound, together with tail escape, creates a canonical cross-section.

Lemma 8.3 — Last-height crossing.

For every sufficiently large integer H, the sequence (h_k) has a last occurrence of height H.

If this last occurrence is at index k, then:

1. $$h_{k+1} = H + 1;$$
2. the transition
$$c_{k+1} \xrightarrow{0} c_k$$
is a pure one-symbol pop;
3. every later configuration on that residue-class tail has height $> H$;
4. all later configurations share the same bottom-H work-stack suffix.

Proof.

Tail escape implies that sufficiently large heights are eventually exceeded forever.

Because

$$h_{k+1} \le h_k + 1,$$

the forward sequence cannot jump upward over a height H.

Thus every sufficiently large H is attained.

Choose its last occurrence.

The next height cannot be at most H, otherwise either H occurs again immediately or a later climb to values above H must pass through H once more.

Therefore

$$h_{k+1} = H + 1.$$

A one-step PDA transition that lowers height from $H + 1$ to H can only delete the visible work symbol. The lower H work symbols are untouched.

After the last height-H occurrence, every later configuration stays above H. Since subsequent transitions rewrite only the visible top region, that bottom-H suffix can never again be exposed or altered.

Hence it stabilizes forever. □

This is the key physical lemma.

No regular tail shape was assumed.

A stable suffix emerged from injectivity and one-top locality alone.

8.4 Positive residue classes synchronize

For each positive residue class

$$s \in \mathbb{Z}_p,$$

choose the last-height-H cross-section.

Let

$$\beta_s \in \Delta^H$$

be the resulting work-stack word above Z_0.

Lemma 8.4 — Positive synchronization.

For sufficiently large H,

$$\boxed{\beta_s = \beta_{s+q}}$$

for every residue class $s \bmod p$.

Consequently all p positive residue classes have the same last-height-H stack word, and their control states must be pairwise distinct.

Hence

$$\boxed{|Q| \geq p}$$

on the positive side.

Proof.

Take sufficiently deep positive configurations so that both source and target lie beyond their last-height-H crossings.

Input 1 maps residual class

$$s$$

to

$$s + q \bmod p.$$

A single top rewrite cannot modify the already stabilized bottom-H suffix.

Therefore

$$\beta_s = \beta_{s+q}.$$

Because

$$\gcd(p, q) = 1,$$

addition by q is one cycle on

$$\mathbb{Z}_p.$$

Thus all β_s are equal.

At the selected height-H cross-sections, the complete stack word is now identical across all p classes.

Since the residuals are distinct and η is injective, their control states must be distinct. □

The negative side is symmetric.

Lemma 8.5 — Negative synchronization.

At sufficiently large synchronized cross-sections on the negative tail,

$$\boxed{|Q| \geq q.}$$

Therefore every nonempty work alphabet already forces

$$\boxed{|Q| \geq \max(p, q).}$$

This is the first shape-free lower bound matching the Chapter 7 construction.

8.5 Why one work symbol is worse

Suppose now

$$g = 1.$$

Write

$$\Delta = \{A\}.$$

At a fixed height H, every work stack is literally

$$A^H Z_0.$$

The positive and negative synchronized cross-sections can therefore be chosen at the same height with exactly the same physical stack.

Lemma 8.6 — Cross-sign unary separation.

If

$$g = 1,$$

the p synchronized positive states and the q synchronized negative states must all be distinct.

Thus

$$\boxed{|Q| \geq p + q.}$$

Proof.

Choose the synchronized positive and negative cross-sections sufficiently far from zero.

All selected configurations have identical stack

$$A^H Z_0.$$

They represent

$$p + q$$

distinct residuals.

Injectivity can distinguish them only through control state.

Hence all $p + q$ states are distinct. □

The reason the second work symbol is powerful is now exact.

With one work symbol, sign cannot be carried by the tail symbol, so positive and negative residue roles cannot share states.

With two work symbols, the stack itself can separate the ends, allowing state reuse.

8.6 The complete raw frontier

Theorem 8.7 — Exact raw residual-faithful frontier.

For every coprime

$$p, q \geq 1,$$

$$\boxed{S_{\mathrm{raw}}(g) = \begin{cases} \infty, & g = 0, \\ p + q, & g = 1, \\ \max(p, q), & g \geq 2. \end{cases}}$$

Proof.

If

$$g = 0,$$

the only stack is the permanent bottom marker. With finitely many control states, only finitely many physical configurations exist, so no injective encoding of all

$$r \in \mathbb{Z}$$

is possible.

For

$$g \geq 1,$$

positive and negative synchronization give

$$|Q| \geq \max(p, q).$$

For

$$g = 1,$$

cross-sign unary separation strengthens the bound to

$$p + q.$$

The matching upper constructions are exactly the Chapter 7 machines:

- one work symbol with $p + q$ states;
- two work symbols with $\max(p, q)$ states.

Extra work symbols may simply remain unused. □

This theorem changes the status of several earlier results.

The Chapter 7 homogeneous-tail lower bound was not merely an artifact of homogeneous tails.

The exact same resource points survive after all such physical-shape assumptions are removed.

Thus the improved three-symbol machine discovered from the original DPDA lies on the complete raw residual-faithful frontier.

8.7 Scope of the theorem

The theorem is strong, but its category remains explicit.

It proves exact resource complexity for **raw residual conjugacy**.

It does not prove unrestricted recognition complexity for every DPDA recognizing

$$L_{p,q}.$$

A completely different recognizer may:

- fail to give every residual one stable configuration;
- encode several residuals indirectly through longer physical histories;
- use normalization;
- use a different semantic quotient;
- leave Model R entirely.

Therefore the correct statement is:

$$\boxed{\text{the raw residual-faithful frontier is exact.}}$$

Not:

$$\boxed{\text{the unrestricted DPDA state/stack frontier is solved.}}$$

This distinction becomes even more important in Chapter 9, where the analysis moves beyond residual-faithful representation to a language-level hierarchy problem.

8.8 Acceptance without a right endmarker

In the raw right-endmarker constructions, residual zero is accepted only when the endmarker arrives.

Without a right endmarker, ordinary final-state acceptance requires a state that is final exactly at residual zero.

The synchronized lower-bound states all occur on nonzero residuals, so none may be final.

Corollary 8.8 — Exact no-endmarker raw cost.

Under standard final-state acceptance without a right endmarker,

$$\boxed{S_{\mathrm{raw}}^{\mathrm{noend}}(g) = \begin{cases} \infty, & g = 0, \\ p + q + 1, & g = 1, \\ \max(p, q) + 1, & g \geq 2. \end{cases}}$$

The extra state is an acceptance-wrapper cost.

It does not change the nonzero residual geometry.

This is another place where the two original Chapter 1 variants become conceptually useful: acceptance conventions and semantic core complexity should not be conflated.

8.9 Can the optimum survive bounded pushes?

The raw three-symbol construction may use fixed replacement words longer than two symbols.

Suppose we impose the standard bounded-push convention

$$\max |\alpha| \leq 2.$$

A naive compilation could add many states.
The remarkable fact is that no new state is necessary.
Assume

$$p \geq q.$$

The positive-side writes are already of length at most two because

$$c_s \in \{0, 1\}.$$

Only some negative unary writes are long.
For each negative target phase v, define a family of suffix roles

$$H_{v,j},$$

meaning:

append exactly j additional unary 1's, then enter stable state R_v.

The total number of such transient roles is exactly

$$\boxed{p - q.}$$

Now inspect the stable three-symbol construction.
Negative configurations use top 1 only in states

$$R_0, \ldots, R_{q-1}.$$

Therefore the state/top cells

$$(R_i, 1), \qquad q \leq i < p,$$

are unused by stable negative semantics.
There are exactly

$$p - q$$

such cells.
The transient suffix roles fit into them perfectly.

Theorem 8.9 — State-preserving bounded-push compiler.

For every coprime p, q, there is a deterministic right-endmarker DPDA whose stable configurations realize the residual graph and satisfy

$$\boxed{|Q| = \max(p, q), \qquad |\Gamma| = 3, \qquad \max|\alpha| \leq 2.}$$

The only ε-moves are uniquely forced terminating suffix-chain moves.

Proof idea. Split every long unary write into:

1. one input-consuming push of length at most two;
2. a strictly decreasing suffix-completion chain.

The total transient role count is

$$|p - q|.$$

Exactly that many stable state/top cells are unused.
Overlay the transient roles on those cells.

Since the base machine has no consuming transition on the overlaid cells, the forced ε-moves introduce no input/ε conflict.

Every chain terminates and reconstructs exactly the macro-result of the original raw transition.

The case

$$q > p$$

is symmetric. □

This theorem is a useful warning against counting "temporary states" too early.

A transient normalization role need not require a new control state if an unused state/top cell already exists.

8.10 A stronger semantic-visibility budget

Model R asks only for exact residual faithfulness.

We now intentionally demand more.

Fix

$$B \geq \max(p, q)$$

and write every residual as

$$r = s + Bz, \qquad 0 \leq s < B.$$

Define the positive and negative phase periods

$$h_+(B) = \frac{p}{\gcd(B, p)},$$

$$h_-(B) = \frac{q}{\gcd(B, q)},$$

and

$$\boxed{H(B) = h_+(B) + h_-(B).}$$

In the **phase-local architecture**, each remainder s must expose its currently active signed quotient phase through:

- the control state;
- one visible work symbol.

Let

$$C$$

be the number of available work symbols.

For one fixed remainder, one state can provide at most C distinct state/top cells.

Therefore $H(B)$ phase roles require at least

$$\left\lceil \frac{H(B)}{C} \right\rceil$$

states.

Lemma 8.10 — Phase capacity.

For each remainder class s,

$$\boxed{|Q_s| \geq \left\lceil \frac{H(B)}{C} \right\rceil .}$$

The B remainder classes are semantically distinct in this architecture.
Summing gives the exact global frontier.

Theorem 8.11 — Exact normalized phase-local frontier.

For

$$B \geq \max(p, q),$$

$$\boxed{Q_{\min}^{\text{phase}}(B, C) = B \left\lceil \frac{\frac{p}{\gcd(B,p)} + \frac{q}{\gcd(B,q)}}{C} \right\rceil .}$$

For

$$C \geq 2,$$

the matching construction can be ε-free and raw.
For

$$C = 1,$$

the same state bound is attained with one forced bottom-cell normalization at the negative-to-zero crossing.

This theorem should not be confused with Theorem 8.7.
Theorem 8.7 asks for exact residual faithfulness only.
Theorem 8.11 additionally requires a chosen signed phase to be locally visible.
A stronger representation contract can require more states.
This is not a contradiction.
It is the beginning of a general principle:

> Resource complexity depends on what physical information the representation is required to expose locally.

8.11 A superlinear exact family

Take

$$p = k^2, \qquad q = k^2 - 1, \qquad B = k(k+1).$$

Then

$$H(B) = 2k - 1.$$

Hence

$$\boxed{Q_{\min}^{\text{phase}}(B, C) = k(k+1) \left\lceil \frac{2k-1}{C} \right\rceil .}$$

For fixed C,

$$Q_{\min}^{\text{phase}} = \Theta(k^3).$$

Since

$$\max(p, q) = \Theta(k^2),$$

this is

$$\boxed{\Theta\big(\max(p, q)^{3/2}\big)\,.}$$

The raw residual-faithful machine still needs only

$$\Theta(\max(p, q))$$

states.

The superlinear growth comes entirely from the extra phase-local visibility requirement.

This example makes category-relative resource claims impossible to ignore.

8.12 Direct lossless factorization: counting semantic cells

A still more structured model keeps the residual split into explicit semantic roles.

The stable roles are:

1. $p + q - 1$ boundary/bottom roles;
2. p deep positive roles with visible top 0;
3. q deep negative roles with visible top 1.

These role classes are disjoint.

Theorem 8.12 — Exact semantic-cell count.

Every exact implementation of this direct lossless unary-tail factorization requires

$$\boxed{2p + 2q - 1}$$

reachable stable state/top cells.

The canonical factorization attains this bound exactly.

The theorem counts semantic cells, not merely control states.

This is the correct unit when the representation contract explicitly requires different local roles to remain distinguishable.

8.13 Exact normalization-rule cost under push-at-most-two

Assume

$$p > q.$$

For each deep negative remainder t, the bounded-push normalization required by input 0 has a net stack-height increase

$$D_t = \left\lfloor \frac{p + t}{q} \right\rfloor - 1.$$

A terminating deterministic ε-computation that gains D persistent stack levels needs at least D distinct persistent-push rules.

Why?

If the same state/top rule created two different persistent levels, the intervening positive-drift ε-macro could repeat forever, contradicting termination.

Different source remainders also cannot share the same persistent rule, because after the common rule the lower context remains hidden and deterministic continuation would be identical, while the semantic target remainder is different.

The arithmetic sum is exact:

$$\sum_{t=0}^{q-1} D_t = p - q.$$

Theorem 8.13 — Exact ε-transition count.

In the direct lossless unary-tail factorization with:

- push-at-most-two;
- right endmarker;

the minimum number of forced normalization rules is

$$\boxed{|\delta_\varepsilon|_{\min} = |p - q|.}$$

Theorem 8.14 — Exact total transition count.

Under the same model,

$$\boxed{|\delta|_{\min} = 4(p + q) - 1 + |p - q|.}$$

Reason. The stable semantic graph forces:

$$4(p + q) - 2$$

binary input rules.
The endmarker contributes one accepting rule.
The bounded-push normalization requires exactly

$$|p - q|$$

forced ε-rules.
All three lower bounds are attained simultaneously.
This is the most explicit descriptional-complexity layer so far.
We are no longer counting only states or stack symbols.
We are counting:

- semantic state/top cells;
- binary rules;
- forced normalization rules;
- total transition description size.

8.14 A top-window implementation lemma

For completeness, the following finite-control compilation records how a bounded semantic visibility window can be represented physically when only the current abstract top symbol is rewritten.

Proposition 8.14A — Top-λ annotation compiler.

Suppose an abstract pushdown system inspects the top λ work symbols but rewrites only the current top abstract symbol. If the abstract work alphabet has size g, the same one-step dynamics can be implemented with the same control states by annotating physical stack symbols, using at most

$$\boxed{\sum_{d=1}^{\lambda} g^d}$$

work annotations, plus the bottom marker.

Proof. Annotate each physical symbol by the abstract symbol at that level together with the next at most $\lambda-1$ abstract symbols below it. The visible physical top then contains the entire abstract window needed to choose the transition. Because only the current abstract top is rewritten, the new annotation can be computed locally from the old annotation and the replacement word. The number of nonempty words of lengths 1 through λ over a g-symbol alphabet gives the stated bound. □

Scope. The proposition does not cover an abstract operation that rewrites or pops an entire visible λ-window at once; that stronger operation requires a different compiler.

8.15 Four resource questions, four different answers

We can now place the resource results in a hierarchy.

Level 1 — Raw residual conjugacy

Only the residual semantics is fixed.

$$\boxed{S_{\mathrm{raw}}(g) = \infty,\ p+q,\ \max(p,q).}$$

Level 2 — Bounded-push normalized realization

The optimal three-symbol/state point survives:

$$\boxed{|Q| = \max(p,q), \quad |\Gamma| = 3, \quad \max|\alpha| \le 2.}$$

The price is uniquely forced normalization.

Level 3 — Phase-local architecture

A selected signed phase must be visible from one state/top cell.

$$\boxed{Q_{\min}^{\mathrm{phase}}(B,C) = B\left\lceil \frac{H(B)}{C} \right\rceil.}$$

Level 4 — Direct lossless unary-tail factorization

The semantic roles themselves are fixed and counted.

$$\boxed{C_{\mathrm{sem,min}} = 2p + 2q - 1,}$$

$$\boxed{|\delta_\varepsilon|_{\min} = |p-q|,}$$

$$\boxed{|\delta|_{\min} = 4(p+q) - 1 + |p-q|.}$$

These theorems do not compete.
They answer increasingly structured physical questions.

8.16 What is now genuinely settled

The early DPDA analysis has produced a complete statement:

> Within raw residual conjugacy, the improved three-symbol machine is not merely good—it is on the exact frontier.

The discovery sequence is now complete:

original machine

↓

completion semantics

↓

finite-frontier coordinates

↓

four-symbol normalized lift

↓

exact boundary defect

↓

coordinate shift

↓

three-symbol raw DPDA

↓

complete raw residual-faithful frontier.

At this point the natural resource question becomes more radical.

The entire analysis still assumes that a competing machine faithfully represents our residual semantics.

But a DPDA recognizing the same language has no obligation to use our residual coordinate at all.

It may use a completely different stack representation.

It may use deterministic ε-moves.

It may hide the arithmetic in a way our residual-faithful lower bounds do not see.

So we now ask:

> Can a stack-alphabet lower bound survive against **arbitrary equivalent deterministic pushdown representations**?

That question takes the monograph outside the original counting family and into a previously stated hierarchy problem.

That is the subject of the next research branch.

Chapter 9

From Residual Frontiers to a Representation-Independent Hierarchy Theorem

Research branch — When the Resource Question Leaves Our Representation. *This branch develops the representation-independent hierarchy question that emerges when the raw residual-faithful representation is no longer imposed; its scope is deliberately kept separate from the frontiers of the preceding resource-frontier branch. Related descriptional-complexity work on pushdown alphabets and determinization provides the surrounding context [16, 15, 9].*

The resource story of the previous chapters has reached a natural stopping point.

Within raw residual conjugacy, the frontier is exact:

$$S_{\mathrm{raw}}(g) = \begin{cases} \infty, & g = 0, \\ p + q, & g = 1, \\ \max(p, q), & g \geq 2. \end{cases}$$

The result is strong because the lower bound no longer assumes homogeneous tails, sign-separated coordinates, or a particular stack formula.

But it still assumes something fundamental:

the competing machine is required to realize the residual graph itself.

A DPDA recognizing the same language need not do that.

It may encode the future language in a completely different way. It may normalize internally. It may use deterministic ε-moves between input symbols. It may never expose anything resembling our residual coordinate.

The next step in the research program was therefore not another attempt to optimize the same representation.

It was to remove the representation from the lower bound.

9.1 The methodological jump

The first two resource stages asked:

> Given the semantic object we have discovered, how efficiently can it be represented?

The new question is different:

> Can the **continuation behavior of the language itself** force a state–stack lower bound, even if an equivalent DPDA chooses an unrelated internal representation?

This distinction is the reason Chapter 8 and the present research branch must not be conflated.

Chapter 8 is about **representation-faithful optimality**.

This research branch is about **arbitrary equivalent recognizers**.

The lower-bound invariant can therefore no longer refer to:

- our residual stack shape;
- our canonical completion word;
- our sign symbol;
- our chosen state phases.

It must be externally visible through the accepted continuation language.

Completion lengths provide exactly such an observable.

9.2 The external hierarchy question

Let

$$\mathcal{D}_{m,k}$$

denote the class of languages recognized by deterministic pushdown automata with at most:

$$m$$

control states and at most

$$k$$

pushdown symbols, in the general deterministic model where deterministic ε-moves are allowed.

The inclusion

$$\mathcal{D}_{m,k} \subseteq \mathcal{D}_{m,k+1}$$

is trivial: an extra stack symbol may simply remain unused.

The question is whether the inclusion is strict.

Masopust established a binary stack-alphabet hierarchy for stateless DPDAs and extended the separation to m-state **realtime** DPDAs for every $m \geq 1$ [16].

The remaining general-model case allows deterministic ε-moves. The theorem proved in this branch establishes the strict hierarchy

$$\boxed{\mathcal{D}_{m,k} \subsetneq \mathcal{D}_{m,k+1}}$$

over a fixed binary input alphabet for

$$m \geq 2, \qquad k \geq 1,$$

under the general single-initial-symbol convention used in this branch.

The stateless case $m = 1$ is kept separate because the earlier stateless formulation uses a slightly different initial-store convention.

That model distinction must remain explicit.

9.3 Why realtime separation does not automatically survive ε-moves

A realtime DPDA processes one input symbol per transition.

Its visible state/top pair is therefore synchronized with the input position.

A general DPDA can insert a deterministic ε-computation between two input symbols.

Such an internal computation can:

- rewrite the stack;
- move information between stack and control;
- expose a deeper symbol;
- normalize a temporary encoding;
- alter the local mode before the next symbol is consumed.

Therefore a lower bound based only on what a realtime machine must expose at input boundaries may collapse when the adversary is allowed internal stack work.

This is precisely why the lower bound must reason about **raw computation positions**, including positions inside forced ε-blocks.

The proof developed in this branch will do that.

9.4 Lineage from the original DPDA

The hierarchy theorem is formally self-contained.

It does not depend on the original 2:1 transition table.

Nevertheless, its mathematical route is directly connected to the earlier program.

The progression was:

> original completion-aware DPDA $\rightarrow$ shortest continuation semantics $\rightarrow$ exact state/stack tradeoffs $\rightarrow$ question of arbitrary alternative representations.

The key methodological lesson from the earlier machines was:

> Stack contents can be probed indirectly by the continuations they require.

The hierarchy witness turns that observation into an adversarial lower-bound tool.

Instead of assuming that a competitor stores a residual, we construct many prefix families whose required unary completion lengths grow at pairwise different rates.

Those rates are visible in the language itself.

The competitor may encode them however it wants.

9.5 The improved 2:1 raw DPDA as the immediate precursor

For the motivating ratio $(2, 1)$, Chapter 7 obtained a raw residual-conjugate construction with:

$$|Q| = 2, \qquad \Gamma = \{0, 1, Z_0\},$$

and no ε-normalization on binary residual edges.

This representation is more compact and more literal than the original center/left/right realization for the residual-conjugacy objective.

But that result still does not imply an unrestricted recognition lower bound.

An adversarial DPDA recognizing the same language is not required to use:

- the same two states;
- the same three stack symbols;
- the same residual coordinate;
- any stable residual encoding at all.

This is the conceptual bridge into the hierarchy problem.

9.6 Why the witness development did not end with a direct encoding trick

The route to the final binary hierarchy witness passed through two intermediate prototypes.

Four-letter prototype

The first general prototype used semantic modes

$$(i, t) \in \mathbb{Z}_m \times \mathbb{Z}_k$$

with distinct weights

$$\omega(i, t) = tm + i + 1 \in \{1, \ldots, mk\}.$$

Its purpose was to isolate the product phenomenon:

many externally different completion rates
$\Longrightarrow$ many recurrent state/top roles.

Three-letter prototype

A mixed-radix control symbol merged two semantic navigation coordinates.

This reduced the input alphabet but still retained a separate delimiter/cleanup phase.

Binary witness

A direct binary code of the three-letter system would have required decoder memory.

Under an exact m-state upper budget, that decoder memory is itself a resource and can ruin the comparison.

The final construction therefore changes the invariant instead of coding the alphabet.

Rather than forcing

$$mk \text{ slopes} + 1 \text{ cleanup role},$$

it forces directly

$$\boxed{mk + 1 \text{ distinct completion slopes.}}$$

The cleanup role disappears as a separate semantic mode.

The witness becomes binary without needing a phase decoder.

This is an important design lesson in its own right:

When an encoding adds resource overhead, change the observable rather than merely recoding symbols.

9.7 The general DPDA model used in the lower bound

A DPDA is

$$M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F),$$

with partial transition function

$$\delta : Q \times \Gamma \times (\Sigma \cup \{\varepsilon\}) \to Q \times \Gamma^*.$$

We write the stack top on the right.
Thus if

$$\delta(q, X, a) = (p, \beta),$$

then

$$(q, \gamma X, av) \vdash (p, \gamma\beta, v).$$

No transition is available from an empty stack.
Determinism requires that for each state/top pair:

- there is at most one transition for each input symbol;
- if an ε-transition is enabled, then no input-consuming transition is enabled from that same state/top pair.

Acceptance is by:

$$\boxed{\text{all input consumed} + \text{final state} + \text{empty stack.}}$$

A realtime DPDA is the special case with no ε-moves.
The upper witness will be realtime.
The lower-bound opponent is allowed to be fully general.

9.8 Modes

A nonempty control-store configuration has the form

$$(q, \eta X),$$

where X is the visible top symbol.
Its **mode** is the pair

$$\boxed{[q, X].}$$

A DPDA has at most

$$\boxed{|Q||\Gamma|}$$

nonempty modes.
The product lower bound will ultimately be a mode-counting theorem.
But merely counting modes is not enough.
We must first prove that different externally visible completion slopes cannot share one recurrent mode.
That is the hard part.

9.9 The lower-bound target

For every

$$m \geq 2, \qquad k \geq 1,$$

we will construct a binary language

$$B_{m,k}$$

such that:

1. $B_{m,k}$ is recognized by a realtime DPDA with exactly

   $$m$$

   states and

   $$k + 1$$

   stack symbols;

2. every general DPDA recognizing $B_{m,k}$, even with arbitrary deterministic ε-moves, satisfies

   $$\boxed{|Q||\Gamma| \geq mk + 1.}$$

The hierarchy corollary will then be immediate.
If a machine used at most m states and at most k stack symbols, then

$$|Q||\Gamma| \leq mk,$$

contradicting the product theorem.
The real work is therefore to build a binary witness exposing exactly

$$mk + 1$$

pairwise incompatible continuation slopes.
That construction is the subject of Chapter 10.

Chapter 10

Completion Slopes and ε-Robust Pumping

Fix

$$m \geq 2, \qquad k \geq 1,$$

and write

$$\boxed{\ell = mk + 1.}$$

The witness will have ℓ distinguished live prefix families.

Each family will admit exactly one all-zero completion length after an accumulation block of 1's.

Those lengths will be affine functions with slopes

$$1, 2, \ldots, \ell.$$

The upper machine uses exactly:

$$m \text{ states}, \qquad k + 1 \text{ stack symbols}.$$

The lower bound then asks whether an arbitrary equivalent DPDA can hide those ℓ externally different rates inside fewer than ℓ state/top modes.

It cannot.

10.1 The binary realtime witness

Let

$$Q = \{q_0, \ldots, q_{m-1}\}$$

and

$$\Gamma = \{A, M_0, \ldots, M_{k-1}\}.$$

Hence

$$|Q| = m, \qquad |\Gamma| = k + 1.$$

Define the permutation

$$\pi(0) = 1, \qquad \pi(1) = 0,$$

and

$$\pi(r) = r \qquad (2 \le r < m).$$

Choose ℓ pairwise distinct slope modes

$$x_s = (r_s, X_s), \qquad 0 \le s < \ell,$$

by

$$x_0 = (q_1, A),$$

and, for

$$0 \le r < m, \qquad 0 \le t < k,$$

$$x_{1+rk+t} = (q_{\pi(r)}, M_t).$$

Thus

$$\{x_1, \ldots, x_{\ell-1}\} = Q \times \{M_0, \ldots, M_{k-1}\}.$$

The initial pair

$$(q_0, A)$$

is deliberately not a slope mode.
Define the fixed table word

$$\boxed{T = X_{\ell-1} X_{\ell-2} \cdots X_1 X_0,}$$

with top on the right, so X_0 is initially visible.

10.2 Four rule families

The witness DPDA $U_{m,k}$ is realtime.

Header

$$\boxed{\delta(q_0, A, 1) = (q_1, T).}$$

The first 1 installs the entire finite mode table.

Generic zero-pop on counter symbol

For every state q_i,

$$\boxed{\delta(q_i, A, 0) = (q_i, \varepsilon).}$$

Selector transitions

For

$$0 \le s < \ell - 1,$$

$$\boxed{\delta(r_s, X_s, 0) = (r_{s+1}, \varepsilon).}$$

At the last table mode,

$$\boxed{\delta(r_{\ell-1}, X_{\ell-1}, 0) = (q_0, \varepsilon).}$$

Thus a zero can walk down the installed table.

Slope transitions

For every

$$0 \le s < \ell,$$

$$\boxed{\delta(r_s, X_s, 1) = (r_s, A^{s+1}X_s).}$$

One accumulation 1 inserts exactly $s+1$ copies of A immediately below the visible mode symbol.

The same mode remains visible.

This is the entire slope mechanism.

10.3 Upper-bound theorem

Proposition 10.1 — Realtime upper construction.

$U_{m,k}$ is a deterministic realtime DPDA with exactly

$$\boxed{m \text{ states}}$$

and

$$\boxed{k+1 \text{ stack symbols.}}$$

Proof.

There are no ε-moves.

The slope modes are pairwise distinct, so their 1-rules and selector 0-rules do not conflict.

The only apparent overlap is at

$$x_0 = (q_1, A),$$

where the selector rule agrees exactly with the generic A-pop rule.

The header uses

$$(q_0, A, 1),$$

which is not one of the slope modes.

Thus the transition function is deterministic. □

Define

$$\boxed{B_{m,k} = L(U_{m,k}).}$$

The lower bound will not require a closed combinatorial description of the entire language.

One exact slice is enough.

10.4 The exact completion-slope slice

For

$$0 \le s < \ell,$$

define

$$\boxed{N_s(n) = \ell - s + (s+1)n.}$$

Let

$$p_s = 10^s.$$

Lemma 10.2 — Exact pure-zero slice.

For every

$$0 \le s < \ell$$

and every

$$n, r \ge 0,$$

$$\boxed{10^s 1^n 0^r \in B_{m,k} \iff r = N_s(n).}$$

Proof.

After the initial 1, the machine is in mode x_0 with table T.

Each of the next s zeros removes the current table symbol and exposes the next designated slope mode.

Thus after

$$10^s$$

the visible mode is

$$x_s$$

and the remaining table contains exactly

$$\ell - s$$

symbols.

Each subsequent 1 inserts

$$s + 1$$

copies of A below the visible mode symbol.

After n such ones, the total stack height is therefore

$$\ell - s + (s + 1)n.$$

Now read zeros.

The first zero removes the visible mode symbol.

Inserted A's are then popped one by one, and whenever the next table symbol is exposed the machine is already in its designated state.

Thus every remaining zero removes exactly one stack symbol.

Because acceptance requires empty stack after the input has been exhausted, the all-zero suffix is accepted exactly when its length equals the current stack height. □

Hence the ℓ prefix families

$$p_s 1^n$$

have unique pure-zero completion slopes

$$\boxed{w_s = s + 1.}$$

So the language exposes:

$$\boxed{1, 2, \ldots, mk + 1}$$

as pairwise distinct affine continuation rates.

10.5 The smallest new case

Take

$$m = 2, \qquad k = 1.$$

Then

$$\ell = 3.$$

The slope modes are

$$x_0 = (q_1, A),$$

$$x_1 = (q_1, M_0),$$

$$x_2 = (q_0, M_0).$$

The exact slice becomes:

$$1\,1^n 0^{3+n},$$

$$10\,1^n 0^{2+2n},$$

$$100\,1^n 0^{1+3n}.$$

The three slopes are:

$$1, \qquad 2, \qquad 3.$$

The theorem of the next chapter will imply that no DPDA with:

$$2 \text{ states}$$

and

$$1 \text{ stack symbol}$$

can recognize the same language—even if that competitor uses deterministic ε-moves.

10.6 The lower-bound opponent is arbitrary

Let

$$M = (Q, \{0, 1\}, \Gamma, \delta, q_0, Z_0, F)$$

be any general DPDA with

$$L(M) = B_{m,k}.$$

Fix one slope index s. Each finite prefix

$$p_s 1^n$$

is live because it has the accepted continuation

$$0^{N_s(n)}.$$

Define the auxiliary *s-analysis path* of control-store configurations, rather than treating $p_s 1^\omega$ as an accepted infinite input. Begin immediately after the last symbol of p_s is consumed. At every subsequent transition boundary, follow the forced ε-move if one exists; otherwise follow the transition labeled 1. The path extends indefinitely: for each $n \geq 0$, the finite word $p_s 1^{n+1} 0^{N_s(n+1)}$ is accepted, so the deterministic run cannot get stuck or remain in an infinite ε-block before the next 1. Each finite path segment therefore appears in a computation on a sufficiently long accepted finite word.

Its **raw positions** are all individual transition boundaries, including those inside forced ε-blocks. At raw position i, write $C_i = (q_i, \gamma_i)$, let $a(i)$ be the number of accumulation ones consumed after p_s, and put $h(i) = |\gamma_i|$. Every γ_i is nonempty, since the empty stack admits no further transitions. The informal term *slope ray* below always refers to this auxiliary path.

The proof must not look only at input-boundary configurations.

10.7 Continuation determinism

Auxiliary lemma (continuation determinism). At any raw control-store configuration

$$(q, \gamma),$$

and for any finite unread input word, the next transition (if any) is uniquely determined. An enabled ε-move is forced irrespective of the next unread symbol; otherwise the first unread symbol determines at most one consuming transition. Iteration therefore gives the same entire continuation from the same control-store configuration and the same unread suffix, until halting or divergence.

Proof. This follows inductively from deterministic exclusion of a consuming transition whenever an ε-transition is enabled and from uniqueness of the transition on the next unread symbol. In particular, a raw configuration inside a forced ε-block after a consumed prefix is also reached when that prefix is followed by a different finite unread suffix. □

This is the bridge from equality of physical configurations to equality of continuation behavior.

10.8 No raw configuration can repeat on a live slope ray

Lemma 10.3 — No raw configuration repetition.

Along the fixed s-analysis path, the same full control-store configuration cannot occur twice at distinct raw positions.

Proof.

Suppose first that the two occurrences have consumed different numbers

$$n < n'$$

of accumulation ones.

The later prefix has accepted continuation

$$0^{N_s(n')}.$$

If the earlier and later raw control-store configurations are identical, determinism gives the same future computation from both.

Thus M would also accept

$$p_s 1^n 0^{N_s(n')}.$$

But the exact slice says that after

$$p_s 1^n$$

the unique all-zero completion length is

$$N_s(n),$$

and

$$N_s(n') \neq N_s(n).$$

Contradiction.

If the two repeated raw configurations occur after the same amount of input has been consumed, then the segment between them contains only ε-moves.

Determinism would repeat the same ε-cycle forever, preventing the next live input symbol from being read.

Again contradiction. □

This lemma absorbs arbitrary deterministic internal normalization into the argument.

Repeated physical configurations are impossible because the language exposes a unique continuation length.

10.9 Raw stack height must escape

Lemma 10.4 — Raw-height escape.

Along every slope analysis path,

$$\boxed{\text{raw stack height tends to infinity.}}$$

More explicitly, for every H, after some raw position the stack height is always greater than H.

Proof.

There are only finitely many control-store configurations of stack height at most H.

By Lemma 10.3, none can occur twice.

Therefore only finitely many raw positions have height at most H. □

The adversary may use any representation it likes.

But the infinitely many distinguishable live prefixes force its physical store to escape every bounded region.

10.10 Suffix-minimum positions

A raw position t of a fixed analysis path is a **step** if its stack height is a minimum over the entire future suffix of that path:

$$\boxed{h(t) \leq h(t') \quad \text{for every } t' \geq t.}$$

Lemma 10.5 — Infinitely many steps.

Every slope analysis path contains infinitely many step positions.

Proof.

The height sequence tends to infinity.

For every starting position, the future tail has a minimum.

Choose a position where that minimum occurs.

As the starting point moves arbitrarily far to the right, these minima cannot all lie in a fixed finite set. □

Step positions are the correct pumping locations because the computation never descends below their initial stack height.

Therefore the lower stack context remains hidden.

10.11 The ε-robust step pump

Lemma 10.6 — Step-pump lemma.

On every fixed slope analysis path there exist two step positions $t_1 < t_2$ with the same mode and the following associated data:

- a mode
$$[q, X];$$
- an integer
$$c \geq 1;$$
- a nonempty word
$$\rho \in \Gamma^+;$$

such that for every lower context

$$\eta \in \Gamma^*,$$

$$\boxed{(q, \eta X) \xRightarrow{1^c} (q, \eta\rho X),}$$

where the macro-computation:

1. consumes exactly c input symbols 1;
2. may contain forced ε-moves;
3. never goes below the initial height
$$|\eta X|.$$

For this fixed analysis path, the mode $[q, X]$ is called an **up-mode of the path** precisely when it occurs at the two witnessing step positions. The definition is path-relative; the witnessing positions supply the pump data c and ρ.

Proof.

There are infinitely many step positions but only

$$|Q||\Gamma|$$

possible modes.

Choose two step positions

$$t_1 < t_2$$

with the same mode

$$[q, X].$$

Write the earlier stack as

$$\eta X.$$

Because t_1 is a suffix minimum, the computation between t_1 and t_2 never goes below height

$$|\eta X|.$$

Therefore the lower prefix η is never exposed.

The later stack has form

$$\eta\rho X.$$

If

$$\rho = \varepsilon,$$

the two positions would have the same full configuration, contradicting Lemma 10.3.
Hence

$$\rho \in \Gamma^{+}.$$

Let c be the number of input 1's consumed between the two steps.
Because the lower context is never inspected, the same deterministic transition sequence works for every replacement lower context.
Thus the pump is context-independent.
Finally,

$$c \neq 0.$$

If it consumed no input, it would be a height-increasing ε-macro from the same mode back to itself. Determinism would repeat it forever and prevent the next live symbol from being consumed.
Therefore

$$c \geq 1.$$

□
The pump iterates:

$$\boxed{(q, \eta X) \xrightarrow{1^{jc}} (q, \eta \rho^{j} X) \qquad (j \geq 0).}$$

This is the technical point at which the lower bound becomes genuinely ε-robust.
The proof did not forbid internal normalization.
It found a recurrent growth mechanism inside it.

10.12 The remaining proof obligation

Every slope ray now has an up-mode.
To obtain the product lower bound, we need one final structural fact:

$$\boxed{\text{one up-mode cannot serve two different completion slopes.}}$$

If that holds, then the $\ell = mk + 1$ slopes force ℓ different modes.
The proof requires one more determinism lemma about unary completion.
That is where the next chapter begins.

Chapter 11

One Mode, One Slope

The upper witness exposes

$$\ell = mk + 1$$

different affine completion slopes.

Chapter 10 showed that every arbitrary general DPDA recognizing the witness must, along each live slope ray, develop a context-independent height-increasing up-mode.

We now prove the separation theorem:

$$\boxed{\text{one up-mode} \Longrightarrow \text{at most one completion slope.}}$$

Once this is established, the product bound and the hierarchy follow immediately.

11.1 Unary completion uniqueness

The witness slice uses all-zero continuations.

Determinism makes those continuations rigid.

Lemma 11.1 — Unary completion uniqueness.

Fix any raw control-store configuration

$$(p, \gamma)$$

of a deterministic pushdown automaton in the present model.

There is at most one integer

$$r \geq 0$$

such that input

$$0^r$$

leads from (p, γ) to acceptance.

Proof.

Suppose

$$r < r'$$

and both

$$0^r$$

and

$$0^{r'}$$

were accepted.

The two deterministic computations are identical until the shorter input has consumed all r zeros, including every forced ε-move enabled independently of the presence of further input.

The accepting run on 0^r eventually reaches:

- empty unread input;
- a final state;
- empty stack.

The run on $0^{r'}$ reaches the same empty-stack configuration while

$$r' - r > 0$$

zeros remain unread.

No transition is available from the empty stack.

Thus the longer word cannot be accepted.

Contradiction. □

This is a very small lemma, but it is exactly what converts a recurrent physical mode into a constraint on completion length.

11.2 Slope separation

Lemma 11.2 — Slope separation.

Let

$$s \neq t.$$

The s- and t-analysis paths cannot have the same up-mode in any DPDA recognizing

$$B_{m,k}.$$

Proof.

Assume $[q, X]$ is an up-mode of both analysis paths. Take the context-independent pump from two witnessing step positions on the s-path:

$$(q, \eta X) \xRightarrow{1^c} (q, \eta\rho X),$$

with

$$c \geq 1, \qquad \rho \in \Gamma^+.$$

Use the earlier witnessing step position on the s-path after n_s accumulation ones, with lower context γ_s, and a witnessing step position on the t-path after n_t accumulation ones, with lower context γ_t. Context independence makes the same pump available from either stack.

After j pump iterations the two rays reach:

$$(q, \gamma_s\rho^j X)$$

and

$$(q, \gamma_t\rho^j X)$$

after respectively

$$n_s + jc$$

and

$$n_t + jc$$

accumulation ones.

Continuation determinism identifies these pumped control-store configurations with the respective analysis paths at their accumulation counts. The exact witness slice provides accepted zero completions of lengths

$$N_s(n_s + jc)$$

and

$$N_t(n_t + jc).$$

Now run those two zero computations in parallel.

As long as the common upper segment

$$\rho^j X$$

has not been completely removed, the two computations have:

- the same state;
- the same visible active stack suffix;
- the same deterministic local behavior.

The lower contexts γ_s and γ_t remain hidden.

Thus the common upper segment is removed after the same number

$$e_j$$

of consumed zeros.

Immediately after the final active symbol of that segment disappears, the two raw configurations have the form

$$(p_j, \gamma_s) \qquad \text{and} \qquad (p_j, \gamma_t)$$

for the same state p_j.

Because Q is finite, there is an infinite set of indices J on which

$$p_j = p$$

is constant.

For $j \in J$, the remaining accepted zero lengths are

$$R_s(j) = N_s(n_s + jc) - e_j$$

from the fixed raw configuration

$$(p, \gamma_s),$$

and

$$R_t(j) = N_t(n_t + jc) - e_j$$

from

$$(p, \gamma_t).$$

Unary completion uniqueness forces each of these two quantities to be constant on J.

Therefore

$$N_s(n_s + jc) - N_t(n_t + jc)$$

must be constant on J.

But

$$N_s(n) = \ell - s + (s+1)n,$$

so

$$\begin{aligned} &N_s(n_s + jc) - N_t(n_t + jc) \\ &= (s-t)cj + O(1). \end{aligned}$$

Because

$$s \neq t$$

and

$$c \geq 1,$$

this expression is unbounded along the infinite set J.

Contradiction. □

This lemma is the structural center of the hierarchy proof.

It says that a recurrent state/top mode has an externally measurable asymptotic identity:

$$\boxed{\text{its continuation slope.}}$$

Two different slopes cannot be hidden in the same up-mode.

11.3 The product lower bound

There are

$$\ell = mk + 1$$

slope rays.

Every slope ray has an up-mode.

Distinct slopes require distinct up-modes.

A DPDA with state set Q and stack alphabet Γ has at most

$$|Q||\Gamma|$$

nonempty modes.

Everything is now ready.

Theorem 11.3 — Product lower bound.

Let

$$m \geq 2, \qquad k \geq 1.$$

If a general deterministic pushdown automaton M, with arbitrary deterministic ε-moves allowed, recognizes

$$B_{m,k},$$

then

$$\boxed{|Q_M||\Gamma_M| \geq mk + 1.}$$

Proof.
There are

$$mk + 1$$

slope rays.
By the step-pump lemma, each has an up-mode.
By slope separation, no two slope rays share an up-mode.
Therefore M has at least

$$mk + 1$$

distinct nonempty state/top-symbol modes.
But there are at most

$$|Q_M||\Gamma_M|$$

such modes.
Hence

$$|Q_M||\Gamma_M| \geq mk + 1.$$

□
This is stronger than the adjacent hierarchy statement.
It excludes an entire hyperbolic resource region:

$$\boxed{|Q||\Gamma| \leq mk.}$$

11.4 The hierarchy theorem

Corollary 11.4 — Binary adjacent stack-alphabet hierarchy.
For every

$$m \geq 2, \qquad k \geq 1,$$

$$\boxed{\mathcal{D}_{m,k} \subsetneq \mathcal{D}_{m,k+1}.}$$

The strictness is witnessed over the binary alphabet

$$\{0, 1\}.$$

Proof.
The realtime witness $U_{m,k}$ uses exactly:

$$m$$

states and

$$k + 1$$

stack symbols.
Therefore

$$B_{m,k} \in \mathcal{D}_{m,k+1}.$$

If

$$B_{m,k} \in \mathcal{D}_{m,k},$$

some recognizing DPDA would satisfy

$$|Q| \leq m, \qquad |\Gamma| \leq k.$$

Hence

$$|Q||\Gamma| \leq mk,$$

contradicting Theorem 11.3.

The non-strict inclusion follows by allowing an unused extra stack symbol. □

Thus deterministic ε-moves do not collapse the fixed-state stack-alphabet hierarchy in the stated general model for

$$m \geq 2.$$

11.5 What exactly has been resolved

The bibliographic claim must remain narrow.

The theorem closes the fixed-state stack-alphabet hierarchy case left open after Masopust's realtime result [16], under:

- the general deterministic model of this branch;
- the single-initial-symbol convention;
- fixed state budget
$$m \geq 2;$$
- binary witness alphabet.

The stateless case is not silently folded into the same theorem because its earlier formulation uses a different initial-store convention.

Thus the monograph should say:

$$\boxed{\text{the general-model fixed-state hierarchy is resolved here for } m \geq 2}$$

and separately acknowledge the known stateless result.

11.6 Why this is not a realtime lower bound in disguise

The upper witness is realtime.

The lower-bound opponent is not.

The arbitrary competitor may use deterministic ε-moves throughout.

The proof survives because it deliberately reasons at raw positions.

The chain is:

$$\boxed{\text{exact continuation slice}}$$

$$\Downarrow$$

$$\boxed{\text{no raw configuration repetition}}$$

$$\Downarrow$$

$$\boxed{\text{raw-height escape}}$$

$$\Downarrow$$

$$\boxed{\text{suffix-minimum step pump}}$$

$$\Downarrow$$

$$\boxed{\text{slope separation}}$$

$$\Downarrow$$

$$\boxed{|Q||\Gamma| \geq mk + 1.}$$

The first box is a property of the explicit binary witness.

Every later box concerns an arbitrary equivalent DPDA.

That is why the theorem is representation-independent in a way the earlier residual-frontier results were not.

11.7 Why the product, rather than states or symbols separately, appears

The proof never forces a slope into a unique state alone.

It forces it into a unique **mode**:

$$[q, X].$$

Therefore the natural capacity is:

$$\boxed{|Q||\Gamma|.}$$

This is exactly the resource unit the proof can count.

A machine may trade states against stack symbols.

The product theorem allows that trade.

What it forbids is packing more than one distinct completion slope into one recurrent up-mode.

Thus the proof yields a quantitative state–stack tradeoff rather than only a yes/no alphabet separation.

11.8 Scope and excluded claims

The theorem is deliberately scoped and does not assert the following stronger statements.

Not a full Pareto frontier

The bound

$$|Q||\Gamma| \geq mk + 1$$

alone does not prove that every point above the hyperbola is attainable.

It excludes one region.

It does not determine the complete state/stack Pareto frontier of the witness.

Not a bounded-push hierarchy theorem

The upper witness installs a fixed table word of length

$$\ell = mk + 1$$

in one header transition.

The general PDA model permits replacement by an arbitrary fixed stack word.

If one imposed a push-length bound, compiling the header would change the resource accounting.

No bounded-push hierarchy claim is made in this theorem; that model is outside the present scope.

Not an unrestricted lower bound for $L_{p,q}$

The witness was engineered so that its continuation slopes are maximally separable.

The theorem does not automatically transfer to the original rational counting languages.

The representation-independent state question for the natural languages $L_{p,q}$ is resolved later by protected phase pumping, which yields the exact unrestricted state bound.

Not a claim that mode arguments are themselves new

State/top modes and state–stack-symbol tradeoffs have a pre-existing literature.

The contribution here is the particular binary witness, the ε-robust slope argument, the product theorem, and the resulting resolution in the stated model.

11.9 The methodological meaning for the monograph

This branch began with a problem created by our own DPDA analysis.

The original machine and its successors showed that the same semantic quantity can be redistributed between:

- control state;
- stack symbol;
- stack height;
- normalization.

That made representation-scoped lower bounds suspicious.

The hierarchy proof answers that suspicion by abandoning the representation.

It asks only what continuations the language forces.

This produces a new lower-bound pattern:

> externally visible continuation rates $\rightarrow$ recurrent physical modes
> $\rightarrow$ representation-independent resource count.

This method is conceptually distinct from the raw residual frontier.

Both belong in the monograph because they answer different questions.

The two resulting boundaries. The representation-independent product obstruction has two distinct successors. The first asks for *exact attainable coordinates and the width of their Pareto frontier* against arbitrary deterministic ε-PDA representations; the second asks for a *decision method to synthesize or refute* epsilon-free realizations under fixed budgets. The former is not an epsilon-elimination theorem, and the latter does not resolve the unrestricted resource-existence problem. In the next two sections, proofs already given in the hierarchy chapters are cited rather than repeated; new resource and compactness arguments are proved in place.

11.10 Exact Pareto Geometry and Resource Thresholds for the Binary Hierarchy

The hierarchy established above gives more than a strict inclusion: its binary witness $B_{m,k}$ has a representation-independent lower bound $|Q||\Gamma| \geq mk + 1$ in the full deterministic ε-DPDA model. We now ask a sharper coordinatewise question. With the same *single physical initial symbol, no transition from empty store, and final-state-plus-empty-store* convention, call (q, g) attainable if a recognizer has at most q controls and at most g physical stack letters. Write $\mathrm{Pareto}(B_{m,k})$ for its coordinatewise minimal attainable points and

$$b_p(B_{m,k}) = \min\{g : (p, g) \text{ is attainable}\}, \qquad a_g(B_{m,k}) = \min\{p : (p, g) \text{ is attainable}\}.$$

The canonical realtime source and the explicit completion modes are exactly those already constructed in the preceding chapters; no second witness family or changed acceptance convention is introduced here. The following lemma records the quantitative profile that will be used by the new resource arguments. It is an immediate specialization of the earlier completion-ray computation; including its direct argument makes the subsequent one-state and corner estimates transparent.

Lemma 11.1 (Exact completion profiles)**.** *For $1 \leq j \leq \ell$ and $n, t \geq 0$, letting $p_j = 10^{j-1}$, we have*

$$p_j 1^n 0^t \in B_{m,k} \quad \Longleftrightarrow \quad t = N_j(n) := \ell - j + 1 + jn.$$

Proof. After p_j, the current visible symbol is X_j and $\ell - j + 1$ source-table letters remain. Each additional input 1 deposits j ordinary counter letters below the same visible mode. Subsequent zeros each remove precisely one physical symbol until the stack becomes empty. Every control state is final, so the exact completion length is the resulting height. A shorter suffix leaves a nonempty stack; a longer suffix tries to read input from the empty stack. □

11.10.1 Representation-independent lower bound and its arithmetic consequence

Theorem 11.2 (Product obstruction recalled)**.** *Every deterministic ε-DPDA recognizing $B_{m,k}$ has $|Q||\Gamma| \geq mk + 1$.*

Proof. This is the product theorem of the preceding hierarchy chapter (the product theorem established there). The proof there treats raw ε-steps, protected suffix-minimum positions, context-independent pumps, and pairwise distinct completion slopes. It applies without any restriction on the representation of the competing machine. We reuse that established proof rather than reproduce it. □

Lemma 11.3 (A convenient arithmetic minimality test)**.** *Suppose every recognizer of L satisfies $qg \geq C$. If (q, g) is attainable and*

$$q(g - 1) < C, \qquad (q - 1)g < C,$$

then $(q, g) \in \mathrm{Pareto}(L)$.

Proof. Each strict coordinatewise improvement is dominated by one of the immediate neighbors $(q, g - 1)$ or $(q - 1, g)$, both forbidden by the product bound. For $q = 1$ or $g = 1$, omit the nonexistent neighbor. □

Lemma 11.4 (Unary physical stack alphabets are insufficient)**.** *For every $m \geq 2$, $k \geq 1$, no deterministic ε-PDA with one physical stack letter and finitely many states recognizes $B_{m,k}$.*

Proof. We show that $B_{m,k}$ has exponentially many distinguishable right quotients represented by words of linear length. Here use *top-first* notation just within this proof. Let $x_s = (r_s, X_s)$, $0 \le s < \ell$, denote the zero-based canonical slope-mode list; after reading the header 1 the stack is $T = X_0X_1 \cdots X_{\ell-1}$ in the source state r_0, and $X_0 = A$. Write

$$v_A = 0^k 101, \qquad v_B = 0^{\ell-1}101.$$

The zero-pop, slope-one, and header rules show that either word, starting with a header T on top of an arbitrary lower stack R, re-creates that header and inserts one of the two blocks

$$W_A = A^k X_{k+1} \cdots X_{\ell-1}, \qquad W_B = A^{\ell-1}$$

beneath it, returning to the source header's post-1 state. These blocks both have length $\ell-1$ but differ: following their first k occurrences of A, W_A has the non-A symbol $X_{k+1} = M_1$, while W_B still has A. (Since $m \ge 2$, $\ell-1 = mk > k$.) Iterating gives, for every $b \in \{A, B\}^n$, a live prefix

$$P_b = 1v_{b_1} \cdots v_{b_n}$$

whose source stack is $TW_{b_n} \cdots W_{b_1}$ in the same source state. All such prefixes have length at most $1 + n(\ell + 2)$ and their stacks have equal height $\ell + n(\ell - 1)$.

These stacks also have *different continuation languages.* For two different b, b', erase the common full top blocks by zeros and then erase the next k occurrences of A in their first differing block. Each complete block has the same length, and after popping it the source control has returned to the header state, so the two erased prefixes stay synchronized. The same number of zeros is read in both runs, and both source controls are in the header state. The tops are respectively A and $M_1 = X_{k+1}$; all zero erasures of complete blocks reset the header state. One following 1 increases stack height by $\ell - 1$ at the header A, whereas at M_1 it increases height by $k + 2$. If these increments coincide, supply *two* following ones instead: their total increments become ℓ and $2(k+2)$, which cannot coincide simultaneously with $\ell - 1 = k + 2$. Therefore some common continuation of the form 1 or 11 produces different source stack heights. Supplying exactly enough zeros to empty the first of these stacks accepts on that side (every source zero rule pops one letter and all source controls are final), but not on the other. Hence the 2^n prefixes P_b have pairwise distinct right quotients.

Now suppose an equivalent deterministic ε-PDA uses only the stack letter X, with q control states and maximal replacement length $R < \infty$. Along every *live* prefix and its continuation, the forced ε-closure at a completed input boundary cannot diverge or empty the store while further input is needed. A finite forced ε-segment from positive stack height to a positive-height settled configuration has at most q steps: on a unary stack each ε-enabled control prescribes the same height increment whenever it recurs at positive height. Repeating that control would either continue the same forced cycle without a settled positive-height exit, or eventually empty the stack; neither is possible at a live boundary needing a further real input. Thus the positive-height enabled controls cannot recur before settlement. Thus after at most h consumed input symbols, the normalized unary height is bounded by

$$1 + [h + q(h + 1)] \max\{R - 1, 0\} = O(h).$$

There are at most q control states at each such height and hence only $O(h)$ possible normalized configurations. We need this estimate only for live prefixes admitting nonempty extensions: on these runs an empty stack or an infinite forced-ε suffix before the next genuine input is impossible. In a deterministic machine each normalized configuration determines one right quotient. Taking $h \le 1 + n(\ell + 2)$ contradicts the 2^n distinct right quotients constructed above as $n \to \infty$. □

11.10.2 Constructive resource conversions

Proposition 11.5 (Typed state erasure for the witness)**.** *For $\chi = \min(m, 3)$, $B_{m,k}$ admits a one-state realtime recognizer with $d = mk + \chi$ physical stack symbols and the same singleton initial stack convention.*

Proof. Replace each source stack mode by a typed letter (q, X) recording the source state in which the letter is to be exposed. Each typed letter also has a uniquely prescribed state after its next zero-pop. Maintain the invariant that the state attached to each lower typed symbol equals the prescribed return state of the symbol directly above it. The initial letter is typed, and the long header rewrite and all slope self-rewrites preserve this return invariant. Each zero removes the top typed symbol and exposes the next one in exactly the state required by the source computation. The unique control state of the typed recognizer may therefore be fixed, with state changes encoded in these return types. This is a step-by-step simulation on all words, not a finite-word test.

There are mk marker-mode types. Among the letters of the form (q, A), only three transition roles occur: header, first-slope, and ordinary counter; the last role is absent for $m = 2$. Types sharing the ordinary-counter role have identical transitions and can be identified. The number of remaining physical letters is $mk + \min(m, 3)$, and the initial physical stack is still a single header letter. □

Theorem 11.6 (g-ary stack alphabet recoding)**.** *If a deterministic ε-PDA has n control states and $d \geq 2$ stack symbols, then for any $g \geq 2$ an equivalent machine exists with at most*

$$n \left\lceil \frac{d-1}{g-1} \right\rceil \quad \text{states and} \quad g \text{ stack symbols.}$$

The recoding preserves final-and-empty acceptance and the single physical initial stack letter, but not necessarily realtime behavior or bounded rewrite lengths.

Proof. Put $t = \lceil (d-1)/(g-1) \rceil$. Construct a prefix-free code of d source letters on g target letters using a caterpillar prefix tree with exactly t internal nodes: from each of its first $t-1$ internal nodes, expose $g-1$ leaf children and one continuing child; the last internal node has the remaining leaf children. One source letter can be assigned a one-letter root leaf labelled 0; assign it to the *actual* singleton initial source letter. Each source letter has a unique leaf code, and exactly t code prefixes need to be remembered by the finite control.

The target states are pairs (q, u), with q a source control state and u an internal tree node. At a non-leaf code continuation, a forced ε-move pops the physical code digit and changes u. At the final digit, the simulator executes the corresponding source move and replaces the decoded source symbol with the concatenated codes of the source replacement word, resetting u to the root. For right-top stack notation, stored codewords are reversed so digits are inspected in prefix order. Every configuration at a fully decoded boundary represents precisely the corresponding source configuration. At non-boundary decoder sites only an ε-move exists, so the target remains deterministic. The exact simulation preserves physical emptiness and final state at the end of input. The initial source letter's one-digit code ensures the initial physical stack has just one symbol. □

Corollary 11.7 (Universal resource bounds)**.** *Let $L = mk + 1$, $\chi = \min(m, 3)$, and $D = mk + \chi - 1$. For $p \geq 1$ and $g \geq 2$, respectively,*

$$\left\lceil \frac{L}{p} \right\rceil \leq b_p(B_{m,k}) \leq 1 + \left\lceil \frac{D}{p} \right\rceil, \tag{11.1}$$

$$\left\lceil \frac{L}{g} \right\rceil \leq a_g(B_{m,k}) \leq \left\lceil \frac{D}{g-1} \right\rceil. \tag{11.2}$$

A second bound from the canonical machine is

$$a_g(B_{m,k}) \leq m \left\lceil \frac{k}{g-1} \right\rceil.$$

Proof. The lower bounds follow from Theorem 11.2. Apply Theorem 11.6 to the one-state source from Proposition 11.5 and to the canonical $(m, k+1)$ source for the upper bounds. To obtain (11.1), choose the target alphabet size $g = 1 + \lceil D/p \rceil$; then $\lceil D/(g-1) \rceil \leq p$. □

11.10.3 The exact one-state endpoint

For a word u with a unique accepting zero-only continuation, let $z(u^{-1}B_{m,k})$ denote the length of that continuation; when no such continuation exists we leave z undefined. We use the same notation $z(K)$ for a language K with a unique pure-zero word.

Lemma 11.8 (Direct canonical quotient profiles)**.** *Put $B = B_{m,k}$, $\ell = mk + 1$ and $P_h = (1^{h+1})^{-1}B$ for $h \geq 0$. Then*

$$B \cap 0\{0,1\}^* = \{0\}, \qquad \varepsilon, 1, 10 \notin B,$$

and

$$\begin{aligned} z(0^{-1}P_h) &= \ell + h - 1, \\ z((10)^{-1}P_h) &= \ell + h, \\ z((10)^{-1}B) &= \ell - 1. \end{aligned}$$

For every $n \geq 1$, the unique pure-zero completion of 1^n in B has length $\ell + n - 1$.

Proof. The source initial letter A has its sole enabled zero move to the empty stack, so $0 \in B$ and no longer word beginning with 0 belongs to B. Its header move on 1 installs a stack of exactly $\ell \geq 3$ letters. Hence $\varepsilon, 1, 10 \notin B$. After 1^{h+1}, the source has taken the header move and applied the first-mode slope-one self-rewrite h times; it still exposes $X_1 = A$ in state q_2, with stack height $\ell + h$. Its zero move removes X_1 and leaves a height $\ell + h - 1$ stack that is completely drained by exactly that many zeros. If instead another 1 is read before that zero, the slope-one move inserts one additional A, yielding after the zero a height $\ell + h$ stack; it is drained by exactly $\ell + h$ zeros. These prove the first two displayed z identities. Setting $h = 0$ in the first gives the $(10)^{-1}B$ identity. The all-ones completion is the first-mode affine profile in Lemma 11.1. □

Lemma 11.9 (One-state stack concatenation)**.** *Let M be any one-state deterministic ε-PDA, with one final control state and physical-empty acceptance. If stacks are written* top first*, then for all stack words U, V,*

$$L_M(UV) = L_M(U)\, L_M(V).$$

Each nonempty $L_M(W)$ is prefix-free; in particular it contains at most one pure-zero word. If two such languages have pure-zero words, their concatenation has exactly the sum of their pure-zero lengths.

Proof. In any accepting computation from UV, the first time the symbols descending from the upper block U have all been erased separates the input into a word emptying U and a word emptying V. Until that first erasure the lower block is neither inspected nor modified, even in the presence of forced ε moves or arbitrary finite replacements. The split may occur between real inputs or during a forced ε closure; it still leaves the same sole control state and the unaltered physical lower stack. Because that state is final, both subcomputations independently accept from U and V under the same final-plus-empty convention. Conversely, they concatenate, since the lower block remains invisible until the upper block is gone. This gives equality. For prefix-freeness, an accepting computation on a shorter word empties the stack, leaving no operation available to consume additional input; determinism precludes another accepting computation from the same stack on its proper extension. □

Theorem 11.10 (Exact one-state stack threshold)**.** *For all $m \geq 2, k \geq 1$,*

$$b_1(B_{m,k}) = mk + \min(m, 3).$$

Consequently $(1, mk + \min(m, 3))$ is Pareto-minimal.

Proof. The upper bound is Proposition 11.5. For the converse let M be any one-state recognizer, with initial physical letter H. By Lemma 11.9, $L_M(H) = B := B_{m,k}$. Theorem 11.2 produces ℓ distinct protected up-mode *letters* $X_1, \ldots, X_\ell$ on the ℓ affine-slope rays; one state makes the mode-letter correspondence injective.

Every protected X_j has an accepting continuation beginning in 1. Indeed, its context-independent protected pump consumes a positive number of ones before its upper layer can be erased. Thus $\varepsilon \notin L_M(X_j)$; otherwise determinism would force the isolated X_j-layer to be erased on ε before the pump consumed input. At the live protected occurrence, an accepted continuation begins with 1. Factorization in Lemma 11.9 then forces an accepted word of $L_M(X_j)$ beginning with 1. The header H is likewise 1-live because B contains words beginning with 1.

We first show $H \neq X_j$ for all j. The required distinction concerns the physical *initial* letter, not merely a source-level header label; equality would make its full future language B available whenever that letter became exposed. On a higher-slope ray ($j \geq 2$), an occurrence of $X_j = H$ above a fixed lower stack W would give the continuation language $BL_M(W)$. The accepted zero continuations along a one-pumping run would have slope 1, since the initial 1^n cannot finish the leading B factor, and the only pure-zero completion after 1^n inside B has length $\ell + n - 1$ by Lemma 11.8; the lower contribution is constant. This contradicts the ray's slope $j \geq 2$.

For the slope-one ray, suppose its protected letter also equals H. At an occurrence after 1^{h+1} real input ones (possibly inside the forced ε-segment at that input boundary), with lower stack W, the residual language is

$$P_h = (1^{h+1})^{-1}B = BL_M(W).$$

Since B has only the zero-starting word 0, taking a left quotient by 0 yields

$$L_M(W) = 0^{-1}P_h, \qquad z(L_M(W)) = \ell + h - 1.$$

Neither 1 nor 10 is a complete word of B, so taking the left quotient by 10 in $BL_M(W)$ does not split that prefix across the two factors:

$$(10)^{-1}P_h = ((10)^{-1}B)L_M(W).$$

The right-hand side has a pure-zero word of length

$$(\ell - 1) + (\ell + h - 1) = 2\ell + h - 2.$$

Lemma 11.8, however, gives $z((10)^{-1}P_h) = \ell + h$. The difference is $\ell - 2 > 0$, so $H \neq X_1$. Altogether M requires $\ell + 1 = mk + 2$ physical letters.

For $m \geq 3$ there is one further unavoidable letter. Put $u = 10^{2k}10$. In the source machine, the first $2k$ zeros after the header select the final marker of the second group of k markers. Its input-1 rule inserts $2k + 1$ counter letters. The next zero pops the marker and enters source state q_3, exposing a counter A for which input 1 is undefined. The exposed A is a genuine zero-live but input-1-dead physical obligation. By Lemma 11.9, a top letter with an accepted continuation beginning with 1 could not represent such a stack: an accepted lower-stack suffix could be appended. The resulting stack height is exactly ℓ, so

$$u0^\ell \in B, \qquad u1v \notin B \quad (v \in \{0,1\}^*).$$

The continuation language after u is live, contains a pure-zero word, and contains no word starting in 1. The latter is a property of the entire right quotient, not merely of a convenient canonical path: every rival machine recognizing B has the same residual language after the input prefix u. At the corresponding configuration of M, let Y be the upper physical letter. By Lemma 11.9, $L_M(Y)$ is nonempty and has no word starting in 1; otherwise one could append an accepted lower-stack continuation to obtain an accepted word starting in 1. Thus Y cannot be the 1-live letter H or any X_j. It supplies an additional distinct physical letter, making at least $\ell + 2 = mk + 3$ in all $m \geq 3$ cases. Since $\min(m, 3) = 2$ for $m = 2$ and 3 otherwise, the result follows. □

11.10.4 Exact resource thresholds and Pareto corners

Theorem 11.11 (Exact state threshold strips)**.** *Fix $m \geq 2, r \geq 1, g \geq 2$. If*

$$rg - \left\lfloor \frac{g}{m} \right\rfloor \leq k \leq r(g-1),$$

then $g \geq mr$ and

$$a_g(B_{m,k}) = mr.$$

Proof. The canonical compiler from Corollary 11.7 yields $a_g \leq m\lceil k/(g-1) \rceil \leq mr$. The lower end of the interval is precisely $mk \geq (mr-1)g$; hence $(mr-1)g < mk+1$, so Theorem 11.2 gives $a_g \geq mr$. The interval is nonempty only if $\lfloor g/m \rfloor \geq r$, equivalently $g \geq mr$. □

Theorem 11.12 (Exact divisor-corner Pareto points)**.** *For any integers $m \geq 2$, $r \geq 1$ and $g \geq mr$, set $k = r(g-1)$. Then*

$$\boxed{(mr, g) \in \mathrm{Pareto}(B_{m,k}).}$$

Equivalently, for every positive divisor $r \mid k$ such that $k \geq mr^2 - r$, the pair

$$\left(mr, \frac{k}{r} + 1\right)$$

is a genuine globally Pareto-minimal point of $B_{m,k}$.

Proof. The canonical source recoding gives a recognizer using $m\lceil k/(g-1) \rceil = mr$ states and g stack letters. Here $C = mk + 1 = mr(g-1) + 1$. Writing $q = mr$, we have

$$q(g-1) = C - 1 < C, \qquad (q-1)g = qg - g < C$$

because $g \geq q$. Lemma 11.3 therefore proves global Pareto minimality. The divisor formula follows by putting $g = k/r + 1$ and rearranging $g \geq mr$. □

Corollary 11.13 (Arbitrarily many explicitly located points)**.** *For each fixed $m \geq 2$ and $t \geq 1$ there exist infinitely many k with at least t* explicitly located*, pairwise distinct elements of* $\mathrm{Pareto}(B_{m,k})$*.*

Proof. Choose k divisible by $\mathrm{lcm}(1, \ldots, t)$ and with $k \geq mt^2$. Every $r \in \{1, \ldots, t\}$ divides k and satisfies $k \geq mr^2 - r$, so Theorem 11.12 applies. As r increases, the state count mr increases and the stack-letter count $k/r + 1$ decreases. □

Remark 11.14 (Exact points versus exact thresholds)*.* An equality $a_g = q$ does not by itself imply (q, g) is Pareto-minimal: there may be a plateau $a_{g-1} = a_g$. Theorem 11.12 proves both-coordinate minimality, while Theorem 11.11 asserts the exact threshold without claiming every interior point is on the frontier.

11.10.5 Uniform order of Pareto-frontier width

Theorem 11.15 (Uniform width bound)**.** *For every $m \geq 2, k \geq 1$, set $L = mk + 1$, $D = mk + \min(m, 3) - 1$ and $T = \lfloor \sqrt{L} \rfloor$. Then*

$$\boxed{\left\lceil \frac{T}{2} \right\rceil \leq |\,\mathrm{Pareto}(B_{m,k})| \leq 2\left\lceil \sqrt{D} \right\rceil - 1.}$$

In particular $|\,\mathrm{Pareto}(B_{m,k})| = \Theta(\sqrt{mk})$ with absolute constants independent of m, k.

Proof. For the state axis, (11.1) shows that for $3 \le p \le \min(m, T)$,

$$b_{p-2} \ge \left\lceil \frac{L}{p-2} \right\rceil > 1 + \left\lceil \frac{D}{p} \right\rceil \ge b_p.$$

Indeed, $L \ge p^2$ and $\chi = \min(m, 3) \le 3$ imply $2L > (p-2)(2p + \chi - 2)$ and thus $L/(p-2) > 2 + D/p$. The strict ceiling step uses $y > x + 2 \Rightarrow \lceil y \rceil > 1 + \lceil x \rceil$; each disjoint two-position block must therefore contain at least one strict decrease and hence a distinct state-wing Pareto point. If $m \ge k$, then $T \le m$, so this gives at least $\lceil T/2 \rceil$ points including the $q = 1$ endpoint.

For the alphabet axis, let $4 \le g \le \min(k, T)$. From (11.2),

$$\frac{L}{g-2} - \frac{D}{g-1} = \frac{L - (\chi - 2)(g-2)}{(g-2)(g-1)} > 1,$$

using $g^2 \le L$ and $\chi \le 3$. Hence $a_{g-2} > a_g$. Together with the attainable $g = 2$ endpoint and Lemma 11.4, this gives at least $\lfloor \min(k, T)/2 \rfloor$ distinct alphabet-wing points. When $k \ge m$, $T \le k$, and the $q = 1$ endpoint lies outside this wing because its alphabet budget exceeds k. Therefore $|\operatorname{Pareto}(B_{m,k})| \ge 1 + \lfloor T/2 \rfloor \ge \lceil T/2 \rceil$.

For the upper bound write $t = \lceil \sqrt{D} \rceil$. Every Pareto point belongs to the state wing $q \le m$ or the alphabet wing $2 \le g \le k$, since the canonical pair $(m, k+1)$ dominates any pair strictly outside these wings. If $t \le k - 1$, count at most t possible strict alphabet thresholds at $g = 2, \ldots, t+1$. Every subsequent strict drop consumes at least one unit of a_g, and therefore there can be at most $a_{t+1} - a_k$ additional drops. The state wing has at most m points, while $a_k \ge m + 1$ by the product bound. It follows that

$$|\operatorname{Pareto}(B_{m,k})| \le t + a_{t+1} - a_k + m \le t + \left\lceil \frac{D}{t} \right\rceil - 1 \le 2t - 1.$$

If instead $t \le m-1$, the symmetric state-wing argument gives at most $t + b_t - b_m$ state-wing points; using $b_t \le 1 + \lceil D/t \rceil$ and $b_m = k + 1$, and allowing at most $k - 1$ alphabet-wing points, we again obtain at most $2t-1$. Otherwise $t \ge m, k$, and the trivial $m+k-1 \le 2t-1$ bound suffices. The three possibilities are exhaustive: either $t \le k - 1$, or $t \le m - 1$, or both fail, in which case the integer t satisfies $t \ge k$ and $t \ge m$. This covers every parameter pair. □

11.10.6 Full state-coordinate support in a square-root wedge

Our uniform cardinality theorem does not say which state budgets support a minimal pair. The next theorem gives a complete state-*coordinate* support region, while deliberately making no claim to know the corresponding numeric stack minima b_p.

Theorem 11.16 (Every state coordinate is supported)**.** *Let $m \ge 4$ and put*

$$K_m = 2m + 2 - \left\lceil \sqrt{8(m+1)} \right\rceil.$$

If $k \ge K_m$, then

$$b_1(B_{m,k}) > b_2(B_{m,k}) > \cdots > b_m(B_{m,k}) = k + 1.$$

Consequently the m pairs (p, b_p) for $p = 1, \ldots, m$ are all members of the full unrestricted Pareto frontier.

Proof. Write $L = mk+1$, $N = mk+2$, and $\chi = \min(m, 3) = 3$. The product lower bound and the one-state-source compiler provide

$$b_{p-1} \ge \left\lceil \frac{L}{p-1} \right\rceil, \qquad b_p \le 1 + \left\lceil \frac{N}{p} \right\rceil \quad (2 \le p < m),$$

while the canonical source realizes $b_m = k+1$. It is therefore sufficient to verify

$$\left\lceil \frac{L}{p-1} \right\rceil > 1 + \left\lceil \frac{N}{p} \right\rceil \quad (2 \le p < m), \qquad \left\lceil \frac{L}{m-1} \right\rceil > k+1. \tag{11.3}$$

If $k \ge 2m-4$, then for $2 \le p < m$,

$$L \ge m(2m-4)+1 \ge (p-1)(2p+1).$$

Thus $L/(p-1) - N/p = [L-(p-1)]/[p(p-1)] \ge 2$, and the first test follows. Moreover $L-(m-1)(k+1) = k-m+2 > 0$, so the last test follows.

It remains to consider $K_m \le k \le 2m-5$. Set

$$c = 2m-k, \quad d = \lceil \sqrt{8(m+1)} \rceil, \quad c_* = d-2.$$

Then $5 \le c \le c_*$ and $(c+1)^2 \le (d-1)^2 < 8(m+1)$. For $2 \le p < m$ set $a = m-p+1$ ($2 \le a \le m-1$). Direct division gives

$$\frac{L}{p-1} = 2m + 2a - c + \frac{R(a)}{m-a}, \qquad R(a) = 2a^2 - ca + 1,$$
$$\frac{N}{p} = 2m + 2a - c - 2 + \frac{U(a-1)}{m-a+1}, \qquad U(b) = 2b^2 - cb + 2.$$

Therefore the first test in (11.3) is equivalent to

$$\left\lceil \frac{R(a)}{m-a} \right\rceil \ge \left\lceil \frac{U(a-1)}{m-a+1} \right\rceil. \tag{11.4}$$

By completing the square,

$$R(a) + (m-a) = 2a^2 - (c+1)a + (m+1) \ge m + 1 - \frac{(c+1)^2}{8} > 0,$$

so the left-hand ceiling in (11.4) is at least zero. If $U(a-1) \le 0$, the right-hand ceiling is at most zero. Otherwise put $b = a-1 \ge 1$. The inequality $U(b) > 0$ implies $b > (c-2)/2$, since otherwise $U(b) \le -2b+2 \le 0$. Hence $a > c/2$ and

$$R(a) - U(a-1) = 4a - c - 3 > c - 3 \ge 2.$$

The numerator $R(a)$ is strictly larger than the positive numerator $U(a-1)$, while its denominator is smaller, so (11.4) again follows. Finally $c \le c_* \le m+1$ gives $L-(m-1)(k+1) = m-c+2 > 0$. All comparisons hold, and each strict drop yields a Pareto point by coordinatewise minimality. □

The threshold above is not asserted to be the earliest k at which the *actual* DPDA state wing becomes fully supported. The following optimality statement concerns exactly the numerical sufficient comparisons used in the proof, a distinction important for avoiding unwarranted optimality claims.

Proposition 11.17 (Exact threshold for this arithmetic certificate)**.** *For each $m \ge 4$, the conjunction of all inequalities in* (11.3) *holds if and only if $k \ge K_m$.*

Proof. The forward direction for $k \ge K_m$ was proved above. Suppose $k < K_m$ and put $c = 2m-k$. If $k \le m-2$, then $L-(m-1)(k+1) = k-m+2 \le 0$, so the last test fails. We may therefore assume $m-1 \le k < K_m$, equivalently $c_*+1 \le c \le m+1$.

Let $n = 8(m+1)$ and $d = \lceil \sqrt{n} \rceil = c_*+2$. Choose a_0 to be a nearest integer to $d/4$; for $m \ge 4$ one has $2 \le a_0 \le m-1$. Put $e = 4a_0 - d$. The value $e^2 \in \{0,1,4\}$ is the least nonnegative integer congruent to d^2 modulo 8: taking a_0 nearest to $d/4$ makes $|4a_0 - d|$

minimal, and the square of an integer congruent to d modulo 4 has this least residue. Since $d^2 - n \geq 0$ and n is divisible by 8, we have $d^2 - n \geq e^2$, and consequently

$$8\big(2a_0^2 - da_0 + m + 1\big) = e^2 - (d^2 - n) \leq 0.$$

Define $T_c(a) = 2a^2 - (c+1)a + m + 1$. As $c + 1 \geq d$, the preceding inequality gives $T_c(a_0) \leq 0$, whereas $T_c(1) = m + 2 - c \geq 1$. Choose the first integer $a \in \{2, \ldots, a_0\}$ with $T_c(a) \leq 0$. Then $T_c(a-1) > 0$, and

$$R(a) + (m - a) = T_c(a) \leq 0, \qquad U(a-1) + (m - a + 1) = T_c(a-1) + 1 > 0.$$

Thus the left ceiling in (11.4) is at most -1 and the right ceiling at least 0, so the corresponding test fails. This proves the equivalence. □

Corollary 11.18 (Sharper width on the supported-state region)**.** *For $m \geq 4$, $k \geq K_m$, and $T = \lfloor \sqrt{mk+1} \rfloor$,*

$$|\operatorname{Pareto}(B_{m,k})| \geq m + \left\lfloor \frac{\min(k, T)}{2} \right\rfloor.$$

Proof. Theorem 11.16 provides m distinct Pareto minima with at most m states and hence at least $k + 1$ stack letters. The alphabet-wing two-step argument in the proof of Theorem 11.15, together with the unary-alphabet impossibility lemma, supplies $\lfloor \min(k, T)/2 \rfloor$ distinct minima with at most k stack letters. The two groups are disjoint. □

Remark 11.19 (Weaker and stronger numerical certificates). The sufficient certificate established here uses only $qg \geq mk + 1$. A stronger product bound, if separately established, may yield further numerical Pareto points; no such strengthening is used in this paper.

Remark 11.20 (Model-dependent literature comparison). The classical recodings cited in the introduction do not directly state the exact corner theorem or the uniform frontier-width theorem for the concrete $B_{m,k}$ witness. The present results depend on the precise deterministic and simultaneous final-and-empty model. We do not claim novelty for generic state–stack alphabet recodings.

11.10.7 Illustrative exact points and scope

For $B_{3,60}$, Theorem 11.12 certifies $(3, 61)$, $(6, 31)$, $(9, 21)$, and $(12, 16)$. For $B_{5,120}$ it certifies $(5, 121)$, $(10, 61)$, $(15, 41)$, $(20, 31)$, and $(25, 25)$. These are subsets of the full frontier, not complete coordinate lists. Theorem 11.16 also guarantees full state-coordinate support for $B_{12,15}$ and for $B_{100,173}$, implying respectively at least 18 and 165 Pareto points by Corollary 11.18; these are cardinality lower bounds, not complete coordinate sets.

The product inequality used here is $qg \geq mk + 1$. A potentially stronger product inequality would alter some small arithmetic corners, but is not required for the stated results and is not invoked.

11.10.8 Scope and remaining resource questions

The exact all-parameter two-resource Pareto classification of $B_{m,k}$ remains open. The current results identify infinite exact subfamilies, prove the uniform cardinality order, certify every state coordinate in a large parameter region, and specify some exact thresholds. The conventional state-alphabet exchange mechanisms used constructively here are not asserted to be new. The 1982 and 1993 state/stack conversion papers and the 2020 complexity study are central comparators. The exact coordinates of every Pareto point for arbitrary (m, k) remain unknown. Related open questions include the complete characterization of plateaux in a_g and b_q, as well as extensions to restricted replacement lengths.

11.11 Epsilon-Free Fixed-Resource Existence and Zero-Drain Compactness

The previous Pareto geometry allows deterministic ε-moves. A different, algorithmic resource question arises when *only ε-free competing machines* are admitted: for fixed m, k, q, g, can any equivalent recognizer of $B_{m,k}$ be built with at most q controls and g physical stack letters when each real-input transition may replace its top by an arbitrarily long finite word? Although each concrete transition table is finite, fixed (q, g) alone does not bound its replacement lengths. This question is therefore not settled by a naive finite enumeration. We retain the canonical $U_{m,k}$ and its mode order (r_j, X_j) from the earlier hierarchy construction, with $\ell = mk + 1$ and all source controls final. For fixed budgets define

$$\mathsf{SAT}_{\mathrm{rt}}(m, k, q, g) \iff \text{there is an equivalent } \varepsilon\text{-free DPDA at resources} \le (q, g).$$

All physical-empty/final-state and initial-singleton conventions continue unchanged. In particular, *competitor* transitions on input 0 need not be pops, and no bounded-push hypothesis is used. The source family and its general ε-enabled lower bound are not new assumptions.

11.11.1 Finite-ones language calibration

For each $H \ge 0$, let

$$R_H = \{w \in \{0,1\}^* : \#_1(w) \le H\}, \qquad S_H = B_{m,k} \cap R_H, \qquad K_H = \max(1, \ell H).$$

The set R_H is regular and infinite, whereas the positive language slice S_H will be finite.

Lemma 11.21 (Zero completions of live source prefixes)**.** *For every prefix u of any word accepted by $U_{m,k}$, its unique reached source configuration has some stack height $a(u) \ge 0$ and*

$$u0^{a(u)} \in B_{m,k}, \qquad a(u) \le K_H \quad \text{if } \#_1(u) \le H.$$

For $H \ge 1$, every $v \in S_H$ has at most ℓH zeros and therefore $|v| \le H(\ell + 1)$. For $H = 0$, $S_0 = \{0\}$.

Proof. Every defined source input-0 move removes precisely one stack letter, and every source control state is final. Therefore $a(u)$ zeros drain exactly the reached stack, and no shorter or longer pure-zero suffix can be accepted. Initially $a(\varepsilon) = 1$. The initial input 1, if used, changes the height from 1 to ℓ. Every later defined 1-move increases height by some $j \in \{1, \dots, \ell\}$; a 0-move decreases it by one. Thus for any live prefix containing $h \ge 1$ ones, $a(u) \le \ell h$. If $h = 0$, the only live prefixes are the empty word and 0, with heights 1 and 0. The terminal stack of an accepted word is empty, so the number of zero moves does not exceed the initial height plus all increases from ones, which is at most ℓH for $H \ge 1$. The no-one language consists only of the singleton word 0. □

Lemma 11.22 (Finite positive slices)**.** *Each S_H is finite and effectively enumerable by testing the finitely many words of length at most $H(\ell + 1)$ for $H \ge 1$ and by $S_0 = \{0\}$ for $H = 0$.*

Proof. Immediate from Lemma 11.21 and effective membership in $U_{m,k}$. □

11.11.2 Deletion-only normalization

Given an epsilon-free table M and a positive integer K, let $\pi_K(M)$ be the same machine with the same physical state and alphabet labels, initial site and final set, but *with every real-input transition of replacement-word length greater than K deleted.* No other transition is modified.

Lemma 11.23 (Finite-horizon push normalization)**.** *Suppose an epsilon-free deterministic PDA M agrees with $B_{m,k}$ on R_H. Then $\pi_{K_H}(M)$ still agrees with $B_{m,k}$ on R_H:*

$$\mathcal{L}(\pi_{K_H}(M)) \cap R_H = S_H.$$

This does not require M to pop on input 0*.*

Proof. Removing transitions cannot create an accepting computation. For preservation of positive words, take any $v \in S_H$ and any prefix $u \preceq v$. By Lemma 11.21, the accepted target word $u0^{a(u)} \in S_H$ has $a(u) \le K_H$. Thus M accepts that word, and because it is deterministic and epsilon-free, the configuration of M after u in this computation is exactly the configuration reached after u in its accepting computation on v.

In any epsilon-free PDA, consuming one 0 can decrease the physical stack height by at most one: it replaces one visible symbol with a word of nonnegative length. Accepting the all-zero suffix $0^{a(u)}$ by a physically empty stack therefore implies

$$\mathrm{height}_M(u) \le a(u) \le K_H.$$

Consider a real input transition used on the accepting computation of v, and let u denote the prefix *after* that transition. Its replacement word is a contiguous top block of the new physical stack, so its length is at most $\mathrm{height}_M(u) \le K_H$. None of the transitions of the accepting computation on v is deleted. Therefore all positive words in S_H are preserved, while no new negative word can be accepted. □

Remark 11.24 (Why epsilon-free matters)*.* With silent transitions, the machine might erase an arbitrarily large stack before consuming the first zero of a correct zero-only completion. Its height at u would then not be bounded by $a(u)$, and the preceding deletion argument would fail. No fixed-resource epsilon-elimination construction is supplied here.

11.11.3 Finite-level decision and compactness

Write $\mathsf{FEAS}_H(q, g)$ if there is an epsilon-free at-most-(q, g) table M such that

$$\mathcal{L}(M) \cap R_H = S_H.$$

We may pad controls and alphabet letters that are unused, so it suffices to consider tables over fixed labeled sets of exactly q controls and g letters.

Lemma 11.25 (Effective finite-horizon consistency)**.** *The Boolean value $\mathsf{FEAS}_H(q, g)$ is decidable for each fixed H, m, k, q, g.*

Proof. By Lemma 11.23, if FEAS_H holds then a witnessing table has no replacement word longer than K_H. There are only finitely many such tables: at each of the $2qg$ input/state/top slots we choose either an undefined transition or one of $q\sum_{j=0}^{K_H} g^j$ possible state/replacement pairs. There are also finitely many choices of final subset and initial labels.

For each candidate, acceptance of every word in the finite set S_H is effectively checked. To exclude all false positives without enumerating the infinite set $R_H \setminus S_H$, construct a DFA for that regular language and decide whether the intersection of its language with $\mathcal{L}(M)$ is empty. Intersection of a PDA language with a regular language is effectively context-free, and emptiness of a context-free language is decidable. Every candidate can thus be tested exactly, and only finitely many candidates exist. □

For $H \ge 0$, denote by T_H the *finite* family of all labeled tables with pushes of length at most K_H and with the correct R_H-behavior. The next statement is the compactness core.

Theorem 11.26 (Deletion compactness)**.** *For all fixed m, k, q, g,*

$$\mathsf{SAT}_{\mathrm{rt}}(m, k, q, g) \iff \forall H \ge 0\ \mathsf{FEAS}_H(q, g).$$

Proof. The forward implication is immediate. For the converse, assume every T_H is nonempty. If a table $M \in T_{H+1}$, applying the deletion map π_{K_H} yields an element of T_H: indeed M agrees with $B_{m,k}$ on R_{H+1} and hence on R_H, and Lemma 11.23 applies. Therefore the maps

$$\pi_{K_H} : T_{H+1} \longrightarrow T_H$$

are well defined. All T_H are finite. Form the forest whose vertices at level H are tables in T_H, with each vertex at $H+1$ joined to its image at H. Since T_0 is finite, this is a finitely branching forest with vertices at arbitrarily large levels; Koenig's lemma gives an infinite compatible branch

$$M_0, M_1, \ldots, \quad \pi_{K_H}(M_{H+1}) = M_H.$$

Each of the $2qg$ possible real-input transition slots has a simple behavior along this branch. If a transition is defined at level H, its replacement word has length at most K_H and therefore survives every subsequent deletion projection. It must remain *the same transition* at all later levels. Conversely an undefined slot may become defined, but once defined cannot change or be removed. It can make this change at most once. Administrative choices (initial site, final-state set) are preserved by every projection. Thus there exists an index H_* after which the entire labeled table is constant, say $M_H = M_*$ for every $H \geq H_*$. Because M_H agrees with $B_{m,k}$ on R_H, the fixed M_* agrees with the source on *every* R_H. Their union is $\{0,1\}^*$, so $\mathcal{L}(M_*) = B_{m,k}$. □

11.11.4 A terminating resource-existence decision procedure

Lemma 11.27 (Effective conversion of the acceptance convention)**.** *Given a deterministic PDA in the present model, one can effectively construct an ordinary final-state accepting deterministic PDA for the same language. This auxiliary conversion need not preserve either resource budget.*

Proof. Insert a fresh permanent bottom marker beneath the original singleton initial symbol. Simulate every original transition above this marker. For each original final control, when the marker becomes exposed, allow a single forced ε-move into a designated accepting sink without input-consuming transitions. If the marker is reached in an original nonfinal control, no such move is provided. No original transition inspects an empty physical store, so no ordinary action competes at the marker, and determinism is preserved. If unread input remains when the accepting sink is entered, that input cannot be consumed and hence the word is not accepted. Thus exactly the original input words ending in a final control with an empty physical store are accepted in the transformed convention. □

Theorem 11.28 (Epsilon-free fixed-resource decidability)**.** *For every $m \geq 2, k \geq 1$ and every $q, g \geq 1$, the statement $\mathsf{SAT}_{\mathrm{rt}}(m, k, q, g)$ is decidable. Competitor zero-transitions may be arbitrary finite replacements, and no common push-length bound is supplied.*

Proof. Run two effective processes in parallel.

1. *Positive semidecision:* enumerate all finite labeled epsilon-free at-most-(q, g) tables in increasing order of maximum replacement length. Use the decidability of deterministic PDA language equivalence [21] to compare each candidate against the explicit canonical source $U_{m,k}$ (using Lemma 11.27 to match the acceptance convention of the equivalence algorithm; this conversion need not preserve resource counts). If an equivalent machine exists, eventually find one and return YES together with its transition table.
2. *Negative semidecision:* for $H = 0, 1, 2, \ldots$ decide $\mathsf{FEAS}_H(q, g)$ using Lemma 11.25. At the first negative level return NO with the finite-level table-exhaustion certificate.

If a correct epsilon-free table exists, the positive process terminates. Otherwise Theorem 11.26 implies that some finite horizon is infeasible, so the negative process terminates. Exactly one of these outcomes is correct. The dovetail thus terminates in all cases. □

Corollary 11.29 (Some computable replacement bound exists)**.** *There is a total computable function $F(m, k, q, g)$ such that whenever an epsilon-free (q, g) recognizer for $B_{m,k}$ exists, one exists with all replacement lengths at most $F(m, k, q, g)$.*

Proof. Run the decision procedure. On a negative instance return 0. On a positive instance, enumerate until a globally equivalent table is found and return its maximum replacement length. The value is total and has the claimed existential property. No useful elementary or complexity estimate is asserted. □

11.11.5 Finite refutation packets

The negative part of Theorem 11.28 can be made independently checkable after a finite horizon is found. This is useful because a report that "all tables failed" is not, by itself, a reproducible mathematical certificate.

Definition 11.30 (Horizon-refutation packet)**.** For fixed integers (m, k, q, g, H), let $\mathcal{C}_H$ be the explicit finite set of *labeled* epsilon-free tables with the specified budgets and replacement words of length at most K_H. A *horizon-refutation packet* is a finite set of binary words $W \subseteq R_H$ such that for each $M \in \mathcal{C}_H$ there exists a word $w \in W$ on which M and $U_{m,k}$ disagree. An equivalent indexed representation stores one witness w_M for every table M.

Theorem 11.31 (Finite verifiable negative certificates)**.** *For every fixed (m, k, q, g) the following statements are equivalent:*

(i) no epsilon-free machine with the prescribed resources recognizes $B_{m,k}$;

(ii) for some $H \geq 0$ there exists a horizon-refutation packet (H, W).

Given such a packet, its validity can be verified by finite enumeration and ordinary finite-word simulation; a deterministic-PDA equivalence oracle is not *needed for certificate checking. One can choose $|W| \leq |\mathcal{C}_H| \leq qg\, 2^q \big(1 + q \sum_{j=0}^{K_H} g^j\big)^{2qg}$ by assigning at most one counterexample per table.*

Proof. Assume (i). By Theorem 11.26, for some H the finite table family T_H is empty. Thus every table $M \in \mathcal{C}_H$ disagrees with $U_{m,k}$ on at least one word $w_M \in R_H$. Because $\mathcal{C}_H$ is finite, selecting one such word per table produces a finite packet. The witnesses are effective: if a candidate misses a word of finite S_H, use that word; otherwise intersect the candidate language with the regular set $R_H \setminus S_H$, and extract an accepted word from a nonempty PDA–DFA intersection.

Conversely, let (H, W) be a valid packet and suppose a globally equivalent epsilon-free M existed. Since M agrees with the source throughout R_H, Lemma 11.23 would give a bounded table $\pi_{K_H}(M) \in \mathcal{C}_H$ agreeing on R_H, including every word of W. This contradicts the packet. To verify a claimed packet, an independent checker enumerates every table of $\mathcal{C}_H$, recomputes membership of each finitely listed word in the explicit source, and checks that each candidate fails on at least one word. No infinite-input or global-equivalence assertion is needed in this final verification step. The mathematical reduction from arbitrary rewrite lengths still rests on Lemma 11.23; the packet *alone*, divorced from that lemma, must not be mistaken for an exhaustion of unbounded tables. □

Remark 11.32 (Finite negative witnesses versus a cutoff)*.* The packet may contain relatively long words or many separate counterexamples; no efficient size or complexity upper bound is claimed. It is a proof-carrying *negative* output. A successful bounded table for one finite horizon is *not* a global positive certificate: the positive output is a genuinely equivalent finite machine, established using deterministic-PDA equivalence.

Corollary 11.33 (Constructive yes/no outcomes)**.** *For the family $B_{m,k}$ and fixed q, g, the resource-existence decision can be organized to return either (i) a concrete equivalent epsilon-free transition table, or (ii) a finite horizon H and a finite input-word packet $W \subseteq R_H$*

refuting every candidate transition table bounded by K_H. The correctness of a NO *output is checkable using only finite-word simulation and the normalization theorem; checking a* YES *output uses deterministic-PDA equivalence.*

Proof. On a positive instance, enumerate candidate tables and stop at the first one certified equivalent by the deterministic-PDA equivalence procedure. On a negative instance, compactness gives a first infeasible horizon; the construction in Theorem 11.31 extracts one finite counterexample for every bounded table and packages them as W. □

11.11.6 A calibrated zero-drain principle

The proof can be separated from the exact grammar of $B_{m,k}$.

Definition 11.34 (Effective zero-drain calibration)**.** Let $T \subseteq \{0,1\}^*$ be a language given by an effective deterministic pushdown automaton under the same physical empty-store/final-state acceptance convention. A *zero-drain calibration* is an algorithm which on input H computes a positive integer K_H such that $(K_H)_{H\geq 0}$ is nondecreasing and has the following property: for every $H \geq 0$, every accepted word $v \in T \cap R_H$, and every prefix $u \preceq v$, there is some $a \in \{0, \ldots, K_H\}$ with $u0^a \in T$.

Lemma 11.35 (Drain calibration already implies finite sections)**.** *Every effectively zero-drain calibrated language T in this model has finite, effectively enumerable sections $T \cap R_H$. In fact, every $v \in T \cap R_H$ satisfies*

$$|v| \leq H + (H+1)K_H.$$

Proof. Every language accepted by this physically-empty-stack/no-empty-move model is prefix-free: if an input word has drained the store and been accepted, no proper extension can also be accepted. Decompose an accepted word v with $h \leq H$ ones into its $h+1$ maximal blocks of zeros separated by ones. Immediately before any zero block, at a prefix $u \preceq v$, there is by calibration an accepted completion $u0^a$ with $a \leq K_H$. If the block contains more than a zeros, then $u0^a$ is a proper accepted prefix of v, impossible. Thus every zero block has length at most K_H and $|v| \leq h + (h+1)K_H \leq H + (H+1)K_H$. Testing the finitely many words of that length against the given target DPDA enumerates $T \cap R_H$. □

Theorem 11.36 (General calibrated resource decision)**.** *Given a deterministic PDA for T with the acceptance convention of Section 11.11,* together with an algorithm computing a valid zero-drain calibration (K_H)*, and any $q, g \geq 1$, one can decide whether T has an equivalent epsilon-free recognizer with at most q states and g stack letters, with arbitrary finite replacement words. In particular, for every fixed effectively calibrated T, this resource-existence predicate is decidable.*

Proof. In Lemma 11.23 use the accepted drain $u0^a \in T$ supplied by calibration in place of the canonical source height $a(u)$. The height inequality for every epsilon-free competitor remains $\mathrm{height}_M(u) \leq a \leq K_H$. Truncation, finiteness of the candidate tables, the finite positive set from Lemma 11.35, and regular-language emptiness for all false positives give decidable H-consistency. The deletion-only compactness proof is unchanged. Finally, the given target is a deterministic PDA, so the positive enumeration can use deterministic PDA equivalence. Dovetail the two semidecision procedures as before. □

Remark 11.37 (Status and limits of the general criterion). Theorem 11.36 is a logical abstraction of the proof for $B_{m,k}$, not a claim of priority or of a classification of all languages admitting such calibrations. Its hypotheses impose a strong *accepted-prefix* zero-completion property, not simply a short zero-completion from a few chosen source rays. The calibration algorithm is part of the *effective promise*: the theorem does not assert an algorithm detecting this property from an arbitrary given target DPDA without a supplied calibration. There is no claim about epsilon-enabled competitors. Sénizergues' theorem is used for the positive existence branch, while all finite negative-horizon checks require only ordinary PDA emptiness in the calibrated setting.

11.11.7 A transparent finite infeasibility certificate

An explicit small example illustrates why a negative H-test can be decisive without being a global machine enumeration.

Proposition 11.38. *No epsilon-free deterministic PDA with one state and one physical stack letter recognizes $B_{2,1}$. More precisely, $H = 2$ is its* first infeasible ones-horizon*: $\mathsf{FEAS}_0(1,1) = \mathsf{FEAS}_1(1,1) = \mathsf{true}$ and $\mathsf{FEAS}_2(1,1) = \mathsf{false}$.*

Proof. Here $\ell = 3$. By direct evaluation of $U_{2,1}$,

$$0 \in B_{2,1}, \qquad 1000 \in B_{2,1}, \qquad 110000 \in B_{2,1}.$$

Any epsilon-free one-control/one-letter table agreeing with these words must mark its sole control final. Accepting 0 forces its input-0 transition to replace the single symbol by ε. Accepting 1000 then forces the input-1 transition to replace the single symbol by three copies of itself. On 110000 the two ones leave five stack copies (one initial letter, plus two increments of two); four zeros remove only four copies, so this word is rejected. All three displayed positive words belong to R_2, contradicting R_2-agreement. The argument is independent of any finite push cutoff beyond those forced transitions. Indeed the three-word set $W = \{0, 1000, 110000\}$ is an $H = 2$ horizon-refutation packet: all three words are source-positive, and they jointly contradict every one-state/one-letter transition table. Under $K_2 = 6$, exactly $2(1+7)^2 = 128$ labeled tables are present (two possible final-state sets and eight choices for each real-input rule, including undefined); exhaustive simulation confirms that none accepts all three words.

For comparison, at $H = 1$ the one-state/one-letter epsilon-free machine with its sole state final, $X \xrightarrow{0} \varepsilon$ and $X \xrightarrow{1} X^3$, recognizes precisely the two source words 0 and 1000 among all words in R_1. Indeed an accepted word can contain no ones or exactly one one; the latter must be followed by exactly three zeros. Therefore $\mathsf{FEAS}_1(1,1)$ holds, and the same table works at $H = 0$. This proves the claimed minimality of the obstructing horizon. □

The proposition is deliberately pedagogical: it is not a new optimal resource threshold and does not establish the general $g = 1$ impossibility for all control budgets.

11.11.8 Computational Pareto case study for $B_{3,1}$

The abstract finite-ones decision procedure does not by itself identify a practically small obstruction horizon. A separate archived line of structural reductions and executable certificates (G72–G89) supplies a more demanding example. Its mathematical scope is the *epsilon-free* resource class, not arbitrary epsilon-enabled DPDAs. We now isolate three key necessary steps, including their exact logical and computational dependencies.

Root separation without a normal-form assumption

In the next two lemmas stack tops are written on the left. Put $\ell = mk + 1$, and $P_j = 10^j$ for $0 \le j < \ell$. The canonical $B_{m,k}$ source has the exact completion profiles

$$P_j 1^n 0^a \in B_{m,k} \quad \Longleftrightarrow \quad a = \ell - j + (j+1)n \qquad (n, a \ge 0).$$

The statement concerns actual physical control/top pairs of any competing machine; it does not assume source-aligned physical stack letters.

Lemma 11.39 (G72: physical root exclusion)**.** *In every equivalent epsilon-free deterministic pushdown automaton in the stated one-initial-letter final-and-empty model, the initial physical control/top site is different from the physical site exposed immediately after each P_j, $0 \le j < \ell$. The target may have any finite number of controls or stack symbols and arbitrary finite* 0- *and* 1*-rewrites.*

Proof. Write the initial configuration as (p, A), and suppose a clean-prefix configuration exposes the same site (p, A) above a fixed buried word β. For $j \geq 1$, consider the singleton-root computations on $1^n 0^{\ell+n-1}$, $n \geq 1$. Just after the 1^n part, their stacks are nonempty, or the subsequent zero suffix could never be consumed. Therefore, by deterministic top-only locality, starting instead at $(p, A\beta)$ performs the identical upper-layer 1^n computation without touching β. The subsequent $\ell + n - 1$ zero moves erase this common upper layer and enter some control f_n, which is final in the singleton-root computation. For the clean-prefix completion there must then be $\ell - j + (j+1)n - (\ell + n - 1) = j(n-1) + 1$ further zero moves emptying the *fixed* buried word β from control f_n. At a fixed control and a fixed physical stack there is at most one accepted pure-zero length. Since f_n ranges over finitely many controls, the possible lengths are bounded, contradicting $j(n-1) + 1 \to \infty$.

For $j = 0$, assume instead that the machine again exposes (p, A) after the initial input 1. Its first 1 rule must write $A\beta$ and return to p. The word 0 is accepted from (p, A), so the 0 rule pops A and enters some final state s. Here $\beta \neq \varepsilon$, because otherwise the configurations before and after reading 1 would coincide, whereas their unique accepted zero-completion lengths are 1 and ℓ. The physical configurations after 10 and 110 therefore have the same control s and stacks β and $\beta\beta$, respectively. The genuine source accepts the two words

$$10\,11\,0^{\ell+3} \qquad \text{and} \qquad 110\,11\,0^{\ell+2}.$$

From the first configuration, the physical 11 computation must retain a nonempty upper stack word W and then erase W in exactly $\ell + 3$ zeros. The same 11 steps from $(s, \beta\beta)$ cannot touch the buried second copy of β and therefore leave $W\beta$ in the same control. Its zero-only computation cannot erase the common W upper layer in merely $\ell + 2$ zeros. This contradicts the second genuine acceptance, completing the root exclusion. □

Two controls and the first shared-source input

Lemma 11.40 (G74: immediate POP at a shared clean site)**.** *Suppose an epsilon-free two-control DPDA correctly recognizes $B_{m,k}$. If two distinct clean prefixes P_i, P_j expose the same physical control/top site, then the input-*1 *rewrite at that site is the empty stack word. No global* 0*-POP hypothesis is required.*

Proof. The configurations have the form $(p, A\beta)$ and $(p, A\gamma)$. Suppose the site's first input-1 rule writes a nonempty physical word V. At counts $n = 0, 1, 2$ of further input ones, the upper-layer computations from the two configurations remain identical: at $n = 0$ the common upper word is A, at $n = 1$ it is V, and at $n = 2$ the second 1 rewrites the top of V in exactly the same way on both sides. The resulting upper word at $n = 2$ may be empty; that does not affect this argument. Denote the common upper word by W_n and the common control by s_n. An accepted all-zero suffix must remove W_n before inspecting either buried tail. The first-return zero cost and exit control e_n of this upper erasure are identical for both runs. Subtracting the unique accepted zero-completion lengths leaves

$$\tau_{e_n}(\beta) - \tau_{e_n}(\gamma) = \big[\ell - i + (i+1)n\big] - \big[\ell - j + (j+1)n\big] = (j - i)(1 - n), \quad n = 0, 1, 2,$$

where $\tau_e(\xi)$ is the successful zero-emptying length of fixed physical tail ξ from control e. Each fixed e determines at most one such difference. Since the machine has only two controls, at most two distinct differences can occur, whereas the three displayed values are distinct because $i \neq j$. This contradiction forces the first 1 rewrite to be a physical POP. □

G71/G76: complete finite clean-zero interfaces

The following enumeration accounts for the precise class entering the computational certificates; it makes no claim that the enumerated partial tables recognize the whole language.

Lemma 11.41 (Complete clean-zero skeleton bound)**.** *Normalize the unique initial physical site of a putative epsilon-free* $(q, g) = (2, 3)$ *recognizer of* $B_{3,1}$ *to* $(0, 0)$*. Every accepting computation on* 0 *and* 10000 *is represented by one of finitely many* clean-zero skeletons*, including non-POP real-*0 *rules and either possible cardinality of the final-state set. The four zero-step replacement lengths have precisely* 14 *possible ordered-forest shapes, partitioned by the first input-*1 *header length* $d = 1, 2, 3, 4$ *into* $5, 5, 3, 1$ *shapes.*

Proof. Acceptance of 0 forces the root's real-0 rule to POP its only physical symbol and to enter a final control. Let the first real-1 write $d \geq 1$ symbols, and let U_j be the word written by the j-th real-0 rule along the accepting run on 10000, $0 \leq j \leq 3$. With $h_0 = d$ and $h_{j+1} = h_j - 1 + |U_j|$, each of the first four heights is positive and $h_4 = 0$. Therefore $d \leq 4$ and

$$\sum_{j=0}^{3} |U_j| = 4 - d, \qquad |U_3| = 0, \qquad 1 \leq h_j \leq 4 - j.$$

Every such length vector is exactly the preorder outdegree sequence of a plane ordered forest with four nodes. Adjoining a single auxiliary root gives one of $C_4 = 14$ plane trees. Conversely the fourteen legal shape vectors satisfy the necessary positive-height and final-empty conditions. For any chosen vector, all two-control destinations and three-letter replacement words can be enumerated, assigning a transition only when its physical site is first used; a reused site must keep the same rule. This tests every deterministic partial table on the required clean scan, without constraining any unvisited site. □

An independent *forest-first* enumerator, which fixes one of these fourteen length vectors before choosing any state/letter labels, reproduced 9,107 one-final and 15,885 both-final raw clean skeletons, for 24,992 in total. The repaired G71 root/singleton-source filters leave $3{,}808 + 10{,}332 = 14{,}140$ necessary raw interfaces. Replacing the weaker G71 two-final root filter by Lemma 11.39 leaves 11,424 raw interfaces; quotienting only by the swap of the two noninitial physical stack-letter names gives 5,712 orbits. Exactly 5,040 orbits have four distinct clean source sites, disjoint from the root; the other 672 have a shared source site. Each of the fourteen clean-zero forest shapes contributes *exactly 360* of those 5,040 source-distinct orbits. These are exact finite enumeration counts, not an assumption that the input-0 transition always POPs.

G75: sound pruning of source-sharing interfaces

If a physical singleton site accepts $1^n 0^{a+bn}$ for all sufficiently large n, then the same exposed site with a fixed buried tail cannot have an accepted eventually affine zero-completion profile with a different slope: the singleton upper run shields the tail, and after its upper zero-drain the possible tail completion lengths belong to a finite set indexed by the finite control states. For $B_{3,1}$ the physical root is a singleton of slope one, whereas the physical clean site at 1000 is a singleton of slope four.

Lemma 11.40 first forces an immediate real-input-1 POP whenever two of P_0, P_1, P_2 share a clean physical top site. The resulting lower tails must each accept a five-zero continuation from one common control. When a currently undefined physical input-0 rule is first used with t zeros left and a buried tail of length e, any replacement U on an accepting computation must satisfy

$$|U| + e \leq t - 1.$$

Indeed only $t - 1$ real zero moves remain after that transition, and one real move reduces physical height by at most one. Thus the finite backtracking of new zero rules along *these accepted runs* is complete even with arbitrary non-POP rules off the runs. Together with the singleton-slope constraints and the authentic D-debt one-dead prefixes, this yields the necessary-orbit counts

$$672 \xrightarrow{\text{G74}} 126 \xrightarrow{\text{G75}} 110, \qquad 110 = 36 \text{ (one-final)} + 74 \text{ (both-final)}.$$

The 110 are a finite *superset of necessary interfaces*, not a list of complete correct machines. Their remaining physical zero/one rules are subsequently handled by the G89 computation.

G78: soundness of the zero-nonproductive exception gate

The 5,040 source-distinct clean interfaces occupy five of the six physical state/top-letter sites. Write E for the sixth. If E has no finite zero first-return, it cannot occur on any accepting run of a genuinely correct recognizer: every live source prefix admits a pure-zero accepting completion, which would necessarily remove its currently exposed top-origin layer. Hence *both* real input rules at E may be deleted without affecting any successful computation. This is an exact normalization for this branch, not a general zero-POP normal form.

For each of the four clean sites Z_j the real input-1 rule is required by the accepted word $P_j 10^5$. If its current physical stack has height h_j, its replacement word V_j must satisfy

$$|V_j| + h_j - 1 \le 5 \quad \Longrightarrow \quad |V_j| \le 6 - h_j \le 5.$$

Thus every source-1 rule needed in the finite contradiction has a mathematically proved finite search domain. The remaining root-1 rule and the five productive zero rules were fixed by the accepted clean scan. The archived G78 certificate imposes the authentic seventeen three-1 accepted completions and the three D-debt one-dead tests. Every surviving candidate must be supported in each single-source domain and each cross-source necessary binary constraint; arc consistency can only *delete* a choice when it lacks such support. The four nonempty arc-consistent interfaces then admit exactly ten complete source-rule tuples. Direct physical simulation excludes two by those same tests; all eight others reject the authentic accepted word 10101100000000. The independent reproduction found precisely

$$5{,}040 \longrightarrow 4 \longrightarrow 10 \longrightarrow 8 \longrightarrow 0.$$

The code's direct last-stage simulation is essential: successful local constraint propagation alone is not claimed to prove UNSAT. The normalization, upper bounds, and monotonicity arguments establish why G78 tests a necessary superset of *all* zero-nonproductive-E competitors. The numerical exhaustion remains a computational premise and depends on archived finite-search code that is not part of this PDF-only submission. The numerical exhaustion is therefore not independently executable from the submitted PDF alone.

G79: where the numerical cutoff uses computation

For $B_{3,1}$ and resource budget $(2, 3)$, G72 and the clean-prefix singleton and transition-count constraints yield a finite necessary set of source interfaces. In the source-distinct branch, five distinct physical sites are the initial root and the four clean sites; denote the sixth physical site by E. The archived G78 algorithm supplies the following *computational premise*: no such candidate in which both input transitions at E are deleted can simultaneously accept the specified 25 positive source words and reject the three prescribed zero-continued D-debt negative families. This premise was reproduced by executing its original certificate; it is not a consequence of Lemma 11.39 or Lemma 11.40 alone.

Lemma 11.42 (G79: productive exceptional-site bound)**.** *Conditional on the soundness and exhaustive necessary-interface coverage of that G78 computational premise, any globally correct source-distinct* $(2, 3)$ *epsilon-free recognizer of* $B_{3,1}$ *has a finite zero first-return cost* $1 \le c_E \le 9$ *at the exceptional site* E*. Its physical input-*0 *replacement word has length at most* 8*.*

Proof. If a globally correct candidate never exposed E at any prefix of the 25 accepted test words, deleting both E transitions would preserve all 25 accepted runs. It would also preserve every required rejection, since removing transitions cannot introduce acceptance in a deterministic machine. This would contradict the G78 finite premise. Hence some live

test prefix w exposes E. An independent literal canonical-source calculation shows that each such prefix has an accepted pure-zero completion of length $h(w) \leq 9$; the maximum nine occurs at the prefix 100011. From the competitor's physical configuration after w, those $h(w)$ zero transitions must empty its full stack. In particular, they must erase the currently exposed E-origin layer, so its context-independent first zero-return cost satisfies $1 \leq c_E \leq 9$. If the actual real input-0 rule at E writes a physical word U_E, its inserted layer needs at least $|U_E|$ further zero moves to be erased, hence $1 + |U_E| \leq c_E \leq 9$. □

Remark 11.43 (Do not confuse a finite-corpus cutoff with normal form). A physical input-1 rule at E with a replacement longer than nine cannot occur in any of the above accepted test computations. For the *finite* necessary G78 tests only, such a rule may be omitted. This does not impose a global nine-letter bound on arbitrary correct machines' input-1 rules; it is also not an epsilon-elimination theorem. The numerical G79 cutoff is downstream of G78, whereas G72 and G74 are standalone structural statements independent of that computation.

G83–G88: an exhaustive exceptional-return cost partition

The certificate-based exclusion of the source-distinct branch is not a single unrestricted finite-table enumeration. It is a disjoint and exhaustive partition by final-state cardinality and by the *first zero-return cost* of the sixth physical site E. We spell out the resource-model arguments that justify that partition.

Lemma 11.44 (Physical completeness of short exceptional returns)**.** *In the source-distinct, one-final* $(2, 3)$ *epsilon-free class, suppose E has a finite zero first-return cost $c_E \in \{1, 2, 3, 4, 5\}$. For each fixed clean zero table, its physical input-*0 *rule writes a word V of length at most $c_E - 1$. Moreover, a zero computation exposing the original buried tail cannot visit E again before doing so. Consequently, using the already fixed five ordinary zero sites, one can enumerate every physical rule for $E0$ by examining finite words V with the genuine ordinary-site zero-drain cost $c_E - 1$. Across all* 1,680 *alphabet-normalized one-final, source-distinct clean interfaces, the numbers of distinct* (interface, $E0$ rule) *pairs are*

c_E	1	2	3	4	5	total
#	3360	5880	12048	26700	60372	108360.

Proof. One real input-0 transition removes its original top symbol and writes V in its place. Each of the $|V|$ inserted symbols must later be removed, with at most one removed per zero, before the older tail is exposed. Hence $c_E \geq 1+|V|$. If the first-return computation exposed the site E again while the original E-origin layer was still above its tail, determinism would repeat the same $E0$ action on the new upper layer, never reach the older tail, and contradict finite first return. Therefore after the first $E0$ step only the five fixed ordinary zero sites are visited until the inserted word drains. The preceding upper bound limits V to lengths at most four. Enumerating its two possible target controls and all such words, computing their exact zero drain through the fixed ordinary rules, and retaining exactly those of cost $c_E - 1$ is exhaustive. A second implementation built the clean interfaces by enumerating the fourteen preorder plane-forest shapes *before* assigning control and stack-letter labels; its physical first-return simulation reproduced all displayed counts, independently of the G84–G87 E-rule search implementations. The finite enumeration establishes the counts; the preceding locality argument establishes their completeness. □

Lemma 11.45 (G84: reduction for two final controls)**.** *Assume the source-distinct epsilon-free recognizer has both control states final and every physical top site has finite zero first return. Then each of the three authentic zero-live but one-dead D-debt prefixes* 10010, 100100, *and* 1001000 *exposes the sole exceptional site E, whose input-*1 *rule is undefined. Necessarily its input-*0 *rule is a physical POP returning to the E control state; thus $c_E = 1$.*

Proof. With all zero returns finite and both states final, any defined input-1 transition from a nonempty stack is followed by some accepting pure-zero continuation, possibly of length zero. Such a transition is therefore forbidden at the three D-debt sites. The initial root and all four clean sites have mandatory input-1 continuations, so each D-debt site must be E. The first two D-debt configurations are separated by one consumed zero yet expose the same physical E site. If the $E0$ replacement were nonempty, its successor would again expose the identical site and could never complete a finite zero return. Thus the replacement is empty, and the successor control must equal the control component of E, as required by the second exposed D-debt site. This proves the claim. □

Lemma 11.46 (G83: safe high-cost relaxation)**.** *For $6 \le c_E \le 9$, every input-1 rule at a clean source P_j can be selected within the physically necessary length bound $|V_j| \le 6-h_j \le 5$, using only the five already known zero sites for its mandatory P_j10^5 completion. At all required positive three-one test positions, the third input-1 transition also uses an ordinary site, never E. Thus replacing the physical $E0$ rule by an ideal first-return macro of cost c_E and either exit control, and leaving $E1$ unconstrained, is a* necessary overapproximation *of all correct competitors in this cost range.*

Proof. A zero computation with at most five steps cannot return from a currently exposed E layer whose first return costs at least six. The clean P_j10^5 computations therefore use no E on their mandatory successful returns. Likewise, immediately before the third input 1 in any of the genuine accepted tests $P_j10^z10^v$, the intermediate stack is zero-completable in $5 - z \le 5$ zeros, so its exposed site is ordinary. The ideal macro retains every actual first-zero-return cost and exit control while dropping restrictions from the internal $E0$ computation. Hence no correct physical machine is excluded by that relaxation. □

The finite-search checks apply the stated mathematical reductions to labelled partial transition tables. At each stage the admissible physical control/top pairs and source-return conditions determine a finite rule domain; inconsistent partial assignments are discarded, and sound local constraints are propagated before any remaining full tables are evaluated on specified positive and negative source words. Where a search omits a transition, the text explicitly identifies the necessary overapproximation that makes the omission safe. The resulting counts describe the internally executed enumeration, not a substitute for justifying that every globally correct machine enters the enumerated classes. The following is a *complete case partition*, not a claim that every certificate has been freshly re-executed in this manuscript preparation session:

Exceptional zero mode	Final set	Exclusion source
Zero-nonproductive	one or two	G78 finite corpus
$c_E = 6, 7, 8, 9$	one or two	G83 ideal-macro UNSAT
$c_E = 1, 2, 3, 4, 5$	both final	G84 symbolic $c_E = 1$ reduction + finite UNSAT
$c_E = 5$	one final	G85 physical E0 refinement
$c_E = 2, 3$	one final	G86 physical E0 and bounded necessary E1
$c_E = 4$	one final	G87 E1-free overapproximation
$c_E = 1$	one final	G88 full last case

The upper cutoff $c_E \le 9$ is Lemma 11.42; its computational G78 dependency remains explicit. G83's certificate examines all 5,040 clean source-distinct interfaces with both choices for the ideal E-exit control at each high cost; G84 covers all 3,360 both-final clean interfaces and directly rejects all 163 locally compatible full source-rule combinations. For the one-final physical E0 cases, G85 reduces 1,372 refined cost-five combinations to four D-consistent candidates, which all reject a required true word. G86 reports zero cost-two survivors; its unique cost-three relaxed survivor is discharged by bounded actual E1 choices and explicit positive/negative counterexamples. G87 independently reports 26,700 cost-four E0 pairs and no survivors after necessary unary and binary checks. G88 handles cost one.

Current audit distinction. The original G85 and G87 complete certificates were re-executed with successful expected assertions; G85 additionally has a four-trace check proving its last contradiction independent of an unassigned exceptional input-1 action. The G83 costs-six/seven and G84 all-final certificates also passed earlier reruns. In a further 25 October 2026 audit, the entire archived G86 cost-two/cost-three program was re-executed successfully on its original source: it checked all 1,680 normalized one-final clean layouts, 5,880 physical cost-two $E0$ pairs and 12,048 cost-three pairs. The cost-two side had zero survivors. The cost-three side had a single necessary interface with two known-one rule tuples; one misaccepts 10010001000 regardless of the exceptional one-rule. The other tuple required consideration of exactly 242 bounded exceptional-one replacements, 45 of which accepted the first mandatory positive word and seven of which passed the stated 25 positive and D-debt necessary tests. Of these final seven, six reject the genuine positive word 11100100000 and one accepts the genuine negative word 1001000110001.

The exact physical tables of those seven final options were exported from a separately instrumented replay and checked by an independent right-top-stack simulator against a separate canonical $B_{3,1}$ interpreter. A subsequent independent four-part G86 constraint solver also reproduced the *entire* cost-two/cost-three physical $E0$ enumeration: $5{,}880 = 1{,}936 + 3{,}944$ cost-two pairs all fail; $12{,}048 = 3{,}908 + 8{,}139 + 1$ cost-three pairs leave just one necessary interface and its two source-one assignments. Those physical assignments coincide with the archived tables up to renaming stack letters. This new solver covers the finite constraint enumeration, whereas the seven-witness interpreter independently checks its terminal rejections. The earlier G101 full-chain audit remains historical, not a fresh clean-room mathematical proof. A forest-first $E0$ counter also independently confirms all five physical return-cost-domain sizes. These computational agreements do not certify the mathematical necessity of every upstream interface reduction or constitute a proof-assistant verification of the complete case split.

Referee audit of source-one projection and exceptional-rule independence

The exceptional physical site E is the single state/top-letter site outside the root and four pairwise-distinct clean sites. We record why its possibly very long input-1 rule does *not* invalidate the negative finite-domain checks in the cost-four and cost-five cases.

Lemma 11.47 (Safe projection of an unassigned exceptional input rule)**.** *Fix a clean-distinct, one-final, epsilon-free candidate with all six real-*0 *rules having finite zero-first-return. At a clean prefix $P_j = 10^j$ suppose the physical stack has height h_j and the input-*1 *rule at its exposed clean site writes a word V_j. Every globally correct candidate satisfies*

$$|V_j| \le 6 - h_j \le 5.$$

*Consider a test of the prescribed continuation $P_j 1\, 0^z 1\, 0^v$, or its D-debt forbidden variant. Simulate the intermediate z zeros using the actual six input-*0 *rules. If the pending input-*1 *site is E,* discard this particular test constraint*; otherwise impose only its genuine positive/negative zero-completion obligation at the root or one of the four clean sites. All such retained constraints are necessary for a globally correct machine, independently of the value, definedness or output length of its E input-*1 *transition. Deleting source-rule options by repeated binary arc consistency (only when no partner option supports a retained obligation) cannot eliminate a globally correct machine.*

Proof. The source accepts $P_j 1\, 0^5$. The physical height immediately after $P_j 1$ is $|V_j| + h_j - 1$, and each of the following five input-0 steps can decrease height by at most one. Physical empty-store acceptance therefore forces the bound. The real input-0 trace before the pending input-1 is determined by the fixed zero table. If its top site is E, omitting the condition is an *existential relaxation*: every choice of $E1$ is admitted, not just short writes or undefined rules. If it is another known site, the deterministic input-1 action is the root rule, the current clean rule or some other selected clean rule; zero-only completion from the resulting stack is fully determined by the six finite-return zero rules. Each retained

positive or negative constraint is consequently necessary. A valid complete machine's four selected clean actions give a simultaneously consistent witness for every binary dependency; a rule can never lack a supporting partner in the domains containing that complete tuple. Induction through each monotone deletion round proves the claim. □

At exceptional cost $c_E = 5$, the initial relaxation requires a little more care. A clean first-1 output that reaches *singleton* E can use the five-zero return exactly, and all later mixed-1 tests of that source are deliberately bypassed at this stage. For any other valid first-1 output, the next accepted five-zero drain cannot expose E: its return would already consume five zeros, leaving no room for a prior zero step or a buried tail. Thus the subsequent constraints only read the five known one-sites, and the actual $E0$ rule is filled in and tested separately. This is a stronger relaxation than deciding or globally bounding $E1$.

Proposition 11.48 (An $E1$-independent final G85 witness)**.** *For the four physical candidates surviving G85's cost-five, one-final and source-distinct D-debt tests, the fully determined physical computation after input* $u = 100100001$ *is in control* 1 *with an* empty physical stack*. No input-*1 *transition at* E *is read along any of these four computations. On the other hand,*

$$u0^5 = 10010000100000 \in B_{3,1}.$$

Consequently none of these four candidates can be repaired by assigning an arbitrary $E1$ *transition.*

Finite trace certificate. The four traces were replayed with the six literal physical real-0 rules and the five fixed root/clean real-1 rules. Their final configurations are all $(1, \varepsilon)$; the asserted absence of any $E1$ visit is checked at each input-1 step. An independent literal source simulation verifies the required positive completion. Since no operation is available on the physically empty stack, the five remaining zeros cannot be consumed. The exact recorded trace and its executable assertions are bundled with the v0.7 audit. □

For $c_E = 4$, the literal G87 executable was independently rerun to completion: among its 26,700 possible physical $E0$ rules over all 1,680 normalized one-final clean layouts, 7,998 leave an empty unary clean-rule domain and the remaining 18,702 lose support under monotone binary arc consistency. No option survives even when every test whose pending input-1 site is E is omitted. This proves UNSAT of this *necessary overapproximation*, not an assumed normal form for $E1$. For G86 at exceptional costs two and three, the original source-level projection argument was inspected and the complete program rerun on 25 October 2026. An independently written four-part constraint solver reconstructed all 5,880 cost-two and 12,048 cost-three physical $E0$ alternatives, including the unique surviving relaxed interface at cost three and its two source-one assignments. A separate simulator distinguished the last seven $E1$ options. The finite enumeration is independently reproduced, but the mathematical necessity of the source-interface reduction remains a distinct proof-to-code obligation.

Proposition 11.49 (Computational lower certificate)**.** *Assuming the remaining necessary-interface reductions G75 and the computationally exhaustive exceptional-site case split G78–G87, with G72 and G74 justified independently in Lemmas 11.39 and 11.40, the G88/G89 exhaustive certificates exclude every epsilon-free DPDA for* $B_{3,1}$ *with at most two controls and three physical stack symbols.*

Certificate decomposition and audit scope. Normalize the initial physical site to $(0, 0)$ and interchange the two noninitial stack-letter labels as needed. Let $P_j = 10^j$ for $j = 0, 1, 2, 3$ and write Z_j for the physical control/top-symbol site after P_j. The archived necessary interface lemmas exclude equality between the initial site and any Z_j, exclude sharing by Z_3, and prove that when two other Z_j share a site their input-1 transition must pop immediately. This partitions all candidate machines into pairwise-distinct and sharing-source interfaces.

For the source-distinct class G78–G87 exclude all exceptional-site zero-return patterns except the final $c_E = 1$ one-final case; the only necessary global finite-cost cutoff $c_E \leq 9$ is justified by Lemma 11.42, explicitly conditional on G78. G88 exhausts 1,680 initial interfaces and tests 188,089 physically realizable remaining $E1$-transition choices; of the candidates meeting earlier positive obligations, the final genuine positive word 101100100000000 rejects the remaining 32 cases. For the sharing class, G75 leaves 110 necessary interfaces. G89 enumerates 866 one-final and 6,981 both-final complete physical zero-rule tables. The former are reduced to 18 source-one assignments, each of which incorrectly accepts the genuine source-negative word 10010100000000; the latter all violate a D-debt one-dead constraint. Thus neither side admits a globally correct candidate.

The G72, G74, G78, and G79 scripts as well as the G88/G89 terminal scripts were rerun from archival copies, with successful completion and matching assertions. Independent follow-up solvers reproduced the entire G86 finite constraint scan, the G88 count of 385 compatible source/$E1$ summaries and 188,089 physical $E1$ options (including the last 32 tables), and the G89 count of $7{,}847 = 866 + 6{,}981$ complete zero-rule tables. The 32 final G88 rejects and 18 final G89 erroneous acceptances were checked by separately implemented physical simulators. In the independent G89 audit, an overstrong D-debt filter inherited from an earlier auxiliary check was corrected; the original G89 program was not affected. An independent canonical-source interpreter also checked the 25 positive corpus words, 165 distinct live test prefixes, and the unique prefix of semantic height nine. The displayed computations are exhaustive *within the necessary reductions*; the mathematics of those reductions is a separate proof obligation. In particular, a successful script execution alone is not a formal machine-checked proof of G72–G87. The finite-search computations and their dependency records are preserved in unpublished working records and are not submitted as code, datasets, or a technical supplement. The numerical case study consequently cannot be independently re-executed from the submitted article alone. The reductions, case partition, test-word obligations, survivor counts, and limits of their interpretation are stated here, but external verification of exhaustive coverage remains a separate obligation. □

Proposition 11.50 (Explicit epsilon-free two-control, four-letter source)**.** *There is a deterministic epsilon-free automaton recognizing $B_{3,1}$ with exactly two controls and four physical stack letters, one physical initial letter, and final-and-empty acceptance.*

Proof. Give a complete positive table directly. Take $Q = \{0, 1\}$, $\Gamma = \{a, b, c, d\}$, initial site $(0, b)$, both controls final, and top written on the *left*. All unlisted transitions are undefined:

site	input	(next state, replacement)	site	input	(next state, replacement)
$(0, a)$	0	$(0, \varepsilon)$	$(0, b)$	0	$(0, \varepsilon)$
$(0, b)$	1	$(1, accd)$	$(0, c)$	0	$(0, \varepsilon)$
$(0, c)$	1	$(0, caaa)$	$(0, d)$	0	$(0, \varepsilon)$
$(0, d)$	1	$(0, dbbbb)$	$(1, a)$	0	$(1, \varepsilon)$
$(1, a)$	1	$(1, aa)$	$(1, c)$	0	$(0, \varepsilon)$
$(1, c)$	1	$(1, cbb)$			

These are eleven transition sites. This is the two-physical-row return-typed compilation of the canonical $m = 3, k = 1$ automaton under the row partition $(0, 1, 0)$. More explicitly, the original four slope modes are $(1, A), (1, M), (0, M), (2, M)$, and the physical source-state and stack-letter roles are encoded by the displayed table. Each source input-0 step removes its top letter and transfers control to the row prescribed by the exposed next tag. Each input-1 step inserts the exact counter debt above the unchanged buried stack, or expands the initial header as prescribed. The typed zero-return adjacency ensures that a newly exposed lower symbol is read in the same source control mode as in the canonical machine. The six accessible physical control/top types are $(0, b), (1, a), (0, a), (1, c), (0, c), (0, d)$. Each input-0 action removes its top letter and exposes the prescribed return type; each of the five input-1 rewrites installs a typed block whose zero-drain return control agrees with that of the replaced symbol, independently

of any buried tail. The relation between source stack tags and physical future-exposure types is therefore invariant under every input step. Induction proves equivalence on all finite words, including undefined actions and the final-and-physically-empty acceptance condition. The independent literal table was also compared against the canonical source on all 16,383 binary inputs of length at most 13 and 1,200 longer random inputs, with no mismatch. □

Corollary 11.51 (Conditional complete epsilon-free frontier)**.** *Subject to the full mathematical validity of the archived G72–G89 certificate chain and the representation-independent one-state and one-letter lower lemmas for this witness, the epsilon-free Pareto frontier is*

$$\boxed{\mathrm{Pareto}_{\mathrm{rt}}(B_{3,1}) = \{(1,6),(2,4),(3,2)\}.}$$

In particular the exact epsilon-free two-state stack-alphabet threshold is $b_2^{\mathrm{rt}}(B_{3,1}) = 4$.

Proof. The one-control lower bound $b_1 \geq 6$ and a matching one-control realtime typed construction give $(1,6)$. Proposition 11.49 excludes $(2,3)$, and Proposition 11.50 attains $(2,4)$. The canonical three-control/two-letter source attains $(3,2)$, and the independent unary-stack impossibility theorem excludes all $(q,1)$. Upward closure of resource attainability therefore leaves precisely the three listed coordinatewise minimal points. The negative $(2,3)$ conclusion is *not* transferred to epsilon-enabled machines. □

Remark 11.52 (Certificate provenance and reproducibility). The terminal G88 source-distinct and G89 source-sharing UNSAT programs were replayed and independently enumerated in separate implementations. The G86 cost-two/cost-three scan was independently reproduced in full; G88 produced 385 compatible summaries, 188,089 physical exceptional input-one options, and the same last 32 tables; G89 produced 7,847 complete physical zero-rule tables and the same last 18 contradictions. The independent $(2,4)$ checker verifies the 11-transition future-exposure invariant without calling the archived source compiler. The G101 checkpoint additionally records successful replay of all 15 predecessor and terminal scripts. Executable copies, logs and digests are retained in an internal research archive and are *not* submitted with this PDF-only article. Reproduced finite counts do not formally certify the necessary-reduction lemmas; the conditional status of Corollary 11.51 remains.

11.11.9 Discussion and conclusion

The principal achievement is the fixed-resource *existence decision* for the entire binary hierarchy $B_{m,k}$ in the epsilon-free model, rather than an equivalence test for two specified machines. A language-derived zero-completion bound, deletion-only projection and stabilization of finitely many transition slots are what overcome the unbounded-replacement obstacle. The separate negative-certificate theorem makes resource nonexistence witnessable by finitely many words; the generalized calibration theorem identifies a larger, explicitly promised class of target languages to which the mechanism applies. No practicable complexity or small universal bound on the first infeasible horizon is claimed.

The additional computational $B_{3,1}$ example is recorded in the preceding case study with its hypotheses and verification boundary, not as a dependency of the preceding theorems. The epsilon-enabled fixed-resource existence question, as well as a closed-form complete state–stack Pareto classification of all $B_{m,k}$, remains open in the scope of this work.

11.12 Transition to higher-dimensional completion geometry

We have now completed the first major resource arc, including the Pareto-coordinate consequences and the separate epsilon-free fixed-resource existence decision.

Starting from the 2023 2:1 machine, we obtained:

completion semantics

↓

geodesic geometry

↓

better coordinates

↓

exact residual-faithful frontier

↓

representation-independent hierarchy theorem and Pareto geometry, with separately calibrated epsilon-free resource synthesis

The next research branch changes direction.

Instead of asking how many resources are needed to realize a one-dimensional residual geometry, we ask what happens when the counting constraint itself becomes higher-dimensional.

The binary rectangle will become an optimal Parikh fiber.

The absence of equal-length trade directions will become a kernel-rank condition.

And the first nontrivial equal-length trade will create topology.

We now leave resource complexity and return to completion geometry.

Part II

Completion Geometry, Geodesic Compilation, and Universal Compatibility

---

Part guide. **Question.** What remains of shortest completion after the scalar residual is replaced by a higher-dimensional demand fiber? **Main result.** Completion fibers become finite geodesic objects with exact local compatibility complexes, culminating in full simplicial universality, exact Boolean-lattice-width row complexity, and a common-distance Bellman lift. **Why here.** This is the geometric enlargement of the Part I invariant before the book returns to concrete pushdown normalization. **Dependency.** It uses shortest-completion semantics from Part I and is logically independent of the later WSC machinery.

Chapter 12

When Minimum Demand Becomes a Fiber

The first half of the monograph was built around one scalar residual:

$$\rho_{p,q}(w) = qN_1(w) - pN_0(w).$$

That scalar residual had a unique minimum completion Parikh vector

$$\mu(r) = (x_r, y_r).$$

As a consequence, every shortest-completion interval was a rectangle.
The natural higher-dimensional question is:

> What survives if the counting constraint is no longer one scalar equation?

The answer preserves the completion viewpoint but changes the geometry fundamentally.
The minimum demand need no longer be one vector.
It can become a finite **fiber of equally short Parikh vectors**.
That is where higher-dimensional completion geometry begins.

12.1 Matrix counting languages

Let

$$\Sigma = \{a_1, \ldots, a_d\}$$

and let

$$\Psi : \Sigma^* \to \mathbb{N}^d$$

be the Parikh map.
Fix an integer matrix

$$A \in \mathbb{Z}^{k \times d}.$$

Define the vector residual

$$\boxed{\rho(w) = A\Psi(w)}$$

and the matrix counting language

$$\boxed{L_A = \{w : \rho(w) = 0\}.}$$

Residuals lie in the lattice

$$A\mathbb{Z}^d.$$

The binary rational counting family is a rank-one special case.
For

$$L_{p,q},$$

take

$$A = \begin{bmatrix} -p & q \end{bmatrix}.$$

Then

$$A\Psi(w) = qN_1(w) - pN_0(w).$$

Thus the higher-dimensional theory genuinely contains the earlier one.

12.2 Completion totality

A matrix counting language need not allow every residual to be completed by appending letters.

We therefore isolate a convenient hypothesis.

Definition 12.1 — Completion totality.

We call A **completion-total** if there exists

$$h \in \mathbb{N}_{>0}^d$$

such that

$$\boxed{Ah = 0.}$$

The vector h is a strictly positive balanced Parikh vector.
It plays the role of the primitive balanced packet

$$(q, p)$$

from the binary theory, except that it need not generate the entire integer kernel.

Lemma 12.2 — Every lattice residual is completable.

Assume completion totality.
For every

$$r \in A\mathbb{Z}^d,$$

there exists

$$x \in \mathbb{N}^d$$

such that

$$\boxed{Ax = -r.}$$

Proof.
Write

$$r = Az$$

with

$$z \in \mathbb{Z}^d.$$

Because every coordinate of h is positive, for sufficiently large t,

$$th - z \in \mathbb{N}^d.$$

Since

$$Ah = 0,$$

$$A(th - z) = -Az = -r.$$

Thus $th - z$ is a nonnegative completion vector. □

This hypothesis is deliberately strong enough to make completion semantics total on the entire residual lattice.

It lets us generalize the quotient theory cleanly.

12.3 Residuals still classify right quotients

Proposition 12.3 — Matrix residual quotient classification.

Under completion totality, for all prefixes

$$u, v \in \Sigma^*,$$

$$\boxed{u^{-1}L_A = v^{-1}L_A \iff \rho(u) = \rho(v).}$$

Proof.

A continuation z completes u exactly when

$$A\Psi(z) = -\rho(u).$$

Therefore equal residuals give equal quotients.

Conversely, suppose

$$\rho(u) \neq \rho(v).$$

By Lemma 12.2, choose

$$x \in \mathbb{N}^d$$

with

$$Ax = -\rho(u).$$

Choose a word z with

$$\Psi(z) = x.$$

Then

$$uz \in L_A.$$

But if

$$vz \in L_A,$$

we would have

$$Ax = -\rho(v)$$

as well, forcing

$$\rho(u) = \rho(v).$$

Contradiction. □

The semantic lesson from Chapter 4 survives unchanged:

$$\boxed{\text{the residual is an exact coordinate for future right behavior.}}$$

What changes is the geometry of shortest return to residual zero.

12.4 Minimum completion length

For residual

$$r \in A\mathbb{Z}^d,$$

define

$$\boxed{\ell_A(r) = \min\{\mathbf{1}^\top x : x \in \mathbb{N}^d,\ Ax = -r\}.}$$

This is the shortest number of appended letters required to cancel r.

In the binary case,

$$\ell_A(r) = \Delta_{p,q}(r).$$

So far the generalization looks formal.

The decisive change appears when we ask which Parikh vectors attain this minimum.

12.5 The optimal Parikh fiber

Definition 12.4 — Optimal Parikh fiber.

Define

$$\boxed{O_A(r) = \{x \in \mathbb{N}^d : Ax = -r,\ \mathbf{1}^\top x = \ell_A(r)\}.}$$

In the binary p:q theory,

$$O_A(r)$$

was always a singleton:

$$O_A(r) = \{\mu(r)\}.$$

In higher dimensions, several distinct nonnegative Parikh vectors can complete the same residual at the same minimum length.

Therefore the semantic object

$$\mu(r)$$

must be replaced by the entire set

$$\boxed{O_A(r).}$$

This is the first major qualitative change.

12.6 Equal-length trades

Introduce the augmented matrix

$$\boxed{C = \begin{bmatrix} A \\ \mathbf{1}^\top \end{bmatrix}.}$$

Define the integer equal-length kernel

$$\boxed{K_A = \ker_{\mathbb{Z}} C = \left\{ g \in \mathbb{Z}^d : Ag = 0,\ \mathbf{1}^\top g = 0 \right\}.}$$

A vector

$$g \in K_A$$

changes one completion Parikh vector into another while preserving:

1. residual contribution;
2. total completion length.

Thus K_A is the lattice of **equal-length trades**.

This is the higher-dimensional replacement for the binary kernel direction.

But there is a crucial difference.

For the binary minimum demand, subtracting one nonzero kernel packet always shortened the completion, so two distinct minimum Parikh vectors could not coexist.

In higher dimensions, a nonzero vector can have positive and negative coordinates while still satisfying

$$\mathbf{1}^\top g = 0.$$

Such a trade changes the distribution of letters without changing total length.

That is how an optimal fiber acquires multiple points.

12.7 Affine-semigroup viewpoint

The columns

$$C_i = \begin{bmatrix} A_i \\ 1 \end{bmatrix}$$

generate an affine semigroup.

For residual r, define the augmented optimum element

$$\boxed{b_r = (-r, \ell_A(r)).}$$

Then

$$O_A(r)$$

is exactly the factorization fiber of b_r into the generators C_i.

Every such factorization has the same last coordinate:

$$\ell_A(r).$$

This places the theory next to established affine-semigroup and factorization geometry.

The monograph uses that literature as background.

The new object of interest remains specifically the geometry of **minimum-completion schedules**.

12.8 Continuous optimum geometry is not the same as integer multiplicity

Define the continuous optimum polytope

$$P_r^A = \{x \in \mathbb{R}^d_{\geq 0} : Ax = -r,\ \mathbf{1}^\top x = \ell_A(r)\}.$$

Let

$$S(r) = \{i : \exists x \in P_r^A \text{ with } x_i > 0\}$$

be its active support.

Proposition 12.5 — Dimension of the continuous optimum polytope.

If

$$P_r^A \neq \varnothing,$$

then

$$\boxed{\dim P_r^A = |S(r)| - \operatorname{rank}_{\mathbb{R}} \begin{bmatrix} A_{S(r)} \\ \mathbf{1}^\top \end{bmatrix}.}$$

Proof.

For every active coordinate choose a feasible point positive in that coordinate.

The average of these points is positive in every active coordinate and lies in the relative interior of the nonnegative orthant restricted to $S(r)$.

Thus no additional coordinate inequality is active in the relative interior.

The dimension is therefore the number of active variables minus the rank of the affine equality system on those variables. □

The proposition is useful mainly because it prevents a common confusion.

A positive-dimensional real optimum polytope does not automatically imply many integer optimum points.

Conversely, discrete equal-length trade structure can be significant even when a simple continuous dimension count does not capture its combinatorics.

The integer fiber

$$O_A(r)$$

is the primary shortest-completion object.

12.9 The shortest-prefix down-set

In the binary rectangle, a shortest-completion prefix was described by how many required zeros and ones had already been consumed.

For a higher-dimensional optimum vector

$$x \in O_A(r),$$

every coordinatewise subvector

$$0 \leq u \leq x$$

is the Parikh vector of some prefix of an ordering of that optimum completion.

Define

$$B_x = \{u \in \mathbb{N}^d : 0 \leq u \leq x\}.$$

Now take the union over all optimum vectors.

Definition 12.6 — Shortest-prefix down-set.

$$\boxed{\mathcal{C}_A(r) =\downarrow O_A(r) = \bigcup_{x\in O_A(r)} B_x.}$$

This is the higher-dimensional replacement for one rectangle.

If the optimum is unique,

$$O_A(r) = \{x\},$$

then

$$\mathcal{C}_A(r) = B_x,$$

a literal coordinate box.

If the optimum fiber contains several vectors, the shortest-prefix region becomes a union of overlapping boxes.

The next theorem says this down-set is not merely a convenient overapproximation.

It is exact.

12.10 Exact geodesic-prefix criterion

Theorem 12.7 — Exact geodesic-prefix criterion.

For every

$$u \in \mathbb{N}^d,$$

$$\boxed{u \in \mathcal{C}_A(r) \iff \ell_A(r + Au) = \ell_A(r) - |u|_1.}$$

Equivalently,

$$\mathcal{C}_A(r)$$

is exactly the set of Parikh vectors that occur as prefixes of shortest completions of residual r.

Proof.

Suppose

$$u \in \mathcal{C}_A(r).$$

Then

$$u \le x$$

for some

$$x \in O_A(r).$$

The vector

$$x - u$$

completes residual

$$r + Au$$

and has length

$$\ell_A(r) - |u|_1.$$

If a shorter completion existed, appending it after u would produce a completion of r shorter than $\ell_A(r)$.

Therefore

$$\ell_A(r + Au) = \ell_A(r) - |u|_1.$$

Conversely, assume the displayed equality.

Let y be an optimal completion of

$$r + Au.$$

Then

$$A(u + y) = -r,$$

and

$$|u + y|_1 = |u|_1 + \ell_A(r + Au) = \ell_A(r).$$

Thus

$$u + y \in O_A(r),$$

so

$$u \le u + y.$$

Hence

$$u \in \mathcal{C}_A(r).$$

□

This theorem is the higher-dimensional form of minimum-demand descent.

The difference is that there may be several optimal endpoints x, so the geodesic-prefix set is a union of coordinate boxes rather than one box.

12.11 First actions and compatible action sets

Let e_i denote the i-th unit vector.

Corollary 12.8 — First actions.

Letter type i can begin a shortest completion exactly when

$$\boxed{\ell_A(r + Ae_i) = \ell_A(r) - 1.}$$

More generally, for

$$J \subseteq [d],$$

there exists an optimal completion using every letter type in J at least once if and only if

$$\boxed{\ell_A(r + A\mathbf{1}_J) = \ell_A(r) - |J|.}$$

Thus shortest-action compatibility can be tested intrinsically from the distance function.

This is the higher-dimensional analogue of the binary rule:

$$x_r > 0 \iff 0 \text{ is geodesic},$$

$$y_r > 0 \iff 1 \text{ is geodesic}.$$

12.12 The conceptual transition

The completion object has now changed from

$$\boxed{\text{one minimum vector}}$$

to

$$\boxed{\text{a finite optimum fiber.}}$$

Accordingly, the shortest-prefix geometry has changed from

$$\boxed{\text{one rectangle}}$$

to

$$\boxed{\text{a union of boxes.}}$$

But the residual quotient still identifies prefixes that reach the same semantic state.

If two different shortest-prefix Parikh vectors reach the same residual at the same geodesic depth, they must differ by an equal-length trade in

$$K_A.$$

That observation leads to the next object:

$$\boxed{|\mathcal{C}_A(r)|/K_A.}$$

The next chapter will show that this quotient is exactly the filled shortest-path interval.

Chapter 13

The Filled Shortest-Prefix Quotient

Chapter 12 produced two objects:

$$\mathcal{C}_A(r) =\downarrow O_A(r),$$

the exact Parikh down-set of shortest prefixes, and

$$K_A = \ker_{\mathbb{Z}} \begin{bmatrix} A \\ \mathbf{1}^\top \end{bmatrix},$$

the lattice of equal-length residual-preserving trades.

We now combine them.

The result is the higher-dimensional completion object that replaces the binary shortest rectangle.

13.1 Why vertices alone are not enough

A shortest prefix is not only a point.

If several distinct next letters can be taken in any order while remaining geodesic, the geometry contains a commuting square, cube, or higher-dimensional cell.

The binary rectangle already had this property.

A pair of actions 0 and 1 produced an elementary commuting square.

In d letters, a compatible set

$$J \subseteq [d]$$

should produce a $|J|$-dimensional cube.

Thus we should fill the shortest-prefix down-set cubically.

13.2 Cubical realization of a down-set

Let

$$D \subseteq \mathbb{N}^d$$

be a finite coordinatewise down-set.

Define

$$|D|$$

to be the cubical complex whose vertices are the elements of D.

For

$$J \subseteq [d],$$

an elementary J-oriented cube based at u is present exactly when

$$\boxed{u + \mathbf{1}_J \in D.}$$

Because D is a down-set, all intermediate vertices of that cube are automatically present.

Applied to

$$D = \mathcal{C}_A(r),$$

every cube records a set of distinct letter actions that can be executed in any order while remaining on shortest paths.

This is the filled schedule geometry before residual identifications.

13.3 Equal-length identifications

Two shortest-prefix Parikh vectors can be different while representing the same reached residual.

Let

$$u, v \in \mathcal{C}_A(r).$$

They reach the same residual exactly when

$$A(u - v) = 0.$$

But because both are shortest prefixes reaching the same residual, they also have the same geodesic depth.

Thus

$$\mathbf{1}^\top (u - v) = 0.$$

Therefore

$$u - v \in K_A.$$

This gives the correct equivalence relation.

Two equally oriented cells are identified when their base vertices differ by an element of

$$K_A.$$

Definition 13.1 — Filled shortest-geodesic quotient.

Define

$$\boxed{\mathfrak{I}_A(r) = |\mathcal{C}_A(r)|/K_A.}$$

This quotient keeps:

- every shortest prefix;
- every compatible commuting cell;

but identifies prefixes that reach the same residual at the same shortest depth.

13.4 The quotient theorem

Theorem 13.2 — Shortest-prefix quotient theorem.

The 0-skeleton of

$$\mathfrak{I}_A(r)$$

is label-preserving isomorphic to the complete directed shortest-path interval from residual r to residual 0.

Under this identification, elementary cubes are exactly compatible commuting sets of geodesic one-letter actions.

Proof.

Define

$$\pi_r : \mathcal{C}_A(r) \to A\mathbb{Z}^d$$

by

$$\boxed{\pi_r(u) = r + Au.}$$

By Theorem 12.7, every $u \in \mathcal{C}_A(r)$ is a geodesic prefix, so every image vertex lies on a shortest path from r to 0.

Conversely, every shortest path prefix has some Parikh vector u, and Theorem 12.7 places that vector in $\mathcal{C}_A(r)$.

Now suppose

$$\pi_r(u) = \pi_r(v).$$

Then

$$A(u - v) = 0.$$

Because u and v reach the same residual and both are geodesic,

$$\ell_A(r) - |u|_1 = \ell_A(\pi_r(u)) = \ell_A(\pi_r(v)) = \ell_A(r) - |v|_1.$$

Hence

$$|u|_1 = |v|_1,$$

so

$$\mathbf{1}^\top(u - v) = 0.$$

Therefore

$$u - v \in K_A.$$

Conversely, if

$$u - v \in K_A,$$

then

$$A(u - v) = 0,$$

so the two vertices have the same residual label.

Thus the vertex fibers of π_r are exactly the K_A-classes.

Finally, a J-cube based at u exists exactly when

$$u + \mathbf{1}_J \in \mathcal{C}_A(r).$$

By the geodesic-prefix criterion, every subcollection of the actions in J remains shortest, and coordinate addition commutes.

Passing to the equal-length kernel quotient therefore gives exactly the filled commuting shortest-path structure. □

This theorem is the central bridge from integer completion arithmetic to topology.

13.5 Recovery of the binary rectangle

For the binary scalar system

$$A = \begin{bmatrix} -p & q \end{bmatrix},$$

the augmented matrix is

$$\begin{bmatrix} -p & q \\ 1 & 1 \end{bmatrix}.$$

Its determinant is

$$-(p+q) \neq 0.$$

Therefore

$$\boxed{K_A = \{0\}.}$$

There are no nontrivial equal-length trades.

The optimum fiber is a singleton:

$$O_A(r) = \{\mu(r)\}.$$

Hence

$$\mathcal{C}_A(r) = [0, x_r] \times [0, y_r]$$

and no quotient identifications occur.

Thus

$$\boxed{\mathfrak{I}_A(r) = [0, x_r] \times [0, y_r].}$$

The binary rectangle is therefore the **rank-zero member** of the general theory.

This is an important reinterpretation.

The rectangle did not disappear in higher dimensions.

It became the simplest possible fiber-and-quotient case.

13.6 When is the quotient literally a box?

Theorem 13.3 — Literal-box criterion.

$$\boxed{|O_A(r)| = 1 \iff \mathfrak{I}_A(r) \text{ is a literal coordinate box.}}$$

Here "literal coordinate box" means that the quotient contains no nontrivial kernel identifications among the vertices and cells of the prequotient box.

Proof.

Suppose

$$O_A(r) = \{x\}.$$

Then

$$\mathcal{C}_A(r) = B_x.$$

Assume distinct

$$u, v \leq x$$

were identified.
Then

$$u - v \in K_A.$$

Consider

$$x - u + v.$$

It is nonnegative.
Because

$$A(u - v) = 0,$$

it has the same residual contribution as x.
Because

$$\mathbf{1}^\top (u - v) = 0,$$

it has the same length as x.
It is different from x.
Therefore it would be a second element of

$$O_A(r),$$

contradicting uniqueness.
Hence no nontrivial kernel collision occurs inside the box.
Conversely, suppose

$$x \neq y$$

both belong to

$$O_A(r).$$

Then x and y are distinct maximal vertices of $\mathcal{C}_A(r)$, both reach residual zero, and

$$x - y \in K_A.$$

Thus the quotient identifies distinct shortest schedules.
It cannot be the original literal coordinate box with all coordinate vertices distinct. □
This theorem identifies the first source of global complexity:

nonunique optimum $\Longleftrightarrow$ nontrivial equal-length identification.

13.7 Unique optimum versus unique shortest word

The phrase "unique optimum" refers to the Parikh vector.

It does not mean there is only one shortest word.

Even when

$$O_A(r) = \{x\},$$

the letters represented by x can be ordered in many ways.

The literal box already records those schedules.

What uniqueness eliminates is a stronger ambiguity:

two different minimum Parikh endpoints.

Thus there are two levels of shortest-path multiplicity.

Ordering multiplicity

One Parikh vector, many permutations.

This produces a box.

Factorization multiplicity

Several distinct minimum Parikh vectors.

This creates kernel identifications among different boxes.

The second phenomenon is new beyond the binary line.

13.8 Local compatibility and global identification are different

The cubical complex

$$|\mathcal{C}_A(r)|$$

records which letter actions can coexist in one optimum and commute as schedule choices.

The quotient

$$\mathfrak{I}_A(r)$$

adds a different phenomenon:

globally different shortest prefixes can reach the same residual.

A later chapter will show that these two notions have different topology.

In rank one:

- every connected component of the local action-compatibility complex is contractible;
- the full global quotient can nevertheless be a circle.

Thus local coexistence and global schedule identification must not be conflated.

We are now ready to study that distinction.

13.9 The rank of equal-length trade

The integer lattice

$$K_A$$

measures the number of independent directions in which one optimal Parikh vector can trade letters for others without changing:

- residual;
- completion length.

Three regimes will become especially important.

Rank zero

$$K_A = \{0\}.$$

No equal-length trade exists.
Every optimum is unique.
The shortest interval is a literal box.
This is the binary p:q regime.

Rank one

One primitive equal-length trade direction exists.
The optimum fiber becomes an arithmetic interval.
The global quotient may acquire its first cycle.

Rank at least two

Several independent trade directions exist.
Even the local action complex can develop holes.
The next chapters will make this hierarchy exact.

Chapter 14

Local and Global Topology

Chapter 13 produced the filled shortest-prefix quotient

$$\mathfrak{I}_A(r) = |\mathcal{C}_A(r)|/K_A.$$

Two different kinds of structure are present inside this object.
The first is **local**:

> Which distinct letter types can coexist inside one shortest completion?

The second is **global**:

> After all shortest prefixes are filled cubically and equal-length residual-preserving schedules are identified, what is the topology of the whole shortest interval?

These questions need not have the same answer.
The first genuinely new phenomenon occurs already when the equal-length kernel has rank one.
Locally, nothing topologically complicated happens.
Globally, a circle can appear.
That separation is one of the principal structural results of the higher-dimensional branch.

14.1 The local action complex

Recall the optimal Parikh fiber

$$O_A(r) = \{x \in \mathbb{N}^d : Ax = -r,\ |x|_1 = \ell_A(r)\}.$$

Define the simplicial complex

$$\boxed{\mathcal{H}_A(r) = \{J \subseteq [d] : \exists x \in O_A(r) \text{ with } x_j > 0 \text{ for every } j \in J\}.}$$

A face

$$J \in \mathcal{H}_A(r)$$

means that some shortest completion contains at least one copy of every action in J.
Equivalently, by Chapter 12,

$$J \in \mathcal{H}_A(r) \iff \ell_A(r + A\mathbf{1}_J) = \ell_A(r) - |J|.$$

Thus $\mathcal{H}_A(r)$ is the local compatibility complex of geodesic letter types.
It deliberately forgets:

- how many times an action occurs;
- the order in which actions occur;
- which shortest prefixes are globally identified.

It records only coexistence.

14.2 Squarefree-divisor interpretation

Let

$$C_i = \begin{bmatrix} A_i \\ 1 \end{bmatrix}$$

and let

$$S_C = \langle C_1, \dots, C_d \rangle$$

be the affine semigroup generated by the augmented columns.
For residual r, recall

$$b_r = (-r, \ell_A(r)).$$

Proposition 14.1 — Squarefree divisor identification.

$$\boxed{\mathcal{H}_A(r) = \left\{ J \subseteq [d] : b_r - \sum_{j \in J} C_j \in S_C \right\}.}$$

Proof.
The semigroup condition means that there exists

$$y \in \mathbb{N}^d$$

with

$$Cy = b_r - \sum_{j \in J} C_j.$$

Then

$$x = y + \mathbf{1}_J$$

satisfies

$$Cx = b_r.$$

Hence

$$x \in O_A(r)$$

and every coordinate indexed by J is positive.
The converse is obtained by subtracting $\mathbf{1}_J$ from any optimum vector supporting J. □
Squarefree divisor complexes are established objects in affine-semigroup topology [3].
Their role here is specific:

they are exactly the local shortest-action complexes of the completion problem.

The topology of $\mathcal{H}_A(r)$ therefore measures local compatibility among geodesic actions.

14.3 Rank zero: one simplex

If

$$K_A = \{0\},$$

the optimum fiber is a singleton

$$O_A(r) = \{x\}.$$

Then

$$\mathcal{H}_A(r)$$

is simply the simplex on

$$\mathrm{supp}(x).$$

Thus rank zero has no nontrivial local topology.
This includes every binary p:q completion problem.
The local action set may have one or two symbols, but it never develops a hole.

14.4 Rank one: the optimum fiber is an arithmetic interval

Assume now

$$K_A = \mathbb{Z}g$$

for a primitive nonzero vector

$$g \in \mathbb{Z}^d.$$

Because the optimum fiber is finite, it has the form

$$\boxed{O_A(r) = \{x_t = x_0 + tg : 0 \le t \le N\}.}$$

For each letter type i, define

$$T_i = \{t \in [0, N] \cap \mathbb{Z} : x_t(i) > 0\}.$$

Since

$$x_t(i) = x_0(i) + tg_i$$

is affine in t, the set T_i is an integer interval, possibly empty.
A set of actions

$$J$$

is locally compatible exactly when

$$\bigcap_{j \in J} T_j \neq \varnothing.$$

Thus the local action complex is the nerve of a finite family of intervals.

Theorem 14.2 — Rank-one local action topology.

If

$$\mathrm{rank}_{\mathbb{Z}}\, K_A \le 1,$$

then every connected component of

$$\mathcal{H}_A(r)$$

is contractible.

In particular,

$$\boxed{\widetilde{H}_j(\mathcal{H}_A(r);\mathbb{Z}) = 0 \qquad (j \geq 1).}$$

Proof.

The rank-zero case was just described.

In rank one, $\mathcal{H}_A(r)$ is the nerve of the interval family

$$\{T_i\}.$$

Every connected component of the union of finitely many intervals is itself an interval.

By the nerve theorem [14], each connected component of the nerve is homotopy equivalent to the corresponding connected component of the union.

Hence every connected component is contractible. □

This theorem says that one equal-length trade direction is not enough to create a local homology hole.

Yet it is enough to create a global cycle.

Before proving that, we show that rank two changes the local picture immediately.

14.5 Rank two already permits a local hole

Consider the scalar weight vector

$$\boxed{\omega = (-11, -2, 1, 10)}$$

and residual

$$r = 7.$$

The minimum completion length is 5, and the complete optimum fiber is

$$\boxed{(0,4,1,0), \qquad (1,0,4,0), \qquad (1,3,0,1).}$$

The maximal local action faces are

$$\{2,3\}, \qquad \{1,3\}, \qquad \{1,2,4\}.$$

Therefore

$$f_0 = 4, \qquad f_1 = 5, \qquad f_2 = 1.$$

There is one filled triangle, but one independent graph cycle remains unfilled.

Hence

$$\boxed{H_1(\mathcal{H}_\omega(7);\mathbb{Z}) \cong \mathbb{Z}.}$$

The augmented equal-length kernel has rank two.

This example proves that the rank-one local theorem is sharp.

The first nontrivial local H_1 can appear as soon as two independent equal-length trades are available.

The figure should make the distinction visual:

- one triangular face is filled;
- another cycle is not.

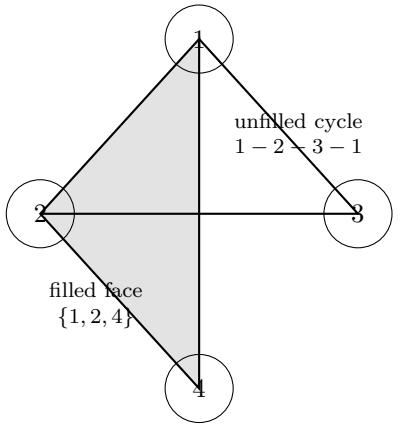


Figure 14.1: The local action complex for $\omega = (-11, -2, 1, 10)$ at residual 7. Its maximal faces are $\{2, 3\}$, $\{1, 3\}$, and the filled triangle $\{1, 2, 4\}$. The cycle $1 - 2 - 3 - 1$ remains unfilled, giving $H_1 \cong \mathbb{Z}$.

This is not a global schedule-identification phenomenon.
The hole is already present in the local coexistence complex.

14.6 Rank-one global geometry

We now return to

$$K_A = \mathbb{Z}g.$$

Write the primitive generator as

$$g = g^+ - g^-,$$

where

$$g^+, g^- \in \mathbb{N}^d$$

have disjoint supports.
Let

$$O_A(r) = \{x_t = x_0 + tg : 0 \le t \le N\}.$$

Define the prequotient shortest-prefix complex

$$X = |\mathcal{C}_A(r)| = \left| \bigcup_{t=0}^{N} B_{x_t} \right|.$$

Because $\mathcal{C}_A(r)$ is a finite coordinate down-set,

$$\boxed{X \simeq *.}$$

So the topology of the final quotient cannot come from a hole already present in the prequotient region.
It must come from the kernel identification.

14.7 Overlap complexes

For

$$m \geq 1,$$

define

$$P_m = \{\sigma \subseteq X : \sigma + mg \subseteq X\},$$

where translation preserves cube orientation.

Lemma 14.3 — Exact m-step overlap.

For

$$1 \leq m \leq N,$$

translation by

$$-mg^-$$

identifies P_m with the anchored cubical down-set

$$\boxed{\bigcup_{t=0}^{N-m} B_{x_t - mg^-}.}$$

Hence P_m is nonempty and contractible.

For

$$m > N,$$

$$\boxed{P_m = \varnothing.}$$

Interpretation. The complete optimum fiber has parameter span

$$0, \ldots, N.$$

Two shortest-prefix cells can overlap after an mg translation exactly while that translation stays inside the span.

Thus the global quotient glues only finitely many neighboring translates.

The overlap itself remains topologically simple.

14.8 Periodic saturation

Define the periodic saturation

$$\boxed{\widetilde{X} = \bigcup_{k \in \mathbb{Z}} (X + kg).}$$

Let

$$X_k = X + kg.$$

Translation by g acts freely on $\widetilde{X}$.

Moreover,

$$\boxed{\widetilde{X}/\mathbb{Z} \cong \mathfrak{I}_A(r).}$$

The quotient by the integer translation action performs exactly the rank-one equal-length kernel identification.

To understand $\widetilde{X}$, we study the cover

$$\{X_k\}_{k\in\mathbb{Z}}.$$

Lemma 14.4 — Good-cover intersections.

Let

$$k_0 < \cdots < k_s.$$

Then

$$\boxed{X_{k_0} \cap \cdots \cap X_{k_s} \neq \varnothing \iff k_s - k_0 \leq N.}$$

Every nonempty finite intersection is contractible.

Proof idea. For any fixed cell, the set of translate indices k for which the cell lies in X_k is an integer interval.

Thus membership in the two extreme translates implies membership in all intermediate translates.

After translating by $-k_0 g$, the intersection reduces to the $(k_s - k_0)$-step overlap from Lemma 14.3.

That overlap is contractible exactly when the index gap is at most N. □

The nerve of this cover has vertices $\mathbb{Z}$.

A finite set of vertices spans a simplex exactly when its diameter is at most N.

For

$$N \geq 1,$$

this is the nerve of the interval cover

$$\{[k, k+N]\}_{k\in\mathbb{Z}}$$

of the real line.

Hence the nerve is contractible.

Theorem 14.5 — Contractible rank-one saturation.

If

$$N \geq 1,$$

then

$$\boxed{\widetilde{X} \simeq *.}$$

The periodic saturation is therefore a contractible universal-cover candidate.

14.9 The global circle

Theorem 14.6 — Rank-one global homotopy dichotomy.

If

$$\operatorname{rank}_{\mathbb{Z}} K_A = 1,$$

then

$$\boxed{\mathfrak{I}_A(r) \simeq \begin{cases} *, & |O_A(r)| = 1, \\ S^1, & |O_A(r)| > 1. \end{cases}}$$

Proof.

If the optimum is unique, Chapter 13 gives a literal coordinate box, hence a contractible complex.

Now suppose

$$|O_A(r)| > 1.$$

Then

$$N \geq 1.$$

By Theorem 14.5,

$$\widetilde{X}$$

is contractible.
The action of

$$\mathbb{Z}$$

by translation through g is free, properly discontinuous, and cellular.
Therefore

$$\widetilde{X} \to \widetilde{X}/\mathbb{Z} \cong \mathfrak{I}_A(r)$$

is a universal covering with deck group

$$\mathbb{Z}.$$

Hence

$$\pi_1(\mathfrak{I}_A(r)) \cong \mathbb{Z}$$

and all higher homotopy groups vanish.
Thus

$$\mathfrak{I}_A(r)$$

is a $K(\mathbb{Z}, 1)$, hence homotopy equivalent to

$$S^1.$$

□

This gives the first exact topological transition in the completion theory:

$$\boxed{\text{rank one + nonunique optimum} \Longrightarrow \text{global circle.}}$$

14.10 Local triviality, global nontriviality

Combining Theorems 14.2 and 14.6 gives the sharp distinction:

rank-one local action complex: every component contractible
rank-one global shortest-prefix quotient: possibly S^1.

The circle is not caused by a local incompatibility hole.

It is caused by globally identifying different shortest schedules through an equal-length trade.

This difference will matter later when the monograph returns to normalization schedules.

Local feasibility and global identification are fundamentally different layers of structure.

14.11 What the topology means operationally

A point of

$$\mathcal{C}_A(r)$$

records a shortest-prefix Parikh vector.

A path through the cubical complex records gradual execution of a shortest schedule.

Translation by g changes the Parikh bookkeeping without changing:

- residual;
- geodesic depth.

The quotient therefore says that two apparently different shortest schedule histories have reached the same semantic point.

When one primitive trade direction exists, those identifications wrap the contractible periodic saturation around one generator.

That is the geometric origin of the circle.

The result is not merely "some complex has homology."

It is a statement about the topology created by schedule equivalence.

The next chapter turns kernel rank into an exact alphabet-size hierarchy for scalar counting languages.

Chapter 15

Scalar Rank Hierarchy and Exact Three-Letter Geometry

The previous chapter expressed the new topology in terms of

$$\operatorname{rank}_{\mathbb{Z}} K_A.$$

For a general matrix A, this rank is an independent structural parameter.
For a scalar counting language, it has a remarkably simple form.
This produces a sharp hierarchy by alphabet size.

15.1 Scalar counting systems

Let

$$\omega = (\omega_1, \dots, \omega_d) \in \mathbb{Z}^d$$

have at least two distinct entries.
Treat

$$A = \omega$$

as a $1 \times d$ matrix.
The equal-length kernel is

$$K_\omega = \ker_{\mathbb{Z}} \begin{bmatrix} \omega \\ \mathbf{1}^\top \end{bmatrix}.$$

The two rows are linearly independent over $\mathbb{R}$.
Therefore:

$$\boxed{\operatorname{rank}_{\mathbb{Z}} K_\omega = d - 2.}$$

This single formula turns the topology of Chapter 14 into an alphabet-size threshold.

15.2 The $d = 2, 3, 4+$ hierarchy

Two letters

If

$$d = 2,$$

then

$$\operatorname{rank} K_\omega = 0.$$

Therefore:

- every optimum Parikh vector is unique;
- every shortest interval is a literal coordinate box;
- no equal-length schedule identification occurs.

This is exactly the binary theory of Chapters 2–5.

Three letters

If

$$d = 3,$$

then

$$\operatorname{rank} K_\omega = 1.$$

Therefore:

- the optimum fiber is an arithmetic interval;
- every connected component of the local action complex is contractible;
- the global quotient is either contractible or homotopy equivalent to

$$S^1.$$

Thus three letters are the first scalar alphabet size capable of creating a global shortest-schedule cycle.

Four or more letters

If

$$d \geq 4,$$

then

$$\operatorname{rank} K_\omega \geq 2.$$

Higher-rank equal-length trade geometry becomes possible.
The four-letter example of Chapter 14 already gives

$$H_1(\mathcal{H}; \mathbb{Z}) \cong \mathbb{Z}.$$

Thus four letters are the first scalar alphabet size at which a local shortest-action hole can occur.

The hierarchy is therefore:

d	shortest-completion geometry
2	rank 0: literal boxes
3	rank 1: first global S^1
≥ 4	higher rank: local holes possible

This gives a structural interpretation of alphabet size that is invisible in the original binary problem.

15.3 Three-letter normal form

Assume

$$\omega_1 < \omega_2 < \omega_3.$$

Define the adjacent gaps

$$a = \omega_3 - \omega_2, \qquad b = \omega_2 - \omega_1.$$

Let

$$h = \gcd(a, b),$$

and set

$$\alpha = \frac{a}{h}, \qquad \beta = \frac{b}{h}, \qquad m = \alpha + \beta.$$

A primitive equal-length kernel generator is

$$\boxed{g = (\alpha, -m, \beta).}$$

Indeed,

$$\alpha - m + \beta = 0,$$

and

$$\alpha\omega_1 - m\omega_2 + \beta\omega_3 = 0.$$

Thus one equal-length trade replaces m copies of the middle letter by:

- α copies of the first;
- β copies of the third.

Or vice versa.
This is the exact three-letter analogue of a primitive commuting schedule trade.

15.4 Parameterizing the optimum fiber

Write the complete optimum fiber as

$$\boxed{O_\omega(r) = \{x(t) = y + tg : 0 \le t \le N\},}$$

where

$$y = (y_1, y_2, y_3).$$

Thus

$$x(t) = (y_1 + \alpha t,\ y_2 - mt,\ y_3 + \beta t).$$

Let

$$z = y_2 - mN, \qquad 0 \le z < m.$$

Define

$$P_t = (y_1 + \alpha t + 1)(y_3 + \beta t + 1).$$

Slicing the prequotient down-set by the middle coordinate gives an exact vertex count before quotienting:

$$\boxed{|\mathcal{C}_\omega(r)| = m \sum_{t=0}^{N-1} P_t + (z+1)P_N.}$$

The kernel quotient then collapses contiguous segments along the g-direction.

Because every rank-one orbit intersects the down-set in an integer interval, the quotient can be counted by:

$$\text{vertices} - \text{adjacent orbit pairs}.$$

15.5 Exact three-letter quotient vertex count

Proposition 15.1 — Exact rank-one quotient vertex count.

Let

$$\Omega(y, N)$$

be the number of vertex orbits in

$$\mathcal{C}_\omega(r)/\mathbb{Z}g.$$

If

$$N = 0,$$

then

$$\boxed{\Omega(y, 0) = (y_1 + 1)(y_2 + 1)(y_3 + 1).}$$

If

$$N \geq 1,$$

then

$$\boxed{\Omega(y, N) = mP_{N-1} + (z+1)(P_N - P_{N-1}).}$$

Proof idea. Every kernel orbit meets the down-set in a finite integer interval.

An orbit containing s vertices contributes exactly $s - 1$ adjacent pairs

$$u,\ u + g.$$

Therefore:

$$\#\text{orbits} = \#\text{vertices} - \#\text{adjacent pairs}.$$

The same middle-coordinate slicing gives the adjacent-pair count, and subtraction yields the formula. □

The result is an exact finite formula for the number of semantic shortest-prefix states after equal-length schedule identification.

15.6 Exact cubical f-vector

The same reduction works separately for every cube orientation.

For

$$J \subseteq \{1,2,3\},$$

define

$$T_J = \{t : x_i(t) \geq 1 \text{ for every } i \in J\}.$$

Because each coordinate is affine in t,

$$T_J$$

is an integer interval, possibly empty.

If nonempty, write

$$T_J = [L_J, U_J] \cap \mathbb{Z},$$

set

$$N_J = U_J - L_J,$$

and define

$$y^{(J)} = x(L_J) - \mathbf{1}_J.$$

Proposition 15.2 — Exact rank-one cubical f-vector.

The number of quotient cubes oriented by J is

$$\boxed{f_J(r) = \begin{cases} 0, & T_J = \varnothing, \\ \Omega(y^{(J)}, N_J), & T_J \neq \varnothing. \end{cases}}$$

Hence

$$\boxed{f_j(r) = \sum_{|J|=j} f_J(r), \qquad 0 \leq j \leq 3.}$$

Thus the entire finite cubical shortest-interval geometry is computable from one rank-one orbit formula.

15.7 Worked example

Take

$$\boxed{\omega = (-5,-3,1), \qquad r = 9.}$$

Then:

$$a = 4, \qquad b = 2, \qquad h = 2,$$

so

$$\alpha = 2, \qquad \beta = 1, \qquad m = 3.$$

The primitive equal-length trade is

$$\boxed{g = (2, -3, 1).}$$

The complete optimum fiber is

$$\boxed{O_\omega(9) = \{(0, 3, 0), (2, 0, 1)\}.}$$

Thus the minimum completion has two distinct Parikh realizations of the same length:

$$(0, 3, 0)$$

and

$$(2, 0, 1).$$

The quotient cubical f-vector is

$$\boxed{(f_0, f_1, f_2, f_3) = (8, 10, 2, 0).}$$

Therefore the filled quotient contains:

- 8 vertices;
- 10 labeled edges;
- 2 elementary commuting squares;
- no 3-cube.

Yet by Theorem 14.6,

$$\boxed{\mathfrak{I}_\omega(9) \simeq S^1.}$$

This is an ideal example for the monograph because the topology is nontrivial while the finite complex remains small enough to draw completely.

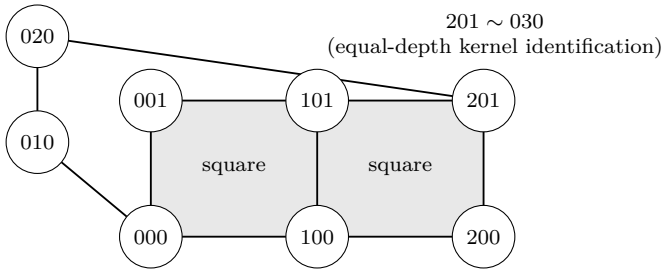


Figure 15.1: The complete rank-one quotient for $\omega = (-5, -3, 1)$ and $r = 9$. The down-set below $(2, 0, 1)$ contributes two filled squares; the path below $(0, 3, 0)$ shares the origin, and its far endpoint is identified with $(2, 0, 1)$ by the equal-length kernel translation. The quotient has $(f_0, f_1, f_2, f_3) = (8, 10, 2, 0)$ and is homotopy equivalent to S^1.

15.8 What changed relative to the binary rectangle?

In two letters, the minimum completion is one Parikh vector.

Different shortest words differ only by ordering.

In three letters, there can be two different minimum Parikh vectors:

$$x_0 \qquad \text{and} \qquad x_0 + g.$$

The prequotient geometry contains boxes beneath both.

The quotient then identifies equal-depth prefixes differing by g.

That is what creates the global circle.
Thus the transition

$$d = 2 \to d = 3$$

does not merely add another coordinate.
It adds the first equal-length trade direction.
And the transition

$$d = 3 \to d = 4$$

does not merely add one more trade.
It permits two independent trades, enough for local action topology itself to develop a hole.

15.9 Why this hierarchy belongs in the same monograph

The scalar alphabet-size hierarchy may look distant from the original 2:1 DPDA.
Its origin is nevertheless direct.
The original machine led to:

$$\text{shortest completion} \to \text{minimum Parikh demand} \to \text{rectangle}.$$

The higher-dimensional question was:

> Was the rectangle an accident of the binary alphabet?

The answer is no.
It was the rank-zero case of a general equal-length trade geometry.
The binary theory therefore sits at the first level of a topological hierarchy.
The next chapter studies a different direction: not how geometry changes with alphabet size, but how it behaves far out along a residual ray.

Chapter 16

Eventual Geodesic Arithmetic

The previous chapters studied the exact shortest-completion geometry of one fixed residual.

We now vary the residual systematically.

Fix two lattice residuals

$$r, b \in A\mathbb{Z}^d$$

and consider the ray

$$\boxed{r_n = r + nb, \qquad n \in \mathbb{N}.}$$

The minimum completion length

$$\ell_A(r_n)$$

need not be exactly linear in n.

Integer constraints can create periodic corrections.

The main result of this chapter is that those corrections eventually stabilize into a finite defect profile.

The same finite mechanism controls both:

- numerical shortest-completion distance;
- local shortest-action geometry.

16.1 The continuous directional slope

Consider the linear program

$$\boxed{\lambda_b = \min\{\mathbf{1}^\top x : Ax = -b,\ x \in \mathbb{R}^d_{\geq 0}\}.}$$

Completion totality guarantees feasibility.

Because the data are rational,

$$\lambda_b \in \mathbb{Q}_{\geq 0}.$$

The number

$$\lambda_b$$

is the continuous asymptotic completion cost per unit step in direction b.

If integer effects disappeared completely, one would expect

$$\ell_A(r + nb) \approx n\lambda_b + \text{constant}.$$

The eventual defect theory measures the exact failure of this naive linear law.

16.2 Dual certificate and reduced costs

Choose a rational dual optimum

$$y^*$$

for

$$\max\{-b^\top y : A^\top y \le \mathbf{1}\}.$$

Thus

$$-b^\top y^* = \lambda_b.$$

Choose an integer

$$L > 0$$

clearing all denominators needed below.
Define the reduced costs

$$\boxed{\delta_i = L(1 - A_i^\top y^*).}$$

Dual feasibility gives

$$\delta_i \in \mathbb{N}.$$

If

$$x_n$$

is any optimal integer completion of r_n, then

$$Ax_n = -r_n.$$

Therefore

$$\begin{aligned} L\big(\ell_A(r_n) + (y^*)^\top r - n\lambda_b\big) &= L\left(\mathbf{1}^\top x_n - (y^*)^\top A x_n\right) \\ &= \sum_i \delta_i x_n(i). \end{aligned}$$

This motivates the exact integer defect.

Definition 16.1 — Scaled integrality defect.

$$\boxed{\Delta_r(n) = L\big(\ell_A(r_n) + (y^*)^\top r - n\lambda_b\big).}$$

Then

$$\boxed{\Delta_r(n) \in \mathbb{N}.}$$

The defect is not an asymptotic error term.
It is an exact nonnegative integer reduced-cost total.
That arithmetic discreteness is what makes eventual stabilization possible.

16.3 Zero-defect shifts

Define

$$\boxed{\mathcal{S}_b = \left\{t \geq 1 : \exists z \in \mathbb{N}^d,\ Az = -tb,\ |z|_1 = t\lambda_b\right\}.}$$

An element

$$t \in \mathcal{S}_b$$

is a direction length for which the continuous optimum is attained integrally with zero defect.

Lemma 16.2 — Zero-defect semigroup.

$$\mathcal{S}_b$$

is nonempty and closed under addition.

Proof sketch. A rational primal optimum for direction b becomes integral after clearing denominators, giving at least one zero-defect multiple.

If

$$t_1, t_2 \in \mathcal{S}_b$$

with witnesses z_1, z_2, then

$$z_1 + z_2$$

satisfies

$$A(z_1 + z_2) = -(t_1 + t_2)b$$

and

$$|z_1 + z_2|_1 = (t_1 + t_2)\lambda_b.$$

Thus

$$t_1 + t_2 \in \mathcal{S}_b.$$

□

Define the canonical arithmetic period

$$\boxed{\tau_b = \gcd \mathcal{S}_b.}$$

A zero-defect shift by any $t \in \mathcal{S}_b$ can be appended to an existing completion without increasing the defect.

Thus $\mathcal{S}_b$ transports good integer behavior forward along the ray.

16.4 Stabilized defect profile

For each residue class

$$s \in \mathbb{Z}/\tau_b\mathbb{Z},$$

define

$$\boxed{M_s = \min\{\Delta_r(n) : n \geq 0,\ n \equiv s \pmod{\tau_b}\}.}$$

Theorem 16.3 — Stabilized defect profile.

For every residue class s, there exists

$$N_s$$

such that

$$\boxed{\Delta_r(n) = M_s}$$

whenever

$$n \geq N_s$$

and

$$n \equiv s \pmod{\tau_b}.$$

Consequently, eventually,

$$\boxed{\ell_A(r_n) = -(y^*)^\top r + n\lambda_b + \frac{M_{n \bmod \tau_b}}{L}.}$$

Proof.

Choose finitely many elements

$$t_1, \ldots, t_m \in \mathcal{S}_b$$

whose gcd is

$$\tau_b.$$

The additive numerical semigroup they generate contains every sufficiently large multiple of τ_b.

For each residue class s, choose an index n_s attaining the minimum defect M_s.

If

$$n \equiv s \pmod{\tau_b}$$

is sufficiently large, then

$$n - n_s$$

is a nonnegative combination of the t_i.

Append the corresponding zero-defect completions to an optimum at n_s.

This gives a feasible completion at n with the same defect M_s.

By definition of M_s, the defect cannot be smaller.

Hence it is exactly M_s. □

The formula has a useful interpretation:

$$\boxed{\text{eventual distance} = \text{rational linear drift} + \text{finite periodic integer defect.}}$$

The eventual arithmetic is therefore controlled by a finite cyclic table.

16.5 Why τ_b already contains the slope denominator

Write

$$\lambda_b = \frac{a}{q}$$

in lowest terms.
For every

$$t \in \mathcal{S}_b,$$

the zero-defect completion has integer length

$$t\lambda_b.$$

Hence

$$q \mid t.$$

Therefore

$$\boxed{q \mid \tau_b.}$$

The canonical zero-defect period is thus automatically compatible with the denominator of the asymptotic slope.
But the **minimal eventual affine period** can be smaller than τ_b.
The remaining reduction comes from symmetry of the stabilized defect table.

16.6 Fundamental defect period

View

$$M = (M_0, \dots, M_{\tau_b - 1})$$

as a cyclic word.
Define its fundamental circular period

$$\boxed{d_M = \min\{p \geq 1 : M_{s+p} = M_s \text{ for every } s\}.}$$

Then

$$d_M \mid \tau_b.$$

We can now determine the exact eventual distance period.

Theorem 16.4 — Exact minimal eventual distance period.
Let

$$\lambda_b = \frac{a}{q}$$

in lowest terms.
The smallest positive integer p for which there exists an integer D_p satisfying

$$\ell_A(r_{n+p}) = \ell_A(r_n) + D_p$$

for all sufficiently large n is

$$\boxed{p_{\min}(r, b) = \operatorname{lcm}(q, d_M).}$$

Moreover,

$$\boxed{p_{\min}(r,b) \mid \tau_b,}$$

and

$$\boxed{D_p = p\lambda_b.}$$

Proof.

Suppose p is an eventual affine period.
Comparing the linear terms in Theorem 16.3 forces

$$D_p = p\lambda_b.$$

Since D_p is integral,

$$q \mid p.$$

The periodic correction term must also be invariant under shift by p, so

$$d_M \mid p.$$

Therefore

$$\operatorname{lcm}(q, d_M) \mid p.$$

Conversely, let

$$p_0 = \operatorname{lcm}(q, d_M).$$

Then

$$p_0\lambda_b$$

is integral and the stabilized defect table is invariant under shift by p_0.
The formula of Theorem 16.3 gives

$$\ell_A(r_{n+p_0}) = \ell_A(r_n) + p_0\lambda_b$$

for all sufficiently large n.
Thus p_0 is the minimal eventual affine period. □
This theorem separates two arithmetic sources of periodicity:

1. the denominator of the continuous slope;
2. the intrinsic symmetry period of the integer defect profile.

The actual distance period is their least common multiple.

16.7 Eventual local action geometry

The same mechanism controls local shortest-action compatibility.
For

$$J \subseteq [d],$$

let

$$c_J = A\mathbf{1}_J.$$

Recall:

$$J \in \mathcal{H}_A(r_n) \iff \ell_A(r_n + c_J) = \ell_A(r_n) - |J|.$$

Apply Theorem 16.3 not only to the ray

$$r_n$$

but also to every finitely many shifted rays

$$r_n + c_J.$$

The same:

- dual optimum;
- zero-defect semigroup;
- canonical period τ_b;

can be used.
Only the stabilized defect table changes with J.
Therefore the truth value of

$$J \in \mathcal{H}_A(r_n)$$

is eventually determined by

$$n \bmod \tau_b.$$

Let

$$\Theta_s$$

be the stabilized local action complex on residue class s.
Define

$$\boxed{d_{\mathcal{H}} = \min\{p \geq 1 : \Theta_{s+p} = \Theta_s \text{ for every } s\}.}$$

Corollary 16.5 — Exact minimal local-geometry period.

The smallest eventual period of the local action complex is

$$\boxed{p^{\mathcal{H}}_{\min}(r, b) = d_{\mathcal{H}}.}$$

Moreover,

$$\boxed{d_{\mathcal{H}} \mid \tau_b.}$$

The first-action set is eventually periodic for the same reason.
Thus one finite defect mechanism governs both:

$$\boxed{\text{numerical shortest distance}}$$

and

$$\boxed{\text{local shortest-action geometry.}}$$

16.8 What is and is not being claimed

Eventual periodicity in semilinear and parametric integer settings is not new in itself.

The point here is more specific.

The completion program produces one exact object:

$$\Delta_r(n).$$

From it we derive:

- eventual affine distance;
- exact minimal distance period;
- eventual local action complex;
- exact minimal local-geometry period.

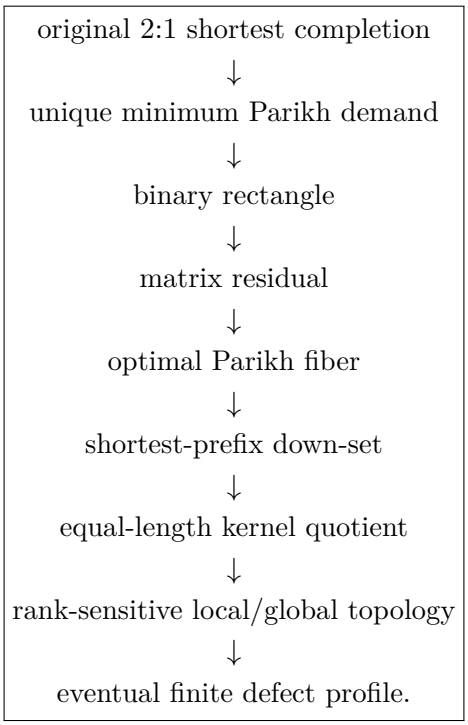
Thus the numerical and geometric observables are not proved periodic by unrelated arguments.

They descend from the same minimum-completion defect profile.

This is the structural contribution relevant to the monograph.

16.9 The higher-dimensional branch in one picture

The path from the original binary machine to the present theorem is now complete:



The binary rectangle was therefore not a small isolated trick.

It was the rank-zero member of a much larger completion geometry.

16.10 From completion geometry to finite compilation

The higher-dimensional branch has now identified optimal fibers, shortest-prefix down-sets, local action complexes, quotient topology, and eventual arithmetic. The next step is not another special counting matrix. It is to ask which part of this shortest-path structure can be compiled from finite periodic data and how much information is carried by higher-order compatibility among commuting optimal actions.

This leads to a general finite-phase periodic model. The metric backend is classical periodic shortest-path arithmetic; the new layer developed in the next chapters compiles exact optimal compatibility of every structural order and then proves that no fixed lower-order truncation universally reconstructs the full local compatibility complex.

Chapter 17

From Completion Geometry to Finite Geodesic Compilers

Research branch — Geodesic Compilers and Higher-Order Optimal Structure. *This research branch abstracts the shortest-completion geometry of the preceding chapters to positive-cost finite-phase periodic action systems. Periodic minimum-cost paths, semilinear and Presburger descriptions, shortest-path preprocessing, cubical state complexes, and squarefree divisor complexes are established background; the contribution is an exact finite compiler for all-order optimal compatibility together with prescribed-order separation, Boolean amplification, and universal k-fiber realization.*

The preceding completion-geometry chapters started from a fixed linear counting system and studied the geometry of its shortest completions. We now isolate the mechanism that survived those examples. The resulting model is a finite-phase periodic shortest-path system: a finite phase graph carries integral translations and positive costs, while the lattice coordinate remains unbounded. This abstraction retains the arithmetic structure needed for exact completion queries but no longer assumes a particular pushdown implementation or a particular counting matrix.

The background/contribution boundary matters here. Minimum-cost paths in periodic graphs, semilinear and Presburger descriptions, and shortest-path preprocessing are established theories [70, 68, 69, 73, 76]. The contribution of this branch is their organization into an exact finite compiler for geodesic branching and all-order optimal compatibility, followed by a structure theory showing that the compiled compatibility hierarchy is universally non-collapsing across the class.

In the dependency chain of this monograph, the abstraction is reached from the earlier chapters as

$$\begin{array}{c}\text{completion-aware 2:1 DPDA} \longrightarrow \rho_{p,q} = qN_1 - pN_0 \\ \longrightarrow \text{minimum completion} \longrightarrow \text{geodesic rectangles and optimal fibers} \\ \longrightarrow \text{finite-phase periodic geodesic compilation.}\end{array}$$

The original machine is therefore not a hypothesis of the results below; it records the concrete route by which the compiler abstraction was found.

17.1 Finite-phase periodic graphs

Let

$$\mathcal{B} = (\Phi, E)$$

be a finite directed graph. Every edge

$$e : \phi \to \psi$$

carries an integral displacement

$$g_e \in \mathbb{Z}^n$$

and a positive rational cost

$$c_e \in \mathbb{Q}_{>0}.$$

Its periodic lift has vertex set

$$\mathbb{Z}^n \times \Phi$$

and edges

$$(z, \phi) \xrightarrow{e} (z + g_e, \psi).$$

Fix a target phase $\phi_\star \in \Phi$ and target $t = (0, \phi_\star)$. Multiplying all costs by a common denominator preserves all shortest paths and scales all distances uniformly. Hence we assume

$$c_e \in \mathbb{N}_{>0}.$$

A state $v = (z, \phi)$ is *target-reachable* if there exists a path from v to t. For target-reachable states, let

$$d(v) = d_\phi(z)$$

be the minimum path cost to t. When $v = (z, \phi)$, we write $\mathrm{Reach}(v)$ as shorthand for $\mathrm{Reach}_\phi(z)$. The same convention is used for translated successor states.

Minimum-cost path problems in periodic graphs with integral gains and rational costs are classical; see Höfting and Wanke [70]. The role of the present section is to extract a finite arithmetic representation used by the compatibility compiler.

17.2 Gain-cost relations

Let

$$L_{\phi,\phi_\star} \subseteq E^*$$

be the regular language of base-graph walks from ϕ to $\phi_\star$. For $w = e_1 \cdots e_q$, define

$$G(w) = \sum_{j=1}^{q} g_{e_j}, \qquad C(w) = \sum_{j=1}^{q} c_{e_j}.$$

Set

$$R_\phi = \{(G(w), C(w)) : w \in L_{\phi,\phi_\star}\}.$$

Proposition 17.1 (Semilinear path relation)**.** *For every phase ϕ, the relation*

$$R_\phi \subseteq \mathbb{Z}^n \times \mathbb{N}$$

is effectively Presburger-definable.

Proof. The Parikh image of a regular language is effectively semilinear [68, 77]. Total gain and total cost are integral linear functions of the edge-count vector. Semilinear sets are closed under integral linear images and coincide with Presburger-definable sets [69].

We use the standard extension of semilinearity to subsets of $\mathbb{Z}^n \times \mathbb{N}$. Equivalently, signed coordinates may be represented by nonnegative coordinate pairs and projected by the integral map $x = x^+ - x^-$; this does not change Presburger definability. □

Define

$$\mathrm{Reach}_\phi(z) \iff \exists N \in \mathbb{N}\ R_\phi(-z, N).$$

Proposition 17.2 (Arithmetic distance predicate)**.** *There is an effectively constructible Presburger predicate*

$$\mathrm{Dist}_\phi(z, N)$$

such that

$$\mathrm{Dist}_\phi(z, N) \iff N = d_\phi(z)$$

for every target-reachable (z, ϕ)*.*

Proof. Define $\mathrm{Dist}_\phi(z, N)$ by

$$R_\phi(-z, N) \wedge \neg\exists N' \, (0 \le N' < N \wedge R_\phi(-z, N')) \, .$$

For a target-reachable state, the feasible costs form a nonempty subset of $\mathbb{N}$, hence possess a least element. □

The finite family

$$\mathfrak{C}_{\mathrm{dist}} = \{\mathrm{Reach}_\phi, \mathrm{Dist}_\phi : \phi \in \Phi\}$$

will be called the *arithmetic distance compiler*. "Finite" refers to the description; the coordinate $z \in \mathbb{Z}^n$ remains unbounded.

17.3 Bellman guards

For a lifted edge $v \xrightarrow{e} T_e v$, define $e \in \mathrm{First}(v)$ when e is the first edge of some shortest path from v to t.

Lemma 17.3 (Optimal-edge criterion)**.** *For target-reachable* v*,*

$$\boxed{e \in \mathrm{First}(v) \iff \mathrm{Reach}(T_e v) \text{ and } c_e + d(T_e v) = d(v).}$$

Proof. If a shortest path begins with e, its remaining suffix must be shortest from $T_e v$. Conversely, if the equality holds, append any shortest suffix from $T_e v$. □

Lemma 17.4 (Prefix criterion)**.** *Let* $w = e_1 \cdots e_q$ *be executable from a target-reachable state* v*. Then* w *is a prefix of some shortest path from* v *if and only if*

$$\boxed{\mathrm{Reach}(T_w v) \quad \text{and} \quad C(w) + d(T_w v) = d(v).}$$

Proof. If w occurs on a shortest path, replacing its suffix by a shorter one would shorten the full path, so the suffix is shortest. Conversely, append a shortest suffix from $T_w v$. The displayed equality makes the resulting total cost equal to $d(v)$. □

17.4 Optimal unfolding

For a target-reachable state v, let $\mathrm{OptUnf}(v)$ be the directed subgraph reachable from v using only edges satisfying the Bellman guard of Theorem 17.3.

Theorem 17.5 (Finite geodesic compiler)**.** *For every fixed positive-cost finite-phase periodic graph, one can effectively construct a finite Presburger-guarded translation program such that, from every target-reachable state* v*,*

(i) it recovers $d(v)$*;*

(ii) it recovers every optimal first edge;

(iii) it decides shortest-prefix membership for every finite executable word; and

(iv) its optimal unfolding is exactly the union of all shortest v*-to-*t *paths.*

Proof. Items (i)–(iii) follow from Theorems 17.2 to 17.4. Suppose

$$v = v_0 \to v_1 \to \cdots \to v_q$$

is a path in the optimal unfolding. Each edge satisfies

$$d(v_{j-1}) = c_j + d(v_j),$$

hence

$$d(v_q) = d(v) - \sum_{j=1}^{q} c_j.$$

Every guarded successor is target-reachable, so a shortest suffix from v_q exists. Appending it yields a shortest path containing the given prefix. Conversely, every edge of a shortest path satisfies the Bellman equality by optimal substructure. □

Corollary 17.6 (Canonical and complete generation)**.** *Fix a total order on outgoing base edges. At every non-target target-reachable state, at least one optimal first edge exists. Choosing the least one gives a deterministic canonical shortest route.*

Moreover, because every cost is positive,

$$v \to v' \text{ optimal} \quad \Longrightarrow \quad d(v') < d(v).$$

Hence d is a ranking function on the optimal unfolding. Recursive branching over all optimal first edges generates all and only shortest paths and terminates on every branch.

Chapter 18

Cubical Geodesic Compilation

The local action complexes developed earlier in the higher-dimensional completion branch are recovered here as special cases. In a one-phase translation system whose structural action family is the previously fixed completion action set, the complex defined below is exactly the earlier shortest-completion local action complex. The new notation separates two layers that were intertwined in the special counting model: $\mathcal{S}_\phi$ records structural commutation, whereas $\mathcal{H}(v)$ records the faces selected by shortest-path optimality.

18.1 Periodic action systems

Let

$$I = \{1, \dots, s\}$$

be a finite action alphabet. For every $i \in I$, let

$$\tau_i : \Phi \rightharpoonup \Phi$$

be a partial phase transition. Whenever $\tau_i(\phi)$ is defined, associate

$$g_i(\phi) \in \mathbb{Z}^n, \qquad c_i(\phi) \in \mathbb{Q}_{>0}.$$

The lifted action is

$$T_i(z, \phi) = \big(z + g_i(\phi), \tau_i(\phi)\big).$$

For an executable action word $w = i_1 \cdots i_q$, T_w denotes the corresponding composition of lifted partial actions. If J is a structural cube, $T_J v$ denotes the common endpoint obtained from any permutation of the actions in J.

18.2 Structural cubes

Definition 18.1 (Structural cube)**.** Let $J \subseteq I$. We call J a *structural cube at phase* ϕ if, for every $K \subseteq J$, every permutation of the actions in K

(i) is executable from ϕ;

(ii) reaches the same final phase;

(iii) produces the same total displacement;

(iv) has the same total cost;

and, additionally,

$$K \neq L \quad \Longrightarrow \quad (\tau_K(\phi), g_K(\phi)) \neq (\tau_L(\phi), g_L(\phi)).$$

Write $\tau_K(\phi)$, $g_K(\phi)$, and $c_K(\phi)$ for the common data.

The final injectivity condition makes the $2^{|J|}$ subset vertices distinct, so the terminology "cube" is nondegenerate. Let

$$\mathcal{S}_\phi = \{J \subseteq I : J \text{ is a structural cube at } \phi\}.$$

Lemma 18.2 (Structural simpliciality)**.** *For every ϕ, $\mathcal{S}_\phi$ is a simplicial complex.*

Proof. Every defining condition is inherited by subsets, including injectivity of the subset-endpoint map. □

Because Φ and I are finite, each $\mathcal{S}_\phi$ is finite and can be computed offline from the finite action table. Cubical state complexes generated by independent local moves are classical in reconfiguration theory; see Ghrist and Peterson [72]. Our additional object is the subcomplex selected by shortest-path optimality.

18.3 Optimal compatibility

Definition 18.3 (Optimal-compatibility complex)**.** For a target-reachable state $v = (z, \phi)$ and $J \in \mathcal{S}_\phi$, define

$$J \in \mathcal{H}(v)$$

if

$$\mathrm{Reach}(T_J v)$$

and

$$\boxed{c_J(\phi) + d(T_J v) = d(v).}$$

For $h \geq 0$, define

$$\mathcal{H}_{\leq h}(v) = \{J \in \mathcal{H}(v) : |J| \leq h\}, \qquad \mathcal{H}_{=h}(v) = \{J \in \mathcal{H}(v) : |J| = h\}.$$

For later comparison across structural orders, write

$$\Gamma_h(v) = \big(d(v), \mathcal{H}_{\leq h}(v)\big).$$

The empty set belongs to both $\mathcal{S}_\phi$ and $\mathcal{H}(v)$, with $T_\varnothing v = v$ and $c_\varnothing = 0$. Hence $\mathcal{H}_{\leq 0}(v) = \{\varnothing\}$.

Proposition 18.4 (Cubical Bellman equivalence)**.** *Let $J \in \mathcal{S}_\phi$. For target-reachable $v = (z, \phi)$, the following are equivalent:*

(i) $J \in \mathcal{H}(v)$*;*

(ii) one permutation of the actions in J is a shortest prefix from v;

(iii) every permutation of the actions in J is a shortest prefix from v.

Proof. Every permutation of a structural cube has the same cost and endpoint. Hence the prefix criterion of Theorem 17.4 either holds for all permutations or for none. □

Proposition 18.5 (Optimal simpliciality)**.** *For every target-reachable v, $\mathcal{H}(v)$ is a simplicial subcomplex of $\mathcal{S}_\phi$.*

Proof. Let $J \in \mathcal{H}(v)$ and $K \subseteq J$. Order the actions of J so that the actions of K occur first. By Theorem 18.4 this permutation is a shortest prefix, and hence its initial K-segment is a shortest prefix. Therefore $K \in \mathcal{H}(v)$. □

18.4 Arithmetic face guards

For each defined action i at phase ϕ, let $\mathrm{Opt}_{\phi,i}(z)$ denote the Presburger Bellman guard supplied by Theorem 17.3. For $J \in \mathcal{S}_\phi$, define $\mathrm{Face}_{\phi,J}(z)$ by

$$\exists N, N' \left[\mathrm{Dist}_\phi(z, N) \wedge \mathrm{Dist}_{\tau_J(\phi)}(z + g_J(\phi), N') \wedge N = c_J(\phi) + N'\right].$$

Thus

$$J \in \mathcal{H}((z, \phi)) \iff \mathrm{Face}_{\phi,J}(z).$$

Definition 18.6 (Cubical geodesic compiler)**.** Let

$$\mathfrak{C}_{\mathrm{cube}} = \big(\{\mathrm{Reach}_\phi, \mathrm{Dist}_\phi\}, \{\mathrm{Opt}_{\phi,i}\}, \{\mathrm{Face}_{\phi,J}\}\big)$$

range over all phases, defined actions, and structural faces.

Theorem 18.7 (Finite cubical geodesic compilation)**.** *For every fixed positive-cost finite-phase periodic action system, the finite arithmetic compiler $\mathfrak{C}_{\mathrm{cube}}$ is effectively constructible and determines, at every target-reachable state,*

- *exact shortest distance;*
- *all optimal first actions;*
- *shortest-prefix membership for arbitrary finite executable words;*
- *the complete local optimal-compatibility complex;*
- *a canonical shortest route under fixed tie breaking; and*
- *an implicit exact representation of all shortest paths.*

Proof. The metric, first-action, prefix, canonical, and all-geodesic claims follow from Theorems 17.5 and 17.6. For each phase, the structural complex is finite. Membership of every structural face in $\mathcal{H}(v)$ is exactly one arithmetic face guard. Evaluating all such guards therefore recovers the complete local complex. □

18.5 Signature regions

For a subcomplex $K \subseteq \mathcal{S}_\phi$, define

$$\Omega_{\phi,K} = \{z : \mathrm{Reach}_\phi(z) \text{ and } \mathcal{H}((z, \phi)) = K\}.$$

Every such set is Presburger-definable, and

$$\boxed{\{z : \mathrm{Reach}_\phi(z)\} = \bigsqcup_K \Omega_{\phi,K}.}$$

Thus the target-reachable region admits a finite arithmetic partition by local optimal-compatibility type.

18.6 Affine-semigroup interpretation

In the one-phase unit-cost translation case, write $\mathcal{S}$ for the unique phase structural complex. The compatibility complex admits an exact divisor-complex interpretation. Let the distinct action translations be

$$a_1, \ldots, a_s \in \mathbb{Z}^n$$

and let

$$D = d(r).$$

Define

$$b_i = (-a_i, 1) \in \mathbb{Z}^{n+1}, \qquad \beta_r = (r, D),$$

and the semigroup

$$\mathsf{M} = \langle b_1, \dots, b_s \rangle_{\mathbb{N}}.$$

Define the labeled squarefree divisor complex

$$\Delta_{\beta_r} = \left\{ J \subseteq [s] : \beta_r - \sum_{i \in J} b_i \in \mathsf{M} \right\}.$$

Proposition 18.8 (Homogenized semigroup correspondence)**.** *For every $J \subseteq [s]$,*

$$J \in \Delta_{\beta_r}$$

if and only if

$$|J| + d\left(r + \sum_{i \in J} a_i \right) = d(r).$$

Consequently,

$$\boxed{\mathcal{H}(r) = \mathcal{S} \cap \Delta_{\beta_r}.}$$

If every action subset is structurally nondegenerate, then $\mathcal{H}(r) = \Delta_{\beta_r}$.

Proof. Indeed,

$$\beta_r - \sum_{i \in J} b_i = \left(r + \sum_{i \in J} a_i, D - |J| \right).$$

Membership in M is therefore equivalent to the existence of a suffix consisting of exactly $D - |J|$ unit-cost actions that completes the state

$$r + \sum_{i \in J} a_i$$

to the target.

If the same state admitted a suffix of length strictly smaller than $D - |J|$, then preceding it by the J-prefix would give a completion of r of total length strictly smaller than

$$|J| + (D - |J|) = D = d(r),$$

a contradiction. Hence

$$d\left(r + \sum_{i \in J} a_i \right) = D - |J|,$$

which is equivalent to

$$|J| + d\left(r + \sum_{i \in J} a_i \right) = d(r).$$

Intersecting with the structural complex gives $\mathcal{H}(r) = \mathcal{S} \cap \Delta_{\beta_r}$. □

Squarefree divisor complexes are classical in affine-semigroup theory [71]; parametric Presburger methods can distinguish full divisor-complex types in affine-semigroup families [75]. We therefore do not claim novelty for divisor complexes or arithmetic definability in isolation.

18.7 Complexity of the Compiled Representation

The compiler has two distinct computational stages:

offline arithmetic compilation + online querying.

No polynomial-time claim is made for the general offline stage.

18.7.1 Query parameters

For $v = (z, \phi)$, let

$$\lambda(v) = 1 + \sum_{j=1}^{n} \lceil \log_2(1 + |z_j|) \rceil$$

be the coordinate encoding length.

Let $Q_{\mathrm{reach}}(\lambda)$ denote the cost of evaluating target reachability and let $Q_{\mathrm{dist}}(\lambda)$ denote the additional cost of recovering the exact distance once reachability has been established. We write

$$Q_{\mathrm{met}}(\lambda) = Q_{\mathrm{reach}}(\lambda) + Q_{\mathrm{dist}}(\lambda)$$

for a complete metric query. This convention is necessary because $d(v)$ is defined only at target-reachable states. In particular, successor states appearing in optimal-action, prefix, and face queries are tested for reachability before their distance values are evaluated.

Let $Q_{\mathrm{opt}}(\lambda)$ and $Q_{\mathrm{face}}(\lambda)$ denote the costs of evaluating one compiled optimal-action guard and one face guard, respectively. Let $Q_{\mathrm{first}}(\lambda)$ denote the cost of recovering the complete set of optimal first actions at one target-reachable state. Let $\deg^{+}_{\max}$ be the maximum outgoing base degree.

Proposition 18.9 (Distance and first-action queries)**.** *Exact distance at a target-reachable state is recovered in $Q_{\mathrm{dist}}(\lambda(v))$. All optimal first actions are recovered in*

$$O\big(\deg^{+}_{\max} Q_{\mathrm{opt}}(\lambda_*)\big)\,.$$

If optimal guards are evaluated through metric queries instead, then

$$O\big(\deg^{+}_{\max} Q_{\mathrm{met}}(\lambda_*) + Q_{\mathrm{dist}}(\lambda(v))\big)$$

suffices.

18.7.2 Prefix queries

Let w be an explicit executable word of length q. A single scan computes its endpoint phase, accumulated gain, and accumulated cost. For a fixed system,

$$\lambda(T_w v) = O(\lambda(v) + n \log(1 + q))\,.$$

Proposition 18.10 (Prefix-query complexity)**.** *Shortest-prefix membership is decidable in*

$$O(q\ \mathrm{poly}(\lambda_w) + Q_{\mathrm{met}}(\lambda_w) + Q_{\mathrm{dist}}(\lambda(v)))\,.$$

The linear dependence on q is unavoidable up to arithmetic costs because the explicit word must be read.

18.7.3 Face queries

A fixed structural face has precomputed phase, displacement, and cost data.

Proposition 18.11 (Face-query complexity)**.** *A specified structural face can be tested in* $Q_{\mathrm{face}}(\lambda_*)$*. Using only reachability and distance recovery gives*

$$O(Q_{\mathrm{dist}}(\lambda(v)) + Q_{\mathrm{met}}(\lambda(T_J v)))\,.$$

Let

$$F_h(\phi) = |\{J \in \mathcal{S}_\phi : |J| \le h\}|.$$

Then

$$F_h(\phi) \le \sum_{j=0}^{h} \binom{s}{j}.$$

Thus the complete explicit truncation $\mathcal{H}_{\le h}(v)$ *can be reported in*

$$O(F_h(\phi) Q_{\mathrm{face}}(\lambda_*) + |\mathcal{H}_{\le h}(v)|)\,.$$

Explicit recovery of the full complex can require exponential output, since even a full simplex has 2^s faces.

18.7.4 Canonical routes

Let

$$c_{\min} = \min_e c_e > 0.$$

Every optimal edge decreases distance by at least $c_{\min}$. Therefore any shortest route has at most

$$\left\lfloor \frac{d(v)}{c_{\min}} \right\rfloor$$

edges. If $\ell_{\mathrm{can}}(v)$ is the canonical route length and Q_{canon} is the cost of selecting the canonical enabled optimal action, then route generation takes

$$O(\ell_{\mathrm{can}}(v) Q_{\mathrm{canon}}(\lambda_{\mathrm{path}}))\,.$$

This is path-length-sensitive. With binary-encoded coordinates, an explicitly printed shortest path may itself be exponentially long in the input bit length.

18.7.5 Enumeration of all shortest paths

Let $\mathrm{Geo}(v)$ be the finite set of shortest paths from v, and define

$$\mathrm{OUT}(v) = 1 + \sum_{\gamma \in \mathrm{Geo}(v)} (|\gamma| + 1).$$

The shortest-prefix tree has at most $O(\mathrm{OUT}(v))$ nodes.

Theorem 18.12 (Output-sensitive geodesic enumeration)**.** *All shortest paths from* v *can be enumerated in*

$$\boxed{O\big(\mathrm{OUT}(v)\big[Q_{\mathrm{first}}(\lambda_{\mathrm{path}}) + 1\big]\big)\,.}$$

The recursion depth is at most

$$\left\lfloor \frac{d(v)}{c_{\min}} \right\rfloor.$$

We describe this as output-sensitive and path-length-sensitive rather than polynomial-delay enumeration, because a single shortest path may already be exponentially long in the binary coordinate encoding.

18.7.6 Normalized arithmetic compilers

A stronger online statement is available under an explicit normalization assumption. Suppose offline preprocessing produces

(i) a finite decision list of quantifier-free Presburger regions;

(ii) an explicit affine distance output on each region; and

(iii) quantifier-free optimal-action and face guards.

Let $\|\mathfrak{C}\|$ be the encoded compiler size and $L_{\mathfrak{C}}$ the maximum coefficient bit length.

Corollary 18.13 (Online bit complexity after normalization)**.** *Under this explicit normalization assumption,* DIST*,* FIRST*, and* FACE *are computable in*

$$\mathrm{poly}\big(\|\mathfrak{C}\|, L_{\mathfrak{C}}, \lambda(v)\big).$$

For a fixed compiled system, these queries are therefore polynomial in the queried coordinate bit length.

This is a conditional normalized-compiler result; the monograph does not claim that the required normalization is polynomial-time constructible in general. General Presburger elimination can incur substantial blow-up. Recent work obtains singly exponential elimination for an existential block, but this does not imply polynomial complexity for the complete compilation pipeline [78].

Chapter 19

Prescribed-Order Separation, Boolean Amplification, and Universal k-Fibers

We now show that the higher-order compatibility compiled in Chapter 18 cannot universally be reconstructed from any fixed lower-order truncation, then sharpen that nonreconstruction into a universal realization theorem. The first construction forces one discrepancy to appear at a prescribed order; independent products amplify such discrepancies Booleanly. The final construction asks whether that freedom is intrinsic to one lower-order signature fiber and proves that, on a fixed structural simplex, the entire next-order layer can vary arbitrarily.

Fix $k \geq 1$ and define

$$M = 2^{k+1}, \qquad \delta = 2^k - 1, \qquad N = 2k.$$

Consider the one-phase unit-cost translation actions

$$a_0 = -M,$$

$$a_i = -M + 2^{i-1}, \qquad 1 \leq i \leq k,$$

and

$$a_+ = 1.$$

Let

$$A_- = \{a_0, a_1, \ldots, a_k\}.$$

All action maps are translations, so they commute globally.

19.1 Completion and nondegeneracy

Lemma 19.1 (Completion-totality)**.** *Every integer state admits a finite path to* 0*.*

Proof. If $x \leq 0$, repeatedly apply $a_+ = 1$. If $x > 0$, apply $a_0 = -M$ until the state becomes nonpositive and then apply a_+. □

Lemma 19.2 (Nondegeneracy of the negative family)**.** *Distinct subsets of A_- have distinct total translations.*

Proof. For $K \subseteq A_-$, define

$$B(K) = \sum_{\substack{1 \leq i \leq k \\ a_i \in K}} 2^{i-1}.$$

Then

$$\sum_{a\in K} a = -|K|M + B(K).$$

If two subsets K, L have the same total translation, then

$$(|K| - |L|)M = B(K) - B(L).$$

The right side has absolute value $< M$, hence $|K| = |L|$. Then $B(K) = B(L)$, and uniqueness of binary expansion identifies all special actions. Cardinality then determines the presence of a_0. Thus $K = L$. □

In particular,

$$J_\star = \{a_1, \dots, a_k\}$$

is a genuine nondegenerate structural k-cube.

19.2 Binary-deficit distance

Lemma 19.3 (Binary-deficit distance lemma)**.** *If*

$$0 \le D < 2^k$$

and

$$\ell \ge k,$$

then

$$\boxed{d(\ell M - D) = \ell.}$$

Proof. Write

$$D = \sum_{i\in S} 2^{i-1}.$$

Apply a_i once for every $i \in S$ and apply a_0 another $\ell - |S|$ times. The total translation is $-\ell M + D$, so $d(\ell M - D) \le \ell$.

On the other hand, $D < 2^k < M$, so

$$\ell M - D > (\ell - 1)M.$$

One action decreases the state by at most M, so $\ell - 1$ actions cannot suffice. □

Action labels used in a cubical prefix may be reused later in the completion suffix; this is ordinary path semantics.

19.3 Adjacent equal-distance states

Define

$$r = NM - \delta$$

and

$$r' = r + 1 = NM - (\delta - 1).$$

Then

$$r \xrightarrow{a_+} r'.$$

Proposition 19.4 (Equal distance)**.**

$$\boxed{d(r) = d(r') = N.}$$

Proof. Apply Theorem 19.3 with deficits δ and $\delta - 1$. □

For a negative action set J, define

$$B(J) = \sum_{\substack{1 \le i \le k \\ a_i \in J}} 2^{i-1}.$$

Proposition 19.5 (Exact common lower-order truncation)**.** *For every action set J satisfying $|J| < k$,*

$$\boxed{J \in \mathcal{H}(r) \iff J \in \mathcal{H}(r') \iff J \subseteq A_-.}$$

Consequently,

$$\boxed{\mathcal{H}_{\le k-1}(r) = \mathcal{H}_{\le k-1}(r').}$$

Proof. Suppose first that $J \subseteq A_-$ and let $q = |J| < k$. After the prefix,

$$r_J = (N-q)M - (\delta - B(J)),$$

and

$$r'_J = (N-q)M - (\delta - 1 - B(J)).$$

Since $q < k$, not all k positive binary deficits occur, so $B(J) \le \delta - 1$. Thus both remaining deficits belong to $[0, 2^k)$, while $N - q \ge k + 1$. Theorem 19.3 gives remaining distance $N - q$ at both states. Hence J is optimal at both.

Now suppose $a_+ \in J$. Let q be the number of negative actions in J. Then $|J| = q+1 < k$. After the prefix from r, the residual exceeds $(N - q - 1)M$, because

$$M - \delta + B(J) + 1 > 0.$$

Yet an optimal suffix would have only $N - q - 1$ unit-cost steps available, each decreasing by at most M. This is impossible. The same argument at r' has an extra 1 of residual and is again impossible. If such a set is structurally degenerate, it is excluded from $\mathcal{H}$ already by definition; in either case it is not an optimal face. □

19.4 First discrepancy at order k

Proposition 19.6 (Distinguished separating face)**.** *For*

$$J_\star = \{a_1, \dots, a_k\},$$

$$\boxed{J_\star \in \mathcal{H}(r), \qquad J_\star \notin \mathcal{H}(r').}$$

Proof. Because

$$\sum_{i=1}^{k} 2^{i-1} = \delta,$$

after applying $J_\star$ at r,

$$r + \sum_{i=1}^{k} a_i = (N-k)M = kM.$$

Theorem 19.3 with $D = 0$ gives $d(kM) = k = N - k$. Hence $J_\star$ is optimal.

At r', the same prefix leaves $kM + 1$. The required suffix would have k actions, which can decrease the residual by at most kM. Hence no such optimal suffix exists. □

Proposition 19.7 (Single-face difference through order k)**.**

$$\boxed{\mathcal{H}_{\le k}(r) = \mathcal{H}_{\le k}(r') \cup \{J_\star\}.}$$

Proof. The complexes agree below order k by Theorem 19.5. Any k-set containing a_+ fails the same maximum-descent test at both states.

A negative k-set other than $J_\star$ contains a_0 and omits exactly one special action a_i. Its deficit is $\delta - 2^{i-1}$. After the prefix, the two residuals are

$$kM - 2^{i-1}$$

and

$$kM - (2^{i-1} - 1),$$

respectively. Both deficits lie in $[0, 2^k)$, so Theorem 19.3 gives remaining distance k in both cases. □

Theorem 19.8 (Prescribed-order adjacent-state non-factorization)**.** *For every prescribed $k \geq 1$, there exists a completion-total one-phase unit-cost system of globally commuting translation actions and adjacent states $r \to r'$ such that*

$$\boxed{d(r) = d(r'),}$$

$$\boxed{\mathcal{H}_{\leq k-1}(r) = \mathcal{H}_{\leq k-1}(r'),}$$

and

$$\boxed{\mathcal{H}_{\leq k}(r) = \mathcal{H}_{\leq k}(r') \cup \{J_\star\}.}$$

Equivalently,

$$\Gamma_{k-1}(r) = \Gamma_{k-1}(r')$$

but

$$\Gamma_k(r) \neq \Gamma_k(r').$$

Thus the first compatibility discrepancy occurs exactly at the prescribed order k.

Corollary 19.9 (No fixed lower-order reconstruction)**.** *For every fixed $h \geq 0$, the data*

$$\boxed{\big(d(v), \mathcal{H}_{\leq h}(v)\big)}$$

do not universally determine $\mathcal{H}(v)$.

Proof. Apply Theorem 19.8 with $k = h + 1$. □

The claim concerns this exact lower-order truncation. It does not assert insufficiency of an arbitrary auxiliary state description that might retain additional information such as the raw coordinates.

19.5 Exact full complexes

Proposition 19.10 (Full-complex description)**.**

$$\boxed{\mathcal{H}(r) = 2^{A_-}}$$

and

$$\boxed{\mathcal{H}(r') = 2^{A_-} \setminus \{J_\star, A_-\}.}$$

Proof. Every proper negative subset of size at most k is handled by Theorems 19.3, 19.5 and 19.7. For the full negative set,

$$r + \sum_{a \in A_-} a = (k-1)M.$$

If $k = 1$, this is already the target. If $k \geq 2$, its distance is $k - 1$ by maximal descent and the obvious a_0-completion. Hence $A_- \in \mathcal{H}(r)$.

At r', the same prefix leaves $(k-1)M + 1$, which cannot be completed in the required $k - 1$ steps. No face containing a_+ is optimal. If its prefix cost already exceeds N, this is immediate; otherwise the same maximum-descent argument used above applies. □

Accordingly,

$$\boxed{\text{through order } k \text{ the complexes differ by one } k\text{-face,}}$$

whereas

$$\boxed{\text{the full complexes differ by two faces.}}$$

This distinction is retained throughout this branch.

19.6 Large optimization-induced minimal nonfaces

For $k \geq 2$, on the distinguished vertex set $J_\star$,

$$\mathcal{H}(r)|_{J_\star} = \Delta^{k-1},$$

while

$$\mathcal{H}(r')|_{J_\star} = \partial\Delta^{k-1}.$$

For $k = 1$, the distinguished vertex is present at r and absent at r'. Thus shortest-path optimality creates minimal nonfaces of arbitrarily large cardinality inside a globally commuting translation system. For $k \geq 3$, the resulting restricted complex is nonflag.

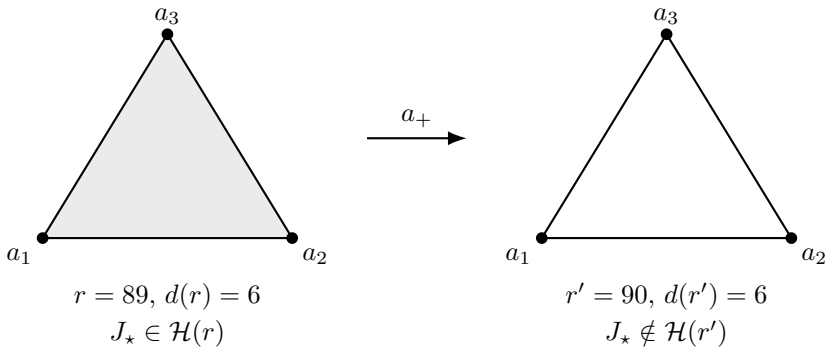


Figure 19.1: Prescribed-order separation for $k = 3$. The adjacent states $r = 89$ and $r' = 90$ have equal shortest distance and identical optimal-compatibility data through order 2. On the distinguished action set $J_\star = \{a_1, a_2, a_3\}$, the restriction of the optimal-compatibility complex changes from a filled triangle to its boundary. Thus the first discrepancy occurs exactly at order 3.

The novelty claim is not the existence of nonflag complexes or simplex boundaries in isolation; such phenomena are classical in semigroup topology [71]. The relevant constraint package is adjacency, equal shortest distance, one fixed action system, agreement of all lower compatibility orders, and first discrepancy at a prescribed order.

Remark 19.11 (Computational verification). As an independent regression check, the binary-deficit family was exhaustively verified for $1 \le k \le 10$. The distance formula, the exact lower-order agreement, and the full-complex identities of Theorem 19.10 matched the symbolic formulas in every case. Independent product-graph checks were also performed for representative small (k, m) pairs and recovered exactly the Boolean signatures of Theorem 19.16. These checks are not premises of any theorem.

19.7 Boolean Amplification

We now take independent products of the separating block.

19.7.1 Product formulas

Consider m periodic action systems with disjoint action alphabets.

Lemma 19.12 (Product distance)**.** *For*

$$V = (v_1, \dots, v_m),$$

$$\boxed{d(V) = \sum_{t=1}^{m} d_t(v_t).}$$

Proof. Every product path projects to a path in each factor, giving the lower bound. Interleaving shortest factor paths gives equality. □

For $v_t = (z_t, \phi_t)$, write $\mathcal{S}_t(\phi_t)$ for the structural complex in factor t, and write $\mathcal{S}(V)$ for the product structural complex at V. For an action set

$$J = J_1 \sqcup \cdots \sqcup J_m,$$

structural compatibility factors:

$$J \in \mathcal{S}(V) \iff J_t \in \mathcal{S}_t(\phi_t) \quad \forall t.$$

Distinct factor-subset tuples give distinct product endpoints, so nondegeneracy also factors.

Lemma 19.13 (Product-join lemma)**.**

$$\boxed{\mathcal{H}(V) = \mathcal{H}_1(v_1) * \cdots * \mathcal{H}_m(v_m).}$$

Proof. First, $T_J V$ is target-reachable exactly when every $T_{J_t} v_t$ is target-reachable. In that case define

$$\sigma_t = c_{J_t} + d_t(T_{J_t} v_t) - d_t(v_t) \ge 0.$$

Then

$$c_J + d(T_J V) - d(V) = \sum_{t=1}^{m} \sigma_t.$$

The sum vanishes exactly when every σ_t vanishes. □

19.7.2 Product family

Take m independent copies of the Chapter 19 block. Write

$$r^{(0)} = r, \qquad r^{(1)} = r',$$

and for $\varepsilon \in \{0,1\}^m$ define

$$R^{\varepsilon} = \big(r^{(\varepsilon_1)}, \dots, r^{(\varepsilon_m)}\big).$$

Let $J_\star^{(t)}$ be the distinguished k-face in factor t.

Proposition 19.14 (Constant lower-order signature)**.** *For every $\varepsilon, \eta \in \{0,1\}^m$,*

$$d(R^{\varepsilon}) = d(R^{\eta}) = mN$$

and

$$\boxed{\Gamma_{k-1}(R^{\varepsilon}) = \Gamma_{k-1}(R^{\eta}).}$$

Proof. Distance additivity gives the first assertion. Every factor component of an action set of total size at most $k-1$ has size at most $k-1$, where the two base states have identical compatibility. Apply Theorem 19.13. □

Proposition 19.15 (Exact order-k variation)**.** *There exists a fixed complex $\mathcal{K}_{k,m}$ such that*

$$\boxed{\mathcal{H}_{\le k}(R^{\varepsilon}) = \mathcal{K}_{k,m} \cup \{J_\star^{(t)} : \varepsilon_t = 0\}.}$$

Proof. If an order-$\le k$ face uses more than one factor, then every factor component has size $< k$, so its membership is determined entirely by the common lower-order base data. If all k actions lie in one factor, Theorem 19.7 says the only variable face is that factor's distinguished $J_\star$. □

Theorem 19.16 (Boolean fiber amplification)**.** *For every $k, m \ge 1$, there exists a completion-total one-phase unit-cost translation system containing 2^m states with one common order-$(k-1)$ signature and 2^m pairwise distinct order-k signatures.*

The variable part of the signature is precisely

$$\{J_\star^{(t)} : \varepsilon_t = 0\},$$

so the constructed order-k signatures form a Boolean lattice of rank m.

The states themselves form a directed m-cube in the underlying state graph, because changing $\varepsilon_t : 0 \to 1$ is exactly the factor action $a_+^{(t)}$. Throughout this state cube,

$$d(R^{\varepsilon}) = mN$$

and $\Gamma_{k-1}(R^{\varepsilon})$ remain constant, whereas Γ_k distinguishes every vertex.

19.7.3 Exponential fibers

Each factor has $k+2$ action labels. Thus the product has

$$s = m(k+2)$$

actions.

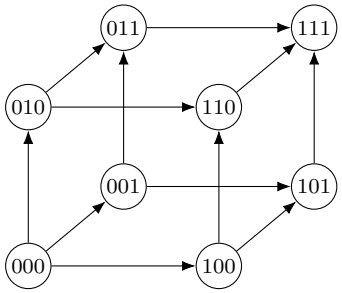
Corollary 19.17 (Exponential signature fibers)**.** *For every fixed $k \ge 1$ and every $m \ge 1$, there is a system with*

$$s = m(k+2)$$

actions for which one order-$(k-1)$ signature supports at least

$$\boxed{2^m = 2^{s/(k+2)}}$$

distinct order-k signatures. Thus exponentially large fibers occur along the infinite sequence of action dimensions divisible by $k+2$.



$d(R^{\varepsilon}) = 3N$ for all ε

$\Gamma_{k-1}(R^{\varepsilon})$ is constant

Γ_k distinguishes all 8 vertices

coordinate t toggles $J_{\star}^{(t)}$

Figure 19.2: Boolean amplification for three independent factors. The eight product states form a directed state cube. Exact distance and the complete order-$(k-1)$ signature are constant throughout the cube. Moving in coordinate t toggles exactly the distinguished order-k face $J_{\star}^{(t)}$, producing eight distinct order-k signatures.

19.7.4 Conditional discrimination

Suppose an auxiliary summary $C(v)$ is required to support an exact decoder

$$\Gamma_k(v) = F\big(\Gamma_{k-1}(v), C(v)\big)$$

on the constructed family, with no raw state coordinate supplied separately.

Corollary 19.18 (Conditional discrimination bound)**.**

$$\boxed{|\operatorname{im} C| \geq 2^m}$$

and therefore

$$\boxed{\log_2 |\operatorname{im} C| \geq m = \frac{s}{k+2}.}$$

Proof. All R^{ε} have the same lower-order signature but pairwise distinct order-k signatures. Hence two distinct states cannot receive the same summary value. □

This is a discrimination bound for exact summaries under the stated decoder model. It is not a general RAM-space lower bound.

19.8 From Boolean amplification to k-fibers

The product theorem proves that a common order-$(k-1)$ signature can support many different order-k extensions, but its independent bits live in disjoint factors. This leaves a sharper structural question: after exact distance and the *entire* lower-order compatibility complex have been frozen, how much freedom can remain among overlapping k-faces on one structural simplex?

Definition 19.19 (k-fiber)**.** Fix a phase ϕ, let $k \geq 1$, and let γ be a realizable value of Γ_{k-1} at phase ϕ. The *k-fiber above γ* is

$$\mathcal{F}_k(\phi, \gamma) = \{\mathcal{H}_{=k}(v) : v \text{ is in phase } \phi,\ \Gamma_{k-1}(v) = \gamma\}.$$

The word *fiber* here refers to a fiber of the truncation map on optimal-compatibility signatures. It is distinct from the optimal Parikh fibers of Chapter 12, which collect minimum completion vectors rather than higher-order compatibility layers.

For $A \subseteq \{J \in \mathcal{S}_{\phi} : |J| = k\}$ define the exact realization region

$$\Omega^{(k)}_{\phi,\gamma,A} = \{z : \Gamma_{k-1}((z,\phi)) = \gamma,\ \mathcal{H}_{=k}((z,\phi)) = A\}.$$

Theorem 19.20 (Finite k-fiber compilation)**.** *For every fixed positive-cost finite-phase periodic action system, every phase ϕ, every $k \geq 1$, and every realizable order-$(k-1)$ signature γ:*

(i) every region $\Omega^{(k)}_{\phi,\gamma,A}$ is effectively Presburger-definable;

(ii) nonemptiness of every such region is decidable;

(iii) the complete finite fiber $\mathcal{F}_k(\phi,\gamma)$ is effectively computable.

Proof. The structural complex $\mathcal{S}_\phi$ is finite. Exact equality $\Gamma_{k-1}((z,\phi)) = \gamma$ is a finite Boolean combination of the distance predicate and the compiled face guards for structural faces of size at most $k-1$. Exact equality $\mathcal{H}_{=k}((z,\phi)) = A$ is a finite conjunction of positive guards for $J \in A$ and negative guards for structural k-faces not in A. Hence every exact realization region is Presburger-definable. Presburger decidability determines which of the finitely many candidate patterns occur, and those patterns are exactly the k-fiber. □

This theorem is an effective classification statement rather than a polynomial-time claim. It converts the statewise compiler of Chapter 18 into a finite classifier for all possible next-order extensions of a fixed lower-order signature.

19.9 The universal two-phase construction

Fix integers

$$s \geq k \geq 2, \qquad [s] = \{1,\dots,s\}, \qquad \mathcal{E} = \binom{[s]}{k}, \qquad m = |\mathcal{E}| = \binom{s}{k},$$

and put

$$R = \binom{s-1}{k-1}.$$

We construct one fixed two-phase periodic action system $\mathcal{U}_{s,k}$.

Let the phases be G and C, with target $(0, C)$. The coordinate set is

$$Q = \{b\} \sqcup \{p_1,\dots,p_s\} \sqcup \{h_E : E \in \mathcal{E}\},$$

so the lattice dimension is $1+s+\binom{s}{k}$. The coordinate b balances distances across witnesses, the private coordinates p_i force subset-endpoint injectivity, and the incidence coordinates h_E carry the next-order compatibility bits.

For each $i \in [s]$, introduce a distinguished action

$$a_i : G \to G$$

with translation

$$u_i = e_{p_i} + \sum_{\substack{E \in \mathcal{E} \\ i \in E}} e_{h_E}$$

and cost

$$c(a_i) = R + 1 = \|u_i\|_1.$$

Write

$$I_\star = \{a_1,\dots,a_s\}.$$

Completion is isolated in the second phase. Introduce a switch

$$\sigma : G \to C$$

with zero translation and cost 1. For every $q \in Q$, introduce unit-cost completion actions

$$\xi_q^+, \xi_q^- : C \to C$$

with translations $+e_q$ and $-e_q$, respectively. No completion action is defined in phase G, and no distinguished action is defined in phase C.

Lemma 19.21 (Completion-totality)**.** *Every state of $\mathcal{U}_{s,k}$ is target-reachable.*

Proof. From phase C, move coordinatewise to the origin. From phase G, apply σ and then complete coordinatewise in phase C. □

19.10 Exact metric and structural isolation

Lemma 19.22 (Completion-phase metric)**.** *For every $x \in \mathbb{Z}^Q$,*

$$d(x, C) = \|x\|_1.$$

Proof. Coordinate unit moves attain $\|x\|_1$. Conversely, the total cost of any completion path is the sum of the ℓ_1-norms of its unit translations and is therefore at least the norm of the total displacement $-x$. □

Lemma 19.23 (Geodesic-phase metric)**.** *For every $x \in \mathbb{Z}^Q$,*

$$d(x, G) = 1 + \|x\|_1.$$

Proof. Switching immediately and completing coordinatewise gives the upper bound. Conversely, suppose a path first performs a distinguished prefix with total displacement U and cost C_U, then switches to phase C. Every distinguished displacement is coordinatewise nonnegative and has cost equal to its ℓ_1-norm, so $C_U = \|U\|_1$. By Lemma 19.22, the remaining cost is at least $\|x + U\|_1$. Hence the total cost is at least

$$\|U\|_1 + 1 + \|x + U\|_1 \geq 1 + \|x\|_1.$$

□

Lemma 19.24 (Distinguished structural simplex)**.** *The action set $I_\star$ spans a structural simplex in phase G.*

Proof. The distinguished actions are translations that preserve phase G, so every ordering of every subset is executable with the same displacement and cost. The p_i-coordinate of the total displacement of a subset is 1 exactly when a_i is present; therefore distinct subsets have distinct endpoints. □

Lemma 19.25 (Auxiliary structural isolation)**.** *The structural complex in phase G is*

$$\mathcal{S}_G = 2^{I_\star} \cup \{\{\sigma\}\}.$$

In particular, no structural face contains both σ and a distinguished action.

Proof. The singleton $\{\sigma\}$ is structural. If a set contains σ and some a_i, the ordering that executes σ first cannot execute a_i afterward because a_i is undefined in phase C. No completion action is defined in phase G. □

19.11 Hypergraph-coded witness states

For each

$$A \subseteq \mathcal{E},$$

let $L = m + 1$ and define $v_A \in \mathbb{Z}^Q$ by

$$\begin{aligned} (v_A)_b &= -(L - |A|), \\ (v_A)_{p_i} &= -1 \qquad (i \in [s]), \\ (v_A)_{h_E} &= -\big(k - 1 + \mathbf{1}_{E \in A}\big) \qquad (E \in \mathcal{E}). \end{aligned}$$

Thus $(v_A)_{h_E} = -k$ when $E \in A$ and $-(k-1)$ otherwise. We regard (v_A, G) as the witness state encoding the k-uniform hypergraph $([s], A)$.

Lemma 19.26 (Equal witness distance)**.** *All witness states have the same exact shortest distance*

$$D_{s,k} = s + 2 + k\binom{s}{k}.$$

Proof. Since $L = m + 1$,

$$\|v_A\|_1 = (L - |A|) + s + m(k-1) + |A| = s + 1 + mk.$$

Apply Lemma 19.23. □

For $J \subseteq [s]$, write

$$U_J = \sum_{i \in J} u_i.$$

The incidence identity driving the construction is

$$(U_J)_{h_E} = |J \cap E| \qquad (E \in \mathcal{E}). \tag{19.1}$$

Also $(U_J)_{p_i} = \mathbf{1}_{i \in J}$, $(U_J)_b = 0$, and

$$\|U_J\|_1 = |J|(R+1).$$

Lemma 19.27 (Coordinatewise geodesic criterion)**.** *For every $A \subseteq \mathcal{E}$ and every $J \subseteq [s]$,*

$$\{a_i : i \in J\} \in \mathcal{H}(v_A, G)$$

if and only if

$$U_J \leq -v_A$$

coordinatewise.

Proof. By Lemma 19.23, the Bellman equality for the distinguished prefix is

$$\|U_J\|_1 + 1 + \|v_A + U_J\|_1 = 1 + \|v_A\|_1.$$

Every coordinate of v_A is nonpositive and every coordinate of U_J is nonnegative. Equality therefore holds exactly when no coordinate overshoots zero, equivalently $U_J \leq -v_A$ coordinatewise. □

19.12 Universal k-fiber realization

Lemma 19.28 (Complete lower-order invariance)**.** *For every $A, B \subseteq \mathcal{E}$,*

$$\mathcal{H}_{\leq k-1}(v_A, G) = \mathcal{H}_{\leq k-1}(v_B, G).$$

More precisely, every distinguished subset of cardinality at most $k-1$ is optimal at every witness state, the switch singleton $\{\sigma\}$ is optimal at every witness state, and these are all structural faces of phase G through order $k-1$.

Proof. Let $J \subseteq [s]$ with $|J| \leq k-1$. At a private coordinate,

$$(U_J)_{p_i} \leq 1 = -(v_A)_{p_i}.$$

At an incidence coordinate, (19.1) gives

$$(U_J)_{h_E} = |J \cap E| \leq |J| \leq k-1 \leq -(v_A)_{h_E}.$$

The balance coordinate receives zero distinguished displacement, so Lemma 19.27 makes every such J optimal for every A. The switch singleton is optimal because

$$1 + d(v_A, C) = 1 + \|v_A\|_1 = d(v_A, G).$$

By Lemma 19.25, there are no other structural faces in phase G. □

Let

$$K_{s,k} = \mathcal{H}_{\le k-1}(v_A, G), \qquad \gamma_{s,k} = (D_{s,k}, K_{s,k});$$

these are independent of A.

Lemma 19.29 (Exact k-face decoder)**.** *For every $A \subseteq \mathcal{E}$ and every $J \in \mathcal{E}$,*

$$\{a_i : i \in J\} \in \mathcal{H}(v_A, G) \quad \Longleftrightarrow \quad J \in A.$$

Proof. At the coordinate h_J, (19.1) gives $(U_J)_{h_J} = k$. If $J \in A$, the threshold is $-(v_A)_{h_J} = k$ and is saturated exactly. If $J \notin A$, the threshold is only $k-1$, so the prefix overshoots and cannot be geodesic. For $E \neq J$ with $|E| = |J| = k$,

$$|J \cap E| \le k - 1 \le -(v_A)_{h_E},$$

while private coordinates are saturated at most once and the balance coordinate is unchanged. Apply Lemma 19.27. □

Theorem 19.30 (Universal k-Fiber Theorem)**.** *For every $s \ge k \ge 2$, there exists a fixed completion-total positive-cost two-phase periodic action system $\mathcal{U}_{s,k}$, a phase G, a distinguished structural s-simplex $I_\star$, and a realizable order-$(k-1)$ signature $\gamma_{s,k}$ whose complete k-fiber is the full power set of the distinguished k-faces:*

$$\boxed{\mathcal{F}_k(G, \gamma_{s,k}) = 2^{\binom{I_\star}{k}}.}$$

Indeed, the witness family

$$\{(v_A, G) : A \subseteq \binom{[s]}{k}\}$$

contains $2^{\binom{s}{k}}$ equal-distance states with one common complete lower-order compatibility truncation, and

$$\mathcal{H}_{=k}(v_A, G) = \big\{\{a_i : i \in J\} : J \in A\big\}.$$

Proof. Equal distance and the common order-$(k-1)$ signature follow from Lemmas 19.26 and 19.28. Since $k \ge 2$, Lemma 19.25 implies that every structural k-face in phase G lies entirely in the distinguished simplex. Hence every member of the fiber is a subset of $\binom{I_\star}{k}$. The exact decoder of Lemma 19.29 realizes every such subset as A ranges over $2^{\binom{[s]}{k}}$. □

Corollary 19.31 (Uniform-hypergraph realization)**.** *For every $s \ge k \ge 2$, every k-uniform hypergraph on $[s]$ occurs as the complete order-k optimal-compatibility layer on one fixed distinguished structural simplex, while exact shortest distance and the complete optimal-compatibility structure through order $k-1$ remain unchanged.*

Corollary 19.32 (Graphs inside one 2-fiber)**.** *For every $s \ge 2$, one fixed periodic shortest-path system has a common order-1 signature whose order-2 fiber realizes every simple graph on s distinguished vertices.*

Corollary 19.33 (Overlapping-face independence)**.** *Let $\mathcal{A} \subseteq \binom{[s]}{k}$ have arbitrary intersections among its members. Inside one common order-$(k-1)$ fiber, the membership bits of all faces in $\mathcal{A}$ can be prescribed independently.*

19.13 A worked universal fiber: $s = 3$, $k = 2$

Take

$$s = 3, \qquad k = 2, \qquad \mathcal{E} = \{12, 13, 23\}.$$

Then $m = 3$, $R = 2$, and $L = 4$. In the coordinate order

$$(b, p_1, p_2, p_3, h_{12}, h_{13}, h_{23}),$$

the distinguished translations are

$$\begin{aligned} u_1 &= e_{p_1} + e_{h_{12}} + e_{h_{13}}, \\ u_2 &= e_{p_2} + e_{h_{12}} + e_{h_{23}}, \\ u_3 &= e_{p_3} + e_{h_{13}} + e_{h_{23}}, \end{aligned}$$

all of cost 3.

For

$$A = \{12, 23\},$$

the witness vector is

$$v_A = (-2, -1, -1, -1, -2, -1, -2).$$

Its ℓ_1-norm is 10, so $d(v_A, G) = 11$; the same distance holds for all eight choices of $A \subseteq \{12, 13, 23\}$. Every distinguished singleton and the switch singleton are optimal at every witness, so

$$\mathcal{H}_{\leq 1} = \{\varnothing, \{\sigma\}, \{a_1\}, \{a_2\}, \{a_3\}\}$$

is fixed. At order two, the pairs 12 and 23 saturate their threshold-2 coordinates, while 13 overshoots the threshold-1 coordinate h_{13}. Hence

$$\mathcal{H}_{=2}(v_A, G) = \big\{\{a_1, a_2\}, \{a_2, a_3\}\big\}.$$

As A ranges over all subsets of $\{12, 13, 23\}$, the common order-1 signature remains fixed while the order-2 layer runs through all eight simple graphs on the three distinguished vertices.

19.14 Exact next-order capacity

Definition 19.34 (Distinguished k-fiber capacity)**.** For $s \geq k \geq 2$, define $\mathrm{Cap}_k(s)$ to be the supremum, over all positive-cost finite-phase periodic action systems, phases ϕ, realizable order-$(k-1)$ signatures γ, and distinguished action sets $I_\star$ with $|I_\star| = s$ and $2^{I_\star} \subseteq \mathcal{S}_\phi$, of

$$\left| \left\{ \mathcal{H}_{=k}(v) \cap \binom{I_\star}{k} : \Gamma_{k-1}(v) = \gamma \right\} \right|.$$

Theorem 19.35 (Exact distinguished-layer capacity)**.** *For every $s \geq k \geq 2$,*

$$\boxed{\mathrm{Cap}_k(s) = 2^{\binom{s}{k}}.}$$

Proof. An s-vertex simplex has exactly $\binom{s}{k}$ possible k-faces, so no fiber can induce more than $2^{\binom{s}{k}}$ patterns on them. Theorem 19.30 realizes every such pattern inside one common fiber. □

Thus the universal theorem does not merely give an exponential lower bound: relative to a fixed distinguished structural simplex, it attains the maximum combinatorially possible next-order freedom.

19.15 From separation to universality

The progression inside this branch is now exact. Prescribed-order threshold separation arranges that every prefix of size at most $k-1$ lies below two nearby thresholds while one selected k-prefix distinguishes them. Boolean amplification places independent copies in disjoint factors. Universal k-fiber realization removes the product mechanism: all variable k-faces live on one fixed set of s distinguished actions and may overlap arbitrarily.

The arithmetic replacement for factor independence is the incidence identity

$$(U_J)_{h_E} = |J \cap E|.$$

Every proper lower-order prefix sees at most $k-1$ in every incidence coordinate, while a k-set reaches the value k only in its own coordinate. The architecture separates four jobs: h_E coordinates carry the variable next-order bits, p_i coordinates enforce structural injectivity, the balance coordinate b fixes the common distance, and the completion phase C supplies completion-totality without creating mixed structural faces.

Accordingly, the progression is

prescribed-order threshold separation
$\Downarrow$
Boolean amplification in disjoint factors
$\Downarrow$
universal overlapping k-fiber realization.

19.16 Relation to realization theories and scope

The surrounding ingredients remain classical: periodic shortest paths, semilinear and Presburger descriptions, cubical state complexes, and squarefree divisor complexes were already separated from the contribution in Chapters 17–18. Shortest-path computation on a *given* metric simplicial complex is another established setting [74]; here the simplicial object is itself selected by joint shortest-prefix optimality in an infinite periodic graph. Broad realization behavior is also known for squarefree divisor complexes [71, 3], and parametric Presburger arithmetic controls complex types in affine-semigroup families [75].

No novelty is claimed for simplicial or hypergraph realization in isolation, for commuting actions forming cubes, or for Presburger definability. The structural constraint package is the point: one fixed periodic shortest-path system, one fixed phase, one fixed exact distance, one fixed complete compatibility truncation through order $k-1$, one fixed structural simplex, and every possible order-k pattern on that simplex. The capacity theorem is deliberately stated relative to the distinguished structural simplex; it makes no claim in terms of the total number of auxiliary action labels in the completion-total system.

19.17 Synthesis: finite compilation and universal next-order freedom

The compiler and structure theorems now give three levels of conclusion. First, every fixed positive-cost finite-phase periodic action system admits a finite arithmetic description of exact metric and local optimal-compatibility data. Second, the hierarchy does not collapse: for every prescribed k, a first discrepancy can occur exactly at order k, and independent products amplify such discrepancies Booleanly. Third, the ambiguity left after fixing all lower-order data can be maximal rather than sparse: one common lower-order signature can support every order-k pattern on a distinguished s-simplex.

Thus

shortest-path geometry is finite in description

while

optimal compatibility is unbounded in informational order
and can be universally free at the next order.

Equivalently, fixing exact distance and the entire compatibility hierarchy through order $k-1$ imposes no universal restriction on the next-order layer. The $\binom{s}{k}$ possible distinguished k-faces can carry exactly $\binom{s}{k}$ independent binary bits, yielding the full pattern count $2^{\binom{s}{k}}$.

The symbolic proofs above are primary. Appendix D records independent small-parameter regression checks of common distance, complete lower-order invariance, and exhaustive realization of all prescribed k-face patterns.

Research branch — Full simplicial universality. *The universal k-fiber theorem varies one distinguished compatibility layer while freezing the lower layers. The next four chapters remove that distinguished-rank restriction: one fixed coefficient bank and one fixed Bellman system are required to realize every labeled simplicial complex. Discovery chronology is recorded in the Preface; here the branch is placed directly beside its geodesic prerequisites.*

Chapter 20

Universal Coefficient Banks and Simultaneous Simplicial Representation

The preceding k-fiber theorem isolates one rank at a time. The present chapter changes the quantifier order: the coefficient system is fixed before the target complex is known, and only the quota vector is allowed to vary. This produces a simultaneous representation problem rather than a target-dependent threshold-dimension problem.

20.1 Definitions and Main Results

Let $[s] = \{1, \dots, s\}$, and let $\mathrm{Simp}([s])$ denote the nonvoid downward-closed families of subsets of $[s]$. Ghost vertices are allowed unless stated otherwise. Throughout this branch, the rank of a face J means its cardinality $|J|$; thus a rank-k face has simplicial dimension $k-1$.

For

$$A \in \mathbb{R}^{d\times s}_{\geq 0}, \qquad b \in \mathbb{R}^{d}_{\geq 0},$$

define

$$\mathcal{K}(A,b) = \{J \subseteq [s] : A\mathbf{1}_J \leq b\}.$$

Every such family is simplicial because $A \geq 0$.

Definition 20.1. The *simultaneous universal row complexity* is

$$u(s) = \min\left\{d : \exists A \in \mathbb{R}^{d\times s}_{\geq 0}\ \forall K \in \mathrm{Simp}([s])\ \exists b_K \geq 0 :\ K = \mathcal{K}(A, b_K)\right\}.$$

The *common-shell universal row complexity* is

$$u_\Sigma(s) = \min\Big\{d : \begin{array}{l}\exists A \in \mathbb{R}^{d\times s}_{\geq 0},\ \exists R \geq 0,\\ \forall K \in \mathrm{Simp}([s])\ \exists b_K \geq 0 :\ K = \mathcal{K}(A, b_K),\ \|b_K\|_1 = R\end{array}\Big\}.$$

Put

$$w_s = \binom{s}{\lfloor s/2 \rfloor}, \qquad W_s = 2^s - 1, \qquad R_s = w_s W_s.$$

Theorem 20.2 (Exact Universal Row Complexity and Zero Shell Defect)**.** *For every $s \geq 2$,*

$$u(s) = u_\Sigma(s) = w_s.$$

Moreover, there exists

$$A_s \in \mathbb{Z}^{w_s\times s}_{>0}$$

such that for every $K \in \mathrm{Simp}([s])$ *there is* $b_K \in \mathbb{Z}_{\geq 0}^{w_s}$ *with*

$$K = \mathcal{K}(A_s, b_K), \qquad \|b_K\|_1 = R_s.$$

No nonnegative real coefficient matrix with fewer than w_s *rows is universal. Consequently* $\delta_\Sigma(s) = 0$.

This result is proved as theorem 21.5.

The notation $H(v, G)|_{I_\star}$, defined formally in Section 22, denotes the family of distinguished action subsets satisfying the corresponding Bellman optimality equality at v.

Theorem 20.3 (Universal Bellman Compatibility)**.** *For every* $s \geq 2$*, there exists one positive-cost translation-periodic shortest-path system* $\mathcal{U}_s$ *with distinguished action set* $I_\star = \{a_1, \ldots, a_s\}$ *such that for every* $K \in \mathrm{Simp}([s])$ *there exists a state* v_K *satisfying*

$$H(v_K, G)|_{I_\star} = K.$$

The witness may be chosen so that

$$d((v_K, \mathsf{P}), G) = 1 + R_s$$

for every K.

This result is proved as theorem 22.5.

Theorem 20.4 (Universal Extension-Fiber Completeness)**.** *Fix* $s \geq 2$, $0 \leq r \leq s$, *and a downward-closed lower truncation* $L \subseteq 2^{[s]}_{\leq r}$. *Let* $P_{s,r}(L)$ *be the poset of higher faces compatible with* L. *Then*

$$\mathrm{ExtFib}_r(L) \cong \mathcal{J}(P_{s,r}(L)),$$

and the same fixed Bellman system realizes the entire fiber:

$$\mathrm{ExtFib}_r^{\mathcal{U}_s}(L) = \mathrm{ExtFib}_r(L).$$

Every member admits a witness at the same canonical distance $1 + R_s$.

This result is proved as theorem 23.2.

Corollary 20.5 (Simultaneous All-Rank Saturation)**.** *For every* $2 \leq k \leq s$*, the same system* $\mathcal{U}_s$ *realizes every subset of the rank-*k *layer while the complete lower truncation through rank* $k-1$ *is fixed. Hence*

$$|\,\mathrm{RankFib}_k\,| = 2^{\binom{s}{k}}.$$

This result appears as theorem 23.3.

20.2 Universal Coefficient Construction

Fix $s \geq 2$. Choose a symmetric chain decomposition

$$2^{[s]} = C_1 \sqcup \cdots \sqcup C_{w_s},$$

whose classical existence goes back to de Bruijn, van Ebbenhorst Tengbergen, and Kruyswijk [92]. For each chain C_q, choose an ordering $i_{q,1}, \ldots, i_{q,s}$ of $[s]$ such that every member of C_q is an initial segment for the corresponding admissible cardinalities. Define the qth row of A_s by

$$a_{q,i_{q,j}} = 2^{s-j}.$$

Thus every row is a permutation of $2^{s-1}, \ldots, 1$, and every row has total sum W_s.

20.2.1 Exact one-row separation

Let $E \in C_q$ with $|E| = m$, so

$$E = \{i_{q,1}, \dots, i_{q,m}\}.$$

Define

$$\tau_q(E) = W_s - 2^{s-m}.$$

Then $A_q \mathbf{1}_E = \tau_q(E) + 1$.

Lemma 20.6 (Exact chain separator)**.** *For every $J \subseteq [s]$,*

$$A_q \mathbf{1}_J \le \tau_q(E) \quad \Longleftrightarrow \quad E \nsubseteq J.$$

Proof. If $E \subseteq J$, then $A_q \mathbf{1}_J \ge A_q \mathbf{1}_E = \tau_q(E) + 1$. Conversely, suppose $E \nsubseteq J$. At least one element of E is omitted. The smallest weight carried by an element of E is 2^{s-m}, whereas the total weight of all positions outside E is $2^{s-m} - 1$. Hence the largest score of a set omitting an element of E is at most

$$A_q \mathbf{1}_E - 2^{s-m} + (2^{s-m} - 1) = \tau_q(E).$$

□

20.2.2 Minimal nonfaces and canonical quotas

For $K \in \mathrm{Simp}([s])$, let $\mathrm{MinNF}(K)$ denote its set of minimal nonfaces. This is an antichain, so each chain C_q contains at most one minimal nonface. Define

$$b^0_{K,q} = \begin{cases} \tau_q(E), & C_q \cap \mathrm{MinNF}(K) = \{E\}, \\ W_s, & C_q \cap \mathrm{MinNF}(K) = \varnothing. \end{cases}$$

Proposition 20.7 (Universal realization)**.** *For every $K \in \mathrm{Simp}([s])$,*

$$K = \{J \subseteq [s] : A_s \mathbf{1}_J \le b^0_K\}.$$

Proof. If $J \in K$, no minimal nonface is contained in J, so every active row accepts J by theorem 20.6; inactive rows have quota W_s and are automatic. If $J \notin K$, then J contains some minimal nonface E, and the row corresponding to the chain of E rejects J by theorem 20.6. □

Thus $u(s) \le w_s$.

20.2.3 Canonical quota mass

The canonical quota total is

$$\|b^0_K\|_1 = w_s W_s - \sum_{E \in \mathrm{MinNF}(K)} 2^{s-|E|}.$$

Define the canonical shell deficit by

$$\mathrm{def}(K) := R_s - \|b^0_K\|_1.$$

Then

$$\mathrm{def}(K) = \sum_{E \in \mathrm{MinNF}(K)} 2^{s-|E|}.$$

If $|\,\mathrm{MinNF}(K)| < w_s$, some row is inactive. Increasing its quota from W_s to $W_s + \mathrm{def}(K)$ leaves the represented family unchanged and brings the total quota to R_s. Thus only the extremal case

$$|\,\mathrm{MinNF}(K)| = w_s$$

remains.

20.2.4 Injectivity of subset scores

For every row q, the map $J \mapsto A_q \mathbf{1}_J$ is injective because the row entries are distinct powers of two. Hence

$$J \longmapsto A_s \mathbf{1}_J$$

is injective on $2^{[s]}$.

Chapter 21

Exact Row Complexity and Common-Shell Compression

The previous chapter gives a universal construction. We now determine its exact row cost and show that synchronizing all target quota vectors to one common ℓ_1 shell does not increase that optimum.

21.1 Maximum antichains

Suppose $|\operatorname{MinNF}(K)| = w_s$. Since $\operatorname{MinNF}(K)$ is an antichain, equality holds in Sperner's theorem [91]. We use the LYM inequality in the form proved by Lubell [93].

Proposition 21.1 (Maximum antichains are middle levels)**.** *If $\mathcal{A} \subseteq 2^{[s]}$ is an antichain with $|\mathcal{A}| = w_s$, then*

$$\mathcal{A} = \binom{[s]}{r}$$

for some middle rank r.

Proof. The LYM inequality gives

$$\sum_{E \in \mathcal{A}} \frac{1}{\binom{s}{|E|}} \leq 1.$$

Since $|\mathcal{A}| = w_s$ and $\binom{s}{|E|} \leq w_s$, equality must hold throughout, so every member lies on a maximum level. If s is even this level is unique.

Let $s = 2m + 1$. The two maximum levels are $\mathcal{L}_m$ and $\mathcal{L}_{m+1}$. Consider the bipartite inclusion graph between them; it is regular and connected. Put $X = \mathcal{A} \cap \mathcal{L}_m$. Since $\mathcal{A}$ is an antichain,

$$\mathcal{A} \cap \mathcal{L}_{m+1} \subseteq \mathcal{L}_{m+1} \setminus N(X).$$

For nonempty proper X, regularity gives $|N(X)| \geq |X|$. Equality would force $X \cup N(X)$ to be a union of connected components, contradicting connectedness. Hence $|N(X)| > |X|$, which forces $|\mathcal{A}| < w_s$. Therefore a maximum antichain cannot mix the two middle levels. □

Consequently, in the extremal case

$$\operatorname{MinNF}(K) = \binom{[s]}{r}, \qquad K = 2^{[s]}_{\leq r-1}.$$

Every canonical row then has quota

$$\tau = W_s - 2^{s-r},$$

and

$$\operatorname{def}(K) = w_s 2^{s-r}.$$

21.2 Two-for-one middle-level compression

Choose a chain C_{q_1} and write its rank-r member as

$$E = \{i_{q_1,1}, \dots, i_{q_1,r}\}.$$

Let

$$H = \{i_{q_1,1}, \dots, i_{q_1,r-1}\}, \qquad \beta = A_{q_1}\mathbf{1}_H = 2^s - 2^{s-r+1}.$$

Lemma 21.2 (Compressed row)**.** *Every J with $|J| \le r-1$ satisfies $A_{q_1}\mathbf{1}_J \le \beta$, while every r-set F containing H violates this inequality.*

Proof. H contains the $r-1$ largest row weights, so it maximizes the score among subsets of cardinality at most $r-1$. If $F = H \cup \{x\}$, then positivity gives $A_{q_1}\mathbf{1}_F = \beta + a_{q_1,x} > \beta$. □

There are $s - r + 1 \ge 2$ distinct r-sets containing H. Choose one $F \ne E$, and let q_2 be its chain. Modify the quotas by

$$\widetilde{b}_{K,q_1} = \beta, \qquad \widetilde{b}_{K,q_2} = W_s,$$

and keep all other rows at the canonical value τ.

Proposition 21.3 (Extremal compression preserves the complex)**.**

$$K = \{J \subseteq [s] : A_s\mathbf{1}_J \le \widetilde{b}_K\}.$$

Proof. If $|J| \le r-1$, the compressed row accepts by theorem 21.2, and every other active row is a canonical separator of an r-set, so all such J are accepted.

Now let $|J| \ge r$, and choose an r-subset $G \subseteq J$. If $G \notin \{E, F\}$, the canonical separator row belonging to the chain of G is unchanged and rejects J. If $G = E$ or $G = F$, then $H \subseteq G \subseteq J$, so the compressed row q_1 rejects J. Hence every set of cardinality at least r is rejected. □

The compressed row changes from τ to β, so

$$\beta - \tau = 1 - 2^{s-r}.$$

Making row q_2 inactive adds 2^{s-r}, for a net gain of 1. Therefore

$$\|\widetilde{b}_K\|_1 = w_s W_s + 1 - w_s 2^{s-r}.$$

The remaining deficit is

$$\Delta = w_s 2^{s-r} - 1 \ge 0.$$

Since row q_2 is inactive, increasing its quota by Δ preserves the represented family and yields total quota R_s. Therefore

$$u_\Sigma(s) \le w_s.$$

21.3 Lower-orthant shattering

For $b \in \mathbb{R}^d$, write

$$O_b = \{x \in \mathbb{R}^d : x \le b\}.$$

Lemma 21.4 (Lower-orthant shattering bound)**.** *If $x_1, \dots, x_n \in \mathbb{R}^d$ are shattered by lower orthants, then $n \le d$.*

Proof. For each i, shattering provides a lower orthant containing every point except x_i. Hence in some coordinate c_i, the point x_i is the unique strict maximum among all x_j. Two distinct points cannot be the unique strict maximum in the same coordinate, so $i \mapsto c_i$ is injective and $n \le d$. □

Let $A \in \mathbb{R}_{\geq 0}^{d \times s}$ be any simultaneous universal matrix. Choose a maximum antichain $\mathcal{A}$ with $|\mathcal{A}| = w_s$, and define score points

$$x_E = A\mathbf{1}_E, \qquad E \in \mathcal{A}.$$

For any $S \subseteq \mathcal{A}$, set $K_S = {\downarrow}S$ if $S \neq \varnothing$, and $K_\varnothing = \{\varnothing\}$. Since $\mathcal{A}$ is an antichain,

$$K_S \cap \mathcal{A} = S.$$

Universality of A gives a lower orthant cutting out exactly S from the score points. Thus the score points are shattered. They are also pairwise distinct: if $x_E = x_F$ for distinct $E, F \in \mathcal{A}$, no lower orthant could contain one without the other. By theorem 21.4, $d \geq w_s$. Since every common-shell universal matrix is, after forgetting the shell constraint, an ordinary universal matrix, we also have

$$u_\Sigma(s) \geq u(s).$$

Combining these lower bounds with the constructions above gives equality in both models.

Theorem 21.5 (Exact Universal Row Complexity and Zero Shell Defect)**.** *For every $s \geq 2$,*

$$u(s) = u_\Sigma(s) = w_s = \binom{s}{\lfloor s/2 \rfloor}.$$

Moreover, one positive integer matrix $A_s \in \mathbb{Z}_{>0}^{w_s \times s}$ simultaneously realizes every labeled simplicial complex on $[s]$, and every target admits a quota vector with

$$\|b_K\|_1 = w_s(2^s - 1).$$

Thus $\delta_\Sigma(s) = 0$. The row count is exact even against arbitrary nonnegative real coefficient matrices.

21.4 Full-vertex and one-vertex boundary cases

If ghost vertices are forbidden, then for $s \geq 3$ the same maximum-antichain shattering argument works using a middle level of rank at least two, so

$$u^{\mathrm{fv}}(s) = u_\Sigma^{\mathrm{fv}}(s) = w_s.$$

For $s = 2$,

$$u^{\mathrm{fv}}(2) = 1, \qquad u_\Sigma^{\mathrm{fv}}(2) = 2.$$

Indeed, the only full-vertex complexes are $\{\varnothing, \{1\}, \{2\}\}$ and $2^{[2]}$. A single row $(1, 1)$ with thresholds 1 and 2 realizes the two targets. A one-row common-shell realization is impossible because equal ℓ_1-mass would force the scalar thresholds to coincide. Two identical rows $(1, 1)$ with quota vectors $(1, 3)$ and $(2, 2)$ realize the two targets at common total 4.

For completeness, if $s = 1$ and ghosts are allowed,

$$u(1) = 1, \qquad u_\Sigma(1) = 2.$$

A common-shell realization is given by

$$A = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad b_{\{\varnothing\}} = (0, 2)^T, \qquad b_{2^{[1]}} = (1, 1)^T.$$

One common-shell row is impossible because the two scalar quota values would have to be equal.

Chapter 22

Periodic Shortest-Path Lift and Exact Bellman Universality

The coefficient theorem is now returned to the shortest-completion language of Part II. The same fixed additive data are lifted to one positive-cost periodic system in which every target complex appears literally as a distinguished Bellman-optimal compatibility complex.

Write

$$A_s = [u_1 \ \cdots \ u_s], \qquad u_i \in \mathbb{Z}_{>0}^{w_s}.$$

Define the state space

$$V_s = \mathbb{Z}^{w_s} \times \{\mathsf{P}, \mathsf{C}\},$$

with goal $G = (0, \mathsf{C})$. For each $i \in [s]$, introduce a distinguished action

$$a_i : (x, \mathsf{P}) \to (x + u_i, \mathsf{P}), \qquad c(a_i) = \|u_i\|_1.$$

Introduce a commit action

$$\gamma : (x, \mathsf{P}) \to (x, \mathsf{C}), \qquad c(\gamma) = 1,$$

and completion moves

$$(x, \mathsf{C}) \to (x \pm e_q, \mathsf{C}), \qquad q = 1, \dots, w_s,$$

each of cost 1. There is no return from C to P. Denote the system by $\mathcal{U}_s$.

Proposition 22.1 (Exact completion metric)**.** *For every $x \in \mathbb{Z}^{w_s}$,*

$$d((x, \mathsf{C}), G) = \|x\|_1.$$

Proof. Coordinatewise cancellation gives a path of length $\|x\|_1$, and every unit move changes the ℓ_1-distance to the origin by at most one. □

Proposition 22.2 (Exact prefix metric)**.** *For every $x \in \mathbb{Z}^{w_s}$,*

$$d((x, \mathsf{P}), G) = 1 + \|x\|_1.$$

Proof. Committing immediately gives the upper bound. For an arbitrary path, suppose distinguished actions are used with multiplicities $n_i \geq 0$ before commit, and set $U = \sum_i n_i u_i \geq 0$. Their total cost equals $\|U\|_1$. The remaining path costs at least $1 + \|x + U\|_1$, so the total is at least

$$\|U\|_1 + 1 + \|x + U\|_1 \geq 1 + \|x\|_1$$

by the triangle inequality. □

Thus no one-use gadget is required: distinguished actions may repeat arbitrarily.
Let

$$I_\star = \{a_1, \dots, a_s\}.$$

For $J \subseteq [s]$, define

$$U_J = \sum_{i \in J} u_i = A_s \mathbf{1}_J, \qquad c(J) = \sum_{i \in J} \|u_i\|_1 = \|U_J\|_1.$$

We declare J Bellman-optimal at (v, P) when

$$c(J) + d((v + U_J, \mathsf{P}), G) = d((v, \mathsf{P}), G).$$

The family of all such J is denoted $H(v, G)|_{I_\star}$. By theorem 22.2, this is equivalent to

$$\|U_J\|_1 + \|v + U_J\|_1 = \|v\|_1.$$

Lemma 22.3 (Coordinate identity)**.** *For $u, b \in \mathbb{R}^{w_s}_{\geq 0}$,*

$$\|u\|_1 + \|u - b\|_1 = \|b\|_1 \quad \iff \quad u \leq b.$$

Proof. Coordinatewise, for $u_q, b_q \geq 0$,

$$u_q + |u_q - b_q| = b_q$$

if and only if $u_q \leq b_q$. Summing gives the claim. □

Proposition 22.4 (Exact Bellman encoding)**.** *For every $b \in \mathbb{R}^{w_s}_{\geq 0}$,*

$$H(-b, G)|_{I_\star} = \{J \subseteq [s] : A_s \mathbf{1}_J \leq b\}.$$

Proof. At $v = -b$, the Bellman equality becomes

$$\|U_J\|_1 + \|U_J - b\|_1 = \|b\|_1,$$

which is equivalent to $U_J \leq b$ by theorem 22.3. □

If J is Bellman-optimal at $-b$, then every $I \subseteq J$ is Bellman-optimal because $U_I \leq U_J \leq b$. Therefore every ordering of the actions in J appears as an initial segment of a shortest path.

Theorem 22.5 (Universal Optimal-Compatibility Theorem)**.** *For every $s \geq 2$, the fixed system $\mathcal{U}_s$ realizes every labeled simplicial complex on $[s]$: for every $K \in \mathrm{Simp}([s])$ there exists $v_K \in \mathbb{Z}^{w_s}$ with*

$$H(v_K, G)|_{I_\star} = K.$$

The canonical choice is $v_K = -b_K$, where b_K is the common-shell quota vector of theorem 21.5. Every canonical witness satisfies

$$d((v_K, \mathsf{P}), G) = 1 + w_s(2^s - 1).$$

Proof. By theorem 21.5, choose b_K with $K = \{J : A_s \mathbf{1}_J \leq b_K\}$ and $\|b_K\|_1 = R_s$. Then theorem 22.4 gives the exact equality of complexes, and theorem 22.2 gives the common distance. □

The complete equivalence chain is

$$\begin{aligned} J \in K &\iff A_s \mathbf{1}_J \leq b_K \\ &\iff \|U_J\|_1 + \|U_J - b_K\|_1 = \|b_K\|_1 \\ &\iff J \in H(v_K, G)|_{I_\star}. \end{aligned}$$

Chapter 23

Extension Fibers, Full Metric-Shell Capacity, and Topological Universality

The final step removes any remaining rankwise viewpoint. Lower truncations are classified by their complete extension fibers, all ranks saturate simultaneously, and the common distance shell realizes the full labeled simplicial universe.

23.1 Extension fibers and simultaneous all-rank saturation

For $0 \le r \le s$, write

$$2^{[s]}_{\le r} = \{J \subseteq [s] : |J| \le r\}.$$

A lower truncation through rank r is a downward-closed family $L \subseteq 2^{[s]}_{\le r}$ containing $\varnothing$.

Define the compatible higher-face poset

$$P_{s,r}(L) = \{E \subseteq [s] : |E| > r,\ \forall F \subseteq E\ (|F| \le r \Rightarrow F \in L)\},$$

ordered by inclusion, and the extension fiber

$$\mathrm{ExtFib}_r(L) = \{K \in \mathrm{Simp}([s]) : K \cap 2^{[s]}_{\le r} = L\}.$$

Theorem 23.1 (Extension-Ideal Classification)**.** *For every lower truncation L,*

$$\mathrm{ExtFib}_r(L) \cong \mathcal{J}(P_{s,r}(L)).$$

The bijection is $K \mapsto K \cap P_{s,r}(L)$, with inverse $I \mapsto L \cup I$.

Proof. If $K \in \mathrm{ExtFib}_r(L)$, then every higher face of K lies in $P_{s,r}(L)$, and the higher part is downward closed inside that poset. Conversely, if I is an order ideal of $P_{s,r}(L)$, then $L \cup I$ is simplicial: a subface of size at most r lies in L by compatibility, while a higher subface lies in $P_{s,r}(L)$ and hence in I by ideality. The two maps are inverse. □

The fiber is therefore a finite distributive lattice with meet and join given by intersection and union.

Define the system extension fiber

$$\mathrm{ExtFib}^{\mathcal{U}_s}_r(L) = \{H(v,G)|_{I_\star} : H(v,G)|_{I_\star} \cap 2^{[s]}_{\le r} = L\}.$$

Theorem 23.2 (Universal Extension-Fiber Completeness)**.** *For every $s \ge 2$, every $0 \le r \le s$, and every lower truncation L,*

$$\mathrm{ExtFib}^{\mathcal{U}_s}_r(L) = \mathrm{ExtFib}_r(L) \cong \mathcal{J}(P_{s,r}(L)).$$

Every member admits a canonical witness at distance

$$D_s = 1 + w_s(2^s - 1).$$

Proof. Every system compatibility family is simplicial. Conversely, every $K \in \mathrm{ExtFib}_r(L)$ is realized by theorem 22.5, with canonical witness at distance D_s. □

Thus Bellman optimality imposes no additional higher-order restriction beyond simplicial downward closure in the constructed system.

For $2 \le k \le s$, let

$$L_{k-1} = 2^{[s]}_{\le k-1},$$

and define

$$\mathrm{RankFib}_k = \left\{ K \cap \binom{[s]}{k} : K \in \mathrm{ExtFib}_{k-1}(L_{k-1}) \right\}.$$

Every $A \subseteq \binom{[s]}{k}$ is realized by

$$K_A = L_{k-1} \cup A,$$

with no faces above rank k.

Corollary 23.3 (Simultaneous All-Rank Saturation)**.** *For every $2 \le k \le s$,*

$$\mathrm{RankFib}_k = 2^{\binom{[s]}{k}}, \qquad |\,\mathrm{RankFib}_k\,| = 2^{\binom{s}{k}}.$$

The same system $\mathcal{U}_s$ works for every k.

If k_s is a middle rank, then

$$\log_2 |\,\mathrm{RankFib}_{k_s}\,| = w_s.$$

23.2 Full Metric-Shell Capacity and Entropy

Let

$$R_s = w_s(2^s - 1), \qquad D_s = 1 + R_s,$$

and define the full prefix-phase metric shell

$$\mathbb{S}_s = \{v \in \mathbb{Z}^{w_s} : d((v, \mathsf{P}), G) = D_s\} = \{v : \|v\|_1 = R_s\}.$$

Proposition 23.4 (Arbitrary-state classification)**.** *If $v \le 0$ coordinatewise, then*

$$H(v, G)|_{I_\star} = \{J \subseteq [s] : A_s \mathbf{1}_J \le -v\}.$$

If v has at least one positive coordinate, then

$$H(v, G)|_{I_\star} = \{\varnothing\}.$$

In particular, every prefix-phase state produces a simplicial compatibility complex.

Proof. The Bellman equality is coordinatewise. If $v \le 0$, it holds exactly when $U_J \le -v$. If some coordinate of v is positive, then every nonempty U_J is strictly positive in that coordinate, so equality fails there; only $J = \varnothing$ remains. □

Theorem 23.5 (Full-Shell Universality)**.** *For every $s \ge 2$,*

$$\{H(v, G)|_{I_\star} : v \in \mathbb{S}_s\} = \mathrm{Simp}([s]).$$

Proof. By theorem 23.4, every shell state produces a simplicial complex. Conversely, every $K \in \mathrm{Simp}([s])$ has a canonical witness $v_K \in \mathbb{S}_s$ by theorem 22.5. □

Let $M(s)$ denote the Dedekind number, equivalently the number of antichains in $2^{[s]}$. Since we count nonvoid simplicial complexes, their number is $M(s)-1$. Define the semantic shell capacity by

$$\mathrm{Cap}(\mathbb{S}_s) = \left|\{H(v,G)|_{I_\star} : v \in \mathbb{S}_s\}\right|.$$

Then

$$\mathrm{Cap}(\mathbb{S}_s) = M(s) - 1.$$

This is semantic capacity, not geometric shell cardinality.

We use the phrase *semantic entropy* only as shorthand for the base-two logarithm of the number of distinct compatibility complexes; no probabilistic or Shannon-theoretic structure is intended. Kleitman's classical asymptotic solution of Dedekind's problem gives [94]

$$\log_2 M(s) = (1 + o(1))w_s.$$

Hence

$$\log_2 \mathrm{Cap}(\mathbb{S}_s) = (1 + o(1))w_s$$

and therefore

$$u(s) \sim \log_2 \mathrm{Cap}(\mathbb{S}_s).$$

The same middle-layer width controls both quantities, but no generic dimension–entropy law is claimed.

Moreover, for a middle rank k_s,

$$\frac{\log_2 |\,\mathrm{RankFib}_{k_s}\,|}{\log_2 \mathrm{Cap}(\mathbb{S}_s)} \to 1.$$

Thus complete all-order universality strengthens exact structure without changing the leading first-order information scale already visible at the middle layer.

Let C_s be the number of labeled simplicial complexes on $[s]$ containing every singleton. Inclusion–exclusion over ghost vertices gives

$$C_s = \sum_{j=0}^{s} (-1)^j \binom{s}{j} (M(s-j) - 1).$$

The number with at least one ghost vertex is at most $s(M(s-1)-1)$. Since $w_{s-1} = (1/2 + o(1))w_s$ and $\log_2 M(t) = (1 + o(1))w_t$, we have $M(s-1)/M(s) \to 0$, hence

$$C_s = (1 - o(1))M(s), \qquad \log_2 C_s = (1 + o(1))w_s.$$

23.3 Topological Universality and Effective Consequences

Every finite labeled simplicial complex occurs literally as a distinguished compatibility complex. Consequently, across the uniform family $\{\mathcal{U}_s\}_{s\geq 2}$, every value of simplicial dimension, f-vector, Euler characteristic, homology or cohomology group, Betti-number pattern, torsion pattern, and finite simplicial homotopy type that is attained by a finite simplicial complex is also attained by a compatibility complex. In particular, every finitely presented group occurs as a fundamental group, as recorded below.

Corollary 23.6 (Distance does not determine topology)**.** *Let $\mathcal{I}$ be any invariant nonconstant on* $\mathrm{Simp}([s])$*. Then there exist $v_1, v_2 \in \mathbb{S}_s$ with equal distance D_s from the goal but*

$$\mathcal{I}(H(v_1,G)|_{I_\star}) \neq \mathcal{I}(H(v_2,G)|_{I_\star}).$$

Every finitely presented group occurs as the fundamental group of a finite two-dimensional simplicial complex; see, for example, standard presentation-complex treatments such as [95].

Corollary 23.7 (Fundamental-group universality)**.** *For every finitely presented group* Γ*, there exist* $s \geq 2$ *and* $v \in \mathbb{Z}^{w_s}$ *such that*

$$\pi_1\big(|H(v,G)|_{I_\star}|\big) \cong \Gamma.$$

The realizing compatibility complex may be chosen two-dimensional, and v *may be chosen as a canonical common-shell witness.*

For algorithmic consequences, fix for each s a standard recursive symmetric-chain decomposition. Then $s \mapsto A_s$ is computable, and from a finite labeled simplicial complex K one can effectively enumerate minimal nonfaces, assign canonical quotas, perform ordinary shell padding or the explicit extremal compression, compute b_K, and output $v_K = -b_K$.

Proposition 23.8 (Effective universality)**.** *There is an effective transformation*

$$K \longmapsto (s, v_K)$$

such that

$$H(v_K, G)|_{I_\star} = K, \qquad d((v_K, \mathsf{P}), G) = D_s.$$

For fixed s, the semantic universe is finite, with $M(s) - 1$ objects, so no fixed-s undecidability claim is made. Undecidability can arise only when s varies over the uniform family.

Theorem 23.9 (Effective transfer principle)**.** *Let* $\mathcal{P}$ *be a property of finite simplicial complexes. If deciding* $\mathcal{P}$ *for finite simplicial complexes is undecidable, then deciding whether* $H(v,G)|_{I_\star}$ *has property* $\mathcal{P}$ *is undecidable across the uniform family* $\{\mathcal{U}_s\}_{s\geq 2}$*.*

Proof. Given K, compute (s, v_K) using theorem 23.8. A decision procedure for the Bellman formulation would decide $\mathcal{P}$ for K. □

Triviality of the fundamental group is undecidable for finite simplicial complexes, and this remains true in dimension two [96, 97]. Hence simple connectivity is undecidable across the uniform Bellman family. The reduction already uses canonical witnesses on their prescribed common-distance shells.

These topological and algorithmic statements are consequences of exact effective universality, not the structural center of this branch.

Bridge back to the main pushdown spine. The four chapters above complete the geodesic-compiler branch at the full simplicial level. They are logically independent of the cancellation and pushdown constructions that follow, so the monograph now returns to the original DPDA and resumes the pushdown-normalization spine.

23.4 Returning to the original machine from the opposite direction

The compiler and full-simplicial universality Part completes the outward abstraction of shortest-completion geometry. We now zoom back in.

In Chapter 1 we deliberately marked one local transition of the original 2:1 DPDA:

$$\boxed{(L, 0S) \xrightarrow{0} (L, S).}$$

So far that transition has been interpreted as:

- a stack cancellation;
- one unit of residual correction;

- one edge of a shortest-completion ladder.

But it contains another structure that the completion and compiler theories do not use. When the nonnegative 2:1 computations are drawn as lattice paths, the cancellation occurs at a precise arithmetic family of landing heights. That observation leads to an independent branch:

pushdown cancellation → correction signatures → intrinsic defect → Catalan geometry.

We now return to the 2023 machine and follow that transition.

Part III

Cancellation Geometry, Chronology, and the General Real-Time p:q Machine

---

Part guide. **Question.** How does the original cancellation mechanism generalize from the historical 2:1 machine to every coprime positive ratio, and what arithmetic structure is exposed on the way? **Main result.** Defect and chronological-carry analysis leads to the centered real-time literal-demand p:q normalizer with exact WRITE, SEARCH, and CANCEL roles. **Why here.** It closes the constructional loop between scalar completion semantics and an explicit general machine. **Dependency.** It uses the residual/completion coordinates of Part I; the cancellation-enumeration offshoot is a consequence rather than a prerequisite.

Chapter 24

Zooming Back into the Original DPDA

The preceding geodesic-compiler branch generalized shortest completion outward.

We now move in the opposite direction.

Return to the original 2:1 DPDA of Chapter 1 and isolate one local transition:

$$\boxed{(L, 0S) \xrightarrow{0} (L, S).}$$

At the machine level, this is a cancellation.

At the residual level, it decreases

$$\rho = N_1 - 2N_0$$

by 2.

At the geodesic level, Chapter 3 showed that it consumes one required zero along a shortest-completion path.

There is, however, another structure hidden in the same event.

If the input word is drawn as a nonnegative 2-Dyck path, the cancellation occurs at a precise arithmetic set of landing heights.

That observation starts a second geometry.

24.1 The machine facts we reuse, not re-prove

The cancellation analysis needs only two facts already established in Chapter 1:

Reachable-store normal form

$$\mathcal{S}_C = 0^* \cup 1^* \cup \{10\},$$

$$\mathcal{S}_L = 0^*,$$

$$\mathcal{S}_R = 1^* \cup \{0\}.$$

State-potential invariant

For every consumed prefix,

$$\boxed{\rho(w) = D(S) + \Phi(q),}$$

with

$$D(S) = 2N_0(S) - N_1(S),$$

$$\Phi(C) = 0, \qquad \Phi(L) = 3, \qquad \Phi(R) = -3.$$

We do not duplicate those proofs here.

Their consequence for positive odd residuals is immediate.

Corollary 24.1 — Positive odd residuals.

Every reachable core configuration with positive odd residual at least 3 is uniquely of the form

$$\boxed{(L, 0^k)}$$

for some

$$k \geq 0,$$

and its residual is

$$\boxed{2k + 3.}$$

The left state therefore has a very precise path-height meaning.

24.2 The nonnegative path sector

Encode input symbols as steps:

$$1 \longleftrightarrow U \quad \text{of height } +1,$$

$$0 \longleftrightarrow D \quad \text{of height } -2.$$

For a prefix w, the path height is

$$\boxed{h(w) = N_1(w) - 2N_0(w) = \rho(w).}$$

A word whose prefix heights are all nonnegative and whose final height is 0 is a 2-Dyck excursion.

The cancellation statistic will first be discovered inside this nonnegative sector.

This restriction matters.

The later combinatorial theory will be defined directly on generalized Dyck paths and will no longer depend on the remaining DPDA transition table.

24.3 Exact landing-height correspondence

Immediately before a distinguished cancellation,

$$(L, 0^k) \xrightarrow{0} (L, 0^{k-1}),$$

the logical top 0 must exist, so

$$k \geq 1.$$

The pre-step residual is

$$2k + 3.$$

Reading 0 decreases the height by 2, so the down-step lands at

$$2k + 1.$$

Therefore every cancellation lands at one of:

$$3, 5, 7, \ldots$$

The converse is equally rigid.

Proposition 24.2 — Motivating $d = 2$ correspondence.

Within the nonnegative 2-Dyck sector of the original computation, the transition

$$\boxed{0, 0/\varepsilon}$$

occurs if and only if the corresponding down-step lands at height

$$\boxed{3, 5, 7, \ldots}$$

Proof.

The forward direction was just computed.

Conversely, suppose a down-step lands at

$$2k + 1 \geq 3.$$

Its pre-step height is

$$2k + 3 \geq 5.$$

This is a positive odd residual.

Corollary 24.1 therefore forces the pre-step configuration to be

$$(L, 0^k)$$

with

$$k \geq 1.$$

The enabled input-0 transition is exactly the distinguished left cancellation. □

This theorem is the only point at which the full original machine is needed to motivate the correction statistic.

Everything that follows will be stated intrinsically for paths.

24.4 From one machine event to a d-Dyck statistic

For

$$d \geq 2,$$

a d-Dyck excursion has:

- up-step U of height $+1$;
- down-step D of height $-d$;
- nonnegative prefix heights;
- final height 0.

A size-n excursion has:

$$n$$

down-steps and

$$dn$$

up-steps.
The original $d = 2$ cancellation landing heights were

$$3, 5, 7, \ldots$$

or equivalently

$$h_{\text{end}} \equiv 1 \pmod 2, \qquad h_{\text{end}} \geq 3.$$

This suggests the intrinsic generalization.

Definition 24.3 — Correction.
A down-step is a **correction** if

$$\boxed{h_{\text{end}} \equiv 1 \pmod d, \qquad h_{\text{end}} \geq d + 1.}$$

Equivalently,

$$h_{\text{end}} = d + 1 + dr$$

for a unique

$$r \in \mathbb{N}.$$

The parameter d now belongs to the combinatorial path system.
For $d > 2$, we are **not** claiming that the original DPDA itself implements this generalized rule.
The correct causal statement is:

original $d = 2$ cancellation $\rightarrow$ intrinsic landing-height condition
$\rightarrow$ combinatorial d-generalization.

24.5 Chronological correction signatures

Suppose a path contains $m \geq 1$ corrections, in chronological order, landing at heights

$$c_1, \ldots, c_m.$$

Write

$$c_i = d + 1 + dr_i.$$

Definition 24.4 — Normalized correction signature.
The ordered tuple

$$\boxed{r(P) = (r_1, \ldots, r_m)}$$

is the normalized correction signature.
The order is essential.
Two paths can have the same number of corrections and even the same multiset of correction heights while having different ordered signatures.
This branch therefore remembers more than the scalar count statistic.

24.6 Path defect

If a size-n path has $m \geq 1$ corrections, define

$$\boxed{\delta(P) = n - m - 2.}$$

The form may look artificial before the realization theorem.
It is chosen so that:

- maximal-correction paths have defect 0;
- every proposed ordered signature has an exact minimum defect;
- the excess above that minimum can be increased freely by zero-to-zero padding.

The theorem in Chapter 25 will prove in particular that

$$\delta(P) \geq 0.$$

24.7 The new inverse problem

The original question was forward:

> Given a path, where do cancellations/corrections occur?

The signature theory asks the inverse question:

> Given an ordered sequence
>
> $$r = (r_1, \ldots, r_m),$$
>
> can it occur as a correction history?

And, if so:

> What is the smallest path defect at which it can occur?

This is much stronger than counting corrections.
The answer is one explicit functional

$$\Delta(r).$$

To derive it, we first need the exact local price of moving from one correction height to the next.

Chapter 25

Correction Signatures and Minimum Defect

Let

$$r = (r_1, \dots, r_m) \in \mathbb{N}^m, \qquad m \geq 1.$$

The goal is to determine exactly when r is realizable as a normalized correction signature.

The answer is built from two local costs:

1. the price of moving between consecutive corrections;
2. the price of terminating after the final correction.

The first cost contains the key nonlinearity.

25.1 The threshold function

Define

$$\boxed{\psi(a) = \begin{cases} 0, & a \leq 1, \\ a, & a \geq 2. \end{cases}}$$

Why is there a jump from zero to a?

Suppose consecutive normalized correction heights are

$$r \quad \text{and} \quad r',$$

and set

$$a = r - r'.$$

Let b be the number of non-correction down-steps strictly between the two corrections, and let u be the number of intervening up-steps.

Height balance gives

$$\boxed{u = d(b + 1 - a).}$$

This one equation explains the threshold.

25.2 Local Gap Cost

Lemma 25.1 — Local Gap Cost.

For two consecutive normalized correction heights r, r', with

$$a = r - r',$$

the number b of intervening non-correction down-steps satisfies

$$\boxed{b \geq \psi(a).}$$

The bound is sharp.

Proof.

If

$$a \leq 1,$$

choose

$$b = 0.$$

Then

$$u = d(1 - a) \geq 0.$$

The block

$$\boxed{U^{d(1-a)}D}$$

moves directly to the next correction without an intervening down-step.

Now assume

$$a \geq 2.$$

The balance equation gives

$$b \geq a - 1.$$

If

$$b = a - 1,$$

then

$$u = 0.$$

But the first intervening down-step would land again in the correction residue class, creating an unintended correction between the two prescribed ones.

Therefore

$$b \geq a.$$

Equality is attainable by the detour block

$$\boxed{UD^aU^{d-1}D.}$$

The intermediate down-steps lie one residue class away from the correction class, so they are not corrections. The final D returns exactly to the prescribed correction height. □

This theorem is the geometric origin of ψ.

A downward change of size at most 1 is free.

A drop of size at least 2 cannot be taken directly because the naive descent would create an unwanted correction.

The path must leave the correction residue class and return.

That detour costs exactly the full drop a.

25.3 Terminal Cost

Lemma 25.2 — Terminal Cost.

If the final correction has normalized height r, then at least

$$\boxed{r+2}$$

non-correction down-steps are required after it.

The bound is sharp.

Proof.

Immediately after the final correction, the height is

$$d+1+dr.$$

If b down-steps and u up-steps remain, then

$$d+1+dr+u-db=0.$$

Hence

$$u=d(b-r-1)-1.$$

Since

$$u\geq 0,$$

we must have

$$b\geq r+2.$$

Equality is attained by

$$\boxed{U^{d-1}D^{r+2}.}$$

All its down-steps land at multiples of d, so none is a correction. □

The final signature entry therefore contributes directly to the minimum defect.

25.4 Intrinsic signature defect

Define

$$\boxed{\Delta(r)=r_m+\sum_{i=1}^{m-1}\psi(r_i-r_{i+1}).}$$

This quantity contains no d.

That universality is already striking.

The physical correction heights scale as

$$d+1+dr_i,$$

but the normalized realizability cost is independent of the Dyck parameter.

25.5 Canonical blocks

To prove sufficiency constructively, define:

Initial block

$$\boxed{I_r = U^{2d+1+dr}D.}$$

Its unique down-step lands at correction height

$$d + 1 + dr.$$

Gap block

For $a \in \mathbb{Z}$,

$$\boxed{G_a = \begin{cases} U^{d(1-a)}D, & a \le 1, \\ UD^aU^{d-1}D, & a \ge 2. \end{cases}}$$

Terminal block

$$\boxed{T_r = U^{d-1}D^{r+2}.}$$

Padding block

$$\boxed{P = U^dD.}$$

The padding block starts and ends at height zero and creates no correction.

Proposition 25.3 — Block Purity.

Each canonical block creates exactly the advertised correction behavior and no unintended correction.

Moreover, when started at its prescribed height, every intermediate height remains nonnegative.

This proposition is what makes the global realization theorem compositional.

25.6 Defect–Signature Characterization

Theorem 25.4 — Defect–Signature Characterization.

For every

$$d \ge 2$$

and every normalized signature

$$r = (r_1, \dots, r_m),$$

the following are equivalent:

1. r is realized by a d-Dyck excursion of defect δ;
2. $$\boxed{\Delta(r) \le \delta.}$$

Equivalently,

$$\boxed{\delta_{\min}(r) = \Delta(r).}$$

If the signature has length m, then its minimum path size is

$$\boxed{n_{\min}(r) = m + 2 + \Delta(r).}$$

Proof — lower bound. A path realizing r contains:

$$n - m$$

non-correction down-steps.
Between correction i and correction $i+1$, Lemma 25.1 forces at least

$$\psi(r_i - r_{i+1})$$

such steps.
After the last correction, Lemma 25.2 forces at least

$$r_m + 2.$$

Any non-correction down-steps before the first correction only increase the count.
Therefore

$$n - m \geq 2 + r_m + \sum_{i=1}^{m-1} \psi(r_i - r_{i+1}) = 2 + \Delta(r).$$

Hence

$$\delta(P) = n - m - 2 \geq \Delta(r).$$

Proof — construction. Let

$$a_i = r_i - r_{i+1}.$$

For

$$s \geq 0,$$

define

$$\boxed{W_d(r; s) = I_{r_1} G_{a_1} \cdots G_{a_{m-1}} T_{r_m} P^s.}$$

Block Purity shows that this is a nonnegative d-Dyck excursion whose normalized correction signature is exactly r.
Its number of down-steps is

$$m + 2 + \Delta(r) + s.$$

Thus

$$\delta(W_d(r; s)) = \Delta(r) + s.$$

Taking $s = 0$ gives a minimum realization.
Taking

$$s = \delta - \Delta(r)$$

realizes every larger defect. □

This is the structural center of the cancellation branch.
It solves both directions:

$$\boxed{\text{path} \to \text{signature}}$$

and

$$\boxed{\text{signature} \to \text{minimum realizing path.}}$$

25.7 Canonical does not mean unique

Corollary 25.5 — Canonical minimum realization.
Every signature r has the explicit minimum witness

$$\boxed{W_d^{\min}(r) = W_d(r; 0).}$$

The word **canonical** means:

> the distinguished block construction selected by the theory.

It does **not** mean:

> every minimum realizing path is identical to this word.

This scope matters.
The theorem determines the minimum defect uniquely, not the complete set of minimum path orderings.

25.8 Signature persistence

Corollary 25.6 — Signature Persistence.
A signature r is realizable at exactly the defect levels

$$\boxed{\Delta(r), \Delta(r)+1, \Delta(r)+2, \dots}$$

because each copy of

$$P = U^d D$$

adds one defect unit without changing the correction signature.
Therefore the intrinsic defect is a **birth level**.
A signature appears for the first time at $\Delta(r)$, then persists forever.

25.9 Maximum number of corrections

Since

$$\Delta(r) \geq 0,$$

every path with $m \geq 1$ corrections satisfies

$$\boxed{m \leq n - 2.}$$

Thus defect zero is exactly the maximal-correction regime.
That boundary turns out to be Catalan.

25.10 Zero defect

Corollary 25.7 — Zero-defect signatures.

$$\boxed{\Delta(r) = 0}$$

if and only if

$$\boxed{r_m = 0}$$

and

$$\boxed{r_i - r_{i+1} \le 1 \qquad (1 \le i < m).}$$

Reverse the sequence:

$$a_i = r_{m+1-i}.$$

Then

$$a_1 = 0$$

and

$$\boxed{a_{i+1} \le a_i + 1.}$$

These are exactly the usual Dyck area-sequence inequalities.
Therefore the number of zero-defect signatures of length m is

$$\boxed{C_m.}$$

The Catalan family is not the whole correction theory.
It is its exact zero-cost boundary.

25.11 Defect as penalized overshoot

After reversal,

$$a_i = r_{m+1-i},$$

the defect becomes

$$\boxed{\Delta^{\leftarrow}(a) = a_1 + \sum_{\substack{1 \le i < m \\ a_{i+1} - a_i \ge 2}} (a_{i+1} - a_i).}$$

Thus ordinary Dyck area sequences are precisely the nonnegative sequences with zero cost.

Positive defect allows overshoots above the Catalan nearest-neighbor condition, but every overshoot of size at least 2 is charged by its full magnitude.

This makes the geometry intuitive:

$$\boxed{\text{Catalan behavior} = \text{zero-defect regime;}}$$

$$\boxed{\text{non-Catalan jumps} = \text{quantified correction debt.}}$$

The next chapter shows that the Catalan boundary is not only enumerative.
Its maximal-correction paths themselves contain a full binary tree.

Chapter 26

The Catalan Boundary and the Binary Cancellation Skeleton

At defect zero,

$$m = n - 2.$$

Thus a size-n path contains the maximum possible number of corrections.
The signature count is Catalan:

$$C_{n-2}.$$

But the path count contains an additional factor d:

$$dC_{n-2}.$$

This chapter explains both factors structurally.
The intrinsic correction structure is a full binary tree.
The factor d records only how the path enters that binary regime.

26.1 Two unavoidable unmarked nodes

Use the standard first-return decomposition of a nonempty d-Dyck excursion:

$$\boxed{P = UP_1UP_2 \cdots UP_dDP_{d+1}.}$$

Each internal node corresponds to one down-step.
A node is marked exactly when its down-step is a correction.
A maximal-correction size-n path has:

$$n - 2$$

marked internal nodes and therefore exactly two unmarked internal nodes.
The root is necessarily unmarked because its baseline is zero.
Before a correction baseline

$$d + 1$$

can be reached, at least one further unmarked internal node is necessary.
In the maximal-correction regime, these are the only two unmarked nodes.
That rigid entrance forces the rest of the tree into a binary skeleton.

26.2 The d entrance choices

Choose the unique nonempty pre-down-step child of the root.

If it is slot

$$j \in [d],$$

its baseline is j.

Every other root child must be empty, otherwise another unmarked internal node would be created.

At this second unmarked node, the unique child that reaches correction baseline $d+1$ without introducing an additional unmarked node is slot

$$d+1-j.$$

Thus there are exactly

$$\boxed{d}$$

possible entrance choices.

After those two forced unmarked nodes, the remaining structure is entirely inside the correction residue class.

26.3 Why correction nodes become binary

Suppose a correction node has baseline

$$h \equiv 1 \pmod{d}, \qquad h \geq d+1.$$

Its pre-down-step child baselines are

$$h+1, \ldots, h+d,$$

and its trailing child has baseline h.

Among these, the only baselines that remain in the correction class are:

$$h+d$$

and

$$h.$$

Therefore, once all unmarked nodes have been used up, a correction node has exactly two recursive correction slots:

- slot d;
- slot $d+1$.

All other child slots must be empty.

The correction structure is therefore binary.

26.4 Binary Cancellation Skeleton

Let

$$\mathcal{B}_m$$

be the set of plane full binary trees with m internal nodes.
Let

$$\mathcal{M}_{d,n}$$

be the class of size-n maximally corrected d-Dyck excursions.

Theorem 26.1 — Binary Cancellation Skeleton.
For every

$$n \geq 3,$$

$$\boxed{\mathcal{M}_{d,n} \cong [d] \times \mathcal{B}_{n-2}.}$$

Consequently,

$$\boxed{Q_d(n, n-2) = dC_{n-2}.}$$

Proof.
From a maximal-correction path:

1. record the root entrance slot
$$j \in [d];$$
2. delete the two forced unmarked entrance nodes;
3. retain the remaining marked subtree;
4. reinterpret its recursive slots d and $d+1$ as the two ordered binary children.

This gives an element of

$$[d] \times \mathcal{B}_{n-2}.$$

Conversely, given:

- an entrance choice j;
- a plane full binary tree;

reinsert the two forced unmarked nodes and place the binary tree recursively in the only two correction-preserving child slots.
The constructions are inverse. □
The Catalan factor is therefore not a generating-function accident.
It is the count of an actual binary cancellation skeleton.

26.5 Tree coordinates

Assign child weights

$$\omega(j) = j \qquad (1 \leq j \leq d),$$

and

$$\omega(d+1) = 0.$$

Lemma 26.2 — Tree-coordinate lemma.

If an internal node has root-to-node child address

$$i_1 \cdots i_s,$$

then the baseline height of its associated down-step is

$$\boxed{h(v) = \sum_{\nu=1}^{s} \omega(i_\nu).}$$

For a correction node, the two recursive binary directions are:

- **high**: slot d, increment d;
- **low**: slot $d+1$, increment 0.

The two entrance weights always sum to

$$d + 1.$$

Therefore the binary root always begins at correction height

$$d + 1,$$

independently of the entrance choice.

26.6 Correction height from the binary address

Let

$$L(v)$$

be the number of high edges on the path from the binary root to a correction node v. Then:

$$\boxed{c(v) = d + 1 + dL(v),}$$

and hence its normalized correction height is

$$\boxed{r(v) = L(v).}$$

Thus the normalized signature has a purely binary-tree interpretation.
The parameter d only rescales the physical path heights.
The normalized correction values themselves count high edges.
This is another manifestation of the universality first seen in Chapter 25.

26.7 Inorder chronology

In the d-ary path traversal:

- the slot-d correction subtree is visited before the node's closing down-step;
- the slot-$(d+1)$ correction subtree is visited after it.

Therefore correction nodes appear in ordinary binary inorder:

$$\boxed{\text{high subtree} \to \text{node} \to \text{low subtree.}}$$

Let

$$v_1, \ldots, v_m$$

be the internal binary nodes in inorder.
Then the normalized signature is

$$\boxed{r_i = L(v_i).}$$

So the chronological correction signature is exactly an inorder depth statistic on the binary skeleton.

26.8 The signature is lossless

A Catalan-counted statistic need not normally determine the full tree.
Here it does.

Theorem 26.3 — Lossless Signature Theorem.

In the zero-defect regime, the normalized correction signature uniquely determines the binary cancellation skeleton.

Proof.

Let

$$r = (r_1, \ldots, r_m)$$

satisfy the zero-defect conditions.
Let j be the first index such that

$$r_j = 0.$$

In binary inorder, the first zero is the root.
Entries before j lie in the high subtree and each carries one additional high edge.
Therefore the normalized code of the high subtree is

$$\boxed{(r_1 - 1, \ldots, r_{j-1} - 1).}$$

Entries after the root lie in the low subtree, whose code is

$$\boxed{(r_{j+1}, \ldots, r_m).}$$

Both subsequences satisfy the same zero-defect conditions.
Recursively apply the same decomposition.
This reconstructs the tree uniquely. □
Hence:

$$\boxed{\begin{array}{c}\text{binary cancellation skeleton} \longleftrightarrow \text{zero-defect signature} \\ \longleftrightarrow \text{Dyck area sequence.}\end{array}}$$

The factor d is external to this intrinsic Catalan structure.
It remembers only the entrance choice.

26.9 Area statistic

If the zero-defect signature has length m, then

$$c_i = d + 1 + dr_i.$$

Therefore

$$\boxed{\sum_{i=1}^{m} c_i = m(d+1) + d\sum_{i=1}^{m} r_i.}$$

After reversal,

$$\sum_i r_i$$

is the ordinary area statistic of the corresponding Dyck path.
If

$$\mathcal{A}_m(v) = \sum_D v^{\mathrm{area}(D)}$$

is the Dyck area polynomial, then the maximal-correction paths satisfy

$$\boxed{\sum_{P \in \mathcal{M}_{d,m+2}} v^{\sum_i c_i} = d\, v^{m(d+1)} \mathcal{A}_m(v^d).}$$

Thus the Catalan boundary is structural at three levels:

1. number of signatures;
2. binary cancellation skeleton;
3. correction-height distribution.

26.10 What the original transition has become

We began with:

$$(L, 0S) \xrightarrow{0} (L, S).$$

At defect zero, the ordered history of occurrences of that transition becomes:

a full binary tree encoded losslessly by its normalized correction signature.

This is a much stronger statement than "Catalan numbers appear."
The original transition has generated a complete combinatorial coordinate system.
The next chapter moves away from individual signatures and counts:

- paths by number of corrections;
- signatures by exact intrinsic defect.

Those are different arrays and must remain separate.

Chapter 27

Exact Correction and Defect Enumeration

The structural theory has produced three different kinds of objects.
They must not be conflated.

$$\boxed{K_d \quad \text{auxiliary kernel coefficients,}}$$

$$\boxed{Q_d(n,k) \quad \text{paths with exactly } k \text{ corrections,}}$$

and

$$\boxed{N_{m,\ell} \quad \text{normalized signatures born at exact defect } \ell.}$$

The first appears naturally inside the scalar generating-function grammar.
The second is the actual path correction statistic.
The third belongs to the intrinsic signature geometry and is independent of d.

27.1 Scalar correction generating function

Let

$$G_h(z,t)$$

count d-Dyck excursions that begin and end at baseline h, never go below h, with:

- z marking down-steps;
- t marking corrections.

A down-step landing at baseline h has weight

$$\tau_h = \begin{cases} t, & h \in \{d+1, 2d+1, \ldots\}, \\ 1, & \text{otherwise.} \end{cases}$$

The standard first-return decomposition gives

$$\boxed{G_h = 1 + z\tau_h G_{h+1} \cdots G_{h+d} G_h.}$$

Above the exceptional low boundary, the marking pattern is periodic modulo d.
This collapses the infinite family to three formal series:

- F: baseline zero;
- A: unmarked tail residue;

- B: marked correction residue.

They satisfy

$$\boxed{F = 1 + zA^d F,}$$

$$\boxed{A = 1 + zA^d B,}$$

$$\boxed{B = 1 + ztA^{d-1}B^2.}$$

Define

$$\boxed{Q_d(n,k) = [z^n t^k]F(z,t).}$$

Thus $Q_d(n,k)$ is the actual correction triangle.

27.2 The auxiliary kernel and A371395

Set

$$Y = zA^{d-1}B.$$

Then

$$A = \frac{1}{1-Y}, \qquad B = \frac{1}{1-tY},$$

and

$$\boxed{Y(1-Y)^{d-1}(1-tY) = z.}$$

Define

$$\boxed{K_d = \frac{Y}{z}.}$$

Lagrange inversion gives

$$\boxed{[z^n t^k]K_d = \frac{1}{n+1}\binom{n+k}{k}\binom{dn+d-k-2}{n-k}.}$$

For

$$d = 2,$$

this becomes

$$\boxed{[z^n t^k]K_2 = \frac{1}{n+1}\binom{n+k}{k}\binom{2n-k}{n-k}.}$$

This is OEIS A371395 [19].
But K_2 is not the correction distribution.
The actual correction generating function is

$$\boxed{F_d(z,t) = \frac{1-Y}{1-2Y+tY^2}.}$$

Proposition 27.1 — Kernel–correction separation.

For $d = 2$, A371395 is the coefficient triangle of the auxiliary kernel K_2, not the path correction triangle Q_2.

This distinction is important because the two arrays diverge immediately.

A371395 is a useful algebraic waypoint.

It is not the final statistic generated by the DPDA-derived correction event.

27.3 Exact correction triangle

Eliminating B from the three-class system and writing

$$U = A - 1$$

gives

$$\boxed{U\big(1 + (1-t)U\big) = z(1+U)^{d+1}.}$$

Lagrange–Bürmann inversion yields the exact path count.

Theorem 27.2 — Exact correction triangle.

For

$$n \geq 1, \qquad k \geq 0,$$

$$\boxed{Q_d(n,k) = \mathbf{1}_{\{k=0\}} + \sum_{j=k+1}^{n-1} \frac{d(n-j)}{j} \binom{j+k-1}{k} \binom{dn-k-1}{j-k-1}.}$$

In particular,

$$\boxed{Q_d(n,k) = 0 \qquad (k > n-2).}$$

The complete Lagrange–Bürmann coefficient derivation is given in Appendix E.

The theorem statement and its structural consequences remain in the main text.

27.4 Row sums and boundary columns

Setting

$$t = 1$$

forgets the correction marking.

The row sum is the ordinary Fuss–Catalan number:

$$\boxed{\sum_{k \geq 0} Q_d(n,k) = \frac{1}{dn+1} \binom{(d+1)n}{n}.}$$

At the zero-correction boundary,

$$\boxed{Q_d(n,0) = \frac{1}{n} \sum_{j=0}^{n-1} (n-j) \binom{dn}{j}.}$$

For

$$d = 2,$$

this simplifies to

$$\boxed{Q_2(n,0) = \binom{2n-1}{n-1}.}$$

At the maximal-correction boundary,

$$\boxed{Q_d(n,n-2) = dC_{n-2}.}$$

Chapter 26 already explained the last identity bijectively through the binary cancellation skeleton.

This is a useful example of two proof styles meeting:

- algebraic coefficient extraction;
- structural combinatorial decomposition.

27.5 Intrinsic signature enumeration

Now forget the paths and enumerate normalized signatures directly.

Define

$$\boxed{N_{m,\ell} = \#\{r \in \mathbb{N}^m : \Delta(r) = \ell\}.}$$

Let

$$\boxed{S(x,q) = \sum_{m\geq 1}\sum_{\ell\geq 0} N_{m,\ell} x^m q^\ell.}$$

Here:

- x marks signature length;
- q marks exact intrinsic minimum defect.

This enumeration is fundamentally different from $Q_d(n,k)$.

A signature of defect ℓ can be realized at every larger path defect.

Thus $N_{m,\ell}$ records the level at which a signature is **born**.

27.6 Translation and first-zero decomposition

Let $Z(x,q)$ count nonempty signatures whose minimum entry is zero.

Every nonempty signature has a unique decomposition

$$r = \bar{r} + s\mathbf{1},$$

where

$$s = \min_i r_i$$

and

$$\min_i \bar{r}_i = 0.$$

Adding 1 to every entry leaves all adjacent differences unchanged and raises the terminal term by 1.

Therefore

$$\boxed{\Delta(r+\mathbf{1}) = \Delta(r) + 1.}$$

This gives

$$\boxed{Z = (1-q)S.}$$

Now cut a minimum-zero signature at its first zero.

A second auxiliary series E counts nonempty signatures ending in zero.

The first-zero decomposition yields a closed algebraic system, which simplifies to the master equation.

27.7 Defect–Signature Master Equation

Theorem 27.3 — Defect–Signature Master Equation.

The formal power series $S(x,q)$ with zero constant term is uniquely determined by

$$\boxed{(1-q)S(1+qS) = x(1+S)^2(1+q^2S).}$$

Equivalently,

$$\boxed{S = x\frac{(1+S)^2(1+q^2S)}{(1-q)(1+qS)}.}$$

The equation contains no d.

Thus the universality of the Defect–Signature theorem persists at the enumerative level.

The path parameter d changes physical height scaling and path multiplicity.

It does not change the intrinsic normalized signature birth counts.

27.8 Exact defect coefficients

Lagrange inversion gives a finite coefficient formula for every

$$N_{m,\ell}.$$

The first four defect layers simplify to:

$$\boxed{N_{m,0} = C_m,}$$

$$\boxed{N_{m,1} = \frac{3m}{m+2}C_m = C_{m+1} - C_m,}$$

$$\boxed{N_{m,2} = \frac{m(m^2+10m+1)}{(m+2)(m+3)}C_m,}$$

and

$$\boxed{N_{m,3} = \frac{5m(m^3+7m^2+2m+2)}{(m+2)(m+3)(m+4)}C_m.}$$

The zero layer is exactly Catalan.

The positive layers quantify how the signature class expands as non-Catalan overshoots are admitted.

27.9 Cumulative realizability

Define

$$\boxed{\widehat{N}_{m,\delta} = \#\{r \in \mathbb{N}^m : r \text{ is realizable at path defect } \delta\}.}$$

By signature persistence,

$$\boxed{\widehat{N}_{m,\delta} = \sum_{\ell=0}^{\delta} N_{m,\ell}.}$$

Thus:

- $N_{m,\ell}$ counts signatures born at exact intrinsic defect ℓ;
- $\widehat{N}_{m,\delta}$ counts all signatures visible by defect δ.

At defect one,

$$\boxed{\widehat{N}_{m,1} = C_{m+1}.}$$

The first cumulative layer is still exactly Catalan, one size higher.
Defect two is the first level at which a genuinely non-Catalan local overshoot such as

$$(2, 0)$$

can appear.

27.10 Three arrays, three questions

The distinction can now be stated cleanly.

K_d

An auxiliary kernel series produced by the scalar grammar.
Question:

> What coefficient array appears in the algebraic series-reversion core?

$Q_d(n, k)$

Path correction distribution.
Question:

> How many size-n d-Dyck excursions contain exactly k correction events?

$N_{m,\ell}$

Intrinsic ordered signature distribution.
Question:

> How many length-m correction histories are born at exact minimum defect ℓ?

The DPDA-derived project passes through all three, but they are not interchangeable.
The next chapter studies the asymptotic hierarchy of the third array.

Chapter 28

The Asymptotic Defect Hierarchy

The master equation

$$(1-q)S(1+qS) = x(1+S)^2(1+q^2S)$$

provides exact coefficients.

It also reveals a surprisingly rigid asymptotic structure.

For fixed defect ℓ, the number of length-m signatures grows on the Catalan exponential scale:

$$4^m.$$

The defect level changes only the polynomial factor.

More precisely, defect levels occur in pairs:

$$(0,1), \quad (2,3), \quad (4,5), \ldots$$

and every two defect units increase the polynomial exponent by one.

28.1 Fixed-defect generating functions

Define

$$\boxed{S_\ell(x) = [q^\ell]S(x,q) = \sum_{m\geq 1} N_{m,\ell}x^m.}$$

Set

$$\boxed{t = \sqrt{1-4x}.}$$

At defect zero, the master equation gives

$$S_0(x) = C(x) - 1,$$

hence

$$\boxed{S_0(x) = \frac{1-t}{1+t}.}$$

Thus the dominant singularity is the Catalan point

$$x = \frac{1}{4}.$$

The same singular location controls every fixed defect layer.

28.2 Algebraicity and singularity location

Proposition 28.1 — Fixed-defect algebraicity.

For every fixed

$$\ell \geq 0,$$

$$\boxed{S_\ell(x) \in \mathbb{Q}(t), \qquad t = \sqrt{1-4x}.}$$

More precisely,

$$\boxed{S_\ell(x) \in \mathbb{Q}[t, t^{-1}, (1+t)^{-1}].}$$

Therefore, on the branch $t(0) = 1$, the only possible dominant singularity is

$$\boxed{x = \frac{1}{4}.}$$

Proof idea. Let

$$\mathcal{F}(S, q, x) = (1-q)S(1+qS) - x(1+S)^2(1+q^2S).$$

At

$$q = 0,$$

the coefficient of the unknown S_ℓ in the extracted q^ℓ equation is

$$\frac{\partial \mathcal{F}}{\partial S} = 1 - 2x(1+S_0) = t.$$

Thus each new fixed-defect layer is obtained rationally from the earlier ones by dividing by powers of t.

Induction gives the claimed field. □

This proposition justifies using singularity analysis uniformly across all fixed defect levels.

28.3 Critical scaling

To extract the leading singular behavior of all layers simultaneously, set

$$\boxed{q = ty.}$$

Seek an expansion

$$\boxed{S(x, ty) = 1 + tf(y) + t^2 g(y) + O(t^3).}$$

The branch compatible with

$$S(x, 0) = \frac{1-t}{1+t}$$

must satisfy

$$f(0) = -2.$$

Substitution into the master equation gives

$$f^2 - 4yf + 8y^2 - 4 = 0.$$

The correct branch is

$$\boxed{f(y) = 2y - 2\sqrt{1-y^2}.}$$

At the next order,

$$\boxed{g(y) = 2\frac{4y^3 - 3y + \sqrt{1-y^2}}{\sqrt{1-y^2}}.}$$

The even coefficients of f and odd coefficients of g determine the dominant singular terms of successive defect layers.

This is the analytic origin of the pair structure.

28.4 Pairwise leading singularities

For

$$j \geq 1,$$

one obtains

$$S_{2j}(x) = A_j t^{1-2j} + O(t^{2-2j}),$$

and

$$S_{2j+1}(x) = B_j t^{1-2j} + O(t^{2-2j}),$$

with

$$A_j = \frac{C_{j-1}}{4^{j-1}},$$

and

$$B_j = (2j+3)A_j.$$

The key point is that defects $2j$ and $2j+1$ have the **same singular exponent**.

Their leading constants differ, but their polynomial growth degree is identical.

28.5 Asymptotic Defect Hierarchy

Theorem 28.2 — Asymptotic Defect Hierarchy.

For every fixed

$$j \geq 0,$$

$$\boxed{N_{m,2j} \sim \frac{m^j}{j!} C_m,}$$

and

$$\boxed{N_{m,2j+1} \sim \frac{2j+3}{j!} m^j C_m.}$$

Equivalently,

$$\boxed{N_{m,2j} \sim \frac{4^m}{j!\sqrt{\pi}} m^{j-\frac{3}{2}},}$$

and

$$\boxed{N_{m,2j+1} \sim \frac{2j+3}{j!\sqrt{\pi}} 4^m m^{j-\frac{3}{2}}.}$$

Proof.

Apply the standard transfer theorem to the leading singular terms in powers of

$$t = \sqrt{1-4x}.$$

For $j \geq 1$, the singular exponent is

$$t^{1-2j} = (1-4x)^{-(j-\frac{1}{2})}.$$

The transfer coefficient, together with the explicit constants A_j, B_j, gives the displayed formulas.

The boundary case

$$j = 0$$

uses the exact identities

$$N_{m,0} = C_m$$

and

$$N_{m,1} = C_{m+1} - C_m \sim 3C_m.$$

□

Thus:

$$\boxed{\frac{N_{m,\ell}}{C_m} = \Theta\left(m^{\lfloor \ell/2 \rfloor}\right)}$$

for fixed ℓ.

28.6 The defect pairs

The first leading ratios to the Catalan scale are:

$$1, \quad 3, \quad m, \quad 5m, \quad \frac{1}{2}m^2, \quad \frac{7}{2}m^2, \quad \frac{1}{6}m^3, \quad \frac{3}{2}m^3, \ldots$$

The hierarchy is therefore:

$$\boxed{(0,1)}$$

on the Catalan polynomial scale,
then

$$\boxed{(2,3)}$$

one power of m higher,
then

$$\boxed{(4,5)}$$

two powers higher,
and so on.
Every two defect units add one asymptotically free polynomial degree.

28.7 Geometric interpretation

The theorem is analytic.
The following interpretation comes afterward.
The zero-defect Catalan condition permits adjacent reversed-signature rises of at most 1.
A true violation requires a rise of at least 2.
Such an overshoot costs at least two defect units.
Therefore the pair structure

$$(2j, 2j+1)$$

is consistent with j asymptotically free non-Catalan defect locations among m signature positions.
The factor

$$\frac{m^j}{j!}$$

has exactly the scale expected from choosing j widely separated defect locations.
This is an interpretation of the proven singularity result, not a replacement for its proof.
The monograph should keep that distinction explicit.

28.8 Cumulative asymptotics

Since

$$\widehat{N}_{m,\delta} = \sum_{\ell \le \delta} N_{m,\ell},$$

the top defect layer dominates within each fixed cumulative cutoff.
Thus:

$$\boxed{\widehat{N}_{m,2j} \sim \frac{m^j}{j!} C_m,}$$

and

$$\boxed{\widehat{N}_{m,2j+1} \sim \frac{2j+4}{j!} m^j C_m.}$$

At

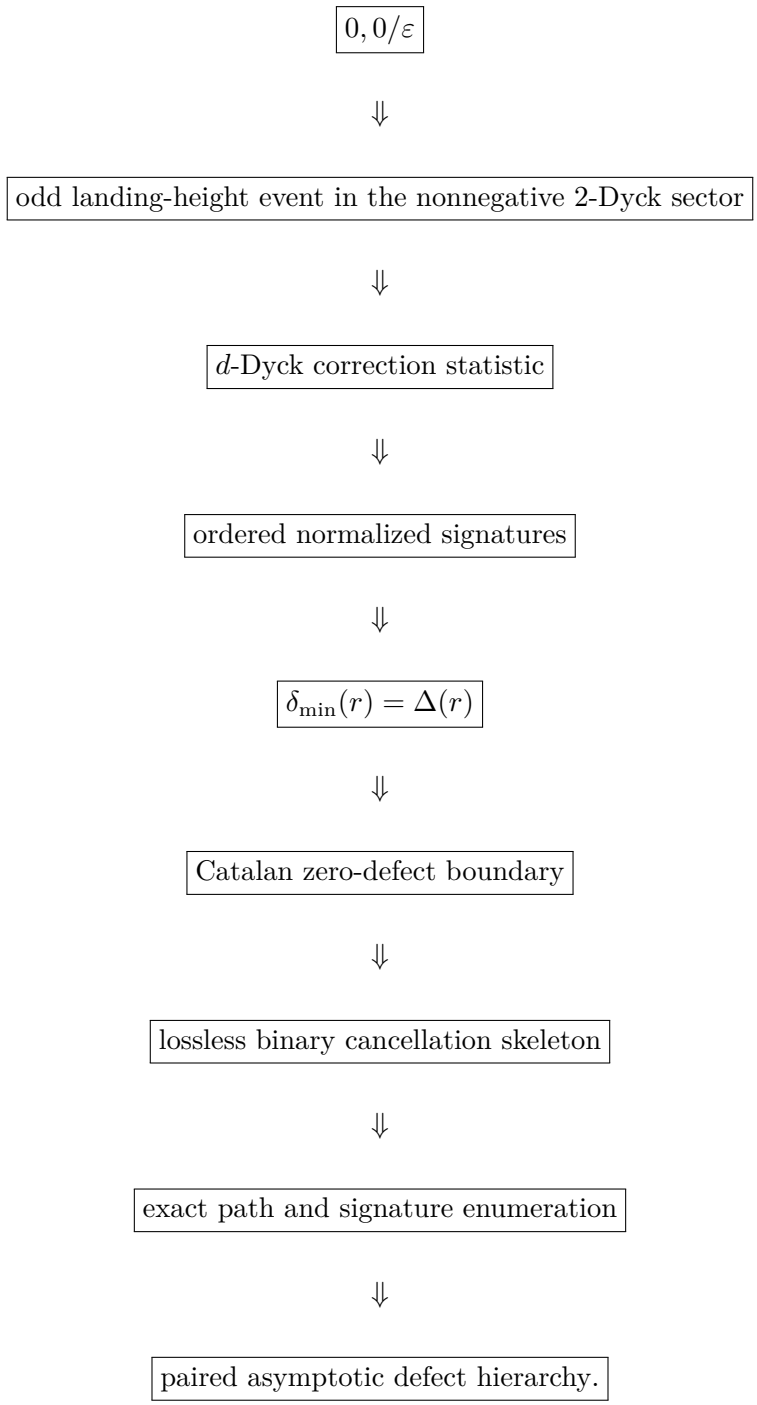
$$j = 0,$$

this agrees with the exact identity

$$\widehat{N}_{m,1} = C_{m+1}.$$

28.9 What the cancellation branch has established

We can now summarize the complete branch generated by one transition of the original DPDA:



The DPDA was the discovery mechanism.

The final normalized signature theory is intrinsic and no longer depends on the machine's remaining states or stack alphabet.

28.10 Chronological carry as an arithmetic offshoot

The ordered correction signature remembers chronology.

Pushing that chronology one step further attaches an explicit third-order arithmetic interaction to:

- singleton correction positions;
- adjacent pairs;
- connected triples.

That construction produces the CCR-3 recurrence.

It then develops a modular Stirling shadow, a simple Hensel branch, singular extinction, and ramified p-adic recovery. The subsequent fourth-order chapter isolates the first independent higher connected coordinate, and the CCR-k chapter closes the same terminal-block mechanism at arbitrary finite order. A later active-observability chapter studies the inverse problem of recovering masked chronological sources by controlled probes, with exact source contracts and sharp adaptive/predetermined costs.

This arithmetic refinement is a genuine continuation of the correction branch, but it is not a prerequisite for the later general real-time p:q DPDA theory.

Accordingly, the following research branch is presented as a specialized offshoot.

Readers interested only in the main normalization spine may skip it and rejoin when the pushdown-machine line resumes.

Reader's guide.
Chapters 29–35 are a specialized arithmetic offshoot. They continue the ordered-correction branch of Chapters 24–28, but they are not prerequisites for the later real-time p:q pushdown-normalization theory. Readers following only the main DPDA spine may skip directly to Chapter 38.

Chapter 29

From Ordered Corrections to CCR-3

Research branch — Chronological Carry and the Arithmetic Offshoot. *This branch develops the chronological-carry refinement and p-adic root analysis that emerge from the cancellation branch. The surrounding arithmetic literature includes carries and the p-adic behavior of Stirling numbers [7, 1, 12, 18].*

The preceding cancellation branch ended with an intrinsic correction-signature theory.
The original DPDA supplied a distinguished cancellation event.
The path interpretation ordered those events chronologically.
The Defect–Signature theorem then showed that the realization cost is local along that chronology:

$$\Delta(r) = r_m + \sum_{i=1}^{m-1} \psi(r_i - r_{i+1}).$$

Thus the correction history carries a natural path graph

$$1 - 2 - \cdots - n.$$

This is exactly the amount of pushdown-derived structure needed for the next step.
It is also the boundary of what the DPDA itself forces.

29.1 Scientific boundary

The arithmetic refinement introduced in this branch must be stated carefully.
What is inherited from the earlier DPDA/correction branch is:

1. a distinguished cancellation/correction event;
2. chronological ordering of those events;
3. nearest-neighbor locality along the ordered positions.

What is **introduced here** is a new arithmetic polarization:

1. a numerical weight for a singleton chronological slot;
2. a weight for an adjacent pair interaction;
3. a weight for the first connected triple collision.

Therefore the correct implication is

DPDA-derived chronological locality + specified arithmetic polarization $\Longrightarrow$ CCR-3.

The stronger statement

$$\text{original DPDA} \Longrightarrow \text{unique CCR-3 coefficients}$$

is not claimed.

This distinction is not editorial caution; it marks the exact scientific scope of the arithmetic branch.

29.2 Connected supports through order three

A connected subset of a path graph is an interval.

Therefore, up to size three, there are only three local connected support types.

Lemma 29.1 — Third-order connected support.

Every connected chronological support of cardinality at most three is one of

$$\boxed{\{j\}, \qquad \{j, j+1\}, \qquad \{j, j+1, j+2\}.}$$

These are, respectively:

- a singleton slot;
- an adjacent pair;
- a consecutive triple.

The lemma is elementary, but its role is structural: no other connected third-order motif exists on a one-dimensional chronology.

29.3 The CCR-3 recurrence

Fix a prime

$$p \geq 5$$

and define

$$\boxed{\ell_n = (p+1)n - p.}$$

Work in the truncated ring

$$\mathbb{Z}[u]/(u^4).$$

Set

$$P^{[3]}_{p,0}(u) = 1,$$

and

$$P^{[3]}_{p,-1}(u) = P^{[3]}_{p,-2}(u) = 0.$$

Definition 29.2 — CCR-3.

For

$$n \geq 1,$$

define

$$\boxed{\begin{aligned} P^{[3]}_{p,n}(u) &= (1+\ell_n u)P^{[3]}_{p,n-1}(u) \\ &\quad + p(n-1)\ell_{n-1}u^2 P^{[3]}_{p,n-2}(u) \\ &\quad - p(n-2)(n-1)u^3 P^{[3]}_{p,n-3}(u) \pmod{u^4}. \end{aligned}}$$

The three nontrivial sources represent:

$$\boxed{\text{singleton } + \text{ adjacent pair carry } + \text{ connected triple collision.}}$$

The first rows are

$$P^{[3]}_{p,0} = 1,$$

$$P^{[3]}_{p,1} = 1 + u,$$

$$P^{[3]}_{p,2} = 1 + (p+3)u + 2(p+1)u^2,$$

and

$$P^{[3]}_{p,3} = 1 + (3p+6)u + (4p^2+15p+11)u^2 + 6(p+1)^2u^3.$$

The superscript [3] is important.

The recurrence is intentionally third-order.

It should not be extrapolated blindly into an all-column theory: at order four a genuinely new connected motif may enter.

29.4 Hessenberg realization

Let

$$H_{p,n}$$

be the $n \times n$ leading principal truncation of the infinite lower-Hessenberg matrix defined by

$$(H_p)_{i,i} = \ell_i,$$

$$(H_p)_{i,i+1} = 1,$$

$$(H_p)_{i+1,i} = -pi\ell_i,$$

$$(H_p)_{i+2,i} = -pi(i+1),$$

with all entries below the second subdiagonal equal to zero.

Define

$$D_{p,n}(u) = \det(I_n + uH_{p,n}).$$

Proposition 29.3 — Hessenberg realization.

$$\boxed{\begin{aligned} D_{p,n} &= (1+\ell_n u)D_{p,n-1} \\ &\quad + p(n-1)\ell_{n-1}u^2 D_{p,n-2} \\ &\quad - p(n-2)(n-1)u^3 D_{p,n-3}. \end{aligned}}$$

Hence

$$\boxed{P^{[3]}_{p,n}(u) \equiv D_{p,n}(u) \pmod{u^4}.}$$

Proof.

Expand the determinant by the last row.

The diagonal contribution is

$$(1 + \ell_n u) D_{p,n-1}.$$

The first subdiagonal closes with one superdiagonal entry and contributes

$$+p(n-1)\ell_{n-1} u^2 D_{p,n-2}.$$

The second subdiagonal closes with two consecutive superdiagonal entries and contributes

$$-p(n-2)(n-1) u^3 D_{p,n-3}.$$

No deeper band is present. □

Thus the three chronological interaction orders are encoded by a three-band Hessenberg structure.

29.5 Operator form

Use the exponential basis

$$e_i(x) = \frac{x^{i-1}}{(i-1)!}, \qquad i \geq 1.$$

Let

$$D = \partial_x, \qquad Xf = xf, \qquad N = XD,$$

and define

$$L_p = I + (p+1)N.$$

Then

$$De_i = e_{i-1},$$

$$Xe_i = ie_{i+1},$$

$$X^2 e_i = i(i+1)e_{i+2},$$

and

$$L_p e_i = \ell_i e_i.$$

Corollary 29.4 — Operator form.

The Hessenberg matrix H_p is the matrix, in the basis (e_i), of

$$\boxed{\mathcal{H}_p = D + L_p - pXL_p - pX^2.}$$

Equivalently,

$$\boxed{\mathcal{H}_p = \left[1 + (p+1)x - p(p+1)x^2\right]\partial_x + 1 - px - px^2.}$$

The determinant recurrence and differential-operator form are therefore two representations of the same CCR-3 dynamics.

29.6 What this chapter has and has not established

We have not derived CCR-3 from the original DPDA transition table.

We have established the narrower chain

> ordered correction chronology → path-local connected supports
> → specified third-order polarization → Hessenberg/operator recurrence.

The next question is arithmetic.

What do the first three fixed columns of this recurrence look like?

The answer is unexpectedly rigid: they are polynomials, and modulo p the entire third jet collapses to a universal first-kind Stirling shadow.

Chapter 30

Fixed Columns and the Universal Stirling Shadow

The CCR-3 recurrence is nonlinear in its dependence on the chronological index but finite in interaction order.

To expose its arithmetic, we now freeze the power of u and let the row index vary.

This produces a transport system for the first three fixed columns.

30.1 Column coefficients

Define

$$\boxed{C^{(p)}_{n,k} = [u^k]P^{[3]}_{p,n}(u), \qquad 0 \le k \le 3.}$$

For

$$m \ge k + 1,$$

normalize by the falling factorial:

$$\boxed{\Pi_{p,k}(m) = \frac{C^{(p)}_{m-1,k}}{(m-1)_{\underline{k}}}.}$$

The normalization removes the obvious chronological placement multiplicity and reveals polynomial structure.

30.2 Transport equations

Proposition 30.1 — Transport system.

For the indicated ranges,

$$\boxed{m\Pi_{p,1}(m+1) = (m-1)\Pi_{p,1}(m) + \ell_m,}$$

$$\boxed{m\Pi_{p,2}(m+1) = (m-2)\Pi_{p,2}(m) + \ell_m\Pi_{p,1}(m) + p\ell_{m-1},}$$

and

$$\boxed{\begin{aligned} m\Pi_{p,3}(m+1) = (m-3)\Pi_{p,3}(m) + \ell_m\Pi_{p,2}(m) \\ + p\ell_{m-1}\Pi_{p,1}(m-1) - p. \end{aligned}}$$

Proof.

Extract the coefficients of

$$u, \quad u^2, \quad u^3$$

from CCR-3.

Then substitute

$$n = m$$

and divide by the corresponding falling factorial.

The chronological factors cancel exactly, leaving the displayed first-order transport equations. □

The third source can be separated into a connected form.

Since

$$\Pi_{p,1}(m-1) = \frac{\ell_{m-1} - p}{2},$$

the genuinely new third-order source is

$$\boxed{\mathfrak{T}_{p,3}(m) = p\left[\frac{1}{2}\ell_{m-1}(\ell_{m-1} - p) - 1\right].}$$

This isolates the connected triple correction from propagated lower-order contributions.

30.3 First three fixed columns

Theorem 30.2 — First three fixed columns.

For

$$p \geq 5,$$

$$\boxed{\Pi_{p,1}(m) = \frac{(p+1)m - 2p}{2},}$$

$$\boxed{\Pi_{p,2}(m) = \frac{p+1}{24}\left[3(p+1)m^2 - (5p+1)m - 12p\right],}$$

and

$$\boxed{\Pi_{p,3}(m) = \frac{p+1}{48}F_p(m),}$$

where

$$\boxed{\begin{aligned} F_p(m) = (p+1)^2 m^3 + (p^2-1)m^2 \\ - (34p^2 + 18p)m + 56p^2 - 8p. \end{aligned}}$$

The boundary values are

$$\Pi_{p,1}(2) = 1,$$

$$\Pi_{p,2}(3) = p + 1,$$

$$\Pi_{p,3}(4) = (p+1)^2.$$

Proof.

The first transport equation is solved directly by the displayed linear polynomial.

Substituting that result into the second equation gives a first-order recurrence with polynomial source; the displayed quadratic satisfies both the recurrence and its boundary value.

Substituting the first two solutions into the third transport equation gives a cubic source. The displayed cubic satisfies the recurrence and boundary value.

Uniqueness follows at each stage from the first-order recurrence. □

The key methodological point is that these formulas arise from transport.

They should not be presented as unexplained computer-algebra output.

30.4 Reduction modulo p

Modulo p,

$$\ell_n = (p+1)n - p \equiv n \pmod{p}.$$

Every pair and triple interaction source carries an explicit factor of p.

Therefore those interaction terms vanish modulo p.

CCR-3 degenerates to the free recurrence

$$P^{[3]}_{p,n}(u) \equiv (1+nu)P^{[3]}_{p,n-1}(u) \pmod{(p,u^4)}.$$

Iteration gives the universal product.

Theorem 30.3 — Stirling shadow.

For every

$$n \geq 0,$$

$$\boxed{P^{[3]}_{p,n}(u) \equiv \prod_{j=1}^{n}(1+ju) \pmod{(p,u^4)}.}$$

Consequently, for $k \leq 3$,

$$\boxed{C^{(p)}_{m-1,k} \equiv e_k(1,\dots,m-1) \equiv \begin{bmatrix} m \\ m-k \end{bmatrix} \pmod{p},}$$

where the bracket denotes the unsigned Stirling number of the first kind.

Thus the arithmetic interaction deformation has a universal free shadow.

30.5 Normalized shadow polynomials

The normalized fixed columns reduce to:

$$\boxed{S_1(m) = \frac{m}{2},}$$

$$\boxed{S_2(m) = \frac{m(3m-1)}{24},}$$

$$\boxed{S_3(m) = \frac{m^2(m-1)}{48}.}$$

So, as polynomial congruences,

$$\boxed{\Pi_{p,k}(m) \equiv S_k(m) \pmod{p} \qquad (k=1,2,3).}$$

Important notation point

One must not derive these congruences by naively cancelling the falling factorial

$$(m-1)_{\underline{k}}$$

inside $\mathbb{F}_p$.
That denominator may vanish at exceptional residues.
The safe argument is:

1. derive the exact polynomial formulas over $\mathbb{Q}[m]$;
2. reduce those polynomial formulas modulo p.

Because

$$2, \quad 24, \quad 48$$

are units for $p \geq 5$, the reduction is legitimate.
This avoids illegal pointwise modular division.

30.6 Why the third column is the first singular one

The third normalized shadow is

$$S_3(m) = \frac{m^2(m-1)}{48}.$$

Therefore modulo p, the third-column zero set contains:

- a simple root at
$$m \equiv 1 \pmod{p};$$
- a double root at
$$m \equiv 0 \pmod{p}.$$

The exact cubic deformation $F_p(m)$ must decide what happens to these two shadow branches.
The next chapter shows that they behave completely differently.
The simple root survives integrally.
The double root becomes extinct in $\mathbb{Z}_p$.

Chapter 31

The Third-Column p-Adic Split

The third normalized column is

$$\Pi_{p,3}(m) = \frac{p+1}{48} F_p(m),$$

where

$$F_p(m) = (p+1)^2 m^3 + (p^2-1)m^2 - (34p^2+18p)m + 56p^2 - 8p.$$

Modulo p,

$$\boxed{F_p(m) \equiv m^2(m-1) \pmod{p}.}$$

Thus the shadow has two different types of roots:

$$m \equiv 1 \pmod{p}$$

is simple, while

$$m \equiv 0 \pmod{p}$$

is double.
The exact deformation separates them sharply.

31.1 The simple branch

At

$$m = 1,$$

$$F_p(1) \equiv 0 \pmod{p},$$

and

$$\boxed{F_p'(1) \equiv 1 \pmod{p}.}$$

The derivative is a unit.
Hensel's lemma therefore applies in its simplest form.

Theorem 31.1 — Simple Hensel tower.

There exists a unique

$$\boxed{\alpha_p \in \mathbb{Z}_p}$$

such that

$$\alpha_p \equiv 1 \pmod{p}$$

and

$$F_p(\alpha_p) = 0.$$

Its expansion begins

$$\boxed{\begin{aligned}\alpha_p = 1 + 24p - 888p^2 + 58008p^3 \\ - 4715256p^4 + 428583192p^5 + O(p^6).\end{aligned}}$$

Moreover, throughout the residue class

$$m \equiv 1 \pmod{p},$$

$$\boxed{\nu_p(F_p(m)) = \nu_p(m - \alpha_p).}$$

Proof.

Hensel gives existence and uniqueness.
Factor in $\mathbb{Z}_p[m]$:

$$F_p(m) = (m - \alpha_p)G_p(m).$$

Since

$$G_p(\alpha_p) = F_p'(\alpha_p)$$

is a unit, $G_p(m)$ remains a unit throughout the residue class

$$m \equiv \alpha_p \equiv 1 \pmod{p}.$$

Therefore the valuation is exactly the distance to the Hensel root. □
Thus the simple shadow branch survives exactly as expected.

31.2 The double shadow branch

Now set

$$m = pu.$$

Direct substitution gives

$$\boxed{\frac{F_p(pu)}{p} \equiv -8 \pmod{p}.}$$

Since $p \geq 5$,

$$-8$$

is nonzero modulo p.
Therefore:

Theorem 31.2 — Singular extinction.

For every

$$u \in \mathbb{Z}_p,$$

$$\boxed{\nu_p(F_p(pu)) = 1.}$$

Consequently, although

$$m \equiv 0 \pmod{p}$$

is a double root of the Stirling shadow, it has **no lift to a root modulo** p^2.
We call this phenomenon

$$\boxed{\textbf{singular extinction}.}$$

The term means exactly:

$$\boxed{\text{multiple root mod } p \quad \text{but no integral branch survives mod } p^2.}$$

It does not mean that the corresponding algebraic roots disappear from the algebraic closure.

Chapter 32 will show where they go.

31.3 Generic residue classes

If

$$m \not\equiv 0, 1 \pmod{p},$$

then

$$F_p(m) \not\equiv 0 \pmod{p}.$$

Hence

$$\boxed{\nu_p(F_p(m)) = 0.}$$

Thus the residue classes split into three exact arithmetic regimes:

1. generic;
2. simple Hensel;
3. singularly extinct.

31.4 The scaled third column

Define

$$\boxed{Q_{m,3}^{\mathrm{CCR}} = p^{m-4}\Pi_{p,3}(m), \qquad m \geq 4.}$$

Because

$$\frac{p+1}{48}$$

is a p-adic unit for $p \geq 5$,

$$\nu_p(\Pi_{p,3}(m)) = \nu_p(F_p(m)).$$

The previous three cases therefore give the complete valuation law.

Theorem 31.3 — Complete third-column valuation law.

For

$$m \geq 4,$$

$$\boxed{\nu_p(Q_{m,3}^{\mathrm{CCR}}) = \begin{cases} m-4, & m \not\equiv 0,1 \pmod{p}, \\ m-3, & m \equiv 0 \pmod{p}, \\ m-4+\nu_p(m-\alpha_p), & m \equiv 1 \pmod{p}. \end{cases}}$$

This theorem is best read after the branch analysis.
The three cases are not an arbitrary valuation table.
They are exactly:

$$\boxed{\text{generic unit}}$$

$$\boxed{\text{one forced } p\text{-factor from singular extinction}}$$

$$\boxed{\text{Hensel distance to the unique integral root.}}$$

31.5 Independent consistency checks

The monograph audit independently checked the arithmetic at several levels:

- the fixed-column formulas were regenerated symbolically through several rows;
- the displayed formulas matched 15 tested symbolic column/boundary instances;
- substituting the displayed Hensel expansion into F_p cancels every coefficient through p^5;
- digitwise Hensel lifting for
$$p = 5, 7, 11, 13, 17, 19$$
agrees through modulus p^6;
- hundreds of sample valuation checks agree with both the simple-branch and extinction laws.

These computations support the proof chain.
They do not replace it.

31.6 Ramified location of the remaining algebraic roots

The double shadow root has now disappeared from the integral p-adic locus:

$$0 \bmod p \longrightarrow \varnothing \bmod p^2.$$

But F_p is cubic.
One root has become the integral Hensel root α_p.
Two algebraic roots remain.
The discriminant will show that they have not disappeared.
They have moved into a ramified quadratic extension, with half-integral valuation.

Chapter 32

Ramified Recovery

The simple shadow root has survived inside

$$\mathbb{Z}_p.$$

The double shadow root has undergone singular extinction.

The two remaining algebraic roots are recovered in a ramified quadratic extension. Their location is encoded by the cubic discriminant.

32.1 Exact discriminant factorization

Theorem 32.1 — Discriminant and ramified field.

The discriminant of F_p factors as

$$\boxed{\mathrm{Disc}(F_p) = 4p(p+1)^2 R_p,}$$

where

$$\boxed{R_p = 9801p^5 + 22672p^4 + 32502p^3 + 14360p^2 - 927p - 8.}$$

Modulo p,

$$\boxed{R_p \equiv -8 \pmod{p}.}$$

Therefore

$$\boxed{\nu_p(\mathrm{Disc}(F_p)) = 1.}$$

The discriminant has odd valuation, so the residual quadratic factor after removing the simple Hensel root is ramified.

32.2 Removing the integral root

Factor

$$F_p(m) = (m - \alpha_p)Q_p(m)$$

over $\mathbb{Z}_p$.

The root α_p is simple and integral.

The remaining quadratic Q_p has discriminant with the same square class as the cubic discriminant, because the resultant contribution from the linear/quadratic splitting is a square.

Now

$$R_p/(-8) \equiv 1 \pmod{p}.$$

By Hensel's lemma,

$$R_p/(-8)$$

is a square in

$$\mathbb{Z}_p^\times.$$

The factors

$$4 \qquad \text{and} \qquad (p+1)^2$$

are also squares.
Hence the discriminant square class is

$$\boxed{[-2p].}$$

The quadratic splitting field is therefore

$$\boxed{\mathbb{Q}_p(\sqrt{-2p}).}$$

32.3 Half-integral roots

The quadratic factor is Eisenstein after the integral branch is removed.
Its two roots

$$\beta_{p,+}, \qquad \beta_{p,-}$$

therefore satisfy

$$\boxed{\nu_p(\beta_{p,\pm}) = \frac{1}{2}.}$$

Thus the double shadow root has not vanished algebraically.
Instead:

$$\boxed{\text{double root mod } p}$$

splits in the exact deformation into

$$\boxed{\text{two ramified roots of valuation } 1/2.}$$

This is the conceptual resolution of singular extinction.
Integral lifting fails because the two branches leave the integral valuation lattice.

32.4 Ramified square class by $p \bmod 8$

Corollary 32.2 — Ramified square class.
The ramified field is exactly

$$\boxed{\mathbb{Q}_p(\sqrt{-2p}).}$$

Its standard square-class description depends on

$$p \bmod 8.$$

For the residue classes in which -2 is a square unit, this is the standard p-ramified class.

For the remaining residue classes, it is the other ramified quadratic square class represented by a nonsquare unit times p.

This corollary is classical square-class arithmetic applied to the exact discriminant already proved.

32.5 The complete third-order arithmetic picture

The results of Chapters 29–32 can now be summarized without adding a new proof.

Corollary 32.3 — CCR-3 arithmetic geometry.

The third-order chronological carry refinement has the following structure.

1. Chronological locality allows only singleton, adjacent-pair, and consecutive-triple connected supports through order three.
2. The chosen arithmetic polarization produces the CCR-3 recurrence and its Hessenberg/operator realization.
3. The first three normalized columns are explicit polynomials.
4. Modulo p, all interaction terms disappear and

$$P_{p,n}^{[3]}(u) \equiv \prod_{j=1}^{n}(1+ju) \pmod{(p,u^4)}.$$

5. The normalized columns reduce to near-diagonal first-kind Stirling polynomials.
6. In the third column:
   - the simple shadow root
   $$1 \pmod{p}$$
   lifts uniquely to
   $$\alpha_p \in \mathbb{Z}_p;$$
   - the double shadow root
   $$0 \pmod{p}$$
   undergoes singular extinction;
   - the two missing algebraic branches survive with valuation
   $$1/2$$
   over
   $$\mathbb{Q}_p(\sqrt{-2p}).$$

This is the complete third-order arithmetic offshoot.

32.6 Novelty boundary

Nothing in this chapter requires claiming novelty for the ingredients individually.

The following are classical or established areas:

- carry processes;
- Holte-type carry matrices;
- Stirling numbers;
- p-adic valuations of Stirling numbers;
- Hensel lifting;
- Eisenstein irreducibility;
- discriminants;
- Newton polygons;
- ramified quadratic extensions;
- lower-Hessenberg production matrices.

The family-specific contribution is the chain

chronological local interactions → fixed-column polynomialization → Stirling shadow → simple lifting / singular extinction → exact ramified recovery.

The bibliographic claim is restricted to the proved family-specific chain stated here; no broader priority claim is made.

Chapter 33

CCR-4: Fourth-Order Chronological Carry, Connected Quadruples, and the Stirling Shadow

The preceding arithmetic Part established CCR-3 and its third-jet Stirling shadow. The fourth-order problem requires a scope correction: the complete third jet and chronological path-locality determine the new connected support type, but not its numerical weight. The new datum is therefore introduced as an explicit fourth-order polarization and then identified from the normalized fourth column.

33.1 The CCR-3 baseline

Fix a prime $p \geq 5$ and define

$$\ell_n = (p+1)n - p.$$

The third-order arithmetic polarization is taken as the baseline object of this chapter.

Definition 33.1 (CCR-3 baseline)**.** Work in $\mathbb{Z}[u]/(u^4)$ and set

$$P^{[3]}_{p,0}(u) = 1, \qquad P^{[3]}_{p,-1}(u) = P^{[3]}_{p,-2}(u) = 0.$$

For $n \geq 1$, define

$$\begin{aligned} P^{[3]}_{p,n}(u) = (1+\ell_n u)P^{[3]}_{p,n-1}(u) + p(n-1)\ell_{n-1}u^2 P^{[3]}_{p,n-2}(u) \\ - p(n-2)(n-1)u^3 P^{[3]}_{p,n-3}(u) \pmod{u^4}. \end{aligned} \tag{33.1}$$

The three sources are called the singleton slot, adjacent-pair interaction, and connected triple interaction.

For $0 \leq k \leq 3$ put

$$C^{(p)}_{n,k} = [u^k]P^{[3]}_{p,n}(u), \qquad \Pi_{p,k}(m) = \frac{C^{(p)}_{m-1,k}}{(m-1)_{\underline{k}}} \quad (m \geq k+1).$$

Coefficient extraction from (33.1) gives the transport equations

$$\begin{aligned} m\Pi_{p,1}(m+1) &= (m-1)\Pi_{p,1}(m) + \ell_m, \\ m\Pi_{p,2}(m+1) &= (m-2)\Pi_{p,2}(m) + \ell_m\Pi_{p,1}(m) + p\ell_{m-1}, \\ m\Pi_{p,3}(m+1) &= (m-3)\Pi_{p,3}(m) + \ell_m\Pi_{p,2}(m) \\ &\qquad + p\ell_{m-1}\Pi_{p,1}(m-1) - p. \end{aligned} \tag{33.2}$$

Proposition 33.2 (First three normalized columns)**.** *For the indicated ranges,*

$$\Pi_{p,1}(m) = \frac{(p+1)m - 2p}{2},$$

$$\Pi_{p,2}(m) = \frac{p+1}{24}\left[3(p+1)m^2 - (5p+1)m - 12p\right],$$

and

$$\Pi_{p,3}(m) = \frac{p+1}{48}F_p(m),$$

where

$$F_p(m) = (p+1)^2m^3 + (p^2-1)m^2 - (34p^2+18p)m + 56p^2 - 8p.$$

The boundary values are

$$\Pi_{p,1}(2) = 1, \qquad \Pi_{p,2}(3) = p+1, \qquad \Pi_{p,3}(4) = (p+1)^2.$$

Proof. Substitution into (33.2) verifies the three displayed polynomials. Each transport equation is first order in its current column, so the corresponding boundary value gives uniqueness. No computer-algebra interpolation is used as a proof premise. □

Modulo p, every non-singleton source in (33.1) disappears and $\ell_n \equiv n$. Hence

$$P^{[3]}_{p,n}(u) \equiv \prod_{j=1}^{n}(1+ju) \pmod{(p, u^4)}.$$

The purpose of this chapter is to determine exactly what survives, and what becomes new, when the truncation is raised to u^5.

33.2 What is genuinely new at order four

Definition 33.3 (Connected chronological support)**.** A set of correction positions is *connected* if it induces a connected subgraph of the chronological path $1 - 2 - \cdots - n$.

Lemma 33.4 (Unique connected quadruple)**.** *Every connected chronological support of cardinality four is an interval of the form*

$$\{j, j+1, j+2, j+3\}.$$

Thus there is exactly one new connected support type at order four.

Proof. Every connected subset of a path graph is an interval. An interval with four vertices has the displayed form. □

The structural issue is not the shape of the new support but the arithmetic weight assigned to it. The recurrence itself gives a clean way to separate propagated lower-order terms from the genuinely new fourth-order source.

Definition 33.5 (Terminal block form through order four)**.** A path-local fourth-order extension of CCR-3 is a recurrence in $\mathbb{Z}[u]/(u^5)$ of the form

$$\begin{aligned} P_{p,n}(u) = (1+\ell_n u)P_{p,n-1}(u) + p(n-1)\ell_{n-1}u^2P_{p,n-2}(u) \\ - p(n-2)(n-1)u^3P_{p,n-3}(u) + q_{p,n}u^4P_{p,n-4}(u) \pmod{u^5}, \end{aligned} \tag{33.3}$$

with $P_{p,0} = 1$ and $P_{p,n} = 0$ for negative n. The first three terminal block weights are fixed by CCR-3; $q_{p,n}$ is the connected quadruple source.

Proposition 33.6 (Lower-order block separation)**.** *In* (33.3)*, every contribution to* $[u^4]P_{p,n}$ *obtained by appending a terminal connected block of size one, two, or three is already determined by CCR-3. The only term in the fourth coefficient not generated by those lower bands is* $q_{p,n}$ *itself.*

Proof. Extracting u^4 from (33.3) gives

$$\begin{aligned}[u^4]P_{p,n} = {} & [u^4]P_{p,n-1} + \ell_n[u^3]P_{p,n-1} \\ & + p(n-1)\ell_{n-1}[u^2]P_{p,n-2} \\ & - p(n-2)(n-1)[u]P_{p,n-3} + q_{p,n}.\end{aligned} \tag{33.4}$$

The middle three lines attach a terminal block of size one, two, or three to a previously generated lower-order history. Iterating those terms produces exactly the degree-four histories decomposable into lower connected block sizes, including the patterns $1+1+1+1$, $2+1+1$, $2+2$, and $3+1$ in chronological order. The final scalar $q_{p,n}$ is the contribution of a single terminal block of size four. By theorem 33.4, that block is the unique consecutive quadruple support. □

Remark 33.7 (Relation with cumulant language)*.* Classical cumulants isolate connected information by Möbius inversion on the lattice of set partitions [65, 64]; mixed cumulants vanish when the variables split into independent families. That theory motivates the word "connected," but theorem 33.6 does not identify the CCR coefficients with probabilistic cumulants. It is an internal recurrence statement about terminal interval blocks on a path.

Theorem 33.8 (Fourth-order non-identifiability from CCR-3)**.** *CCR-3 and chronological path-locality do not determine a unique fourth-order numerical weight. More precisely, for any integer sequence* $(q_{p,n})_{n\geq 4}$*, recurrence* (33.3) *has exactly the same coefficients through* u^3 *as CCR-3.*

Proof. The new term is divisible by u^4, so it cannot affect any coefficient of degree at most three. Induction on n therefore shows that every choice of $q_{p,n}$ has the same third jet as (33.1). The fourth coefficient is the first place where the choices can differ. □

This theorem is the central scope correction at order four: a numerical CCR-4 coefficient must come from an additional arithmetic polarization, not from third-order data alone.

33.2.1 Exact reconstruction of a general p-divisible fourth source

Suppose the connected source is p-divisible after the natural size-four placement factor:

$$q_{p,m} = p(m-1)_{\underline{3}}\eta_m, \qquad \eta_m \in \mathbb{Z}, \qquad m \geq 4.$$

For the corresponding solution of (33.3), write

$$C^{(q)}_{n,k} = [u^k]P_{p,n}(u), \qquad \Pi^{(q)}_{p,4}(m) = \frac{C^{(q)}_{m-1,4}}{(m-1)_{\underline{4}}}, \qquad m \geq 5.$$

Let $\Pi^{(0)}_{p,4}$ denote the zero-fourth-source solution and put

$$\Delta_m = \Pi^{(q)}_{p,4}(m) - \Pi^{(0)}_{p,4}(m).$$

Theorem 33.9 (Fourth-source reconstruction)**.** *For every* $m \geq 5$*,*

$$\boxed{m\Delta_{m+1} = (m-4)\Delta_m + p\eta_m.} \tag{33.5}$$

At the boundary,

$$\boxed{\Delta_5 = \frac{p\eta_4}{4}.} \tag{33.6}$$

Consequently the connected source is exactly identifiable from the fourth normalized column:

$$\boxed{\eta_m = \frac{m\Delta_{m+1} - (m-4)\Delta_m}{p} \qquad (m \geq 5),} \tag{33.7}$$

with $\eta_4 = 4\Delta_5/p$.

Proof. Extract u^4 from (33.3), put $n = m$, and divide by $(m)_{\underline{4}} = m(m-1)(m-2)(m-3)$. The lower three source bands are identical for the given source and for the zero-source recurrence because all columns of degree at most three agree by theorem 33.8. Their difference therefore cancels. The remaining fourth source contributes

$$\frac{q_{p,m}}{(m)_{\underline{4}}} = \frac{p(m-1)_{\underline{3}}\eta_m}{(m)_{\underline{4}}} = \frac{p\eta_m}{m},$$

while the preceding fourth coefficient contributes $(m-4)\Delta_m/m$. Multiplying by m gives (33.5). At $m = 4$, the only difference in $C_{4,4}$ is $q_{p,4} = 6p\eta_4$; division by $(4)_{\underline{4}} = 24$ gives (33.6). Solving (33.5) for η_m yields (33.7). □

Thus fourth-order identifiability is not restricted to the stationary model: the entire p-divisible connected chronology can be read back from the fourth column. Stationarity becomes a testable property of that column rather than merely an imposed coefficient ansatz.

33.3 The stationary p-divisible fourth-order family

We now specify one natural class in which the new fourth-order datum can be classified exactly.

Definition 33.10 (Stationary p-divisible fourth polarization)**.** A recurrence of the form (33.3) is *stationary p-divisible* if, for every $n \geq 4$,

$$p(n-1)_{\underline{3}} \;\big|\; q_{p,n},$$

and the normalized integer quotient

$$\eta_n = \frac{q_{p,n}}{p(n-1)_{\underline{3}}}$$

is independent of n. Thus stationarity is imposed only after removing the inherited factor p and the falling-factorial normalization of a size-four terminal block.

Theorem 33.11 (Stationarity and one-parameter classification)**.** *For every prime $p \geq 5$ and every p-divisible fourth-order extension, the following are equivalent:*

(i) *the normalized connected source η_n is independent of n;*

(ii) *the fourth-column difference $\Delta_m = \Pi^{(q)}_{p,4}(m) - \Pi^{(0)}_{p,4}(m)$ is independent of $m \geq 5$;*

(iii) *for one integer λ, the recurrence is exactly*

$$\boxed{\begin{aligned} P^{[4,\lambda]}_{p,n}(u) = (1+\ell_n u)P^{[4,\lambda]}_{p,n-1}(u) + p(n-1)\ell_{n-1}u^2P^{[4,\lambda]}_{p,n-2}(u) \\ - p(n-2)(n-1)u^3P^{[4,\lambda]}_{p,n-3}(u) \\ + \lambda p(n-3)(n-2)(n-1)u^4P^{[4,\lambda]}_{p,n-4}(u) \end{aligned}} \pmod{u^5}, \tag{33.8}$$

with $\lambda \in \mathbb{Z}$. In that case

$$\Delta_m = \frac{\lambda p}{4} \qquad (m \geq 5).$$

The member $\lambda = 0$ has no new connected size-four source, whereas every $\lambda \neq 0$ has a genuine connected quadruple contribution.

Proof. The equivalence of (i) and (iii) is the definition of the normalized source: $\eta_n \equiv \lambda$ is exactly $q_{p,n} = \lambda p(n-1)_{\underline{3}}$. If (i) holds, theorem 33.9 has boundary value $\Delta_5 = \lambda p/4$, and the constant sequence $\Delta_m = \lambda p/4$ satisfies (33.5); first-order uniqueness gives (ii) and the displayed constant. Conversely, if $\Delta_m \equiv c$, then (33.7) gives $\eta_m = 4c/p$ for every $m \geq 5$, while (33.6) gives the same value for $m = 4$. Hence (ii) implies (i), completing the equivalence. □

Definition 33.12 (Primitive same-sign CCR-4)**.** The choice

$$\lambda = 1$$

is called the *primitive same-sign fourth polarization*. It is the smallest positive integral extension compatible with the sign convention for the lower Hessenberg interaction bands introduced in section 33.4.

Remark 33.13 (What $\lambda = 1$ means)*.* Definition 33.12 is a normalization choice. It gives one concrete CCR-4 recurrence, but it is not asserted that the original DPDA uniquely forces $\lambda = 1$. The theorem-level object is the full λ-family and the exact way in which λ is visible.

33.4 Four-band Hessenberg and operator realization

The extension parameter has a direct linear-algebraic realization.

Definition 33.14 (Fourth-band Hessenberg matrix)**.** Let $H_{p,\lambda}$ be the infinite lower-Hessenberg matrix with

$$(H_{p,\lambda})_{i,i} = \ell_i, \qquad (H_{p,\lambda})_{i,i+1} = 1,$$

$$(H_{p,\lambda})_{i+1,i} = -pi\ell_i, \qquad (H_{p,\lambda})_{i+2,i} = -pi(i+1),$$

$$(H_{p,\lambda})_{i+3,i} = -\lambda pi(i+1)(i+2),$$

and all entries below the third subdiagonal equal to zero. Write $H_{p,\lambda,n}$ for its $n \times n$ leading principal truncation and

$$D^{(\lambda)}_{p,n}(u) = \det(I_n + uH_{p,\lambda,n}).$$

Theorem 33.15 (Four-band Hessenberg realization)**.** *The determinants $D^{(\lambda)}_{p,n}$ satisfy recurrence* (33.8) *exactly. Consequently*

$$P^{[4,\lambda]}_{p,n}(u) \equiv D^{(\lambda)}_{p,n}(u) \pmod{u^5}.$$

Proof. Expand $\det(I_n + uH_{p,\lambda,n})$ by the last row. The diagonal term gives

$$(1 + \ell_n u)D^{(\lambda)}_{p,n-1}.$$

The first subdiagonal entry closes with one superdiagonal entry and gives

$$+p(n-1)\ell_{n-1}u^2 D^{(\lambda)}_{p,n-2}.$$

The second subdiagonal closes with two consecutive superdiagonal entries and gives

$$-p(n-2)(n-1)u^3 D^{(\lambda)}_{p,n-3}.$$

The third subdiagonal closes with three consecutive superdiagonal entries. Its cofactor sign reverses the negative matrix entry, producing

$$+\lambda p(n-3)(n-2)(n-1)u^4 D^{(\lambda)}_{p,n-4}.$$

No deeper band is present. This is exactly (33.8). □

Use the exponential basis

$$e_i(x) = \frac{x^{i-1}}{(i-1)!}, \qquad i \geq 1.$$

Let $D = \partial_x$, let $Xf = xf$, set $N = XD$, and define

$$L_p = I + (p+1)N.$$

Then

$$De_i = e_{i-1}, \qquad Xe_i = ie_{i+1}, \qquad X^k e_i = i(i+1)\cdots(i+k-1)e_{i+k},$$

and $L_p e_i = \ell_i e_i$.

Corollary 33.16 (Operator form)**.** *The matrix $H_{p,\lambda}$ is the matrix, in the basis (e_i), of*

$$\boxed{\mathcal{H}_{p,\lambda} = D + L_p - pXL_p - pX^2 - \lambda pX^3.} \tag{33.9}$$

Equivalently,

$$\boxed{\mathcal{H}_{p,\lambda} = \left[1 + (p+1)x - p(p+1)x^2\right]\partial_x + 1 - px - px^2 - \lambda px^3.} \tag{33.10}$$

Proof. The actions of D, L_p, XL_p, X^2, and X^3 on e_i reproduce respectively the superdiagonal, diagonal, first subdiagonal, second subdiagonal, and third subdiagonal of $H_{p,\lambda}$. Expanding $L_p = I + (p+1)XD$ gives (33.10). □

Banded Hessenberg determinants and their finite-term recurrences are classical; see, for example, [66]. The point here is the exact identification of the fourth chronological parameter with the third subdiagonal and the X^3 term.

33.5 Fourth-column transport and exact identifiability

For the fourth-order family define

$$C_{n,k}^{(p,\lambda)} = [u^k]P_{p,n}^{[4,\lambda]}(u), \qquad 0 \leq k \leq 4,$$

and, for $m \geq k+1$,

$$\Pi_{p,k}^{(\lambda)}(m) = \frac{C_{m-1,k}^{(p,\lambda)}}{(m-1)_{\underline{k}}}.$$

Because the λ-term starts in degree four, theorem 33.8 implies

$$\Pi_{p,k}^{(\lambda)} = \Pi_{p,k} \qquad (k = 1, 2, 3).$$

Theorem 33.17 (Fourth-column transport)**.** *For $m \geq 5$,*

$$\boxed{\begin{aligned} m\Pi_{p,4}^{(\lambda)}(m+1) &= (m-4)\Pi_{p,4}^{(\lambda)}(m) + \ell_m \Pi_{p,3}(m) \\ &\quad + p\ell_{m-1}\Pi_{p,2}(m-1) - p\Pi_{p,1}(m-2) + \lambda p. \end{aligned}} \tag{33.11}$$

The last term is exactly the normalized connected quadruple source.

Proof. Extract u^4 from (33.8) and put $n = m$. Divide by

$$(m)_{\underline{4}} = m(m-1)(m-2)(m-3).$$

The four source terms simplify using

$$\frac{(m-1)_{\underline{3}}}{(m)_{\underline{4}}} = \frac{1}{m}, \qquad \frac{(m-1)(m-2)_{\underline{2}}}{(m)_{\underline{4}}} = \frac{1}{m},$$

$$\frac{(m-2)(m-1)(m-3)_{\underline{1}}}{(m)_{\underline{4}}} = \frac{1}{m}, \qquad \frac{(m-3)(m-2)(m-1)}{(m)_{\underline{4}}} = \frac{1}{m}.$$

Multiplying by m gives (33.11). □

The key payoff is that the new connected source does not distort the shape of the fourth normalized column; it translates it.

Theorem 33.18 (Connected-quadruple translation law)**.** *For every $\lambda, \mu \in \mathbb{Z}$ and every $m \geq 5$,*

$$\boxed{\Pi_{p,4}^{(\lambda)}(m) - \Pi_{p,4}^{(\mu)}(m) = \frac{p(\lambda - \mu)}{4}.} \tag{33.12}$$

Equivalently,

$$\boxed{C_{m-1,4}^{(p,\lambda)} - C_{m-1,4}^{(p,\mu)} = \frac{p(\lambda - \mu)}{4}(m-1)_{\underline{4}}.} \tag{33.13}$$

Proof. Subtract the two instances of (33.11). With

$$\Delta_m = \Pi_{p,4}^{(\lambda)}(m) - \Pi_{p,4}^{(\mu)}(m),$$

one obtains

$$m\Delta_{m+1} = (m-4)\Delta_m + p(\lambda - \mu).$$

The constant sequence

$$\Delta_m = \frac{p(\lambda - \mu)}{4}$$

satisfies this recurrence. At the first admissible row $m = 5$, direct extraction from $P_{p,4}^{[4,\lambda]}$ gives the same difference. Uniqueness of the first-order recurrence proves (33.12); multiplying by $(m-1)_{\underline{4}}$ gives (33.13). □

Corollary 33.19 (One-row identifiability)**.** *Once the lower-order polarization is fixed, any single normalized fourth-column value at an admissible row identifies λ exactly:*

$$\boxed{\lambda = \frac{4}{p}\left(\Pi_{p,4}^{(\lambda)}(m) - \Pi_{p,4}^{(0)}(m)\right).}$$

At the first raw row $n = 4$, changing λ by one changes $[u^4]P_{p,4}$ by exactly $6p$.

Thus order four is simultaneously non-identifiable from the entire third jet and exactly identifiable from one fourth-column observation.

Theorem 33.20 (Explicit normalized fourth column)**.** *For $m \geq 5$,*

$$\boxed{\Pi_{p,4}^{(\lambda)}(m) = \frac{G_{p,\lambda}(m)}{5760},} \tag{33.14}$$

where

$$\begin{aligned} G_{p,\lambda}(m) = {} & 15(p+1)^4 m^4 + 30(p+1)^3(3p-1)m^3 \\ & - 5(p+1)^2(287p^2 + 158p - 1)m^2 \\ & + 2(p+1)^2(1645p^2 - 50p + 1)m \\ & + 240p(6\lambda - 5p^3 - 5p^2 + 11p + 5). \end{aligned} \tag{33.15}$$

In particular,

$$\Pi_{p,4}^{(\lambda)}(5) = \frac{\lambda p + 5p^3 + 15p^2 + 13p + 4}{4}. \tag{33.16}$$

Proof. Substitute the three explicit polynomials of theorem 33.2 into the source of (33.11). The resulting source is polynomial in m of degree four. Direct expansion verifies that (33.14) satisfies (33.11); evaluation at $m = 5$ gives (33.16). The first-order transport equation then gives uniqueness. The λ-dependent part of (33.15) is $1440\lambda p$, which is exactly the translation $\lambda p/4$ of theorem 33.18. □

33.6 The fourth-jet Stirling shadow

The new connected parameter is exact information, but it disappears modulo p.

Theorem 33.21 (Fourth-jet Stirling shadow)**.** *For every prime $p \geq 5$, every $\lambda \in \mathbb{Z}$, and every $n \geq 0$,*

$$\boxed{P_{p,n}^{[4,\lambda]}(u) \equiv \prod_{j=1}^{n}(1+ju) \pmod{(p,u^5)}.} \tag{33.17}$$

Consequently,

$$\boxed{C_{m-1,4}^{(p,\lambda)} \equiv e_4(1,\dots,m-1) \equiv \begin{bmatrix} m \\ m-4 \end{bmatrix} \pmod{p},} \tag{33.18}$$

where the bracket is the unsigned Stirling number of the first kind.

Proof. Modulo p, one has $\ell_n \equiv n$. Every pair, triple, and quadruple interaction source in (33.8) contains an explicit factor of p, so the recurrence reduces to

$$P_{p,n}^{[4,\lambda]}(u) \equiv (1+nu)P_{p,n-1}^{[4,\lambda]}(u) \pmod{(p,u^5)}.$$

Iterating gives (33.17). The coefficient of u^4 in the product is the fourth elementary symmetric function of $1,\dots,m-1$, which is the near-diagonal unsigned Stirling number shown in (33.18); see, for example, [67]. □

Over $\mathbb{Q}[m]$ the normalized fourth shadow is

$$\boxed{S_4(m) = \frac{m(15m^3-30m^2+5m+2)}{5760},} \tag{33.19}$$

because

$$e_4(1,\dots,m-1) = (m-1)_{\underline{4}}S_4(m).$$

Corollary 33.22 (Normalized shadow away from $p=5$)**.** *For every prime $p \geq 7$,*

$$\boxed{\Pi_{p,4}^{(\lambda)}(m) \equiv S_4(m) \pmod{p}}$$

as a polynomial congruence. The congruence is independent of λ.

Proof. Subtract (33.19) from the exact quartic (33.14). The resulting rational polynomial has an explicit factor of p. For $p \geq 7$, the denominator

$$5760 = 2^7 \cdot 3^2 \cdot 5$$

is invertible modulo p, so reduction is legitimate. □

Remark 33.23 ($p=5$ denominator resonance). The raw coefficient congruence (33.18) remains valid at $p=5$ because it is obtained before any division. However, 5760 is divisible by 5, so one must not obtain a normalized polynomial congruence by cancelling $(m-1)_{\underline{4}}$ or reducing (33.19) coefficientwise modulo 5. This is a normalization-denominator issue, not a failure of the fourth-jet Stirling shadow.

Corollary 33.24 (Shadow blindness versus exact identifiability)**.** *All stationary p-divisible fourth polarizations have the same mod-p fourth jet, but distinct parameters have distinct exact fourth columns. Thus the Stirling shadow forgets λ, while theorem 33.19 recovers it from one exact row.*

Chapter 34

CCR-k General Theory: Higher-Order Chronological Carry, Hessenberg Hierarchies, and Universal Stirling Shadows

Research branch — The all-order chronological-carry closure. *The preceding arithmetic branch established the third-order CCR theory and then isolated the genuinely new fourth-order coordinate. This chapter keeps those low-order chapters intact and passes to the all-order terminal-block hierarchy. The geometric input remains chronological path locality; the numerical connected sources are explicitly typed arithmetic data.*

Chapters 29–33 identify the first two nontrivial stages of chronological carry. The point of the present chapter is not to replace those stages, but to explain the general mechanism they reveal. At every order there is one connected interval support type, one new typed source, one corresponding Hessenberg band, and one new source term in normalized fixed-column transport. The exact source remains visible integrally and reconstructible from the normalized column, while the entire p-divisible hierarchy collapses to a common Stirling shadow modulo p.

Throughout this chapter fix a prime $p \geq 5$ unless stated otherwise and write

$$\ell_n = (p+1)n - p.$$

The determinant and transport identities are polynomial identities and make sense more generally; the prime hypothesis is needed for the mod-p arithmetic statements.

CCR-3 $\longrightarrow$ CCR-4 $\longrightarrow$ CCR-k:
low-order discovery $\longrightarrow$ first independent higher coordinate $\longrightarrow$ all-order source–band–transport–shadow theory.

34.1 Chronological support geometry and scope

The correction chronology is represented by the path graph on consecutive correction positions. The following elementary fact is the geometric reason that exactly one connected support *type* appears at each order.

Lemma 34.1 (Connected interval support)**.** *Every connected subset of a path having cardinality r is a consecutive interval*

$$\{j, j+1, \ldots, j+r-1\}.$$

Hence there is exactly one connected support type of size r.

Proof. If $a < c$ lie in a connected subset S, then the unique path from a to c contains every integer between them. Connectivity forces each of those intermediate vertices into S. Thus S is an interval, and its cardinality determines its length. □

Remark 34.2 (Geometry does not determine arithmetic). The lemma determines the support of a new connected interaction, not its numerical weight. The fourth-order chapter already exhibits this distinction: the complete third jet and chronological locality determine the consecutive-quadruple support but leave a genuinely new arithmetic coordinate. Numerical connected weights are therefore treated below as explicitly typed arithmetic data.

The structural dictionary is

$$\boxed{\begin{aligned}&\text{connected } r\text{-interval} \Longleftrightarrow \text{source } a_{r,n} \Longleftrightarrow \text{subdiagonal } r-1\\ &\qquad\qquad \Longleftrightarrow \text{column-}r \text{ transport source}\end{aligned}}$$

The first arrow is a modeling interface: path geometry fixes the support type, while arithmetic polarization fixes its value.

34.2 General terminal-block CCR-k

Fix $k \geq 3$ and work in the truncated ring

$$R_k = \mathbb{Z}[u]/(u^{k+1}).$$

For $r \geq 3$, let $a_{r,n} \in \mathbb{Z}$ be the normalized connected source at order r, after removal of the chronological placement factor

$$n - 1_{\underline{r-1}} = (n-1)(n-2)\cdots(n-r+1).$$

The third-order baseline fixes

$$a_{3,n} = -p. \tag{34.1}$$

Definition 34.3 (Typed terminal-block CCR-k)**.** A typed terminal-block CCR-k family is a sequence $P^{[k]}_{p,n}(u) \in R_k$ satisfying

$$\boxed{\begin{aligned}P^{[k]}_{p,n}(u) &= (1+\ell_n u)P^{[k]}_{p,n-1}(u)\\ &\quad + p(n-1)\ell_{n-1}u^2 P^{[k]}_{p,n-2}(u)\\ &\quad + \sum_{r=3}^{k} n - 1_{\underline{r-1}}\, a_{r,n} u^r P^{[k]}_{p,n-r}(u) \pmod{u^{k+1}},\end{aligned}} \tag{34.2}$$

with $P_{p,0} = 1$, $P_{p,n} = 0$ for $n < 0$, and (34.1).

The definition is intentionally explicit: it is the class of chronological recurrences that factor by the size of the terminal connected interval block. It does not assert that every conceivable path-local model must have this factorization.

Write

$$C_{n,j} = [u^j]P^{[k]}_{p,n}(u).$$

Theorem 34.4 (All-order terminal-block separation)**.** *For every $r \leq k$,*

$$\boxed{\begin{aligned}C_{n,r} &= C_{n-1,r} + \ell_n C_{n-1,r-1} + p(n-1)\ell_{n-1}C_{n-2,r-2}\\ &\quad + \sum_{s=3}^{r-1} n - 1_{\underline{s-1}}\, a_{s,n} C_{n-s,r-s} + n - 1_{\underline{r-1}}\, a_{r,n}.\end{aligned}} \tag{34.3}$$

After all sources of orders below r have been fixed, the only new recurrence datum entering degree r is $a_{r,n}$.

Proof. Extract u^r from (34.2). A terminal source of size s contributes

$$n - 1_{\underline{s-1}} a_{s,n} [u^{r-s}] P^{[k]}_{p,n-s}(u).$$

When $s < r$, this depends only on a coefficient of degree $r-s < r$, hence only on lower-order data. For the $s = r$ term there are two boundary regimes. If $1 \le n < r$, then the placement factor $n - 1_{\underline{r-1}}$ contains a zero factor, so the entire term vanishes. If $n \ge r$, then $P_{p,n-r}$ has constant term one (inductively from $P_{p,0} = 1$ and the recurrence), so $[u^0]P_{p,n-r} = 1$ and the final term in (34.3) follows. Sources of order greater than r are divisible by a power of u too large to contribute. □

Corollary 34.5 (All-order non-identifiability)**.** *Two typed terminal-block families that agree through order $r-1$ but differ in $a_{r,n}$ have identical $(r-1)$-jets. Thus lower-order data do not identify the next connected arithmetic source.*

Proof. The first term involving $a_{r,n}$ has degree r, so all lower coefficients are independent of it. □

34.3 Higher-band lower-Hessenberg realization

For $i \ge 1$ and $m \ge 0$, write

$$i^{\overline{m}} = i(i+1)\cdots(i+m-1), \qquad i^{\overline{0}} = 1.$$

Definition 34.6 (General Hessenberg matrix)**.** Let $H^{[k]}$ be the infinite lower-Hessenberg matrix with

$$H_{i,i} = \ell_i, \qquad H_{i,i+1} = 1, \tag{34.4}$$

$$H_{i+1,i} = -pi\ell_i, \tag{34.5}$$

and, for $3 \le r \le k$,

$$\boxed{H_{i+r-1,i} = (-1)^{r-1} i^{\overline{r-1}}\, a_{r,i+r-1}.} \tag{34.6}$$

All entries below the $(k-1)$-st subdiagonal are zero. Let $H^{[k]}_n$ denote the leading $n \times n$ principal truncation.

Theorem 34.7 (Exact Hessenberg realization)**.** *The determinant polynomials*

$$D^{[k]}_n(u) = \det(I_n + uH^{[k]}_n)$$

satisfy (34.2) *exactly. Therefore*

$$\boxed{D^{[k]}_n(u) = P^{[k]}_{p,n}(u) \quad \text{in } R_k.}$$

Proof. Expand $\det(I_n + uH^{[k]}_n)$ along the last row. The diagonal term contributes

$$(1 + \ell_n u) D^{[k]}_{n-1}.$$

The first subdiagonal entry is $-p(n-1)\ell_{n-1}$. Its cofactor sign is negative, and the remaining forced superdiagonal closure contributes one factor of u, so this term is

$$+p(n-1)\ell_{n-1} u^2 D^{[k]}_{n-2}.$$

Now choose the entry in column $n-r+1$, where $r \ge 3$. By (34.6),

$$H_{n,n-r+1} = (-1)^{r-1} n - r + 1^{\overline{r-1}} a_{r,n}.$$

The cofactor sign is $(-1)^{r-1}$. Lower-Hessenberg structure forces the last $r-1$ columns of the cofactor to close through the consecutive unit superdiagonal entries, contributing u^{r-1}; the selected last-row entry contributes the remaining factor u. The two signs cancel, and

$$n - r + 1^{\overline{r-1}} = n - 1_{\underline{r-1}}.$$

The untouched leading block is $I_{n-r} + uH_{n-r}^{[k]}$. Hence the contribution is

$$n - 1_{\underline{r-1}} a_{r,n} u^r D_{n-r}^{[k]}.$$

Summing the available bands gives (34.2). The initial conditions agree, so the two families coincide. □

The theorem turns the chronological order filtration into an exact band filtration. The next result shows that this filtration is also visible coefficientwise.

34.4 The subdiagonal light cone

Theorem 34.8 (Subdiagonal light cone)**.** *Let $M = (m_{ij})$ be any lower-Hessenberg matrix, so $m_{ij} = 0$ for $j > i + 1$, and let*

$$Q_n(u) = \det(I_n + uM_n).$$

Then $[u^j]Q_n(u)$ depends only on the diagonal, superdiagonal, and first $j-1$ subdiagonals. Equivalently, an entry on the s-th subdiagonal can first affect degree $s+1$.

Proof. The coefficient $[u^j]Q_n(u)$ is the sum of the principal $j \times j$ minors of M_n. Suppose a permutation term in such a minor uses an entry on the s-th subdiagonal. Relative to the diagonal, that move has displacement $-s$. Because M is lower-Hessenberg, every positive displacement in the same permutation is at most $+1$. Since the total permutation displacement is zero, compensating $-s$ requires at least s positive unit displacements. The permutation therefore moves at least $s+1$ indices and cannot occur in a minor smaller than $(s+1) \times (s+1)$. Thus the s-th subdiagonal is invisible below degree $s+1$. □

Corollary 34.9 (Exact order filtration)**.** *In the typed CCR hierarchy, the order-r source affects no fixed column below r, while every band deeper than $r-1$ is invisible to the r-th fixed column.*

This provides a matrix-theoretic proof that higher interactions cannot retroactively change lower jets.

34.5 Normalized fixed-column transport and inverse reconstruction

For $j \geq 0$, set

$$C_{n,j} = [u^j]P_{p,n}^{[k]}(u).$$

For $m \geq j+1$, define the falling-factorial normalized column

$$\boxed{\Pi_{p,j}(m) = \frac{C_{m-1,j}}{m - 1_{\underline{j}}}.} \tag{34.7}$$

Also set $\Pi_{p,0}(m) = 1$.

The first admissible row $m = j$ must be treated as a boundary equation; this avoids the meaningless expression $0 \cdot \Pi_{p,j}(j)$.

Theorem 34.10 (Boundary and transport equations)**.** *For $1 \le j \le k$, the first normalized row is determined by*

$$\boxed{\begin{aligned} j\Pi_{p,j}(j+1) = \ell_j \Pi_{p,j-1}(j) + p\ell_{j-1}\Pi_{p,j-2}(j-1) \\ + \sum_{r=3}^{j} a_{r,j}\Pi_{p,j-r}(j-r+1), \end{aligned}} \tag{34.8}$$

where terms with negative column index are omitted. For every $m \ge j+1$,

$$\boxed{\begin{aligned} m\Pi_{p,j}(m+1) = (m-j)\Pi_{p,j}(m) + \ell_m \Pi_{p,j-1}(m) \\ + p\ell_{m-1}\Pi_{p,j-2}(m-1) \\ + \sum_{r=3}^{j} a_{r,m}\Pi_{p,j-r}(m-r+1). \end{aligned}} \tag{34.9}$$

Proof. Extract u^j from (34.2) with $n = m$:

$$\begin{aligned} C_{m,j} = C_{m-1,j} + \ell_m C_{m-1,j-1} + p(m-1)\ell_{m-1}C_{m-2,j-2} \\ + \sum_{r=3}^{j} m - 1_{\underline{r-1}} a_{r,m} C_{m-r,j-r}. \end{aligned}$$

Use

$$C_{m,j} = m_{\underline{j}}\Pi_{p,j}(m+1) = mm - 1_{\underline{j-1}}\Pi_{p,j}(m+1),$$
$$C_{m-1,j} = m - 1_{\underline{j}}\Pi_{p,j}(m) = (m-j)m - 1_{\underline{j-1}}\Pi_{p,j}(m),$$

and

$$m - 1_{\underline{r-1}} m - r_{\underline{j-r}} = m - 1_{\underline{j-1}}.$$

For $m \ge j+1$, division by $m - 1_{\underline{j-1}}$ gives (34.9). At $m = j$, the term $C_{j-1,j}$ is zero because its degree exceeds its row size, so the same coefficient extraction yields (34.8) without ever defining $\Pi_{p,j}(j)$. □

The transport equation is triangular in the interaction order and therefore invertible.

Theorem 34.11 (Exact source reconstruction)**.** *Fix $r \ge 3$ and suppose all sources of orders below r are known. The first source value is recovered from the boundary row by*

$$\boxed{\begin{aligned} a_{r,r} = r\Pi_{p,r}(r+1) - \ell_r \Pi_{p,r-1}(r) - p\ell_{r-1}\Pi_{p,r-2}(r-1) \\ - \sum_{s=3}^{r-1} a_{s,r}\Pi_{p,r-s}(r-s+1). \end{aligned}} \tag{34.10}$$

For every $m \ge r+1$,

$$\boxed{\begin{aligned} a_{r,m} = m\Pi_{p,r}(m+1) - (m-r)\Pi_{p,r}(m) - \ell_m \Pi_{p,r-1}(m) \\ - p\ell_{m-1}\Pi_{p,r-2}(m-1) \\ - \sum_{s=3}^{r-1} a_{s,m}\Pi_{p,r-s}(m-s+1). \end{aligned}} \tag{34.11}$$

Thus the complete normalized source sequence is identifiable from the r-th column together with the already known lower-order data.

Proof. Set $j = r$ in theorem 34.10. The term with source order r is multiplied by $\Pi_{p,0} = 1$. Solving the boundary and transport equations for that source gives (34.10) and (34.11). □

34.6 Stationary p-divisible extensions

The preceding theory allows arbitrary row-dependent source sequences. A particularly rigid and useful class is obtained by removing that row dependence after the chronological placement factor.

Definition 34.12 (Stationary p-divisible class)**.** For $r \geq 4$, a source is stationary p-divisible if

$$a_{r,n} = (-1)^r p \lambda_r$$

for an integer λ_r independent of n. We set $\lambda_3 = 1$, so that $a_{3,n} = -p$ agrees with CCR-3.

Theorem 34.13 (Stationary extension classification)**.** *A stationary p-divisible extension through order k is uniquely determined by the parameter vector*

$$\boxed{(\lambda_4, \lambda_5, \ldots, \lambda_k) \in \mathbb{Z}^{k-3}.}$$

Its recurrence is

$$\boxed{\begin{aligned} P_{p,n}^{[k,\boldsymbol{\lambda}]}(u) &= (1 + \ell_n u) P_{p,n-1} + p(n-1)\ell_{n-1} u^2 P_{p,n-2} \\ &\quad + p \sum_{r=3}^{k} (-1)^r \lambda_r \underline{n - 1_{r-1}} u^r P_{p,n-r} \pmod{u^{k+1}}, \end{aligned}} \tag{34.12}$$

with $\lambda_3 = 1$.

Proof. By Lemma 34.1, there is one connected support type at each order. By theorem 34.4, a typed terminal-block model introduces one new normalized source sequence at that order. Stationarity makes the normalized source independent of n, and p-divisibility expresses it uniquely as p times an integer. The factor $(-1)^r$ is a sign convention chosen so that positive λ_r produces a common negative sign on the corresponding Hessenberg subdiagonal; see equation (34.14) below. □

Definition 34.14 (Primitive same-sign hierarchy)**.** The primitive same-sign normalization is the stationary point

$$\boxed{\lambda_r = 1 \qquad (3 \leq r \leq k).}$$

Thus

$$\boxed{\begin{aligned} P_{p,n}^{[k],\mathrm{prim}}(u) &= (1 + \ell_n u) P_{p,n-1} + p(n-1)\ell_{n-1} u^2 P_{p,n-2} \\ &\quad + p \sum_{r=3}^{k} (-1)^r \underline{n - 1_{r-1}} u^r P_{p,n-r} \pmod{u^{k+1}}. \end{aligned}} \tag{34.13}$$

The word *primitive* here names a selected arithmetic normalization. It is not a claim that the underlying pushdown transition table forces $\lambda_r = 1$.

34.7 Operator realization

Under Definition 34.12, the deeper Hessenberg bands simplify to

$$\boxed{H_{i+r-1,i} = -\lambda_r p \, \overline{i^{r-1}}, \qquad 3 \leq r \leq k.} \tag{34.14}$$

All higher connected bands therefore have the same matrix sign when $\lambda_r > 0$.

In the divided-power basis $e_i = x^{i-1}/(i-1)!$, one has

$$X^s e_i = \overline{i^s} \, e_{i+s}.$$

Theorem 34.15 (General differential-operator realization)**.** *The stationary* CCR-k *Hessenberg matrix is the matrix of*

$$\boxed{\mathcal{H}^{[k]}_{p,\boldsymbol{\lambda}} = D + L_p - pXL_p - p\sum_{r=3}^{k} \lambda_r X^{r-1}.} \tag{34.15}$$

For the primitive same-sign hierarchy,

$$\boxed{\mathcal{H}^{[k]}_{p,\mathrm{prim}} = D + L_p - pXL_p - p(X^2 + X^3 + \cdots + X^{k-1}).} \tag{34.16}$$

Proof. The terms D, L_p, and $-pXL_p$ reproduce the unit superdiagonal, diagonal ℓ_i, and first lower band $-pi\ell_i$. For $r \geq 3$,

$$-p\lambda_r X^{r-1} e_i = -p\lambda_r i^{\overline{r-1}} e_{i+r-1},$$

which is exactly (34.14). □

34.8 Polynomial fixed columns

The normalized transport system has a useful consequence not visible from the determinant alone: stationary fixed columns are polynomial at every order.

For $j \geq 1$, define the linear operator on $\mathbb{Q}[m]$

$$(\mathcal{T}_j f)(m) = mf(m+1) - (m-j)f(m). \tag{34.17}$$

If $f(m) = c_d m^d + \cdots$, then

$$\mathcal{T}_j f = (d+j)c_d m^d + \text{lower-degree terms}. \tag{34.18}$$

Thus $\mathcal{T}_j$ preserves degree and is an automorphism of $\mathbb{Q}[m]$.

Theorem 34.16 (Arbitrary-order fixed-column polynomiality)**.** *For every stationary p-divisible* CCR-k *family and every $0 \leq j \leq k$, the normalized column $\Pi_{p,j}(m)$ is the restriction, for $m \geq j+1$, of a unique polynomial in*

$$\mathbb{Q}[p, \lambda_4, \ldots, \lambda_j][m].$$

For $j \geq 1$, its degree is exactly j, and its leading coefficient is

$$\boxed{[m^j]\Pi_{p,j}(m) = \frac{(p+1)^j}{2^j j!}.} \tag{34.19}$$

Proof. Proceed by induction on j. The case $j = 0$ is $\Pi_{p,0} = 1$. Suppose the assertion holds below j. Under stationarity, the right-hand side of (34.9) excluding the term $(m-j)\Pi_{p,j}(m)$ is a polynomial $G_j(m)$. Its only degree-j contribution comes from

$$\ell_m \Pi_{p,j-1}(m),$$

because the pair term has degree at most $j-1$, while the source term of order r multiplies a column of degree $j-r$. Hence G_j has degree j.

Equation (34.9) is precisely

$$\mathcal{T}_j \Pi_{p,j} = G_j.$$

By (34.18), $\mathcal{T}_j$ is triangular with nonzero diagonal $d+j$ on the monomial basis, so it is invertible on $\mathbb{Q}[m]$. There is therefore a unique polynomial solution of degree j. At $m = j$, the polynomial identity reduces to the boundary equation (34.8), so the polynomial solution

has the same first admissible value as the normalized column. The first-order recurrence then identifies the two for all $m \geq j+1$.

If c_{j-1} is the leading coefficient of $\Pi_{p,j-1}$, the leading coefficient of G_j is $(p+1)c_{j-1}$. Since $\mathcal{T}_j$ multiplies the leading coefficient of a degree-j polynomial by $2j$,

$$c_j = \frac{p+1}{2j} c_{j-1}.$$

Starting from $c_0 = 1$ gives (34.19). □

The leading term is independent of every higher connected parameter. Those parameters change lower-degree structure while leaving the universal top-degree growth fixed.

34.9 Exact parameter translation and identifiability

Suppose two stationary families agree at every order below r but have order-r parameters λ_r and μ_r. Let

$$\Delta_m = \Pi_{p,r}^{(\lambda_r)}(m) - \Pi_{p,r}^{(\mu_r)}(m).$$

Theorem 34.17 (Stationary translation law)**.** *The first admissible row satisfies*

$$\boxed{\Delta_{r+1} = \frac{(-1)^r p(\lambda_r - \mu_r)}{r}.} \tag{34.20}$$

For $m \geq r+1$,

$$\boxed{m\Delta_{m+1} = (m-r)\Delta_m + (-1)^r p(\lambda_r - \mu_r).} \tag{34.21}$$

The unique solution is the constant translation

$$\boxed{\Delta_m = \frac{(-1)^r p(\lambda_r - \mu_r)}{r} \qquad (m \geq r+1).} \tag{34.22}$$

Proof. Subtract the two order-r instances of theorem 34.10. Every lower-order term cancels, leaving only the difference

$$(-1)^r p(\lambda_r - \mu_r).$$

The boundary equation gives (34.20); the ordinary transport rows give (34.21). The constant in (34.22) satisfies both equations, and the first-order recurrence has a unique solution after its first value is fixed. □

Corollary 34.18 (One-row identifiability)**.** *Once lower parameters are known, a single exact normalized observation in column r identifies the stationary parameter λ_r.*

At $r = 4$, equation (34.22) becomes the recovered fourth-order translation

$$\Pi_{p,4}^{(\lambda)}(m) - \Pi_{p,4}^{(\mu)}(m) = \frac{p(\lambda - \mu)}{4}.$$

At $r = 5$, it predicts the alternating next case

$$\Pi_{p,5}^{(\lambda_5)}(m) - \Pi_{p,5}^{(\mu_5)}(m) = -\frac{p(\lambda_5 - \mu_5)}{5}.$$

34.10 Universal Stirling shadows

Assume now that every connected source is divisible by p; in particular this holds throughout the stationary class. Since

$$\ell_n = n + p(n-1) \equiv n \pmod{p},$$

and every interaction term beyond the singleton term has an explicit factor of p, the full recurrence collapses modulo p.

Theorem 34.19 (Arbitrary-order raw Stirling shadow)**.** *For every $k \geq 3$,*

$$\boxed{P_{p,n}^{[k]}(u) \equiv \prod_{j=1}^{n}(1+ju) \pmod{(p, u^{k+1})}.} \tag{34.23}$$

Consequently, for $0 \leq r \leq k$,

$$\boxed{[u^r]P_{p,n}^{[k]}(u) \equiv e_r(1,2,\ldots,n) = \begin{bmatrix} n+1 \\ n+1-r \end{bmatrix} \pmod{p},} \tag{34.24}$$

where the bracket denotes an unsigned Stirling number of the first kind.

Proof. Modulo p, recurrence (34.2) becomes

$$P_{p,n}^{[k]}(u) \equiv (1+nu)P_{p,n-1}^{[k]}(u) \pmod{(p, u^{k+1})}.$$

Iterating from $P_{p,0} = 1$ gives (34.23). Coefficient extraction gives the elementary symmetric function in $1, \ldots, n$, which is the displayed near-diagonal unsigned Stirling number. □

The raw form (34.24) is the safest universal theorem because it requires no modular division. The normalized shadow admits a precise rational-polynomial formulation.

Proposition 34.20 (Normalized shadow and denominator resonances)**.** *For fixed r, define*

$$S_r(m) = \frac{e_r(1,2,\ldots,m-1)}{m-1_{\underline{r}}}. \tag{34.25}$$

Then $S_r(m) \in \mathbb{Q}[m]$ has degree r. In the stationary p-divisible hierarchy,

$$\Pi_{p,r}(m) - S_r(m) \in p\,\mathbb{Q}[p, \lambda_4, \ldots, \lambda_r, m]. \tag{34.26}$$

Hence there exists an integer $D_r > 0$, depending only on r, such that for every prime $p \nmid D_r$ the normalized polynomial congruence

$$\Pi_{p,r}(m) \equiv S_r(m) \pmod{p} \tag{34.27}$$

is legitimate coefficientwise after denominators are cleared. For primes dividing the chosen denominator, the raw congruence (34.24) *remains valid and no cancellation should be performed.*

Proof. For fixed r, the quantity $e_r(1, \ldots, m-1)$ is a polynomial in m of degree $2r$. One direct justification uses Newton's identities

$$re_r = \sum_{j=1}^{r}(-1)^{j-1}e_{r-j}\sum_{t=1}^{m-1}t^j,$$

together with Faulhaber's polynomial formulas for the power sums; induction on r gives polynomiality, and the leading term is $m^{2r}/(2^r r!)$. This polynomial vanishes at $m =$

$1, 2, \ldots, r$, because fewer than r nonzero slots are present. Hence it is divisible in $\mathbb{Q}[m]$ by $m - 1_{\underline{r}}$, proving that S_r is a polynomial of degree r.

By theorem 34.16, $\Pi_{p,r}$ is polynomial over $\mathbb{Q}[p, \boldsymbol{\lambda}]$. Specializing the recurrence at the formal value $p = 0$ removes every interaction term and sends ℓ_n to n, so the family becomes exactly the free product $\prod_{j=1}^{n}(1 + ju)$. Therefore $\Pi_{0,r} = S_r$. Polynomial division in the indeterminate p gives (34.26). Clearing rational denominators yields some integer D_r, and coefficientwise reduction modulo primes not dividing D_r gives (34.27). If $p \mid D_r$, only the pre-division raw theorem is automatic. □

Remark 34.21 (Exact information versus shadow information). The same parameter that is recovered exactly from one normalized row by Corollary 34.18 disappears from the entire raw mod-p jet by theorem 34.19. Thus the hierarchy exhibits a systematic exact-versus-shadow information loss at every order.

34.11 The all-order closure theorem

We can now package the preceding results into one statement.

Theorem 34.22 (General higher-order chronological-carry theorem)**.** *Fix $k \geq 3$. Within the typed terminal-block class of Definition 34.3:*

(i) *at every order $r \leq k$, the chronological path has exactly one connected support type, namely the consecutive r-block;*

(ii) *after all lower sources are fixed, the sole new degree-r recurrence datum is the source sequence $a_{r,n}$;*

(iii) *the recurrence has the exact lower-Hessenberg realization* (34.4)–(34.6)*;*

(iv) *the r-th fixed column is insensitive to every subdiagonal deeper than $r - 1$;*

(v) *normalized columns satisfy the boundary/transport system* (34.8)–(34.9)*;*

(vi) *every connected source sequence is reconstructed exactly by* (34.10)–(34.11)*;*

(vii) *in the stationary p-divisible subclass, the extension space through order k is $\mathbb{Z}^{k-3}$, with operator* (34.15)*;*

(viii) *stationary normalized columns are polynomials of exact degree equal to their column index;*

(ix) *every p-divisible member has the universal raw Stirling shadow* (34.23)*.*

No additional connected support type or deeper band datum is required to determine the k-jet.

Proof. Part (i) is Lemma 34.1. Parts (ii) and the absence of an additional order-r source are theorem 34.4. Part (iii) is theorem 34.7. Part (iv) is theorem 34.8. Parts (v) and (vi) are theorems 34.10 and 34.11. Part (vii) is theorems 34.13 and 34.15. Part (viii) is theorem 34.16. Part (ix) is theorem 34.19. Finally, a new connected support type cannot appear because connected subsets of the chronological path are already classified by Lemma 34.1, and a deeper subdiagonal cannot alter the k-jet by theorem 34.8. This closes the order induction. □

34.12 The first genuinely new checkpoint beyond CCR-4: order five

The all-order theorem contains the preceding CCR-3 and CCR-4 chapters as its first two nontrivial levels. The first new checkpoint not already developed there is order five, where a second independent stationary parameter appears. Its recurrence is

$$\begin{aligned} P^{[5]}_{p,n}(u) = (1+\ell_n u)P_{p,n-1} + p(n-1)\ell_{n-1}u^2P_{p,n-2} - pn - 1_{\underline{2}}u^3P_{p,n-3} \\ + \lambda_4 pn - 1_{\underline{3}}u^4P_{p,n-4} - \lambda_5 pn - 1_{\underline{4}}u^5P_{p,n-5} \pmod{u^6}. \end{aligned} \tag{34.28}$$

The new fifth-order source occupies the fourth subdiagonal

$$H_{i+4,i} = -\lambda_5 p\, i(i+1)(i+2)(i+3)$$

and contributes the operator term $-\lambda_5 pX^4$. Changing only λ_5 translates the fifth normalized column by

$$-\frac{p(\lambda_5-\mu_5)}{5}.$$

The sign alternation in the recurrence is therefore converted into a common negative sign in the stationary Hessenberg bands.

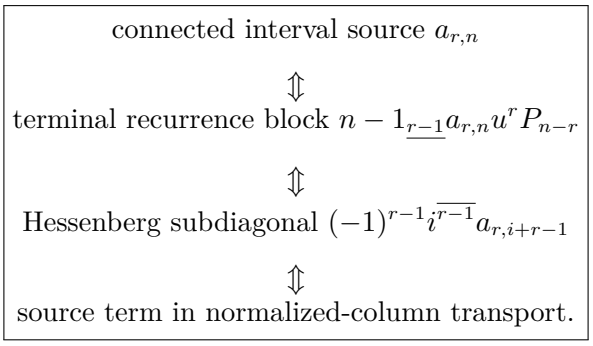
Table 34.1: The first orders of the stationary hierarchy. The numerical source is additional arithmetic data; only its support type is fixed by the path geometry.

order r	recurrence source a_r	Hessenberg band	normalized translation per unit λ_r
3	$-p$	$-p\, i^{\overline{2}}$	fixed baseline
4	$+p\lambda_4$	$-p\lambda_4\, i^{\overline{3}}$	$+p/4$
5	$-p\lambda_5$	$-p\lambda_5\, i^{\overline{4}}$	$-p/5$
6	$+p\lambda_6$	$-p\lambda_6\, i^{\overline{5}}$	$+p/6$

34.13 Synthesis

The CCR-k General Theory has a simple structural message. Chronological path geometry is rigid enough to determine where a new connected interaction can live, but not strong enough to determine its numerical weight. Once that weight is admitted as typed arithmetic data, the entire hierarchy becomes exact and recursively observable.

At every order r, four descriptions of the same new datum coexist:



In the stationary p-divisible class, this reduces to one integer λ_r per order. The exact normalized column sees λ_r immediately through the constant translation $(-1)^r p\lambda_r/r$, while reduction modulo p sees none of it. The free Stirling shadow is therefore universal precisely because it is an information-losing projection.

The resulting all-order spine is

$$\boxed{\begin{array}{c}\text{chronological locality + typed connected arithmetic data}\\ \Longrightarrow \text{ higher-band Hessenberg transport}\\ \Longrightarrow \text{ exact identifiability + universal Stirling shadow.}\end{array}}$$

34.14 Proof-dependency ledger

For audit purposes, the logical dependencies of the main results are summarized in table 34.2.

Table 34.2: Dependency ledger for the all-order theorem package.

Result	Direct inputs	Role
Connected interval support	path graph	one support type per order
Terminal-block separation	Definition 34.3	isolates the new order-r source
Hessenberg realization	source normalization + last-row determinant expansion	recurrence/band equivalence
Light cone	lower-Hessenberg displacement bound	excludes deeper-band contamination
Normalized transport	recurrence coefficient extraction + falling-factorial cancellation	row evolution of fixed columns
Source reconstruction	normalized transport	inverse identifiability
Stationary classification	one source type/order + stationarity + p-divisibility	parameter lattice $\mathbb{Z}^{k-3}$
Operator realization	divided-power basis identities	compact all-band representation
Polynomiality	stationary transport + invertibility of $\mathcal{T}_j$	exact degree and leading coefficient
Translation law	subtraction of two stationary transports	one-row parameter identifiability
Stirling shadow	p-divisibility + $\ell_n \equiv n \pmod p$	universal information-losing projection
Main theorem	all preceding rows	all-order closure

34.15 Scope safeguards and nonclaims

The following statements are part of the mathematical scope of this chapter.

1. The original pushdown correction process supplies a motivating chronology and path-local support geometry. The numerical CCR sources are an added arithmetic polarization.

2. The typed terminal-block factorization (34.2) is an explicit model class. We do not assert that every possible path-local deformation must factor in this form.

3. The stationary p-divisible classification is a classification *inside that stated class.*

4. The primitive same-sign point $\lambda_r = 1$ is a normalization convention, not a theorem of machine-forced uniqueness.

5. The universal raw mod-p shadow is stronger and safer than any normalized congruence. Normalized polynomial reduction requires denominator control as in Proposition 34.20.

6. Classical results on cumulants, Möbius inversion, carry matrices, Stirling numbers, continued fractions, and lower-Hessenberg recurrences are used only as context or standard tools.

34.16 Arithmetic stratification of the normalized CCR columns

The terminal-block source hierarchy and the universal Stirling shadow are established in Chapter 34. Here the normalized fixed-column polynomial is studied over local and global arithmetic fields. The fourth *column* is the index $r = 4$ in the full CCR-k hierarchy, not a fourth-order replacement for that hierarchy. Algebraic sources and physical chronological realizations remain distinct.

34.16.1 Fourth-column normalization and the integral numerator

For a prime $p \geq 5$ and a stationary integer parameter λ, retain the integral quartic numerator $G_{p,\lambda}$ from (33.15). Its normalized fourth column satisfies

$$G_{p,\lambda}(m) = 5760\Pi_{p,4}^{(\lambda)}(m). \tag{34.29}$$

The roots of the normalized column in $\mathbb{Q}_p$ are those of $G_{p,\lambda}$. Crucially, at $p = 5$ one works with the *integral numerator* G, not a coefficientwise reduction of Π. Write

$$H(m) = 15m^3 - 30m^2 + 5m + 2, \qquad G_{p,\lambda}(m) \equiv mH(m) \pmod{p}. \tag{34.30}$$

The parameter λ disappears in this reduction. Direct computation gives

$$\operatorname{disc}(H) = 125700 = 10^2 \cdot 3 \cdot 419, \qquad \operatorname{disc}(mH) = 502800 = 2^4 \cdot 3 \cdot 5^2 \cdot 419. \tag{34.31}$$

34.17 The complete local factorization classification

Theorem 34.23 (All-prime root and factor stratification)**.** *For each integer λ and prime $p \geq 5$:*

1. *If $p \geq 7$ and $p \neq 419$, the roots of $G_{p,\lambda}$ in $\mathbb{Q}_p$ are precisely the unique lifts of the roots of $mH(m)$ in $\mathbb{F}_p$. Their number is 1, 2, or 4, independent of λ, and all residual irreducible factors lift unramified.*

2. *If $p = 5$, there is exactly one root in $\mathbb{Q}_5$. For $\lambda \neq 115$ its valuation is $2 + v_5(\lambda - 115)$, and at $\lambda = 115$ the root is 0. The remaining degree-three factor is irreducible and totally ramified over $\mathbb{Q}_5$.*

3. *If $p = 419$, the simple residues $m \equiv 0, 20 \pmod{419}$ always lift to two distinct roots. A further quadratic pair in $m \equiv -9 \pmod{419}$ is controlled by a unique 419-adic critical parameter Λ. Its exact splitting and ramification rule is stated in Theorem 34.25 below for all integer λ.*

Proof. For $p \geq 7$, $p \neq 419$, the coefficient of m^4 in G is a unit; the reduction mH is squarefree by (34.31) (and $H(0) = 2$). Ordinary Hensel lifting gives a bijection of linear residue factors with $\mathbb{Q}_p$-roots. Hensel factorization also lifts each separable irreducible factor of the reduction, yielding unramified extensions.

At $p = 5$, direct specialization is

$$G_{5,\lambda}(m) = 19440m^4 + 90720m^3 - 1433520m^2 + 2943072m + 7200(\lambda - 115).$$

The coefficients of m, m^2, m^3, m^4 have valuations $0, 1, 1, 1$, respectively. For $\lambda \neq 115$ the constant term has valuation $2+d$, where $d = v_5(\lambda-115)$. Its Newton polygon has segments $(0, 2 + d) \to (1, 0) \to (4, 1)$: the first slope gives one integral root of valuation $2 + d$; the second gives three conjugate roots of valuation $-1/3$. A degree-three factor whose root valuation has denominator 3 is totally ramified and irreducible. When $\lambda = 115$, factor out m and apply the same final segment to the cubic.

At $p = 419$, one checks the residue factorization

$$G_{419,\lambda}(m) \equiv 15m(m - 20)(m + 9)^2 \pmod{419}.$$

The two simple residues lift. The double residue is analyzed below with the stationary derivative and a p-adic quadratic normal form. □

34.18 The 419-adic critical coordinate

Let $f_\lambda(m) = G_{419,\lambda}(m)$. Its derivative is independent of λ and satisfies

$$f'_\lambda(-9) \equiv 0, \qquad \frac{1}{2}f''_\lambda(-9) \equiv 144 \pmod{419}.$$

Hensel's lemma yields a unique $c \in \mathbb{Z}_{419}$ such that $c \equiv -9 \pmod{419}$ and $f'_\lambda(c) = 0$. Since the parameter occurs as $1440 \cdot 419\lambda$, there is a unique $\Lambda \in \mathbb{Z}_{419}$ satisfying $f_\Lambda(c) = 0$.

Lemma 34.24 (Exact quadratic normal form)**.** *There is an analytic invertible change of variable near c, defined over $\mathbb{Z}_{419}$, under which the quadratic branch of $f_\lambda(m) = 0$ becomes*

$$z^2 = -10 \cdot 419\,(\lambda - \Lambda). \tag{34.32}$$

For $\lambda \neq \Lambda$, put $h = v_{419}(\lambda - \Lambda)$. The quadratic is totally ramified if h is even; if h is odd, it splits over $\mathbb{Q}_{419}$ exactly when the leading unit of $-10(\lambda - \Lambda)/419^h$ is a square modulo 419, and otherwise it is an unramified irreducible quadratic.

Proof. For $m = c + y$, stationary Taylor expansion yields

$$f_\lambda(c + y) = 1440 \cdot 419(\lambda - \Lambda) + y^2U(y), \quad U(0) \equiv 144 \pmod{419}.$$

For $y \in 419\mathbb{Z}_{419}$, the unit $U(y)/144$ is congruent to 1 modulo 419. Its unique square root congruent to 1 defines the analytic coordinate $z = y\sqrt{U(y)/144}$; moreover it is an isometry of the entire residue disk, not merely a local map with unit derivative. Indeed, for $s(y) = \sqrt{U(y)/144}$, the integral convergent power series on $419\mathbb{Z}_{419}$ satisfies $s(y) \equiv 1 \pmod{419}$ and $s(y) - s(y') \in (y - y')\mathbb{Z}_{419}$. Thus

$$z(y) - z(y') = (y - y')\left(s(y) + y'\frac{s(y) - s(y')}{y - y'}\right),$$

and the factor in parentheses is congruent to 1 modulo 419. Hensel lifting from $y = z = 0 \pmod{419}$ gives surjectivity and a unique inverse modulo every 419^n, hence a bi-analytic disk isometry. It gives (34.32) exactly on that disk, and therefore all integral-root and finite-congruence assertions. In the nonresonant case $v_{419}(\lambda - \Lambda) = 0$, however, the two algebraic zeros have $v_{419}(y) = 1/2$ and *lie outside* the integral residue disk: their irreducibility and total ramification follow separately from the translated polynomial's Newton segment joining $(0, 1)$ to $(2, 0)$, using $f'(c) = 0$ and $U(0)$ a unit. The same Newton-polygon argument handles all odd critical-value valuations; even critical-value valuations give the split/unramified cases by the local square-unit criterion. This separation of the integral disk from extension-valued roots is essential, and is consistent with the established p-adic Morse/inverse-function framework [175]. □

34.19 An exact integer discriminant certificate

The preceding description refers to the non-rational critical coordinate Λ. Our next result eliminates it from every integer-parameter decision.

$$\boxed{\begin{aligned} F(T) = {} & 127983969T^3 - 773766474332943627T^2 \\ & + 19464840642527447214601687T \\ & + 25531064848926331469188878661677 79. \end{aligned}} \tag{34.33}$$

Theorem 34.25 (Integer discriminant certificate and effective resonance bound)**.** *The polynomial $F(T) \in \mathbb{Z}[T]$ is irreducible over $\mathbb{Q}$, and its unique zero in $\mathbb{Z}_{419}$ is Λ. Consequently, $\Lambda \notin \mathbb{Q}$; in particular, $\lambda \neq \Lambda$ for every integer λ.*

For any $\lambda \in \mathbb{Z}$, calculate the nonzero integer $F(\lambda)$ *and set*

$$k = v_{419}(F(\lambda)), \qquad u = 419^{-k} F(\lambda) \pmod{419} \in \mathbb{F}_{419}^{\times}.$$

The full local factorization of $G_{419,\lambda}$ is given by the following finite certificate:

Criterion	*Additional quadratic factor*	$\mathbb{Q}_{419}$*-roots*
k *even*	*irreducible, totally ramified*	2
k *odd,* $\left(\frac{374u}{419}\right) = +1$	*splits into linear factors*	4
k *odd,* $\left(\frac{374u}{419}\right) = -1$	*irreducible, unramified*	2

In particular $G_{419,\lambda}$ is separable for every integer λ and the exceptional repeated-root branch of Lemma 34.24 is not attained by integer sources. *Moreover*

$$k \leq \log_{419} |F(\lambda)| \leq \log_{419} C + 3 \log_{419}(1 + |\lambda|), \quad C = \sum_{j=0}^{3} |[T^j]F|. \tag{34.34}$$

This is an explicit input-height bound on resonance depth and a terminating integral decision algorithm.

Proof. Independently calculate the discriminant of the quartic (34.29) with $p = 419$ as a polynomial in the parameter:

$$\begin{aligned} \operatorname{disc}_m G_{419,T} &= C_0 F(T), \\ C_0 &= 44676505705166237495036215296 \cdot 10^{18} \neq 0. \end{aligned} \tag{34.35}$$

Modulo 23, we have

$$F(T) \equiv 9(T^3 + 5T^2 + 2T - 6) \pmod{23}.$$

The displayed cubic has no root in $\mathbb{F}_{23}$; as its degree is three, it is irreducible. Thus F is irreducible in $\mathbb{Q}[T]$. Modulo 419,

$$F(T) \equiv 233(T - 132) \pmod{419},$$

and 233 is a unit. Hensel gives a unique zero $\Lambda_0 \in \mathbb{Z}_{419}$. But $G_{419,\Lambda}$ has a multiple root c by construction, so (34.35) implies $F(\Lambda) = 0$. Therefore $\Lambda_0 = \Lambda$.

The exact difference identity

$$F(T) - F(\Lambda) = (T - \Lambda) W(T, \Lambda), \qquad W(T, \Lambda) \equiv 233 \pmod{419}$$

for all $T \in \mathbb{Z}_{419}$ yields

$$v_{419}(F(\lambda)) = v_{419}(\lambda - \Lambda), \qquad \frac{\lambda - \Lambda}{419^k} \equiv 233^{-1} u \pmod{419}.$$

Substitute these equalities into Lemma 34.24. The quadratic has valuation $1 + k$ on the right of (34.32). For even k, it is totally ramified. For odd k, it has even valuation and splits precisely when

$$\left(\frac{-10 \cdot 233^{-1} u}{419}\right) = \left(\frac{374u}{419}\right) = +1;$$

otherwise it is irreducible unramified. Since F is irreducible cubic, it has no rational zero: its critical repeated-root equality is impossible for integer λ. Finally $419^k \mid F(\lambda) \neq 0$, and the triangle inequality gives $|F(\lambda)| \leq C(1 + |\lambda|)^3$, proving (34.34). □

Remark 34.26. For example, $\lambda = 132$ gives $k = 1$ and an unramified quadratic, whereas $\lambda = 551 = 132 + 419$ gives $k = 1$ and a split quadratic. The deeper class $\lambda = 38261$ gives $k = 2$ and a totally ramified quadratic. The theorem needs no precomputed digits of Λ.

34.20 Prime-axis density: S_3 Chebotarev law

Theorem 34.27 (Uniform prime-density law)**.** *Fix any integer λ. Among primes $p \geq 7$, excluding the single prime* 419*, the natural densities of those for which $G_{p,\lambda}$ has exactly* 1*,* 2*, or* 4 *roots in $\mathbb{Q}_p$ are, respectively,*

$$\boxed{\delta_1 = \frac{1}{3}, \qquad \delta_2 = \frac{1}{2}, \qquad \delta_4 = \frac{1}{6}.} \tag{34.36}$$

The classification at each individual generic prime, not merely the densities, is independent of λ. In particular the average number of $\mathbb{Q}_p$-roots across primes tends to 2*.*

Proof. The reduction of H modulo 7 is $m^3 - 2m^2 - 2m + 2$ and has no root in $\mathbb{F}_7$, so the cubic H is irreducible over $\mathbb{Q}$. Its nonsquare discriminant $125700 = 100 \cdot 3 \cdot 419$ shows that its splitting-field Galois group is S_3. Frobenius cycle types in S_3 produce 0, 1, or 3 distinct roots of H modulo almost every prime with natural densities 2/6, 3/6, or 1/6, respectively, by the Chebotarev theorem [178]. Equation (34.30) adds exactly the simple root $m = 0$ and Theorem 34.23 lifts the roots. The weighted mean is $(1/3) + 2(1/2) + 4(1/6) = 2$. □

34.21 Parameter-axis density and precision law

Endow $\mathbb{Z}_{419}$ with normalized additive Haar measure μ; we may extend the quartic formally to all 419-adic parameters. The following distributions also hold as natural densities of integer parameters $0 \leq \lambda < X$ when $X \to \infty$.

Theorem 34.28 (Exact parameter density)**.** *For the quadratic pair at the exceptional prime* 419*,*

$$\boxed{\begin{aligned} \mu(\textit{totally ramified}) &= \frac{419}{420}, \\ \mu(\textit{unramified irreducible}) &= \frac{1}{840}, \\ \mu(\textit{split into two linear factors}) &= \frac{1}{840}. \end{aligned}} \tag{34.37}$$

The critical equality $\lambda = \Lambda$ has measure zero and never occurs for integer λ. The mean number of distinct $\mathbb{Q}_{419}$-roots under Haar-random parameter is 841/420*.*

Proof. Since translation preserves Haar measure, $\lambda - \Lambda$ is Haar-uniform. For every $j \geq 0$, $\mu(v_{419}(\lambda - \Lambda) = j) = 418/419^{j+1}$. The normal form (34.32) is totally ramified for j even, yielding the geometric sum

$$\sum_{a \geq 0} \frac{418}{419^{2a+1}} = \frac{419}{420}.$$

For j odd the valuation is even and the leading unit is uniformly random; precisely half are squares. Hence each of the split and unramified cases has measure

$$\frac{1}{2}\sum_{a\geq 0}\frac{418}{419^{2a+2}}=\frac{1}{840}.$$

At finite precision all decided branches are unions of congruence classes, while the single undecided congruence class has density tending to zero. This establishes the same natural densities for integer λ. The mean is $2+2(1/840)=841/420$. □

Proposition 34.29 (Exact finite-precision residue counts)**.** *For each $N\geq 1$ there are exactly the following numbers of parameter classes modulo 419^N whose quadratic factorization is decided for* every *lift to $\mathbb{Z}_{419}$:*

$$S_N=U_N=209\sum_{j=0}^{\lfloor N/2\rfloor-1}419^{N-2j-2},$$
$$R_N=418\sum_{j=0}^{\lceil N/2\rceil-1}419^{N-2j-1}.$$

Here S_N,U_N,R_N denote split, unramified and totally ramified classes, respectively. Exactly one class, $\lambda\equiv\Lambda \pmod{419^N}$, remains undecided, and $S_N+U_N+R_N+1=419^N$. For $N=2$, the counts are $(209,209,175142,1)$.

Proof. Among classes with exact $j=v_{419}(\lambda-\Lambda)<N$ there are $418\cdot 419^{N-j-1}$ choices. If j is even all are ramified; if j is odd half split and half are unramified, by equidistribution of their leading units. Sum over the allowed parity classes. The only class with valuation $\geq N$ is $\lambda\equiv\Lambda \pmod{419^N}$. □

34.22 Prime-power congruence fibers and a resonance moment law

The number of distinct $\mathbb{Q}_{419}$-roots does not determine the number of roots modulo 419^n: singular residue classes can persist for several precision levels before disappearing. To capture this finer arithmetic, define

$$N_n(\lambda)=\#\{m \bmod 419^n : G_{419,\lambda}(m)\equiv 0 \pmod{419^n}\},\qquad n\geq 1. \tag{34.38}$$

Write $P=419$ for brevity, and retain $k=v_P(F(\lambda))$, $u=P^{-k}F(\lambda) \bmod P$ and $a=k+1$ from Theorem 34.25.

Theorem 34.30 (Exact root counts at every prime-power precision)**.** *For every integer λ and $n\geq 1$, let $S_n(\lambda)=N_n(\lambda)-2$. Then*

$$\boxed{S_n(\lambda)=\begin{cases}P^{\lfloor n/2\rfloor}, & 1\leq n\leq a,\\ 2P^{a/2}, & n>a,\ a\ \text{even},\ \left(\frac{374u}{P}\right)=1,\\ 0, & \text{otherwise.}\end{cases}} \tag{34.39}$$

In particular, $N_n(\lambda)$ is determined by finite integer arithmetic applied to $F(\lambda)$, for arbitrary n. Writing $N_0=1$, the unnormalized root-count series is rational and explicit:

$$\sum_{n\geq 0}N_n(\lambda)T^n=1+\frac{2T}{1-T}+\sum_{n=1}^{a}P^{\lfloor n/2\rfloor}T^n+\mathbf{1}_{\text{split}}\frac{2P^{a/2}T^{a+1}}{1-T}, \tag{34.40}$$

where $\mathbf{1}_{\text{split}}$ is one exactly in the second case of (34.39)*.*

Proof. The reduction of $G_{419,\lambda}$ has two simple roots at $0, 20$ and one double root at -9. The first two contribute exactly two roots modulo each P^n by Hensel's lemma. In the disk $m \equiv c \equiv -9 \pmod P$, the coordinate $z = (m-c)\sqrt{U(m-c)/144}$ from Lemma 34.24 is an integral analytic isometry: its derivative is a unit congruent to one modulo P, and it induces a bijection on residue classes modulo P^n. Thus the singular contribution S_n equals the number of z mod P^n with $z \equiv 0 \pmod P$ and $z^2 \equiv B \pmod{P^n}$, where $B = -10P(\lambda-\Lambda)$ and $v_P(B) = a$.

If $n \le a$, this is $z^2 \equiv 0 \pmod{P^n}$ and has $P^{\lfloor n/2 \rfloor}$ solutions. If $n > a$, a solution requires $a = 2h$ even and $z = P^h w$. The reduced unit equation $w^2 \equiv B/P^{2h} \pmod{P^{n-2h}}$ has two Hensel roots exactly when its unit is a square modulo P; each gives P^h choices of z modulo P^n. The square criterion is precisely $\left(\frac{374u}{P}\right) = 1$ by Theorem 34.25. This proves the formula, and summing its eventually constant terms proves (34.40). □

Theorem 34.31 (Exact parameter moments and rare-resonance limit)**.** *Let λ be Haar-uniform in $\mathbb{Z}_P$. For every $n \ge 1$,*

$$\boxed{\mathbb{E}_\lambda N_n = 3, \qquad \mathrm{Var}_\lambda(N_n) = (P-1)(n-1) = 418(n-1).} \tag{34.41}$$

More precisely, with $S_n = N_n - 2$,

$$\Pr(S_n = P^{\lfloor n/2 \rfloor}) = P^{-(n-1)},$$
$$\Pr(S_n = 2P^h) = \frac{P-1}{2P^{2h}}, \quad 1 \le h \le \left\lfloor \frac{n-1}{2} \right\rfloor,$$

and the remaining probability is at $S_n = 0$. Moreover $N_n(\lambda)$ stabilizes almost surely as $n \to \infty$ to a finite N_∞, with

$$\Pr(N_\infty = 2 + 2P^h) = \frac{P-1}{2P^{2h}} \quad (h \ge 1), \qquad \Pr(N_\infty = 2) = \frac{839}{840}. \tag{34.42}$$

The limit has $\mathbb{E}N_\infty = 3$ but $\mathbb{E}N_\infty^2 = +\infty$, whereas the finite-$n$ variances in (34.41) *grow exactly linearly.*

Proof. Translation by the critical parameter Λ preserves Haar measure, so $\Pr(v_P(\lambda-\Lambda) \ge n-1) = P^{-(n-1)}$. The first atom follows from (34.39). The second occurs when $v_P(\lambda - \Lambda) = 2h-1$ and the leading unit has the required quadratic character; its probability is $(P-1)/(2P^{2h})$. Since P is odd, the displayed positive values of S_n are distinct.

One may sum these atoms to obtain the first moment. Alternatively, for each m mod P^n reducing to one of the three zeros $0, 20, -9$ modulo P, the congruence $G_{P,\lambda}(m) \equiv 0 \pmod{P^n}$ has Haar probability $P^{-(n-1)}$: the coefficient of λ is $1440P$ with $1440 \in \mathbb{Z}_P^\times$. Exactly $3P^{n-1}$ choices of m are eligible, whence $\mathbb{E}N_n = 3$. The second moment of the singular contribution is

$$\mathbb{E}S_n^2 = P^{2\lfloor n/2 \rfloor - (n-1)} + 2(P-1)\left\lfloor \frac{n-1}{2} \right\rfloor = 1 + (P-1)(n-1).$$

Since $\mathbb{E}S_n = 1$, the variance identity follows. Except on the singleton $\lambda = \Lambda$, the singular contribution eventually vanishes or stabilizes at $2P^h$, proving (34.42). Finally $\sum_{h\ge1} 2P^h(P-1)/(2P^{2h}) = 1$, but $\sum_{h\ge1}(2P^h)^2(P-1)/(2P^{2h}) = +\infty$. □

Remark 34.32 (Distinguishing local roots from congruence fibers). The average number of *distinct* $\mathbb{Q}_{419}$-roots is $841/420$ by Theorem 34.28, not 3. An actual 419-adic root with derivative divisible by 419^h can account for 419^h residue solutions at every sufficiently high precision. The general theory and rationality of univariate local zeta/Poincaré functions are prior work; the present formulas are a CCR-specialized calculation, not a claim of inventing the generic method [176].

Corollary 34.33 (The exceptional prime 5 over parameters)**.** *For Haar-random $\lambda \in \mathbb{Z}_5$ the unique $\mathbb{Q}_5$-root α satisfies, for all $d \geq 0$,*

$$\Pr(v_5(\alpha) = 2 + d) = \frac{4}{5^{d+1}}, \quad \mathbb{E}\, v_5(\alpha) = \frac{9}{4}.$$

The other cubic is totally ramified for all parameters.

Proof. Use Theorem 34.23 and the geometric distribution of $v_5(\lambda - 115)$; the singleton parameter 115 has measure zero. □

34.23 Universal odd-column fold and p-adic source resonance

This section derives an extension to *all odd normalized CCR column orders* $r \geq 5$. Throughout, $\Pi_{p,r}(m)$ denotes the stationary normalized column supplied by the established higher-order CCR transport, not the d-orthogonal characteristic polynomial Q_r. The source parameters below r are fixed integers and the top source λ_r varies. The series identity used below belongs to classical Nörlund theory [172]; its role in the CCR deformation and the ensuing p-adic counts are the application here.

34.23.1 The all-order Stirling generating function

Let $S_r(m) = \Pi_{0,r}(m)$ be the formal $p = 0$ specialization and $S_0 = 1$. The original normalized polynomial transport yields

$$T_r S_r(m) = m S_{r-1}(m), \qquad (T_r f)(m) = m f(m+1) - (m-r) f(m). \tag{34.43}$$

Lemma 34.34 (Nörlund identity and universal double root)**.** *As an identity in $\mathbb{Q}[m][[t]]$,*

$$\sum_{r \geq 0} S_r(m) t^r = \left(\frac{t}{1 - e^{-t}}\right)^m = e^{mt/2} \left(\frac{t}{2\sinh(t/2)}\right)^m. \tag{34.44}$$

For every $s \geq 1$, the polynomial S_{2s+1} has an exact double *root at zero:*

$$S_{2s+1}(m) = c_s m^2 + O(m^3), \qquad c_s = -\frac{B_{2s}}{4s(2s)!} \neq 0. \tag{34.45}$$

Here B_{2s} is the classical Bernoulli number, and $O(m^3)$ is a polynomial divisible by m^3.

Proof. Let $A(t) = t/(1 - e^{-t})$. The generating form of (34.43) is $m(F(m+1,t) - F(m,t)) + t\partial_t F = mtF$. The proposed $F = A(t)^m$ satisfies this because $A - 1 + tA'/A = t$. Uniqueness follows since $T_r(m^d)$ is triangular with diagonal coefficient $r + d \neq 0$. The logarithm identity

$$\log A(t) = \frac{t}{2} - \sum_{j \geq 1} \frac{B_{2j}}{2j(2j)!} t^{2j}$$

shows that the coefficient of mt^{2s+1} vanishes and the coefficient of $m^2 t^{2s+1}$ is half the coefficient of t^{2s} in $\log A(t)$, giving (34.45). Classical Bernoulli numbers B_{2s} are nonzero. □

34.23.2 An all-odd-order arithmetic fold theorem

By stationary polynomiality, for each fixed r there exists an integer $D_r \geq 1$ which clears all rational coefficients of S_r and $(\Pi_{p,r} - S_r)/p$ in their independent formal variables. Define a *good prime at order* $r = 2s + 1$ as an odd prime p not dividing rD_r and for which c_s in (34.45) is a p-adic unit. Only finitely many primes are excluded for each fixed r.

Theorem 34.35 (Universal odd-column CCR p-adic fold)**.** *Fix any odd $r = 2s+1 \geq 5$, any good prime p, and integer lower source parameters $\lambda_4, \ldots, \lambda_{r-1}$. The dependence of the stationary normalized polynomial on its top source is* exactly

$$\Pi_{p,r}^{(\lambda_r)}(m) = \Pi_{p,r}^{(0)}(m) - \frac{p}{r}\lambda_r. \tag{34.46}$$

There is a unique critical center $c \in p\mathbb{Z}_p$, independent of λ_r, and a unique critical parameter $\Lambda_r \in \mathbb{Z}_p$ for which $\Pi_{p,r}^{(\Lambda_r)}(c) = 0$. An integral analytic isometric coordinate z on $m \equiv 0 \pmod p$ converts the local root equation exactly into

$$\boxed{z^2 = K_r p(\lambda_r - \Lambda_r), \qquad K_r = \frac{1}{rc_s} = -\frac{4s(2s)!}{rB_{2s}} \in \mathbb{Z}_p^\times.} \tag{34.47}$$

If $k = v_p(\lambda_r - \Lambda_r) < \infty$, the local degree-two factor is totally ramified and irreducible when k is even. When k is odd, it splits into two $\mathbb{Q}_p$-linear factors precisely when $K_r(\lambda_r - \Lambda_r)/p^k$ is a square unit modulo p, and otherwise is an unramified irreducible quadratic. At $\lambda_r = \Lambda_r$, the local factor has a repeated root. No nonattainment claim for integer parameters is made in this generality.

Proof. In the order-r transport, the source λ_r first appears as the constant $(-1)^r p\lambda_r = -p\lambda_r$. Since T_r maps constants to r times themselves, subtracting the two transport equations gives (34.46) independently of the lower parameters. The good-prime condition ensures $\Pi_{p,r} \equiv S_r \pmod p$ coefficientwise, and Lemma 34.34 gives a nondegenerate double root at zero. Thus Hensel's lemma applied to the derivative of $\Pi_{p,r}^{(0)}$ yields the unique $c \in p\mathbb{Z}_p$. Because $\Pi_{p,r}^{(0)}(c) \equiv 0 \pmod p$ and p/r is a p times a unit, there is exactly one $\Lambda_r \in \mathbb{Z}_p$ canceling its critical value. Taylor expansion at c is

$$\Pi_{p,r}^{(\lambda_r)}(c+y) = -\frac{p}{r}(\lambda_r - \Lambda_r) + y^2 U(y), \qquad U(0) \equiv c_s \pmod p.$$

For $y \in p\mathbb{Z}_p$, the unit $U(y)/c_s$ has a unique analytic square root congruent to one modulo p, and $z = y\sqrt{U(y)/c_s}$ induces an isometric bijection of the disk. This proves (34.47). Odd-valuation critical values have quadratic Newton slope with denominator two, giving an Eisenstein/total-ramification conclusion; at even valuation the local equation lies within the integral disk and the p-adic square-unit criterion distinguishes split from unramified quadratic cases. The nonsingular complementary factor is separated by Hensel factorization. □

34.23.3 Universal local prime-power fiber and moment laws

For $n \geq 1$, count *only* the zero residue disk, rather than assuming all other residues are simple:

$$M_{p,r,n}(\lambda_r) = \#\{m \bmod p^n : m \equiv 0 \pmod p, \quad \Pi_{p,r}^{(\lambda_r)}(m) \equiv 0 \pmod{p^n}\}.$$

Theorem 34.36 (All-odd-order rare-resonance statistics)**.** *Under Theorem 34.35, write $a = 1 + v_p(\lambda_r - \Lambda_r)$ for a noncritical parameter. If a is even let ε be the Legendre symbol of the unit $K_r(\lambda_r - \Lambda_r)/p^{a-1}$. Then*

$$M_{p,r,n}(\lambda_r) = \begin{cases} p^{\lfloor n/2 \rfloor}, & n \leq a, \\ 2p^{a/2}, & n > a,\ a \text{ even},\ \varepsilon = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{34.48}$$

At the critical singleton, $M_{p,r,n} = p^{\lfloor n/2 \rfloor}$ for every n. For Haar-uniform $\lambda_r \in \mathbb{Z}_p$, with all lower parameters fixed, the exact local moments are

$$\boxed{\mathbb{E}M_{p,r,n} = 1, \qquad \operatorname{Var}(M_{p,r,n}) = (p-1)(n-1).} \tag{34.49}$$

The almost-sure limiting count M_∞ has mass $(p-1)/(2p^{2h})$ at $2p^h$ for every $h \geq 1$, and remaining mass $1 - 1/[2(p+1)]$ at zero; it has mean one and infinite second moment.

Proof. The local coordinate of Theorem 34.35 bijects every finite residue disk, so the count equals the solutions of $z^2 \equiv K_r p(\lambda_r - \Lambda_r) \pmod{p^n}$ with $z \equiv 0 \pmod{p}$. The same valuation and Hensel count as in Theorem 34.30 gives (34.48). Haar translation makes $\lambda_r - \Lambda_r$ uniform. The positive atoms have probabilities $p^{-(n-1)}$ at $p^{\lfloor n/2 \rfloor}$ and $(p-1)/(2p^{2h})$ at $2p^h$, for $1 \le h \le \lfloor (n-1)/2 \rfloor$. Their first moment is one; their second moment is $1 + (p-1)(n-1)$. The limit follows by summing the stable split atoms; its second moment diverges by constant contributions over infinitely many values of h. □

34.23.4 The entire fifth column away from $p = 73$

For $r = 5$, the sharp denominator $D_5 = 11520$ is an earlier fixed-order CCR result and $c_2 = 1/5760$. Thus *every prime $p \ge 7$ is good for the local fifth-column fold*, with $K_5 = 1152$. The inherited Stirling polynomial is

$$S_5(m) = \frac{m^2(m-1)(3m^2 - 7m - 2)}{11520}. \tag{34.50}$$

Moreover, writing $N_5 = 11520\Pi_{p,5}$, direct use of the original transport yields

$$\begin{aligned}
[m^5]N_5 &= 3(p+1)^5, \\
[m^4]N_5 &= 10(p+1)^4(4p-1), \\
[m^3]N_5 &= -5(p+1)^3(107p^2 + 74p - 1), \\
[m^2]N_5 &= 2(p+1)^2(280p^3 + 705p^2 + 106p + 1), \\
[m]N_5 &= 4p(p+1)(360\lambda_4 + 1183p^3 + 1157p^2 + 629p + 295), \\
[1]N_5 &= -96p(66\lambda_4 p + 36\lambda_4 + 24\lambda_5 + 85p^4 + 138p^3 + 76p^2 + 30p + 13).
\end{aligned}$$

Consequently

$$\Lambda_5(p, \lambda_4) \equiv -\frac{36\lambda_4 + 13}{24} \pmod{p}. \tag{34.51}$$

Corollary 34.37 (Global fifth-column congruence moments)**.** *Fix $p \ge 7$, $p \ne 73$, and integer λ_4. Let $N_{p,n}(\lambda_5)$ count every root of $\Pi_{p,5}$ modulo p^n. With $\chi_p = \left(\frac{73}{p}\right) \in \{-1, 1\}$, the complete count is*

$$N_{p,n}(\lambda_5) = \underbrace{(2 + \chi_p)}_{\text{simple residual roots}} + M_{p,5,n}(\lambda_5). \tag{34.52}$$

For Haar-uniform $\lambda_5 \in \mathbb{Z}_p$,

$$\boxed{\mathbb{E}N_{p,n} = 3 + \chi_p, \qquad \operatorname{Var}(N_{p,n}) = (p-1)(n-1).} \tag{34.53}$$

Proof. After excluding $p = 73$, the reduction (34.50) has exactly one double zero at 0 and all other zeros are simple. The simple zero at $m = 1$ always lifts. The quadratic $3m^2 - 7m - 2$ has discriminant 73, hence has exactly $1 + \chi_p \in \{0, 2\}$ simple $\mathbb{F}_p$-roots. These yield $2 + \chi_p$ lifts at all precisions. Add the singular contribution of Theorem 34.36. □

Remark 34.38 (Source and model scope). The exceptional prime 73 has a second repeated residue; its complete global classification is proved separately in Section 34.24. For $r \ge 7$ the all-odd-order theorem is intentionally local to the zero residue disk: other singular residues are not assumed away. The Nörlund identity [172], the standard analytic square coordinate and the moment-summation method are classical; the qualified research advance is this exact CCR-specific source deformation for every odd order.

34.24 The exceptional fifth-column prime 73: two coupled folds

The discriminant of $3m^2-7m-2$ is 73. Thus the prime excluded from Corollary 34.37 is not merely a bad denominator: it creates a *second* nondegenerate critical residue. Throughout this section put $p = 73$, $L = \lambda_4$, $T = \lambda_5$, and denote the integral fifth-column numerator by $N_{L,T}(m) = 11520\Pi_{73,5}^{(L,T)}(m)$. The coefficient formula above shows that the dependence on T is exactly $-2304 \cdot 73T$ and

$$N_{L,T}(m) \equiv 9m^2(m-1)(m-62)^2 \pmod{73}. \tag{34.54}$$

The residue $m = 1$ is simple; the two repeated residues 0 and 62 have nonzero quadratic Taylor coefficients 2 and 24, respectively, modulo 73.

Theorem 34.39 (Two-fold normal form and critical-value collision)**.** *For every $L \in \mathbb{Z}_{73}$ there are unique critical centers $c_0(L) \equiv 0$ and $c_{62}(L) \equiv 62 \pmod{73}$, and two critical source values $\Lambda_0(L), \Lambda_{62}(L) \in \mathbb{Z}_{73}$. The two residue disks admit integral analytic isometric coordinates in which their equations are, respectively,*

$$z_0^2 = 1152 \cdot 73\,[T - \Lambda_0(L)], \qquad z_{62}^2 = 96 \cdot 73\,[T - \Lambda_{62}(L)]. \tag{34.55}$$

Their critical parameters have first digits

$$\Lambda_0(L) \equiv 39 + 35L, \qquad \Lambda_{62}(L) \equiv 37 + 19L \pmod{73}. \tag{34.56}$$

In particular, their reductions coincide precisely when $L \equiv 9 \pmod{73}$. There is exactly one $L_* \in \mathbb{Z}_{73}$ for which $\Lambda_0(L_*) = \Lambda_{62}(L_*)$; its first four base-73 digits are

$$L_* \equiv 9 + 11 \cdot 73 + 30 \cdot 73^2 + 35 \cdot 73^3 \pmod{73^4}. \tag{34.57}$$

At the pair $\big(L_*, \Lambda_0(L_*)\big)$, the quintic has two distinct double roots and one simple root in $\mathbb{Q}_{73}$.

Proof. Apply Hensel's lemma to $\partial_m N_{L,T}$ in the two disks: its reduction vanishes simply at both residue centers by the nonzero second derivatives. The derivative does not depend on T, hence has unique 73-adic centers $c_i(L)$ independent of T. Put $\Lambda_i(L) = N_{L,0}(c_i)/(2304 \cdot 73)$; integrality follows from (34.54). In each disk the exact Taylor expansion is

$$N_{L,T}(c_i + y) = -2304 \cdot 73(T - \Lambda_i) + y^2 U_i(y),$$

where $U_0(y) \equiv 2$ and $U_{62}(y) \equiv 24 \pmod{73}$. Therefore $z_i = y\sqrt{U_i(y)/u_i}$, $u_0 = 2$, $u_{62} = 24$, is integral analytic and induces an isometric bijection on every congruence quotient of its residue disk. To see the global assertion, write $s_i(y) = \sqrt{U_i(y)/u_i}$. On $73\mathbb{Z}_{73}$, $s_i(y) \equiv 1 \pmod{73}$ and $s_i(y) - s_i(y') \in (y - y')\mathbb{Z}_{73}$. Consequently $z_i(y) - z_i(y') = (y - y')\big(s_i(y) + y'\,(s_i(y) - s_i(y'))/(y - y')\big)$, whose second factor is a 73-adic unit. The inverse is obtained by successive unique Hensel lifts. The equations become (34.55) because $2304/2 = 1152$ and $2304/24 = 96$.

Since $c_i \equiv i \pmod{73}$ and $N'_{L,0}(c_i) = 0$, the first digit of Λ_i is obtained by evaluating $N_{L,0}(i)/(2304 \cdot 73)$ modulo 73. The displayed exact coefficient formula yields (34.56). Differentiating through the critical center,

$$\Lambda_i'(L) = \frac{\partial_L N_{L,0}(c_i(L))}{2304 \cdot 73},$$

so $\Lambda_0' - \Lambda_{62}' \equiv 35 - 19 = 16 \pmod{73}$ is a unit for all L. Analytic Hensel lifting of $\Lambda_0(L) - \Lambda_{62}(L)$ gives a unique zero in the residue class $L \equiv 9 \pmod{73}$. Successive exact derivative-center lifts give (34.57). The residue disks remain distinct, so at the simultaneous critical parameter there are two double roots, plus the simple lift of 1. □

For $i \in \{0, 62\}$ and $n \geq 1$, let $M_{i,n}(T)$ count roots modulo 73^n in residue class i of $N_{L,T}$. These are the separate quadratic-fiber counts from Theorem 34.36, with constants $K_0 = 1152$, $K_{62} = 96$ and critical sources Λ_i. The ratio $K_0/K_{62} = 12$ is a square in $\mathbb{Z}_{73}^{\times}$: indeed $31^2 \equiv 12 \pmod{73}$.

Theorem 34.40 (Exact 73-adic global count and covariance law)**.** *For fixed $L \in \mathbb{Z}_{73}$ and any $T \in \mathbb{Z}_{73}$, the entire quintic root count at every precision $n \geq 1$ is*

$$N_n(L, T) = 1 + M_{0,n}(T) + M_{62,n}(T). \tag{34.58}$$

For Haar-uniform T and fixed L, $\mathbb{E}_T N_n = 3$ for every $n \geq 1$. Put $d = v_{73}(\Lambda_{62}(L) - \Lambda_0(L)) \in \{0, 1, 2, \ldots, \infty\}$, with $v_{73}(0) = \infty$. At $n = 1$, $N_1 = 3$ deterministically. For $n \geq 2$, the exact mixed moment of the two singular contributions is

$$\mathbb{E}_T[M_{0,n} M_{62,n}] = \begin{cases} 0, & d = 0, \\ 1 + 72(n-1), & d \geq 1, \quad n \leq d+1, \\ 72d, & 1 \leq d < \infty, \quad n \geq d+2. \end{cases} \tag{34.59}$$

Consequently, with $\mathrm{Var}(N_1) = 0$*,*

$$\boxed{\mathrm{Var}_T(N_n) = \begin{cases} 144(n-1) - 2, & d = 0, \\ 288(n-1), & d \geq 1, \quad n \leq d+1, \\ 144(n-1) + 144d - 2, & 1 \leq d < \infty, \quad n \geq d+2. \end{cases}} \tag{34.60}$$

For $d = \infty$ the second line applies at all $n \geq 2$, reflecting perfectly correlated fold counts.

Proof. The factorization (34.54) and Hensel's lemma give one root in the simple disk at every 73^n, establishing (34.58). Each quadratic disk has mean one and variance $72(n-1)$ by Theorem 34.36.

For $n \geq 2$ write the local coordinate $z_i = 73a_i$ with $a_i \pmod{73^{n-1}}$. Equation (34.55) becomes

$$T \equiv \Lambda_i + \frac{73}{K_i} a_i^2 \pmod{73^{n-1}}.$$

Let $\delta = \Lambda_{62} - \Lambda_0$. Expanding the product of the finite solution counts and averaging over uniform $T \pmod{73^{n-1}}$ gives

$$\mathbb{E}[M_{0,n} M_{62,n}] = 73^{-(n-1)} \#\Big\{(a, b) \pmod{73^{n-1}} : 73\Big(\frac{a^2}{1152} - \frac{b^2}{96}\Big) \equiv \delta \pmod{73^{n-1}}\Big\}.$$

If $d = 0$, this set is empty. For $d \geq 1$, put $t = n - 2$ and divide by 73. Each variable has one additional free base-73 digit, so the last display equals

$$73^{1-t} \#\{(a, b) \pmod{73^t} : a^2 - 12b^2 \equiv 1152\delta/73 \pmod{73^t}\}.$$

Choose $s \in \mathbb{Z}_{73}^{\times}$ with $s^2 = 12$. The invertible linear transformation $(a, b) \mapsto (u, v) = (a - sb, a + sb)$ reduces the count to $uv \equiv 1152\delta/73 \pmod{73^t}$. If $0 \leq d - 1 < t$, each valuation $v_{73}(u) = 0, \ldots, d-1$ supplies $72 \cdot 73^{t-1}$ pairs, giving $72d$ after normalization. If $d - 1 \geq t$ (including $d = \infty$), the zero-product congruence has $73^t + t \cdot 72 \cdot 73^{t-1}$ pairs, giving $1 + 72(n-1)$. The case $t = 0$ is the single trivial residue class, with the same count by direct interpretation. This proves (34.59). Since each local mean is one, the variance of the sum of the two folds is $2 \cdot 72(n-1) + 2(\mathbb{E}[M_{0,n} M_{62,n}] - 1)$, proving (34.60). □

Corollary 34.41 (Attainability of complete local splitting)**.** *For every fixed ordinary integer L, there exists an ordinary integer T for which $N_{L,T}$ has five distinct $\mathbb{Q}_{73}$ roots if and only if*

$$L \equiv 9 \pmod{73}. \tag{34.61}$$

In particular, $L = 9$, $T = 135$ *supplies a witness with*

$$(N_1, N_2, N_3, N_4) = (3, 147, 293, 293),$$

and both quadratic factors split over $\mathbb{Q}_{73}$.

Proof. If $L \not\equiv 9 \pmod{73}$, the critical parameter residues are distinct. Splitting of a fold requires T to lie in its critical residue class modulo 73, so both cannot split. If $L \equiv 9$, put $d = v_{73}(\Lambda_{62} - \Lambda_0) \geq 1$. For $d \geq 2$, take T with $v_{73}(T - \Lambda_0) = 1$ and with the square leading unit specified by K_0; then the second fold has the same valuation and compatible square character, because $K_0/K_{62} = 12$ is a 73-adic square. Such T is available as an ordinary integer by choosing its residue modulo 73^2. For $d = 1$, write $T = \Lambda_0 + 73u$ and $\delta = 73\Delta$ with Δ a unit. Let χ be the quadratic character modulo 73 extended by $\chi(0) = 0$. Both split if $\chi(K_0 u) = \chi(K_{62}(u - \Delta)) = 1$. Since $\chi(K_0) = \chi(K_{62})$ and $\chi(-1) = 1$, a standard quadratic-character correlation gives exactly

$$\frac{73 - 3 - 2\chi(K_0\Delta)}{4} \in \{17, 18\}$$

admissible residues $u \pmod{73}$, hence an integer T exists. Direct evaluation at $L = 9$ yields $\Lambda_0 \equiv 5026$, $\Lambda_{62} \equiv 1887 \pmod{73^2}$; $T = 135$ is one of the admissible residues. The displayed modular counts are independently verified by direct integer-polynomial Hensel lifting, and the split local normal forms prove the assertion about five $\mathbb{Q}_{73}$-roots. □

Remark 34.42 (Scope of the collision phenomenon). The parameter L_* is a 73-adic analytic critical-value coincidence, not a claim that L_* is an ordinary integer. The local analytic folds and character-sum/product-congruence counts are standard methods; the explicit two-fold coupling and the exact finite-precision covariance stratification belong to the concrete CCR fifth-column family. They do not establish global all-order physical DPDA realization or a new generic p-adic Morse theorem.

34.25 Joint source equidistribution and the exact five-root density at 73

The preceding section fixed the lower stationary source $L = \lambda_4$ and varied $T = \lambda_5$. We now vary both L, T independently according to normalized Haar measure on $\mathbb{Z}_{73}^2$. This change reveals a sharp distinction between *conditional* dependence and *unconditional* independence of the two singular residue disks.

Theorem 34.43 (Two-source critical-value isometry)**.** *Retain the two critical parameter functions* $A(L) = \Lambda_0(L)$ *and* $B(L) = \Lambda_{62}(L)$ *of Section 34.24. The difference* $D(L) = B(L) - A(L)$ *is an isometric bijection* $\mathbb{Z}_{73} \to \mathbb{Z}_{73}$. *In fact, for every* $L \neq L'$ *in* $\mathbb{Z}_{73}$,

$$\frac{D(L) - D(L')}{L - L'} \equiv 19 - 35 = -16 \pmod{73}. \tag{34.62}$$

Consequently the map

$$\Phi : (L, T) \longmapsto \big(T - A(L),\ T - B(L)\big)$$

is a measure-preserving bijection of $\mathbb{Z}_{73}^2$. *Thus its two output coordinates are independent Haar-uniform* 73*-adic integers when* (L, T) *is Haar-uniform.*

Proof. The established quintic numerator $N_{L,T}(m)$ is affine in L and T, with L- and T-coefficients divisible by 73; moreover the T-coefficient is the constant $-2304 \cdot 73$. Let $c_i(L)$ be the stationary derivative-zero center for $i = 0, 62$. Since $\partial_m^2 N_{L,0}(c_i(L))$ is a unit modulo

73, implicit Hensel lifting makes c_i an integral analytic function of L. As $\partial_L \partial_m N_{L,0}$ is divisible by 73,

$$c_i(L) - c_i(L') \in 73(L - L')\mathbb{Z}_{73}.$$

By definition $A(L) = N_{L,0}(c_0(L))/(2304 \cdot 73)$ and likewise $B(L) = N_{L,0}(c_{62}(L))/(2304 \cdot 73)$. The vanishing of the m-derivative at each center, combined with the preceding Lipschitz estimate, shows that the critical-value difference quotients reduce modulo 73 to the explicit L-coefficient at the corresponding residue center. The exact coefficient formula in the previous section gives

$$\frac{A(L) - A(L')}{L - L'} \equiv 35, \qquad \frac{B(L) - B(L')}{L - L'} \equiv 19 \pmod{73}.$$

This proves (34.62). It also proves that D is an isometry. Since $D(L) \equiv -2 - 16L \pmod{73}$ is a permutation, for each target $t \in \mathbb{Z}_{73}$ and every solution L_j of $D(L_j) \equiv t \pmod{73^j}$ there is exactly one next digit $h \pmod{73}$ such that $D(L_j + h73^j) \equiv t \pmod{73^{j+1}}$: the difference quotient above makes the digit coefficient -16, a unit. Induction proves that D is bijective modulo every 73^j and preserves Haar measure.

Now put $(U, V) = \Phi(L, T)$. The inverse is explicit:

$$L = D^{-1}(U - V), \qquad T = U + A(L).$$

Both the forward and inverse maps preserve congruence classes modulo every 73^j and give bijections of their finite residue grids; therefore Φ preserves product Haar measure. In particular U, V are independent Haar variables. □

Theorem 34.44 (Complete joint law for the fifth column)**.** *For Haar-independent $L, T \in \mathbb{Z}_{73}$ and every $n \geq 1$, the two singular-disk solution counts $M_{0,n}, M_{62,n}$ are independent and identically distributed. The complete fifth-column congruence count $N_n = 1 + M_{0,n} + M_{62,n}$ obeys*

$$\mathbb{E}_{L,T} N_n = 3, \qquad \mathrm{Var}_{L,T}(N_n) = 144(n - 1). \tag{34.63}$$

More precisely, write $h = \lfloor (n-1)/2 \rfloor$, $q_n = 73^{-(n-1)}$, $a_n = 73^{\lfloor n/2 \rfloor}$, and $w_j = 36/73^{2j}$. The exact probability generating polynomial is

$$P_n(X) = \left(1 - q_n - \sum_{j=1}^{h} w_j\right) + q_n X^{a_n} + \sum_{j=1}^{h} w_j X^{2 \cdot 73^j}, \tag{34.64}$$
$$\mathbb{E}_{L,T}[X^{N_n}] = X P_n(X)^2.$$

Let R be the number of distinct roots in $\mathbb{Q}_{73}$ of the fifth-column quintic. Then

$$\begin{aligned} \Pr(R = 1) &= \frac{21609}{21904}, & \Pr(R = 3) &= \frac{294}{21904}, \\ \Pr(R = 5) &= \boxed{\frac{1}{21904}}, & \mathbb{E}R &= \frac{38}{37}. \end{aligned} \tag{34.65}$$

These probabilities are also the natural densities of ordinary integer source pairs (L, T) in expanding integer squares. The almost-surely stabilized congruence count has expectation 3 but infinite variance.

Proof. The local formulas of Section 34.23 depend on (L, T) only through the appropriate difference $U = T - A(L)$ or $V = T - B(L)$. Theorem 34.43 makes these independent Haar inputs. Each local count has mean 1, variance $72(n-1)$ and probability polynomial $P_n(X)$, as proved previously; their sum gives (34.63) and (34.64). Each quadratic splits over $\mathbb{Q}_{73}$ with probability $s = 1/(2 \cdot 74) = 1/148$, independently. The simple residue root $m \equiv 1$ always survives; therefore $R = 1 + 2I_0 + 2I_{62}$, where I_0, I_{62} are independent Bernoulli variables with parameter s, outside a Haar-null repeated-root locus. This yields (34.65).

The sets specifying splitting are unions of 73-adic cylinders up to null critical singletons. Approximating by finite cylinders proves equality of Haar probabilities with the natural densities among integer pairs. The limit-moment statements follow from the independent heavy-tailed local limits already proved for one fold. □

Proposition 34.45 (Finite two-digit certificate)**.** *Modulo* 73^2*, the critical gap map* $L \mapsto D(L)$ *permutes all* 5329 *residue classes. Among the* $73^4 = 28\,398\,241$ *source pairs* (L, T) *modulo* 73^2*, exactly*

$$\left(\frac{73-1}{2}\right)^2 = \boxed{1296}$$

classes certify that both folds split at their first possible resonance depth $v_{73}(T - \Lambda_i) = 1$*. For the* 73 *values of* $L \pmod{73^2}$ *with* $L \equiv 9 \pmod{73}$*, precisely* 36 *admit* 18 *such* T*,* 36 *admit* 17*, and one admits* 36*. Moreover the full* N_2 *distribution on* $(L, T) \pmod{73}$ *is*

$$\#\{N_2 = 1\} = 5184, \qquad \#\{N_2 = 74\} = 144, \qquad \#\{N_2 = 147\} = 1.$$

Proof. By Theorem 34.43, (U, V) ranges bijectively over the pair of residue classes modulo 73^2. A first-depth split in each coordinate requires a nonzero square leading unit, for which there are $(73 - 1)/2 = 36$ possibilities. This gives 1296. The finer $36, 36, 1$ census follows from the two explicit critical functions and a quadratic character sum (or direct enumeration of all 73 coincident lower-source residues). At precision $n = 2$, each disk contributes 73 solutions iff its critical source residue matches $T \pmod{73}$, otherwise none. The bijection $\Phi \pmod{73}$ therefore gives the stated binomial census. □

Remark 34.46 (Conditional and unconditional source correlations). The independent global law does not negate the sharply nonzero *conditional* covariance of Theorem 34.40: fixing L creates dependence, which cancels exactly on averaging through the measure-preserving lower-source coordinate. The isometry/Haar principle is standard in p-adic dynamics; the specific CCR two-critical-value map, explicit simultaneous-splitting density, and joint source distribution are the concrete arithmetic statements established here. The conclusions do not include a generic isometry theorem or a theorem on physical source realizability.

34.26 Fixed integer sources: finite resonance support across primes

The preceding root-density results distinguish two different sampling schemes: variation of a stationary source in $\mathbb{Z}_p$ with p fixed, versus variation of the prime while the sources are fixed *ordinary integers*. The latter exhibits a rigid arithmetic obstruction not apparent from the Haar probability of splitting at any one prime.

Theorem 34.47 (Fixed-source prime support and exact two-digit stability)**.** *Fix* $L = \lambda_4, T = \lambda_5 \in \mathbb{Z}$ *and put*

$$A(L, T) = 24T + 36L + 13 = 12(2T + 3L + 1) + 1 \neq 0.$$

For every rational prime $p \geq 7$ *satisfying* $p \neq 73$ *and* $p \nmid A(L, T)$*, the integer fifth-column numerator* $N_{p,L,T}(m) = 11520\Pi_{p,5}^{(L,T)}(m)$ *has exactly*

$$b_p = 2 + \left(\frac{73}{p}\right) \in \{1, 3\}$$

distinct roots in $\mathbb{Q}_p$*. More precisely, letting* $C_{p,n}$ *count solutions of* $N_{p,L,T}(m) \equiv 0 \pmod{p^n}$*,*

$$C_{p,1} = 3 + \left(\frac{73}{p}\right), \qquad C_{p,n} = 2 + \left(\frac{73}{p}\right) \quad (n \geq 2).$$

All surviving roots arise from simple residues, while the double residue $m \equiv 0 \pmod{p}$ *yields a totally ramified irreducible quadratic over* $\mathbb{Q}_p$ *and disappears already modulo* p^2*.*

Proof. The exact coefficient formula of the previous section gives

$$N_{p,L,T}(m) \equiv m^2(m-1)(3m^2-7m-2) \pmod{p}.$$

For $p \geq 7$, $p \neq 73$, the nonzero roots of the right side are simple: $m = 1$, plus two roots of its quadratic factor exactly when $\left(\frac{73}{p}\right) = 1$. Hensel's lemma lifts each such root uniquely through all p^n and to $\mathbb{Q}_p$.

The exact constant and linear coefficients satisfy

$$N_{p,L,T}(0) \equiv -96p(24T + 36L + 13) \pmod{p^2}, \qquad [m]N_{p,L,T} \equiv 0 \pmod{p}.$$

Consequently, for every $z \in \mathbb{Z}_p$,

$$N_{p,L,T}(pz) \equiv -96pA(L,T) \pmod{p^2}.$$

The right side is nonzero modulo p^2 when $p \nmid A$, as $-96 \in \mathbb{Z}_p^\times$. Thus *no* zero-disk solution survives even to p^2. Its local quadratic factor has Newton slope $-1/2$ (nonzero constant valuation one, unit quadratic coefficient), hence is irreducible and totally ramified. This proves the exact congruence and field-root counts. Finally $A(L,T) \equiv 1 \pmod{12}$, so no integral source pair makes $A = 0$. □

Corollary 34.48 (Source-independent prime-axis densities)**.** *For every fixed integer pair (L,T) the fifth-column quintic has exactly one $\mathbb{Q}_p$-root at a natural density $1/2$ of rational primes, and exactly three at density $1/2$; primes exhibiting five roots are contained in the* finite *exceptional set*

$$\{73\} \cup \{p \geq 7 : \ p \mid (24T + 36L + 13)\}$$

(aside from the excluded small primes). The average local-root count over primes tends to 2. Moreover, for any $B \geq 0$, all integer sources $|L|, |T| \leq B$ satisfy Theorem 34.47 simultaneously for every $p > \max\{73, 60B + 13\}$.

Proof. Since $73 \equiv 1 \pmod 4$, quadratic reciprocity gives $\left(\frac{73}{p}\right) = \left(\frac{p}{73}\right)$. Exactly 36 of the 72 nonzero residues modulo 73 are squares; Dirichlet's theorem on primes in arithmetic progressions assigns density $1/72$ to each residue class, so the two root counts each have density $1/2$. Only finitely many primes divide the nonzero integer A. Finally $|A| \leq 24|T| + 36|L| + 13 \leq 60B + 13$, which yields the uniform threshold. □

Remark 34.49 (Comparison with source-Haar statistics)*.* The density-$1/2$ prime dichotomy concerns *fixed integer* sources; it is not the source-Haar distribution of Theorem 34.36, and it does not override the exceptional prime 73 two-fold model. Standard Dirichlet and Hensel methods are used; the new CCR-specific content is the exact integer resonance obstruction $A(L,T)$ and its two-digit criterion.

34.27 All odd orders: a Stirling source obstruction and finite prime support

The fifth-column obstruction of the previous section is not isolated. In this section we derive a closed source-obstruction formula *for every odd column*, using the pre-existing normalized transport and the classical Nörlund generating function. The odd-index double-zero property itself has appeared in the Stirling-polynomial literature [167]; the CCR source obstruction and its prime-support application are the subject here.

Write $\left\{ {n \atop k} \right\}$ for a Stirling number of the second kind. Throughout, $r \geq 3$, $\lambda_3 = 1$, and all λ_a are independent stationary source parameters unless explicitly specialized to integers. Denote by $S_r(m) = \Pi_{0,r}(m)$ the Stirling specialization and set $E_r(p, m, \boldsymbol{\lambda}) = (\Pi_{p,r}(m) - S_r(m))/p$ in the rational coefficient ring supplied by the original polynomial transport.

Theorem 34.50 (Exact first-order source obstruction)**.** *For every $r \geq 3$ one has*

$$A_r(\boldsymbol{\lambda}) := \sum_{a=3}^{r} (a-1)! \left\{ {r-1 \atop a-1} \right\} \lambda_a - 1, \qquad E_r(0,0,\boldsymbol{\lambda}) = \frac{(-1)^{r-2}}{r!} A_r(\boldsymbol{\lambda}). \tag{34.66}$$

For integer stationary sources, A_r is always odd. For every odd $r \geq 5$, the stronger arithmetic rigidity holds:

$$A_r = 2^{r-1} - 3 + \sum_{a=4}^{r} (a-1)! \left\{ {r-1 \atop a-1} \right\} \lambda_a \equiv 1 \pmod 6. \tag{34.67}$$

In particular $A_r(\boldsymbol{\lambda}) \neq 0$ for every *integer vector $\boldsymbol{\lambda}$.*

Proof. The original transport with $f_j = \Pi_{p,j}$ is

$$T_j f_j(m) = \ell_m f_{j-1}(m) + p\ell_{m-1} f_{j-2}(m-1) + \sum_{a=3}^{j} (-1)^a p \lambda_a f_{j-a}(m-a+1),$$

where $\ell_m = m + p(m-1)$ and $T_j f(m) = m f(m+1) - (m-j) f(m)$. Differentiate formally at $p = 0$, put $m = 0$, use $S_j(0) = 0$ for $j \geq 1$, and note $T_j f(0) = j f(0)$. This gives

$$j E_j(0,0) = -S_{j-2}(-1) + \sum_{a=3}^{j} (-1)^a \lambda_a S_{j-a}(1-a). \tag{34.68}$$

The established Nörlund identity $\sum_{q \geq 0} S_q(m) t^q = (t/(1-e^{-t}))^m$ yields

$$S_{j-2}(-1) = \frac{(-1)^{j-2}}{(j-1)!}, \qquad S_{j-a}(1-a) = \frac{(-1)^{j+a-2}(a-1)!}{(j-1)!} \left\{ {j-1 \atop a-1} \right\}.$$

The second formula follows by expanding $(1-e^{-t})^{a-1}/t^{a-1}$ and using the standard finite-difference identity defining Stirling numbers. Substituting these coefficients into (34.68) proves (34.66). All $(a-1)!$ are even for $a \geq 3$, so the sum minus one is odd. Since $2! \left\{ {r-1 \atop 2} \right\} = 2(2^{r-2} - 1)$ and $(a-1)!$ is divisible by 6 for $a \geq 4$, equation (34.67) follows when $r - 1$ is even and at least four. □

Theorem 34.51 (Exact CCR source-obstruction lattice and coherent integral witness)**.** *For every $r \geq 4$ put $n = r - 1$ and define the positive integer*

$$d_r = \gcd_{3 \leq j \leq n} \left(j! \left\{ {n \atop j} \right\} \right).$$

With $\lambda_3 = 1$ and all subsequent stationary sources ranging freely over $\mathbb{Z}$, the exact image *of the first-order source obstruction is*

$$\left\{ A_r(1, \lambda_4, \ldots, \lambda_r) : \lambda_a \in \mathbb{Z} \right\} = (-1)^{r-1} + d_r \mathbb{Z}. \tag{34.69}$$

There is a single coherent choice of integer sources for every order,

$$\lambda_a^* = (-1)^{a-1} \quad (a \geq 3), \qquad A_r(\boldsymbol{\lambda}^*) = (-1)^{r-1} \quad (r \geq 3). \tag{34.70}$$

For every odd $r \geq 5$, the image in (34.69) *is $1 + d_r\mathbb{Z}$, and $12 \mid d_r$. In the first three odd cases,*

$$d_5 = 12, \qquad d_7 = 60, \qquad d_9 = 252.$$

Thus (34.67) *may be strengthened to the exact lattice image, not just a necessary congruence.*

Proof. The established Stirling finite-difference identity [171, Eq. 26.8.36] is $\sum_{j=0}^{n}(-1)^j j!\left\{{n \atop j}\right\} = (-1)^n$. At $n \geq 2$ the $j = 0$ term vanishes and the $j = 1$ term equals -1. Substituting $\lambda^*_{j+1} = (-1)^j$ into (34.66) therefore yields $A_r(\boldsymbol{\lambda}^*) = (-1)^n$ exactly. Subtracting the canonical-source equality from the general source gives

$$A_r(\boldsymbol{\lambda}) - (-1)^n = \sum_{j=3}^{n} j!\left\{{n \atop j}\right\}\big(\lambda_{j+1} - (-1)^j\big).$$

Every right-hand term is divisible by d_r. Conversely, Bézout's identity shows that arbitrary integer values of the freely chosen source differences generate *every* element of $d_r\mathbb{Z}$. This proves the exact image rather than only a containment. For odd $r \geq 5$, $n \geq 4$ is even. The terms $j!$ for $j \geq 4$ are multiples of 24. The remaining generator satisfies

$$3!\left\{{n \atop 3}\right\} = 3^n - 3 \cdot 2^n + 3 \equiv 0 \pmod{12},$$

since $3^n \equiv 9 \pmod{12}$ and $3 \cdot 2^n \equiv 0 \pmod{12}$. Hence $12 \mid d_r$. Exact Stirling arithmetic gives the three displayed small-order gcd values. □

Proposition 34.52 (Closed form of the odd-order source-lattice index)**.** *Let $r \geq 5$ be odd, and let d_r be the index in Theorem 34.51. Then*

$$\boxed{d_r = 4 \prod_{\substack{\ell \geq 3 \text{ prime} \\ \ell - 1 \mid r-3}} \ell^{1+\mathbf{1}_{\ell \mid r-3}}.} \tag{34.71}$$

Equivalently, for every odd prime ℓ,

$$\ell \mid d_r \iff \ell - 1 \mid r - 3, \qquad \ell^2 \mid d_r \iff \ell(\ell-1) \mid r - 3, \qquad \ell^3 \nmid d_r,$$

and $v_2(d_r) = 2$. In particular, the exact source image $1 + d_r\mathbb{Z}$ is determined solely by the elementary prime factorizations of the divisors of $r - 3$; its first values are $12, 60, 252, 60, 132, 16380$ at $r = 5, 7, 9, 11, 13, 15$.

Proof. Write $n = r - 1 \geq 4$ (even), set $a = 2^{n-1} - 1$, and introduce the integer polynomial

$$P_n(X) = X^n - X - aX(X-1).$$

By $X^n = \sum_{j=0}^{n} j!\left\{{n \atop j}\right\}\binom{X}{j}$ and $2!\left\{{n \atop 2}\right\} = 2^n - 2 = 2a$, its binomial-basis expansion is

$$P_n(X) = \sum_{j=3}^{n} j!\left\{{n \atop j}\right\}\binom{X}{j}.$$

Integral forward differences and binomial inversion therefore identify d_r with the fixed divisor $\gcd\{P_n(x) : x \in \mathbb{Z}\}$: taking differences recovers every binomial coefficient, and conversely every integer polynomial value is an integral combination of them.

Let $\ell \geq 5$ be prime. The condition $\ell \mid d_r$ means that P_n vanishes as a function on $\mathbb{F}_\ell$. At nonzero x this requires

$$x^{n-1} = 1 + a(x-1) \quad (x \in \mathbb{F}_\ell^\times).$$

Reducing X^{n-1} modulo $X^{\ell-1} - 1$, the left side is a single monomial of degree $0, \ldots, \ell - 2$, while the right side is a linear polynomial. Since evaluation on $\mathbb{F}_\ell^\times$ is injective for polynomials of degree at most $\ell - 2$, the exponent must reduce to 0 or 1 modulo $\ell - 1$. The case 0 is impossible because $n - 1$ is odd and $\ell - 1$ is even. Thus $\ell - 1 \mid n - 2$. Conversely

that divisibility gives $x^{n-1} = x$ for every nonzero $x \in \mathbb{F}_\ell$ and $a \equiv 1 \pmod{\ell}$, so P_n vanishes. For $\ell = 3$ the same criterion holds directly: $2 \mid n-2$ and $a \equiv 1 \pmod{3}$.

Suppose an odd ℓ divides d_r. Evaluating the polynomial at $1+\ell$ yields

$$P_n(1+\ell) \equiv \ell\,(n-2) \pmod{\ell^2},$$

so $\ell^2 \mid d_r$ forces $\ell \mid n-2$. Conversely, if both $\ell - 1$ and ℓ divide $n-2$, Euler's theorem modulo ℓ^2 implies $x^{n-2} \equiv 1$ for every ℓ-adic unit, and also $a \equiv 1 \pmod{\ell^2}$. Consequently

$$P_n(x) = x^2(x^{n-2}-1) - (a-1)x(x-1) \equiv 0 \pmod{\ell^2}$$

for units, and separately for multiples of ℓ; hence $\ell^2 \mid d_r$. No odd ℓ^3 can divide d_r, because

$$P_n(2\ell) - 2P_n(\ell) \equiv -2a\ell^2 \not\equiv 0 \pmod{\ell^3},$$

where $a \equiv 1 \pmod{\ell}$. Finally $4 \mid P_n(x)$ for every integer x since n is even, $n \geq 4$, and $a \equiv -1 \pmod 4$; but $3!\left\{ {n \atop 3} \right\} = 3^n - 3 \cdot 2^n + 3 \equiv 4 \pmod 8$, so $v_2(d_r) = 2$. Combining these local valuations proves (34.71). □

Remark 34.53 (Classical Stirling divisibility versus the CCR application). The fixed-divisor object $\Delta_{n,3} = \gcd\{j!\left\{ {n \atop j} \right\} : 3 \leq j \leq n\}$ is *not* introduced here: Lundell studied prime-power divisibility of precisely $\Delta_{n,m}$ in 1978 [168]. Formula (34.71) is included with a self-contained elementary proof as a useful classical specialization, *not claimed as an independently original Stirling theorem.* The CCR-specific interpretation uses the previously established first-order source transform to identify the exact source image and its permitted-prime resonance consequences. In particular, $\ell \mid d_r$ occurs only when $\ell \leq r-2$; these potentially small primes need not satisfy the independent p-integrality hypotheses of Theorem 34.55, and no local splitting claim is inferred there merely from this divisibility statement.

Corollary 34.54 (All-odd-order fixed-source nonresonance)**.** *Choose the coherent integral stationary source family $\lambda_a = (-1)^{a-1}$ for every $a \geq 3$. At each fixed odd $r \geq 5$, and for* every *prime satisfying the order-r good-prime hypotheses of Theorem 34.55, the zero-residue fold has no roots modulo p^2 or over $\mathbb{Q}_p$ and is an irreducible totally ramified quadratic factor. There are no additional* source-dependent *exceptional primes for this family.*

Proof. Theorem 34.51 gives $A_r = 1$, which is a unit at every prime. The proof of Theorem 34.55 applies verbatim. □

The first new higher-order example beyond the fifth column is

$$\begin{aligned} A_5 &= 13 + 36\lambda_4 + 24\lambda_5, \\ A_7 &= 61 + 540\lambda_4 + 1560\lambda_5 + 1800\lambda_6 + 720\lambda_7. \end{aligned} \tag{34.72}$$

At order nine one obtains $A_9 = 253 + 5796\lambda_4 + 40824\lambda_5 + 126000\lambda_6 + 191520\lambda_7 + 141120\lambda_8 + 40320\lambda_9$.

Theorem 34.55 (Universal fixed-source finite prime-support theorem)**.** *Let $r = 2s+1 \geq 5$ be fixed and $\lambda_4, \ldots, \lambda_r \in \mathbb{Z}$. Choose a positive integer D_r clearing the rational coefficients of both S_r and E_r as polynomials in their independent variables, as guaranteed by stationary polynomiality. Put $c_s = -B_{2s}/(4s(2s)!) \neq 0$. At every prime p such that*

$$p > r, \qquad p \nmid D_r, \qquad v_p(c_s) = 0, \qquad p \nmid A_r(\boldsymbol{\lambda}),$$

the disk $m \equiv 0 \pmod p$ contributes exactly one double residual root modulo p, *but* no solutions modulo p^2, *no solutions modulo higher powers, and no $\mathbb{Q}_p$ roots. Its degree-two local factor is irreducible and totally ramified. Consequently the primes at which this CCR odd-column zero-disk fold can split form a* finite *set, contained in the union of the finite order-dependent bad-prime set and the prime divisors of the explicit nonzero integer $A_r(\boldsymbol{\lambda})$.*

Proof. At the stated primes, the established integral coefficientwise reduction gives $\Pi_{p,r} \equiv S_r \pmod p$. The classical Nörlund coefficient identity gives $S_r(m) = c_s m^2 + O(m^3)$, with c_s a p-unit, so the disk is a nondegenerate double residual root. For $m = pz$, $z \in \mathbb{Z}_p$, we have $S_r(pz) \equiv 0 \pmod{p^2}$, and polynomial integrality yields

$$\Pi_{p,r}(pz) \equiv pE_r(0, 0, \boldsymbol{\lambda}) = -\frac{pA_r(\boldsymbol{\lambda})}{r!} \not\equiv 0 \pmod{p^2}.$$

The stated conditions make $r!$ invertible and A_r a unit modulo p. Thus the whole disk vanishes at the next digit. The derivative-zero center in that disk exists uniquely by Hensel's lemma applied to the derivative; its critical value has p-valuation exactly one. Its quadratic Newton polygon therefore has slope $-1/2$, proving irreducibility and total ramification. The nonvanishing congruence (34.67) makes A_r a nonzero integer, hence it has finitely many prime divisors. □

Corollary 34.56 (Finite-prime simultaneous tuning versus fixed-source rigidity)**.** *Fix an odd order $r \geq 5$, fixed lower integer sources $\lambda_4, \ldots, \lambda_{r-1}$, and a finite set $\mathcal{P}$ of pairwise distinct order-r good primes. There exist* ordinary integer *top source values λ_r for which the zero-disk quadratic factor splits into two distinct $\mathbb{Q}_p$-linear factors* simultaneously for all $p \in \mathcal{P}$. *Precisely, $\prod_{p\in\mathcal{P}}(p-1)/2$ residue classes of λ_r modulo $\prod_{p\in\mathcal{P}} p^2$ realize splitting at the first resonance depth $v_p(\lambda_r - \Lambda_r) = 1$ at every such prime. Nevertheless for every fixed integral source vector the number of primes at which this quadratic factor can split is finite.*

Proof. At each good prime, the top-source translation $\Pi_{p,r}^{(\lambda_r)} = \Pi_{p,r}^{(0)} - p\lambda_r/r$ and the local normal form in Theorem 34.35 give a unique critical $\Lambda_r \in \mathbb{Z}_p$ and a unit K_r with $z^2 = K_r p(\lambda_r - \Lambda_r)$. Exactly one class modulo p satisfies $\lambda_r \equiv \Lambda_r \pmod p$. Of its p lifts modulo p^2, exactly $(p-1)/2$ satisfy $v_p(\lambda_r - \Lambda_r) = 1$ and $\left(\frac{K_r(\lambda_r-\Lambda_r)/p}{p}\right) = 1$. These lifts split as claimed. The Chinese remainder theorem combines independent prime-square conditions, giving the exact product count and integer representatives. For any one such integer representative, Theorem 34.55 limits the split primes to the fixed finite order-dependent bad-prime set and prime divisors of the nonzero integer A_r. □

Remark 34.57 (Exact finite example, order five). With $\lambda_4 = 0$ and $\lambda_5 = 774982$, both zero-disk roots split at $p = 7, 11, 13$. The corresponding local field-root counts are three at each prime, whereas the unnormalized polynomial's full congruence-root counts for $n = 1, 2, 3, 4$ are, respectively,

$$p = 7 : \ (2, 8, 15, 15), \quad p = 11 : \ (2, 12, 23, 23), \quad p = 13 : \ (2, 14, 27, 27).$$

The difference between the field-root number and stable congruence-fiber cardinality is essential: a split local root with nonunit derivative represents p distinct classes modulo each sufficiently high power. The exact integer witness is a finite consistency check, not a substitute for the CRT proof.

Corollary 34.58 (Uniform source-height threshold)**.** *If $|\lambda_a| \leq B$ for all $4 \leq a \leq r$, then*

$$|A_r| \leq C_r(B) := 2^{r-1} - 3 + B\sum_{a=4}^{r}(a-1)!\left\{{r-1 \atop a-1}\right\}.$$

For every prime meeting the order-r good-prime conditions and satisfying $p > C_r(B)$, the zero-residue disk disappears at precision p^2 uniformly for every integral source vector of that height. *For $r = 7$, $C_7(B) = 61 + 4620B$; the old $r = 5$ bound is $13 + 60B$.*

Remark 34.59 (Precise global scope and mathematical antecedents). Theorem 34.55 concerns the CCR-specific zero-residue quadratic factor. It does *not* assert an all-prime factorization of the remaining degree-$r-2$ factor for every $r \geq 7$. If $S_r(m)/m^2$ is squarefree

modulo a particular good prime p, then all nonzero residual roots are simple and the complete root count modulo every p^n, $n \geq 2$, equals the number of nonzero roots of $S_r \pmod p$; in characteristic zero the quotient is explicitly squarefree for $r = 5, 7, 9, 11$ by finite symbolic calculation. The alternating Stirling sum and Bézout argument in Theorem 34.51 are classical combinatorial facts, not an assertion of new Stirling-number identities. Their interpretation as an exact CCR source-obstruction lattice and simultaneous p-adic splitting criterion is the qualified specialized contribution. The Nörlund identity, and the odd-index double-zero factor itself, are established independently: see Bencherif–Garici [167]. In the CCR hierarchy, the source-deformation formula (34.66), together with parity and the fixed-source condition, yields the exact obstruction lattice and finite-prime-support conclusion.

34.28 Chronological integer-source observations and their limits

The preceding stationary theory concerns a single vector of source parameters. It is important to ask exactly which *terminal raw coefficient* constraints survive when only the top order-r source varies in time. This section gives finite, closed criteria; its assertions are *not* identities of entire normalized polynomial columns, and there is no claim that the prescribed source amplitudes arise from a physical DPDA transition system.

Fix an odd integer $r \geq 5$. Retain the stationary lower integer sources $\lambda_3 = 1, \lambda_4, \ldots, \lambda_{r-1}$ and let the top source at temporal row $t \geq r$ be an independent integer $\eta_{r,t}$. Write

$$w_t = \binom{t-1}{r-1}, \quad B_N = \binom{N}{r} = \sum_{t=r}^{N} w_t, \quad Z_N = \sum_{t=r}^{N} w_t \eta_{r,t}, \quad \lambda_{\mathrm{eff}}(N) = \frac{Z_N}{B_N}. \tag{34.73}$$

The r-th connected source enters the raw CCR recurrence in degree r, so changes to its temporal sequence affect the degree-r coefficient linearly through the terminal-block weights w_t; no lower-degree coefficient is affected. Thus $\lambda_{\mathrm{eff}}(N)$ is the *unique stationary top-source shadow* of that single raw terminal coefficient. From Theorem 34.50 the first-order obstruction of this shadow is

$$A_r^{\mathrm{shadow}}(N) = L_r + (r-1)!\lambda_{\mathrm{eff}}(N), \quad L_r = 2^{r-1} - 3 + \sum_{a=4}^{r-1} (a-1)! \left\{ {r-1 \atop a-1} \right\} \lambda_a. \tag{34.74}$$

Because $L_r \equiv 1 \pmod 6$ and $(r-1)!$ is divisible by 6, the cancellation target $c_r = -L_r/(r-1)!$ is a noninteger rational.

Theorem 34.60 (Signed integer multisnapshot criterion)**.** *Let $r \leq N_1 < \cdots < N_m$, put $N_0 = r-1$, and fix $c \in \mathbb{Q}$. With unrestricted signed integer sources $\eta_{r,t}$, one history satisfies $\lambda_{\mathrm{eff}}(N_i) = c$ for all i if and only if*

$$c\left(\binom{N_i}{r} - \binom{N_{i-1}}{r} \right) \in g_i \mathbb{Z}, \qquad g_i = \gcd\{ w_t : N_{i-1} < t \leq N_i \}, \quad 1 \leq i \leq m. \tag{34.75}$$

In particular no integer history can cancel A_r^{shadow} at two consecutive terminal rows, but there are single signed integer histories attaining the cancellation at infinitely many suitably separated rows.

Proof. The independent increment $Z_{N_i} - Z_{N_{i-1}}$ is an integer linear combination of the weights in that interval. Bézout's identity characterizes all such increments as $g_i\mathbb{Z}$; separate intervals do not share source cells, giving necessity and sufficiency. For consecutive snapshots subtraction in (34.73) forces $\eta_{r,N+1} = c_r$, impossible since $c_r \notin \mathbb{Z}$. Conversely, the $(r-1)$-st finite difference gives $\gcd(w_{N+1}, \ldots, w_{N+r}) = 1$. Writing $c_r = a/D$ in lowest terms, the sequence $N_j = j\, r!D$ has integral $c_r \binom{N_j}{r}$ and gaps at least r. Formula (34.75) then produces the claimed infinite history. □

Theorem 34.61 (Nonnegative integer sources and exact reachability)**.** *Keep the same lower stationary parameters, which may be signed, but impose $\eta_{r,t} \in \mathbb{Z}_{\geq 0}$. For the snapshot rows of Theorem 34.60, let $\mathcal{S}_i = \langle w_t : N_{i-1} < t \leq N_i \rangle_{\mathbb{Z}_{\geq 0}}$. One nonnegative integer history realizes the rational target c at all snapshots if and only if*

$$c\left(\binom{N_i}{r} - \binom{N_{i-1}}{r}\right) \in \mathcal{S}_i \qquad (1 \leq i \leq m). \tag{34.76}$$

For $c < 0$ no snapshot is attainable; for $c = 0$ all are attainable using zero sources; for every $c > 0$ one history attains infinitely many sparse snapshots, without a uniform amplitude or gap bound.

Proof. Equation (34.76) says exactly that each independent nonnegative weighted increment is representable by source pulses in its time interval. Negative increments are impossible and zero increments are attained by the zero history. For $c = a/D > 0$, choose an initial large multiple N_1 of $r!D$ and place the integral amount cB_{N_1} at weight $w_r = 1$. Given an attained N_j, the first r weights past N_j have gcd one by finite differences. Their nonnegative numerical semigroup has a finite complement in $\mathbb{Z}_{\geq 0}$ [184]. Choose the next sufficiently large multiple N_{j+1} of $r!D$ so that the integral required increment $c(B_{N_{j+1}} - B_{N_j})$ lies beyond the conductor. Represent it using those first r weights and set all other intervening sources to zero. This proves induction. □

Theorem 34.62 (Sharp mixed-sign lower-band threshold)**.** *Let all $\eta_{r,t} \in \mathbb{Z}_{\geq 0}$ remain unrestricted in amplitude. Define positive Stirling weights and the signed lower-source budgets*

$$K_{r,a} = (a-1)!\left\{ {r-1 \atop a-1} \right\} > 0,$$
$$P_r = \sum_{a=4}^{r-1} K_{r,a} \max(\lambda_a, 0),$$
$$D_r = \sum_{a=4}^{r-1} K_{r,a} \max(-\lambda_a, 0).$$

Then $A_r^{\mathrm{shadow}}(N) = 0$ is attainable for some terminal row N if and only if

$$\boxed{D_r \geq 2^{r-1} + 2 + P_r.} \tag{34.77}$$

For each odd $r \geq 5$, the separate *choice $\lambda_4 = -1$ and $\lambda_5 = \cdots = \lambda_{r-1} = 0$ satisfies this threshold. In the typed higher-band Hessenberg realization, writing $a_{r,t} = (-1)^r p \eta_{r,t}$ gives*

$$H_{i+r-1,i} = -p\, i(i+1)\cdots(i+r-2)\eta_{r,i+r-1};$$

thus a negative lower stationary λ_a is equivalent to a positive *corresponding lower Hessenberg entry. This equivalence is formal and does not assert DPDA realizability.*

Proof. By (34.74), $L_r = 2^{r-1} - 3 + P_r - D_r \equiv 1 \pmod 6$. Nonnegative top sources imply $\lambda_{\mathrm{eff}}(N) \geq 0$, so cancellation requires $L_r < 0$, equivalently (34.77); equality $L_r = 0$ is excluded by its congruence. Conversely if $L_r < 0$, set $c = -L_r/(r-1)! = a/D > 0$ in lowest terms and choose $N = r!D$. Since $D \mid \binom{N}{r}$, the single nonnegative source pulse $\eta_{r,r} = c\binom{N}{r}$ attains the required shadow. Finally $K_{r,4} = 3^{r-1} - 3 \cdot 2^{r-1} + 3$, so the stated lower choice gives $L_r = 4 \cdot 2^{r-1} - 3^{r-1} - 6 < 0$ for odd $r \geq 5$. The Hessenberg identity is the explicit signed source realization of the established typed CCR recurrence. □

Remark 34.63 (Interpretation and a checkable example). At $r = 5$, $\lambda_4 = -1$ gives $L_5 = -23$ and $c_5 = 23/24$. The source $\eta_{5,5} = 759$ yields the target at $N = 12$, as $759/\binom{12}{5} = 23/24$. It cannot simultaneously be attained at $N = 16$ with nonnegative pulses confined to the interval $(12, 16]$, although that interval has weights of gcd one; 3427 is outside the semigroup generated by $495, 715, 1001, 1365$. At $(12, 18]$ the pulses $\eta_{5,13} = 3$, $\eta_{5,14} = 3$, $\eta_{5,15} = 2$, and $\eta_{5,17} = 1$ give the required increment 7452. This separates signed Bézout reachability from nonnegative semigroup reachability. The comparison remains at *one terminal raw degree-r coefficient*, not at the level of full p-adic normalized columns.

34.29 Explicit Galois laws at the seventh and ninth columns

The odd-column result above removes the double zero residue disk, but it leaves the nonzero residual roots to be determined. Orders seven and nine provide complete global root distributions at almost every prime, as an arithmetic consequence of the *previously known* Stirling polynomials. Bencherif–Garici [167] explicitly list the underlying quartic and sextic (with the opposite overall sign). We neither claim their polynomial formulas nor general Chebotarev theory as new. The present statement connects their Frobenius cycle types to *fixed-source CCR* prime-power root counts by Theorem 34.55.

Write

$$q_7(m) = 9m^4 - 54m^3 + 51m^2 + 58m + 16, \tag{34.78}$$

$$q_9(m) = 15m^6 - 165m^5 + 465m^4 + 17m^3 - 648m^2 - 548m - 144. \tag{34.79}$$

The Nörlund identities already used in Section 34.23 give

$$S_7(m) = \frac{m^2(m-1)q_7(m)}{5806080}, \qquad S_9(m) = \frac{m^2(m-1)q_9(m)}{2786918400}. \tag{34.80}$$

The factors $m-1$ are simple, and the two quotients are squarefree over $\mathbb{Q}$: directly,

$$\operatorname{disc}(q_7) = -2^4 3^5 \cdot 83 \cdot 97 \cdot 271,$$
$$\operatorname{disc}(q_9) = -2^{16} 3^3 5^2 \cdot 7 \cdot 263 \cdot 67891 \cdot 63049079.$$

Furthermore $q_7(0) = 16$, $q_7(1) = 80$, $q_9(0) = -144$ and $q_9(1) = -1008$, so these extra roots do not collide with 0 or 1 for sufficiently large primes away from the displayed discriminants.

Theorem 34.64 (Exact small-order Galois groups)**.** *For the explicit polynomials* (34.78)–(34.79)*,*

$$\operatorname{Gal}(q_7/\mathbb{Q}) \cong S_4, \qquad \operatorname{Gal}(q_9/\mathbb{Q}) \cong S_6.$$

Proof. The squarefree modular factor-degree certificates for q_7 are (4) at $p = 13$ and $(1, 3)$ at $p = 17$. Thus the Galois group is transitive and contains both a four-cycle and a three-cycle. The only transitive subgroup of S_4 with both types is S_4.

For q_9, the factor-degree certificates are (6) at $p = 29$, $(1, 5)$ at $p = 17$, and $(1, 1, 1, 1, 2)$ at $p = 607$. None of these primes divides $\operatorname{disc}(q_9)$. Consequently the Galois group $G \leq S_6$ is transitive and contains a five-cycle and a transposition. It is primitive: a nontrivial system of blocks in a transitive degree-six action has block size two or three, and a five-cycle cannot act nontrivially on either the blocks or any such block. Finally, the graph on six letters whose edges are the G-conjugates of the supplied transposition has a G-invariant component partition. Primitivity forces that graph to be connected; its edge transpositions generate S_6. Hence $G = S_6$. The stated factor degrees are determined by factoring the explicit integer polynomials (34.78)–(34.79) in the indicated finite fields. The discriminants follow by exact resultant evaluation, $\operatorname{disc}(q) = (-1)^{d(d-1)/2} \operatorname{Res}(q, q')/\operatorname{lc}(q)$ for $d = \deg q$. Thus the cycle-type witnesses and discriminant assertions are finite, reproducible algebraic certificates rather than additional hypotheses. □

For each odd r, let $A_r(\boldsymbol{\lambda})$ be the nonzero integer in (34.66). Define the two explicit finite bad-prime sets

$$\mathcal{B}_7(\boldsymbol{\lambda}) = \{p \text{ prime} : p \leq 14 \text{ or } p \mid A_7(\boldsymbol{\lambda})\, 83 \cdot 97 \cdot 271\},$$
$$\mathcal{B}_9(\boldsymbol{\lambda}) = \{p \text{ prime} : p \leq 18 \text{ or } p \mid A_9(\boldsymbol{\lambda})\, 263 \cdot 67891 \cdot 63049079\}.$$

Here the small-prime cutoffs also ensure that every denominator in the order-r normalized transport is p-invertible. Indeed the triangular operator T_j in Section 34.27 has diagonal entries $j + d$ for $0 \leq d \leq j$, all at most $2r$, and its right-hand sides have integral coefficients once the earlier columns are integral. Thus, by induction, the coefficients of S_r and its p-deformation are p-integral for all $p > 2r$. No claim is made at the explicitly excluded primes.

Theorem 34.65 (Fixed-source seventh/ninth prime-axis laws)**.** *Fix a stationary integer CCR source vector $\lambda_3 = 1, \lambda_4, \ldots, \lambda_r \in \mathbb{Z}$ and $r \in \{7, 9\}$. For any rational prime $p \notin \mathcal{B}_r$, and every $n \geq 2$, the number of solutions of the normalized r-th CCR column modulo p^n equals*

$$1 + \#\{u \in \mathbb{F}_p : q_r(u) = 0\},$$

and is also the number of distinct roots in $\mathbb{Q}_p$. Their natural densities, among all rational primes, are

root count $(r = 7)$	1	2	3	5
density	9/24	8/24	6/24	1/24

root count $(r = 9)$	1	2	3	4	5	7
density	265/720	264/720	135/720	40/720	15/720	1/720.

In both cases the prime-axis mean number of distinct roots converges to 2, independently of the fixed integral source vector.

Proof. For $p > 2r$, the original normalized transport reduces coefficientwise to S_r modulo p. By (34.80) and the exclusion of $\mathcal{B}_r$, every nonzero residual root comes either from $m = 1$ or a simple root of q_r and lifts uniquely modulo all p^n by Hensel's lemma. The double root at zero contributes no solution modulo p^2, by Theorem 34.55, because $p \nmid A_r$. This proves the exact count for every such prime, not merely a statistical limit.

Apply Chebotarev to the groups in Theorem 34.64. The number of roots of q_r modulo an unramified prime is the number of fixed points of the corresponding permutation. For S_4, the permutation counts with $0, 1, 2, 4$ fixed points are $9, 8, 6, 1$ out of $4! = 24$. For S_6, the counts with $0, 1, 2, 3, 4, 6$ fixed points are $265, 264, 135, 40, 15, 1$ out of $6! = 720$ (the classical rencontres numbers $\binom{6}{k} D_{6-k}$). The unavoidable root $m = 1$ adds one to each. The finite excluded sets do not change the natural densities. In both symmetric groups the mean number of fixed points is one, so the total mean is two. □

Corollary 34.66 (Conditional all-odd orbit-count mean)**.** *Fix any odd $r \geq 5$. Over $\mathbb{Q}$, factor $S_r(m) = m^2(m-1)q_r(m)$ with $q_r \in \mathbb{Q}[m]$. Assume q_r is squarefree and $q_r(1) \neq 0$, and let t_r be its number of distinct irreducible factors over $\mathbb{Q}$. For any fixed integral stationary source vector, the mean number of $\mathbb{Q}_p$ roots of the normalized r-th column, averaged over rational primes $p \leq X$, tends to $1 + t_r$ as $X \to \infty$. The same mean holds for the number of roots modulo p^n for every fixed $n \geq 2$ outside finitely many source-dependent primes. In particular, if q_r is irreducible, the mean is exactly 2.*

Proof. The classical odd Nörlund factorization and its exact double zero at the origin are already cited in Section 34.23; the factor $m-1$ follows from vanishing odd Bernoulli numbers at $m = 1$. For a fixed r, away from finitely many coefficient-denominator, discriminant, resultant and A_r prime divisors, all nonzero residual roots are simple and lift, whereas the zero disk disappears modulo p^2 by Theorem 34.55. The remaining roots are the single root at $m = 1$ and the fixed points of Frobenius acting on the roots of q_r. By the Chebotarev density theorem the limiting average number of such fixed points is the group-average number of fixed points. Burnside's lemma identifies it with the number of Galois orbits, namely t_r, the number of irreducible factors of the squarefree rational polynomial q_r. The finite excluded prime set has zero density. □

Remark 34.67 (Exactly what is new here)*.* The explicit q_7, q_9 appeared in the 2017 Stirling-polynomial literature [167]. Their algebraic Galois computations and the Frobenius fixed-point technique are standard mathematical approaches. The *CCR-specific arithmetic result* of this section is the combination with the previously established integral source obstruction A_r, producing source-independent global p-adic root-count laws at all but finitely many primes. This is an arithmetic specialization, not a theorem about all odd r or actual DPDA source realizability; the underlying unperturbed polynomial identities are properly attributed to the Stirling literature.

34.30 Exact certificates, higher odd columns, and adelic laws

The following finite-depth congruence certificates are obtained by reducing the displayed integer coefficient formulas modulo prime powers, lifting derivative-critical residues by Hensel's lemma, and counting the resulting congruence classes. They supplement, but do not replace, the exact all-depth proofs in Lemma 34.24 and Theorem 34.40.

The 419-adic coordinate and its domain. A digitwise Hensel lift of the derivative-zero center followed by evaluation of the critical value gives the first six base-419 digits of the critical parameter as

$$\Lambda \equiv 132 + 91 \cdot 419 + 73 \cdot 419^2 + 157 \cdot 419^3 + 402 \cdot 419^4 + 44 \cdot 419^5 \pmod{419^6}.$$

For $\lambda = 0, 132, 551, 38261$, direct congruence lifting from the integer quartic, without using the closed solution-count formula, gives the following root counts:

λ	N_1	N_2	N_3
0	3	2	2
132	3	421	2
551	3	421	840
38261	3	421	421

The isometry claim concerns integral residue classes. As clarified in Lemma 34.24, half-integral-valuation algebraic roots are instead classified by the Newton polygon of the translated integral polynomial.

The 73-adic covariance. For the fifth-column quintic, reducing the original integer polynomial modulo 73, 73^2, and 73^3 and averaging over all $73^2 = 5329$ source residues $T \pmod{73^2}$ for fixed L gives

L	$d = v_{73}(\Lambda_{62} - \Lambda_0)$	$\mathbb{E}_T[M_{0,3}M_{62,3}]$	$\mathrm{Var}_T(N_3)$
0	0	0	286
9	1	72	430
812	2	145	576

Each individual fold has mean 1. These counts independently validate all three regimes of Theorem 34.40 at precision $n = 3$. For clarity, the *ordered* critical gap throughout this calculation is $\delta = \Lambda_{62} - \Lambda_0$, and its first residue is $\delta \equiv -2 - 16L \pmod{73}$. The opposite sign refers to the reverse ordered gap and must not be substituted into the product congruence without also reversing its sign. All formulas and independent counts below use this ordered gap convention; reversing the order also reverses the sign in the corresponding product congruence.

Methodological boundary. Analytic Morse coordinates, Hensel lifting, and such finite modular counts are established tools, not claimed as new [175]. These finite congruence identities apply at the stated precisions and under the given coordinate hypotheses; the general conclusions follow from the preceding symbolic proofs.

34.31 Haar-product independence across different odd CCR columns

The preceding fifth-column two-fold theorem couples two critical disks *inside one column* at the exceptional prime 73. We now consider one distinguished zero-residue fold in *each of several different odd columns*, all at the same prime. This is a separate assertion whose proof uses the original CCR-k triangular source structure [166], not the special 73-adic two-curve isometry.

Normalized p-integrality at large primes. The good-prime hypothesis below can be checked directly from the original stationary transport, rather than assumed implicitly. For $1 \le j \le R$ and $0 \le k \le j$, its triangular operator satisfies

$$T_j(m^k) = (j+k)m^k + \sum_{h=0}^{k-2} \binom{k}{h} m^{h+1},$$

where an empty sum means zero. On polynomials of degree at most j, T_j has integral triangular matrix and diagonal $j, j+1, \dots, 2j$. If $p > 2R$, all these diagonal entries are p-adic units, so T_j^{-1} preserves the module of p-integral polynomials of degree at most j. The original normalized CCR recurrence has p-integral polynomial right-hand side whenever the preceding normalized columns and the stationary sources are p-integral. Induction from $\Pi_{p,0} = 1$ therefore shows that every $\Pi_{p,j}$, $j \le R$, is coefficientwise p-integral for every source vector in $\mathbb{Z}_p$; specialization at $p = 0$ gives $\Pi_{p,j} \equiv S_j \pmod p$. The additional unit-curvature condition stated below ensures that the residual zero-disk double root is nondegenerate. No assertion is made for the finitely many smaller or curvature-bad primes.

Theorem 34.68 (Odd-column Haar product law)**.** *Fix a finite set $I \subseteq \{5, 7, 9, \dots, R\}$ of odd indices and put $d = |I|$. Let $p > 2R$ be an odd prime such that, for every $r = 2s+1 \in I$, the curvature*

$$c_s = -\frac{B_{2s}}{4s(2s)!} \in \mathbb{Z}_p^\times.$$

Work with the stationary normalized CCR polynomial extension, $\lambda_3 = 1$, and regard all source parameters with indices outside I as arbitrary fixed elements of $\mathbb{Z}_p$. Choose $\{\lambda_r : r \in I\}$ independently with normalized Haar measure on $\mathbb{Z}_p$. For $n \ge 1$, denote by $M_{r,n}$ the number of zeros of $\Pi_{p,r}(m)$ modulo p^n with $m \equiv 0 \pmod p$.

Then the variables $M_{r,n}$, $r \in I$, are independent and have the same single-fold law. More precisely, define

$$h = \left\lfloor \frac{n-1}{2} \right\rfloor, \quad q = p^{-(n-1)}, \quad a = p^{\lfloor n/2 \rfloor}, \quad w_j = \frac{p-1}{2p^{2j}}$$

and the probability polynomial

$$P_{p,n}(X) = \left(1 - q - \sum_{j=1}^{h} w_j\right) + qX^a + \sum_{j=1}^{h} w_j X^{2p^j}. \tag{34.81}$$

For $S_n = \sum_{r \in I} M_{r,n}$,

$$\mathbb{E}[X^{S_n}] = P_{p,n}(X)^d, \qquad \mathbb{E}[S_n] = d, \qquad \mathrm{Var}(S_n) = d(p-1)(n-1). \tag{34.82}$$

The number B of indices in I whose zero-residue quadratic factor has two distinct $\mathbb{Q}_p$*-roots satisfies*

$$B \sim \mathrm{Binomial}\left(d, \frac{1}{2(p+1)}\right). \tag{34.83}$$

The possible repeated-root critical parameters form a Haar-null set.

Proof. For each $r \in I$, the established normalized shadow has $S_r(m) = c_s m^2 + O(m^3)$ and a unit quadratic coefficient modulo p. The p-integral transport admits a unique derivative-critical center $c_r(\lambda_{<r}) \in p\mathbb{Z}_p$. This center depends only on the lower-order sources, because the already established exact stationary translation law for the top source is

$$\Pi_{p,r}^{(\lambda_r)}(m) = \Pi_{p,r}^{(0)}(m) - \frac{p}{r}\lambda_r$$

for odd r, with all lower sources fixed. Define $\Lambda_r(\lambda_{<r}) \in \mathbb{Z}_p$ by $\Pi_{p,r}^{(\Lambda_r)}(c_r) = 0$. Since the critical value of $\Pi^{(0)}$ is divisible by p and r is invertible, such a parameter exists uniquely.

The elementary p-adic quadratic coordinate in the disk $m \in p\mathbb{Z}_p$ produces an isometric coordinate $z_r \in p\mathbb{Z}_p$ satisfying

$$z_r^2 = K_r p U_r, \qquad U_r = \lambda_r - \Lambda_r(\lambda_{<r}), \qquad K_r = \frac{1}{rc_s} \in \mathbb{Z}_p^\times. \tag{34.84}$$

Here the analytic square-root normalization is applied only on the *integral residue disk*; field-factor ramification outside this disk, if any, follows separately from its Newton polygon.

The source substitution

$$(\lambda_4, \dots, \lambda_R) \longmapsto (\widetilde{\lambda}_4, \dots, \widetilde{\lambda}_R), \quad \widetilde{\lambda}_r = \begin{cases} \lambda_r - \Lambda_r(\lambda_{<r}), & r \in I, \\ \lambda_r, & r \notin I \end{cases}$$

is triangular with unit diagonal, and is invertible recursively: recover $\lambda_4, \lambda_5, \dots$ in increasing order. All critical functions are p-integral and compatible with reduction modulo every p^a by Hensel's implicit-function theorem. Hence this map is a bijection on each finite residue vector space and preserves product Haar measure. With parameters outside I fixed, the same recursive argument gives independent Haar U_r for $r \in I$.

The zero-disk count $M_{r,n}$ is now a function only of U_r via (34.84). Multiplication by each fixed unit K_r does not change a Haar-uniform distribution. Elementary quadratic valuation counting gives probability q for the fully resonant count a, probability w_j for a split count $2p^j$ at each $1 \le j \le h$, and the remaining mass at zero. This is (34.81). In particular, $\mathbb{E}M_{r,n} = 1$ and $\mathrm{Var}(M_{r,n}) = (p-1)(n-1)$, the universal one-fold moment formulas already established in Section 34.23. Independence proves (34.82).

Finally, a quadratic branch splits with two distinct $\mathbb{Q}_p$ roots exactly when $v_p(U_r) = 2j-1$ for some $j \ge 1$, and the leading unit of $K_r U_r$ is a square. Its probability is

$$\sum_{j \ge 1} \frac{p-1}{2p^{2j}} = \frac{1}{2(p+1)}.$$

The branch events are independent, proving (34.83). □

Remark 34.69 (What is and is not independent). Even when the lower sources are correlated through the higher transport, the upper-source coordinates give exact independence of these *distinguished local folds*. Nothing here asserts independence of the *full* root counts from all other residue disks, or of arbitrary nonstationary/physically realized source data. Triangular Haar changes and one-dimensional quadratic congruence counting are classical methods; the precise CCR application follows from the pre-existing top-source translation theorem.

Finite-precision two-column illustration. Direct polynomial reduction using the full normalized CCR transport (rather than the local normal form (34.84)) fixes $\lambda_4 = \lambda_6 = 0$ and enumerates every $(\lambda_5, \lambda_7) \in \mathbb{F}_p^2$ for $p = 17, 19, 23$. For every $m \in p\mathbb{Z}/p^2\mathbb{Z}$ the resulting joint distribution shows that $(M_{5,2}, M_{7,2})$ takes values $(0,0), (p,0), (0,p), (p,p)$ with multiplicities $(p-1)^2, p-1, p-1, 1$, respectively. These three complete finite-precision distributions are consistent with the independent all-depth triangular/Hensel proof.

34.32 Adelic resonance dichotomy: fixed integer sources versus profinite Haar sources

The all-odd fixed-integer source obstruction and the multicolumn Haar-product theorem have complementary quantifiers. The former fixes an integer vector and varies the prime; the latter varies source coordinates in a single p-adic space. They admit an exact common adelic formulation, provided these two ambient spaces are not conflated.

Fix a nonempty finite set of odd column orders $I \subseteq \{5, 7, \dots, R\}$ and write $d = |I|$. Let $\mathcal{G}_R$ be the cofinite set of primes $p > 2R$ at which all selected normalized columns are

coefficientwise p-integral and their quadratic curvatures $c_{(r-1)/2}$ are p-units. This exclusion set depends on R, not on the source parameters. Define the profinite source space

$$\widehat{\mathbb{Z}}^{R-3} = \prod_p \mathbb{Z}_p^{R-3},$$

with independent normalized Haar factors and $\lambda_3 = 1$. For $p \in \mathcal{G}_R$ and $r \in I$, let $E_{p,r}$ be the event that the distinguished $m \equiv 0 \pmod p$ fold of the r-th CCR column has two distinct $\mathbb{Q}_p$ roots. No condition is placed on other root residue disks.

Theorem 34.70 (Adelic CCR prime-by-column product law)**.** *On the above profinite Haar source space, the events $\{E_{p,r} : p \in \mathcal{G}_R,\ r \in I\}$ are mutually independent, with*

$$\mathbb{P}(E_{p,r}) = q_p := \frac{1}{2(p+1)}. \tag{34.85}$$

Writing $N_I(X) = \sum_{\substack{p \le X \\ p \in \mathcal{G}_R}} \sum_{r \in I} \mathbf{1}_{E_{p,r}}$, we have

$$\mu_I(X) := \mathbb{E}N_I(X) = \frac{d}{2} \log\log X + O_R(1), \tag{34.86}$$

$$\mathrm{Var}(N_I(X)) = \frac{d}{2} \log\log X + O_R(1). \tag{34.87}$$

Moreover, almost surely,

$$\frac{N_I(X)}{\frac{d}{2}\log\log X} \longrightarrow 1, \tag{34.88}$$

and the centered count satisfies the standard central limit theorem

$$\frac{N_I(X) - \mu_I(X)}{\sqrt{\mathrm{Var}(N_I(X))}} \implies \mathcal{N}(0,1).$$

For some positive constant $C_{I,R}$ (depending on the finite excluded prime set),

$$\mathbb{P}(N_I(X) = 0) = \prod_{\substack{p \le X \\ p \in \mathcal{G}_R}} (1 - q_p)^d \sim \frac{C_{I,R}}{(\log X)^{d/2}}. \tag{34.89}$$

In particular, almost every profinite source vector has infinitely many split zero-disk CCR folds in each *chosen odd column.*

Proof. At each fixed good prime, the multicolumn triangular coordinates of the construction in Section 34.31 express the critical source difference as $U_r = \lambda_r - \Lambda_r(\lambda_{<r})$. As the chosen source coordinates vary independently in $\mathbb{Z}_p$, the U_r are independent Haar-uniform. The local quadratic equation $z_r^2 = K_r p U_r$ has two distinct $\mathbb{Q}_p$ roots precisely when $v_p(U_r)$ is positive odd and the corresponding leading unit satisfies the quadratic-character condition. Since every K_r is a unit, the exact probability is

$$\sum_{j \ge 1} \frac{p-1}{2p^{2j}} = \frac{1}{2(p+1)}.$$

The profinite Haar space is the product of the local source spaces *over primes*, so the already-proved within-prime independence extends to the entire prime-by-column family.

The reciprocal-prime Mertens estimate [177] and $q_p = \frac{1}{2p} + O(p^{-2})$ imply $\sum_{p \in \mathcal{G}_R,\ p \le X} q_p = \frac{1}{2}\log\log X + O_R(1)$. The sum of q_p^2 converges, proving the first two asymptotics. For completeness, enumerate the independent indicators as Y_j, put $\mu_n = \sum_{j \le n} \mathbb{E}Y_j \to \infty$, and observe

$$\sum_j \frac{\mathrm{Var}(Y_j)}{\mu_j^2} \le \sum_j \frac{\mathbb{E}Y_j}{\mu_j^2} < \infty,$$

where the initial finite terms are ignored and the final convergence follows by comparing increments of μ_j with $u^{-2}du$. Kolmogorov's convergence theorem and Kronecker's lemma then give $\sum_{j\le n}(Y_j-\mathbb{E}Y_j)/\mu_n\to 0$ almost surely. Because the centered indicators satisfy $|Y_j-\mathbb{E}Y_j|\le 1$ while their cumulative variance tends to infinity, Lindeberg's condition eventually has no nonzero summands, so the stated central limit theorem follows.

The constant in the no-splitting asymptotic has an explicit convergent Euler-product expression. Put $\mathcal{B}_R=\{p:\ p\notin\mathcal{G}_R\}$ and $a_p=(1-q_p)/(1-1/p)^{1/2}$ for $p\in\mathcal{G}_R$. Since $\log a_p=O(p^{-2})$, the product $\prod_{p\in\mathcal{G}_R}a_p$ converges to a strictly positive value. The prime-product form of Mertens' theorem gives

$$C_{I,R}=e^{-d\gamma/2}\prod_{p\in\mathcal{B}_R}(1-p^{-1})^{-d/2}\prod_{p\in\mathcal{G}_R}\left(\frac{1-1/(2(p+1))}{\sqrt{1-p^{-1}}}\right)^{d}>0,$$

where γ is Euler's constant. This supplies (34.89) without leaving the positive constant implicit. Applying the almost-sure conclusion to each of the finitely many individual columns proves the final assertion. □

Theorem 34.71 (Integral-source finiteness and the height-limit dichotomy)**.** *Fix all unselected source coordinates as ordinary integers, and let the selected coordinates λ_r range over ordinary integers. For* every fixed *integer source vector, only finitely many pairs (p,r) with $p\in\mathcal{G}_R$, $r\in I$ can satisfy $E_{p,r}$. In sharp contrast, let selected integer coordinates be chosen independently and uniformly in $[-H,H]\cap\mathbb{Z}$ and write $N_I(\infty)$ for their total number of split good-prime folds. Then, for every fixed nonnegative integer K,*

$$\lim_{H\to\infty}\mathbb{P}_H\big(N_I(\infty)\le K\big)=0. \tag{34.90}$$

Nevertheless $N_I(\infty)$ is finite for every *integer source vector, including all those sampled at height H.*

Proof. For any odd order $r\in I$, the CCR first-order source obstruction $A_r(\boldsymbol{\lambda})$ belongs to $1+d_r\mathbb{Z}$, so it cannot vanish for integral sources (in particular $A_r\equiv 1 \pmod 6$). The fixed-source theorem proves that a good-prime splitting implies $p\mid A_r(\boldsymbol{\lambda})$, giving finitely many such primes for each fixed integral source vector.

For a *fixed finite* collection of good primes and column orders, Chinese-remainder equidistribution of integer source vectors in large boxes gives the same joint limiting distribution as the product Haar source measure. Indeed, each local split event is a union of p-adic residue cylinders, and its only possible boundary lies on the critical-value equality locus $U_r=0$, of Haar measure zero. Thus its frequency among integer sources can be approximated from above and below by finite residue conditions. For fixed X, the law of $N_I(X)$ under the integer-box model therefore converges to the finite-prime law in Theorem 34.70. Since always $N_I(\infty)\ge N_I(X)$,

$$\limsup_{H\to\infty}\mathbb{P}_H(N_I(\infty)\le K)\le\mathbb{P}_{\mathrm{Haar}}(N_I(X)\le K).$$

In fact this limit can be obtained without invoking an additional almost-sure theorem: whenever $\mu_I(X)>K$, Chebyshev's inequality and the exact mean/variance estimates yield

$$\mathbb{P}_{\mathrm{Haar}}(N_I(X)\le K)\le\frac{\mathrm{Var}(N_I(X))}{(\mu_I(X)-K)^2}=O_{I,R,K}\left(\frac{1}{\log\log X}\right).$$

Because X is arbitrary after taking the height limit, the preceding lim sup is zero, proving (34.90). This does not claim any uniform bound in H and X simultaneously. □

Remark 34.72 (Quantifiers and historical scope). The integer lattice is countable and Haar-null inside the profinite source space. It is therefore essential *not* to conclude from the adelic almost-sure statement that any fixed ordinary integer source has infinitely many resonance

primes. The derived zero–one, strong-law, central-limit, and Mertens-product arguments use classical probability and analytic number theory; their proposed CCR contribution is solely the specific source-dependent resonance events to which the product laws apply. Neither a general full-root-count independence claim nor a theorem of physical DPDA source realization is intended.

Exact small-prime CRT certificate. For the fifth column with $\lambda_4 = 0$, reduce the integer numerator modulo p^3 and classify all $\lambda_5 \pmod{p^2}$ for $p = 7, 11, 13$ by the number of roots with $m \equiv 0 \pmod{p}$. The number of first-depth splitting source classes is $(p-1)/2 = 3, 5, 6$, respectively, with the unique critical class giving p roots and the nonsquare classes giving none. CRT therefore gives $3 \cdot 5 \cdot 6 = 90$ simultaneously splitting classes modulo $(7 \cdot 11 \cdot 13)^2 = 1002001$, an exact integer-source density $90/1002001$. In contrast, allowing *all* higher odd splitting valuations under profinite Haar gives the larger simultaneous probability $1/(16 \cdot 24 \cdot 28) = 1/10752$. The ordinary integer source $\lambda_5 = 774982$ is one explicit CRT witness. These finite certificates illustrate, but do not establish, the infinite-prime probability theorems.

34.33 Boundary of arithmetic specialization

The polynomial, stationary-source, and Haar-source conclusions do not identify arbitrary p-adic roots with physically realizable chronological machine histories. In particular, the results do not classify all nonstationary raw sources or unbounded physical realization constraints. Hessenberg principal-determinant polynomials Q_n and fixed normalized column polynomials $\Pi_{p,r}$ are different objects.

The preceding arithmetic theory supplies the stationary transport and its source hierarchy. We now sharpen one near-leading coefficient without repeating the general CCR-k construction.

34.34 Sharp p-adic rigidity for the third near-leading coefficient

For the coefficient $[X^{r-3}]F_r(X)$ we prove a sharp good-prime valuation bound at $\lambda_3 = 1$ and an exact valuation law over the full third-source fiber $\lambda_3 \in \mathbb{Z}_p$. These coefficient-specific statements refine the all-odd-order source obstruction above and retain the same formal stationary transport setting.

Near-diagonal Stirling arithmetic is classical, and the Nörlund and Bernoulli polynomial formalisms provide a powerful setting for its congruences; see, e.g., [196, 169, 170, 198, 1]. The present section concerns a different, explicitly specified p-dependent *stationary CCR transport*. Its fixed-column polynomials retain an ordinary Stirling shadow modulo p, but contain additional connected-source corrections with nontrivial higher p-adic behavior. Our goal is a precise theorem for one particular high coefficient of this family, not a new general theorem for ordinary Stirling numbers.

For a prime p and stationary parameters $\lambda_a \in \mathbb{Z}_p$ $(a \geq 3)$, define $F_0(X) = 1$ and, for $j \geq 1$, define $F_j(X)$ by

$$\begin{aligned} T_j F_j(X) = {} & ((p+1)X - p)F_{j-1}(X) \\ & + p\big((p+1)(X-1) - p\big)F_{j-2}(X-1) \\ & + p\sum_{a=3}^{j}(-1)^a \lambda_a F_{j-a}(X-a+1), \end{aligned} \tag{34.91}$$

with missing negative-index terms omitted and

$$(T_j f)(X) = Xf(X+1) - (X-j)f(X).$$

The operator T_j is triangular with nonzero diagonal on $\mathbb{Q}_p[X]$ and hence invertible. Thus (34.91) is a self-contained definition of the polynomial family studied here; it is the same stationary normalized transport already used earlier in this arithmetic branch. Write

$$C_{r,p}^{(\theta)} = [X^{r-3}]F_r, \qquad \lambda_3 = \theta,$$

and suppress θ when it equals 1. Higher sources λ_a, $a \geq 4$, will be arbitrary.

Theorem 34.73 (Sharp rigidity at the canonical source)**.** *For every odd integer $r \geq 5$, every prime $p > 2r$, and every choice of higher stationary sources $\lambda_a \in \mathbb{Z}_p$ ($a \geq 4$), the normalization $\lambda_3 = 1$ satisfies*

$$\boxed{0 \leq v_p(C_{r,p}) \leq 2.} \tag{34.92}$$

The bound is sharp: $v_p(C_{7019,21569}) = 2$.

For the second result, set

$$P(r) = 5r^2 - 45r + 64$$

and let $H(r,p)$ be the explicit integer polynomial in (34.95) below. The source fiber exhibits a striking contrast with Theorem 34.73.

Theorem 34.74 (Complete third-source valuation law)**.** *Fix odd $r \geq 5$ and prime $p > 2r$. Allow $\lambda_3 = \theta$ to range over $\mathbb{Z}_p$, with arbitrary fixed higher sources. Then*

(i) *If $p \nmid P(r)$, then $v_p(C_{r,p}^{(\theta)}) = 0$ for every $\theta \in \mathbb{Z}_p$.*

(ii) *If $p \mid P(r)$, the element*

$$\theta_* = 1 + \frac{H(r,p)}{17280p} \in \mathbb{Z}_p \tag{34.93}$$

is well defined and unique, and

$$\boxed{v_p(C_{r,p}^{(\theta)}) = 1 + v_p(\theta - \theta_*),} \tag{34.94}$$

with the convention $v_p(0) = +\infty$. The affine source map $\theta \mapsto C_{r,p}^{(\theta)}$ is a bijection $\mathbb{Z}_p \to p\mathbb{Z}_p$. Every positive finite valuation is attained by infinitely many ordinary integer values of θ; $\theta = \theta_*$ makes the coefficient vanish.

The uniform exclusion uses an exact finite computation, independently verified below. As in the preceding CCR arithmetic, these conclusions are claims about the formal stationary polynomial family, not about arbitrary physical automaton realizations or general Stirling-number valuations.

34.35 The high-jet formula and source translation

Define

$$\begin{aligned} U_1(r) &= 35r^3 + 997r - 6816, \\ U_2(r) &= 49r^3 - 180r^2 - 925r - 384, \\ U_3(r) &= 49r^3 - 1128r^2 + 6047r - 7584, \end{aligned}$$

and let

$$H(r,p) = -(r-3)P(r) + 3pU_1(r) - 15p^2U_2(r) + 35p^3U_3(r). \tag{34.95}$$

Proposition 34.75 (Exact normalized high coefficient)**.** *For $r \geq 3$ and $\lambda_3 = 1$,*

$$C_{r,p} = E_{r,p}\, H(r,p), \qquad E_{r,p} = \frac{r(r-1)(r-2)(p+1)^{r-3}}{6480\, 2^r r!}. \tag{34.96}$$

The expression is independent of λ_a for $a \ge 4$. For $p > 2r$, the factor $E_{r,p}$ is a p-adic unit, whence

$$v_p(C_{r,p}) = v_p(H(r,p)).$$

More generally, for $\lambda_3 = \theta$ one has the exact formula

$$C_{r,p}^{(\theta)} = E_{r,p}\big[H(r,p) - 17280p(\theta-1)\big]. \tag{34.97}$$

Proof. Source order a enters polynomial degrees at most $r-a$, so the coefficient of X^{r-3} is independent of λ_a for $a \ge 4$. To make the coefficient induction explicit, write $F_j(X) = \sum_{k=0}^{j} f_{j,k} X^k$ and denote the right-hand side of (34.91) by $\mathcal{R}_j(X)$. Direct expansion of T_j gives, for $0 \le k \le j$,

$$(j+k) f_{j,k} + \mathbf{1}_{k\ge 1} \sum_{d=k+1}^{j} \binom{d}{k-1} f_{j,d} = [X^k]\mathcal{R}_j(X).$$

This triangular identity, together with $F_0 = 1$, determines each column from its highest coefficient downward. For $h = 0,1,2,3$, normalize the four top jets by

$$J_h(j,p) = \frac{2^j j!}{(p+1)^{j-h}} [X^{j-h}] F_j.$$

Comparing powers of X in (34.91) gives, with $\lambda_3 = 1$,

$$\begin{aligned}
J_0 &= 1,\\
J_1 &= \frac{j}{6}\big[p(7j-19) - (j-1)\big],\\
J_2 &= \frac{j(j-1)}{72}\big[(j-3)(j-2) - 2p(7j^2 + j + 42)\\
&\qquad\qquad + p^2(49j^2 - 509j + 678)\big],\\
J_3 &= \frac{j(j-1)(j-2)}{6480} H(j,p).
\end{aligned}$$

Each is an identity of rational polynomials in the independent symbols (j,p); substitution into the transport recurrence, together with the initial cases, proves them by induction on j. These polynomial identities have also been checked by exact symbolic expansion. If λ_3 changes from 1 to θ, the J_0, J_1, J_2 jets are unchanged, while the difference $\delta J_3(j)$ satisfies

$$(2j-3)\delta J_3(j) - 2j\delta J_3(j-1) = -8j(j-1)(j-2)p(\theta-1).$$

Its initial value at $j=3$ gives $\delta J_3(j) = -(8/3)j(j-1)(j-2)p(\theta-1)$, which is precisely the shift in (34.97). Finally, if $p > 2r$, neither $6480\,2^r r!$ nor $r(r-1)(r-2)(p+1)$ is divisible by p; the factor $E_{r,p}$ is therefore a unit. □

34.36 A bounded cofactor obstruction

This section supplies the algebraic reduction that turns an *a priori* infinite collection of possible depth-three counterexamples into a finite integer certificate.

Suppose $p^3 \mid H(r,p)$ for an odd $r \ge 5$ and prime $p > 2r$. Reduction of (34.95) modulo p gives $p \mid P(r)$ since $0 < r-3 < p$. For $r = 5, 7, 9$, the values $P(r) = -36, -6, 64$ admit no such prime; hence $r \ge 11$. Put

$$m = \frac{P(r)}{p} \in 2\mathbb{Z}_{>0}, \qquad 0 < m < \frac{5p}{4}.$$

The parity follows directly from oddness of r, and the inequality from $r < p/2$.

Introduce

$$\begin{aligned} K(m) &= 13m^2 - 279612m + 1487901312, \\ V(m) &= -m^3 + 17616m^2 - 82615752m + 9313947648, \\ R(m) &= 76867815144960 - 8768830680m, \\ G(m,p) &= 338K(m) - pR(m). \end{aligned}$$

Lemma 34.76 (Two necessary cofactor digits)**.** *Under the depth-three hypothesis, $p \mid K(m)$ and $p^2 \mid G(m,p)$.*

Proof. Divide the original polynomial $H(X,p)$ by $P(X)-mp$. An exact integral polynomial remainder calculation gives

$$H(X,p) \equiv p\big(A(m,p)X + B(m,p)\big) \pmod{P(X) - mp},$$

where $A, B \in \mathbb{Z}[m,p]$ are printed in Section 34.42. Define

$$N(m,p) = 5B^2 + 45AB + (64 - mp)A^2.$$

The Vieta identity for the roots X and $9 - X$ of $P(X) - mp$ states

$$5(AX + B)(A(9 - X) + B) - N(m,p) = -A^2(P(X) - mp).$$

At $X = r$, the congruence becomes $H(r,p) = p(Ar+B)$. Hence $p^3 \mid H$ implies $p^2 \mid (Ar+B)$, and therefore $p^2 \mid N$, without needing any assumption on the conjugate root. The exactly computed expansion

$$N(m,p) = -2K(m) + pV(m) + p^2W(m,p), \qquad W \in \mathbb{Z}[m,p],$$

implies $p \mid K(m)$ and $p^2 \mid -2K(m) + pV(m)$. Polynomial division gives

$$169V(m) = (-13m - 50604)K(m) + R(m).$$

Multiplying by 169 and using $p \mid K$ now yields $p^2 \mid 338K(m) - pR(m) = G(m,p)$. □

Lemma 34.77 (Bounded integer and square obstruction)**.** *Under the same hypothesis there is an integer u such that*

$$-7667 \le u \le 2626, \tag{34.98}$$

and an admissible prime divisor $p \ge 23$, $p \notin \{29, 149\}$, of

$$Q(u) = 169u^2 + 4455261180u - 81299205874753920, \tag{34.99}$$

for which

$$\Delta(u,p) = (108993600 + 1040u)p^2 + 145398240p + 503620 \tag{34.100}$$

is a square in $\mathbb{Z}$.

Proof. We use the exact identity

$$(13m - 5220r - 116316)(13m + 5220r - 163296) = 13K(m) - 5449680P(r). \tag{34.101}$$

Since $p \mid P(r)$ and $p \mid K(m)$, it divides at least one factor. Let $\sigma = +1$ select the first or $\sigma = -1$ the second factor, and let u be that factor divided by p. Writing $x = 10r - 45$ gives

$$13m = up + \sigma 522x + 139806, \qquad x^2 - 745 = 20mp. \tag{34.102}$$

Using $0 < m/p < 5/4$, $0 < r/p < 1/2$ and $p \ge 23$ yields $-7667 \le u \le 16$ when $\sigma = +1$ and $-7099 \le u \le 2626$ when $\sigma = -1$. Thus (34.98) is universal.

Set

$$T_\sigma = 13(x^2 - 745) - 20p(up + \sigma 522x + 139806),$$
$$A_u = 352872u + 4651292671920, \qquad D_0 = 246462892177680.$$

A direct polynomial identity is

$$13G\left(\frac{up + \sigma 522x + 139806}{13}, p\right) = 7084584T_\sigma + p\big[p(338u^2 + 8910522360u) + \sigma A_u x + D_0\big]. \tag{34.103}$$

At an actual pair $T_\sigma = 0$, and Lemma 34.76 gives $p^2 \mid G$. Thus $p \mid \sigma A_u x + D_0$. But $x^2 \equiv 745 \pmod p$ by (34.102), so

$$p \mid D_0^2 - 745A_u^2 = -548913568320\,Q(u). \tag{34.104}$$

The fixed factor is $2^6 \cdot 3^4 \cdot 5 \cdot 13^2 \cdot 29^2 \cdot 149$. Every possible prime $p \geq 23$ dividing it is 29 or 149. Both are excluded separately: at $p = 29$ the only eligible odd base root is $r = 11$ and $p^2 \nmid H(11, 29)$; at $p = 149$ no eligible odd base root satisfies $r < p/2$. Thus $p \mid Q(u)$.

Finally, $T_\sigma = 0$ is the integer quadratic equation

$$13x^2 - \sigma 10440px - 20up^2 - 2796120p - 9685 = 0.$$

Its discriminant is precisely (34.100), which must be a square for integral x. □

34.37 Exhaustive finite exclusion and sharpness

Lemma 34.78 (Exhaustive arithmetic certificate)**.** *No integer $u \in [-7667, 2626]$ and prime $p \geq 23$, $p \notin \{29, 149\}$, satisfy both $p \mid Q(u)$ and the condition that $\Delta(u, p)$ be a perfect square.*

Computer-assisted proof. The convex quadratic $Q(u)$ is strictly negative on the stated interval (its two endpoint values are negative). For every u in this interval, the nonzero integer $|Q(u)|$ has

$$0 < |Q(u)| \leq 81{,}333{,}354{,}427{,}912{,}739 < 2^{57}.$$

Factor all 10,294 such integers, check the primality and product of their prime factors, and test (34.100) by exact integer square roots for every admissible prime divisor. Independent exact-arithmetic checks, using complete integer factorization, primality verification and arbitrary-precision integer square roots, agree on the following counts:

Finite-certificate quantity	Exact count
Integer parameters u	10,294
Distinct prime-factor occurrences	37,539
Eligible (u, p) records	24,248
Square discriminants among eligible records	0

The finite calculation is exhaustive because every putative counterexample must produce a prime divisor of one of the explicitly specified nonzero integers $Q(u)$; there is no empirical bound on p. The canonical serialization and SHA-256 fingerprint of the complete eligible-pair enumeration are specified in Section 34.43. □

Proof of Theorem 34.73. If $v_p(C_{r,p}) \geq 3$, Proposition 34.75 gives $p^3 \mid H(r, p)$. Lemmas 34.76 and 34.77 produce (u, p) satisfying the two necessary conditions contradicted by Lemma 34.78. Hence $v_p(C_{r,p}) \leq 2$. Nonnegativity follows from (34.96), since $E_{r,p}$ is a p-adic unit and H is integral. At $r = 7019$ and $p = 21569$, direct evaluation of (34.95) satisfies $p^2 \mid H$ and $p^3 \nmid H$, and $p > 2r$. Thus the upper bound is attained. □

34.38 Exact source fibers and resonance engineering

Proof of Theorem 34.74. Proposition 34.75 yields

$$C_{r,p}^{(\theta)} = E_{r,p}\big[H(r,p) - 17280p(\theta-1)\big],$$

with $E_{r,p} \in \mathbb{Z}_p^\times$. As $p > 2r \geq 10$, the integer 17280 is also a p-adic unit. Furthermore,

$$H(r,p) \equiv -(r-3)P(r) \pmod{p},$$

and $r-3$ is a unit. If $p \nmid P(r)$, the coefficient is a unit regardless of θ. If $p \mid P(r)$, then $p \mid H$ and θ_* of (34.93) belongs to $\mathbb{Z}_p$. Substitution gives

$$C_{r,p}^{(\theta)} = -17280pE_{r,p}(\theta - \theta_*). \tag{34.105}$$

Multiplication by $-17280E_{r,p} \in \mathbb{Z}_p^\times$ proves the bijection and valuation formula. For any $k \geq 1$, choose integer $\theta \equiv \theta_* + p^{k-1} \pmod{p^k}$; then $v_p(C_{r,p}^{(\theta)}) = k$. Infinitely many such ordinary integers exist. Choosing the p-adic value $\theta = \theta_*$ makes (34.105) zero. □

Corollary 34.79 (Geometric depth distribution)**.** *For a fixed order–prime pair with $p \mid P(r)$, if θ is sampled according to normalized Haar measure on $\mathbb{Z}_p$, then*

$$\Pr\big(v_p(C_{r,p}^{(\theta)}) = k\big) = \frac{p-1}{p^k}, \qquad k = 1, 2, 3, \ldots.$$

The infinite-depth fiber is a singleton of Haar measure zero.

Proof. By (34.94), the event of depth k is $\theta - \theta_* \in p^{k-1}\mathbb{Z}_p \setminus p^k\mathbb{Z}_p$, whose Haar measure is $p^{-(k-1)} - p^{-k}$. □

Remark 34.80 (A fixed arithmetic witness). For $(r,p) = (7019, 21569)$ the canonical choice $\theta = 1$ has depth two, but the *same pair* admits every prescribed larger depth by altering θ. For example, $\theta = 281712710$ produces depth three and $\theta = 2554814402361$ produces depth four; both are checked from the exact integer polynomial H. The freedom in θ does not contradict Theorem 34.73: the canonical frozen source is its essential hypothesis.

34.39 Sheffer structure and classical comparison

The fixed-column generating function admits an exact formal description which permits a precise, albeit limited, comparison with classical Bernoulli and Nörlund templates.

Proposition 34.81 (Stationary Sheffer factorization)**.** *Let $\Phi(X,t) = \sum_{j\geq 0} F_j(X)t^j$. There are unique formal power series $A(t)$ and $B(t)$ with $A(0) = 1$ and $B(0) = 0$ such that*

$$\Phi(X,t) = A(t)e^{XB(t)}.$$

They satisfy

$$e^B - 1 + tB' = (p+1)t + p(p+1)t^2e^{-B}, \tag{34.106}$$

$$t(\log A)' = -pt - p(2p+1)t^2e^{-B} + p\sum_{a\geq 3}(-1)^a\lambda_a t^a e^{(1-a)B}. \tag{34.107}$$

In particular, B is independent of the connected-source parameters.

Proof. Multiply (34.91) by t^j, sum over $j \geq 1$, and use the initial column $F_0 = 1$. One obtains

$$\begin{aligned} X(\Phi(X+1,t) - \Phi(X,t)) + t\partial_t \Phi(X,t) = {} & ((p+1)X - p)t\Phi(X,t) \\ & + p((p+1)(X-1) - p)t^2\Phi(X-1,t) \\ & + p\sum_{a\geq 3}(-1)^a \lambda_a t^a \Phi(X-a+1,t). \end{aligned}$$

Insert $\Phi = Ae^{XB}$, divide by Φ, and equate coefficients of the independent variable X; this gives (34.106)–(34.107). Both equations determine their coefficients uniquely by formal recursion; invertibility of each T_j ensures they reproduce the column recurrence. □

Writing $s = B(t)$, which is an invertible formal coordinate since $[t]B = (p+1)/2$, the source-normalized model $\lambda_3 = 1$ has

$$\log A(t(s)) = c_1 s + c_2 s^2 + c_3 s^3 + \cdots, \quad c_2 = -\frac{p(5p+7)}{3(p+1)^2}, \quad c_3 = \frac{p(35p^2+56p-19)}{9(p+1)^3}. \tag{34.108}$$

Both c_2 and c_3 are nonzero for positive p. A pure binomial-type family $\exp((aX+b)g(t))$ has vanishing s^2 coefficient in this coordinate, even after an invertible formal change of t and an invertible affine change of X. Likewise the standard generalized Bernoulli family $(s/(e^s-1))^\alpha e^{Xs}$, with fixed order α, has zero cubic coefficient in $\log(s/(e^s-1))$. Consequently (34.108) rules out these *specific direct forms* for the CCR columns, under the indicated transformations.

These distinctions concern the stated transformations only; they are not a classification of all possible umbral equivalences or implications of earlier arithmetic theorems. In particular, Adelberg's finite-difference and Nörlund works [169, 170] are acknowledged as important related theory. We do not claim that Sheffer structure, finite-difference techniques, or Nörlund congruences originated here. Instead, the present arithmetic theorem concerns the explicit p-dependent lift defined by (34.91) and the unusually restrictive canonical-source depth bound.

34.40 Computational reproducibility and scope

The computer-assisted part of the proof is only Lemma 34.78. Its finite input is defined by (34.99) over all integers $-7667 \leq u \leq 2626$. Each nonzero value $|Q(u)|$ is factored exactly; every factor is checked for primality and the factorization product is reconstructed. For each eligible prime divisor, the discriminant is tested using an exact integer square root. Independent exact-integer calculations reproduce the counts in the lemma. Neither floating-point tolerances, random-prime sampling nor an unproved cutoff for counterexample primes enters the argument.

The theorem concerns formal stationary polynomial columns with specified source restrictions. It neither classifies all higher-source laws nor asserts physical automaton realizability or settles cross-order Stirling polynomial common-root problems. These distinctions are essential when comparing the present result to classical Stirling and generalized Bernoulli literature [196, 198, 197, 12, 1].

34.41 Novelty boundary and related arithmetic literature

The cited finite-difference, Nörlund and p-adic Stirling literature supplies classical techniques, not the coefficient-specific bound of Theorem 34.73. The refinement concerns the p-dependent stationary CCR transport in (34.91); the Sheffer calculation gives a structural comparison, not a classification of all umbral equivalences.

Contribution and novelty statement. The coefficient-specific contributions here are the uniform, sharp inequality $0 \le v_p([X^{r-3}]F_r) \le 2$ for odd $r \ge 5$, $p > 2r$, and $\lambda_3 = 1$, independent of the higher stationary sources; the witness $(r,p) = (7019, 21569)$; and the exact affine source-fiber valuation law in (34.94). A finite arithmetic exclusion, with an explicitly specified exhaustive verification procedure and a reproducibility fingerprint, supports the uniform upper bound. No broader optimal valuation theorem for all CCR coefficients or for classical Stirling polynomials is claimed.

34.42 Technical identities for the coefficient bound

In Lemma 34.76, division of $H(X,p)$ by $P(X)-mp$ yields the linear remainder $p(AX+B)$, where

$$A(m,p) = 343mp^3 - 147mp^2 + 21mp - m - 26712p^2 - 11952p + 10152,$$
$$B(m,p) = -3\big(1603mp^3 + 261mp^2 - 63mp - m - 14112p^2 - 18624p + 10848\big).$$

The two further identities

$$(13m - 5220r - 116316)(13m + 5220r - 163296) = 13K(m) - 5449680P(r)$$

and

$$D_0^2 - 745A_u^2 = -548913568320Q(u)$$

are integral polynomial identities; direct expansion verifies both, and the identities have additionally been checked by exact symbolic algebra.

34.43 Exact reproduction of the finite exclusion

The exact finite verification can be reconstructed solely from the integer formulas in the proof. For each integer $u = -7667, -7666, \dots, 2626$, compute $|Q(u)|$ from (34.99), factor it into positive prime factors, verify primality and that the product of the prime powers equals $|Q(u)|$, and retain the distinct prime divisors $p \ge 23$ other than 29 and 149. For each retained pair (u,p), calculate $\Delta(u,p)$ from (34.100) and check whether the nonnegative integer is a square by comparing it with the square of its integer square root. The independently reproduced totals agree with all four exact counts in Lemma 34.78. For an exact reproducibility fingerprint, order the eligible pairs first by increasing u and, for each fixed u, by increasing p. Serialize them as UTF-8 text with an initial line `u,p`, followed by one `u,p` line per pair, with no spaces, a comma delimiter and an LF newline after every line. The SHA-256 digest of precisely this canonical serialization is (concatenate the two lines):

```
09ffb1e53a700529e73f1d9be145ad8b
794a1b0856522166d64cd1d9abc37c4c
```

The fingerprint provides an additional equality check on independently reconstructed enumerations; the exhaustiveness argument remains the exact finite reduction in Lemma 34.78.

The next refinement uses a different polynomial family: primitive Nörlund–Stirling quotients, with no connected-source parameters. Its finite-root and density results belong to the same arithmetic narrative but do not follow by identifying these quotients with the CCR transport columns.

34.44 Prime-indexed Nörlund–Stirling quotient pairs and single-site rigidity

Put

$$S_n(X) = [z^n]\Big(\frac{z}{1-e^{-z}}\Big)^X, \qquad S_n(X) = \frac{X^2(X-1)}{D_n} q_n(X) \quad (n \ge 5 \text{ odd}),$$

where $q_n \in \mathbb{Z}[X]$ is primitive and D_n is the corresponding positive integer normalizer. The good-prime normalization needed below is proved directly: if $p > n+1$, then $p \nmid D_n$, as explained after the definition of the prime-indexed pair. For clarity our prime-indexed pair is

$$p \equiv 1 \pmod 8, \qquad p \geq 41, \qquad h = (p-1)/4,$$
$$f_p = q_{h-1} = q_{(p-5)/4}, \qquad g_p = q_{2h-1} = q_{(p-3)/2}.$$

The indices are odd and satisfy the good-prime inequality $p > 2(2h-1) = p-3$. For a fixed negative centre $-a$ we impose $2 \leq a < (p+3)/2$; this implies $(2h-1+a)!$ is a p-unit, and $a < p$, so the negative-centre Stirling identity transfers to the primitive quotients without hidden divisions by p.

To justify the normalization, write $H(z) = (1-e^{-z})/z = \sum_{k\geq 0}(-1)^k z^k/(k+1)!$. If $p > n+1$, then H, its inverse $A = 1/H$, $\log A$ and $\exp(X \log A)$ are coefficientwise in $\mathbb{Z}_{(p)}[X]$ through degree n: every denominator introduced by the truncated logarithmic and exponential series is a product of integers smaller than p. Thus $S_n(X) \in \mathbb{Z}_{(p)}[X]$. For odd $n \geq 5$, the identities $\log A(z) = z/2 +$ even powers of z and $A(z) = 1 + z/2 +$ even powers of z show that S_n has a double zero at $X = 0$ and a zero at $X = 1$. Dividing by the monic $X^2(X-1)$ preserves p-integrality. Since q_n is primitive over $\mathbb{Z}$ and $S_n/[X^2(X-1)] = q_n/D_n$, Gauss's lemma implies $p \nmid D_n$.

Stirling exponential sums, their congruences, quartic power residues, and Chebotarev's theorem are classical; see [171, 172, 169, 170, 198, 201]. The assertion studied here is more specific: *does a common root at one prescribed site force the entire preceding consecutive block?* The answer is asymptotically yes for an explicit finite range of sites, but not literally always: we prove that a unique exceptional pair occurs for $2 \leq a \leq 12$. In contrast with the preceding source-dependent CCR transport, the primitive quotients here are source-independent and independently defined; none of their proofs identifies the two polynomial families.

The distinction from prescribed-prefix results is logical: the hypothesis is a single common root at $-a$, not a preassigned prefix of common roots. The analysis establishes an effective no-cancellation criterion and verifies it for $2 \leq a \leq 28$, classifies every eligible prime for $2 \leq a \leq 12$, derives exact fixed-site densities, and separates resultant multiplicity from lifting depth in a local example. The unique sporadic failure is $(a,p) = (10, 3617329)$. The Gaussian, Kummer and finite-difference inputs remain classical; the specialized single-site implication and its exception are the claimed new results.

34.45 Finite-difference reduction

For integers $a \geq 2$, put

$$w_{a,b} = (-1)^{a-b}\binom{a}{b} b^{a-1} \quad (1 \leq b \leq a), \qquad \sum_{b=1}^{a} w_{a,b} = 0.$$

The last equality is the a-th finite difference of a polynomial of degree $a-1$. Write $\left\{{N \atop a}\right\}$ for the Stirling number of the second kind.

Lemma 34.82 (Exact negative-centre identity)**.** *For every odd $n \geq 5$,*

$$-(n+a)!S_n(-a) = a!\left\{{n+a \atop a}\right\} = \sum_{b=1}^{a}(-1)^{a-b}\binom{a}{b} b^{n+a}.$$

Consequently, for $n_j = jh-1$ and $j = 1, 2$,

$$-(n_j+a)!S_{n_j}(-a) = \sum_{b=1}^{a} w_{a,b}(b^h)^j.$$

Proof. Write $S_n(-a) = [z^n]((1-e^{-z})/z)^a$ and expand the numerator binomially; for odd n the sign is negative. The second equality is the standard Stirling exponential sum [171, 172, §26.8]. Substitute $n_j + a = jh + a - 1$ for the final identity. □

Let $\mu_4 = \{1, -1, i, -i\} \subset \mathbb{Z}[i]$ and suppose $p \equiv 1 \pmod 8$. For each prime $r \le a$, choose $e_r \in \{0, 1, 2, 3\}$ such that $r^h = i_p^{e_r}$ for a fixed primitive fourth root $i_p \in \mathbb{F}_p$. Extend the digits additively to every $b \le a$ by prime factorization, giving $b^h = i_p^{e_b}$. Form the two Gaussian moments

$$E_j(e) = \sum_{b=1}^{a} w_{a,b} i^{je_b} \in \mathbb{Z}[i], \qquad j = 1, 2.$$

If $f_p(-a) = g_p(-a) = 0 \pmod p$, then $E_1(e), E_2(e)$ vanish under $\mathbb{Z}[i] \to \mathbb{F}_p$ sending $i \mapsto i_p$, and hence p divides their respective Gaussian norms. This implication is elementary, but the use of *two* moments is essential.

Lemma 34.83 (No-cancellation criterion)**.** *Suppose no nonempty proper subset $I \subset \{1, \dots, a\}$ has $\sum_{b \in I} w_{a,b} = 0$ in $\mathbb{Z}$. Then the only assignment of fourth-root digits for prime bases $r \le a$ for which $E_1(e) = E_2(e) = 0$ as Gaussian integers is the all-zero assignment.*

Proof. Let A, B, C, D be the sums of weights in the four classes where i^{e_b} respectively equals $1, -1, i, -i$. Then $E_1 = (A - B) + i(C - D)$ and $E_2 = A + B - C - D$. With $A + B + C + D = 0$, the equations $E_1 = E_2 = 0$ imply $A = B = C = D = 0$. Every nonempty class is therefore a subset of zero weight. The hypothesis permits only one nonempty class; because $b = 1$ occurs with $1^h = 1$, this is the class A, and all $b^h = 1$, hence all prime-base digits vanish. □

34.46 An effective single-site rigidity theorem

Define the integer

$$T_a = \sum_{b=1}^{a} |w_{a,b}| = \sum_{b=1}^{a} \binom{a}{b} b^{a-1}.$$

Theorem 34.84 (Effective single-site rigidity)**.** *For any fixed $a \ge 2$ satisfying the no-cancellation hypothesis of Lemma 34.83, every prime $p \equiv 1 \pmod 8$ with*

$$p > \max\{41,\, 2a + 1,\, T_a^2\}$$

satisfies the equivalences

$$\begin{aligned} f_p(-a) \equiv g_p(-a) \equiv 0 \pmod p \\ \iff r^{(p-1)/4} \equiv 1 \pmod p \quad \text{for every prime } r \le a \\ \iff f_p(-b) \equiv g_p(-b) \equiv 0 \pmod p \quad \text{for all } 2 \le b \le a. \end{aligned}$$

In particular the exceptional primes are not merely finite: each is bounded above by the explicit number $\max\{41, 2a + 1, T_a^2\}$ (outside the stated normalization eligibility).

Proof. Suppose the two values at $-a$ vanish, and encode the actual quartic residues as above. If the digit assignment is nonzero, Lemma 34.83 ensures at least one E_j is nonzero as a Gaussian integer. Its norm is a positive integer at most T_a^2, because $|E_j| \le T_a$. But p divides that norm, contradicting $p > T_a^2$. Thus all prime bases $r \le a$ are fourth powers mod p.

Conversely, if all the relevant prime bases are fourth powers, then $b^h = 1$ for every $1 \le b \le a$; Lemma 34.82 and $\sum_b w_{a,b} = 0$ show both values vanish at $-a$. The same argument applies to each earlier $-b$. Finally, simultaneous vanishing on the entire prefix implies fourth-power residuosity of all $b \le a$, by applying the first-polynomial identities successively for $b = 2, 3, \dots, a$: the resulting lower triangular matrix has diagonal entries b^{b-1}, units modulo p. □

Proposition 34.85 (Certified no-cancellation range)**.** *The hypothesis of Lemma 34.83 holds for every integer* $2 \le a \le 28$*.*

Exact finite certificate. For each a split $\{1, \dots, a\}$ into sets of sizes $\lfloor a/2 \rfloor$ and $\lceil a/2 \rceil$. Enumerate the integer subset sums of the left and right halves. An exact match $u + v = 0$, other than the empty subset and the full subset, would violate the assertion. An exact finite enumeration of these two labelled subset-sum families for all 27 values of a finds no prohibited match. The enumeration protocol, including checks against the two trivial subsets, is specified in Section 34.47. This is a finite computational proof, not an assumed infinite family statement. □

34.47 Computational proof certificates

The finite calculations are small enough to be specified exactly and rerun without a computer algebra system. They are part of the proof of Proposition 34.85 and of the complete exceptional-prime theorem below, not evidence for an unbounded extrapolation.

Certificate I: integer subset-sum exclusion. For fixed a let $w_b = w_{a,b}$. Divide the indices into two halves L and R and compute the two sets of labelled subset sums

$$\mathcal{S}_L = \{(\sum_{b \in I} w_b, I) : I \subseteq L\}, \qquad \mathcal{S}_R = \{(\sum_{b \in J} w_b, J) : J \subseteq R\}.$$

Match $u \in \mathcal{S}_L$ against $-u \in \mathcal{S}_R$. Accept only the two trivial zero-sum subsets, the empty set and the full index set. All operations are on unbounded Python integers. The enumeration was performed for every $a = 2, \dots, 28$. The labels I, J are retained, so coincident sums from different subsets are not accidentally discarded.

Certificate II: exception exhaustiveness. For each $a = 2, \dots, 12$ enumerate all $4^{\pi(a)} - 1$ nontrivial vectors $e = (e_r)_{r \le a}$ with $e_r \in \{0, 1, 2, 3\}$. Multiplicativity supplies e_b for each $b \le a$. Compute the pairs $(E_1(e), E_2(e)) \in \mathbb{Z}[i]^2$, then

$$G(a, e) = \gcd\big(N(E_1(e)), N(E_2(e))\big), \qquad N(x + iy) = x^2 + y^2.$$

Factor every positive $G(a, e)$ into prime factors and certify the factorization by exact division with a primality test. Retain only $p \equiv 1 \pmod 8$ satisfying the size and normalization conditions; recompute actual residues $r^{(p-1)/4} \pmod p$ and both original Stirling sums for each surviving pair (a, p). This final step is crucial: dividing a Gaussian norm is only a *necessary* test, since it does not prescribe the correct choice of fourth root modulo p. The same exact-integer procedure was run for every site $2 \le a \le 12$, including separate implementations of the Gaussian moment calculation and the final modular test.

Independent arithmetic audit. A second implementation evaluates the second-kind Stirling numbers by the exact recurrence

$$\left\{ {N \atop k} \right\} = k \left\{ {N-1 \atop k} \right\} + \left\{ {N-1 \atop k-1} \right\}$$

modulo the selected prime. It agrees with the finite exponential sums on the independently checked sample cases. The candidate-prime factor lists are given in Table 34.3; the finite enumeration is completely determined by the mathematical procedure above.

Validation scope and checks. For each a, the number of nontrivial fourth-root assignments is exactly $4^{\pi(a)} - 1$, as reported in Table 34.3. The finite exceptional-prime search is therefore *complete by construction*: every potential failure must divide at least one positive Gaussian norm gcd, and every such gcd is factored completely and reconstructed from its prime factors before the actual modular moments are checked. The tests use exact unbounded integers; no numerical tolerance, random prime sample, or preselected prime cutoff enters the exceptional-prime classification. A separately implemented Gray-code subset-sum search and independent Gaussian-moment factorization reproduced the $a \le 28$ no-cancellation and $a \le 12$ exception outcomes. The elementary finite-field and Stirling calculations were also checked independently at selected values. The exact arithmetic methodology and its verification scope are stated here.

34.48 Exact exceptional-prime classification through twelve sites

Theorem 34.86 (Complete small-centre classification)**.** *For every integer* $2 \le a \le 12$ *and every prime* $p \equiv 1 \pmod 8$ *with* $p \ge 41$*, the following assertions are equivalent,* except *for the unique parameter pair* $(a, p) = (10, 3617329)$*:*

1. $f_p(-a) \equiv g_p(-a) \equiv 0 \pmod p$*;*
2. $r^{(p-1)/4} \equiv 1 \pmod p$ *for every rational prime* $r \le a$*;*
3. $f_p(-b) \equiv g_p(-b) \equiv 0 \pmod p$ *for every* $2 \le b \le a$*.*

At $(a, p) = (10, 3617329)$*, assertion (1) holds whereas (2) and (3) fail. In particular, this exception is genuine and cannot be removed from the theorem.*

Exhaustive finite certificate. The implication (2)⇔(3) follows from the lower-triangular Stirling system, and (2)⇒(1) is immediate from the finite-difference formula. It remains to classify violations of (1)⇒(2). For a fixed a enumerate the $4^{\pi(a)} - 1$ nonzero fourth-root digit assignments on primes $r \le a$. For each, compute the exact Gaussian moment pair (E_1, E_2) and

$$G(a, e) = \gcd\big(N(E_1(e)), N(E_2(e))\big) > 0.$$

The positivity follows from Proposition 34.85 and Lemma 34.83. A counterexample prime necessarily divides one of these finitely many positive integers. The exact finite computation lists all distinct prime factors, tests their primality, and verifies that repeated division by the listed factors leaves quotient one *for every assignment.* It then checks each eligible $p \ge 41$, $p \equiv 1 \pmod 8$ by modular exponentiation and the two explicit Stirling sums. For $a = 2, \dots, 8$ the previous verified certificate returns no exceptions. The new certificate for $a = 9, \dots, 12$ yields Table 34.3; the sole true counterexample is $(10, 3617329)$. No untested prime can be an exception because the norm-gcd divisors are exhaustively certified. □

Proposition 34.87 (An explicit non-prefix simultaneous zero)**.** *Let* $p = 3617329$ *(prime),* $h = (p-1)/4 = 904332$*, and*

$$f_p = q_{904331}, \qquad g_p = q_{1808663}.$$

Then $f_p(-10) \equiv g_p(-10) \equiv 0 \pmod p$*, but* $f_p(-2) \not\equiv 0 \pmod p$*.*

Proof. Direct modular exponentiation gives

$$2^h \equiv -1, \quad 3^h \equiv 1, \quad 5^h \equiv -1, \quad 7^h \equiv -1 \pmod p.$$

a	Eligible Gaussian-norm candidate primes	Genuine exceptions	Assignments
2–4	none	none	$3, 15, 15$
5	41	none	63
6	$73, 257$	none	63
7	$233, 1873, 2953$	none	255
8	$73, 7457$	none	255
9	$52249, 132929$	none	255
10	$241, 313, 6569, 32993,$ $489001, 1097513, 3617329, 9803873$	3617329	255
11	$281, 337, 433, 2161, 8353, 40193,$ $83497, 375049, 400601, 504457,$ $12119761, 42074561$	none	1023
12	$89, 137, 353, 617, 641, 1033,$ $1409, 2273, 409217, 1424809,$ $2186113, 3448561, 59951393, 182991329$	none	1023

Table 34.3: Complete eligible exception-candidate lists, not a bounded brute-force search over primes. For each fixed a the assignment count is exactly $4^{\pi(a)} - 1$; certification is by exact norm divisibility.

Consequently $b^h \equiv \pm 1 \pmod{p}$ for every $1 \le b \le 10$. Applying the exact weights $w_{10,b}$ and replacing these residues by their integer signs, the two finite Gaussian-moment representatives are

$$E_1 = \sum_{b=1}^{10} w_{10,b} \operatorname{sgn}(b^h) = -5643033240 = -1560p, \qquad E_2 = \sum_{b=1}^{10} w_{10,b} = 0.$$

Here $\operatorname{sgn}(b^h)$ means the representative $+1$ or -1 of $b^h \pmod{p}$; E_j is a *finite representative modulo p*, not the original large Stirling integer. The reduction in Lemma 34.82 makes both quotient values zero. But at -2 that lemma gives $-(n_1+2)!S_{n_1}(-2) = 2(2^h-1) \equiv -4 \not\equiv 0 \pmod{p}$. All normalizations and factorials are p-units, so $f_p(-2)$ is nonzero as asserted. Primality of p is independently certified by deterministic trial division (up to $\lfloor\sqrt{p}\rfloor$). □

34.49 Exact natural densities and first failure

Let $\pi(a)$ count rational primes $\le a$, and let

$$K_a = \mathbb{Q}\big(\zeta_8,\ r^{1/4} : r \le a \text{ rational prime}\big).$$

Lemma 34.88 (Quartic Kummer degree)**.** *For every $a \ge 2$, $[K_a : \mathbb{Q}] = 2 \cdot 4^{\pi(a)}$.*

Proof. The cyclotomic field $F = \mathbb{Q}(\zeta_8) = \mathbb{Q}(i, \sqrt{2})$ has degree 4. Adjoining $2^{1/4}$ has relative degree 2: $\sqrt{2} \in F$, whereas $2^{1/4} \notin F$, since the nonnormal quartic field $\mathbb{Q}(2^{1/4})$ cannot lie in the abelian Galois field $F/\mathbb{Q}$. The resulting field ramifies only at 2. On successively adjoining $r^{1/4}$ for distinct odd primes r, the intermediate field is unramified at r: its generators have discriminants supported on 2 and previously adjoined primes. At a prime above r, $v(r) = 1$, so $X^4 - r$ is Eisenstein. Each odd radical therefore raises the degree by 4, giving $4 \cdot 2 \cdot 4^{\pi(a)-1}$. □

Corollary 34.89 (Density of a single prescribed common root)**.** *For each fixed $2 \le a \le 28$, the set of primes $p \equiv 1 \pmod 8$ for which $f_p(-a) \equiv g_p(-a) \equiv 0 \pmod{p}$ has natural density*

$$\boxed{\delta_a = \frac{1}{2 \cdot 4^{\pi(a)}}.}$$

Moreover, the first missing root in the initial consecutive block $-2, -3, \ldots$ can occur only at a rational-prime centre, apart from finitely many ineligible small primes; for odd prime r the exact absolute density of primes whose first missing centre is $-r$ equals $3/(8 \cdot 4^{\pi(r-1)})$.

Proof. By Theorem 34.84 and Proposition 34.85, except for finitely many primes the single-site common-root property is equivalent to every rational prime $r \le a$ being a fourth power modulo p. This holds exactly when p splits completely in K_a, which has Chebotarev density $1/[K_a : \mathbb{Q}]$. For the prefix the same equivalence holds for every fixed endpoint, even without the no-cancellation hypothesis, by the triangular Stirling system. A composite m cannot be the first failure: all prime factors of m are smaller and would already be fourth powers. The density at an odd prime r is $\delta_{r-1} - \delta_r = \frac{3}{4}\delta_{r-1}$; here δ_m denotes prefix density, defined at all m, not just $m \le 28$. □

Remark 34.90 (Scope of the result). The single-site theorem applies to each fixed $a \le 28$ outside an explicit finite prime set, and is exception-free for $a \le 12$ except for the one explicitly certified pair $(a, p) = (10, 3617329)$. It makes no statement about a growing with p. The prime-jump distribution is related to the classical least quartic nonresidue problem; the proposed specialized contribution here is the implication from a *single prescribed common-root site* to an entire root prefix, together with the exact sporadic failure, not the classical quartic-residue distribution itself. The methods used are standard Gaussian linear algebra and Kummer–Chebotarev theory. No novelty is claimed for the underlying classical power-residue distribution or Kummer-theoretic methods.

34.50 A sharp local resultant example and scope

For $p = 577$ the two indices are 143 and 287, and both polynomials have the three common roots $-2, -3, -4$ modulo 577. A complete modular coefficient computation gives

$$\gcd_{\mathbb{F}_{577}[X]}(\bar{q}_{143}, \bar{q}_{287}) = (X+2)(X+3)(X+4).$$

Each of these three roots is simple in both reductions. Direct first-order Hensel calculations show that the lifts for q_{143} and q_{287} differ at every one of the three centres; the six respective first digits are $(514, 219, 156)$ and $(412, 319, 171)$. Hence each of the three common residual factors contributes exactly one to the local resultant valuation; all other factors contribute zero, because the displayed gcd is the entire common residual factor. Since the leading coefficients are 577-adic units, this proves

$$v_{577}(\mathrm{Res}(q_{143}, q_{287})) = 3.$$

These exact modular gcd and first-Hensel-digit calculations were independently recomputed, using integer or finite-field arithmetic. The values stated above specify their outputs. This example distinguishes three depth-one collisions from a single root of depth three.

The finite verification has three parts: exact subset-sum enumeration for $a \le 28$, complete Gaussian-norm prime-candidate enumeration and modular filtering for $a \le 12$, and independent Stirling-recurrence checks at sample parameters. These calculations are described in Section 34.47. Independent exact-arithmetic implementations reproduce the first two enumerations and the $p = 577$ Hensel calculation. No finite computation is claimed to extrapolate beyond the stated range.

34.51 Open Problems and Future Directions

Unsolved problem. For distinct odd $5 \le r < s$, does there exist a good prime $p > 2s$ and an $X \in \mathbb{Z}/p^4\mathbb{Z}$ such that $q_r(X) = q_s(X) = 0 \pmod{p^4}$? The present mod-$p$ work does not settle this question. A separate nonvanishing hypothesis for first Hensel derivatives is also deliberately left as a conjecture, not used here.

34.52 Novelty boundary and related arithmetic literature

The identity of Lemma 34.82, congruence periodicity of Stirling sequences, Gaussian integers, Kummer extensions, and Chebotarev are standard tools. Earlier work by Adelberg [169, 170] studies finite differences, factorization and arithmetic of Nörlund polynomials, while Young [198] develops Kummer-type congruences involving Stirling numbers. Kwong [199] studies periodicities modulo integers, and Sun [200] establishes broad combinatorial congruences for Stirling numbers. None of these classical ingredients alone is claimed as new.

Contribution and novelty statement. The claimed contribution is the single-site-to-prefix forcing theorem for $q_{(p-5)/4}$ and $q_{(p-3)/2}$, with its effective Gaussian-norm bound and the complete $a \leq 12$ classification including the unique non-prefix exception $(10, 3617329)$. The exception rules out an unconditional all-prime form of the forcing assertion. To the best of the author's knowledge these specific results have not appeared together in the cited literature. The Stirling exponential sum, fourth-power-residue conditions, Gaussian norms, Kummer extensions, Chebotarev theorem, and the resulting classical power-residue density mechanism are not claimed as new.

34.53 Fixed-centre and finite-gap Nörlund–Stirling exclusions

The prime-indexed Nörlund–Stirling single-site analysis above concerns a source-independent quotient family. We next impose three equally spaced odd indices at the fixed negative centre -3 and on a shifted p-adic disk. The fixed-centre two-base Wieferich obstruction, the bounded-gap all-prime exclusion, and their exact distinct precision hypotheses are kept separate throughout.

Contributions and novelty

The paper establishes five related results for Nörlund–Stirling quotient polynomials, with their hypotheses kept separate. First, it derives an exact prime-power criterion and divisibility depth for three equally spaced odd indices at the fixed centre $X = -3$, including propagation along the index progression. Second, it proves a two-base Wieferich necessary condition for triple depth at least three and its prime-bounded exclusion for $p \leq 10^7$. Third, it obtains an interpolation obstruction and an explicit bound on eligible primes and depth for fixed negative integer centres. Fourth, it gives an exact first-Hensel-digit compatibility criterion for a shifted triple in the specified residue disk. Fifth, a finite arithmetic certificate combined with the first-digit obstruction rules out all such triple roots modulo p^2 for every even gap $2 \leq h \leq 120$ and *every* prime satisfying the stated good-prime inequality.

The finite-difference identities, lifting-the-exponent arguments, Hensel expansion, and finite-field linear algebra themselves are classical. The contribution is the specified collection of exact congruence and exclusion results, not a new general method of Hensel lifting or an assertion of historical priority. The finite-gap theorem uses an explicit finite computational certificate as part of its proof; the prime-bounded Wieferich sieve is a separate calculation. The conclusions do not extend to arbitrary residue disks, unrestricted gaps, or pairs of orders.

34.53.1 Notation and exact negative-integer coefficients

For odd $n \geq 5$ set

$$S_n(X) = [t^n] \left(\frac{t}{1 - e^{-t}} \right)^X = \frac{X^2(X-1)}{D_n} q_n(X),$$

where D_n is the canonical integer normalization of the primitive quotient polynomial q_n. The required good-prime normalization property is proved below, in the slightly stronger range $p > n+1$. Since $p \geq 5$ makes $(-3)^2(-4)$ a unit, zeros of $q_n(-3)$ and $S_n(-3)$ modulo p^k agree at each good prime.

Proposition 34.91 (Integral normalization at good primes)**.** *For odd $n \geq 5$, write*

$$S_n(X) = \frac{X^2(X-1)}{D_n} q_n(X),$$

where $q_n \in \mathbb{Z}[X]$ is primitive and D_n is the positive integer normalizing denominator. For every prime $p > n+1$,

$$p \nmid D_n.$$

More precisely, the polynomial $S_n(X)/(X^2(X-1))$ belongs to $\mathbb{Z}_{(p)}[X]$ and has leading coefficient $1/(2^n n!)$, a p-adic unit.

Proof. Set $H(t) = t/(1-e^{-t})$. Its reciprocal is

$$H(t)^{-1} = \sum_{j \geq 0} \frac{(-1)^j}{(j+1)!} t^j.$$

For $p > n+1$ all denominators up to degree n are p-adic units, so $H(t) \in \mathbb{Z}_{(p)}[[t]]$ through degree n. The binomial expansion

$$H(t)^X = \sum_{j=0}^{n} \binom{X}{j} (H(t)-1)^j + O(t^{n+1})$$

therefore proves $S_n(X) \in \mathbb{Z}_{(p)}[X]$, because $j!$ is a p-unit for $j \leq n$.

The identities $H(-t) = H(t) - t$ and $\log H(-t) = \log H(t) - t$ show that both $H(t) - t/2$ and $\log H(t) - t/2$ are even power series. Hence for odd $n \geq 5$,

$$S_n(0) = 0, \quad S_n'(0) = [t^n] \log H(t) = 0, \quad S_n(1) = [t^n] H(t) = 0.$$

Thus $X^2(X-1)$ divides S_n over $\mathbb{Q}[X]$. Since it is monic, polynomial division over $\mathbb{Z}_{(p)}[X]$ shows that the quotient is p-integral. Finally $\log H(t) = t/2 + O(t^2)$, so the coefficient of X^n in S_n (and the leading coefficient of the quotient) is

$$\frac{1}{n!}[t^n](\log H(t))^n = \frac{1}{2^n n!},$$

a p-unit. If p divided D_n, every coefficient of the integer polynomial $q_n = D_n S_n/(X^2(X-1))$ would be divisible by p, contradicting primitiveness. Therefore $p \nmid D_n$. □

For $m \geq 2$, expansion of $((1-e^{-t})/t)^m$ gives the exact identity

$$-(n+m)! S_n(-m) = \sum_{a=1}^{m} (-1)^{m-a} \binom{m}{a} a^{n+m}, \quad n \geq 5 \text{ odd}. \tag{34.109}$$

Take even $E \geq 6$, even $h \geq 2$, and $n_j = E + jh - 1$. Equation (34.109) makes the dependence on the progression variable j a finite exponential sum.

34.53.2 An exact criterion for three odd orders at -3

Put

$$C_j = 3\cdot 3^{E+jh} - 4\cdot 2^{E+jh} + 1 \quad (j \ge 0), \qquad L = 4\cdot 2^E,\ M = 3\cdot 3^E,\ x = 2^h - 1,\ y = 3^h - 1.$$

The normalization in (34.109) gives $-(n_j+3)!S_{n_j}(-3) = 3C_j$. Write

$$D_1 = C_1 - C_0 = My - Lx, \qquad D_2 = C_2 - 2C_1 + C_0 = My^2 - Lx^2, \qquad C_0 = M - L + 1.$$

A direct expansion yields the identity

$$MD_2 = D_1^2 + 2LxD_1 + L(1 - C_0)x^2. \tag{34.110}$$

Theorem 34.92 (Exact fixed-centre triple criterion)**.** *For every prime $p \ge 5$, integers $E \ge 6$ and $h \ge 2$ even, and $k \ge 1$, the congruences $C_0 \equiv C_1 \equiv C_2 \equiv 0 \pmod{p^k}$ are equivalent to*

$$C_0 \equiv D_1 \equiv 0 \pmod{p^k}, \qquad (2^h - 1)^2 \equiv 0 \pmod{p^k}.$$

Equivalently, in valuation form,

$$K_p(E,h) = \min\bigl\{v_p(C_0), v_p(D_1), 2v_p(2^h - 1)\bigr\} \tag{34.111}$$

is the exact simultaneous divisibility depth of C_0, C_1, C_2. Moreover, divisibility modulo p^k forces $p^{\lceil k/2\rceil}$ to divide both $2^h - 1$ and $3^h - 1$.

Proof. The triple congruence is equivalent to $C_0 \equiv D_1 \equiv D_2 \equiv 0 \pmod{p^k}$. Under $C_0 \equiv 0 \pmod p$, both L and M are p-adic units, as is $1 - C_0$. Formula (34.110) makes $D_2 \equiv 0$ equivalent to $x^2 \equiv 0 \pmod{p^k}$ if $D_1 \equiv 0$. This proves the iff statement and (34.111). Finally $My \equiv Lx \pmod{p^k}$ transfers the lower valuation bound from x to y. □

Proposition 34.93 (Propagation along the entire index progression)**.** *In the hypotheses of Theorem 34.92, if $C_0 = C_1 = C_2 = 0 \pmod{p^k}$, then $C_j = 0 \pmod{p^k}$ for every $j \ge 0$. Consequently, whenever the good-prime inequality $p > 2(E + jh - 1)$ holds, the corresponding primitive quotient q_{E+jh-1} also vanishes at $X = -3$ modulo p^k.*

Proof. Put $x = 2^h - 1$ and $y = 3^h - 1$. Theorem 34.92 gives $x^2 = y^2 = 0$ modulo p^k, and $C_0 = D_1 = 0$ in the same ring. The binomial identity yields, for each integer $j \ge 0$,

$$C_j = M(1+y)^j - L(1+x)^j + 1 \equiv C_0 + j(My - Lx) = C_0 + jD_1 = 0 \pmod{p^k}.$$

The final claim follows from the stated integral normalization and the factorial-unit condition at a good prime. □

34.53.3 The two-base Wieferich obstruction

Theorem 34.94. *Let $E \ge 6$ and $h \ge 2$ be even, and let $p > 2(E + 2h - 1)$ be prime. If the three quotient polynomials q_{E-1}, q_{E+h-1}, q_{E+2h-1} vanish at $X = -3$ modulo p^k, then for $b = 2$ and $b = 3$,*

$$v_p(b^{p-1} - 1) \ge \lceil k/2\rceil.$$

In particular, if $k \ge 3$ then

$$2^{p-1} \equiv 3^{p-1} \equiv 1 \pmod{p^2}. \tag{34.112}$$

Proof. By Theorem 34.92, $p^{\lceil k/2\rceil} \mid b^h - 1$ for each $b \in \{2,3\}$. The good-prime inequality implies $h < p$. Set $d = \operatorname{ord}_p(b)$. Then $d \mid h$ and $d \mid p - 1$, while p divides neither h/d nor $(p-1)/d$. The lifting-the-exponent lemma gives

$$v_p(b^h - 1) = v_p(b^d - 1) = v_p(b^{p-1} - 1).$$

The claim follows. □

Corollary 34.95. *For all even $E \geq 6$, even $h \geq 2$, and prime $2(E+2h-1) < p \leq 10^7$, the three quotient polynomials with indices $E-1, E+h-1, E+2h-1$ cannot all vanish at $X = -3$ modulo p^3.*

Proof. A complete deterministic sieve lists the 664,577 primes $5 \leq p \leq 10^7$. Among these, the primes satisfying $2^{p-1} \equiv 1 \pmod{p^2}$ are $1093, 3511$; the primes satisfying $3^{p-1} \equiv 1 \pmod{p^2}$ are $11, 1006003$. Their intersection is empty. Theorem 34.94 now applies, with no bound on h other than the good-prime restriction. □

34.53.4 General negative integer centres

Proposition 34.96. *Fix $m \geq 2$, even $E \geq 6$, even $h \geq 2$, and $p > m$. If for $j = 0, \ldots, m-1$ one has $-(n_j+m)!S_{n_j}(-m) \equiv 0 \pmod{p^k}$, then for each $1 \leq a \leq m$,*

$$p^k \;\Big|\; \prod_{\substack{1 \leq b \leq m \\ b \neq a}} (a^h - b^h).$$

Consequently all possible prime divisors for a fixed pair (m, h) divide an explicit integer independent of E, and the p-adic depth obeys

$$k \leq \min_{1 \leq a \leq m} \sum_{\substack{1 \leq b \leq m \\ b \neq a}} v_p(a^h - b^h).$$

Proof. Define $\lambda_a = a^h$, $U_a = (-1)^{m-a}\binom{m}{a}a^{E+m-1}$, and $Z_j = \sum_a U_a \lambda_a^j$ using (34.109). For fixed a, write $P_a(T) = \prod_{b \neq a}(T - \lambda_b) = \sum_{j=0}^{m-1} c_j T^j \in \mathbb{Z}[T]$. Then

$$\sum_{j=0}^{m-1} c_j Z_j = \sum_{b=1}^{m} U_b P_a(\lambda_b) = U_a \prod_{b \neq a}(\lambda_a - \lambda_b).$$

The left side is divisible by p^k, whereas U_a is a p-adic unit because $p > m$. Divide by U_a. □

34.53.5 The first shifted Hensel jet at -3

This section concerns a common root $X \equiv -3 \pmod{p}$, which is allowed to shift at higher precision. It does *not* identify or restrict common roots in other residue disks. Let

$$F_j(X) = -(n_j+3)!S_{n_j}(X), \qquad P_j = F_j'(-3), \quad j = 0, 1, 2.$$

At a good prime $p > 2n_2$ these have p-integral polynomial coefficients and $F_j(-3) = 3C_j$. In particular their Taylor expansions at -3 make sense modulo p^2.

Proposition 34.97 (Finite-prime obstruction for shifted triples)**.** *Suppose E, h are even, $E \geq 6, h \geq 2$, and $p > 2n_2$ is prime. If there is an $X \equiv -3 \pmod{p}$ such that $q_{n_j}(X) \equiv 0 \pmod{p}$ for all $j = 0, 1, 2$, then*

$$p \mid \gcd(2^h - 1, 3^h - 1).$$

Consequently for each fixed gap h there are only finitely many eligible primes and triples (p, E, h) with such a shifted residue disk, regardless of the precision to which it lifts.

Proof. Reduction $X \equiv -3 \pmod{p}$ and the good-prime normalizer give $C_0 \equiv C_1 \equiv C_2 \equiv 0 \pmod{p}$. Theorem 34.92 with $k = 1$ gives $2^h \equiv 3^h \equiv 1 \pmod{p}$. For fixed h, the gcd is a fixed nonzero integer. Each eligible p limits E by $2(E+2h-1) < p$. □

Theorem 34.98 (First-order shift-curvature obstruction)**.** *Assume $C_0 \equiv C_1 \equiv C_2 \equiv 0 \pmod{p}$, where $p > 2n_2$. If an integer u satisfies*

$$q_{n_j}(-3+pu) \equiv 0 \pmod{p^2} \qquad (j=0,1,2),$$

then

$$\boxed{u\big(P_2 - 2P_1 + P_0\big) \equiv 0 \pmod{p}.} \tag{34.113}$$

In particular, if $P_2 - 2P_1 + P_0$ is a unit modulo p, every such common lift must have $u \equiv 0 \pmod{p}$.

Proof. Theorem 34.92 with $k=1$ shows that $x = 2^h - 1$ and $y = 3^h - 1$ are both divisible by p. Thus the exact second difference at the fixed centre is

$$F_2(-3) - 2F_1(-3) + F_0(-3) = 3\big(My^2 - Lx^2\big) \equiv 0 \pmod{p^2}.$$

Taylor expansion modulo p^2 gives $F_j(-3+pu) \equiv F_j(-3) + puP_j \pmod{p^2}$. Subtract twice the middle congruence from the first and third. The three shifted values vanish modulo p^2, so $pu(P_2 - 2P_1 + P_0) \equiv 0 \pmod{p^2}$. Cancel p to obtain (34.113). □

Theorem 34.99 (Exact first-digit compatibility for shifted triples)**.** *Retain the hypotheses of Theorem 34.98. Define residues in $\mathbb{F}_p$ by*

$$A = F_0(-3)/p, \qquad B = (F_1(-3) - F_0(-3))/p, \qquad d_j = F_j'(-3) \quad (j=0,1,2).$$

These quotients are integral because all three fixed-centre values vanish modulo p. Put $e = d_1 - d_0$ and $D = d_2 - 2d_1 + d_0$. There exists a common lifted root $X = -3 + pu$ modulo p^2 if and only if the following finite-field conditions hold:

1. *If $D \neq 0$, then $A = B = 0$. In this case $u = 0$ is the only possible first digit.*
2. *If $D = 0$ but $(d_0, e) \neq (0,0)$, then $Ae - Bd_0 = 0$. In this case a unique u satisfies $A + ud_0 = B + ue = 0$.*
3. *If $D = 0$ and $(d_0, e) = (0,0)$, then $A = B = 0$. In this case every $u \in \mathbb{F}_p$ is a common lift at this precision.*

The conditions concern only the specified disk $X \equiv -3 \pmod{p}$.

Proof. The fixed-centre triple modulo p implies $2^h \equiv 3^h \equiv 1 \pmod{p}$, so the exact difference identity gives

$$F_2(-3) - 2F_1(-3) + F_0(-3) \equiv 0 \pmod{p^2}.$$

Therefore $F_j(-3)/p = A + jB$ in $\mathbb{F}_p$ for $j = 0,1,2$. Taylor expansion around -3 gives

$$F_j(-3+pu)/p \equiv A + jB + ud_j \pmod{p}.$$

The second difference of the three right sides is uD. When $D \neq 0$ this forces $u = 0$, leaving $A = B = 0$. When $D = 0$ the derivative vector is affine, $d_j = d_0 + je$, and the three equations reduce to $A + ud_0 = 0$ and $B + ue = 0$. An elementary two-dimensional linear-dependence test gives the remaining two alternatives, including the degenerate zero vector. □

Remark 34.100. The exact first-digit test combines a CCR-specific affine fixed-centre identity with elementary linear Hensel algebra. It is not a depth-four existence theorem, and we do not assert that the underlying linear algebra is a new technique.

Proposition 34.101 (The $p = 431$ triple is obstructed at depth two)**.** *For $(p, E, h) = (431, 42, 86)$ and $(n_0, n_1, n_2) = (41, 127, 213)$, there is no common root $X \equiv -3 \pmod{431}$ of the three quotient polynomials modulo 431^2.*

Proof. Exact finite-field coefficient arithmetic gives

$$(P_0, P_1, P_2) \equiv (62, 191, 86) \pmod{431}, \qquad P_2 - 2P_1 + P_0 \equiv 197 \not\equiv 0 \pmod{431}.$$

In the notation of Theorem 34.99, the fixed-centre first quotients are $(A, B) = (274, 117)$ modulo 431. The curvature $D = 197$ is a unit, but $A \neq 0$; hence the exact first-digit compatibility criterion excludes a common lift. In fact the individual required digits are 79, 359, and 385. These residues can be obtained by exact coefficient extraction from $H(t)^3$ and $-H(t)^3 \log H(t)$ with $H(t) = (1 - e^{-t})/t$, using the reciprocal-series identity and the coefficient recurrence derived from $A(t)f'(t) = XA'(t)f(t)$ for $A(t) = t/(1 - e^{-t})$. The values displayed above give the finite-field certificate needed for the stated first-digit obstruction. □

Remark 34.102 (Scope and originality). The proof of Theorem 34.98 uses elementary Taylor expansion and a special fixed-centre second-difference identity. It is a useful *specialized consequence*, not a claim that Hensel's lemma or the jet method is new. It does not constrain an arbitrary pair (q_r, q_s) in a residue disk other than -3.

34.53.6 All-prime finite-gap exclusion of shifted triples

The fixed-centre classification and the first-digit shift criterion combine to exclude *shifted* triples uniformly over all primes within a nontrivial range of index gaps. This is an explicitly certified finite-gap theorem; it is not a universal statement over unbounded h.

Theorem 34.103 (Complete $h \leq 120$ shifted-triple exclusion)**.** *Let $E \geq 6$ and $2 \leq h \leq 120$ be even, put $n_j = E + jh - 1$ $(j = 0, 1, 2)$, and let $p > 2n_2$ be a prime. There is* no *integer $X \equiv -3 \pmod{p}$ for which*

$$q_{n_0}(X) \equiv q_{n_1}(X) \equiv q_{n_2}(X) \equiv 0 \pmod{p^2}.$$

The conclusion has no upper bound on p. In particular no such disk has simultaneous depth $k \geq 2$ for three equally spaced orders in this gap range.

Proof. If the displayed congruences held, the three values at -3 would vanish modulo p. By Theorem 34.92,

$$p \mid \gcd(2^h - 1, 3^h - 1), \qquad p > 4h + 10.$$

A self-contained deterministic certificate for the 60 even gaps uses a shorter complete test than factoring all 60 gcd values: for each gap h, set $g_h = \gcd(2^h - 1, 3^h - 1)$ and divide out *every prime* $\ell \leq 4h + 10$ to its full multiplicity. If the remainder is 1, no prime factor of g_h exceeds the necessary goodness threshold; if the remainder is larger, all its prime factors exceed the threshold. Eratosthenes sieving up to 490 and exact integer division give remainder 1 for 59 gaps and remainder 431 for $h = 86$. Trial division through $\lfloor\sqrt{431}\rfloor$ confirms that 431 is prime. Thus the *only* pair surviving $p > 4h + 10$ is

$$(h, p) = (86, 431).$$

The goodness inequality then leaves finitely many even E; checking $3 \cdot 3^E - 4 \cdot 2^E + 1 \equiv 0 \pmod{431}$ gives only $E = 42$. Consequently the *only* first-level triple in this entire range is $(p; n_0, n_1, n_2) = (431; 41, 127, 213)$.

At this triple, all three roots are simple modulo 431 in the -3 disk. Independent modular series coefficient arithmetic gives the unique first Hensel digits of the three polynomial roots:

$$u_{41} = 79, \qquad u_{127} = 359, \qquad u_{213} = 385.$$

Because these residues are pairwise distinct, no shared u exists with $X = -3 + 431u \pmod{431^2}$. Thus the remaining case is also excluded. This finishes the finite certificate: for each of the 60 even gaps, form the exact integer gcd and remove all prime powers at or below $4h + 10$; the sole nonunit remainder is 431 at $h = 86$. The good-prime inequality

then reduces the surviving case to $E = 42$, and the displayed distinct first Hensel digits exclude its only possible common lift. The certificate requires exact integer and finite-field computations, but no further assumption about an untested prime range. □

Remark 34.104 (Computational scope and logical strength). The certificate is finite *in the gap*, not in the prime: a fixed nonzero integer $\gcd(2^h - 1, 3^h - 1)$ forces all possible primes into a finite list for each h. Certifying every gap up to 120 therefore yields a genuine all-prime result in that range, unlike an exhaustive search over primes with a fixed numeric bound. No assertion is made for a different residue disk or for a pair of orders alone. This result combines earlier exact project identities and computation; the calculation itself is part of the supporting finite certificate.

34.53.7 Finite verification, certification and relation to earlier results

The exact computations play two different roles in the results above. For Corollary 34.95, a complete sieve of the 664,577 primes $5 \le p \le 10^7$ finds only $1093, 3511$ for the base-2 Wieferich condition and only $11, 1006003$ for the base-3 condition, with empty intersection. This is a *prime-bounded* exclusion and does not assert that no larger simultaneous Wieferich prime exists.

Theorem 34.103 instead has a *finite-gap, all-prime* computer-assisted component: for each of the 60 even gaps $2, 4, \ldots, 120$, exact gcd and prime-stripping calculations give remainder 1 in 59 cases and 431 only for $h = 86$. At that gap, the good-prime bound and the exact fixed-centre congruence leave only $E = 42$, with indices $(41, 127, 213)$. The three required first Hensel digits are $(79, 359, 385)$ modulo 431, hence no common lifted root exists. This is a finite and fully specified arithmetic certificate for the theorem's unlimited-prime conclusion *within that gap range*; the theorem is not being presented as a computation-free symbolic proof.

Additional bounded consistency checks include 30 low-degree exact-rational instances of (34.109), 180 interpolation identities, an exhaustive $5^5 = 3125$ five-field-instance check of the affine-jet criterion, and an independent exact modular-coefficient computation of the $(p, E, h) = (431, 42, 86)$ residues. These finite checks corroborate the stated identities and the exceptional certificate; they do not replace the symbolic arguments in the general results or justify extending any theorem beyond its hypotheses. No verification program, data archive, or supplementary code is claimed to be distributed with this article.

Classical finite-difference and Nörlund methods appear in Adelberg [169, 170, 220], and general Bernoulli–Stirling congruences in Young [198]. Results on prime divisors of resultants, relevant to broader collision searches, were proved by Gómez-Pérez et al. [221]. The algebraic principles used here are elementary and established; our conclusions are the stated fixed-centre and finite-gap results, not new general Hensel or finite-difference methods.

The previously prepared CCR works [164, 165, 166, 195] provide the chronological-carry and Stirling-polynomial provenance. A separate existing article by the same author [219] studies single-site-to-prefix rigidity for two *prime-indexed* quotient polynomials, whereas this article treats three equal-gap indices, simultaneous fixed-centre depth and the first shifted lift. Both use classical Stirling identities and the same good-prime normalization; neither article treats its shared preliminary lemmas as an independent new contribution.

The exact first-digit criterion and the finite-gap exclusion here cover three equal-gap orders only in the residue disk $X \equiv -3 \pmod{p}$; it does not generalize the Wieferich conclusion to shifted pairs. The known depth-three example of two distinct orders at a shifted half-integer centre lies outside these hypotheses. The fixed-centre and triple-shifted statements hold precisely in the residue disks and order configurations specified in their hypotheses.

34.54 Integral terminal-source completion and finite local–global decision

The earlier CCR coefficient and source-translation formulas naturally lead to a simultaneous, rather than single-jet, completion problem. We first solve the terminal three-source system over $\mathbb{Q}$, then characterize the integral lattice of solutions and explain why a fixed-parameter local–global test is finite. The order-77 denominator certificates specify decision precision; they do not establish existence or nonexistence of an integer-source triple zero at that order.

Contributions and novelty

The main contributions concern the explicit *formal stationary CCR source family*, not new general forms of Smith normal form, polynomial denominator clearing, or the Chinese remainder theorem. First, an exact terminal-source triangularity theorem produces the unique rational three-source completion and explicit integrality conditions for the bottom three polynomial coefficients. Second, source-weight filtration shows that the upper half of the sources acts affinely, giving a concrete integral image-lattice criterion. Third, coefficientwise denominator control turns the unrestricted integer-source completion question, at each fixed order and transport prime, into a finite local–global decision problem; under the specified Bernoulli-admissibility condition only auxiliary primes at most the order are needed. Fourth, the order-77 calculations give exact sharp coefficient-denominator exponents for the two Bernoulli-admissible transport primes, without claiming an actual global YES/NO decision there.

The CCR Sheffer representation and the elementary lattice and CRT principles are acknowledged as prior structural tools. The novelty claimed here lies in the simultaneous three-jet completion map and its application to this precisely stated integral-source problem; no all-order triple-zero impossibility or physical realization theorem is asserted.

34.54.1 Problem and precise scope

For a parameter P and source variables λ_s, let $F_0(X) = 1$, $F_j = 0$ for $j < 0$, and define $F_j \in \mathbb{Q}[P, \boldsymbol{\lambda}, X]$ by

$$\begin{aligned} T_jF_j(X) &= ((P+1)X - P)F_{j-1}(X) + P((P+1)(X-1) - P)F_{j-2}(X-1) \\ &\quad + P\sum_{s=3}^{j}(-1)^s\lambda_s F_{j-s}(X-s+1), \\ T_jG(X) &= XG(X+1) - (X-j)G(X). \end{aligned} \tag{34.114}$$

As $T_j(X^k) = (j+k)X^k + \text{lower-degree terms}$, the transport has a unique solution over $\mathbb{Q}$. Fix odd $n \geq 5$ and a prime $P > 2n$. Our problem is

$$\exists\,(\lambda_3, \ldots, \lambda_n) \in \mathbb{Z}^{n-2}: \quad [X^0]F_n = [X^1]F_n = [X^2]F_n = 0. \tag{34.115}$$

We distinguish ordinary-integer, local integral, and rational local sources. The results below give an exact finite decision reduction for each fixed (n, P) without asserting a uniform bound on complexity across orders.

34.54.2 CCR response series and high-source filtration

Write $G(t, X) = \sum_{j\geq 0} F_j(X)t^j$. Comparing the X-linear and constant terms after summing the recurrence gives a uniquely normalized Sheffer representation

$$G(t,X) = H_0(t)\exp(Xf(t))\exp\left(\sum_{s\geq 3}\lambda_s K_s(t)\right), \qquad f = \log Y, \quad H_0(0) = Y(0) = 1. \tag{34.116}$$

The source-response kernel is the exact rational series

$$K_s(t) = P(-1)^s \int_0^t z^{s-1} Y(z)^{-(s-1)}\, dz = \frac{P(-1)^s}{s} t^s + O(t^{s+1}). \tag{34.117}$$

With $a = (P+1)/2$ and $D = (P+1)(5P+1)/4$, the source-free series is

$$Y(t) = 1 + at + \sum_{k\geq 1} \frac{2^{2k} B_{2k} D^k}{(2k)!}\, t^{2k}. \tag{34.118}$$

This is the stationary CCR deformation: the mere Sheffer form is classical, not new. In particular

$$[X^j]F_n = [t^n]H_0(t)\frac{f(t)^j}{j!} \exp\left(\sum_{s=3}^{n} \lambda_s K_s(t)\right). \tag{34.119}$$

Since $\operatorname{ord}_t K_s = s$ and $\operatorname{ord}_t f = 1$, any source monomial $\prod_s \lambda_s^{e_s}$ occurring in the j-th coefficient necessarily satisfies

$$\sum_{s\geq 3} s e_s \leq n - j. \tag{34.120}$$

This elementary filtration is the key extra structural input.

Theorem 34.105 (Terminal jet triangularity)**.** *For $1 \leq r \leq n-2$, put $d_{n,j} = a^{-j}[X^j]F_n$ for $0 \leq j < r$. With the first $n-r-2$ source coordinates fixed, the map from $(\lambda_{n-r+1}, \ldots, \lambda_n)$ to $(d_{n,r-1}, \ldots, d_{n,0})$ is affine triangular with constant diagonal entries*

$$\gamma_{n,j} = \frac{P(-1)^{n-j}}{(n-j)j!}, \quad j = r-1, \ldots, 0.$$

Consequently its Jacobian determinant is

$$\det M_{n,r} = \frac{(-1)^{rn - r(r-1)/2} P^r}{\prod_{j=0}^{r-1}(n-j)j!} \neq 0.$$

For $r = 3$ and odd n this specializes to $P^3/(2n(n-1)(n-2))$.

Proof. Since $f(t) = at + O(t^2)$, the leading term of the λ_{n-j} contribution to (34.119) is obtained uniquely from $t^{n-j} \cdot (at)^j/j!$. By (34.117) its coefficient is $P(-1)^{n-j}a^j/((n-j)j!)$. Sources of index greater than $n-j$ do not occur. Multiplication of the triangular diagonal entries gives the determinant. The bound $P > 2n$ ensures $a \neq 0$ and permits the normalization. □

Corollary 34.106 (Unique rational completion)**.** *Fix any earlier sources $u = (\lambda_3, \ldots, \lambda_{n-3}) \in \mathbb{Q}^{n-5}$. There exists a unique triple $W(u) \in \mathbb{Q}^3$ for the final sources $(\lambda_{n-2}, \lambda_{n-1}, \lambda_n)$ that makes all three lowest jets zero. Each coordinate of W is a rational-coefficient polynomial in u.*

Proof. By Theorem 34.105, the final three columns form a constant invertible rational triangular matrix and the remaining jets are rational polynomials in u. Solving that matrix equation proves both assertions. □

34.54.3 Explicit three-source integrality formulas

Let $s = n-2$, $t = n-1$, and set the last three sources to zero temporarily. Write $b_j = a^{-j}[X^j]F_n\big|_{\lambda_s=\lambda_t=\lambda_n=0}$, for $j = 0, 1, 2$. Set $\mu_k = (-1)^k \lambda_k$ for the three terminal sources and define the integer polynomials

$$\begin{aligned} A &= 6Ps^2 - Ps + 5P + 6s^2 - 5s + 1, \\ C &= Ps^2 + 3Ps + 4P + s^2 + s, \\ D_0 &= 3P^2s^3 + 12P^2s^2 + 17P^2s + 44P^2 \\ &\quad + 6Ps^3 + 12Ps^2 + 8Ps + 28P + 3s^3 - s. \end{aligned}$$

(The term $8Ps$ is important; changing it to $8P$ is incorrect.)

Theorem 34.107 (Pivot-free integrality criterion)**.** *The unique terminal signed sources satisfying the three-jet equation are*

$$\boxed{\begin{aligned}\mu_s &= -\frac{2s}{P}b_2,\\ \mu_t &= -\frac{t}{P}b_1 - \frac{A}{6P}b_2,\\ \mu_n &= -\frac{n}{P}b_0 - \frac{C}{2P}b_1 - \frac{D_0}{12P}b_2.\end{aligned}} \tag{34.121}$$

For the fixed early vector, ordinary-integer completion exists if and only if all three displayed rational values are integers. Equivalently, for every prime q the three numerator forms have the valuations needed to cancel their denominators P, $6P$ and $12P$, respectively.

Proof. For clarity, the *entire* constant terminal response matrix can be written explicitly. Put

$$\begin{aligned}B = (P+1)\big(&3Ps^4 + 2Ps^3 - 3Ps^2 - 50Ps - 24P\\ &+ 3s^4 - 2s^3 - 3s^2 + 2s\big).\end{aligned}$$

In decreasing jet order, and with $t = s+1$ and $n = s+2$, the exact normalized response is

$$\begin{pmatrix} d_{n,2}\\ d_{n,1}\\ d_{n,0}\end{pmatrix} = \begin{pmatrix} b_2\\ b_1\\ b_0\end{pmatrix} + P\begin{pmatrix} \dfrac{1}{2s} & 0 & 0\\ -\dfrac{A}{12st} & \dfrac{1}{t} & 0\\ \dfrac{B}{24stn} & -\dfrac{C}{2tn} & \dfrac{1}{n}\end{pmatrix}\begin{pmatrix}\mu_s\\ \mu_t\\ \mu_n\end{pmatrix}.$$

The first row gives $\mu_s = -2sb_2/P$; substitution into the second gives $\mu_t = -tb_1/P - Ab_2/(6P)$. In the third row the remaining b_2 coefficient is $(B - AC)/(12tP)$. Direct expansion of the displayed integer polynomials gives the exact identity

$$B - AC = -tD_0 \qquad \text{in } \mathbb{Z}[P, s],$$

which yields the last expression in (34.121). The triangular matrix has nonzero diagonal, so the rational completion is unique. Because changing each λ_k to $\mu_k = (-1)^k\lambda_k$ preserves integrality, an integer completion exists precisely when all three rational expressions are integers. □

34.54.4 Half-source affine elimination and a lattice criterion

Put $h = \lfloor n/2 \rfloor$ and split

$$u = (\lambda_3, \dots, \lambda_h), \qquad v = (\lambda_{h+1}, \dots, \lambda_n).$$

Theorem 34.108 (Source-half elimination)**.** *For $j = 0, 1, 2$ there are rational polynomials $b_j(u)$ and $A_{js}(u)$ such that*

$$[X^j]F_n = b_j(u) + \sum_{s=h+1}^{n} A_{js}(u)\lambda_s. \tag{34.122}$$

Moreover A_{js} only depends on low sources of indices $i \le n-s-j$, and is zero for $s > n-j$.

Proof. Two occurrences of a high source (possibly with different indices) have t-order at least $2(h+1) > n$ and cannot contribute to the coefficient of t^n. The support refinement is the degree budget (34.120). Hence the dependence is affine in all the high-source variables simultaneously, including mixed terms with arbitrary low sources. □

For a fixed *integer* low-source vector u, select positive row multipliers D_j that clear the rational denominators of the coefficients of every b_j and A_{js} as polynomials in u. Let $M(u) \in \mathbb{Z}^{3\times(n-h)}$ and $B(u) \in \mathbb{Z}^3$ denote the resulting integer matrix and constant vector. The terminal three-source minor is invertible by Theorem 34.105, so $M(u)$ has rank three.

Theorem 34.109 (Determinantal integer completion)**.** *Write $g(M)$ for the positive gcd of all 3×3 minors of a full-row-rank integer matrix. Then, for every fixed $u \in \mathbb{Z}^{h-2}$,*

$$\exists v \in \mathbb{Z}^{n-h} \text{ with } B(u) + M(u)v = 0 \quad \Longleftrightarrow \quad \boxed{g(M(u)) = g([M(u) \mid B(u)])}.$$

Proof. The image $L = M\mathbb{Z}^{n-h} \subseteq \mathbb{Z}^3$ has index $g(M)$, by Smith normal form. The enlarged lattice $L' = L + B\mathbb{Z}$ has index $g([M \mid B])$. Since $L \subseteq L'$, the indices are equal if and only if $L = L'$, equivalently $B \in L$. The sign is immaterial. □

Remark 34.110. Smith normal form and lattice-index equalities are classical. The CCR-specific content is the exact filtration guaranteeing a *constant-rank high-source affine matrix* at every order; (34.122) and the terminal pivot give a finite-dimensional concrete instance. We make no general novelty claim for the lattice theorem absent CCR structure.

34.54.5 Effective finite local–global principle

Write the signed or unsigned terminal triple in Corollary 34.106 as $W = (W_1, W_2, W_3)$, polynomials in the $n - 5$ early sources u. Define

$$E^*(n, P) = \operatorname{lcm}\{\operatorname{den}(w_{i,\alpha}) : W_i(u) = \sum_{\alpha} w_{i,\alpha} u^{\alpha},\ 1 \le i \le 3\}. \tag{34.123}$$

The denominators are positive integers in reduced terms. Notice that E^* is *the least positive integer clearing every rational polynomial coefficient*; we do not assert that it is the least period of the Boolean integer-solvability predicate.

Lemma 34.111 (Exact completion periodicity)**.** *For every $u, u' \in \mathbb{Z}^{n-5}$ with $u \equiv u' \pmod{E^*}$ coordinatewise,*

$$W(u') - W(u) \in \mathbb{Z}^3, \quad \textit{hence} \quad W(u') \in \mathbb{Z}^3 \iff W(u) \in \mathbb{Z}^3.$$

Proof. Each E^*W_i has integral polynomial coefficients. Therefore its values at congruent inputs are congruent modulo E^*, and division by E^* gives the integer difference. □

Theorem 34.112 (Finite local–global decision for fixed n, P)**.** *The following assertions are equivalent:*

1. *Some ordinary integer source vector has $X^3 \mid F_n$ over $\mathbb{Q}[X]$.*
2. *For every prime q, some q-adic integral source vector has $X^3 \mid F_n$ over $\mathbb{Q}_q[X]$.*
3. *For each prime $q \mid E^*$, there exists an early source vector $u_q \in \mathbb{Z}_q^{n-5}$ with $W(u_q) \in \mathbb{Z}_q^3$.*
4. *There exists an early residue tuple $u \in (\mathbb{Z}/E^*\mathbb{Z})^{n-5}$ with $W(u) \in \mathbb{Z}^3$.*

For fixed (n, P), the test in (4) is finite, although its cost may be enormous.

Proof. (1)⇒(2) is immediate, and (2)⇒(3) uses the uniqueness of the terminal tuple in $\mathbb{Q}_q$. For (3)⇒(4), factor $E^* = \prod_{q \mid E^*} q^{e_q}$ and, for each $q \mid E^*$, approximate u_q by integer residues modulo q^{e_q}. Combine those residues coordinatewise by CRT to get an integer u. Since $q^{e_q} W_i \in \mathbb{Z}_q[u]$, the difference $W_i(u) - W_i(u_q)$ lies in $\mathbb{Z}_q$. Thus $W_i(u)$ is q-adically integral at all primes dividing E^*; at primes not dividing E^*, all coefficients of W_i are already integral. A rational number integral at every rational prime is an integer, so $W(u) \in \mathbb{Z}^3$. Taking the class modulo E^* gives (4). Finally (4)⇒(1) by substituting the unique terminal triple. The periodicity lemma makes the residue formulation independent of the chosen representatives. □

Remark 34.113. The theorem is an effective restricted local–global principle for this *polynomially parametrized, terminally invertible* problem. It is a consequence of denominator clearing and CRT, not a general local–global theorem for arbitrary polynomial equations. Existing general local–global results about polynomial maps must be compared before claiming priority for the abstract mechanism.

Theorem 34.114 (Only small auxiliary primes can obstruct)**.** *Assume n odd and $P > 2n$ prime, and that the Bernoulli-admissibility condition*

$$P \mid \mathrm{num}(B_{n-1})$$

holds. Then $q \nmid E^*(n, P)$ for every rational prime $q > n$. In particular an ordinary-integer triple-zero completion exists if and only if *local $\mathbb{Z}_q$-integral completions exist for each prime $q \le n$.*

Proof. Fix a prime $q > n$ with $q \neq P$. We work with series truncated through degree n over $\mathbb{Z}_q$. Division by each integer $1, \dots, n$ is allowed. The denominators of the Bernoulli terms occurring in (34.118) are q-adic units: by von Staudt–Clausen, a prime q in the denominator of B_{2k} would require $q - 1 \mid 2k$; but $2k \le n$ and, since $n \ge 5$ is odd, every prime $q > n$ has $q - 1 > n$. Set $a = (P + 1)/2$; the identity $D = a(5P + 1)/2$ therefore implies

$$Y(t) - 1 \in a\, t\mathbb{Z}_q[t]/(t^{n+1}), \qquad f(t) = \log Y(t) \in a\, t\mathbb{Z}_q[t]/(t^{n+1}).$$

Even when $q \mid (P+1)$, the quotient f/a consequently has q-integral coefficients and leading term t. The source-free relation

$$(\log H_0)' = -P - P(2P + 1)t/Y$$

and the kernel formula (34.117) show that H_0 and each K_s, truncated through degree n, have q-integral coefficients; exponentials preserve this property to degree n because $1, \dots, n$ are units. Hence the *normalized* jets are coefficientwise q-integral polynomials in the sources:

$$d_{n,j} = [t^n]H_0(t)\frac{(f(t)/a)^j}{j!}\exp\left(\sum_{s=3}^{n}\lambda_s K_s(t)\right) \in \mathbb{Z}_q[\lambda_3, \dots, \lambda_n] \quad (j = 0, 1, 2).$$

The constant triangular terminal pivots $\gamma_{n,j} = P(-1)^{n-j}/((n - j)j!)$ are q-adic units for $q > n$, $q \neq P$; recursive elimination therefore gives $W \in \mathbb{Z}_q[u]^3$. In particular, no division by the potentially nonunit a is needed after normalization.

At $q = P$, the reduction of the CCR recurrence modulo P is source-independent and equals the classical Nörlund–Stirling shadow. Its first two jets vanish at odd order, and the third vanishes by the assumed $P \mid \mathrm{num}(B_{n-1})$. Thus the low-jet source polynomials have all coefficients divisible by P over $\mathbb{Z}_P$. After factoring P from the terminal response matrix, the remaining triangular diagonal entries are P-adic units (since $P > 2n$), so $W \in \mathbb{Z}_P[u]^3$. Therefore no prime above n divides a coefficient denominator of W. The local–global equivalence follows from Theorem 34.112. □

Remark 34.115. The Bernoulli condition does not say that an *integer* completion exists. It says that the transport prime itself causes no local obstruction; the small auxiliary primes still need to be tested. The equivalence is fixed-parameter and does not assert a uniform conclusion over unbounded orders.

34.54.6 Exact examples and decision precision

n	P	Role of example	E from RN177	E^*
5	11	Formal nonadmissible test	5940	1485
7	17	Formal nonadmissible test	1156680	385560
9	23	Formal nonadmissible test	10432800	1304100
13	691	Actual Bernoulli-admissible candidate	563415288259824000	6810804000

These coefficientwise periods are computed with exact rational polynomial arithmetic and independently checked against the original transport recurrence. In the admissible $(n, P) = (13, 691)$ case,

$$E^* = 2^5 \cdot 3^5 \cdot 5^3 \cdot 7^2 \cdot 11 \cdot 13,$$

so no prime beyond $n = 13$ can obstruct and the transport prime 691 does not occur. These figures are *not* claims about minimal periods of integer-valued Boolean predicates.

For the explicit higher-order case $n = 77$, Bernoulli admissibility leaves

$$P_1 = 58231, \qquad P_2 = 22284285930116236430122855560372707885169924709.$$

The exact coefficientwise 2- and 3-adic denominator exponents of the three unique signed terminal polynomials are:

P	$(e_2(W_1), e_2(W_2), e_2(W_3))$	$(e_3(W_1), e_3(W_2), e_3(W_3))$
P_1	$(52, 52, 52)$	$(34, 34, 35)$
P_2	$(54, 52, 55)$	$(34, 34, 35)$

These are exact *coefficient-denominator* valuations, proved by complete min-plus bounds over all 72 early sources and sharp rational coefficient witnesses, rather than statements about the least root-lifting depth. They give sufficient finite test moduli 2^{52}, 3^{35} at P_1 and 2^{55}, 3^{35} at P_2. The high-index sources cannot simply be discarded: already λ_{74} can alter the relevant local integrality. The upper half $\lambda_{39}, \ldots, \lambda_{74}$ can instead be handled by the exact affine lattice test.

Additional archived computations give restricted local source certificates at order 77, but they are not used to establish the global theorems of this article.

34.54.7 Relation to prior papers and novelty boundary

Earlier CCR-3, CCR-4 and CCR-k studies [164, 165, 166] introduce the chronological-carry setting. The existing articles *CCR-k Arithmetic Refinement: p-Adic Resonance, Ramification, and Arithmetic Density Laws* and *Sharp p-Adic Rigidity and Source-Tunable Resonance in Stationary CCR Polynomial Transports* [195, 218] study, respectively, first-order Stirling-source obstruction lattices and near-leading coefficients, with root-disk deformations and resonance bounds. The present work concerns simultaneous vanishing of the *three lowest coefficients* and ordinary-integer completion of *all* sources. It does not reuse their high-jet resonance theorem as a new contribution, and it cites the previously established Sheffer decomposition instead of claiming its discovery.

The finite-denominator local–global mechanism, Smith index theorem, CRT, and the general theory of rational-polynomial integral values are standard tools. The cited Nörlund denominator results and general CRT/Smith principles remain prior art; the distinctions from the CCR-specific new results are summarized in the opening Contributions and novelty section.

34.54.8 Exact computational verification

The symbolic theorems above use explicit triangular elimination, source-weight filtration, integer-lattice indices, and coefficientwise denominator clearing. Independent exact-rational consistency checks previously recorded for the formal transport include 24 terminal matrices (with 216 checked entries), 126 multiplicity tests, and 46 direct terminal-completion substitutions in the original stationary CCR recurrence. Four fixed-order examples, including $(n, P) = (13, 691)$, were checked by coefficientwise rational-polynomial denominator reconstruction and direct recurrence substitution. Such finite example checks corroborate the derivations; they do not establish the universal theorems by sampling.

The order-77 exponent table has a different proof status: it uses finite but exhaustive *coefficient-level* arithmetic for all 72 early-source coordinates. Each source monomial is

included in a min-plus valuation bound obtained from exact truncated series; coefficients attaining the reported 2- and 3-adic exponents certify sharpness. In the recorded exact certificate, the 2-adic exponents are attained by constant coefficients, while the 3-adic bounds have constant or λ_3 coefficient witnesses. This is a computer-assisted finite certificate for those exact exponent vectors, not a random sample or a solution of the remaining local existence tests at order 77. The complete high-dimensional coefficient calculation has not been independently rerun in the independent audit described here; its archived mathematical outcomes, not an additional external verification claim, are reported here.

No verification code, executable files, or supplementary data archive is included or promised as part of this publication. The mathematical claims and the stated finite computation outcomes, including their respective roles and limitations, are presented in the article.

34.55 Finite-range stationary CCR triple-zero impossibility

The finite local–global completion criterion above leaves open whether any admissible ordinary-integer source vector actually achieves a threefold zero. For the first thirty-six odd orders, the following independent obstruction theorem answers this negatively for every transport prime above twice the order. The argument combines Bernoulli transport-prime sieves with uniform finite-ring certificates; local q-adic source solvability is not confused with global ordinary-integer completion.

34.55.1 Introduction and stationary transport

Relation to preceding CCR research. The chronological-carry-refinement transport originates in the author's third-order, fourth-order, and general higher-order CCR studies [164, 165, 166]. Earlier arithmetic refinement classified selected source lattices, local ramification phenomena, and p-adic densities [195]. A related treatment of stationary transports established a sharp valuation law for a specific high-degree coefficient [218], whereas the author's single-site paper addresses prime-indexed Nörlund–Stirling root rigidity [219]. The result here concerns a different question: the *simultaneous* vanishing of the three lowest coefficients for *every* integer-source choice at all odd orders in a prescribed finite interval. We do not claim that classical Bernoulli congruences, p-adic lifting, or finite-field ideal elimination are novel in isolation. Relevant prior arithmetic literature includes [170, 198].

For an integer parameter P and source variables $\lambda_3, \lambda_4, \ldots$, put $F_0(X) = 1$ and $F_j(X) = 0$ for $j < 0$. For $j \geq 1$ define the rational polynomial F_j by

$$\begin{aligned} T_j F_j(X) &= ((P+1)X - P)F_{j-1}(X) \\ &\quad + P\big((P+1)(X-1) - P\big)F_{j-2}(X-1) \\ &\quad + P\sum_{s=3}^{j}(-1)^s \lambda_s F_{j-s}(X-s+1), \qquad (34.124) \\ T_j h(X) &= Xh(X+1) - (X-j)h(X). \end{aligned}$$

The operator T_j is upper triangular on coefficients in the monomial basis, with nonzero diagonal $j, j+1, \ldots$, so this determines F_j uniquely over $\mathbb{Q}$.

A *triple zero at the origin* means

$$X^3 \mid F_n(X) \quad \Longleftrightarrow \quad [X^0]F_n = [X^1]F_n = [X^2]F_n = 0.$$

Here "integer sources" always means *all* $\lambda_3, \ldots, \lambda_n \in \mathbb{Z}$; it does not refer to arbitrary rational or merely locally integral sources. This distinction is essential: as we recall below, suitable local source vectors often *can* produce triple zeros.

Theorem 34.116 (Certified finite-range exclusion)**.** *For each odd integer n with $5 \le n \le 75$, each prime $P > 2n$, and every source vector $(\lambda_3, \dots, \lambda_n) \in \mathbb{Z}^{n-2}$, the polynomial F_n defined by* (34.124) *satisfies*

$$X^3 \nmid F_n(X).$$

This theorem is *finite in n* but uniform in the transport prime and in *all* integer sources. The proof combines general source-independent congruences with exact coefficientwise certificates at the exceptional orders. No finite enumeration of integer-source vectors is assumed.

34.55.2 Contributions and novelty

This paper makes three specific contributions relative to the preceding CCR source literature. First, it gives an *all-integer-source* nonexistence theorem at each of the 36 odd orders from 5 to 75, for *every* transport prime $P > 2n$; this is stronger than excluding sampled source vectors, but deliberately does not assert a result at every odd order. Second, the Bernoulli transport filter is combined with source-independent prime-order and prime-plus-two obstructions and with explicit exceptional-order finite-ring compatibility identities. Third, at order 65 the argument eliminates all high-index sources by a uniform 5-adic coefficient-valuation bound and obtains a nonzero *function-level* Frobenius obstruction from the remaining 1,013 low-source monomials.

The Bernoulli arithmetic, finite-field elimination and p-adic methods are standard tools, not claimed as individually new. The contribution claimed here is the particular source-uniform finite-range exclusion and its specified combination of exact symbolic and computer-assisted certificates within the formal stationary CCR model. No priority claim against independently developed results, or conclusion about the physical realizability of arbitrary source vectors, is made.

34.55.3 Exact generating representation

Set $G(t, X) = \sum_{j \ge 0} F_j(X) t^j$ and write $G(t, X) = H(t) e^{X f(t)}$. Define $Y = e^f$. Directly summing (34.124) and comparing the coefficients of X and the constant term gives

$$tf'(t) + Y(t) - 1 = (P+1)t + P(P+1)t^2/Y(t), \tag{34.125}$$

$$(\log H)'(t) = -P - P(2P+1)t/Y(t) + P\sum_{s \ge 3} (-1)^s \lambda_s t^{s-1}/Y(t)^{s-1}. \tag{34.126}$$

They uniquely determine $f \in t\mathbb{Q}[P][[t]]$ and $H \in 1 + t\mathbb{Q}[P, \lambda][[t]]$. Writing

$$a = \frac{P+1}{2}, \qquad d^2 = \frac{(P+1)(5P+1)}{4}$$

solves (34.125) as a *formal* series:

$$Y(t) = at + dt \coth(dt) = 1 + at + \sum_{m \ge 1} \frac{2^{2m} B_{2m}}{(2m)!} d^{2m} t^{2m}. \tag{34.127}$$

Only even powers of d occur, so no quadratic extension is needed. A convenient exact consequence is

$$[X^k] F_n(X) = [t^n]\ H(t) \frac{f(t)^k}{k!}. \tag{34.128}$$

This is a source-sensitive Sheffer representation, not an identification of the entire family with classical source-free Nörlund polynomials.

34.55.4 Bernoulli transport condition and auxiliary-prime sieves

A necessary condition at the transport prime

Proposition 34.117 (Transport-prime Bernoulli filter)**.** *Let $n \geq 5$ be odd, $P > 2n$ be prime, and $\lambda_3, \dots, \lambda_n \in \mathbb{Z}$. If $X^3 \mid F_n(X)$ then*

$$P \mid \mathrm{num}(B_{n-1}), \tag{34.129}$$

where Bernoulli numbers are reduced rational fractions.

Proof. The inverse T_j^{-1} only divides by $j, j+1, \dots, 2j$ in the degrees of interest; these are P-units for $j \leq n$ and $P > 2n$. By induction the coefficients of F_j are P-integral. Mod P the source terms and the extra transport corrections vanish, leaving the ordinary Nörlund–Stirling shadow

$$S_j(X) = [t^j] \left(\frac{t}{1 - e^{-t}} \right)^X .$$

For odd $n \geq 5$, $[X^0]S_n = [X^1]S_n = 0$ and

$$[X^2]S_n = -\frac{B_{n-1}}{2(n-1)(n-1)!}. \tag{34.130}$$

These rational denominators are P-units because $P > 2n$. Thus $[X^2]F_n = 0$ implies $B_{n-1} = 0$ modulo P, proving (34.129). □

Remark 34.118. For P-adic *sources*, the corresponding condition is not an obstruction at the same prime: the final three source variables act via an affine triangular P-adic unit matrix on the three lowest jets. Consequently Bernoulli irregularity permits a uniquely tuned P-adic terminal triple after earlier sources are fixed. This does not imply existence of an ordinary-integer terminal triple, because integrality at other primes must also hold. The finite-range theorem exploits exactly this distinction.

Prime-order source-independent residues

Let $\ell \geq 5$ be an odd prime, and put $a = (P+1)/2$ and $D = (P+1)(5P+1)/4$. For $c_j = [X^j]F_\ell(X)$, a formal Bernoulli-pole computation using (34.127), (34.126), and von Staudt–Clausen gives

$$\ell c_2 \equiv a D^{(\ell-1)/2}, \qquad \ell c_1 \equiv a^\ell - (a+P) D^{(\ell-1)/2} \pmod{\ell}. \tag{34.131}$$

Both residues hold for *all* ℓ-integral source vectors. The apparent products ℓc_j are defined by first taking rational coefficients: the pole is at most simple at ℓ. The coefficient of $t^{\ell-1}$ of Y has residue $\ell[t^{\ell-1}]Y = D^{(\ell-1)/2}$. The unique pole terms in f_ℓ and $H_1 f_{\ell-1}$ then yield (34.131); all remaining contributions are ℓ-integral.

Proposition 34.119 (Prime-order obstruction)**.** *For any odd prime $\ell \geq 5$, any integer P, and integer sources,*

$$X^3 \mid F_\ell(X) \quad \Longrightarrow \quad P \equiv -1 \pmod{\ell}.$$

Proof. Eliminating $D^{(\ell-1)/2}$ from (34.131) yields

$$\ell\big((P+1)c_1 + (3P+1)c_2\big) \equiv \frac{(P+1)^2}{2} \pmod{\ell}.$$

If both c_1 and c_2 vanish, the right-hand side vanishes. Since ℓ is odd, this forces $P = -1$ modulo ℓ. □

Orders two above a prime

Let $n = \ell + 2$, $\ell \geq 5$ an odd prime, and set

$$a = \frac{P+1}{2}, \quad d^2 = \frac{(P+1)(5P+1)}{4}, \quad b = \frac{(P+1)(7P-1)}{24}, \quad \Delta = (d^2)^{(\ell-1)/2}.$$

Define the finite-field polynomial

$$I_\ell(P) = -\frac{(P+1)^2}{1152}\Big[(162P^2+100P+2)a^\ell + (27P^3-269P^2-171P-3)\Delta\Big] \quad \in \mathbb{F}_\ell. \tag{34.132}$$

Proposition 34.120 (Prime-plus-two obstruction)**.** *For $n = \ell+2$ and arbitrary ℓ-integral sources,*

$$X^3 \mid F_n(X) \quad \Longrightarrow \quad I_\ell(P) = 0.$$

Proof. The Sheffer representation in Subsection 34.55.3 gives the exact prime-pole coefficient calculation. Write $H = \sum h_j t^j$, $f = \sum f_j t^j$, $y_2 = d^2/3$, $b = y_2 - a^2/2$, and $\Theta = -d^2\Delta/3$. Von Staudt–Clausen and Kummer congruences give

$$\ell[t^{\ell-1}]Y = \Delta, \qquad \ell[t^{\ell+1}]Y = \Theta \pmod{\ell}.$$

Every other relevant coefficient of Y is ℓ-integral, and two pole factors cannot enter a coefficient below order $\ell+2$, since $2(\ell-1) > \ell+2$ for $\ell \geq 5$. Set, in $\mathbb{F}_\ell$,

$$E = a^\ell - a\Delta, \qquad U = \Delta(a^2 - y_2) + \Theta,$$
$$V = \Delta(-a^3 + 2ay_2) - a\Theta, \quad s = a^3/3 - ay_2.$$

With $q = \ell h_\ell$, direct extraction from $H' = H(\log H)'$ gives

$$\ell h_{\ell+1} = -Pq + P(2P+1)\Delta,$$
$$\ell h_{\ell+2} = -Paq + P\big(\lambda_3 - (P+a)(2P+1)\big)\Delta.$$

These expressions include *all* possible source contributions to the pole residues, not just a chosen source specialization. Using $h_1 = -P$, $h_2 = -Pa$ and $h_3 = (P^2a + P^2(2P+1) + P(2P+1)a - P\lambda_3)/3$, the three scaled terminal jets satisfy

$$\begin{aligned} \ell c_0 &= \ell h_{\ell+2}, \\ \ell c_1 &= V + h_1 U + h_2 E + h_3\Delta + bq + a\ell h_{\ell+1}, \\ \ell c_2 &= aU + bE + s\Delta + h_1(aE + b\Delta) + h_2 a\Delta + (a^2/2)q, \end{aligned} \qquad (\mathrm{mod}\ell).$$

The linear combination $\ell[(a^2/6)c_0 + (a^2/2)c_1 + (4Pa/3 - b)c_2]$ cancels both q and λ_3. Substituting the displayed values of U, V, E, Θ, y_2 and simplifying gives exactly $I_\ell(P)$ in (34.132). Because 2 and 3 are ℓ-units, this is valid for all ℓ-integral sources. If $c_0 = c_1 = c_2 = 0$, then necessarily $I_\ell(P) = 0$. □

34.55.5 Certified order reductions below 57

A finite-order theorem demands a *complete* enumeration of the eligible transport primes after Proposition 34.117, with proof of primality and a source-independent obstruction for every survivor. The first 26 odd orders divide naturally into three classes:

Class	Orders	Certificate family
Odd-prime orders	$5, 11, 17, 23, 29, 37, 41, 47, 53$	Bernoulli + prime-order sieve
Prime-plus-two orders	$7, 9, 13, 15, 19, 21, 25, 31, 33, 39, 43, 45, 49, 55$	Exact shifted-order sieve and exceptional auxiliary-prime test
Composite exceptions	$27, 35, 51$	Exact finite-ring polynomial identities below

For the prime-plus-two class, the exact finite-field census checks (34.132) at every eligible Bernoulli transport prime. Its single zero case is $(n, P) = (21, 283)$, which is removed by a separate source-independent 5-adic coefficient congruence: for the specified integer denominator D_{21},

$$D_{21}[X^2]F_{21}(X) \equiv 2(P+1)^{11} \pmod 5;$$

at $P = 283$ this is nonzero. The remaining three composite orders are settled by:

1. $n = 27$: a second-level 13-adic polynomial ideal identity valid for all source residues.
2. $n = 35$: a second-level 17-adic polynomial ideal identity valid for all source residues.
3. $n = 51$: a complete 17-adic two-coordinate nonsquare criterion, with zero common residue roots.

The decisive finite-ring identities for these orders are recorded below; their source-polynomial coefficients are obtained by exact rational arithmetic over all bounded-weight source monomials, not by sampling integer-source vectors.

Explicit finite-ring certificates for the three composite exceptions

The following equations describe the finite checks used in the three remaining composite orders. The preliminary exact rational expansion includes *every* source monomial of admissible weight; the formulas below are the resulting small finite-field certificates, not specializations obtained by setting higher sources to zero.

Order 27. The Bernoulli filter leaves only $P = 657931$. For arbitrary early sources $z = (\lambda_3, \dots, \lambda_{24})$, let $R_{25}(z), R_{26}(z), R_{27}(z)$ be the uniquely determined *rational* terminal sources making $c_2 = c_1 = c_0 = 0$. Their exact triangular diagonal coefficients are $-P(P+1)^2/(8 \cdot 25)$, $P(P+1)/(2 \cdot 26)$ and $-P/27$. Coefficientwise exact expansion through weight 27 involves 1,049 early-source monomials. Over $\mathbb{F}_{13}$ set

$$A = \overline{13^2 R_{25}}, \qquad B = \overline{13^2 R_{27}}, \qquad T = \overline{13(R_{26} - 2R_{25})}.$$

The coefficientwise integrality calculation yields the *polynomial ideal identity* $T = 1 + Q_A A + Q_B B$, with the following explicit affine polynomials (all coefficients reduced modulo 13, $x_j = \bar{\lambda}_j$):

$$\begin{aligned}
A &= 3 + 2x_3 + 6x_4 + 9x_5 + 12x_6 + 10x_7 + 3x_8 + 8x_9 + 3x_{10} + 7x_{11} + 2x_{12} + 12x_{13},\\
B &= 3x_3 + 4x_4 + 6x_5 + 8x_6 + 11x_7 + 2x_8 + x_9 + 2x_{10} + 9x_{11} + 10x_{12} + 8x_{13},\\
Q_A &= -5 + 4x_3 + 5x_4 + x_5 - 3x_6 + 4x_7 - 4x_8 - 2x_9 - 4x_{10} - 5x_{11} + 6x_{12} - 3x_{13},\\
Q_B &= -3 - 6x_3 + 3x_4 - 2x_5 + 6x_6 + 5x_7 - 5x_8 + 4x_9 - 5x_{10} - 3x_{11} + x_{12} + 6x_{13}.
\end{aligned}$$

An integral terminal triple forces $A = B = T = 0$ and contradicts $T = 1 + Q_A A + Q_B B$. The algebraic multiplication of the displayed certificate is a finite check; deriving T from the rational CCR source expansion remains the computer-assisted coefficient step.

Order 35**.** The Bernoulli filter leaves only $P = 151628697551$. Let R_{33}, R_{34}, R_{35} denote the rational terminal-source polynomials, and set $A = \overline{17^2 R_{33}}$, $B = \overline{17^2 R_{35}}$, $T = \overline{17(R_{34} - 2R_{33})}$. The coefficient audit includes 4,737 source monomials. Write $\mathbf{x} = (x_3, \ldots, x_{17})$ over $\mathbb{F}_{17}$; the coefficient of x_9 is zero. The exact reduction is

$$\begin{aligned} A &= 7 + (7, 7, 16, 11, 2, 16, 0, 8, 8, 12, 16, 5, 13, 7, 5) \cdot \mathbf{x}, \\ B &= 15 + (9, 2, 7, 8, 3, 7, 0, 12, 12, 1, 7, 16, 11, 2, 16) \cdot \mathbf{x}, \\ Q_A &= 6 + (4, 3, 2, 12, 13, 2, 0, 1, 1, 10, 2, 7, 8, 3, 7) \cdot \mathbf{x}, \\ Q_B &= 5 + (16, 13, 3, 1, 11, 3, 0, 10, 10, 15, 3, 2, 12, 13, 2) \cdot \mathbf{x}. \end{aligned}$$

The degree-two reduced polynomial satisfies $T = 10 + Q_A A + Q_B B$ over $\mathbb{F}_{17}[\mathbf{x}]$. Consequently $A = B = T = 0$ is impossible. These short ideal identities are recorded explicitly; the exhaustive coefficient reductions used to obtain them are computer-assisted.

Order 51**.** The Bernoulli filter leaves exactly $P_1 = 417202699$ and $P_2 = 47464429777438199$. For $u = \bar{\lambda}_3$ and $v = \bar{\lambda}_4 + \sum_{s=5}^{17} b_s \bar{\lambda}_s$, the vectors $(b_5, \ldots, b_{17})$ are respectively

$$\begin{aligned} P_1 &: (10, 0, 9, 15, 11, 12, 1, 0, 6, 10, 13, 8, 12), \\ P_2 &: (3, 5, 12, 10, 15, 9, 14, 14, 15, 3, 1, 9, 1). \end{aligned}$$

The exact 17-adic reduction of the two lowest nonconstant jets, after scaling by 17^3, is

$$\begin{aligned} P_1: \quad E_1 &= -2u^2 + 8uv + 4u - 8v^2 - 5v - 6, \\ E_2 &= -7u^2 - 6uv - 7u + 6v^2 - 2, \\ P_2: \quad E_1 &= 3u^2 - 4uv + 5u + 7v^2 + 2v - 3, \\ E_2 &= 8u^2 - 5uv - 6u - 4v^2 - 2v + 5. \end{aligned}$$

All 66,418 possible weighted source monomials are included in the finite coefficient audit for each prime; none of the omitted higher-source terms contributes modulo 17 after scaling. Lexicographic elimination in $\mathbb{F}_{17}[u, v]$ yields

$$\begin{array}{c|l} P_1 & (u + 2v + 8,\ v^2 + 6v + 2), \\ P_2 & (u + 8v - 8,\ v^2 - 8v - 6). \end{array}$$

The respective discriminants are 11 and 3 modulo 17, neither a square. Thus these pairs have no common $\mathbb{F}_{17}$-point, eliminating all integral sources.

34.55.6 Exceptional orders 57 through 75

Order 57

At $n = 57$, the exact Bernoulli numerator is

$$|\mathrm{num}(B_{56})| = 7 \cdot 113161 \cdot 163979 \cdot 19088082706840550550313.$$

The largest eligible factor is removed by a source-uniform 19-adic nonsquare obstruction computed from the exact source-polynomial jets. For the remaining two factors, let

$$E_i = \overline{19^3 [X^i] F_{57}} \in \mathbb{F}_{19}[\lambda_3, \ldots, \lambda_{57}], \qquad i = 0, 1, 2.$$

Exact coefficient enumeration proves that each E_i depends only on two effective linear source coordinates $U = \overline{\lambda_3}$ and $V = \overline{\lambda_4} + \sum_{s=5}^{19} b_s \overline{\lambda_s}$, with b_s depending on the transport prime. In index order $s = 5, \ldots, 19$, the two coefficient vectors are, respectively,

$$\begin{aligned} P = 113161: &\quad (15, 11, 2, 17, 1, 12, 12, 17, 9, 3, 10, 13, 10, 2, 1), \\ P = 163979: &\quad (0, 15, 8, 0, 6, 7, 0, 10, 18, 0, 4, 11, 0, 13, 12). \end{aligned}$$

For $P = 113161$, elimination of $E_1 = E_2 = 0$ gives

$$U - 4V + 1 = 0, \qquad V^2 - 6V - 7 = 0,$$

whereas reduction of E_0 by these two equations yields $V - 9$. At $V = 9$ the displayed quadratic equals 1 modulo 19. For $P = 163979$ the corresponding equations are

$$U + 8V + 1 = 0, \qquad V^2 - 4V - 7 = 0,$$

and the reduction of E_0 is $-V - 4$; its zero $V = 15$ leaves quadratic residue $6 \neq 0$. Thus the ideal (E_0, E_1, E_2) contains 1 in both cases. The source-coordinate reduction is an *exact coefficientwise identity*, derived by an exhaustive finite-ring coefficient reduction, not an assumption that higher sources may simply be set to zero.

Orders 59 **through** 69

At $n \in \{59, 61, 63, 67, 69\}$ the Bernoulli admissibility sieve followed by Proposition 34.119 or 34.120 has been checked against every certified eligible transport prime by exact rational and modular arithmetic. The remaining order $n = 65$ is qualitatively different.

Theorem 34.121 (Source-uniform 5-adic obstruction at $n = 65$)**.** *For every prime $P >$ 130 and every integral source vector, $X^3 \nmid F_{65}(X)$.*

Proof. The Bernoulli numerator of B_{64} has exactly two eligible transport prime factors:

$$P_1 = 1226592271, \qquad P_2 = 870573153545221791849896997917 27.$$

Put $c_i = [X^i]F_{65}(X)$ and $E_1 = \overline{5^{17}c_1}$, $E_2 = \overline{5^{18}c_2}$ in $\mathbb{F}_5$. A complete min-plus source-weight bound for the coefficients of the Sheffer expansion proves that every monomial involving some λ_s with $s \geq 6$ vanishes from both E_1 and E_2. This coefficientwise bound holds *uniformly* for all integer sources, including arbitrary high-index ones. The exact min-plus convolution yields the following lower bounds on 5-adic coefficient valuations:

	$\lambda_3, \lambda_4, \lambda_5$ only	at least one λ_s, $s \geq 6$
c_0	-16	-14
c_1	-17	-15
c_2	-18	-16

In particular $5^{17}c_1$ and $5^{18}c_2$ are integral coefficientwise, and terms involving a source index at least six reduce to zero modulo 5.

Let $a = \bar{\lambda}_3$, $b = \bar{\lambda}_4$ and $c = \bar{\lambda}_5$. The exact surviving low-source polynomials for $P = P_1$ are

$$E_1 = 1 + 4a + 2b + 4c + 3a^5 + 4b^5 + 3c^5, \qquad E_2 = 3 + 3a^5 + 4b^5 + 3c^5,$$

and for $P = P_2$ they are

$$E_1 = b + 3c + 2b^5 + c^5, \qquad E_2 = 3 + 2b^5 + c^5.$$

Reducing modulo the *function* ideal $(a^5 - a, b^5 - b, c^5 - c)$ gives

$$E_1 - 4E_2 = \begin{cases} 4, & P = P_1, \\ 3, & P = P_2. \end{cases} \qquad \text{on } \mathbb{F}_5^3.$$

Both right-hand sides are nonzero. Thus c_1 and c_2 cannot both vanish for any integer source vector. The computer-assisted finite input is the exhaustive source-weight valuation bound and exact 1,013-monomial low-source coefficient calculation, summarized quantitatively below. These computations are not replaced by substituting finitely many source values. □

Orders 71, 73, and 75

The exact Bernoulli numerators and complete prime certificates leave six eligible transport-prime cases in these three orders. Applying Proposition 34.119 at $n = 71$ and Proposition 34.120 at $n = 73, 75$ gives the following reproducible table:

n	Eligible transport prime P	Test	Value
71	688531	$P \bmod 71$	44
71	20210499584198062453	$P \bmod 71$	12
71	3090850068576441179447	$P \bmod 71$	18
73	3112655297839	$I_{71}(P)$	10
73	18723419087606889767942264996363043575678 11	$I_{71}(P)$	68
75	92303830511408562200892091166142257261319750 7651	$I_{73}(P)$	35

At order 71 the required residue is 70; at orders 73 and 75 the invariant is required to vanish. None of the six cases satisfies its necessary condition. The displayed candidates arise from exact Bernoulli fractions, complete numerator factorizations and recursive primality certificates for the corresponding factor trees. Its reproduced final assertion is "orders 71, 73, 75 globally excluded for all integer sources; PASS."

34.55.7 Completion of the finite-range proof

Proof of Theorem 34.116. For $5 \le n \le 55$ the odd-order set is partitioned by the three mutually exclusive rows in Subsection 34.55.5: $9 + 14 + 3 = 26$ orders, each treated by the cited Bernoulli and auxiliary-prime certificates. Order 57 is covered by the 19-adic three-jet computation of Subsection 34.55.6. Of the six odd orders $59, 61, 63, 65, 67, 69$, five are covered by the exact transport-prime sieve and order 65 by Theorem 34.121. The three remaining odd orders $71, 73, 75$ are covered by the explicit table in Subsection 34.55.6. Thus every one of the $26 + 1 + 6 + 3 = 36$ odd orders in the interval is excluded, with all prime transport parameters and all ordinary integer source vectors quantified universally. □

Remark 34.122 (Quantifier firewall). A source vector in $\mathbb{Q}_P$, a vector in $\mathbb{Z}_P$, and an ordinary integer vector are different objects. Local solvability at the transport prime does not contradict any theorem in this paper. The universal quantifiers in this theorem are over all transport primes and ordinary integer sources for the stated finite set of orders; no claim about physical machines follows.

Exact verification and finite-certificate scope

The theorem combines symbolic arguments with exact, exhaustive computer-assisted finite arithmetic. The computations are part of the proof at the exceptional degrees; they are not merely randomized consistency experiments and do not replace the universal symbolic reductions to finite residue rings. Specifically:

Orders / stage	Exact finite assertion and scope
$5 \le n \le 55$	Complete Bernoulli transport-prime factorizations and prime/shifted-order congruences; the single $(21, 283)$ residual case is excluded by the 5-adic congruence displayed above.
$27, 35$	All 1,049 / 4,737 early-source monomials in the terminal-source expansion; the resulting explicit polynomial ideal identities with residues 1 / 10.
51	All 66,418 possible weighted source monomials for each eligible transport prime; exact two-coordinate reduction and the two nonsquare discriminants.
57	All 162,878 possible weighted monomials through source weight 57, and two 19-adic, three-jet unit-ideal tests; the third eligible prime is separately excluded.
65	All-index min-plus valuation bounds and all 1,013 low-source monomials for each of the two eligible primes; function-level Frobenius obstruction in $\mathbb{F}_5^3$.
$71, 73, 75$	Complete Bernoulli numerator factorization and primality checks for eligible factors, followed by six exact finite-field congruence tests in the displayed table.

The finite arithmetic uses exact rational coefficients, prime certificates and modular polynomial identities, with no floating-point tolerance. The displayed formulas record the mathematical outcomes used by the proof. The underlying full coefficient enumerations, although internally audited, are not reproduced line by line in this article, and no source-code repository or downloadable verification supplement is represented as part of this publication. A complete independent replication of those computer-assisted stages would require regenerating the stated finite expansions; the symbolic lemmas alone do not establish those finite assertions.

Conclusion

Theorem 34.116 excludes a triple zero at the origin for all 36 odd orders from 5 to 75, every eligible prime transport and all integral source choices. It separates a general Bernoulli transport filter and auxiliary-prime criteria from the genuinely source-polynomial obstructions required at exceptional composite orders. All stated claims are about the explicitly defined stationary CCR polynomial recurrence and the specified finite range.

34.56 Primitive-orbit rigidity and predecessor noncoverage

The stationary transport also possesses a source-independent Nörlund–Stirling shadow. Having separated rational source completion from ordinary-integer obstructions, we now ask a different question: when does a target shadow root or an irreducible Frobenius orbit first appear? The distinctions between rational nonsharing, finite-field polynomial noncoverage, geometric root support, and physical machine realizability remain explicit.

Contributions and novelty

The distinct contributions of this article, relative to the cited classical Nörlund–Stirling literature and the authorś earlier stationary CCR work, are as follows.

1. A source-uniform interpolation versus fixed-shadow rigidity dichotomy, including the precise upper limit on simultaneous anchoring of all split shadow-root positions.

2. The exact adjacent-column shift-companion identity, its multiplicity-sensitive polynomial-gcd formulation, and the resulting degree-pairing obstruction.

3. A family of primitive long-Frobenius-orbit criteria in four explicit factor degrees, and an all-odd-order rational predecessor noncoverage theorem derived using the cited large-Zsigmondy result.

4. An effective coefficient-content criterion for each fixed odd order, together with an exact finite, computer-assisted noncoverage certificate at every good prime for odd orders through 71.

5. A Bernoulli integer-coordinate ladder and explicit p-adic first-occurrence examples, with singular-root limitations kept separate from simple-root lifting.

The generating functions, classical number-theoretic theorems, and general computational-algebra tools are prior work. We do not claim an all-order, all-good-prime noncoverage proof or physical realization by a pushdown automaton.

34.56.1 Introduction and provenance

The Nörlund coefficients

$$S_j(X) = [t^j] \left(\frac{t}{1 - e^{-t}} \right)^X \tag{34.133}$$

are classical polynomials. Their arithmetic normalization and their relation to Stirling coefficients are established topics; see Adelberg [170], Bencherif–Garici [167], McCarthy [225], and the NIST reference [172]. We make no claim to introducing this generating series or classical Hensel, resultant, subresultant, or Chebotarev methods. The stationary chronological-carry-refinement (CCR) recurrence and its source interpretation originate in the author's earlier CCR-3, CCR-4, CCR-k, and arithmetic-refinement studies [164, 165, 166, 195]. This article instead concerns whether an entire Frobenius orbit of a later column can have occurred in an earlier column, *uniformly across stationary source choices*.

The contribution combines several distinct, logically separated pieces. An exact *shift-companion characterization* converts an adjacent-column common factor into a self-translate common factor of one polynomial; it is proved without a finite calculation. A range of *large-factor primitive-orbit criteria* follows from a small number of leading coefficients together with exact Euclidean certificates for the harder codimensions. These tools complement, rather than imply, the global rational and Bernoulli congruence arguments developed below. A finite predecessor-product certificate through order 71 is reported separately; its finite range must not be confused with the uniform irreducible-orbit theorems.

34.56.2 The stationary model and good-prime normalization

For a prime p and a source vector $\boldsymbol{\lambda} = (\lambda_3, \lambda_4, \ldots)$, define $F_0 = 1$, $F_j = 0$ for $j < 0$, and

$$\begin{aligned} T_j F_j(X) &= ((p+1)X - p)F_{j-1}(X) \\ &\quad + p((p+1)(X-1) - p)F_{j-2}(X-1) \\ &\quad + p\sum_{s=3}^{j} (-1)^s \lambda_s F_{j-s}(X - s + 1), \\ T_j h(X) &= Xh(X+1) - (X-j)h(X). \end{aligned} \tag{34.134}$$

The diagonal of T_j on polynomials of degree at most j is $j, j+1, \ldots, 2j$. Fix an integer N and assume throughout a *good prime* $p > 2N$. Then every inverse T_j^{-1}, $j \le N$, is p-integral, and

$$F_j(X; p, \boldsymbol{\lambda}) \equiv S_j(X) \pmod{p} \qquad (j \le N,\ \lambda_s \in \mathbb{Z}_p). \tag{34.135}$$

This is coefficientwise congruence. No statement here is made about physical machines.

For odd $n \geq 5$ the shadow has the rational factorization

$$S_n(X) = \frac{X^2(X-1)}{2^n n!}\, f_n(X), \qquad f_n \in \mathbb{Q}[X] \text{ monic}, \quad \deg f_n = n-3. \tag{34.136}$$

The polynomial f_n is the monic normalization of the primitive integral Stirling quotient q_n used in the CCR literature. The normalizing leading coefficients are p-units in the good-prime range. We always use gcd to mean the *monic* polynomial greatest common divisor over the relevant field.

An irreducible monic factor $g \mid \overline{f}_n$ over $\mathbb{F}_p$ is called *primitive relative to the preceding columns* when

$$\gcd(g, \overline{S_m/X}) = 1 \qquad (1 \leq m < n). \tag{34.137}$$

A bar indicates coefficientwise reduction. The condition is unaffected by the choice of a common integral source vector because of (34.135). For $\deg g \geq 2$ it is equivalent to nonsharing against the full earlier shadows S_m.

34.56.3 Source interpolation and positional rigidity

A source-vector obstruction and a root-location obstruction are not the same. Even when the shadow factorization is fixed, the stationary source parameters can be used to move the lifted roots. The following two theorems give a sharp control/rigidity distinction; neither should be read as a statement about the number of roots of the deformed target polynomial.

Theorem 34.123 (Universal interpolation by a trailing source block)**.** *Fix $n \geq 5$, a prime $p > 2n$, and $1 \leq d \leq n-2$. Fix arbitrary sources $\lambda_3, \dots, \lambda_{n-d} \in \mathbb{Z}_p$ (when present). Let $a_1, \dots, a_d$ be pairwise distinct zeros of $\overline{S}_n$ and let $\alpha_i \in \mathbb{Z}_p$ satisfy $\alpha_i \equiv a_i \pmod p$. There exists a* unique *tail tuple $(\lambda_{n-d+1}, \dots, \lambda_n) \in \mathbb{Z}_p^d$ for which*

$$F_n(\alpha_1) = \cdots = F_n(\alpha_d) = 0.$$

The a_i need not be simple target-shadow roots. If the prescribed points are roots of fixed earlier columns of order at most $n-d$, those parent columns remain unchanged.

Proof. Set $C_{n,s}(X) = p^{-1}\partial F_n/\partial\lambda_s \pmod p$. Differentiation of the stationary recurrence, or extraction of the linear source contribution from its generating series, gives

$$C_{n,s}(X) = (-1)^s [t^n] A(t)^X \int_0^t z^{s-1} A(z)^{-(s-1)}\, dz \pmod p.$$

Since $A(t) = 1 + t/2 + O(t^2)$, the polynomial $C_{n,s}$ has degree exactly $n-s$, with leading coefficient

$$[X^{n-s}] C_{n,s} = \frac{(-1)^s}{s\, 2^{n-s}(n-s)!} \quad \text{in } \mathbb{F}_p^\times.$$

Thus $C_{n,n}, C_{n,n-1}, \dots, C_{n,n-d+1}$ form a polynomial basis of degrees $0, 1, \dots, d-1$. Their evaluation matrix at the distinct a_i has a nonzero Vandermonde determinant, multiplied by nonzero leading coefficients. Because each $\overline{S}_n(a_i)$ vanishes, the vector map $(F_n(\alpha_i)/p)_i$ is integral and affine modulo p in the tail variables, with exactly this invertible Jacobian. Nonlinear source terms contribute at least p^2 before the division by p. The multivariate Hensel lemma produces the unique $\mathbb{Z}_p$ solution, to all source digits. A tail source of order above a parent column cannot modify that parent. □

A complementary source-independent coefficient survives *all* stationary source deformations. Put $R_n = S_n/X$, and in the zero-source transport allow a formal transport parameter P in place of the prime p; denote $E_n = \partial_P F_n(X; P, 0)|_{P=0}$.

Lemma 34.124 (The uncorrectable transport drift)**.** *For every odd $n \geq 5$,*

$$\boxed{[X^{n-2}]\,\mathrm{rem}_{R_n} E_n = -\frac{4(n+3)}{9\,2^n(n-2)!}.} \tag{34.138}$$

Moreover, every source-dependent summand of F_n has X-degree at most $n-3$, even when several sources interact nonlinearly.

Proof. The second statement follows by induction in (34.134): the source insertion at order $s \geq 3$ starts at degree at most $n-s$, and T_n^{-1} preserves the degree filtration. For the first statement, extract the three highest X-coefficients from the zero-source generating series $H_P(t)\exp(Xf_P(t))$ and its derivative at $P=0$; the exact series expansion at $P=0$ gives $f_0(t) = t/2 - t^2/24 + O(t^4)$ and derivative coefficients $a' = 1/2, b' = 1/4, c' = -1/6$. The polynomial division by R_n has quotient $nX + n(4n-10)/3$ at its top two coefficients. Substituting the extracted coefficients yields (34.138). The equality is obtained by symbolic coefficient extraction and polynomial division, not by extrapolation from finitely many numerical orders. Its top coefficient is the displayed nonzero rational function of n. □

Theorem 34.125 (Sharp fixed-shadow anchor capacity)**.** *Let $n \geq 5$ be odd, $p > 2n$, and suppose $R_n \bmod p$ splits into $n-1$ distinct roots, with exact p-adic shadow-root positions $\beta_1, \ldots, \beta_{n-1}$. Then any selected $n-2$ of these positions can be made simultaneous zeros of F_n by a unique integral source vector $\lambda_3, \ldots, \lambda_n$, but* all *$n-1$* positions cannot *be made zeros even modulo p^2. When $n-2$ are anchored, the evaluation at the remaining exact shadow position has p-adic valuation exactly one.*

Proof. Existence and uniqueness follow from Theorem 34.123 with $d = n-2$. For impossibility, reduce $p^{-1}(F_n - S_n)$ modulo p and divide it by R_n. The source contribution has degree at most $n-3$, whereas Lemma 34.124 shows the remainder has exact degree $n-2$ and p-unit leading coefficient, independently of all sources. It cannot vanish at $n-1$ distinct residue classes. If it vanishes at any $n-2$ classes, it is a unit multiple of their monic root polynomial; its value in the last residue class is a unit, so the corresponding evaluation of F_n has valuation exactly one. □

Corollary 34.126 (Source-independent weighted first motion)**.** *Retain the complete splitting hypothesis of Theorem 34.125. Write $a_i = \overline{\beta}_i$ and allow moving targets $\gamma_i = \beta_i + p\delta_i$. If all $F_n(\gamma_i) \equiv 0 \pmod{p^2}$, then*

$$\boxed{\sum_{i=1}^{n-1} a_i\delta_i \equiv \frac{4n(n-1)(n+3)}{9} \pmod{p}.} \tag{34.139}$$

Proof. For $H = (F_n - S_n)/p \pmod{p}$, its remainder modulo R_n has top coefficient κ_n from (34.138). Lagrange interpolation on the distinct β_i gives $\sum_i H(\beta_i)/R_n'(\beta_i) = \kappa_n/L_n$, where $L_n = 1/(2^n n!)$ is the leading coefficient of R_n. Because $S_n = XR_n$, Taylor expansion at $\gamma_i = \beta_i + p\delta_i$ gives $F_n(\gamma_i)/p \equiv H(\beta_i) + a_i\delta_i R_n'(\beta_i) \pmod{p}$. Sum after dividing by $R_n'(\beta_i)$ and insert $\kappa_n/L_n = -4n(n-1)(n+3)/9$. □

The fixed-position capacity concerns preassigned exact shadow zeros, not freely moving roots; in particular it does not contradict the source-universal primitive residue-orbit results below. It also does not imply a no-go for every possible assignment of roots from earlier *moving* CCR columns, which can depend on some of the same tail sources.

34.56.4 The exact shift-companion identity

Theorem 34.127 (Shift-companion theorem)**.** *For every odd $n \geq 5$ the following identities hold over $\mathbb{Q}[X]$:*

$$\frac{S_{n-1}(X)}{X} = \frac{(X+1)^2 f_n(X+1) - (X-n)(X-1)f_n(X)}{2^n n!}, \tag{34.140}$$

$$f_n(-1) = \frac{2^{n-1}}{n+1}. \tag{34.141}$$

Consequently, for each prime $p > 2n$,

$$\boxed{\gcd\left(\overline{f}_n, \overline{S_{n-1}/X}\right) = \gcd\left(\overline{f}_n(X), \overline{f}_n(X+1)\right).} \tag{34.142}$$

The equality is valid with multiplicities and does not require $\overline{f}_n$ to be squarefree.

Proof. The shadow recurrence obtained from (34.133) is $T_n S_n(X) = X S_{n-1}(X)$. Substitute (34.136), expand T_n, and divide by X twice to obtain (34.140). At $X = -1$, direct coefficient extraction yields $S_n(-1) = [t^n](1-e^{-t})/t = (-1)^n/(n+1)! = -1/(n+1)!$. Using (34.136) at -1 gives (34.141). Because $p > 2n$, $(n+1)2^n n!$ is invertible modulo p, so $\overline{f}_n(-1)$ is nonzero. Reducing (34.140) modulo $\overline{f}_n$ gives $\overline{S_{n-1}/X} \equiv (X+1)^2 \overline{f}_n(X+1)/(2^n n!)$. The factor $(X+1)^2$ is relatively prime to $\overline{f}_n$ because $\overline{f}_n(-1) \neq 0$. Taking monic gcds proves (34.142). □

Corollary 34.128 (Degree pairing of adjacent-column Frobenius orbits)**.** *Suppose $p > 2n$ and $g \mid \overline{f}_n$ is irreducible of degree k and also divides $\overline{S_{n-1}/X}$. Then the distinct irreducible translate $g(X-1)$ also divides $\overline{f}_n$. Therefore*

$$\boxed{2k \leq n - 3.} \tag{34.143}$$

In particular, every irreducible factor of $\overline{f}_n$ with degree greater than $(n-3)/2$ is absent from the immediate *preceding shadow column.*

Proof. By (34.142), $g(X)$ divides $f_n(X+1)$; equivalently $g(X-1)$ divides $f_n(X)$. The polynomials $g(X)$ and $g(X-1)$ cannot coincide: for a monic degree-k polynomial their X^{k-1} coefficients differ by $-k$, which is nonzero because $1 \leq k \leq n-3 < p$. Thus two distinct monic irreducible factors, each of degree k, divide $\overline{f}_n$; their product divides it, giving $2k \leq \deg f_n = n - 3$. □

Corollary 34.129 (Exact exceptional-prime test)**.** *If a good prime supports any common factor of $\overline{f}_n$ and $\overline{S_{n-1}/X}$, it divides the numerator of the reduced rational resultant*

$$\mathrm{Res}_X(f_n(X), f_n(X+1)). \tag{34.144}$$

When this resultant is a nonzero rational number, the converse holds away from its denominators: its good-prime divisors are exactly the adjacent-column common-factor primes.

Proof. The statement is the standard resultant criterion applied to (34.142); good primes do not divide any leading coefficient or denominator used in normalization. □

For $(n, p) = (9, 23)$, the consecutive quotient roots are $19, 20 \pmod{23}$; the common root with order 8 is 19. Table 34.4 is a small exact illustration, not a claim about the factorization of all self-shift resultants. The general identities above do not depend on the table.

Table 34.4: Exact self-shift resultant prime support for primitive integral q_n (not for monic f_n).

n	$\lvert\operatorname{Res}(q_n(X), q_n(X+1))\rvert$	eligible good-prime support
5	$2^6 3^2$	none
7	$2^{27} 3^9$	none
9	$2^{54} 3^{15} 5^4 \cdot 23$	23

34.56.5 Primitive long Frobenius factors at every odd order

The next theorem organizes the uniform long-orbit statements obtained from the leading coefficients of the shadows. The first two cases admit short direct coefficient proofs; the last two are supported by symbolic Euclidean certificates whose exceptional branches are described below.

Theorem 34.130 (Prime-indexed primitive long-orbit theorem)**.** *Let $n \geq 5$ be odd, $p > 2n$, and $d = n-3$. Suppose $\overline{f}_n$ has an irreducible factor g of one of the following degrees, within the stated range:*

$\deg g$	*valid orders*
d	$n \geq 5$
$d-1$	$n \geq 7$
$d-2$	$n \geq 9$
$d-3$	$n \geq 11$.

(34.145)

Then g is primitive relative to every *earlier shadow column:* $\gcd(g, \overline{S_m/X}) = 1$ *for* $1 \leq m < n$.

Proof and symbolic certificate. Put $L_j = 1/(2^j j!)$ and write the monic top coefficients of S_j as

$$\frac{S_j(X)}{L_j} = X^j + a_j X^{j-1} + b_j X^{j-2} + \cdots, \quad a_j = -\frac{j(j-1)}{6}, \quad b_j = \frac{j(j-1)(j-2)(j-3)}{72}. \tag{34.146}$$

For $Q = f_n$ (monic) the first coefficients are

$$Q = X^d + AX^{d-1} + BX^{d-2} + \cdots, \quad A = 1 - \frac{n(n-1)}{6}, \quad B = A + \frac{n(n-1)(n-2)(n-3)}{72}. \tag{34.147}$$

The case $\deg g = d$ follows by comparing degrees against each S_m/X. At $m = n-2$ the next-to-leading discrepancy is $2(n-3)/3$; at $m = n-1$ the forced linear quotient leaves discrepancy $2(n-3)(n-1)/9$. Both are units in $\mathbb{F}_p$. This is also a direct proof that Q cannot divide a *single* earlier column.

For $\deg g = d-1$ only $m = n-3, n-2, n-1$ require attention; smaller orders have insufficient degree. The $m = n-3$ forced linear quotient has top remainder $-2(n-3)/3$, a unit. At $m = n-2$ the odd-column quotient factorization reduces the potentially shared long factor to one of degree $d-2$, too small. At $m = n-1$ the top remainder of S_{n-1}/X upon division by Q has degree $d-2$ and leading coefficient $2(n-3)(n-1)/9$, excluding a shared degree-$(d-1)$ factor.

For the remaining degrees we use *exact algebraic certificates in the variable n*, not sampling over a bounded set of n. The coefficient-extraction input, which is reproducible from (34.133), is

$$\sigma_k(j) = 2^k (j)_k [z^k] \left(1 - \frac{z}{12} + \frac{z^3}{1440} - \frac{z^5}{90720} + \frac{z^7}{4838400} - \frac{z^9}{239500800} + \cdots\right)^{j-k}. \tag{34.148}$$

Here $(j)_k$ is the falling factorial, and only terms of degree at most k are used. The monic coefficients of Q are $Q_k(n) = \sum_{i=0}^{k} \sigma_i(n)$. In any quotient-by-predecessor calculation, the monic coefficients of S_m/X are $\sigma_k(m)$. We apply the ordinary Euclidean algorithm symbolically, splitting off any vanishing-leading-coefficient cases *before* normalizing a remainder.

For $\deg g = d-2$ and $d-3$, the possible predecessor orders are confined to $m \geq n-4$ and $m \geq n-5$, respectively, by degree and the odd predecessor's known factors $X(X-1)q_m$. The symbolic reductions and their prime filters are summarized in Table 34.5. For the $m = n-5$ case, the two necessary remainder numerators (after removing factors that are units in the good-prime range) are

$$n^2 + 3n + 18, \qquad 11n^3 + 11n^2 + 492n - 3132.$$

Their exact integer resultant is $2^{11}3^4 \cdot 7 \cdot 11$, leaving no prime $p > 2n$ in this branch. For $m = n-2$, the corresponding necessary numerators are

$$27n - 56, \qquad 405n^2 - 2330n + 2064,$$

whose resultant is $-2^6 3^3 \cdot 433$. At $p = 433$, the required residue is $n \equiv 403$, incompatible with $2n < p$. These are exact polynomial resultant identities, not conclusions from a bounded scan. For the degree-$(d-2)$ case, use the *corrected* last-predecessor Euclidean chain from the degree-$(d-3)$ certificate; the incorrectly shifted quadratic-quotient computation is not used.

Table 34.5: Degree-$(d-3)$ common-factor exclusions, $n \geq 11$ odd. An *eligible* branch means $p > 2n$ and all division leading coefficients are units.

Predecessor	Exact obstruction/certified elimination
$m \leq n-6$	$\deg(S_m/X) = m-1 < d-3$.
$m = n-4$ (odd)	S_m/X factors as $X(X-1)q_m$, with $\deg q_m = n-7 < d-3$.
$m = n-5$	Two cubic-division conditions have resultant supported only on $2, 3, 7, 11$.
$m = n-2$	Two successive subresultant conditions yield $p = 433$ only; the forced residue $n \equiv 403 \pmod{433}$ violates $p > 2n$.
$m = n-3$	Three Euclid conditions leave $23, 1063, 3862141$; the admissible candidates are eliminated by parity, no common residue, or an earlier degree drop.
$m = n-1$	The correctly indexed chain has unit $r_0 = 2(n-3)(n-1)/9$ and unit $u_0 = -(n-2)(3n-11)/75$. The last conditions leave $151, 4967, 236479$, eliminated by parity, no common residue, or degree drop.

All certificates are polynomial identities in n and integer resultant factorizations. At the two exceptional branches where the next leading coefficient vanishes, the *degree drops below the proposed irreducible factor degree*, rather than creating an additional solution. Thus no good-prime pair (n, p) survives in any of the predecessor classes. For $m = n-3$, the reduced resultant-prime support is $\{2, 3, 5, 7, 23, 1063, 3862141\}$; at 23 the only common residue is even, at 1063 there is no common residue, and the residual candidate at 3862141 makes the next leading coefficient vanish, dropping the gcd degree. For $m = n-1$, the corresponding support is $\{2, 3, 5, 7, 151, 4967, 236479\}$; the candidates are excluded, respectively, by parity, absence of a common root, or a degree drop. These latter resultant supports and exceptional-residue eliminations are recorded exact symbolic certificate outcomes, not independently reconstructed from every coefficient in the present independent audit. The proof remains computer-assisted at those steps and does not infer the all-order result from a finite scan. □

Corollary 34.131 (Source-universal unramified factors)**.** *Under the hypotheses of Theorem 34.130, for every common stationary source vector $\boldsymbol{\lambda} \in \mathbb{Z}_p^{n-2}$ the target polynomial F_n has a unique monic p-integral factor $G_{n,g}$ reducing to g. It is irreducible over $\mathbb{Q}_p$, defines the unramified extension of degree* $\deg g$*, and shares no root with any F_m, $m < n$, under the* same *source vector.*

Proof. An irreducible factor of a finite-field polynomial is separable; in the four degree ranges of Theorem 34.130 the residual total degree $d - \deg g \le 3$ is strictly smaller than $\deg g$ except at the small excluded orders, so g occurs with multiplicity one. The remaining trivial factors $X^2(X-1)$ are relatively prime to g because $\deg g \ge 2$. Apply Hensel's coprime-factor lifting over $\mathbb{Z}_p$ to (34.135). The lifted factor defines an unramified extension, and any shared root with earlier F_m would reduce to a forbidden common residue root of g and S_m. □

34.56.6 Every odd order has a new rational irreducible factor

Write $\mathcal{B}_n = \prod_{m=2}^{n-1} P_m/X$ for the primitive integral predecessor product, where P_m denotes the primitive positive-leading integral numerator of S_m. The distinction between a *rational factor missing from every predecessor* and the stronger *absence of all possible mod-p coverage* is essential.

Lemma 34.132 (Negative integer specialization)**.** *For integers $a, n \ge 1$,*

$$S_n(-a) = \frac{(-1)^n a!}{(n+a)!} \left\{ {n+a \atop a} \right\}, \quad \textit{in particular} \quad S_n(-2) = \frac{2(-1)^n(2^{n+1}-1)}{(n+2)!}. \tag{34.149}$$

Proof. Since $A(t)^{-a} = t^{-a}(1-e^{-t})^a$, apply the binomial theorem and the ordinary finite-difference representation of the Stirling numbers of the second kind. The special case $a = 2$ then follows from $\left\{ {n+2 \atop 2} \right\} = 2^{n+1} - 1$. □

Lemma 34.133 (Boundary-prime integral shadows)**.** *If $r \ge n+2$ is prime, all coefficients of S_j, $0 \le j \le n$, belong to $\mathbb{Z}_{(r)}$, and their leading coefficients $1/(2^j j!)$ are units there. In particular, primitive normalization and polynomial evaluation commute with reduction modulo r.*

Proof. The coefficients of $(1-e^{-t})/t$ through t^n have denominators dividing $(n+1)!$. Its formal inverse $A(t)$ is therefore integral at r. Expand $A(t)^X = (1 + (A(t)-1))^X$ in binomial polynomials $\binom{X}{k}$, whose denominators $k!$ are r-units for $k \le n$. The claimed leading coefficient is obtained from $A(t) = 1 + t/2 + O(t^2)$. □

Theorem 34.134 (Global rational predecessor noncoverage)**.** *For every odd integer $n \ge 5$, the polynomial $q_n \in \mathbb{Z}[X]$ has an irreducible factor $g_n \in \mathbb{Q}[X]$ that does* not *divide $\mathcal{B}_n$. In particular,*

$$\boxed{\operatorname{rad}(q_n) \nmid \mathcal{B}_n \quad \textit{and} \quad q_n \nmid \mathcal{B}_n \quad \textit{over } \mathbb{Q}[X].} \tag{34.150}$$

Proof. The decisive external input is Feit's *large Zsigmondy prime theorem* in the form verified by Roitman [223, 224]. For any $m > 2$ outside $\{4, 6, 10, 12, 18\}$ there exists a prime r such that $\operatorname{ord}_r(2) = m$ and either $r > m+1$ or $r^2 \mid 2^m - 1$. Put $m = n+1$ (even). If $r > m+1 = n+2$, the congruence $r \equiv 1 \pmod m$ gives $r \ge 2m+1 > 2n$. By Lemma 34.132, $S_n(-2) = 0 \pmod r$, while $S_j(-2) \ne 0 \pmod r$ for every $1 \le j < n$ since $\operatorname{ord}_r(2) = n+1$ and all relevant denominators are r-units.

If the large divisor is instead $r = m+1 = n+2$ and $r^2 \mid 2^m - 1$, the denominator $(n+2)!$ in (34.149) has r-valuation exactly one, and the numerator has valuation at least two. Thus $S_n(-2) \equiv 0 \pmod r$ once again. For all $j < n$ the denominator $(j+2)!$ is an r-unit and the multiplicative order prevents a zero. Lemma 34.133 legitimizes every reduction at this borderline prime. This step is a *shadow-polynomial statement only*: the stationary recurrence need not admit source-independent reduction at $r \le 2n$.

The four excluded even exponents $m = 6, 10, 12, 18$ correspond to target orders $n = 5, 9, 11, 17$. Their exact witnesses are, respectively, $(a, r) = (3, 23), (3, 19), (4, 29), (4, 3169)$ at the coordinate $-a$. Equation (34.149) and independent integer Stirling recurrences certify target vanishing and all preceding nonvanishing; the values are tabulated and reproducible in the supplement. In every case we have a prime $r \ge n+2$ and $a \in \mathbb{F}_r$ with

$$\overline{q}_n(a) = 0, \qquad \overline{\mathcal{B}}_n(a) \ne 0. \tag{34.151}$$

Factor the primitive integral q_n into primitive irreducible factors over $\mathbb{Z}[X]$. Because the leading coefficient of q_n is an r-unit, at least one factor g_n vanishes at a modulo r. If that factor divided $\mathcal{B}_n$ over $\mathbb{Q}[X]$, Gauss's lemma would give integral divisibility up to a unit and therefore $\overline{\mathcal{B}}_n(a) = 0$, contradicting (34.151). Thus g_n is missing from the predecessor product, proving the asserted radical nondivisibility. □

Corollary 34.135 (Cofinite good-prime noncoverage)**.** *For every fixed odd $n \geq 5$, one has $q_n \nmid \mathcal{B}_n$ over $\mathbb{F}_p[X]$ for all but finitely many primes $p > 2n$.*

Proof. Divide $\mathcal{B}_n$ by q_n over $\mathbb{Q}[X]$ and reduce the nonzero remainder from Theorem 34.134 modulo p. Only primes dividing a coefficient denominator, a leading coefficient or the content of its nonzero integer numerator can make the remainder vanish. □

Corollary 34.136 (A positive-density class with actual new residues)**.** *For each fixed odd $n \geq 5$, the set of primes $p > 2n$ admitting $a_p \in \mathbb{F}_p$ with $\overline{q}_n(a_p) = 0$ and $\overline{\mathcal{B}}_n(a_p) \neq 0$ has natural density at least $1/(n-3)!$.*

Proof. Choose the missing irreducible factor g_n of Theorem 34.134. Its resultant with $\mathcal{B}_n$ is nonzero over $\mathbb{Z}$, hence their reductions are coprime outside finitely many primes. The rational splitting field K of g_n has degree at most $(\deg g_n)! \leq (n-3)!$. By the Chebotarev density theorem [226], the primes splitting completely in K have density $1/[K : \mathbb{Q}]$. At every such prime away from the finitely many exclusions, g_n has a linear root absent from all predecessor polynomials. □

Remark 34.137 (Why this does not settle the universal congruence). For a fixed order, the nonzero rational remainder may vanish modulo finitely many good primes. Our results do not exclude *all* such exceptional primes, and the auxiliary boundary primes used in the proof of Theorem 34.134 are not necessarily good CCR transport primes. The density conclusion concerns linear *finite-field* roots; exact $\mathbb{Q}_p$-rational lifts for every source choice require a separate simple-root hypothesis.

An explicit infinite family independent of factorization

A particularly transparent subfamily needs no large-Zsigmondy classification. For every $t \geq 3$ let $n = 2^t - 1$ and let p be any prime divisor of the Fermat number $2^{2^{t-1}} + 1$. Then $\mathrm{ord}_p(2) = 2^t = n + 1$, and $p \equiv 1 \pmod{2^{t+1}}$, so $p > 2n$. Equation (34.149) gives $q_n(-2) = 0$ while $\mathcal{B}_n(-2) \neq 0$ modulo p. Distinct Fermat numbers are pairwise coprime, so this yields infinitely many *distinct* good primes and target orders, with a new source-independent residue root -2. No Fermat number is assumed prime.

34.56.7 Chebotarev densities from long irreducible factors

Assume for a fixed odd n that $\mathrm{Gal}(f_n/\mathbb{Q}) = S_d$. At an unramified prime, a degree-k irreducible factor corresponds to a k-cycle in Frobenius. For $k \in \{d, d-1, d-2, d-3\}$ and $d \geq 8$, the types are disjoint: two such long cycles cannot coexist. The proportion of permutations containing a k-cycle is $1/k$, because the remaining $d - k < k$ points may be permuted freely. Consequently, the primes on which Theorem 34.130 guarantees a source-universal, previously unseen unramified orbit have density *at least*

$$\frac{1}{d} + \frac{1}{d-1} + \frac{1}{d-2} + \frac{1}{d-3} \quad (d \geq 8), \tag{34.152}$$

subject to the stated verified Galois hypothesis. The bounds for the first four columns with established symmetric groups are:

n	d	$\mathrm{Gal}(f_n/\mathbb{Q})$	sufficient-prime density
5	2	S_2	1/2
7	4	S_4	7/12
9	6	S_6	37/60
11	8	S_8	533/840.

(34.153)

For $n = 5, 7, 9$ only the available degree cases are summed. These are *lower densities for a sufficient class*, not the density of every prime with a primitive orbit; no S_d hypothesis is assumed at orders where it has not been audited.

34.56.8 Exact content obstructions and an unconditional finite range

Proposition 34.138 (The fixed-order coefficient-content criterion)**.** *For odd $n \geq 5$, define*

$$\begin{aligned} H_n(X) &= \mathrm{rem}_{\mathbb{Q}[X]}(\mathcal{B}_n, q_n), \\ D_n &= \mathrm{lcm}\{\textit{coefficient denominators of } H_n\}, \\ C_n &= \mathrm{cont}(D_n H_n). \end{aligned}$$

Here $D_n > 0$ is minimal, the content is the positive gcd of the integer coefficients, and $C_n = 0$ if $H_n = 0$. Then for every prime $p > 2n$,

$$\boxed{\overline{q}_n \mid \overline{\mathcal{B}}_n \iff p \mid C_n.} \tag{34.154}$$

Moreover, Theorem 34.134 shows $C_n > 0$ for every *odd $n \geq 5$.*

Proof. The same p-integrality argument as in Lemma 34.133 (now $p > 2n$) shows the leading coefficient of q_n is a p-unit, and all coefficients of $\mathcal{B}_n$ and the quotient and remainder of Euclidean division lie in $\mathbb{Z}_{(p)}$. Reducing that division modulo p therefore gives the correct polynomial remainder over $\mathbb{F}_p$. It vanishes exactly when every coefficient of the integer polynomial $D_n H_n$ vanishes, or equivalently p divides its content. By Theorem 34.134, $H_n \neq 0$, so C_n is positive. □

The content criterion is an elementary *exact decision reduction*, rather than a claim that the content has a known closed form. Computation of H_n uses exact rational polynomial arithmetic and requires no search over primes. In a finite but nontrivial order range it produces a complete unconditional certificate.

Theorem 34.139 (All-good-prime noncoverage through order 71)**.** *For every odd $5 \leq n \leq 71$ and* every *prime $p > 2n$,*

$$\boxed{\overline{q}_n \nmid \overline{\mathcal{B}}_n \quad \textit{in } \mathbb{F}_p[X].} \tag{34.155}$$

Proof. The exact finite coefficient-content calculation gives $C_n = 2^{e_n}$ in all 34 odd orders; its complete exponent table is displayed below. One exact construction, using the Bernoulli numbers B_{2r}, is the recurrence

$$jS_j(X) = X\left(\frac{1}{2}S_{j-1}(X) - \sum_{r=1}^{\lfloor j/2 \rfloor} \frac{\mathsf{B}_{2r}}{(2r)!} S_{j-2r}(X)\right), \qquad S_0 = 1.$$

Construct the primitive positive-leading integral numerators P_m, form q_n, and successively take the exact rational remainder of $\prod_{m=2}^{n-1}(P_m/X)$ modulo q_n. The least coefficient-denominator multiple D_n and the gcd of the resulting integral coefficients yield C_n. The full recorded certificate covers all 34 rows and cross-checks shadow construction by an independent transport recurrence. An independent exact-rational implementation reproduced the displayed contents through $n = 61$; the last five rows $(63, 65, 67, 69, 71)$ retain their earlier full-certificate provenance rather than being claimed as newly reexecuted. By Proposition 34.138, each recorded identity $C_n = 2^{e_n}$ excludes every prime $p > 2n$. The theorem is a finite computer-assisted result, not an induction over arbitrary odd n. □

Remark 34.140 (Geometric support versus polynomial divisibility). If $\overline{q}_n$ is squarefree, the failure of divisibility in (34.155) ensures at least one geometric root of $\overline{q}_n$ is absent from all predecessors over an algebraic closure. In the presence of repeated factors, *polynomial* nondivisibility might result solely from multiplicity; the theorem is stated at its actually certified strength. The universal open problem is exactly that C_n has no prime divisor greater than $2n$ for *every* odd n.

Table 34.6: Exact integer-content exponents $C_n = 2^{e_n}$ for the 34 certified odd orders.

n	e_n	n	e_n	n	e_n	n	e_n
5	5	7	10	9	20	11	33
13	51	15	73	17	95	19	126
21	155	23	196	25	230	27	279
29	322	31	377	33	432	35	499
37	556	39	629	41	696	43	781
45	860	47	944	49	1023	51	1120
53	1211	55	1314	57	1425	59	1527
61	1656	63	1753	65	1873	67	2004
69	2125	71	2262				

34.56.9 Bernoulli anchors and primitive integer-coordinate ladders

Let B_k denote the ordinary Bernoulli numbers with $\mathsf{B}_2 = 1/6$, and let $Q_n = f_n$ denote the *monic* odd-order quotient, to distinguish it from the predecessor product $\mathcal{B}_n$.

Proposition 34.141 (Four Bernoulli-controlled anchor evaluations)**.** *For every odd $n \geq 5$,*

$$\begin{aligned} Q_n(0) &= \frac{2^{n-1}n}{n-1}\mathsf{B}_{n-1}, \quad Q_n(1) = (n-2)Q_n(0), \\ Q_n(2) &= \frac{n-1}{2}Q_n(0), \quad Q_n(3) = -\frac{(n-1)(n-2)}{6}Q_n(0). \end{aligned} \tag{34.156}$$

If $p > 2n$ divides the numerator of B_{n-1}, then $0, 1, 2, 3$ are four pairwise distinct roots of $\overline{Q}_n$.

Proof. Put $A(t) = t/(1 - e^{-t})$ and $L(t) = \log A(t)$. The identities

$$A(t) - A(-t) = t, \quad L(t) - L(-t) = t, \quad tA'(t) = (1+t)A(t) - A(t)^2$$

permit extraction of the odd coefficients of $L(t)^2/2$, $A(t)L(t)$, $A(t)^2$, and $A(t)^3$. In particular $[X^2]S_n = [t^n]L^2/2 = -\mathsf{B}_{n-1}/(2(n-1)(n-1)!)$, which after $S_n = X^2(X-1)Q_n/(2^n n!)$ gives the first identity. Evaluating the corresponding derivative at $X = 1$ and the evaluations at $X = 2, 3$ gives the remaining rational multiples in (34.156). The denominators are p-units when $p > 2n$. □

The four residues are *target* roots, not necessarily new ones. Indeed $0, 1$ are already present in early columns, and 2 occurs in the immediate predecessor for the same Bernoulli irregularity. A genuine *primitive* residue at 3 requires the additional absence of zeros in every preceding column.

Proposition 34.142 (Exact first occurrence at coordinate 3)**.** *For $j \geq 5$ odd and $j \geq 4$ even, respectively,*

$$\begin{aligned} S_j(3) &= -\frac{3(j-2)}{2(j-1)!}\mathsf{B}_{j-1}, \\ S_j(3) &= \frac{j-1}{2j!}\big((j-2)\mathsf{B}_j + 2j\mathsf{B}_{j-2}\big). \end{aligned} \tag{34.157}$$

The initial values are $S_1(3) = 3/2$, $S_2(3) = 1$, $S_3(3) = 3/8$. At a good irregular-index pair (n, p), the coordinate 3 first occurs at n precisely when the odd Bernoulli values and the even Bernoulli linear combinations on the right of (34.157) *are all nonzero modulo p at prior orders.*

Proof. Iterate $tA' = (1+t)A - A^2$ to express A^3 as a combination of A, A', A'' with polynomial coefficients in t, then extract coefficients of each parity. In the good-prime range all displayed factorial and Bernoulli denominators are p-units, proving the necessary-and-sufficient first-occurrence criterion. □

Theorem 34.143 (Integer-coordinate Bernoulli staircase)**.** *For all positive integers r, n,*

$$\boxed{S_n(r+1) = \frac{r-n}{r} S_n(r) + S_{n-1}(r).} \tag{34.158}$$

If $n \geq 5$ is odd, $p > 2n$, and $h \geq 0$ satisfies $2h+3 \leq n-2$, then

$$\begin{aligned} S_n(2) \equiv S_n(3) \equiv \cdots \equiv S_n(2h+3) \equiv 0 \pmod{p} \\ \iff \quad \mathsf{B}_{n-1} \equiv \mathsf{B}_{n-3} \equiv \cdots \equiv \mathsf{B}_{n-2h-1} \equiv 0 \pmod{p}. \end{aligned} \tag{34.159}$$

In that case $\prod_{a=0}^{2h+3}(X-a)$ divides $\overline{Q}_n(X)$.

Proof. Multiply the differential identity for A by rA^{r-1} to obtain $A^{r+1} = (1+t)A^r - (t/r)(A^r)'$ and take the t^n-coefficient, giving (34.158). Write $S_n(r)$ as a linear combination $\sum_{j=0}^{r-1} c_{r,j}(n) S_{n-j}(1)$; induction from (34.158) gives $c_{r,r-1}(n) = 1$. Since n is odd and $S_k(1) = \mathsf{B}_k/k!$ for even $k \geq 2$, whereas $S_k(1) = 0$ for odd $k \geq 3$, the equations for $S_n(2), S_n(4), \ldots, S_n(2h+2)$ form a triangular system in the ordered Bernoulli values $\mathsf{B}_{n-1}, \mathsf{B}_{n-3}, \ldots, \mathsf{B}_{n-2h-1}$ with unit diagonal after dividing by factorials. The intervening odd integer evaluations involve only the same Bernoulli values and therefore vanish simultaneously. The factors $a^2(a-1)/(2^n n!)$ are units at the integer coordinates $2 \leq a \leq 2h+3$, and (34.156) provides the roots 0 and 1. This proves the factor divisibility. □

Example 34.144 (An uncovered simple 587-adic disk beyond the finite content range)**.** For $(n,p) = (93, 587)$, exact arithmetic gives $\mathsf{B}_{90} \equiv \mathsf{B}_{92} \equiv 0 \pmod{587}$. Theorem 34.143 produces the six target roots $0, 1, 2, 3, 4, 5$ modulo 587. Yet the certificate checks

$$\begin{aligned} &S_{93}(5) = 0, \quad S_{92}(5) = 179, \quad S'_{93}(5) = 498 \pmod{587}, \\ &S_m(5) \not\equiv 0 \pmod{587} \qquad (1 \leq m < 93). \end{aligned}$$

Therefore the root residue 5 is new. Since the derivative is nonzero, Hensel's lemma gives *for every common* 587*-integral stationary source vector* a unique root of F_{93} inside $5 + 587\mathbb{Z}_{587}$, with none of the earlier columns vanishing in that disk. A double Bernoulli zero thus produces a *primitive* target root even though coordinate 3 is already present at orders 91 and 92.

Remark 34.145 (A singular residue cannot be lifted automatically). At $(n,p,a) = (91, 277, -2)$ one has $S_{91}(-2) = S'_{91}(-2) = 0$ modulo 277, while $S''_{91}(-2) = 125 \not\equiv 0$. With zero sources, $F_{91}(-2) \equiv 277 \cdot 24 \pmod{277^2}$, so no root exists in the entire disk $-2 + 277\mathbb{Z}_{277}$ modulo 277^2. This does not contradict the mod-p primitive-root theorem; the elementary simple-root Hensel hypothesis is genuinely necessary.

34.56.10 Proof provenance, scope and the uniform open problem

The proofs in this paper have distinct levels of logical dependence. The exact shift-companion identity and the Bernoulli staircase are direct symbolic identities; the all-order rational noncoverage theorem applies the *published* Feit–Roitman large-Zsigmondy theorem; the long-orbit exclusions of degrees $d-2, d-3$ use exact symbolic Euclidean and resultant identities in the order variable plus finite exceptional-prime checks. The content statement through $n = 71$ is a finite exact computer-assisted certificate; it does not imply its continuation to higher orders. Source-universal assertions rest only on coefficientwise $F_j \equiv S_j \pmod{p}$ for $p > 2n$ and on standard Hensel lifting when the necessary separability holds.

Every odd $n \geq 5$ has a nonzero rational remainder H_n by Theorem 34.134; whether its integer content has a prime divisor greater than $2n$ is not established. We therefore retain the following open problem.

Open Problem 34.146 (Uniform predecessor noncoverage)**.**

$$\boxed{\textit{For all odd } n \geq 5 \textit{ and every prime } p > 2n, \quad \overline{q}_n \nmid \overline{\mathcal{B}}_n \textit{ in } \mathbb{F}_p[X].} \tag{34.160}$$

Equivalently, C_n has no prime factor greater than $2n$ at every odd order. The claim has been proved in this manuscript only for $n \leq 71$ and for the explicit infinite families and special prime patterns covered by the preceding theorems.

The open problem must not be replaced by the stronger assertion that every geometric target root is absent from every preceding column: actual common roots occur, and in nonsquarefree characteristic polynomial nondivisibility need not detect an entirely new root position. A separate physical-realizability theorem for deterministic pushdown automata is likewise not implied by the formal stationary source setting. The CCR interpretation is conditional on the algebraic model (34.134) and integral source choice.

Finite certificate scope and verification

The results above have different forms of justification. The shift-companion and Bernoulli-ladder identities are symbolic derivations; the rational all-order noncoverage theorem relies on the published Feit–Roitman input and the explicit exceptional-order witnesses. The long-orbit degree-$(d-2)$ and degree-$(d-3)$ exclusions require the exact Euclidean identities in the order parameter. Their first two obstruction resultants are given in the proof, while the higher Euclidean coefficient and exceptional-residue checks are finite symbolic certificate components. The degree-$(d-2)$ argument uses the corrected linear-quotient chain in the $m = n - 1$ case, not the superseded quadratic-quotient calculation.

The finite predecessor-content theorem uses exact rational coefficient arithmetic, primitive-part normalization, successive polynomial remaindering, denominator clearing and an integer coefficient gcd. All 34 exponents are stated in Table 34.6; they are not sampled statistics. An independently implemented Bernoulli recurrence and remainder calculation has reproduced the first 29 table rows, through odd order 61. The five remaining rows (63 through 71) are retained on the basis of the original complete finite certificate; their full high-order coefficient computations have not been repeated as part of this editorial audit. Small-order self-shift resultant checks, Feit-exception witnesses, and the explicit $n = 93$, $p = 587$ and $n = 91$, $p = 277$ residue calculations provide separate consistency checks, not substitutes for those exhaustive steps.

No executable source, verification script or supplementary data archive is part of, or promised with, this paper. The displayed formulae and certified finite outcomes carry the stated mathematical claims; for the heavy symbolic and full-content calculations, independent end-to-end external checking remains an appropriate publication review step. We make no assertion of first-in-world priority for the classical ingredients or of physical-machine realizability from the formal stationary CCR model.

Chapter 35

Active Observability of Chronological Carry Recurrences: Masking, Probing, and Sharp Intervention Costs

Research branch — Active observability of chronological carry. *The terminal-block CCR-k source recurrence from Chapter 34 supports a distinct inverse problem: identify masked old sources by nonresetting interventions under explicitly restricted later unknown inputs. The full-row equivalence and pulse-cost bounds below are relative to this source-and-observation contract; they do not establish observability for unrestricted CCR sources or arbitrary physical machines.*

The preceding CCR source theory characterizes the chronological coefficient transport and its arithmetic constraints. This chapter moves from forward source construction to *active source identification*: which concealed source coordinates can be recovered when the observer is permitted to select later interventions? The work is integrated from the separately authored study [256]. Its article-level mathematical statements and proofs are retained below, with book-wide theorem, citation, and label conventions. No auxiliary software files are required to read the proofs.

35.1 Introduction

Chronological carry recurrences separate elementary source inputs from transported coefficient histories. Earlier CCR–3, CCR–4, and CCR–k studies developed the source reconstruction and connected-block formulation [164, 165, 166]; the separate arithmetic-refinement paper concerns p-adic resonance, ramification and density laws [195]. Here we examine a different question: whether a hidden old source can be identified with a limited number of active interventions despite unknown contemporaneous higher-order sources.

The distinction between adaptive and preset experiments is familiar in finite-state checking sequences [257], and observability with unmeasured inputs is a developed topic for nonlinear discrete-time systems [260]. Hence neither adaptive decision trees nor affine-output collision tests are claimed as new. Our focus is the specific source–time intervention normal form and the sharp pulse gap in an exact CCR recurrence.

Two features distinguish this inverse problem from reconstructing sources when complete chronological input data are given. Distinct old histories may lead to the same recent coefficient rows, so passive observation alone need not recover the discrepancy. Moreover, unknown higher-rank inputs can conceal the response to a controlled pulse. It is therefore essential to specify which source terms are allowed to vary, what the observer can see, and whether a diagnostic reading captures the information in a complete coefficient row.

35.2 Contributions

We connect (i) the CCR–k chronology and its finite masking horizon, (ii) rank–time-normalized controlled interrogation with bounded unknown high-degree sources, (iii) a sharp adaptive-versus-predetermined pulse count separation, and (iv) an exact worst-case versus expected pulse-cost comparison. Classical tools (triangular elimination, finite-field difference sets and Bellman recursion) are used explicitly and are not claimed as new in isolation.

35.3 Chronological recurrence and normalized readouts

Fix an integer p, a cutoff k, and write $C_{n,j} = [u^j]P_n(u)$. The typed source system is

$$P_n(u) = (1+\ell_n u)P_{n-1}(u) + p(n-1)\ell_{n-1}u^2P_{n-2}(u) + \sum_{r=3}^{k}(n-1)_{r-1}a_{r,n}u^rP_{n-r}(u) \pmod{u^{k+1}}, \tag{35.1}$$

where $\ell_n = (p+1)n - p$, $a_{3,n} = -p$, $P_0 = 1$ and $P_n = 0$ for $n < 0$. The falling factorial $(m)_r$ denotes $m(m-1)\cdots(m-r+1)$. The source inputs $a_{r,n}$ for $r \geq 4$ are treated as part of the explicit model, not automatically identified with arbitrary external events.

For times s and degrees j, let $R_{s,j}$ be the coefficient obtained from the three preceding complete observed rows by deleting the fresh higher-source summands in (35.1). The key observation is that current source rank r has a diagonal coefficient $(s-1)_{r-1}$, and the higher current sources can be solved recursively whenever these factors are units in the field. This lets us compare full-row collisions rather than appealing only to a heuristic scalar readout.

35.4 Passive masking and the need for active interrogation

The source separation theorem of CCR–k makes new rank-r sources triangular in the observed coefficient of degree r. Earlier source variables interact once their chronological blocks can occupy disjoint intervals. Starting at rank eight, two independent rank-four source blocks can contribute multiplicatively. In particular, the effect of an old rank-four source on the current rank-eight diagnostic may become visible only after a controlled rank-four pulse.

The three-lag baseline in (35.1) also gives a sharp form of persistent passive masking. Take two old histories that differ only at rank four, a prime field $\mathbb{F}_q$ with $q > n_0 + 3$, and let the fresh source of rank eight be fixed to zero during times n_0, n_0+1, n_0+2, while other high-rank sources can vary with the old history. The fresh-source map to a full output row is triangular, with nonzero diagonal. Fixing rank eight restricts the row image to an affine hyperplane. The two histories change the corresponding normal in the rank-four direction, and the two hyperplanes meet in q^{k-5} output rows. Three consecutive common rows can therefore be achieved. If all later high sources vanish, the common output rows perpetuate indefinitely because the remaining recurrence has three delays. This fact is not a claim that the system is unobservable under all future controlled inputs.

Under a known rank-four pulse at time $n_0 + 3$, with all other fresh high sources zero, the old source difference reappears at degree eight:

$$\Delta C_{n_0+3,8} = \alpha(n_0+2)_3(t-1)_3\Delta h \pmod{q}.$$

This calculation is a special case of the rank–time interrogation normal form below and motivates the bounded-nuisance experiment.

35.4.1 A sharp adaptive–predetermined separation in CCR–10

Let $\mathbb{F} = \mathbb{F}_{17}$ and use the typed terminal-block CCR–k recurrence with $(p, k, n_0, t) = (5, 10, 11, 4)$, where all identities are in $\mathbb{F}[u]/(u^{11})$. The only varying old source before time 11 is $h = a_{4,4}$ from $H = \{3, 5, 6, 11, 16\}$. At times $11, 12, 13$ all higher sources $r = 4, \dots, 10$ may depend on h and are chosen to yield the same three full coefficient rows for every h. The designated nonresetting pulse opportunities are $(14, 4), (15, 5), (16, 6)$, where a rank-$r$ pulse $\alpha_r \neq 0$ and independent diagnostic nuisance $z_r = a_{r+4,10+r} \in E = \{0, 1, 2, 3, 4\}$ are allowed. All fresh sources above rank $r+4$ are unrestricted; all other fresh high sources are zero. At unused opportunities all fresh high sources are zero. The observer sees complete coefficient rows. Cost is the number of nonzero chosen pulses.

Write $B = C_{10,4}(h = 0)$, and let $R_{s,j}$ be the baseline obtained from the three previous observed full rows with fresh high sources zero. For $s_r = 10 + r$ define

$$w_r = (C_{s_r,r+4} - R_{s_r,r+4} - (s_r - 1)_{r-1}\alpha_r B)\left[(s_r - 1)_{r+3}\right]^{-1}.$$

The rank–time reduction from the original CCR recurrence gives

$$w_r = c\alpha_r h + z_r, \qquad c = (3)_3\left[(10)_4\right]^{-1} = 5 \pmod{17}. \tag{35.2}$$

The source–to–row triangular pivots $(s-1)_{j-1}$ are units in $\mathbb{F}_{17}$ through time 16. Therefore any two histories with identical previous three full rows and the same w_r can match the *entire* next full row by choosing unrestricted sources of degrees $r + 5, \dots, 10$ successively.

Theorem 35.1 (Strict adaptive advantage with no reset)**.** *In this source and interrogation contract, the minimum worst-case number of active pulses is* 2 *for an adaptive controller but* 3 *for a controller whose pulse times and nonzero amplitudes must be predetermined without seeing intermediate coefficient observations. Both bounds are attained.*

Proof. At time 14 use $\alpha_4 = 14$, corresponding to normalized slope $a_4 = c\alpha_4 = 2$. The possible first normalized outputs $y = 2h + z_1$ leave candidate sets

y	$0, 1, 2$	5	$6, 7, 8, 9$	10	11	$12, 13, 14$	$15, 16$
H_y	$\{16\}$	$\{11\}$	$\{3, 11\}$	$\{3, 5\}$	$\{5\}$	$\{5, 6\}$	$\{6, 16\}$

with other residues impossible. If H_y is a singleton, stop. Otherwise pulse at time 15 with a normalized slope a_5 equal to 1 for $H_y = \{3, 11\}, \{6, 16\}$, equal to 3 for $H_y = \{3, 5\}$, and equal to 5 for $H_y = \{5, 6\}$. The actual rank-five amplitude is $\alpha_5 = 7a_5$ in $\mathbb{F}$. Since the nonzero pair differences, multiplied by the chosen slopes, are respectively $\pm 8, \pm 10, \pm 6, \pm 5$, none lies in $E - E = \{0, \pm 1, \pm 2, \pm 3, \pm 4\}$, hence the two normalized readouts uniquely determine h for every allowed nuisance history. The triangular completion argument makes the conclusion valid for full coefficient trajectories.

One pulse cannot identify all five h robustly: each generates five diagnostic residues, so $25 > 17$ prohibits pairwise disjoint one-pulse attainable-output sets. Thus the adaptive optimum is 2.

For the predetermined lower bound, note that $H - H = \mathbb{F}$. Set $D = (E - E) \setminus \{0\} = \{\pm 1, \pm 2, \pm 3, \pm 4\}$. At a nonzero slope a a difference d is unseparated precisely when $ad \in D$. The element $g = 3$ generates $\mathbb{F}^\times$; modulo the sign subgroup $\{\pm 1\}$, D has exponent classes $S = \{0, 1, 4, 6\}$ in $\mathbb{Z}/8\mathbb{Z}$. A direct calculation gives $S - S = \mathbb{Z}/8\mathbb{Z}$, so for any two nonzero slopes a, b, the sets $a^{-1}D$ and $b^{-1}D$ intersect. Every such nonzero difference occurs between two old values in H. Hence no predetermined two-pulse schedule separates all old histories; unrestricted higher sources can complete a common full trajectory.

Finally, the three predetermined slopes $(2, 1, 5)$ have $2^{-1}D \cap 1^{-1}D \cap 5^{-1}D = \varnothing$; at the three opportunities they correspond to amplitudes $(14, 7, 1)$. Thus three predetermined pulses suffice and are necessary. □

Remark 35.2 (Finite adaptive decision rule). For any finite old-source set S, nuisance set E, and normalized slope a, the posterior candidate set after observed y is $S_{a,y} = \{h \in S :$

$y - ah \in E\}$. A finite-horizon minimax recursion over these candidate sets gives the exact adaptive worst-case pulse cost. The decision-tree principle itself is classical; the point here is its exact realization by the nonresetting typed CCR–k recurrence under the specified higher-source contract.

35.5 Sharp expected intervention cost

The preceding sharp separation compares *worst-case* counts. We now consider the same three nonresetting opportunities, but assume that the old source is uniform on $H = \{3, 5, 6, 11, 16\}$ and each diagnostic nuisance z_i is independently uniform on $E = \{0, 1, 2, 3, 4\}$, conditioned only on an intervention being used. All other unknown high sources remain arbitrary and have no effect on the normalized diagnostics. The cost of a policy is its number of activated slots. Skipping an opportunity does not add an observation; the history still advances under the specified zero-source contract.

For a current candidate set $S \subseteq H$ define

$$S_{a,y} = \{h \in S : y - ah \in E\}, \qquad a \in \mathbb{F}_{17}^{\times}.$$

Writing $V_i(S)$ for minimum expected *remaining* pulse cost at opportunity $i \in \{14, 15, 16\}$ yields the exact finite recursion

$$V_i(S) = \min \left\{ V_{i+1}(S), \quad 1 + \min_{a \in \mathbb{F}_{17}^{\times}} \sum_{y : S_{a,y} \neq \varnothing} \frac{|S_{a,y}|}{5|S|} V_{i+1}(S_{a,y}) \right\}, \tag{35.3}$$

with $V_i(S) = 0$ for $|S| \leq 1$ and $V_{17}(S) = +\infty$ for $|S| > 1$. The same recursion with a restriction on the number of remaining pulses characterizes expected cost under a worst-case pulse budget. These are classical finite-state decision-tree recursions; here they are applied to the exact CCR–10 additive probe normal form.

Theorem 35.3 (Sharp worst-case/mean-cost tradeoff for CCR–10)**.** *Under the CCR–10 source contract of the preceding section and the specified independent uniform prior and diagnostic nuisances:*

1. *The minimum worst-case pulse count is* 2*, while the minimum expected count among strategies that* never *use more than two pulses is exactly* $44/25 = 1.76$*.*
2. *If all three opportunities may be used on exceptional trajectories, the minimum expected pulse count is exactly*

$$\boxed{V_{14}(H) = \frac{199}{125} = 1.592.}$$

This optimum is attained by a policy whose worst-case count is 3*.*

Proof. The robust worst-case optimum 2 was established in the preceding section. For expected-cost optimality, the finite recursion (35.3) is an exact Bellman equation: conditional on a residue y, each $h \in S_{a,y}$ is equiprobable, since z_i is uniform on E and $y = ah + z_i$ has one admissible noise value per candidate. Thus $\Pr(y \mid S, a) = |S_{a,y}|/(5|S|)$. Exhausting the 16 nonzero slopes at each of the three slots gives the following eight pairs of symmetric first slopes a and $17 - a$, with the optimum over all remaining decisions:

a (and $17 - a$)	1	2	3	4	5	6	7	8
expected pulses	199/125	9/5	44/25	219/125	209/125	237/125	44/25	206/125

The option of skipping the first slot has exact value $44/25$. Consequently the global three-slot minimum is $199/125$.

Here is an explicit optimal decision tree. Use normalized slope $a = 1$ (rank-four pulse amplitude 7) at time 14. If the first observation y leaves a singleton, stop. The ambiguous posterior sets and a valid rank-five second slope at time 15 are

y	$S_{1,y}$	second slope a_2
3	$\{3, 16\}$	2
5	$\{3, 5\}$	3
6, 7	$\{3, 5, 6\}$	4
8, 9	$\{5, 6\}$	5

All pairs are resolved by the second pulse. For a triple $S = \{3, 5, 6\}$, slope 4 resolves every second observation except $y_2 = 7$, which leaves $\{5, 6\}$; there a final rank-six slope 5 resolves the pair. Actual pulse amplitudes are $7a_i \pmod{17}$. The first observation determines a singleton in 11 of the 25 equally likely (h, z_{14}) combinations. Of the 14 remaining combinations, exactly 4/125 of all source/noise trajectories require a third pulse. Therefore the expected number is

$$1 + \frac{14}{25} + \frac{4}{125} = \frac{199}{125}.$$

For the stricter two-pulse budget, the same exhaustive finite Bellman calculation with the pulse budget as an additional state finds a minimum 44/25 (for example, with first slope 7 or 10). The result is a finite exact certificate, not an asymptotic theorem. The finite calculation uses exact rational probabilities and exhaustively enumerates the actions, residues and posterior candidate sets. The resulting policies were also checked directly against all $5 \cdot 5^3 = 625$ possible CCR–10 source/noise trajectories. These finite checks follow from the stated Bellman recursion and source alphabets. □

Remark 35.4 (Interpretation). The exact cost gap is a property of the *specified* finite source alphabet, independent uniform diagnostic nuisance, available rank–time slots and pulse-count cost. It is not a universal probability law for CCR–k, nor does it show that adaptive decision trees are novel. The new check distinguishes a worst-case-optimal policy from an expected-cost-optimal policy, even within this one CCR–10 realization.

35.6 A finite-field criterion and a cross-prime family

The previous results used $q = 17$. We now isolate the exact finite-field condition behind the adaptive separation and verify its CCR realization at several different prime moduli. Let $q > 16$ be prime and let $H, E \subset \mathbb{F}_q$ be nonempty finite sets. At a nonzero normalized slope a, a controlled measurement returns

$$y = ah + z, \qquad h \in H, \quad z \in E.$$

The same unknown h persists, whereas z may be chosen independently and adversarially on each used opportunity. Write

$$D = E - E, \quad S_{a,y} = \{h \in H : y - ah \in E\}, \quad \mathcal{B}(S) = \{b \in \mathbb{F}_q^\times : b((S-S)\setminus\{0\}) \cap D = \varnothing\}.$$

Theorem 35.5 (Exact two-stage criteria)**.** *A predetermined two-measurement design exists if and only if there are $a, b \in \mathbb{F}_q^\times$ such that*

$$\big((a^{-1}D) \cap (b^{-1}D) \cap ((H - H) \setminus \{0\})\big) = \varnothing.$$

An adaptive controller using at most two measurements identifies h for all allowed nuisance values if and only if there is $a \in \mathbb{F}_q^\times$ such that

$$\mathcal{B}(S_{a,y}) \neq \varnothing \quad \textit{for every } y \textit{ with } |S_{a,y}| \geq 2.$$

Thus a strict adaptive/predetermined worst-case gap at two measurements occurs exactly when the second criterion holds and the first fails, provided no one-measurement design succeeds. If a predetermined three-measurement design exists, its optimum is then exactly three.

Proof. For a pair $h \neq h'$, the observation sets $ah + E$ and $ah' + E$ intersect if and only if $a(h - h') \in E - E = D$. Two fixed observations cannot distinguish that pair precisely when both scaled differences belong to D, proving the first equivalence. Conditional on an observed first residue y, the exact remaining candidate set is $S_{a,y}$. One more measurement can separate all candidates precisely when a nonzero slope b avoids D on all of its nonzero pairwise differences, which means $b \in \mathcal{B}(S_{a,y})$. Choosing such a b after seeing y proves both necessity and sufficiency. The last statement follows by comparison of the two criteria and the existence of a three-test witness. □

Proposition 35.6 (Exact nonresetting CCR–10 embedding)**.** *Fix $p = 5$, $k = 10$, an old source $h = a_{4,4} \in H$, and prime $q > 16$. The high sources at times $11, 12, 13$ can be chosen, depending on h, to make all three full coefficient rows independent of h. At rank–time opportunities $(14, 4), (15, 5), (16, 6)$, fix all fresh high sources to zero except a controlled nonzero source of rank r, a diagnostic nuisance source of rank $r + 4$ in E, and optionally unrestricted sources above rank $r + 4$. Then the normalized diagnostic is exactly*

$$w_r = c_q \alpha_r h + z_r, \qquad c_q = (3)_3[(10)_4]^{-1} \in \mathbb{F}_q^\times.$$

Every slope in $\mathbb{F}_q^\times$ is attainable. Thus the criteria in Theorem 35.5 apply to a single CCR–10 system without resetting between observations. If two hypotheses have identical normalized diagnostics, unrestricted higher sources can match the remaining entries of the entire observed row by chronological triangular elimination. The claim concerns this specified source and observation contract, not arbitrary unknown CCR sources.

Proof. The diagonal coefficient of fresh source $a_{j,n}$ in coefficient $C_{n,j}$ equals $(n - 1)_{j-1}$. The needed factors at times 11–16 and degrees at most ten are units modulo any prime $q > 16$. Consequently fresh ranks $4, \dots, 10$ at each masking time can be solved successively to reproduce any given reference row. At the opportunity $s = 10 + r$, the previous row indexed $s - r$ is row ten, and its degree-four coefficient is $B + (3)_3 h$, with B independent of h. Subtracting the known three-lag baseline and the controlled contribution $(s-1)_{r-1}\alpha_r B$, then dividing by $(s-1)_{r+3}$, gives $c_q \alpha_r h + z_r$. The free sources of greater rank can then match subsequent coefficients without altering the diagnostic, by the same triangular argument. □

Proposition 35.7 (Affine invariance)**.** *For nonzero $u, w \in \mathbb{F}_q$ and any $v, d \in \mathbb{F}_q$, replacing H by $uH+v$ and E by $wE+d$ preserves the optimal adaptive and predetermined intervention counts. Under corresponding uniform priors and nuisance laws, it preserves the optimal expected intervention counts as well.*

Proof. Under the substitution $h' = uh + v$, $z' = wz + d$, choose $a' = wa/u$. Then $a'h' + z' = w(ah + z) + a'v + d$. The two sides have mutually invertible, known output changes and the choice $a \leftrightarrow a'$ is a bijection on nonzero slopes. Thus every fixed or adaptive decision tree transports bijectively, including its probability law and cost. □

Proposition 35.8 (Four certified prime instances)**.** *For each row of the following table, with $E = \{0, \dots, b - 1\}$, there is a two-pulse adaptive strategy; every predetermined two-pulse schedule fails; and a predetermined three-pulse strategy succeeds. All instances are realized by Proposition 35.6.*

q	H	b	*first adaptive slope*	*fixed triple*
17	$\{3, 5, 6, 11, 16\}$	5	2	$(1, 2, 5)$
19	$\{2, 5, 7, 17, 18\}$	5	1	$(1, 2, 5)$
23	$\{0, 10, 13, 16, 18, 22\}$	5	1	$(1, 2, 5)$
29	$\{8, 10, 18, 19, 21, 25\}$	7	1	$(1, 2, 7)$

The exact optimal expected pulse counts with three opportunities under uniform h and independent uniform nuisances are respectively $199/125, 199/125, 22/15, 11/7$. *If worst-case pulse count is bounded by two, the expected optima are* $44/25, 42/25, 22/15, 11/7$.

Proof. The first slope in each row satisfies the second condition of Theorem 35.5, as verified by its finite table of nonempty posterior sets and their separating slopes. For every ordered nonzero slope pair, the first condition fails: exhaustive finite enumeration of the $(q-1)^2$ slope pairs produces a colliding source pair for each. The displayed fixed triple passes the same exact pair-difference test. Since $|H||E| > q$ in each case, one test cannot separate all h by the pigeonhole principle. Exact rational Bellman recursions over all 3 available opportunities, including skipped slots, give the expected costs; complete source/diagnostic trajectories are independently checked in the CCR–10 recurrence. This is a finite reproducible certificate for four concrete instances, not an assertion for all primes. □

Proposition 35.9 (Large-prime obstruction for fixed alphabets)**.** *Write* $\Delta = ((H - H) \setminus \{0\})$ *and* $D = E - E$. *If*

$$q - 1 > |\Delta|\,(|D| - 1),$$

then some nonzero normalized slope identifies h with a single measurement despite arbitrary $z \in E$. In particular, for any fixed finite sets of distinct integers H, E reduced modulo primes q sufficiently large, the adaptive-versus-predetermined pulse-count gap must disappear. More explicitly it suffices (when residues remain distinct) that

$$q - 1 > |H|(|H| - 1)|E|(|E| - 1).$$

Proof. For each nonzero old-source difference $d \in \Delta$, a slope a causes a collision only if $ad \in D \setminus \{0\}$. Since d is invertible over $\mathbb{F}_q$, there are exactly $|D| - 1$ bad slopes for this d. The union over d contains at most $|\Delta|(|D| - 1)$ slopes. If this is smaller than $q - 1$, at least one nonzero slope avoids every collision. Finally $|\Delta| \le |H|(|H| - 1)$ and $|D| - 1 \le |E|(|E| - 1)$, yielding the stated uniform bound. □

Remark 35.10 (Interpretation)*.* The difference-set and decision-tree equivalences are standard finite identification tools. The CCR-specific construction consists of their exact nonresetting embedding in the CCR chronology together with multiple prime-field instances and explicit costs. Affine changes create isomorphic instances, not fundamentally different arithmetic theories. The table does not establish an asymptotic density of suitable primes.

35.7 An explicit infinite-prime adaptive gap

The preceding finite examples may be replaced by a genuine arithmetic family. The difference-set argument below is elementary; its significance here is the realization of a fixed-versus-adaptive gap at *every sufficiently large prime characteristic* inside a nonresetting CCR–10 chronology with the specified restricted-input observation contract.

For each prime $q \ge 41$, let

$$H = \{0, 2, 7, 8\} \subset \mathbb{F}_q, \qquad E_q = \{0, 1, \ldots, \lfloor q/3 \rfloor\} \subset \mathbb{F}_q.$$

A pulse of nonzero normalized slope a produces $y = ah + z \pmod q$ with $h \in H$ fixed throughout the experiment and a new independent adversarial $z \in E_q$ at each pulse. We analyze these normalized diagnostic residues first. Under the permitted unrestricted higher-rank nuisance inputs, the subsequent full-row lifting lemma proves that additional coefficient entries do not weaken the robust lower bound. For $h \ne h'$, the corresponding possible diagnostic outputs overlap exactly when

$$a(h - h') \in E_q - E_q = \{t \in \mathbb{F}_q : \|t/q\| \le \lfloor q/3 \rfloor / q\}.$$

Here $\|x\|$ denotes distance to the nearest integer. Because $q > 3$ is prime, none of the nonzero residues ad with $d \in \{1, 2, 5, 6, 7, 8\}$ lies on a boundary $\|ad/q\| = 1/3$.

Lemma 35.11 (Circle difference certificate)**.** *Let $\mathcal{D} = \{1, 2, 5, 6, 7, 8\}$, and for $x \in \mathbb{R}/\mathbb{Z}$ away from the finitely many threshold points define*

$$B(x) = \{d \in \mathcal{D} : \|dx\| < 1/3\}.$$

Then $B(x) \cap B(y) \neq \varnothing$ for all such x, y.

Proof. By the symmetry $B(x) = B(1-x)$, it suffices to inspect $0 < x < 1/2$. The indicator of each d changes only at $x = (j \pm 1/3)/d$. Sorting all these rational breakpoints for $d \in \mathcal{D}$ produces 25 open subintervals of $(0, 1/2)$. Direct evaluation at one rational midpoint of each subinterval shows that every possible $B(x)$ contains at least one of the following seven sets:

$$\{1, 2\}, \quad \{1, 5, 6\}, \quad \{1, 5, 8\}, \quad \{1, 6, 7\},$$
$$\{1, 7, 8\}, \quad \{2, 5, 7\}, \quad \{2, 6, 8\}.$$

These seven sets are pairwise intersecting (which is checked immediately from the displayed list). Hence every two possible collision sets intersect. The precise rational breakpoint partition and an exact-integer checker are included with the accompanying reproducibility archive; the statement for prime residues does not require evaluating threshold points. □

Theorem 35.12 (Infinite-prime sharp adaptive separation)**.** *For every prime $q \geq 41$ in the above diagnostic experiment:*

1. *One pulse cannot identify h for every nuisance value.*
2. *No two predetermined nonzero slopes identify h for every pair of nuisance values.*
3. *There exists a predetermined three-pulse schedule that identifies h.*
4. *An adaptive strategy identifies h using at most two pulses.*

Therefore the optimal worst-case pulse counts are exactly three for predetermined schedules and exactly two for adaptive schedules. The same experiment is realizable without resetting between pulses in the CCR–10 source chronology at times $14, 15, 16$, under the explicit source and readout restrictions of the previous chronological-embedding theorem.

Proof. (1) If one pulse separated every source, the four sets $ah + E_q$ would be pairwise disjoint in $\mathbb{F}_q$. But $4|E_q| = 4(\lfloor q/3 \rfloor + 1) > q$.

(2) The positive source differences are precisely $\mathcal{D}$. For any two fixed slopes a, b, the collision sets $B(a/q)$ and $B(b/q)$ intersect by Lemma 35.11. Thus there is a nonzero source difference $d \in \mathcal{D}$ that can be masked by nuisance differences at both interventions. Independent nuisances produce a common diagnostic pair, so no fixed pair succeeds.

(3) Set $a_1 = \mathrm{nint}(3q/40)$, $a_2 = \mathrm{nint}(q/5)$ and $a_3 = \mathrm{nint}(3q/7)$, where nint denotes nearest-integer rounding. For every prime $q \geq 61$ the fractions a_i/q lie respectively in the open intervals

$$(1/15, 1/12), \qquad (4/21, 5/24), \qquad (5/12, 4/9),$$

because their target fractions lie in these intervals with margin at least $1/120$, and nearest rounding has error at most $1/(2q)$. The three corresponding collision patterns are $\{1, 2\}$, $\{1, 5, 6\}$ and $\{2, 5, 7\}$, whose intersection is empty. For the five smaller eligible primes $41, 43, 47, 53, 59$, direct residue checks with the same formulas likewise give an empty triple intersection. Consequently the three measurements distinguish every pair of hidden sources.

(4) Choose the first slope $a = \mathrm{nint}(5q/16)$. The four circular interval start points $ah/q \pmod 1$ for $h \in H$ approximate $0, 3/16, 1/2, 5/8$. Every three of the four ideal start points require a circular arc of length at least $7/16$ to contain them. Nearest rounding changes the circular separation of any pair by at most $4/q$ (the diameter of H is eight). Thus for $q \geq 41$ no three actual start points fit inside an arc of length $1/3$, since $7/16 - 4/q > 1/3$. As every translate $ah + E_q$ is a circular interval of length at most $1/3$, each first residue y

leaves at most two candidates. If only one remains, stop. Otherwise let the two candidates have nonzero difference $d = h - h'$. Choose the second slope

$$b = \frac{q-1}{2}\, d^{-1} \pmod{q}.$$

Then $bd \equiv (q-1)/2 \pmod{q}$, which does not belong to $E_q - E_q$ since $\lfloor q/3 \rfloor < (q-1)/2$ for $q \geq 41$. The second observation consequently separates the pair even under an adversarial fresh nuisance.

Finally, for $p = 5, k = 10$ the source coefficients at times $11, 12, 13$ can be chosen recursively by the invertible triangular pivots $(n-1)_{r-1}$ to prepare identical full output rows for every $h \in H$. At times $14, 15, 16$ use rank–$4, 5, 6$ pulses with the predetermined or adaptively chosen normalized slopes, and independent nuisance inputs at ranks $8, 9, 10$, respectively. The normalization of the selected rank–$(r+4)$ diagnostic is $y = ah + z$, with all required denominators invertible because $q \geq 41 > 16$. This is the chronological embedding established before this section, and requires no system reset. □

Remark 35.13 (Interpretation). This theorem is an infinite family, unlike an affine relabeling of one finite example. The source alphabet H remains fixed, but the nuisance alphabet E_q grows linearly with q, so it does not contradict the previous eventual-one-pulse bound for *both* alphabets fixed. The circle difference-set calculation and adaptive elimination are classical-style combinatorial arguments, not standalone priority claims. The CCR-specific result is the exact nonresetting chronological realization under a transparent restricted diagnostic observation contract. Under the robust source contract of Section 35.8, additional complete coefficient rows do not remove the lower bound; this conclusion is not asserted for other source contracts or for disclosure of the unmasked pre-preparation history.

35.8 Full-row robust observability under restricted fresh-source freedom

The distinction between a scalar diagnostic and disclosure of the full CCR coefficient row requires care. We record an explicit lifting statement. The observer is shown the *prepared rows from time 11 onward*, not the unmasked rows at times $0, \ldots, 10$; the value of the hidden old source is not disclosed during preparation. These observation-window restrictions are part of the experiment, not consequences of the CCR recurrence.

Fix $p = 5$, $k = 10$ and a prime $q \geq 41$, and let $h = a_{4,4} \in H = \{0, 2, 7, 8\}$. All high sources prior to time 11 apart from h vanish. The fresh ranks $4, \ldots, 10$ at times $11, 12, 13$ may be chosen depending on h so that the three entire coefficient rows are identical for all $h \in H$. The observer knows the rule defining the admissible family, but observes the resulting rows rather than the source choices.

There are three designated pulse opportunities $(s, r) = (14, 4), (15, 5), (16, 6)$. At a selected opportunity the controlled rank-r source has a known nonzero amplitude α, the unknown rank-$(r+4)$ source lies in $E_q = \{0, \ldots, \lfloor q/3 \rfloor\}$, all other new source ranks below $r + 4$ vanish, and ranks strictly above $r + 4$ may take unrestricted values in $\mathbb{F}_q$. If an opportunity is skipped, every fresh high source at that time is zero. This is the *robust full-row source contract*.

Lemma 35.14 (Lifting scalar collisions to entire CCR rows)**.** *Suppose two distinct old sources $h, h' \in H$ have produced the same three prepared rows. At one designated pulse opportunity with controlled normalized slope*

$$a = c\alpha, \qquad c = (3)_3 \left[(10)_4\right]^{-1} \in \mathbb{F}_q^*,$$

the two histories can produce the same entire *current coefficient row with permitted fresh nuisance inputs if and only if*

$$a(h - h') \in E_q - E_q.$$

If they produce a common row, the same assertion may be applied at every subsequent opportunity without restarting either history. Skipped opportunities preserve equality of rows.

Proof. Write $s = 10 + r$ with $r \in \{4, 5, 6\}$. The contribution to the current row not involving fresh high sources is identical because the preceding three full rows agree. The chosen rank-r input contributes $(s-1)_{r-1}\alpha u^r P_{10}$; the old histories have identical coefficients of P_{10} below degree four and

$$C_{10,4}(h) - C_{10,4}(h') = (3)_3(h - h').$$

Consequently, all current coefficients below degree $r+4$ agree before choosing a rank-$(r+4)$ input. In degree $r + 4$, equality holds exactly if

$$(s-1)_{r-1}\alpha(3)_3(h - h') + (s-1)_{r+3}(z - z') = 0 \pmod{q}$$

for some $z, z' \in E_q$. Since $(s-1)_{r+3} = (s-1)_{r-1}(s-r)_4$ and $s - r = 10$, this condition is exactly $c\alpha(h - h') \in E_q - E_q$ (the sign is immaterial because $E_q - E_q$ is symmetric).

At each higher degree $j > r + 4$, the fresh rank-j source has coefficient $(s-1)_{j-1}$ in the j-th observed coefficient. This is a unit in $\mathbb{F}_q$, because $s \le 16 < q$. Thus, proceeding from degree $r + 5$ to 10, choose the unrestricted source in the second history to match the first history's current coefficient, without disturbing lower degrees. Both entire rows now agree. The argument also proves necessity, since no source of rank higher than $r + 4$ can affect degree $r + 4$.

At a skipped time, all fresh high sources are zero and the CCR recurrence only uses the preceding three full rows, which are identical; so equality persists. This supplies induction through all three opportunities. □

Corollary 35.15 (Sharp full-row gap for the infinite prime family)**.** *For every prime $q \ge 41$, in this source contract, observing* all *coefficient rows at times $11, 12, \dots, 16$ does not reduce the worst-case number of predetermined pulse opportunities below three. An adaptive strategy needs at most two pulses even with the full rows disclosed, and a single pulse cannot succeed. Thus the worst-case optima remain three for predetermined schedules and two for adaptive schedules.*

Proof. The scalar circle-difference theorem in Section 35.7 gives a common source difference for any two predetermined normalized pulse slopes. The lemma lifts the two scalar collisions to identical complete coefficient rows and propagates equality through skipped time slots. It also lifts the one-pulse pigeonhole collision. Conversely, the scalar projection of a complete row is available to the observer, so the existing adaptive two-pulse construction and predetermined three-pulse construction retain their upper bounds. All preparation rows were equalized before the first pulse. □

Remark 35.16 (Source-contract dependence). No lower bound is claimed if the unmasked rows before time 11, the hidden source inputs themselves, or the unrestricted high-rank nuisance values are revealed, or if the source contract forbids choosing higher-degree nuisances to equalize the rows. The controller may choose its next pulse using the previously observed full coefficient rows, but all comparisons are between histories with the same observable prefix. The theorem is therefore a robust identifiability result for the explicitly allowed input family, not an unconditional claim about every CCR system.

35.8.1 Exact rational certificate for the circle lemma

For completeness, the 25 open cells on $0 < x < 1/2$ and their full collision sets $B(x)$ are listed below. This table is obtained by sorting all rational boundaries $(j \pm 1/3)/d$ for $d \in \{1, 2, 5, 6, 7, 8\}$. Each row can be verified using the midpoint of the indicated interval. No residue a/q for primes $q \ge 41$ can be one of these boundaries, since $3ad = 3jq \pm q$ would force $q \mid d$.

No.	Interval	Collision set
1	$(0, 1/24)$	$\{1, 2, 5, 6, 7, 8\}$
2	$(1/24, 1/21)$	$\{1, 2, 5, 6, 7\}$
3	$(1/21, 1/18)$	$\{1, 2, 5, 6\}$
4	$(1/18, 1/15)$	$\{1, 2, 5\}$
5	$(1/15, 1/12)$	$\{1, 2\}$
6	$(1/12, 2/21)$	$\{1, 2, 8\}$
7	$(2/21, 1/9)$	$\{1, 2, 7, 8\}$
8	$(1/9, 2/15)$	$\{1, 2, 6, 7, 8\}$
9	$(2/15, 1/6)$	$\{1, 2, 5, 6, 7, 8\}$
10	$(1/6, 4/21)$	$\{1, 5, 6, 7\}$
11	$(4/21, 5/24)$	$\{1, 5, 6\}$
12	$(5/24, 2/9)$	$\{1, 5, 6, 8\}$
13	$(2/9, 5/21)$	$\{1, 5, 8\}$
14	$(5/21, 4/15)$	$\{1, 5, 7, 8\}$
15	$(4/15, 5/18)$	$\{1, 7, 8\}$
16	$(5/18, 7/24)$	$\{1, 6, 7, 8\}$
17	$(7/24, 1/3)$	$\{1, 6, 7\}$
18	$(1/3, 8/21)$	$\{2, 5, 6, 8\}$
19	$(8/21, 7/18)$	$\{2, 5, 6, 7, 8\}$
20	$(7/18, 5/12)$	$\{2, 5, 7, 8\}$
21	$(5/12, 4/9)$	$\{2, 5, 7\}$
22	$(4/9, 11/24)$	$\{2, 5, 6, 7\}$
23	$(11/24, 7/15)$	$\{2, 5, 6, 7, 8\}$
24	$(7/15, 10/21)$	$\{2, 6, 7, 8\}$
25	$(10/21, 1/2)$	$\{2, 6, 8\}$

Finite verification. Exact arithmetic checks cover all 625 source–nuisance trajectories and 256 nonzero slope pairs for $q = 17$, and all 16,440 two-slot schedules over $q = 41, 43, 47$. These finite calculations corroborate the constructions; the infinite-prime conclusion follows from the proofs above.

Chapter 36

The Computational Power of Typed Chronological Carry Refinement: Sharp Forecast-Information Bounds, Fast Exact Algorithms, Finite-Ring Profiles, and Lossless Source Coding

Research branch — Sharp information bounds and exact algorithms for typed CCR-k. *Following the active-observability chapter, the focus turns to the information required to answer arbitrary future-source queries, exact finite-ring response families, selective evaluation, and lossless source-mask coding. The results separate a sharp mathematical archive optimum from task-dependent computation and application measurements.*

The results in this chapter are developed in the independent CCR-k computational study [261]. The statements, proofs, exact computational measurements, technical supplements, and finite-certificate summaries are integrated here with book-wide numbering and references. The original article source is preserved separately.

36.1 Introduction

Typed CCR-k is more than a formal coefficient recurrence: its delayed-source chronology determines what can be observed, remembered, compressed, and evaluated efficiently. This paper presents its *complete resource story* for a range of exact tasks. We prove an optimal five-step forecast archive, construct an online realization of that bound, compute exact finite-ring response families, and give high-throughput implementations whose gains are measured against explicitly identified alternatives. We also derive an optimal lossless encoding law for event-equality source signatures. Together, these results supply quantitative, implementable consequences of the CCR-k theory, rather than an isolated collection of numerical coincidences.

The central strength is a tight coupling between mathematical structure and code: the same source-delay restrictions yield algebraic rank bounds, sparse future-response representations, and verifiable computational reuse. We evaluate both its favorable regimes and the regimes in which another exact method is better. A task-specific speedup is meaningful

because it comes with a reproducible input/output contract, validated coefficients, and an honest accounting of the work being compared.

CCR-k is defined here by its complete typed polynomial recurrence, including all mandatory and optional terms. This makes the claims reproducible without requiring a companion paper. The symbols p, k, J, T, H, the modulus and the source family are retained explicitly whenever they affect a theorem. In particular, a freely assignable algebraic source program is *not* interchangeable with a binary event-equality feature extracted from an external trace; this distinction is critical in the finite-field archive theorems and the real-data studies.

36.1.1 Contributions

The specific contributions assembled and cross-checked in this paper are:

1. an exact *source-aware* CCR recurrence, weighted-DAG representation, and an account of the precise failure of a simpler independent-event polynomial representation;

2. mathematically certified low-degree and interaction-channel monitors, accompanied by identifiability and passive-observation counterexamples;

3. a hierarchy of sharp response-information results: one-step read-coordinate necessity, general response rank, the first four future horizons, and the five-step archive theorem with a factored determinant certificate;

4. a constructive chronological minimum-message archive and a sparse future-response polynomial whose structurally allowed monomial count is $6J - 21$ at horizon five;

5. exact finite-ring profile images with appropriate Smith/Hermite-form arithmetic, including a task-equal faster counting experiment;

6. a quantitative algorithm-selection framework for pruned dynamic programming, compiled response evaluation, low-storage checkpoints, vectorized batches and finite lookup, with complete cost qualifications;

7. a graph characterization of event-equality source masks and sharp lossless payload bits, plus finite CCR-coordinate injectivity certificates and round-trip real-log coding;

8. two independent real-data studies that separate *correct CCR computation* from *unproven downstream predictive benefit*, supported by explicit mathematical and empirical validation.

36.1.2 Main achievements in one view

The following distinct statements constitute the principal mathematically or experimentally certified findings. Their hypotheses and cost models are specified at the appropriate theorem and experiment.

CCR-k contribution	Certified consequence
Sharp future-response information	$m(T, J, 5) = 5J - 31$ over the stated prime fields; exact lower bound applies to nonlinear fixed-length encoders as well as linear summaries.
Constructive exact computation	A causal minimum-message update and a five-step future-response polynomial with at most $6J - 21$ source monomials.
Finite-ring arithmetic	Exact response-profile cardinalities through accessible source lattices and Smith/Hermite reduction; 1,953,125 profiles in a recorded mod-25 example.
Measured accelerator	8.62× for an exact 100-checkpoint, 512-futures/checkpoint CCR workload versus task-pruned Python suffix DP; alternative NumPy baselines explicitly tested.
Measured memory saving	Up to 4.40× less `tracemalloc` peak allocation in a specified low-query multi-checkpoint regime.
Optimal event-source encoding	$\lceil \log_2 C_N \rceil = \Theta(N \log N)$ bits for realizable equality masks, versus $\Theta(N^2)$ raw indicator bits; 26.67% smaller actual packed payload on BPI2013 prefixes.

36.1.3 Relationship to the author's CCR series

The chronological carry program was developed in the author's third-order arithmetic-lift study, its fourth-order connected-quadruple extension, and the subsequent *CCR-k General Theory* [164, 165, 166]. Those antecedents supply the mathematical lineage of higher-order source transport and source-indexed coefficients used in the present paper. The related *CCR-k Arithmetic Refinement* manuscript studies stationary normalized columns and local arithmetic geometry [195]; the present work instead develops the *nonstationary, finite-horizon forecast-information and exact-computation* interface. This separation matters: prior source-structure results are acknowledged, while the exact five-step archive contract, its minimal online encoder and the execution experiments are explicitly stated and proved or measured here. Another related stationary resonance study concerns a different high-coefficient p-adic task [218]; none of its rigidity theorems is needed to derive the present resource bounds.

36.1.4 What counts as an advantage

We use four distinct notions: an *exact structural theorem* (about a well-defined CCR family), a *message-information bound* (field symbols or payload bits), a *measured implementation advantage* (specific machine, code and competitor), and a *downstream external benefit* (better real predictive or operational performance on held-out cases). The first three are established for several distinct contracts below; the fourth has not been established. The distinction preserves the actual value of the results while preventing a local benchmark from being misread as a universal deployment guarantee.

General-purpose semiring algorithms [266], weighted-automata minimization [265], connected partitions [267], and Hermite/Smith methods [263] are existing frameworks. The exact source geometry and response formulas derived here are specialized to CCR; they should not be mistaken for inventions of those general mathematical tools. The chronology-specific statements are separated explicitly from the source and stationary-column results in the author's earlier CCR papers [164, 165, 166, 195].

36.2 CCR recurrence, source interfaces, and query tasks

Let R be a commutative ring and $u^{J+1} = 0$. Define $P_0 = 1$, $P_t = 0$ for $t < 0$, and

$$P_n = (1 + (4n-3)u)P_{n-1} + 3(n-1)(4n-7)u^2 P_{n-2} \tag{36.1}$$
$$- 3(n-1)(n-2)u^3 P_{n-3} + \sum_{r=4}^{\min(J,n)} \frac{(n-1)!}{(n-r)!} a_{r,n} u^r P_{n-r} \pmod{u^{J+1}}.$$

Here $p = 3$ and the source positions (r, n) have rank/length $r \geq 4$. Write $q_d(n) = [u^d]P_n$. A checkpoint occurs at time $T \geq J$, after which the archive is frozen. Later, an arbitrary five-step future source program $A = (a_{r,T+h})$ is announced, and the decoder must return $(q_J(T+1), \ldots, q_J(T+5))$ exactly.

The mathematical encoder may read the historical CCR source stream and all its accessible high-degree coefficients before finalizing the message. The fixed-length cost is the number of $\mathbb{F}_\ell$ symbols retained *at the checkpoint*, not the peak RAM, number of input sensors, or total polynomial coefficients processed along the way.

36.2.1 General parameterization and source distinctions

The more general typed recurrence replaces $4n-3$ by $\ell_n = (p+1)n - p$, the quadratic coefficient by $p(n-1)\ell_{n-1}$ and the cubic term by $-p(n-1)(n-2)$, retaining the same falling-factorial weights $w_{r,n} = (n-1)!/(n-r)!$. The particularly detailed information-rank theorems below take $p = 3$. Exact numerical modular experiments use the residue ring $\mathbb{Z}/M\mathbb{Z}$; it is a field only if M is prime. We write $\lambda_n = 4n-3$ interchangeably with the earlier ℓ_n when convenient, but distinguish the coefficient sequence from the field prime ℓ.

A *free-source* instance supplies each $a_{r,n}$ independently as a ring element. An *event-derived* instance first observes a sequence $e_0, \ldots, e_{N-1}$ and sets $a_{r,n} = \mathbf{1}\{e_{n-1} = e_{n-r}\}$. The first class supports the full-reachability lower bounds; the second has stringent combinatorial restrictions, derived later. Evaluating CCR on an external event log is a chosen feature transformation; it does not mean the original process follows CCR dynamics.

36.3 An exact weighted-path interpretation

Take the $p = 3$ typed recurrence defined above, with $\ell_n = 4n-3$ and optional source weights $w_{r,n}a_{r,n}u^r$. Over $R[u]/(u^{J+1})$, create time vertices $0, \ldots, N$. Connect $n-1 \to n$ by parallel edges of weights 1 and $\ell_n u$; connect $n-2 \to n$ with $3(n-1)\ell_{n-1}u^2$; connect $n-3 \to n$ with $-3(n-1)(n-2)u^3$; and add an optional $n-r \to n$ edge of weight $w_{r,n}a_{r,n}u^r$. Splitting all $0 \to n$ paths by their last edge gives precisely P_n.

Proposition 36.1 (Weighted-path identity)**.** *The sum of weights of all paths from* 0 *to* n *in the above DAG equals* P_n *in the truncated polynomial ring.*

Proof. The empty path at vertex zero has weight one. For $n > 0$, partition all paths by their last edge and apply the induction hypothesis at its predecessor. The resulting path sum is exactly the typed CCR recurrence. □

This identity supplies a strong correctness baseline and also prevents us from claiming that path-sum evaluation itself is an original general-purpose paradigm.

Remark 36.2 (Methodological consequence). The DAG equivalence is exact for any commutative coefficient ring once edge products are interpreted in the truncated polynomial ring. It therefore provides both an independent implementation check and an appropriate strong baseline. It does *not* imply that each explicit CCR archive-rank formula or terminal-coordinate separation result has already appeared in general weighted-automata literature.

36.4 Exact source diagnosis and observability obstructions

36.4.1 The independent-event factorization gate

An external event log does not come with a native CCR polynomial state. For example, an ordinary independent-event representation is $E_n(u) = \prod_{i=1}^{n}(1 + w_i u)$. Setting all optional CCR sources to zero and matching linear coefficients forces $w_1 = 1$ and $w_2 = p + 2$; hence $[u^2]E_2 = p + 2$. The typed CCR recurrence instead gives $[u^2]P_2 = 2p + 2$. The representations differ by p for every $p \geq 1$. This exact counterexample rules out *that simple factorization*, not every possible external realization. It motivates the explicit distinction between naturally generated CCR states and deliberately constructed CCR features.

36.4.2 Two terminal coefficients can recover a limited parity load

A useful algebraic diagnostic prototype occurs over $R = \mathbb{Z}/8\mathbb{Z}$ with $p = 3$, $N = k = 8$. Write $x = a_{4,4}$, $z = a_{4,6}$ and $y = a_{4,8}$, and allow the other optional sources to vary freely. For the terminal differences $\Delta_j = [u^j](P_8 - B_8)$ against the zero-optional-source reference, the complete modular identities are

$$\Delta_4 = 6x + 4z + 2y, \qquad \Delta_5 = \Delta_6 = \Delta_7 = 0, \qquad \Delta_8 = 4(x + y + xy) \pmod 8. \quad (36.2)$$

Here the other source weights vanish modulo eight under the stated horizon. Therefore $\Delta_4/2 \bmod 2$ recovers the XOR of the parities of x and y, and $\Delta_8/4 \bmod 2$ recovers their OR. Together these determine the exact integer load

$$W = (x \bmod 2) + (y \bmod 2) \in \{0, 1, 2\}.$$

The proof reduces the recurrence to the three nonzero source positions; the only compatible nonlinear pair is $(4, 4)$ and $(4, 8)$, with integer product coefficient $6 \cdot 210 = 1260 \equiv 4 \pmod 8$. The corresponding numerical audits cover all $8^3 = 512$ effective (x, z, y) assignments and additional full-source checks. This is a complete mathematical diagnostic for the *specified CCR coefficients*; it is not an improved error-correcting code: directly transmitting the two source parity bits already uses two bits, and the tested one-bit modifications often changed W silently.

The exact diagnostic tests distinguish successful identification from robustness. All 64 choices of the two active mod-eight sources, and all 512 effective three-source assignments, satisfied the formulas; additional tests varied all 15 optional source positions. The two observed parity quantities classify the three possible loads without ambiguity, yet under 1,024 deliberate single-coordinate changes, 384 altered signatures were rejected and 640 led to an undetected wrong class. Thus exact identification is established in the stated noiseless model, not automatic detection or correction of corrupted records.

36.4.3 A passive terminal jet can conceal an interaction

For an illustrative source-channel obstruction, take $p = 3$, $N = k = 8$, $M = 101$ and monitor only $q_8(8)$. The zero-source program has $q_8(8) = 79$ and a quadratic degree-four interaction channel $H_2 = 0$. A second program with $a_{4,4} = a_{4,8} = 1$ and $a_{8,8} = 91$ has the same terminal $q_8(8) = 79$, while its isolated quadratic channel is $H_2 = 1260 \equiv 48 \pmod{101}$. The final source compensates the terminal scalar but not the formal quadratic coefficient. More generally, identical passive traces of one high-order coefficient can admit different future source-sensitive responses. Such obstructions make additional archived information or an explicit future-source interface mathematically necessary; they are not claims about physically measured systems.

36.4.4 Information cannot be generated by deterministic features

If a raw event history is X, its CCR-derived feature is $S = \phi(X)$ and the later label is Y, then $Y - X - S$ is a Markov chain and

$$I(Y;S) \le I(Y;X).$$

A deterministic CCR feature may make a restricted learner more effective, or make query evaluation cheaper, but cannot create statistical information absent from a complete raw history. Accordingly, a real-log claim of independent predictive value requires strong raw, interaction and task-specific baselines, not merely a comparison with an impoverished input model.

36.5 Selective monitors and source-polynomial compilation

36.5.1 Selective monitoring

Let $b_d(n)$ be the degree-d coefficient when every optional source amplitude is zero and $D_d(n) = q_d(n) - b_d(n)$. For every p, $k \ge 4$ and coefficient ring,

$$D_4(n) = \sum_{t=4}^{n} (t-1)(t-2)(t-3)a_{4,t}.$$

Thus a single source-dependent register suffices for the degree-four correction. For degree five, two registers suffice:

$$\begin{aligned} D_5(n) &= D_5(n-1) + \ell_n D_4(n-1) + w_{4,n}a_{4,n}b_1(n-4) + w_{5,n}a_{5,n}, \\ D_4(n) &= D_4(n-1) + w_{4,n}a_{4,n}. \end{aligned}$$

Source-free $b_d(n)$ are integer-valued polynomials of degree at most $2d$, so Newton-binomial bases permit exact evaluation even over composite rings, without dividing by nonunits. In a Python benchmark the degree-four monitor ran roughly 2–3 times faster than a task-pruned degree-four DP; at degree five the advantage fell to 1.08–1.63 times. A source-monitor for a high jet was approximately 3–4 times *slower* than a strong full-task DP. We report that negative test because it defines a real boundary.

36.5.2 Full source expansion is not always a good compiler

Optional interventions correspond to non-overlapping chronological blocks of length at least four. If $C(n,d)$ counts compatible block selections of total degree d, then

$$C(n,d) = C(n-1,d) + \sum_{r=4}^{\min(n,d)} C(n-r,d-r).$$

At full degree $k = N$ the number A_N of potential source monomials has generating function

$$\sum_{N\ge 0} A_N z^N = \frac{1-z}{(1-z)^2 - z^4}, \qquad A_N = \Theta(\varphi^N),$$

where $\varphi = (1+\sqrt{5})/2$. Thus full explicit polynomial expansion becomes exponentially large as $N = k$ grows, although the direct numeric recurrence can be evaluated in $O(Nk^2)$ ring operations with $O(k^2)$ rolling state. In the 48-program test the compiled explicit form was 3.52 times faster at $(N,k) = (8,8)$ excluding its setup, but slower than numeric CCR DP by $(16,16)$. The output-size bound applies to explicit monomial expansion, not to all CCR computation.

36.5.3 Compiling future response at a checkpoint

A checkpoint stores a set of historical coordinates or a minimal linear response sketch. When future source values are supplied, a response polynomial in these future amplitudes can be evaluated repeatedly. For a horizon of five, the structural future monomial support has size

$$K(J,5) = 1 + 5(J-3) + (J-7) = 6J - 21 \qquad (J \geq 8).$$

The term of degree two corresponds to the possible disjoint future pair at the first and fifth steps. This is a *structural support* count; individual coefficients can vanish in particular rings or histories. Compilation and batched polynomial evaluation are classical tools; the CCR-specific contribution is the explicit available support, parameter accounting and its validated implementation.

36.5.4 Homogeneous degree-four interaction channels

For $m \geq 1$, introduce a bookkeeping variable v that multiplies only the optional rank-four sources and define $H_m(N) = [u^{4m}v^m]P_N(u;v)$. If $c_n = w_{4,n}a_{4,n} \pmod M$, disjoint length-four source blocks give the exact identity

$$H_m(N) = \sum_{\substack{4\leq t_1<\cdots<t_m\leq N\\ t_{i+1}-t_i\geq 4}} \prod_{i=1}^{m} c_{t_i}, \qquad H_m(n) = H_m(n-1) + c_n H_{m-1}(n-4). \tag{36.3}$$

Take $H_0(n) = 1$ and $H_m(n) = 0$ when $n < 4m$. The second equation partitions valid block families according to whether the last block ends at n. It provides $O(Nm)$ ring operations and $O(m)$ fixed-delay storage. For $m = 3$ and $N = 32768$, the recorded special Python monitor took 37.90 ms, versus 46.76 ms for a full-history interval DP and 138.47 ms for the specified four-phase weighted-automata implementation. A well-optimized general interval DP has the *same* asymptotic cost, so these ratios characterize the tested implementations, not an independent complexity-class separation. Equation (36.3) computes a source-aware formal channel and must not be confused with passively reading one scalar q_{12}.

36.6 Forecast-response information and sharp archives

36.6.1 One-step read coordinates versus encoded archives

Two resource contracts must be distinguished. In a *coordinate-read* contract, the recorder is charged for how many distinct historical coefficient–time cells it reads. In a *joint message* contract, it may store arbitrary (possibly nonlinear) combinations of the complete accessible history. The minima need not agree.

For $T = J$ and known but arbitrarily selectable one-step optional future sources, the direct coefficient identity is

$$\begin{aligned} q_J(J+1) = {} & q_J(J) + \ell_{J+1}q_{J-1}(J) + pJ\ell_J q_{J-2}(J-1) - pJ(J-1)q_{J-3}(J-2) \\ & + \sum_{r=4}^{J} w_{r,J+1}a_{r,J+1}q_{J-r}(J+1-r). \end{aligned} \tag{36.4}$$

The source-dependent low jets $q_0, \ldots, q_3$ are known from the mandatory recurrence. Thus the coordinate set

$$\mathcal{S}_J = \{q_J(J)\} \cup \{q_d(d+1) : 4 \leq d \leq J-1\}$$

is always sufficient, with $J-3$ scalar reads. Under the free-history, large-field and coefficient-unit conditions in the original one-step result, every cell of $\mathcal{S}_J$ is *necessary* within the preselected-coordinate class. If all future optional sources are zero, the corresponding minimum becomes $\min(4, J-3)$. This is not a lower bound against arbitrary linear sketches and does not count physical sensors.

36.6.2 General response-profile principle

With historical high-jet vector x and a future source program A, the exact output has the form $Y_A(x) = c_A + L_A x$ whenever the mandatory low jets are treated as known. Although $A \mapsto L_A$ can be polynomial, the dependence on the *old high jets* remains affine. Over a field, if every historical high-jet vector is reachable, the span of all row functionals occurring among the matrices L_A has rank r, and *exactly* r field symbols suffice for the joint message. Necessity follows from the $|\mathbb{F}|^r$ distinct answer profiles, not merely from a restriction to linear encoders. This general rank principle is classical; calculating r for the CCR chronology is the substantive problem.

36.6.3 The first four future horizons

For a fixed future program A, the target is affine in the earlier high-jet vector x:

$$Y_A(x) = c_A + L_A x.$$

The rank of the row span of all L_A determines the linear archive cost, provided the high-jet historical coordinates are independently reachable. In a finite field with $\ell > T + 5$, the source coefficient $w_{r,n} = (n-1)!/(n-r)!$ is a nonzero unit at every relevant historical position; increasing-degree source choices give the required triangular accessibility.

For horizons $H \le 4$, two future optional source blocks cannot be disjoint in a single chronological path because each requires at least four time positions. Therefore the future response is affine in future-source amplitudes.

Proposition 36.3 (First four horizons)**.** *At checkpoint $T = J \ge 8$, with freely assignable historical sources over $\mathbb{F}_\ell$, $\ell > J + H$, and with the $H \times H$ mandatory-response minor T_H of full rank H over that field, the minimal fixed-length archive for the H terminal outputs is $H(J-6)$ field symbols.*

Proof. The zero-source future contributes H response functionals. The possible single-source pulses read mutually distinct historical high-jet coordinates whose count is $H(J-7)$ for the indicated range. Each pulse has a nonzero scalar coefficient, so the corresponding coordinate functionals are independently recoverable. The mandatory-transfer responses contribute the remaining H independent directions, giving $H(J-6)$. No higher future source monomials arise at these horizons. □

The support formula is not a claim that all mandatory-transfer determinants stay nonzero over every ring; the field nondegeneracy hypothesis is essential.

36.6.4 The sharp five-step theorem

Set

$$D_5(T) = 192T(T+1)\big(64T^6 + 1312T^5 + 11120T^4 + 49565T^3 + 121776T^2 + 155655T + 80706\big).$$

Theorem 36.4 (Five-step forecast archive)**.** *Let $p = 3$, $T \ge J \ge 8$, $R = \mathbb{F}_\ell$ with ℓ prime, $\ell > T + 5$, and $\ell \nmid D_5(T)$. Suppose historical sources are freely chosen and accessible chronologically, whereas future source programs are disclosed only after the archive is fixed. Every deterministic fixed-length $\mathbb{F}_\ell$-alphabet archive returning $q_J(T+1), \dots, q_J(T+5)$ for* every *future source program requires at least $5J - 31$ field symbols. A causal archive with exactly that many symbols exists. Thus*

$$\boxed{m(T, J, 5) = 5J - 31}.$$

Proof. Write the historical high-jet coordinates read by future single pulses as

$$X_{d,s} = q_d(T - J + s + d), \quad 4 \le d \le J - 4,\ 1 \le s \le 4,$$

and additionally $X_{d,5}$ for $4 \le d \le J - 5$. Their total is $4(J - 7) + (J - 8) = 5J - 36$. Let $Z_1, \dots, Z_5$ be the five outputs when all future optional source amplitudes vanish. We first show the entire future response polynomial is determined by the X's and Z's.

A future optional block has length at least four. At horizon five, the only way two such blocks can appear in one chronological path is to place the first block at $T + 1$ and a rank-four block at $T + 5$. For first rank r, their interaction reads

$$w_{r,T+1} w_{4,T+5}\, q_{J-r-4}(T + 1 - r).$$

But this is the very historical coordinate read by the single pulse $(r + 4, T + 5)$, as $T + 5 - (r + 4) = T + 1 - r$. Mandatory transfers and individual pulses likewise read linear combinations of the listed Z and X coordinates. Therefore no new historical direction is created by the only possible future source product, proving the $5J - 31$ upper bound.

For the lower bound, switch on one future source pulse at $(J - d, T + s)$ (and all others off). The difference between its response and the zero-source response isolates $X_{d,s}$ up to the invertible factor $w_{J-d,T+s}$; $\ell > T + 5$ ensures invertibility. These give $5J - 36$ independent historical directions. Modulo their span, consider the five history coordinates

$$q_J(T), \quad q_{J-1}(T), \quad q_{J-2}(T), \quad q_{J-3}(T), \quad q_{J-3}(T - 1).$$

The 5×5 mandatory-transfer response minor is $D_5(T)$. Its nonzero residue modulo ℓ makes the five zero-source responses independent of the X directions. Thus the full response rank is $5J - 31$.

The triangular source-accessibility argument implies that all ℓ^{5J-31} combinations of these independent response coordinates can arise from past source assignments. Two distinct response profiles cannot be given the same archive message, since some later future program separates them. An encoder of b symbols has at most ℓ^b messages, and hence $b \ge 5J - 31$, even if the encoder is nonlinear. Retaining the $5J - 31$ linear forms proves equality. □

Remark 36.5. The determinant factor and historical accessibility hypotheses cannot be silently dropped. The theorem does not assert the same minimum for a fixed binary equality-source family, a composite modulus, or a code whose encoder is denied access to intermediate CCR coefficients.

36.6.5 An exact future-source support budget

The same chronological restriction gives an independent combinatorial quantity relevant to a compiled evaluator.

Proposition 36.6 (Structural future support)**.** *For $J \ge 8$, $T \ge J$ and horizon five, the joint response polynomial in the future optional source amplitudes has no source monomials of degree greater than two. Its possible support contains exactly*

$$K(J, 5) = 1 + 5(J - 3) + (J - 7) = 6J - 21$$

monomials: one constant, $5(J - 3)$ singleton future-source positions, and $J - 7$ allowable products.

Proof. There are $J - 3$ optional source ranks $4, \dots, J$ at each of five future steps, hence $5(J - 3)$ singleton positions. Every product requires two non-overlapping chronological blocks, each of length at least four. In the five-step interval this forces the later block to have rank four and to finish at $T + 5$; the earlier block must finish at $T + 1$ with rank $r \le J - 4$. This leaves $J - 7$ choices $r = 4, \dots, J - 4$. Three future blocks cannot fit. The constant monomial corresponds to the all-zero future source program. □

The proposition counts *structurally allowed* monomials. It does not assert that every coefficient is nonzero after specialization to a particular past history or a ring with zero divisors. The count supports exact sparse compilation and is used as a workload contract in the accompanying computational study.

36.6.6 A causal realization with the minimum retained message

Let $B_1, \dots, B_m$ be a basis of the historical linear functionals underlying all future-response coefficients, $m = 5J - 31$. Express each as $B_i \cdot x$. When historical coordinate $x_{t,d}$ is computed, update

$$S_i \leftarrow S_i + B_{i,(t,d)} x_{t,d} \pmod{\ell}.$$

At the checkpoint, retain $S \in \mathbb{F}_\ell^m$ only. Every future program A admits $L_A = \Lambda_A B$, and its result is $c_A + \Lambda_A S$. This is a constructive causal linear sketch of the optimal fixed-length *archive*; its builder still uses ordinary rolling CCR state and a shared basis table while the past is processed.

36.6.7 Composite moduli and exact response-profile cardinalities

Over $\mathbb{Z}/M\mathbb{Z}$ the field-rank formula may be wrong: factorial weights can be zero divisors, reachable high jets need not fill the entire free module, and answer-profile fibers can change. Given an integral matrix K describing the reachable response lattice (after restricting it to truly accessible histories), Hermite and Smith normal forms yield exact image sizes. For a free-domain integer matrix with Smith diagonal d_i, the image modulo M has size $\prod_i M/\gcd(M, d_i)$ over its nonzero invariant factors; if the accessible domain is a proper submodule, its own generators must be included first [263].

In the exact CCR experiment $J = 8, H = 5, M = 25$, the exact number of answer profiles is 1,953,125. For $J = 8, H = 6, M = 8$, an HNF/SNF-based profile counter with its self-checks took 38.8–40.9 ms against 519.8–563.8 ms for an optimized exhaustive counter; this is a specific *profile-counting* comparison, not a streaming inference speedup. For $H = 6$, formal finite-parameter ranks are known at selected J values but no all-J closed-form theorem is claimed here.

Table 36.1: Exact formal answer ranks at $p = 3$, $\ell = 101$ or 103 for representative order and future horizons. These are finite-family certificates; the $H = 6$ column is not asserted as a universal formula.

J	$H = 4$	$H = 5$	$H = 6$
8	8	9	9
10	16	19	20
12	24	29	32
16	40	49	56

Table 36.2: Exact counts of distinct response profiles over composite rings, $p = 3$, $J = 8$, with freely variable historical sources.

Modulus M	$H = 4$	$H = 5$	$H = 6$
8	3	3	3
12	1	1	1
16	10	10	10
25	78,125	1,953,125	9,765,625

These counts are exact algebraic image cardinalities, not estimated prediction scores. They illustrate the influence of nonunit source weights and cannot be extrapolated from the field-rank theorem.

36.6.8 Why the information optimum does not imply a working-memory optimum

For $J = 8$ and $J = 12$, the theorem gives nine and 29 field symbols, compared with 11 and 31 visibly relevant raw high-jet coordinates. At $T = 16384$, example serialized single archives occupied 58 versus 367 bytes and 154 versus 777 bytes against the much fuller rolling checkpoint; however the minimal compiled method used *more* total Python allocations during one-query execution. For $J = 12$, the recorded peak allocations were 191,992 bytes for minimal compilation and 13,972 bytes for rolling DP. The encoder updates its small eventual message while a rolling CCR state and basis data are alive. These are different physical resources, and the theorem concerns only the former.

36.7 Event-equality signatures: graph geometry and coding

For a length-N trace $e_0, \dots, e_{N-1}$ over an unrestricted symbol alphabet, define the source family

$$a_{r,n} = \mathbf{1}\{e_{n-1} = e_{n-r}\}, \qquad 4 \le r \le n \le N.$$

There are $D_N = (N-3)(N-2)/2$ such bits. Construct the graph G_N on vertices $0, \dots, N-1$, with an edge (i, j) if and only if $|i - j| \ge 3$, so $G_N = \overline{P_N^2}$. The source mask specifies which edges have endpoints carrying the same symbol.

Theorem 36.7 (Equality masks and connected partitions)**.** *The distinct realizable source masks from all event traces of length N are in bijection with partitions of $V(G_N)$ into blocks that induce connected subgraphs of G_N.*

Proof. Given a mask, take the graph consisting of the positive equality edges. Each of its connected components is monochromatic, since equality propagates along a path. If two vertices of a component are adjacent in G_N, their edge is necessarily positive; therefore the mask is completely determined by these connected components. Conversely, for any partition into connected induced subgraphs, assign a distinct symbol to every block. Then precisely the within-block edges of G_N are positive, and the components of the positive graph are exactly those blocks. The two constructions are inverse. □

Let C_N count these partitions. With $F(\varnothing) = 1$, for every nonempty $S \subseteq V$ and $v = \min S$,

$$F(S) = \sum_{\substack{T \subseteq S,\ v \in T \\ G_N[T] \text{ connected}}} F(S \setminus T), \qquad C_N = F(V).$$

This is an exact subset recurrence with a naive $O(3^N)$ arithmetic-work bound and $O(2^N)$ table storage. It produces $C_8 = 1419$, $C_9 = 7953$, $C_{10} = 46610$, $C_{11} = 287134$, $C_{12} = 1860478$, and $C_{16} = 5226655287$. Independent restricted-growth-string enumeration matches the results through $N = 11$.

36.7.1 Sharp lossless payload bits

Theorem 36.8 (Fixed-length source-signature code)**.** *A fixed-length lossless code for every realizable length-N equality mask needs exactly*

$$b_N = \lceil \log_2 C_N \rceil$$

bits. Moreover $b_N = \Theta(N \log N)$, whereas the raw edge-mask dimension is $D_N = \Theta(N^2)$.

Proof. There are C_N distinct messages to distinguish, which requires at least $\lceil \log_2 C_N \rceil$ bits. Enumerating the connected partitions, assigning their ranks $0, \dots, C_N - 1$, and decoding by unranking achieves this bound. For the asymptotic upper bound, $C_N \le B_N \le N^N$,

where B_N is the Bell number. For a lower bound, every third vertex of the path forms a clique in G_N of size $\lceil N/3 \rceil$. Fix approximately half of those vertices as distinct anchors and independently attach each remaining clique vertex to one anchor, leaving all non-clique vertices singleton. This yields at least m^{K-m} connected partitions, where $K = \lceil N/3 \rceil$ and $m = \lfloor K/2 \rfloor$. Their logarithm is $\Omega(N \log N)$. □

Table 36.3: Exact lossless source-mask payload bits.

N	Raw D_N	Connected masks C_N	Optimal b_N
8	15	1,419	11
9	21	7,953	13
10	28	46,610	16
11	36	287,134	19
12	45	1,860,478	21
16	91	5,226,655,287	33
17	105	42,283,431,279	36

36.7.2 What the terminal CCR outputs retain

Set $p = 3$ and $k = N$ in the typed CCR recurrence, with the binary source family above and terminal vector $(q_4(N), \ldots, q_N(N)) \bmod M$. This is a deterministic transform of the source mask, and may merge different masks.

Proposition 36.9 (Finite-coordinate certificates)**.** *For $N = 8, 9, 10, 11$, the full terminal CCR output is injective on the C_N realizable masks at $M = 31$ and at $M = 101$. For $M = 101$, the minimum number of output coordinates sufficient to retain injectivity is $N - 5$ in these four enumerated instances (respectively $3, 4, 5, 6$); at $M = 31$ the minimum counts are $4, 5, 6, 7$.*

Computer-assisted finite proof protocol. Generate each connected-partition mask without omission, evaluate the terminal CCR recursion for that mask with exact modular arithmetic, and group the resulting vectors. All classes are singletons. To prove minimality of the stated coordinate sizes, enumerate every subset of the output coordinate set of smaller size and find at least one collision in each, while checking the displayed witness-sized subsets are injective. Independent recurrence and C++/NumPy subset implementations are archived. The claim is limited to the four enumerated values of N and two moduli. □

The following explicit subsets witness all reported finite-coordinate minima, with each displayed subset evaluated at terminal time N. For $N = 8$, $M = 101$, one minimum

Table 36.4: Minimum terminal CCR coordinate witnesses on all realizable equality masks, at $p = 3$, $k = N$.

N	Modulo 31	Modulo 101
8	(q_4, q_6, q_7, q_8)	(q_4, q_7, q_8)
9	$(q_4, q_5, q_7, q_8, q_9)$	(q_4, q_5, q_8, q_9)
10	$(q_4, q_5, q_6, q_7, q_9, q_{10})$	$(q_4, q_5, q_7, q_9, q_{10})$
11	$(q_4, q_6, q_7, q_8, q_9, q_{10}, q_{11})$	$(q_4, q_5, q_7, q_8, q_{10}, q_{11})$

separating triple is (q_4, q_7, q_8), whereas no pair works. Three elements of $\mathbb{Z}/101\mathbb{Z}$ require at least 21 fixed bits if stored as uncompressed field symbols; this is *not* smaller than the 11-bit optimal connected-mask rank. With unrestricted binary amplitudes at $N = 8$, rather than only realizable equality masks, the full terminal vector is first injective at integer modulus $M = 31$ among moduli $2, \ldots, 31$ by exhaustive checking.

Corollary 36.10 (Inevitable collisions at fixed modulus)**.** *For every fixed modulus M, terminal CCR outputs modulo M cannot distinguish all realizable equality masks for all sufficiently large N.*

Proof. The terminal output space has size at most $M^{N-3} = 2^{O(N)}$, while the number of realizable source masks is $C_N = 2^{\Theta(N \log N)}$, which eventually exceeds M^{N-3}. □

36.7.3 A real-log round-trip coding check

In BPI Challenge 2013 incident data [262], 3,175 cases had at least eight activities. Their source masks occupied 191 distinct realizable signatures. A *universal* rank code supporting all 1,419 realizable eight-event masks mapped 15 raw bits per case to 11 bits without loss. Concatenated payloads consumed 5,954 and 4,366 bytes, respectively, a reduction of 26.67%. Every original mask was recovered.

However the experimental dense mask-to-rank table used 65,536 bytes and its inverse rank-to-mask array 2,838 bytes before Python overhead. The cold codebook build required approximately 74.6 ms and did not improve total RAM or one-shot end-to-end throughput. Thus the result is a rigorous *payload information bound*, with an explicit implementation and its overhead, rather than a universal low-memory codec claim.

36.7.4 Relation to exact terminal diagnosis

The terminal CCR mapping can also be useful for highly specific diagnosis even when full-mask reconstruction is not possible. At $p = 3, N = k = 8$ over $\mathbb{Z}/8\mathbb{Z}$, two source positions $x = a_{4,4}$ and $y = a_{4,8}$ produce a nonlinear product in the terminal degree-eight coefficient. For a parity-load task $W = (x \bmod 2) + (y \bmod 2)$, appropriately normalized degree-four and degree-eight residues yield an XOR and an OR bit, determining W exactly even when other listed high-rank sources are free under the relevant recurrence restrictions. This diagnostic example has no proven telemetry-bit advantage over directly transmitting the two raw parity bits and no demonstrated error-correction guarantee; it belongs here as a mathematically certified case study, not as a physical device claim.

36.8 Algorithm contracts, equality checks, and resources

36.8.1 Four exact evaluators for four kinds of work

The central choice is an algorithmic task, not a different definition of CCR. The direct polynomial recurrence retains $J+1$ coefficient columns per current polynomial and uses rolling time slices; for unrestricted sources with $r \leq J$, a straightforward evaluation is polynomial in the horizon, unlike an explicit source-monom expansion. A *degree-pruned* evaluator computes only coefficients required for the query, precomputing source-independent low degrees. A *response compiler* constructs a sparse polynomial in delayed future-source amplitudes after one checkpoint and then evaluates it over many futures. A *table evaluator* precomputes outputs for every small input mask, trading startup and storage for constant indexed lookup. These evaluators may implement equivalent path sums, but the work they reuse is different.

Mode	Reused state	Main cost to charge
Streaming rolling DP	Last $J+1$ polynomial time slices	Per-step updates and all required degrees
Task-pruned suffix DP	Fixed low jets and targeted high jets	Suffix evaluation per future
Compiled response	Past-dependent response polynomial	Build/compile once, then each future evaluation
Precomputed finite-mask LUT	Full table or reachable-mask dictionary	Table construction, memory, lookup, and persistence

36.8.2 Verifiable equality and modular safety

All reported speedups correspond to methods that were checked against the same typed recurrence, coefficient by coefficient. The source-polynomial implementation was also checked against a weighted-path evaluator and direct recurrence on the entire set of 1,419 realizable $N = 8$ equality masks. In the real BPIC13 case, 3,175 prefixes times five high jets yielded 15,875 exact coefficient matches across the principal evaluators; a later three-channel source experiment checked 47,625 further equalities. Large-modulus NumPy batch products must be split into overflow-safe groups before modular reduction: exact modular arithmetic, not silent signed-integer wraparound, is the contract. The validation records include bounded products for $M = 1,000,000,007$ and $M = 2,147,483,647$.

36.8.3 Benchmark taxonomy

The reported *hot kernel* excludes initial source extraction, the input XML parser and sometimes basis preparation. A *cold compiler* counts its basis preparation. *End-to-end synthetic CCR computation* includes history processing, compilation and all follow-up queries but excludes the common cost of constructing synthetic future-input dictionaries. *Real file end-to-end* additionally includes parsing, reading and construction of source indicators. Python `tracemalloc` estimates tracked Python allocation, not process RSS; serialized checkpoint bytes are smaller still and must not be substituted for the complete working set. Baselines should be labeled in each table rather than all called simply "DP."

36.9 Reproducible timing and memory evidence

Every benchmark checks exact coefficient agreement. One method is a *degree-pruned suffix DP* that precomputes source-independent low jets and truncates the remaining state to the coefficients that the query genuinely needs. This is substantially stronger than recalculating the entire historical CCR trajectory per query. The compiler is benchmarked with its history and setup cost when specified. Different CPU libraries and source densities preclude a single universal speed ratio.

Table 36.5: Synthetic multi-checkpoint task: $T = 4096$, $C = 100$, $M = 1{,}000{,}003$, horizon five; times include streaming and compiler preparation.

J	Futures/checkpoint	Pruned DP (ms)	Compiled batch (ms)	Speedup
8	128	261.31	66.41	3.93×
12	128	500.63	108.77	4.60×
8	512	869.00	114.36	7.60×
12	512	1614.63	187.35	8.62×

The large advantage is relative to a strong *Python* DP whose suffix operation has not been batched through the same NumPy kernel. The study includes an equally vectorized per-checkpoint comparator, against which the large factor does not generally persist. The measured improvement is consequently an executable CCR workload optimization, not a proof of a new complexity class.

The same workload design remained effective at higher jet orders. At $T = 4096$, $C = 100$ checkpoints and 128 futures per checkpoint, an exact compiled evaluation of $J = 16$ used 209.58 ms versus 670.27 ms for the degree-pruned DP (3.20-fold), while $J = 20$ used 286.29 ms versus 902.14 ms (3.15-fold). Streaming, model preparation and future evaluation were included in these matched synthetic CCR tasks. The results extend the practical range of the measured optimization without claiming hardware-independent acceleration.

In the $J = 12$ row the raw-archive method was *slower*: 477.87 versus 399.60 ms. When $Q = 512$ the compiled batch can itself consume *more* Python allocation than pruned DP. In the same multi-query task, a separate 16-workload heuristic selected the faster measured

Table 36.6: Long stream, many checkpoints, $Q = 16$: peak Python `tracemalloc` allocations, not operating-system RSS.

J	T	Checkpoints	DP (bytes)	Raw archive (bytes)	Reduction
8	16384	512	1,774,584	554,892	3.20×
12	16384	512	3,534,688	802,960	4.40×

policy in 15 cases; a miss caused about 10.5% time regret. This heuristic has no cross-machine optimality guarantee.

At 1,000 checkpoints and $Q = 16$, the corresponding pipelined evaluator used 1.52 MB rather than 3.36 MB of tracked Python allocation for $J = 8$ (approximately 380 versus 423 ms); for $J = 12$ it used 2.64 MB rather than 6.80 MB but was slower (911 versus 729 ms). With $J = 12$, 100 checkpoints and $Q = 128$, changing the future-source density from 0.03 to 0.90 left the vectorized pipeline near 132 and 131 ms, while the pruned DP shifted from about 288 to 661 ms. The method selector was checked on 16 held-out synthetic workloads with separate input and future seeds, selected the measured faster method in 15, and preserved exact arithmetic outputs in all 16. For moduli 1,000,000,007 and 2,147,483,647, safe integer batch lengths of nine and two terms respectively were established to avoid signed-64-bit overflow.

36.9.1 Long-stream and real-prefix checks

For $T = 16384$ and 2048 distinct future programs, including the entire historical stream and response setup, the scalar compiler was 1.28 times faster than the strong degree-pruned DP at $J = 8$, and 1.11 times faster at $J = 12$. For a single or a small number of futures the margin vanished, and at $J = 12$ the compiler could be slower. Minimum archive size (9 or 29 field symbols in the theoretical five-step setting) should never be confused with total live RAM: the expanded response polynomial and compiler basis may dominate.

The user-provided BPI Challenge 2013 incident log has 7,554 cases and 65,533 events. Its 3,175 length-eight prefixes generated 15,875 exact CCR coefficient matches across five outputs. On these same source masks, a scalar compiled feature kernel was about 2.5–2.9 times faster than a task-pruned Python kernel, and a matched NumPy compiled kernel had a roughly 2.9–3.1-times hot-kernel advantage. When the NumPy basis was rebuilt on every trial, the difference was much smaller (0.930 versus 1.079 ms for all 3,175 prefixes). XML parsing alone took around 1.57 seconds; no major whole-pipeline speedup is established.

36.9.2 Precomputed small-mask lookup

At $N = 8$ there are 15 source equality bits but only 1,419 realizable masks from actual equality sequences. A dense $2^{15} \times 5$ `uint8` table (163,840 bytes of value storage) can answer the five CCR output coordinates by direct indexing. In three separate processes, warm queries for 3,175 masks required roughly 0.036 ms for the full table, 0.369–0.493 ms for the compiled polynomial, and 1.33–1.53 ms for vectorized pruned DP. The lookup therefore attained about 10–13-times the warm speed of the polynomial but consumed far more model storage and 5–7 ms of preparation. In fully charged repeated-work experiments, the table lost at 16 repeats, but the full table won in all three trials at 24 or more repeats. At 16 repeated batches the compiled polynomial still won after charged setup (about 6.49 versus 7.09 ms for the dense lookup); at 24 repeats the dense table won (9.73 versus 7.58 ms), and at 32 repeats both the dense and reachable-mask tables won (13.71 versus 8.44 and 9.37 ms). A reachable-mask-only value table used approximately 72.6 kB rather than the 163.8 kB dense table, subject to its own indexing cost. No claim is made that lookups are new in computing.

36.9.3 External prediction: no established added benefit

At first-eight-event prediction on BPI2013, a separate *exploratory* temporal split gave held-out macro-F1 of 0.92536 for the base features and 0.93122 after adding CCR; the paired bootstrap interval for the difference included zero, and a raw-plus-pairwise model obtained 0.93616. The original preregistered chronological split was not evaluable after prospective label purging. A second independent, SHA-verified Helpdesk log contains 3,804 cases and 13,710 events [264]. It is too short for a powered eight-event CCR interaction test; in a separately frozen four-event closure-proxy experiment, CCR did not change the baseline classifications (macro-F1 0.4949 for both). These are genuine real-log computations but *not* validation of general predictive superiority. The CCR features are deterministic functions of the raw trace and cannot create information absent from that trace.

36.9.4 Why repeated forecasting is an accelerator rather than a mere compression trick

Let H_T be the common cost of reading the historical stream, B the source-independent basis and response-polynomial preparation cost, c_D the cost of one exact degree-pruned suffix query, and c_C the cost of evaluating the already compiled response. Under a fixed engine, source density, ring and horizon, the two complete compute costs satisfy

$$\mathsf{Cost}_{\mathrm{DP}}(Q) = H_T + Qc_D, \qquad \mathsf{Cost}_{\mathrm{compiled}}(Q) = H_T + B + Qc_C.$$

Hence whenever $c_D > c_C$, compilation is provably favored by this measured cost model for

$$Q > Q_{\mathrm{cross}} := \frac{B}{c_D - c_C}.$$

This is an arithmetic break-even criterion rather than a new complexity lower bound: the actual quantities depend on implementation and input class. Crucially, the model explains why a compiler can lose for one query while winning on thousands of futures from the same CCR checkpoint. The archive theorem controls how much *historical response information* must be retained; the speed gain comes from *factoring and reusing* the same source response, which can also be applied to an uncompressed visible-coordinate archive. These are complementary, not synonymous, advances.

With C checkpoints, replace B by the sum of per-checkpoint preparation expenses and model-evaluation reuse; for a single shared relative-future design, NumPy can batch all $C \times Q$ evaluations. The large speed factors in Table 36.7 describe this exact workload specialization, and not an assertion that merely choosing the CCR recurrence improves the asymptotic cost of every weighted automaton.

36.9.5 Executable architecture: minimal message, fast decoder, explicit cost switches

For a fixed prime field and five-step query interface satisfying Theorem 36.4, the archival stage computes a basis of historical response functionals *once* from the fixed parameter contract. During the source stream it accumulates their values; at checkpoint time the stored message consists of exactly $5J-31$ field symbols. The on-demand stage then chooses between two equivalent exact engines: a five-row degree-pruned suffix update for low Q, or a prepared sparse future-response polynomial for high Q. If equality-source masks from a small, fixed domain are repeatedly evaluated at the same model parameters, a fully built finite lookup provides a third, memory-heavy route. Every route outputs the *same* terminal CCR coefficients.

Stage	What is retained / reused	What is measured
Online source processing	Rolling coefficients plus fixed linear-response basis; update m accumulators	Stream CPU and temporary peak memory
Checkpoint archiving	Exactly $m = 5J - 31$ field symbols in the theorem's contract	Persistent message bytes, separately from model RAM
Repeated future queries	Sparse response coefficients or degree-pruned suffix state	All Q replies, compilation and numeric exactness
Finite-mask service	Value table over all 2^{15} masks or reachable-mask index	Cold build, value bytes, hot access and amortization

36.9.6 Evidence-based method selection

Workload	Method supported by available evidence
One q_4 coefficient at long horizon	Use one-register selective monitor; compare with degree-four pruned DP.
Few future programs	Use pruned suffix DP; the compiler's setup can dominate.
Many futures at one checkpoint	Compile the response polynomial and amortize setup; compare against the best matched vectorization.
Many checkpoints, small futures	Use compact checkpoint archives if peak Python allocations matter; elapsed time can be worse.
Repeated small finite masks	Consider a prebuilt lookup; include storage and cold build in the cost model.
Downstream classification	Include raw and pairwise baselines; currently no validated CCR-specific predictive gain.

36.9.7 Measured favorable results in separate tasks

A concise results inventory, retaining the original comparator for each number, is as follows:

Table 36.7: Positive CCR computational outcomes. Ratios are *not* interchangeable across rows.

Stage	Precisely tested task	Recorded result	Comparison caveat
APP-RN006	$N = J = 8$, five terminal source differences, 1,200 synthetic traces	about 5.2× faster	Full CCR DP, not strongest pruned compiler
APP-RN008	48 short source programs, $(N, J) = (8, 8)$	3.52× faster	Offline compilation excluded; advantage reverses at large N
APP-RN008	$N = 2048$, $J = 8$; chronological state only	919 KB → 6.2 KB	Generic ring-buffer, input storage excluded
APP-RN009	Selective q_4 monitor	typically 2–3× faster	Task-matched Python DP; q_5 margin smaller
APP-RN010	Degree-four homogeneous interaction at long horizons	3.7–5.1× versus named WFA implementation	Same $O(Nm)$ complexity, Python implementation dependent
APP-RN015	$J = 8, H = 6, M = 8$ exact response-profile counting	12.7–14.5× faster	HNF/SNF versus exhaustive counting; not online monitoring

Stage	Precisely tested task	Recorded result	Comparison caveat
APP-RN016	One checkpoint, 2,048 distinct futures, $J = 8/12$	2.58–3.58× faster	Full-stream/suffix baseline, compiler preparation included
APP-RN017	$T = 16384$, 2,048 futures, degree-pruned DP	1.28× at $J = 8$; 1.11× at $J = 12$	Stronger competitor; small margins not universal
APP-RN018	$T = 4096$, 100 checkpoints, 512 futures each	7.60× / 8.62× at $J = 8/12$	Python pruned DP; matched NumPy may be similar
APP-RN018	$T = 16384$, 512 checkpoints, $Q = 16$	3.20× / 4.40× less Python peak allocation at $J = 8/12$	$J = 12$ is slower in time; not OS RSS
APP-RN020	3,175 real length-eight activity prefixes	about 2.5–3.1× faster hot CCR feature kernel	Parsing and basis setup separate
APP-RN022	3,175 real prefixes, five outputs, repeated lookup	about 10–13× faster warm table vs compiled polynomial	164 KB value table, 5–7 ms build
APP-RN023	Fixed-length event-source payload	5,954 → 4,366 bytes	26.67% smaller payload; codebook excluded

36.9.8 The strength of the measured cases

The two largest non-table acceleration findings arise in *different exact algorithmic problems*. The finite-ring example accelerates a complete cardinality computation by replacing an optimized enumerator with an arithmetic image calculation. The forecasting example accelerates repeated exact terminal coefficients by compiling a sparse source response and batching it over histories and queries. Neither calculation changes the CCR mathematical output. Thus their gains provide independent evidence of two productive CCR-k interfaces: *symbolic response geometry* and *amortized exact evaluation*. The strongest headline cases are reproduced below with absolute times rather than speed factors alone.

Exact task	**Comparator**	**CCR method**	**Result**
100 checkpoints, $J = 12$, $Q = 512$	1614.63 ms	187.35 ms	8.62× speedup, complete stream/setup included
100 checkpoints, $J = 8$, $Q = 512$	869.00 ms	114.36 ms	7.60× speedup, same task
$M = 8$, $J = 8$, $H = 6$ profile image	563.8 ms	38.8 ms	14.5× speedup, separate counting task
$J = 12$, $Q = 2048$ future queries	59.397 ms	23.061 ms	2.58× including compiler and history

The first two comparisons use a task-pruned *Python* DP, and their headline factors shrink against the strongest equally vectorized NumPy alternatives. The third concerns an algebraic exact-counting task, not online prediction. These qualifications specify where a practitioner can reproduce the gain; they do not diminish its measured magnitude under the stated conditions.

36.9.9 Negative findings that control the deployment claim

The same experimental corpus contains meaningful counterexamples: a ragged high-jet monitor at q_8/q_{12} was about 3–4× *slower* than strong task-pruned DP; with one future the compiled approach was slower owing to its setup; a future-specific coefficient matrix recompiled for every input was slower at $Q = 256$; a low-query $J = 12$ memory-saving archive lost in elapsed time; the full source expansion became counterproductive when (N, J) grew; the one-shot full LUT lost after paying its build; and external classification experiments did not confirm CCR-specific predictive benefit. A serious engineering evaluation must keep those counterexamples, because they determine the method-selection rule rather than contradicting the exact mathematics.

36.10 Independent event logs: exact features versus predictive utility

36.10.1 Primary data identity and validity gates

The BPI Challenge 2013 Incidents log [262] was processed from a user-supplied `.xes.gz` file: 7,554 cases, 65,533 events, 3,175 cases with at least eight events, and SHA-256 `6ec2136c607648608655802486df97a866f351668751b7ecbca51e1f8715d522`. The record counts match the public dataset description; byte-for-byte equality against an official 4TU checksum was not independently established. The independent Helpdesk dataset [264] was supplied as a ZIP with published SHA-256 `21111bd002278a0b9a721afa12b43ff595eb5a525fbcd266bb61c48c4dc3d107`, which the uploaded ZIP matched. It contains 3,804 cases, 13,710 events and nine activity types. These are real process data, but they do not natively contain the CCR polynomial state.

The initial BPI2013 chronology protocol divided 3,175 eligible cases by start time into 1,905/635/635. Its strict retrospective-label availability purge left only ten training cases before the validation boundary. Thus the registered split *failed*, and there is no valid preregistered prediction test score from that design. Separately, a post-hoc time-of-eighth-event split with explicit label boundaries retained 448/435/351 cases and excluded 1,941 eligible cases. Its class proportions shifted substantially. This second split can inform an exploratory computation check, not a preregistered performance claim.

Table 36.8: Exploratory BPI2013 later-activity-repetition proxy: same held-out test cases.

Feature family	Test macro-F1	Interpretative status
Activity baseline	0.92536	Strong baseline
Baseline + CCR features	0.93122	Difference not significant
Baseline + raw + CCR	0.93122	Weaker than pairwise baseline
Baseline + raw + pairwise	**0.93616**	Best listed model
CCR only	0.42108	Poor independent performance

The paired 2,000-resample bootstrap estimated $\Delta\text{macroF1} = 0.005858$ for baseline-plus-CCR minus baseline with percentile interval $[-0.014242, 0.028426]$, containing zero. The task target—later repetition of an earlier activity—is an operational proxy, not an adjudicated incident failure label. In separate group-handoff experiments, the first frozen three-step task had only 29 test cases versus a required minimum 30 and produced no scores. An exploratory one-step target had 73 test cases but only *two positive* handoffs; a reported baseline score of 0.68936 was unchanged by adding CCR, and no reliable predictive gain followed.

36.10.2 The independent Helpdesk test

Helpdesk cases are typically short: 45 contain at least eight events, and only 26 contain nine or more. Thus a suitably powered eight-event CCR interaction test with a future target cannot be run on this log. A separate four-event protocol, frozen before target-label/model-score inspection after schema checks, predicted seven-day ticket closure. Its chronological eligibility/purging procedure yielded 372 training, 124 validation and 126 test cases (82 test positives and 44 negatives). The base model and base-plus-CCR model both achieved test macro-F1 0.49490 and made identical class predictions; CCR only scored 0.39423. The outcome is a seven-day closure *proxy*, not a certified contractual service-level label. At four events there is only one source equality bit, and

$$q_4(4) - q_4(4;0) \equiv 6a_{4,4} \pmod{101}.$$

The CCR feature is just a fixed multiple of one raw bit in this task. Direct checks covered 627 eligible cases.

Table 36.9: All seven fixed classifier variants on the independent Helpdesk seven-day closure proxy, evaluated on the same 126 chronological test cases.

Feature family	Test macro-F1
Baseline	0.49490
Baseline and raw source	0.49490
Baseline and CCR	0.49490
Baseline, raw source and CCR	0.49490
Baseline and pairwise features	0.47670
Baseline, pairwise features and CCR	0.47670
CCR only	0.39423

The external result is a genuine negative for this limited feature/learner configuration; it cannot validate or refute an eight-event interaction hypothesis for which there are too few long cases.

36.10.3 Measured real computational work and reproducibility boundaries

In the 3,175 eight-event BPI prefixes, complete CCR recurrence, degree-pruned DP, scalar source compiler, NumPy compiler and NumPy degree-pruned DP agreed exactly on all 15,875 terminal coefficients. XML parsing itself took around 1.57 s, whereas the compiled mathematical kernel was sub-millisecond. It would therefore be incorrect to translate a 3× kernel gain into a 3× file-processing gain. Similarly, a payload bit reduction cannot be equated with an equal reduction in whole-process RAM. Real deployment value should be tested with a fixed workload specification, truly matched competitors and end-to-end measurements on the target hardware.

36.11 Interpreting CCR computational strength

36.11.1 What follows mathematically

The main positive mathematical claim is the exact dependence of forecast-response information on chronological source structure. The sharp five-step result is a property of the defined CCR model under field-unit and nondegeneracy conditions. The typed recurrence also supports exact low-degree selective statistics, source interactions, finite-ring cardinalities, and event-equality output certificates. These results are useful because they specify *which information is sufficient, which information is impossible to omit, and which compressed representations are algebraically sound.*

36.11.2 What follows computationally

A working engineer can choose and implement a correct method from the reported regimes: pruned dynamic programming for few forecasts, compiled response polynomials for repeated counterfactuals, batched evaluation for many checkpoints, compact archive storage when query volume is small, or a prepared lookup for repeated small finite-input tasks. The largest reported speedups are relative to explicitly named Python implementations. Their portability requires a shared time budget for setup, an end-to-end benchmark and a similarly optimized rival. No theorem here states that CCR is faster than every general weighted-DAG or semiring implementation.

36.11.3 What requires further validation before an application claim

Neither BPI2013 nor Helpdesk establishes a naturally occurring physical CCR process, new statistical information in CCR features, robust downstream predictive gain or a general production-RSS reduction. In a designed CCR engine, exact output equivalence and task-specific speed are genuine technical gains; in an independent domain, additional value must be demonstrated separately. This distinction is not a deficiency of the mathematical archive theorem: it is the boundary between a model theorem, an implemented algorithm and an external application.

36.12 From mathematical theorem to reproducible software system

36.12.1 A precision-preserving implementation contract

The inputs are (i) an explicit modulus or field, (ii) the order J, (iii) a historical chronological source stream, (iv) a checkpoint and future horizon, and (v) later supplied future-source programs. The outputs are the terminal jets and, when requested, a persistent checkpoint summary. The typed recurrence is authoritative: an optimized implementation must agree with it exactly, coefficient by coefficient. For composite rings, the modular arithmetic must respect zero divisors; for large residue representatives in machine integer arrays, the implementation must constrain multiplication groups before reduction to avoid overflow. This input/output contract is sufficient to reproduce the *mathematical* outputs with any suitable exact modular backend.

The developed implementations are not black-box estimators trained on data. They are exact arithmetic procedures for explicitly supplied CCR source inputs. If a user maps an external event sequence to source flags, the mapping itself is part of the application design, and its operational meaning must be validated separately. A software deployment should therefore publish both a mathematical correctness test suite and an application-domain validation protocol, with the former being a prerequisite for the latter.

36.12.2 Portable benchmark protocol for an independent implementation

To reproduce the resource cases, an implementer should use the specified parameter grids, execute independent exact recurrence checks, and report full initialization and evaluation times alongside matched dynamic-programming baselines. In particular, a high-query test should process one $T = 4096$ history, form $C = 100$ checkpoints, supply $Q = 512$ distinct futures at each checkpoint, set $H = 5$, $p = 3$ and $M = 1{,}000{,}003$, and compare the compiled pipeline to both task-pruned suffix CCR and a matched vectorized per-checkpoint evaluator. A low-query memory test should use the recorded $T = 16384$, $C = 512$, $Q = 16$ configuration, report Python allocation *and* operating-system RSS separately, and disclose whether shared input tables are counted. A finite-profile test uses the specified $M = 8$, $J =$

$8, H = 6$ image-count task and must compare exactly the same set of reachable response functions.

Independent implementations should run different histories, future programs, query densities and resource constraints without changing the definition of the mathematical target. Relative speed can move with vectorization and caching; exact CCR coefficients cannot. For the event-source code, evaluation must additionally count both the encoded payload and the encoder/decoder model tables. A bit-optimal representation does not become a low-total-RAM codec unless that full cost is paid.

36.12.3 An application-oriented significance statement

The strongest use case for this theory is a system that must repeatedly evaluate *many counterfactual future programs on a shared historical CCR state*. In such an environment the exact archive theorem identifies what must be preserved, sparse response compilation identifies what is inexpensive to reuse, and the benchmark grid identifies when the cost of preparing that reuse is amortized. An application that instead needs one answer or operates on short unrelated traces should select the direct pruned recurrence or a raw-feature baseline as warranted. Both choices arise from the same self-contained CCR theory. The value is an *implementable, workload-specific resource calculus* backed by exactness certificates, not dependence on a universally faster primitive.

36.13 Conclusion

The chronological source geometry of typed CCR-k supports a substantial family of *exact mathematical and computational advances*. We have established a sharp five-step forecast archive law $m(T, J, 5) = 5J - 31$ with a constructive online realization and a lower bound against arbitrary fixed-length deterministic encoders; a sparse support theorem for future-source responses; exact source-sensitive monitors and diagnoses; and complete finite-ring answer-image calculations in specified contracts. The results identify *which historical information is essential* and turn that information into working exact algorithms.

Their computational impact is measurable. Separate validated CCR computations realize 8.62× improvement in a defined multi-checkpoint forecast workload, 12.7–14.5× in finite-profile image counting, and up to 4.40× lower peak tracked allocation in a low-query checkpoint workload. The same programme supplies $\Theta(N \log N)$ optimal coding of event-equality masks, demonstrated by a 26.67% smaller real-log source payload, and a high-throughput precomputed query engine when finite-mask inputs recur. The results are accompanied by direct-recursion equality checks, finite exact certificates and stronger comparator experiments that show how to select the right method.

These findings establish that CCR-k is a *mathematically structured and computationally exploitable framework* for exact chronological source processing. They do not require a claim of superiority for every modulus, hardware backend, query density, or statistical prediction target. Instead, they give practitioners sufficient mathematics, implementations and measured workloads to obtain the demonstrated advantages where the stated resource contracts apply, while retaining fully reproducible failure cases where direct computation or matched vectorization is preferable.

36.14 Further derivations and mathematical safeguards

36.14.1 The coefficient-wise recurrence

Writing $q_d(n) = [u^d]P_n$ with $q_d(n) = 0$ when the degree is negative or $n < 0$, the typed recurrence gives for each $0 \le d \le J$:

$$\begin{aligned} q_d(n) = q_d(n-1) + \ell_n q_{d-1}(n-1) + p(n-1)\ell_{n-1}q_{d-2}(n-2) \\ - p(n-1)(n-2)q_{d-3}(n-3) + \sum_{r=4}^{\min(J,n,d)} w_{r,n}a_{r,n}q_{d-r}(n-r). \end{aligned}$$

This equation is a complete independent implementation specification. By setting all optional sources zero one obtains the source-free coefficients, which can be prepared once and shared by task-pruned evaluators. All relevant sums are computed in the chosen ring; field division or use of floating point in exact modular recurrences is not permitted unless separately justified.

36.14.2 Mandatory-response determinant and its hypotheses

The sharp five-step theorem uses the precise polynomial

$$\begin{aligned} D_5(T) = 192T(T+1)\big(64T^6 + 1312T^5 + 11120T^4 + 49565T^3 \\ + 121776T^2 + 155655T + 80706\big). \end{aligned}$$

It is the determinant of the five mandatory-transfer responses restricted to the historical coordinates $q_J(T)$, $q_{J-1}(T)$, $q_{J-2}(T)$, $q_{J-3}(T)$ and $q_{J-3}(T-1)$, after quotienting by the single-pulse read coordinates. Two independent symbolic determinant derivations returned the same exact factorization. Their function is a reproducibility check, not a substitute for the displayed algebraic proof or a license to omit the invertibility hypothesis $\ell \nmid D_5(T)$. For the prime 1,000,003 and checkpoints $T = 1024, 4096, 16384$, the recorded nonzero determinant residues are 641603, 206172 and 130774, respectively.

36.14.3 Composite-ring profiles and finite-field rank are different

For $\mathbb{Z}/M\mathbb{Z}$ with composite M, a source coefficient can be a zero divisor. Distinct old source programs need not yield distinct high jets, and accessible profiles need not fill a free module. Let Φ map the *accessible* old histories to their complete answer functions on future source programs. Then an arbitrary fixed-length M-ary code needs exactly $\lceil \log_M |\Phi(S)| \rceil$ symbols. When an integral generator matrix of the accessible response group has Smith diagonal d_i, its image size can be computed from the products $\prod_i M/\gcd(M, d_i)$ only after identifying the correct accessible source module. For $M = 8$, $J = 8$, $H = 4$, a complete enumeration gave three response profiles, so one mod-eight symbol suffices. For $M = 12$ in the same J, H contract, the recorded answer profile was constant across accessible old programs and required zero mod-twelve symbols. These are both consistent with the profile-cardinality formulation and inconsistent with indiscriminate use of a field-rank formula.

36.14.4 Formal and empirical certification are different kinds of proof

A symbolic determinant identity, a witness-family argument and a full finite enumeration can certify exact mathematical statements in their stated ranges. Benchmark timing can never prove a complexity lower bound. Conversely, a complexity result does not guarantee a speed factor in a particular Python/NumPy build. The audit classifies the reported evidence as (i) algebraic proof, (ii) computer-assisted finite proof, (iii) exact recurrence implementation equivalence, (iv) measured resource comparison and (v) real-log held-out evaluation. These categories are not interchangeable.

36.15 Detailed resource measurements and failure cases

36.15.1 Cold forecasting at low and high query volumes

Table 36.10: End-to-end synthetic CCR forecasts, $H = 5$, $M = 101$, with preparation and the full single historical stream charged. Both the exact pruned suffix DP and the minimum-message compiler compute all requested future coefficients.

J	Different futures Q	Pruned DP (ms)	Compiled archive (ms)	DP/compiled
8	1	0.0478	0.5547	0.09
8	16	0.2847	0.6948	0.41
8	128	2.1666	1.3263	1.63
8	2048	34.9560	12.0545	2.90
12	1	0.1269	2.5992	0.05
12	16	0.5199	2.6163	0.20
12	128	4.0370	3.6888	1.09
12	2048	59.3973	23.0613	2.58

The separate compiled-but-uncompressed historical archive has essentially the same fast query kernel; the smaller *persistent* message is therefore an information improvement, not the sole cause of acceleration. The table shows the preparation crossover and prevents the high-query gain from being misapplied to one forecast.

36.15.2 A chronological performance ledger

Table 36.11: Confirmed CCR computational gains and adverse cases, with each task and its comparator stated.

Task	Defined task and observed values	Required qualification
Small-source compiler	Compiled $N = 8$ source signature: about 5.2× over full CCR DP	1,200 synthetic traces; setup/better pruned comparator absent
Source expansion	Explicit expansion 3.52× at $N = 8$, but scaling reverses by $N = 16$	The polynomial *output* may be exponential; numeric CCR DP remains polynomial
Source expansion	Ring state 919 KB → 6.2 KB for $N = 2048, k = 8$	Chronological state allocation only; generic buffering
Selective monitors	Single q_4 monitor 2–3×, $q_4 + q_5$ 1.08–1.63×	Same target low-degree DP, Python-dependent
H_3 monitor	Homogeneous H_3 37.90 ms versus interval DP 46.76 ms and named WFA 138.47 ms at $N = 32768$	Source precomputation excluded; other optimized weighted algorithms can match complexity
High-jet monitor	Full q_8/q_{12} ragged monitor 3–4× slower	Negative result; no high-jet general monitor superiority
Finite-ring counts	$M = 8, J = 8, H = 6$ image count 38.8–40.9 ms versus 519.8–563.8 ms	Entirely different task from online forecasting
Forecast compiler	Q=1: compiler 0.555 versus 0.048 ms ($J = 8$), 2.599 versus 0.127 ms ($J = 12$)	Setup dominates, compiler loses decisively

Task	Defined task and observed values	Required qualification
Forecast compiler	Q=2048: compiled 12.05 versus 34.96 ms ($J = 8$), 23.06 versus 59.40 ms ($J = 12$)	History/compile/query inclusive; synthetic sources
Long streams	$T = 16384, Q = 2048$: 102.73 vs 131.98 ms ($J = 8$), 168.89 vs 187.26 ($J = 12$)	Stronger degree-pruned comparator; smaller gains
Long streams	One archived minimal message: 58 vs 367 B ($J = 8$), 154 vs 777 B ($J = 12$)	Serialized checkpoint; compiler working RAM higher
Many checkpoints	100 checkpoints ×512 futures: $J = 8$, 114.36 versus 869.00 ms	Batch versus scalar pruned suffix DP; not same NumPy backend
Many checkpoints	Same at $J = 12$, 187.35 versus 1614.63 ms	8.62-fold work-specific gain; high-query allocation may be worse
Many checkpoints	512 checkpoints ×16 futures: $J = 12$, 802,960 versus 3,534,688 bytes	Python tracked allocation, not RSS; time is 477.87 vs 399.60 ms (slower)
Real BPI prefixes	Real BPI eight-event prefixes: hot 2.5–3.1-fold compiled feature advantage	Original XES parse 1.57 seconds, no large end-to-end gain
Group-handoff test	Group handoff: only two positive test cases in exploratory short horizon	No statistically strong advantage claim
Warm lookup	Warm LUT 0.036 ms versus compiler 0.369–0.493 ms	163,840-byte value table, 5–7 ms build
Lossless masks	Packed source mask 5,954 to 4,366 bytes on BPI data	26.67% payload saving, codebook excluded
Helpdesk study	Helpdesk closure-proxy base and base+CCR macro-F1 0.49490	No added prediction performance, N8 insufficient cases

36.15.3 A fully charged algorithm selection criterion

For a fixed model/workload instance let B denote initial basis preparation, L the finite table build, $E_D(Q)$ the cumulative pruned DP suffix cost and $E_C(Q)$ the compiled evaluator cost. Compiled evaluation is favorable only if $B + E_C(Q) < E_D(Q)$ (after common history charges are accounted for); a table is favorable only if $L + E_L(Q) < B + E_C(Q)$. When the history itself dominates, the proportional gain shrinks even if future-query kernels improve. A method that reduces archive bytes but increases active response-polynomial size can violate a strict peak-RAM constraint. This explains why a single universal method cannot be extracted from the recorded results.

36.16 Coverage of mathematical structures, algorithms, and evidence

The following index identifies the distinct substantive conclusions represented within this integrated standalone article. Every item is supported by the definitions, proof statements, computational comparisons or empirical validity checks in the relevant preceding section; these are mathematical topics, not identifiers of private research files.

Table 36.12: Coverage of the distinct source, information, computation and evaluation questions in the paper.

Scientific question	Established finding or documented boundary
Exact terminal reconstruction	The typed recurrence and exact coefficient computations determine the selected terminal objects.
Two-source modular diagnosis	Two terminal coordinates recover a three-valued parity load for the specified $\mathbb{Z}/8\mathbb{Z}$ source model.
Unknown-source robustness	The diagnosis remains valid under the other optional sources, but it does not improve equal-bit raw parity recording or error correction.
Independent-event model	A simple independent-event factorization cannot reproduce the typed CCR recurrence.
Statistical information limit	A deterministic CCR feature cannot increase the information in the full raw event trace.
Finite signature compilation	Exact polynomial compilation accelerates particular bounded source families with a setup and scaling tradeoff.
Independent empirical task validity	Chronological and label-availability restrictions govern which held-out prediction claims can be made.
Stream complexity	Explicit source expansion can grow exponentially while direct truncated CCR remains polynomial-time.
Low-degree selective computation	Exact degree-four and degree-five registers give task-dependent acceleration.
Pure source interaction	Finite-delay nonoverlapping-block recurrences compute homogeneous interaction channels.
Passive observation	A terminal scalar can conceal a nonzero source interaction and a restricted high-jet monitor may be slower.
Direct coordinate necessity	One-step future responses have exact coordinate-read needs under the stated access contract.
Joint response-profile rank	The observable row span yields a sharp finite-field message information bound, including nonlinear encoders.
Short future horizons	The first four future horizons admit exact source-access formulas, whereas composite moduli can collapse profiles.
Five-step information threshold	The sharp $5J - 31$ rank, explicit determinant and finite-ring profile counting reveal first nontrivial future overlap.
Causal archive computation	A chronological linear sketch attains the minimum retained message and supports compiled future evaluation.
Long checkpoint histories	The same sharp five-step archive law holds under explicit unit and determinant conditions even for long historical streams.
Sparse batch forecasting	The $6J - 21$ support, higher-degree compiled evaluation, memory Pareto and method selector are tested quantitatively.
Label timing and leakage	A registered temporal dataset protocol can become infeasible after legitimate label availability filtering.
Real incident-log computation	Thousands of real prefixes validate exact coefficients and hot-kernel improvements without implying a full-parser speedup.
Real group handoff	Positives are too sparse for a reliable claimed predictive improvement.

Scientific question	Established finding or documented boundary
Weighted-path and lookup checks	Equivalent weighted-path outputs, finite-coordinate separability and repeated lookups admit exact validation.
Equality source coding	Connected graph partitions imply sharp bit costs, full finite-mask tests and real lossless payload reduction.
Independent helpdesk log	Official input integrity and all fixed classifier variants support a limited negative external prediction result.

36.17 Worked certificates and executable design templates

36.17.1 A concrete one-step coordinate menu

Take $J = 8$ at checkpoint $T = J$. For all subsequently disclosed single-step future sources, a sufficient fixed read-coordinate menu is

$$\mathcal{S}_8 = \{q_8(8), q_7(8), q_6(7), q_5(6), q_4(5)\}.$$

Under the stated freely varying historical-source field hypothesis and unit conditions, each of these five cells is necessary in the *preselected coordinate-read* model. If all new optional sources are zero, the last cell $q_4(5)$ becomes unnecessary, leaving four. For example, when the new source $a_{4,9} = 1$, the old $q_4(5)$ enters $q_8(9)$ with coefficient $w_{4,9} = 8 \cdot 7 \cdot 6 = 336$ (33 mod 101). This makes the additional read requirement transparent and illustrates why zero future and arbitrary future must not be conflated.

36.17.2 The first interaction at future horizon five

With $J = 8$, the structural five-step response polynomial supports exactly

$$6J - 21 = 27$$

future-source monomials: one constant, 25 individual source amplitudes and one possible quadratic pair. For $J = 12$, the same count is 51. The first quadratic contribution requires a source of rank r ending at $T+1$ and a rank-four source ending at $T+5$. The old historical jet read by their product is also read by a single allowable longer future pulse. This reuse, rather than the mere existence of quadratic terms, is what makes the rank $5J - 31$ and not the naive extrapolation $5J - 30$ from the first four horizons.

36.17.3 An exact interaction-channel test case

For the homogeneous channel of Eq. (36.3), let the only rank-four inputs be $a_{4,4} = a_{4,8} = a_{4,12} = 1$, with $p = 3$. The three disjoint chronological intervals then contribute exactly

$$H_3(12) = w_{4,4}w_{4,8}w_{4,12} = 6 \cdot 210 \cdot 990 = 1{,}247{,}400.$$

Reducing modulo 1,000,003 gives 247,397. Additional higher-rank sources cannot change the formal coefficient $[u^{12}v^3]$ because that coefficient already saturates the minimum degree of three rank-four blocks. The contrast with a terminal q_{12} measurement is intentional: the channel requires the source-gating definition.

36.17.4 An explicit finite-ring non-identifiability profile

For $p = 3$, $J = 8$, $H = 4$, $M = 8$, only three historical source locations survive after modular annihilation. Exhaustive evaluation of the $8^3 = 512$ effective old programs against eight effective future choices produces exactly three response functions, indexed by the parity load W from Eq. (36.2). They have old-source multiplicities $128, 256, 128$, so a

fixed-size binary message needs two bits and a mod-eight message needs one symbol. At $M = 12$ for the same J, H, a different complete enumeration yields a single future-response function (constant over the accessible old histories). The two counts demonstrate why replacing a finite-ring image by a generic field rank can be wrong even when the formal recurrence is identical.

36.17.5 Source-mask counts and terminal coordinate tests

A complete table for the event-equality graph specialization is given here both to document the exact count and to make the compression law independently checkable:

N	Raw bits D_N	Valid masks C_N	Minimum packed bits
4	1	2	1
5	3	8	3
6	6	48	6
7	10	260	9
8	15	1,419	11
9	21	7,953	13
10	28	46,610	16
11	36	287,134	19
12	45	1,860,478	21
13	55	12,658,618	24
14	66	90,241,142	27
15	78	672,467,210	30
16	91	5,226,655,287	33
17	105	42,283,431,279	36

For $N = 8$ and modulus 101, three terminal coordinates (q_4, q_7, q_8) distinguish all 1,419 valid masks and no pair of the five high jets does so. These three *field symbols* require 21 conventional fixed bits; they are not the eleven-bit minimum source-mask code. Under unrestricted independent 15-bit source assignments, one must likewise not substitute the reachable-mask counts above for the full binary cube of size 32,768.

36.17.6 Reference pseudocode for a fixed-query compiler

Step	Contract-preserving action
1	Parse or accept the historical source program and initialize all low jets in the chosen exact ring.
2	Process chronological time slices once, retaining a rolling full CCR state while updating the sufficient archive linear functionals.
3	At the checkpoint, store the m historical message elements; if repeated future queries are expected, build the sparse response polynomial using the exact $K(J, 5)$ structural support.
4	For each future program, evaluate the response polynomial using modular arithmetic and return the five requested q_J values.
5	If Q is small, bypass compilation and use a degree-pruned suffix recurrence; if the input mask domain is small and repeated, compare a prebuilt exact lookup including its build cost.
6	Assert equality against direct typed CCR for test histories and futures; log both elapsed time and live allocation with the proper task labels.

This is deliberately *not* a new machine-independent complexity theorem. It is a reproducible selection recipe based on the exact mathematical identities established in the preceding sections and the measured regimes of the source programme.

36.18 Reproducibility, exact certificates, and empirical scope

36.18.1 A self-contained mathematical record

The exact mathematical results in this article are determined by the typed CCR-k recurrence, its chronological source restrictions, and the field or ring parameters stated with each result. In particular, the five-step mandatory-transfer determinant, response ranks, finite-ring profile counts, connected-partition counts, and terminal-coordinate distinctions concern fully specified finite or symbolic objects. They require no external event logs, proprietary inputs, or separately distributed certificate files. The role of an *exact certificate* here is a compact identity, rank, or finite-domain cardinality that makes a mathematical assertion independently checkable from the definitions. Such checks support, but do not replace, the proofs of universally quantified statements.

The distinction is particularly important for the sharp archive theorem. Under the field-unit, historical-accessibility, and nonvanishing conditions of Theorem 36.4, its minimum retained five-step response information is $m(T, J, 5) = 5J - 31$ field symbols. The determinant below certifies a crucial mandatory-transfer block; the theorem also uses the historical-accessibility and lower-bound arguments given in the forecast-information section. The finite rank examples are illustrations of the theorem's regime, not a proof by sampling.

36.18.2 Algebraic and finite-ring certificate summary

For $p = 3$, the five historical directions $q_J(T)$, $q_{J-1}(T)$, $q_{J-2}(T)$, $q_{J-3}(T)$, and $q_{J-3}(T-1)$ define the five-step mandatory-transfer block after the one-pulse read coordinates are removed. Its exact determinant is

$$\begin{aligned} D_5(T) = 192T(T+1)\big(&64T^6 + 1312T^5 + 11120T^4 + 49565T^3 \\ &+ 121776T^2 + 155655T + 80706\big). \end{aligned} \tag{36.5}$$

For the prime 1,000,003, the residues of this polynomial at $T = 1024, 4096, 16384$ are $641603, 206172, 130774$, respectively. They are nonzero for these checkpoints, and therefore certify the required invertibility in those particular instances, not for arbitrary moduli or checkpoints.

Table 36.13 collects other exact, task-specific values determined by the source interfaces and parameter choices of the paper. The distinction between finite-field ranks and composite-ring image sizes is deliberate: a rank computed over a field must not be substituted for an image cardinality over a ring with zero divisors.

Table 36.13: Finite mathematical certificates and representative exact values.

Object	Parameters	Exact result
Five-step response rank	$\mathbb{F}_{101}$; $J = T$; $J = 8, 10, 12, 16$	Ranks $9, 19, 29, 49$, respectively, agreeing with $5J - 31$ in these cases.
Composite-ring images	$T = J = 8$, $H = 4, 5, 6$	3 profiles modulo 8, 1 modulo 12, and 10 modulo 16, at each listed horizon. The modulo-16 finite check covers 65,536 effective histories.
Modulo-25 images	$T = J = 8$, $H = 4, 5, 6$	$5^7 = 78{,}125$, $5^9 = 1{,}953{,}125$, and $5^{10} = 9{,}765{,}625$ profiles, respectively.
Source-mask geometry	$4 \le N \le 17$	Connected-partition counts C_N and optimal $\lceil \log_2 C_N \rceil$ payloads are given in Section 36.17; e.g., $C_8 = 1{,}419$, requiring 11 instead of 15 raw bits.
Terminal distinction	$N = 8, 9, 10, 11$; moduli $31, 101$	Eight finite minimal terminal-coordinate tests, with witness sets specified in the event-equality section and Section 36.17.

The modulus-25 image sizes also admit a structural explanation rather than requiring enumeration of every historical program. In the indicated setting, the relevant quadratic

historical product is $a_{4,4}a_{4,8}$; its response column equals four times the $a_{8,8}$ column modulo 25. Consequently, the triangular change $a_{8,8} \mapsto a_{8,8} + 4a_{4,4}a_{4,8}$ preserves the attainable response set, and the image cardinality reduces to an exact modular linear-image calculation. The connected-partition characterization similarly determines the lossless coding counts without dependence on a particular observed event log.

36.18.3 Computational verification and measured applications

The computational study evaluates the typed CCR-k recurrence on specified source histories, moduli, forecast horizons, and query workloads. The optimized evaluators were checked against direct recurrence outputs on matching inputs, confirming exact coefficient equality alongside the measured computational costs. Selective monitoring, future-response compilation, checkpoint reuse, finite-ring profile calculation, and small-domain lookup were tested in the regimes described in the main text.

Several tasks yielded substantial gains. Multi-checkpoint five-step forecasting reached an 8.62× speedup over the specified degree-pruned Python baseline; a separate exact finite-profile counting method improved by 12.7–14.5×. Selected low-query experiments reduced peak tracked Python allocations by factors of 3.20–4.40, while repeated warm finite-mask lookups were 10–13× faster than the stated compiled-polynomial comparison. These results concern distinct workloads, and their measured gains are not multiplied together. Query counts, output-equality checks, and relevant preparation costs are given with the experiments in the main text.

The finite calculations furnish direct numerical checks on the theory. At $T = J = 8$, the modulo-8, modulo-12, and modulo-16 response families have exactly 3, 1, and 10 profiles for the listed horizons; the modulo-16 check covers 65,536 effective histories. The modulo-25 horizons give 78,125, 1,953,125, and 9,765,625 profiles, matching the structural image calculations. Connected-partition counts and minimal terminal-coordinate tests supply further finite checks computed directly from the recurrence and source geometry.

The event-log experiments test an additional application. The BPI Challenge 2013 Incidents record contains 7,554 cases and 65,533 events, with 3,175 eligible eight-event prefixes. On these prefixes, the exact lossless source-mask code reduced payload by 26.67%. The independent Helpdesk record comprises 3,804 cases and 13,710 events. The study includes round-trip source reconstruction and the application-level tests reported in the main text. Both public datasets are identified in [262, 264]; the mathematical coding and counting results themselves do not require external event records.

Together, the evaluations show how the CCR-k source chronology leads to concrete algorithm choices: direct computation for isolated queries, compilation and checkpoints for repeated futures, and lookup for frequently reused small input domains. Exact algebraic identities, verified finite counts, equality-checked computations, and measured implementation gains provide complementary evidence for the computational contributions. The recurrence, proofs, algorithms, and parameters reported here also supply the information needed to reconstruct the mathematical computations.

Chapter 37

Exact Future-Archive Complexity and Exceptional-Characteristic Recovery in Typed Chronological Carry Refinement

Research branch — CCR-k: all-horizon information and exceptional characteristics. *This chapter continues the preceding five-step forecast-archive result with an exact all-horizon rank formula for free-source typed CCR over admissible prime fields. It distinguishes direct-read historical coordinates from a five-column mandatory transfer, classifies characteristic-dependent rank losses, and proves finite-horizon recovery and saturation. It does not claim machine-wide memory optimality, unrestricted event-derived sources, composite-modulus extensions, or a fixed-field infinite-horizon theorem.*

Integration and source boundary. This chapter contains the mathematical results of the independent Article 66 [305]. The earlier five-step minimum and its computational experiments belong to Chapter 36; the new claims concern the exact free-source finite-field response-profile archive at arbitrary admissible horizon, including exceptional characteristics. Its results are separate from the general weighted-automata and LPV realization principles recalled below.

Principal contributions. Theorem 37.3 gives an exact minimal checkpoint archive and an equivalence test for historical responses; Theorem 37.5 classifies all admissible horizon-dependent ranks; and Corollary 37.6 identifies the seventh-horizon recovery and eventual archive saturation. The hypotheses $T \geq J \geq 8$, $\ell > T + H$ prime, and freely selectable historical sources are part of these statements.

37.1 Introduction

A future-query interface must preserve exactly the aspects of a past that can affect subsequent outputs when later source inputs become known. For typed chronological carry refinement (CCR), the relevant past is a source-indexed polynomial recurrence with delayed influences. A previous study [261] established the exact five-step minimum $m(T, J, 5) = 5J - 31$ under a nonvanishing-determinant hypothesis, along with a constructive archive and several distinct computational applications. The present work asks the different, general question: *what is the exact minimum archive for any prescribed horizon, including exceptional field characteristics?*

37.1.1 Contributions

The results are as follows.

1. A self-contained decomposition of all future response profiles into individually readable historic jet coordinates and a mandatory-only transfer quotient of dimension at most five (Theorem 37.3). It yields an exact equivalence test for complete histories, and a lower bound against nonlinear fixed-length encoders.
2. A complete characteristic-dependent rank classification for every horizon (Theorem 37.5), including two exceptional pairs at horizon four, a degree-six criterion at horizon five, and three exceptional pairs at horizon six.
3. A seventh-horizon recovery result and an explicit eventual saturation law for the archive size (Corollary 37.6). The term *saturation* is relative to each admissible horizon/field pair; one prime cannot satisfy $\ell > T + H$ for arbitrarily large H.
4. Exact determinant and resultant certificates for the exceptional-characteristic classifications, together with a paper-contained exact-arithmetic verification procedure. No external code, data file, or supplementary document is needed to reconstruct the finite checks.

37.1.2 Relationship to realization theory and earlier CCR work

Field-valued Hankel-rank and minimal weighted-automaton principles are classical; see [265]. Minimality and finite Hankel rank for affine LPV systems have a well-developed theory [306]. Sontag's polynomial input/output realization results [307] provide a related algebraic perspective over infinite fields. None of those general rank principles is claimed as new here. We instead evaluate the *specific time-dependent delayed CCR source-response map* exactly and classify its finite-characteristic rank defects. This paper concerns minimum checkpoint *message symbols*, not smallest internal implementation state, transient working RAM, or end-to-end execution time. In particular, the historical sources here are algebraically free, not constrained event-equality masks. The earlier five-step theorem and the CCR recurrence originate in [261, 166] and are used with attribution rather than presented as new discoveries.

37.2 The recurrence and archive contract

Fix integers

$$J \geq 8, \qquad T \geq J, \qquad H \geq 1, \qquad \ell > T + H \quad \text{with } \ell \text{ prime},$$

and work over $K = \mathbb{F}_\ell$ and $K[u]/(u^{J+1})$. Write $P_n = 0$ for $n < 0$, $P_0 = 1$, and for $n \geq 1$ impose the typed $p = 3$ recurrence

$$\begin{aligned} P_n = {} & \big(1 + (4n-3)u\big)P_{n-1} + 3(n-1)(4n-7)u^2 P_{n-2} - 3(n-1)(n-2)u^3 P_{n-3} \\ & + \sum_{r=4}^{\min(J,n)} \mathrm{w}_{r,n} a_{r,n} u^r P_{n-r} \pmod{u^{J+1}}, \\ & \mathrm{w}_{r,n} = \frac{(n-1)!}{(n-r)!} = \prod_{i=1}^{r-1}(n-i). \end{aligned} \tag{37.1}$$

Set $q_d(n) = [u^d]P_n$. For times $n \leq T$ the optional inputs $a_{r,n}$ are independently selectable in K; a history is their complete assignment. At checkpoint T, an encoder that has observed the historical source stream and its derived jets writes *exactly b symbols of K*. Only after this checkpoint is a freely chosen future source program $(a_{r,T+s})_{1 \leq s \leq H}$ revealed. A deterministic decoder, given the archive and this program, must return

$$\big(q_J(T+1), \ldots, q_J(T+H)\big)$$

for every history and every future program. Encoder and decoder may otherwise be nonlinear; their transient internal computation is not charged. Let $m(T, J, H; \ell)$ be the smallest possible integer b.

The first four low-jet coefficients $q_0(n), \dots, q_3(n)$ depend only on the mandatory part of (37.1); hence they are fixed, known values, independent of the optional sources.

37.3 Historical accessibility and direct-read information

Write

$$\mathcal{J} = \big\{(t, d) : 1 \le t \le T,\ 4 \le d \le \min(J, t)\big\}.$$

For each $(t, d) \in \mathcal{J}$ the value $q_d(t)$ is a historical high-jet coordinate.

Lemma 37.1 (Triangular accessibility)**.** *The map from freely chosen historical source amplitudes $(a_{d,t})_{(t,d)\in\mathcal{J}}$ to the tuple $(q_d(t))_{(t,d)\in\mathcal{J}}$ is a bijection $K^{\mathcal{J}} \to K^{\mathcal{J}}$.*

Proof. Proceed first by increasing time t, and at fixed t by increasing degree d. In $q_d(t)$ the *current* source $a_{d,t}$ occurs only in the summand with $r = d$, with multiplier $\mathrm{w}_{d,t} q_0(t - d) = \mathrm{w}_{d,t}$. Every other current-time source has rank smaller than d; all mandatory terms and earlier-time source contributions have already been determined. Thus

$$q_d(t) = c_{t,d} + \mathrm{w}_{d,t} a_{d,t},$$

where $c_{t,d}$ is determined by previously fixed source amplitudes. Every factor in $\mathrm{w}_{d,t}$ is a positive integer at most $T - 1$, hence is nonzero in K because $\ell > T + H$. Choosing $a_{d,t} = \mathrm{w}_{d,t}^{-1}(q_d(t) - c_{t,d})$ realizes any target high-jet tuple uniquely. □

For $1 \le s \le H$ define the *direct-read* historical coordinates

$$X_{d,s} := q_d(T - J + s + d), \qquad 4 \le d \le \min(J - 4, J - s). \tag{37.2}$$

They are historical because the defining restriction gives $T - J + s + d \le T$, and they are mutually distinct since the pair (d, s) is recoverable from the time and degree. Define

$$x(J, H) := \sum_{s=1}^{H} \max\big(0, \min(J - 4, J - s) - 3\big). \tag{37.3}$$

Terms for $s > J - 4$ vanish.

Lemma 37.2 (Complete future-dependent coordinate span)**.** *For the family of outputs $q_J(T + h)$, $1 \le h \le H$, all history-dependent coefficients of nonconstant monomials in the future source amplitudes belong to the span of the coordinate functionals $X_{d,s}$. Conversely, every $X_{d,s}$ can be recovered from the response profile of a suitable one-pulse future program.*

Proof. Use the usual weighted-path expansion of (37.1). Its mandatory edges have (time length, polynomial degree) $(1, 0)$, $(1, 1)$, $(2, 2)$, $(3, 3)$, and an optional source edge of rank r has (r, r). Thus each edge has time length at least its degree. On any path contributing to a future output and containing an optional future source edge, cut at the *first edge that crosses time T*. Write its old endpoint as (t, d), with $t \le T$. Because the path uses a future source edge of degree at least four, $d \le J - 4$. The suffix from (t, d) to $(T + h, J)$ obeys $(T + h) - t \ge J - d$, giving

$$s := t - (T - J) - d \le h.$$

It remains to see $s \ge 1$. If the crossing edge itself is optional, with rank r and future endpoint $T + v$, the residual degree after that edge is $\delta = J - d - r \ge 0$; the predecessor has $t = T + v - r$ and hence $s = v + \delta \ge 1$. If the crossing edge is mandatory, then its predecessor has $t \ge T - 2$, while $J - d \ge 4$, so $s \ge 2$. This puts the historical high-jet

predecessor in (37.2); contributions from degrees below four are known constants. Summing paths proves the inclusion.

For the reverse inclusion, fix one index (d, s) in (37.2) and turn on only the optional future source $(r, n) = (J - d, T + s)$. In the output at the same time $T + s$ the response minus the all-zero-source response is precisely

$$\mathrm{w}_{J-d,T+s}\, q_d(T + s - (J - d)) = \mathrm{w}_{J-d,T+s} X_{d,s}.$$

This multiplier is a unit in K. Hence every $X_{d,s}$ is separately recoverable. Finally, outputs are multilinear in each future optional source variable along any chronological path. Evaluations on binary source assignments in $\{0, 1\}$ recover all formal monomial coefficients by inclusion–exclusion, which is valid in every field. This identifies the full span of history-dependent nonconstant response coefficients. □

37.4 The five-dimensional mandatory boundary

Let Z_h denote $q_J(T+h)$ evaluated with *all* future optional amplitudes zero. A first-crossing argument without future optional pulses shows that, modulo $\mathrm{span}(X)$, these mandatory-only outputs depend on just the following seven end-history cells:

$$\begin{gathered} q_J(T),\ q_{J-1}(T),\ q_{J-2}(T),\ q_{J-3}(T), \\ q_{J-2}(T-1),\ q_{J-3}(T-1),\ q_{J-3}(T-2). \end{gathered}$$

The cells $q_J(T)$, $q_{J-2}(T-1)$, $q_{J-3}(T-2)$ have mandatory output coefficients independent of h, respectively 1, $3T(4T-3)$, and $-3T(T-1)$: after crossing T, they can only be followed by zero-degree one-step edges. Put

$$\begin{aligned} &U_0 = q_J(T) + 3T(4T-3)q_{J-2}(T-1) - 3T(T-1)q_{J-3}(T-2), \\ &U_1 = q_{J-1}(T), \quad U_2 = q_{J-2}(T), \quad U_3 = q_{J-3}(T), \\ &U_4 = q_{J-3}(T-1). \end{aligned} \tag{37.4}$$

These five coordinates are independent of X, by Lemma 37.1 and because the coefficient of $q_J(T)$ in U_0 equals one.

Define $B_H(T)$ to be the $H \times 5$ matrix of coefficients of the five original unit boundary seeds

$$q_J(T), \quad q_{J-1}(T), \quad q_{J-2}(T), \quad q_{J-3}(T), \quad q_{J-3}(T-1) \tag{37.5}$$

in the outputs $Z_1, \dots, Z_H$, with all other formal historical high jets put to zero. The recurrence used in this calculation is the mandatory part of (37.1); its entries depend on T but not on J (for $J \geq 8$). The first column is identically one. The two discarded seed columns are proportional to it as indicated above. Hence

$$(Z_1, \dots, Z_H)^{\mathsf{t}} = \text{known constant} + C_H X + B_H(T)U, \tag{37.6}$$

for an explicitly determined matrix C_H; all components are linear in the old high jets.

Theorem 37.3 (Response equivalence and optimal archive)**.** *Under the hypotheses of Section 37.2, two historical high-jet assignments are indistinguishable by* every *future source program through horizon H if and only if their $X_{d,s}$ values agree and*

$$B_H(T)(U - U') = 0.$$

Consequently the exact minimum fixed-length field-symbol archive is

$$\boxed{m(T, J, H; \ell) = x(J, H) + \mathrm{rank}_{\mathbb{F}_\ell}\, B_H(T).} \tag{37.7}$$

This lower bound holds against all deterministic encoders, linear or nonlinear; equality is attained by a causal checkpoint archive.

Proof. The nonconstant future-source coefficients of every output have, by Lemma 37.2, precisely the history-dependent span generated by the independently accessible X coordinates. Equality of all future-program responses therefore forces equality of X. With X fixed, equation (37.6) shows that equality for the all-zero future program is exactly the condition $B_H(U - U') = 0$. Conversely these two conditions force equality of the zero-source outputs and every nonconstant future-source coefficient, hence of all program evaluations.

By Lemma 37.1 the $x(J, H)$ direct-read cells and five boundary coordinates U range freely and independently. Profile equivalence therefore has exactly $\ell^{x(J,H)+\operatorname{rank} B_H(T)}$ classes. A deterministic message of b field symbols has no more than ℓ^b values, so it must have $b \geq x(J, H) + \operatorname{rank} B_H(T)$ even for a nonlinear encoder. Retaining the X values and any $\operatorname{rank} B_H$ independent linear forms of $B_H U$ gives a matching message, from which every future response can be reconstructed by the known recurrence and its coefficient map. The archive uses only information already available at checkpoint T and may be updated causally as the historical source stream arrives. Its minimality concerns stored field symbols, not temporary execution memory. □

Remark 37.4. The matrix B_H can be generated without enumerating historical source assignments: initialize the five seed coefficients from (37.5), use the mandatory recurrence for $n = T+1, \ldots, T+H$, and read $q_J(T+h)$. Thus (37.7) is an explicit finite computation even in exceptional characteristics.

37.5 Complete characteristic-dependent rank classification

Define

$$F(T) = 64T^6 + 1312T^5 + 11120T^4 + 49565T^3 + 121776T^2 + 155655T + 80706, \tag{37.8}$$

$$G(T) = 256T^6 + 5632T^5 + 51040T^4 + 242500T^3 + 633339T^2 + 858576T + 471216. \tag{37.9}$$

Define finite sets of ordered pairs (ℓ, T):

$$E_4 = \{(73, 13), (701, 70)\}, \qquad E_6 = \{(31, 23), (199, 54), (3594541, 1230238)\}. \tag{37.10}$$

The conditions $T \geq J \geq 8$ and $\ell > T + H$ are understood throughout the following theorem.

Theorem 37.5 (All admissible horizons and exceptional characteristics)**.** *Write* $b_H(T; \ell) = \operatorname{rank}_{\mathbb{F}_\ell} B_H(T)$. *Then*

$$b_H(T;\ell) = \begin{cases} H, & H = 1, 2, 3, \\ 3, & H = 4 \text{ and } (\ell, T) \in E_4, \\ 4, & H = 4 \text{ and } (\ell, T) \notin E_4, \\ 4, & H = 5 \text{ and } F(T) \equiv 0 \pmod{\ell}, \\ 5, & H = 5 \text{ and } F(T) \not\equiv 0 \pmod{\ell}, \\ 4, & H = 6 \text{ and } (\ell, T) \in E_6, \\ 5, & H = 6 \text{ and } (\ell, T) \notin E_6, \\ 5, & H \geq 7. \end{cases} \tag{37.11}$$

In particular, every exceptional boundary-rank deficiency disappears by horizon seven.

Proof. All minors below are determinants of the formal-seed matrix defined before Theorem 37.3. The displayed polynomials, factorizations and modular witnesses constitute the finite algebraic certificates. Section 37.7 specifies how to reconstruct them directly from the recurrence without any separately supplied files.

For $H = 1$ the first row has first entry one. For $H = 2$, two 2×2 minors are $4T + 5$ and $(4T+1)(7T+8)$, whose resultant as polynomials in T is, up to sign, $48 = 2^4 \cdot 3$. Since $\ell > T + 2 \geq 10$, the two minors cannot vanish together. For $H = 3$, two 3×3 minors are

$$4(16T^3 + 88T^2 + 157T + 87), \qquad 3T(T + 1)(4T + 9).$$

The resultant of the nonunit factors of interest has absolute value $192 = 2^6 \cdot 3$, and $T(T+1)$ is nonzero modulo ℓ. Hence rank three is unavoidable.

For $H = 4$, two 4×4 minors are $4f_4(T)$ and $-12T(T + 1)g_4(T)$, where

$$\begin{aligned} f_4(T) &= 1024T^6 + 14896T^5 + 88776T^4 + 272351T^3 \\ &\quad + 444443T^2 + 360948T + 114144, \\ g_4(T) &= 16T^3 + 136T^2 + 381T + 348. \end{aligned}$$

Their resultant is

$$|\operatorname{Res}(f_4, g_4)| = 2^{29}3^3 \cdot 29 \cdot 73 \cdot 701.$$

The only joint modular roots at these candidate primes are $T \equiv 0 \pmod{29}$, $T \equiv 13 \pmod{73}$, and $T \equiv 70 \pmod{701}$. Because $8 \leq T < \ell$, the first is impossible. For the remaining two pairs, all five maximal 4×4 minors vanish; rank is exactly three by the $H = 3$ result. This proves the horizon-four classification.

For $H = 5$, the square determinant is the previously identified CCR mandatory-transfer determinant [261]:

$$\det B_5(T) = 192T(T + 1)F(T).$$

Thus its vanishing is equivalent to $F(T) \equiv 0 \pmod{\ell}$. Whenever the first four rows have rank four, B_5 has rank at least four. At the two exceptional horizon-four pairs, the 4×4 minor with rows $1, 2, 3, 5$ and columns $1, 2, 3, 4$ has residues 12 modulo 73 and 575 modulo 701, respectively. Therefore rank cannot be smaller than four, proving the exact horizon-five criterion.

For $H = 6$, two of the six maximal 5×5 row-minors are

$$192T(T + 1)F(T), \qquad 192T(T + 1)G(T).$$

An independently calculated integer Sylvester resultant gives

$$\big|\operatorname{Res}(F, G)\big| = 2^{26}3^4 5^5 \cdot 31 \cdot 73 \cdot 199 \cdot 701 \cdot 3594541.$$

The admissible characteristic eliminates $2, 3, 5$. Modular polynomial gcd calculations yield exactly the five candidate pairs

$$(31, 23), \quad (73, 13), \quad (199, 54), \quad (701, 70), \quad (3594541, 1230238).$$

Checking the *remaining* maximal row-minors excludes $(73, 13)$ and $(701, 70)$; all six vanish for the three pairs in E_6. A 4×4 leading minor is nonzero at each remaining pair, with residues 17, 20 and 1907807 in the same order, so the rank is exactly four there. The appended seventh row restores rank five: the 5×5 determinant using rows $1, 2, 3, 4, 7$ and all five columns has residues 26, 171, and 1970365 modulo 31, 199, and 3594541, respectively. These are nonzero exact certificates. At every other admissible parameter, the first six rows already have rank five. Therefore $H \geq 7$ always has boundary rank five, completing the proof. □

Corollary 37.6 (Exact saturation and recovery)**.** *For every admissible parameter set, the exact archive size is obtained by inserting* (37.11) *into* (37.7)*. In particular, for $H \geq 7$,*

$$m(T, J, H; \ell) = 5 + \sum_{s=1}^{H} \max\big(0, \min(J - 4, J - s) - 3\big). \tag{37.12}$$

For $H \geq \max(7, J - 4)$, this simplifies to

$$\boxed{m(T, J, H; \ell) = 5 + \frac{J(J - 7)}{2}.} \tag{37.13}$$

Proof. Theorem 37.5 gives $b_H = 5$ for $H \geq 7$. In (37.3), for every $d = 4, \dots, J-4$ the allowable values of s are $1, \dots, J-d$. The sum of their counts is $\sum_{d=4}^{J-4}(J-d) = J(J-7)/2$; when $J = 8$ the same equality holds with one term. Thus the direct-read contribution ceases growing by horizon $J-4$. □

Remark 37.7 (No fixed-field infinite-horizon assertion). The statements hold for each instance with $\ell > T + H$. If T, J, ℓ are fixed, admissible horizons have $H < \ell - T$. The phrase *eventual saturation* means that once the threshold lies within this admissible range, extending the horizon while preserving the stated assumptions requires no further field symbols. It does not assert one fixed field can support arbitrarily large H under the same accessibility proof.

37.6 Examples and information interpretation

For $J = 12$ and generic admissible characteristics, formula (37.7) gives

Horizon H	4	5	6	7	8 and higher
Archive symbols	24	29	32	34	35

Here $H = 5$ reproduces the prior five-step theorem; the new result explains the subsequent growth and the exact plateau. For the exceptional case $(T, J, \ell) = (23, 8, 31)$, admissible horizons $5, 6, 7$ have costs $(8, 8, 9)$ rather than the corresponding generic values $(9, 9, 9)$. Thus an exceptional characteristic can temporarily remove a response direction, and an additional future observation can restore it.

The response-equivalence description is more informative than a mere number of symbols. It says exactly which differences in an old history can never be resolved by future programs through horizon H: they must vanish on all $X_{d,s}$ and lie in the kernel of the mandatory transfer. This finite characteristic kernel may depend on (ℓ, T, H); the direct-read coordinate family does not.

37.7 Reconstruction of the algebraic certificates

All algebraic inputs required for an independent check are stated in this article. In particular, no executable code or separate certificate file is needed. The following exact-arithmetic procedure reconstructs the finite computations used in Theorem 37.5.

1. Work over $\mathbb{Z}[T]$ and initialize five formal boundary seeds corresponding, in order, to the cells in (37.5), with all other historical high-jet seeds equal to zero. Use the mandatory three-lag recurrence in (37.1) to propagate the five response columns. Extract the coefficient of degree J at times $T+1, \dots, T+7$. Because only offsets from the terminal degree occur, the resulting rows of $B_7(T)$ do not depend on $J \geq 8$.
2. Compute the maximal minors of $B_4(T)$ and $B_6(T)$ by exact determinant expansion, fraction-free elimination, or another exact method. Check the two displayed fourth-horizon minor factors $4f_4(T)$ and $-12T(T+1)g_4(T)$ and the first two sixth-horizon factors $192T(T+1)F(T)$ and $192T(T+1)G(T)$. All the defining polynomials and their coefficients appear in Section 37.5.
3. Form the Sylvester matrices for (f_4, g_4) and (F, G) over $\mathbb{Z}$. Their determinants have the prime factorizations shown in the proof. For each candidate prime, reduce both polynomials in $\mathbb{F}_\ell[T]$ and compute their polynomial gcd. The common roots are explicitly enumerated in the proof, so no numerical root approximation is necessary.
4. At every surviving prime–checkpoint pair, reduce the entire transfer matrix modulo ℓ and use exact row reduction to test all maximal minors. The stated nonzero lower-order minor residues certify the precise rank drops. For each of the three sixth-horizon

exceptions, the nonzero determinant residues 26, 171, and 1970365 for rows $1, 2, 3, 4, 7$ certify the seventh-horizon rank recovery.

5. Finally, obtain any archive size by computing the finite integer sum (37.3) and adding the exact modular rank specified by (37.11). The examples of Section 37.5 and the final section can be checked by this rule without reconstructing source-history data or using approximate arithmetic.

The all-horizon archive theorem is proved by the accessibility, path decomposition, and profile-counting arguments of Sections 37.3–37.4; finite determinant checks are used only to evaluate the exceptional-characteristic rank profile. No external datasets were used.

37.8 Discussion

The exact rank profile shows that CCR's source chronology can force a sharp separation between individually interrogable historical coordinates and the small mandatory transfer left after quotienting them out. The resulting five-column matrix makes exceptional field behavior explicit and supplies an optimal checkpoint message for the stated free-source model. These results extend the earlier five-step theory, but they do not establish optimal peak RAM, universal algorithmic speedups, optimal archives for event-derived constrained sources, or any statement about composite moduli. Extending accessibility beyond the characteristic constraint is a separate question and cannot be assumed without proof.

37.9 Re-entry to the main DPDA story

The arithmetic offshoot has developed connected CCR-k source transport, stationary arithmetic constraints, Nörlund–Stirling rigidity, and primitive-orbit noncoverage. Chapter 35 adds a distinct inverse problem: masked old sources can be recovered by nonresetting interventions, with a sharp three-versus-two preset/adaptive pulse distinction for an explicitly restricted-input CCR-10 family over primes $q \geq 41$. Chapter 36 establishes sharp five-step forecast information requirements, a matching causal archive, exact finite-ring response profiles, source-mask coding bounds, and verified algorithmic comparisons for typed chronological carry. Chapter 37 extends the same free-source forecast interface to every admissible horizon: its exact information cost separates independently read historical coordinates from a five-column mandatory transfer and determines characteristic-dependent rank losses and subsequent recovery. These information and computational results concern their exact source/query contracts; physical machine-cost bounds require separate realization arguments.

We now return to the original 2:1 center/left/right DPDA. The earlier shortest-demand analysis in Chapters 2–8 produced compact residual-faithful representations, but these altered the physical stack encoding. The next chapter recovers a general centered real-time construction for all coprime p:q; from that point the main line proceeds through Structure Theory, Uniqueness/Lift Theory, and the General Theory.

Chapter 38

Why Static Two-Visibility Fails

Research branch — Returning to the Machine: General Real-Time p:q. *This branch develops the uniform centered real-time literal-demand construction for all coprime positive p, q.*

We now return to pushdown machines.
This is not another residual-counter construction.
The goal is more specific:

> Reconstruct, for every coprime p:q, the operational idea hidden in the original center/left/right machine.

The original 2:1 design had three striking features:

1. a literal completion demand was physically visible in the stack;
2. a mismatch could temporarily move information into finite control;
3. a later long replacement could rematerialize that information.

The general construction will preserve this style rather than replace it by the compact Euclidean residual encoding of Chapters 7–8.
That is why its resource count is different.

38.1 We do not re-prove minimum demand

The semantic foundation is already complete.
For

$$L_{p,q} = \{w : qN_1(w) = pN_0(w)\},$$

the residual is

$$\rho = qN_1 - pN_0.$$

For every residual r, the unique minimum completion demand is

$$\mu(r) = (x_r, y_r),$$

and the primitive zero-weight kernel is

$$\boxed{K = (q, p).}$$

Every nonnegative vector of the same weight differs from the minimum demand by a nonnegative multiple of K.
Those results are imported from the earlier chapters.
The new problem is physical ordering.

38.2 A two-visible canonical word

For a demand vector

$$(x, y),$$

define

$$\boxed{\eta(x,y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & y > x. \end{cases}}$$

This word has an attractive property.
If both symbols occur, the stack begins with

$$10.$$

More generally, every suffix that remains mixed begins with either

$$10$$

or

$$01.$$

Thus the presence of both demand types is always visible within two stack cells.
This is the general literal-demand analogue of the original 2:1 depth-two phenomenon.
But static visibility is not enough.

38.3 The 3:2 obstruction

Consider the canonical stack

$$\boxed{1010.}$$

Both demand symbols are visible immediately.
Suppose the next input is 0, but the required zero is the **second** visible cell.
A naive LIFO strategy could:

1. pop the top 1;
2. delete the 0;
3. immediately restore the displaced 1.

This produces

$$1010 \longrightarrow 110.$$

The surviving zero has moved to depth three.
Two-cell visibility has been destroyed.
The problem is not lack of static information.
It is chronology.

38.4 Static two-visible obstruction

Lemma 38.1 — Static two-visible obstruction.

Suppose a mixed binary stack contains at least two zeros and two ones and begins with

10.

No purely static strategy can support both deletion directions while:

1. deleting either visible demand type;
2. immediately restoring a displaced first-cell blocker;
3. guaranteeing that a surviving mixed suffix still exposes both symbol types in its first two cells.

Proof.

If deleting the top 1 must leave a mixed two-visible suffix, then the next two surviving symbols must differ. Therefore the third original symbol must be 1.

If instead the second-cell 0 is deleted and the first 1 is immediately restored, preserving two-visibility requires the next surviving symbol beneath it to be 0. Therefore the third original symbol must be 0.

The same third cell cannot satisfy both conditions. □

This lemma isolates the exact reason the original architecture needs side-state memory.

A LIFO blocker sometimes must leave the stack temporarily.

38.5 SEARCH

The repair is:

SEARCH = move one displaced blocker into finite control.

The blocker is **not** immediately restored.

Its semantic effect is carried as finite-control debt while the remaining physical stack continues to be popped.

A later WRITE will materialize that debt.

This is the general form of the role played by L and R in the original machine.

The crucial design question becomes:

> How much signed debt must finite control be able to carry before WRITE becomes safe?

That will produce the centered state rail.

Before building the rail, we need to formalize what a debt state means semantically.

Chapter 39

Debt, Virtual Demand, and Minimal WRITE

Let

$$m = p + q.$$

A one-cell mismatch changes the relationship between the physical stack and the semantic demand by one kernel-length unit m.

This motivates a centered integer debt coordinate.

39.1 Virtual demands

For

$$d \in \mathbb{Z},$$

define the virtual demand

$$\boxed{V_d = \mu(dm).}$$

Its residual weight is

$$W(V_d) = dm.$$

A debt state D_d therefore carries finite-control potential

$$\boxed{\Phi(D_d) = dm.}$$

Let a physical logical stack S have Parikh vector

$$P(S) = (N_0(S), N_1(S)).$$

The stack and state together may contain more material than the minimum demand. The semantic projection removes any complete kernel packets.

39.2 Virtual-demand projection

Define kernel reduction

$$\mathrm{red}_{p,q}(x, y)$$

to mean repeated subtraction of

$$K = (q, p)$$

while both coordinates remain nonnegative.

The result is exactly the unique minimum demand of the same residual weight.

Definition 39.1 — Rich-to-demand projection.

For configuration

$$(D_d, S),$$

define

$$\boxed{\Theta(D_d, S) = \mathrm{red}_{p,q}\big(P(S) + V_d\big).}$$

Theorem 39.2 — Virtual-demand projection.

If

$$\boxed{\rho = W(P(S)) + dm,}$$

then

$$\boxed{\Theta(D_d, S) = \mu(\rho).}$$

Proof.

The nonnegative vector

$$P(S) + V_d$$

has weight

$$W(P(S)) + W(V_d) = W(P(S)) + dm = \rho.$$

Kernel reduction of any nonnegative vector of weight ρ is the unique minimum demand $\mu(\rho)$. □

This theorem is one of the main semantic bridges in the monograph.

It says:

$$\boxed{\text{rich state+stack configuration} \longrightarrow \text{unique shortest-demand semantics.}}$$

The rich machine may retain normalization history.

Its semantic projection does not.

39.3 The WRITE equation

Suppose one transition:

- consumes input a;
- removes visible top symbol z;
- changes debt state from D_s to D_t;
- writes replacement word u.

Let

$$e_0 = (1, 0), \qquad e_1 = (0, 1).$$

Ignoring the unchanged stack suffix, residual preservation forces

$$\boxed{P(u) = e_z - e_a + V_s - V_t + \kappa K}$$

for some integer κ.

At the bottom marker, the term e_z is omitted.

Thus every invariant-preserving rewrite lies on one affine kernel ray.

39.4 Minimal kernel carry

Let

$$B = e_z - e_a + V_s - V_t.$$

The first kernel shift making both coordinates nonnegative is

$$\boxed{\kappa_* = \max\left(\left\lceil -\frac{B_0}{q} \right\rceil, \left\lceil -\frac{B_1}{p} \right\rceil\right).}$$

Theorem 39.3 — Minimal kernel-carry rewrite.

Among all nonnegative invariant-preserving replacement vectors, the unique shortest one is

$$\boxed{R_* = B + \kappa_* K.}$$

Every other solution has the form

$$\boxed{R_* + \ell K, \qquad \ell \geq 1,}$$

and is longer by exactly

$$\ell(p + q)$$

symbols.

Proof.

All integer solutions differ by a multiple of the primitive kernel K.

The displayed κ_* is the least shift that places both coordinates in the nonnegative quadrant.

Every further kernel packet adds

$$q + p = m$$

symbols. □

When a physical word is needed, we write the vector in canonical order using η.

39.5 Why the historical long writes existed

The original 2:1 table contained replacements such as:

$$10, \qquad 11, \qquad 000, \qquad 111111.$$

Before the general theory, these can look ad hoc.

The WRITE equation changes their interpretation.

A long replacement is determined by:

1. the consumed input symbol;
2. the visible stack symbol removed;
3. the source debt potential;

4. the target debt potential;
5. the least kernel carry needed to make the replacement vector nonnegative.

Thus the long words are not arbitrary repair strings.
They are shortest solutions of a constrained residual-preservation equation.
This is one of the main retrospective explanations of the original machine.

39.6 From arbitrary debt to a finite rail

The virtual-demand definition allows every

$$d \in \mathbb{Z}.$$

A finite-state DPDA cannot.
The construction therefore needs a bounded debt interval.
Set

$$\boxed{h_{\mathrm{ctr}} = \left\lfloor \frac{m-1}{2} \right\rfloor.}$$

The centered core will use exactly

$$\boxed{D_{-h_{\mathrm{ctr}}}, \ldots, D_{-1}, D_0, D_1, \ldots, D_{h_{\mathrm{ctr}}}.}$$

with

$$D_0 = C.$$

Why is this width sufficient?
Because the canonical word created at every WRITE has bounded minority count.
Why is it exact in the deferred architecture?
Because one common canonical witness can exhaust the physical stack while leaving every debt value in this interval purely in finite control.
Those are the two halves of Chapters 40 and 41.

Chapter 40

The Centered Real-Time p:q DPDA

Set

$$m = p + q, \qquad h_{\mathrm{ctr}} = \left\lfloor \frac{m-1}{2} \right\rfloor.$$

The centered state set is

$$\boxed{Q = \{D_{-h_{\mathrm{ctr}}}, \ldots, D_{-1}, D_0, D_1, \ldots, D_{h_{\mathrm{ctr}}}\},}$$

with

$$D_0 = C.$$

The stack alphabet is

$$\boxed{\{0, 1, Z_0\}.}$$

The bottom marker Z_0 remains fixed throughout binary-input processing and is removed only by the final right-endmarker acceptance move.

The construction has four binary transition roles:

$$\boxed{\text{MATCH,} \qquad \text{SEARCH,} \qquad \text{outer WRITE,} \qquad \text{bottom WRITE.}}$$

Every binary move consumes exactly one input symbol.

There are no ε-moves during binary processing.

40.1 The centered rail

MATCH

For every

$$|d| \le h_{\mathrm{ctr}}$$

and $z \in \{0, 1\}$,

$$\boxed{(D_d, zT) \xrightarrow{z} (D_d, T).}$$

A matching input simply cancels one physically present demand cell.

Interior SEARCH

For

$$d < h_{\mathrm{ctr}},$$

$$\boxed{(D_d, 0T) \xrightarrow{1} (D_{d+1}, T).}$$

For

$$d > -h_{\mathrm{ctr}},$$

$$\boxed{(D_d, 1T) \xrightarrow{0} (D_{d-1}, T).}$$

A mismatch still pops the visible cell, but instead of restoring it the state debt moves one step.

Outer-boundary WRITE

Define

$$U_+ = \eta\big(\mu((h_{\mathrm{ctr}} + 1)m)\big),$$

$$U_- = \eta\big(\mu(-(h_{\mathrm{ctr}} + 1)m)\big).$$

Then

$$\boxed{(D_{h_{\mathrm{ctr}}}, 0T) \xrightarrow{1} (C, U_+T),}$$

and

$$\boxed{(D_{-h_{\mathrm{ctr}}}, 1T) \xrightarrow{0} (C, U_-T).}$$

At the outer debt boundary, another outward mismatch triggers physical rematerialization and resets debt to zero.

Bottom WRITE

For every

$$|d| \le h_{\mathrm{ctr}},$$

$$\boxed{(D_d, Z_0) \xrightarrow{1} (C, \eta(\mu(dm + q))Z_0)\,,}$$

and

$$\boxed{(D_d, Z_0) \xrightarrow{0} (C, \eta(\mu(dm - p))Z_0)\,.}$$

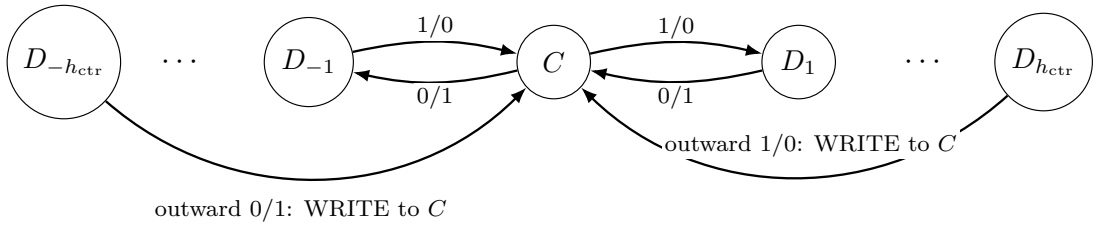
When no pre-existing demand remains, the next input directly creates the exact new minimum-demand word.

Acceptance

A mandatory right endmarker is allowed only at

$$\boxed{(C, Z_0).}$$

It removes Z_0 and accepts by empty stack.
No other reachable configuration has a right-endmarker acceptance move.



Adjacent mismatch rules repeat along the rail.
Matching 0/0 or 1/1 pops and keeps the same debt state.

Figure 40.1: Schematic centered debt rail. Labels 1/0 and 0/1 denote input/top mismatches. Interior mismatches pop the visible cell and move debt by one; an outward mismatch at $\pm h_{\text{ctr}}$ triggers WRITE back to the center.

The figure should visually emphasize:

- adjacent mismatch SEARCH moves;
- center state C;
- outer WRITE arcs back to C;
- MATCH as state-preserving pops.

40.2 Residual invariant

Let the logical stack be S.

Theorem 40.1 — Residual invariant.
After every processed binary prefix, if the centered configuration is

$$(D_d, SZ_0),$$

then

$$\boxed{\rho = W(P(S)) + dm.}$$

Proof.
Initially,

$$(C, Z_0)$$

has residual zero.
A matching 0-pop decreases stack weight by p, exactly matching the input-0 residual update.
A matching 1-pop increases stack weight by q.
For mismatch 1/0, popping one physical zero changes stack weight by $-p$, while moving

$$d \to d + 1$$

adds state potential

$$m = p + q.$$

The net change is

$$-p + m = q,$$

exactly the input-1 residual update.
The 0/1 case is symmetric:

$$q - m = -p.$$

At an outer WRITE, the replacement word is defined to carry exactly the missing boundary potential.

At bottom WRITE, the target demand is defined directly from the new residual.

Therefore every binary transition preserves the invariant. □

Combined with Chapter 39:

$$\boxed{\Theta(D_d, S) = \mu(\rho)}$$

for every reachable configuration.

40.3 Why the debt rail can stop at h_{ctr}

The key is the **minority size**

$$\mathrm{minc}(x, y) = \min(x, y).$$

Lemma 40.2 — WRITE-frontier minority bound.

Let

$$h_{\mathrm{ctr}} = \left\lfloor \frac{m-1}{2} \right\rfloor.$$

Then:

1. for every $|d| \le h_{\mathrm{ctr}}$, both bottom-WRITE demands

   $$\mu(dm + q)$$

   and

   $$\mu(dm - p)$$

   have minority size at most h_{ctr};

2. if

   $$\mu((h_{\mathrm{ctr}} + 1)m) = (x_+, y_+),$$

   then

   $$\boxed{y_+ \le h_{\mathrm{ctr}} < x_+;}$$

3. if

$$\mu(-(h_{\mathrm{ctr}}+1)m) = (x_-, y_-),$$

then

$$\boxed{x_- \le h_{\mathrm{ctr}} < y_-.}$$

Proof.

For the bottom-WRITE family, both residual forms can be written

$$r = jm + q$$

with

$$-h_{\mathrm{ctr}} - 1 \le j \le h_{\mathrm{ctr}}.$$

Let

$$\mu(r) = (x, y), \qquad \ell = x + y.$$

Modulo m, since

$$p \equiv -q \pmod{m},$$

the completion equation gives

$$-q\ell \equiv q \pmod{m}.$$

Because

$$\gcd(q, m) = 1,$$

$$\ell \equiv -1 \pmod{m}.$$

Hence

$$\ell = km - 1.$$

If both x and y were at least $h_{\mathrm{ctr}} + 1$, then $\ell > m - 1$, so $k \ge 2$.

But $\mu(r)$ is kernel-free, so either

$$x < q$$

or

$$y < p.$$

Each alternative, together with $x, y \ge h_{\mathrm{ctr}} + 1$, forces the quotient index j outside the allowed interval.

Contradiction.

For the outer positive boundary, write $D = h_{\mathrm{ctr}} + 1$. Integer solutions to weight Dm have form

$$x = D + qk, \qquad y = pk - D.$$

The minimum nonnegative solution uses the first feasible k, giving

$$y \le h_{\mathrm{ctr}}$$

and

$$x \geq h_{\mathrm{ctr}} + 1.$$

The negative case is symmetric. □

This lemma explains the centered width geometrically.

Every freshly written canonical word contains at most h_{ctr} copies of its minority symbol.

The debt rail is exactly wide enough to search through that minority frontier.

40.4 Boundary guard

The centered construction also requires a boundary guarantee at $D_{h_{\mathrm{ctr}}}$. An outward mismatch must not occur while the surviving suffix is still mixed, because writing a new canonical prefix in front of such a suffix could destroy the normal form.

The minority bound provides exactly this guarantee.

Lemma 40.3 — Boundary guard.

Assume the current stack is a suffix of a canonical word

$$\eta(x, y)$$

with

$$\min(x, y) \leq h_{\mathrm{ctr}}.$$

Then:

1. if the state is $D_{h_{\mathrm{ctr}}}$ and the stack begins with 0, an outward mismatch 1/0 can occur only on a unary suffix

$$\boxed{0^{t+1};}$$

2. symmetrically, at $D_{-h_{\mathrm{ctr}}}$, an outward 0/1 mismatch can occur only on a unary suffix

$$\boxed{1^{t+1}.}$$

Proof.

Consider the positive case.

Reaching debt h_{ctr} requires at least h_{ctr} positive mismatch events.

Each such mismatch has popped one zero from the original canonical stack.

Suppose the surviving suffix begins with 0 but still contains a later 1.

Then the current zero lies inside the alternating part of $\eta(x, y)$, not in its final unary tail.

Every previously popped zero in that alternating region has a distinct preceding one that must also have been popped.

Together with:

- the current surviving zero;
- the later surviving one;

the original word would contain at least

$$h_{\mathrm{ctr}} + 1$$

zeros and at least

$$h_{\mathrm{ctr}} + 1$$

ones.

This contradicts

$$\min(x, y) \le h_{\mathrm{ctr}}.$$

Therefore the suffix must be unary.

The negative case is symmetric. □

This is the LIFO safety theorem of the construction.

40.5 Canonical segments

Theorem 40.4 — Canonical-segment invariant.

Immediately after every WRITE, the machine is in the center state with stack

$$\boxed{\eta(\mu(r))Z_0}$$

for its current residual r.

The corresponding demand has minority size at most h_{ctr}.

Between two WRITEs, the logical stack is a suffix of that canonical word.

Proof.

Initially the logical stack is empty.

A bottom WRITE is canonical by definition, and Lemma 40.2 gives the minority bound.

Between WRITEs, MATCH and interior SEARCH only pop one pre-existing work symbol, so the stack remains a suffix of the last canonical WRITE word.

Now consider a positive outer WRITE.

By Lemma 40.3, its source suffix is

$$0^{t+1}.$$

Write

$$\mu((h_{\mathrm{ctr}} + 1)m) = (x, y).$$

Lemma 40.2 gives

$$y \le h_{\mathrm{ctr}} < x.$$

Thus the canonical word $\eta(x, y)$ ends in 0, and the WRITE produces

$$\eta(x, y)0^t = \eta(x + t, y).$$

The residual invariant shows that this vector has exactly the new residual weight.

Moreover, $y < p$, so no kernel packet can be subtracted; the vector is therefore already the minimum demand.

The negative boundary is symmetric. □

The entire reachable-store theorem has now been reduced to episodes:

$$\boxed{\text{WRITE creates a canonical word;}}$$

$$\boxed{\text{MATCH/SEARCH consume a suffix of it;}}$$

$$\boxed{\text{boundary WRITE can occur only after the mixed part is gone.}}$$

40.6 Two-cell qualitative visibility

Corollary 40.5 — Two-cell visibility.

For every reachable logical stack S,

$$\boxed{\mathrm{alph}(S) = \mathrm{alph}(\mathrm{pref}_2(S)).}$$

Equivalently, if both 0 and 1 occur anywhere in S, then both occur among the first two cells.

Proof.

Every canonical word $\eta(x, y)$, and every suffix of it, avoids

$$001 \qquad \text{and} \qquad 110.$$

Therefore a mixed suffix cannot begin with two equal symbols.

It begins with

$$10 \qquad \text{or} \qquad 01.$$

Apply Theorem 40.4. □

This is qualitative, not quantitative.

The first two cells reveal which demand types remain.

They do not reveal the unbounded magnitudes.

40.7 Exact WRITE normalization

Theorem 40.6 — Exact WRITE normalization.

After every bottom or outer-boundary WRITE,

$$\boxed{(D_d, S) = (C, \eta(\mu(\rho)))\,,}$$

where ρ is the residual after the consumed binary symbol.

WRITE therefore does more than preserve the invariant.

It returns the physical stack to the exact minimum-demand normal form.

This result is the normalization analogue of the original endmarker materialization theorem, but now it occurs online.

40.8 Zero residual

Theorem 40.7 — Zero-residual correctness.

At the end of every binary prefix,

$$\boxed{\rho = 0 \iff (D_d, S) = (C, \varepsilon).}$$

Proof.

The reverse implication follows from the invariant.

For the forward direction, consider a nonempty processed prefix.

Immediately after the most recent WRITE, the machine was at

$$(C, \eta(\mu(r_0)))$$

for some residual r_0.

Write

$$\mu(r_0) = (x, y).$$

Since then, each MATCH or SEARCH step has popped exactly one pre-existing demand cell.

Suppose the intervening input contains:

$$u$$

zeros and

$$v$$

ones, so

$$u + v \leq x + y.$$

If the current residual is zero, this segment completes r_0.

By minimality of $\mu(r_0)$, every completion requires length at least

$$x + y.$$

Hence

$$u + v = x + y.$$

The entire canonical stack has been consumed.

So

$$S = \varepsilon.$$

The residual invariant then gives

$$dm = 0,$$

hence

$$d = 0.$$

Therefore the state is C. $\square$

This proof is important because acceptance is obtained from shortest-demand minimality, not from a large case analysis of reachable stacks.

40.9 Main construction theorem

Theorem 40.8 — Real-time centered rich DPDA.

For every coprime

$$p, q \geq 1,$$

the centered schema defines a deterministic pushdown machine with:

$$\boxed{|Q| = 2\left\lfloor \frac{p+q-1}{2} \right\rfloor + 1}$$

semantic core states and stack alphabet

$$\boxed{\{0, 1, Z_0\}.}$$

Every binary transition consumes exactly one symbol.

No ε-transition is used during binary processing.

With the mandatory right endmarker, the machine recognizes

$$\boxed{L_{p,q}.}$$

Furthermore:

1. every WRITE returns to the exact minimum-demand normal form;
2. every reachable stack has two-cell qualitative visibility;
3. the residual is exactly decoded by the state-plus-stack invariant.

The state count is

$$\boxed{|Q| = \begin{cases} p+q, & p+q \text{ odd}, \\ p+q-1, & p+q \text{ even}. \end{cases}}$$

This is the exact size of the centered construction.

This is a construction-size statement rather than an unrestricted state lower bound. The representation-independent state theorem is established in the later physical-complexity macro-Part.

40.10 Shortest-demand quotient

The centered machine is physically richer than a residual counter.

Two configurations can store the same semantic demand while distributing it differently between:

- physical stack;
- finite-control debt.

Corollary 40.9 — Shortest-demand quotient.

For reachable configurations, define

$$c_1 \sim_\Theta c_2 \iff \Theta(c_1) = \Theta(c_2).$$

Then the residual/minimum-demand transition system is a semantic quotient of the rich configuration graph.

The quotient forgets normalization chronology.

The physical fiber may remember it.

That observation will later become a central theorem family in the Uniqueness/Lift Theory.

For now, it is already visible concretely.

Chapter 41

What "Optimal" Means in the Deferred-SEARCH Architecture

The centered construction uses

$$2h_{\mathrm{ctr}} + 1$$

debt states, where

$$h_{\mathrm{ctr}} = \left\lfloor \frac{p+q-1}{2} \right\rfloor .$$

Is that number necessary?
The answer is yes—but only for a precisely stated physical interface.
This theorem must not be confused with the raw residual-faithful frontier of Chapter 8.
The two results optimize different representation categories.

41.1 Canonical deferred-SEARCH model

Definition 41.1 — Canonical deferred-SEARCH realization.
A realization belongs to the canonical deferred-SEARCH model if:

1. the normalized minimum demand
$$\mu(r)$$
is physically represented by the canonical word
$$\eta(\mu(r));$$

2. during a SEARCH episode, each binary transition consuming a pre-existing demand cell removes exactly one such cell;

3. a mismatch is stored only in finite control—no auxiliary debt marker is inserted into the remaining pre-existing word;

4. in the witness episode used below, the transition consuming the final pre-existing demand cell performs no materializing push and leaves the same bottom marker Z_0;

5. debt materialization is deferred to a later transition.

The last condition is what makes the model **deferred**.
An eager implementation lies outside this model.

41.2 Residual distinguishability

Lemma 41.2 — Residual distinguishability.

If two prefixes have distinct residuals

$$r_1 \neq r_2,$$

then a common binary continuation followed by the right endmarker distinguishes them.

Proof.

Choose a word z whose Parikh vector is

$$\mu(r_1).$$

Appending z sends the first residual to zero.

The second residual becomes

$$r_2 - r_1 \neq 0.$$

The right endmarker is accepted after the first prefix and rejected after the second. □

This is a language property.

It is not assumed by the architecture.

41.3 One witness containing enough of both symbols

Define

$$\boxed{\nu_{h_{\mathrm{ctr}}} = \begin{cases} (h_{\mathrm{ctr}}, h_{\mathrm{ctr}}), & m = p + q \text{ odd}, \\ (h_{\mathrm{ctr}}, h_{\mathrm{ctr}} + 1), & m \text{ even}. \end{cases}}$$

Its canonical word is

$$(10)^{h_{\mathrm{ctr}}}$$

in the odd case and

$$(10)^{h_{\mathrm{ctr}}}1$$

in the even case.

Lemma 41.3 — Canonical witness.

$$\boxed{\nu_{h_{\mathrm{ctr}}} = \mu(W(\nu_{h_{\mathrm{ctr}}})).}$$

Thus $\nu_{h_{\mathrm{ctr}}}$ is kernel-free and contains at least h_{ctr} zeros and h_{ctr} ones.

Proof.

If m is odd and a kernel packet

$$(q, p)$$

could be subtracted from $(h_{\mathrm{ctr}}, h_{\mathrm{ctr}})$, then

$$p \leq h_{\mathrm{ctr}}, \qquad q \leq h_{\mathrm{ctr}},$$

so

$$m = p + q \leq 2h_{\mathrm{ctr}} = m - 1,$$

impossible.

If m is even, subtraction from $(h_{\mathrm{ctr}}, h_{\mathrm{ctr}} + 1)$ would require

$$q \le h_{\mathrm{ctr}}, \qquad p \le h_{\mathrm{ctr}} + 1,$$

giving again

$$m \le 2h_{\mathrm{ctr}} + 1 = m - 1.$$

Contradiction. □

This single canonical block contains enough of both symbol types to generate every debt in the centered interval.

41.4 Exhaust the same stack, leave different debts

Lemma 41.4 — Debt distinguishability witness.

Starting from the same canonical witness word $\eta(\nu_{h_{\mathrm{ctr}}})$, a deferred SEARCH episode can consume the entire pre-existing work stack while leaving any debt

$$\boxed{d \in \{-h_{\mathrm{ctr}}, -h_{\mathrm{ctr}} + 1, \dots, h_{\mathrm{ctr}}\}.}$$

The resulting residual is

$$\boxed{dm.}$$

Proof.

For

$$d \ge 0,$$

choose exactly d witness zero-cells and mismatch them by reading 1 instead of 0.

Match every other pre-existing cell.

Debt increases monotonically from 0 to d.

For

$$d < 0,$$

mismatch exactly $|d|$ witness one-cells by reading 0.

Because the witness contains at least h_{ctr} copies of both symbols, every debt in the interval is possible.

Let the witness contain x zeros and y ones.

If A zero-cells are mismatched by input 1 and B one-cells are mismatched by input 0, then

$$d = A - B.$$

The consumed input contains

$$x - d$$

zeros and

$$y + d$$

ones.

Starting from residual

$$px - qy,$$

the final residual is

$$px - qy + q(y + d) - p(x - d) = d(p + q) = dm.$$

□

By the deferred-model assumption, after the last witness cell is consumed, every case has the **same physical stack**:

$$\boxed{Z_0.}$$

Only finite control can distinguish the residuals.

41.5 Exact deferred width

Theorem 41.5 — Exact deferred-SEARCH control width.

Every canonical deferred-SEARCH realization requires at least

$$\boxed{2h_{\mathrm{ctr}} + 1}$$

finite-control states.

The centered construction uses exactly $2h_{\mathrm{ctr}} + 1$.

Therefore

$$\boxed{|Q|_{\min}^{\mathrm{deferred}} = 2\left\lfloor \frac{p + q - 1}{2} \right\rfloor + 1}$$

or equivalently

$$\boxed{|Q|_{\min}^{\mathrm{deferred}} = \begin{cases} p + q, & p + q \text{ odd}, \\ p + q - 1, & p + q \text{ even}. \end{cases}}$$

Proof.

Lemma 41.4 produces

$$2h_{\mathrm{ctr}} + 1$$

different residuals:

$$-hm, \ldots, -m, 0, m, \ldots, hm.$$

In every case the work stack has been exhausted and the remaining physical stack is the same bottom marker Z_0.

If two distinct residuals shared the same finite-control state, the two instantaneous configurations would be physically identical.

But Lemma 41.2 gives a common continuation that must distinguish the residuals.

Determinism from identical configurations would force identical behavior.

Contradiction.

Thus the $2h_{\mathrm{ctr}} + 1$ residual debts require $2h_{\mathrm{ctr}} + 1$ distinct states.

The centered rail attains the bound. □

41.6 Scope — place this beside the theorem

The theorem does **not** say:

$$\boxed{\text{every DPDA recognizing } L_{p,q} \text{ needs } 2h_{\mathrm{ctr}} + 1 \text{ states.}}$$

A smaller machine may:

- abandon literal-demand storage;
- use another stack encoding;
- materialize debt eagerly;
- insert auxiliary physical markers;
- use another normalization schedule;
- leave the deferred-SEARCH interface entirely.

This scope is part of the theorem.
It is not a technical footnote.

41.7 Why Chapter 8 gives a different optimum

Chapter 8 proved the raw residual-faithful frontier:

$$S_{\mathrm{raw}}(g) = \begin{cases} \infty, & g = 0, \\ p + q, & g = 1, \\ \max(p, q), & g \geq 2. \end{cases}$$

With two work symbols, the raw residual graph can therefore be realized using only

$$\max(p, q)$$

states.
The centered deferred machine can require

$$p + q \quad \text{or} \quad p + q - 1.$$

There is no contradiction.

Raw residual-faithful question

> Give every residual one stable physical configuration and make every residual edge one raw input transition.

Deferred literal-demand question

> Store the actual canonical minimum-demand word, search it by pure pops, and keep displaced mismatch information only in finite control until later materialization.

These are different representation contracts.
The correct conclusion is:

$$\boxed{\text{resource optimality is representation-category relative.}}$$

This observation is more important than either number individually.
It is one of the direct motivations for the later Structure and Uniqueness theories.

41.8 Why this chapter belongs before the 2:1 recovery

The width theorem tells us what the centered architecture itself costs.

Only after its category has been made explicit should we specialize back to the historical 2:1 machine.

Otherwise the reader could incorrectly interpret the three-state historical architecture as globally state-optimal.

The next chapter will instead say something more precise and more interesting:

> The general centered theory reconstructs the unusual architecture of the original machine.

And the remaining transition differences will turn out to be schedule choices.

Chapter 42

Closing the First Loop: Recovering the Original 2:1 Machine

The strongest test of a proposed generalization is not that it accepts the right family.

It is that, when specialized back to the motivating case, it explains the machine from which the theory arose.

Set

$$\boxed{p = 2, \qquad q = 1.}$$

Then

$$m = p + q = 3,$$

and

$$h_{\mathrm{ctr}} = \left\lfloor \frac{3-1}{2} \right\rfloor = 1.$$

The centered rail has exactly three states:

$$D_{-1}, \qquad D_0, \qquad D_1.$$

Identify them as

$$\boxed{D_{-1} = R, \qquad D_0 = C, \qquad D_1 = L.}$$

The state potentials become

$$\boxed{-3, \qquad 0, \qquad +3.}$$

These are exactly the state potentials discovered directly in the original DPDA.

The coincidence is not imposed.

It falls out of the general debt formula

$$\Phi(D_d) = d(p + q).$$

42.1 Center mismatch directions

From $C = D_0$:

- a mismatch $1/0$ moves to $D_1 = L$;
- a mismatch $0/1$ moves to $D_{-1} = R$.

Thus the general SEARCH rail reproduces the original side-state directions.
The original left state is the positive debt state.
The original right state is the negative debt state.
This gives the old L/R asymmetry a uniform arithmetic interpretation.

42.2 Left cancellation rail

At $L = D_1$, matching $0/0$ is simply MATCH:

$$\boxed{(L, 0T) \xrightarrow{0} (L, T).}$$

This is exactly the original cancellation transition that generated the entire Fuss–Catalan branch.

Thus the transition that later produced correction geometry is also a literal specialization of the general MATCH rule inside positive debt.

42.3 Right locality-repair rail

At $R = D_{-1}$, the general rules reproduce the right-side locality repair behavior.

The state is not an unrelated special-purpose gadget.

It is the negative SEARCH-debt sheet required when a one-cell blocker has been displaced from the canonical demand stack.

This is the general explanation of why the original machine needed a right side at all.

42.4 Historical WRITE arithmetic

The primitive kernel is

$$\boxed{K = (1, 2).}$$

The minimal kernel-carry WRITE equation now reproduces the characteristic replacement Parikh vectors of the historical table.

Depending on:

- source debt;
- target debt;
- input;
- removed top symbol;

the first nonnegative kernel shift produces the historical families represented by words such as

10,

11,

00,

$$000,$$

$$11111,$$

$$111111,$$

together with the corresponding bottom-WRITE forms.

The precise ordering is the selected canonical literal-demand order.

Thus the long WRITE strings are recovered from the general arithmetic compiler rather than copied into the generalization by hand.

42.5 Specialization and recovery theorem

Theorem 42.1 — 2:1 **specialization and recovery.**

Specializing the centered real-time p:q construction to

$$(p, q) = (2, 1)$$

recovers:

1. the three-state
$$C/L/R$$
skeleton;
2. the state potentials
$$0, +3, -3;$$
3. the center mismatch directions;
4. the left cancellation rail;
5. the right locality-repair rail;
6. the arithmetic of the historical WRITE words.

The only reachable transition families not reproduced literally are two right-state matching families.

Those differences are purely normalization-schedule differences.

This theorem closes the first large loop of the monograph:

$$\boxed{\text{original machine} \to \text{general theory} \to \text{original architecture reconstructed.}}$$

42.6 Lazy versus eager discharge

Consider the reachable configuration

$$(R, 0).$$

The centered specialization performs the lazy matching pop

$$\boxed{(R, 0) \xrightarrow{0}_{\text{lazy}} (R, \varepsilon).}$$

The historical machine instead immediately materializes the represented demand and moves to the center:

$$\boxed{(R,0) \xrightarrow{0}_{\text{eager}} (C,111).}$$

Both configurations represent the same minimum demand.
Likewise, for a unary right stack

$$(R,1^k),$$

the centered specialization uses

$$\boxed{(R,1^k) \xrightarrow{1}_{\text{lazy}} (R,1^{k-1}),}$$

whereas the historical machine uses

$$\boxed{(R,1^k) \xrightarrow{1}_{\text{eager}} (C,1^{k+2}).}$$

Again, the semantic demand agrees.
Only the time at which virtual debt is rematerialized differs.

42.7 Canonical materialization map

Define

$$\boxed{\mathcal{N}(c) = (C, \eta(\Theta(c)))\,.}$$

This map forgets the physical distribution of demand between state and stack and returns the canonical center representation of the same shortest demand.

The historical eager successor is exactly the canonical materialization of the centered lazy successor on the two differing right-state families.

42.8 Eager–lazy semantic equivalence

Theorem 42.2 — Eager–lazy semantic equivalence.

Let

$$c_{\text{hist}}(w)$$

and

$$c_{\text{cent}}(w)$$

be the core configurations reached after the same binary prefix w by:

- the historical 2:1 machine;
- the centered 2:1 specialization.

Then

$$\boxed{\Theta(c_{\text{hist}}(w)) = \Theta(c_{\text{cent}}(w))}$$

for every prefix w.
On every transition family where the two tables differ,

$$\boxed{\delta_{\text{hist}}(c,a) = \mathcal{N}\big(\delta_{\text{cent}}(c,a)\big).}$$

Thus:

$$\boxed{\text{historical 2:1 = eager discharge,}}$$

while

$$\boxed{\text{centered 2:1 = lazy discharge.}}$$

The semantics is the same.
The normalization chronology is different.

42.9 Historical machine as an eager refinement

Corollary 42.3 — Historical eager refinement.

Start with the centered 2:1 specialization.

Retain every specialized transition except on the two reachable right-state matching families.

On exactly those families, replace the lazy successor by its canonical materialization $\mathcal{N}$.

The resulting core is the historical 2:1 machine.

Therefore the return from the general p:q construction is exact at the level of:

- control architecture;
- state potentials;
- WRITE arithmetic;
- shortest-demand semantics;

and becomes literal transition-table identity after choosing the historical eager normalization schedule.

Direct specialization alone is not claimed to be literal table identity.

The remaining difference is precisely the eager/lazy schedule choice.

42.10 Computational audit

The centered family was independently checked by enumerating its reachable transitions for every ordered coprime pair

$$1 \le p, q \le 20,$$

including

$$(1, 1).$$

The exact finite enumeration covers:

$$\boxed{\text{255 ratios,}}$$

$$\boxed{\text{1,909,191 distinct reached configurations,}}$$

and

$$\boxed{\text{3,738,864 binary edges.}}$$

It checked:

1. residual invariant;
2. residual-collision consistency;

3. two-cell qualitative visibility;
4. zero-residual correctness.

Result:

$$\boxed{\text{PASS.}}$$

These computations are consistency checks.
The symbolic theorems remain the proof.

42.11 The structural consequence of the $p{:}q$ generalization

The general machine has done more than extend 2:1.

It has exposed a structural nonuniqueness.

The semantic evolution is rigid:

$$\boxed{\text{one residual} \to \text{one minimum demand.}}$$

But the physical normalization chronology is not rigid:

$$\boxed{\text{same semantics} \not\Rightarrow \text{unique physical schedule.}}$$

This is the hinge of the remainder of the monograph.

The Structure Theory developed in the following Part explains why WRITE, SEARCH, and CANCEL are the primitive operational roles and separates the components forced by semantic arithmetic from those forced by one-top LIFO locality.

That is the beginning of the **Structure Theory**.

From this point onward the main sequence is uninterrupted:

$$\boxed{\text{Structure Theory} \to \text{Uniqueness/Lift Theory} \to \text{General Theory.}}$$

Part IV

Structural Pushdown Normalization: WSC, Lift Freedom, Causality, and Confluence

---

Part guide. **Question.** Which features of a correct normalizer are forced by semantics, which by one-top LIFO locality, and which are merely choices of physical schedule? **Main result.** WRITE–SEARCH–CANCEL factorization separates the unique semantic quotient from nonunique physical lifts, classifies causal schedules, protected interfaces, bounded recodings, and typed confluence. **Why here.** This Part converts one family of machines into structural objects suitable for abstraction. **Dependency.** It starts from the general $p{:}q$ machine of Part III and provides the concrete model for the General Theory of Part V.

Chapter 43

Semantic Arithmetic Before Representation

Chapter 38 ended with a fact that the original 2:1 machine could not reveal by inspection alone:

the same semantic demand can be normalized on different physical schedules.

The centered generalization is lazy on two historical right-state families.
The original machine is eager.
Both are correct.
The next task is therefore no longer to construct a machine.
It is to separate what is forced by arithmetic from what is forced by LIFO locality and what is merely a normalization choice.
This is the beginning of the Structure Theory.

43.1 Dominating kernel level

Let a prefix contain

$$a = N_0(w), \qquad b = N_1(w).$$

The primitive balanced kernel is

$$K = (q, p).$$

Define

$$\boxed{\kappa(w) = \max\left(\left\lceil \frac{a}{q} \right\rceil, \left\lceil \frac{b}{p} \right\rceil\right).}$$

Theorem 43.1 — Dominating kernel level.
The unique minimum completion demand is

$$\boxed{\mu(w) = \big(q\kappa(w) - a,\ p\kappa(w) - b\big).}$$

Equivalently,

$$\boxed{(a, b) + \mu(w) = \kappa(w)K.}$$

Proof.

If (x, y) completes w, then

$$q(b + y) = p(a + x).$$

Coprimality gives an integer k such that

$$a + x = qk, \qquad b + y = pk.$$

Thus

$$x = qk - a, \qquad y = pk - b.$$

Nonnegativity requires

$$k \geq \left\lceil \frac{a}{q} \right\rceil$$

and

$$k \geq \left\lceil \frac{b}{p} \right\rceil.$$

The least feasible k is therefore $\kappa(w)$.
The completion length is

$$(p + q)k - (a + b),$$

which is strictly increasing in k.
Hence the first dominating kernel level gives the unique minimum demand. □
Geometrically, κ is the first point

$$kK = (kq, kp)$$

on the kernel ray that dominates the current count point (a, b) coordinatewise.
The minimum demand is the slack vector from the current count point to that first dominating kernel point.

43.2 One input symbol: CANCEL plus optional carry

Let

$$e_0 = (1, 0), \qquad e_1 = (0, 1).$$

Write

$$\mu_t$$

and

$$\kappa_t$$

for the minimum demand and dominating kernel level after t input symbols.

Theorem 43.2 — Unique semantic CANCEL–carry factorization.
If the next input symbol is

$$a \in \{0, 1\},$$

then

$$\boxed{\mu_{t+1} = \mu_t - e_a + \chi_t K,}$$

where

$$\boxed{\chi_t = \kappa_{t+1} - \kappa_t \in \{0, 1\}.}$$

Moreover:

- on input 0,
$$\chi_t = 1 \iff \mu_t^{(0)} = 0;$$
- on input 1,
$$\chi_t = 1 \iff \mu_t^{(1)} = 0.$$

Proof.
Reading one symbol increments exactly one prefix-count coordinate.
Each ceiling in the formula for κ therefore changes by at most one.
Hence

$$\chi_t \in \{0, 1\}.$$

Before the input,

$$(N_0, N_1) + \mu_t = \kappa_t K.$$

After the input,

$$(N_0, N_1) + e_a + \mu_{t+1} = \kappa_{t+1} K.$$

Subtracting gives the factorization.
For input 0, the zero slack is

$$q\kappa_t - N_0.$$

The kernel level rises precisely when that coordinate is already saturated, i.e. when the zero-demand coordinate is 0.
The input-1 case is symmetric. □
Thus every semantic input step has exactly two roles:

$$\boxed{\text{CANCEL one unit of the incoming symbol}}$$

and, if the relevant demand coordinate would become negative,

$$\boxed{\text{add one primitive kernel packet.}}$$

This is realization-independent.
It is the semantic reason WRITE exists at all.

43.3 Semantic WRITE count

Every semantic carry increments κ by one.

Corollary 43.3 — Semantic WRITE count.
For every prefix,

$$\boxed{W_{\text{sem}}(w) = \kappa(w).}$$

If

$$w \in L_{p,q}$$

and

$$N_0(w) = qn, \qquad N_1(w) = pn,$$

then

$$\boxed{W_{\mathrm{sem}}(w) = n.}$$

The number of semantic kernel arrivals is fixed by the input.
What can vary is when those arrivals are physically materialized.

43.4 Demand-length clock

Let

$$r = p + q.$$

Corollary 43.4 — Demand-length clock.
At prefix length t,

$$\boxed{|\mu_t| = r\kappa_t - t.}$$

Hence

$$\boxed{|\mu_t| \equiv -t \pmod{r}.}$$

The minimum-demand length carries a rigid modular clock.
This clock will later have a physical analogue.

43.5 Semantic chronology is unique

Corollary 43.5 — Semantic chronology uniqueness.
For a fixed input word, the entire sequences

$$(\rho_t), \qquad (\kappa_t), \qquad (\mu_t), \qquad (\chi_t)$$

are uniquely determined by the input.
They do not depend on which correct pushdown representation is used.
This is the semantic backbone of the Structure Theory.
It says:

$$\boxed{\text{the semantic chronology is unique}}$$

before we say anything about:

- stack order;
- side states;
- SEARCH debt;
- eager/lazy materialization.

To see which part of this chronology survives one-top execution, we next impose an exact literal-demand lift and isolate the suffix obstruction created by LIFO locality.

Chapter 44

What One-Top LIFO Locality Forbids

We now formalize a physical realization of the unique demand dynamics.

The abstraction appears here, not at the beginning of the monograph, because the reader has already seen why it is needed.

44.1 Exact literal-demand lift

Let

$$\mathcal{M}_{p,q} = \{\mu(\rho) : \rho \in \mathbb{Z}\}$$

be the semantic minimum-demand system.

Definition 44.1 — Exact real-time literal-demand lift.

An exact literal-demand lift consists of:

- a reachable physical configuration set
$$\mathcal{C};$$
- deterministic binary transitions
$$\delta_a;$$
- a semantic projection
$$\Theta : \mathcal{C} \to \mathcal{M}_{p,q};$$
- a canonical section
$$\sigma : \mathcal{M}_{p,q} \to \mathcal{C};$$

such that

$$\boxed{\Theta \circ \sigma = \mathrm{id}}$$

and

$$\boxed{\Theta(\delta_a(c)) = U_a(\Theta(c)),}$$

where U_a is the unique semantic update from Chapter 39.
The section chooses one preferred physical representative of each semantic demand.
The other physical representatives form the semantic fiber.
That vocabulary will become central later.

44.2 Fixed block orientation

For the present Structure Theory, choose the literal canonical encoding

$$\boxed{\eta(x, y) = 1^y 0^x Z_0,}$$

with stack top on the left.
A one-top real-time transition has the form

$$\boxed{XT \longrightarrow \gamma T.}$$

The suffix T is immutable during that transition.
This one fact gives an exact direct-realizability criterion.

44.3 Top-rewrite suffix criterion

Theorem 44.2 — Top-rewrite suffix criterion.
Suppose a one-top transition begins from

$$XT.$$

A desired target word S^* can be reached in that one transition if and only if

$$\boxed{T \text{ is a suffix of } S^*.}$$

Proof.
Every one-top successor has the form

$$\gamma T.$$

Therefore direct realization requires T to be a suffix of the target.
Conversely, if

$$S^* = \gamma T,$$

replace X by γ. □
This theorem is one of the simplest and most important results in the entire structural line.
It converts one-top LIFO locality into an exact string condition.

44.4 Complete obstruction classification

Apply the suffix criterion to the canonical word

$$1^y 0^x Z_0.$$

Theorem 44.3 — Canonical direct-realizability classification.
For the fixed orientation $1^y 0^x Z_0$:

1. input 1 is directly canonical-realizable for every canonical demand;
2. input 0 fails to be directly canonical-realizable exactly in two families:

Buried-CANCEL obstruction

$$\boxed{x > 0, \qquad y > 0.}$$

Carry-order obstruction

$$\boxed{x = 0, \qquad y \geq 2, \qquad q \geq 2.}$$

For the reverse block orientation, exchange 0 and 1.

Proof — input 1. If $y > 0$, pop the top 1.
If $y = 0$ and $x > 0$, the semantic target is

$$1^{p-1}0^{x+q}Z_0.$$

The untouched suffix after removing the top zero is

$$0^{x-1}Z_0,$$

which is a suffix of the target.
Thus a single top replacement suffices.
The bottom case is direct as well.

Proof — input 0. If $y = 0$ and $x > 0$, pop the top 0.
Now let $x > 0, y > 0$.
The semantic target is

$$1^{y}0^{x-1}Z_0.$$

But after removing the top 1, the untouched suffix is

$$1^{y-1}0^{x}Z_0.$$

That suffix is not a suffix of the target.
Direct canonicalization is impossible.
Finally let $x = 0$.
The carry target is

$$1^{y+p}0^{q-1}Z_0.$$

For $y = 0$, write at the bottom.
For $y = 1$, the untouched suffix is empty and direct replacement is possible.
For $y \geq 2$, the untouched suffix is

$$1^{y-1}Z_0.$$

It is a suffix of the target exactly when

$$q - 1 = 0,$$

that is,

$$q = 1.$$

So for $q \geq 2$ the second obstruction family is exact. □

44.5 Two fundamentally different failures

The classification reveals that “cannot stay canonical” has two different causes.

Buried cancellation

The symbol that semantic CANCEL wants to remove exists in the physical word, but a blocker lies above it.

This is a visibility/locality failure.

Carry-order obstruction

A semantic carry can create the correct **number** of zeros and ones in one transition, yet the immutable suffix forces those letters into the wrong order.

This is not a quantity failure.

It is an ordering failure.

Thus:

$$\boxed{\text{count-exact} \not\Rightarrow \text{canonical.}}$$

This failure of direct canonicalization is the source of representation debt; the following analysis separates that debt into quantity, order, and finite-control channels.

Chapter 45

Quantity, Order, and Control Debt

A configuration may leave the canonical section in more than one way.
Parikh counts see only one of them.
The Structure Theory therefore distinguishes:

1. quantity debt;
2. order debt;
3. control/sheet debt.

These channels will remain separate through the Uniqueness and General Theory Parts.

45.1 Quantity defect

Let $S(c)$ be the literal work-stack word of a rich configuration c.
Define its physical Parikh vector

$$P(c) = \mathrm{Par}(S(c)).$$

Definition 45.1 — Quantity defect.

$$\boxed{Q(c) = \Theta(c) - P(c).}$$

A canonical configuration has

$$Q = 0.$$

But $Q = 0$ does not imply canonicality because Parikh counts forget literal order.

45.2 Universal representation conservation

Proposition 45.2 — Universal representation conservation.
For every exact transition on input a,

$$\boxed{\Delta P + \Delta Q = -e_a + \chi K.}$$

Proof.
Since

$$Q = \Theta - P,$$

$$\Delta Q = \Delta\Theta - \Delta P.$$

The semantic update is

$$\Delta\Theta = -e_a + \chi K.$$

Rearrange. □
This equation holds before we impose the later WSC lattice restrictions.
It is the universal accounting identity.

45.3 Order defect

For literal word

$$S = S_1 \cdots S_m,$$

define

$$\boxed{\Omega(S) = \#\{(i,j) : i < j,\ S_i = 0,\ S_j = 1\}.}$$

Lemma 45.3 — Block-order test.

$$\boxed{\Omega(S) = 0 \iff S \in 1^*0^*.}$$

Thus Ω measures departure from the chosen canonical block order.

45.4 Three-channel canonicality

A third channel is finite control.
A configuration can have:

- correct counts;
- correct literal order;

but still lie on a noncanonical control sheet.

Proposition 45.4 — Three-channel canonicality.
A rich configuration is canonical exactly when:

1. quantity defect is zero;
2. order defect is zero;
3. the control state is the canonical section state.

Thus:

$$\boxed{\text{canonicality} = \text{quantity} + \text{order} + \text{sheet/control}.}$$

This is the first mature answer to the question:

Where can normalization debt live?

Proposition 45.4A — Configuration-level separation of the three defect channels.

At the level of exact semantic representations used in this theory, none of the three canonicality channels is implied by the other two.

Each channel has a witness for which the other two vanish.

- **Quantity-only defect.** In the 2:1 schedule interaction studied later, the kernel-excess configuration
$$(C, 1110)$$
is compared with canonical
$$(C, 1).$$
The control sheet and block order are canonical, but the quantity defect is one nonzero kernel packet.

- **Order-only defect.** The carry-order repair of Theorem 45.6 preserves the exact canonical Parikh vector while forcing positive inversion count
$$\Omega > 0.$$

- **Control-only defect.** Take any exact literal lift and adjoin a semantically ignored two-sheet control decoration. A balanced input can return the base machine to a canonical stack while leaving the auxiliary sheet noncanonical. Quantity and order are then canonical while control defect remains nonzero.

Therefore
$$\boxed{Q = 0, \quad \Omega = 0, \quad \text{canonical control}}$$
are genuinely independent physical requirements.

This point becomes important in the Uniqueness Theory: a count-shadow invariant alone cannot reconstruct the full physical semantic fiber.

45.5 Kernel economy

Define a literal-demand representation to be **kernel-economical** if
$$\boxed{|P(c)| \le |\Theta(c)|}$$
for every reachable configuration.

This prohibits adding complete zero-residual kernel packets merely as redundant workspace.

Under economy, the buried-CANCEL obstruction forces an exact quantity debt.

45.6 Forced unit SEARCH

Theorem 45.5 — Forced unit SEARCH for buried cancellation.

Assume:

- one-top real-time locality;
- kernel economy;
- canonical mixed demand
$$\Theta(c) = (x, y)$$
with
$$x > 0, \quad y > 0;$$
- literal encoding
$$1^y 0^x Z_0.$$

On input 0, the one-step successor has

$$\boxed{Q' = (-1, 1) = -(1, -1).}$$

Proof.

The semantic target is

$$(x - 1, y),$$

whose length is

$$x + y - 1.$$

A one-top transition can replace only the blocking top 1 by some word γ.
The successor physical length is

$$x + y - 1 + |\gamma|.$$

Kernel economy requires this to be at most the semantic target length.
Hence

$$|\gamma| = 0.$$

The top 1 must be popped.
The physical successor counts are

$$(x, y - 1),$$

while the semantic target is

$$(x - 1, y).$$

Therefore

$$Q' = (-1, 1).$$

□

Thus SEARCH is not merely one convenient implementation trick.

Under the stated locality and economy assumptions, the buried cancellation produces exactly one forced exchange defect.

45.7 Carry obstruction with zero quantity debt

The second obstruction behaves differently.

Assume:

$$x = 0, \qquad y \geq 2, \qquad q \geq 2.$$

On input 0, the semantic target is

$$1^{y+p} 0^{q-1} Z_0.$$

A one-top transition can replace the top 1 by

$$\gamma = 1^{p+1} 0^{q-1}.$$

The resulting word is

$$1^{p+1} 0^{q-1} 1^{y-1} Z_0.$$

Theorem 45.6 — Count-exact carry-order repair.

The successor has exactly the target Parikh counts:

$$\boxed{Q' = 0.}$$

But it has nonzero order defect.
Thus the representation can be semantically count-exact and still noncanonical.

45.8 Minimum order debt

Theorem 45.7 — Minimum carry-order debt.

Under the same hypotheses, among all one-top successors with exact target counts,

$$\boxed{\Omega_{\min} = (q-1)(y-1).}$$

Proof.
The untouched suffix is

$$1^{y-1}Z_0.$$

Exact target counts force the replacement word to contain:

$$q-1$$

zeros and

$$p+1$$

ones.
Every one of those $q-1$ zeros lies before all $y-1$ untouched suffix ones.
Therefore at least

$$(q-1)(y-1)$$

inversions are forced.
The replacement

$$1^{p+1}0^{q-1}$$

introduces no additional inversions inside the replacement and attains the bound. □
The obstruction/debt principle is now clear:

$$\boxed{\text{LIFO obstruction forces departure from the canonical section,}}$$

but the debt may be stored in:

- quantity;
- order;
- finite control;
- or several channels at once.

With the three debt channels separated, we can now freeze a count-shadow policy and ask which part of the physical arithmetic becomes rigid.

Chapter 46

The WSC Count Shadow

The preceding debt theory was general within the literal-demand setting.

We now impose an explicit representation policy to obtain a sharper arithmetic structure.

This policy is not claimed to describe every correct DPDA.

The scope is part of the theorem.

Set

$$\boxed{\mathbf{h} = (1, -1), \qquad K = (q, p), \qquad r = p + q.}$$

46.1 Phase coherence

Definition 46.1 — Phase coherence.

A reachable configuration is phase-coherent if

$$\boxed{|P(c)| \equiv |\Theta(c)| \pmod{r}.}$$

A realization is phase-coherent if every reachable configuration satisfies this congruence.

Since

$$Q = \Theta - P,$$

phase coherence is equivalent to

$$Q_0 + Q_1 \equiv 0 \pmod{r}.$$

This congruence produces the WSC lattice.

46.2 WSC defect lattice

Define

$$\boxed{\mathcal{D}_{p,q} = \mathbb{Z}\mathbf{h} + \mathbb{Z}K.}$$

Theorem 46.2 — WSC defect lattice.

$$\boxed{\mathcal{D}_{p,q} = \{(u, v) \in \mathbb{Z}^2 : u + v \equiv 0 \pmod{r}\}.}$$

Consequently every phase-coherent quantity defect has a unique decomposition

$$\boxed{Q = s\mathbf{h} + wK, \qquad s, w \in \mathbb{Z}.}$$

Proof.

Both generators have coordinate sum divisible by r:

$$(\mathbf{h})_0 + (\mathbf{h})_1 = 0,$$

$$K_0 + K_1 = r.$$

Conversely, if

$$u + v \equiv 0 \pmod{r},$$

define

$$\boxed{w = \frac{u+v}{r},}$$

and

$$\boxed{s = \frac{pu - qv}{r}.}$$

The first is integral by assumption.
Since

$$p \equiv -q \pmod{r},$$

the numerator of s is also divisible by r.
Direct substitution gives

$$(u, v) = s\mathbf{h} + wK.$$

Uniqueness follows because

$$\det(\mathbf{h}, K) = p + q = r \neq 0.$$

□

46.3 Defect coordinates

Define

$$\Phi(Q) = pQ_0 - qQ_1,$$

$$\Lambda(Q) = Q_0 + Q_1.$$

Corollary 46.3 — Defect coordinates.

For

$$Q = s\mathbf{h} + wK,$$

$$\boxed{\Phi(Q) = rs,}$$

$$\boxed{\Lambda(Q) = rw.}$$

Hence

$$\boxed{s = \frac{\Phi(Q)}{r}, \qquad w = \frac{\Lambda(Q)}{r}.}$$

The coordinate s measures the SEARCH/exchange sheet in the quantity shadow.
The coordinate w measures deferred kernel quantity.

46.4 Defect quotient

Corollary 46.4 — Defect quotient.

$$\boxed{\mathbb{Z}^2/\mathcal{D}_{p,q} \cong \mathbb{Z}_r.}$$

The homomorphism is

$$(u, v) \mapsto u + v \pmod{r}.$$

This quotient is a useful arithmetic shadow.
It is not the physical semantic fiber.

46.5 WSC lattice is not the full LIFO fiber

The map

$$S \mapsto \mathrm{Par}(S)$$

forgets stack order.
The quantity defect also forgets finite-control sheets.
Therefore two physical configurations can have the same coordinates

$$(s, w)$$

while having:

- different literal stack words;
- different control states;
- different future locality properties.

Thus:

$$\boxed{\text{WSC defect lattice} \neq \text{full physical representation fiber.}}$$

This warning is essential.
Chapter 46 onward will develop the missing lift/fiber theory.

46.6 Kernel economy in lattice coordinates

Since

$$|\Theta| - |P| = Q_0 + Q_1 = rw,$$

we get:

Corollary 46.5 — Kernel economy.

Within the phase-coherent class,

$$\boxed{\text{kernel economy} \iff w \geq 0.}$$

Thus the deferred-kernel coordinate acquires a direct nonnegative backlog interpretation under economy.

46.7 Abstract count-shadow fiber region

Suppose the semantic demand is

$$\Theta(c) = \mu = (x, y)$$

and

$$Q = s\mathbf{h} + wK.$$

Then

$$P = \mu - Q = (x, y) - s(1, -1) - w(q, p).$$

So

$$P_0 = x - s - qw,$$

$$P_1 = y + s - pw.$$

Proposition 46.6 — Count-shadow fiber region.

Physical nonnegativity is equivalent to

$$\boxed{pw - y \le s \le x - qw,}$$

with

$$\boxed{0 \le w \le \left\lfloor \frac{x+y}{r} \right\rfloor.}$$

For fixed w, the number of integer s-positions is

$$\boxed{x + y - rw + 1.}$$

The count-shadow fiber therefore has a finite triangular/layered shape over each semantic demand.

Again, this is an abelian shadow of the full physical fiber, not the complete fiber itself.

These coordinates determine how much kernel mass exists, but not when it is physically written. Materialization timing is therefore the remaining variable.

Chapter 47

Physical Materialization, Backlog, and WSC Factorization

The semantic carry schedule

$$\chi_t$$

is unique.
The physical machine may materialize those kernel packets at different times.
We now make that chronology explicit.
Let

$$Q_t = s_t h + w_t K$$

along a phase-coherent run.

47.1 Materialization level

Definition 47.1 — Physical materialization level.
Define

$$\boxed{\lambda_t = \kappa_t - w_t.}$$

The semantic system has generated κ_t kernel packets.
The quantity shadow still defers w_t packets.
Therefore λ_t is the number of kernel packets already materialized physically at the count-shadow level.

47.2 Physical length formula

Theorem 47.2 — Physical length formula.
At prefix length t,

$$\boxed{|P_t| = r\lambda_t - t.}$$

Proof.

$$|P_t| = |\Theta_t| - (Q_{t,0} + Q_{t,1}).$$

Chapter 39 gives

$$|\Theta_t| = r\kappa_t - t.$$

Chapter 42 gives

$$Q_{t,0} + Q_{t,1} = rw_t.$$

Subtract and use

$$\lambda_t = \kappa_t - w_t.$$

□

The physical stack length therefore has its own modular clock:

$$\boxed{|P_t| \equiv -t \pmod{r}.}$$

47.3 Materialization amount

Define

$$\boxed{M_t = \lambda_{t+1} - \lambda_t.}$$

Theorem 47.3 — Materialization amount from stack length.

For every phase-coherent one-top transition,

$$\boxed{M_t = \frac{\Delta|P_t| + 1}{r}.}$$

In particular,

$$\boxed{M_t \in \mathbb{N}.}$$

Proof.

Subtract the physical length formula at t and $t+1$:

$$\Delta|P_t| = rM_t - 1.$$

Thus the formula follows.

Because a one-top rewrite can delete at most the visible literal cell,

$$\Delta|P_t| \geq -1.$$

Hence

$$M_t \geq 0.$$

□

The number of kernel packets physically materialized by a transition can therefore be reconstructed from the physical stack-length change alone.

47.4 Backlog law

Theorem 47.4 — Backlog law.

$$\boxed{w_{t+1} - w_t = \chi_t - M_t.}$$

Equivalently, if

$$A_t = \kappa_t$$

is cumulative semantic arrival and

$$B_t = \lambda_t$$

is cumulative physical materialization, then

$$\boxed{w_t = A_t - B_t.}$$

This is a queue-like law:

$$\boxed{\text{backlog change} = \text{semantic arrivals} - \text{physical service.}}$$

The analogy is structural, not stochastic.

47.5 Canonical carry dichotomy

Assume phase coherence and kernel economy, and begin a transition from canonical quantity state

$$w_t = 0.$$

Theorem 47.5 — Canonical carry dichotomy.

If

$$\chi_t = 0,$$

then

$$\boxed{M_t = 0.}$$

If

$$\chi_t = 1,$$

then

$$\boxed{M_t \in \{0, 1\}.}$$

Thus a semantic carry is either:

$$\boxed{\text{eagerly materialized}}$$

or

$$\boxed{\text{deferred as exactly one kernel packet of backlog.}}$$

This is the structural generalization of the eager/lazy distinction discovered concretely in Chapter 38.

47.6 Unique physical WSC role factorization

Let

$$\alpha_t = s_{t+1} - s_t.$$

Theorem 47.6 — Unique physical WSC count-shadow factorization.

Every phase-coherent one-top transition on input a satisfies

$$\boxed{\Delta P = -e_a - \alpha_t \mathbf{h} + M_t K.}$$

The coefficients

$$\alpha_t \in \mathbb{Z}$$

and

$$M_t \in \mathbb{N}$$

are unique.

Proof.

The defect change is

$$\Delta Q = \alpha_t \mathbf{h} + (w_{t+1} - w_t)K.$$

Use the backlog law:

$$\Delta Q = \alpha_t \mathbf{h} + (\chi_t - M_t)K.$$

Insert this into the universal conservation identity

$$\Delta P + \Delta Q = -e_a + \chi_t K.$$

The $\chi_t K$ terms cancel:

$$\Delta P = -e_a - \alpha_t \mathbf{h} + M_t K.$$

Uniqueness follows from linear independence of $\mathbf{h}$ and K. □

Meaning

This is a **role decomposition**, not a syntactic transition taxonomy.

The terms mean:

$$-e_a$$

= semantic CANCEL forced by the input;

$$-\alpha \mathbf{h}$$

= SEARCH/exchange effect in the abelian quantity shadow;

$$MK$$

= physically materialized kernel packets.

One concrete PDA transition may combine several of these roles.

47.7 Reconstructing the charges

Corollary 47.7 — Reconstruction.

$$\boxed{M = \frac{\Delta|P| + 1}{r},}$$

and

$$\boxed{\alpha = -\frac{p(\Delta P_0 + e_{a,0}) - q(\Delta P_1 + e_{a,1})}{r}.}$$

Thus the WSC charges can be recovered from the physical count effect.

Corollary 47.7A — General physical phase clock.

Every phase-coherent run satisfies

$$\boxed{|P_t| \equiv -t \pmod{r}.}$$

This follows immediately from

$$|P_t| = r\lambda_t - t.$$

The semantic demand clock of Chapter 39 and the physical stack clock therefore have the same modulus $r = p + q$, even though the physical materialization schedule may be eager, lazy, or mixed.

47.8 Global materialization budget

Let an accepted word have

$$N_0 = qn, \qquad N_1 = pn.$$

Theorem 47.8 — Global materialization budget.

For a phase-coherent kernel-economical run starting canonically,

$$\boxed{\sum_t M_t = n.}$$

This equals the total semantic carry count:

$$\boxed{\sum_t \chi_t = n.}$$

Proof.

The accepted input length is

$$rn.$$

At acceptance, semantic demand is zero.
Kernel economy forces the physical literal count to zero as well.
The physical length formula gives

$$0 = r\lambda_{rn} - rn.$$

Hence

$$\lambda_{rn} = n.$$

Since $\lambda_0 = 0$,

$$\sum_t M_t = n.$$

The semantic equality follows from Chapter 39. □
The schedule may move materialization earlier or later.
The total materialized kernel budget cannot change.

47.9 Deferral capacity

Corollary 47.9 — Deferral capacity bound.

At prefix length t,

$$\boxed{\left\lceil \frac{t}{r} \right\rceil \leq \lambda_t \leq \kappa_t,}$$

and

$$\boxed{0 \leq w_t \leq \kappa_t - \left\lceil \frac{t}{r} \right\rceil .}$$

Thus the physical representation has an exact finite capacity for delaying kernel materialization at a given semantic prefix.

Once materialization timing has been separated from semantic count, the natural question is uniqueness: which coordinates belong to the semantic quotient, and which belong only to a chosen physical presentation?

Chapter 48

Unique Semantic Quotient, Nonunique Physical Presentation

The Structure Theory has now separated:

- the unique semantic demand trajectory;
- the physical normalization trajectory.

We can state that distinction as a quotient.

48.1 Semantic fibers

For semantic demand

$$m \in \mathcal{M}_{p,q},$$

define

$$\boxed{F_m = \Theta^{-1}(m).}$$

Two rich configurations are semantically equivalent when

$$\boxed{c \sim c' \iff \Theta(c) = \Theta(c').}$$

A fiber can contain several physical configurations representing the same minimum demand.

48.2 Fiber congruence

Theorem 48.1 — Fiber congruence and semantic quotient.
In every exact literal-demand lift, the relation $\sim$ is preserved by binary input transitions. Hence

$$\boxed{\mathcal{C}/\sim \cong \mathcal{M}_{p,q}.}$$

Proof.
If

$$\Theta(c) = \Theta(c'),$$

then exactness gives

$$\Theta(\delta_a(c)) = U_a(\Theta(c)) = U_a(\Theta(c')) = \Theta(\delta_a(c')).$$

Thus successors remain in the same semantic fiber.

The quotient state is therefore exactly the common semantic demand, and the quotient transitions are the canonical demand updates. □

This is a concrete precursor of the machine-independent semantic quotient theorem developed below.

The stronger language-level statement will come in the Uniqueness Theory.

48.3 Canonical retraction

Define

$$\boxed{N = \sigma \circ \Theta.}$$

This sends every physical configuration to the canonical section representative of its semantic state.

Proposition 48.2 — Canonical absorption.

$$\boxed{N^2 = N.}$$

Moreover, for every input word w,

$$\boxed{N \circ \delta_w = N \circ \delta_w \circ N.}$$

Proof.

Since

$$\Theta \circ \sigma = \mathrm{id},$$

$$N^2 = \sigma\Theta\sigma\Theta = \sigma\Theta = N.$$

For one symbol,

$$N\delta_a = \sigma\Theta\delta_a = \sigma U_a\Theta.$$

Also,

$$N\delta_a N = \sigma\Theta\delta_a\sigma\Theta = \sigma U_a\Theta\sigma\Theta = \sigma U_a\Theta.$$

Induct on $|w|$. □

Interpretation:

> Once we normalize to the canonical section, earlier physical normalization history is erased.

This is a structural form of the eager/lazy semantic equivalence.

48.4 What is actually unique

The following are semantic and unique:

- residual;
- minimum demand;
- dominating kernel level;
- semantic carry indicator;
- canonical quotient path.

The following need not be unique:

- physical stack word;
- finite-control sheet;
- timing of materialization;
- normalization schedule;
- full pushdown presentation.

Thus the correct statement is:

unique semantic core; nonunique physical normalization lift.

48.5 Cheap nonuniqueness already exists

Proposition 48.3 — Unrestricted physical presentation nonuniqueness.

The canonical semantic quotient does not determine a unique deterministic pushdown presentation.

Proof.

Start with any exact deterministic lift with finite control Q.

For integer

$$m \geq 2,$$

adjoin an independent cyclic coordinate

$$j \in \mathbb{Z}_m.$$

On every consumed input symbol, update

$$j \mapsto j + 1 \pmod{m}.$$

Let:

- stack dynamics;
- semantic projection;
- acceptance

ignore the extra coordinate.

The result is still an exact deterministic lift.

Its control set is

$$Q \times \mathbb{Z}_m.$$

For infinitely many suitable pairwise distinct m, these presentations have different finite-control cardinalities and are therefore non-isomorphic, while their semantic quotient is unchanged. □

This proposition proves physical presentation nonuniqueness.

But it is deliberately cheap.

It uses redundant control decoration.

It does **not** by itself distinguish genuine reduced presentations from redundant decoration. Chapters 61–62 supply the protected quotient/recoding framework that removes this ambiguity and proves existence of minimal protected cores. The conclusion used here is existence and minimality under the protected quotient/recoding framework, not a Church–Rosser assertion for arbitrary terminal protected cores.

48.6 Scope of the confluence claim

Canonical absorption says that normalizing after a fixed input word forgets earlier physical history.

It establishes endpoint coherence for the chosen normalizer. A Church–Rosser theorem for arbitrary rewrite schedules would be a different claim because the present framework fixes a deterministic normalizer rather than an arbitrary-choice rewrite relation.

Therefore the safe conclusion is:

canonical endpoint coherence under the chosen normalizer

not

global confluence.

This scope distinction is used immediately in the lift-comparison chapters below and again in the General Theory of Part V.

Chapter 49

The Original 2:1 Machine as WSC

The Structure Theory now returns to the original machine for a second reinterpretation.

Chapter 38 explained the historical transition table as an eager schedule of the centered 2:1 construction.

The present chapter goes deeper.

It interprets:

- the historical side states as SEARCH sheets;
- the historical stack-length changes as materialization bursts;
- an old transition-count identity as a global kernel budget law.

Set

$$\boxed{p = 2, \qquad q = 1, \qquad r = 3, \qquad K = (1, 2).}$$

49.1 Historical states as SEARCH sheets

The original state potentials were

$$\Phi(C) = 0,$$

$$\Phi(L) = 3,$$

$$\Phi(R) = -3.$$

Since the WSC sheet coordinate s contributes residual quantity

$$rs = 3s,$$

we obtain:

Corollary 49.1 — Historical states as SEARCH sheets.

$$\boxed{s(C) = 0, \qquad s(L) = 1, \qquad s(R) = -1.}$$

Thus the original $C/L/R$ architecture is exactly a three-sheet realization of the WSC exchange coordinate.

The side states are not merely "left" and "right."

They are the ± 1 SEARCH sheets.

49.2 Historical stack-length changes

For $r = 3$, the physical materialization formula is

$$\Delta|P| = 3M - 1.$$

Therefore:

$$M = 0 \Longrightarrow \Delta|P| = -1,$$

$$M = 1 \Longrightarrow \Delta|P| = +2,$$

$$M = 2 \Longrightarrow \Delta|P| = +5.$$

Corollary 49.2 — Historical stack actions as materialization bursts.
The characteristic historical stack-length changes

$$\boxed{-1, \qquad +2, \qquad +5}$$

are exactly the WSC materialization levels

$$\boxed{M = 0, \qquad M = 1, \qquad M = 2.}$$

Therefore a $+5$ transition is not an anomalously long push.
It is a two-kernel-packet materialization burst.
This explains why some original transition replacements were much longer than others.

49.3 The global identity $b + 2c = n$

Suppose an accepted 2:1 run has:

- b transitions with stack-length change $+2$;
- c transitions with stack-length change $+5$.

Those transitions carry materialization amounts:

$$1 \qquad \text{and} \qquad 2.$$

The global materialization budget says that an accepted word with

$$N_0 = n, \qquad N_1 = 2n$$

must satisfy

$$\sum_t M_t = n.$$

Hence:

Corollary 49.3 — Structural derivation of the historical identity.

$$\boxed{b + 2c = n.}$$

The identity is no longer a transition-table counting curiosity.
It is the 2:1 specialization of the universal materialization budget.

49.4 What Structure Theory has accomplished

The generalization and Structure Theory now give three successive interpretations of the same original machine.

Chapter 1

$$\boxed{\text{state+stack residual invariant.}}$$

Chapter 38

$$\boxed{C/L/R = \text{centered } p{:}q \text{ debt rail at } (2,1),}$$

with historical behavior as eager discharge.

Chapter 45

$$\boxed{C/L/R = \text{WSC SEARCH sheets,}}$$

and the physical stack actions are kernel-materialization bursts.
This repeated return to the original machine is intentional.
Each return explains something that was previously only observed.

49.5 The Structure Theory endpoint

The strongest conclusion is:

$$\boxed{\text{CANCEL and semantic WRITE are forced by arithmetic;}}$$

$$\boxed{\begin{array}{c}\text{SEARCH and normalization debt arise when that semantic update meets}\\ \text{one-top LIFO locality.}\end{array}}$$

The Structure Theory has shown that physical presentations are nonunique, but the examples so far do not yet separate genuine lift freedom from redundant decoration.
The next branch therefore asks the sharper question:

> After fixing the semantic quotient, the canonical section, and an admissible LIFO morphism category, can two **reduced** physical lifts still be genuinely different?

The answer is yes. The next four chapters make that statement precise: they derive the semantic core from the language, define exact sectioned lifts and category-relative minimality, isolate forced fiber branching, and exhibit genuinely distinct reduced 2:1 lifts. Only after that reduction step does the schedule enumeration begin.
That is the Uniqueness/Lift Theory.

Chapter 50

The Semantic Core Belongs to the Language

Research branch — Semantic Uniqueness and Physical Lift Freedom. *This branch develops the language-level semantic quotient, exact sectioned lifts, reduced-lift nonuniqueness, and causal schedule analysis. The terminology is kept separate from classical VPL congruence/minimization and DPDA equivalence results [2, 8, 21].*

The Structure Theory established a canonical semantic quotient inside exact literal-demand lifts.

The Uniqueness Theory strengthens that result.

The semantic core will now be derived from the **language itself**, not from a chosen WSC construction.

50.1 Right residuals

For a prefix u, define the right-continuation language

$$u^{-1}L_{p,q} = \{z : uz \in L_{p,q}\}.$$

Chapter 4 already proved that this continuation language depends only on the arithmetic residual

$$\rho(u) = qN_1(u) - pN_0(u).$$

We now promote that result to the canonical semantic theorem.

Theorem 50.1 — Right-residual classification.

For all prefixes u, v,

$$\boxed{u^{-1}L_{p,q} = v^{-1}L_{p,q} \iff \rho(u) = \rho(v).}$$

Every integer residual occurs.

Thus the reachable right-residual classes are canonically indexed by

$$\mathbb{Z}.$$

The semantic transition system is therefore

$$\boxed{\mathcal{R}_{p,q}: \qquad r \xrightarrow{0} r - p, \qquad r \xrightarrow{1} r + q.}$$

This is the language-determined behavioral core.

Corollary 50.1A — Canonical behavioral core.

The reachable right-residual classes of $L_{p,q}$ are canonically identified with

$$\boxed{\mathbb{Z}},$$

and their induced labeled transition system is exactly

$$\boxed{\mathcal{R}_{p,q}: \quad r \xrightarrow{0} r-p, \qquad r \xrightarrow{1} r+q.}$$

Thus the integer residual line is not a coordinate convention inherited from a particular DPDA. It is the canonical continuation-behavior core of the language.

50.2 Universal semantic quotient

Consider any correct deterministic real-time recognizer of $L_{p,q}$.

Look only at reachable configurations at input boundaries.

Map every such physical configuration c, reached by prefix u, to

$$\boxed{\Theta_{\mathrm{res}}(c) = \rho(u).}$$

Determinism and correctness make this map well defined on behavioral equivalence classes.

Theorem 50.2 — Universal semantic quotient.

Every correct deterministic real-time recognizer of $L_{p,q}$ admits a unique surjective transition-preserving quotient onto

$$\boxed{\mathcal{R}_{p,q}.}$$

Thus the residual system is not merely the semantic quotient of our chosen constructions.

It is the canonical behavioral quotient forced by the language.

This strengthens Chapter 44.

50.3 Minimum-demand semantics is a coordinate change

The minimum-demand system

$$\mathcal{M}_{p,q}$$

has one state

$$\mu(r)$$

for every residual r.

The map

$$\boxed{r \longmapsto \mu(r)}$$

is bijective.

It also intertwines the input transitions.

Theorem 50.3 — WSC semantic core equals the residual core.

$$\boxed{\mathcal{R}_{p,q} \cong \mathcal{M}_{p,q}.}$$

Therefore:

$$\boxed{\text{language semantics} = \text{right-residual system},}$$

while

$$\boxed{\text{completion semantics} = \text{a useful isomorphic coordinate system.}}$$

This distinction is conceptually important.

The minimum-demand representation was discovered first because it grew naturally out of the original DPDA.

But the language-level semantic object is the residual quotient.

50.4 Three levels of description

We can now separate:

Language level

$$\mathcal{R}_{p,q}.$$

This is canonical.

Semantic coordinate level

$$\mathcal{M}_{p,q}.$$

This is isomorphic to the residual system.

Physical lift level

A DPDA realization mapping onto that semantic system.

This level is not unique.

The rest of the Uniqueness Theory asks how to compare such lifts without quotienting away the very pushdown structure we want to study.

Chapter 51

Exact Sectioned Lifts and Category-Relative Minimality

If arbitrary behavioral quotienting is allowed, every correct machine collapses to the same residual core.

That is too coarse for physical comparison.

We need a category that remembers:

- the chosen semantic projection;
- the canonical section;
- the LIFO coordinates.

51.1 Reachable versus ambient configurations

Let

$$C^{\mathrm{amb}}$$

be all syntactically possible physical configurations.

Let

$$\boxed{C^{\mathrm{reach}} \subseteq C^{\mathrm{amb}}}$$

be the configurations actually reachable from the initial condition.

All minimality, fiber, and schedule claims in this branch concern reachable configurations unless explicitly stated otherwise.

This prevents unreachable syntactic artifacts from distorting the theory.

51.2 Exact sectioned lift

Definition 51.1 — Exact sectioned lift.

An exact sectioned lift is

$$\boxed{\mathcal{L} = (C^{\mathrm{reach}}, \delta, \Theta, \sigma)}$$

such that:

1. Θ is a surjective semantic projection;
2. σ is a reachable section;
3. $$\Theta \circ \sigma = \mathrm{id};$$

4. physical transitions commute exactly with semantic transitions.

The section is part of the data.

Two lifts with the same quotient but different preferred physical representatives are therefore not automatically identified.

51.3 Section-preserving morphisms

A morphism

$$F : \mathcal{L}_1 \to \mathcal{L}_2$$

is section-preserving when

$$\boxed{\Theta_2 F = \Theta_1,}$$

$$\boxed{F\delta_a^{(1)} = \delta_a^{(2)} F,}$$

and

$$\boxed{F\sigma_1 = \sigma_2.}$$

For concrete pushdown comparison, a **coordinate LIFO morphism** must also respect the finite-control and stack coordinates letterwise rather than replacing the pushdown by an arbitrary behavioral quotient.

This is the correct category for asking whether two physical lifts are genuinely the same representation.

51.4 Minimality is category-relative

Proposition 51.2 — Category-relative minimality.

There is no meaningful representation-independent notion of a "minimal physical lift" without specifying:

- the admissible physical policy;
- the allowed morphisms;
- whether the canonical section is preserved;
- which stack/control coordinates may be identified.

The same semantic system can have different minima in different categories.

This is not a weakness of the theory.

It is the same lesson already encountered in the resource frontiers:

physical optimality is meaningful only relative to a representation category.

With the comparison category fixed, we can now ask what one-top locality forces inside one literal policy rather than across arbitrary presentations.

Chapter 52

When LIFO Locality Forces Fiber Branching

The Structure Theory proved the top-rewrite suffix criterion.

We now use it at the level of an entire observation fiber.

52.1 Uniform local rule

A deterministic one-top transition cannot inspect the hidden stack suffix.

Therefore all reachable configurations sharing the same locally visible cell must be handled by the same rule.

Theorem 52.1 — Uniform one-top rewrite criterion.

Fix one deterministic local observation class.

A single one-top rule can send every source in that class to its canonical semantic target if and only if the untouched suffix of every source is compatible with that source's canonical target under the suffix criterion.

This theorem is the fiberwise form of Chapter 40.

The obstruction is no longer one bad stack.

It is incompatibility across a family of sources that the local rule cannot distinguish.

52.2 Canonical-section closure dichotomy

Use the fixed literal section

$$\boxed{\sigma(x, y) = (C, 1^y 0^x Z_0).}$$

Theorem 52.2 — Canonical-section closure dichotomy.

The canonical literal section is transition-closed under one-top real-time updates if and only if

$$\boxed{(p, q) = (1, 1).}$$

Meaning. For the balanced one-to-one language, the literal section can remain canonical after every input step.

For every nontrivial coprime ratio, some exact semantic successor cannot be realized while staying on the canonical section.

Thus the use of noncanonical physical representatives is not merely a feature of our centered construction.

Inside this literal one-top policy, it is unavoidable.

52.3 Essential physical fiber branching

Corollary 52.3 — Essential branching.

For every nontrivial ratio

$$(p, q) \neq (1, 1),$$

every exact one-top realization of the fixed literal policy must reach some noncanonical physical configuration in the same semantic fiber as a canonical one.

Therefore:

nontrivial semantic fibers are forced physically

inside the policy.

This is a much stronger statement than:

> "our machine happens to use side states."

It says the canonical section itself cannot be closed.

52.4 Quotient-protected branching

Could one simply identify the noncanonical representative with the canonical one?

Not if the quotient must preserve legal one-top transitions in the same policy.

Corollary 52.4 — Quotient-protected branching.

The forced branch cannot be quotiented away while preserving:

- the fixed literal section;
- the one-top LIFO policy;
- exact transition behavior.

Thus some physical branching is structurally protected inside the representation category.

But the claim remains policy-relative.

Chapter 56 will show that a broader residual re-coordination can remove this branching entirely.

Chapter 53

Two Reduced 2:1 Lifts

We now revisit the eager and lazy 2:1 machines.

Chapter 38 proved semantic equivalence.

The Structure Theory explained their chronology.

The Uniqueness Theory asks whether one is merely a redundant presentation of the other.

The answer is no.

53.1 Historical eager lift

The historical machine eagerly materializes two reachable right-state matching families.

It returns immediately to the canonical center representation.

53.2 Centered lazy lift

The centered specialization leaves those same semantic debts on the right sheet and postpones materialization.

Both lifts use:

- the same semantic residual core;
- the same three control labels C, L, R;
- the same two work symbols;
- the same canonical section.

Their difference is temporal.

53.3 Section incomparability

Theorem 53.1 — Eager–lazy section incomparability.

There is no section-preserving transition morphism

$$\mathcal{L}_{\text{eager}} \to \mathcal{L}_{\text{lazy}}$$

and no such morphism

$$\mathcal{L}_{\text{lazy}} \to \mathcal{L}_{\text{eager}}.$$

The two lifts are therefore genuinely incomparable in the chosen category.

Their semantic quotient is identical.

Their physical transition structures are not.

53.4 Coordinate reducedness

Reachable witnesses force all three control states to remain distinct.

Likewise, the two work symbols cannot be merged without collapsing physically distinguishable reachable behavior.

Proposition 53.2 — Coordinate reducedness.

Both eager and lazy lifts are reduced inside the stated:

three-state / two-work-symbol coordinate category.

Thus their nonisomorphism is not caused by obvious redundant states or symbols.

53.5 Morphism-relative nonuniqueness

Corollary 53.3 — Reduced lift nonuniqueness.

The same semantic system admits multiple:

- coordinate-reduced;
- section-preserving;
- non-isomorphic

physical lifts.

Therefore:

same semantics + same coordinate budget + reducedness $\not\Rightarrow$ unique physical lift.

This strengthens Chapter 44's cheap cyclic-decoration nonuniqueness into a substantive structural theorem.

These two reduced lifts are not isolated examples. The next chapter enumerates the complete selected three-site schedule family in which they sit.

There are exactly six.

Chapter 54

The Six 2:1 Normalization Schedules

The historical and centered machines differ at specific reachable sites.

To understand the space of schedule choices, isolate three eager-discharge sites.

54.1 Sites

Site A

Left-state 0/0 matching family.

Site B

Right-state, top 0, input 0 family.

Site C

Right-state, top 1, input 1 family.

A schedule is a bit vector

$$\boxed{(a, b, c) \in \{0, 1\}^3,}$$

where 1 means that the corresponding eager rule is enabled.

There are eight possible schedules.

Exactly six are admissible.

54.2 Exact schedule classification

Theorem 54.1 — Exact three-site schedule classification.

$$\boxed{\mathrm{Adm}_{2,1} = \{(a, b, c) \in \{0, 1\}^3 : ab = 0\}.}$$

Thus the six admissible schedules are

$$\boxed{000,\ 001,\ 010,\ 011,\ 100,\ 101.}$$

The forbidden schedules are

$$\boxed{110,\ 111.}$$

The unique minimal conflict is

$$\boxed{\{A, B\}.}$$

Site C is compatible with either side of that conflict.

54.3 Schedule geometry

Order schedules by inclusion of eager sites.

Corollary 54.2 — Schedule poset.

The admissible schedule poset is not a lattice.
Its two maximal elements are

$$\boxed{\{A, C\}}$$

and

$$\boxed{\{B, C\}.}$$

Their union

$$\{A, B, C\}$$

is inadmissible.
So no join exists inside the admissible poset.

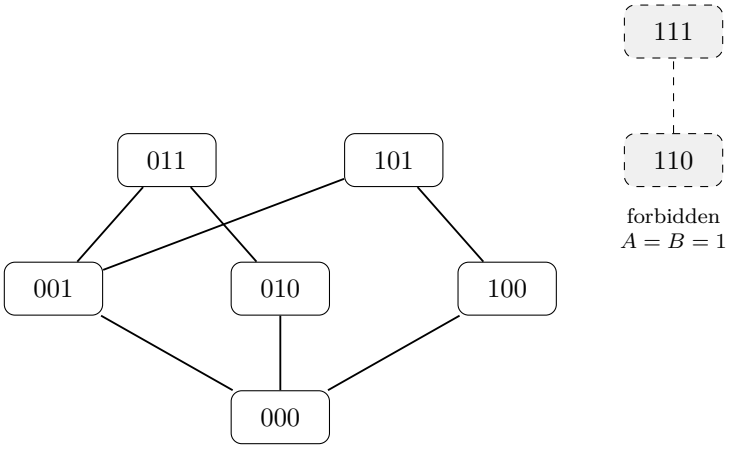


Figure 54.1: Hasse diagram of the six admissible 2:1 schedules. The schedule bits are ordered by inclusion of eager sites A, B, C. The two omitted cube vertices 110 and 111 are shown separately as forbidden because A and B cannot be simultaneously eager.

The missing 110 and 111 vertices should be shown separately as forbidden.

54.4 Six reduced lifts

Proposition 54.3 — Six coordinate-reduced schedule lifts.

All six admissible schedules define pairwise non-isomorphic coordinate-reduced lifts in the selected 2:1 category.

Thus schedule nonuniqueness is not merely eager versus lazy.

The physical realization space already contains a finite family of six reduced points.

54.5 Terminal normalization coherence

Let

$$\mathfrak{N}$$

be the stated terminal canonical normalizer.

Theorem 54.4 — Six-schedule terminal coherence.

For every admissible schedule and every common processed prefix, applying $\mathfrak{N}$ yields the same canonical semantic endpoint.

Thus all six systems have common normalized endpoint behavior.

Terminology boundary

This is **not** a Church–Rosser theorem.

The safe term is:

$$\boxed{\text{normalizer-relative terminal schedule coherence.}}$$

The six lifts are physically non-isomorphic.

They nevertheless agree after the specified terminal normalization.

Corollary 54.4A — Reduced non-isomorphism with common normalized behavior.

There exist coordinate-reduced, pairwise non-isomorphic physical lifts with identical terminal-normalization behavior on every processed prefix.

Thus even after obvious coordinate redundancy is removed,

$$\boxed{\text{common normalized semantics} \not\Rightarrow \text{unique reduced physical schedule.}}$$

Part V will abstract precisely this separation between a fixed semantic update and several physically admissible realizations.

Chapter 55

Why A Causes B

The schedule classification tells us that

$$A + B$$

is forbidden.

Its directionality is supplied by one of the clearest causal mechanisms in the corpus.

55.1 Local carry visibility

A fixed eager rule is chosen from a bounded local observation.

If two sources share that observation but require different semantic carry behavior, the same eager rule cannot be backlog-neutral on both.

Theorem 55.1 — Local carry visibility criterion.

A fixed eager local rule can remain backlog-neutral throughout an observation fiber only if the semantic carry indicator is constant on that fiber.

Thus carry aliasing is a local source of schedule failure.

55.2 Portability

For an eager rule r, define its portability set

$$\mathrm{Port}(r)$$

to be the physical source configurations on which applying that rule is safe relative to the selected canonical policy.

If F_o^α is the observation fiber exposed by schedule α, then:

Proposition 55.2 — Eager-rule portability criterion.

The eager rule is portable over that observation class exactly when

$$\boxed{F_o^\alpha \subseteq \mathrm{Port}(r).}$$

Schedule interactions therefore arise when enabling one rule changes which configurations another rule encounters.

55.3 A expands the B-fiber

Without A, site B sees only its baseline portable contexts.

Enabling A changes earlier normalization chronology.

That change exposes to B the infinite family

$$\boxed{(R, 0^m), \qquad m \geq 2.}$$

Lemma 55.3 — Schedule-induced fiber expansion.

Enabling A enlarges the first-hit observation fiber of B by the family $(R, 0^m)$, $m \geq 2$.
This is the causal mechanism.
A does not make B's local rule intrinsically different.
It changes which hidden suffixes reach the same local B-observation.

55.4 Direct canonicalization becomes impossible

For source

$$(R, 0^m)$$

on input 0, the canonical semantic target is

$$\boxed{10^{m-2}.}$$

A one-top rewrite must preserve the untouched suffix

$$\boxed{0^{m-1}.}$$

But 0^{m-1} is not a suffix of 10^{m-2}.

Theorem 55.4 — Schedule-induced direct-canonicality obstruction.

No one-top rule at site B can directly canonicalize every newly exposed source $(R, 0^m)$, $m \geq 2$.
This is the suffix-theorem diagnosis.

55.5 Forced kernel excess

Suppose the B-rule nevertheless insists on returning directly to the center while preserving residual.
Then it must create a nonminimal amount of literal demand.

Theorem 55.5 — Forced kernel-excess return.

Every such center-return repair introduces at least one extra kernel packet:

$$\boxed{(t+1)K}$$

relative to the minimum demand.
The historical replacement

$$\boxed{111}$$

realizes the minimum one-packet excess.
Thus the same conflict has three exact readings.

Corollary 55.6 — Three-way diagnosis.

Within this $A \to B$ family,

$$\boxed{\text{suffix failure}}$$

$$\Longleftrightarrow$$

$$\boxed{\text{carry aliasing}}$$

$$\Longleftrightarrow$$

$$\boxed{\text{forced kernel excess.}}$$

This triangle is important because the three notions are not globally identical in arbitrary representations.

They coincide here because of the specific one-top literal policy and source family.

The next chapter turns this concrete explanation into a directed causal relation.

Chapter 56

Directed Causal Scheduling

A minimal conflict such as

$$\{A, B\}$$

is set-theoretic.
It does not identify which rule caused the other to become unsafe.
The first-hit fiber expansion gives a directed notion.

56.1 First-hit exposure

For target site b and previously enabled schedule set H, let

$$\boxed{\mathcal{E}_b(H)}$$

be the first-hit exposure fiber: the configurations at which a computation first reaches site b under the schedule H.
A directed cause

$$H \rightsquigarrow b$$

occurs when H is minimal among schedule sets whose first-hit exposure contains a b-source outside the portability set of the b-rule.
Thus causality is about **newly exposed unsafe contexts**.

56.2 Exact 2:1 primary dependency

Theorem 56.1 — Exact primary dependency classification.
For the selected 2:1 three-site family,

$$\boxed{\mathcal{D}_{2,1}^{\mathrm{prim}} = \{\{A\} \rightsquigarrow B\}.}$$

So the directed causal graph is

$$\boxed{A \longrightarrow B \qquad \sqcup \qquad C.}$$

There is:

- no $B \to A$;
- no edge involving C.

Thus the forbidden pair $\{A, B\}$ has a unique primary direction.

Corollary 56.2 — Directed cause recovers the conflict.

Forgetting orientation gives the unique minimal schedule conflict:

$$\boxed{A \to B \quad \mapsto \quad \{A, B\}.}$$

56.3 Concrete causal certificate

One first-hit witness is

$$\boxed{1111100 \rightsquigarrow (R, 00).}$$

Applying the B-rule yields

$$\boxed{(C, 1110)}$$

instead of canonical

$$\boxed{(C, 1).}$$

The quantity excess is one kernel packet in the relevant sign convention:

$$\boxed{-K.}$$

Appending one more 1 reaches residual zero at the noncanonical configuration

$$\boxed{(C, 110).}$$

This gives a finite causal certificate:

1. enabling A creates a new first hit at B;
2. B's eager rule is not portable there;
3. the resulting debt is behaviorally exposed by a short continuation.

Proposition 56.2A — 2:1 causal certificate.

There exists a finite word, beginning with the witness prefix

$$\boxed{1111100,}$$

whose first use of the B-site after enabling A reaches

$$(R, 00)$$

and whose eager B-successor is the noncanonical kernel-excess configuration

$$(C, 1110).$$

A finite continuation then distinguishes this successor behaviorally from the canonical semantic successor.

Hence the directed dependency

$$\boxed{A \rightsquigarrow B}$$

admits a finite first-hit certificate.

This concrete witness is the prototype for the Fatal Frontier certificates of the General Theory.

56.4 Coordinate invariance

Suppose two lift presentations are related by a site-preserving coordinate LIFO isomorphism.

Then:

- first-hit exposure fibers correspond;
- portability sets correspond;
- directed causal hyperedges correspond.

Theorem 56.3 — Causal hypergraph invariance.

The directed causal schedule structure is invariant under site-preserving coordinate isomorphism.

Thus the $A \to B$ graph is not an artifact of renaming states or stack symbols inside the same coordinate category.

56.5 But semantics does not determine causality

A raw residual-conjugacy machine can realize the same language semantic quotient without any literal A, B, C normalization sites.

Theorem 56.4 — Semantic non-determination of causal structure.

The same semantic quotient can admit physical lifts with different causal schedule structures.

Therefore:

$$\boxed{\text{semantic quotient} \not\Rightarrow \text{causal schedule signature.}}$$

Corollary 56.5 — Coordinate/semantic separation principle.

A property can be invariant under coordinate isomorphism and still fail to be language-intrinsic.

Thus:

$$\boxed{\text{coordinate-intrinsic} \not\Rightarrow \text{semantic-intrinsic.}}$$

This distinction is central to the obstruction theorems that follow.

Chapter 57

Which Obstructions Are Really Forced?

The Uniqueness Theory has proved several kinds of necessity.

But necessity depends on the class of allowed representations.

This final chapter of the branch makes that dependence explicit.

57.1 Nested representation classes

Consider nested classes

$$\boxed{\mathfrak{C}_0 \subseteq \mathfrak{C}_1 \subseteq \mathfrak{C}_2.}$$

A typical interpretation is:

- $\mathfrak{C}_0$: a narrow coordinate category;
- $\mathfrak{C}_1$: a fixed literal-demand policy with more presentation freedom;
- $\mathfrak{C}_2$: a broader realization class allowing re-coordination.

For an obstruction $\mathcal{O}$, ask whether $\mathcal{O}$ is unavoidable in each class.

57.2 Obstruction monotonicity

Theorem 57.1 — Obstruction monotonicity.

If the realization class is enlarged, an essential obstruction can disappear but cannot newly become essential.

Thus:

$$\boxed{\mathrm{Ess}(\mathfrak{C}_2) \subseteq \mathrm{Ess}(\mathfrak{C}_1) \subseteq \mathrm{Ess}(\mathfrak{C}_0).}$$

For fully resolved three-level binary profiles, only

$$\boxed{111, \quad 110, \quad 100, \quad 000}$$

are possible.

57.3 Branching profile

For nontrivial semantic-fiber branching in the stated nested classes:

Theorem 57.2 — Policy-essential but broad-class removable branching.

$$\boxed{\Pi(\mathcal{O}_{\text{branch}}) = 110.}$$

Thus branching is:

- essential in the narrow coordinate class;
- essential in the fixed literal policy;
- removable in a broader class containing raw residual re-coordination.

This theorem connects the early raw DPDA constructions to the later lift theory.
The compact raw representation did not contradict forced branching.
It left the literal-demand policy in which branching was forced.

57.4 Category-relative status of $A \to B$

The specific causal obstruction has an exact category-relative profile once the schedule presentation is typed separately from the underlying lift.

Proposition 57.3 — Exact category bifurcation for the classical $A \to B$ edge.

In the site-flexible fixed-literal-policy presentation class, the optional menu may vary while the baseline literal-demand lift is held fixed. The named B site can then be omitted, so

$$\boxed{\Pi_{\text{flex}}(A \to B) = (1, 0, 0).}$$

If instead the designated A, B cells and the finite first-hit causal certificate are part of the preserved interface, the dependency persists:

$$\boxed{\Pi_{\text{cert}}(A \to B) = (1, 1, 0).}$$

Broad exact re-coordination can remove the named phenomenon in either case. Chapter 60 gives the full presentation typing and the arbitrary-coprime extension.

57.5 Representation–obstruction separation

Theorem 57.4 — Representation-obstruction separation.

Every theorem claiming that a physical obstruction is "forced" must specify the representation class in which the necessity is proved.

This is one of the main methodological conclusions of the project. It applies to state lower bounds, stack-symbol lower bounds, fiber branching, normalization debt, schedule conflicts, and causal edges.

57.6 Computational validation note

The associated semantic-uniqueness/lift validation study reports

$$\boxed{\text{458,776 assertion-level checks}},$$

including exhaustive words through length 14 for all eight schedules and the first common forbidden-schedule failure.

This historical count is recorded as a reported finite consistency check, not as a separately established theorem. The enumeration ranges and schedule conditions specify what was checked. The mathematical assertions above rest on their symbolic proofs and do not use the reported count as an assumption.

57.7 Transition to General Theory

The Uniqueness Theory has now produced a family of concepts that are no longer specific to weighted binary balance:

- semantic transition system;
- exact physical lift;
- canonical section;
- semantic fiber;
- bounded local observation;
- forced branching;
- portability;
- causal schedule structure;
- representation-relative obstruction.

Before the abstraction to General Theory, Part IV must close three scope questions created by the 2:1 analysis: whether the schedule geometry persists for arbitrary coprime ratios; which coordinate quotients and bounded recodings preserve the named physical interface; and what kind of confluence is actually justified at each normalization layer. The next structural branches answer these questions in turn.

Once those distinctions are fixed, Part V asks:

Which of these notions survive for a general semantic system and a bounded-local physical normalization policy?

The abstract chain will then organize them into:

locality → debt → behavioral safety → causality → synthesis → verification.

Chapter 58

From the Six 2:1 Schedules to the Arbitrary-Coprime Compiler

The preceding chapters isolated the six reduced schedules of the original 2:1 literal-demand lift and their directed causal relation. We now keep that same shortest-completion semantics but remove the special ratio. The result is a uniform compiler and a finite arithmetic site universe for every coprime p:q.

58.1 Centered literal-demand normalization

Put

$$m = p + q, \qquad h = \left\lfloor \frac{m-1}{2} \right\rfloor, \qquad K = (q, p).$$

The centered literal-demand lift has finite-control debt states

$$D_{-h}, D_{-h+1}, \ldots, D_h$$

and a center state C. The semantic residual is represented by a canonical shortest nonnegative demand vector

$$\mu(r) = \arg\min\{x + y : x, y \geq 0,\ px - qy = r\},$$

and the chosen literal stack order is denoted by $\eta(x, y)$.

For debt state D_d define the virtual demand

$$V_d = \mu(dm).$$

A local work-top observation is a triple

$$o = (d, a, z), \qquad |d| \leq h, \quad a, z \in \{0, 1\},$$

meaning that the current state is D_d, the next input letter is a, and the visible stack symbol is z.

Definition 58.1 (Schedule presentation)**.** A schedule presentation consists of a fixed mandatory centered baseline policy together with a finite optional-site menu and, for each site, a designated eager-return rule. A schedule is a subset of the menu; enabled optional rules may pre-empt the corresponding baseline transition.

Definition 58.2 (Admissibility and fatal supports)**.** A schedule H is *admissible* if the resulting deterministic machine recognizes exactly $L_{p,q}$. A support F is *fatal* if every schedule containing F is inadmissible, and is *minimal fatal* if no proper subset is fatal.

The distinction between a lift and a schedule presentation will be essential in Section 60.2: the physical transition system does not determine which optional sites are named or offered.

58.2 The uniform eager-return compiler

Let $e_0 = (1, 0)$ and $e_1 = (0, 1)$. For an observation $o = (d, a, z)$ define

$$B_o = e_z - e_a + V_d$$

and

$$\kappa(o) = \max\left\{\left\lceil -\frac{(B_o)_0}{q}\right\rceil, \left\lceil -\frac{(B_o)_1}{p}\right\rceil\right\}.$$

Set

$$R_o = B_o + \kappa(o)K.$$

The associated local eager rule is

$$E_o : (D_d, zT) \xrightarrow{a} (C, \eta(R_o)T).$$

Theorem 58.3 (Uniform centered eager-return compiler)**.** *For every local observation o, the vector R_o is the unique shortest nonnegative replacement vector that preserves the residual invariant and returns directly to the center. Hence E_o is the canonical direct eager-return candidate at o.*

Proof. Any direct return to the center must compensate the change in the visible stack cell and the consumed input, hence its replacement vector lies in the affine lattice

$$B_o + \mathbb{Z}K.$$

The only admissible replacements are nonnegative vectors. Adding one copy of $K = (q, p)$ increases both coordinates, so the smallest integer carry making both coordinates nonnegative is exactly $\kappa(o)$. Any smaller carry has a negative coordinate, while any larger carry increases total length by $p + q$. Therefore R_o is feasible and uniquely length-minimal. □

This formula recovers, for example, the six explicit $3 : 2$ eager words used in the finite stress test. The important point here is not those special values but the existence of one ratio-uniform compiler from which the whole site universe can be derived.

58.2.1 The mandatory reset envelope

Write

$$\mu((h+1)m) = (x_+, y_+), \qquad \mu(-(h+1)m) = (x_-, y_-).$$

Define

$$\begin{aligned}\mathcal{R}^0_{p,q} = {} & \{\mu(jm+q) : -h-1 \le j \le h\} \\ & \cup \{(x_+ + n, y_+) : n \ge 0\} \\ & \cup \{(x_-, y_- + n) : n \ge 0\}.\end{aligned}$$

Lemma 58.4 (Mandatory-WRITE reset envelope)**.** *Every mandatory WRITE in the centered $p : q$ baseline lands at a canonical center pair belonging to $\mathcal{R}^0_{p,q}$.*

Proof. A bottom WRITE from D_d on input 1 produces $\mu(dm+q)$, whereas input 0 produces

$$\mu(dm - p) = \mu((d-1)m + q),$$

which gives the finite seed range $-h - 1 \le j \le h$. At the positive outer boundary, the untouched suffix is a unary 0-tail, so the reset ray is $(x_+ + n, y_+)$. The negative side is symmetric. □

Definition 58.5 (Baseline-portable site)**.** A baseline episode starts at a center configuration $(C, \eta(x, y))$ with $(x, y) \in \mathcal{R}^0_{p,q}$, follows only mandatory MATCH/SEARCH transitions, and stops before the next mandatory bottom or outer WRITE. A compiler site is *baseline-portable* if its eager rule lands on the canonical center representative for every source exposed in every baseline episode.

Baseline portability is deliberately weaker than singleton admissibility: an eager reset can be safe on all baseline sources and later become unsafe on a source created by its own earlier activation. This self-exposure mechanism is responsible for the extra singleton-fatal vertices below.

58.3 The finite baseline-portable site universe

Assume first that $p \ge q \ge 2$. For $j \ge 0$, the bottom-seed formula is

$$\mu(jm+q) = (j+q, p-j-1).$$

For $j = -t < 0$, write $t = kq + s$, $0 \le s < q$; then

$$\mu(-tm+q) = \begin{cases} (0, km-1), & s = 0, \\ (q-s, p+km+s-1), & 1 \le s < q. \end{cases}$$

For $1 \le d \le h$,

$$V_d = (d+q, p-d).$$

The three nontrivial positive-debt compiler count vectors are

$$M_d = (d+q, p-d), \qquad P_d = (d+q+1, p-d-1), \qquad N_d = (d+q-1, p-d+1),$$

with imbalances

$$\Delta(M_d) = 2d+q-p,$$
$$\Delta(P_d) = 2d+q-p+2,$$
$$\Delta(N_d) = 2d+q-p-2.$$

Negative-debt compiler words are one-heavy except for the trivial forced return that is already part of the baseline.

A central reset word excludes almost every interior observation. If m is odd, the diagnostic seed is (h, h) with word $(10)^h$; if m is even, it is $(h, h+1)$ with word $(10)^h 1$. Tracking untouched suffixes shows that all negative-interior candidates fail portability and leaves at most two positive-interior survivors.

Theorem 58.6 (Eight-Site Theorem)**.** *Let $p \ge q \ge 2$ be coprime and let $h = \lfloor (p+q-1)/2 \rfloor$.*

If $p+q$ is odd, the baseline-portable site universe consists exactly of the six core sites

$$(-h, 0, 0),\ (-h, 1, 0),\ (-h, 1, 1),\ (h-1, 0, 0),\ (h-1, 1, 0),\ (h, 0, 0),$$

together with

$$X = (h-q+1, 0, 1) \iff p > 5q-5,$$

and

$$Y = (h-q, 1, 1) \iff p > 5q-1.$$

If $p+q$ is even, the universe consists exactly of the four core sites

$$(-h, 1, 1),\ (h, 0, 0),\ (h, 0, 1),\ (h, 1, 1),$$

together with

$$X = (h-q+1, 0, 0) \iff p > 5q-5,$$

and

$$Y = (h-q, 1, 0) \iff p > 5q.$$

In particular,

$$|\mathcal{S}^{\mathrm{bp}}_{p,q}| \le 8.$$

For $q > p \ge 2$, the complete list is obtained by the physical duality

$$(p, q, d, a, z) \longleftrightarrow (q, p, -d, 1-a, 1-z).$$

Proof. The proof is an exact arithmetic elimination. The central diagnostic removes every negative-interior observation and all but two positive-interior observations; exact boundary fibers give the core; the two survivors are decided by primitive-kernel minimality. All formulas, including the negative-seed decomposition and the parity-specific threshold simplifications, are given in Section 59.4. □

Chapter 59

Exact Causal-Hypergraph Classification for All Coprime Ratios

The compiler reduces schedule admissibility to a finite obstruction problem. The classification separates the stable high-minimum regime from the two low-minimum lines where finite fork phenomena survive.

59.1 The high-minimum regime: every core schedule is safe

Let $\mathrm{Core}_{p,q}$ denote the parity-dependent core in Theorem 58.6. For a named site A, write $\mathcal{E}_A$ for the set of visible source stack words that can occur at that site during an episode under the chosen core schedule. These source envelopes are physical reachability sets, not new semantic states.

Theorem 59.1 (Core downward closure)**.** *If* $\min(p,q) \geq 3$*, then every subset* $H \subseteq \mathrm{Core}_{p,q}$ *is admissible. Equivalently,*

$$2^{\mathrm{Core}_{p,q}} \subseteq \mathrm{Adm}_{p,q}\,.$$

No minimal fatal support is contained entirely in the core.

Proof. By duality assume $p \geq q \geq 3$. Induct over reset episodes. Every enabled core activation returns to a canonical minimum-demand word. It therefore suffices to show that the source fibers of core sites stay within protected unary envelopes after any sequence of core activations.

If $m = 2h+1$, label the six core sites

$$A = (-h, 0, 0),\ B = (-h, 1, 0),\ C = (-h, 1, 1),$$

$$D = (h-1, 0, 0),\ E = (h-1, 1, 0),\ F = (h, 0, 0).$$

Uniformly over all core schedules,

$$\mathcal{E}_A = \mathcal{E}_B = \{0\}, \qquad \mathcal{E}_C \subseteq 1^+,$$

and

$$\mathcal{E}_D, \mathcal{E}_E, \mathcal{E}_F \subseteq 0^+.$$

Negative-core resets contain fewer than $q \leq h$ zeros, so they cannot create a new negative-boundary top-0 source. Positive-core outputs have one-count at most $h-1$, so their mixed part is exhausted before the next positive-boundary source. The eager compiler has the same tail orientation as the protected source, and kernel minimality is preserved.

If $m = 2h + 2$, the four core sites satisfy

$$\mathcal{E}_A \subseteq 1^+, \qquad \mathcal{E}_B \subseteq 0^+, \qquad \mathcal{E}_C = \mathcal{E}_D = \{1\}.$$

The same minority-count argument prevents a negative reset from generating a new positive-boundary source and prevents a positive reset from generating an additional top-1 source at D_h. Hence every optional activation is canonical and kernel-reduced. The residual invariant and the baseline terminal argument then imply correctness. □

59.2 Extra-site self-exposure for $\min(p, q) \geq 3$

The extra vertices X, Y are baseline-portable precisely because their first activation stays below the primitive kernel threshold. Repeating the same activation increases a hidden alternating prefix until one full kernel packet becomes trapped.

For odd $m = 2h + 1$, define

$$S_k^X = (10)^{k(q-1)}1^{2q-2}, \qquad S_k^Y = (10)^{kq}1^{2q}.$$

For even $m = 2h + 2$, define

$$S_k^X = 0(10)^{k(q-1)}1^{2q-3}, \qquad S_k^Y = 0(10)^{kq}1^{2q-1}.$$

The first source S_0 is reached by an explicit monotone baseline prefix. Between consecutive X-activations one reads $1^{p-q+1}0$; between consecutive Y-activations one reads 1^{p-q}. Thus the alternating exponent grows by $q - 1$ or q respectively.

Lemma 59.2 (One-kernel trapping)**.** *Let*

$$k_X = \left\lceil \frac{p - (h + 2q - 2)}{q - 1} \right\rceil, \qquad k_Y = \left\lceil \frac{p - (h + 2q)}{q} \right\rceil.$$

At the first nonminimal repeated activation, the physical target differs from the canonical minimum demand by exactly one primitive kernel packet $K = (q, p)$.

Proof. At the fatal activation the one-count is $p + e$ with $0 \leq e \leq q - 1$ (and $e \leq q - 2$ in the X case), while the zero-count is at least q. Subtracting one K therefore remains nonnegative, while subtracting two cannot. Thus the target is exactly $\mu(r) + K$. □

Feeding the shortest semantic completion of r removes the minimal component and leaves exactly one balanced kernel packet physically present. The final semantic residual is zero but the machine is not in the balanced accepting configuration.

Theorem 59.3 (Complete schedule classification for $\min(p, q) \geq 3$)**.** *For every coprime p, q with $\min(p, q) \geq 3$,*

$$\boxed{\mathrm{Adm}_{p,q} = 2^{\mathrm{Core}_{p,q}}.}$$

Every present extra site is an upward-fatal singleton conflict, and these are the only minimal fatal supports.

Proof. Every core subset is admissible by Theorem 59.1. The Y self-exposure witness meets no other optional site, hence every schedule containing Y is bad. The X witness meets no core site and at most the neighboring Y observation; if Y is disabled it proves X fatal, while if Y is enabled the Y witness already proves failure. Thus every schedule containing either extra is inadmissible. □

59.3 Low-minimum regimes and arbitrary-coprime classification

59.3.1 The line $\min(p, q) = 2$

By duality it suffices to take $q = 2$, so $p \geq 3$ is odd. The six-site core has exactly two minimal fatal supports,

$$\{A, D\}, \qquad \{B, D\}.$$

The two possible extra sites are

$$X_1 = (h - 1, 0, 1) \quad (p \geq 7),$$
$$X_2 = (h - 2, 1, 1) \quad (p \geq 11).$$

and both are singleton-fatal.

Theorem 59.4 (Exact $\min(p, q) = 2$ classification)**.** *For $q = 2$,*

$$\mathrm{Adm}_{p,2} = \{H \subseteq \mathcal{S}_{p,2}^{\mathrm{bp}} : X_1, X_2 \notin H,\ \{A, D\} \not\subseteq H,\ \{B, D\} \not\subseteq H\},$$

with absent extras omitted. Exactly 40 *schedules are admissible.*

Proof. Section 59.5.1 gives a self-contained reset-closure proof of sufficiency and explicit robust witnesses for the two fork pairs $\{A, D\}$ and $\{B, D\}$; the six-site core therefore has exactly 40 safe schedules. Each present extra site is an upward-fatal singleton by the one-kernel trapping certificate, so every admissible expanded schedule excludes all extras. Conversely, after the extras are excluded the machine is exactly one of the safe core schedules. Thus the admissible count remains 40 independently of how many extras are present. □

59.3.2 The exceptional line $q = 1$

For $q = 1$, negative minimum demands are unary, so the negative corridor has no order freedom. The complete baseline-portable site universe is exceptional.

For $p = 1$ it is empty. For $p = 2$ the four sites are

$$X = (0, 1, 0), \quad A = (-1, 1, 1), \quad B = (-1, 0, 0), \quad G = (1, 0, 0).$$

For odd $p = 2h + 1 \geq 3$, the always-present sites are

$$A = (-h, 1, 1), \qquad D = (h, 1, 1), \qquad E = (h, 0, 0),$$

and for $p \geq 5$ also

$$B = (h - 1, 1, 0), \qquad C = (h, 0, 1).$$

For even $p = 2h \geq 4$, the sites

$$A = (-h, 1, 1),\ B = (-h, 0, 0),\ C = (-h, 1, 0),$$

$$E = (h - 1, 0, 0),\ F = (h - 1, 1, 0),\ G = (h, 0, 0)$$

are always present, and $D = (h - 1, 1, 1)$ appears for $p \geq 6$.

Theorem 59.5 (Exact $q = 1$ schedules)**.** *Up to physical duality:*

1. $(p, q) = (1, 1)$*: the site universe is empty and there is one admissible schedule.*
2. $(2, 1)$*: the minimal fatal supports are $\{B, X\}$ and $\{B, G\}$; exactly* 10 *schedules are admissible.*

3. odd $p \geq 3$:

$$\mathrm{Adm}_{p,1} = \{\varnothing, \{A\}\}.$$

4. even $p \geq 4$: the maximal safe sectors are

$$\mathcal{N} = \{A, B, C\}, \qquad \mathcal{P} = \{A, F, G\},$$

while D and E are singleton-fatal when present and the cross-pairs

$$\{B, F\}, \{B, G\}, \{C, F\}, \{C, G\}$$

are fatal. Exactly 14 schedules are admissible.

Proof. The site-universe elimination and the balanced fatal certificates are recorded in Section 59.5. The safe cases are proved by episode-closed reset families: for odd p, only the unary negative ray generated by A is added; for even p, every safe schedule lies in one of the two closed sectors $2^{\mathcal{N}}$ or $2^{\mathcal{P}}$. The listed singleton and cross-pair witnesses trap a primitive kernel packet at residual zero. The small case $(2, 1)$ is verified by its four-site transition geometry, with explicit witnesses 111001 and 111110001. □

Theorem 59.6 (Arbitrary-coprime causal-hypergraph classification)**.** *For every coprime positive pair (p, q), the canonical baseline-portable weighted-balance schedule family has an explicit finite set of minimal fatal supports given by Theorems 59.3, the* min $= 2$ *theorem, and Theorem 59.5, together with physical duality. Consequently*

$$H \in \mathrm{Adm}_{p,q}, \quad H' \subseteq H \quad \Longrightarrow \quad H' \in \mathrm{Adm}_{p,q}\,.$$

Equivalently, $\mathrm{Adm}_{p,q}$ *is the independence complex of the exact fatal-support hypergraph, in the standard simplicial-complex sense [48].*

59.4 Arithmetic details for the Eight-Site Theorem

This appendix records the elimination argument behind Theorem 58.6; it is included so that the arbitrary-coprime vertex classification can be checked without relying on a finite search.

Put $m = p + q$ and $h = \lfloor (m-1)/2 \rfloor$, and assume $p \geq q \geq 2$. For the bottom-WRITE seed family $B_j = \mu(jm + q)$ one has, for $j \geq 0$,

$$B_j = (j + q, p - j - 1).$$

For $j = -t < 0$, write $t = kq + s$, $0 \leq s < q$. Then

$$B_{-t} = \begin{cases} (0, km - 1), & s = 0, \\ (q - s, p + km + s - 1), & 1 \leq s < q. \end{cases}$$

For $1 \leq d \leq h$,

$$V_d = \mu(dm) = (d + q, p - d).$$

Thus the positive-debt compiler count vectors are

$$M_d = (d + q, p - d), \quad P_d = (d + q + 1, p - d - 1), \quad N_d = (d + q - 1, p - d + 1),$$

with imbalances

$$2d + q - p, \qquad 2d + q - p + 2, \qquad 2d + q - p - 2.$$

Every nontrivial negative-debt compiler word is one-heavy; the only exception is the forced $d = -1$, input/top $1/0$ return already contained in the baseline.

Lemma 59.7 (Central diagnostic exclusion)**.** *For $p \geq q \geq 2$, no negative-interior observation is baseline-portable. If m is odd, the only possible positive-interior survivors are*

$$(h - q + 1, 0, 1), \qquad (h - q, 1, 1),$$

and if m is even the only possible positive-interior survivors are

$$(h - q + 1, 0, 0), \qquad (h - q, 1, 0).$$

Proof. If $m = 2h+1$, the seed $B_{h-q} = (h, h)$ has canonical word $(10)^h$. At an interior debt d, the untouched suffix is $(10)^{h-d-1}$ under a visible top 0 and $0(10)^{h-d-1}$ under a visible top 1. A top-0 eager concatenation is canonical only when its compiler word is balanced; all relevant interior imbalances are odd. A top-1 concatenation is canonical only when the compiler has exactly one excess 1, which gives precisely the two displayed equations for d.

If $m = 2h + 2$, the central seed is $(h, h + 1)$ with word $(10)^h1$. The corresponding suffixes are $(10)^{h-d-1}1$ and $0(10)^{h-d-1}1$. A top-0 candidate must have zero imbalance, while the parity of the positive-debt compiler imbalances excludes every top-1 candidate. Solving the zero-imbalance equations gives the two even-parity survivors. For negative debt, the compiler has a terminal 1-tail whereas the central diagnostic exposes an untouched 0 beyond the visible cell; the eager target is therefore not in canonical alternating-tail form. This excludes all negative-interior sites. □

Lemma 59.8 (Boundary fibers)**.** *The always-present boundary sites are exactly the parity-dependent cores in Theorem 58.6.*

Proof. For odd m, the negative boundary has a singleton top-0 source and unary top-1 sources; at $d = h - 1$ every top-0 source is unary, and at $d = h$ every source is unary top-0. A source 11 at $d = h - 1$ eliminates the remaining top-1 candidates. For even m, the negative boundary has unary top-1 sources but also the diagnostic source 00, while at $d = h$ the top-0 sources are unary and the only top-1 source is 1; the central sources 01 and 101 eliminate the remaining $d = h - 1$ candidates. Direct substitution into the compiler gives exactly the core lists stated in Theorem 58.6. □

For the two odd-parity survivors the common compiler word is $(10)^h1$. Their exposed source one-counts are bounded by $2q-2$ and $2q$, with equality attained by baseline sources. Kernel reduction is therefore equivalent to

$$h + 2q - 2 < p, \qquad h + 2q < p,$$

which simplify to $p > 5q - 5$ and $p > 5q - 1$. For even m the common compiler is $(10)^{h+1}$, the extremal untouched one-counts are $2q-3$ and $2q-1$, and the same calculation gives $p > 5q - 5$ and, using coprimality of odd p, q, $p > 5q$. This completes the proof of Theorem 58.6.

59.5 Low-minimum certificates

The low-minimum cases are included explicitly because they are the only places where non-singleton fatal supports survive.

59.5.1 Self-contained odd-p:2 fork certificate

Fix $q = 2$ and write

$$p = 2h - 1, \qquad m = p + q = 2h + 1, \qquad h \geq 2.$$

For $t \geq 1$ put

$$\epsilon_t = t \bmod 2, \qquad k_t = \left\lceil \frac{t}{2} \right\rceil, \qquad \beta_t = t + pk_t, \qquad \Gamma_t = \eta(\epsilon_t, \beta_t).$$

The six core rules are, in literal stack notation,

$$\begin{array}{c|c} A : (D_{-h}, 0T) \xrightarrow{0} (C, \Gamma_h T) & B : (D_{-h}, 0T) \xrightarrow{1} (C, \Gamma_{h-1} T) \\ C : (D_{-h}, 1T) \xrightarrow{1} (C, \Gamma_h T) & D : (D_{h-1}, 0T) \xrightarrow{0} (C, (10)^h 0T) \\ E : (D_{h-1}, 0T) \xrightarrow{1} (C, (10)^{h-1}000T) & F : (D_h, 0T) \xrightarrow{0} (C, (10)^{h-1}000T). \end{array}$$

The arithmetic identities needed below are

$$\mu(-tm) = (\epsilon_t, \beta_t), \qquad \mu(jm+2) = (j+2, p-j-1) \quad (0 \le j \le h),$$

$$\mu(-tm+2) = (\epsilon_t, \beta_t - 1) \quad (1 \le t \le h+1), \qquad \mu((h+1)m) = (h+3, h-2).$$

They follow by direct substitution in $px - 2y = r$ and kernel reduction by $(2, p)$.

Define the canonical baseline reset family

$$\begin{aligned} \mathcal{R}_h^{\text{base}} = {} & \{\mu(jm+2) : -h-1 \le j \le h\} \\ & \cup \{(x, h-2) : x \ge h+3\} \\ & \cup \{(\epsilon_{h+1}, y) : y \ge \beta_{h+1}\}. \end{aligned}$$

Two episode-closed repertoires separate the safe schedule regimes:

$$\begin{aligned} \mathcal{R}_{\neg D,h} = {} & \mathcal{R}_h^{\text{base}} \cup \{(\epsilon_h, y) : y \ge \beta_h\} \\ & \cup \{(\epsilon_{h-1}, \beta_{h-1})\} \cup \{(x, h-1) : x \ge h+2\}, \\ \mathcal{R}_{D,h} = {} & \mathcal{R}_h^{\text{base}} \cup \{(\epsilon_h, y) : y \ge \beta_h\} \\ & \cup \{(x, h-1) : x \ge h+2\} \cup \{(x, h) : x \ge h+1\}. \end{aligned}$$

Every pair in either repertoire has minority coordinate at most h. Consequently an episode beginning in one of these repertoires has the following exact boundary-source geometry: in $\mathcal{R}_{\neg D,h}$, a D_{-h} source with visible 0 has remaining stack exactly 0; in $\mathcal{R}_{D,h}$ it has remaining stack 0^n, $n \ge 1$, with $n \ge 2$ arising only from the D-ray. In either repertoire, D_{-h} with visible 1, D_{h-1} with visible 0, and D_h with visible 0 have unary remaining stacks. This follows because MATCH and interior SEARCH only delete the visible top cell, so every pre-reset stack is a suffix of a canonical reset word; an opposite symbol beyond the displayed unary suffix would force minority size at least $h+1$.

For a unary source of length $n \ge 1$, the six eager targets have canonical count pairs

$$\begin{array}{c|c} A : 0 & (\epsilon_h, \beta_h) \\ B : 0 & (\epsilon_{h-1}, \beta_{h-1}) \\ C : 1^n & (\epsilon_h, \beta_h + n - 1) \\ D : 0^n & (n+h, h) \\ E : 0^n & (n+h+1, h-1) \\ F : 0^n & (n+h+1, h-1). \end{array}$$

Hence baseline resets together with any enabled subset of A, B, C, E, F preserve $\mathcal{R}_{\neg D,h}$ when D is disabled; when D is enabled but A, B are disabled, baseline resets together with C, D, E, F preserve $\mathcal{R}_{D,h}$. Therefore all $2^5 = 32$ schedules with $D = 0$ are safe, and with $D = 1$ precisely the $2^3 = 8$ schedules having $A = B = 0$ are safe.

For necessity, fix any $x \ge h+1$. Then

$$(C, \eta(x,h)) \xrightarrow{0^{2h-1}} (D_{-h}, 0^{x-h+1}),$$

so enabling D expands the common A/B source fiber from the singleton $\{0\}$ to the entire unary ray $\{0^n : n \ge 1\}$. There is a uniform prefix u_h reaching $(D_{-h}, 000)$ and meeting no optional site except D. Put

$$v_h = \begin{cases} 1^{(h-1)m/2}, & h \text{ odd}, \\ 0\,1^{2h+(h-2)m/2}, & h \text{ even}, \end{cases} \qquad u_h = v_h 1^{2h-1} 0^{2h}.$$

Then u_h has the stated property: the increment block from an empty stack advances the centered debt by two, the middle block reaches $(D_{h-1}, 00)$, the next zero invokes D, and the final zeros expose $(D_{-h}, 000)$ without meeting A, B, C, E, F earlier.

Finally, Γ_t has a suffix 1^p for every $t \geq 1$; write $\Gamma_t = z_t 1^p$. At the common source $(D_{-h}, 000)$,

$$A: \ (D_{-h}, 000) \xrightarrow{0} (C, z_h 1^p 00), \qquad B: \ (D_{-h}, 000) \xrightarrow{1} (C, z_{h-1} 1^p 00).$$

Reading z_h or z_{h-1} by MATCHes leaves $(C, 1^p 00)$. This word contains p ones and two zeros, so its residual is zero, but it is a nonempty primitive kernel packet and is therefore nonaccepting. The complete fatal witnesses

$$w_A^{(h)} = u_h 0 z_h, \qquad w_B^{(h)} = u_h 1 z_{h-1}$$

show that $\{A, D\}$ and $\{B, D\}$ are robust fatal supports. Thus the six-site core has exactly 40 admissible schedules and no other minimal fatal support.

The extra baseline-portable sites

$$X_1 = (h-1, 0, 1) \quad (p \geq 7), \qquad X_2 = (h-2, 1, 1) \quad (p \geq 11)$$

are individually upward-fatal by the one-kernel trapping mechanism proved above. Excluding them reduces the expanded problem to the proved six-site fork classification, so the admissible count remains exactly 40.

For $q = 1$ the full vertex sets are as follows. At $(2, 1)$,

$$\mathcal{S}_{2,1}^{\mathrm{bp}} = \{X = (0, 1, 0),\ A = (-1, 1, 1),\ B = (-1, 0, 0),\ G = (1, 0, 0)\}.$$

For odd $p = 2h + 1 \geq 3$, $A = (-h, 1, 1), D = (h, 1, 1), E = (h, 0, 0)$ are always present and $B = (h-1, 1, 0), C = (h, 0, 1)$ occur for $p \geq 5$. For even $p = 2h \geq 4$, the always-present sites are

$$A = (-h, 1, 1), \quad B = (-h, 0, 0), \quad C = (-h, 1, 0),$$
$$E = (h-1, 0, 0), \quad F = (h-1, 1, 0), \quad G = (h, 0, 0).$$

and $D = (h-1, 1, 1)$ is present iff $p \geq 6$.

Lemma 59.9 (Explicit $q = 1$ fatal certificates)**.** *The fatal supports in Theorem 59.5 admit balanced witnesses that end at residual zero in a nonaccepting physical configuration.*

Proof. For odd $p = 2h + 1$, the positive sites are witnessed by

$$w_B = 1^{p(p-2)} 0^{p-2},$$
$$w_C = 1^{2h(h+1)-1} (0\, 1^{p-1})^{h-1} 0^h,$$
$$w_D = 1^{2h(h+1)} 0^p 1^{2h(h+1)+1}, \qquad w_E = 1^{2h^2+4h+1} 0^{p+1} 1^{2h(h+1)+1}.$$

The B-witness meets no other optional site; with B disabled the C- and D-witnesses are unpreempted, and with B, D disabled the E-witness is unpreempted. Site A acts only on the unary negative corridor and preserves an episode-closed unary reset family. Hence only $\varnothing$ and $\{A\}$ are admissible.

For even $p = 2h$, the two safe sectors are $\mathcal{N} = \{A, B, C\}$ and $\mathcal{P} = \{A, F, G\}$. The individually fatal sites have witnesses

$$w_D = 1^{p(p-3)} 0^{p-3} \quad (h \geq 3),$$
$$w_E = 1^{(h+1)(p-1)} (0\, 1^{p-1})^{h-1} 0^h.$$

The four cross-pair certificates are

$$w_{BF} = 1^{h(p+1)} 0^p 1^{h(p-1)}, \qquad w_{CF} = 1^{h(p+1)} 0^{p-1} 1^{h(p-3)},$$
$$w_{BG} = 1^{h(p+3)} 0^{p+1} 1^{h(p-1)}, \quad w_{CG} = 1^{h(p+3)} 0^p 1^{h(p-3)}.$$

They trap exactly one primitive kernel packet after the indicated mixed pair has been activated. Finally, for $(p, q) = (2, 1)$ the two balanced words 111001 and 111110001 witness $\{B, X\}$ and $\{B, G\}$ and both end at $(C, 110)$, a nonaccepting ordering of the primitive kernel packet. These certificates, together with the episode-closed safe sectors, prove the stated low-minimum classification. □

Chapter 60

Downward Closure and Category-Relative Causal Persistence

The exact weighted-balance family is downward closed, but that fact is not a theorem of exact lifts in general. The correct boundary statement is obtained by separating the lift from the schedule presentation and by typing the category in which a named causal edge is preserved.

60.1 Downward closure is not universal

Theorem 59.6 is a theorem about the canonical weighted-balance family, not a theorem about arbitrary exact lifts. The distinction is real.

Consider one semantic state m with two semantic letters x, y, both acting as the identity. Let the physical states be c, d, b, all projecting to m. Assign terminal values

$$\alpha(c) = \alpha(d) = 0, \qquad \alpha(b) = 1.$$

The baseline transitions are

$$c \xrightarrow{x} c, \quad c \xrightarrow{y} c,$$
$$d \xrightarrow{x} c, \quad d \xrightarrow{y} b,$$

with b absorbing. Add two optional sites:

$$A : c \xrightarrow{x} d, \qquad B : d \xrightarrow{y} c.$$

Theorem 60.1 (Two-site failure of universal downward closure)**.** *For this exact schedule environment,*

$$\mathrm{Adm} = \{\varnothing, \{B\}, \{A, B\}\}.$$

In particular $\{A, B\}$ is admissible but $\{A\}$ is not.

Proof. The empty schedule and $\{B\}$ never leave c. Under $\{A\}$, the word xy follows $c \xrightarrow{x} d \xrightarrow{y} b$ and fails. Under $\{A, B\}$, the same first step reaches d but the enabled B rule repairs the second step and returns to c. Thus adding an optional rule can repair a failure created by another optional rule. □

This two-site example is minimal among counterexamples with an admissible empty schedule: with only one optional site, the only possible failure of downward closure would require the empty schedule to be bad and the singleton schedule good.

60.2 Causal persistence depends on the presentation category

A named edge such as $A \to B$ refers to two designated optional sites and to a first-hit or pre-emption relation between them. This data is not contained in the underlying lift alone.

Proposition 60.2 (Typing of named causal edges)**.** *A fixed exact lift does not determine a named causal edge. The optional-site menu and the local alternative rules are part of the schedule presentation, so forgetting the presentation can change the causal graph without changing the baseline lift.*

For a named edge e, let $\Pi(e) = (\pi_0, \pi_1, \pi_2)$ record persistence under (i) site-preserving coordinate isomorphism, (ii) a specified fixed-literal-policy presentation class, and (iii) broad exact physical re-coordination. Because the middle coordinate depends on what presentation data are frozen, its class is written as a subscript.

For the classical $2:1$ literal one-top policy, consider two presentation categories. In the first, the baseline policy is fixed but the optional-site menu may vary. In the second, the designated A, B cells and the finite causal certificate are preserved.

Theorem 60.3 (Exact $A \to B$ category bifurcation)**.** *For the classical* $2:1$ *baseline literal one-top policy,*

$$\Pi_{\text{flex}}(A \to B) = (1, 0, 0)$$

in the site-flexible fixed-policy class, while

$$\Pi_{\text{cert}}(A \to B) = (1, 1, 0)$$

in the certificate/site-preserving class. Hence the middle persistence bit has no category-independent value.

Proof. The same all-lazy baseline lift admits two presentations. With menu $\{A, B, C\}$ the classical first-hit certificate yields $A \to B$; with menu $\{A, C\}$ the named B site is absent and the directed edge cannot be formed. Hence the middle bit is 0 in the site-flexible class. In the certificate-preserving class the cells

$$1111100 \rightsquigarrow (R, 00), \qquad (R, 00) \xrightarrow[B]{0} (C, 1110)$$

and the distinguishing continuation 1 are part of the morphism data; the same noncanonical first-hit fiber therefore survives every allowed change. Broad re-coordination can remove the named phenomenon. Section 60.3 gives the typed formulation. □

60.3 Persistence signature and category typing

A schedule presentation is a triple

$$\mathfrak{P} = (\mathcal{L}_0, E, \{\delta^e\}_{e \in E}),$$

where $\mathcal{L}_0$ is the fixed baseline lift, E is the named optional-site menu, and each e carries its alternative local rule. A named directed edge therefore belongs to $\mathfrak{P}$, not to $\mathcal{L}_0$ alone.

For a named edge e define its three-level persistence signature

$$\Pi(e) = (\pi_0, \pi_1, \pi_2) \in \{0, 1\}^3,$$

where π_0 records persistence under site-preserving coordinate isomorphism, π_1 records persistence in a specified fixed-literal-policy presentation class, and π_2 records persistence under broad exact physical re-coordination. The middle class must therefore be named as part of the datum.

For the classical 2:1 edge $A \to B$, the all-lazy baseline admits both the menu $\{A, B, C\}$, where the directed dependency is present, and the menu $\{A, C\}$, where it is absent. This proves the middle bit is 0 in the site-flexible class. If instead the designated A, B cells and the finite first-hit certificate

$$1111100 \rightsquigarrow (R, 00), \qquad (R, 00) \xrightarrow[B]{0} (C, 1110)$$

with distinguishing continuation 1 are preserved, the same unsafe first-hit fiber survives every allowed presentation change; the middle bit is then 1. Broad re-coordination can eliminate the named phenomenon. Thus

$$\Pi_{\text{flex}}(A \to B) = (1, 0, 0), \qquad \Pi_{\text{cert}}(A \to B) = (1, 1, 0).$$

60.4 Monograph synthesis

The 2:1 schedule geometry is therefore not an isolated finite accident. It sits inside an arbitrary-coprime finite causal-hypergraph classification. What is special to the canonical weighted-balance family is the downward-closed independence-complex structure; arbitrary exact-lift schedule families need not share it. Likewise, the edge $A \to B$ is coordinate-stable but not language-intrinsic: its middle persistence bit depends on whether the optional-site presentation is held fixed. These conclusions close the schedule questions generated in the preceding lift chapters and prepare the protected comparison theory that follows.

Chapter 61

Strict Coordinate Quotients and Protected Interfaces

Exact sectioned lifts admit both genuine reductions and vacuous semantic collapses. To compare physical presentations without erasing the LIFO information under study, we first restrict reduction to coordinate quotients that preserve the exact section and the permanent bottom marker, and then declare explicitly which finite physical interface is protected.

61.1 Exact sectioned lifts and used coordinates

Let M be a semantic transition system over input alphabet Σ. An exact sectioned pushdown lift is a tuple

$$\mathcal{L} = (C^{\mathrm{reach}}, \delta, \Theta, \sigma),$$

where C^{reach} is the reachable physical configuration set, δ is the one-top pushdown transition structure, $\Theta : C^{\mathrm{reach}} \to M$ is the semantic projection, and $\sigma : M \to C^{\mathrm{reach}}$ is a chosen section satisfying

$$\Theta\sigma = \mathrm{id}_M.$$

Exactness means that the physical transitions project to the semantic transitions.

For a finite-coordinate lift, define

$$Q_{\mathrm{use}} = \{q : \exists S\ (q, S) \in C^{\mathrm{reach}}\}$$

and let Γ_{use} be the stack symbols appearing in reachable stacks, including exactly one permanent bottom marker Z_0.

The basic resource vector is

$$\mathbf{R}(\mathcal{L}) = \big(|Q_{\mathrm{use}}|, |\Gamma_{\mathrm{use}}|\big).$$

Unreachable ambient states and symbols are ignored.

61.2 Strict coordinate quotients

Definition 61.1 (Strict coordinate quotient)**.** Let $\mathcal{L}_1, \mathcal{L}_2$ be exact sectioned one-top lifts of the same semantic system. A strict coordinate quotient (SCQ)

$$F : \mathcal{L}_1 \twoheadrightarrow_{\mathrm{SCQ}} \mathcal{L}_2$$

consists of surjections

$$f_Q : Q^{(1)}_{\mathrm{use}} \twoheadrightarrow Q^{(2)}_{\mathrm{use}}, \qquad f_\Gamma : \Gamma^{(1)}_{\mathrm{use}} \twoheadrightarrow \Gamma^{(2)}_{\mathrm{use}}$$

such that

$$f_\Gamma^{-1}(Z_0) = \{Z_0\},$$

and such that the induced stack map is letterwise and non-erasing:

$$F(q, X_1 \cdots X_k Z_0) = \big(f_Q(q), f_\Gamma(X_1) \cdots f_\Gamma(X_k) Z_0\big).$$

It must satisfy

$$F(C_1^{\mathrm{reach}}) = C_2^{\mathrm{reach}}, \quad \Theta_2 F = \Theta_1, \quad F\sigma_1 = \sigma_2,$$

and

$$F\delta_a^{(1)} = \delta_a^{(2)} F$$

on every reachable transition.

The bottom-marker condition is essential: a work symbol may not be merged with the unique permanent bottom cell.

Theorem 61.2 (SCQ reduction theorem)**.** *SCQ morphisms compose and define a preorder. Mutual SCQ reducibility is exactly SCQ isomorphism. If*

$$\mathcal{L}_1 \twoheadrightarrow_{\mathrm{SCQ}} \mathcal{L}_2,$$

then

$$\mathbf{R}(\mathcal{L}_2) \le \mathbf{R}(\mathcal{L}_1)$$

componentwise; if the quotient is not an isomorphism, at least one coordinate decreases strictly.

Proof. Composition is coordinatewise and preserves the transition, semantic, section, and bottom-marker conditions. If SCQ morphisms exist in both directions, finiteness and surjectivity force equality of the used state and symbol cardinalities; all coordinate maps are bijective and their inverses preserve the same structure. For monotonicity, each coordinate map is surjective. Equality in both resource coordinates forces both maps to be bijections, hence an isomorphism. □

Corollary 61.3 (Finite minimal-core existence)**.** *Every finite-coordinate exact sectioned lift has at least one SCQ-minimal quotient descendant.*

Proof. Each strict SCQ step strictly decreases a point of $\mathbb{N}^2$. No infinite strictly descending chain exists. □

Cheap finite-control decorations are automatically removed: if a redundant cyclic coordinate $j \in \mathbb{Z}_m$ is added to control but ignored by semantics and stack dynamics, projection $(q, j) \mapsto q$ is a strict SCQ quotient. On the other hand, an arbitrary behavioral quotient that erases the LIFO store is excluded by the letterwise non-erasing coordinate requirement.

61.3 Protected interfaces and maximal safe quotients

SCQ is intentionally rigid. To formulate which physical distinctions a quotient is allowed to erase, we attach a finite interface signature.

Definition 61.4 (Protected-interface lift)**.** A protected-interface lift is a pair

$$(\mathcal{L}, \lambda), \qquad \lambda : C^{\mathrm{reach}} \to \Lambda,$$

where Λ is finite and λ records physical information declared to be part of the comparison contract.

For a literal one-top policy, a useful interface is

$$\lambda_{\mathrm{lit}}(c) = \big(o_{\mathrm{lit}}(c), \chi_\sigma(c)\big),$$

where o_{lit} is the locally visible state/top observation and

$$\chi_\sigma(c) = \mathbf{1}_{\{c \in \sigma(M)\}}$$

records whether the configuration lies on the chosen canonical section.

Theorem 61.5 (Maximal interface-safe quotient criterion)**.** *Let $F : C_1^{\mathrm{reach}} \twoheadrightarrow C_2^{\mathrm{reach}}$ be surjective and let $\lambda_1 : C_1^{\mathrm{reach}} \to \Lambda$. The following are equivalent:*

1. *there is a unique $\lambda_2 : C_2^{\mathrm{reach}} \to \Lambda$ such that*

$$\lambda_1 = \lambda_2 \circ F;$$

2. *λ_1 is constant on every fiber of F.*

Hence λ-separated configurations can never be identified by an interface-safe quotient.

Proof. Necessity is immediate. Conversely define $\lambda_2(F(c)) = \lambda_1(c)$; fiber constancy makes the definition well posed, and surjectivity gives uniqueness. □

This criterion is maximal at the configuration level: it imposes exactly the identifications forbidden by the protected observable and no others.

Chapter 62

Bounded Bi-Recoding and Minimal Protected Cores

Coordinate quotients reduce resources, whereas bounded invertible recodings merely change the physical alphabet in a locally recoverable way. Treating these operations separately produces recoding-invariant minimal cores and finite protected Pareto envelopes.

62.1 Bounded bi-recoding and recoding-invariant resource spectra

A bounded invertible change of stack code should not be counted as a reduction.

Definition 62.1 (Bounded bi-recoding)**.** A bounded bi-recoding (BBR) between protected-interface lifts consists of a bijective control recoding and a non-erasing work-stack block code

$$\varphi : \Gamma_{1,\mathrm{use}} \setminus \{Z_0\} \to (\Gamma_{2,\mathrm{use}} \setminus \{Z_0\})^+$$

extended multiplicatively above the fixed bottom marker, such that:

1. $1 \le |\varphi(X)| \le B$ for a uniform finite B;
2. an inverse bounded non-erasing block code exists on the reachable stack languages;
3. the induced configuration bijection intertwines semantics, section, transitions, and the protected interface.

BBR is therefore an actual conjugacy of reachable dynamics. It preserves every obstruction expressible in semantic fibers, paths, section membership, and protected-interface values.

The identity code, the stipulated bounded inverse, and composition of bounded block codes show that BBR is an equivalence relation; see Section 62.2.

Define the protected recoding spectrum

$$\mathcal{R}_\lambda(\mathcal{L}) = \{(|Q_{\mathrm{use}}(\mathcal{L}')|, |\Gamma_{\mathrm{use}}(\mathcal{L}')|) : \mathcal{L}' \simeq_{\mathrm{BBR},\lambda} \mathcal{L}\}.$$

Its minimal antichain is

$$\mathcal{P}_\lambda(\mathcal{L}) = \operatorname{Min} \mathcal{R}_\lambda(\mathcal{L}).$$

Theorem 62.2 (Finite protected Pareto envelope)**.** *For every finite-coordinate protected lift, $\mathcal{P}_\lambda(\mathcal{L})$ is nonempty and finite and depends only on the BBR-equivalence class.*

Proof. The original presentation belongs to its spectrum, so the set is nonempty. Every nonempty subset of $\mathbb{N}^2$ has minimal elements, and Dickson's lemma [49] implies that the antichain of all componentwise-minimal elements is finite. BBR-equivalent lifts have the same spectrum by transitivity. □

A normalized protected core step first chooses a representative whose resource vector lies in $\mathcal{P}_\lambda$, then takes a strict λ-preserving SCQ quotient, and finally passes to the target BBR class. By Theorem 61.2, every such step strictly decreases the normalized resource pair. Hence terminal protected BBR-classes exist. Uniqueness is not asserted.

62.2 Typed protected contracts and BBR equivalence

For complete type correctness, a protected representation contract is written

$$\mathfrak{K} = \big(\mathfrak{C}, \{\lambda_{\mathcal{L}}\}_{\mathcal{L}\in\mathfrak{C}}\big),$$

where $\mathfrak{C}$ is a BBR-closed class of finite-coordinate exact sectioned lifts of one semantic system and each $\lambda_{\mathcal{L}} : C_{\mathcal{L}}^{\mathrm{reach}} \to \Lambda$ is a protected interface. The family is required to be compatible with BBR conjugacies: if $F : \mathcal{L} \to \mathcal{L}'$ is a BBR equivalence, then $\lambda_{\mathcal{L}} = \lambda_{\mathcal{L}'} \circ F$. In the body we suppress the object subscript when no ambiguity is possible.

Proposition 62.3 (BBR is an equivalence relation)**.** *Bounded bi-recoding is reflexive, symmetric, and transitive on finite-coordinate protected lifts.*

Proof. The identity code gives reflexivity. The defining bounded non-erasing inverse on reachable stack languages gives symmetry. If block codes φ and ψ have maximum block lengths B_φ, B_ψ, then $\psi \circ \varphi$ has maximum block length at most $B_\varphi B_\psi$; the inverse is the reverse composition of the bounded inverses. The induced configuration bijections compose and continue to intertwine semantics, section, transitions, and the protected-interface family. Hence transitivity holds. □

Under contract refinement $\mathfrak{K}' \succeq \mathfrak{K}$, the representation class is included and the protected interfaces refine objectwise. Therefore the resource-set inclusion used in Theorem 103.4 is literal, rather than an informal comparison between unrelated coordinate systems.

62.3 Interface with General Theory

The outcome is a comparison category strong enough to exclude semantic trivialization and weak enough to identify harmless bounded recodings. Terminal protected cores exist in every finite-coordinate class, but no uniqueness or Church–Rosser statement is asserted here. The remaining confluence chapter completes Part IV by comparing exact normalization layers under this protected interface; Part V can then formulate General Theory over a cleaner physical comparison layer without conflating exact semantics, coordinate quotient, and recoding equivalence.

Chapter 63

Confluence Across Exact Pushdown Normalization Layers

Research branch — Confluence and Normalization Choice. *The preceding causal-schedule and protected-interface theory distinguishes semantic correctness from physical presentation. This branch makes the rewrite layer explicit: schedule activation, full completion-fiber normalization, finite-control sheets, and protected-core descent are treated as different confluence problems rather than conflated under one endpoint-coherence statement.*

The exact-lift, causal-schedule, SCQ, and BBR objects used here have already been developed in the preceding chapters. The purpose of this chapter is to add the missing rewrite-theoretic layer and to separate four questions: canonical endpoint coherence, schedule-presentation confluence, physical completion-fiber confluence, and confluence of protected quotient descent.

63.1 Exact lifts and four notions of normalization coherence

Definition 63.1 (Exact sectioned lift)**.** Let M be a deterministic semantic transition system over input alphabet Σ, with deterministic partial semantic transitions $U_a : M \rightharpoonup M$. An *exact sectioned pushdown lift* is a reachable physical transition system

$$\mathcal{L} = (C^{\mathrm{reach}}, \delta, \Theta, \sigma)$$

with semantic projection $\Theta : C^{\mathrm{reach}} \to M$, partial physical settled macro-transitions δ_a, and section $\sigma : M \to C^{\mathrm{reach}}$ satisfying

$$\Theta\sigma = \mathrm{id}_M, \qquad c \in \mathrm{Dom}(\delta_a) \iff \Theta(c) \in \mathrm{Dom}(U_a), \qquad \Theta\delta_a = U_a\Theta$$

on the common defined domain. The canonical retraction is $N = \sigma\Theta$.

Proposition 63.2 (Canonical retraction and absorption; baseline)**.** *The canonical retraction satisfies*

$$N^2 = N, \qquad \Theta N = \Theta,$$

and for every input word w for which the transition is defined,

$$N\delta_w = N\delta_w N = \sigma U_w \Theta.$$

Thus the canonical endpoint after reading w depends only on the initial semantic state. This is endpoint coherence, not a Church–Rosser theorem for an internal normalization rewrite.

Proof. Since $\Theta\sigma = \mathrm{id}_M$, one has $N^2 = \sigma\Theta\sigma\Theta = N$ and $\Theta N = \Theta$. Exactness gives $\Theta\delta_w = U_w\Theta$, hence

$$N\delta_w = \sigma U_w\Theta = \sigma U_w\Theta\sigma\Theta = N\delta_w N.$$

□

The following distinctions will be used throughout.

Definition 63.3 (Coherence notions)**.** Let $\Rightarrow$ be an internal normalization relation, distinct from the ordinary input transition δ.

1. *Endpoint coherence*: the designated normalizer sends every reachable configuration to the chosen section representative of its semantic value.
2. *Physical confluence*: every fork under $\Rightarrow$ joins to one physical configuration.
3. *Confluence modulo* $\approx$: every fork joins to configurations equivalent under a declared relation $\approx$.
4. *Section-complete normalization*: every configuration in a semantic fiber reduces to the chosen section point.

Proposition 63.4 (Bare semantic modulo-confluence is vacuous)**.** *Suppose*

$$c \Rightarrow d \quad\Longrightarrow\quad \Theta(c) = \Theta(d).$$

Then every fork $c \Rightarrow^* u$, $c \Rightarrow^* v$ already satisfies $\Theta(u) = \Theta(v)$. Thus confluence modulo the equivalence $u \equiv_\Theta v$ iff $\Theta(u) = \Theta(v)$ is automatic.

Proof. Semantic preservation is inherited by the reflexive-transitive closure. Both descendants have semantic value $\Theta(c)$. □

Consequently, a substantive modulo-confluence theorem must protect more physical information than Θ alone.

Definition 63.5 (Strictly canonicalizing physical rewrite)**.** A tuple $(C, \Rightarrow, \Theta, \sigma, D)$ is *strictly canonicalizing* if D takes values in a well-founded ordered set and:

1. $c \Rightarrow d$ implies $\Theta(c) = \Theta(d)$;
2. every rewrite strictly decreases D;
3. $D(c) = 0$ iff $c = \sigma(\Theta(c))$;
4. every noncanonical c admits a rewrite step.

Proposition 63.6 (Canonical-normal-form criterion)**.** *Every strictly canonicalizing physical rewrite is strongly normalizing and confluent. The unique normal form in the fiber $\Theta^{-1}(m)$ is $\sigma(m)$.*

Proof. Well-founded strict descent forbids infinite chains. Every maximal chain terminates, and the last two conditions imply that a terminal point is exactly the section point in its semantic fiber. Semantic preservation keeps every descendant of a fork in the same fiber, so both branches terminate at the same section representative. □

63.2 Finite schedule-activation rewriting

Let E be a finite set of optional eager-normalization sites. A schedule is a subset $H \subseteq E$, and $\mathrm{Adm}(E) \subseteq 2^E$ denotes the schedules preserving the intended recognition/normalization contract.

Definition 63.7 (Schedule activation)**.** Define

$$H \rightsquigarrow_E H \cup \{e\}$$

when $e \notin H$ and $H \cup \{e\} \in \mathrm{Adm}(E)$. A schedule is normal when no further one-site activation is admissible.

Proposition 63.8 (Termination)**.** *Schedule activation terminates for every finite E.*

Proof. The rank $|E| - |H|$ decreases by one at every step. □

Theorem 63.9 (Exact criterion for downward-closed schedule families)**.** *Assume* $\mathrm{Adm}(E)$ *is nonempty and downward closed. The following are equivalent:*

1. $(\mathrm{Adm}(E), \rightsquigarrow_E)$ *is confluent;*
2. *there is a unique normal form reachable from* $\varnothing$*;*
3. $\mathrm{Adm}(E)$ *has a unique inclusion-maximal member.*

Proof. In a downward-closed family, if $H \subseteq K$ are admissible then every intermediate set obtained by adding elements of $K \setminus H$ remains admissible. Hence rewrite normal forms are exactly the inclusion-maximal schedules and every admissible schedule reaches every maximal extension containing it. A unique maximal member therefore joins every branch. Two distinct maximal members are distinct terminal descendants of $\varnothing$ and cannot join. □

Suppose now that admissibility is the independence complex of a finite hypergraph $\mathcal{F}$ of minimal fatal supports:

$$\mathrm{Adm}(E) = \mathrm{Ind}(\mathcal{F}) = \{H \subseteq E : \text{ no } F \in \mathcal{F} \text{ is contained in } H\}.$$

Theorem 63.10 (Fatal-support criterion)**.** *Schedule activation is confluent iff every minimal fatal support is a singleton. If*

$$S_1 = \{e \in E : \{e\} \in \mathcal{F}\},$$

then, in the confluent case, the unique normal form is $E \setminus S_1$.

Proof. If every fatal support is singleton, admissibility is exactly $2^{E \setminus S_1}$. Conversely, let $F \in \mathcal{F}$ contain distinct x, y. Minimality makes $F \setminus \{x\}$ and $F \setminus \{y\}$ admissible. Extend each to a maximal admissible schedule. They cannot have the same maximal extension, since such an extension would contain their union F. Thus there are at least two normal forms, and theorem 63.9 gives nonconfluence. □

63.3 Rational schedule confluence

The canonical baseline-portable literal-demand presentation of rational $p{:}q$ normalization has a finite optional-site family whose minimal fatal supports are summarized in table 63.1. Appendix 63.7 records the schedule data and the low-width certificates needed for the present rewrite application. The important point here is not the site coordinates themselves but the size of the minimal supports.

Regime	minimal fatal-support structure	maximal-schedule count
$\min(p,q) \geq 3$	all present fatal extras are singleton; the canonical core is free	1
$\min(p,q) = 2$	two non-singleton fork supports $\{A,D\},\{B,D\}$, plus singleton extras	2
$(p,q) = (1,1)$	no optional site	1
$\min(p,q) = 1$, odd maximum ≥ 3	all positive optional sites singleton-fatal; one negative-boundary site is free	1
$(2,1)$ up to duality	$\{B,X\},\{B,G\}$	2
$\min(p,q) = 1$, even maximum ≥ 4	singleton fatal sites plus the four pairs between two negative- and two positive-sector sites	2

Table 63.1: Fatal-support geometry needed for the rational schedule-confluence theorem. Physical duality exchanges p and q.

Corollary 63.11 (The canonical 2:1 fork)**.** *For the complete four-site baseline-portable family*

$$E_{2,1} = \{X, A, B, G\},$$

one has

$$\mathrm{Adm}_{2,1} = \{H \subseteq E_{2,1} : \{B,X\} \not\subseteq H,\ \{B,G\} \not\subseteq H\}.$$

Schedule activation terminates but is not confluent. Its two normal forms are

$$\{A,B\}, \qquad \{A,X,G\}.$$

Proof. Appendix 63.7 gives the exact minimal fatal supports $\{B,X\}$ and $\{B,G\}$; the site A is free. Hence the maximal admissible schedules are exactly the two displayed sets, and theorem 63.9 gives nonconfluence. □

Theorem 63.12 (Exact arbitrary-coprime schedule-confluence dichotomy)**.** *For every coprime $p,q \geq 1$, activation in the canonical baseline-portable rational schedule family is confluent exactly when*

$$\boxed{\min(p,q) \geq 3 \quad \textit{or} \quad \min(p,q) = 1 \ \textit{and}\ \max(p,q)\ \textit{is odd.}}$$

It is nonconfluent exactly when

$$\boxed{\min(p,q) = 2 \quad \textit{or} \quad \min(p,q) = 1 \ \textit{and}\ \max(p,q)\ \textit{is even.}}$$

Proof. Read the minimal fatal supports from table 63.1 and apply theorem 63.10. In the high-minimum and odd boundary regimes all minimal supports are singleton. In the width-two, $(2,1)$, and even boundary regimes a non-singleton fatal pair is present. □

Corollary 63.13 (One-or-two normal-form law)**.** *Every canonical rational schedule-activation system has exactly one or two normal forms. It has one precisely in the confluent regimes of theorem 63.12 and two precisely in the nonconfluent regimes.*

Proof. The maximal independent sets are visible in table 63.1. For $\min = 2$, either the fork vertex D is chosen and A, B are excluded, or D is excluded and A, B are chosen, with all free sites added. The $(2,1)$ and even boundary cases similarly split into exactly two safe sectors. The remaining regimes have one maximal admissible schedule. □

This classification is a presentation-level result. It says that some valid ways of adding eager normalization sites cannot be reconciled by further admissible activations. It does not yet describe rewriting inside a fixed physical semantic fiber.

63.4 The full rational completion fiber

Fix coprime $p, q \geq 1$ and use the completion equation

$$pN_0(z) - qN_1(z) = r.$$

For each $r \in \mathbb{Z}$, let

$$\mu(r) = (x_r, y_r) \in \mathbb{N}_0^2$$

be the unique minimum-length nonnegative solution. The elementary one-dimensional kernel structure gives

$$(x, y) = \mu(r) + k(q, p), \qquad k \in \mathbb{N}_0, \tag{63.1}$$

for every other nonnegative solution (x, y). Thus $K = (q, p)$ is the primitive redundant completion packet.

Define the canonical shortest completion

$$C(r) = \begin{cases} 1^{y_r}0^{x_r}, & r \geq 0, \\ 0^{x_r}1^{y_r}, & r < 0. \end{cases}$$

For $r = 0$ this is ε.

Definition 63.14 (Full completion-fiber rewrite)**.** Let

$$\mathcal{C}_{p,q} = \{(r, z) : r \in \mathbb{Z},\ z \in \{0,1\}^*,\ pN_0(z) - qN_1(z) = r\}.$$

Within a fixed residual fiber, generate $\Rightarrow_{p,q}$ by

$$\begin{aligned} r \geq 0: &\quad 01 \Rightarrow 10, \quad 1^p0^q \Rightarrow \varepsilon, \\ r < 0: &\quad 10 \Rightarrow 01, \quad 0^q1^p \Rightarrow \varepsilon. \end{aligned}$$

The swap discharges boundary-order debt and the block deletion removes one primitive kernel packet.

Lemma 63.15 (Semantic preservation and exact kernel debt)**.** *Every rewrite in theorem 63.14 preserves r. If $\mathrm{Par}(z) = (x, y)$, then the integer k in* (63.1) *is unique and*

$$\mu(r) - \mathrm{Par}(z) = -k(q, p).$$

Thus k is the exact kernel-excess quantity-debt level.

Proof. Swaps preserve Parikh counts. A deletion removes q zeros and p ones, so the value $pN_0 - qN_1$ changes by $pq - qp = 0$. The solution lattice of the coprime equation is generated by (q, p), and minimality of $\mu(r)$ forces the coefficient to be nonnegative. □

Theorem 63.16 (Full completion-fiber confluence)**.** *For every coprime $p, q \geq 1$ and every residual r, the rewrite of theorem 63.14 is strongly normalizing and confluent on the entire completion fiber. Its unique normal form is $C(r)$.*

Proof. Define the sign-adapted inversion number

$$\Omega_r(z) = \begin{cases} \#\{i < j : z_i = 0, z_j = 1\}, & r \geq 0, \\ \#\{i < j : z_i = 1, z_j = 0\}, & r < 0, \end{cases}$$

and let $k(z)$ be the kernel level from theorem 63.15. Use lexicographic order on

$$D_r(z) = (k(z), \Omega_r(z)) \in \mathbb{N}_0^2.$$

A swap fixes k and decreases Ω_r by one. A kernel deletion decreases k by one. Hence every rewrite strictly decreases D_r, so the system is strongly normalizing.

Let z be irreducible. If $r \geq 0$, absence of 01 forces $z = 1^y 0^x$; if $r < 0$, absence of 10 forces $z = 0^x 1^y$. Write $(x, y) = \mu(r) + k(q, p)$. If $k \geq 1$, then the block junction contains $1^p 0^q$ in the nonnegative orientation, respectively $0^q 1^p$ in the negative orientation, contradicting irreducibility. Therefore $k = 0$ and $z = C(r)$. Conversely $C(r)$ has neither an order redex nor a removable kernel packet. Thus every fiber has one normal form, and termination implies confluence (equivalently, apply theorem 63.6). □

Corollary 63.17 (Shortest-completion stratum)**.** *On the stratum $k = 0$, kernel deletion is unavailable and theorem 63.16 reduces to the boundary-oriented adjacent-swap normalization of shortest completions. The theorem therefore extends geodesic normalization to every nonminimal kernel level.*

Corollary 63.18 (Two-channel physical debt)**.** *The pair*

$$(k(z), \Omega_r(z))$$

is an exact normalization debt on completion words: its first coordinate measures redundant primitive-kernel quantity and its second coordinate measures word-order debt.

The point of theorem 63.16 is physical, not merely semantic. The local relation has many possible reduction orders; the theorem proves that all of them reach the same literal word. This is strictly stronger than applying the retraction $N = \sigma\Theta$ after the fact.

63.5 Finite-control sheets and three-channel confluence

Completion words have no independent finite-control coordinate. To model the third debt channel, let W_r be the word fiber above and let $(S_r, \to_r)$ be a separately declared control-sheet rewrite system with distinguished sheet s_r^*.

Definition 63.19 (Sheet-separable normalization)**.** Set

$$\widehat{C}_r = W_r \times S_r, \qquad \widehat{\sigma}(r) = (C(r), s_r^*).$$

The combined rewrite is generated by

$$(z, s) \Longrightarrow_r (z', s) \quad \text{if } z \Rightarrow_{p,q} z',$$

and

$$(z, s) \Longrightarrow_r (z, s') \quad \text{if } s \to_r s'.$$

The two move types commute exactly because they act on disjoint coordinates.

Theorem 63.20 (Exact product reduction)**.** *For every residual r:*

1. *$\Longrightarrow_r$ terminates iff $\to_r$ terminates;*
2. *$\Longrightarrow_r$ is confluent iff $\to_r$ is confluent;*
3. $$\mathrm{NF}(\widehat{C}_r) = \{C(r)\} \times \mathrm{NF}(S_r);$$
4. *every point of $\widehat{C}_r$ reduces to $(C(r), s_r^*)$ iff every sheet reduces to s_r^*.*

Proof. The word relation is terminating and confluent with one normal form by theorem 63.16. An infinite combined chain would contain infinitely many moves in one coordinate; this is possible exactly when the sheet relation has an infinite chain. For confluence, project a combined fork to the two coordinates. Word projections join by theorem 63.16, sheet projections join exactly when $\to_r$ is confluent, and exact cross-commutation combines the joins. Conversely, any sheet fork lifts to a fork from $(C(r), s)$, so combined confluence projects back to sheet confluence. Normal forms and section reachability are coordinatewise. □

Corollary 63.21 (Confluence does not imply section-complete normalization)**.** *Let S_r contain at least two sheets and let $\to_r$ be empty. Then the product rewrite is terminating and confluent but has one normal form $(C(r), s)$ for each sheet.*

Proof. The empty sheet relation is terminating and confluent. Apply theorem 63.20. □

Thus

branching coherence $\not\Rightarrow$ collapse of the whole semantic fiber to the section.

Theorem 63.22 (Rooted-sheet canonicalization)**.** *Suppose there is a well-founded rank $\eta_r : S_r \to W_r^{\mathrm{sh}}$ such that every sheet move strictly decreases η_r, $\eta_r(s) = 0$ iff $s = s_r^*$, and every noncanonical sheet admits a move. Then the combined rewrite is strictly canonicalizing with lexicographic debt*

$$\widehat{D}_r(z, s) = (k(z), \Omega_r(z), \eta_r(s)).$$

It is therefore strongly normalizing and confluent, with unique normal form $(C(r), s_r^*)$.

Proof. Kernel deletion decreases the first coordinate, sorting decreases the second while fixing the first, and sheet discharge decreases the third while fixing the first two. The debt is zero exactly at the section point. Apply theorem 63.6. □

The separability hypothesis is deliberate. We do not assert that an arbitrary DPDA configuration fiber decomposes as word × sheet. The theorem identifies a natural class in which the independent control channel can be handled without semantically erasing it.

63.6 Protected confluence and canonical Pareto anchoring

We finally allow harmless changes of physical code while retaining declared interface information. The minimum structure needed for the argument is stated abstractly so that the protected confluence theorem is self-contained.

Definition 63.23 (Protected exact-lift class)**.** Fix a class of finite-coordinate exact sectioned lifts of one semantic system. Each lift $\mathcal{L}$ carries a finite protected interface

$$\lambda_{\mathcal{L}} : C_{\mathcal{L}}^{\mathrm{reach}} \to \Lambda.$$

A *bounded bi-recoding* (BBR) is an invertible bounded physical recoding of finite control and reachable stack blocks that intertwines semantics, section, input transitions, and λ. We write $[\mathcal{L}]$ for its BBR class.

A *strict protected coordinate quotient* (SCQ) is a surjective coordinate quotient preserving semantics, transitions, section, the unique bottom marker, and λ, and inducing a componentwise nonincreasing map on the used resource coordinates, with at least one strict decrease:

$$\mathbf{R}(\mathcal{K}) < \mathbf{R}(\mathcal{L}) \quad \text{componentwise, with at least one strict coordinate,}$$

where $\mathbf{R}(\mathcal{L}) = (|Q_{\mathrm{use}}|, |\Gamma_{\mathrm{use}}|)$.

For a BBR class $X = [\mathcal{L}]$, define its protected recoding spectrum

$$\mathcal{R}_\lambda(X) = \{\mathbf{R}(\mathcal{L}') : \mathcal{L}' \simeq_{\mathrm{BBR},\lambda} \mathcal{L}\},$$

and its Pareto envelope

$$\mathcal{P}_\lambda(X) = \operatorname{Min} \mathcal{R}_\lambda(X) \subset \mathbb{N}^2.$$

Dickson's lemma implies that this antichain is finite. Every member is realized by an actual representative because it is a spectrum point.

Definition 63.24 (Protected modulo-equivalence)**.** Let X be the state set of a normalization rewrite and let

$$\tau : X \to \Lambda^\sharp$$

be a declared protected signature. An equivalence $\approx_\tau$ is *protected* when

$$x \approx_\tau y \quad \Longrightarrow \quad \tau(x) = \tau(y).$$

At configuration level one may take $\tau = \lambda$. When the rewrite states are BBR classes, τ may be any BBR-invariant signature induced by the protected interface, such as the isomorphism type of a specified finite interface-labelled observable. Passing from representatives to BBR classes already quotients out BBR equivalence itself.

Proposition 63.25 (Terminal protected fork)**.** *In a terminating rewrite presentation equipped with a protected signature τ, if a source has terminal descendants u, v with $\tau(u) \neq \tau(v)$, then the system is not confluent modulo any protected equivalence $\approx_\tau$.*

Proof. Terminality leaves no further reduction. A modulo-join would therefore force $u \approx_\tau v$, contradicting preservation of τ. □

A subtlety arises if each reduction step is allowed to choose an arbitrary incomparable Pareto point of the current BBR class: one no longer has a single resource pair that visibly decreases along every possible path. We remove this hidden source choice canonically.

Definition 63.26 (Lex-normalized protected-core rewrite)**.** Let

$$\nu_\lambda(X) = \min_{\text{lex}} \mathcal{P}_\lambda(X).$$

A step

$$X \Longrightarrow_\lambda^{\text{lex}} Y$$

chooses a BBR-equivalent representative of X attaining the actual resource pair $\nu_\lambda(X)$, takes a strict protected SCQ quotient to a lift $\mathcal{K}$, and sets $Y = [\mathcal{K}]$.

Theorem 63.27 (Strong normalization of canonical protected-core descent)**.** *Every lex-normalized protected-core step strictly decreases ν_λ in lexicographic order on $\mathbb{N}^2$. Hence $\Longrightarrow_\lambda^{\text{lex}}$ is strongly normalizing.*

Proof. Put $r = \nu_\lambda(X)$. A strict protected SCQ quotient from a representative attaining r has raw resource vector $r' < r$ componentwise, with at least one strict coordinate. The quotient is a representative of the target BBR class Y. Hence some Pareto point $r^\sharp \in \mathcal{P}_\lambda(Y)$ satisfies $r^\sharp \leq r'$. By lexicographic minimality,

$$\nu_\lambda(Y) \leq_{\text{lex}} r^\sharp \leq_{\text{lex}} r' <_{\text{lex}} r.$$

Lexicographic order on $\mathbb{N}^2$ is well-founded. □

Theorem 63.28 (Protected-core confluence criterion)**.** *On the descendant cone of any starting BBR class under $\Longrightarrow_\lambda^{\text{lex}}$, the following are equivalent:*

1. *confluence;*
2. *local confluence;*
3. *exactly one terminal BBR class is reachable from the starting class.*

Proof. Termination is theorem 63.27, so Newman's lemma gives equivalence of confluence and local confluence [62, 61]. Every descendant of a terminating system reaches a terminal class. Confluence forces all terminal descendants to coincide. Conversely, if one terminal class T is reachable from the source and is the only terminal descendant of that source, then every descendant also terminates at T, because any terminal descendant of a descendant is a terminal descendant of the source. Thus any two descendants join at T. □

Corollary 63.29 (Protected signature certificate)**.** *If two terminal lex-normalized descendants have different values of a BBR-invariant protected signature τ, then the protected-core rewrite is nonconfluent from that source (and cannot be repaired by passing to any equivalence that preserves τ).*

Proof. Apply theorem 63.25. □

The canonical anchor is essential to the theorem as stated. We do not claim that a larger unanchored relation allowing arbitrary re-selection of incomparable Pareto points is strongly normalizing or confluent.

63.7 Rational fatal-support data used in Section 63.3

This appendix makes the rational schedule application self-contained at the level needed by the confluence theorem: it specifies the optional-site contract, the complete site universe, and the exact minimal-fatal-support data. No theorem in the abstract rewrite sections depends on these coordinates.

Put

$$m = p + q, \qquad h = \left\lfloor \frac{m-1}{2} \right\rfloor, \qquad K = (q, p),$$

and let

$$\mu(r) = \arg\min\{x + y : x, y \in \mathbb{N},\ px - qy = r\}$$

be the unique minimum nonnegative demand vector. Write $\eta(x, y)$ for the fixed canonical literal stack word representing the demand pair (x, y). The centered literal-demand lift has debt states $D_{-h}, \dots, D_h$ and center state C; write $V_d = \mu(dm)$. A local work-top observation is a triple

$$o = (d, a, z), \qquad |d| \le h, \quad a, z \in \{0, 1\},$$

meaning that the current state is D_d, the next input letter is a, and the visible stack symbol is z. With $e_0 = (1, 0)$, $e_1 = (0, 1)$, define

$$\begin{aligned} B_o &= e_z - e_a + V_d, \\ \kappa(o) &= \max\left\{ \left\lceil -\frac{(B_o)_0}{q} \right\rceil, \left\lceil -\frac{(B_o)_1}{p} \right\rceil \right\}, \\ R_o &= B_o + \kappa(o)K. \end{aligned}$$

The optional eager candidate at o consumes a, replaces the visible cell by the literal minimum-demand word representing R_o, and returns directly to C. This is the unique shortest nonnegative direct return preserving the residual invariant, because every other direct-center replacement belongs to the affine line $B_o + \mathbb{Z}K$.

A *baseline episode* begins at a canonical center demand reached by a mandatory WRITE, then follows only mandatory MATCH/SEARCH transitions, and ends before the next mandatory bottom or outer WRITE. A site is *baseline-portable* when its eager rule lands on the canonical center representative for every source exposed in every baseline episode. A schedule is a subset of the resulting finite menu; it is admissible when the deterministic machine still recognizes exactly $L_{p,q}$. A set F of sites is *minimal fatal* when every schedule containing F is bad but no proper subset has that property. Thus the terms used in table 63.1 refer to a completely specified finite presentation rather than to unnamed abstract vertices.

63.7.1 Complete site universe for $\min(p,q) \ge 2$

By physical duality it suffices to take $p \ge q \ge 2$. If m is odd, the complete baseline-portable universe consists of the six core sites

$$(-h,0,0),\ (-h,1,0),\ (-h,1,1),\ (h-1,0,0),\ (h-1,1,0),\ (h,0,0),$$

together with

$$X = (h-q+1,0,1) \quad\Longleftrightarrow\quad p > 5q-5,$$
$$Y = (h-q,1,1) \quad\Longleftrightarrow\quad p > 5q-1.$$

If m is even, the complete universe consists of the four core sites

$$(-h,1,1),\ (h,0,0),\ (h,0,1),\ (h,1,1),$$

together with

$$X = (h-q+1,0,0) \quad\Longleftrightarrow\quad p > 5q-5,$$
$$Y = (h-q,1,0) \quad\Longleftrightarrow\quad p > 5q.$$

Hence the menu has at most eight sites. For $q > p$, apply

$$(p,q,d,a,z) \longleftrightarrow (q,p,-d,1-a,1-z).$$

For completeness, the exclusion mechanism is exact rather than computational. For the bottom-WRITE seed family

$$B_j = \mu(jm+q)$$

one has, for $j \ge 0$,

$$B_j = (j+q, p-j-1),$$

while for $j = -t < 0$, writing $t = kq+s$ with $0 \le s < q$,

$$B_{-t} = \begin{cases} (0, km-1), & s = 0, \\ (q-s, p+km+s-1), & 1 \le s < q. \end{cases}$$

For $1 \le d \le h$, $V_d = (d+q, p-d)$ and the three nontrivial positive-debt compiler count vectors have imbalances

$$2d+q-p, \qquad 2d+q-p+2, \qquad 2d+q-p-2.$$

If $m = 2h+1$, the central seed $B_{h-q} = (h,h)$ has word $(10)^h$. At an interior debt d, the untouched suffix is $(10)^{h-d-1}$ below a visible 0 and $0(10)^{h-d-1}$ below a visible 1. A top-0 eager concatenation can be canonical only when its compiler word is balanced, while a top-1 concatenation requires exactly one excess 1; solving these conditions leaves precisely the two odd-parity candidates displayed above. If $m = 2h+2$, the central seed is $(h, h+1)$ with word $(10)^h 1$; the corresponding parity calculation eliminates every top-1 interior candidate and leaves precisely the two displayed top-0 candidates. On the negative side, every nontrivial compiler has a terminal 1-tail whereas the diagnostic exposes an untouched 0, so no negative-interior observation is portable.

The boundary source fibers give exactly the stated cores. For the two odd-parity survivors the common compiler word is $(10)^h 1$ and the extremal exposed one-counts are $2q-2$ and $2q$, so kernel reduction is equivalent to

$$h+2q-2 < p, \qquad h+2q < p,$$

which simplify to $p > 5q-5$ and $p > 5q-1$. In even parity the common compiler is $(10)^{h+1}$, the extremal counts are $2q-3$ and $2q-1$, and the same calculation gives $p > 5q-5$ and, using coprimality of odd p, q, $p > 5q$. Thus no unlisted optional site is being silently ignored.

63.7.2 High-minimum fatal supports

Assume $\min(p, q) \geq 3$. Every subset of the parity-dependent core above is admissible. Indeed, induction over reset episodes keeps every core source inside a protected unary envelope, and every enabled core activation returns to a canonical minimum-demand word. Minority counts stay below the opposite boundary threshold, so a core reset cannot create a new incompatible source on the other side.

Each present extra site X or Y is singleton-fatal. Repeated self-activation increases a hidden alternating prefix until the first nonminimal target is reached. At that first failure the target is exactly

$$\mu(r) + K,$$

not $\mu(r) + jK$ with $j \geq 2$. Feeding a shortest semantic completion for r removes the minimum component and leaves exactly one balanced primitive-kernel packet physically present at semantic residual zero. The witness for Y meets no other optional site; the witness for X meets no core site and at most the neighboring Y observation. Therefore the singleton fatality is robust under every larger schedule. Consequently the only minimal fatal supports in this regime are the present singleton extras.

63.7.3 Width two

By duality take $q = 2$ and odd $p \geq 3$, with the six odd-parity core sites labelled

$$\begin{aligned} A &= (-h, 0, 0), \quad & B &= (-h, 1, 0), \quad & C &= (-h, 1, 1), \\ D &= (h-1, 0, 0), \quad & E &= (h-1, 1, 0), \quad & F &= (h, 0, 0). \end{aligned}$$

The core has exactly the two non-singleton minimal supports

$$\{A, D\}, \qquad \{B, D\}.$$

The possible extras are

$$X_1 = (h-1, 0, 1) \quad (p \geq 7), \qquad X_2 = (h-2, 1, 1) \quad (p \geq 11),$$

and each present extra is singleton-fatal. Hence

$$\mathrm{Adm}_{p,2} = \{H : X_1, X_2 \notin H,\ \{A, D\} \not\subseteq H,\ \{B, D\} \not\subseteq H\},$$

with absent extras omitted.

Here the sufficiency and necessity can be checked directly. Write $p = 2h-1$, $m = 2h+1$, and for $t \geq 1$ put

$$\epsilon_t = t \bmod 2, \qquad k_t = \left\lceil \frac{t}{2} \right\rceil, \qquad \beta_t = t + pk_t, \qquad \Gamma_t = \eta(\epsilon_t, \beta_t).$$

The mandatory reset family is explicitly

$$\begin{aligned} \mathcal{R}_h^{\mathrm{base}} = {} & \{\mu(jm+2) : -h-1 \leq j \leq h\} \\ & \cup \{(x, h-2) : x \geq h+3\} \\ & \cup \{(\epsilon_{h+1}, y) : y \geq \beta_{h+1}\}. \end{aligned}$$

Two episode-closed repertoires are

$$\begin{aligned} \mathcal{R}_{\neg D,h} = {} & \mathcal{R}_h^{\mathrm{base}} \cup \{(\epsilon_h, y) : y \geq \beta_h\} \\ & \cup \{(\epsilon_{h-1}, \beta_{h-1})\} \cup \{(x, h-1) : x \geq h+2\}, \\ \mathcal{R}_{D,h} = {} & \mathcal{R}_h^{\mathrm{base}} \cup \{(\epsilon_h, y) : y \geq \beta_h\} \\ & \cup \{(x, h-1) : x \geq h+2\} \cup \{(x, h) : x \geq h+1\}. \end{aligned}$$

Every pair in either repertoire has minority coordinate at most h. Before the first reset, MATCH and interior SEARCH only delete the visible top cell, so a boundary source is a suffix of its starting canonical word. This forces the exact source geometry: with D disabled, the common A/B source at D_{-h} is the one-cell stack 0; with D enabled it is an arbitrary 0^n, $n \geq 1$; all C sources are unary 1^n and all D, E, F sources are unary 0^n.

The eager targets are therefore exactly

site	canonical target pair
A	(ϵ_h, β_h)
B	$(\epsilon_{h-1}, \beta_{h-1})$
C	$(\epsilon_h, \beta_h + n - 1)$
D	$(n + h, h)$
E, F	$(n + h + 1, h - 1)$.

Thus $\mathcal{R}_{\neg D,h}$ is closed under every subset of A, B, C, E, F, giving $2^5 = 32$ safe schedules with $D = 0$, while $\mathcal{R}_{D,h}$ is closed when $D = 1$ and $A = B = 0$, giving another $2^3 = 8$. This proves the 40 safe schedules without enumeration.

For necessity, the D-ray gives, for every $x \geq h + 1$,

$$(C, \eta(x, h)) \xrightarrow{0^{2h-1}} (D_{-h}, 0^{x-h+1}),$$

so enabling D expands the A/B source fiber from $\{0\}$ to the whole unary ray. A uniform prefix reaches $(D_{-h}, 000)$ while meeting no optional site except D. At that common source

$$A : (D_{-h}, 000) \xrightarrow{0} (C, \Gamma_h 00), \qquad B : (D_{-h}, 000) \xrightarrow{1} (C, \Gamma_{h-1} 00).$$

Each Γ_t ends in at least p consecutive ones; writing $\Gamma_t = z_t 1^p$, subsequent MATCHes along z_t leave the nonempty stack $1^p 00$. Its residual is $2p - 2p = 0$, so it is one primitive kernel packet at semantic zero and is nonaccepting. Hence $\{A, D\}$ and $\{B, D\}$ are robust minimal fatal supports. The two maximal core schedules are $\{A, B, C, E, F\}$ and $\{C, D, E, F\}$; the extra sites are independently singleton-fatal.

63.7.4 The boundary line $q = 1$

For $(p, q) = (1, 1)$ the optional-site universe is empty. For $(2, 1)$ the complete four-site universe is

$$X = (0, 1, 0), \qquad A = (-1, 1, 1), \qquad B = (-1, 0, 0), \qquad G = (1, 0, 0),$$

and the minimal fatal supports are

$$\{B, X\}, \qquad \{B, G\}.$$

The balanced words 111001 and 111110001 are explicit fatal certificates; they end at semantic residual zero with a nonaccepting ordering of one primitive kernel packet. There are exactly ten admissible schedules and two maximal ones.

For odd $p = 2h + 1 \geq 3$, the always-present sites are

$$A = (-h, 1, 1), \qquad D = (h, 1, 1), \qquad E = (h, 0, 0),$$

and for $p \geq 5$ also

$$B = (h - 1, 1, 0), \qquad C = (h, 0, 1).$$

Every positive optional site is singleton-fatal, while A alone generates an episode-closed unary negative reset corridor. For $h \geq 2$ one may use

$$w_B = 1^{p(p-2)} 0^{p-2}, \qquad w_C = 1^{2h(h+1)-1} (0\, 1^{p-1})^{h-1} 0^h,$$

and for every $h \geq 1$,

$$w_D = 1^{2h(h+1)}0^p1^{2h(h+1)+1}, \qquad w_E = 1^{2h^2+4h+1}0^{p+1}1^{2h(h+1)+1}.$$

The B-witness meets no other optional site; after B is excluded the C- and D-witnesses are unpreempted; after B, D are excluded the E-witness is unpreempted. The last two end at the noncanonical zero-residual kernel word $(C, 1^p0)$, and the first two likewise end at explicit nonaccepting zero-residual configurations. Hence no positive-site failure can be repaired by enlarging the schedule, whereas the unary corridor generated by A stays reset-closed. Thus

$$\mathrm{Adm}_{p,1} = \{\varnothing, \{A\}\},$$

so there is one maximal schedule.

For even $p = 2h \geq 4$, the always-present sites are

$$\begin{aligned} A &= (-h, 1, 1), \quad & B &= (-h, 0, 0), \quad & C &= (-h, 1, 0), \\ E &= (h-1, 0, 0), \quad & F &= (h-1, 1, 0), \quad & G &= (h, 0, 0), \end{aligned}$$

and $D = (h-1, 1, 1)$ is also present when $p \geq 6$. The vertices D and E are singleton-fatal when present, and the non-singleton minimal supports are exactly

$$\{B, F\}, \quad \{B, G\}, \quad \{C, F\}, \quad \{C, G\}.$$

After singleton-fatal sites are removed, every safe schedule lies in one of the two closed sectors

$$\mathcal{N} = \{A, B, C\}, \qquad \mathcal{P} = \{A, F, G\}.$$

For subsets of $\mathcal{N}$, the only new optional reset words are unary-one words and the source fibers satisfy $\mathcal{E}_A \subseteq 1^+$ and $\mathcal{E}_B = \mathcal{E}_C = \{0\}$; the mandatory reset family plus this unary ray is episode-closed. For subsets of $\mathcal{P}$, the only new positive reset ray is $(10)^h0^n$, $n \geq 1$, together with the unary-one negative ray from A; all positive sources remain unary zero and all targets remain canonical and kernel-reduced. Thus every subset of either sector is safe.

Necessity has uniform balanced certificates. For $h \geq 3$ one may take

$$w_D = 1^{p(p-3)}0^{p-3},$$

and for $h \geq 2$,

$$w_E = 1^{(h+1)(p-1)}(0\,1^{p-1})^{h-1}0^h.$$

The four cross-pair supports are certified by

$$\begin{aligned} w_{BF} &= 1^{h(p+1)}0^p1^{h(p-1)}, \quad & w_{CF} &= 1^{h(p+1)}0^{p-1}1^{h(p-3)}, \\ w_{BG} &= 1^{h(p+3)}0^{p+1}1^{h(p-1)}, \quad & w_{CG} &= 1^{h(p+3)}0^p1^{h(p-3)}. \end{aligned}$$

In each cross-pair certificate the positive site first expands the negative-boundary top-0 source from 0 to 00, after which the indicated negative site traps exactly one primitive kernel packet. Consequently the only maximal schedules are $\mathcal{N}$ and $\mathcal{P}$, and the union of their subset families has $8 + 8 - 2 = 14$ schedules because the overlap is $2^{\{A\}}$. Physical duality gives the cases $p = 1$.

Combining the four regimes proves the exact fatal-support table used in table 63.1. In particular, its use in theorem 63.12 is no longer an appeal to an unnamed external classification: all vertices, the admissibility contract, the exhaustive universe, and the fatal-support mechanisms required by the confluence argument have been specified here.

63.8 Interface with the General Theory

The structural program of Part IV is now closed at the level needed for abstraction. Exact semantics has been separated from schedule choice; arbitrary-coprime causal geometry has been typed by presentation category; protected quotients and bounded recodings specify which physical information is retained; and confluence has been split into distinct endpoint, schedule, fiber, sheet, and protected-descent questions.

Part V can therefore remove the weighted-balance arithmetic without erasing the physical distinctions that made the preceding theorems meaningful. Its starting data are the reusable objects isolated here: a semantic transition system, an exact physical lift, a canonical section, bounded local observation, and an explicitly chosen comparison contract.

Part V

General Theory of Exact Pushdown Normalization

Part guide. **Question.** Which conclusions of the WSC analysis survive after weighted-balance arithmetic is removed? **Main result.** Rooted exact semantic lifts, canonical sections, bounded-local realizability, collapse capacity, representation debt, behavioral safety, conflict complexes, synthesis, re-coordination, and verification are developed as an abstract framework. **Why here.** This is the conceptual center of the monograph: it states the reusable theory rather than another special-purpose automaton. **Dependency.** Part IV supplies the motivating structures; subsequent Parts reuse the framework but do not define its scope.

Chapter 64

Semantic Fibers and Canonical Retraction

Parts I–IV progressively separated semantic evolution from physical normalization. Part IV completed that separation by distinguishing schedule causality, protected comparison, and confluence at different physical layers. The reusable conclusion is

semantic evolution can be unique while physical normalization is not.

The General Theory now removes the weighted-balance specifics.
Its objects are:

- an abstract semantic transition system;
- an exact physical lift;
- a chosen section;
- bounded-local physical update rules;
- explicitly stated representation categories.

The word **General** is scoped accordingly.
This macro-Part does not claim:

- a theory of every pushdown automaton;
- existence of a global section for every correct DPDA;
- unrestricted DPDA minimization;
- unrestricted infinite-lift synthesis;
- universal schedule downward closure;
- universal confluence.

64.1 Rooted exact semantic lifts and defined domains

Let M be a semantic state set with a distinguished root m_0 and deterministic partial actions

$$U_a : M \rightharpoonup M, \qquad a \in \Sigma.$$

Let C be a physical policy domain with root c_0 and deterministic partial settled-to-settled macro-updates

$$\delta_a : C \rightharpoonup C.$$

A macro-update may stand for one input-consuming raw transition together with uniquely forced finite internal normalization before or after that input. Thus the abstract theory is compatible both with real-time lifts and with deterministic ε-pushdown realizations viewed at settled boundaries.

Write M^{reach} and C^{reach} for the states and physical configurations reachable from the two roots by words whose successive actions are defined. An *exact rooted lift* is a surjective projection

$$\boxed{\Theta : C^{\mathrm{reach}} \to M^{\mathrm{reach}}}$$

such that, for every reachable c and symbol a,

$$\boxed{c \in \mathrm{Dom}(\delta_a) \iff \Theta(c) \in \mathrm{Dom}(U_a)}$$

and, on that common defined domain,

$$\boxed{\Theta(\delta_a(c)) = U_a(\Theta(c)).}$$

This domain-reflection clause prevents a physical policy from silently disabling a semantic action or inventing one that the semantic system does not possess.

Choose a reachable section

$$\boxed{\sigma : M^{\mathrm{reach}} \to C^{\mathrm{reach}}}, \qquad \Theta\sigma = \mathrm{id}.$$

When zero representation debt from the root is discussed we additionally assume

$$\boxed{c_0 = \sigma(m_0).}$$

Define the canonical retraction

$$\boxed{N = \sigma\Theta.}$$

For readability, the superscript "reach" is suppressed in the remainder of the General Theory: M and C henceforth mean these rooted reachable domains unless an ambient domain is explicitly named.

64.2 Canonical fibers

Theorem 64.1 — Canonical fiber and retraction.

The map N satisfies

$$\boxed{N^2 = N,}$$

$$\boxed{\Theta N = \Theta,}$$

and

$$\boxed{\mathrm{Fix}(N) = \mathrm{im}\,\sigma.}$$

Moreover,

$$\boxed{\Theta(c) = \Theta(c') \iff N(c) = N(c').}$$

Thus the fibers of the canonical retraction are exactly the semantic fibers.

Proof.

Idempotence follows from

$$\Theta\sigma = \mathrm{id}.$$

If $N(c) = c$, then $c = \sigma(\Theta(c))$, so c lies in the section image.

Conversely every section point is fixed.

Finally, equal semantic values clearly give equal canonical representatives, and equal canonical representatives project to equal semantic states. □

This is the abstract form of the canonical-fiber idea first seen concretely in WSC.

64.3 Canonical absorption

Theorem 64.2 — Canonical absorption.

For every input word w for which the rooted transition is defined,

$$\boxed{N\delta_w = N\delta_w N = \sigma U_w \Theta.}$$

Thus the canonical endpoint after reading w depends only on the initial semantic state. All prior physical normalization history is forgotten by the final retraction.
This theorem is the general form of:

- eager/lazy semantic equivalence;
- WSC canonical absorption;
- terminal normalization coherence.

It remains weaker than an unrestricted confluence theorem.

64.4 Zero debt and raw conjugacy

A reachable section is **transition-closed** when, for every $m \in M$ and every symbol a with $U_a(m)$ defined,

$$\boxed{\delta_a(\sigma(m)) = \sigma(U_a(m)).}$$

Domain reflection guarantees that the left-hand side is defined whenever the right-hand side is.

Theorem 64.3 — Rooted zero-debt/raw-conjugacy equivalence.

Assume $c_0 = \sigma(m_0)$ and that M, C denote the rooted reachable domains just defined. Then the following are equivalent:

1. the chosen section is transition-closed;
2. every reachable physical configuration lies on the section, equivalently

$$\boxed{C = \sigma(M).}$$

Proof. If the section is transition-closed, induction on the consumed word gives $\delta_w(c_0) = \sigma(U_w(m_0))$ whenever the word is defined, so every reachable physical configuration lies on the section. Conversely, let $m \in M$ and let a be defined at m. Because m is rooted-reachable, some word w satisfies $U_w(m_0) = m$. By the second hypothesis the corresponding physical endpoint lies on the section; exactness forces that endpoint to be the unique section point $\sigma(m)$. The next physical macro-update is reachable and again lies on the section, while exactness gives semantic value $U_a(m)$. Hence it must equal $\sigma(U_a(m))$. Thus the section is transition-closed. □

The rooted hypotheses are essential: bijectivity of Θ on some unrelated reachable representatives does not imply closure of a separately chosen section.

64.5 Three levels now separated permanently

From this point onward we distinguish:

$$\boxed{\text{semantic state}}$$

$$\boxed{\text{canonical section representative}}$$

$$\boxed{\text{arbitrary reachable physical representative in the same fiber.}}$$

The next chapter asks whether a bounded-local physical policy can keep every semantic transition on the chosen section.

Chapter 65

Bounded-Local Realizability, Collapse Capacity, and the Three-Stage Obstruction

The top-rewrite suffix criterion was discovered in the literal WSC setting. The General Theory now separates the part that depends on a pushdown stack from the more abstract information-capacity statement.

The fundamental question is:

> Can many semantically distinct sources that share one next semantic state be represented by canonical physical sources whose one-symbol macro-updates all hit one chosen canonical target without using more local information, exposing hidden source information, or retaining representation debt?

65.1 Collapse interfaces and retained information

Fix a symbol $a \in \Sigma$ and a reachable semantic target $m^* \in M$.

Collapse family. A nonempty finite set

$$\mathcal{S} \subseteq U_a^{-1}(m^*)$$

is an a-collapse family with target m^*. Put

$$N = |\mathcal{S}|, \qquad I_{\mathrm{ret}} = \log_2 N.$$

The quantity I_{ret} measures distinctions present immediately before the common semantic action; it is not the size of an earlier ambient fiber.

For $x \in \mathcal{S}$, exactness and domain reflection make the physical macro-update

$$\delta_a(\sigma(x)) \in \Theta^{-1}(m^*)$$

well defined. Call x a *canonical hit* when

$$\delta_a(\sigma(x)) = \sigma(m^*),$$

and a *debt source* otherwise. Write

$$D = \#\{x \in \mathcal{S} : x \text{ is a debt source}\}, \qquad K = N - D.$$

65.2 Observation filtrations and hidden information

Let

$$\omega_h : \sigma(\mathcal{S}) \longrightarrow \Omega_h, \qquad h = 0, 1, 2, \ldots,$$

be a nested family of finite source observations: ω_{h+1} refines ω_h. Put

$$V(h) = |\Omega_h|.$$

The index h is an abstract information depth. In a pushdown specialization it will be the number of source-stack cells made available; in another physical model it may describe a different nested observation window.

A canonical source computation is called *h-confined* when, before its first canonical hit, it never accesses source information beyond that represented by ω_h. Let

$$X_h = \#\{x \in \mathcal{S} : x \text{ is a canonical hit but not } h\text{-confined}\}.$$

Hidden-tail noncoalescence. The filtration has hidden-tail noncoalescence on this collapse interface if two distinct canonical sources with the same ω_h-observation cannot both remain h-confined and reach the same chosen physical target $\sigma(m^*)$. This is the precise hypothesis needed by the abstract counting argument. It is not assumed merely because the physical device is called a pushdown automaton; the later pushdown-exposure chapter proves it from untouched-source-cell provenance.

Theorem 65.1 — Filtered collapse-capacity theorem.

If hidden-tail noncoalescence holds, then for every $h \geq 0$,

$$\boxed{D + X_h + V(h) \geq N.}$$

Equivalently,

$$\boxed{X_h \geq (K - V(h))_+.}$$

Proof. Among the K canonical sources, exactly $K - X_h$ are h-confined. Hidden-tail noncoalescence makes ω_h injective on this subset, so

$$K - X_h \leq V(h).$$

Substituting $K = N - D$ gives the claim. □

Corollary 65.2 — Exposure-rank form.

Assume zero debt and that every canonical computation is h-confined for some finite h. Let $e(x)$ be the least such depth and sort the values as

$$e_{(1)} \leq \cdots \leq e_{(N)}.$$

With generalized inverse

$$V^{-1}(r) = \min\{h \geq 0 : V(h) \geq r\},$$

one has

$$\boxed{e_{(r)} \geq V^{-1}(r)} \qquad (1 \leq r \leq N).$$

Corollary 65.3 — Bit strong converse.

Let

$$H_{\text{phys}}(h) = \log_2 V(h).$$

Then

$$\boxed{\frac{D + X_h}{N} \geq \left(1 - 2^{H_{\text{phys}}(h) - I_{\text{ret}}}\right)_+.}$$

In particular, if $H_{\text{phys}}(h) \leq I_{\text{ret}} - \lambda$, then

$$\boxed{\frac{D + X_h}{N} \geq 1 - 2^{-\lambda}.}$$

65.3 Uniform one-top portability

For ordinary one-top rewriting, sources with the same local observation must share one rewrite rule. The concrete model-specific alignment test is already proved in Chapter 40: after the common local rewrite, every part of the source below the inspected top position is untouched, so direct canonical realization requires the corresponding untouched suffix to occur as the required suffix of the chosen canonical target. We use that established suffix criterion here as the one-top specialization of Stage III rather than promoting a second, abstractly worded portability lemma.

65.4 Finite observation capacity

Corollary 65.5 — Zero-debt finite-observation obstruction.

Assume hidden-tail noncoalescence and fix h. If all sources are canonical hits and h-confined, then

$$\boxed{N \le V(h).}$$

Consequently, a finite observation regime cannot keep a transition-closed canonical section across a collapse family larger than its observation capacity. If one target has arbitrarily large finite collapse subfamilies while $V(h)$ is fixed, total h-confined transition closure is impossible.

65.5 Bounded-d-top pushdown capacity

For a top-access pushdown interface that can distinguish a control state and at most d source-stack symbols, one may take the source signature consisting of that state and the complete shorter stack or the top d symbols. Hence a coarse universal bound is

$$\boxed{V(d) \le |Q| \sum_{r=0}^{d} |\Gamma|^r.}$$

The later source-exposure specialization refines this count by charging only states and work symbols that occur on the source interface.

Corollary 65.6 — Bounded-d-top capacity.

Under hidden-tail noncoalescence, a zero-debt d-confined collapse family must satisfy

$$\boxed{N \le |Q| \sum_{r=0}^{d} |\Gamma|^r.}$$

Thus semantic collapse can force control/alphabet capacity even when no particular residual encoding is assumed.

65.6 The three stages

Synthesis Principle 65.7 — Three-stage bounded-local obstruction decomposition.

Failure of direct canonical bounded-local realizability can be localized to one of three ordered tests:

Stage I — semantic capacity overflow

The collapse load exceeds the observation capacity allowed by the physical interface.

Stage II — chosen-section observation collision

Capacity is sufficient in principle, but the chosen section assigns sources to one observation class even though their canonical updates require physically incompatible behavior.

Stage III — hidden-tail/action alignment failure

The sources are locally separated, but the available local action cannot transform the hidden remainder into the chosen canonical target. For one-top pushdown rewriting this becomes the suffix/tail obstruction.

The three stages are a diagnostic organization, not three new independent lower-bound theorems. The quantitative theorem is the filtered collapse-capacity inequality above; the later model-specific chapters supply the alignment tests.

65.7 One-top specialization

Corollary 65.8 — One-top specialization.

In the literal one-top setting, the Stage III test is exactly

$$\boxed{\text{untouched suffix is a suffix of the canonical target.}}$$

Thus the theorem of Chapter 40 is the one-top boundary case of the general locality decomposition.

65.8 Locality–debt–recoordination principle

Synthesis Principle 65.9 — Locality–debt–recoordination trilemma.

When direct canonical realization fails, an exact implementation cannot keep all three of the following fixed:

1. the local observation regime;
2. the chosen canonical section;
3. direct canonical successor behavior.

At least one response is necessary:

$$\boxed{\text{refine observation}} \quad \text{or} \quad \boxed{\text{allow representation debt}}$$

$$\text{or} \quad \boxed{\text{re-coordinate the section}}.$$

This is a synthesis principle extracted from the exact alternatives already proved; it is intentionally not presented at the same theorem level as the collapse-capacity inequality.

The next chapter asks an equally important question:

> If debt is forced, is it necessarily dangerous?

The answer is no.

Chapter 66

Representation Debt and Behavioral Safety

A physical configuration can be noncanonical and still behave perfectly.

The General Theory therefore separates:

$$\boxed{\text{representation debt}}$$

from

$$\boxed{\text{behavioral incorrectness.}}$$

This distinction prevents a common mistake:

> "The machine left the canonical section, therefore it is already wrong."

That implication is false.

66.1 Terminal semantics

Let

$$f : M \to O$$

be the semantic terminal observable and

$$\alpha : C \to O$$

the physical terminal observable.

A physical configuration c is behaviorally correct for all continuations if

$$\boxed{\alpha(\delta_w(c)) = f(U_w(\Theta(c)))}$$

for every continuation w.

66.2 Immediate disagreement

Define the immediate disagreement set

$$\boxed{\mathcal{D} = \{c : \alpha(c) \neq f(\Theta(c))\}.}$$

A configuration can be outside $\mathcal{D}$ now but still have a future continuation reaching $\mathcal{D}$.

Thus immediate agreement is not enough.

Proposition 66.2A — Terminal factorization.

The identity

$$\boxed{\alpha = f\Theta}$$

holds on all physical configurations if and only if:

1. α is constant on every semantic fiber of Θ; and
2. $$\boxed{\alpha\sigma = f.}$$

Proof.

If $\alpha = f\Theta$, equal semantic projections have equal terminal values, and

$$\alpha\sigma = f\Theta\sigma = f.$$

Conversely, let c be any physical configuration. Fiber constancy gives

$$\alpha(c) = \alpha(\sigma(\Theta(c))).$$

Using $\alpha\sigma = f$,

$$\alpha(c) = f(\Theta(c)).$$

□

This proposition separates two requirements that are easy to conflate:

$$\boxed{\text{terminal correctness of the chosen section}}$$

and

$$\boxed{\text{terminal constancy across every physical representative of a semantic fiber.}}$$

66.3 Behavioral hull

Define

$$\boxed{\mathcal{B} = \{c : \alpha(\delta_w(c)) = f(U_w(\Theta(c))) \text{ for all } w\}.}$$

Theorem 66.1 — Behavioral hull characterization.

$$\boxed{\mathcal{B} = \nu X.\Phi(X) = C \setminus \mathrm{Pre}^*(\mathcal{D}).}$$

That is:

- $\mathcal{B}$ is the greatest forward-safe fixed point;
- a configuration is behaviorally safe exactly when no continuation reaches immediate disagreement.

This is the exact physical correctness object.

Corollary 66.2 — Root correctness.

A fixed policy is correct from root c_0 exactly when

$$\boxed{c_0 \in \mathcal{B}.}$$

66.4 Forward invariance

Theorem 66.3 — Forward invariance of behavioral safety.

If

$$c \in \mathcal{B},$$

then every reachable successor

$$\delta_w(c)$$

also lies in $\mathcal{B}$.

Thus safe debt remains safe under the fixed policy.

66.5 Debt birth dichotomy

Observation 66.4 — Debt birth dichotomy.

When a transition first leaves the canonical section, the resulting debt is born in exactly one of two regimes:

1. $\boxed{c \in \mathcal{B}}$

   — behaviorally harmless debt;

2. $\boxed{c \notin \mathcal{B}}$

   — debt that is already behaviorally unsafe.

A later continuation does not transform safe debt into unsafe debt.

If a continuation eventually reveals an error, the configuration was outside $\mathcal{B}$ from the moment the unsafe debt was created.

This corrects the tempting narrative:

"debt becomes fatal later."

The correct statement is:

$$\boxed{\text{a later suffix reveals pre-existing unsafety.}}$$

66.6 Two opposite examples

Weighted-balance schedule conflict

Some noncanonical debts produced by forbidden eager schedules lie outside the behavioral hull and are exposed by short continuations.

Multi-type Dyck collapse

At a rejecting collapse boundary, bounded-locality may force noncanonical physical representatives, yet every such state can lie in an absorbing rejecting semantic fiber.

The debt is forced but harmless.

Thus:

$$\boxed{\text{forced debt} \not\Rightarrow \text{fatal debt.}}$$

The next chapter develops semantic certificates for proving harmlessness without constructing the exact physical behavioral hull.

Chapter 67

Semantic Containment Cores

The behavioral hull

$$\mathcal{B}$$

is exact.

But it lives in the full physical configuration space.

A coarser proof certificate can sometimes be obtained entirely at the semantic level.

67.1 Semantic trap containment

Let

$$S \subseteq M$$

be forward invariant:

$$U_a(S) \subseteq S.$$

Suppose every physical representative in every fiber over S is immediately terminally correct.

Theorem 67.1 — Semantic trap containment.

Under those hypotheses,

$$\boxed{\Theta^{-1}(S) \subseteq \mathcal{B}.}$$

Proof idea. A run beginning over S remains over S semantically.

Every reached physical representative lies in a fiber whose entire ambient fiber is terminally correct.

Therefore no continuation can reach disagreement. □

This is conservative because it requires **every** representative in the full fiber to be safe.

67.2 Absorbing-fiber harmlessness

Corollary 67.2 — Absorbing-fiber harmlessness.

If a semantic state is absorbing and all physical representatives in its fiber give the correct terminal observation, then the entire fiber lies in the behavioral hull.

This is the clean explanation of multi-type Dyck collapse debt:

- the debt can be forced physically;
- the semantic state is already an absorbing reject;
- the entire fiber is harmless.

67.3 Maximal full-fiber containment core

Define the semantic good set of states whose full fibers are terminally correct.

Iteratively remove any semantic state whose successor leaves the current set.

Theorem 67.3 — Maximal fiberwise semantic containment core.

There exists a greatest forward-invariant semantic set

$$\boxed{\mathcal{T}}$$

such that

$$\boxed{\Theta^{-1}(\mathcal{T}) \subseteq \mathcal{B}.}$$

This is the maximal **ambient full-fiber** semantic containment certificate.

It is deliberately conservative.

An unreachable bad representative can remove an otherwise useful semantic state from $\mathcal{T}$.

67.4 Reachable-fiber refinement

Let

$$F_m^{\text{reach}}$$

be only the representatives actually reachable under the policy.

Use those reachable fibers in the containment construction.

Theorem 67.4 — Reachable semantic containment.

There is a greatest reachable-fiber containment core

$$\boxed{\mathcal{T}^{\text{Reach}}}$$

whose reachable physical representatives are behaviorally safe.

It satisfies the conceptual hierarchy

$$\boxed{\mathcal{T} \subseteq \mathcal{T}^{\text{Reach}}.}$$

The reachable core removes ambient syntactic contamination.

67.5 Root correctness via containment

Theorem 67.5 — Root correctness via reachable containment.

If the root semantic state belongs to

$$\mathcal{T}^{\text{Reach}},$$

then the physical root lies in

$$\mathcal{B}.$$

Thus the three correctness scales are:

$$\boxed{\mathcal{T} \longrightarrow \mathcal{T}^{\text{Reach}} \longrightarrow \mathcal{B}.}$$

They answer increasingly exact questions:

- ambient semantic certificate;

- reachable semantic certificate;
- exact physical behavioral safety.

The next chapter returns to normalization schedules and uses the behavioral hull to distinguish merely structural unsafe targets from genuinely fatal ones.

Chapter 68

Safe Reset and the Fatal Frontier

The 2:1 causal theory introduced first-hit exposure and portability.

The General Theory now separates two notions:

1. a structural reset certificate;
2. exact behavioral safety.

This separation is essential because leaving a convenient reset domain need not be fatal.

68.1 Wordwise safe-reset domain

Let $\mathcal{R} \subseteq C$ be a set of physical configurations.

Informally, $\mathcal{R}$ is a safe-reset domain if every run that stays within the certified reset behavior can be normalized correctly word by word.

Proposition 68.1 — Safe-reset certificate.

If $\mathcal{R}$ is a wordwise safe-reset domain for a schedule family $\mathfrak{H}$, then every schedule whose eager activations remain within the certified reset behavior is correct.

The domain is a proof certificate.

It need not equal the exact behavioral hull.

68.2 Structural unsafe targets

For schedule H, let

$$U_{\mathcal{R}}(H)$$

be the set of actual eager activation targets that lie outside $\mathcal{R}$.

These are structurally uncertified targets.

Some may still lie in the behavioral hull.

Define that difference as behavioral slack.

The exact badness criterion therefore needs both layers.

68.3 Fatal Frontier

Theorem 68.2 — Fatal Frontier.

Assume $\mathcal{R}$ is a wordwise safe-reset domain for the full schedule family.

Then

$$\boxed{H \notin \mathrm{Adm} \iff U_{\mathcal{R}}(H) \setminus \mathcal{B}_H \neq \varnothing.}$$

Thus a schedule is bad exactly when it actually produces an eager target that is:

1. outside the reset certificate;
2. outside the exact behavioral hull.

This is the abstract form of what Chapter 56 saw concretely.

A structural deviation becomes a **fatal frontier event** only when no behavioral slack saves it.

68.4 Activation tightness

Corollary 68.3 — Activation tightness.

If every actual eager target outside $\mathcal{R}$ is behaviorally unsafe, then

$$\boxed{H \notin \mathrm{Adm} \iff U_{\mathcal{R}}(H) \neq \varnothing.}$$

In that case the structural unsafe frontier is exact.

Without activation tightness, structural detection is only sufficient, not necessary.

68.5 Site-free diagnostics

Sometimes one can certify fatality by a continuation that uses no additional optional eager site.

Theorem 68.4 — Site-free diagnostic frontier.

Under the stated safe-reset hypotheses, a finite site-free continuation can certify that an activation target is already outside the behavioral hull.

This gives a causal witness local to the activation event.

The 2:1 certificate

$$1111100 \rightsquigarrow (R, 00)$$

followed by the short exposing suffix is the motivating example.

68.6 Fatal causality

Proposition 68.5 — Fatal causality implies conflict.

If a minimal enabled site set H creates a fatal first-hit target at site b, then

$$H \cup \{b\}$$

is an inadmissible schedule set.

Thus directed fatal cause induces set-theoretic schedule conflict.

The next chapter develops the combinatorics of those conflicts and stress-tests the theory on 3:2 and an infinite odd-p:2 family.

Chapter 69

Conflict Complexes and Weighted-Balance Stress Tests

The Fatal Frontier tells us when one schedule is wrong.

We now ask how the whole family of schedules is organized.

69.1 Minimal conflicts

A conflict is a set of optional sites that cannot be enabled together.

A conflict is minimal if all of its proper subsets are admissible.

Lemma 69.1 — Minimal-conflict participation.

Every site belonging to a minimal conflict participates in some fatal mechanism for that conflict.

No member can be completely irrelevant.

Otherwise the conflict would not be minimal.

69.2 Robust fatal supports

Suppose a family of minimal site sets has fatal certificates that remain valid whenever additional sites outside the support are enabled.

These are robust fatal supports.

Theorem 69.2 — Robust fatal supports give an independence complex.

Under the robust-certificate hypotheses, the admissible schedule family is exactly the family of site subsets containing no minimal fatal support.

Therefore it is the independence complex of the hypergraph of minimal fatal supports.

The phrase **independence complex** is combinatorial packaging.

The project-specific content is the derivation of the fatal supports from physical normalization causality.

Corollary 69.3 — Finite diagnostic basis.

If the robust minimal fatal support family is finite, the entire finite schedule classification has a finite diagnostic basis.

69.3 Reset-ray fork

Proposition 69.4 — Reset-ray fork template.

Suppose one site D changes the reachable reset-ray geometry so that it exposes fatal first hits at two victim sites A and B, while schedules not containing D remain reset-safe and schedules containing D but neither victim remain safe.

Then the minimal fatal supports are

$$\boxed{\{A, D\}, \qquad \{B, D\}.}$$

The admissible schedules are the independence complex avoiding those two pairs.
This template abstracts the directed fork mechanism later verified uniformly.

69.4 Imported weighted-balance foundations

The General Theory does **not** re-prove:

- primitive-kernel minimum demand;
- centered residual exactness;
- centered canonical-episode terminal correctness.

Those results were already established in the p:q construction.
Here they are used as a stress-test environment for the new causal machinery.

69.5 Exact 3:2 case

A finite reset analysis produces an exact certificate for the centered 3:2 schedule family.

Lemma 69.5A — Exact 3:2 finite reset certificate.

For the centered 3:2 schedule family there is an explicit finite reset repertoire whose episode closure certifies every schedule avoiding the two fatal pairs

$$\boxed{\{A, D\}, \qquad \{B, D\}.}$$

The same finite analysis produces first-hit fatal witnesses when either forbidden pair is enabled.

This lemma is the finite reset engine behind the exact schedule count; the detailed reset catalogue is retained for the technical appendix rather than repeated in the narrative.

Theorem 69.5 — Exact 3:2 schedule classification.

There are

$$\boxed{\text{40 admissible schedules}}$$

and

$$\boxed{\text{24 inadmissible schedules.}}$$

The minimal fatal fork is

$$\boxed{D \to A, \qquad D \to B.}$$

Equivalently, the minimal fatal supports are

$$\boxed{\{A, D\}, \qquad \{B, D\}.}$$

The technical finite reset catalogue supporting this result belongs in the appendix; the classification remains in the main narrative.

69.6 Infinite odd-p:2 family

Now let

$$q = 2$$

and

$$p \geq 3$$

be odd.
The underlying calculation gives:

- a detailed arithmetic catalogue;
- source geometry;
- reset closure;
- uniform D-induced fiber expansion;
- reachability of a common length-three fatal frontier;
- kernel-suffix fatality.

The long arithmetic/source catalogues are appendix material.
The structural conclusion is main-text material.

Lemma 69.6A — Odd-p:2 reset closure.
Let

$$q = 2, \qquad p = 2h - 1 \geq 3.$$

There are two explicit safe reset repertoires:

- $\mathcal{R}_{\neg D}$, used when site D is absent;
- $\mathcal{R}_D$, used when D is present but A and B are absent.

They satisfy

$$\boxed{D \notin H \Longrightarrow H \in \mathrm{Adm}_{p,2},}$$

and

$$\boxed{D \in H, \quad A, B \notin H \Longrightarrow H \in \mathrm{Adm}_{p,2}\,.}$$

This accounts for

$$32 + 8 = 40$$

safe schedules before the fatal pairs are analyzed.

Lemma 69.6B — Uniform D-induced fiber expansion.
For every

$$x \geq h + 1,$$

the D-activated canonical ray satisfies

$$\boxed{(C, \eta(x,h)) \xrightarrow{0^{2h-1}} (D_{-h}, 0^{x-h+1}).}$$

Therefore enabling D expands the common A/B observation fiber from the singleton unary source

$$\{0\}$$

to

$$\boxed{\{0, 00, 000, \ldots\}.}$$

This is the uniform directed mechanism

$$\boxed{D \rightsquigarrow A, \qquad D \rightsquigarrow B.}$$

Lemma 69.6C — Uniform reachability of the length-three frontier.

For every

$$h \geq 2,$$

there is an explicit prefix u_h, using no optional eager site except D, that reaches the common fatal-frontier source

$$\boxed{(D_{-h}, 000).}$$

One convenient construction is

$$\boxed{u_h = v_h\, 1^{\,2h-1} 0^{\,2h},}$$

where v_h is the parity-dependent prefix that first reaches

$$(D_{h-1}, \varepsilon).$$

Thus the same length-three unary source is not merely syntactically possible; it is uniformly reachable across the entire odd-p:2 family.

Lemma 69.6D — Uniform kernel-suffix fatality.

For

$$p{:}2,$$

the primitive kernel is

$$\boxed{K = (2, p).}$$

The zero-weight physical kernel word

$$\boxed{1^p 00}$$

appears as a suffix in the eager A- and B-targets generated from the common source

$$(D_{-h}, 000).$$

A site-free continuation removes the non-kernel prefix and exposes this residual-zero kernel excess as behavioral disagreement.

Hence both A and B are uniformly fatal after the D-induced fiber expansion.

Appendix migration note

Two long supporting results are intentionally not expanded here:

- the **odd-p:2 arithmetic catalogue**;
- the **odd-p:2 source geometry catalogue**.

They are retained as appendix-level proof infrastructure. Their purpose is to justify the closed forms and source restrictions used in Lemmas 61.6A–61.6D, not to create a second narrative inside the main theorem.

Theorem 69.6 — Odd-p:2 fatal-fork family.

For every odd

$$p \geq 3,$$

$$\boxed{\mathrm{Adm}_{p,2} = \mathrm{Ind}\{\{A, D\}, \{B, D\}\}.}$$

Thus every member of the infinite family has:

$$\boxed{40 \text{ admissible}, \qquad 24 \text{ inadmissible}}$$

schedules and the same directed fatal fork

$$\boxed{D \to A, \qquad D \to B.}$$

This is an important stress test.
The General Theory is not merely repackaging the 2:1 example.
Its causal mechanism persists uniformly across an infinite parameter family.
The next chapter changes direction from diagnosis to construction:

Can a transition-closed section be synthesized when one exists?

Chapter 70

Rooted Section Synthesis

The theory has so far explained why canonical sections fail.

It now becomes constructive.

Given:

- a semantic system;
- a physical transition template;
- a chosen physical representative of the root;

can we synthesize a transition-closed section?

The answer is an exact path-outcome coherence criterion.

70.1 Rooted path-outcome fibers

Fix semantic root

$$m_0$$

and physical representative

$$c_0$$

with

$$\Theta(c_0) = m_0.$$

For semantic state m, define

$$\boxed{\mathcal{H}_{c_0}(m)}$$

to be the set of all physical outcomes obtained by following physical transitions along semantic words that take m_0 to m.

Different semantic paths to the same state can produce different physical endpoints.

That is exactly the synthesis obstruction.

70.2 Path-outcome coherence

Theorem 70.1 — Rooted path-outcome coherence criterion.

A rooted transition-closed section through c_0 exists if and only if

$$\boxed{|\mathcal{H}_{c_0}(m)| = 1}$$

for every reachable semantic state m.

When it exists, the section is uniquely determined by the root representative.

Proof.

If a transition-closed section exists, every semantic path to m must end at the single section point $\sigma(m)$.

Conversely, suppose every path-outcome fiber is a singleton.

Define $\sigma(m)$ to be that unique physical outcome.

If $m \xrightarrow{a} U_a(m)$, append a to any path reaching m.

Path-outcome uniqueness gives

$$\delta_a(\sigma(m)) = \sigma(U_a(m)).$$

Thus the section is transition-closed. □

Corollary 70.2 — One root representative determines the section.

For a fixed physical root candidate, there is at most one rooted transition-closed section.

70.3 Finite synthesis algorithm

For finite semantic systems, the theorem yields an exact propagation procedure.

Start from a root candidate.

Propagate its physical image along semantic edges.

Whenever a semantic state is reached again:

- if the physical endpoint agrees, continue;
- if it differs, synthesis fails.

Theorem 70.3 — Finite rooted synthesis.

For a finite semantic transition graph, this propagation decides exactly whether a rooted transition-closed section exists and constructs it when it does.

The algorithm is graph-theoretic once the physical transition template is fixed.

70.4 Short obstruction certificate

Theorem 70.4 — Short parallel-path obstruction certificate.

If finite rooted synthesis fails, there exist two semantic paths from the root to the same semantic state whose physical endpoints differ.

A shortest such pair gives a finite bounded certificate of failure.

Thus synthesis failure is witnessed by path disagreement.

70.5 Counting sections

Corollary 70.5 — Counting synthesized sections.

If the physical root fiber is finite, the number of rooted transition-closed sections equals the number of root representatives that pass the path-outcome coherence test.

The theory therefore turns section existence into a concrete search over root candidates.

70.6 Infinite semantic cores require symbolic synthesis

The coherence theorem itself does not require finiteness.

But brute-force propagation over an infinite semantic graph is not an algorithm.

The next chapter shows one important infinite case where a finite symbolic rule skeleton can be synthesized exactly.

That construction also demonstrates again that re-coordination can remove debt.

Chapter 71

Symbolic Re-Coordination and Forced-Debt Frontiers

The locality trilemma gave three responses to a failed canonical section:

refine observation, allow debt, re-coordinate.

This chapter gives a constructive example of the third option, then quantifies what remains unavoidable when only a partial section can be maintained.

71.1 Modular quotient/remainder synthesis

Consider integer translation semantics generated by fixed signed weights.

Choose a modulus large enough to absorb every one-step weight change.

Represent each semantic integer by:

- a finite sign/remainder state;
- a unary quotient stack.

Theorem 71.1 — Modular quotient/remainder synthesis.

There exists an explicit one-top zero-debt lift in which:

- the finite control stores the signed remainder;
- the stack stores the unbounded quotient;
- every semantic translation is realized directly;
- the chosen section is transition-closed.

This is a general constructive example of:

re-coordination removes normalization debt.

Scope

The construction is:

- pseudo-polynomial in the numerical weight bound;
- not polynomial in the bit length of weights;
- not a state-minimality theorem;
- not a general algorithm for arbitrary infinite lifts.

71.2 Finite symbolic skeleton

Theorem 71.2 — Finite symbolic skeleton and global path-outcome coherence.

The modular representation is governed by a finite set of symbolic local rules.

Those rules satisfy global rooted path-outcome coherence on the infinite semantic core.

Thus an infinite transition-closed section can sometimes be certified by finite symbolic arithmetic.

Corollary 71.3 — Zero-weight loop coherence.

Every semantic zero-weight loop returns the synthesized physical representation to the same section point.

This is an immediate path-outcome consequence, not a separate confluence theorem.

71.3 Partial canonical sections

Total transition closure may be impossible.

Let

$$G \subseteq M$$

be a semantic region over which a chosen section is transition-closed.

Proposition 71.4 — Canonical interior.

As long as the semantic run remains inside G, the physical run remains exactly on the chosen canonical section.

Representation debt is born only on an edge leaving the canonical interior.

This turns debt creation into a boundary phenomenon.

71.4 Canonical-hit capacity

Suppose many semantic source states share one target under the same input.

Only finitely many source representatives can land directly on one canonical target if the local observation space is finite.

Theorem 71.5 — Canonical-hit capacity.

For a fixed bounded-local policy, the number of source edges that can hit one chosen canonical target directly is bounded by the available local observation capacity.

This theorem no longer assumes total transition closure.

It applies specifically at the boundary of a partial section.

71.5 Quantitative forced debt

Let $\mathcal{S}$ be a finite set of relevant same-target source states and let Ω_a be the input-a observation set.

Corollary 71.6 — Quantitative forced-debt frontier.

At least

$$\boxed{|\mathcal{S}| - |\Omega_a|}$$

source edges are forced to leave the chosen section.

If $\mathcal{S}$ is infinite and Ω_a is finite, infinitely many debt-producing edges are forced.

This is the quantitative form of the earlier fiber-branching theorem.

71.6 Multi-type Dyck frontier

For one bracket type,

$$\boxed{\mathrm{Dyck}_1 : \text{ total canonical synthesis.}}$$

For

$$k \geq 2,$$

the multi-type Dyck semantic system has an infinite collapse boundary that exceeds bounded local observation capacity.

Theorem 71.7 — Dyck forced-debt frontier.

For

$$k \geq 2,$$

there is:

$$\boxed{\text{zero-debt live interior} + \text{infinite forced-debt collapse boundary.}}$$

Yet that boundary lies in an absorbing rejecting semantic fiber.
Therefore the forced debt is behaviorally harmless.
This example is crucial because it separates:

$$\boxed{\text{forced physical noncanonicality}}$$

from

$$\boxed{\text{behavioral failure.}}$$

The theory can therefore quantify unavoidable debt without equating debt with error.

Chapter 72

Effective Verification Hierarchy

The General Theory ends with algorithms.

The objective is not to claim new classical automata algorithms.

The contribution is a hierarchy of reductions from normalization correctness to established decision problems.

There are three principal verification levels.

72.1 Level I — Structural safe-reset saturation

Suppose a regular set Bad_{str} contains every structurally unsafe eager target outside a certified safe-reset domain.

Theorem 72.1 — Regular structural safe-reset certification.

If pushdown reachability saturation proves that no configuration in Bad_{str} is reachable under the schedule, then the safe-reset certificate proves policy correctness.

This test is:

$$\boxed{\text{sufficient in general.}}$$

Under activation tightness, it becomes exact.

This is the cheapest level when a good structural reset certificate is available.

72.2 Level II — Exact immediate-disagreement saturation

Suppose the exact immediate disagreement set

$$\mathcal{D}_H$$

is regular and effectively represented.

Theorem 72.2 — Regular immediate-disagreement saturation.

Then

$$\boxed{\mathcal{B}_H = C \setminus \mathrm{Pre}_H^*(\mathcal{D}_H)}$$

is regular and effectively constructible by pushdown predecessor saturation.

Thus the exact behavioral hull can be computed.

Corollary 72.3 — Exact root classification.

The fixed-policy root is correct exactly when it lies outside the saturated predecessor set of $\mathcal{D}_H$.

Corollary 72.4 — Exact finite schedule classification.

For a finite optional-site family, if the regular disagreement construction is effective for every schedule, the full schedule family is exactly decidable by enumerating schedules and applying the saturation test.

This turns the Fatal Frontier theory into a complete finite classifier in the regular subclass.

72.3 Reachable-containment shortcut

Sometimes one only needs to prove correctness of one root, not construct the entire exact hull.

Corollary 72.5 — Reachable-containment root shortcut.

A root can be certified through the reachable semantic containment core without explicitly constructing the whole physical behavioral hull.

This can be substantially cheaper when semantic containment is small.

72.4 Finite semantic containment computation

If the semantic system M is finite and the initial semantically good set G_0 is known:

Theorem 72.6 — Finite containment-core computation.

The maximal forward-invariant semantic containment core is computed by iterative removal of states with outgoing edges leaving the current set.

Once G_0 is known, the edge-processing complexity is

$$\boxed{O(|M||\Sigma|).}$$

Corollary 72.7 — Finite containment-failure certificate.

If a semantic root is removed, there is a semantic witness path of length at most

$$\boxed{|M| - 1}$$

leading from the root to a containment failure.

Thus semantic containment failure has a short certificate.

72.5 Level III — DPDA-equivalence fallback

Regular immediate disagreement is powerful but not universal.

When no useful regular representation of $\mathcal{D}_H$ is available, compare continuation languages directly.

For physical configuration c, construct:

- the physical continuation language from c;
- the semantic continuation language from $\Theta(c)$.

If both have effective deterministic pushdown presentations, classical deterministic pushdown equivalence can decide equality [21, 13].

Theorem 72.8 — DPDA-equivalence behavioral fallback.

Behavioral safety of a fixed physical root reduces exactly to equivalence between its physical and semantic deterministic pushdown continuation languages.

Corollary 72.9 — Exact fixed-policy root verification.

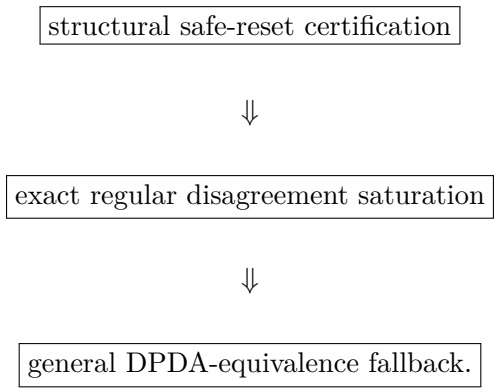
Whenever those two continuation languages are effectively represented as DPDAs, fixed-policy root correctness is decidable.

The imported equivalence algorithm is classical.

The contribution here is the reduction from normalization safety to that equivalence problem.

72.6 Verification hierarchy

The three levels are therefore:

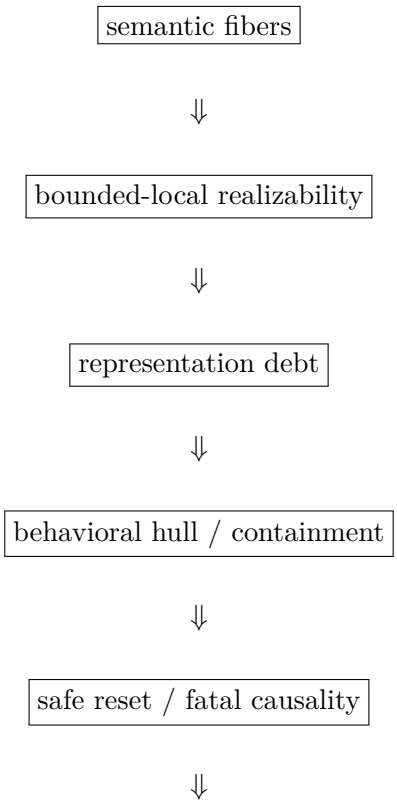


They trade:

- structural specificity;
- computational convenience;
- generality.

72.7 General Theory synthesis

The final theoretical chain is now complete:



conflict complexes

$\Downarrow$

section synthesis / forced debt

$\Downarrow$

effective verification.

But the monograph should not end on an abstract verification theorem. The post-theory branches below first record the enumerative, arithmetic, probabilistic, topological, and unrestricted-complexity consequences that emerged from the completed framework.

The final Part then returns one last time to the two 2023 DPDA figures.

The point of the General Theory is not to replace the original machine.

It is to explain it.

Part VI

Enumerative and Arithmetic Consequences of Completion Semantics

Part guide. **Question.** What exact combinatorial and arithmetic structures are encoded by the same residual and shortest-completion semantics? **Main result.** The Part derives operational event enumerations, solves the Dai–Fu–Qiu run–return and ratio-run–return problems, develops contact-preserving barrier compression, and obtains Farey-recursive and Cohn–Markoff completion structures. **Why here.** These are consequence branches sharing semantic coordinates, not additional axioms of the General Theory. **Dependency.** The branches reuse Parts I–V selectively and are intentionally separable from one another.

Chapter 73

Primitive Operational Statistics and Schubert Cancellation Geometry

The original 2:1 construction has already supplied the provenance for CANCEL, SEARCH, WRITE, and shortest-completion demand. We therefore begin directly with the centered $p{:}q$ event process rather than repeating the automaton construction.

73.1 The centered $p{:}q$ event process

Fix coprime positive integers p, q, put

$$m = p + q, \qquad h = \left\lfloor \frac{m-1}{2} \right\rfloor,$$

and write D_d for the centered debt state with $-h \le d \le h$. For an integer residual r, let

$$\mu(r) = (x, y) = \arg\min\{x + y : x, y \in \mathbb{Z}_{\ge 0},\ px - qy = r\},$$

which is unique after restricting to the standard nonnegative kernel strip. The canonical binary realization of a demand pair is

$$\eta(x, y) = \begin{cases} (10)^y 0^{x-y}, & x \ge y, \\ (10)^x 1^{y-x}, & x < y. \end{cases}$$

The centered real-time process has four event classes. If the stack top matches the next input bit, the top cell is popped (MATCH/CANCEL). A mismatch is absorbed by changing d by one while $|d| < h$ (SEARCH$\pm$). At the outer boundary a further same-direction mismatch performs OUTER-WRITE and returns to debt 0. If the stack is empty, the next symbol performs BOTTOM-WRITE using the shortest-demand map μ.

The exact transition rule is the following. With input bit a and stack word S whose leftmost symbol is the top,

$$\begin{array}{ll} \text{top } z = a: & (d, zT) \mapsto (d, T), \\ (a, z) = (1, 0),\ d < h: & (d, 0T) \mapsto (d + 1, T), \\ (a, z) = (0, 1),\ d > -h: & (d, 1T) \mapsto (d - 1, T), \\ (a, z) = (1, 0),\ d = h: & (h, 0T) \mapsto (0, \eta(\mu((h + 1)m))T), \\ (a, z) = (0, 1),\ d = -h: & (-h, 1T) \mapsto (0, \eta(\mu(-(h + 1)m))T), \\ S = \varepsilon: & (d, \varepsilon) \mapsto (0, \eta(\mu(dm + w(a)))), \end{array}$$

where $w(1) = q$ and $w(0) = -p$. The language-recognition correctness of this centered normalization is background for the present event-enumeration problem; all statistics below refer to these displayed transitions.

A word is *rational-Dyck* when every prefix has

$$\rho = qN_1 - pN_0 \geq 0.$$

A scale-k balanced word has kp ones and kq zeros. Let $C(P)$ be its number of MATCH/CANCEL events and $O(P)$ its number of OUTER-WRITE events.

73.2 Primitive balanced words: fixed-stack Hamming geometry

A primitive balanced word has p ones and q zeros.

Lemma 73.1 (No later WRITE in the primitive sector)**.** *After the first input symbol of a primitive balanced word, the shortest-demand stack has length $m-1$. Exactly $m-1$ input symbols remain. Hence no later WRITE occurs: every remaining transition is SEARCH or CANCEL.*

Proof. The first BOTTOM-WRITE realizes the unique shortest completion of the first-symbol residual. Because the completed word has total composition (p, q), the demand length is $m-1$. Every subsequent non-WRITE transition consumes one input symbol and one stack cell. A new WRITE before the end would create more stack cells than remaining input can remove, contradicting balanced acceptance. □

Thus $C(P)$ is exactly Hamming agreement between the suffix of P and its fixed canonical stack reference.

Theorem 73.2 (Primitive balanced CANCEL distribution)**.** *For every coprime $p, q > 0$ with $p \neq q$,*

$$\boxed{c_{\min}(p, q) = |p - q| - 1.}$$

The number of primitive balanced words attaining this minimum is

$$\boxed{\binom{\max(p,q)-1}{|p-q|-1} = \binom{\max(p,q)-1}{\min(p,q)}.}$$

In the unique coprime diagonal case $(p, q) = (1, 1)$, both primitive balanced words 10 *and* 01 *have exactly one CANCEL, so*

$$\boxed{c_{\min}(1, 1) = 1}$$

with multiplicity 2*. More generally, for $j \geq 0$,*

$$\boxed{\#\{P : C(P) = m - 1 - 2j\} = \binom{p-1}{j}\binom{q}{j} + \binom{p}{j}\binom{q-1}{j},}$$

with all other values zero.

Proof. Condition on the first bit. Lemma 73.1 reduces the suffix to comparison of two binary strings of equal length and fixed composition. If the first bit is 1, the input suffix contains $p-1$ ones and q zeros; if it is 0, it contains p ones and $q-1$ zeros. A Hamming disagreement swaps one selected 1-position with one selected 0-position, so disagreements occur in pairs. Choosing j swapped positions in the two composition classes gives the two binomial products. For $p \neq q$, maximizing the number of paired disagreements gives $|p-q|-1$ agreements and the stated multiplicity. In the remaining coprime case $(1, 1)$ the distribution formula itself gives two words at CANCEL value 1, yielding the separate diagonal statement. □

Corollary 73.3 (Cancellation-free primitive orbit)**.** *A cancellation-free primitive balanced word exists if and only if* $|p-q|=1$*. It is unique: for* $p=q+1$ *it is* $(10)^q1$*, while for* $q=p+1$ *it is* $0(01)^p$*.*

This is the first operational appearance of the Christoffel/mechanical-word boundary: the extremal word is a cyclic representative of the corresponding balanced mechanical pattern, but no Christoffel theorem is needed in the proof.

73.3 Primitive rational-Dyck paths and the Schubert statistic

Assume first $p=q+d$ with $d\geq 1$; the opposite inequality follows by the exact bit/physical duality. Define

$$R_{q,d}=1(10)^q1^{d-1},\qquad A_q=\{3,5,\dots,2q+1\}.$$

For a primitive rational-Dyck path P, let $Z(P)$ be the set of zero positions.

Theorem 73.4 (Canonical-demand agreement theorem)**.** *For every primitive rational-Dyck* P *of type* $(q+d):q$*,*

$$\boxed{C(P)=d-1+2|Z(P)\cap A_q|.}$$

Moreover, writing $Z(P)=\{z_1<\cdots<z_q\}$*, rational-Dyck nonnegativity is equivalent to*

$$\boxed{z_i\geq\left\lceil\frac{(2q+d)i}{q}\right\rceil,\qquad 1\leq i\leq q.}$$

Hence the zero sets form the bases of a Schubert, equivalently lattice-path, matroid.

Proof. The first BOTTOM-WRITE fixes the entire remaining demand by Lemma 73.1. Direct comparison with that demand yields the reference word $R_{q,d}$ and therefore the displayed agreement formula. At the position z_i of the ith zero, the prefix has z_i-i ones and i zeros; $q(z_i-i)-pi\geq 0$ is exactly the stated lower flag. Such coordinatewise lower flags define the standard Schubert/lattice-path basis family [55]. □

Define the normalized polynomial

$$\mathcal{R}_{q,d}(x)=\sum_B x^{|B\cap A_q|},$$

where B runs over these bases.

Corollary 73.5 (Ultra-log-concavity)**.** *If* $\mathcal{R}_{q,d}(x)=\sum_j r_jx^j$*, then*

$$\boxed{\left(\frac{r_j}{\binom{q}{j}}\right)^2\geq\frac{r_{j-1}}{\binom{q}{j-1}}\frac{r_{j+1}}{\binom{q}{j+1}}.}$$

Thus the primitive normalized CANCEL coefficients are ultra-log-concave, and hence log-concave and unimodal with no internal zero.

Proof. The multivariate basis polynomial of a matroid is Lorentzian; nonnegative specialization to a marked subset preserves the required Lorentzian consequences. Apply the strongest Mason inequality of Brändén–Huh [56]. □

The support is exact:

$$j_{\max} = \left\lfloor \frac{q(2q+1)}{2q+d} \right\rfloor, \qquad j_{\min} = \begin{cases} 1, & d = 1, \\ 0, & d \geq 2. \end{cases}$$

After reversal to an upper flag, put

$$s_i = 1 + \left\lfloor \frac{(2q+d)(i-1)}{q} \right\rfloor, \qquad \bar{A}_q = \{d, d+2, \ldots, d+2q-2\}.$$

If $u_i = |\bar{A}_q \cap [1, s_i]|$, $v_i = s_i - u_i$, and

$$\sum_{r \geq 0} H_{i,r}(x) z^r = (1 - xz)^{-u_i} (1 - z)^{-v_i},$$

then the standard flagged Jacobi–Trudi/Lindström–Gessel–Viennot specialization [60, 24] gives

$$\boxed{\mathcal{R}_{q,d}(x) = \det[H_{i,1-i+j}(x)]_{1 \leq i,j \leq q}.}$$

Because the determinant is upper Hessenberg it also has the matrix-free recurrence

$$D_{q+1} = 1, \qquad D_i = \sum_{r=1}^{q-i+1} (-1)^{r-1} H_{i,r} D_{i+r}, \qquad \mathcal{R}_{q,d} = D_1.$$

73.3.1 Narayana windows

The near-diagonal cases recover classical distributions after the operational identification. If $d = 1$, deleting the initial 1 gives an ordinary Dyck path of semilength q, and

$$\#\{P : C(P) = 2j\} = \frac{1}{q} \binom{q}{j} \binom{q}{j-1}.$$

If $d = 2$ (so q is odd by coprimality), a last-level decomposition $P = UAUB$ gives

$$\boxed{\#\{P : C(P) = 2j-1\} = \frac{1}{j} \binom{q+1}{j-1} \binom{q-1}{j-1}.}$$

These are ordinary/rational Narayana distributions; the second equality is distributional, not pathwise equality with the classical peak statistic [51].

Chapter 74

Repeated Kernels, Carry Histories, and Algebraic Closure

Primitive normalization has a fixed demand stack. Repeated kernels introduce physical resets, so the event law becomes history-sensitive. The correct decomposition is a Bizley factorization refined by operational events, followed by a Raney carry skeleton and a stationary shifted-Schubert phase kernel.

74.1 Repeated kernels and operational Bizley factorization

For scale k, let $\mathcal{D}_k$ be the rational-Dyck paths with kp ones and kq zeros, and let $\mathcal{I}_k$ be those with no proper zero-residual contact. Put

$$D_k(x,y) = \sum_{P \in \mathcal{D}_k} x^{C(P)} y^{O(P)}, \qquad I_k(x,y) = \sum_{P \in \mathcal{I}_k} x^{C(P)} y^{O(P)}.$$

A zero-residual contact returns the centered machine to the canonical accepting configuration. Both statistics are additive under concatenation, hence

$$\boxed{D(z;x,y) = 1 + \sum_{k \geq 1} D_k(x,y) z^k = \frac{1}{1 - I(z;x,y)}.}$$

Define operational Bizley cumulants by

$$\boxed{\log D(z;x,y) = \sum_{k \geq 1} \frac{A_k(x,y)}{k} z^k.}$$

At $x = y = 1$ this reduces to the classical rational-Dyck/Bizley decomposition; in particular

$$\boxed{A_k(1,1) = \frac{1}{p+q} \binom{k(p+q)}{kp}.}$$

The unweighted factor-free structure is classical [54, 57, 53]; the refinement here is by actual normalization events.

74.2 Physical carry hierarchy and WRITE-weight conservation

Set

$$a = 1 + \left\lfloor \frac{h}{q} \right\rfloor, \qquad c = 1 + \left\lfloor \frac{h}{p} \right\rfloor.$$

A positive OUTER-WRITE is called a *carry*.

Theorem 74.1 (Carry-onset hierarchy)**.** *The first scale at which $r \geq 1$ carries are possible is*

$$\boxed{\kappa_r(p,q) = a + rc.}$$

Equivalently,

$$\boxed{O_{\max}(k) = \max\left\{0, \left\lfloor \frac{k-a}{c} \right\rfloor\right\}.}$$

At $k = \kappa_r$, every path realizing r carries is factor-free, so

$$\boxed{[y^r]A_{\kappa_r}(x,y) = \kappa_r[y^r]D_{\kappa_r}(x,y).}$$

Proof. Before the first positive outer carry, SEARCH+ must accumulate enough positive centered debt to cross the outer boundary. Among rational-Dyck prefixes with a fixed number of completed scale blocks, replacing a zero by a one cannot delay that positive boundary: a zero lowers the rational height and never supplies positive mismatch budget. Hence the all-ones trajectory is extremal for earliest carry onset. Writing $h = q(a-1)+s$ with $0 \leq s < q$ shows that $a = 1 + \lfloor h/q \rfloor$ scale units are necessary and sufficient before the first OUTER+ trigger.

After an OUTER+ reset the written frame has length $cm - 1$, where $c = 1 + \lfloor h/p \rfloor$. The WRITE-weight invariant of Theorem 74.2 assigns this reset weight c. Between two consecutive carries no event can create scale budget; SEARCH and CANCEL only consume existing cells. Thus each further carry costs at least an additional c units of total scale, giving the lower bound $k \geq a + rc$ for r carries. The same extremal positive-boundary trajectory, followed at each stage by its shortest balancing continuation, realizes equality, so $\kappa_r = a + rc$ and the displayed formula for $O_{\max}(k)$ follows.

At $k = \kappa_r$, suppose a path with r carries had a proper zero-residual factorization into positive scales $k_1 + \cdots + k_s = k$, $s \geq 2$, with carry counts $r_1 + \cdots + r_s = r$. Every factor with $r_i > 0$ satisfies $k_i \geq a + r_i c$, while a factor with $r_i = 0$ still has $k_i \geq 1$. If carries occur in two or more factors, summing the onset bounds contributes at least one extra copy of a; if all carries occur in one factor, every remaining factor contributes at least one extra unit of scale. In either case $k > a + rc = \kappa_r$, a contradiction. Hence every onset path with r carries is factor-free. The standard logarithmic relation between all paths and factor-free factors then gives $[y^r]A_{\kappa_r} = \kappa_r[y^r]D_{\kappa_r}$. □

The hierarchy is a consequence of a stronger pathwise invariant.

Theorem 74.2 (WRITE-weight conservation)**.** *Assign every WRITE event a positive integer scale weight ω: a written frame of length $\omega m - 1$ at an empty stack has weight ω, while OUTER+ has fixed weight c and OUTER− fixed weight a. For a balanced accepted scale-k computation,*

$$\boxed{\sum_{\text{WRITE } j} \omega_j = k.}$$

Proof. Every WRITE of weight ω changes physical stack height by $\omega m - 1$, while every SEARCH or CANCEL decreases height by one. If W is the number of WRITEs and N the number of non-WRITEs, empty-to-empty height balance gives

$$N = m\sum_j \omega_j - W.$$

Since every input symbol produces exactly one event, $W + N = km$, and the identity follows. □

For BOTTOM-WRITE at debt d the exact weight is

$$\omega_B(d,1) = \max\left\{1, \left\lceil -\frac{d}{q} \right\rceil, \left\lceil \frac{d+1}{p} \right\rceil\right\},$$

$$\omega_B(d,0) = \max\left\{1, \left\lceil \frac{1-d}{q} \right\rceil, \left\lceil \frac{d}{p} \right\rceil \right\}.$$

Thus scale is literally the total physical shortest-demand WRITE budget.

74.3 Minimal multi-carry paths: Raney signatures and Schubert edge weights

Define the first-carry reserve

$$\boxed{R_0 = aq - h}, \qquad Q = qc.$$

At minimal r-carry onset let z_j be the number of zeros read before the jth carry. The feasible signatures are exactly

$$0 \le z_1 \le \cdots \le z_r, \qquad z_j \le R_0 - 1 + (j-1)Q.$$

Introduce the zero-tail slack

$$\lambda_j = R_0 - 1 + (j-1)Q - z_j.$$

Then

$$\boxed{0 \le \lambda_1 \le R_0 - 1, \qquad 0 \le \lambda_j \le \lambda_{j-1} + Q.}$$

Theorem 74.3 (Raney signature law)**.** *The number of feasible r-carry signatures is*

$$\boxed{|\mathcal{Z}_r| = \frac{R_0}{(Q+1)r + R_0} \binom{(Q+1)r + R_0}{r}.}$$

Hence the signature skeleton is the Raney number $R_r^{(Q+1,R_0)}$.

Proof. Interpret z_j as the number of horizontal steps before the jth vertical step. The displayed bounds are exactly the strict ballot boundary $x < R_0 + Qy$. The classical Raney/cycle lemma gives the formula [59]. □

Before a carry with slack λ, the state is $(D_h, 0^{\lambda+1})$. After the carry, define

$$X = h + 1 + Q, \qquad Y = pc - h - 1.$$

The written frame is

$$W_+ = (10)^Y 0^{X-Y}.$$

For a transition $\lambda \to \mu$ put

$$\delta = Q + \lambda - \mu, \qquad n = pc - 1 + \delta.$$

The no-WRITE phase contains $pc - 1$ ones and δ zeros and is compared with

$$Q_{\lambda,\mu} = (10)^Y 0^{n-2Y}.$$

If $B = \{b_1 < \cdots < b_\delta\}$ is the input-zero set and

$$H_\lambda = (h+1)m + p\lambda,$$

then rational nonnegativity is the lower flag

$$\boxed{b_i \ge \ell_i(\lambda) = \max\left\{i, \left\lceil \frac{mi - H_\lambda}{q} \right\rceil \right\}.}$$

Let $A_{\lambda,\mu}$ be the zero-position set of $Q_{\lambda,\mu}$. The inter-carry polynomial is

$$\boxed{M_{\lambda,\mu}(x) = \sum_{B} x^{n-|A_{\lambda,\mu}|-\delta+2|A_{\lambda,\mu}\cap B|}.}$$

Thus each edge of the Raney signature skeleton is a shifted-Schubert Hamming enumerator.

After reversal, $M_{\lambda,\mu}$ has the standard flagged determinant and upper-Hessenberg recurrence. This gives an exact finite product/sum over signatures for every minimal-onset polynomial $G^{(r)}_{p,q}(x)$.

74.3.1 Stationary bulk kernel and finite wall

When the rational lower flag is inactive, $M_{\lambda,\mu}$ depends only on $\delta = Q + \lambda - \mu$:

$$\boxed{\widehat{M}_\delta(x) = \sum_{j=\max(0,\delta-Y)}^{\delta} \binom{h+\delta}{j}\binom{Y}{\delta-j} x^{Y-\delta+2j}.}$$

Its generating function is the rational symbol

$$\boxed{\widehat{K}(v;x) = \sum_{\delta\ge 0} \widehat{M}_\delta(x) v^\delta = \frac{(x+(1-x^2)v)^Y}{(1-xv)^{pc}}.}$$

For every $p > q$ the wall is absent. For steep $q > p$, let

$$\Theta = pQ - m(h+1), \qquad b = \left\lceil \frac{\Theta}{p} \right\rceil.$$

If $\Theta > 0$, then

$$\boxed{b = Q - \left\lfloor \frac{m(h+1)}{p} \right\rfloor, \qquad 1 \le b \le Q-1,}$$

and only target slacks $0, \dots, b-1$ differ from the Toeplitz bulk. The exact operator is therefore Toeplitz-Hessenberg bulk plus a finite-rank rational-wall correction.

74.4 Algebraic closure of the minimal-onset family

The bulk kernel equation is

$$\boxed{v^Q = t\widehat{K}(v;x),}$$

or

$$\boxed{v^Q(1-xv)^{pc} = t(x+(1-x^2)v)^Y.}$$

Let $v_1, \dots, v_Q$ be the Q small Puiseux branches, taken in an algebraic Puiseux extension of $\mathbb{Q}(x)((t^{1/Q}))$. The symmetric Green-function expressions below descend to algebraic formal series over $\mathbb{Q}(x,t)$; see Section 74.6. The bulk excursion Green function is

$$\boxed{\mathcal{E}_Q(t;x) = \frac{(-1)^{Q-1}}{tx^Y}\prod_{i=1}^{Q} v_i,}$$

and from slack 0 to slack μ,

$$\boxed{\mathcal{R}_\mu(t;x) = \mathcal{E}_Q(t;x) h_\mu(v_1,\dots,v_Q),}$$

where h_μ is the complete homogeneous symmetric polynomial. These are the standard small-root half-line formulas specialized to the DPDA-derived symbol [52].

For a general start $s < Q$, the Green row has the finite form

$$\mathcal{R}_\mu^{(s)} = \sum_{i=1}^{Q} c_i^{(s)} v_i^\mu.$$

The coefficients are determined by a $Q \times Q$ boundary matrix. Writing

$$K_{<n}(v;x) = \sum_{d=0}^{n-1} \widehat{M}_d(x) v^d, \qquad \beta_n(v) = t v^{-n} K_{<n}(v;x),$$

define

$$\mathsf{C}_{\mu i} = \beta_{Q-\mu}(v_i), \qquad 0 \le \mu < Q.$$

Then

$$\boxed{\mathsf{C} c^{(s)} = e_s.}$$

The first-carry prefix vector has finite support $0 \le s < R_0 \le Q$, while the terminal slack generating series is rational. For $p > q$, this boundary system closes the entire minimal-onset family. For $q > p$, the finite wall is added by the matrix inversion lemma; hence:

Theorem 74.4 (Complete minimal-onset algebraicity)**.** *For every fixed coprime p, q, the generating series*

$$\mathcal{G}_{p,q}(t;x) = \sum_{r \ge 1} G_{p,q}^{(r)}(x) t^{r-1}$$

is algebraic over $\mathbb{Q}(x,t)$ and is given by a finite small-root/boundary system. No infinite-dimensional boundary problem remains.

For $q = 1$, $Q = 1$ and ordinary Lagrange inversion gives a scalar coefficient theorem. At 2:1 this simplifies further: if $W = 1 + xtW^3$, then

$$\boxed{\mathcal{G}_{2,1}(t;x) = x^4 W^3,}$$

so at the minimal r-carry scale $\kappa_r = r + 2$,

$$\boxed{G_{2,1}^{(r)}(x) = \frac{1}{r}\binom{3r}{r-1} x^{r+3}.}$$

74.5 Carry-onset lower bound and wall-rank lemma

Let

$$a = 1 + \left\lfloor \frac{h}{q} \right\rfloor, \qquad c = 1 + \left\lfloor \frac{h}{p} \right\rfloor.$$

Before the first positive outer carry, the maximal all-1 trajectory is the cheapest way to accumulate the required positive mismatch debt. It consumes exactly a units of scale before the first carry. After each OUTER+ reset, the written frame has length $cm - 1$, hence, by WRITE-weight conservation, each additional carry consumes at least one further block of scale weight c. Zeros cannot reduce either requirement because they decrease the rational height available to sustain the positive boundary trajectory. The all-1 extremal prefix followed by the unique shortest balancing continuation realizes equality at every level. Consequently the first scale supporting r carries is $\kappa_r = a + rc$.

At $k = \kappa_r$, a proper zero-residual factor would split the total scale into two positive balanced blocks. The carry-onset bound applied separately to the factors would then leave

strictly less than the scale required for r carries in the factor containing them, contradicting minimality. Thus every r-carry path at first onset is factor-free.

For steep $q > p$, recall

$$\Theta = pQ - m(h+1) > 0, \qquad b = \left\lceil \frac{\Theta}{p} \right\rceil .$$

Then

$$b = Q - \left\lfloor \frac{m(h+1)}{p} \right\rfloor ,$$

and because $m(h+1)/p > 1$ in every genuine wall case,

$$1 \le b \le Q - 1.$$

Hence every row needed by the finite Woodbury wall correction lies inside the already solved $Q \times Q$ boundary Green block. No additional start-state formula is required.

74.6 Formal domain of the small-root formulas

The Q small roots of

$$v^Q = tK_c(v; x)$$

are naturally Puiseux series in an algebraic extension of $\mathbb{Q}(x)((t^{1/Q}))$. The symmetric combinations used for the Green functions descend to algebraic formal series over $\mathbb{Q}(x, t)$; the underlying coefficient recurrence has polynomial coefficients in x. Formulas containing x^{-Y} are therefore interpreted first over $\mathbb{Q}(x)$ and specialized at singular values such as $x = 0$ only through the polynomial recurrence or by a removable-limit argument. Likewise, Vandermonde-denominator expressions are interpreted for distinct generic roots and extended to collisions by symmetric continuation.

Chapter 75

Phase Normality and Exact One-Counter Compression

Despite the history dependence created by WRITE resets, the centered rational chamber has a much smaller exact state description than the literal binary stack suggests. A phase-normal reconstruction theorem reduces the full reachable demand to one signed residual together with a finite phase bit, and a fixed shift makes the residual a nonnegative counter.

75.1 Centered phase normal form and arbitrary-scale compression

Minimal onset still uses a distinguished reset skeleton. At arbitrary scale, BOTTOM and OUTER frames can be nested. The decisive fact is that the apparent binary-stack complexity collapses on the rational chamber.

For a stack word S, write

$$\rho(S) = qN_1(S) - pN_0(S), \qquad R = -\rho(S) = pN_0(S) - qN_1(S).$$

The variable R is signed; on the rational chamber it is bounded below by $-hm$. The shifted counter $J = R + hm$ is therefore nonnegative. Let $A_{10}(S)$ and $A_{01}(S)$ denote the numbers of adjacent factors 10 and 01. A word is called *phase normal* if it consists of an alternating prefix followed by a monochromatic tail. The following theorem is proved by simultaneous induction on the physical transitions; the induction constructs two canonical reconstruction maps ν_0, ν_1 on the reachable shortest-demand pairs.

Theorem 75.1 (Centered phase-normal form)**.** *For every configuration reachable from a rational-Dyck prefix:*

1. *the stack Parikh vector is the shortest-demand pair $\mu(R)$;*
2. *there is a phase bit $b \in \{0, 1\}$ such that*

$$\boxed{S = \nu_b(\mu(R));}$$

   thus the complete bit stack is reconstructible from (R, b);

3. *the centered debt satisfies*

$$\boxed{A_{10}(S) - h \le d \le h - A_{01}(S).}$$

The three assertions are transition-closed only when proved simultaneously.

Proof. Define the two reconstruction maps explicitly by

$$\nu_0(x,y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & x < y, \end{cases} \qquad \nu_1(x,y) = \begin{cases} (01)^y 0^{x-y}, & x \geq y, \\ (01)^x 1^{y-x}, & x < y. \end{cases}$$

They have the same Parikh vector (x, y) and are exactly the two phase-normal orderings compatible with a shortest-demand pair. We prove simultaneously by induction over the displayed centered transitions that (i) the current Parikh vector is $\mu(R)$, (ii) the stack equals one of these two words, and (iii) the debt inequality holds.

For a BOTTOM-WRITE the exact shortest-demand formulas are

$$\begin{aligned} (x,y) &= (d + q\omega,\ p\omega - d - 1) \quad \text{on input 1}, \\ (x,y) &= (d + q\omega - 1,\ p\omega - d) \quad \text{on input 0}. \end{aligned}$$

with the minimal positive weight ω of Theorem 74.2. Minimality of ω makes the minority coordinate at most h, so the canonical frame lies in the centered alternating interval and satisfies the debt bounds.

For CANCEL, deleting the visible matching cell decreases exactly one Parikh coordinate. Because the pre-transition pair is kernel-reduced and the deleted letter is visible in a phase-normal representative, the remaining pair is again the unique shortest pair for the updated signed demand; deleting the first letter of ν_b leaves a phase-normal word and changes A_{10}, A_{01} by at most one in the direction compensated by the unchanged debt.

For an interior SEARCH, the visible mismatching cell is consumed while d moves by one. If a SEARCH+ sees a stack beginning 00, phase normality forces the stack to be monochromatic from that point onward; hence $A_{01} = 0$ and the upper debt inequality remains strict enough after $d \mapsto d + 1$. The SEARCH− case beginning 11 is the physical dual. All alternating-prefix SEARCH cases simply shorten the alternating prefix and preserve both inequalities.

At OUTER+, the new frame has

$$X_+ = h + 1 + qc, \qquad Y_+ = pc - h - 1,$$

with $Y_+ \leq h$ and a terminal zero-run, so it is one of the displayed ν_b words and resets d to 0 inside the required interval. OUTER− is the dual transition at the level of the induction; Theorem 75.2 below then excludes it on the rational-Dyck chamber. These cases exhaust the transition table, proving the three assertions simultaneously. The explicit reconstruction and compressed successor table are collected in Section 75.5. □

Theorem 75.2 (No negative outer carry)**.** *OUTER− never occurs on a rational-Dyck prefix.*

Proof. At a hypothetical OUTER− trigger, $d = -h$ and the top symbol is 1. The lower centered inequality in Theorem 75.1 forces $A_{10}(S) = 0$. Phase normality therefore makes the entire nonempty stack 1^r. Hence the global rational height is

$$\rho_{\text{global}} = dm - \rho(S) = -hm - rq < 0,$$

contradicting rational-Dyck nonnegativity. □

75.1.1 Exact one-counter recurrence

Theorem 75.1 makes the physical stack a derived object. The exact compressed state is

$$\boxed{\sigma = (d, J, b), \qquad J = R + hm,}$$

where $d \in [-h, h]$, the signed demand residual $R = pN_0(S) - qN_1(S)$ satisfies $R \geq -hm$ on the rational chamber, hence $J \in \mathbb{Z}_{\geq 0}$ is a genuine one-counter coordinate, and b is the phase

bit. When evaluating transitions we recover $R = J - hm$ and reconstruct $S = \nu_b(\mu(R))$; Section 75.5 gives the exact case table. If $\mu(R) = (x, y)$, popping a top 1 sends

$$\boxed{R \mapsto R + q,}$$

while popping a top 0 sends

$$\boxed{R \mapsto R - p.}$$

The next phase bit is uniquely determined by the reconstructed remaining shortest-demand word. OUTER+ sends

$$\boxed{R' = (R - p) + (h + 1)m, \qquad d' = 0,}$$

and an empty-stack BOTTOM-WRITE on bit a sends

$$\boxed{R' = dm + w(a), \qquad d' = 0.}$$

By Theorem 75.2, no OUTER− transition is present in the rational chamber.

For each positive-height compressed state $\sigma = (d, J, b)$, let $F_\sigma(z, x, y)$ be the generating series of factor-free accepting continuations, with x marking CANCEL, y marking OUTER+, and z marking WRITE scale weight. Section 75.5 gives, for each next bit a, an explicit deterministic successor $T_a(\sigma)$ and monomial $M(\sigma, a)$. Therefore

$$\boxed{F_\sigma = \sum_{a \in \{0,1\}} M(\sigma, a)\, \Xi(T_a(\sigma)),}$$

where Ξ returns 1 at the first legal empty-stack terminal state, 0 if the rational height becomes negative or a proper zero-residual contact occurs, and otherwise returns the corresponding successor series. The initial factor-free series $I(z; x, y)$ is obtained by the two possible first BOTTOM transitions, and

$$\boxed{I_{k,r}(x) = [z^k y^r] I(z; x, y).}$$

The only unbounded coordinate is the nonnegative integer J; (d, b) is finite control. This is thus an exact weighted one-counter recurrence, to which standard one-counter/PDA generating-function methods apply [58, 50].

Corollary 75.3 (Arbitrary-scale operational enumeration)**.** *For every coprime p, q and every scale k, the factor-free bivariate polynomial $I_k(x, y)$ is determined exactly by the compressed one-counter recurrence above. All rational-Dyck paths and operational Bizley cumulants follow from*

$$\boxed{D(z; x, y) = \frac{1}{1 - I(z; x, y)}, \qquad \log D(z; x, y) = \sum_{k \geq 1} \frac{A_k(x, y)}{k} z^k.}$$

The primitive and minimal-onset formulas are specializations of the same compressed dynamics.

75.2 The original 2:1 family at arbitrary scale

The route returns to the motivating ratio with a particularly rigid all-scale law. In a factor-free scale-k 2:1 path, each WRITE has scale weight 1, every nonterminal WRITE episode contains exactly one SEARCH+, and the terminal episode contains none. If W, S, C are the numbers of WRITEs, SEARCHes, and CANCELs, stack-height and input-length balance give

$$C + S = 2W, \qquad W + C + S = 3k.$$

Hence $W = k$, $S = k - 1$, and

$$\boxed{C = k + 1}$$

for every factor-free path.

If an arbitrary rational-Dyck path of scale k has $s(P)$ zero-residual factors, additivity therefore gives the pathwise identity

$$\boxed{C(P) = k + s(P).}$$

This is the operational contribution; the factor counts themselves belong to the classical Fuss–Catalan/factor-free theory.

The number of paths with s factors is

$$\boxed{\#\{P : C(P) = k + s\} = \frac{s}{k}\binom{3k - s - 1}{k - s}, \qquad 1 \le s \le k,}$$

so

$$\boxed{D_k(x, 1) = \sum_{s=1}^{k} \frac{s}{k}\binom{3k - s - 1}{k - s} x^{k+s}.}$$

The operational Bizley cumulants simplify to

$$\boxed{A_k(x, 1) = x^k \sum_{s=1}^{k} \binom{3k - s - 1}{k - s} x^s,}$$

and $A_k(1, 1) = \frac{1}{3}\binom{3k}{k}$ as required.

75.3 Computational validation and structural checks

The proofs above are symbolic. Exact computation was used independently as regression testing while the formulas were developed. The carry-onset hierarchy was checked on coprime $p, q \le 7$ through scales up to 10 (350 scale tests and 138 onset tests for the first four carry layers) with no discrepancy. The stationary inter-carry determinant/Hessenberg formula was compared with direct local-word enumeration on 481 small parameter/slack pairs, and the rational Toeplitz bulk formula on 660 tested transitions, again with zero discrepancies. The scalar p:1 formulas were compared polynomial-by-polynomial with direct centered-machine enumeration in representative 2:1, 3:1, and 4:1 cases. Before Theorem 75.2 was proved, exact state-DP searches over 34 ordered coprime ratios with $p, q \le 7$ and 176 ratio/scale instances found no accepted rational-Dyck example containing OUTER−; the theorem now explains that observation.

These computations are not used as proof. They serve to detect sign, parity, boundary, and reset errors in formulas whose symbolic proofs are independent.

75.4 Operational synthesis: the loss of primitive log-concavity

The event statistics exhibit a sharp structural transition. Primitive normalization has one fixed reference word; CANCEL is therefore a single Hamming/Schubert statistic and inherits the corresponding matroid convexity. A physical WRITE reset replaces that static geometry by sums and products of Schubert pieces. For example, the first-carry polynomial at 5:3 is

$$36x^{10} + 67x^{11} + 153x^{12} + 180x^{13} + 135x^{14} + 39x^{15} + 18x^{16} + 2x^{17},$$

whose coefficients are not log-concave. Thus the loss of primitive Lorentzian behavior is not an artifact of proof technique; it records genuine history sensitivity introduced by a reset.

The repeated-kernel theory can be summarized as

physical WRITE budget + Raney carry skeleton +Schubert cancellation phases + Bizley zero-contact factorization.

The phase-normal theorem shows that this history-sensitive system is nevertheless much smaller than the literal binary stack suggests: on the rational chamber it is exactly a one-counter process with finite phase control.

75.5 Phase reconstruction and exact compressed transition table

For a shortest-demand pair (x, y) define the two phase-normal words

$$\nu_0(x,y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & x < y, \end{cases}$$

$$\nu_1(x,y) = \begin{cases} (01)^y 0^{x-y}, & x \geq y, \\ (01)^x 1^{y-x}, & x < y. \end{cases}$$

When one of the formulas coincides with the other, the phase bit is immaterial. The simultaneous induction in Theorem 75.1 proves that every stack reachable from a rational-Dyck prefix is one of these two words for its shortest-demand pair.

It is important that the signed stack-demand residual need not itself be nonnegative. Put

$$R = pN_0(S) - qN_1(S) = -\rho(S).$$

The rational-chamber condition is

$$dm + R \geq 0.$$

Since $-h \leq d \leq h$, it follows that $R \geq -hm$. Thus the genuine one-counter coordinate is the shifted variable

$$J := R + hm \in \mathbb{Z}_{\geq 0}.$$

This shift corrects the misleading shorthand $R \geq 0$: R is signed, whereas J is the nonnegative counter.

Given a compressed state (d, J, b), recover the signed value $R = J - hm$ and reconstruct $S = \nu_b(\mu(R))$ (with the empty-stack case treated separately by the BOTTOM rule). The exact transition and its weight monomial are:

condition	(d', R')	weight
$S = \zeta T,\ \zeta = a$	$(d, R + w_S(a))$	x
$S = 0T,\ a = 1,\ d < h$	$(d+1, R-p)$	1
$S = 1T,\ a = 0,\ d > -h$	$(d-1, R+q)$	1
$S = 0T,\ a = 1,\ d = h$	$(0, (R-p) + (h+1)m)$	yz^c
$S = 1T,\ a = 0,\ d = -h$	excluded on the rational chamber	$--$
$S = \varepsilon$	$(0, dm + w(a))$	$z^{\omega_B(d,a)}$

where $w(1) = q$, $w(0) = -p$, $w_S(1) = q$, $w_S(0) = -p$ records the change in R under a pop, $c = 1 + \lfloor h/p \rfloor$, and ω_B is the exact bottom-WRITE weight from Theorem 74.2. In the first row, equivalently, popping a 1 sends $R \mapsto R + q$ and popping a 0 sends $R \mapsto R - p$. After computing the successor stack by the centered rule, b' is the unique phase for which $S' = \nu_{b'}(\mu(R'))$ (or is ignored when the stack is empty). This gives an

executable deterministic recurrence on the finite control (d, b) and the single nonnegative counter $J = R + hm$; after each row one stores $J' = R' + hm$.

For a factor-free positive-height state, the corresponding generating series therefore satisfies

$$F_{d,J,b} = \sum_{a\in\{0,1\}} M(d, J, b; a)\, \Xi(T_a(d, J, b)),$$

where M is the monomial in the table and Ξ is 1 on the first legal empty-stack terminal state, 0 after a negative rational height or a proper zero-residual contact, and otherwise the successor F-series. This is the precise one-counter recurrence used in Corollary 75.3.

75.6 Operational-to-intrinsic bridge

This research branch has counted statistics of a chosen physical normalization process. The next enumerative branch changes the viewpoint: the residual becomes the intrinsic rank of a rational path, returns become barrier contacts, and the counting problem is expressed without reference to a physical execution history. The two branches share the same scalar residual but answer different questions—operational event history here, intrinsic path statistics there.

Chapter 76

From Residual Rank to Explicit Run–Return Enumeration

Research branch — From Residual Contacts to Refined Rational-Path Enumeration. *This branch first develops the explicit run–return solution and its determinant reduction, then extracts the contact-preserving barrier-compression framework, its four fixed-composition rational-Dyck enumerations, and its arbitrary-boundary and parking-function applications. The determinant technology itself is classical [25, 24, 26]; the contribution is scoped to the contact-sensitive reduction and the resulting refined enumerations.*

The preceding macro-Parts developed the residual

$$\rho_{p,q} = qN_1 - pN_0$$

from the completion semantics of the original 2:1 machine and then from the centered real-time p:q family. The same scalar invariant also applies directly to a rational Dyck rectangle. Write

$$m = gq, \qquad n = gp.$$

At a lattice point (x, y) the usual rational-path rank is

$$R(x, y) = my - nx = g(qy - px) = g\rho_{p,q}.$$

Thus the rank is exactly the pushdown residual up to the dilation factor g. At a nonterminal return position the rank condition $0 \le R < n$ is therefore equivalent to the low-residual band

$$0 \le \rho_{p,q} < p.$$

This identity is the semantic bridge; none of the determinant proofs below depends logically on an automaton construction.

Dai, Fu, and Qiu introduced the rational run and ratio-run statistics and proved composition-preserving symmetry results with the return statistic [23]. Their concluding remarks posed explicit enumeration of rational Dyck paths with prescribed ordinary run and returns, and with prescribed ratio-run and returns. This chapter identifies the return/contact bridge; Chapter 77 gives explicit finite determinant solutions to both questions.

The key step was not the determinant technology itself. Binomial barrier determinants and the Lindström–Gessel–Viennot method are classical [25, 24, 26]. The new step in the solution was to turn run constraints into avoidance of selected coarea values while *retaining return positions as contacts of the compressed upper barrier.* The resulting structural chain is

residual → rational rank → return as contact → forbidden-value compression → weighted determinant

The specialized reduction below solves the two posed enumeration questions; the subsequent chapters then extract the reusable barrier-compression principle and its refinements.

76.1 Rational Dyck paths and sampled runs

Throughout, m, n are positive integers with $m \geq n$. A path consists of north steps $N = (0, 1)$ and east steps $E = (1, 0)$.

Definition 76.1 — Rational Dyck path.

An $m \times n$ *rational Dyck path* is a path from $(0, 0)$ to (m, n) that remains weakly above $y = nx/m$. Its set is denoted by $\mathcal{D}_{m,n}$.

For $P \in \mathcal{D}_{m,n}$, let

$$\mathcal{U}(P) = (u_1, \ldots, u_n)$$

be its coarea sequence, where u_i is the horizontal coordinate of the ith north step, equivalently the number of cells to its left. Define

$$b_i := \left\lfloor \frac{m(i-1)}{n} \right\rfloor, \qquad 1 \leq i \leq n. \tag{76.1}$$

Proposition 76.2 — Coarea characterization.

A sequence $u = (u_1, \ldots, u_n)$ is the coarea sequence of a path in $\mathcal{D}_{m,n}$ if and only if

$$0 = u_1 \leq u_2 \leq \cdots \leq u_n, \qquad u_i \leq b_i \quad (1 \leq i \leq n). \tag{76.2}$$

Proof. The weak inequalities express the fact that successive north steps occur in nondecreasing columns. Immediately before the ith north step the path is at $(u_i, i-1)$, which lies weakly above the diagonal precisely when

$$i - 1 \geq \frac{n}{m} u_i,$$

or equivalently $u_i \leq m(i-1)/n$. Since u_i is integral, this is $u_i \leq b_i$. Finally $b_1 = 0$, so $u_1 = 0$. □

Dai–Fu–Qiu define a return position as a nonterminal lattice point on the path having rank smaller than n, where the rank of (x, y) is $my - nx$ [23]. Their return statistic counts the corresponding north steps.

Definition 76.3 — Sampled run.

For $d \geq 1$ and $P \in \mathcal{D}_{m,n}$, define

$$\mathrm{run}_d(P) := \min\{a \in [n] : da \notin \{u_1, \ldots, u_n\}\}. \tag{76.3}$$

The minimum exists because the sequence contains only n entries, one of which is 0.

For $d = 1$, Definition 76.3 is the ordinary rational run of [23]. If

$$\kappa = \left\lfloor \frac{m}{n} \right\rfloor,$$

then $d = \kappa$ gives their ratio-run $\widetilde{\mathrm{run}}$.

76.2 Returns are coarea contacts

The following observation is the conceptual bridge used throughout this branch.

Theorem 76.4 — Return–contact correspondence.

For every $P \in \mathcal{D}_{m,n}$ with coarea sequence $(u_1, \ldots, u_n)$,

$$\boxed{\mathrm{ret}(P) = \#\{i \in [n] : u_i = b_i\}.} \tag{76.4}$$

Proof. Immediately before the ith north step the path is at $(u_i, i-1)$ and hence has rank

$$R_i = m(i-1) - nu_i.$$

Write

$$m(i-1) = nb_i + r_i, \qquad 0 \le r_i < n.$$

Then

$$R_i = r_i + n(b_i - u_i). \tag{76.5}$$

By Proposition 76.2, $b_i - u_i \ge 0$. Therefore

$$R_i < n \iff b_i - u_i = 0 \iff u_i = b_i.$$

Every nonterminal path point of rank $< n$ must be followed by a north step: an east step would decrease the rank by n and make it negative, contradicting the rational-Dyck condition. Hence the north steps whose initial points have rank $< n$ are precisely the returns of [23]; the terminal endpoint is not an initial point of a north step. Summing over i gives (76.4). □

Remark 76.R1 — Pushdown residual interpretation.

Let $g = \gcd(m, n)$ and write $m = gq$, $n = gp$ with $\gcd(p, q) = 1$. Under the step encoding $0 = E$, $1 = N$, a prefix containing y north steps and x east steps has the weighted-balance residual

$$\rho = qy - px,$$

which is the scalar residual obtained by generalizing the arithmetic carried by the original 2:1 pushdown machine. The rational-path rank is $R = g\rho$, so a return position satisfies

$$0 \le \rho < p.$$

Thus Theorem 76.4 may equivalently be read as the statement that the finite low-residual band singled out by the pushdown semantics becomes upper-barrier contact in coarea coordinates. This remark records the origin of the viewpoint; the theorem itself depends only on the rank calculation above.

76.3 Deleting forbidden coarea values

Let $F \subset \mathbb{Z}_{>0}$ be finite. We wish to enumerate paths whose coarea sequence avoids F, while retaining the return weight from Theorem 76.4.

Define

$$\phi_F(t) := t - \#(F \cap [1, t]), \qquad t \in \mathbb{Z}_{\ge 0}, \tag{76.6}$$

and

$$c_i(F) := \phi_F(b_i) = b_i - \#(F \cap [1, b_i]). \tag{76.7}$$

Also put

$$\epsilon_i(F) := \mathbf{1}_{\{b_i \notin F\}}. \tag{76.8}$$

Lemma 76.5 — Forbidden-value compression.

The map

$$(u_1, \dots, u_n) \longmapsto (v_1, \dots, v_n), \qquad v_i = \phi_F(u_i), \tag{76.9}$$

is a bijection between

$$\left\{ \begin{array}{c} 0 = u_1 \le \cdots \le u_n, \quad u_i \le b_i, \\ \{u_1, \dots, u_n\} \cap F = \varnothing \end{array} \right\} \tag{76.10}$$

and

$$\{0 = v_1 \le \cdots \le v_n : \ v_i \le c_i(F)\}. \tag{76.11}$$

Moreover, whenever $b_i \notin F$,

$$u_i = b_i \iff v_i = c_i(F). \tag{76.12}$$

If $b_i \in F$, the equality $u_i = b_i$ is impossible on the domain.

Proof. For each $B \geq 0$, deletion of the elements of $F \cap [1, B]$ makes ϕ_F an order-preserving bijection

$$\{0, 1, \ldots, B\} \setminus F \longrightarrow \{0, 1, \ldots, \phi_F(B)\}.$$

Applying this coordinatewise with $B = b_i$ proves that (76.9) is well-defined, preserves weak order, and has a unique inverse obtained by reinserting the deleted values. This proves the bijection. If $b_i \notin F$, the maximum allowed value b_i maps to the maximum compressed value $c_i(F)$, proving (76.12). The final assertion is immediate. □

Chapter 77

Weighted Determinants and Explicit Run–Return Enumeration

Chapter 76 converted returns into upper-barrier contacts and isolated the specialized forbidden-value compression that arose in the explicit-enumeration solution. We now retain the complete determinant and inclusion–exclusion argument, including the coefficient-free form and the independent exact checks. The sampled statistic run_d is used only as a unifying notation: $d = 1$ is ordinary run and $d = \lfloor m/n \rfloor$ is the Dai–Fu–Qiu ratio-run. No independent novelty claim is made for intermediate sampling widths.

77.1 A weighted binomial determinant

We use the convention

$$\binom{a}{b} = 0 \quad \text{if } b < 0 \text{ or } b > a,$$

for integers $a \geq 0$.

Definition 77.1 — Weighted forbidden-value determinant polynomial.
For a finite $F \subset \mathbb{Z}_{>0}$, define

$$\boxed{D_F(z) := \det_{1 \leq i,j \leq n} \left[\binom{c_i(F)+1}{j-i+1} + (z-1)\epsilon_i(F) \binom{c_i(F)}{j-i} \right].} \tag{77.1}$$

Theorem 77.2 — Weighted forbidden-value determinant.
For every finite $F \subset \mathbb{Z}_{>0}$,

$$\boxed{D_F(z) = \sum_{\substack{P \in \mathcal{D}_{m,n} \\ \mathcal{U}(P) \cap F = \varnothing}} z^{\mathrm{ret}(P)}.} \tag{77.2}$$

Proof. Write $c_i = c_i(F)$ and $\epsilon_i = \epsilon_i(F)$. Consider the directed acyclic lattice network with east steps $(1, 0)$ and northeast steps $(1, 1)$. For $1 \leq i, j \leq n$, set

$$S_i = (-c_i - 1, i), \qquad T_j = (0, j+1).$$

A path from S_i to T_j has horizontal displacement $c_i + 1$ and vertical displacement $j - i + 1$, hence there are

$$\binom{c_i+1}{j-i+1}$$

such paths.

If $\epsilon_i = 1$, give weight z to the northeast edge leaving S_i and weight 1 to all other edges. If $\epsilon_i = 0$, all edges leaving S_i have weight 1. The number of $S_i \to T_j$ paths using that first northeast edge is

$$\binom{c_i}{j-i},$$

so the weighted single-path generating function is exactly the (i,j) entry of (77.1).

By the Lindström–Gessel–Viennot lemma [24], the determinant is the signed generating function of vertex-disjoint path families connecting the sources to the targets. The source–target configuration is nonpermutable. Indeed, since c_i is nondecreasing, for $i < k$ we have $x(S_i) \geq x(S_k)$. On the common horizontal range from $x(S_i)$ to 0, an S_i-path starts strictly below an S_k-path. If their assigned targets were inverted, say T_j and T_ℓ with $j > \ell$, their vertical order at $x = 0$ would be reversed. Because every east or northeast step advances the horizontal coordinate by one, each path has one vertex at every integer abscissa in this common range; the change of vertical order therefore forces a common vertex. Thus a vertex-disjoint family has no inversion, and only the identity matching $S_i \to T_i$ contributes.

An identity path $\pi_i : S_i \to T_i$ contains exactly one northeast step and c_i east steps. Let $e_i \in \{0, \dots, c_i\}$ be the number of east steps occurring before its unique northeast step, and set

$$v_i := c_i - e_i. \tag{77.3}$$

For adjacent paths, vertex-disjointness is equivalent to

$$-c_{i+1} - 1 + e_{i+1} < -c_i + e_i,$$

which, since all quantities are integral, is equivalent to

$$c_i - e_i \leq c_{i+1} - e_{i+1},$$

that is, $v_i \leq v_{i+1}$. Thus identity nonintersecting families are in bijection with

$$0 \leq v_1 \leq \cdots \leq v_n, \qquad v_i \leq c_i.$$

This is precisely the compressed sequence class in Lemma 76.5.

Finally, π_i uses its first northeast edge if and only if $e_i = 0$, equivalently $v_i = c_i$. Such an edge is weighted by z exactly when $\epsilon_i = 1$. By Lemma 76.5, these weighted events correspond exactly to the original contacts $u_i = b_i$, and Theorem 76.4 identifies their number with $\mathrm{ret}(P)$. Therefore each family has weight $z^{\mathrm{ret}(P)}$, proving (77.2). □

Corollary 77.3 — Return polynomial.

For $F = \varnothing$,

$$\sum_{P \in \mathcal{D}_{m,n}} z^{\mathrm{ret}(P)} = \det_{1 \leq i,j \leq n} \left[\binom{b_i}{j-i+1} + z \binom{b_i}{j-i} \right]. \tag{77.4}$$

Proof. When $F = \varnothing$, $c_i = b_i$ and $\epsilon_i = 1$. Pascal's identity gives

$$\binom{b_i+1}{j-i+1} + (z-1)\binom{b_i}{j-i} = \binom{b_i}{j-i+1} + z\binom{b_i}{j-i}.$$

Apply Theorem 77.2. □

77.2 Explicit run–return enumeration

For $d \geq 1$ and $a \geq 1$, define the required sampled values

$$S_{d,a-1} := \{d, 2d, \ldots, (a-1)d\}, \tag{77.5}$$

with $S_{d,0} = \varnothing$.

Theorem 77.4 — Main enumeration theorem.
Let $m \geq n \geq 1$, $d \geq 1$, and $a, b \in [n]$. Then

$$\boxed{\begin{aligned} N_{m,n}^{(d)}(a,b) &:= \#\{P \in \mathcal{D}_{m,n} : \mathrm{run}_d(P) = a,\ \mathrm{ret}(P) = b\} \\ &= [z^b] \sum_{T \subseteq S_{d,a-1}} (-1)^{|T|} D_{T \cup \{ad\}}(z) \end{aligned}} \tag{77.6}$$

Proof. By Definition 76.3, $\mathrm{run}_d(P) = a$ if and only if

$$S_{d,a-1} \subseteq \mathcal{U}(P) \qquad \text{and} \qquad ad \notin \mathcal{U}(P). \tag{77.7}$$

First restrict to paths avoiding ad. To force every value of $S_{d,a-1}$ to occur, apply inclusion–exclusion to the events that a specified value of $S_{d,a-1}$ is missing. For $T \subseteq S_{d,a-1}$, the corresponding class avoids the forbidden set $T \cup \{ad\}$ and therefore has return polynomial $D_{T \cup \{ad\}}(z)$ by Theorem 77.2. Thus the return polynomial of paths satisfying (77.7) is

$$\sum_{T \subseteq S_{d,a-1}} (-1)^{|T|} D_{T \cup \{ad\}}(z).$$

Extracting $[z^b]$ gives (77.6). □

Corollary 77.5 — Ordinary run–return.
For $a, b \in [n]$,

$$\#\{P \in \mathcal{D}_{m,n} : \mathrm{run}(P) = a,\ \mathrm{ret}(P) = b\} = [z^b] \sum_{T \subseteq \{1,\ldots,a-1\}} (-1)^{|T|} D_{T \cup \{a\}}(z). \tag{77.8}$$

Corollary 77.6 — Ratio-run–return.
Let $\kappa = \lfloor m/n \rfloor$. For $a, b \in [n]$,

$$\boxed{\begin{aligned} &\#\{P \in \mathcal{D}_{m,n} : \widetilde{\mathrm{run}}(P) = a,\ \mathrm{ret}(P) = b\} \\ &\quad = [z^b] \sum_{T \subseteq \{\kappa, 2\kappa, \ldots, (a-1)\kappa\}} (-1)^{|T|} D_{T \cup \{a\kappa\}}(z) \end{aligned}} \tag{77.9}$$

Corollaries 77.5 and 77.6 give explicit finite formulas for the two joint enumeration problems posed in [23].

77.2.1 A coefficient-free form

Formula (77.6) is already finite, but the coefficient extraction can be removed. For a finite forbidden set F, define $n \times n$ matrices A^F and B^F by

$$A_{ij}^F := \binom{c_i(F) + 1}{j - i + 1} - \epsilon_i(F) \binom{c_i(F)}{j - i}, \tag{77.10}$$

$$B_{ij}^F := \epsilon_i(F) \binom{c_i(F)}{j - i}. \tag{77.11}$$

Then

$$D_F(z) = \det(A^F + zB^F). \tag{77.12}$$

For $I \subseteq [n]$, let M_I^F be the matrix whose ith row is the ith row of B^F if $i \in I$, and the ith row of A^F otherwise.

Theorem 77.7 — Coefficient-free explicit formula.

For $a, b \in [n]$,

$$\boxed{N_{m,n}^{(d)}(a,b) = \sum_{T \subseteq S_{d,a-1}} (-1)^{|T|} \sum_{\substack{I \subseteq [n] \\ |I|=b}} \det M_I^{T \cup \{ad\}}.} \tag{77.13}$$

In particular, the joint count is expressed using only finite subset sums, binomial coefficients, and integer determinants.

Proof. The determinant in (77.12) is multilinear in its rows. Choosing the zB^F contribution in exactly the rows indexed by I and the A^F contribution in all remaining rows gives

$$[z^b]D_F(z) = \sum_{\substack{I \subseteq [n] \\ |I|=b}} \det M_I^F.$$

Substitute this identity into Theorem 77.4. □

Remark 77.R1 — Coefficient-free scope.

The formula is effective without symbolic polynomial expansion. For fixed m, n, d, a, b, every determinant in (77.13) is an $n \times n$ integer determinant. The two subset sums have respectively 2^{a-1} and $\binom{n}{b}$ terms before cancellations or structural simplifications.

77.3 Examples

We display joint matrices with rows indexed by the run value and columns by the return value.

Example 77.E1 — Ordinary run, $(m, n) = (12, 4)$.

With $d = 1$, Theorem 77.4 gives

$$\left(N_{12,4}^{(1)}(a,b)\right)_{1 \le a,b \le 4} = \begin{pmatrix} 52 & 30 & 8 & 1 \\ 30 & 8 & 1 & 0 \\ 8 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}. \tag{77.14}$$

This is the ordinary run–return matrix displayed in [23].

Example 77.E2 — Ordinary run, $(m, n) = (13, 5)$.

The formula gives

$$\begin{pmatrix} 106 & 106 & 39 & 9 & 1 \\ 106 & 39 & 9 & 1 & 0 \\ 39 & 9 & 1 & 0 & 0 \\ 9 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}, \tag{77.15}$$

again reproducing the published example [23].

Example 77.E3 — Ratio-run, $(m, n) = (11, 5)$.

Here $\kappa = \lfloor 11/5 \rfloor = 2$. Corollary 77.6 gives

$$\boxed{\begin{pmatrix} 85 & 43 & 13 & 2 & 0 \\ 43 & 32 & 11 & 2 & 0 \\ 13 & 11 & 7 & 2 & 0 \\ 2 & 2 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}}. \tag{77.16}$$

For instance, there are exactly 11 paths with $\widetilde{\mathrm{run}} = 2$ and ret $= 3$. The sum of all entries is 273.

Example 77.E4 — Ratio-run, $(m, n) = (10, 4)$.

Here $\kappa = 2$ and

$$\begin{pmatrix} 30 & 11 & 2 & 0 \\ 11 & 9 & 4 & 1 \\ 2 & 4 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}. \tag{77.17}$$

The total number of paths is 76.

The symmetry of the displayed matrices is guaranteed independently by the involutions of Dai–Fu–Qiu. In the present approach, that symmetry becomes a family of identities between the determinant sums in Theorem 77.4.

77.4 Computational verification

The proofs above are purely combinatorial and do not depend on computation. We nevertheless used exact computation as an audit of indexing, endpoint conventions, and the inclusion–exclusion signs. Two independent exact checks were performed:

1. direct enumeration of all weakly increasing coarea sequences satisfying (76.2), followed by direct computation of run_d and ret;
2. independent evaluation of Theorem 77.4 from the forbidden-value determinants.

The displayed matrices in Section 77.3 agree entry by entry under these calculations. A small exhaustive determinant audit covered 33 sampled-run cases with no mismatches. An independent audit of the coefficient-free formula covered 86 cases with no mismatches. These computations are consistency checks only and are not used in place of the symbolic proofs.

77.5 Interpretation and scope

The main formula separates the solution into three transparent operations:

$$\boxed{\text{return} = \text{barrier contact}} \longrightarrow \boxed{\text{delete forbidden values}}$$

$$\longrightarrow \boxed{\text{weighted determinant} + \text{inclusion–exclusion}}.$$

The determinant technology itself is classical: the binomial determinant belongs to a well-established family associated with nondecreasing sequences below barriers [25, 26, 24]. The new step in the present solution is the reduction of the run constraints to forbidden-value avoidance while retaining return positions as weighted contacts of the compressed upper barrier; inclusion–exclusion then converts that refinement into the requested joint enumeration.

The sampled statistic run_d is used here as a unifying notation: it packages the ordinary case $d = 1$ and the ratio case $d = \lfloor m/n \rfloor$ into one theorem. No additional structural claim about intermediate sampling widths is needed for the explicit enumeration established here.

The automata-theoretic origin and the combinatorial proof play different roles in this branch. The completion-aware DPDA suggested that a weighted residual, rather than a particular stack representation, is the stable semantic quantity. Under the rational-path encoding this residual becomes the usual rank up to scale, and the low-residual band becomes the contact statistic of Theorem 76.4. No automata construction is needed after this bridge: forbidden-value compression, the weighted LGV determinant, and inclusion–exclusion provide the complete enumeration.

Chapter 78

Contact-Preserving Barrier Compression

Chapters 76–77 resolve the two unrestricted joint enumeration questions. The next step is to separate what was problem-specific from what is a reusable reduction principle. The answer is *contact-preserving barrier compression*.

The novelty boundary is deliberately narrow. Lindström–Gessel–Viennot determinants, Kreweras-type barrier determinants, order-preserving relabeling, and inclusion–exclusion are classical. What is used here as one reduction is the coupling

value deletion + induced barrier compression
+exact contact marking + presence/absence constraints

This coupling permits arbitrary required and forbidden values, exact contact sets, and simultaneous first-missing statistics without losing the original boundary-contact information. The theory from this point is self-contained combinatorics; the residual-to-rank bridge explains its discovery but is not a proof hypothesis.

Throughout this section let

$$\mathbf{b} = (b_1, \dots, b_n), \qquad 0 \le b_1 \le \cdots \le b_n,$$

and define

$$\mathcal{A}_{\mathbf{b}} = \{u \in \mathbb{Z}^n_{\ge 0} : 0 \le u_1 \le \cdots \le u_n,\ u_i \le b_i\}.$$

For $u \in \mathcal{A}_{\mathbf{b}}$, set

$$\mathrm{Val}(u) = \{u_1, \dots, u_n\}, \qquad C_{\mathbf{b}}(u) = \{i \in [n] : u_i = b_i\}.$$

Presence and absence concern values of the sequence; contacts concern row positions. Keeping these two kinds of information separate is the central point of the method. We use the convention $\binom{a}{b} = 0$ whenever $b < 0$ or $b > a$.

78.0.1 Deleting forbidden values

Let $F \subset \mathbb{Z}_{>0}$ be finite and define

$$\phi_F(t) = t - \#(F \cap [1, t]), \qquad c_i(F) = \phi_F(b_i), \qquad \epsilon_i(F) = \mathbf{1}_{\{b_i \notin F\}}. \tag{78.1}$$

Lemma 78.1 — Contact-preserving value compression.

The coordinatewise map

$$\Phi_F(u_1, \dots, u_n) = (\phi_F(u_1), \dots, \phi_F(u_n))$$

is a bijection

$$\{u \in \mathcal{A}_{\mathbf{b}} : \mathrm{Val}(u) \cap F = \varnothing\} \longrightarrow \mathcal{A}_{\mathbf{c}(F)}.$$

Moreover, if $b_i \notin F$, then

$$u_i = b_i \iff \phi_F(u_i) = c_i(F),$$

whereas if $b_i \in F$ then no F-avoiding sequence can contact the original barrier in row i.

Proof. The restriction of ϕ_F to $\mathbb{Z}_{\geq 0} \setminus F$ is the unique order-preserving bijection from the allowed nonnegative integers onto $\mathbb{Z}_{\geq 0}$. Hence an F-avoiding weakly increasing sequence remains weakly increasing after applying ϕ_F, and $u_i \leq b_i$ implies $\phi_F(u_i) \leq \phi_F(b_i) = c_i(F)$. Conversely, applying the inverse order isomorphism coordinatewise to a sequence below $\mathbf{c}(F)$ gives an F-avoiding weakly increasing sequence below $\mathbf{b}$. If $b_i \notin F$, injectivity on the allowed alphabet gives $\phi_F(u_i) = \phi_F(b_i)$ if and only if $u_i = b_i$. If $b_i \in F$, that original contact is forbidden. □

78.0.2 Multivariate contacts

Define

$$D_{\mathbf{b},F}(z_1, \dots, z_n) = \det_{1 \leq i,j \leq n} \left[\binom{c_i(F) + 1}{j - i + 1} + (z_i - 1)\epsilon_i(F) \binom{c_i(F)}{j - i} \right]. \tag{78.2}$$

Theorem 78.2 — Multivariate contact determinant.

For every finite $F \subset \mathbb{Z}_{>0}$,

$$\boxed{D_{\mathbf{b},F}(\mathbf{z}) = \sum_{\substack{u \in \mathcal{A}_{\mathbf{b}} \\ \mathrm{Val}(u) \cap F = \varnothing}} \prod_{i \in C_{\mathbf{b}}(u)} z_i.} \tag{78.3}$$

Proof. Write $c_i = c_i(F)$ and $\epsilon_i = \epsilon_i(F)$. Consider the directed acyclic lattice network with east steps $(1, 0)$ and northeast steps $(1, 1)$, sources

$$S_i = (-c_i - 1, i),$$

and targets

$$T_j = (0, j + 1).$$

A path from S_i to T_j has $c_i + 1$ horizontal steps and $j - i + 1$ northeast steps, so there are $\binom{c_i+1}{j-i+1}$ such paths. If $\epsilon_i = 1$, give weight z_i to the northeast edge leaving S_i and weight 1 to all other edges; if $\epsilon_i = 0$, give all initial edges weight 1. The number of S_i–T_j paths using the initial northeast edge is $\binom{c_i}{j-i}$, giving exactly the matrix entry in (78.2).

By the Lindström–Gessel–Viennot lemma [24], the determinant is the signed generating function of vertex-disjoint path families. Since c_i is nondecreasing, the source–target configuration is nonpermutable: an inversion of targets would reverse vertical order over a common horizontal interval and force an intersection. Thus only the identity matching contributes.

An identity path $\pi_i : S_i \to T_i$ has one northeast step and c_i east steps. Let e_i be the number of east steps before that northeast step and set $v_i = c_i - e_i$. Adjacent paths are vertex-disjoint exactly when $v_i \leq v_{i+1}$. Hence nonintersecting identity families are in bijection with

$$0 \leq v_1 \leq \cdots \leq v_n, \qquad v_i \leq c_i.$$

Moreover, π_i uses its initial northeast edge exactly when $v_i = c_i$. Lemma 78.1 identifies the weighted instances of these compressed contacts with the original contacts $u_i = b_i$. Therefore the family corresponding to u has weight $\prod_{i \in C_{\mathbf{b}}(u)} z_i$. □

For exact contacts define

$$B_{ij}^{F} = \epsilon_i(F)\binom{c_i(F)}{j-i}, \qquad A_{ij}^{F} = \binom{c_i(F)+1}{j-i+1} - B_{ij}^{F}. \tag{78.4}$$

For $J \subseteq [n]$, let $\mathsf{M}_{J}^{\mathbf{b},F}$ be the matrix whose ith row is B_i^F for $i \in J$ and A_i^F for $i \notin J$.

Corollary 78.3 — Exact contact-set determinant.

For every $J \subseteq [n]$,

$$\boxed{\#\{u \in \mathcal{A}_{\mathbf{b}} : \mathrm{Val}(u) \cap F = \varnothing,\ C_{\mathbf{b}}(u) = J\} = \det \mathsf{M}_{J}^{\mathbf{b},F}.}$$

Proof. The ith row in (78.2) is $A_i^F + z_i B_i^F$. Expand by row multilinearity and compare the coefficient of $\prod_{i\in J} z_i$ with Theorem 78.2. □

78.0.3 Presence, absence, and simultaneous first-missing statistics

Let $R, F \subset \mathbb{Z}_{>0}$ be finite and disjoint. Set

$$G_{\mathbf{b};R,F}(\mathbf{z}) = \sum_{T \subseteq R} (-1)^{|T|} D_{\mathbf{b},F\cup T}(\mathbf{z}). \tag{78.5}$$

Theorem 78.4 — Presence–absence contact calculus.

For finite disjoint $R, F \subset \mathbb{Z}_{>0}$,

$$\boxed{G_{\mathbf{b};R,F}(\mathbf{z}) = \sum_{\substack{u\in\mathcal{A}_{\mathbf{b}} \\ R\subseteq \mathrm{Val}(u),\ \mathrm{Val}(u)\cap F=\varnothing}} \prod_{i\in C_{\mathbf{b}}(u)} z_i.}$$

Consequently, for every $J \subseteq [n]$,

$$\boxed{\begin{aligned}\#\{u \in \mathcal{A}_{\mathbf{b}} : R \subseteq \mathrm{Val}(u),\ \mathrm{Val}(u) \cap F = \varnothing,\\ C_{\mathbf{b}}(u) = J\} = \sum_{T\subseteq R} (-1)^{|T|} \det \mathsf{M}_{J}^{\mathbf{b},F\cup T}\end{aligned}} \tag{78.6}$$

Proof. Among the objects avoiding F, require every $r \in R$ to occur. Inclusion–exclusion over the events $r \notin \mathrm{Val}(u)$ gives (78.5); Theorem 78.2 supplies the weighted determinant, and Corollary 78.3 gives (78.6). □

Let $s^{(1)}, \dots, s^{(q)}$ be injective sequences of positive integers and define

$$\mathrm{fm}_{s^{(h)}}(u) = \min\{a \geq 1 : s_a^{(h)} \notin \mathrm{Val}(u)\}.$$

For prescribed $a_1, \dots, a_q$, set

$$R(\mathbf{a}) = \bigcup_{h=1}^{q} \{s_1^{(h)}, \dots, s_{a_h-1}^{(h)}\}, \qquad F(\mathbf{a}) = \{s_{a_1}^{(1)}, \dots, s_{a_q}^{(q)}\}.$$

Corollary 78.5 — Simultaneous first-missing statistics.

If $R(\mathbf{a}) \cap F(\mathbf{a}) \neq \varnothing$, no sequence satisfies all prescribed first-missing conditions. Otherwise, for every $J \subseteq [n]$,

$$\boxed{\#\{u \in \mathcal{A}_{\mathbf{b}} : \mathrm{fm}_{s^{(h)}}(u) = a_h\ \forall h,\ C_{\mathbf{b}}(u) = J\} = \sum_{T\subseteq R(\mathbf{a})} (-1)^{|T|} \det \mathsf{M}_{J}^{\mathbf{b},F(\mathbf{a})\cup T}.}$$

Proof. Each condition $\mathrm{fm}_{s^{(h)}}(u) = a_h$ requires $s_1^{(h)}, \dots, s_{a_h-1}^{(h)}$ to be present and $s_{a_h}^{(h)}$ to be absent. Combine these conditions and apply Theorem 78.4. □

Remark 78.R1 — Exact-contact specialization.

The ordinary probe $s_a = a$ and the sampled probe $s_a = \delta a$ are immediate special cases. No rational-slope, Dyck-path, or automata assumption occurs anywhere in this section.

78.1 Fixed-composition rational Dyck paths

Assume henceforth that $m > n \geq 1$ and let

$$\alpha = (\alpha_1, \ldots, \alpha_d) \vDash n, \qquad A_0 = 0, \quad A_i = \alpha_1 + \cdots + \alpha_i.$$

Let $\mathcal{D}_{m,\alpha}$ be the set of $m \times n$ rational Dyck paths with ordered vertical-run composition α.

78.1.1 Component-column coordinates and returns

For $P \in \mathcal{D}_{m,\alpha}$, let x_i be the horizontal coordinate of the initial vertex of the ith vertical component and set

$$\beta_i = \left\lfloor \frac{mA_{i-1}}{n} \right\rfloor. \tag{78.7}$$

Lemma 78.6 — Fixed-composition component encoding.
The map $P \mapsto (x_1, \ldots, x_d)$ is a bijection from $\mathcal{D}_{m,\alpha}$ onto

$$\mathcal{X}_{m,\alpha} = \{(x_1, \ldots, x_d) : 0 = x_1 < x_2 < \cdots < x_d,\ x_i \leq \beta_i\}.$$

Proof. The ith component starts at (x_i, A_{i-1}). The Dyck condition gives $A_{i-1} \geq nx_i/m$, hence $x_i \leq \beta_i$. Distinct vertical components are separated by east steps, so their columns are strictly increasing. Conversely, from such a column sequence reconstruct

$$N^{\alpha_1} E^{x_2 - x_1} N^{\alpha_2} \cdots E^{x_d - x_{d-1}} N^{\alpha_d} E^{m - x_d}.$$

Each horizontal segment stays above the diagonal because its right endpoint satisfies the corresponding bound $x_{i+1} \leq \lfloor mA_i/n \rfloor$. □

Let $\mathrm{RetComp}(P)$ be the set of component indices whose first north step is a return.

Lemma 78.7 — Return components are barrier contacts.
For $P \leftrightarrow (x_1, \ldots, x_d)$,

$$\boxed{i \in \mathrm{RetComp}(P) \iff x_i = \beta_i.}$$

Consequently $\mathrm{ret}(P) = |\,\mathrm{RetComp}(P)|$.

Proof. At the start of component i, the rank is

$$R_i = mA_{i-1} - nx_i.$$

Write $mA_{i-1} = n\beta_i + r_i$ with $0 \leq r_i < n$. Then

$$R_i = r_i + n(\beta_i - x_i).$$

Since $x_i \leq \beta_i$, one has $R_i < n$ exactly when $x_i = \beta_i$. This is precisely the return condition of Dai–Fu–Qiu [23]. A later north step in the same component has rank at least $R_i + m \geq n$, so a component contributes at most one return. □

Corollary 78.8 — Exact north-step return positions.
If $J = \mathrm{RetComp}(P)$, then

$$\boxed{\mathrm{ReturnSet}(P) = \{A_{i-1} + 1 : i \in J\}.}$$

78.1.2 Forbidden columns and the effective barrier

For finite $F \subset \mathbb{Z}_{>0}$, set

$$c_i(F) = \beta_i - \#(F \cap [1, \beta_i]), \qquad g_i(F) = c_i(F) - (i-1),$$

and define the suffix-minimum effective barrier

$$\boxed{h_i(F) = \min_{j \geq i} g_j(F).} \tag{78.8}$$

Then $h_1(F) \leq \cdots \leq h_d(F)$.

Lemma 78.9 — Effective-barrier regularization.

Let

$$\mathcal{X}_{m,\alpha}(F) = \{x \in \mathcal{X}_{m,\alpha} : \{x_1, \ldots, x_d\} \cap F = \varnothing\}.$$

If $h_1(F) < 0$, the class is empty. If $h_1(F) \geq 0$, the maps

$$y_i = \phi_F(x_i), \qquad v_i = y_i - (i-1)$$

give a bijection

$$\mathcal{X}_{m,\alpha}(F) \longleftrightarrow \{0 \leq v_1 \leq \cdots \leq v_d : v_i \leq h_i(F)\}.$$

Proof. Because ϕ_F is strictly increasing on the allowed alphabet, strict x-inequalities are equivalent to strict y-inequalities. After the shift $v_i = y_i - (i-1)$, strictness becomes weak monotonicity and the raw upper bound is $v_i \leq g_i(F)$. For a weakly increasing sequence, the full system $v_j \leq g_j(F)$ is equivalent to $v_i \leq \min_{j \geq i} g_j(F) = h_i(F)$ for every i. Since $\beta_1 = 0$, feasibility forces $h_1(F) = 0$; if $h_1(F) < 0$ no nonnegative sequence exists. □

Define the surviving-contact indicator

$$\eta_i(F) = \mathbf{1}_{\{\beta_i \notin F,\ h_i(F) = g_i(F)\}}. \tag{78.9}$$

Lemma 78.10 — Surviving return contacts.

Under Lemma 78.9,

$$\boxed{x_i = \beta_i \iff \eta_i(F) = 1 \text{ and } v_i = h_i(F).}$$

Proof. If $x_i = \beta_i$, then $\beta_i \notin F$ and $v_i = c_i(F) - (i-1) = g_i(F)$. Since $v_i \leq h_i(F) \leq g_i(F)$, equality is forced. Conversely, if $\eta_i(F) = 1$ and $v_i = h_i(F) = g_i(F)$, then $y_i = c_i(F) = \phi_F(\beta_i)$; injectivity on the allowed alphabet gives $x_i = \beta_i$. □

Define

$$D_{\alpha,F}(z_1, \ldots, z_d) = \det_{1 \leq i,j \leq d} \left[\binom{h_i(F)+1}{j-i+1} + (z_i - 1)\eta_i(F) \binom{h_i(F)}{j-i} \right] \tag{78.10}$$

when $h_1(F) \geq 0$, and set it equal to zero otherwise. Also define

$$B_{ij}^{\alpha,F} = \eta_i(F) \binom{h_i(F)}{j-i}, \qquad A_{ij}^{\alpha,F} = \binom{h_i(F)+1}{j-i+1} - B_{ij}^{\alpha,F},$$

and let $M_J^{\alpha,F}$ use row $B_i^{\alpha,F}$ when $i \in J$ and $A_i^{\alpha,F}$ otherwise.

Theorem 78.11 — Fixed-composition forbidden-column determinant.

For finite $F \subset \mathbb{Z}_{>0}$,

$$\boxed{D_{\alpha,F}(\mathbf{z}) = \sum_{\substack{P \in \mathcal{D}_{m,\alpha} \\ \{x_i(P)\} \cap F = \varnothing}} \prod_{i \in \mathrm{RetComp}(P)} z_i.}$$

Consequently,

$$\boxed{\#\{P \in \mathcal{D}_{m,\alpha} : \{x_i(P)\} \cap F = \varnothing,\ \mathrm{RetComp}(P) = J\} = \det M_J^{\alpha,F}.} \tag{78.11}$$

Proof. Lemma 78.9 gives a weakly increasing sequence below $\mathbf{h}(F)$. Apply Theorem 78.2 to that effective barrier, but mark row i only when $\eta_i(F) = 1$. Lemma 78.10 identifies exactly those marked effective contacts with the original return components. The exact-set formula follows by row multilinearity. □

Remark 78.R2 — Effective-barrier interpretation.

For $F = \varnothing$, $g_i = \beta_i - (i-1)$ is already nondecreasing because $m > n$ and every part of α is positive. Thus $h_i = g_i$ and $\eta_i = 1$. The suffix-minimum regularization is needed only after forbidden values are deleted.

Chapter 79

Four Composition-Refined Enumeration Problems

With the general contact calculus and the fixed-composition effective barrier in place, the framework yields four distinct rational-Dyck enumeration results at fixed ordered vertical-run composition α:

1. ordinary run together with returns;
2. ratio-run together with the exact return components;
3. simultaneous ordinary run, ratio-run, and exact returns;
4. fixed ratio-signature, where every class on the associated skew diagonal has one common determinant cardinality.

The fourth result also yields a positive complete-homogeneous expansion of the fixed-signature polynomial. The fixed-signature discussion retains an important scope correction: a ratio-signature is *not* generally an initial component prefix; that prefix interpretation becomes valid only after passage to the minimal-return representative used in the proof.

79.1 Explicit fixed-composition run–return enumeration

Dai–Fu–Qiu define a statistic $\mathrm{rr}(\alpha, m)$ depending only on the composition and m; if $I_1(\alpha)$ is the length of the maximal initial string of parts equal to 1, then [23]

$$r_\alpha := \mathrm{rr}(\alpha, m) = \min\left\{ I_1(\alpha) + 1, \left\lceil \frac{n}{m-n} \right\rceil \right\}. \tag{79.1}$$

Every path of composition α has $\mathrm{run}(P), \mathrm{ret}(P) \geq r_\alpha$. Their fixed-composition bijections imply that for $K = k + \ell - r_\alpha$,

$$|\mathcal{D}_{m,\alpha}^{(k,\ell)}| = |\mathcal{D}_{m,\alpha}^{(K,r_\alpha)}|, \tag{79.2}$$

whenever the class is nonempty [23, Theorem 3.2 and Corollary 3.5].

For $K < d$, put $p = d - K$ and define

$$B_i^{(\alpha,K)} = \beta_{K+i} - (K+i) - 1 = \left\lfloor \frac{mA_{K+i-1}}{n} \right\rfloor - (K+i) - 1, \qquad 1 \leq i \leq p. \tag{79.3}$$

Lemma 79.1 — Minimal-return suffix barrier.

For a path in $\mathcal{D}_{m,\alpha}^{(K,r_\alpha)}$, the first K component columns are forced:

$$x_i = i - 1 \qquad (1 \leq i \leq K).$$

If $K < d$, then after the shift

$$v_i = x_{K+i} - (K + i),$$

the free suffix is in bijection with

$$0 \le v_1 \le \cdots \le v_p, \qquad v_i \le B_i^{(\alpha,K)}.$$

Moreover $B_1^{(\alpha,K)} \le \cdots \le B_p^{(\alpha,K)}$.

Proof. For the minimal-return representative, Dai–Fu–Qiu use $M_1 \cdots M_K ET$, where $M_i = N^{\alpha_i} E$ [23, Corollary 3.5]. Thus the first K component columns are $0, 1, \ldots, K-1$, while the forced separator gives $x_{K+1} \ge K + 1$. Since the suffix contributes no further returns, Lemma 78.7 gives $x_i \le \beta_i - 1$ for $i > K$. The stated shift yields the weak sequence and its barrier. Finally

$$B_{i+1}^{(\alpha,K)} - B_i^{(\alpha,K)} = \beta_{K+i+1} - \beta_{K+i} - 1 \ge 0,$$

because $m > n$ and $\alpha_{K+i} \ge 1$. □

Theorem 79.2 — Explicit composition-refined run–return enumeration.
Let $k, \ell \ge 1$ and set $K = k + \ell - r_\alpha$. If any of

$$k < r_\alpha, \qquad \ell < r_\alpha, \qquad K > d$$

holds, the class is empty. If $K < d$ and $B_1^{(\alpha,K)} < 0$, it is likewise empty. Otherwise,

$$\boxed{|\mathcal{D}_{m,\alpha}^{(k,\ell)}| = \det_{1 \le i,j \le d-K} \left[\binom{B_i^{(\alpha,K)} + 1}{j - i + 1} \right].} \tag{79.4}$$

For $K = d$, the empty determinant is interpreted as 1.

Proof. Use the Dai–Fu–Qiu reduction (79.2), then apply the classical bounded weak-sequence determinant to the suffix barrier in Lemma 79.1. □

Set

$$C_{\alpha,K} = \det_{1 \le i,j \le d-K} \left[\binom{B_i^{(\alpha,K)} + 1}{j - i + 1} \right].$$

Corollary 79.3 — Positive complete-homogeneous expansion.
With

$$F_{m,\alpha}(u, v) = \sum_{P \in \mathcal{D}_{m,\alpha}} u^{\mathrm{run}(P)} v^{\mathrm{ret}(P)},$$

one has

$$\boxed{F_{m,\alpha}(u, v) = (uv)^{r_\alpha} \sum_{K=r_\alpha}^{d} C_{\alpha,K} h_{K-r_\alpha}(u, v),} \tag{79.5}$$

where $h_j(u, v) = u^j + u^{j-1}v + \cdots + v^j$.

Proof. For fixed K, the feasible statistic pairs are $(r_\alpha + a, K - a)$ for $0 \le a \le K - r_\alpha$, and Theorem 79.2 gives the common cardinality $C_{\alpha,K}$. Summing their monomials gives $(uv)^{r_\alpha} h_{K-r_\alpha}(u, v)$. □

Example 79.E1 — Example.

Let $(m,n) = (11,5)$ and $\alpha = (2,1,1,1)$. Then $\beta = (0,4,6,8)$ and $r_\alpha = 1$. For $(\mathrm{run},\mathrm{ret}) = (1,2)$, $K = 2$ and the suffix barrier is $(2,3)$. Hence

$$|\mathcal{D}^{(1,2)}_{11,(2,1,1,1)}| = \det\begin{pmatrix} 3 & 3 \\ 1 & 4 \end{pmatrix} = 9.$$

Remark 79.R1 — Ordinary-run scope.

The determinant in Theorem 79.2 is classical. The new enumerative step is the reduction from a fixed ordered composition and prescribed $(\mathrm{run},\mathrm{ret})$ to a single bounded monotone suffix. In particular, it makes the cardinality in the partial-path characterization of Dai–Fu–Qiu Corollary 3.5 explicit.

79.2 Composition-refined ratio-run and returns

Set

$$\kappa = \left\lfloor \frac{m}{n} \right\rfloor.$$

For $P \in \mathcal{D}_{m,\alpha}$, let $X(P) = \{x_1(P), \dots, x_d(P)\}$.

Lemma 79.4 — Coarea values and component columns.

The set of distinct north-step coarea values of P is $X(P)$. Consequently

$$\boxed{\widetilde{\mathrm{run}}(P) = \min\{a \geq 1 : a\kappa \notin X(P)\}.} \tag{79.6}$$

Proof. Every north step in one vertical component has the same horizontal coordinate, and every component contains at least one north step. Thus the distinct coarea values are exactly the component columns. The second statement is the definition of ratio-run [23]. □

For $a \geq 1$, define

$$R_{\kappa,a} = \{\kappa, 2\kappa, \dots, (a-1)\kappa\}, \qquad F_{\kappa,a} = \{a\kappa\}.$$

Then $\widetilde{\mathrm{run}}(P) = a$ exactly when $R_{\kappa,a} \subseteq X(P)$ and $X(P) \cap F_{\kappa,a} = \varnothing$.

Theorem 79.5 — Composition-refined ratio-run with exact returns.

For every $a \geq 1$ and $J \subseteq [d]$,

$$\boxed{\begin{aligned} &\#\{P \in \mathcal{D}_{m,\alpha} : \widetilde{\mathrm{run}}(P) = a,\ \mathrm{RetComp}(P) = J\} \\ &\qquad = \sum_{T \subseteq R_{\kappa,a}} (-1)^{|T|} \det M_J^{\alpha, F_{\kappa,a} \cup T} \end{aligned}} \tag{79.7}$$

Proof. The ratio-run condition is exactly the presence–absence condition above. Inclusion–exclusion over the required values, followed by Theorem 78.11, gives the formula. □

Corollary 79.6 — Composition-refined ratio-run–return enumeration.

For $a, b \geq 1$,

$$\boxed{\#\{P \in \mathcal{D}_{m,\alpha} : \widetilde{\mathrm{run}}(P) = a,\ \mathrm{ret}(P) = b\} = \sum_{\substack{J \subseteq [d] \\ |J| = b}} \sum_{T \subseteq R_{\kappa,a}} (-1)^{|T|} \det M_J^{\alpha, F_{\kappa,a} \cup T}.}$$

Example 79.E2 — Example.

For $(m,n)=(11,5)$ and $\alpha=(2,1,1,1)$, one has $\beta=(0,4,6,8)$ and $\kappa=2$. Prescribe $\widetilde{\mathrm{run}}=2$ and $J=\{1,4\}$. Then $R_{2,2}=\{2\}$ and $F_{2,2}=\{4\}$. The two inclusion–exclusion determinants are

$$\det M^{\alpha,\{4\}}_{\{1,4\}}=6, \qquad \det M^{\alpha,\{2,4\}}_{\{1,4\}}=3,$$

so the required number is $6-3=3$. The three component-column sequences are

$$(0,1,2,8), \qquad (0,2,3,8), \qquad (0,2,5,8).$$

79.3 Simultaneous run, ratio-run, and exact returns

Lemma 79.7 — Ordinary run as a first-missing component column.

For every $P\in\mathcal{D}_{m,\alpha}$,

$$\boxed{\mathrm{run}(P)=\min\{a\geq 1 : a\notin X(P)\}.}$$

Proof. By Lemma 79.4, $X(P)$ is the set of distinct coarea values. The ordinary run statistic is the first missing positive coarea value [23]. □

For positive integers a,c, define

$$R_{a,c}=\{1,\dots,a-1\}\cup\{\kappa,2\kappa,\dots,(c-1)\kappa\}, \qquad F_{a,c}=\{a,c\kappa\}. \tag{79.8}$$

Theorem 79.8 — Simultaneous run–ratio-run–return-set enumeration.

For $J\subseteq[d]$, if $R_{a,c}\cap F_{a,c}\neq\varnothing$, then

$$\#\{P\in\mathcal{D}_{m,\alpha} : \mathrm{run}(P)=a,\ \widetilde{\mathrm{run}}(P)=c,\ \mathrm{RetComp}(P)=J\}=0.$$

Otherwise,

$$\boxed{\begin{aligned}&\#\{P\in\mathcal{D}_{m,\alpha} : \mathrm{run}(P)=a,\ \widetilde{\mathrm{run}}(P)=c,\ \mathrm{RetComp}(P)=J\}\\ &\qquad=\sum_{T\subseteq R_{a,c}}(-1)^{|T|}\det M_J^{\alpha,F_{a,c}\cup T}\end{aligned}} \tag{79.9}$$

Proof. The two first-missing conditions together are precisely $R_{a,c}\subseteq X(P)$ and $X(P)\cap F_{a,c}=\varnothing$. If the sets intersect, the requirements are inconsistent. Otherwise apply inclusion–exclusion and the exact-return determinant (78.11). □

Corollary 79.9 — Exact north-step return set.

If $S\subseteq[n]$ is not of the form $S=\{A_{i-1}+1 : i\in J\}$ for some $J\subseteq[d]$, then no path of composition α has $\mathrm{ReturnSet}(P)=S$. For such a set S and disjoint $R_{a,c},F_{a,c}$, the number of paths with

$$\mathrm{run}(P)=a, \qquad \widetilde{\mathrm{run}}(P)=c, \qquad \mathrm{ReturnSet}(P)=S$$

is the right-hand side of (79.9).

Example 79.E3 — Simultaneous run and ratio-run.

Again let $(m,n)=(11,5)$ and $\alpha=(2,1,1,1)$, so $\beta=(0,4,6,8)$ and $\kappa=2$. Prescribe $\mathrm{run}=3$ and $\widetilde{\mathrm{run}}=2$. Then

$$R_{3,2}=\{1,2\}, \qquad F_{3,2}=\{3,4\}.$$

The admissible component-column sequences are

$$(0,1,2,5),\ (0,1,2,6),\ (0,1,2,7),\ (0,1,2,8).$$

The first three have $\mathrm{RetComp}=\{1\}$, while the last has $\mathrm{RetComp}=\{1,4\}$. Hence the corresponding exact counts are 3 and 1.

Remark 79.R2 — Automatic compatibility detection.

The condition $R_{a,c}\cap F_{a,c}=\varnothing$ automatically detects incompatibilities. For example, if $\kappa=2$, then $\mathrm{run}=2$ requires column 2 to be absent while $\widetilde{\mathrm{run}}=2$ requires it to be present, so the simultaneous class is empty.

79.4 Explicit enumeration at fixed ratio-signature

This section uses the first ratio-signature $\mathrm{sign}_1(P)$ of Dai–Fu–Qiu [23]. Their auxiliary path P_1 is formed by scanning P and collecting each κ-east-step block contributing to ratio-run or return. Thus P_1 is generally a shuffle-selected subpath rather than an initial segment. Let

$$\mathcal{D}_{m,\alpha,s} = \{P \in \mathcal{D}_{m,\alpha} : \mathrm{sign}_1(P) = s\},$$

and assume this class is nonempty. Write

$$r_{\alpha,s} = \widetilde{\mathrm{rr}}(\alpha, m, s),$$

for the minimum value attained by either $\widetilde{\mathrm{run}}$ or ret on this fixed-signature class in the notation of Dai–Fu–Qiu.

Remark 79.R3 — The signature is not generally a prefix.

No argument below identifies a general P_1 with a prefix. The prefix interpretation arises only after passing, via the fixed-signature bijections of Dai–Fu–Qiu Theorem 3.7, to the representative with minimal return.

Lemma 79.10 — Minimal-return prefix collapse.

Suppose $P \in \mathcal{D}_{m,\alpha,s}$ has $\widetilde{\mathrm{run}}(P) = a$ and $\mathrm{ret}(P) = b$. Set

$$h = a + b - r_{\alpha,s}.$$

The fixed-signature bijections of Dai–Fu–Qiu reduce the class to a representative Q with

$$\widetilde{\mathrm{run}}(Q) = h, \qquad \mathrm{ret}(Q) = r_{\alpha,s}, \qquad \mathrm{sign}_1(Q) = s.$$

For this minimal-return representative, Q_1 is exactly the initial segment occupying the first $h\kappa$ east-step columns. Consequently

$$\boxed{|s| = h\kappa.} \tag{79.10}$$

Proof. Dai–Fu–Qiu Theorem 3.7 constructs fixed-signature bijections along the skew diagonal and states that the corresponding minimal value is $r_{\alpha,s}$ [23]. In the minimal-return representative, every return block already belongs to the initial family of blocks contributing simultaneously to ratio-run and return; there are no additional return-only blocks. The ratio-run value h contributes the first h sampled κ-blocks, so the union defining Q_1 is precisely the initial $h\kappa$ east-step segment. Since s is the ordinary east-step signature of Q_1, its length is $h\kappa$. □

Corollary 79.11 — Fixed-signature skew diagonal.

If $\mathcal{D}_{m,\alpha,s} \neq \varnothing$, then $\kappa \mid |s|$. With

$$h_s = \frac{|s|}{\kappa},$$

every nonempty fixed-signature statistic class satisfies

$$\boxed{\widetilde{\mathrm{run}}(P) + \mathrm{ret}(P) = h_s + r_{\alpha,s}.}$$

Let the positions of the 1's in s be

$$1 \le p_1 < \cdots < p_t \le |s|, \qquad t = |s|_1.$$

Lemma 79.12 — Prefix components determined by the signature.

For a minimal-return representative Q as above, the first t component columns are forced:

$$\boxed{x_i = p_i - 1 \qquad (1 \le i \le t).}$$

Proof. By Lemma 79.10, s is the ordinary east-step signature of the first $|s|$ east steps of Q. An east step has signature 1 exactly when it immediately follows a north step. The east step in signature position p_i therefore closes the ith vertical component, whose horizontal coordinate is $p_i - 1$. □

Corollary 79.13 — Reading the minimal return from the signature.

For an admissible signature,

$$\boxed{r_{\alpha,s} = \#\{i \in [t] : p_i - 1 = \beta_i\}.}$$

Proof. In the minimal-return representative, all returns occur among the forced prefix components. Apply Lemma 78.7 and Lemma 79.12. □

Lemma 79.14 — Forced separator and free suffix.

For a minimal-return representative Q,

$$x_{t+1} \geq h_s\kappa + 1 = |s| + 1$$

when $t < d$, and

$$x_{t+j} \leq \beta_{t+j} - 1 \qquad (1 \leq j \leq d - t).$$

Proof. Because $\widetilde{\mathrm{run}}(Q) = h_s$, the sampled column $h_s\kappa$ is missing, so the next component after the prefix starts at least one column later. Since $\mathrm{ret}(Q) = r_{\alpha,s}$ and the prefix already accounts for all returns, no suffix component is a return; Lemma 78.7 gives the strict upper bound. □

Put $p = d - t$ and define

$$c_j^{(\alpha,s)} = \beta_{t+j} - |s| - j - 1, \qquad 1 \leq j \leq p. \tag{79.11}$$

Then $c_1^{(\alpha,s)} \leq \cdots \leq c_p^{(\alpha,s)}$ because $\beta_{i+1} - \beta_i \geq 1$.

Theorem 79.15 — Explicit fixed-ratio-signature cardinality.

Let s be admissible and use the notation above. Then

$$\boxed{\begin{aligned} C_{m,\alpha,s} &:= |\{P \in \mathcal{D}_{m,\alpha,s} : \widetilde{\mathrm{run}}(P) = h_s,\ \mathrm{ret}(P) = r_{\alpha,s}\}| \\ &= \det_{1 \leq i,j \leq d-t} \left[\binom{c_i^{(\alpha,s)} + 1}{j - i + 1} \right] \end{aligned}} \tag{79.12}$$

The empty determinant is 1.

Proof. The signature fixes the prefix columns by Lemma 79.12. For the free suffix set

$$v_j = x_{t+j} - (|s| + j).$$

Lemma 79.14 gives

$$0 \leq v_1 \leq \cdots \leq v_p, \qquad v_j \leq c_j^{(\alpha,s)}.$$

The barrier is nondecreasing, so the classical bounded-sequence determinant gives (79.12). □

Corollary 79.16 — Common cardinality on the fixed-signature skew diagonal.

For every $0 \leq j \leq h_s - r_{\alpha,s}$,

$$\boxed{|\{P \in \mathcal{D}_{m,\alpha,s} : \widetilde{\mathrm{run}}(P) = r_{\alpha,s} + j,\ \mathrm{ret}(P) = h_s - j\}| = C_{m,\alpha,s}.} \tag{79.13}$$

All other fixed-signature statistic classes are empty.

Proof. Apply the fixed-signature skew-diagonal bijection of Dai–Fu–Qiu Theorem 3.7 to the minimal-return class, then use Corollary 79.11 to exhaust the possible pairs. □

Corollary 79.17 — Explicit fixed-signature polynomial.

$$\boxed{\sum_{P\in\mathcal{D}_{m,\alpha,s}} u^{\widetilde{\mathrm{run}}(P)}v^{\mathrm{ret}(P)} = C_{m,\alpha,s}(uv)^{r_{\alpha,s}}h_{h_s-r_{\alpha,s}}(u,v).} \tag{79.14}$$

Example 79.E4 — The fixed signature 11111110**.**

Let

$$(m,n)=(23,11),\qquad \alpha=(1,1,2,1,1,1,1,1,1,1),\qquad s=11111110.$$

Then $\kappa=2$, $|s|=8$, $h_s=4$, and $t=7$. The component barrier is

$$\beta=(0,2,4,8,10,12,14,16,18,20),$$

and the signature forces $(x_1,\dots,x_7)=(0,1,2,3,4,5,6)$. Hence $r_{\alpha,s}=1$. The free suffix barrier is $(6,7,8)$, so

$$C_{23,\alpha,s}=\det\begin{pmatrix}7 & 21 & 35\\ 1 & 8 & 28\\ 0 & 1 & 9\end{pmatrix}=154.$$

Therefore each of the four classes

$$(\widetilde{\mathrm{run}},\mathrm{ret})=(1,4),(2,3),(3,2),(4,1)$$

has cardinality 154, and the fixed-signature polynomial is $154\,uv\,h_3(u,v)$.

Chapter 80

Arbitrary Boundaries and Labeled Applications

The framework is not tied to one rational slope. Two application directions of the barrier-compression development are retained here in full. First, the contact determinant works below an arbitrary northeast boundary and therefore specializes to partial Dyck paths. Second, fixed-composition path counts transfer to column-increasing rational parking-function labelings by the standard multinomial labeling factor. These applications are mathematically separate from the four core rational-Dyck specializations and are kept visible as such.

80.1 Paths below an arbitrary fixed boundary

Let B be a northeast lattice path whose north-step starting columns are

$$\mathbf{b} = (b_1, \dots, b_n), \qquad b_1 \le \cdots \le b_n.$$

A path P lying weakly northwest of B has a north-step column sequence $u(P) \in \mathcal{A}_{\mathbf{b}}$. Thus Section 78 applies verbatim.

Corollary 80.1 — Arbitrary-boundary presence and exact contacts.
Let $R, F \subset \mathbb{Z}_{>0}$ be finite and disjoint. For $J \subseteq [n]$,

$$\boxed{\begin{aligned}&\#\{P \le B : R \subseteq \mathrm{Val}(P),\ \mathrm{Val}(P) \cap F = \varnothing,\ C_B(P) = J\} \\ &\qquad = \sum_{T \subseteq R} (-1)^{|T|} \det \mathsf{M}_J^{\mathbf{b}, F \cup T}\end{aligned}}$$

80.1.1 Partial Dyck paths

Let $0 \le m \le n$ and let $\mathcal{B}_{m,n}$ be the paths from $(0,0)$ to (m,n) remaining weakly above $y = x$. Their north-step barrier is

$$\boxed{b_i = \min(i-1, m).}$$

For $i \le m+1$, contact $u_i = b_i = i-1$ is a diagonal contact; for $i > m+1$, contact $u_i = b_i = m$ is contact with the terminal vertical wall. The multivariate variables therefore separate two geometrically different kinds of boundary contact without changing the compression argument.

Define

$$\mathrm{Ret}_\Delta(P) = \{i \in [m+1] : u_i = i-1\}, \qquad \mathrm{ret}_\Delta(P) = |\,\mathrm{Ret}_\Delta(P)|.$$

Specializing $z_i = z$ for $i \le m+1$ and $z_i = 1$ afterward gives a polynomial $D^{\mathrm{par}}_{m,n;F}(z)$ satisfying

$$D^{\mathrm{par}}_{m,n;F}(z) = \sum_{\substack{P \in \mathcal{B}_{m,n} \\ \mathrm{Val}(P) \cap F = \varnothing}} z^{\mathrm{ret}_\Delta(P)}.$$

If

$$\mathrm{fm}(P) = \min\{a \ge 1 : a \notin \mathrm{Val}(P)\},$$

then $\mathrm{fm}(P) = a$ is the required/forbidden condition $R_a = \{1, \dots, a-1\}$ and $F_a = \{a\}$. Hence the framework gives both the return-number polynomial and exact diagonal-return-set determinant sums for partial Dyck paths.

80.2 Column-increasing rational parking-function labelings

Classical rational parking functions are commonly represented as rational Dyck paths whose north steps carry labels increasing within each column; see Armstrong, Loehr, and Warrington [22]. For the present application, we use the same column-increasing labeling convention for the $m \times n$ paths considered here, without requiring coprimality. The path statistics run, $\widetilde{\mathrm{run}}$, and ret are inherited from the underlying path.

Lemma 80.2 — Labelings of a fixed composition.

If $P \in \mathcal{D}_{m,\alpha}$, then the number of column-increasing labelings of its n north steps by $1, \dots, n$ is

$$\boxed{L(\alpha) = \frac{n!}{\alpha_1! \cdots \alpha_d!}.}$$

Proof. Choose which α_i labels occupy each vertical component. Once those sets are chosen, the increasing condition determines their within-column order uniquely. □

Define the fixed-composition labeled run–return polynomial

$$\mathcal{P}_{m,\alpha}(u,v) = \sum_{\pi:\, \mathrm{comp}(\underline{\pi}) = \alpha} u^{\mathrm{run}(\pi)} v^{\mathrm{ret}(\pi)},$$

where $\underline{\pi}$ is the underlying path.

Theorem 80.3 — Composition-refined labeled run–return expansion.

$$\boxed{\mathcal{P}_{m,\alpha}(u,v) = \frac{n!}{\alpha_1! \cdots \alpha_d!} (uv)^{r_\alpha} \sum_{K=r_\alpha}^{d} C_{\alpha,K} h_{K-r_\alpha}(u,v).} \tag{80.1}$$

Proof. Every underlying path of composition α has the same multiplicity $L(\alpha)$ by Lemma 80.2. Multiply Corollary 79.3 by this factor. □

Summing over all compositions gives an explicit labeled polynomial

$$\boxed{\mathcal{P}_{m,n}(u,v) = \sum_{\alpha \vDash n} \frac{n!}{\prod_i \alpha_i!} (uv)^{r_\alpha} \sum_{K=r_\alpha}^{\ell(\alpha)} C_{\alpha,K} h_{K-r_\alpha}(u,v).} \tag{80.2}$$

For ratio-run, let $\Sigma_{m,\alpha}$ be the admissible first ratio-signatures. Corollary 79.17 similarly gives

$$\boxed{\widetilde{\mathcal{P}}_{m,\alpha}(u,v) = \frac{n!}{\prod_i \alpha_i!} \sum_{s \in \Sigma_{m,\alpha}} C_{m,\alpha,s} (uv)^{r_{\alpha,s}} h_{h_s - r_{\alpha,s}}(u,v).} \tag{80.3}$$

Thus both labeled distributions have manifest positive complete-homogeneous expansions once the determinant coefficients are fixed.

Remark 80.R1 — Labeling scope.

No symmetric-function coefficient claim is needed here. The application uses only the standard column-increasing labeling rule and the fact that the statistics under consideration depend on the underlying path.

80.3 Computational validation

All results in this branch are established theoretically. As an independent check, we implemented exhaustive enumeration of the underlying bounded sequences and fixed-composition rational Dyck paths and compared the cardinalities with the determinant formulas.

For the general exact-contact theorem, every nondecreasing barrier of length at most 5 with entries in $\{0, 1, 2, 3, 4\}$ was tested, together with every forbidden subset of $\{1, 2, 3, 4\}$ and every exact contact set. This produced 88,032 determinant–enumeration comparisons.

For the rational-path specializations, the ordinary fixed-composition formula was tested on all $m > n$ with $n \le 7$ and $m \le 12$, every composition of n, and every tested run–return class: 755 parameter instances and 9,983 statistic queries. The ratio-run formula with exact return components was tested on 628 parameter instances and 54,664 exact-return queries. The simultaneous run–ratio-run formula was tested on 501 parameter instances and 246,392 simultaneous queries. No discrepancy was found.

The fixed-ratio-signature construction was tested independently by forming sign_1 directly from the selected κ-column blocks in the Dai–Fu–Qiu definition, rather than by using the determinant reduction. The test covered 2,357 fixed-signature classes over 12,081 underlying paths, with no mismatch.

As a larger stress test, for

$$(m, n) = (23, 11), \qquad \alpha = (1, 1, 2, 1, 1, 1, 1, 1, 1, 1),$$

all 63,920 fixed-composition paths were enumerated. Exactly 616 have first ratio-signature 11111110, split into the four classes

$$(1, 4),\ (2, 3),\ (3, 2),\ (4, 1)$$

with exactly 154 paths in each, agreeing with Example 79.E4. No mismatch occurred in any exhaustive test.

80.4 Conclusion

We developed a contact-preserving barrier-compression framework for refined lattice-path enumeration. The method separates value constraints, upper-bound geometry, and boundary contacts. Deleting forbidden values produces an order-preserving compression of the alphabet and the barrier, while a multivariate Lindström–Gessel–Viennot determinant retains the surviving contact positions. Inclusion–exclusion then incorporates arbitrary required values and simultaneous first-missing statistics.

The framework arose from a concrete arithmetic route. A completion-aware deterministic pushdown automaton for the 2:1 balance relation led to the scalar residual $\rho_{p,q} = qN_1 - pN_0$. For $m = gq$ and $n = gp$, the identity $R = g\rho_{p,q}$ converted the low-residual condition into the rational-path return condition, which in coarea coordinates became an upper-barrier contact. This branch isolates the resulting compression mechanism from that discovery route and develops it as a self-contained enumerative method.

For rational Dyck paths with fixed ordered vertical-run composition, the method gives explicit formulas at several levels of refinement. Ordinary run–return classes reduce to one determinant. Ratio-run is handled by sampled presence–absence constraints and exact

return-component determinants. Ordinary run and ratio-run can be prescribed simultaneously with exact return positions. At fixed ratio-signature, the minimal-return representative converts the signature data into a forced prefix and a bounded free suffix, giving a single determinant for the common cardinality on the associated skew diagonal. The same framework applies to arbitrary northeast boundaries and partial Dyck paths, while standard column-increasing labeling transfers the fixed-composition formulas to labeled rational parking-function enumerations.

All determinant identities are proved combinatorially, and independent exhaustive computation verified the general framework and the four rational-path specializations over broad finite parameter ranges. The method can therefore be summarized as

$$\boxed{\begin{array}{c}\text{pushdown residual} \longrightarrow \text{rank} \longrightarrow \text{boundary contact} \\ \longrightarrow \text{contact-preserving compression} \longrightarrow \text{explicit determinants}\end{array}}$$

Chapter 81

Canonical Completion Profiles and Farey Unit Lifts

Research branch — Farey Arithmetic of Shortest Completion. *This branch develops the arithmetic of residual-indexed shortest-completion minimizers, the cross-parameter Ordered Farey Recursion, and the universal completion spectra. Farey/Stern–Brocot, Beatty, Christoffel/mechanical, digital-line, and Dedekind structures are classical neighboring theories [27, 28, 32, 29, 33, 30, 31]; novelty is not claimed for those objects in isolation.*

The rational-Dyck branch used the residual as a bridge to enumeration. A distinct arithmetic consequence branch studies the shortest-completion map itself. For coprime positive integers p, q and residual r, define

$$\mu_{p,q}(r) = \arg\min_{x,y\in\mathbb{Z}_{\geq 0}} \{x + y : px - qy = r\}.$$

This is the same minimum-demand semantics developed earlier in the book, now studied as an ordered arithmetic profile in r and across different rational slopes.

The operational provenance remains the original completion-aware DPDA. For example, the prefix 111 has $\rho_{2,1} = 3$, and the machine materializes the completion word 100; its Parikh vector $(2, 1)$ is exactly $\mu_{2,1}(3)$. The automaton is not redrawn here because Chapters 1–8 already preserve it in detail. From this point onward the Farey results are self-contained statements about the minimizer map.

Novelty boundary. The local arithmetic has strong classical shadows. We do not claim novelty for Diophantine-strip reduction, Bézout/Farey unit steps, modular representatives or inverses, mechanical/Christoffel coding, standard Christoffel factorization, digital-line Stern–Brocot updates, Beatty partitions, or Dedekind reciprocity in isolation. Indeed, after a unimodular change of coordinates the local completion walk is a classical rational mechanical/Christoffel path. The principal completion-specific statement appears in Chapter 82: the *cross-parameter Ordered Farey Recursion*, which compares the canonical fixed-residual ℓ^1 minimizer sections of a mediant and both Farey parents and identifies two interleaved child residual subsequences with an unshifted parent profile and a fixed translate of the other. This novelty boundary is a cautious literature assessment, not an absolute priority guarantee.

81.1 Shortest-completion profiles and canonical strips

Fix coprime positive integers p, q. Define

$$W_{p,q}(x, y) = px - qy.$$

The primitive positive kernel vector of $W_{p,q}$ is

$$K_{p,q} = (q, p).$$

Thus any two integer solutions of $px - qy = r$ differ by an integral multiple of (q, p).

Definition 81.1 — Shortest completion.

For $r \in \mathbb{Z}$, define

$$\mu_{p,q}(r) = (x_r, y_r) = \operatorname*{arg\,min}_{x,y \in \mathbb{Z}_{\geq 0}} \{x + y : px - qy = r\},$$

and set

$$L_{p,q}(r) = x_r + y_r.$$

When p, q are fixed, we abbreviate these as $\mu(r)$ and $L(r)$.

Lemma 81.2 — Canonical-strip characterization.

For $r \geq 0$, $\mu(r)$ is the unique nonnegative solution of

$$px - qy = r$$

satisfying

$$0 \leq y < p.$$

For $r \leq 0$, $\mu(r)$ is the unique nonnegative solution satisfying

$$0 \leq x < q.$$

Proof. Suppose $r \geq 0$. Since $\gcd(p, q) = 1$, the congruence

$$qy \equiv -r \pmod{p}$$

has a unique solution $y_0 \in \{0, \ldots, p-1\}$. Put

$$x_0 = \frac{r + qy_0}{p} \geq 0.$$

Every integer solution is

$$(x_0, y_0) + k(q, p), \qquad k \in \mathbb{Z}.$$

Because $0 \leq y_0 < p$, nonnegativity forces $k \geq 0$. Every other nonnegative solution therefore has length

$$x_0 + y_0 + k(p + q),$$

which is strictly larger for $k > 0$. Hence (x_0, y_0) is the unique minimizer. The case $r \leq 0$ is dual, using the unique representative $x_0 \in \{0, \ldots, q-1\}$. □

Corollary 81.3 — Kernel reduction.

If (x, y) is any nonnegative solution of residual r, then

$$(x, y) = \mu(r) + k(q, p)$$

for a unique $k \geq 0$, and

$$x + y = L(r) + k(p + q).$$

Proposition 81.4 — Boundary families.

The cases $p = 1$ or $q = 1$ are explicit. If $p = 1$, then

$$\mu_{1,q}(0) = (0, 0), \qquad \mu_{1,q}(1) = (1, 0),$$

and for $1 \leq s \leq q$,

$$\mu_{1,q}(-s) = (q - s, 1).$$

If $q = 1$, then

$$\mu_{p,1}(0) = (0, 0), \qquad \mu_{p,1}(-1) = (0, 1),$$

and for $1 \leq r \leq p$,

$$\mu_{p,1}(r) = (1, p - r).$$

Proof. Each displayed pair solves the relevant equation and lies in the canonical strip of Lemma 81.2. □

81.2 Canonical Farey unit-lift dynamics

Assume throughout this section that $p, q > 1$. Let

$$(p, q) = (c, u) + (v, d) \tag{81.1}$$

be the oriented Farey-parent decomposition characterized by

$$pu - qc = 1, \qquad qv - pd = 1. \tag{81.2}$$

It follows that

$$uv - cd = 1. \tag{81.3}$$

The existence and uniqueness of this oriented parent pair are classical Farey/Stern–Brocot facts; see, for example, [27].

Theorem 81.5 — Canonical Farey Unit-Lift Theorem.

Define

$$a = (u, c), \qquad b = (-d, -v).$$

Then

$$W_{p,q}(a) = W_{p,q}(b) = 1$$

and

$$a - b = (q, p) = K_{p,q}.$$

If

$$\mu(r) = (x_r, y_r), \qquad 0 \le r < p,$$

then

$$\mu(r+1) = \begin{cases} \mu(r) + (u, c), & 0 \le y_r < v, \\ \mu(r) + (-d, -v), & v \le y_r < p. \end{cases} \tag{81.4}$$

Proof. By equation (81.2),

$$W_{p,q}(u, c) = pu - qc = 1$$

and

$$W_{p,q}(-d, -v) = -pd + qv = 1.$$

Moreover,

$$(u, c) - (-d, -v) = (u + d, c + v) = (q, p).$$

Thus the two vectors are kernel-equivalent lifts of residual increment $+1$.

If $y_r < v$, then

$$0 \le y_r + c < v + c = p,$$

so $\mu(r)+(u, c)$ has residual $r+1$ and remains in the positive canonical strip. By Lemma 81.2, it is $\mu(r + 1)$.

If $y_r \ge v$, then adding (u, c) leaves the strip because $y_r + c \ge p$. Subtracting one kernel gives

$$(u, c) - (q, p) = (-d, -v),$$

and the new second coordinate is $y_r - v \in [0, p - 1]$. Hence $\mu(r) + (-d, -v)$ is the unique canonical representative of residual $r + 1$. □

Remark 81.R1 — Canonical unit lifting.

The mechanism in Theorem 81.5 is intrinsic to shortest completion:

$$\text{unit residual lift} \quad + \quad \text{canonical kernel reduction.}$$

The Farey parent vectors appear as the two kernel-equivalent lifts available to this reduction.

Theorem 81.6 — Negative unit-lift theorem.

Define

$$a^- = (d, v), \qquad b^- = (-u, -c).$$

Then

$$W_{p,q}(a^-) = W_{p,q}(b^-) = -1, \qquad a^- - b^- = (q, p).$$

If

$$\mu(-s) = (x_s, y_s), \qquad 0 \le s < q,$$

then

$$\mu(-(s+1)) = \begin{cases} \mu(-s) + (d, v), & 0 \le x_s < u, \\ \mu(-s) + (-u, -c), & u \le x_s < q. \end{cases} \tag{81.5}$$

Proof. The two increments have residual -1 because

$$pd - qv = -1, \qquad -pu + qc = -1,$$

and their difference is (q, p). If $x_s < u$, adding (d, v) keeps the first coordinate below $u + d = q$. If $x_s \ge u$, this lift leaves the negative canonical strip, and subtracting the kernel replaces (d, v) by $(-u, -c)$. The conclusion follows from Lemma 81.2. □

Corollary 81.7 — Closed vector profiles.

For $0 \le r \le p$,

$$\mu_{p,q}(r) = r(u, c) - \left\lfloor \frac{cr}{p} \right\rfloor (q, p). \tag{81.6}$$

For $0 \le s \le q$,

$$\mu_{p,q}(-s) = s(d, v) - \left\lfloor \frac{ds}{q} \right\rfloor (q, p). \tag{81.7}$$

In particular,

$$\mu(p) = (1, 0), \qquad \mu(-q) = (0, 1).$$

Proof. In the positive walk, the kernel correction occurs exactly when the second coordinate wraps under addition of c modulo p. The number of wraps among the first r steps is $\lfloor cr/p \rfloor$, giving equation (81.6). The negative formula is identical with d modulo q. At $r = p$ and $s = q$, use $pu - qc = 1$ and $qv - pd = 1$. □

Example 81.E1 — The pair 7:4.

For $(p, q) = (7, 4)$ the oriented parents are $(c, u) = (5, 3)$ and $(v, d) = (2, 1)$. Hence the two positive unit lifts are $(3, 5)$ and $(-1, -2)$. The positive fundamental profile is

$$(0, 0), (3, 5), (2, 3), (1, 1), (4, 6), (3, 4), (2, 2), (1, 0),$$

whose successive increments are

$$(3, 5), (-1, -2), (-1, -2), (3, 5), (-1, -2), (-1, -2), (-1, -2).$$

The ordered recursion places the left-parent profile at indices

$$\left\lceil \frac{7r}{5} \right\rceil \in \{0, 2, 3, 5, 6, 7\}$$

and the translated right-parent block at indices

$$\left\lfloor \frac{7s}{2} \right\rfloor + 1 \in \{1, 4\}.$$

Thus the local two-jump law and the global floor/ceiling shuffle are visible in the same small example.

Chapter 82

Ordered Farey Recursion and the Completion Tree

The unit-lift laws identify the two possible local jumps. The central result of the Farey development is stronger and cross-parametric: it reconstructs the entire mediant profile from the two parent profiles at complementary interleaved residual positions. Parent recovery and the recursively decorated Stern–Brocot family are retained as mechanisms and consequences of that theorem rather than promoted to separate novelty claims.

82.1 Local Farey recovery and ordered parent recursion

82.1.1 Local encoding and parent recovery

Theorem 82.1 — Vector jump law.

The positive fundamental profile has exactly two local increments,

$$\mu(r+1) - \mu(r) \in \{(u, c), (-d, -v)\}, \qquad 0 \le r < p.$$

The jump (u, c) occurs v times and the jump $(-d, -v)$ occurs c times.

The negative fundamental profile has exactly the two increments

$$(d, v), \qquad (-u, -c),$$

with multiplicities u and d, respectively.

Proof. The two-value statements are Theorems 81.5 and 81.6. In the positive profile, the number of kernel corrections is

$$\sum_{r=0}^{p-1} \left(\left\lfloor \frac{c(r+1)}{p} \right\rfloor - \left\lfloor \frac{cr}{p} \right\rfloor \right) = c.$$

Since there are $p = c + v$ jumps in total, the other jump occurs v times. The negative case is identical. □

Corollary 82.2 — Parent recovery.

For $p, q > 1$, the ordered positive completion profile uniquely determines the oriented Farey parents. If its two local increments are

$$(a, b), \qquad (-e, -f),$$

then

$$(c, u) = (b, a), \qquad (v, d) = (f, e),$$

and consequently

$$p = b + f, \qquad q = a + e.$$

Proof. By Theorem 82.1, $(a, b) = (u, c)$ and $(-e, -f) = (-d, -v)$. □

82.1.2 Ordered parent-to-mediant recursion

Write

$$\mathcal{M}^{+}_{p,q} = (\mu_{p,q}(0), \mu_{p,q}(1), \dots, \mu_{p,q}(p))$$

and

$$\mathcal{M}^{-}_{p,q} = (\mu_{p,q}(0), \mu_{p,q}(-1), \dots, \mu_{p,q}(-q)).$$

Theorem 82.3 — Cross-parameter Ordered Farey Recursion.
Let $(p, q) = (c, u) + (v, d)$ be as in equations (81.1) and (81.2). For $0 \le r \le c$,

$$\mu_{p,q}\Big(\Big\lceil \frac{pr}{c} \Big\rceil\Big) = \mu_{c,u}(r). \tag{82.1}$$

For $0 \le s < v$,

$$\mu_{p,q}\Big(\Big\lfloor \frac{ps}{v} \Big\rfloor + 1\Big) = (u, c) + \mu_{v,d}(s). \tag{82.2}$$

Dually, for $0 \le r \le u$,

$$\mu_{p,q}\Big(-\Big\lfloor \frac{qr}{u} \Big\rfloor\Big) = \mu_{c,u}(-r), \tag{82.3}$$

and for $1 \le s \le d$,

$$\mu_{p,q}\Big(-\Big(\Big\lceil \frac{qs}{d} \Big\rceil - 1\Big)\Big) = (u, c) + \mu_{v,d}(-s). \tag{82.4}$$

Moreover, the positive index families

$$I^{+}_{L} = \Big\{\Big\lceil \frac{pr}{c} \Big\rceil : 0 \le r \le c\Big\}, \qquad I^{+}_{R} = \Big\{\Big\lfloor \frac{ps}{v} \Big\rfloor + 1 : 0 \le s < v\Big\}$$

form a disjoint partition of $\{0, 1, \dots, p\}$. The corresponding negative index families in equations (82.3) and (82.4) form a disjoint partition of $\{0, 1, \dots, q\}$.

Proof. For equation (82.1), let $\mu_{c,u}(r) = (x, y)$. Then

$$cx - uy = r, \qquad 0 \le y < c.$$

Set $R = px - qy$. Since $pu = qc + 1$,

$$\frac{pr}{c} = px - \frac{pu}{c}y = px - qy - \frac{y}{c} = R - \frac{y}{c}.$$

As $0 \le y/c < 1$, we get

$$R = \Big\lceil \frac{pr}{c} \Big\rceil.$$

Since $y < c < p$, the same vector lies in the child canonical strip, proving equation (82.1).

For equation (82.2), let $\mu_{v,d}(s) = (x, y)$, so

$$vx - dy = s, \qquad 0 \le y < v.$$

Since $pd = qv - 1$,

$$\frac{ps}{v} = px - \frac{pd}{v}y = px - qy + \frac{y}{v},$$

and therefore

$$px - qy = \Big\lfloor \frac{ps}{v} \Big\rfloor.$$

Adding (u, c) raises the child residual by one. Moreover $0 \le c + y < c + v = p$, so the translated vector is canonical. This proves equation (82.2).

For equation (82.3), let $\mu_{c,u}(-r) = (x, y)$, so $cx - uy = -r$ and $0 \le x < u$. Then

$$\frac{qr}{u} = qy - \frac{qc}{u}x = qy - px + \frac{x}{u} = -(px - qy) + \frac{x}{u}.$$

Hence $px - qy = -\lfloor qr/u \rfloor$, and the same vector lies in the child negative strip because $x < u < q$.

For equation (82.4), let $\mu_{v,d}(-s) = (x, y)$, so $vx - dy = -s$ and $0 \le x < d$. Using $qv = pd + 1$,

$$\frac{qs}{d} = qy - \frac{qv}{d}x = qy - px - \frac{x}{d} = -(px - qy) - \frac{x}{d}.$$

Thus $\lceil qs/d \rceil = -(px - qy)$. Adding (u, c) raises the residual by one and keeps the first coordinate below $u + d = q$, which gives equation (82.4).

It remains to verify the two index partitions. We give the positive one directly. Both index lists are strictly increasing. For $1 \le n < p$, the number of left indices at most n is

$$\left\lfloor \frac{cn}{p} \right\rfloor + 1,$$

while the number of right indices at most n is

$$\left\lceil \frac{vn}{p} \right\rceil = n - \left\lfloor \frac{cn}{p} \right\rfloor,$$

because $v = p - c$ and cn/p is nonintegral. Their sum is $n + 1$, exactly the number of positions in $\{0, \dots, n\}$. The endpoint counts at $n = 0, p$ are immediate, so every position $0, \dots, p$ occurs exactly once. The negative partition is identical with $q = u + d$. These index sets are the familiar finite rational Beatty partition; the vector labels supplied by equations (82.1) to (82.4) are the completion-specific content. □

Remark 82.R1 — Vector-valued Beatty shuffle.

Theorem 82.3 says more than set inclusion. The entire residual order is reconstructed: the left parent populates the positions I_L^+ unchanged, while the translated right parent populates the complementary positions I_R^+. The same holds on the negative side. The endpoint $\mu_{v,d}(v) = (1, 0)$ is omitted from equation (82.2) because the child endpoint $\mu_{p,q}(p) = (1, 0)$ is already inherited through the left parent.

Corollary 82.4 — Local–global Farey correspondence.

For a proper pair $p, q > 1$, the oriented Farey parents determine both the local jump law and the global ordered recursion. Conversely, the local positive jump set recovers the oriented parents and hence determines the ordered parent recursion.

Proof. Forward determination is Theorems 81.5 and 82.3; reverse determination is Corollary 82.2. □

82.2 The global Farey completion family

The preceding results concern a single Farey mediant. Iteration over the Stern–Brocot tree produces one recursively coherent completion family.

Definition 82.5 — Farey completion tree.

Decorate every coprime node q/p of the Stern–Brocot tree by the pair of ordered profiles

$$\mathfrak{C}(p, q) = \left(\mathcal{M}_{p,q}^{+}, \mathcal{M}_{p,q}^{-}\right).$$

We call the resulting decorated tree the *Farey completion tree.*

Theorem 82.6 — Recursive coherence.

Whenever

$$(p, q) = (c, u) + (v, d)$$

is a Farey mediant, both ordered profiles in $\mathfrak{C}(p,q)$ are determined uniquely from the two parent profiles by Theorem 82.3. Consequently the family

$$\{\mathfrak{C}(p,q) : p,q \geq 1,\ \gcd(p,q) = 1\}$$

is recursively coherent under Farey mediants.

Proof. Apply Theorem 82.3 at every proper interior node and stop when $p = 1$ or $q = 1$, where Proposition 81.4 supplies the explicit boundary profiles. The two index families partition every proper child profile, so the recursion determines every child entry exactly once. □

Remark 82.R2 — Classical tree, new decoration.

The Stern–Brocot tree and its continued-fraction complexity are classical. The point of Definition 82.5 and Theorem 82.6 is not a new rational tree, but a new vector-valued decoration: the shortest-completion profiles form a single parent-generated system rather than an unrelated collection indexed by $p : q$.

Chapter 83

Universal Completion Spectra and Arithmetic Interfaces

The vector profile remembers the ratio $p{:}q$, but scalar projection reveals a universal phenomenon controlled only by $m = p + q$. This chapter retains the fundamental and shell spectra, their moment consequences, the one-sided reflection law, and the precise interfaces with Beatty, mechanical/Christoffel, digital-line, and Dedekind arithmetic. These classical interfaces are included because they explain what the completion theory projects to; they are not used to overstate the novelty of the local coding.

83.1 Scalar projection and universal spectra

For the Farey-coordinate formulas in the first part of this section assume $p, q > 1$; the boundary families are given by Proposition 81.4. Set

$$m = p + q, \qquad A = c + u, \qquad B = v + d.$$

Then $A + B = m$. From equation (81.2),

$$pA - mc = 1, \qquad qB - md = 1. \tag{83.1}$$

Thus $A \equiv p^{-1} \pmod{m}$ and $B \equiv q^{-1} \pmod{m}$.

Proposition 83.1 — Scalar Farey profiles.
For $0 \le r \le p$,

$$L_{p,q}(r) = Ar - m \left\lfloor \frac{cr}{p} \right\rfloor. \tag{83.2}$$

For $0 \le s \le q$,

$$L_{p,q}(-s) = Bs - m \left\lfloor \frac{ds}{q} \right\rfloor. \tag{83.3}$$

Proof. Take coordinate sums in equations (81.6) and (81.7). □

Corollary 83.2 — Scalar jump law and affine periodicity.
For $0 \le r < p$,

$$L(r+1) - L(r) \in \{A, -B\},$$

with multiplicities v and c. Dually,

$$L(-(s+1)) - L(-s) \in \{B, -A\},$$

with multiplicities u and d. Moreover,

$$L(r+p) = L(r) + 1 \qquad (r \ge 0) \tag{83.4}$$

and

$$L(-(s+q)) = L(-s) + 1 \qquad (s \geq 0). \tag{83.5}$$

Proof. The jump statements are the ℓ^1 projections of Theorem 82.1. Their net positive drift is

$$vA - cB = v(c+u) - c(v+d) = uv - cd = 1,$$

and the negative drift is $uB - dA = 1$. Alternatively, equations (83.4) and (83.5) follow directly from the canonical strips by adding $(1,0)$ or $(0,1)$. □

83.1.1 Fundamental spectra

Define

$$S_+(p,q) = \{L(r) : 0 \leq r < p\}, \qquad S_-(p,q) = \{L(-s) : 0 \leq s < q\}.$$

Proposition 83.3 — Completion–Beatty identification.

As sets,

$$S_+(p,q) = \left\{ \left\lceil \frac{mj}{p} \right\rceil : 0 \leq j < p \right\}, \tag{83.6}$$

and

$$S_-(p,q) = \left\{ \left\lceil \frac{mj}{q} \right\rceil : 0 \leq j < q \right\}. \tag{83.7}$$

Proof. For $0 \leq r < p$, the positive canonical coordinate y_r runs through every residue $j \in \{0, \ldots, p-1\}$ exactly once. For fixed j, the least nonnegative first coordinate satisfying $px - qj \geq 0$ is $\lceil qj/p \rceil$, and therefore the corresponding completion length is

$$j + \left\lceil \frac{qj}{p} \right\rceil = \left\lceil \frac{(p+q)j}{p} \right\rceil.$$

The negative case is dual. □

Theorem 83.4 — Universal Fundamental Spectrum.

For every coprime pair p, q with $m = p + q$,

$$S_+(p,q) \uplus S_-(p,q) = \{0, 0, 2, 3, \ldots, m-1\} \tag{83.8}$$

as multisets.

Proof. By Proposition 83.3, it is enough to show that the two nonzero ceiling sets partition $\{2, \ldots, m-1\}$. For $1 \leq \ell \leq m$, define

$$\chi_p(\ell) = \left\lfloor \frac{p\ell}{m} \right\rfloor - \left\lfloor \frac{p(\ell-1)}{m} \right\rfloor, \qquad \chi_q(\ell) = \left\lfloor \frac{q\ell}{m} \right\rfloor - \left\lfloor \frac{q(\ell-1)}{m} \right\rfloor.$$

Each quantity is 0 or 1. The condition $\chi_p(\ell) = 1$ is equivalent to $\ell = \lceil mj/p \rceil$ for some $1 \leq j \leq p-1$ when $2 \leq \ell \leq m-1$, and similarly for q. Since $q = m - p$ and $\gcd(p, m) = 1$, for every $2 \leq \ell \leq m-1$ neither $p\ell/m$ nor $p(\ell-1)/m$ is integral; using $\lfloor \ell - x \rfloor = \ell - \lceil x \rceil$ gives

$$\chi_p(\ell) + \chi_q(\ell) = 1.$$

Thus every integer $2, \ldots, m-1$ belongs to exactly one of the two nonzero spectra. At $\ell = 1$ neither occurs, while both spectra contain 0. This proves equation (83.8). The resulting partition is the finite rational Beatty complementarity seen in Proposition 83.3. □

83.1.2 Universal shell spectra

For $k \geq 0$, define

$$\Sigma_k^+ = \{L(kp + r) : 0 \leq r < p\}, \qquad \Sigma_k^- = \{L(-kq - s) : 0 \leq s < q\}.$$

Theorem 83.5 — Universal Shell Spectrum.
For every $k \geq 0$,

$$\Sigma_k^+ \uplus \Sigma_k^- = \{k, k, k+2, k+3, \ldots, k+m-1\}. \tag{83.9}$$

Proof. By equations (83.4) and (83.5),

$$L(kp + r) = k + L(r), \qquad L(-kq - s) = k + L(-s).$$

Translate Theorem 83.4 by k. □

Corollary 83.6 — Shell generating polynomial.
If

$$C_k(z) = \sum_{r=0}^{p-1} z^{L(kp+r)} + \sum_{s=0}^{q-1} z^{L(-kq-s)},$$

then

$$C_k(z) = z^k\big(2 + z^2 + z^3 + \cdots + z^{m-1}\big),$$

and hence

$$\sum_{k\geq 0} C_k(z)t^k = \frac{2 + z^2 + z^3 + \cdots + z^{m-1}}{1 - tz}.$$

Corollary 83.7 — Universal moment hierarchy.
For every integer $j \geq 1$,

$$M_j(k) := \sum_{\ell \in \Sigma_k^+ \uplus \Sigma_k^-} \ell^j = 2k^j + \sum_{\ell=2}^{m-1} (k + \ell)^j.$$

In particular,

$$M_j(0) = \sum_{\ell=2}^{m-1} \ell^j.$$

Thus all combined shell power moments depend only on $m = p + q$.

83.1.3 Reflection symmetry

Write the nonzero positive spectrum in increasing order as

$$\ell_j^+ = \left\lceil \frac{mj}{p} \right\rceil, \qquad 1 \leq j < p,$$

and define ℓ_j^- analogously with q.

Proposition 83.8 — One-sided reflection.
For $1 \leq j < p$,

$$\ell_j^+ + \ell_{p-j}^+ = m + 1,$$

and for $1 \leq j < q$,

$$\ell_j^- + \ell_{q-j}^- = m + 1.$$

Consequently, with $h = (m + 1)/2$, every odd centered one-sided moment vanishes.

Proof. Because $\gcd(m,p)=1$, mj/p is nonintegral for $1 \le j < p$. Hence

$$\left\lceil \frac{m(p-j)}{p} \right\rceil = m - \left\lfloor \frac{mj}{p} \right\rfloor,$$

which sums with $\lceil mj/p \rceil$ to $m+1$. The negative identity is dual, and centered odd powers cancel in symmetric pairs. □

83.2 Classical arithmetic interfaces

The results through section 83.1 were obtained from shortest-completion minimization, canonical kernel reduction, and Farey unimodularity. We now record how classical arithmetic structures appear as projections. The purpose of this section is interpretive rather than foundational.

83.2.1 Mechanical and Christoffel coding

The positive wrap indicator is

$$\varepsilon_r = \left\lfloor \frac{c(r+1)}{p} \right\rfloor - \left\lfloor \frac{cr}{p} \right\rfloor \in \{0,1\}. \tag{83.10}$$

Then

$$\mu(r+1) - \mu(r) = (u,c) - \varepsilon_r(q,p).$$

Thus $\varepsilon_r = 0$ selects (u,c) and $\varepsilon_r = 1$ selects $(-d,-v)$. The finite binary sequence $(\varepsilon_r)_{r=0}^{p-1}$ is a rational lower mechanical word; under the standard finite conventions it is a Christoffel encoding, up to the usual choice of slope orientation and alphabet. Standard factorization, continued fractions, and the relation with the Stern–Brocot tree are classical; see [27].

The completion interpretation is that the mechanical/Christoffel coding records exactly when canonical kernel reduction occurs. This should not be read as a novelty claim for the coding itself.

83.2.2 Unimodular realization and the distinction from standard factorization

Let

$$k_r = \left\lfloor \frac{cr}{p} \right\rfloor \qquad (0 \le r \le p)$$

and define

$$J_+ = \begin{pmatrix} u & -d \\ c & -v \end{pmatrix}.$$

Since $uv - cd = 1$, one has $\det J_+ = -1$. Using $(q,p) = (u,c) - (-d,-v)$ in equation (81.6) gives

$$\mu_{p,q}(r) = (r-k_r)(u,c) + k_r(-d,-v) = J_+ \begin{pmatrix} r-k_r \\ k_r \end{pmatrix}. \tag{83.11}$$

Therefore

$$J_+^{-1}\mu_{p,q}(r) = \begin{pmatrix} r - \lfloor cr/p \rfloor \\ \lfloor cr/p \rfloor \end{pmatrix},$$

which is the vertex sequence of a rational mechanical lattice path with v steps of one type and c of the other. Thus the local completion walk is a unimodular image of classical Christoffel/mechanical geometry. This is consistent with digital-straight-line theory, where arithmetic remainders, minimal $a+b$ parameters, Bézout vectors and Stern–Brocot slope updates already occur together [29], and with classical Christoffel duality and modular-inverse phenomena [28].

It is important, however, not to identify Theorem 82.3 with standard Christoffel factorization. Standard factorization decomposes a proper Christoffel word into a *contiguous concatenation* of two Christoffel factors associated with the Farey parents [27]. By contrast, Theorem 82.3 compares three different canonical minimizer maps,

$$\mu_{p,q}, \qquad \mu_{c,u}, \qquad \mu_{v,d},$$

and partitions the child residual positions into two *interleaved* sets

$$I_L^+ = \left\{ \left\lceil \frac{pr}{c} \right\rceil : 0 \le r \le c \right\}, \qquad I_R^+ = \left\{ \left\lfloor \frac{ps}{v} \right\rfloor + 1 : 0 \le s < v \right\}.$$

The corresponding child-vector subsequences are exactly the left-parent completion profile and a fixed translate of the right-parent completion profile. Hence standard factorization explains why Farey parents and Christoffel structure are nearby, but it is not the same residual-indexed cross-parameter identity.

The two scalar jump amplitudes are

$$A = c + u, \qquad B = v + d,$$

so, under the standard Christoffel factorization conventions, the corresponding factor lengths also appear as the ℓ^1 projections of the two completion unit lifts.

83.2.3 Dedekind structure of the one-sided second moment

The combined spectrum is universal, but its split into positive and negative parts depends on $p : q$. This ratio dependence is already visible in the centered second moment.

Let

$$((x)) = \begin{cases} \{x\} - \frac{1}{2}, & x \notin \mathbb{Z}, \\ 0, & x \in \mathbb{Z}, \end{cases}$$

and define the classical Dedekind sum

$$s(h,k) = \sum_{j=1}^{k-1} \left(\left(\frac{j}{k} \right) \right) \left(\left(\frac{hj}{k} \right) \right).$$

Put $h_0 = (m+1)/2$. For the positive sorted spectrum,

$$\ell_j^+ - h_0 = m \left(\left(\frac{j}{p} \right) \right) - \left(\left(\frac{qj}{p} \right) \right). \tag{83.12}$$

Since multiplication by q permutes the nonzero residue classes modulo p,

$$\sum_{j=1}^{p-1} \left(\left(\frac{qj}{p} \right) \right)^2 = \sum_{j=1}^{p-1} \left(\left(\frac{j}{p} \right) \right)^2 = \frac{(p-1)(p-2)}{12p}.$$

Squaring equation (83.12) therefore gives

$$D_2^+(p,q) := \sum_{r=1}^{p-1} \left(L(r) - \frac{m+1}{2} \right)^2 = \frac{(m^2+1)(p-1)(p-2)}{12p} - 2m\, s(q,p). \tag{83.13}$$

Similarly,

$$D_2^-(p,q) = \frac{(m^2+1)(q-1)(q-2)}{12q} - 2m\, s(p,q). \tag{83.14}$$

By Theorem 83.4,

$$D_2^+(p,q) + D_2^-(p,q) = \frac{(m-1)(m-2)(m-3)}{12}. \tag{83.15}$$

Substituting equations (83.13) and (83.14) into equation (83.15) recovers the classical Dedekind reciprocity law

$$s(q,p)+s(p,q)=-\frac{1}{4}+\frac{1}{12}\left(\frac{q}{p}+\frac{p}{q}+\frac{1}{pq}\right);$$

see [31]. Dedekind reciprocity is not a new result here. Its completion interpretation is that Dedekind sums quantify the ratio-dependent one-sided splitting of an otherwise universal completion spectrum.

Remark 83.R1 — Higher even moments.

By Proposition 83.8, all odd centered moments vanish. Higher even moments expand into products of periodic Bernoulli functions and hence into generalized Dedekind-type sums. Since such sums and their reciprocity laws have an extensive independent literature, including [34], we do not pursue new reciprocity statements here.

83.3 Computational discovery and validation

The results above are proved symbolically and do not depend on computation. Numerical experimentation served first as a discovery device and later as an independent audit of signs, endpoints, and floor/ceiling conventions.

An initial audit covered every coprime ordered pair $2 \le p, q \le 100$. A second reviewer-oriented audit extended the exhaustive range to

$$2 \le p, q \le 200,$$

covering 24,064 coprime ordered pairs, and added 30,000 randomized coprime pairs with $2 \le p, q \le 5000$. Shortest-completion vectors were constructed independently from Lemma 81.2. For every residual in the positive and negative fundamental ranges, these direct solutions were compared with the formulas of sections 81.2 and 82.1. The audits verified, without exception:

- the positive and negative canonical unit-lift laws;
- the closed vector formulas equations (81.6) and (81.7);
- all four ordered parent-to-mediant identities equations (82.1) to (82.4);
- the corresponding positive and negative index partitions.

No counterexample was found in either the exhaustive or randomized vector audits.

A separate scalar audit tested every coprime ordered pair

$$1 \le p, q \le 150,$$

comprising 13,715 ratios. For each pair, the fundamental spectrum and the translated shells $k = 0, 1, \ldots, 7$ were generated directly from shortest completions. In every case,

$$\Sigma_k^+ \uplus \Sigma_k^- = \{k, k, k+2, k+3, \ldots, k+p+q-1\}.$$

These computations are not proof premises. The direct completion oracle used only the defining equation $px - qy = r$ and the canonical-strip bounds, rather than the closed Farey formulas being tested. The experiments therefore function only as independent consistency checks.

83.4 Conclusion

We have studied the ordered family of shortest nonnegative integer solutions of

$$px - qy = r,$$

with p and q coprime, motivated by shortest completion in rational balance languages.

For a proper Farey decomposition

$$(p, q) = (c, u) + (v, d), \qquad uv - cd = 1,$$

the positive shortest-completion profile advances through two kernel-equivalent unit-residual lifts,

$$(u, c) \qquad \text{and} \qquad (-d, -v).$$

Canonical reduction in the strip $0 \le y < p$ selects between them, and the negative profile has the corresponding dual dynamics. After a unimodular coordinate change this local walk is a classical rational mechanical path, so its Christoffel form is best viewed as the local arithmetic mechanism rather than as the principal novelty.

The principal completion-specific result is the cross-parameter Ordered Farey Recursion Theorem. It reconstructs the entire mediant minimizer section from the two parent minimizer sections through complementary floor/ceiling index maps: the two parent profiles appear as interleaved child subsequences, one after a fixed translation. The local jump vectors recover the oriented parents, and iteration yields a recursively coherent family over the Stern–Brocot tree.

Returning to the motivating automaton, the map $r \mapsto \mu_{2,1}(r)$ is exactly the shortest-demand semantics realized by its distributed state–stack representation; the present results show that this semantic object belongs to a much larger Farey-recursive arithmetic family.

Scalar projection produces a complementary universal phenomenon. Writing $m = p+q$, the combined positive and negative fundamental spectrum is always

$$\{0, 0, 2, 3, \ldots, m - 1\},$$

and every residual shell is an affine translate of this multiset. Hence all combined shell moments depend only on m. Classical Beatty, mechanical/Christoffel, and Dedekind structures arise as arithmetic shadows of this vector theory rather than as proof premises.

Taken together, these results organize the coprime rational balance family as a single Farey-recursive system of shortest-completion profiles, with locally recoverable parent data and universal scalar spectra.

The Farey chapters preceding this branch studied the shortest nonnegative solutions of $px - qy = r$ as the ratio (p, q) varies through the Stern–Brocot/Farey tree. The next question is external rather than internal: whether the same completion coordinates reveal exact structure in a classical unimodular word system. We therefore pass to the Cohn generators

$$X = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad Y = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix},$$

and study the word-state map they generate. No pushdown theorem is used as a hypothesis in the results below; the link to the earlier machine is one of discovery and coordinate choice.

The central identity of the branch is

$$\boxed{Q_P(q(s10)) - Q_P(q(s01)) = 12F(s)R_p(s)},$$

which refines the classical sign/equality picture by giving an exact magnitude factorization through a Farey determinant residual. The zero set is then rigid on the word orbit, and the homogeneous completion kernel is transported to the fundamental quadratic-return translations.

Chapter 84

Cohn Word States and Farey Completion Coordinates

Research branch — From Farey Completion Arithmetic to Cohn–Markoff Geometry. *The preceding arithmetic Part extracted Farey recursion, mechanical coding, and universal spectra from shortest completion. The present Part carries that completion viewpoint into the classical Cohn–Markoff matrix monoid. Cohn matrices, Christoffel words, Markoff triples, and flip methods are classical background; the contribution developed here is the exact completion-coordinate layer, especially the quadratic adjacent-swap magnitude factorization, the completion-kernel/return correspondence, and the centered reversal-defect identities.*

84.1 The Cohn word-state system

Define

$$X = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad Y = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}.$$

Both matrices are symmetric, positive and unimodular. Let

$$\mu : \{0,1\}^* \longrightarrow \mathrm{SL}_2(\mathbb{Z})$$

be the morphism with $\mu(0) = X$, $\mu(1) = Y$. We write words on the right, so

$$\mu(s0) = \mu(s)X, \qquad \mu(s1) = \mu(s)Y.$$

Let

$$v = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad q(s) = \mu(s)^T v.$$

Because X and Y are symmetric,

$$q(s0) = Xq(s), \qquad q(s1) = Yq(s). \tag{84.1}$$

Every $q(s)$ is primitive because $X, Y \in \mathrm{SL}_2(\mathbb{Z})$ and v is primitive.

Lemma 84.1 (Exact terminal decoder)**.** *Every word state has slope in the invariant interval*

$$I = \left(1, \frac{5}{2}\right).$$

For a nonempty word, the last letter is determined by the slope:

$$q(s0): \quad \frac{3}{2} < \frac{q_1}{q_2} < \frac{12}{7},$$

$$q(s1): \quad \frac{7}{3} < \frac{q_1}{q_2} < \frac{29}{12}.$$

These intervals are disjoint. The corresponding inverse maps are

$$T_0(a,b) = (a-b,\, 2b-a), \qquad T_1(a,b) = (a-2b,\, 5b-2a).$$

Consequently the map $s \mapsto q(s)$ is injective.

Proof. The initial vector $v = (2,1)^T$ has slope $2 \in I$. If a positive vector has slope $r \in I$, then multiplication by X and Y gives slopes

$$\phi_0(r) = \frac{2r+1}{r+1}, \qquad \phi_1(r) = \frac{5r+2}{2r+1}.$$

Both maps are increasing on I, and their endpoint images are

$$\phi_0(I) = \left(\frac{3}{2}, \frac{12}{7}\right), \qquad \phi_1(I) = \left(\frac{7}{3}, \frac{29}{12}\right).$$

Thus I is invariant and the last letter is read from the unique terminal interval. Since

$$X^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \qquad Y^{-1} = \begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix},$$

the displayed inverse maps recover the predecessor state. Iteration reaches the unique empty state v, proving injectivity. □

We use the scalar

$$F(s) = v^T q(s). \tag{84.2}$$

If $q(s) = (a,b)^T$, then

$$F(s) = 2a + b.$$

Lemma 84.2 (Reversal transpose)**.** *For every word s,*

$$\mu(s^R) = \mu(s)^T.$$

In particular, if $p = p^R$ is a palindrome, then $P = \mu(p)$ is symmetric.

Proof. Transpose reverses the order of a product, and both generators are symmetric. □

For a palindromic base p, write

$$P = \mu(p) = \begin{pmatrix} A & B \\ B & D \end{pmatrix}, \qquad m = F(p) = v^T P v, \tag{84.3}$$

and

$$x = q_1(p) = (Pv)_1 = 2A + B. \tag{84.4}$$

Then

$$Pv = \begin{pmatrix} x \\ m - 2x \end{pmatrix}.$$

Lemma 84.3 (Characteristic divisibility)**.** *With the notation above,*

$$x^2 + 1 = Am.$$

In particular

$$x^2 \equiv -1 \pmod{m}.$$

Proof. Since $\det P = AD - B^2 = 1$,

$$\begin{aligned} x^2 + 1 &= (2A + B)^2 + 1 \\ &= 4A^2 + 4AB + B^2 + AD - B^2 \\ &= A(4A + 4B + D) \\ &= Am. \end{aligned}$$

□

The congruence in theorem 84.3 is classical in the Markoff setting. We will use it only as an arithmetic identity attached to a palindromic base.

84.2 Farey completion coordinates

Define a new coordinate vector by

$$\Psi(s) = \begin{pmatrix} q_1(s) \\ F(s) \end{pmatrix}. \tag{84.5}$$

If $q(s) = (a, b)^T$, then

$$\Psi(s) = \begin{pmatrix} a \\ 2a + b \end{pmatrix} = Cq(s), \qquad C = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}.$$

The matrix C is unimodular, so $\Psi(s)$ is primitive whenever $q(s)$ is primitive.

Theorem 84.4 (Word-to-Farey coordinate conjugacy)**.** *For every word s, writing $\Psi(s) = (a, f)^T$, one has*

$$\Psi(s0) = \begin{pmatrix} f \\ 3f - a \end{pmatrix}, \tag{84.6}$$

and

$$\Psi(s1) = \begin{pmatrix} a + 2f \\ 2a + 5f \end{pmatrix}. \tag{84.7}$$

Equivalently, for $\rho(s) = a/f$,

$$\rho(s0) = \frac{1}{3 - \rho(s)}, \qquad \rho(s1) = \frac{\rho(s) + 2}{2\rho(s) + 5}. \tag{84.8}$$

Proof. Conjugating the two append maps in equation (84.1) by C gives

$$CXC^{-1} = \begin{pmatrix} 0 & 1 \\ -1 & 3 \end{pmatrix}, \qquad CYC^{-1} = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}.$$

Applying these matrices to $(a, f)^T$ gives equations (84.6) and (84.7); division of the two coordinates gives equation (84.8). □

Remark 84.5 (Relation with the $a = 0$ Cohn normalization)*.* The two conjugated generators

$$\begin{pmatrix} 0 & 1 \\ -1 & 3 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$$

are exactly Aigner's $a = 0$ Cohn generators. Veselov [84] uses this normalization to identify Cohn-matrix indices with Markov fractions. Thus theorem 84.4 is recorded as the coordinate bridge needed for the quadratic results below, not as a new Cohn–Farey correspondence.

Fix from now on a palindromic base p and write

$$\Psi(p) = \begin{pmatrix} x \\ m \end{pmatrix}.$$

Definition 84.6 (Farey residual)**.** For any word s with $\Psi(s) = (a, f)^T$, define

$$R_p(s) = ma - xf. \tag{84.9}$$

Thus

$$R_p(s) = -\det(\Psi(p), \Psi(s)). \tag{84.10}$$

The sign of $R_p(s)$ records on which side of x/m the fraction a/f lies. In the original q-coordinates,

$$R_p(s) = (m - 2x)q_1(s) - xq_2(s). \tag{84.11}$$

This residual is the natural linear coordinate for the Diophantine equation

$$mu - xd = r,$$

whose homogeneous kernel is generated by (x, m). The next section shows that the same completion lattice appears canonically from the quadratic form defined by P.

84.3 Relative completion coordinates

For $u \in \mathbb{Z}^2$, put

$$Q_P(u) = u^T P u.$$

For a vector $q \in \mathbb{Z}^2$, define

$$\ell(q) = v^T P q, \qquad \delta(q) = \det(v, q). \tag{84.12}$$

Since $v = (2, 1)^T$, if $q = (q_1, q_2)^T$ then

$$\delta(q) = 2q_2 - q_1.$$

Theorem 84.7 (Relative completion coordinates)**.** *For every $q \in \mathbb{Z}^2$,*

$$mQ_P(q) = \ell(q)^2 + \delta(q)^2. \tag{84.13}$$

Moreover,

$$\ell(q) + x\delta(q) = mq_2, \tag{84.14}$$

and

$$\ell(q) - (m - x)\delta(q) = m(q_1 - q_2). \tag{84.15}$$

Consequently

$$\ell(q) \equiv -x\delta(q) \pmod{m}. \tag{84.16}$$

Proof. The two-dimensional Gram determinant identity gives

$$(v^T P v)(q^T P q) - (v^T P q)^2 = \det(P) \det(v, q)^2.$$

Since $\det P = 1$, this is exactly equation (84.13).

Using $Pv = (x, m - 2x)^T$ and $\delta(q) = 2q_2 - q_1$,

$$\begin{aligned} \ell(q) + x\delta(q) &= xq_1 + (m - 2x)q_2 + x(2q_2 - q_1) \\ &= mq_2, \end{aligned}$$

which proves equation (84.14). Similarly,

$$\begin{aligned} \ell(q) - (m - x)\delta(q) &= xq_1 + (m - 2x)q_2 - (m - x)(2q_2 - q_1) \\ &= m(q_1 - q_2). \end{aligned}$$

Reducing equation (84.14) modulo m gives equation (84.16). □

Corollary 84.8 (Exact completion lattice)**.** *The image of the integral map $q \mapsto (\ell(q), \delta(q))$ is exactly*

$$\Lambda_{m,x} = \{(\ell, d) \in \mathbb{Z}^2 : \ell + xd \equiv 0 \pmod{m}\}.$$

In particular, this map identifies $\mathbb{Z}^2$ with an index-m sublattice of $\mathbb{Z}^2$.

Proof. In standard coordinates the map has matrix

$$L = \begin{pmatrix} x & m - 2x \\ -1 & 2 \end{pmatrix}, \qquad \det L = m.$$

By equation (84.16), its image is contained in $\Lambda_{m,x}$. The latter has index m, since it is the kernel of the surjective homomorphism $(\ell, d) \mapsto \ell + xd \pmod{m}$. The image of L also has index $|\det L| = m$, so the two lattices coincide. □

Remark 84.9 (Diophantine completion interpretation). For $d \geq 0$, equation (84.14) reads

$$mu - xd = \ell, \qquad u = q_2.$$

For each fixed value of ℓ, the corresponding slice of $\Lambda_{m,x}$ is the integer solution set of this two-variable completion equation, whose homogeneous kernel is generated by (x, m).

The sign of δ also remembers the terminal branch.

Lemma 84.10 (Terminal sign)**.** *For every word s,*

$$\delta(q(s0)) = q_2(s) > 0, \qquad \delta(q(s1)) = -q_1(s) < 0.$$

Proof. Use equation (84.1) and compute

$$\delta(X(a, b)^T) = b, \qquad \delta(Y(a, b)^T) = -a.$$

All word states have positive coordinates. □

Proposition 84.11 (Palindromic sandwich identity)**.** *For every word t and every palindromic base p,*

$$\boxed{Q_P(q(t)) = F(tpt^R).} \tag{84.17}$$

Moreover,

$$F(w) = \mu(0w1)_{12} \tag{84.18}$$

for every word w. Hence the quadratic observable is also an ordinary Cohn matrix-entry statistic of a palindromic sandwich.

Proof. Since $p = p^R$, one has

$$\mu(tpt^R) = \mu(t)P\mu(t)^T,$$

which is symmetric. Therefore

$$F(tpt^R) = v^T\mu(tpt^R)^T v = v^T\mu(t)P\mu(t)^T v = Q_P(q(t)).$$

For equation (84.18), write $\mu(w) = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$. A direct multiplication gives

$$\mu(0w1)_{12} = 4A + 2B + 2C + D = v^T\mu(w)^T v = F(w).$$

□

84.4 Sharp Apéry multiplicities in genuine Farey steps

At the level of original labelled records, optimal fibers may have several minimal representatives. Before comparing synchronized parents, the precise size of these fibers must be quantified. We pass to the Apéry description of the associated numerical semigroup and establish sharp tie bounds, then return to genuine positive Farey parent–child steps. No claim about unrestricted parent-to-daughter optimal-word transfer follows from the displayed family-specific construction.

84.4.1 Introduction and relation to previous work

Let p, q, s be positive integers with $\gcd(p, q, s) = 1$. For each $0 \le r < p$, we study all nonnegative integral solutions of

$$px - qu - sv = r \tag{84.19}$$

that minimize the *literal* length $x + u + v$. Keeping u and v distinct is essential, even if the two negative letters have equal magnitude. In word applications these solutions are original-labelled Parikh records, not interchangeable unlabelled paths.

The embedding of (84.19) into a numerical semigroup is elementary, and general algorithms for three-generator numerical-semigroup factorizations and L-shaped weighted double-loop diagrams already exist; see [208, 209]. Moreover, Autry, Gomes, O'Neill and Ponomarenko [210] explicitly study factorization nonuniqueness within Apéry sets: their Theorem 4 already uses three-generated semigroups with coprime-reduced nontrivial generators $(2, 3)$ and constructs families with many non-half-factorial Apéry elements. Thus neither the L-shape method, the $(2, 3)$ generator pattern, nor its ability to produce many nonunique Apéry factorizations is claimed here as new. Our narrower question counts *all additional factorizations with multiplicity*, sums them over the Apéry set at fixed smallest generator p, and asks for its exact sharp global maximum and Farey parent–child realizability.

Contributions and novelty

The main contributions of this article are the following. First, we give the exact sharp total excess multiplicity for minimally three-generated semigroups at each fixed p, together with an equality characterization. Second, we construct an authentic positive $\mathrm{SL}_3(\mathbb{Z})$ Farey edge for *every* $p \ge 7$ at equality, whose two fully optimized original parent words contain no ties. Third, we give a restricted word reblocking producing the complete daughter optimum fibers from an already optimized original parent word. This is *not* an unrestricted order-only Farey word recursion; the family is special, and its parental input is reconstructible from source coefficients.

We distinguish proved mathematical statements from historical priority. The cited 2010 paper gives complete factorization algorithms, and the 2007 paper studies monomial ideals and minimum-distance diagrams; these precedents must not be represented as novel components here. The exact fixed-p global *total excess factorization count* extremum and the joint authentic-Farey realization have not been located verbatim in the sources checked for this draft; this does not establish historical firstness. In particular, the already-known $(2, 3)$ Apéry extremizing examples in [210] overlap the semigroup family here, but the statistics addressed there concern the proportion and lengths of nonunique factorizations rather than the fixed-p sum of their *numbers*.

84.4.2 Minimum Parikh fibers and the Apéry correspondence

Write $\mathbb{N} = \mathbb{Z}_{\ge 0}$. Let

$$\mathcal{M}_{p,q,s}(r) = \operatorname*{arg\,min}_{\substack{(x,u,v)\in\mathbb{N}^3 \\ px-qu-sv=r}} (x + u + v), \qquad E(p, q, s) = \sum_{r=0}^{p-1} \big(|\mathcal{M}_{p,q,s}(r)| - 1\big).$$

All these sets are nonempty, since $\gcd(p, q, s) = 1$ and the two negative letters generate all residues modulo p. Define $a = p + q$, $b = p + s$ and the numerical semigroup $H = \langle p, a, b\rangle$. Since $a, b > p$, its multiplicity is p.

Proposition 84.12 (Apéry–Parikh bijection)**.** *For every $r \in \{0, \dots, p - 1\}$ let w_r be the least element of H congruent to $-r$ modulo p. Then $\mathcal{M}_{p,q,s}(r)$ is in bijection with the nonnegative (a, b)-factorizations of w_r. Consequently, if $d_H(w)$ is the number of factorizations*

of w using the minimal generating set when $\operatorname{edim}(H) = 3$*, then*

$$E(p,q,s) = \sum_{w \in \operatorname{Ap}(H,p)} \big(d_H(w) - 1\big). \tag{84.20}$$

Proof. The source equation gives the exact identity

$$p(x+u+v) = r + (p+q)u + (p+s)v = r + au + bv.$$

Conversely, if $au+bv \equiv -r \pmod{p}$ and $u, v \geq 0$, then $x = (r+qu+sv)/p$ is a nonnegative integer and yields (84.19). Thus minimizing length is exactly minimizing $au + bv$ in its congruence class. A factorization of an Apéry element cannot use p, since subtracting p would remain inside H. The claimed correspondence and (84.20) follow. □

Remark 84.13. If $q = s$, primitivity forces $\gcd(p,q) = 1$. The unique smallest total negative count k_r ranges over $0, \ldots, p-1$ as r varies, and it has $k_r + 1$ physically distinct splits into (u,v). Hence $E = p(p-1)/2$ in the two-effective-generator symmetric case. The rest of the main extremal theorem concerns genuine embedding dimension three.

84.4.3 The exact two-cost tie count

We give an elementary reduction establishing the exact total count that underlies the extremal theorem. It is a two-dimensional L-shape argument, not a new general factorization algorithm.

After exchanging the negative labels if necessary, write

$$a = gB, \qquad b = gA, \qquad 1 \leq A \leq B, \quad \gcd(A,B) = 1,$$

where $g = \gcd(a,b)$, so the effective weight of (u,v) is $g(Bu+Av)$. The case $A = B = 1$ has already been addressed. For $A < B$, choose the unique integer $h \in \{1, \ldots, A\}$ satisfying

$$Bh \equiv p \pmod{A}, \qquad t = \frac{p - Bh}{A} \in \mathbb{Z}. \tag{84.21}$$

Lemma 84.14 (Exact positive-coin tie mass)**.** *With this notation, if $t \leq 0$ then $E = 0$. If $t \geq 1$, then*

$$p = At + Bh, \qquad E = ht + A\sum_{j \geq 1} (t - jB)_+, \quad (z)_+ = \max(z, 0). \tag{84.22}$$

Moreover, in the positive-t case a canonical set of one cost-minimizing representative per residue is the L-shaped set

$$\mathcal{D} = \big([0,h) \times [0, B+t)\big) \cap \mathbb{Z}^2 \;\cup\; \big([h,A) \times [0,t)\big) \cap \mathbb{Z}^2. \tag{84.23}$$

Every canonical $(u,v) \in \mathcal{D}$ has exactly $1 + \lfloor v/B \rfloor$ tied nonnegative minimum-cost representatives.

Proof. Assume $t \geq 1$. The set (84.23) contains $h(B+t) + (A-h)t = Bh + At = p$ points. Starting with any nonnegative pair (u,v), apply the following moves, preserving the residue of $g(Bu + Av)$ modulo p:

$$\begin{aligned}
&(u,v) \longmapsto (u - A, v + B) && (u \geq A),\\
&(u,v) \longmapsto (u - h, v - t) && (h \leq u < A,\ v \geq t),\\
&(u,v) \longmapsto (u + A - h, v - B - t) && (u < h,\ v \geq B + t).
\end{aligned}$$

The first move preserves weight and reduces u; each other move decreases weight by exactly gp, hence strictly. Iterating terminates with a point of $\mathcal{D}$, and every residue is represented,

since $\gcd(p,a,b)=1$. As $\mathcal{D}$ has exactly p points, its residue representatives are distinct. Since an arbitrary competing pair reduces to a point of $\mathcal{D}$ without an increase in weight, these representatives are globally minimum-cost.

Two nonnegative pairs of the same cost differ by an integer multiple of $(A,-B)$. As a canonical pair has $0 \le u < A$, all other nonnegative representations of that cost are precisely $(u+jA, v-jB)$ for $0 \le j \le \lfloor v/B \rfloor$. Therefore

$$E = h\sum_{v=0}^{B+t-1}\left\lfloor \frac{v}{B}\right\rfloor + (A-h)\sum_{v=0}^{t-1}\left\lfloor \frac{v}{B}\right\rfloor = ht + A\sum_{j\ge 1}(t-jB)_+.$$

Now assume $t=-k\le 0$. A tied Apéry factorization would have two pairs differing by $(A,-B)$, and one of them has $u \ge A \ge h$. If its weight is $w=g(Bu+Av)$, then

$$w-gp = g\big(B(u-h)+A(v+k)\big) \in \langle a,b\rangle.$$

Thus $w-p=(w-gp)+(g-1)p \in H$, contradicting the Apéry property. No ties exist. □

Corollary 84.15 (Tie segments)**.** *For any certified optimum record $z=(x,u,v)$ in a fixed residual fiber, all tied optimum records are exactly the nonnegative integer points in*

$$z + \mathbb{Z}\,\frac{(q-s,\ p+s,\ -(p+q))}{\gcd(p+q,p+s)}. \tag{84.24}$$

They form one consecutive finite interval in the integer parameter.

Proof. Two optimum triples have the same residual and the same literal length. Their difference solves two independent homogeneous integral equations, whose primitive rank-one kernel is the displayed vector. Conversely each nonnegative point along it preserves residual and length. □

84.4.4 Sharp total multiplicity at fixed embedding dimension

Theorem 84.16 (Exact three-generator maximum)**.** *For every integer $p \ge 3$, among all primitive positive sources (p,q,s) for which $H=\langle p, p+q, p+s\rangle$ is minimally three-generated, the largest possible total extra optimum mass is*

$$\boxed{M_3(p) = \begin{cases} \left\lceil \frac{(p-3)^2}{12}\right\rceil, & p \text{ odd},\\ \left\lfloor \frac{(p-3)^2}{12}\right\rfloor, & p \text{ even}. \end{cases}} \tag{84.25}$$

For $p=5$ or $p\ge 7$, equality holds if and only if the unordered coprime-reduced pair of nontrivial costs is $\{2,3\}$:

$$\left\{\frac{p+q}{g}, \frac{p+s}{g}\right\} = \{2,3\}, \qquad g=\gcd(p+q,p+s). \tag{84.26}$$

The exceptional moduli $p=3,4,6$ have $M_3(p)=0$ and nonunique equality descriptions. Every such three-generated source satisfies $E(p,q,s) < p^2/12$, and the constant $1/12$ is asymptotically best possible.

Proof. Minimal generation forces $A \ge 2$: if $A=1$, one nontrivial generator divides the other. With $A<B$ and $\gcd(A,B)=1$, this gives $B\ge 3$. If $t\le 0$, Lemma 84.14 gives $E=0$. On the tie-active branch $t\ge 1$, using the strict comparison of a positive arithmetic sum with the area below its piecewise-linear interpolation yields

$$\sum_{j\ge 1}(t-jB)_+ < \frac{t^2}{2B}.$$

By (84.22), $h = (p - At)/B$, whence

$$E < ht + \frac{At^2}{2B} = \frac{pt - At^2/2}{B} \le \frac{p^2}{2AB} \le \frac{p^2}{12}. \tag{84.27}$$

For $(A, B) = (2, 3)$, the canonical h is 1 when p is odd and 2 when p is even, and $t = (p - 3h)/2$. When $t > 0$, write $t = 3k + \rho$, $0 \le \rho < 3$. The exact sum in (84.22) becomes

$$E = h(3k + \rho) + 3k(k - 1) + 2k\rho. \tag{84.28}$$

Elementary division by 6 in (84.28) gives the two cases in (84.25); when $t \le 0$ the same formula gives the stated zero maxima for the small exceptional cases. The pair $(2, 3)$ is realized by $a = 2(p + 1)$, $b = 3(p + 1)$ for every $p \ge 5$: $\gcd(p, p + 1) = 1$, and $\langle p, a, b\rangle$ is minimally three-generated. For completeness, the exceptional small multiplicities $p = 3, 4$ also admit minimally three-generated examples with zero excess mass, namely $\langle 3, 4, 5\rangle$ and $\langle 4, 5, 6\rangle$; for $p = 6$ one may take $\langle 6, 14, 21\rangle$. Thus the values in (84.25) are attained.

Suppose A, B is any other coprime pair with $2 \le A < B$. If $A = 2$, then $B \ge 5$; otherwise $A \ge 3$ and $B \ge 4$. In either case $AB \ge 10$, so (84.27) gives

$$E < \frac{p^2}{20}. \tag{84.29}$$

For $p \ge 16$ one checks directly that

$$M_3(p) \ge \frac{(p - 3)^2}{12} - 1 > \frac{p^2}{20}.$$

For $p = 13, 14, 15$ the values of M_3 are $9, 10, 12$, respectively, each strictly greater than $p^2/20$. For $3 \le p \le 12$, the only positive-tie pairs obey $p = At + Bh$ with $2 \le A < B$, $\gcd(A, B) = 1$, $1 \le h \le A$ and $t \ge 1$, so there are finitely many possibilities. The exact check is recorded in Table 84.1; it uses only (84.22) and the indicated integer constraints. Hence outside the zero-maximum exceptional cases, only $(A, B) = (2, 3)$ can attain (84.25). □

p	3	4	5	6	7	8	9	10	11	12
$M_3(p)$	0	0	1	0	2	2	3	4	6	6
Best other (A, B)	0	0	0	0	1	1	2	2	3	2

Table 84.1: Finite verification of the equality classification at small multiplicity. A dash-free zero denotes no positive-tie competitor.

84.4.5 An authentic extremal Farey edge at every multiplicity

A genuine directed Farey edge here means: $F \in \mathrm{SL}_3(\mathbb{Z})$ has strictly positive entries, $P = F(1, 1, 1)^{\mathsf{T}}$ is the complete row-sum source, $Q = Fe_1$ is its true inserted-column source, and $C = P + Q$ is the child produced by the legal first-column-to-second-column addition. All source components are positive and every optimum is in the original three physical labels.

Theorem 84.17 (Tie-free parents and maximally tied children)**.** *For every integer $n \ge 0$, put*

$$F_n = \begin{pmatrix} 2 & n + 2 & 1 \\ 1 & n + 2 & 1 \\ 3 & 2n + 3 & 2 \end{pmatrix} \in \mathrm{SL}_3(\mathbb{Z}). \tag{84.30}$$

The authentic parents and child are

$$P_n = (n + 5, n + 4, 2n + 8), \quad Q_n = (2, 1, 3), \quad C_n = (p, p - 2, 2p - 3),\ p = n + 7. \tag{84.31}$$

Both complete original parent optimum words have one record at each fundamental target, whereas

$$E(C_n) = M_3(p) = \Theta(p^2).$$

The projected binary-parent determinant equals $p-4$ *and the joint parent cost determinant equals* $p-1$*. Thus both are positive and unbounded, and the first parent has no* $v=0$ *optimum record at target* 2*.*

Proof. Subtracting the second row of (84.30) from the first gives $(1,0,0)$; the resulting bottom-right minor is $2(n+2)-(2n+3)=1$. This proves positivity and determinant one. Summing rows and selecting the first column gives (84.31). The child effective costs are

$$a_C = p+(p-2) = 2(p-1), \qquad b_C = p+(2p-3) = 3(p-1).$$

Because $\gcd(p, p-1) = 1$ and $p \geq 7$, the three numbers $p, 2(p-1), 3(p-1)$ form a minimal generating set: the two larger generators are not multiples of p, and an expression of $3(p-1)$ as a nonnegative combination of p and $2(p-1)$ would require $p \mid 1$ or $p \mid 3$, an impossibility. Hence $\operatorname{edim}(H) = 3$, its reduced cost pair is $(2,3)$, and Theorem 84.16 gives the asserted equality.

For the first parent set $h = n+5$; then $P = (h, h-1, 2(h-1))$. Put $w = u+2v$. The residual equation $hx-(h-1)w = r$ forces $w \equiv r \pmod{h}$, so $w = r+kh$ with $k \geq 0$ and $x = r+k(h-1)$. For fixed w, the unique shortest split is

$$u = w \bmod 2, \qquad v = \lfloor w/2 \rfloor.$$

Its length strictly increases with k. Consequently the entire actual first-parent optimum word is

$$\mathcal{M}_P(r) = \left\{ \left(r, r \bmod 2, \lfloor r/2 \rfloor\right) \right\}, \qquad 0 \leq r < h. \tag{84.32}$$

In particular $\mathcal{M}_P(2) = \{(2,0,1)\}$, so this parent lacks a binary $v=0$ optimum shadow. For $Q = (2,1,3)$ direct inspection gives

$$\mathcal{M}_Q(0) = \{(0,0,0)\}, \qquad \mathcal{M}_Q(1) = \{(1,1,0)\}.$$

Both parents therefore have exactly one shortest original-labelled record per target. The projected determinant and price determinant calculate as

$$\begin{aligned} 2(n+4)-(n+5) &= n+3 = p-4, \\ (2n+9)\cdot 5 - 3(3n+13) &= n+6 = p-1. \end{aligned}$$

□

84.4.6 Complete physical daughter word and a restricted parent reblocking

Theorem 84.18 (Exact full child fibers)**.** *For* $C = (p, p-2, 2p-3)$*,* $p \geq 7$*, set*

$$t_r = \begin{cases} 0, & r = 0, \\ p+1, & r = 1, \\ r, & 2 \leq r < p. \end{cases} \tag{84.33}$$

Then for each $0 \leq r < p$*, the* entire *set of original-labelled optimum records, including all ties, is*

$$\boxed{\mathcal{M}_C(r) = \left\{ \left(u+2v-\frac{t_r-r}{p}, u, v \right) : u, v \in \mathbb{N},\ 2u+3v = t_r \right\}.} \tag{84.34}$$

Equivalently, writing

$$\varepsilon_r = t_r \bmod 2, \quad k_r = \lfloor t_r/2 \rfloor, \quad u_r^0 = k_r - \varepsilon_r, \quad x_r^0 = k_r + \varepsilon_r - \frac{t_r - r}{p},$$

we have the compressed arithmetic-progression description

$$\mathcal{M}_C(r) = \left\{ (x_r^0 + j, u_r^0 - 3j, \varepsilon_r + 2j) : 0 \le j \le \left\lfloor \frac{u_r^0}{3} \right\rfloor \right\}. \tag{84.35}$$

The optimum length profile is $L_C(0) = 0$, $L_C(1) = p$, $L_C(r) = r$ *for* $2 \le r < p$.

Proof. The child residual decomposes as

$$px - (p-2)u - (2p-3)v = p(x - u - 2v) + (2u + 3v).$$

For a feasible triple let $t = 2u + 3v$. Then $t \equiv r \pmod{p}$ and

$$x = u + 2v - \frac{t - r}{p}, \qquad x + u + v = \frac{r + (p-1)t}{p}.$$

All nonnegative integers except 1 are representable by 2 and 3. Thus the least feasible t in each residue is exactly (84.33); its indicated x is nonnegative. Any larger feasible t strictly raises the length. The full optimum set is therefore (84.34). Solutions of $2u + 3v = t_r$ have $v = \varepsilon_r + 2j$ and $u = k_r - \varepsilon_r - 3j$, proving (84.35) and the length formula. □

Corollary 84.19 (A restricted actual-parent-word transfer)**.** *Let* $Z_P(t) = (t, \varepsilon_t, k_t)$ *denote the consecutive* true original optimized *first-parent marks in* (84.32)*, where* $\varepsilon_t = t \bmod 2$ *and* $k_t = \lfloor t/2 \rfloor$*. Their successive literal increments alternate between* $(1, 1, 0)$ *and* $(1, -1, 1)$*. Extend this alternating word at most four places past its last parent index* $h - 1 = p - 3$*, so that the mark at* $t_r \le p + 1$ *is available. If the resulting mark is* (t_r, ε, k)*, put*

$$z_C^0(r) = \left(k + \varepsilon - \frac{t_r - r}{p},\ k - \varepsilon, \varepsilon \right).$$

Then the complete child fiber follows without a daughter min-plus minimization *as*

$$\mathcal{M}_C(r) = \left\{ z_C^0(r) + j(1, -3, 2) : 0 \le j \le \left\lfloor \frac{k - \varepsilon}{3} \right\rfloor \right\}.$$

The compact form of the full child word is computed with $O(p)$ *integer-arithmetic operations, independent of the number of additional tied optima. Explicitly writing all records requires* $\Theta(p + E(C))$ *output operations.*

Proof. The input parent word supplies $k = \lfloor t_r/2 \rfloor$ and $\varepsilon = t_r \bmod 2$; the alternating increments determine the at-most-four required extensions uniquely. Substitution into (84.35) proves each coordinate of the asserted reblocking, including its nonnegativity and all tie endpoints. This is a direct source-to-child construction on the displayed family, with no optimization step for the daughter. □

Remark 84.20 (Limit of the transfer claim). The two parent words are genuine, independently optimized original-labelled inputs, and the transfer operates on their observed ordered symbols. However, the same alternating pattern can be reconstructed from the explicit source coefficients. Moreover, the second parent is not informationally necessary in this particular family. Hence Corollary 84.19 proves neither information-theoretically essential dependence on *both* parental words nor an unrestricted genuine two-parent optimization-free transfer theorem. The broader question is stated below.

Open question on source-aware Farey transfer

Open problem (unrestricted original-parent-WORD transfer). For an arbitrary genuine, strictly positive unimodular rank-three Farey parent–child step, can the entire original-labelled minimum-length daughter WORD, including every optimal tie, be recovered from the two independently optimized, complete ordered parent WORDs and their source-frame data by a general transfer procedure that does not perform a new daughter minimization? The family-specific bounded-extension reblocking proved in Corollary 84.19 is not such a general theorem; it also does not establish that both parents are informationally necessary.

84.4.7 Verification and scope of the publication claim

All proofs above are symbolic; the reported finite exact-arithmetic checks serve as independent consistency tests, not substitutes for universal proofs. The checks compared complete optimal original-labelled Parikh fibers, including tied records, against the stated formulas using separate nonnegative weighted-residue shortest-path calculations. The reported verification ranges comprise normalized reduced-cost scans for $3 \le p \le 200$, more than 25,000 primitive minimally three-generated direct enumerations, constructive equality checks through $p = 600$, and 150 strictly positive unimodular Farey edges F_n with their complete original parent and child words. These are the stated finite ranges, not an assertion of an exhaustive computational proof; the $p = 3, 4, 6$ exceptions to the equality characterization remain explicit in the theorem.

As to priority, the Apéry description, L-shape basis, and single-element denumerant algorithms are established in earlier literature [208, 209]; the $(2, 3)$ nonunique-Apéry construction also appears in [210, Theorem 4]. Those precursors are treated here as context, not new contributions. What remains to be evaluated by wider specialist literature review is the exact fixed-multiplicity *sum* extremum (84.25), its unique $(2, 3)$ equality characterization, and its genuine positive Farey realization with tie-free parents. An absence of a matching formula in the two directly reviewed papers is not a certificate of historical originality. This is a self-contained *research manuscript*, not a record of journal submission or acceptance; a definitive global literature-priority assessment remains outside the scope of the paper.

84.5 Saturation defects and price-dependent Farey parent synchronization

The preceding Farey-word and lattice-coordinate constructions lead to a source-level question: does synchronizing two already optimized parent records produce the complete optimum structure of a daughter? The following analysis retains the original labelled parent fibers, but distinguishes their raw discrepancy lattice from the translation-invariant difference lattice. This distinction is essential when a cost-sensitive optimum is reconstructed. We use the notation and hypotheses stated below; in particular the parents are genuine, positive unimodular Farey parents rather than abstract independent cyclic words.

Introduction

The usual synchronization question asks when compatible cyclic word structures can be assembled; the arithmetic question considered here is different. We begin with two *already grade-optimal* collections of original Parikh records arising from a single positive unimodular Farey frame. Their synchronized differences are authentic source data, but there is no a priori reason that these differences generate the saturated lattice needed to describe a new child's optimum. The central issue is therefore not the CRT alone: it is how much integral information survives passing from absolute source letters to their translation-invariant

differences, and whether a primitive child direction admits a determinant-minimal complementary source difference.

The results below give both an exact quantitative defect formula and infinite genuine optimized-word obstructions. The abstract group arguments are elementary; the nontrivial source-specific content is the existence of these phenomena with complete optimal parent provenance and, in the index-29 family, a precise price-dependent repair. The latter also shows why retaining the full source letter alphabet without the numerical price cannot determine the new optimal positive ray. Existing Christoffel synchronization work is compared after the proofs; the article does not assert an unrestricted transfer theorem.

84.5.1 Genuine source and CRT synchronization

Let $B \in SL_3(\mathbb{Z})$ have strictly positive integer entries and put $P = B\mathbf{1}$, $Q = Be_j$, and $C = P + Q = (p, q, s)$. For a positive source $R = (p_R, q_R, s_R)$ and a fundamental target $0 \le r < p_R$, an *original physical Parikh record* is a triple $z = (x, u, v) \in \mathbb{Z}^3_{\ge 0}$ satisfying

$$A_R z = p_R x - q_R u - s_R v = r.$$

Its *grade* is $N(z) = x + u + v$. The *complete optimum fiber* at r consists of all grade-minimizing physical records; the complete target-indexed WORD records all these fibers, including ties. Equivalently, minimizing the grade at a fixed r is minimizing the positive defect price

$$(p_R + q_R)u + (p_R + s_R)v = p_R N(z) - r.$$

The two original parent WORDs for P and Q are optimized *independently*, before their Farey sum is formed. The complete optimum fiber is retained as the underlying semantic datum; the cyclic derivative alphabet below is formed from *one consistently selected anchor per residue*, not from the union of differences among every tied optimum. Fix a deterministic lexicographic ordering of the three physical coordinates to select one anchor $z_R(r)$ in each optimum fiber; the general algebraic statements hold for any consistently selected anchors. Denote the parent first coordinates by $c = P_0$ and $w = Q_0$. Each anchor has the canonical periodic extension $z_R(r + p_R) = z_R(r) + e_x$. Write

$$d_R(i) = z_R(i + 1) - z_R(i), \qquad i \pmod{p_R}, \quad R = P, Q.$$

Put $D = \mathrm{diag}(1, -1, -1)$ and interpret the source residual row as $A_R = R^T D$. Since B is unimodular, a primitive simultaneous source normalizer is explicitly $v_0 = DB^{-T}e_j$, which satisfies $A_P v_0 = A_Q v_0 = 1$. The integral parent transporter $T = I - v_0(A_P - A_Q)$ is unimodular: $A_P T = A_Q$ and $\det T = 1 - (A_P - A_Q)v_0 = 1$. Choose any primitive integral basis K_P of $\ker_{\mathbb{Z}} A_P$ (one canonical choice is $DB^{-T}(e_1 - e_3, e_2 - e_3)$); the parent discrepancy coordinates are $\mu(r) = K_P^{-1}(Tz_Q(r) - z_P(r)) \in \mathbb{Z}^2$.

The identification with the daughter lattice also has an explicit unimodular origin. Choose any $j' \ne j$ and set $U = I + e_j e_{j'}^T$, so $U\mathbf{1} = \mathbf{1} + e_j$ and $\det U = 1$. Then $S = DB^{-T}U^{-T}B^T D \in SL_3(\mathbb{Z})$ satisfies $A_C S = A_P$. Consequently S identifies the primitive parent kernel with the primitive daughter kernel, and projection of SK_P onto the (u, v)-coordinates is an integral basis F of $\Lambda(C) = \{(u, v) \in \mathbb{Z}^2 : qu + sv \equiv 0 \pmod{p}\}$, with $|\det F| = p$. The source letters are thus mapped to daughter physical vectors by an explicitly integral lattice isomorphism, not by an assumed real-linear identification.

Lemma 84.21 (Authentic CRT discrepancy)**.** *For all integers r, the synchronized discrepancy increment is*

$$\delta_r = \mu(r + 1) - \mu(r) = K_P^{-1}\big(Td_Q(r \bmod w) - d_P(r \bmod c)\big).$$

Set $g = \gcd(c, w)$ and $L = \mathrm{lcm}(c, w)$. Over one full period $0 \le r < L$, the ordered pair $(r \bmod c, r \bmod w)$ visits each (i, j) with $i \equiv j \pmod{g}$ exactly once and no other pair. Thus the actual physical discrepancy alphabet is precisely $\mathcal{D} = \{\delta_{ij} : i \equiv j \pmod{g}\}$; all letters are source-derived, not hypothetical daughter optimum vectors.

Proof. Subtract the defining equations for $\mu(r+1)$ and $\mu(r)$ and use periodicity of both optimized parent derivatives. The generalized CRT has a solution for $r \equiv i \pmod{c}$, $r \equiv j \pmod{w}$ exactly when $i \equiv j \pmod{g}$ and then a unique solution modulo L. □

On transporting these coordinates through the known full daughter lattice basis, each δ_{ij} is in the integral daughter lattice $\Lambda = \{(x, y) \in \mathbb{Z}^2 : qx + sy \equiv 0 \pmod{p}\}$ of determinant p. The letters may be written in the primitive parent-kernel coordinate lattice $\mathbb{Z}^2$ or, after the integral source-carried basis identification, as physical vectors in Λ; ranks and subgroup indices are unchanged by this lattice isomorphism. We specify the chosen coordinates in each explicit family.

84.5.2 Compressed raw and difference modules

The *raw* parent letter module and its *translation-invariant* difference module are, respectively,

$$L_{\mathrm{raw}} = \langle \delta : \delta \in \mathcal{D} \rangle_{\mathbb{Z}}, \quad L_{\mathrm{diff}} = \langle \delta - \delta' : \delta, \delta' \in \mathcal{D} \rangle_{\mathbb{Z}}.$$

Fix an anchor (c_0, c_0) in every CRT congruence class $c_0 \in \{0, \dots, g-1\}$; denote $a_{c_0} = \delta_{c_0,c_0}$ and put

$$U_i = \delta_{i,c_0} - a_{c_0}, \qquad V_j = \delta_{c_0,j} - a_{c_0} \quad (i \equiv j \equiv c_0 \pmod{g}).$$

Theorem 84.22 (Exact source-sized generator theorem)**.** *The complete full-period translation-invariant discrepancy module satisfies*

$$\boxed{L_{\mathrm{diff}} = \langle U_i, V_j, a_{c_0} - a_0 : \ i, j \ \textit{eligible}, \ 1 \le c_0 < g \rangle_{\mathbb{Z}}.}$$

At most $c + w - g - 1$ nonzero parent-derived integral difference vectors suffice. Hence the rank, Hermite normal form, and exact index (if full rank) can be found using a linear number of integer gcd operations in the two-parent word input size, without expanding L.

Proof. For each class the linearity of K_P^{-1} and the separability of the two parent derivatives imply $\delta_{ij} = a_{c_0} + U_i + V_j$. Every displayed generator is the difference of two actually attainable CRT phase letters. Conversely subtracting two arbitrary letters from classes c_0 and d_0 gives $(a_{c_0} - a_{d_0}) + U_i + V_j - U_{i'} - V_{j'}$, in the subgroup generated on the right. This proves equality. Each of the g classes contributes one zero U and one zero V at its anchor, so there are at most $(c-g)+(w-g)+(g-1) = c+w-g-1$ nonzero listed generators. In rank two, iterative extended gcd produces HNF without ever forming the full CRT letter list. □

Theorem 84.23 (The affine origin and the exact raw-module recovery)**.** *For any finite nonempty integral alphabet $\mathcal{D} \subseteq \Lambda$, choose $a_0 \in \mathcal{D}$. Then*

$$L_{\mathrm{raw}} = L_{\mathrm{diff}} + \mathbb{Z}a_0, \qquad L_{\mathrm{raw}}/L_{\mathrm{diff}} = \langle [a_0] \rangle$$

is cyclic (possibly infinite). In particular, $\mathrm{rank}(L_{\mathrm{raw}}) - \mathrm{rank}(L_{\mathrm{diff}}) \in \{0, 1\}$. *If L_{diff} has rank two and $m = [\Lambda : L_{\mathrm{diff}}]$, then*

$$\boxed{[\Lambda : L_{\mathrm{raw}}] = \frac{m}{\mathrm{ord}_{\Lambda/L_{\mathrm{diff}}}([a_0])}.}$$

Thus one genuine affine anchor supplies exactly the missing cyclic coset of the difference module, and its order measures how much saturation the absolute origin restores.

Proof. Every $a \in \mathcal{D}$ is $(a - a_0) + a_0$ and $a - a_0 \in L_{\mathrm{diff}}$, giving both inclusions for L_{raw}. Therefore the raw quotient is generated by the image of a_0. If the difference module is full rank, use

$$\Lambda/L_{\mathrm{raw}} \cong (\Lambda/L_{\mathrm{diff}})/\langle [a_0] \rangle$$

and count group orders. The rank assertion follows because adjoining one generator raises the rational rank by at most one. □

Remark 84.24. Theorem 84.23 is an elementary group extension, not a novel abstract-group theorem. Its application prevents conflating the signed span of actual parent letters with their translation-invariant differences. The exact quotient of a rank-one difference module by the whole child lattice is not finite, so the finite-order formula cannot be used in that case.

84.5.3 Exact determinant quantization

For any rank-two integral lattice Λ of positive determinant p, every vector z is called Λ-primitive if $\Lambda/\mathbb{Z}z$ is torsion-free (equivalently, z extends to an integral basis of Λ).

Theorem 84.25 (Sharp quotient-order determinant defect)**.** *Let $L \subseteq \Lambda$ be rank two with $m = [\Lambda : L]$, and let $z \in \Lambda$ be Λ-primitive. Put $G = \Lambda/L$ and $o = \operatorname{ord}_G([z])$. Then*

$$\boxed{\det(z, L) = \{\det(z, h) : h \in L\} = p\frac{m}{o}\mathbb{Z}.}$$

In particular the least nonzero determinant attainable with an L-vector is exactly pm/o, and a full determinant-p complement $h \in L$ exists if and only if $[z]$ generates G.

Proof. Primitivity of z gives a basis (z, h_0) of Λ, with $|\det(z, h_0)| = p$. Consequently $\phi_z : \Lambda \to \mathbb{Z}$, $h \mapsto \det(z, h)/p$, is surjective and has kernel $\mathbb{Z}z$. Write $\phi_z(L) = d\mathbb{Z}$ for some $d \geq 1$. By the third isomorphism theorem,

$$\mathbb{Z}/d\mathbb{Z} \simeq \Lambda/(L + \mathbb{Z}z) \simeq (\Lambda/L)/\langle [z] \rangle,$$

whose order equals m/o. Therefore $d = m/o$. All multiples occur because $\phi_z(L)$ is exactly this ideal. Multiplying by p proves the formula and the determinant-p equivalence. □

Corollary 84.26 (HNF and invariant factors)**.** *If L has physical HNF basis $(a, 0), (b, d)$, then for every $z = (x, y)$*

$$\gcd\big(a|y|, |dx - by|\big) = \gcd\{|\det(z, h)| : h \in L\}.$$

If z is Λ-primitive, this number equals pm/o. Further, if $G \cong \mathbb{Z}/d_1\mathbb{Z} \oplus \mathbb{Z}/d_2\mathbb{Z}$ with $d_1 \mid d_2$ and $d_1 > 1$, it is at least pd_1 for every Λ-primitive z. Thus a noncyclic quotient, if it arises in the relevant source category, blocks every *determinant-p complement chosen from parent-letter differences.*

Proof. The integer ideal generated by determinants of z with the two HNF generators has the displayed positive gcd. For the second assertion, $\operatorname{ord}([z]) \leq d_2$, while $m = d_1 d_2$, and Theorem 84.25 gives the inequality. □

Example 84.27 (Eightfold defect on a genuine Farey edge)**.** An authentic positive determinant-one frame is

$$B = \begin{pmatrix} 3 & 6 & 4 \\ 2 & 5 & 3 \\ 8 & 19 & 12 \end{pmatrix}, \quad C = (16, 12, 47).$$

Here the actual parents are $P = (13, 10, 39)$ and $Q = (3, 2, 8)$. Their independently optimized fundamental words are

$$z_P(r) = ((r + 10u_r)/13, u_r, 0), \quad u_r = [9r]_{13}, \qquad z_Q(r) = (r, r, 0) \quad (0 \leq r < 3).$$

For P, the v-coefficient has zero residue modulo 13 and strictly positive price, so $v = 0$ at any optimum; the least u solving $10u \equiv -r \pmod{13}$ is $[9r]_{13}$. For Q, $u + v \equiv r \pmod 3$ and its u price is 5, versus v price 11, making $(r, r, 0)$ optimal for $0 \leq r < 3$. Thus the actual cyclic derivative alphabets are $\{(7, 9, 0), (-3, -4, 0)\}$ and $\{(1, 1, 0), (-1, -2, 0)\}$.

Take $v_0 = (3, -4, 2)^T$, so $A_P v_0 = A_Q v_0 = 1$, and the genuine integer transporter $T = I - v_0(A_P - A_Q)$. In the primitive parent kernel basis $K_P = ((10, 13, 0)^T, (3, 0, 1)^T)$ the compatible original-step CRT calculation gives four physical daughter letters

$$\mathcal{D} = \{(-19, -4), (10, -8), (-16, 0), (13, -4)\}.$$

Their pairwise differences generate exactly

$$L_{\mathrm{diff}} = \langle (32, 0), (3, 4) \rangle_{\mathbb{Z}}, \qquad [\Lambda : L_{\mathrm{diff}}] = 128/16 = 8.$$

For example, $((29, -4), (4, 0))$ is a full daughter lattice basis of determinant 16 giving the same physical alphabet. The primitive positive daughter winner $z_\star = (3, 4)$ has a short certificate within the paper. For $C = (16, 12, 47)$, every strictly positive feasible pair satisfies $v \equiv 12u \pmod{16}$, with price $\Phi(u, v) = 28u + 63v$. For $1 \le u \le 9$, the least positive admissible values of v are, respectively, $12, 8, 4, 16, 12, 8, 4, 16, 12$; direct substitution shows the unique minimum in this range is $(u, v) = (3, 4)$, of price 336. For $u \ge 10$ and $v \ge 1$, the price is at least $28 \cdot 10 + 63 = 343 > 336$. Thus $z_\star$ is the unique strictly positive price minimizer, independently of any finite computational enumeration. It is primitive in the full daughter lattice (in the displayed full basis it has coordinates $(-1, 8)$) and belongs to the difference lattice. Hence its quotient class has order one, and the minimum nonzero determinant with any parent difference is $16 \cdot 8 = 128$; indeed $\gcd(128, 0) = 128$. In contrast, $z' = (4, 0)$ is primitive in Λ, has quotient order eight, and admits determinant 16. This dependence on the quotient class is invisible to the subgroup index alone.

84.5.4 Two authentic infinite obstructions

The next results are derived from fully optimized genuine parent words rather than arbitrarily assigned alphabet vectors. We include the original-parent optimization and exact transporter calculations, so no daughter-optimum oracle or external research note is needed for these proofs.

Theorem 84.28 (Unbounded raw index but rank-one differences)**.** *For every odd $n \ge 1$, let $p = n + 2$ and $k = 4p$, and set*

$$B_n = \begin{pmatrix} p - 2 & 1 & 1 \\ 2p - 5 + k(p - 2) & 2 + k & 2 + k \\ 2p - 5 & 1 & 2 \end{pmatrix} \in SL_3(\mathbb{Z}).$$

Use its authentic second-column parent Q and the original sum parent $P = B_n \mathbf{1}$. Then the complete original parent optimum words give the source-kernel discrepancy alphabet

$$\mathcal{D}_n = \left\{ \left(p - 1, \frac{p - 3}{2} \right), \left(p - 1, \frac{p - 1}{2} \right) \right\}.$$

Both types are attained by genuine CRT phases. Consequently

$$\operatorname{rank} L_{\mathrm{raw},n} = 2, \quad [\mathbb{Z}^2 : L_{\mathrm{raw},n}] = p - 1 = n + 1 \longrightarrow \infty, \quad L_{\mathrm{diff},n} = \mathbb{Z}(0, 1).$$

The latter has rank one, despite the raw module having rank two.

Proof. The displayed matrix is strictly positive and has determinant one by direct expansion (equivalently, integer elementary column and row operations reduce it to the identity frame). Its two parents are

$$P = (p,\ (k + 2)p - 1,\ 2p - 2), \qquad Q = (1, k + 2, 1).$$

As the fundamental period of Q is one, its sole original optimum mark is zero and its cyclic step is $d_Q = e_x$. The first coordinate p of P is odd. Since $s_P = 2p - 2 \equiv -2 \pmod{p}$, for each $0 \le r < p$ the *unique least* nonnegative v with a $u = 0$ feasible mark is

$$v_r = [r \cdot 2^{-1}]_p, \quad z_P(r) = \left(\frac{r + (2p - 2)v_r}{p},\ 0,\ v_r \right).$$

This is genuinely optimal, not a formally chosen parent shadow: the price of one u is $a_P = p+q_P = (k+3)p-1$, while the greatest candidate $u = 0$ price is at most $(p-1)(3p-2)$. For $k = 4p$ their difference is

$$(k+3)p - 1 - (p-1)(3p-2) = p^2 + 8p - 3 > 0.$$

Hence every parental optimum has $u = 0$, and v_r is minimal because each extra p of v strictly raises the cost. Its cyclic differences satisfy

$$\Delta v_r = v_{r+1} - v_r \in \left\{ \frac{p+1}{2}, -\frac{p-1}{2} \right\}, \quad d_P(r) = \left(\frac{1 + (2p-2)\Delta v_r}{p}, 0, \Delta v_r \right);$$

both possibilities occur as the residues v_r form a complete permutation.

We now compute the synchronized *original-parent* transport, rather than choosing daughter rays. Write $A_R = (p_R, -q_R, -s_R)$, take $v = (-k, -1, 1)^T$ (so $A_P v = 1$), and set

$$T = I - v(A_P - A_Q), \qquad A_P T = A_Q.$$

It follows that $Td_Q = Te_x = (1 + k(p-1), p-1, -(p-1))^T$. A primitive parent kernel basis is

$$K_P = (\kappa_1, \kappa_2), \quad \kappa_1 = (k+2-p, 1, -(p+1)/2)^T, \quad \kappa_2 = (2p-2, 0, p)^T.$$

Indeed $A_P\kappa_1 = A_P\kappa_2 = 0$ and their (u, v)-projections have determinant p, the index of the projected kernel congruence; hence they span the full integer parent kernel. Directly solving $K_P\delta = Td_Q - d_P(r)$ gives

$$\delta_r = \begin{cases} (p-1, (p-3)/2)^T, & \Delta v_r = (p+1)/2, \\ (p-1, (p-1)/2)^T, & \Delta v_r = -(p-1)/2. \end{cases}$$

The two vectors differ by $(0, 1)$ and have determinant $p-1$, proving all the stated raw and difference module assertions, with no appeal to a child optimum computation. □

Theorem 84.29 (Rank-one source but a compulsory new daughter letter)**.** *For odd $n \geq 1$, the same $p = n + 2$, $k = 4p$, and the alternative authentic frame*

$$B'_n = \begin{pmatrix} p-2 & 1 & 1 \\ 2p-5+k(p-2) & 2+k & 2+k \\ p-2 & 1 & 2 \end{pmatrix} \in SL_3(\mathbb{Z})$$

with third-column *insertion has two independently optimized parents with $u = 0$ throughout their fundamental words. Nevertheless its entire original-parent discrepancy-step alphabet is only $\mathcal{D}'_n = \{(0, 0), (0, 1)\}$ in primitive kernel coordinates, whereas the genuine child*

$$C = (p+1,\ 2p+1+k(p+1),\ p+3)$$

has exactly one optimum record at every target $0 \leq r < p+1$, namely

$$u_r = r \bmod 2, \quad v_r = \left[-\frac{r-u_r}{2} \right]_{(p+1)/2}, \quad x_r = \frac{r + q_C u_r + s_C v_r}{p+1}.$$

In particular, $u_r = 1$ is compulsory at every odd child target even though the parent optimum words contain only $u = 0$.

Proof. Both strict positivity and determinant one follow by expansion. The original parents are

$$P = (p, (k+2)p - 1, p+1), \qquad Q = (1, k+2, 2).$$

Again the only fundamental Q mark is zero. Because $s_P \equiv 1 \pmod p$, its $u = 0$ optimum at target r is $v_r = [-r]_p$ with $z_P(r) = ((r + (p+1)v_r)/p, 0, v_r)$. Its price is at most $(p-1)(2p+1)$, whereas any $u \geq 1$ has price at least $(k+3)p-1$; their difference at $k = 4p$ is $2p(p+2) > 0$. Thus these are exactly all parent optima.

The cyclic derivative of this complete parental word is

$$d_P(0) = (p, 0, p-1)^T, \qquad d_P(r) = (-1, 0, -1)^T \quad (1 \leq r < p).$$

Indeed $z_P(0) = (0, 0, 0)$, while $z_P(1) = (p, 0, p-1)$; for every subsequent original phase (including the cyclic last-to-first extension) the difference is $(-1, 0, -1)$. The exact positions matter for the ordered WORD, although they do not affect the two-letter set. Take the genuine integral source transport $v = (-1, 0, -1)^T$, which satisfies $A_P v = 1$, and $T = I - v(A_P - A_Q)$. Hence $Td_Q = Te_x = (p, 0, p-1)^T$. One primitive parent kernel basis is

$$K_P = \big((k+3, 1, 1)^T,\ (p+1, 0, p)^T\big).$$

The difference $Td_Q - d_P$ is either the zero vector or its second basis column; hence the *complete genuine* phase alphabet is $(0, 0), (0, 1)$, each type occurring.

For the child write $N = p + 1$, an even number, and observe $q_C \equiv -1 \pmod N$ and $s_C \equiv 2 \pmod N$. Thus a feasible pair of nonnegative letter counts satisfies $u - 2v \equiv r \pmod N$, forcing $u \equiv r \pmod 2$. Choosing the least nonnegative $u_r \in \{0, 1\}$ of this parity and the displayed least $v_r \in [0, N/2 - 1]$ solves the congruence and gives a nonnegative integer x_r. Let $a_C = N + q_C$, $b_C = N + s_C$; the chosen $k = 4p$ gives

$$a_C = 4p^2 + 7p + 2, \quad b_C = 2p + 4, \quad a_C - (N-1)b_C = 2p^2 + 3p + 2 > 0.$$

Increasing u by two incurs at least $2a_C$, more than any possible saving relative to $v_r b_C \leq (N/2 - 1)b_C$. At the fixed minimal $u = u_r$, all other solutions have $v = v_r + jN/2$, $j \geq 0$, and strictly greater price for $j > 0$. Therefore the stated mark is the unique actual daughter optimum, with $u_r = 1$ on odd targets. No unspecified child search or generic optimization theorem is required. □

84.5.5 A genuine fixed-index family with a complete price-wall repair

The preceding obstructions are not impossibility statements about source transfer with access to absolute parent-basis data. We make this distinction exact on an independently optimized infinite family.

Theorem 84.30 (Index 29 and three exact minimizing rays at the wall)**.** *For every integer $t \geq 3$, let*

$$B_t = \begin{pmatrix} 3 & 6 & 4 \\ 2+3t & 5+6t & 3+4t \\ 8 & 19 & 12 \end{pmatrix} \in SL_3(\mathbb{Z})$$

with insertion of the first column. Its genuine parents are $P_t = (13, 10+13t, 39)$ and $Q_t = (3, 2+3t, 8)$, its child is $C_t = (16, 12+16t, 47)$, and its complete original-parent CRT difference lattice has index

$$\boxed{[\Lambda(C_t) : L_{\mathrm{diff},t}] = 29, \qquad L_{\mathrm{raw},t} = \Lambda(C_t).}$$

A full source-frame-defined child lattice basis is

$$f_1 = (29, -4), \qquad f_2 = (4, 0), \qquad \det(f_1, f_2) = 16.$$

Define $z_A = -f_1 + 8f_2 = (3, 4)$ and $z_B = -3f_1 + 22f_2 = (1, 12)$. For the strictly positive primitive child-ray price $\Phi_t(x, y) = (28 + 16t)x + 63y$, the complete set of positive price-minimizing primitive lattice rays is

$$\mathcal{M}_t = \begin{cases} \{(3, 4)\}, & 3 \leq t \leq 13, \\ \{(1, 12), (2, 8), (3, 4)\}, & t = 14, \\ \{(1, 12)\}, & t \geq 15. \end{cases}$$

At $t = 14$ the canonical lex-first ray is $(1, 12)$. Moreover, the full daughter lattice and the complete signed physical CRT letter alphabet are identical for all $t \geq 3$*, even though the optimal ray changes with t.*

Proof. All entries of B_t are positive and its determinant equals one. We first verify that the parent records really are independently optimum. For P_t, the fundamental constraint modulo 13 is $10u \equiv -r \pmod{13}$, so

$$u_r = [9r]_{13}, \quad v_r = 0, \quad x_r = tu_r + \frac{r + 10u_r}{13}, \qquad 0 \leq r < 13$$

is the unique minimum. Indeed every admissible u is $u_r + 13j$ for $j \geq 0$, and $v \geq 0$ contributes an extra $4v$ to the physical grade, while increasing u adds a positive multiple to it. For Q_t the fundamental congruence is $u + v \equiv r \pmod{3}$; the unique optimum is $(x, u, v) = (3r, 0, r)$ for $r = 0, 1, 2$. This follows by comparing the least representatives $u + v = r$ and then observing that each increment by three only raises the price. Since $t \geq 3$, replacing any v by u strictly increases the Q price.

The full P cyclic derivatives have $\Delta u \in \{9, -4\}$ and

$$d_P = (9t + 7, 9, 0)^T \text{ or } (-4t - 3, -4, 0)^T.$$

The full Q derivatives are $(3, 0, 1)^T$ for phases $0, 1$, and $(-5, 0, -2)^T$ for phase 2. Define $v = (3 - 4t, -4, 2)^T$, which obeys $A_P v = 1$, and the authentic integral source transporter $T = I - v(A_P - A_Q)$. Direct multiplication gives

$$Td_Q = \begin{cases} (6 - 4t, -4, 3)^T, & j = 0, 1, \\ (48t - 41, 48, -26)^T, & j = 2. \end{cases}$$

The primitive parent integer kernel basis

$$K_P = \big((10 + 13t, 13, 0)^T,\ (3, 0, 1)^T\big)$$

then yields the four genuine discrepancy-coordinate types

$$(-1, 3), \quad (0, 3), \quad (3, -26), \quad (4, -26).$$

Since $\gcd(13, 3) = 1$, every permitted pairing of source phases occurs in the true CRT period, and all four types occur. Transport into the physical child lattice with the source-frame-defined full basis

$$F = \begin{pmatrix} 29 & 4 \\ -4 & 0 \end{pmatrix},$$

whose determinant is 16 and whose columns satisfy the child congruence. More explicitly, with first-column insertion $j = 1$ and auxiliary index $j' = 2$ in the source-frame map $S = DB_t^{-T}U^{-T}B_t^T D$, direct multiplication gives $\mathrm{pr}_{u,v}(SK_P) = \big((29, -4)^T, (4, 0)^T\big) = F$ for every $t \geq 3$; thus this basis is source-carried rather than chosen after observing the daughter optimum. The complete physical alphabet is exactly

$$\mathcal{D}_t = \{(-17, 4), (12, 0), (-17, -12), (12, -16)\}.$$

Its translation-invariant difference module is therefore

$$L_{\mathrm{diff},t} = \mathbb{Z}(29, -4) + \mathbb{Z}(0, 16), \quad \det L_{\mathrm{diff},t} = 464,$$

whereas $\det \Lambda(C_t) = 16$. This proves the index 29 assertion without computing a daughter optimum. The child congruence $y \equiv 12x \pmod{16}$ is independent of t, because $q_C = 12 + 16t$ and $s_C = 47 \equiv 15 \pmod{16}$; the four physical letters are also independent of t as explicitly calculated above. Furthermore, the genuine letter $(-17, 4)$ has first coordinate not divisible by 29 and therefore lies outside $L_{\mathrm{diff},t}$, whose first coordinates are all multiples

of 29. Since the quotient $\Lambda(C_t)/L_{\mathrm{diff},t}$ has prime order 29, the affine-origin theorem implies $L_{\mathrm{raw},t} = L_{\mathrm{diff},t} + \mathbb{Z}(-17, 4) = \Lambda(C_t)$. Thus a single genuine absolute source letter repairs the full group-theoretic saturation defect, although finding the optimal price chamber still requires the separate certificate below.

It remains to prove the price minimizer assertion using the full source-carried basis, with no ray scan. The physical prices are

$$\Phi_t(z_A) = 336 + 48t, \quad \Phi_t(z_B) = 784 + 16t.$$

For $3 \le t < 14$, the pair (z_A, h_A) with $h_A = f_2 = (4, 0)$ is a full oriented determinant-16 basis. Every positive lattice vector has the form $w = jz_A + mh_A$ with $j, m \in \mathbb{Z}$. Since $w_y = 4j > 0$, necessarily $j \ge 1$. If $m \ge 0$, its price is no smaller than $\Phi_t(z_A)$. If $m = -l < 0$, then $w_x = 3j - 4l > 0$ forces $j \ge l + 1$, while

$$0 < \Phi_t(h_A) = 112 + 64t < 336 + 48t = \Phi_t(z_A),$$

so $\Phi_t(w) \ge (l+1)\Phi_t(z_A) - l\Phi_t(h_A) > \Phi_t(z_A)$. The first chamber therefore has exactly one optimum.

For $t \ge 14$, use instead $h_B = f_1 - 7f_2 = (1, -4)$. The pair (z_B, h_B) has determinant -16, and

$$\Phi_t(h_B) = 16t - 224 \ge 0, \qquad \Phi_t(z_B) - \Phi_t(h_B) = 1008 > 0.$$

Any positive $w = jz_B + mh_B$ has $w_x = j + m > 0$ and $w_y = 12j - 4m > 0$. If $m \ge 0$, the latter inequality implies $j \ge 1$, so $\Phi_t(w) \ge \Phi_t(z_B)$; equality away from $t = 14$ forces $j = 1, m = 0$. If $m = -l < 0$, the first inequality forces $j \ge l + 1$, so $\Phi_t(w) > \Phi_t(z_B)$. For $t > 14$ the unique optimum is z_B. At $t = 14$, $\Phi_t(h_B) = 0$; price equality requires $j = 1$, and positivity permits exactly $m = 0, 1, 2$, yielding $(1, 12), (2, 8), (3, 4)$. Since their first coordinates are $1, 2, 3$, the canonical lex-first point is $(1, 12)$. Every minimizing vector listed is primitive in the full Λ basis (its coefficients are $(1, m)$, or a basis vector). This proves the complete tie-wall classification. □

Remark 84.31. The quotient obstruction in Theorem 84.25 applies only if the complementary direction is required to lie in L_{diff}. The source-derived full basis F has determinant 16, while the difference lattice has determinant 464; this explicit replacement of a non-saturated letter subgroup by the full carried integer basis is the constructive mechanism behind the repair. No unrestricted $P, Q \mapsto C$ optimal-WORD algorithm is asserted.

Corollary 84.32 (Necessity of numerical price data)**.** *No deterministic rule that uses only the complete synchronized* unordered *physical CRT discrepancy alphabet and the full daughter congruence lattice, but ignores the positive price pair* $(p + q, p + s)$*, can return the correct globally minimizing positive child ray for all genuine positive unimodular Farey frames. This remains true even if the rule knows the translation-invariant and raw integer subgroup generated by the letter alphabet, their HNF and the full price-independent convex hull.*

Proof. For the genuine family of Theorem 84.30, all the listed inputs to such a rule remain exactly the same for every $t \ge 3$. In contrast the strictly positive true prices select $(3, 4)$ when $3 \le t \le 13$ and $(1, 12)$ when $t \ge 14$, with three ties at the separating wall $t = 14$. A rule using no numerical price data necessarily returns the same ray on all these frames and fails in at least one chamber. □

Remark 84.33. The rule excluded by Corollary 84.32 is deliberately narrow. It may not use the actual positive price, other changing frame data, nor the ordered coordinates of the original complete parental optimal records. No impossibility for the unrestricted two-parent WORD transfer is claimed.

Finite verification and scope

All universally quantified results are proved symbolically above; finite verification serves only as an independent consistency check. Exact-integer checks covered source/child unimodular transport in 960 genuine positive Farey-frame/flag cases and the determinant-ideal formula in 20,000 independently generated primitive lattice instances. The complete four-letter index-eight example was checked against separately optimized parent words. Further finite checks covered 25 odd parameters in each unbounded family and 37 parameters in the fixed-index family, comparing original parent optimizers and complete child residue prices with modular shortest-path calculations. These are the reported finite validation ranges, not exhaustive checks of every parameter or substitutes for the symbolic proofs. The rank-one parent-difference obstructions do not imply a no-go result for arbitrary source-aware transfer, which is not claimed here.

Contributions and novelty

The paper-specific contributions are the exact source-sized presentation of the synchronized parent-difference module, the quotient-order determinant obstruction in the genuine optimized-Farey setting, the two independently optimized infinite obstruction families, and the fixed-index family with an exact three-way price wall and its restricted price-blind selection impossibility. The underlying generalized CRT, lattice-index, Hermite/Smith normal form, and cyclic quotient arguments are classical and are not claimed as new in isolation.

Classical superimposition and synchronization of Christoffel words concern compatible word assemblies and discrete segments [205, 206]. More recent Cayley-seed synchronization investigates gap-free superposition of equal-length Christoffel conjugates [207]. Our optimized-parent input is a different, price-constrained target-indexed word category. Any broader priority comparison concerning the primitive daughter lattice, subgroup quotient, and essential new letters requires theorem-level comparison with that literature; complete external novelty certification is not asserted here.

Open question on source-aware Farey transfer

It remains open in the scope of this work to develop a general, source-aware optimal-WORD transfer theorem from the complete *ordered* original parent records to the complete daughter optima, retaining the numerical daughter price data. The restricted impossibility in Corollary 84.32 applies only to selectors that omit the numerical price and use the specified unordered source alphabet and price-independent lattice data; it does not answer the unrestricted transfer question. A further investigation of source-aware transfer would therefore have to retain the distinction between complete optimum fibers and selected cyclic anchors.

Chapter 85

Quadratic Swap Curvature and Completion Returns

85.1 Quadratic chronology and Farey curvature

We now compare the two opposite two-letter chronologies 10 and 01. If $q(s) = q$, then

$$q(s10) = XYq, \qquad q(s01) = YXq.$$

Theorem 85.1 (Exact adjacent-swap factorization)**.** *Let p be a palindromic base, $P = \mu(p)$, and define m, x as above. Then for every word s,*

$$\boxed{Q_P(q(s10)) - Q_P(q(s01)) = 12F(s)R_p(s).} \tag{85.1}$$

Equivalently,

$$Q_P(q(s10)) - Q_P(q(s01)) = -12F(s)\det(\Psi(p), \Psi(s)). \tag{85.2}$$

Proof. Write

$$P = \begin{pmatrix} A & B \\ B & D \end{pmatrix}, \qquad q = \begin{pmatrix} q_1 \\ q_2 \end{pmatrix}.$$

A direct multiplication gives

$$(XY)^T P(XY) - (YX)^T P(YX) = \begin{pmatrix} 48B + 24D & -24A + 6D \\ -24A + 6D & -24A - 12B \end{pmatrix}.$$

The associated quadratic polynomial factors as

$$\begin{aligned} q^T\big((XY)^T P(XY) - (YX)^T P(YX)\big)q \\ = 12(2q_1 + q_2)\big((2B + D)q_1 - (2A + B)q_2\big). \end{aligned}$$

Now $2q_1 + q_2 = F(s)$, $x = 2A + B$, and

$$2B + D = m - 2x.$$

Therefore the second factor is

$$(m - 2x)q_1 - xq_2 = mq_1 - xF(s) = R_p(s),$$

which proves equation (85.1). Equation equation (85.2) follows from equation (84.10). □

Corollary 85.2 (Word-level swap factorization)**.** *For every word s,*

$$F(s10p01s^R) - F(s01p10s^R) = 12F(s)R_p(s).$$

Equivalently, after the standard $0(\cdot)1$ wrapping, the same identity is a difference of upper-right Cohn matrix entries.

Proof. Apply theorem 84.11 with $t = s10$ and $t = s01$, noting that $(s10)^R = 01s^R$ and $(s01)^R = 10s^R$. The wrapped statement follows from equation (84.18). □

Remark 85.3 (Relation to flip methods). Lagisquet et al. [82] obtain sign and equality rules for simple and composite flips of a related matrix-word m-value. The flip idea itself is therefore classical background. The point of theorems 85.1 and 85.2 is the closed base-dependent factorization of the palindromic-sandwich defect into $12F(s)$ times a single determinant residual.

Corollary 85.4 (Farey sign of the swap)**.** *The sign of the quadratic chronology defect is the sign of*

$$\frac{q_1(s)}{F(s)} - \frac{x}{m}.$$

More precisely,

$$Q_P(q(s10)) > Q_P(q(s01)) \iff \frac{q_1(s)}{F(s)} > \frac{x}{m},$$

and the inequality reverses on the other side.

Proof. Since $F(s), m > 0$,

$$R_p(s) = mF(s)\left(\frac{q_1(s)}{F(s)} - \frac{x}{m}\right).$$

Apply theorem 85.1. □

The zero set is rigid because both Farey coordinate vectors are primitive.

Theorem 85.5 (Zero-curvature return rigidity)**.** *For every word s,*

$$\boxed{Q_P(q(s10)) = Q_P(q(s01)) \quad \iff \quad s = p.} \tag{85.3}$$

No palindromicity hypothesis on s is required.

Proof. By theorem 85.1, equality is equivalent to $R_p(s) = 0$, since $F(s) > 0$. Thus $\Psi(s)$ is rationally parallel to $\Psi(p)$. Both are primitive integer vectors with positive coordinates. Hence the rational proportionality factor is positive, and primitivity forces it to be 1. Therefore

$$\Psi(s) = \Psi(p).$$

Because C is unimodular, this gives $q(s) = q(p)$. By the exact decoder in theorem 84.1, equality of word states implies equality of the underlying words. Hence $q(s) = q(p)$ implies $s = p$. The converse is immediate from $R_p(p) = 0$. □

Example 85.6 (The base $p = 0$)**.** Take $p = 0$. Then

$$P = X, \qquad (x, m) = (5, 13), \qquad x^2 + 1 = 2m,$$

and

$$R_p(s) = 13q_1(s) - 5F(s).$$

The first few values illustrate the exact sign and zero laws:

s	$F(s)$	$R_p(s)$	$Q_P(q(s10)) - Q_P(q(s01))$
ε	5	1	60
0	13	0	0
1	29	11	3828
00	34	−1	−408

The unique zero in these rows occurs at the base progenitor $s = p = 0$, as theorem 85.5 predicts.

Remark 85.7 (Local, not global, Markoff rigidity). Theorem 85.5 is a same-progenitor statement. It does not assert that two arbitrary distinct Christoffel words with the same Markoff number have a common progenitor to which the theorem can be applied. Thus it is independent of, and does not prove, the Frobenius uniqueness conjecture.

85.2 Kernel lifts and quadratic returns

The preceding two coordinate systems are related more tightly than the common residual notation suggests. We first express the relative quadratic coordinates after the two terminal chronologies directly in terms of $\Psi(s)$.

Theorem 85.8 (Terminal transport formula)**.** *Let*

$$\Psi(s) = \begin{pmatrix} a \\ f \end{pmatrix}, \qquad R = ma - xf.$$

Then

$$(\ell, \delta)(q(s10)) = (3mf + R,\ f), \tag{85.4}$$

and

$$(\ell, \delta)(q(s01)) = (3mf - R,\ -f). \tag{85.5}$$

Proof. Write $q(s) = (a, b)^T$, so $f = 2a + b$. Directly from XYq and YXq one obtains

$$\delta(XYq) = f, \qquad \delta(YXq) = -f.$$

For the first coordinates, substitute $P = \left(\begin{smallmatrix} A & B \\ B & D \end{smallmatrix}\right)$, $m = 4A + 4B + D$ and $x = 2A + B$ into $\ell = v^T P(\cdot)$; simplification gives

$$\ell(XYq) = 3mf + (ma - xf),$$

$$\ell(YXq) = 3mf - (ma - xf).$$

These are equations (85.4) and (85.5). □

Now consider the Diophantine completion equation

$$ma - xf = R. \tag{85.6}$$

Its integer solution set is invariant under the primitive kernel translation

$$(a, f) \longmapsto (a + x, f + m). \tag{85.7}$$

Theorem 85.9 (Kernel lift equals quadratic-return translation)**.** *Fix a residual R. Under the terminal* 10 *transport, the kernel lift equation* (85.7) *induces*

$$(\ell, \delta) \longmapsto (\ell, \delta) + (3m^2, m). \tag{85.8}$$

Under the terminal 01 *transport it induces*

$$(\ell, \delta) \longmapsto (\ell, \delta) + (3m^2, -m). \tag{85.9}$$

Proof. The kernel lift leaves R invariant:

$$m(a + x) - x(f + m) = ma - xf = R.$$

Apply theorem 85.8. On the 10 branch,

$$(3m(f + m) + R, f + m) - (3mf + R, f) = (3m^2, m).$$

On the 01 branch,

$$(3m(f + m) - R, -f - m) - (3mf - R, -f) = (3m^2, -m).$$

□

Remark 85.10 (Ambient lattice versus word orbit). The kernel translate $(a, f) \mapsto (a+x, f+m)$ is an operation on the ambient Diophantine completion lattice. A translated lattice point need not itself belong to the Cohn word orbit. The theorem is therefore an exact lattice-transport identity; theorem 85.11 records the distinguished return points that do arise from the base word.

The base state itself has $\Psi(p) = (x, m)$ and $R_p(p) = 0$; in particular, $\Psi(p)$ is the primitive homogeneous kernel vector. Applying the two terminal transports to this distinguished kernel point gives the fundamental blocks $p10$ and $p01$.

Corollary 85.11 (Fundamental quadratic returns)**.** *For the two blocks*

$$C_1 = p10, \qquad C_0 = p01,$$

one has

$$(\ell, \delta)(q(C_1)) = (3m^2, m), \qquad (\ell, \delta)(q(C_0)) = (3m^2, -m). \tag{85.10}$$

Consequently

$$m \mid \ell(q(C_i)), \qquad m \mid \delta(q(C_i)), \qquad i = 0, 1.$$

Proof. Set $(a, f) = (x, m)$ and $R = 0$ in theorem 85.8. □

This is the promised kernel/return correspondence: a homogeneous translation in the completion lattice becomes a fixed Euclidean translation in the relative quadratic plane.

Chapter 86

Characteristic Parents, Centered Fibers, and Reversal Defects

86.1 Characteristic roots and Farey parents

The characteristic congruence in theorem 84.3 determines the oriented Farey parents of the pair (m, x) explicitly.

Theorem 86.1 (Characteristic root determines the oriented Farey parents)**.** *Assume*

$$m > x > 0, \qquad \gcd(m, x) = 1, \qquad x^2 + 1 = km.$$

Then

$$\boxed{(c, u) = (x, k), \qquad (v, d) = (m - x, x - k)} \tag{86.1}$$

satisfy

$$(m, x) = (c, u) + (v, d),$$

and the oriented unit-determinant relations

$$mu - xc = 1, \qquad xv - md = 1.$$

Moreover, they are the unique such decomposition with $0 < c < m$ *and* $0 < v < m$*. (The boundary case* $(m, x) = (2, 1)$ *has* $d = 0$*; otherwise the displayed parent coordinates are positive.) Hence the two unit-residual lifts are*

$$(k, x), \qquad (-(x - k), -(m - x)). \tag{86.2}$$

Proof. The characteristic identity gives $x^2 - km = -1$. Therefore the displayed first pair satisfies

$$mk - x \cdot x = 1.$$

For the second pair,

$$x(m - x) - m(x - k) = mk - x^2 = 1,$$

and their sum is (m, x). Thus the displayed pairs have the required oriented unit-determinant relations.

For uniqueness, any normalized first parent satisfies $xc \equiv -1 \pmod{m}$. Since $x^2 \equiv -1 \pmod{m}$ and x is invertible modulo m, one gets $c \equiv x \pmod{m}$. The condition $0 < c < m$ forces $c = x$, and then $u = k$. Similarly $xv \equiv 1 \pmod{m}$ gives $v \equiv -x \equiv m - x \pmod{m}$; the condition $0 < v < m$ forces $v = m - x$, and then $d = x - k$. Finally, $k = (x^2+1)/m \leq x$, with equality only for $(m, x) = (2, 1)$, which proves the stated positivity qualification. □

Remark 86.2. The arithmetic congruence $x^2 \equiv -1 \pmod{m}$ and the classical Farey/Cohn correspondence are part of the established Markoff literature. The role of theorem 86.1 here is to identify the characteristic data with the same completion kernel used in theorem 85.9, so that the local Farey and quadratic-return descriptions share one coordinate system.

86.2 Centered fibers and reversal defects

The completion viewpoint also gives a compact reconstruction statement on a fixed F-fiber. This part does not require the base word to be palindromic.

Theorem 86.3 (Centered-fiber reconstruction)**.** *Let s be any word with $F(s) = m$, and write*

$$\mu(s) = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

$$r = q_1(s) = 2A + C, \qquad r' = q_1(s^R) = 2A + B.$$

Then

$$rr' + 1 = Am. \tag{86.3}$$

Thus, with

$$K = \frac{rr' + 1}{m},$$

the full matrix is reconstructed from (m, r, r') as

$$\boxed{\mu(s) = \begin{pmatrix} K & r' - 2K \\ r - 2K & m - 2r - 2r' + 4K \end{pmatrix}.} \tag{86.4}$$

Proof. Since $F(s) = v^T \mu(s)^T v$,

$$m = 4A + 2B + 2C + D.$$

Also $\det \mu(s) = AD - BC = 1$. Hence

$$\begin{aligned} rr' + 1 &= (2A + C)(2A + B) + 1 \\ &= 4A^2 + 2AB + 2AC + BC + AD - BC \\ &= A(4A + 2B + 2C + D) \\ &= Am. \end{aligned}$$

Thus $A = K$. The identities $r = 2A+C$ and $r' = 2A+B$ give $C = r-2K$ and $B = r'-2K$, and the formula for D follows from the equation for m. □

Theorem 86.4 (Wrapped trace defect equals reversal defect)**.** *Under the hypotheses of theorem 86.3, let*

$$H = \mu(0s1) = X\mu(s)Y.$$

Then

$$H_{12} = F(s) = m, \tag{86.5}$$

and

$$\boxed{\mathrm{Tr}(H) - 3H_{12} = r - r'.} \tag{86.6}$$

Consequently

$$\boxed{\mathrm{Tr}(\mu(0s1)) = 3\mu(0s1)_{12} \quad \Longleftrightarrow \quad s = s^R.} \tag{86.7}$$

Proof. A direct multiplication of $X\mu(s)Y$ gives $H_{12} = 4A + 2B + 2C + D = m$ and

$$\mathrm{Tr}(H) - 3H_{12} = C - B.$$

But

$$C - B = (r - 2A) - (r' - 2A) = r - r'.$$

Thus the trace identity is equivalent to $r = r'$. By theorem 86.3, $r = r'$ is equivalent to $B = C$, hence to symmetry of $\mu(s)$. By theorem 84.2,

$$\mu(s)^T = \mu(s^R).$$

Hence $\mu(s) = \mu(s^R)$, so in particular $q(s) = q(s^R)$. Injectivity of the word-state map from theorem 84.1 gives $s = s^R$. □

86.3 Relation to the classical Markoff setting

When p is a central palindrome, the word $0p1$ is the corresponding lower Christoffel word, and the matrices above lie in the standard Cohn–Markoff parametrization initiated in Cohn's work on Markoff forms and primitive words [80, 81]. Modern accounts identify Christoffel/Farey data with Markoff triples and study the resulting order and uniqueness questions [79, 83, 82]. The results of this chapter add a local completion layer to that picture.

There are three distinct structures:

(a) the classical tree coordinate, which locates a Christoffel/Cohn object by a Farey fraction;

(b) the completion residual $R_p(s)$, which measures the determinant displacement of an arbitrary word state from a fixed palindromic base;

(c) the quadratic chronology defect, which compares the two opposite terminal orders 10 and 01.

Theorem 85.1 identifies (b) and (c) exactly. In particular, the swap energy has no additional nonlinear zero set: on the word orbit, zero curvature is exactly the base state itself. Theorem 85.9 then identifies the completion kernel with the quadratic return lattice.

This local rigidity should not be confused with global Markoff injectivity. In a global collision, the two competing branches need not have the same progenitor. The result proved here is deliberately local: once a common progenitor is fixed, the opposite chronology is completely controlled by a single Farey determinant; no global core-identification theorem is claimed.

86.4 Return to the monograph's completion language

The Cohn–Markoff results close an arithmetic loop rather than introduce a new automata hypothesis. In the rational-balance chapters, a residual chose a canonical completion modulo a primitive kernel. Here a determinant residual again measures displacement from a base, relative coordinates again expose a completion lattice, and terminal chronology transports the homogeneous kernel into explicit opposite return vectors. The shared theme is therefore not the particular monoid but the separation of a canonical completion section from its unimodular translation freedom. With the arithmetic consequence branches closed, Part VII returns to fixed semantics and asks what chronology, topology, and physical transport can still vary inside a completion fiber.

86.5 Primitive denominator spectra and quadratic swap returns

The completion coordinates of Cohn words have an arithmetic refinement: a primitive vector together with a unimodular word matrix determines a unique symmetric determinant-one rational form. Its denominator, cancellation spectrum and local densities quantify precisely when the form is integral. Adjacent-swap returns then reconnect this arithmetic structure with the quadratic chronology already developed in this branch.

86.5.1 Introduction

Matrix words, primitive lattice vectors, positive binary quadratic forms, and palindromes occur throughout the classical Cohn–Markoff and Christoffel theories; see Cohn [81] and Reutenauer [83]. Our question is more specific: given an integral unimodular word state

$M^T u$, can one recover a canonical determinant-one symmetric quadratic observable, determine *exactly* when it is integral, and prescribe the unique word at which two adjacent orders have equal quadratic value?

A principal contribution here is the exact finite cancellation spectrum and a universal Euler-product frequency law in the positive primitive cone. Both build on elementary local quadratic-form and lattice counting, whose general theory is classical. We also retain three matrix-word statements that are logically distinct: (i) an integral denominator identity for an arbitrary $SL_2(\mathbb{Z})$ matrix; (ii) a common-factor condition on a whole family of swap quadratic forms; and (iii) a zero-set statement on an injectively decoded *word orbit*. The last does not follow from a discriminant-square test alone. The proofs are elementary and self-contained. The commutator condition belongs to classical Markoff theory, and we distinguish it explicitly from the canonical exact-denominator formula and from the orbit-level return statements. Existing Markoff quadratic-form theory is extensive, including [214, 215]; no claim of historical firstness of the remaining statements is made.

Write $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, so $\det(a, b) = -a^T J b$. For rational symmetric P, put $Q_P(v) = v^T P v$. For a rational matrix P, let $\mathrm{den}(P)$ denote the smallest positive integer D such that DP has integral entries.

Contributions and novelty

The paper develops five connected results with distinct scopes. First, it gives a canonical, unique determinant-one symmetric rational reconstruction and an exact least-denominator formula for any primitive seed with positive pairing. Second, for nonsymmetric nonnegative unimodular matrices it determines every attainable cancellation factor on the strictly positive primitive cone, including the obstruction to primes 3 (mod 4), the bound on the 2-adic factor, and the even-diagonal exception. Third, it computes the full local-density distribution of those factors and an Euler-product natural-density law with an effective lattice-point error term. Fourth, it connects the classical parabolic Fricke commutator condition to a universal common-factor property of swap quadratic observables. Fifth, with positivity and injective word-state decoding, it constructs quadratic observables having one prescribed swap-zero word and retains the integrality–palindrome criterion.

The local splitting arguments, primitive lattice counting, Fricke trace identity, and classical Markoff and palindrome background are not claimed as new techniques or discoveries. The paper's contribution is the stated combination of exact denominator, spectrum, density, common-factor and return statements in their respective models. No historical priority claim beyond the explicitly proved assertions is inferred from a limited literature comparison.

86.5.2 Canonical reconstruction and its exact denominator

Theorem 86.5 (Unique canonical quadratic reconstruction)**.** *Let* $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) \in SL_2(\mathbb{Z})$ *and let* $u \in \mathbb{Z}^2$ *be primitive with*

$$q = M^T u, \qquad f = u^T q > 0, \qquad z = Ju.$$

Then

$$P(M, u) = \frac{qq^T + zz^T}{f} \tag{86.8}$$

is the unique symmetric rational matrix of determinant 1 *satisfying* $Pu = q$*. It is positive definite and its least integral denominator is*

$$\boxed{\mathrm{den}(P(M, u)) = \frac{f}{\gcd(f, B - C)}.} \tag{86.9}$$

Proof. The vectors q and z are independent because $\det[q, z] = q^T u = f > 0$. Equation (86.8) is therefore positive definite and has determinant $\det[q, z]^2 / f^2 = 1$. Since

$z^T u = 0$ and $q^T u = f$, we have $Pu = q$. Any other symmetric solution is of the form $P + tzz^T$: the difference must annihilate u, so its image is in $\mathbb{Q}z$. The rank-one determinant lemma then yields

$$\det(P + tzz^T) = 1 + t\, z^T \operatorname{adj}(P) z.$$

The last scalar is strictly positive, forcing $t = 0$ for determinant 1.

For the denominator calculation, extend the primitive u to an integral unimodular basis $T = [u, t] \in SL_2(\mathbb{Z})$. Set

$$T^T M^T T = \begin{pmatrix} f & h \\ g & k \end{pmatrix}, \qquad T^T P T = \begin{pmatrix} f & g \\ g & \frac{g^2+1}{f} \end{pmatrix}.$$

The second matrix has the stated first column and determinant one. Integral unimodular congruence preserves the least common denominator. Consequently

$$\operatorname{den}(P) = \frac{f}{\gcd(f, g^2 + 1)}.$$

From $fk - gh = 1$ we obtain $g^2 + 1 \equiv g(g - h) \pmod{f}$. Also $\gcd(f, g) = 1$, so $\gcd(f, g^2 + 1) = \gcd(f, g - h)$. Finally,

$$g - h = u^T (M - M^T) t = (B - C) \det(u, t) = B - C.$$

Substitution completes the proof. □

Corollary 86.6 (Integrality is symmetry in the positive cone)**.** *If $M \in SL_2(\mathbb{Z}_{\geq 0})$ and $u = (r, s)^T$ is primitive with $r, s > 0$, then*

$$P(M, u) \in \operatorname{Sym}_2(\mathbb{Z}) \quad \Longleftrightarrow \quad B = C.$$

In the integral case $P(M, u) = M$.

Proof. We have $f = Ar^2 + (B + C)rs + Ds^2 \geq A + B + C + D > |B - C|$, since $A, D \geq 0$ and $AD - BC = 1$ imply $A + D > 0$. Hence $f \mid (B - C)$ exactly when $B = C$. Theorem 86.5 yields integrality. If M is symmetric, it is positive definite (its diagonal entries are positive and its determinant is one), and it satisfies $Mu = M^T u = q$, so uniqueness gives $P = M$. □

Remark 86.7 (Sharpness of the seed assumption). The strict positivity of both seed coordinates cannot be discarded. With $u = (1, 0)^T$ and $M = \left(\begin{smallmatrix} 1 & 1 \\ 2 & 3 \end{smallmatrix}\right)$ one obtains the integral form $P = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ although $B \neq C$.

Corollary 86.8 (Palindrome detection)**.** *Let $G, H \in SL_2(\mathbb{Z}_{\geq 0})$ be symmetric, and suppose the matrix word map $w \mapsto M(w)$ is injective. For a positive primitive u, put $P_w = P(M(w), u)$. Then*

$$P_w \text{ has integral entries} \quad \Longleftrightarrow \quad w = w^R.$$

Proof. The generators are symmetric, hence $M(w)^T = M(w^R)$. By Corollary 86.6, integrality is equivalent to $M(w) = M(w)^T = M(w^R)$, and injectivity finishes the proof. □

86.5.3 The complete primitive cancellation spectrum

For the remainder of this section let

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in SL_2(\mathbb{Z}_{\geq 0}), \qquad B \neq C, \qquad k = |B - C| > 0,$$

$$F_M(X, Y) = AX^2 + (B + C)XY + DY^2.$$

For every strictly positive primitive column $u = (x, y)^T$, put $c_M(u) = \gcd(F_M(x, y), k)$; Theorem 86.5 gives $\operatorname{den} P(M, u) = F_M(x, y)/c_M(u)$. Write v_ℓ for the usual ℓ-adic valuation.

Theorem 86.9 (Attainable cancellation factors)**.** *The exact set of attained cancellation factors is*

$$\left\{ c : \begin{array}{l} c \mid k, \quad v_2(c) \le 1, \\ \ell \mid c,\ \ell \text{ odd prime} \Rightarrow \ell \equiv 1 \pmod 4, \\ A, D \text{ both even} \Rightarrow 2 \mid c \end{array} \right\}. \tag{86.10}$$

Each admissible c is attained by infinitely many strictly positive primitive vectors u. The exceptional even-diagonal case forces $v_2(k) = 1$.

Proof. Choose an integral unimodular basis $T = [u, t]$ and use the proof of Theorem 86.5: in its notation,

$$\gcd(F_M(u), k) = \gcd(f, g^2 + 1).$$

An odd prime divisor of the right-hand side must admit a square root of -1, hence is $1 \pmod 4$. Also $v_2(g^2 + 1) \le 1$: if g is even then $g^2 + 1$ is odd, while if g is odd then $g^2 + 1 \equiv 2 \pmod 8$. Thus no cancellation factor can contain 4. If k is even, then $B + C$ is even and $F_M(x, y) \equiv Ax + Dy \pmod 2$. When A, D are both even, F_M is always even. Since $AD - BC = 1$, in this case B, C are odd and $4 \nmid B - C$: otherwise $BC \equiv 1 \pmod 4$ contradicts $AD - BC = 1$. This proves necessity.

For sufficiency fix $c \mid k$ as in (86.10). At an odd prime power $\ell^e \| k$, the quadratic discriminant

$$\Delta = (B + C)^2 - 4AD = (B - C)^2 - 4 \equiv -4 \pmod{\ell^e}$$

is a unit. If $\ell \equiv 3 \pmod 4$, the form is anisotropic on nonzero vectors modulo ℓ, and its value is a unit on every primitive class. If $\ell \equiv 1 \pmod 4$, its nondegenerate split reduction can be changed by an invertible linear substitution over $\mathbb{Z}/\ell^e$ to a unit times XY. Every value of $v_\ell(F_M)$ between 0 and e (truncated at e) therefore occurs on a primitive class. At 2, either A, D are both even, when $v_2(k) = 1$ and every primitive class gives $v_2(c) = 1$; or the nonzero linear form $Ax + Dy \pmod 2$ has primitive classes of both parities. In the latter case no primitive class can yield $v_2(F_M) \ge 2$ when $4 \mid k$, by the necessary condition just established. Thus every specified admissible local valuation is attained. CRT supplies a primitive residue class $(a, b) \pmod k$ with $\gcd(F_M(a, b), k) = c$.

Finally every class $(a, b) \pmod k$ with $\gcd(a, b, k) = 1$ contains infinitely many positive primitive integer vectors. Fix any positive $x \equiv a \pmod k$. For primes dividing x and k, the prescribed $b \pmod k$ is nonzero; for the remaining finitely many primes dividing x, choose $y \pmod \ell$ to be nonzero, in addition to $y \equiv b \pmod k$. CRT gives infinitely many such positive y, all coprime to x. The cancellation factor is unchanged modulo k. □

86.5.4 Exact local densities and coefficient universality

Let

$$\mathcal{V}_X = \{(x, y) : 1 \le x, y \le X,\ \gcd(x, y) = 1\}, \qquad N_c(X) = \#\{u \in \mathcal{V}_X : c_M(u) = c\}.$$

For each $\ell^e \| k$, let $\pi_{\ell,e}(t)$ be the following probability distribution on the truncated valuation $t = v_\ell(c)$ for a uniformly chosen primitive pair modulo ℓ^e.

local case	$\pi_{\ell,e}(t)$
$\ell \equiv 3 \pmod 4$	$\pi(0) = 1$
$\ell \equiv 1 \pmod 4$	$\pi(0) = \dfrac{\ell - 1}{\ell + 1}, \quad \pi(t) = \dfrac{2(\ell - 1)}{(\ell + 1)\ell^t}\ (1 \le t < e), \quad \pi(e) = \dfrac{2}{(\ell + 1)\ell^{e-1}}$
$\ell = 2,\ A, D$ both even	$e = 1, \quad \pi(1) = 1$
$\ell = 2$, otherwise	$\pi(0) = \dfrac{2}{3}, \quad \pi(1) = \dfrac{1}{3}.$

All unlisted values have probability zero. For $c \mid k$, define

$$\rho_{M,k}(c) = \prod_{\ell^e \| k} \pi_{\ell,e}(v_\ell(c)), \tag{86.11}$$

with value zero when $c \nmid k$. For $k = 1$, the empty product is 1.

Theorem 86.10 (Exact cancellation densities)**.** *For each fixed nonsymmetric $M \in SL_2(\mathbb{Z}_{\geq 0})$ and each positive integer c,*

$$\boxed{N_c(X) = \frac{6}{\pi^2}\rho_{M,k}(c)X^2 + O_{M,k}(X\log(2X)).} \tag{86.12}$$

In particular the relative frequency among positive primitive vectors has limit $\rho_{M,k}(c)$, and this is positive precisely for the factors in Theorem 86.9. Apart from the exceptional case where both diagonal entries are even, the entire distribution depends only on $k = |B - C|$, not on any other entry of M.

Proof. Fix $\ell^e \| k$. The nonsingular discriminant -4 modulo ℓ makes F_M anisotropic when $\ell \equiv 3 \pmod 4$. For $\ell \equiv 1 \pmod 4$, an invertible transformation over $\mathbb{Z}/\ell^e$ takes F_M to a unit times XY. Condition on the pair being primitive modulo ℓ; its probability is $1 - \ell^{-2}$. For $1 \leq t \leq e$ the event $v_\ell(F_M) \geq t$ consists of the two disjoint cases in which one coordinate is a unit and the other is divisible by ℓ^t. Hence

$$\Pr(v_\ell(F_M) \geq t \mid \text{primitive}) = \frac{2\ell^{-t}(1-\ell^{-1})}{1-\ell^{-2}} = \frac{2}{(\ell+1)\ell^{t-1}}.$$

Taking successive differences gives precisely the $\ell \equiv 1$ distribution in the table. At 2, when both diagonal entries are even, the preceding theorem gives a forced 2 and $e = 1$. Otherwise $F_M(x, y) \pmod 2 = Ax + Dy$ is a nonzero linear form. Of the three primitive pairs modulo 2, exactly one makes it zero. By the same theorem $4 \nmid \gcd(F_M, k)$, giving the two probabilities. The Chinese remainder theorem makes these local distributions independent when a primitive residue pair is chosen uniformly modulo k, and proves (86.11).

For completeness, each primitive residue class $(a, b) \pmod k$ occurs with the same asymptotic frequency among $\mathcal{V}_X$. Möbius inversion in the common divisor, using $\gcd(a, b, k) = 1$, gives

$$\#\{(x,y) \in \mathcal{V}_X : (x,y) \equiv (a,b) \pmod k\} = \frac{X^2}{k^2\zeta(2)} \prod_{\ell \mid k} (1-\ell^{-2})^{-1} + O_k(X\log(2X)).$$

Indeed only divisors coprime to k survive; for each such divisor d, each coordinate has $X/(kd) + O(1)$ choices, and summing the resulting errors costs $O_k(X\log(2X))$. There are $k^2 \prod_{\ell \mid k}(1-\ell^{-2})$ primitive residue classes modulo k. Summing over those with the specified cancellation factor yields (86.12). The usual $\#\mathcal{V}_X = X^2/\zeta(2) + O(X\log(2X))$ finishes the relative frequency statement. □

Remark 86.11 (Two exact spectra and probability laws). For $M = \left(\begin{smallmatrix} 1 & 10 \\ 0 & 1 \end{smallmatrix}\right)$, we have $k = 10$ and

$$\rho(1) = 4/9, \qquad \rho(2) = 2/9, \qquad \rho(5) = 2/9, \qquad \rho(10) = 1/9.$$

For $M = \left(\begin{smallmatrix} 1 & 25 \\ 0 & 1 \end{smallmatrix}\right)$, $\rho(1) = 2/3$, $\rho(5) = 4/15$ and $\rho(25) = 1/15$. The exception $M = \left(\begin{smallmatrix} 2 & 1 \\ 3 & 2 \end{smallmatrix}\right)$ has $k = 2$ and $\rho(2) = 1$.

Remark 86.12 (Relation to classical density methods). The split/anisotropic local classification and Möbius inversion are standard ingredients of arithmetic theory. The result here packages them as the *full distribution of exact denominator cancellations* for the canonical unimodular interpolation problem, not as a claim to introduce new local-density tools. It is different from counting distinct integers represented by a quadratic form; compare general local-density frameworks such as [217].

86.5.5 A fixed factor for a family of adjacent-swap defects

Let $G, H \in \mathrm{Sym}_2(\mathbb{Q})$ with $\det G = \det H = 1$. Set

$$A = GH, \quad A^T = HG, \quad S = \tfrac{1}{2}(A + A^T), \quad K = \tfrac{1}{2}(A - A^T), \quad d = A_{12} - A_{21}.$$

For symmetric P define

$$E_P(v) = Q_P(Av) - Q_P(A^T v).$$

Direct polarization gives

$$E_P(v) = 4(Sv)^T P(Kv), \qquad K = -\frac{d}{2}J, \qquad \det S = 1 - \frac{d^2}{4}. \tag{86.13}$$

Proposition 86.13 (Fricke commutator interpretation)**.** *For the symmetric determinant-one pair G, H above, let $[G, H] = GHG^{-1}H^{-1}$. Then*

$$\boxed{\operatorname{tr}[G, H] = 2 - d^2.} \tag{86.14}$$

In particular $|d| = 2$ if and only if $\operatorname{tr}[G, H] = -2$. Consequently, if $x = \operatorname{tr} G$, $y = \operatorname{tr} H$, and $z = \operatorname{tr}(GH)$, this case satisfies the classical Fricke–Markoff equation

$$x^2 + y^2 + z^2 = xyz. \tag{86.15}$$

Proof. Put $U = GH$ and $V = HG$, so $\det U = \det V = 1$ and $U - V = -dJ$. As $\det(U - V) = \det(I - U^{-1}V)$ and $\det(U^{-1}V) = 1$, the 2×2 determinant identity gives

$$d^2 = 2 - \operatorname{tr}(U^{-1}V) = 2 - \operatorname{tr}(GHG^{-1}H^{-1}).$$

The last trace equality follows from invariance under conjugation and inversion for determinant-one 2×2 matrices. The remaining assertion follows from the classical Fricke trace formula $\operatorname{tr}[G, H] = x^2 + y^2 + z^2 - xyz - 2$; see, for instance, [215, 216]. □

Theorem 86.14 (Universal linear-factor criterion)**.** *Assume $GH \neq HG$. A nonzero rational homogeneous linear form $L(v)$ divides $E_P(v)$ for* every *rational symmetric P if and only if $|d| = 2$, equivalently $\operatorname{tr}[G, H] = -2$. When this occurs $S = \lambda z z^T$ has rank one for a rational nonzero vector z and $\lambda \in \mathbb{Q}^\times$, and*

$$E_P(v) = 4\lambda(z^T v)(z^T PKv)$$

for all symmetric P.

Proof. Noncommutativity implies $d \neq 0$, since $A - A^T$ is a 2×2 skew matrix. If $d^2 = 4$ then $\det S = 0$. Moreover $S \neq 0$: real symmetric determinant-one matrices are either positive or negative definite, so GH has nonzero real eigenvalues of the same sign, while a nonzero skew matrix of determinant one has purely imaginary eigenvalues. Thus S has rank one, giving the factorization by (86.13).

Conversely, let L divide all E_P, and let $v \in \mathbb{Q}^2$ be a nonzero vector in $L(v) = 0$. Varying P over all symmetric matrices gives

$$(Av)(Av)^T = (A^T v)(A^T v)^T.$$

Consequently $Av = \pm A^T v$. The plus sign is impossible, since $(A - A^T) = -dJ$ is invertible. Therefore $(A + A^T)v = 0$. It follows that $\det S = 0$, and (86.13) yields $d^2 = 4$. □

Remark 86.15. The condition $\operatorname{tr}[G, H] = -2$ is the classical parabolic commutator locus of Fricke–Markoff theory, not a newly discovered trace locus. What is shown here is its equivalence with the particular *quadratic-observable common-factor property.*

Remark 86.16. This statement concerns a P-independent factor shared by an entire three-dimensional linear family of quadratic forms. It does not assert that an individual E_P factors over $\mathbb{Q}$ only when $|d| = 2$; individual factorization is governed by the usual binary quadratic discriminant.

86.5.6 A structural return theorem for free positive monoids

The common factor in Theorem 86.14 need not be regarded merely as a polynomial curiosity. It singles out a natural positive state vector and hence a family of canonical return observables.

Theorem 86.17 (Rank-one swap criterion implies programmable word returns)**.** *Let $G, H \in \mathrm{Sym}_2(\mathbb{Z})$ have determinant one and strictly positive entries. Assume that G and H do not commute. Put*

$$A = GH, \quad d = A_{12} - A_{21}, \quad S = (A + A^T)/2.$$

Suppose $|d| = 2$. Then $S = \lambda uu^T$ for some $\lambda \in \mathbb{Q}_{>0}$ and some strictly positive primitive vector $u \in \mathbb{Z}^2$. For a word $w \in \{0,1\}^*$, let $M(w)$ be its matrix product (with $0 \mapsto G$, $1 \mapsto H$) and put

$$q(w) = M(w)^T u, \qquad f(w) = u^T q(w), \qquad P_w = P(M(w), u).$$

If the map $w \mapsto q(w)$ is injective, then for all words s, w,

$$Q_{P_w}(GHq(s)) - Q_{P_w}(HGq(s)) = -2\lambda d\, f(s)\, q(w)^T J q(s). \tag{86.16}$$

In particular the quadratic swap defect vanishes precisely when $s = w$, and

$$P_w \in \mathrm{Sym}_2(\mathbb{Z}) \quad \Longleftrightarrow \quad w = w^R.$$

Proof. Theorem 86.14 gives $\det S = 0$ and $S \neq 0$. Because G, H have strictly positive entries, S has strictly positive entries and is thus a positive semidefinite rank-one rational matrix. Its one-dimensional image is generated by a strictly positive primitive integer vector u and $S = \lambda uu^T$ with $\lambda > 0$ rational.

The skew part is $K = -dJ/2$. Polarization (86.13) and $P_w u = q(w)$ show

$$E_{P_w}(q(s)) = 4\lambda(u^T q(s))\, u^T P_w(-dJ/2)q(s) = -2\lambda d f(s) q(w)^T J q(s).$$

All the word states are positive primitive vectors, since u is primitive and $M(w)$ is integral unimodular. Vanishing of the alternating product is therefore equivalent to $q(s) = q(w)$; injectivity gives $s = w$. Finally the matrix word map is also injective (equality of matrices would imply equality of states). Corollary 86.8 applies and gives the integrality–palindrome equivalence. □

Remark 86.18. The hypotheses are substantive. The relation $|d| = 2$ controls a universal factorization, whereas positivity and injectivity turn its determinant zero locus into a single *word* rather than an entire projective rational line. Without an injective state decoder, different words could produce the same primitive state; without the rank-one swap condition, even a fixed common linear factor need not exist.

Remark 86.19 (The balanced pair is a classical Markoff-trace point). For the pair in (86.17), the trace triple is $(3, 3, 6)$, and $3^2+3^2+6^2 = 3\cdot3\cdot6$. The original Cohn pair with $X = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and $Y = \left(\begin{smallmatrix} 5 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ has trace triple $(3, 6, 15)$, which satisfies the same Fricke–Markoff cubic. The two examples are therefore not evidence for a new trace class.

86.5.7 Prescribing a unique return word

We exhibit a concrete pair distinct from the classical Cohn generators:

$$G = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad u = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{86.17}$$

For a word $w \in \{0,1\}^*$ let $M(\varepsilon) = I$, $M(w0) = M(w)G$, and $M(w1) = M(w)H$. Define the positive primitive column $q(w) = M(w)^T u$ and $f(w) = u^T q(w)$. Let $P_w = P(M(w), u)$ be the canonical form of (86.8).

Lemma 86.20 (Exact state decoder)**.** *The map $w \mapsto q(w)$ is injective. A nonempty word ends in* 0 *precisely when $q_1(w) > q_2(w)$, and ends in* 1 *precisely when $q_1(w) < q_2(w)$.*

Proof. If $q = (x, y)^T$ has positive coordinates, then $Gq = (2x+y, x+y)^T$ has first coordinate larger by x, while $Hq = (x + y, x + 2y)^T$ has second coordinate larger by y. Starting at $(1, 1)$, the last letter is uniquely determined by the sign of the coordinate difference. Invert using G^{-1} or H^{-1}; on a genuine state this recovers the positive preceding state and strictly decreases the coordinate sum. Iteration reaches the unique initial state $(1, 1)$, hence uniquely recovers the word. □

Theorem 86.21 (Programmable single-word swap returns)**.** *For the generators* (86.17)*, every word w has a positive-definite determinant-one rational form P_w with*

$$Q_{P_w}(GHq(s)) - Q_{P_w}(HGq(s)) = -12f(s)\, q(w)^T Jq(s) \tag{86.18}$$

for every word s. In particular,

$$Q_{P_w}(GHq(s)) = Q_{P_w}(HGq(s)) \quad \Longleftrightarrow \quad s = w.$$

Furthermore,

$$P_w \text{ is integral} \quad \Longleftrightarrow \quad w = w^R.$$

Proof. Here

$$GH = \begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix}, \qquad HG = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}, \qquad S = 3uu^T, \quad K = -J.$$

Equation (86.13) gives $E_P(v) = -12(u^T v)(u^T PJv)$. Substituting $P = P_w$ and using $P_w u = q(w)$ yields (86.18). Because $f(s) > 0$, a zero is equivalent to $q(w)^T Jq(s) = 0$. Both vectors are positive primitive integral vectors, since they lie in an $SL_2(\mathbb{Z})$ orbit of the primitive seed. Positive collinearity of such vectors forces equality. Lemma 86.20 then gives $s = w$, and the converse is immediate. The integrality criterion follows from Corollary 86.8. □

For example $w = 01$ has $q(w) = (5, 7)^T$ and $f(w) = 12$; consequently

$$P_{01} = \frac{1}{6}\begin{pmatrix} 13 & 17 \\ 17 & 25 \end{pmatrix}, \qquad \operatorname{den}(P_{01}) = 6,$$

with 01 as its only swap-zero word. For the palindromic one-letter words, $P_0 = G$ and $P_1 = H$.

86.5.8 Scope of the conclusions

Theorem 86.5 does not require a word monoid, nonnegative entries of M, or a Markoff condition. Corollary 86.6 additionally uses positivity of both the matrix entries and the seed; Corollary 86.8 uses injectivity of the matrix word representation. Theorem 86.14 classifies a *common* factor for all quadratic observables attached to two symmetric determinant-one generators. Theorem 86.21 is a genuine *orbit* statement for the explicitly decoded generators (86.17), rather than an identity on the entire plane or a global uniqueness claim for Markoff numbers.

The Fricke trace locus, Markoff cubic, and Cohn–Christoffel palindrome descriptions are classical [81, 83, 214, 215, 216]. The distinct arithmetical content in this arrangement is the exact denominator/reversal formula and prescribed-return realization across well-specified matrix classes. These formulas have short elementary proofs and may follow from more general established form-theoretic results; no assertion of novelty priority is made solely on the basis of the present comparison.

86.5.9 Finite verification and its scope

Exact-arithmetic consistency checks evaluated the least-denominator formula on 27,664 positive primitive seed/matrix instances drawn from 304 nonsymmetric nonnegative unimodular matrices with entries between 0 and 16. For all 304 matrices, direct counting of primitive residue classes modulo $|B - C|$ reproduced the predicted local cancellation distributions. In the explicit two-generator word model, all 511 words of length at most 8 had distinct states; the quadratic swap identity and its unique-zero criterion agreed on all 261,121 ordered word pairs, and the integrality–palindrome equivalence held for the same 511 words. No discrepancy was found in these bounded checks. These finite calculations are consistency tests, not substitutes for the symbolic proofs or a certification of the general statements outside the tested ranges. No computational files are required for the proofs in this article.

Part VII

Chronology, Topology, and Higher-Rank Completion Structure

Part guide. **Question.** After the semantic quotient is fixed, what structure can still remain in ordering, optimal fibers, physical transport, and higher rank? **Main result.** Random chronology has distinct LIFO scales; optimal fibers admit lattice-geometric classification; physical holonomy separates topological information from carrier and repair cost; higher-rank realization recovers broad topological freedom. **Why here.** The Part measures structure invisible to the semantic quotient itself. **Dependency.** It builds on completion fibers and semantic/physical separation, while remaining distinct from the resource lower bounds of Part VIII.

Chapter 87

Fixed-Parikh Random Ordering and Direct Obstruction

The centered real-time $p{:}q$ normalizer was developed earlier as a deterministic physical lift of shortest-completion semantics. We now keep that semantic content fixed. For coprime p, q, write $m = p + q$ and sample uniformly from

$$\mathcal{F}_{gq,gp} = \{w \in \{0,1\}^{mg} : N_0(w) = gq,\ N_1(w) = gp\}.$$

Every word in this fiber has the same zero endpoint residual and the same endpoint completion demand. Only chronology varies. The resulting experiment therefore isolates costs created by LIFO order rather than by semantic variation. We restate below only the shortest-demand and transition data needed for the quantitative analysis; the original 2:1 provenance figure and the full construction have already appeared in earlier Parts.

87.1 Shortest completion demand

For $r \in \mathbb{Z}$, let

$$\mu(r) = (x_r, y_r)$$

be the unique nonnegative solution of

$$px - qy = r$$

that minimizes $x + y$. All other nonnegative solutions are obtained by adding copies of the primitive zero-weight kernel

$$K = (q, p).$$

We use the canonical stack ordering

$$\eta(x,y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & y > x. \end{cases} \tag{87.1}$$

The leftmost symbol is the visible stack top.

Let

$$h = \left\lfloor \frac{m-1}{2} \right\rfloor, \qquad D_{-h}, \ldots, D_h$$

be the centered debt states. For a logical stack word S define

$$W(S) = pN_0(S) - qN_1(S).$$

The state–stack residual invariant is

$$\rho = W(S) + dm \qquad \text{at configuration } (D_d, S). \tag{87.2}$$

The semantic projection reduces the physical stack plus virtual debt by complete kernel packets and returns exactly $\mu(\rho)$.

87.2 Transition classes

Suppose the current input is $a \in \{0,1\}$. If $S \neq \varnothing$ and a equals the visible top, the machine performs a *MATCH*: it pops one cell and leaves d unchanged. If the visible top is 0 and $a = 1$, a mismatch increments debt while $d < h$; this is *SEARCH*. At $d = h$ the next such mismatch is an *outer WRITE*: the visible zero is removed, debt is reset to zero, and

$$U_+ = \eta(\mu((h+1)m))$$

is inserted before the surviving suffix. The negative rule is symmetric: input 0 over visible top 1 decrements debt while $d > -h$, and at $d = -h$ an outer WRITE inserts

$$U_- = \eta(\mu(-(h+1)m)).$$

If the logical stack is empty, the next symbol triggers a *bottom WRITE*. For every $|d| \leq h$,

$$(D_d, \varnothing) \xrightarrow{1} \big(D_0, \eta(\mu(dm+q))\big), \tag{87.3}$$

$$(D_d, \varnothing) \xrightarrow{0} \big(D_0, \eta(\mu(dm-p))\big). \tag{87.4}$$

Every WRITE therefore returns the physical stack to the exact canonical shortest-demand section for the new residual.

The construction has the following properties, all consequences of shortest-demand minimality and the centered debt bound.

Lemma 87.1 (WRITE frontier bound)**.** *Every canonical word created by a bottom or outer WRITE has shortest-demand vector (x, y) satisfying*

$$\min(x, y) \leq h.$$

For the outer-WRITE vectors $\mu(\pm(h+1)m)$, the minority coordinate is at most h while the majority coordinate is at least $h+1$; in the positive case the minority one-count is $< p$, and in the negative case the minority zero-count is $< q$.

Proof. For a bottom WRITE the target residual has the form

$$r = jm + q, \qquad -h - 1 \leq j \leq h,$$

because $dm - p = (d-1)m + q$. Write $\mu(r) = (x, y)$ and $\ell = x + y$. Modulo m, the relation $px - qy = r \equiv q$ and $p \equiv -q \pmod m$ give

$$-q\ell \equiv q \pmod m.$$

Since $\gcd(q, m) = 1$, $\ell \equiv -1 \pmod m$, so $\ell = km - 1$. If both $x, y \geq h + 1$, then $k \geq 2$. The minimum demand is kernel-free, hence either $x < q$ or $y < p$. In the first case $q \geq h+2$ and

$$r = m(x - qk) + q$$

forces $j \leq -q - 1 \leq -h - 3$, a contradiction. In the second case $p \geq h + 2$ and

$$j = pk - y - 1 \geq p > h,$$

again a contradiction. Thus $\min(x, y) \leq h$.

For the positive outer WRITE, put $c = h + 1$. The minimum solution is

$$\mu(cm) = (qk + c, pk - c), \qquad k = \left\lceil \frac{c}{p} \right\rceil.$$

Its second coordinate lies in $[0, h]$, is strictly less than p, and its first is at least $c = h + 1$. The negative case is symmetric, with the zero-coordinate strictly less than q. □

Lemma 87.2 (Boundary guard)**.** *Suppose the current stack is a suffix of a canonical word whose minority count is at most h.*

(a) If the current state is D_h and the stack begins with 0, then an outward $1/0$ mismatch can occur only when the surviving stack is unary zero.

(b) If the current state is D_{-h} and the stack begins with 1, then an outward $0/1$ mismatch can occur only when the surviving stack is unary one.

Proof. Consider the positive case. Since the most recent WRITE, debt started at zero. Reaching D_h requires at least h previous positive mismatches, each of which popped a zero from the pre-existing canonical word. If the current visible zero were still in the mixed alternating part and a one survived later in the stack, then the h previously popped zeros would each have a distinct preceding one in the alternating part. Together with the current zero and a later surviving one, the original canonical word would contain at least $h + 1$ zeros and at least $h + 1$ ones, contradicting Lemma 87.1. Hence the surviving suffix is unary zero. The negative case is symmetric. □

Lemma 87.3 (Canonical-segment property)**.** *Immediately after every WRITE the machine is in state D_0 with stack $\eta(\mu(\rho))$, whose minority count is at most h. Until the next WRITE, every transition removes exactly one cell from that pre-existing canonical word. Hence every reachable logical stack is a suffix of the canonical word produced by the most recent WRITE.*

Proof. A bottom WRITE has the displayed canonical form by definition and satisfies Lemma 87.1. Between WRITEs, MATCH and SEARCH only remove the visible pre-existing cell. At an outer WRITE, Lemma 87.2 shows that the surviving suffix is unary. The newly inserted canonical word and that unary suffix concatenate to the canonical ordering of the new shortest demand; kernel-freeness follows from the minority bound in Lemma 87.1. Induction over WRITE episodes proves the claim. □

Lemma 87.4 (Zero-residual correctness)**.** *At the end of every processed binary prefix,*

$$\rho = 0 \quad \Longleftrightarrow \quad (D_d, S) = (D_0, \varnothing).$$

Proof. The reverse implication follows from (87.2). For the forward implication, consider the most recent WRITE. Immediately after it the stack was $\eta(\mu(r_0))$ for some residual r_0. Until the current time no further WRITE has occurred, so every intervening symbol has popped exactly one of those cells. If the current residual is zero, that intervening input is a completion of r_0. By minimality of $\mu(r_0)$, no completion can use fewer symbols than the length of the written stack. Since no more than that many cells can have been popped, equality is forced: the entire stack has been consumed. Equation (87.2) then gives $dm = 0$, hence $d = 0$. □

Operational observables. For a word w, let $S(w), O(w), B(w)$ be the numbers of SEARCH, outer-WRITE, and bottom-WRITE transitions. The primary LIFO-obstruction count is

$$R(w) = S(w) + O(w).$$

A bottom WRITE is not counted as an obstruction: it occurs when the physical stack is already empty and represents rematerialization rather than LIFO search.

87.3 The fixed-Parikh model as a critical bridge

For arbitrary $a, b \geq 0$ define the fixed-Parikh fiber

$$\mathcal{F}_{a,b} = \{w \in \{0,1\}^{a+b} : N_0(w) = a,\ N_1(w) = b\}.$$

Our balanced model is

$$W_g \sim \mathrm{Unif}(\mathcal{F}_{gq,gp}), \qquad n = mg.$$

Introduce i.i.d. symbols X_i under the critical law

$$\mathbb{P}(X_i = 1) = \frac{p}{m}, \qquad \mathbb{P}(X_i = 0) = \frac{q}{m},$$

and increments

$$\xi_i = \begin{cases} q, & X_i = 1, \\ -p, & X_i = 0. \end{cases}$$

Then

$$\mathbb{E}\xi_i = 0, \qquad \mathrm{Var}(\xi_i) = pq.$$

Write $\rho_t = \sum_{i \le t} \xi_i$.

Lemma 87.5 (Canonical critical-bridge representation)**.** *For $n = mg$,*

$$\mathcal{L}(X_1, \dots, X_n \mid \rho_n = 0) = \mathrm{Unif}(\mathcal{F}_{gq,gp}).$$

Proof. The condition $\rho_n = 0$ gives

$$mN_1(n) - pn = 0,$$

so $N_1(n) = pg$ and $N_0(n) = qg$. Every word with these counts has the same i.i.d. probability

$$\left(\frac{p}{m}\right)^{pg} \left(\frac{q}{m}\right)^{qg}.$$

Conditioning therefore makes the fiber uniform. □

At block boundaries set

$$Z_j = \frac{\rho_{mj}}{m}.$$

If K_j is the number of ones in the next m symbols, then under the critical i.i.d. law

$$Z_{j+1} - Z_j = K_j - p, \qquad K_j \sim \mathrm{Bin}\Big(m, \frac{p}{m}\Big). \tag{87.5}$$

Hence

$$\mathbb{E}(Z_{j+1} - Z_j) = 0, \qquad \mathrm{Var}(Z_{j+1} - Z_j) = \frac{pq}{m}. \tag{87.6}$$

The block skeleton is therefore a centered, irreducible, aperiodic, finite-range random walk on $\mathbb{Z}$.

87.4 Direct realization and the first LIFO obstruction

Call a word *direct* if no SEARCH or outer-WRITE transition occurs, equivalently if $R(w) = 0$.

87.4.1 Primitive direct blocks

The shortest demands after the first symbol are

$$\mu(-p) = (q - 1, p), \qquad \mu(q) = (q, p - 1).$$

Define

$$B_0 = 0\,\eta(q - 1, p), \qquad B_1 = 1\,\eta(q, p - 1). \tag{87.7}$$

Both words have length m and Parikh vector (q, p).

Theorem 87.6 (Direct-realization factorization)**.** *A balanced word $w \in \mathcal{F}_{gq,gp}$ satisfies $R(w) = 0$ if and only if*

$$w = B_{\varepsilon_1} B_{\varepsilon_2} \cdots B_{\varepsilon_g}$$

for a unique $(\varepsilon_1, \ldots, \varepsilon_g) \in \{0,1\}^g$. Consequently

$$\#\{w \in \mathcal{F}_{gq,gp} : R(w) = 0\} = 2^g \tag{87.8}$$

and

$$\mathbb{P}(R(W_g) = 0) = \frac{2^g}{\binom{mg}{qg}}. \tag{87.9}$$

Proof. Start from empty stack and zero debt. The first input symbol triggers a bottom WRITE. If it is 0, the newly written stack is $\eta(\mu(-p))$; if it is 1, the stack is $\eta(\mu(q))$. If no LIFO obstruction occurs thereafter, every subsequent input until the stack empties must match the visible top. Thus the entire first episode is forced to be B_0 or B_1. Each block has length m, consumes exactly (q, p) symbols, and returns the machine to $(D_0, \varnothing)$. Iteration gives the factorization. Conversely, each displayed block is realized by one bottom WRITE followed only by MATCH transitions, so any concatenation is direct. Uniqueness follows from the first symbol of each block. □

By Stirling's formula,

$$\mathbb{P}(R(W_g) = 0) \sim \sqrt{\frac{2\pi pqg}{m}} \left(\frac{2p^p q^q}{m^m} \right)^g. \tag{87.10}$$

The base

$$\lambda_{p,q} = \frac{2p^p q^q}{m^m}$$

is strictly less than one.

87.4.2 Exact first-obstruction law

Let

$$\tau_g = \min\{t \geq 1 : \text{the transition at time } t \text{ is SEARCH or outer WRITE}\},$$

with $\tau_g = mg + 1$ if no obstruction occurs. For $\varepsilon \in \{0,1\}$ and $0 \leq s < m$, write

$$z_{\varepsilon,s} = N_0(\mathrm{pref}_s(B_\varepsilon)).$$

Theorem 87.7 (Exact first LIFO-obstruction survival law)**.** *For $0 \leq k \leq g$,*

$$\mathbb{P}(\tau_g > km) = \frac{2^k \binom{m(g-k)}{q(g-k)}}{\binom{mg}{qg}}. \tag{87.11}$$

For $0 \leq k < g$ and $1 \leq s < m$,

$$\mathbb{P}(\tau_g > km + s) = \frac{2^k \displaystyle\sum_{\varepsilon=0}^{1} \binom{m(g-k) - s}{q(g-k) - z_{\varepsilon,s}}}{\binom{mg}{qg}}. \tag{87.12}$$

Proof. The event $\{\tau_g > km\}$ is exactly the event that the first k primitive blocks are direct. By Theorem 87.6, there are 2^k possible direct prefixes, each consuming (kq, kp) symbols. The remaining suffix is an arbitrary balanced fixed-Parikh word of size $g - k$, giving (87.11). For a partial block, after the first k complete direct blocks the next s symbols must equal $\mathrm{pref}_s(B_0)$ or $\mathrm{pref}_s(B_1)$. These two events are disjoint because B_0 and B_1 begin with different symbols. After fixing such a prefix, the remaining number of zeros is $q(g-k) - z_{\varepsilon,s}$, which gives (87.12). □

Define

$$R_{p,q}(0) = 1,$$

and, for $1 \le s < m$,

$$R_{p,q}(s) = \sum_{\varepsilon=0}^{1} \left(\frac{q}{m}\right)^{z_{\varepsilon,s}} \left(\frac{p}{m}\right)^{s-z_{\varepsilon,s}}.$$

Taking $g \to \infty$ with k, s fixed in Theorem 87.7 gives

$$\mathbb{P}(\tau_g > km + s) \longrightarrow \lambda_{p,q}^{k} R_{p,q}(s). \tag{87.13}$$

In particular,

$$\boxed{\tau_g = O_{\mathbb{P}}(1).} \tag{87.14}$$

87.4.3 Primitive balanced SEARCH distribution

At $g = 1$ no outer WRITE is required before return, and SEARCH occurs in mismatch pairs.

Proposition 87.8 (Primitive SEARCH law)**.** *On $\mathcal{F}_{q,p}$,*

$$S(W_1) \in \{0, 2, 4, \ldots, 2\min(p, q)\},$$

and

$$\#\{w \in \mathcal{F}_{q,p} : S(w) = 2j\} = \binom{p}{j}\binom{q-1}{j} + \binom{q}{j}\binom{p-1}{j}. \tag{87.15}$$

Consequently

$$\mathbb{E}S(W_1) = \frac{2pq(m-2)}{m(m-1)}. \tag{87.16}$$

Proof. Condition on the first symbol. If it is 0, the bottom WRITE creates the deterministic target word $\eta(q-1, p)$, with $q-1$ zero-cells and p one-cells. The remaining input suffix has exactly the same Parikh vector. During this primitive episode no outer WRITE can occur: the debt at any prefix is the difference between the number of input ones and target ones seen so far, whose absolute value is at most $\min(p, q-1) \le h$. Thus every step pops one target cell. If exactly j target one-cells are hit by input zero, then conservation of the two symbol counts forces exactly j target zero-cells to be hit by input one, giving $2j$ SEARCH mismatches. The number of such suffixes is

$$\binom{p}{j}\binom{q-1}{j}.$$

If the first symbol is 1, the same argument with target $\eta(q, p-1)$ gives

$$\binom{q}{j}\binom{p-1}{j}.$$

Adding proves (87.15). For the mean, use

$$\sum_{j} j\binom{a}{j}\binom{b}{j} = \frac{ab}{a+b}\binom{a+b}{a}$$

in the two terms and divide by $\binom{m}{p}$. □

Chapter 88

Physical-Lift Excursions and Rayleigh Rematerialization

The first obstruction is local, but rematerialization of the canonical shortest-demand section is diffusive. The block-boundary lift chain below records precisely that phenomenon.

88.1 The block-boundary physical-lift chain

We now pass from exact finite obstruction counts to the recurrent physical structure that governs rematerialization.

Lemma 88.1 (Empty stack only at block boundaries)**.** *If the logical stack is empty after t processed symbols, then $m \mid t$.*

Proof. If $S_t = \varnothing$, the residual invariant (87.2) gives $\rho_t = dm$ for some integer d. On the other hand,

$$\rho_t = mN_1(t) - pt.$$

Hence $m \mid pt$. Since $\gcd(p, m) = \gcd(p, q) = 1$, we obtain $m \mid t$. □

Let X_j be the full physical configuration after jm symbols under the critical i.i.d. input law, and let

$$\pi(X_j) = Z_j = \rho_{jm}/m.$$

Let

$$o = (D_0, \varnothing).$$

By Lemma 87.4, $X_j = o$ exactly when $Z_j = 0$. Since the scalar block walk Z_j is recurrent, so is the reachable class of the physical block chain.

Let

$$T_o^+ = \min\{j \geq 1 : X_j = o\}.$$

Definition 88.2 (Physical-lift invariant measure)**.** For a reachable block-boundary physical configuration x, define

$$\nu(x) = \mathbb{E}_o \left[\sum_{j=0}^{T_o^+ - 1} \mathbf{1}_{\{X_j = x\}} \right]. \tag{88.1}$$

The standard excursion construction for an irreducible recurrent Markov chain makes ν an invariant measure normalized by $\nu(o) = 1$; see, for example, [46, Ch. 1].

Theorem 88.3 (Unit semantic-shell mass)**.** *For every $z \in \mathbb{Z}$,*

$$\boxed{\sum_{x:\pi(x)=z} \nu(x) = 1.} \tag{88.2}$$

Proof. Push ν forward under the factor map π. The result is an invariant measure for the irreducible recurrent random walk Z_j. Such an invariant measure is unique up to a multiplicative constant. Because the block walk is translation-invariant, counting measure on $\mathbb{Z}$ is invariant. At shell zero, Lemma 87.4 shows that the only physical state is o, so the pushed-forward mass equals $\nu(o) = 1$. The normalization therefore fixes the pushed-forward measure to be counting measure, proving (88.2). □

This is the quantitative form of the semantic/physical distinction: the residual quotient assigns one unit of invariant mass to each shell, while physical chronology determines how that unit is split among its lifts.

Definition 88.4 (Physical rematerialization constant)**.** For $-h \le d \le h$, let

$$E_d = (D_d, \varnothing), \qquad \theta_d^{p,q} = \nu(E_d),$$

and define

$$\boxed{\kappa_{p,q} = \sum_{d=-h}^{h} \theta_d^{p,q}.} \tag{88.3}$$

Because E_d belongs to semantic shell d, Theorem 88.3 gives

$$0 \le \theta_d^{p,q} \le 1, \qquad \theta_0^{p,q} = 1, \qquad 1 \le \kappa_{p,q} \le 2h + 1. \tag{88.4}$$

Let $\mathcal{E} = \{E_d : -h \le d \le h\}$ and consider the trace chain obtained by observing X_j only when it enters $\mathcal{E}$. Restriction of an invariant measure to a recurrent trace set is invariant for the trace chain. Therefore its stationary distribution is

$$\pi_d^{\mathcal{E}} = \frac{\theta_d^{p,q}}{\kappa_{p,q}},$$

and in particular

$$\boxed{\pi_0^{\mathcal{E}} = \frac{1}{\kappa_{p,q}}.} \tag{88.5}$$

88.2 Rayleigh law for Bottom-WRITE rematerialization

The fixed-Parikh bridge can return to physical empty states only at block boundaries by Lemma 88.1. This allows an excursion decomposition at the block level.

A *primitive semantic excursion* is a block path beginning at o, avoiding o at positive intermediate block times, and ending at its first return to o. For such an excursion e, let

$$L(e) = \text{its length in blocks}$$

and let

$$R_B(e) = \#\{\text{empty-stack block states visited in } [0, L(e))\}.$$

Every such visit is followed, unless it is the terminal boundary of the full word, by exactly one bottom WRITE. Concatenating primitive semantic excursions therefore adds the rewards R_B exactly.

Lemma 88.5 (Excursion reward mean and exponential moments)**.** *Under the critical i.i.d. law,*

$$\mathbb{E}R_B(e) = \kappa_{p,q}. \tag{88.6}$$

Moreover there exists $s > 1$ such that

$$\mathbb{E}s^{R_B(e)} < \infty. \tag{88.7}$$

Proof. Equation (88.6) is the excursion formula (88.1) summed over the finite empty-state set $\mathcal{E}$. For (88.7), consider the finite trace chain on reachable empty states. From every such state there is a positive-probability finite input continuation that completes its residual and reaches E_0. Finiteness gives constants $L < \infty$ and $\varepsilon > 0$ such that from every trace state the probability of hitting E_0 within at most L trace transitions is at least ε. Hence the trace hitting time has a geometric tail, which implies an exponential moment for $R_B(e)$. □

Let

$$b_g = \mathbb{P}(Z_g = 0) = \binom{mg}{pg} \left(\frac{p}{m}\right)^{pg} \left(\frac{q}{m}\right)^{qg}.$$

Define the bridge return generating function

$$G(z,1) = \sum_{g \geq 0} b_g z^g.$$

Let $F(z,u)$ be the probability generating function of primitive semantic excursions, with z marking block length and u marking R_B. Unique decomposition into primitive excursions gives

$$G(z,u) = \frac{1}{1 - F(z,u)}. \tag{88.8}$$

At $u = 1$, Stirling's formula yields

$$b_g \sim \sqrt{\frac{m}{2\pi pq}}\, g^{-1/2}. \tag{88.9}$$

A standard Abelian/Tauberian transfer therefore gives

$$G(z,1) \sim \sqrt{\frac{m}{2pq}}(1-z)^{-1/2} \qquad (z \uparrow 1), \tag{88.10}$$

and hence

$$1 - F(z,1) \sim \sqrt{\frac{2pq}{m}}\sqrt{1-z}. \tag{88.11}$$

The same square-root singularity is a standard source of Rayleigh local-time laws for zero-drift lattice-path bridges; compare [38]. We use only the elementary regular-variation consequences of this square-root behavior below. Here the reward is not scalar zero-local-time itself but the physical empty-lift reward carried by each primitive semantic excursion.

Theorem 88.6 (Bottom-WRITE Rayleigh law)**.** *Let $B_g = B(W_g)$. Then*

$$\boxed{\frac{B_g}{\sqrt{mg}} \Longrightarrow \mathrm{Rayleigh}\left(\frac{\kappa_{p,q}}{\sqrt{pq}}\right).} \tag{88.12}$$

Equivalently, for every fixed integer $r \geq 1$,

$$\mathbb{E}\left[\left(\frac{B_g}{\sqrt{mg}}\right)^r\right] \longrightarrow \left(\frac{\kappa_{p,q}}{\sqrt{pq}}\right)^r 2^{r/2}\Gamma\left(1 + \frac{r}{2}\right). \tag{88.13}$$

Proof. Write

$$H_j(z) = \partial_u^j F(z,u)\big|_{u=1}.$$

By Lemma 88.5, $H_j(1) < \infty$ for every fixed j, and

$$H_1(1) = \kappa_{p,q}.$$

Differentiating (88.8) r times in u, the term with r factors of H_1 is

$$r! H_1(z)^r G(z,1)^{r+1}. \tag{88.14}$$

Every other Faà di Bruno term contains at most $r-1$ derivative factors of F and hence at most $G(z,1)^r$. More precisely, each such term is a product of some $H_{j_1}(z),\dots,H_{j_k}(z)$ with $k \le r-1$ and a factor $G(z,1)^{k+1}$. Since every H_j has summable nonnegative coefficients by Lemma 88.5, the convolution theorem for summable sequences against regularly varying coefficients shows that its coefficient is $O(g^{(k-1)/2}) = o(g^{(r-1)/2})$. Thus only (88.14) contributes at leading order.

Using (88.10), Karamata coefficient asymptotics, and the same standard convolution lemma (equivalently, the corresponding singularity-transfer calculation; cf. [40]),

$$[z^g]\partial_u^r G(z,1) \sim \frac{r!\kappa_{p,q}^r}{\Gamma((r+1)/2)} \left(\frac{m}{2pq}\right)^{(r+1)/2} g^{(r-1)/2}.$$

Dividing by b_g in (88.9) gives the conditional factorial moment

$$\mathbb{E}[(B_g)_r] \sim \kappa_{p,q}^r \left(\frac{m}{pq}\right)^{r/2} 2^{r/2}\Gamma\Big(1+\frac{r}{2}\Big)\, g^{r/2},$$

where the duplication formula for the Gamma function was used. Ordinary moments have the same leading term. These are exactly the moments of a Rayleigh variable with scale $\kappa_{p,q}\sqrt{m/(pq)}$ for $B_g/\sqrt{g}$, or equivalently scale $\kappa_{p,q}/\sqrt{pq}$ for $B_g/\sqrt{mg}$. The Rayleigh distribution is moment-determinate, so the method of moments proves (88.12). □

Corollary 88.7 (Mean and variance of rematerialization)**.** *With $n = mg$,*

$$\mathbb{E}B_g \sim \kappa_{p,q}\sqrt{\frac{\pi}{2pq}}\sqrt{n}, \tag{88.15}$$

$$\operatorname{Var}(B_g) \sim \kappa_{p,q}^2 \frac{4-\pi}{2pq}\, n. \tag{88.16}$$

Chapter 89

Macroscopic LIFO Cost, Duality, and Finite Kernels

We now pass from the diffusive rematerialization scale to the macroscopic SEARCH/outer-WRITE regime, then return to finite kernels that compute the physical rematerialization constant.

89.1 Two regenerative bulk regimes

We now analyze the n-scale LIFO cost. Put

$$c = h + 1, \qquad k_+ = \left\lceil \frac{c}{p} \right\rceil, \qquad k_- = \left\lceil \frac{c}{q} \right\rceil.$$

The canonical outer-WRITE demands are

$$\mu(cm) = (qk_+ + c,\ pk_+ - c), \tag{89.1}$$

$$\mu(-cm) = (qk_- - c,\ pk_- + c). \tag{89.2}$$

To define the bulk constants, use the translation-invariant positive-tail extension obtained by quotienting complete unary packets (equivalently, adjoin an unbounded unary zero tail and retain the same finite phase dynamics). A finite deep-positive interval in the actual normalizer follows this extension until it hits the central boundary; only its first and last regeneration cycles can be truncated, and those endpoint pieces are handled in Section 89.2. In the homogeneous positive extension, consider the interval from one positive outer WRITE to the next. The mixed head of the newly written word is finite, while the surviving tail is unary zero. The next positive outer WRITE occurs after exactly pk_+ input ones have been read. Under the critical i.i.d. law the cycle length is therefore

$$T_+ \sim \mathrm{NB}\Big(pk_+, \frac{p}{m}\Big), \tag{89.3}$$

where T_+ counts trials through the final success. Similarly,

$$T_- \sim \mathrm{NB}\Big(qk_-, \frac{q}{m}\Big). \tag{89.4}$$

Thus

$$\mathbb{E}T_+ = mk_+, \qquad \mathrm{Var}(T_+) = \frac{k_+ qm}{p}, \tag{89.5}$$

$$\mathbb{E}T_- = mk_-, \qquad \mathrm{Var}(T_-) = \frac{k_- pm}{q}. \tag{89.6}$$

In particular the outer-WRITE rates are

$$o_+ = \frac{1}{mk_+}, \qquad o_- = \frac{1}{mk_-}. \tag{89.7}$$

Theorem 89.1 (Exact bulk LIFO-obstruction rates)**.** *For $R = S + O$, the critical i.i.d. positive- and negative-bulk rates are*

$$\boxed{r_+ = \frac{2pq}{m^2} + \frac{(p-q)c}{m^2 k_+},} \tag{89.8}$$

$$\boxed{r_- = \frac{2pq}{m^2} - \frac{(p-q)c}{m^2 k_-}.} \tag{89.9}$$

Hence the SEARCH rates are

$$s_+ = r_+ - o_+, \qquad s_- = r_- - o_-. \tag{89.10}$$

Proof. We prove the positive case. Write

$$(x, y) = \mu(cm) = (qk_+ + c, pk_+ - c).$$

A positive cycle ends at the pk_+-th input one. All y one-cells in the canonical head are consumed before the outer boundary can be reached, by Lemma 87.2. Let M be the number of those one-cells that are consumed by input zero; each such event is a negative-direction SEARCH mismatch. Since these cells are reached before stopping and the input is i.i.d.,

$$\mathbb{E}M = y\frac{q}{m}.$$

Among the pk_+ input ones in the cycle, exactly $y - M$ match one-cells. Therefore the number of one-over-zero mismatches is

$$pk_+ - (y - M) = c + M.$$

The total number of LIFO obstructions in the cycle is consequently

$$c + 2M,$$

with mean

$$c + 2(pk_+ - c)\frac{q}{m}.$$

Dividing by the mean cycle length mk_+ gives (89.8). The negative calculation is obtained by interchanging zero and one, p and q, and reversing the debt sign; this gives (89.9). □

The difference

$$r_+ - r_- = \frac{(p-q)c}{m^2}\left(\frac{1}{k_+} + \frac{1}{k_-}\right) \tag{89.11}$$

is nonzero whenever $p \neq q$. Since coprime $p = q$ forces $p = q = 1$, the only symmetric ratio is 1:1.

89.2 Finite-to-macroscopic operational transfer

Let f_t be a bounded operational reward determined by the physical transition at time t; examples include indicators of SEARCH, outer WRITE, MATCH, or bounded functions of the debt state. Put

$$C_g(f) = \sum_{t=1}^{n} f_t, \qquad n = mg.$$

The positive-residual occupation count is

$$A_g = \#\{1 \le t \le n : \rho_t > 0\}.$$

Let $v_+(f)$ and $v_-(f)$ denote the corresponding critical i.i.d. positive- and negative-bulk reward rates. For R, these are r_+ and r_- from Theorem 89.1.

The proof uses a symbol-level finite-phase quotient in the two deep tails. We record the required periodicity explicitly because the rewards in this section are accumulated one input symbol at a time.

Lemma 89.2 (Symbol-level affine tail periodicity)**.** *For every $r \geq 0$,*

$$\mu(r + pm) = \mu(r) + (m, 0), \tag{89.12}$$

and for every $r \leq 0$,

$$\mu(r - qm) = \mu(r) + (0, m). \tag{89.13}$$

Consequently, after quotienting complete unary m-packets, the reachable configurations in each sufficiently deep positive or negative residual tail have a finite symbol-level phase space.

Proof. For $r \geq 0$, the minimum solution $\mu(r) = (x, y)$ has $0 \leq y < p$: among the p residue classes for y, exactly one makes $r + qy$ divisible by p, and the resulting x is nonnegative. Hence $(x + m, y)$ is a nonnegative solution of weight $r + pm$ with the same $y < p$, so no kernel packet (q, p) can be subtracted. By uniqueness of the minimum demand it is $\mu(r + pm)$. The negative identity is symmetric, using the representative with $0 \leq x < q$.

By Lemma 87.3, between WRITEs the stack is a suffix of a canonical shortest-demand word and the mixed alternating part has uniformly bounded length. Equations (89.12)–(89.13) show that translating a deep residual tail changes only the number of complete unary m-packets. Thus a phase may be specified by the debt state, the bounded mixed suffix, the unary symbol and unary length modulo m, together with the residual class modulo pm or qm. Only finitely many phases occur. □

Choose once and for all

$$H \geq \max(p, q)$$

large enough that the finite symbol-level quotients of Lemma 89.2 are homogeneous whenever $\rho > H$ or $\rho < -H$. In the positive tail, feeding sufficiently many 1's first consumes the bounded mixed head and then accumulates positive mismatches on the unary zero tail until the positive outer-WRITE reset; for H large enough there are at least $h + 1$ unary zero cells available, and the residual only increases along this forcing word. The negative statement is symmetric, using a run of 0's. Thus every sufficiently deep phase can reach the sign-specific reset phase with positive probability without leaving its tail. Consequently, after transient phases are discarded, the finite phase kernel has a unique recurrent class. Restrict to that class and let P_+ and P_- denote the critical i.i.d. one-symbol phase kernels. For a bounded transition reward f, let

$$\bar{f}_\sigma(x) = \frac{q}{m} f(x, 0) + \frac{p}{m} f(x, 1), \qquad \sigma \in \{+, -\},$$

where $f(x, a)$ denotes the reward of the deterministic transition from phase x on symbol a. If π_σ is the invariant probability of the recurrent phase class, define

$$v_\sigma(f) = \pi_\sigma \bar{f}_\sigma.$$

The finite Poisson equation

$$g_\sigma - P_\sigma g_\sigma = \bar{f}_\sigma - v_\sigma(f) \tag{89.14}$$

has a bounded solution, unique up to an additive constant.

89.2.1 Bridge perturbation

Under the fixed-Parikh bridge, if the current residual is ρ_t, then

$$\mathbb{P}(X_{t+1} = 1 \mid \mathcal{F}_t) = \frac{p}{m} - \frac{\rho_t}{m(n - t)}. \tag{89.15}$$

Thus the deviation from the critical i.i.d. symbol law is

$$\delta_t = -\frac{\rho_t}{m(n - t)}.$$

For a fixed-Parikh bridge,

$$\operatorname{Var}(\rho_t) = pq\frac{t(n-t)}{n-1}. \tag{89.16}$$

Therefore

Lemma 89.3 (Total bridge-kernel perturbation)**.**

$$\mathbb{E}\sum_{t=1}^{n-1}|\delta_t| \le \frac{\pi\sqrt{pq}}{2m}\sqrt{n} + O(1). \tag{89.17}$$

Proof. By Cauchy–Schwarz and (89.16),

$$\mathbb{E}|\delta_t| \le \frac{\sqrt{pq}}{m\sqrt{n-1}}\sqrt{\frac{t}{n-t}}.$$

Summing and comparing with

$$\int_0^1 \sqrt{\frac{x}{1-x}}\,dx = \frac{\pi}{2}$$

gives (89.17). □

89.2.2 The central band

Let

$$L_H = \#\{1 \le t \le n : |\rho_t| \le H\}.$$

Since $N_1(t)$ is hypergeometric with parameters (n, pg, t) and

$$\rho_t = mN_1(t) - pt,$$

a direct Stirling bound on the hypergeometric probability mass function gives, uniformly in admissible r,

$$\mathbb{P}(\rho_t = r) \le C_{p,q}\sqrt{\frac{n}{t(n-t)}}.$$

(The finitely many endpoint values of t are absorbed into the constant.) The band contains only finitely many residual values, hence

$$\boxed{\mathbb{E}L_H = O(\sqrt{n}).} \tag{89.18}$$

Because $H \ge \max(p, q)$, a path cannot jump directly from the deep positive tail $\{\rho > H\}$ to the deep negative tail $\{\rho < -H\}$ in one input step. Therefore the number of maximal deep-tail intervals is at most $L_H + 1$ up to an absolute endpoint constant.

89.2.3 Poisson decomposition under the bridge

At time t in a fixed deep sign, write Q_t for the bridge phase kernel. Since the next-symbol probabilities differ from the critical law only by δ_t,

$$\|(Q_t - P_\sigma)u\|_\infty \le 2\|u\|_\infty|\delta_t| \tag{89.19}$$

for every bounded phase function u, and likewise the bridge conditional reward mean differs from $\bar{f}_\sigma$ by at most $2\|f\|_\infty|\delta_t|$.

For an interior step of a deep-tail interval, combine (89.14) with Q_t to write

$$f_t - v_\sigma(f) = \big(g_\sigma(Y_t) - g_\sigma(Y_{t+1})\big) + \Delta M_t + \varepsilon_t, \tag{89.20}$$

where (ΔM_t) is a martingale-difference sequence with uniformly bounded increments and

$$|\varepsilon_t| \le C_{p,q,f}|\delta_t|. \tag{89.21}$$

Indeed, subtract first the bridge conditional mean of f_t, and then replace both the bridge reward mean and $Q_t g_\sigma$ by their critical counterparts; the two replacement errors are controlled by (89.19). The remaining centered terms form the martingale difference, while the g_σ terms telescope. The at most two transitions adjacent to each deep-tail interval endpoint are charged to the interval endpoint error.

Theorem 89.4 (Finite-to-macroscopic operational transfer)**.** *For every bounded operational reward f there is a constant $K_{p,q,f} < \infty$ such that*

$$\boxed{\mathbb{E}\,|C_g(f) - [nv_-(f) + (v_+(f) - v_-(f))A_g]| \le K_{p,q,f}\sqrt{n}.} \tag{89.22}$$

Proof. Compare f_t with $v_+(f)$ when $\rho_t > 0$ and with $v_-(f)$ when $\rho_t \le 0$. On the central band the absolute discrepancy is bounded by a constant times L_H, hence has expectation $O(\sqrt{n})$ by (89.18).

On every maximal deep positive or negative interval, sum (89.20). The g_σ terms telescope. Since g_σ is bounded, all interval endpoint and omitted boundary-step contributions have total L^1 size $O(\mathbb{E}(L_H+1)) = O(\sqrt{n})$. The accumulated perturbation has expectation

$$O\left(\mathbb{E}\sum_{t<n}|\delta_t|\right) = O(\sqrt{n})$$

by Lemma 89.3. The sum of all martingale differences has bounded increments and predictable quadratic variation $O(n)$, so Cauchy–Schwarz gives L^1 norm $O(\sqrt{n})$.

Finally,

$$\sum_{t=1}^{n}\left[v_-(f) + (v_+(f) - v_-(f))\mathbf{1}_{\{\rho_t>0\}}\right] = nv_-(f) + (v_+(f) - v_-(f))A_g.$$

Combining the central, endpoint, perturbation, and martingale contributions proves (89.22). □

89.3 Macroscopic operational limit laws

The critical bridge invariance principle gives

$$\frac{\rho_{\lfloor nt\rfloor}}{\sqrt{pq\,n}} \Longrightarrow B^{\mathrm{br}}(t)$$

in $C[0,1]$, where B^{br} is a standard Brownian bridge. The zero set of a Brownian bridge has Lebesgue measure zero almost surely, so the positive-occupation functional is almost surely continuous at the limit. The positive occupation time of a Brownian bridge is uniform on $[0,1]$; uniform sojourn laws also admit exchangeable-increment and Lévy-bridge formulations [36, 37]. Hence

$$\frac{A_g}{n} \Longrightarrow U, \qquad U \sim \mathrm{Unif}(0,1). \tag{89.23}$$

Classical fixed-score ballot/election-return results provide a discrete counterpart to this occupation phenomenon [35].

Theorem 89.5 (Fixed-semantics operational limit)**.** *For every bounded operational reward f,*

$$\boxed{\frac{C_g(f)}{n} \Longrightarrow v_-(f) + U\big(v_+(f) - v_-(f)\big), \qquad U \sim \mathrm{Unif}(0,1).} \tag{89.24}$$

Proof. Divide (89.22) by n. The L^1 error tends to zero. Apply (89.23) and Slutsky's theorem. □

Corollary 89.6 (Uniform LIFO-obstruction law)**.** *If $p \neq q$, then*

$$\boxed{\frac{R_g}{n} \Longrightarrow \mathrm{Unif}\big(\min(r_-, r_+), \max(r_-, r_+)\big).} \tag{89.25}$$

If $p = q = 1$, then

$$\frac{R_g}{2g} \longrightarrow \frac{1}{2}$$

in probability.

Corollary 89.7 (Rank-one macroscopic fluctuations)**.** *Let $f_1, \dots, f_k$ be bounded operational rewards and*

$$\boldsymbol{C}_g = (C_g(f_1), \dots, C_g(f_k)).$$

Write

$$V_\pm = (v_\pm(f_1), \dots, v_\pm(f_k)).$$

Then

$$\frac{\boldsymbol{C}_g}{n} \Longrightarrow V_- + U(V_+ - V_-), \tag{89.26}$$

and

$$\boxed{\frac{\mathrm{Cov}(\boldsymbol{C}_g)}{n^2} \longrightarrow \frac{1}{12}(V_+ - V_-)(V_+ - V_-)^T.} \tag{89.27}$$

In particular the limiting covariance matrix has rank at most one.

Proof. Apply Theorem 89.5 componentwise; the same occupation variable U drives every coordinate. Since each normalized count is bounded, second moments converge. The covariance formula follows from $\mathrm{Var}(U) = 1/12$. □

Theorem 89.8 (Three-scale LIFO realization)**.** *For the balanced fixed-Parikh random-ordering model,*

$$\boxed{\begin{aligned} &\tau_g = O_{\mathbb{P}}(1), \\ &\frac{B_g}{\sqrt{n}} \Longrightarrow \mathrm{Rayleigh}\left(\frac{\kappa_{p,q}}{\sqrt{pq}}\right), \\ &\frac{R_g}{n} \Longrightarrow \mathrm{Unif}\big(\min(r_-, r_+), \max(r_-, r_+)\big). \end{aligned}} \tag{89.28}$$

for $p \neq q$, with the last law degenerating to $1/2$ for $p = q = 1$.

89.4 Physical-lift duality

For a word w, let $\bar{w}$ denote the bitwise complement. Interchanging p and q and complementing input reverses the residual:

$$\rho_t^{q,p}(\bar{w}) = -\rho_t^{p,q}(w). \tag{89.29}$$

Moreover shortest demands are coordinate-swapped:

$$\mu_{q,p}(-r) = \mathrm{swap}(\mu_{p,q}(r)). \tag{89.30}$$

The two canonical words $\eta(x, y)$ and $\eta(y, x)$ have the same alternating prefix and opposite unary tails. By the canonical-segment property, outer WRITE cannot occur until the mixed prefix has disappeared. During that common mixed phase, complemented input produces opposite debt. The unary phases are then mirror images.

Theorem 89.9 (Physical-lift duality)**.** *For every finite word w,*

$$B_{p,q}(w) = B_{q,p}(\bar{w}), \qquad O_{p,q}(w) = O_{q,p}(\bar{w}). \tag{89.31}$$

Consequently, for every g,

$$(B_g, O_g)_{p,q} \overset{d}{=} (B_g, O_g)_{q,p}$$

under the corresponding balanced fixed-Parikh laws, and

$$\boxed{\kappa_{p,q} = \kappa_{q,p}.} \tag{89.32}$$

Proof. Couple the $p{:}q$ run on w with the $q{:}p$ run on $\bar{w}$. At the start of every WRITE episode the two canonical stacks have the same length, the same alternating-prefix length $2\min(x, y)$, and complementary unary tails. The boundary guard prevents an outer WRITE while that mixed prefix is still present, so both runs necessarily pop the entire mixed prefix. If v ones occur in the original input during those $2\min(x, y)$ steps, then the original debt is $v - \min(x, y)$, whereas the complemented run has debt $2\min(x, y) - v - \min(x, y)$; the two debts are therefore opposite exactly when the unary tails begin. From that point onward the visible unary symbols are complementary and the coupled input symbols are complementary, so MATCH/mismatch status agrees and the debts remain opposite. Hence a positive outer WRITE in one run is paired with a negative outer WRITE at the same time in the other. If no outer WRITE occurs, the equal-length stacks empty simultaneously, so the next input is a bottom WRITE in both runs. Induction over WRITE episodes proves equality of bottom- and outer-WRITE indicators at every time, yielding (89.31). The complement map is a bijection between $\mathcal{F}_{gq,gp}$ and $\mathcal{F}_{gp,gq}$. Finally, (89.32) follows from the bottom-WRITE Rayleigh scale or directly from the mirrored invariant measures. □

89.5 Near-diagonal universality

Let

$$p = q + 1, \qquad m = 2q + 1, \qquad h = q.$$

The central physical-lift fibers simplify sharply.

Proposition 89.10 (Near-diagonal central fibers)**.** *At block boundaries, for $0 \le z \le q$ the central semantic shell z has only the empty lift*

$$E_z = (D_z, \varnothing).$$

For $z = -r$, $1 \le r \le q$, the only central lifts are

$$E_{-r} = (D_{-r}, \varnothing), \qquad F_{-r} = (D_{q-r}, 1^m). \tag{89.33}$$

Proof. The proof is by most-recent-WRITE analysis. A positive outer WRITE has

$$\mu((q + 1)m) = (m, 0),$$

so it creates only a zero tail. Before the next WRITE, a nonempty block-boundary configuration on that tail has stack 0^{km} for some $k \ge 1$ and nonnegative debt, hence semantic shell at least $(q + 1)k \ge q + 1$. It cannot lie in the central range.

A negative outer WRITE has

$$\mu(-(q + 1)m) = (q - 1, 3q + 3),$$

so its mixed prefix has length $2(q - 1) < m$ and its final tail is unary one. Let a later block-boundary suffix have length km and contain a zeros. Debt was reset to zero at the WRITE, and positive debt can subsequently be created only by one-over-zero mismatches

that pop zero cells. Since the written word contained only $q-1$ zeros, the current debt satisfies

$$d \le (q-1) - a.$$

Hence its semantic shell obeys

$$z = a + d - qk \le q - 1 - qk.$$

If $k \ge 2$, then $z \le -q-1$, outside the central range. Thus a central nonempty suffix forces $k = 1$; since the mixed prefix is shorter than m, the surviving length-m suffix is exactly 1^m.

It remains to inspect bottom WRITEs. Both possible bottom-WRITE residuals have the form

$$r = jm + q, \qquad -q - 1 \le j \le q.$$

If $\mu(r) = (x, y)$ and $\ell = x + y$, then

$$mx - q\ell = jm + q,$$

so $\ell \equiv -1 \pmod m$. Write $\ell = km - 1$. Solving gives

$$x = j + qk, \qquad y = k(q+1) - 1 - j.$$

For $-q \le j \le q$, the minimum is $k = 1$, so the newly written word has length $m-1$ and is completely consumed by the remaining $m-1$ symbols of the block. The single exceptional value $j = -q-1$ requires $k = 2$; it occurs exactly for input zero from E_{-q}, and

$$\mu(r) = (q-1, 3q+2).$$

Its canonical word is $(10)^{q-1}1^{m+2}$, so after the remaining $m-1$ pops the next block boundary contains precisely 1^m. These cases give exactly (89.33). □

Let

$$D_q = \sum_{r=1}^{q} \nu(F_{-r}).$$

Since every central semantic shell has total invariant mass one,

$$\kappa_{q+1,q} = 2q + 1 - D_q. \tag{89.34}$$

Lemma 89.11 (Near-diagonal predecessor bound)**.** *If a block ends in one of the nonempty central lifts F_{-r}, then the preceding semantic shell satisfies*

$$Z_j \le -q.$$

Proof. If the block contains a WRITE, consider its last WRITE. A positive outer WRITE cannot produce a one-tail. A negative outer WRITE inserts a word of length $2m$ before its surviving suffix, so with at most $m-1$ input symbols left in the block there are too few subsequent pops for the final stack to be exactly 1^m. A bottom WRITE can occur only on the first symbol of a block by Lemma 88.1, and the only such case that can end in a nonempty central lift is the transition from E_{-q}. It remains to consider a block with no WRITE. Then every one of the m input steps pops an old cell, so the starting stack has the form $T1^m$ with $|T| = m$. Let a be the number of zeros in this stack, and let x_0 be the zero-count of the canonical word produced by the most recent WRITE. Because at least m trailing ones survive and the demand is kernel-free, $x_0 \le q-1$. Positive debt can have increased only by popping zero cells through one-over-zero mismatches, so if the current debt is d then

$$d \le x_0 - a.$$

The residual shell of a length-$2m$ stack is

$$Z_j = a + d - 2q \le x_0 - 2q \le -q - 1.$$

This proves the claim. □

Let

$$Y_q = K - (q+1), \qquad K \sim \mathrm{Bin}\left(2q+1, \frac{q+1}{2q+1}\right).$$

Then

$$\mathbb{E}Y_q = 0, \qquad \mathrm{Var}(Y_q) = \frac{q(q+1)}{2q+1}.$$

By Lemma 89.11 and unit shell mass,

$$\begin{aligned} D_q &\le \sum_{z \le -q} \mathbb{P}\{z + Y_q \in [-q, -1]\} \\ &\le \mathbb{E}\big[\min(Y_q + 1, q)\mathbf{1}_{\{Y_q \ge 0\}}\big] \\ &\le 1 + \mathbb{E}Y_q^+ \\ &\le 1 + \frac{1}{2}\sqrt{\frac{q(q+1)}{2q+1}}. \end{aligned} \tag{89.35}$$

Theorem 89.12 (Near-diagonal universal Rayleigh scale)**.** *For $p = q + 1$,*

$$\boxed{\frac{\kappa_{q+1,q}}{\sqrt{q(q+1)}} \longrightarrow 2.} \tag{89.36}$$

More quantitatively,

$$2\sqrt{\frac{q}{q+1}} - \frac{1}{2\sqrt{2q+1}} \le \frac{\kappa_{q+1,q}}{\sqrt{q(q+1)}} \le \frac{2q+1}{\sqrt{q(q+1)}}. \tag{89.37}$$

Consequently, after first taking $g \to \infty$ and then $q \to \infty$,

$$\frac{B_g}{\sqrt{(2q+1)g}} \Longrightarrow \mathrm{Rayleigh}(2).$$

Proof. Combine (89.34) with (89.35) and divide by $\sqrt{q(q+1)}$. Both sides of (89.37) tend to two. The final statement follows from Theorem 88.6. □

89.6 Finite physical-lift kernels and explicit computation of $\kappa_{p,q}$

The distributional results above require only the invariant-measure definition of $\kappa_{p,q}$. We now explain why the same constant is effectively computable from a finite matrix-kernel system.

89.6.1 Affine shortest-demand tails

At block boundaries the shortest demands have explicit forms. For $z \ge 0$,

$$\mu(mz) = \left(z + q\left\lceil \frac{z}{p} \right\rceil, p\left\lceil \frac{z}{p} \right\rceil - z\right). \tag{89.38}$$

For $n \ge 0$,

$$\mu(-mn) = \left(q\left\lceil \frac{n}{q} \right\rceil - n, n + p\left\lceil \frac{n}{q} \right\rceil\right). \tag{89.39}$$

Hence

$$\mu(m(z+p)) = \mu(mz) + (m, 0), \tag{89.40}$$

$$\mu(-m(n+q)) = \mu(-mn) + (0, m). \tag{89.41}$$

Thus every p positive semantic shells add one unary packet 0^m, while every q negative shells add one unary packet 1^m.

Proof of (89.38)–(89.41). For $mz > 0$, all integer solutions of $px - qy = mz$ are

$$x = z + qk, \qquad y = pk - z.$$

The least k making both coordinates nonnegative is $k = \lceil z/p \rceil$, giving (89.38). The negative formula is symmetric. The affine identities follow by replacing z by $z+p$ and n by $n+q$. □

By Lemma 87.3, a reachable stack is a suffix of a canonical shortest-demand word. The mixed alternating head has bounded length because every freshly written shortest demand has minority count at most h; the only unbounded part is a unary tail. Write the unary-tail length as

$$L = m\ell + r, \qquad 0 \le r < m.$$

Modulo complete unary packets a^m, the remaining information consists of the debt state, the unary symbol, the bounded mixed head, and r. There are finitely many possibilities.

Theorem 89.13 (Finite physical-lift kernel)**.** *For every coprime p, q, there exist finite positive- and negative-tail phase sets and finite families of rational matrices $\{M_j^+\}$ and $\{M_j^-\}$ such that the invariant shell vectors $v_\ell^\pm$ satisfy, for all sufficiently deep levels,*

$$v_\ell^\pm = \sum_j v_{\ell-j}^\pm M_j^\pm. \tag{89.42}$$

The corresponding matrix Laurent polynomials

$$\mathcal{K}_\pm(\lambda) = I - \sum_j M_j^\pm \lambda^j \tag{89.43}$$

have rational coefficients. Together with finitely many central balance equations and the normalization $\nu(o) = 1$, the bounded nonnegative solutions of (89.42) *uniquely determine $\kappa_{p,q}$.*

Proof. The affine identities (89.40)–(89.41) show that translation in a deep semantic tail adds only complete unary m-packets. The canonical-segment property bounds the mixed head independently of level. Quotienting the unary length modulo m therefore leaves finitely many physical phases and one integer level. During one m-symbol input block, the level can change by only a bounded amount, and once the tail is sufficiently deep the transition probabilities depend only on the phase and the level increment, not on the absolute level. Under the critical law every block word has rational probability $(p/m)^k(q/m)^{m-k}$, so the matrices are rational. The invariant equation in the homogeneous tail is exactly (89.42). Irreducibility and the normalization $\nu(o) = 1$ make the recurrent invariant measure unique, hence the central empty-state coordinates and $\kappa_{p,q}$ are uniquely determined. □

This is a finite-phase, level-homogeneous matrix-analytic structure of the same general type that underlies quasi-birth-and-death and matrix-geometric methods [39]. The matrix-analytic machinery itself is classical; the point here is that the shortest-demand pushdown quotient canonically produces such a kernel whose central boundary values encode physical LIFO rematerialization.

89.6.2 Small explicit kernel examples

The two smallest asymmetric $q = 1$ cases reduce to scalar tail recurrences. We record their eliminations as compact reproducibility checks, and then give the first genuinely multi-phase example at 3:2.

The ratio 2:1**.** In negative shell $-n$ there are two deep phases,

$$A_n = (D_{-1}, 1^{3(n-1)}), \qquad B_n = (D_0, 1^{3n}).$$

Write $a_n = \nu(A_n)$, $b_n = \nu(B_n)$ and $u_n = a_n - b_n$. The unit-shell identity gives $a_n + b_n = 1$. Exact enumeration of the eight critical three-symbol blocks gives

$$u_{n-2} - 6u_{n-1} + 39u_n - 8u_{n+1} = 0 \tag{89.44}$$

in the homogeneous tail, while the finite central balance equations are

$$31u_1 - 8u_2 = 15, \qquad 4u_1 - 39u_2 + 8u_3 = 3. \tag{89.45}$$

The characteristic polynomial is

$$Q_{2,1}(R) = 8R^3 - 39R^2 + 6R - 1.$$

It has a unique real root $R > 1$. Boundedness of the invariant lift profile cancels that mode, which is equivalent to

$$u_1(31R^2 - 4R + 1) = 15R^2 - 3R.$$

With $\theta = (1 + u_1)/2 = a_1$, elimination of R between this relation and $Q_{2,1}(R) = 0$ gives

$$8\theta^3 - 3\theta - 1 = 0. \tag{89.46}$$

Thus

$$\kappa_{2,1} = 2 + \theta = 2.737843258897860\ldots. \tag{89.47}$$

The ratio 3:1**.** The same two negative phases occur with unary packet length four. Writing again $u_n = a_n - b_n$, exact sixteen-block enumeration gives

$$81u_{n+1} - 364u_n + 54u_{n-1} - 12u_{n-2} + u_{n-3} = 0, \tag{89.48}$$

and the finite central equations are

$$283u_1 - 81u_2 = 148, \tag{89.49}$$

$$-27u_1 + 364u_2 - 81u_3 = -40, \tag{89.50}$$

$$9u_1 - 54u_2 + 364u_3 - 81u_4 = 4. \tag{89.51}$$

The characteristic polynomial is

$$Q_{3,1}(R) = 81R^4 - 364R^3 + 54R^2 - 12R + 1.$$

Cancelling its unique unstable real mode gives

$$u_1(81R^3 - 81R^2 + 27R - 3) = 148R^2 - 40R + 4.$$

Eliminating R and setting $\theta = (1 + u_1)/2$ yields

$$54\theta^4 - 18\theta^2 - 8\theta - 1 = 0, \tag{89.52}$$

so

$$\kappa_{3,1} = 2 + \theta = 2.750694578264632\ldots. \tag{89.53}$$

The ratio 3:2**.** Here the deep negative tail has seven phases. Writing the negative semantic shell as $-n$, the four odd-shell phases for $n = 2j + 1$ are

$$\begin{aligned} O_0(j) &= (D_{-1}, 1^{5j}), \\ O_1(j) &= (D_0, 0\,1^{5j+4}), \\ O_2(j) &= (D_0, 10\,1^{5j+3}), \\ O_3(j) &= (D_1, 1^{5(j+1)}), \end{aligned}$$

and the three even-shell phases for $n = 2j$ are

$$\begin{aligned} E_0(j) &= (D_{-2}, 1^{5(j-1)}), \\ E_1(j) &= (D_{-1}, 0\,1^{5j-1}), \\ E_2(j) &= (D_0, 1^{5j}). \end{aligned}$$

Enumerating the 32 five-symbol blocks with their exact critical probabilities $3^k 2^{5-k}/5^5$ gives a rational seven-phase Laurent kernel whose determinant factors as

$$\det \mathcal{K}_{3,2}(\lambda) = \frac{(\lambda - 1)^2 C_3(\lambda) K_5(\lambda)}{5^{20}\lambda^6}, \tag{89.54}$$

where

$$\begin{aligned} C_3(\lambda) &= 59049\lambda^3 - 1531872\lambda^2 + 9472\lambda - 1024, \\ K_5(\lambda) &= 59049\lambda^5 - 1387530\lambda^4 + 16632265\lambda^3 \\ &\quad - 1448160\lambda^2 + 11520\lambda + 1024. \end{aligned}$$

The critical phase kernel has invariant vector

$$\left(\frac{1}{4}, \frac{3}{50}, \frac{1}{10}, \frac{9}{100}, \frac{1}{4}, \frac{1}{25}, \frac{21}{100}\right) \tag{89.55}$$

in the order $(O_0, O_1, O_2, O_3, E_0, E_1, E_2)$. Coupling this exact tail kernel to the finite central balance equations gives numerically

$$\kappa_{3,2} = 4.64089572244120\ldots. \tag{89.56}$$

An independent direct probability propagation, extrapolated from block horizons 100 and 200, gives $4.6408956068\ldots$, independently confirming the first six decimal places. The rematerialization constant need not be rational even though the underlying transition probabilities are rational: already the cubic (89.46) has no rational root, so $\kappa_{2,1}$ is irrational.

Remark 89.14 (Algebraicity and regular kernels). The finite kernels provide an exact rational boundary-value problem for $\kappa_{p,q}$. When the associated matrix Laurent pencil is regular, companion linearization shows that its characteristic modes are algebraic, and the finite central boundary equations then make the normalized bounded solution algebraic. This applies to the explicit kernels above. We do not require, and do not use, an unconditional algebraicity theorem for every coprime p, q in the probabilistic results of this Part.

Chapter 90

Optimal-Fiber Compression and Active Trade Geometry

Research branch — Optimal-Fiber Topology and Physical Holonomy. *Earlier chapters introduced higher-dimensional minimum-completion fibers and showed how equal-length kernel trades create nontrivial quotient topology. This branch sharpens that picture through the complete active-rank-two classification and then carries the topology into a category-relative physical transport theory. Kernel rank controls when higher topology can appear; the global homotopy type is governed by the translation geometry of the active optimal fiber, while the displacement kernel identifies the pure loops on which physical LIFO holonomy can live.*

For an integer counting matrix $A \in \mathbb{Z}^{k\times d}$ and completable residual r, write

$$\ell_A(r) = \min\{\mathbf{1}^\top x : x \in \mathbb{N}^d,\ Ax = -r\},$$
$$\mathcal{O}_A(r) = \{x : Ax = -r,\ \mathbf{1}^\top x = \ell_A(r)\}.$$

The earlier higher-dimensional completion theory supplied the filled shortest-prefix quotient. The purpose here is to compress its global topology to finite optimal-fiber translation data and then determine what active rank two can actually support.

90.1 Minimum-completion geometry

Let $A \in \mathbb{Z}^{k\times d}$. A residual is an element $r \in A\mathbb{Z}^d$ for which the completion equation

$$Ax = -r, \qquad x \in \mathbb{N}^d$$

is feasible. A convenient global hypothesis is the following.

Definition 90.1 (Completion totality)**.** The matrix A is *completion-total* if there exists $h \in \mathbb{N}^d$ with every coordinate positive and $Ah = 0$.

If A is completion-total, every lattice residual is completable: if $r = Az$ with $z \in \mathbb{Z}^d$, then $th - z \in \mathbb{N}^d$ for all sufficiently large t, and $A(th - z) = -r$.

For a completable residual r define

$$\ell_A(r) = \min\{\mathbf{1}^\top x : x \in \mathbb{N}^d,\ Ax = -r\} \tag{90.1}$$

and the optimal Parikh fiber

$$\mathcal{O}_A(r) = \{x \in \mathbb{N}^d : Ax = -r,\ \mathbf{1}^\top x = \ell_A(r)\}. \tag{90.2}$$

The set is finite because all of its points have the same coordinate sum.

The equal-length kernel is

$$K_A = \ker_{\mathbb{Z}} \begin{bmatrix} A \\ \mathbf{1}^\top \end{bmatrix} = \{g \in \mathbb{Z}^d : Ag = 0,\ \mathbf{1}^\top g = 0\}. \tag{90.3}$$

Differences of optimal completions lie in K_A.

90.1.1 The shortest-prefix down-set

For $x \in \mathbb{N}^d$ put

$$B_x = \{u \in \mathbb{N}^d : 0 \le u \le x\},$$

where inequalities are coordinatewise, and define

$$\mathcal{C}_A(r) =\downarrow \mathcal{O}_A(r) = \bigcup_{x \in \mathcal{O}_A(r)} B_x. \tag{90.4}$$

Proposition 90.2 (Exact geodesic-prefix criterion)**.** *For $u \in \mathbb{N}^d$,*

$$u \in \mathcal{C}_A(r) \quad \Longleftrightarrow \quad \ell_A(r + Au) = \ell_A(r) - \mathbf{1}^\top u.$$

Thus $\mathcal{C}_A(r)$ is exactly the set of Parikh vectors that occur as prefixes of shortest completions of r.

Proof. If $u \le x$ for some $x \in \mathcal{O}_A(r)$, then $x-u$ completes $r+Au$ and has length $\ell_A(r)-\mathbf{1}^\top u$. A strictly shorter completion would combine with u to contradict optimality of x.

Conversely, if the displayed equality holds and y is a shortest completion of $r + Au$, then $u + y$ completes r and has length $\ell_A(r)$. Hence $u + y \in \mathcal{O}_A(r)$ and $u \in \mathcal{C}_A(r)$. □

Give every finite down-set $D \subset \mathbb{N}^d$ its standard cubical realization $|D|$: a J-oriented elementary cube based at u is present when $u + \mathbf{1}_J \in D$. The down-set condition supplies all faces automatically.

Two equally oriented cells in $|\mathcal{C}_A(r)|$ are identified when their base vertices differ by an element of K_A. We write

$$I_A(r) = |\mathcal{C}_A(r)|/K_A \tag{90.5}$$

for the resulting filled shortest-prefix quotient. By theorem 90.2, its vertices are precisely shortest-path residual states, and its higher cubes record commuting one-letter shortest moves.

90.2 Active trades and optimal-fiber lattice geometry

The full kernel K_A can contain directions that never occur inside the optimal fiber for a fixed residual. The correct residual-dependent lattice is smaller.

Definition 90.3 (Active trade lattice)**.** Fix a completable residual r. Define

$$L_A(r) = \langle x - y : x, y \in \mathcal{O}_A(r) \rangle_{\mathbb{Z}} \subseteq K_A. \tag{90.6}$$

For $x_0 \in \mathcal{O}_A(r)$ define the normalized optimal fiber

$$P_A(r) = \mathcal{O}_A(r) - x_0 \subseteq L_A(r). \tag{90.7}$$

Its active rank is $\operatorname{rank}_{\mathbb{Z}} L_A(r)$.

Lemma 90.4 (Active collision lemma)**.** *If $u, v \in \mathcal{C}_A(r)$ and $u - v \in K_A$, then*

$$u - v \in \mathcal{O}_A(r) - \mathcal{O}_A(r) \subseteq L_A(r).$$

Proof. The condition $u - v \in K_A$ implies

$$Au = Av, \qquad \mathbf{1}^\top u = \mathbf{1}^\top v.$$

Hence u and v reach the same residual at the same geodesic depth. Choose a shortest completion y of that common residual. By theorem 90.2, both $u+y$ and $v+y$ lie in $\mathcal{O}_A(r)$. Their difference is $u - v$. □

Proposition 90.5 (Active-fiber saturation)**.** *For every $x_0 \in \mathcal{O}_A(r)$,*

$$\mathcal{O}_A(r) = (x_0 + L_A(r)) \cap \mathbb{N}^d. \tag{90.8}$$

Consequently

$$P_A(r) = \operatorname{conv}(P_A(r)) \cap L_A(r). \tag{90.9}$$

Proof. The inclusion $\mathcal{O}_A(r) \subseteq (x_0 + L_A(r)) \cap \mathbb{N}^d$ follows from the definition of $L_A(r)$. Conversely, if $x = x_0 + \lambda \in \mathbb{N}^d$ with $\lambda \in L_A(r) \subseteq K_A$, then

$$Ax = Ax_0 = -r, \qquad \mathbf{1}^\top x = \mathbf{1}^\top x_0 = \ell_A(r),$$

so $x \in \mathcal{O}_A(r)$.

For the second statement, put $P = P_A(r)$. Equation (90.8) says

$$P = L_A(r) \cap \{z \in \mathbb{R}^d : z_i \geq -(x_0)_i \ \forall i\}.$$

The right-hand real region is convex. Since $\operatorname{conv}(P)$ lies inside it, every lattice point of $\operatorname{conv}(P)$ belongs to P. □

Thus every normalized optimal fiber is a finite lattice-convex configuration in its active lattice. This constraint is decisive in active rank two.

90.3 Optimal-fiber nerve compression

Let

$$X = |\mathcal{C}_A(r)|, \qquad L = L_A(r).$$

Form the periodic saturation

$$\widetilde{X} = \bigcup_{\lambda \in L} (X + \lambda). \tag{90.10}$$

By theorem 90.4, the quotient $\widetilde{X}/L$ is canonically the same cell quotient as $I_A(r)$.

Lemma 90.6 (Contractible translate intersections)**.** *Every nonempty finite intersection*

$$\bigcap_{\lambda \in F} (X + \lambda), \qquad F \subset L \text{ finite},$$

is contractible.

Proof. Work first on vertices. Put $C = \mathcal{C}_A(r)$ and

$$D_F = \bigcap_{\lambda \in F} (C + \lambda).$$

Assume $D_F \neq \varnothing$ and set

$$m_i(F) = \max_{\lambda \in F} \lambda_i.$$

Every $u \in D_F$ satisfies $u \geq m(F)$ coordinatewise: for each coordinate choose a translate attaining the maximum. If $0 \leq z \leq u - m(F)$, then for every $\lambda \in F$,

$$0 \leq z + m(F) - \lambda \leq u - \lambda.$$

Since $u - \lambda \in C$ and C is a down-set, $z + m(F) - \lambda \in C$. Thus $D_F - m(F)$ is a finite coordinate down-set in $\mathbb{N}^d$. Its cubical realization is a union of coordinate boxes anchored at the origin and is therefore contractible. Translating back proves the claim. □

Choose $x_0 \in \mathcal{O}_A(r)$ and write $P = P_A(r)$. Define the periodic simplicial complex

$$\mathcal{N}(P) = \{F \subseteq L : \ F \text{ finite and } \exists c \in L \text{ with } c - F \subseteq P\}. \tag{90.11}$$

Equivalently, $\mathcal{N}(P)$ is the nerve of the family of finite convex-lattice sets

$$\{\lambda + P : \lambda \in L\}.$$

The lattice L acts by translation on $\mathcal{N}(P)$.

Theorem 90.7 (Optimal-Fiber Nerve Compression)**.** *For every completable residual r,*

$$I_A(r) \simeq \mathcal{N}(P_A(r))/L_A(r). \tag{90.12}$$

The homotopy type of the global shortest-prefix quotient is therefore determined by the finite optimal fiber and its translation lattice.

Proof. The translates $\{X + \lambda\}_{\lambda \in L}$ cover $\widetilde{X}$ and are permuted by the L-action. By theorem 90.6, every nonempty finite intersection is contractible. The equivariant simplicial nerve theorem [45, Theorem 2.8] gives an L-equivariant homotopy equivalence between $\widetilde{X}$ and the nerve of this cover.

It remains to identify that nerve with (90.11). A finite $F \subset L$ spans a simplex exactly when there exists a common vertex u with

$$u - \lambda \in \mathcal{C}_A(r) \qquad (\lambda \in F).$$

All points $u - \lambda$ have the same residual and the same geodesic depth because their pairwise differences lie in $L \subseteq K_A$. Choose a shortest suffix y from that common residual. Then

$$c = u + y$$

satisfies $c - \lambda \in \mathcal{O}_A(r)$ for every $\lambda \in F$. Conversely, if $c - F \subseteq \mathcal{O}_A(r)$ then c belongs to every translated shortest-prefix set. After translating by x_0, this is exactly the simplex criterion (90.11).

Finally, the L-action is free and cellular, so passage to quotients preserves the equivariant homotopy equivalence. Using theorem 90.4, $\widetilde{X}/L$ is $I_A(r)$. □

Remark 90.8 (Rank-one rigidity). If $\operatorname{rank} L_A(r) = 1$, lattice-convexity forces $P_A(r)$ to be an interval of consecutive points in a rank-one lattice. If the fiber is nontrivial, $\mathcal{N}(P_A(r))$ is contractible and its translation quotient is S^1. Thus the first non-rigid regime is active rank two.

90.4 Finite presentations and first homology

The compressed quotient has one vertex orbit. Its 2-skeleton can be read directly from differences in the optimal fiber.

Proposition 90.9 (Finite fundamental-group presentation)**.** *Assume $0 \in P = P_A(r)$. Then $\pi_1(I_A(r))$ is isomorphic to*

$$G(P) = \big\langle X_p \ (p \in P),\ X_0 = 1 \ \big|\ X_p^{-1} X_q = X_s^{-1} X_t \text{ whenever } q - p = t - s \big\rangle. \tag{90.13}$$

Proof. In $\mathcal{N}(P)/L$, oriented edge orbits are indexed by differences $d = q - p \in P - P$. Write the corresponding edge generator as a_d, with $a_{-d} = a_d^{-1}$. Every triple $p, q, s \in P$ supplies the triangular relation

$$a_{q-p} a_{s-q} = a_{s-p}.$$

Set $X_p = a_p$ for $p \in P$ and $X_0 = 1$. The triangle on $0, p, q$ gives

$$a_{q-p} = X_p^{-1} X_q.$$

Two edges represent the same edge orbit exactly when their differences agree, which gives precisely the relations in (90.13). Higher simplices do not affect the fundamental group. □

There is a canonical displacement homomorphism

$$\delta : G(P) \longrightarrow L, \qquad \delta(X_p) = p. \tag{90.14}$$

It is surjective because $L = \langle P - P \rangle = \langle P \rangle$.

Corollary 90.10 (Parallelogram–commutator principle)**.** *If*

$$p, \quad p+u, \quad p+v, \quad p+u+v \in P,$$

then the corresponding difference generators satisfy

$$[a_u, a_v] = 1.$$

Proof. The two triangles give

$$a_u a_v = a_{u+v} = a_v a_u.$$

□

90.4.1 Freiman dimension and first homology

Let F be a field of characteristic 0. A normalized Freiman 2-homomorphism on P is a function $f : P \to F$ with $f(0) = 0$ such that

$$p + q = s + t \quad \Longrightarrow \quad f(p) + f(q) = f(s) + f(t).$$

Let $\dim_F(P)$ denote the dimension of this vector space. This agrees with the usual normalized Freiman-dimension convention.

Proposition 90.11 (First homology as the Freiman shadow)**.** *For every finite normalized fiber P,*

$$\beta_1(\mathcal{N}(P)/L) = \dim_F(P).$$

Proof. Abelianizing (90.13) replaces X_p by additive variables x_p and the defining relations by

$$x_q - x_p = x_t - x_s \quad \text{whenever} \quad q - p = t - s.$$

Equivalently,

$$x_p + x_t = x_q + x_s \quad \text{whenever} \quad p + t = q + s.$$

After $x_0 = 0$, this is exactly the vector space of normalized Freiman homomorphisms. Tensoring H_1 with F gives the stated equality of ranks. □

Theorem 90.12 (Rank-two first-Betti bound)**.** *If* $\operatorname{rank} L_A(r) = 2$*, then*

$$\beta_1(I_A(r)) \in \{2, 3\}.$$

Proof. Put $P = P_A(r)$, $m = |P|$, and $Q = \operatorname{conv}(P)$ inside $L \otimes \mathbb{R} \cong \mathbb{R}^2$. By theorem 90.5, $P = Q \cap L$. Let B be the number of lattice points on ∂Q. Pick's theorem and the Ehrhart polynomial give

$$|2Q \cap L| = 4m - B - 3 \leq 4m - 6. \tag{90.15}$$

Since $P + P \subseteq 2Q \cap L$,

$$|P + P| \leq 4m - 6.$$

If $d_F = \dim_F(P)$, Freiman's dimension lemma [47, Chapter 5] gives

$$|P + P| \geq (d_F + 1)m - \binom{d_F + 1}{2}.$$

For $d_F \geq 4$ and $m \geq 5$, the right side is at least $5m - 10 > 4m - 6$. If $m \leq 4$, then trivially $d_F \leq m - 1 \leq 3$. Hence $d_F \leq 3$.

On the other hand, the two coordinate linear functionals on the full-rank planar lattice L give two independent normalized Freiman homomorphisms, so $d_F \geq 2$. Apply theorem 90.11 and theorem 90.7. □

90.5 The optimal-fiber Taylor slice

The nerve model has a second description in commutative algebra. This description is classical on the lattice-ideal side; the point here is that the minimum-completion quotient selects a canonical principal slice.

Fix a field $\Bbbk$, put

$$S = \Bbbk[x_1, \dots, x_d],$$

and let $L = L_A(r) \subseteq \mathbb{Z}^d$. Because $L \subseteq K_A$, every $\lambda \in L$ has coordinate sum 0, hence

$$L \cap \mathbb{N}^d = \{0\}.$$

Define the lattice ideal

$$I_L = \langle x^{u^+} - x^{u^-} : u \in L \rangle \subseteq S. \tag{90.16}$$

Give S the $\mathbb{Z}^d/L$ grading $\deg(x^a) = a + L$. For $\alpha \in \mathbb{N}^d/L$ write

$$\mathrm{fiber}(\alpha) = \{x^a : a \in \mathbb{N}^d,\ a + L = \alpha\}.$$

Choose $x_0 \in \mathcal{O}_A(r)$ and set

$$\alpha = x_0 + L.$$

By theorem 90.5,

$$\mathrm{fiber}(\alpha) \quad \text{has exponent set exactly } \mathcal{O}_A(r). \tag{90.17}$$

For each degree β, let $E_i(\beta)$ be the family of i-element subsets of $\mathrm{fiber}(\beta)$ with monomial gcd equal to 1. Bayer–Sturmfels identify these sets as the homogeneous basis of the lattice-ideal Taylor resolution and give the differential explicitly [44, Proposition 3.10].

Write $\beta \preceq \alpha$ when $\alpha - \beta$ is the degree of a monomial.

Theorem 90.13 (Optimal-Fiber Taylor Slice)**.** *Let*

$$Q(P) = \mathcal{N}(P_A(r))/L_A(r).$$

For every $j \geq 0$ there is a canonical basis correspondence

$$C_j(Q(P); \Bbbk) \cong \bigoplus_{\beta \preceq \alpha} \Bbbk\, E_{j+1}(\beta), \tag{90.18}$$

under which the cellular boundary is the Bayer–Sturmfels Taylor differential specialized by

$$\varepsilon : S \to \Bbbk, \qquad x_i \mapsto 1.$$

Equivalently, $C_\bullet(Q(P); \Bbbk)$ is the dehomogenized principal degree slice of the lattice-ideal Taylor complex.

Proof. Represent a j-simplex orbit of $Q(P)$ by a $(j+1)$-element subset

$$F = \{u_0, \dots, u_j\} \subseteq \mathcal{O}_A(r).$$

Let $g = \min(u_0, \dots, u_j)$ coordinatewise. Subtracting g gives a gcd-one set

$$\overline{F} = F - g$$

in the fiber of the degree

$$\beta = \alpha - \deg(x^g) \preceq \alpha.$$

If F is replaced by an L-translate, the gcd changes by the same translate and $\overline{F}$ is unchanged. Hence the construction depends only on the simplex orbit.

Conversely, let $I \in E_{j+1}(\beta)$ with $\beta \preceq \alpha$. Choose $g \in \mathbb{N}^d$ with $\alpha = \beta + \deg(x^g)$. Multiplying every monomial of I by x^g gives a subset of $\mathrm{fiber}(\alpha) = \mathcal{O}_A(r)$. A different

choice g' differs from g by an element of L and therefore gives the same simplex orbit. The two constructions are inverse.

If one vertex is removed, the remaining set acquires a new common gcd h. Bayer–Sturmfels' differential is

$$\partial(I) = \sum_{m \in I} \operatorname{sign}(m, I)\ \gcd(I \setminus \{m\})\, [I \setminus \{m\}],$$

where the bracket denotes removal of that common factor. Under $x_i \mapsto 1$, every monomial coefficient becomes 1, leaving exactly the oriented simplicial boundary of the quotient. The shift $E_{j+1} \leftrightarrow C_j$ is the usual one: E_1 is homological degree 0 in the resolution of S/I_L. □

90.5.1 Principal truncation and character twists

Let $\mathbb{T}_\bullet$ be the full lattice Taylor resolution and $\mathbb{F}_\bullet$ a minimal $\mathbb{Z}^d/L$-graded free resolution of S/I_L.

Lemma 90.14 (Principal truncation lemma)**.** *The full graded chain-homotopy equivalence between $\mathbb{T}_\bullet$ and $\mathbb{F}_\bullet$ restricts to a chain-homotopy equivalence on the principal degree order ideal*

$$\{\beta : \beta \preceq \alpha\}.$$

Proof. Comparison maps and homotopies may be chosen homogeneous of degree 0. A homogeneous map from a basis summand of degree β to one of degree γ has a monomial coefficient of degree $\beta - \gamma$, so it can be nonzero only when $\gamma \preceq \beta$. Hence every map and homotopy preserves every degree down-set, in particular the principal down-set below α. □

Let $\chi : L \to \Bbbk^\times$ be a character. After a faithfully flat extension of the field if necessary, extend χ to a homomorphism

$$\widetilde{\chi} : \mathbb{Z}^d \to \overline{\Bbbk}^\times.$$

Let $\Bbbk_\chi$ be the corresponding rank-one local system on $Q(P)$.

Proposition 90.15 (Character-twisted Taylor slice)**.** *After a degree-dependent rescaling of basis vectors, the complex*

$$C_\bullet(Q(P); \Bbbk_\chi)$$

is isomorphic to the principal Taylor slice evaluated at

$$x_i = \widetilde{\chi}(e_i).$$

Proof. For every $\beta \preceq \alpha$, choose a lift $g_\beta \in \mathbb{N}^d$ with

$$\alpha = \beta + \deg(x^{g_\beta}).$$

Suppose a Taylor boundary term deletes one vertex, produces common gcd exponent h, and passes from degree β to γ. The corresponding face in the periodic nerve differs from the chosen representative in degree γ by the deck translation

$$\tau = h + g_\beta - g_\gamma \in L.$$

After rescaling the degree-β basis by $\widetilde{\chi}(g_\beta)$, the evaluated Taylor coefficient becomes

$$\widetilde{\chi}(h)\frac{\widetilde{\chi}(g_\beta)}{\widetilde{\chi}(g_\gamma)} = \widetilde{\chi}(\tau) = \chi(\tau),$$

which is exactly the local-system coefficient attached to that face. The signs are the common simplicial incidence signs. □

Chapter 91

Lattice-Ideal Slices and Rank-Two Structural Rigidity

The compression theorem admits an algebraic refinement: the cellular chains of the quotient can be read as a principal dehomogenized slice of the Taylor resolution of the active lattice ideal. This bridge controls homology and prepares the planar classification.

91.1 Rank-two homological dimension

Assume from now on that

$$\operatorname{rank}_{\mathbb{Z}} L_A(r) = 2.$$

Peeva–Sturmfels prove the general projective-dimension bound

$$\operatorname{pd}_S(S/I_L) \le 2^{\operatorname{rank} L} - 1,$$

so in rank two the minimal resolution has no term above homological degree 3 [42].

Lemma 91.1 (Torus-point injectivity)**.** *For every field $\Bbbk$ and every character $\chi : L \to \Bbbk^\times$, the third differential of the full minimal resolution, after the corresponding torus-point specialization, is injective.*

Proof. Extend scalars to an algebraic closure and represent χ by a point $\xi \in (\overline{\Bbbk}^\times)^d$.

If χ is nontrivial, then some lattice binomial evaluates nonzero at ξ, so ξ lies outside the support of S/I_L. The specialized full free resolution is therefore acyclic.

If χ is trivial, localize at the algebraic torus. For a lattice basis g_1, g_2 of L, the Laurent lattice ideal is generated by

$$x^{g_1} - 1, \qquad x^{g_2} - 1.$$

These two independent equations form a regular sequence in the regular Laurent ring. Hence the localized module is a codimension-two complete intersection and its Tor vanishes above degree 2. In both cases the third homology of the specialized full minimal resolution is zero. Since there is no fourth term, the third differential is injective. □

Theorem 91.2 (Rank-Two Homological-Dimension Theorem)**.** *If* $\operatorname{rank} L_A(r) = 2$*, then*

$$H_j(I_A(r); \mathbb{Z}) = 0 \qquad (j \ge 3), \tag{91.1}$$

and $H_2(I_A(r); \mathbb{Z})$ is free abelian.

Proof. Over any field $\Bbbk$, theorems 90.13 and 90.14 identify the chain complex of $Q(P)$ up to chain homotopy with the principal slice of a minimal free resolution of length at most 3. Thus homology vanishes above degree 3. By theorem 91.1, the third differential remains injective after restriction to the principal slice, so

$$H_j(Q(P); \Bbbk) = 0 \qquad (j \ge 3).$$

The same holds for every prime field and for $\mathbb{Q}$. Since $Q(P)$ is finite, the universal coefficient theorem implies (91.1). Applying the mod-p universal coefficient sequence in degree 3 gives

$$\operatorname{Tor}_1^{\mathbb{Z}}(H_2(Q(P);\mathbb{Z}),\mathbb{F}_p) = 0$$

for every prime p, so $H_2(Q(P);\mathbb{Z})$ has no torsion. Finally use theorem 90.7. □

91.2 Planar lattice geometry and the fundamental group

By theorem 90.5, a rank-two normalized fiber is a finite lattice-convex set in a rank-two lattice. We now show that a single geometric feature—a nondegenerate parallelogram—separates the abelian and free branches of the fundamental group.

91.2.1 The parallelogram branch

Lemma 91.3 (Minimal parallelogram lemma)**.** *Let $P \subset L \cong \mathbb{Z}^2$ be finite and lattice-convex. If P contains a nondegenerate lattice parallelogram, then it contains one of normalized area 1 or 2 whose edge directions are primitive.*

Proof. Choose a nondegenerate parallelogram

$$R = \{a, a+u, a+v, a+u+v\} \subseteq P$$

of minimum positive normalized area $D = |\det_L(u,v)|$. If u were not primitive, an interior lattice point on each of the two u-parallel edges would belong to P by lattice-convexity and would produce a smaller parallelogram. Thus u and v are primitive. Hence the boundary of $\operatorname{conv}(R)$ contains exactly four lattice points. Pick's theorem gives

$$I(R) = D - 1$$

interior lattice points.

Let c be the center of R. Reflection in c preserves the lattice and the parallelogram. If there is an interior lattice point $z \neq c$, its reflection $z' = 2c - z$ is another interior lattice point. One of the two parallelograms formed by the pair z, z' and one of the two opposite vertex pairs of R is nondegenerate; it lies strictly inside R and has smaller normalized area, contradicting minimality. Hence R has at most one interior lattice point, so $D - 1 \leq 1$. □

A finite lattice-convex set $X \subset L$ will be called *parity-complete* if it meets every coset of $2L$.

Lemma 91.4 (Observer chain)**.** *If $X \subsetneq P$ are finite lattice-convex subsets of L, there exists $z \in P \setminus X$ such that*

$$\operatorname{conv}(X \cup \{z\}) \cap L = X \cup \{z\}.$$

Iterating gives a chain from X to P by one-point lattice-convex extensions.

Proof. Choose $z \in P \setminus X$ minimizing the positive Euclidean distance to $\operatorname{conv}(X)$. If a further lattice point $y \notin X \cup \{z\}$ belonged to $\operatorname{conv}(X \cup \{z\})$, then $y \in P$ by lattice-convexity and, along a segment from z toward $\operatorname{conv}(X)$, would have strictly smaller distance to $\operatorname{conv}(X)$, a contradiction. □

Lemma 91.5 (Parity propagation)**.** *Suppose $X \subseteq P$ is lattice-convex and parity-complete, and suppose that in $G(P)$ all X_x with $x \in X$ lie in the subgroup generated by commuting elements U, V whose displacements form a basis of L. If z is an observer of X, then X_z lies in the same subgroup, with exponent vector equal to the lattice coordinates of z.*

Proof. Choose $x \in X$ with $x \equiv z \pmod{2L}$ and put $y = (x+z)/2 \in L$. The observer property forces $y \in X$. Since

$$z - y = y - x,$$

relation (90.13) gives

$$X_y^{-1} X_z = X_x^{-1} X_y, \qquad X_z = X_y X_x^{-1} X_y.$$

If $X_x = U^a V^b$ and $X_y = U^c V^d$, commutativity gives

$$X_z = U^{2c-a} V^{2d-b},$$

whose displacement is $2y - x = z$. □

Lemma 91.6 (Determinant-two parity completion)**.** *Let*

$$C_2 = \{(0,0), (1,0), (0,1), (-1,2), (0,2)\} \subset \mathbb{Z}^2.$$

If a lattice-convex set $P \supseteq C_2$ contains a point in the missing parity class $(1,1) + 2\mathbb{Z}^2$, then it contains at least one point of

$$M = \{(1,-1), (1,1), (-1,1), (-1,3)\}.$$

For every $m \in M$, the set $C_2 \cup \{m\}$ is lattice-convex and parity-complete, and the generator X_m is already a word in the generators from C_2.

Proof. Consider the linear map

$$\Phi(x,y) = (y, 2x + y).$$

It sends $\operatorname{conv}(C_2)$ to the square $[0,2]^2$. If z lies in the missing parity class, write

$$\Phi(z) = (s,t), \qquad A = s - 1, \quad B = t - 1.$$

Then A, B are even and $B - A \equiv 2 \pmod 4$, so $|A| \neq |B|$ and therefore

$$\big||A| - |B|\big| \geq 2.$$

One of the four inequalities

$$B \geq |A| + 2, \quad -B \geq |A| + 2, \quad A \geq |B| + 2, \quad -A \geq |B| + 2$$

holds. In the first case, the point $(1,3)$ lies in $\operatorname{conv}([0,2]^2 \cup \{\Phi(z)\})$: indeed, with $\theta = 1/(B-1)$, write it as a convex combination of $\Phi(z)$ and a point of the top side using $|A| \leq B - 2$. The other three cases are symmetric. The four resulting points are precisely the images under Φ of the four elements of M. Since P is lattice-convex, the corresponding element of M lies in P.

A direct check of the four small polygons shows

$$\operatorname{conv}(C_2 \cup \{m\}) \cap \mathbb{Z}^2 = C_2 \cup \{m\}$$

for each $m \in M$, and the added point supplies the fourth parity class. Finally, in each case there are $p, a, b \in C_2$ with $m = p + b - a$. The relation

$$X_p^{-1} X_m = X_a^{-1} X_b$$

expresses X_m in the subgroup generated by C_2. □

Theorem 91.7 (Parallelogram–abelian fundamental group)**.** *Let $P = P_A(r)$ have active rank two. If P contains a nondegenerate lattice parallelogram, then*

$$\pi_1(I_A(r)) \cong L_A(r) \cong \mathbb{Z}^2.$$

Proof. By theorem 91.3, choose a minimal parallelogram of normalized area 1 or 2.

If the area is 1, translate one vertex to 0 and let u, v be its primitive edge directions. They form a basis of L, the four vertices represent all four classes in $L/2L$, and theorem 90.10 gives commuting basis generators X_u, X_v. The four-point core is parity-complete. Apply theorems 91.4 and 91.5 inductively to reach all of P.

If the area is 2, let the primitive edge directions be a, b. Modulo $2L$ they are equal, so the center

$$v = \frac{a+b}{2}$$

is a lattice point. With $u = a$, the pair u, v is a lattice basis and the five lattice points of the minimal parallelogram have coordinates

$$C_2 = \{0, u, v, 2v-u, 2v\}.$$

The relations

$$X_{2v} = X_v^2, \qquad X_{2v-u} = X_v X_u^{-1} X_v, \qquad X_u = X_{2v-u}^{-1} X_{2v}$$

force $[X_u, X_v] = 1$. This core represents three parity classes. If P contains only those three classes, theorems 91.4 and 91.5 again propagate the basis expressions to all of P. If the fourth class occurs, theorem 91.6 enlarges the core by one already-dependent generator to a parity-complete lattice-convex core, after which the same induction applies.

Thus $G(P)$ is generated by two commuting elements whose displacements are a basis of L. The displacement map (90.14) is therefore an isomorphism $G(P) \cong L$. Use theorems 90.7 and 90.9. □

91.2.2 The parallelogram-free branch

Call an affine lattice line ℓ *rich* if $|P \cap \ell| \geq 3$. Lattice-convexity implies that $P \cap \ell$ is a consecutive arithmetic progression in the primitive direction of ℓ.

Lemma 91.8 (Line-locality of repeated differences)**.** *If P contains no nondegenerate parallelogram and*

$$q - p = t - s$$

for two distinct ordered pairs of points of P, then the two segments lie on the same affine lattice line.

Proof. If the two segments lie on different parallel lines, the four endpoints form a nondegenerate parallelogram. If the lines are not parallel, equal displacement is impossible. □

A point lying on at least two rich lines will be called *shared*.

Lemma 91.9 (Transverse determinant lemma)**.** *Let P be lattice-convex and parallelogram-free. Suppose*

$$0, u, 2u, -v, v \in P$$

with primitive, nonparallel u, v. Then

$$|\det_L(u, v)| = 1.$$

Proof. Choose coordinates $u = (1, 0)$ and $v = (a, b)$ with $b > 0$ and $\gcd(a, b) = 1$. The triangle

$$T = \operatorname{conv}\{-v, v, 2u\}$$

lies in $\operatorname{conv}(P)$, hence every lattice point of T lies in P.

For $b \geq 3$, the intersection $T \cap (2u - T)$ has, in the u, v coordinate system, the form

$$\{xu + yv : |x - 1| + 2|y| \leq 1\}.$$

Take the ordinary lattice point $z = (m, 1)$ where m is nearest to $1 + a/b$. Then $y = 1/b$ and

$$|x - 1| = \operatorname{dist}(a/b, \mathbb{Z}) \le \frac{\lfloor b/2 \rfloor}{b} \le 1 - \frac{2}{b}.$$

Thus z and $2u - z$ both lie in $T \cap L \subseteq P$. They are not on the u-line, and

$$0 + 2u = z + (2u - z)$$

gives a nondegenerate parallelogram, a contradiction.

If $b = 2$, a lattice shear gives $v = (1, 2)$. Then the four points

$$(1, 2), \quad (0, -1), \quad (0, 0), \quad (1, 1)$$

lie in $T \cap L \subseteq P$ and satisfy

$$(1, 2) + (0, -1) = (0, 0) + (1, 1),$$

again a nondegenerate parallelogram. Hence $b = 1$. □

Lemma 91.10 (Endpoint-bridge exclusion)**.** *Let P be lattice-convex and parallelogram-free. A rich segment of lattice length at least 2 cannot have both endpoints interior to transverse rich lines.*

Proof. Write the rich segment as

$$0, u, 2u, \ldots, ku, \qquad k \ge 2,$$

and suppose the two endpoints are interior to transverse rich lines, so

$$-v, 0, v \in P, \qquad ku - w, ku, ku + w \in P.$$

By theorem 91.9,

$$|\det(u, v)| = |\det(u, w)| = 1.$$

Choose coordinates $u = (1, 0)$, $v = (0, 1)$. Then $w = (a, \varepsilon)$ with $\varepsilon = \pm 1$. Hence at the levels $y = 1$ and $y = -1$ the right transverse line contributes points with horizontal coordinates $k + a$ and $k - a$ (in some order). At least one of these is nonzero. On the same level the left transverse line contributes $(0, \pm 1)$. Convexity therefore forces a horizontal unit edge on that level. The main rich segment contains the horizontal unit edge from $(0, 0)$ to $(1, 0)$. The two equal unit edges form a nondegenerate parallelogram, a contradiction. □

Lemma 91.11 (Rich-line incidence)**.** *If P is lattice-convex and parallelogram-free, then every rich line contains at most two shared points; if it contains two, they are consecutive lattice points on that line.*

Proof. A point cannot be interior to two transverse rich lines: the four neighboring points have the same midpoint and form a nondegenerate parallelogram. Thus if a shared point p is interior to a rich line D, it is an endpoint of every transverse rich line through it. Let u be the primitive direction of D and let the transverse line contain $p, p+v, p+2v$. If D extended two steps from p in the same direction, then $p+2u, p+2v \in P$ and lattice-convexity would give

$$p + u + v = \frac{(p + 2u) + (p + 2v)}{2} \in P,$$

producing the parallelogram $p, p + u, p + v, p + u + v$. Hence a shared interior point can be at most one lattice step from each endpoint of D.

A point cannot be an endpoint of two transverse rich lines either: the two two-step extensions yield the same midpoint argument. Consequently, if a rich line had three shared points, its middle shared point would force the line to consist of exactly three consecutive points, while the two endpoints would be interior to transverse rich lines, contradicting theorem 91.10. If two shared points were nonconsecutive, neither could be an interior point of the rich line, so both would be endpoints and both would be interior to transverse rich lines, again contradicting theorem 91.10. □

Theorem 91.12 (Parallelogram-free homotopy theorem)**.** *Let $P = P_A(r)$ have active rank two and contain no nondegenerate lattice parallelogram. Then*

$$I_A(r) \simeq \bigvee^{\beta_1(I_A(r))} S^1. \tag{91.2}$$

In particular,

$$\pi_1(I_A(r)) \cong F_{\beta_1(I_A(r))}.$$

Proof. By theorem 91.8, every nontrivial positive-dimensional face identification in a fundamental simplex on P is supported inside the simplex on $P \cap \ell$ for some rich line ℓ. Let R be the union of these rich-line simplices together with the full 0-skeleton. Then the quotient $Q(P)$ is the pushout obtained from the full simplex $\Delta(P)$ by replacing R by its translation quotient $\overline{R}$. Since $\Delta(P)$ is contractible and $R \hookrightarrow \Delta(P)$ is a cofibration,

$$Q(P) \simeq \mathrm{Cofib}(R \to \overline{R}).$$

Distinct rich-line simplices meet only at shared vertices. By theorem 91.11, each rich line contains at most two such vertices, and if there are two they are consecutive. Hence R deformation retracts to a finite incidence graph. The quotient of a single rich-line simplex is the rank-one completion quotient, hence a circle; no positive-dimensional face is identified across distinct rich lines by theorem 91.8. Thus $\overline{R}$ retracts to a wedge of one circle for each rich line.

Choose a spanning forest in the incidence graph. In the cellular mapping cone of $R \to \overline{R}$, every non-tree incidence edge gives one 2-cell relation. Because the two shared points on the corresponding rich line are consecutive, its attaching word traverses that rich-line circle generator with coefficient ± 1. Choosing fundamental cycles with respect to the forest makes that circle generator unique to the corresponding non-tree relation. Successive elementary 2-cell/1-cell cancellations remove all 2-cells. The mapping cone therefore collapses to a finite graph. Its first Betti number is the first Betti number of $Q(P)$, so the graph is homotopy equivalent to the wedge in (91.2). Use theorem 90.7. □

Combining theorems 90.12, 91.7 and 91.12 gives the complete rank-two fundamental-group list

$$F_2, \qquad F_3, \qquad \mathbb{Z}^2.$$

91.3 The parallelogram branch and exact homotopy type

Assume $P = P_A(r)$ contains a nondegenerate parallelogram. By theorem 91.7,

$$\pi_1(Q(P)) \cong L \cong \mathbb{Z}^2.$$

Hence the periodic nerve

$$\widetilde{Q} := \mathcal{N}(P)$$

is the universal cover of $Q(P)$.

Let

$$R_L = \mathbb{Z}[L] \cong \mathbb{Z}[s^{\pm 1}, t^{\pm 1}], \qquad M = H_2(\widetilde{Q}; \mathbb{Z}).$$

The next result is the module-theoretic step that turns the homology classification into a homotopy classification.

Lemma 91.13 (Universal-cover higher homology)**.** *For active rank two,*

$$H_j(\mathcal{N}(P); \mathbb{Z}) = 0 \qquad (j \geq 3).$$

Proof. The complex $\mathcal{N}(P)$ is the nerve of the family of translates $\{\lambda + P\}_{\lambda \in L}$. Since $P = \operatorname{conv}(P) \cap L$, these are convex lattice sets in $L \cong \mathbb{Z}^2$. Alon–Kalai–Matoušek–Meshulam prove that the nerve of every finite family of convex lattice sets in $\mathbb{Z}^d$ is $(2^d - 1)$-collapsible [41]; in dimension two it is therefore 3-Leray. Any finite cellular cycle in $\mathcal{N}(P)$ is supported on a finite induced subcomplex, so homology in degree at least 3 vanishes. □

Proposition 91.14 (Projectivity of the π_2 module)**.** *In the parallelogram branch, M is a finitely generated projective R_L-module.*

Proof. The universal-cover cellular chains are finitely generated free R_L-modules in each degree, and R_L is Noetherian, so M is finitely generated.

Let $\mathfrak{m}$ be a maximal ideal of R_L and let $k(\mathfrak{m})$ be its residue field. Since the group elements of L are units in R_L, their images define a character

$$\chi_{\mathfrak{m}} : L \to k(\mathfrak{m})^{\times}.$$

For the free R_L-chain complex $C_\bullet(\widetilde{Q})$, the hyperhomology spectral sequence gives

$$E^2_{p,q} = \operatorname{Tor}_p^{R_L}(k(\mathfrak{m}), H_q(\widetilde{Q}; \mathbb{Z})) \Longrightarrow H_{p+q}(Q(P); k(\mathfrak{m})_{\chi_{\mathfrak{m}}}).$$

Because $\widetilde{Q}$ is simply connected, $H_1(\widetilde{Q}) = 0$, and by theorem 91.13 only the rows $q = 0, 2$ can be nonzero. The $q = 0$ row is the Koszul homology of the two augmentation elements $s - 1, t - 1$ after evaluation at the residue character; it vanishes in degrees above 2, and for a nontrivial character it is acyclic.

On the other hand, theorems 90.15 and 91.1 give

$$H_3(Q(P); k(\mathfrak{m})_{\chi_{\mathfrak{m}}}) = 0.$$

No differential can kill or create the term $E^2_{1,2}$, so

$$\operatorname{Tor}_1^{R_L}(k(\mathfrak{m}), M) = 0$$

for every maximal ideal $\mathfrak{m}$. Localizing, the finite $R_{L,\mathfrak{m}}$-module $M_{\mathfrak{m}}$ has vanishing first Tor with its residue field. A minimal local presentation and Nakayama's lemma therefore show that $M_{\mathfrak{m}}$ is free. Thus M is locally free at every maximal ideal, hence finitely generated projective. □

Theorem 91.15 (Parallelogram-branch homotopy theorem)**.** *Let $P = P_A(r)$ have active rank two and contain a nondegenerate lattice parallelogram. Put*

$$b = \beta_2(I_A(r)).$$

Then $b \geq 1$ and

$$I_A(r) \simeq T^2 \vee \bigvee^{b-1} S^2. \tag{91.3}$$

Proof. By theorem 91.7, $\pi_1(Q(P)) \cong \mathbb{Z}^2$. The Hopf exact sequence therefore gives a surjection

$$H_2(Q(P); \mathbb{Z}) \twoheadrightarrow H_2(\mathbb{Z}^2; \mathbb{Z}) \cong \mathbb{Z},$$

so $b \geq 1$.

By theorem 91.14, M is finitely generated projective over

$$R_L \cong \mathbb{Z}[s^{\pm 1}, t^{\pm 1}].$$

Swan's Laurent polynomial theorem [43] implies that every finitely generated projective module over this ring is free, so

$$M \cong R_L^q$$

for some $q \geq 0$.

To determine q, take any nontrivial character over a field. In the spectral sequence used above, the $q = 0$ row is acyclic and the only nonzero homology of the local-system complex can occur in degree 2. Hence

$$\dim H_2(Q(P); \Bbbk_\chi) = q.$$

The Euler characteristic of a rank-one local-system chain complex is the ordinary Euler characteristic. By theorem 91.2,

$$\chi(Q(P)) = 1 - 2 + b = b - 1.$$

Thus $q = b - 1$, and

$$\pi_2(Q(P)) \cong H_2(\widetilde{Q}) \cong R_L^{b-1}$$

by Hurewicz on the simply connected universal cover.

Let

$$Y = T^2 \vee \bigvee^{b-1} S^2.$$

Choose a map $T^2 \to Q(P)$ inducing the isomorphism on π_1; the torus 2-cell extends because the two basis loops commute. Choose maps of the 2-spheres representing an R_L-basis of $\pi_2(Q(P))$. Together they define

$$f : Y \to Q(P)$$

inducing isomorphisms on π_1 and on the R_L-module π_2. Lift to universal covers. Both universal covers are simply connected; both have zero homology above degree 2 by theorem 91.13, and the lifted map is an isomorphism in degree 2. Hence it is an integral homology equivalence. The simply connected homological Whitehead theorem [11] makes the lifted map a homotopy equivalence, and therefore so is f.

Finally use theorem 90.7. □

Chapter 92

Complete Rank-Two Classification by Lattice Geometry

The remaining distinction is geometric. In active rank two, the presence of a nondegenerate lattice parallelogram separates a toral branch from a free branch. The following results complete the classification and show that fixed rank does not bound second homology.

92.1 Complete active-rank-two classification

We can now combine the geometric and algebraic branches.

Theorem 92.1 (Complete rank-two homotopy classification)**.** *Let r be a completable residual with*

$$\operatorname{rank} L_A(r) = 2.$$

Then exactly one of the following holds.

(a) *The normalized optimal fiber $P_A(r)$ is nondegenerate-parallelogram-free. Then*

$$I_A(r) \simeq \bigvee^{\beta_1} S^1, \qquad \beta_1 \in \{2, 3\}.$$

(b) *The normalized optimal fiber contains a nondegenerate lattice parallelogram. Then*

$$I_A(r) \simeq T^2 \vee \bigvee^{\beta_2 - 1} S^2, \qquad \beta_2 \geq 1.$$

Consequently the possible active-rank-two homotopy types are precisely

$$S^1 \vee S^1, \qquad S^1 \vee S^1 \vee S^1, \qquad T^2 \vee \bigvee^{m} S^2 \quad (m \geq 0). \tag{92.1}$$

Proof. Apply theorems 90.12, 91.12 and 91.15. The two cases are complementary by definition. □

Corollary 92.2 (Integral homology profiles)**.** *In active rank two the integral homology is exactly one of*

$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z}^2, \quad H_j = 0 \ (j \geq 2),$$
$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z}^3, \quad H_j = 0 \ (j \geq 2),$$

or

$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z}^2, \quad H_2 \cong \mathbb{Z}^b, \quad H_j = 0 \ (j \geq 3), \qquad b \geq 1.$$

92.2 Explicit scalar realizations

The classification is not merely formal: all allowed types occur in four-letter scalar completion systems.

92.2.1 A fixed-system residual bifurcation

For an integer $d \geq 5$, define

$$\omega_d = (1-2d,\ 1-d,\ 1,\ 1+d). \tag{92.2}$$

This row vector is completion-total; for example

$$h_d = (1, 1, 2d-3, 1) > 0, \qquad \omega_d h_d^{\top} = 0.$$

The equal-length kernel is

$$K_{\omega_d} = \mathbb{Z}u \oplus \mathbb{Z}v, \qquad u = (1,-2,1,0), \quad v = (2,-3,0,1). \tag{92.3}$$

(We write the first basis vector as u below.)

Theorem 92.3 (Fixed-system rank-two bifurcation)**.** *For the single scalar system* (92.2)*, define*

$$r_W = 5d-4, \qquad r_T = 3d-4.$$

Then

$$\ell_{\omega_d}(r_W) = \ell_{\omega_d}(r_T) = 4,$$

and in both cases the active lattice is the full equal-length kernel $K_{\omega_d} \cong \mathbb{Z}^2$. Nevertheless,

$$I_{\omega_d}(r_W) \simeq S^1 \vee S^1, \qquad I_{\omega_d}(r_T) \simeq T^2.$$

Proof. For any $x \in \mathbb{N}^4$ of length n put

$$s = x_2 + 2x_3 + 3x_4.$$

Then

$$\omega_d \cdot x = (1-2d)n + ds.$$

For both residuals the completion equation implies

$$n \equiv 4 \pmod{d}.$$

Since $d \geq 5$ and length-4 solutions exist, both minimum lengths equal 4.

For r_W, setting $n = 4$ reduces the equation to $s = 3$. Its complete set of nonnegative solutions is

$$\mathcal{O}_W = \{(1,3,0,0), (2,1,1,0), (3,0,0,1)\}. \tag{92.4}$$

Relative to the first point, the two nonzero differences are

$$u = (1,-2,1,0), \qquad v = (2,-3,0,1),$$

which form the kernel basis. The normalized fiber has three affinely independent lattice points and no nondegenerate parallelogram, so theorem 92.1 gives $S^1 \vee S^1$.

For r_T, the length-4 equation is $s = 5$ and the complete optimal fiber is

$$\mathcal{O}_T = \{(0,3,1,0), (1,1,2,0), (1,2,0,1), (2,0,1,1)\}. \tag{92.5}$$

Relative to the first point, let

$$g = u, \qquad h = v - u = (1,-1,-1,1).$$

Then the four normalized points are

$$\{0, g, h, g+h\},$$

a fundamental lattice parallelogram. Its quotient has Euler characteristic 0, so theorems 91.2 and 91.7 give $\beta_2 = 1$, and theorem 91.15 gives T^2. □

Thus even the matrix, the full equal-length kernel, the active lattice, its rank, and the minimum completion length can be held fixed while the global topology changes. The missing datum is the residual-dependent optimal-fiber translation geometry.

92.2.2 Unbounded second homology at fixed rank two

For $m \geq 2$ put

$$\alpha = 2m + 1$$

and define

$$\omega_m = \big(1 - m\alpha,\ 1 - (m-1)\alpha,\ 1,\ 1 + \alpha\big), \tag{92.6}$$

$$r_m = \alpha m(m-1) - 2m. \tag{92.7}$$

The system is completion-total; for example

$$h_m = \big(1, 1, (2m-2)\alpha - 3, 1\big) > 0$$

satisfies $\omega_m h_m^\top = 0$.

Theorem 92.4 (Unbounded rank-two sphere family)**.** *For every $m \geq 2$,*

$$\ell_{\omega_m}(r_m) = 2m.$$

The complete optimal fiber is

$$\mathcal{O}_m = \{A_0, \dots, A_m, B, C\}, \tag{92.8}$$

where

$$A_t = (t, m-t, m-t, t), \qquad 0 \leq t \leq m,$$

$$B = (0, m+1, 0, m-1), \qquad C = (m-1, 0, m+1, 0).$$

Its active lattice equals the full equal-length kernel and has rank two. Moreover

$$I_{\omega_m}(r_m) \simeq T^2 \vee \bigvee^{m-2} S^2, \tag{92.9}$$

so

$$\beta_2(I_{\omega_m}(r_m)) = m - 1.$$

Proof. All four weights in (92.6) are congruent to 1 modulo α. Hence every completion of r_m with length n satisfies

$$n \equiv -r_m \equiv 2m \pmod{2m+1}.$$

The vectors in (92.8) have length $2m$, so $\ell = 2m$.

At length $2m$, eliminating the common congruence term gives

$$x_2 + mx_3 + (m+1)x_4 = m(m+1).$$

Modulo m,

$$x_2 + x_4 \equiv 0 \pmod{m},$$

and $0 \leq x_2 + x_4 \leq 2m$. Thus $x_2 + x_4$ is 0, m, or $2m$. The three cases give respectively C, the family A_t, and B, proving (92.8).

Set

$$g = (1, -1, -1, 1), \qquad h = (0, 1, -m, m-1).$$

Every equal-length kernel vector is uniquely $ag + bh$: indeed the equations $\mathbf{1}^\top z = 0$ and $\omega_m \cdot z = 0$ imply

$$z_3 = -(m+1)z_1 - mz_2, \qquad z_4 = mz_1 + (m-1)z_2,$$

which corresponds to $a = z_1$, $b = z_1 + z_2$. Hence g, h form a basis of the full kernel. Relative to A_0,

$$A_t - A_0 = tg, \qquad B - A_0 = h, \qquad C - A_0 = (m-1)g - h.$$

Thus in active coordinates

$$P_m = \{(t,0) : 0 \le t \le m\} \cup \{(0,1), (m-1,-1)\}. \tag{92.10}$$

It contains a nondegenerate parallelogram, for example the four points corresponding to $(0,1)$, $(m-1,-1)$, $(0,0)$, and $(m-1,0)$.

It remains to compute β_2. Count simplex orbits in the finite quotient using (92.10). The horizontal-only part is the rank-one quotient and has Euler characteristic 0. The total alternating contribution of faces containing exactly one of the two off-line points, excluding the already-counted singleton vertex, is -1 for each off-line point. Faces containing both off-line points have total alternating contribution 0. The only cross-identifications between the two one-off-line families occur for the m edge pairs

$$\{(0,1),(t,0)\} \sim \{(m-1,-1),(m-1-t,0)\}, \qquad 0 \le t \le m-1.$$

No face of size at least 3 has such a cross-identification because the two off-line levels are unique. Each edge identification raises the Euler characteristic by 1. Therefore

$$\chi(I_{\omega_m}(r_m)) = m-2.$$

The parallelogram branch has $\beta_1 = 2$, and theorem 91.2 gives no homology above degree 2. Hence

$$m-2 = 1-2+\beta_2, \qquad \beta_2 = m-1.$$

Apply theorem 91.15 to obtain (92.9). □

Corollary 92.5 (Fixed-rank unbounded complexity)**.** *There exist completion-total four-letter scalar systems with*

$$\operatorname{rank} K = \operatorname{rank} L = 2, \qquad \pi_1 \cong \mathbb{Z}^2,$$

for which β_2 is arbitrarily large.

92.2.3 A free-rank-three example

Example 92.6 (The F_3 branch)**.** Take

$$\omega = (-11,-8,-5,4), \qquad r = 25.$$

All weights are 1 modulo 3, so every completion length satisfies $n \equiv 2 \pmod 3$. Length 2 is impossible because the smallest sum of two weights is $-22 > -25$, while length 5 solutions exist. At length 5 the equation becomes

$$x_2 + 2x_3 + 5x_4 = 10,$$

whose complete nonnegative solution set is

$$\mathcal{O} = \{(0,0,5,0),(0,3,1,1),(1,1,2,1),(3,0,0,2)\}.$$

The active lattice has rank two, and the six unordered pair differences are distinct up to sign. Hence there is no nontrivial repeated-difference relation and no nondegenerate parallelogram. Presentation (90.13) is therefore free on the three nonzero normalized fiber points. By theorem 91.12,

$$I_\omega(25) \simeq S^1 \vee S^1 \vee S^1.$$

Together, theorems 92.3, 92.4 and 92.6 realize every homotopy type in (92.1).

92.3 Computational consistency checks

The theorems above are symbolic and do not depend on computation. Small exhaustive enumerations were used only as independent checks of the fiber formulas, cell-orbit conventions, and the rank-two dichotomy.

For the fixed-system family (92.2), direct enumeration for $5 \leq d \leq 20$ recovers exactly the two fibers in equations (92.4) and (92.5) and minimum length 4 for the stated residuals. For the unbounded family (92.6), direct enumeration for $2 \leq m \leq 12$ recovers exactly (92.8). The first cases are shown in table 92.1.

m	α	r_m	$\lvert\mathcal{O}_m\rvert$	predicted homotopy type
2	5	6	5	T^2
3	7	36	6	$T^2 \vee S^2$
4	9	100	7	$T^2 \vee 2S^2$
5	11	210	8	$T^2 \vee 3S^2$
6	13	378	9	$T^2 \vee 4S^2$

Table 92.1: Initial cases of the explicit unbounded family. The computations are consistency checks only; theorem 92.4 is proved symbolically.

A separate finite-grid audit of planar lattice-convex fibers was used during discovery to test the parallelogram/free-group dichotomy and the Betti patterns. These experiments motivated the geometric lemmas in section 91.2; they are not proof premises and no computational assumption enters theorem 92.1.

Chapter 93

From Displacement Kernel to Physical Completion Holonomy

The preceding rank-two classification determines the intrinsic homotopy type of the filled optimal-fiber quotient. We now ask what survives when loops are transported by an explicitly specified physical LIFO policy. The displacement map separates lattice motion from pure loops; only the latter can carry holonomy invisible to the semantic endpoint.

93.1 Optimal fibers, displacement, and pure loops

Let $\mathcal{O}(r) \subset \mathbb{N}^s$ be the set of minimum-length completion count vectors at residual r. Fix $x_0 \in \mathcal{O}(r)$ and put

$$P = \mathcal{O}(r) - x_0, \qquad L = \langle \mathcal{O}(r) - \mathcal{O}(r) \rangle_{\mathbb{Z}}.$$

Let $I = I(r)$ denote the filled shortest-prefix quotient associated with this optimal fiber. In active rank two, the completion-fiber construction gives a canonical surjection

$$\delta : \pi_1(I) \twoheadrightarrow L \cong \mathbb{Z}^2,$$

which records the lattice displacement of a loop.

Definition 93.1 (Pure-loop sector)**.** Define

$$K_0 = \ker \delta$$

and

$$K_{\mathrm{lin}} = \ker \big(H_1(I; \mathbb{Z}) \to L \big).$$

The first group is the full zero-displacement sector; the second records the part already visible after abelianization.

Theorem 93.2 (Rank-two displacement-kernel visibility trichotomy)**.** *Assume* $\operatorname{rank} L = 2$. *Then the active-rank-two alternatives have the following pure-loop structure.*

1. *If I is toral, $\pi_1(I) \cong \mathbb{Z}^2$ and δ is an isomorphism. Hence*

$$K_0 = 1, \qquad K_{\mathrm{lin}} = 0.$$

2. *If $I \simeq S^1 \vee S^1$, then $\pi_1(I) \cong F_2$ and*

$$K_0 = [F_2, F_2], \qquad K_{\mathrm{lin}} = 0.$$

Thus pure topology exists but is nonabelian-only.

3. *If* $I \simeq S^1 \vee S^1 \vee S^1$, *then* $\pi_1(I) \cong F_3$ *and*

$$K_{\mathrm{lin}} \cong \mathbb{Z}.$$

The full kernel $K_0 = \ker(F_3 \twoheadrightarrow \mathbb{Z}^2)$ *is free of countably infinite rank, but one pure direction is already visible in* H_1.

Proof. In the toral branch the displacement theorem identifies $\pi_1(I)$ with L. In the F_2 branch, abelianization gives a surjection $\mathbb{Z}^2 \to \mathbb{Z}^2$, hence an isomorphism, so the kernel of $F_2 \to \mathbb{Z}^2$ is exactly the commutator subgroup. In the F_3 branch, abelianization gives a surjection $\mathbb{Z}^3 \to \mathbb{Z}^2$ with primitive rank-one kernel. The assertions about the full kernels follow from Nielsen–Schreier and the standard structure of infinite-index normal subgroups of nonabelian free groups. □

Definition 93.3 (Displacement-compatible physical transport)**.** A flat physical transport is a functor from the fundamental groupoid $\Pi_1(I)$ to a groupoid of invertible physical comparison maps. At a basepoint it induces

$$\mathrm{Hol} : \pi_1(I) \to G_{\mathrm{phys}}.$$

It is displacement-compatible if there is a homomorphism $c : G_{\mathrm{phys}} \to L$ such that

$$c \circ \mathrm{Hol} = \delta.$$

Then $\mathrm{Hol}(K_0)$ lies in the count-invisible subgroup $\ker c$.

Proposition 93.4 (Gauge persistence of pure holonomy)**.** *If* $\mathrm{Hol}\,|_{K_0}$ *is nontrivial, no objectwise change of physical trivialization within the same flat invertible transport category can make every pure loop trivial. Such a change only conjugates the holonomy representation.*

This proposition is deliberately category-relative. It does not say that topology forces debt in an arbitrary DPDA.

93.2 A fixed completion system with a free/toral residual bifurcation

Fix $d \geq 5$ and the four weights

$$\omega_d = (1 - 2d,\ 1 - d,\ 1,\ 1 + d).$$

The equal-length kernel contains the basis

$$u = (1, -2, 1, 0), \qquad v = (2, -3, 0, 1).$$

Consider the two residuals

$$r_W = 5d - 4, \qquad r_T = 3d - 4.$$

Both have minimum completion length 4 and the same active rank two.

At r_W the optimal completions are

$$x_0 = (1, 3, 0, 0), \qquad x_u = (2, 1, 1, 0), \qquad x_v = (3, 0, 0, 1),$$

so the normalized fiber is $\{0, u, v\}$ and

$$I(r_W) \simeq S^1 \vee S^1.$$

At r_T the optimal completions are

$$(0, 3, 1, 0), \quad (1, 1, 2, 0), \quad (1, 2, 0, 1), \quad (2, 0, 1, 1),$$

which form a lattice parallelogram; hence

$$I(r_T) \simeq T^2.$$

By Theorem 93.2, the first residual has a nonabelian-only pure sector $[F_2, F_2]$, whereas the second has no pure sector at all.

93.3 Stable-LIFO comparison transport

For an equal-length count vector $x = (x_1, \dots, x_s)$ define the slot word

$$\eta_{\mathrm{slot}}(x) = 1^{x_1} 2^{x_2} \cdots s^{x_s}.$$

Temporarily label its slots from left to right. For two equal-length count vectors x, y, define the Stable-LIFO matching $\tau_{x,y}$ by preserving same-type occurrences stably from left to right and matching all remaining source positions to remaining target positions in reverse order. Then

$$\tau_{y,x} = \tau_{x,y}^{-1}.$$

For the free optimal endpoints, the canonical words are

$$w_0 = 1222, \qquad w_u = 1123, \qquad w_v = 1114.$$

The three comparison permutations are

$$\tau_{0u} = (234), \qquad \tau_{0v} = (24), \qquad \tau_{uv} = (34).$$

On moved slots $\{2, 3, 4\}$ identify

$$r = (012), \qquad s = (02).$$

Then

$$[r, s] = (021),$$

corresponding on the original slots to (243).

Theorem 93.5 (Pure nonabelian Stable-LIFO holonomy)**.** *The Stable-LIFO comparison transport on the free optimal fiber is flat and sends a zero-displacement commutator in $K_0 = [F_2, F_2]$ to the nontrivial 3-cycle*

$$(243).$$

The image is invisible to every abelian permutation observable.

Proof. The triangle relation is

$$\tau_{0u}\tau_{uv} = \tau_{0v}$$

under chronological right action, so the endpoint comparisons define a flat local system on the three-point free quotient. The two displacement generators map to r and s, and their commutator maps to $(021) \neq 1$. Since every homomorphism from S_4 to an abelian group kills commutators, the certificate is genuinely nonabelian. □

93.4 The full free shortest-completion episode

A prefix count vector y determines

$$n = \mathbf{1}^{\top} y, \qquad s = y_2 + 2y_3 + 3y_4.$$

For r_W,

$$r_W + \omega_d \cdot y = r_W + (1 - 2d)n + ds.$$

Since $0 \leq n \leq 4 < d$, residual equality on shortest prefixes is equivalent to equality of (n, s). The full quotient therefore has vertex set

$$\begin{aligned} V_W = {} & \{(0,0)\} \cup \{(1,s) : 0 \leq s \leq 3\} \\ & \cup \{(2,s) : 0 \leq s \leq 3\} \\ & \cup \{(3,s) : 0 \leq s \leq 3\} \cup \{(4,3)\}, \end{aligned}$$

so $|V_W| = 14$. Its cell vector is

$$(f_0, f_1, f_2, f_3) = (14, 28, 15, 2).$$

Theorem 93.6 (Full free-episode realization)**.** *The entire* 14*-vertex shortest-prefix quotient admits a flat* S_3 *connection whose endpoint products are*

$$w_0 \mapsto e, \qquad w_u \mapsto r, \qquad w_v \mapsto s.$$

All square relations and both 3*-cube boundaries are flat. The pure commutator of Theorem 93.5 therefore survives on the actual one-letter shortest-completion episode, not merely on a compressed comparison complex.*

Construction. Assign edge labels by a spanning-tree gauge and extend them over the remaining edges so that each of the 15 squares has trivial boundary product. The two 3-cubes then have flat boundary automatically. The explicit connection uses only the four local values e, r, r^{-1}, s and the transposition $a = (01)$ at the mixed vertices; direct multiplication gives the endpoint products above for all 20 shortest words. □

Chapter 94

Sharp LIFO Carrier Costs and Toral Correction

The free branch supports genuine pure nonabelian holonomy. The toral branch fills the commutator cell, so global descent instead imposes a correction problem. The resulting comparison gives exact carrier, ancestry, and correction costs inside the stated Stable-LIFO/literal-stack categories.

94.1 Sharp literal-stack carrier cost

Before measuring a carrier, we fix the implementation category. A *buffered ancestry refinement* of capacity b allows at most b temporary finite-control handles that remember distinguished stack-cell ancestries during a one-top macro; the handles are cleared at the endpoint. A macro is ancestry-bijective if every initial distinguished ancestry has exactly one final descendant. Section 94.7 proves the general bottom-move lower bound $b \geq n-1$ when the bottom of an n-cell distinguished prefix changes ancestry. We now impose a specific physical persistence category: no epsilon moves, no permanent finite-control sheet register, common endpoint control and zero potential, literal stack alphabet $\{1,2,3,4\}$ only, and identical endpoint Parikh counts across sheets. Persistent monodromy must therefore live in stack order.

For a zero-weight literal count vector $z \in \mathbb{N}^4$, define its order capacity by the multinomial number

$$M(z) = \frac{|z|_1!}{\prod_i z_i!}.$$

For $k \geq 2$, set

$$g_k^{\mathrm{ord}}(\omega_d) = \min\{|z|_1 : z \neq 0,\ \omega_d z = 0,\ M(z) \geq k\}.$$

Lemma 94.1 (Arithmetic carrier girth)**.** *Every nonempty zero-weight literal word has length at least d. Moreover*

$$g_3^{\mathrm{ord}}(\omega_d) = g_6^{\mathrm{ord}}(\omega_d) = d.$$

Proof. Each weight is congruent to 1 modulo d. Hence $\omega_d z = 0$ implies

$$|z|_1 \equiv 0 \pmod{d},$$

so every nonempty zero-weight word has length at least d. Equality is attained by

$$z^{(3)} = (0, 1, d-1, 0)$$

for three orderings and by

$$z^{(6)} = (0, 2, d-3, 1)$$

for six orderings. □

For the six-state carrier, let

$$T_d = 2\,3^{d-4}$$

and index the six permutations P_g of the three letters 234 by $g \in S_3$. Set

$$Z_g = P_g T_d.$$

All Z_g have the same Parikh vector $z^{(6)}$ and zero total weight.

Theorem 94.2 (Sharp free literal carrier)**.** *There is a real-time one-top realization of the full free connection such that*

$$Z_g \xrightarrow{w_0} Z_g, \qquad Z_g \xrightarrow{w_u} Z_{gr}, \qquad Z_g \xrightarrow{w_v} Z_{gs},$$

with no epsilon moves and no permanent sidecar. The minimum persistent literal order-carrier height in the stated category is exactly

$$\boxed{d}.$$

The chosen realization needs exactly two transient ancestry handles. In the deferred-carrier subclass, exactly three transient path states are necessary and sufficient.

Proof. The lower bound on persistent height is Lemma 94.1: every nonempty zero-weight literal carrier has length divisible by d, hence length at least d. For the upper bound use the six same-Parikh words $Z_g = P_g T_d$, $g \in S_3$. Only the distinguished three-cell prefix P_g is changed; the common suffix T_d remains untouched throughout.

For a shortest word w, let $\sigma(w)$ be the endpoint product supplied by the flat connection of Theorem 93.6. During the four input-synchronous steps, finite control records the current quotient vertex and two temporary ancestry handles. The first two distinguished carrier cells are exposed into those handles, the third remains visible, and on the terminal step the visible third cell together with the two handles determines the incoming prefix permutation g. The transition rewrites that visible cell by the three-cell prefix $P_{g\sigma(w)}$. Thus

$$Z_g \xrightarrow{w_0} Z_g, \qquad Z_g \xrightarrow{w_u} Z_{gr}, \qquad Z_g \xrightarrow{w_v} Z_{gs},$$

with one input symbol consumed per transition and no ε-move. The semantic shortest-episode coordinate in control supplies the unique potential needed to verify the residual identity at each step, so this is an exact lift. Section 94.6 gives the full edge certificate for all twenty shortest words.

Necessity of two ancestry handles follows from Lemma 94.6: the generator $s = (02)$ changes the ancestry occupying the bottom of the distinguished three-cell prefix, so any ancestry-bijective top-access macro needs at least $3 - 1 = 2$ handles. The construction above uses two.

Finally restrict to the deferred-carrier subclass, in which the distinguished carrier order is not changed before the terminal rewrite. The three histories 222, 231, and 411 reach the same preterminal semantic vertex with the same next input but with accumulated products s, a, e; their common last input must therefore produce three distinct endpoint multipliers e, r, s. Determinism forces three transient path states. The explicit connection table has at most three accumulated-product classes at every vertex, so three states suffice. Hence both locality bounds are sharp in the stated subclasses. □

94.2 Toral curvature and the difference between local flattening and torus descent

For the toral residual write the four optimal endpoints as $0, g, h, g + h$. Raw Stable-LIFO endpoint comparisons are

$$A = (1243), \qquad B = (24), \qquad C = (1234), \qquad D = (23).$$

The two chronological route products are

$$AB = (143), \qquad CD = (134) = (143)^{-1}.$$

Thus the raw parallelogram has nontrivial 3-cycle curvature.

On one isolated square, one edge change is enough to make the two route products equal. But a torus local system must descend globally. If normalized path products to the three nonzero lattice points are U, V, W, torus descent requires

$$[U, V] = e, \qquad W = UV = VU.$$

Hence the corrected endpoint tuple must have the form

$$(\widetilde{A}, \widetilde{B}, \widetilde{C}, \widetilde{D}) = (U, V, V, U), \qquad UV = VU.$$

Theorem 94.3 (Minimum number of globally altered endpoint comparisons)**.** *Any torus-descending correction of the raw tuple* (A, B, C, D) *changes at least three of the four endpoint comparisons. Three changes suffice.*

Proof. Because $A \neq D$, at least one of the g-pair must change; because $B \neq C$, at least one of the h-pair must change. If exactly two comparisons changed, the retained commuting pair would have to be one of $(A, B), (A, C), (D, B), (D, C)$. Each pair has nontrivial commutator (243), so two changes are impossible. Taking $U = A$ and $V = e$ gives (A, e, e, A) and changes exactly three entries. □

Let d_T be Cayley distance in S_4 with respect to transpositions and define

$$F(U, V) = d_T(A, U) + d_T(D, U) + d_T(B, V) + d_T(C, V)$$

subject to $[U, V] = e$.

Theorem 94.4 (Exact global toral correction Pareto law)**.** *The minimum total correction distance is*

$$\boxed{F_{\min} = 4},$$

attained by

$$U = (243), \qquad V = (234) = U^{-1},$$

with four single-transposition corrections. Among corrections changing exactly three endpoint comparisons, the minimum total transposition length is 6 *and the minimum possible maximum support is* 4*. Hence the exact nondominated profiles*

$$(\#\ \textit{altered},\ \textit{total transposition length}, \max\ \textit{support})$$

are

$$\boxed{(3, 6, 4), \qquad (4, 4, 2).}$$

Proof. For any commuting U, V,

$$F(U, V) \geq d_T(A, D) + d_T(B, C) = 2 + 2 = 4$$

by the triangle inequality. Equality is realized by the displayed U, V, using

$$A(12) = U, \quad B(34) = V, \quad C(12) = V, \quad D(34) = U.$$

For the three-change branch, retain one raw edge and minimize over its centralizer. The four symmetric cases give minimum total length 6 and force one correction to move all four slots. These two profiles are incomparable and every other correction is dominated by one of them. □

94.3 The full toral episode and repair/debt tradeoff

For r_T the residual quotient again uses (n, s), now with vertex set

$$\begin{aligned} V_T = &\{(0,0)\} \cup \{(1,s) : 0 \le s \le 3\} \\ &\cup \{(2,s) : 0 \le s \le 5\} \\ &\cup \{(3,s) : 2 \le s \le 5\} \cup \{(4,5)\}. \end{aligned}$$

Thus

$$|V_T| = 16, \qquad (f_0, f_1, f_2, f_3) = (16, 38, 28, 6).$$

There are 40 shortest completion words.

For the minimum-distance correction of Theorem 94.4, $V = U^{-1}$ and the image is the cyclic group

$$\langle U \rangle \cong C_3.$$

Define on the active lattice

$$\rho_T(ag + bh) = U^{a-b}.$$

Because the toral displacement map is an isomorphism, this defines a flat local system on the entire quotient. The 40 shortest words split into endpoint products

$$16 \text{ words} \mapsto e, \qquad 12 \mapsto U, \qquad 12 \mapsto V.$$

Theorem 94.5 (Full toral realization and exact repair/debt tradeoff)**.** *The minimum-distance toral repair admits a full* 16*-vertex real-time realization with nontrivial displacement-visible C_3 transport. If this transport is stored persistently in literal stack order, the minimum carrier height is d; the three-slot carrier uses two ancestry handles and, in the deferred subclass, three transient path states.*

Complete erasure of transport is possible by setting $U = V = e$. Its total correction distance from the raw tuple is

$$\boxed{8}.$$

Thus toral topology forces no pure debt: one may retain the nearest nontrivial flat transport at distance 4*, or pay distance* 8 *and remove persistent order memory entirely.*

Proof. The corrected image C_3 acts only on three slots, so the free height-d carrier construction applies. The lower bound is again the zero-weight congruence. For complete erasure,

$$\ell_T(A) + \ell_T(B) + \ell_T(C) + \ell_T(D) = 3 + 1 + 3 + 1 = 8.$$

Since $K_0 = 1$ in the toral branch, none of this retained transport is pure zero-displacement holonomy; it factors through the displacement lattice. □

94.4 The free–toral topology-to-LIFO dichotomy

The fixed family gives a clean contrast with every algebraic parameter except the residual held fixed.

free: local flatness permits global pure nonabelian holonomy,
toral: the filled commutator cell forbids pure holonomy
and converts raw noncommutativity into correction cost.

The distinction is not simply "topology requires memory." In the free branch, the missing parallelogram leaves a nonabelian zero-displacement sector in which order memory can live. In the toral branch, the parallelogram fills the commutator relation, so a flat physical representation must commute on the lattice generators. Noncommutativity survives only as the cost of repairing a chosen raw comparison policy.

94.5 Scope and representation dependence

The following distinctions are essential.

- The group K_0, the existence of nontrivial pure holonomy, the free/toral split, and the literal zero-weight height lower bound are intrinsic within the stated completion-fiber and carrier categories.
- The concrete permutations r, s, A, B, C, D and the correction profiles of Theorem 94.4 are relative to the fixed Stable-LIFO endpoint-comparison policy, although they are invariant under simultaneous conjugation of slot labels.
- The two-handle and three-state bounds are exact for the chosen three-slot and deferred-carrier subclasses; they are not claimed as universal lower bounds over all possible pushdown policies.
- No theorem here says that nontrivial topology alone forces physical debt under unrestricted finite control.

94.6 Full free-episode connection certificate

The full shortest-prefix quotient at r_W has vertices

$$V_W = \{(0,0)\} \cup \{(1,s),(2,s),(3,s) : 0 \le s \le 3\} \cup \{(4,3)\}$$

and f-vector $(14, 28, 15, 2)$. A directed edge is written $(n,s) \xrightarrow{i} (n+1, s+i-1)$. With chronological right multiplication, put

$$r = (012), \qquad r^{-1} = (021), \qquad s = (02), \qquad a = (01).$$

The following table is an explicit flat S_3 connection; omitted edges do not exist.

source	i	σ
$(0,0)$	$1,2,3,4$	e
$(1,0)$	$1,2,3,4$	e
$(1,1)$	1	e
$(1,1)$	2	r^{-1}
$(1,1)$	3	a
$(1,2)$	1	e
$(1,2)$	2	a
$(1,3)$	1	e
$(2,0)$	1	e
$(2,0)$	2	e
$(2,0)$	3	r
$(2,0)$	4	e
$(2,1)$	1	e
$(2,1)$	2	e
$(2,1)$	3	a
$(2,2)$	1	r
$(2,2)$	2	a
$(2,3)$	1	e
$(3,0)$	4	s
$(3,1)$	3	r
$(3,2)$	2	e
$(3,3)$	1	s

Substitution into the fifteen square relations

$$\sigma_{v,i}\sigma_{v+e_i,j} = \sigma_{v,j}\sigma_{v+e_j,i}$$

uses only $r^{-1}r = e$ and $as = r$ besides identities with the same factor on both sides. Flatness of all square faces implies compatibility on the two 3-cube boundaries. Endpoint products are e, r, s on the three optimal boxes, so the commutator image is $(021) \neq e$.

94.7 Ancestry model and sharp local-memory bounds

A buffered ancestry refinement of capacity b augments finite control with at most b temporary handles. A one-top microstep may move the visible cell's ancestry into a handle, attach a stored handle to the rewritten visible cell, or push a new decorated cell carrying a stored handle. A closed macro is ancestry-bijective when every initial distinguished ancestry has exactly one final descendant.

Lemma 94.6 (Bottom-move bound)**.** *If an ancestry-bijective top-access macro on n distinguished cells changes the ancestry occupying the bottom distinguished position, then $b \geq n - 1$. The bound is sharp.*

Proof. Before the original bottom cell can be inspected, all $n - 1$ cells above it have been removed from the live stack. Their ancestry classes must remain available for the final bijection, hence all $n - 1$ must be retained externally at that instant. Conversely, pop the upper $n - 1$ ancestries into handles, rewrite the exposed bottom, and rebuild the prefix in the target order. □

For the free three-slot carrier, the generator $s = (02)$ moves the third distinguished cell, so two handles are necessary and sufficient. The strict-streaming no-go on the full interval follows because, before the third ancestry is visible, every second-layer carrier update lies in a common order-two stabilizer. Flatness then forces the two endpoint generators into a common conjugate of that stabilizer, hence forces them to commute. The noncommuting pair r, s therefore requires transient path memory in addition to the two ancestry handles.

In the deferred-carrier subclass, the three histories 222, 231, and 411 reach the common preterminal vertex with accumulated products s, a, e and share the same final input. They must be distinguished before the final rewrite, giving a lower bound of three transient path states. The edge table above has local accumulated-product sets of size at most three, so three states suffice.

94.8 Global torus-descent audit

For the toral residual, a normalized flat endpoint quadruple has the form

$$(U, V, V, U), \qquad UV = VU.$$

Relative to the raw tuple

$$A = (1243), \quad B = (24), \quad C = (1234), \quad D = (23),$$

at least three entries must change: $A \neq D$, $B \neq C$, and every possible retained pair from $\{A, D\} \times \{B, C\}$ has nontrivial commutator. Three changes are attained by (A, e, e, A).

For Cayley distance d_T with all transpositions as generators,

$$F(U, V) = d_T(A, U) + d_T(D, U) + d_T(B, V) + d_T(C, V) \geq 4.$$

Equality is attained by

$$U = (243), \qquad V = (234) = U^{-1},$$

through

$$A(12) = U, \qquad B(34) = V, \qquad C(12) = V, \qquad D(34) = U.$$

If exactly three raw comparisons are altered, one raw edge is retained. The four centralizer cases give:

retained	fixed generator	$\min F$	$\min\max\mathrm{supp}$
A	$U = A$	6	4
D	$U = D$	6	4
B	$V = B$	6	4
C	$V = C$	6	4

so the nondominated profiles are $(3, 6, 4)$ and $(4, 4, 2)$. The minimum-distance pair generates C_3 and acts on only three distinguished slots; therefore it reuses the exact height-d three-slot carrier with two ancestry handles. The full toral quotient has vertex set

$$\{(0,0)\} \cup \{(1,s) : 0 \le s \le 3\} \cup \{(2,s) : 0 \le s \le 5\} \cup \{(3,s) : 2 \le s \le 5\} \cup \{(4,5)\}$$

and f-vector $(16, 38, 28, 6)$. The representation $\rho_T(ag + bh) = U^{a-b}$ therefore gives a flat connection on all 40 shortest words.

94.9 Topology-to-LIFO synthesis

At active rank two the intrinsic topology and the physical transport layer are now cleanly separated. The group π_1 and displacement kernel determine which pure loop sectors exist; the chosen transport category determines whether those sectors appear as nonabelian holonomy, literal carrier depth, transient ancestry cost, or toral correction. No claim is made that topology alone forces a unique physical debt law.

Higher-rank continuation. The present chapter fixes the physical category and establishes the rank-two free/toral transfer in sharp form. The next chapter first raises the intrinsic completion geometry to arbitrary active rank and then returns to this transport layer: the displacement-marked group, exact product/prism/apex kernel calculus, finite physical pure-image transfer, general carrier girth, support-based ancestry law, connection-memory multiplicity, repair optimization, and an all-rank S_3/A_3 carrier family are developed there without changing the category-sensitive interpretation established here.

Chapter 95

Lattice-Convex Realization, Higher-Rank Topology, and Marked Holonomy

Research branch — Higher-Rank Optimal-Fiber Realization, Rank Raising, and Marked Transfer. *The rank-two classification established a rigid planar picture but did not characterize which finite lattice configurations can occur as optimal fibers or how topology and physical transport behave after active rank is raised. This branch supplies the exact lattice-convex realization theorem and the product/apex/prism rank-raising calculus, then upgrades displacement to a marked-group invariant and transfers its pure-loop sector through finite physical holonomy, literal carrier locality, connection memory, and repair cost. The final all-rank family separates active rank from finite holonomy complexity while retaining an exact carrier-height law.*

For a finite spanning lattice-convex configuration $P \subset L$, let $Q(P)$ denote the periodic translation quotient of the optimal-fiber nerve introduced in the preceding rank-two branch. The completion baseline already proved that every normalized optimal fiber is lattice-convex. The first theorem below proves the converse and therefore turns the intrinsic lattice model into an exact realization theorem.

95.1 Intrinsic periodic-fiber model

A finite set $P \subset L$ in a rank-s lattice is *lattice-convex* when $P = \mathrm{conv}(P) \cap L$, and it spans L when $L = \langle P - P \rangle_{\mathbb{Z}}$. After translation assume $0 \in P$. Let $\mathcal{N}(P)$ be the simplicial complex on vertex set L in which a finite $F \subset L$ is a simplex exactly when $c - F \subseteq P$ for some $c \in L$, and set $Q(P) = |\mathcal{N}(P)|/L$.

For $0 \in P$, the quotient has the difference presentation

$$\pi_1(Q(P)) \cong \big\langle X_p \ (p \in P),\ X_0 = 1 \ \big| \ X_p^{-1} X_q = X_s^{-1} X_t \ \text{ whenever } q - p = t - s \big\rangle. \tag{95.1}$$

and a canonical displacement homomorphism

$$\delta : \pi_1(Q(P)) \longrightarrow L, \qquad \delta(X_p) = p. \tag{95.2}$$

This is the presentation established in the preceding optimal-fiber topology branch and will be used below without rederiving the rank-two classification.

95.2 Every lattice-convex configuration is an optimal fiber

The necessity in the completion baseline proved in the preceding branch has a converse.

Definition 95.1 (Completion-total system)**.** An integer counting matrix A is *completion-total on its integer image* if every $b \in A\mathbb{Z}^d$ has a nonnegative preimage. A sufficient condition is the existence of a strictly positive vector $h \in \ker_{\mathbb{Z}} A$: any integer solution can then be translated by a large nonnegative multiple of h.

Theorem 95.2 (Lattice-convex realization)**.** *Let P be a finite lattice-convex configuration spanning a rank-s lattice L. Then there exist an integer matrix A, a completable residual r, and a strictly positive $h \in \ker_{\mathbb{Z}} A$ such that the normalized optimal fiber $P_A(r)$ is affinely lattice-isomorphic to P and its active lattice has rank exactly s.*

Proof. Identify L with $\mathbb{Z}^s$ and translate so that $0 \in P$. Since $\operatorname{conv}(P)$ is a lattice polytope, choose integral affine facet functions

$$\ell_j(z) = b_j - a_j^{\top} z \qquad (1 \le j \le m)$$

such that

$$\operatorname{conv}(P) = \{z \in \mathbb{R}^s : \ell_j(z) \ge 0 \ \forall j\}.$$

Choose $B \in \mathbb{Z}^s$ so large coordinatewise that $z + B \in \mathbb{N}^s$ for every $z \in P$. Define the integral affine embedding

$$E_0(z) = \big(z + B, \ell_1(z), \dots, \ell_m(z)\big) \in \mathbb{Z}^{s+m}.$$

For lattice points $z \in \mathbb{Z}^s$ we have

$$E_0(z) \ge 0 \quad \Longleftrightarrow \quad z \in P.$$

Indeed, points of P satisfy every facet inequality and the chosen coordinate shifts; a lattice point outside $P = \operatorname{conv}(P) \cap \mathbb{Z}^s$ violates at least one facet inequality.

Choose an integer

$$C \ge \max_{z \in P} \mathbf{1}^{\top} E_0(z)$$

and append one balancing coordinate:

$$E(z) = \big(E_0(z),\ C - \mathbf{1}^{\top} E_0(z)\big) \in \mathbb{Z}^n, \qquad n = s + m + 1.$$

Then

$$E(z) \ge 0 \Longleftrightarrow z \in P, \qquad \mathbf{1}^{\top} E(z) = C \quad (z \in \mathbb{Z}^s). \tag{95.3}$$

Let

$$D = \{E(z) - E(0) : z \in \mathbb{Z}^s\} \subset \mathbb{Z}^n.$$

The first s coordinates of $E(z) - E(0)$ are exactly z, so projection onto those coordinates is a left inverse. Hence $D \cong \mathbb{Z}^s$ is a saturated direct summand of $\mathbb{Z}^n$. By (95.3), every $d \in D$ has coordinate sum zero.

Append one final coordinate and set

$$h = (1, \dots, 1) \in \mathbb{Z}^{n+1}, \qquad K = (D \times \{0\}) + \mathbb{Z}h \subset \mathbb{Z}^{n+1}.$$

The last coordinate of h is one, so quotienting first by $\mathbb{Z}h$ identifies $\mathbb{Z}^{n+1}/\mathbb{Z}h$ with $\mathbb{Z}^n$; the image of K is the saturated lattice D. Thus K is saturated. Consequently there is an integer matrix A with

$$\ker_{\mathbb{Z}} A = K.$$

The vector h is strictly positive, so A is completion-total on its integer image.

Put

$$x_0 = (E(0), 0) \in \mathbb{N}^{n+1}, \qquad r = -Ax_0.$$

Any integer point in the fiber of Ax_0 has a unique expression

$$x = x_0 + (d, 0) + th, \qquad d \in D,\ t \in \mathbb{Z}.$$

Its last coordinate is t, so nonnegativity forces $t \geq 0$. Writing $d = E(z) - E(0)$ gives

$$\mathbf{1}^\top x = C + t(n+1).$$

Hence every minimum-length point has $t = 0$. At $t = 0$, nonnegativity is equivalent by (95.3) to $z \in P$. Therefore

$$\mathcal{O}_A(r) = \{(E(z), 0) : z \in P\}.$$

The active difference lattice is $D \times \{0\}$, which has rank s, and the map $z \mapsto (E(z), 0)$ is an affine lattice isomorphism from P onto the normalized optimal fiber. □

Corollary 95.3 (Exact fiber characterization)**.** *Up to affine lattice equivalence, the normalized optimal fibers of completion-total integer counting systems are exactly the finite lattice-convex configurations.*

Proof. Necessity is the completion baseline proved in the preceding branch; sufficiency is theorem 95.2. □

95.3 Minimal fibers and free groups in every rank

A spanning rank-s configuration has at least $s + 1$ points. At equality the quotient is completely rigid.

Theorem 95.4 (Minimal active fiber)**.** *Let $P \subset L$ be a normalized, spanning, lattice-convex configuration with* $\operatorname{rank} L = s$ *and $|P| = s + 1$. Then*

$$Q(P) \simeq \bigvee^{s} S^1, \qquad \pi_1(Q(P)) \cong F_s, \qquad H_j(Q(P); \mathbb{Z}) = 0 \quad (j \geq 2).$$

Moreover the abelianized displacement map

$$H_1(Q(P); \mathbb{Z}) \longrightarrow L$$

is an isomorphism, so $K_{\mathrm{lin}}(P) = 0$.

Proof. The $s + 1$ points are affinely independent. No positive-dimensional subset can be a nontrivial translate of another subset of P: such a translation would give either a repeated nonzero difference among four vertices or three collinear vertices, in either case an affine dependence. Thus the only identifications in a fundamental simplex are the singleton vertices, which form one translation orbit. Consequently

$$Q(P) \cong \Delta^s / P^{(0)}.$$

Collapsing the $s + 1$ vertices of a contractible simplex to one point gives the homotopy cofiber of the inclusion of a discrete $(s + 1)$-point set into Δ^s, hence a wedge of s circles.

After translating one point to 0, the remaining s points generate L by definition. Their affine independence makes them linearly independent, so they form a $\mathbb{Z}$-basis of L. The s circle generators map to this basis under displacement, proving the final assertion. □

Proposition 95.5 (Explicit scalar realization)**.** *For every $s \geq 1$, the scalar weight vector*

$$\omega_s = (\underbrace{1, \ldots, 1}_{s+1 \text{ entries}}, -(s+1))$$

with residual $r = -1$ has minimum completion length one and exactly the first $s + 1$ unit vectors as optimal completions. Its active rank is s, and the all-ones vector is a positive kernel vector.

Proof. The equation $\omega_s x = 1$ has the $s + 1$ positive unit vectors as length-one solutions; the final coordinate cannot occur in a length-one solution because its weight is negative. Thus the minimum is one and the displayed unit vectors are the complete optimum. Their difference lattice has rank s, while the sum of all $s + 2$ columns is zero, so the all-ones vector lies in the kernel. □

At $s = 3$ this gives a genuine active-rank-three quotient with $\pi_1 \cong F_3$ and $K_{\mathrm{lin}} = 0$.

95.4 Products and higher tori

The periodic nerve is compatible with Cartesian products in a form that is especially useful for raising active rank.

Theorem 95.6 (Optimal-fiber product theorem)**.** *Let $P \subset L$ and $R \subset M$ be finite lattice-convex configurations spanning their lattices. Then $P \times R$ is lattice-convex in $L \oplus M$, has active rank* $\operatorname{rank} L + \operatorname{rank} M$*, and*

$$Q(P \times R) \simeq Q(P) \times Q(R).$$

The equivalence is $(L \oplus M)$-equivariant before passage to the quotient.

Proof. Write $K = \mathcal{N}(P)$ and $H = \mathcal{N}(R)$. A finite set $F \subset L \times M$ is a simplex of $\mathcal{N}(P \times R)$ exactly when some $(c, d) \in L \times M$ satisfies

$$(c, d) - F \subset P \times R.$$

This is equivalent to

$$c - \pi_L(F) \subset P, \qquad d - \pi_M(F) \subset R.$$

Hence $\mathcal{N}(P \times R)$ is the simplicial categorical product whose simplices are precisely the finite sets whose two coordinate projections are simplices of K and H.

On the posets of nonempty faces define

$$\phi(F) = (\pi_L F, \pi_M F), \qquad \psi(A, B) = A \times B.$$

Then $\phi\psi = \mathrm{id}$ and

$$F \subseteq \psi\phi(F)$$

for every face F. Pointwise comparable poset maps induce homotopic maps on order complexes [63]. Therefore ϕ and ψ induce mutually inverse homotopy equivalences after barycentric subdivision. The order complex of the product face poset is the standard barycentric triangulation of the product of the two barycentric subdivisions, so its realization is naturally homeomorphic to $|K| \times |H|$. The construction and the comparison $F \subseteq \psi\phi(F)$ commute with translations, so the equivalence is $(L \oplus M)$-equivariant. Quotienting gives the asserted product law.

Finally,

$$\operatorname{conv}(P \times R) = \operatorname{conv}(P) \times \operatorname{conv}(R),$$

so

$$\big(\operatorname{conv}(P \times R)\big) \cap (L \times M) = (\operatorname{conv}(P) \cap L) \times (\operatorname{conv}(R) \cap M) = P \times R.$$

The difference lattice is $\langle P - P \rangle \oplus \langle R - R \rangle$, proving the rank statement. □

Corollary 95.7 (Iterated products)**.** *For finitely many spanning lattice-convex configurations $P_i \subset L_i$,*

$$Q\left(\prod_i P_i\right) \simeq \prod_i Q(P_i).$$

Every such product is realized as a genuine completion-total optimal fiber by theorem 95.2.

The most symmetric instance is the Boolean cube.

Theorem 95.8 (Higher-torus realization)**.** *For every $s \geq 1$ there is a completion-total integer counting system with active rank s and normalized optimal fiber affinely lattice-isomorphic to $\{0, 1\}^s$. Its translation quotient is*

$$Q(\{0, 1\}^s) \simeq T^s,$$

so

$$H_j(Q; \mathbb{Z}) \cong \mathbb{Z}^{\binom{s}{j}} \qquad (0 \leq j \leq s).$$

Proof. The topological statement follows immediately from theorem 95.6: $\{0,1\}^s$ is the product of s rank-one intervals, and $Q(\{0,1\}) \simeq S^1$ by theorem 95.4. Thus its quotient is T^s.

For a direct completion-total realization, use $2s+1$ letters grouped into pairs (a_i, b_i), $1 \le i \le s$, together with a padding letter c. Let A_s have s rows. For $1 \le i < s$, row i has coefficient 1 on a_i, b_i, coefficient -1 on a_s, b_s, and 0 elsewhere. The final row has coefficient 1 on the first $2s$ letters and coefficient $-2s$ on c.

Take residual

$$r = (0, \dots, 0, -s),$$

so the completion equation $A_s x = -r$ has last coordinate s. If $t_i = x_{a_i} + x_{b_i}$ and $z = x_c$, the first $s-1$ equations force

$$t_1 = \cdots = t_s = t,$$

while the last equation gives

$$st - 2sz = s, \qquad t = 1 + 2z.$$

The total length is

$$st + z = s + (2s+1)z,$$

so it is minimized exactly at $z = 0$, $t = 1$. Hence an optimal completion chooses exactly one letter from each pair, giving 2^s optima and normalized cube fiber. The within-pair swaps generate an active lattice of rank s. The all-ones vector is a strictly positive kernel vector because every difference row has total coefficient zero and the last row has $2s - 2s = 0$. □

Corollary 95.9 (First third homology)**.** *Active rank three admits*

$$Q \simeq T^3, \qquad (\beta_0, \beta_1, \beta_2, \beta_3) = (1, 3, 3, 1).$$

This is the first possible active rank for nonzero third homology, because theorem 91.2 gives $H_j = 0$ for $j \ge 3$ in active rank two.

95.5 Unequal simplex-strand product rigidity

The periodic nerve, marked quotient and Cartesian-product results of this chapter apply to lattice-convex optimal fibers. The following family has a toric product homotopy type even when it is not a Cartesian product of lattice configurations. The entire face preimage, including horizontal simplices, is retained in the argument.

95.5.1 Lattice normalization, full faces, and marked displacement

Fix $s \ge 1$, the transverse vertices $q_0 = 0$, $q_j = e_j$ of the unimodular simplex in $\mathbb{Z}^s$, and nondegenerate integral intervals $I_j = [a_j, b_j] \cap \mathbb{Z}$. The configuration is

$$P = \bigcup_{j=0}^{s} (I_j \times \{q_j\}) \subset \mathbb{Z} \oplus \mathbb{Z}^s. \tag{95.4}$$

The widths need not coincide; in particular, P need not be lattice-affinely a product. The periodic nerve and quotient $Q(P)$ retain the definitions established above.

Throughout, $X \simeq Y$ signifies that there exist continuous maps $u : X \to Y$ and $v : Y \to X$ together with continuous one-parameter families of maps connecting $v \circ u$ to id_X and $u \circ v$ to id_Y. This convention fixes precisely what is meant by equivalence under continuous deformation; it is stronger than an equality of fundamental groups or of integral singular groups. Let L be a finitely generated free abelian group and $P \subset L$ a finite set spanning the lattice $\langle P - P \rangle_{\mathbb{Z}} = L$. We call P *lattice-convex* if $P = \mathrm{conv}(P) \cap L$. Define

$$\begin{aligned} \mathcal{N}(P) &= \{F \subset L : F \text{ finite and } c - F \subseteq P \text{ for some } c \in L\}, \\ Q(P) &= |\mathcal{N}(P)|/L. \end{aligned} \tag{95.5}$$

As usual the empty face is included. L acts on $\mathcal{N}(P)$ by translation. A nonzero lattice translation cannot preserve a finite simplex setwise, so the action is free on simplices and $Q(P)$ is a CW complex. A choice of $0 \in P$ leads to the known difference presentation

$$\pi_1 Q(P) = \left\langle X_p \ (p \in P),\ X_0 = 1 \ \middle|\ X_p^{-1} X_q = X_u^{-1} X_v \text{ if } q - p = v - u \right\rangle. \tag{95.6}$$

The displacement group map $\delta_P : \pi_1 Q(P) \to L$ takes X_p to p; the difference presentation and marking are established in [186, 188].

Lemma 95.10 (Lattice-convexity and alignment)**.** *The configuration P from* (95.4) *is lattice-convex and spans $\mathbb{Z}^{s+1}$. There is an affine unimodular change of coordinates mapping it to*

$$P' = \bigcup_{j=0}^{s} \big(\{0, 1, \dots, m_j\} \times \{q_j\}\big), \qquad m_j = b_j - a_j \geq 1.$$

Proof. The projection of $\mathrm{conv}(P)$ to $\mathbb{R}^s$ is the unimodular simplex $\mathrm{conv}\{q_0, \dots, q_s\}$. Its only integer points are the q_j, and the slice above each vertex is exactly $[a_j, b_j]$. Hence $\mathrm{conv}(P) \cap \mathbb{Z}^{s+1} = P$. A horizontal unit difference is present in each strand. Differences between the strand starts, together with that horizontal vector, generate $\mathbb{Z}^{s+1}$, since the $q_j - q_0$ form a $\mathbb{Z}$-basis in the transverse component. Finally the integral shear

$$(t, y_1, \dots, y_s) \mapsto \left(t - a_0 - \sum_{j=1}^{s} (a_j - a_0) y_j, y_1, \dots, y_s \right)$$

has determinant one and aligns all left endpoints to zero. □

From now on assume all strands are aligned, put $m_j \geq 1$ and $M = \max_j m_j$, and write $T = \{q_0, \dots, q_s\} \subset \mathbb{Z}^s$. Let $K = \mathcal{N}(T)$, $H = \mathbb{Z}$ be the horizontal translation subgroup, and $G = \mathbb{Z}^s$ the transverse subgroup. Projection on the transverse coordinate is a simplicial map

$$p : \mathcal{N}(P) \longrightarrow K,$$

with compatible H- and G-translation actions.

95.6 The corrected full-face lemma

For a nonempty simplex σ of K, let X_σ be the *full induced subcomplex* of $\mathcal{N}(P)$ on all vertices whose transverse coordinate belongs to the vertex set of σ. This is the actual geometric inverse image of $|\sigma|$, not a choice of mixed simplices.

Write B_q^m for the horizontal simplicial complex on $\mathbb{Z} \times \{q\}$ whose simplices are the finite sets of horizontal diameter at most m. It is the union of the full translated horizontal simplices of width m.

Lemma 95.11 (Horizontal interval complexes)**.** *For any integer $m \geq 1$, B_q^m is contractible. The quotient $|B_q^m|/H$ is a $K(\mathbb{Z}, 1)$, and the horizontal barycenter modulo $\mathbb{Z}$ is an equivalence under continuous deformation with S^1.*

Proof. Cover B_q^m by full simplices supported on the intervals $[h - m, h] \cap \mathbb{Z}$, $h \in \mathbb{Z}$. Every nonempty finite intersection is a simplex. Their abstract cover nerve consists of finite nonempty $D \subset \mathbb{Z}$ with $\max D - \min D \leq m$. This nerve is also the nerve of the intervals $[h - m, h] \subset \mathbb{R}$, which cover the real line and have contractible intersections. The two applications of the nerve theorem show $|B_q^m| \simeq \mathbb{R}$, hence it is contractible. Horizontal H acts freely; the quotient has universal cover B_q^m, so is a $K(\mathbb{Z}, 1)$. Barycenter changes by 1 under the deck generator and induces an isomorphism on π_1, giving the asserted equivalence under continuous deformation. □

Lemma 95.12 (Full transverse-face preimage)**.** *For every nonempty simplex $\sigma \subset K$ the full complex X_σ is contractible. If σ is a vertex $\{q\}$, then $X_\sigma = B_q^M$. If $|\sigma| \geq 2$ in vertices, there is a contractible mixed-row core C_σ and uniquely assigned indices $j(q) \in \{0, \dots, s\}$ such that*

$$X_\sigma = C_\sigma \cup \bigcup_{q \in \sigma} B_q^M, \qquad C_\sigma \cap B_q^M = B_q^{m_{j(q)}}. \tag{95.7}$$

The distinct B_q^M are disjoint, and all inclusions in (95.7) *are cellular cofibrations.*

Proof. For a singleton transverse vertex, that vertex can assume every role among the translates of T; its horizontal simplices consequently arise from all strand widths, whose union is the width-M complex.

For clarity, the uniqueness claim used for mixed faces follows directly from $T = \{0, e_1, \dots, e_s\}$: every nonzero difference $q_i - q_j$ uniquely determines the ordered pair (i, j). Indeed, vectors of the forms $\pm e_i$ and $e_i - e_j$ are distinct unless their ordered indices agree (the assertion also holds for $s = 1$). Thus two different transverse translates of T cannot contain the same ordered pair of distinct vertices. A mixed transverse simplex has one role assignment, whereas a singleton has several.

Suppose σ contains at least two vertices. A nonzero difference $q_i - q_j$ with $q_i, q_j \in T$ uniquely specifies the ordered pair (i, j): the simplex vertices $\{0, e_1, \dots, e_s\}$ are affinely independent. Therefore σ belongs to a unique maximal translate of the transverse simplex K and assigns a unique row role $j(q)$ to each $q \in \sigma$. Let C_σ be the subcomplex generated by the mixed-row full simplices

$$D_h(\sigma) = \bigcup_{q \in \sigma} \big(([h - m_{j(q)}, h] \cap \mathbb{Z}) \times \{q\} \big), \quad h \in \mathbb{Z}.$$

Their finite intersections are simplices and are nonempty exactly when the horizontal centers have diameter at most $M_\sigma = \max_{q \in \sigma} m_{j(q)} \geq 1$. The same interval-nerve argument as in Lemma 95.11 gives $C_\sigma \simeq *$.

Every face involving at least two distinct transverse vertices belongs to this core by the uniqueness of its transverse maximal translate. A face involving just one transverse vertex may instead be witnessed by another translate of T and need not belong to the mixed core: it lies precisely in the full-width horizontal complex B_q^M. These two cases exhaust the actual simplices of X_σ, proving the union identity. Restricting a core simplex to the strand at q gives horizontal diameter at most $m_{j(q)}$, and every such horizontal simplex is present in the core; thus the stated intersection is exact.

By Lemma 95.11, C_σ, $B_q^{m_{j(q)}}$, and B_q^M are all contractible. A subcomplex inclusion $B_q^{m_{j(q)}} \hookrightarrow B_q^M$ is therefore an equivalence under continuous deformation and a cofibration. Attaching the width-M strand along that inclusion preserves the deformation class of the core by the CW gluing theorem for equivalences along cofibrations. Repeating for the finitely many $q \in \sigma$ proves $X_\sigma \simeq *$. □

Remark 95.13 (Why the full preimage matters). The tempting identification $X_\sigma = C_\sigma$ is false for unequal intervals. For example, take $s = 2$, strand widths $(m_0, m_1, m_2) = (1, 1, 4)$ and transverse face $\{0, -e_1\}$. Its mixed core uses the first two short strands, yet the horizontal edge with endpoints $(0, 0)$ and $(4, 0)$ is a genuine simplex of X_σ witnessed by the third, longer role. Formula (95.7) accounts for that edge. This repair is essential for the proof of Theorem 95.14.

95.7 Product rigidity and marked displacement

Theorem 95.14 (Simplex-strand product rigidity)**.** *For every $s \geq 1$ and every configuration P of the form* (95.4) *with all $|I_j| \geq 2$,*

$$Q(P) \simeq Q(T) \times S^1 \simeq \left(\bigvee_{j=1}^{s} S^1 \right) \times S^1. \tag{95.8}$$

The displacement-marked fundamental group is $(F_s \times \mathbb{Z}, \mathrm{ab})$*, up to the intrinsic integral basis of the active lattice.*

Proof. After Lemma 95.10, write $L = H \oplus G$, with $H = \mathbb{Z}$ horizontal and $G = \mathbb{Z}^s$ transverse, and set

$$Y = |\mathcal{N}(P)|/H.$$

The affine horizontal barycentric coordinate η on every simplex obeys $\eta(hx) = \eta(x) + h$ and is fixed under transverse translations. Consequently the map

$$f = (p, \eta \bmod \mathbb{Z}) : Y \longrightarrow |K| \times S^1$$

is continuous and strictly G-equivariant, with G translating the first factor.

For a transverse face σ, put $Y_\sigma = |X_\sigma|/H$. By Lemma 95.12, X_σ is contractible. Horizontal translation acts freely on it, so Y_σ is a $K(\mathbb{Z}, 1)$. Since a width-one horizontal edge occurs in every X_σ, $\eta \bmod \mathbb{Z}$ maps the deck generator to winding number one. Therefore $\eta : Y_\sigma \to S^1$ is an equivalence under continuous deformation, and hence so is

$$f|_{Y_\sigma} : Y_\sigma \longrightarrow |\sigma| \times S^1.$$

Importantly, these maps are restrictions of the *same* global f; they commute strictly with face inclusions, without any independently chosen facewise contractions.

For a finite transverse subcomplex $K' \subset K$, write $Y[K'] = p^{-1}(|K'|) \subset Y$. The preimage of $|\sigma|$ for any face σ is exactly Y_σ; since it is the full geometric preimage, inverse images of two faces intersect over $\sigma \cap \tau$. When a maximal simplex σ is attached to a finite subcomplex containing its entire boundary, the two squares

$$\begin{array}{ccc} Y[\partial\sigma] & \longrightarrow & Y_\sigma \\ \downarrow & & \downarrow \\ Y[K'_{\mathrm{old}}] & \longrightarrow & Y[K'_{\mathrm{old}} \cup \sigma] \end{array} \qquad \begin{array}{ccc} |\partial\sigma| \times S^1 & \longrightarrow & |\sigma| \times S^1 \\ \downarrow & & \downarrow \\ |K'_{\mathrm{old}}| \times S^1 & \longrightarrow & |K'_{\mathrm{old}} \cup \sigma| \times S^1 \end{array}$$

are cellular pushouts, with horizontal and vertical attaching maps subcomplex inclusions and hence cofibrations. The natural map f induces a map of squares. On the full simplex and, by induction, on the boundary and the previously attached part it is a weak equivalence. The CW gluing principle for weak equivalences makes it so on the new union. Induction over faces of finite K', followed by exhaustion by finite subcomplexes (compact maps from spheres have images in a finite CW subcomplex), gives a weak equivalence

$$f : Y \xrightarrow{\simeq} |K| \times S^1.$$

We still need to descend this statement to the actual periodic quotient. The transverse complex K is locally finite because T is finite. No nonzero element of G preserves a simplex of K, so the translation action of G on $|K|$ is free and properly discontinuous: each point has a neighborhood U whose distinct G-translates are disjoint. Equivariance of $p : Y \to |K|$ gives the same property for $p^{-1}(U)$ in Y. Thus both quotient projections are regular covering maps with deck group G, not merely free cellular quotients. The map f induces the identity on that deck group. Both covering spaces are connected: K is connected because the edge differences of the unimodular simplex generate G, and Y is connected by the horizontal edges and the mixed simplices. Their regular covering sequences therefore give a commuting diagram

$$\begin{array}{ccccccccc} 1 & \to & \pi_1(Y) & \to & \pi_1(Y/G) & \to & G & \to & 1 \\ & & \downarrow\cong & & \downarrow & & \| & & \\ 1 & \to & \pi_1(|K| \times S^1) & \to & \pi_1((|K| \times S^1)/G) & \to & G & \to & 1. \end{array}$$

The middle arrow is an isomorphism by a group-extension diagram chase; the higher groups π_j also agree, because covering spaces preserve them for $j \geq 2$. The quotient map is

therefore a weak equivalence and hence, by the CW Whitehead theorem, an equivalence under continuous deformation:

$$Q(P) = Y/G \simeq (|K|/G) \times S^1 = Q(T) \times S^1.$$

For the primitive simplex T, every maximal simplex of K is a translate of $-T$. Two different translates cannot share an edge: equality of a nonzero ordered edge difference determines its unique ordered pair of vertices of T, and hence determines the translation. More generally, no two distinct non-vertex faces of $-T$ lie in the same translation orbit. Consequently $Q(T)$ is the quotient of one geometric s-simplex by identifying its $s+1$ vertices and making no further identifications. Let A denote this finite vertex subcomplex. Since $A \hookrightarrow \Delta^s$ is a CW cofibration and Δ^s is contractible, its cofiber Δ^s/A has the homotopy type of the unreduced suspension of A, which is a wedge of s circles. Thus $Q(T) \simeq \bigvee^s S^1$; this also recovers the primitive-simplex case of [187].

Finally, the displacement homomorphism $\delta_P : \pi_1 Q(P) \to L$ is surjective: the differences $P - P$ generate the full active lattice L, and the difference presentation (95.6) realizes these differences as displacements of loops. By the product equivalence just proved, $H_1(Q(P);\mathbb{Z}) \cong \mathbb{Z}^{s+1}$, while $L \cong \mathbb{Z}^{s+1}$. Because L is abelian, δ_P factors through a surjection between free abelian groups of the same finite rank; that map is an isomorphism. Hence the marking is exactly abelianization, up to the intrinsic unimodular choice of coordinates on L. □

Corollary 95.15 (Integral invariants)**.** *In the setting of Theorem 95.14,*

$$\pi_1 Q(P) \cong F_s \times \mathbb{Z}, \quad H_0(Q(P);\mathbb{Z}) \cong \mathbb{Z},$$
$$H_1(Q(P);\mathbb{Z}) \cong \mathbb{Z}^{s+1}, \qquad H_2(Q(P);\mathbb{Z}) \cong \mathbb{Z}^s,$$

$H_j(Q(P);\mathbb{Z}) = 0$ *for* $j \geq 3$*, and* $\pi_j Q(P) = 0$ *for all* $j \geq 2$*. In particular* $Q(P)$ *is aspherical and has no torsion in its integral singular groups. The linear displacement kernel* $\ker(H_1 \to L)$ *vanishes; the full zero-displacement subgroup is* $[F_s, F_s] \times \{0\}$*.*

Proof. A wedge of circles is a graph and hence aspherical. Its product with S^1 is aspherical. Integral Künneth gives the stated integral singular groups, and the marking follows from Theorem 95.14. □

95.8 An exact shortest-completion example with a topological transition

Consider the fixed integer source with six genuine unit-cost actions

$$A = \begin{pmatrix} -1 & 2 & 1 & 0 & 3 & -2 \\ -2 & 3 & 0 & 1 & 2 & -1 \end{pmatrix}. \tag{95.9}$$

Its image is $\mathbb{Z}^2$, since columns $(1,0)$ and $(0,1)$ occur; the strictly positive vector $(2,1,1,1,1,2)$ lies in $\ker_{\mathbb{Z}} A$. Consequently every integer residual has a nonnegative source completion: for an integral preimage u one may add a large positive multiple of the strictly positive kernel vector to make u nonnegative. The source is a genuine rank-two counting system, not a globally recognized one-counter language.

For target $r = (x, y)$ set $S = x + y$ and $D = x - y$. The six action vectors in (D, S) coordinates are precisely $\{\pm 1\} \times \{-3, 1, 5\}$.

Proposition 95.16 (Exact length and counting profile)**.** *For any* $r = (x, y) \in \mathbb{Z}^2$ *and* $N \geq 0$*, a source Parikh vector* $z \in \mathbb{N}^6$ *with* $Az = r$ *and* $|z| = N$ *exists exactly when*

$$N \geq |D|, \qquad -3N \leq S \leq 5N, \qquad N \equiv S \pmod 4. \tag{95.10}$$

When this holds, put $P = (N + D)/2$, $R = (N - D)/2$, $K = (S + 3N)/4$ *and*

$$F_j(t) = \sum_{a_0+a_1+a_2=j} t^{a_1+2a_2} = \begin{bmatrix} j+2 \\ 2 \end{bmatrix}_t .$$

Then the number of Parikh vectors *of length* N *at* r *equals*

$$c_A(r, N) = [t^K]F_P(t)F_R(t). \tag{95.11}$$

Thus $L_A(r)$ *is the least integer* $N \geq \max\{|D|, S/5, -S/3, 0\}$ *satisfying* $N \equiv S \pmod 4$.

Proof. Every action has $D = \pm 1$ and $S = -3+4j$, $j = 0, 1, 2$, proving necessity. Conversely, (95.10) makes P, R nonnegative integral, with $P+R = N$, and makes $0 \leq K \leq 2N$ integral. Decompose K into N digits from $\{0, 1, 2\}$ and assign P of them to the $D = +1$ columns and the remaining R to $D = -1$ columns; each sign group offers all three digit choices. This produces the required vector. At fixed P, R the choices of multiplicities of each digit in each sign group are independent, hence the generating-function product and coefficient formula. This counts unordered source-letter multiplicities, not word permutations. □

Let $r_N = (N-1, 1)$ for $N \geq 3$. Then $S = N$, $D = N-2$, and (95.10) gives $L_A(r_N) = N$. The length-N fiber has one $D = -1$ letter and $N - 1$ $D = +1$ letters. Index the chosen negative letter by digit $j \in \{0, 1, 2\}$ and put $t = z_5$, the multiplicity of the original fifth action. Solving the remaining equations gives

$$z_1 = t + j - 1, \qquad z_3 = N - j - 2t, \qquad z_5 = t,$$

with exactly one of z_6, z_4, z_2 equal to one, corresponding to $j = 0, 1, 2$ respectively. Its integral affine fiber coordinates are

row j	allowed t
0	$1 \leq t \leq \lfloor N/2 \rfloor$
1	$0 \leq t \leq \lfloor (N-1)/2 \rfloor$
2	$0 \leq t \leq \lfloor (N-2)/2 \rfloor$.

(95.12)

This bijection preserves the full active difference lattice, and the total number of optimal vectors is $\lfloor 3N/2 \rfloor$. For $N = 3$ the row sizes are $(1, 2, 1)$. The *entire* shortest Parikh fiber, in the original six-action coordinates, consists of

$$(0, 0, 1, 0, 1, 1), \qquad (0, 0, 2, 1, 0, 0),$$
$$(1, 0, 0, 1, 1, 0), \qquad (1, 1, 1, 0, 0, 0).$$

Using the first point as origin, the three difference vectors have a 3×3 minor of determinant $+1$ on original coordinates $1, 2, 3$. They therefore form an active *primitive* three-simplex (the full active lattice has rank three). For every $N \geq 4$, each of the three complete rows in (95.12) contains at least two consecutive points and their transverse coordinates form a primitive triangle.

Corollary 95.17 (Target-indexed topological transition)**.** *For the* same fixed source (95.9)*, the true full shortest-completion fibers satisfy*

$$Q(P_A(r_N)) \simeq \begin{cases} \bigvee^3 S^1, & N = 3, \\ (S^1 \bigvee S^1) \times S^1, & N \geq 4. \end{cases}$$

Thus H_2 *changes from* 0 *at* $N = 3$ *to* $\mathbb{Z}^2$ *for all* $N \geq 4$*, even though active rank is three throughout and the optimal Parikh multiplicity grows as* $\lfloor 3N/2 \rfloor$.

Proof. The $N = 3$ case is the primitive-minimal-fiber theorem. For $N \geq 4$ apply Theorem 95.14 to the three true rows (95.12). The integral-group conclusion follows from Corollary 95.15. □

95.9 Toric-fiber interpretation and theorem hypotheses

Every signed integer counting source can be embedded into a pointed nonnegative semigroup by adjoining a degree coordinate. Given $A = (a_1, \dots, a_n)$, choose $c \in \mathbb{Z}^d$ with $a_i + c \geq 0$ for every i, and form B_c with columns $(a_i + c, 1)$. The tautological identity

$$\{z \in \mathbb{N}^n : Az = r,\ |z| = N\} = \{z \in \mathbb{N}^n : B_c z = (r + Nc, N)\}$$

identifies fixed-length signed Parikh sets with degree-graded pointed toric fibers. The associated scalar counts belong to established vector partition theory [191]; in our example the Gaussian coefficient expression is an explicit specialization of standard fiber-counting methods. Conversely, our periodic nerve $Q(P)$ is built from lattice translations in the *full* minimum-grade fiber and is not, by definition, the indispensable complex attached to a toric ideal [192].

Theorem 95.14 is stated for unimodular transverse simplices and intervals containing at least two consecutive lattice points. The target-indexed formula in Corollary 95.17 concerns the minimum-grade fibers of the fixed counting matrix (95.9); the input source, rather than its parameters, is held constant. The result is an intrinsic statement about the periodic quotient of these exact Parikh fibers, without requiring a physical pushdown realization.

95.10 Two one-rank extensions: apex versus prism

Product with the interval is one natural one-rank extension. A second extension has a different homotopy effect.

Theorem 95.18 (Translation-isolated apex)**.** *Let $P \subset L$ be finite, spanning, and lattice-convex. In $L \oplus \mathbb{Z}e$ put*

$$P^+ = (P \times \{0\}) \cup \{e\},$$

where e has last coordinate one. Then P^+ is lattice-convex, has active rank $\operatorname{rank} L + 1$*, and*

$$Q(P^+) \simeq Q(P) \vee S^1.$$

Proof. A lattice point of the pyramid $\operatorname{conv}(P^+)$ has last coordinate 0 or 1. At height zero the slice is $\operatorname{conv}(P)$, whose lattice points are exactly P; at height one the slice is only the apex. Thus P^+ is lattice-convex, and its difference lattice is $L \oplus \mathbb{Z}e$.

No positive-dimensional face containing the apex has a nontrivial translate inside P^+. Indeed, a translation with last coordinate zero must fix the unique height-one point and is therefore zero; a translation with last coordinate ± 1 sends one of the two height levels outside $\{0, 1\}$. Hence every nontrivial face identification is confined to the base simplex, although all vertices, including the apex, belong to one translation orbit.

Choose the simplex of the periodic nerve corresponding to the translate P^+ itself as a fundamental simplex; it is the cone on the base simplex $\Delta(P)$. Every simplex orbit has a representative in this cone. The preceding height argument has two consequences for its face identifications. First, all identifications between positive-dimensional base faces are exactly those already defining $Q(P)$. Second, a positive-dimensional face containing the apex cannot be identified with a distinct face containing the apex: a translating vector with last coordinate zero would have to fix the unique height-one vertex and hence be zero, while a vector with nonzero last coordinate cannot preserve the two-level pattern of a mixed face. Only the apex vertex itself is identified with the common vertex orbit, because the translation action is transitive on vertices.

Consequently the quotient is obtained from the cone $C\Delta(P)$ by applying the old base quotient

$$q : \Delta(P) \longrightarrow Q(P)$$

and identifying the cone point with the vertex orbit of $Q(P)$. Equivalently it is the homotopy coequalizer of q and the constant map $\Delta(P) \to Q(P)$ at that vertex. Since $\Delta(P)$ is

nonempty and contractible, q is homotopic to the constant map. The homotopy coequalizer of two homotopic maps from a nonempty contractible space is homotopy equivalent to $Q(P) \vee S^1$ (for example by the standard mapping-cylinder model). This proves the theorem. □

Corollary 95.19 (Prism law)**.** *For every finite spanning lattice-convex P,*

$$Q(P \times \{0,1\}) \simeq Q(P) \times S^1.$$

Proof. Apply theorem 95.6 and $Q(\{0,1\}) \simeq S^1$. □

Thus apex and prism both add one active lattice direction but act differently:

$$\pi_1(Q(P^+)) \cong \pi_1(Q(P)) * \mathbb{Z}, \qquad \pi_1(Q(P \times \{0,1\})) \cong \pi_1(Q(P)) \times \mathbb{Z}.$$

For torsion-free homology, the Künneth formula also gives

$$H_j(Q(P) \times S^1) \cong H_j(Q(P)) \oplus H_{j-1}(Q(P)),$$

while apex extension preserves H_j for $j \geq 2$.

Corollary 95.20 (Rank-three branches)**.** *Both operations produce genuine completion-total active-rank-three families.*

(a) *Apexing a rank-two F_2 fiber gives F_3; apexing a rank-two F_3 fiber gives F_4.*

(b) *Apexing a rank-two toral fiber gives fundamental group $\mathbb{Z}^2 * \mathbb{Z}$ and, for the basic torus, homotopy type $T^2 \vee S^1$.*

(c) *Prisming the rank-two minimal F_2 branch gives an aspherical $F_2 \times \mathbb{Z}$ quotient with Betti vector $(1,3,2)$. Prisming a rank-two F_3 wedge branch gives an aspherical $F_3 \times \mathbb{Z}$ quotient with Betti vector $(1,4,3)$.*

95.11 Fixed-rank higher homology and displacement

We now combine the rank-raising calculus with an explicit rank-two seed.

Proposition 95.21 (Rank-two seed family)**.** *For every $m \geq 2$ there is a completion-total scalar system with active rank two and a normalized fiber affinely equivalent to*

$$P_m = \{(t,0) : 0 \leq t \leq m\} \cup \{(0,1), (m-1,-1)\} \subset \mathbb{Z}^2. \tag{95.13}$$

Its quotient satisfies

$$\pi_1(Q(P_m)) \cong \mathbb{Z}^2, \qquad H_1(Q(P_m);\mathbb{Z}) \cong \mathbb{Z}^2, \qquad H_2(Q(P_m);\mathbb{Z}) \cong \mathbb{Z}^{m-1},$$

and has no homology above degree two. Equivalently, its Betti polynomial is

$$1 + 2t + (m-1)t^2.$$

Proof. Take $\alpha = 2m+1$ and the four scalar weights

$$\omega_m = \big(1 - m\alpha,\ 1 - (m-1)\alpha,\ 1,\ 1 + \alpha\big),$$

with residual

$$r_m = \alpha m(m-1) - 2m.$$

The strictly positive vector

$$h_m = \big(1, 1, (2m-2)\alpha - 3, 1\big)$$

lies in the kernel. All four weights are 1 modulo α, so every completion length satisfies

$$n \equiv -r_m \equiv 2m \pmod{2m+1}.$$

At length $2m$, the complete optimum is

$$A_t = (t, m-t, m-t, t) \quad (0 \le t \le m),$$

$$B = (0, m+1, 0, m-1), \qquad C = (m-1, 0, m+1, 0).$$

Indeed, eliminating the common congruence term gives

$$x_2 + mx_3 + (m+1)x_4 = m(m+1),$$

and reduction modulo m forces $x_2 + x_4 \in \{0, m, 2m\}$, yielding exactly C, the A_t, and B.

The equal-length kernel has basis

$$g = (1, -1, -1, 1), \qquad h = (0, 1, -m, m-1),$$

and relative to A_0 the optimal points have coordinates

$$A_t - A_0 = tg, \qquad B - A_0 = h, \qquad C - A_0 = (m-1)g - h,$$

which is (95.13). Its convex hull is the lattice quadrilateral

$$(0,0),\ (m-1,-1),\ (m,0),\ (0,1).$$

All four boundary edges are primitive. The Euclidean area is m, so Pick's theorem gives $m-1$ interior lattice points. These are exactly $(1,0), \dots, (m-1,0)$; together with the four boundary vertices this is precisely P_m. Hence P_m is lattice-convex.

We next compute the fundamental group directly from the difference presentation (95.1), without invoking a general planar classification. Write

$$x = X_{(1,0)}, \qquad y = X_{(0,1)}, \qquad z = X_{(m-1,-1)}.$$

Repeated horizontal differences give $X_{(t,0)} = x^t$. The repeated difference $(0,1)$ between the pairs $((0,0),(0,1))$ and $((m-1,-1),(m-1,0))$ gives

$$y = z^{-1}x^{m-1}, \qquad z = x^{m-1}y^{-1}.$$

For $1 \le t \le m-1$, the repeated difference $(-t,1)$ between

$$((t,0),(0,1)) \quad \text{and} \quad ((m-1,-1),(m-1-t,0))$$

gives

$$x^{-t}y = yx^{-t}.$$

Taking $t = 1$ shows $xy = yx$. Thus the presentation is generated by the commuting pair x, y. The displacement map (95.2) sends them to the standard basis of $\mathbb{Z}^2$, so it rules out any further relation and yields

$$\pi_1(Q(P_m)) \cong \mathbb{Z}^2.$$

In particular $H_1(Q(P_m);\mathbb{Z}) \cong \mathbb{Z}^2$. The rank-two homological baseline theorem 91.2 gives no homology above degree two and makes H_2 free abelian.

It remains to compute the Euler characteristic by simplex orbits. Put $H = \{(t,0) : 0 \le t \le m\}$. A horizontal-only orbit can be normalized to have minimum horizontal coordinate zero, so its alternating contribution is

$$\sum_{k \ge 1} (-1)^{k-1} \binom{m}{k-1} = 0.$$

For either fixed off-line point, faces containing that point and a nonempty horizontal subset contribute

$$\sum_{k=1}^{m+1}(-1)^k\binom{m+1}{k}=-1,$$

while faces containing both off-line points have total alternating contribution

$$\sum_{k=0}^{m+1}(-1)^{k+1}\binom{m+1}{k}=0.$$

The only cross-identifications between the two one-off-line families are the m edge pairs

$$\{(0,1),(t,0)\}\sim\{(m-1,-1),(m-1-t,0)\},\qquad 0\le t\le m-1.$$

A nonzero vertical translation can match at most one horizontal point because each off-line level contains only one point, so there is no higher-dimensional cross-identification. Identifying each duplicate edge orbit raises the alternating count by one. Therefore

$$\chi(Q(P_m))=0-1-1+m=m-2.$$

Since $\beta_0=1$, $\beta_1=2$, and higher homology vanishes,

$$m-2=1-2+\beta_2,$$

so $\beta_2=m-1$. Torsion-freeness has already been supplied by theorem 91.2. □

Corollary 95.22 (Apex branch with unbounded second homology)**.** *Let P_m^+ be the apex extension of the seed in theorem 95.21. Then P_m^+ has active rank three and*

$$Q(P_m^+)\simeq Q(P_m)\vee S^1,\qquad \pi_1(Q(P_m^+))\cong\mathbb{Z}^2*\mathbb{Z}.$$

Moreover

$$H_2(Q(P_m^+);\mathbb{Z})\cong\mathbb{Z}^{m-1},\qquad H_j(Q(P_m^+);\mathbb{Z})=0\quad(j\ge 3).$$

Hence β_2 is unbounded at fixed active rank three and fixed fundamental group $\mathbb{Z}^2*\mathbb{Z}$.

Proof. Apply theorem 95.18 to theorem 95.21. Wedging on a circle changes only H_1 and takes the fundamental group to the free product with $\mathbb{Z}$. □

The prism operation behaves differently: it moves the seed's second homology into degree three. This leads to the main all-rank phenomenon.

Theorem 95.23 (Fixed rank and fixed fundamental group with unbounded higher homology)**.** *For every $s\ge 2$ there is a one-parameter family of completion-total integer counting systems whose normalized optimal fibers have active rank exactly s and whose quotients $Q_{m,s}$ satisfy*

$$\pi_1(Q_{m,s})\cong\mathbb{Z}^s,$$

while every Betti number $\beta_j(Q_{m,s})$, $2\le j\le s$, is unbounded as $m\to\infty$.

More precisely, take

$$P_{m,s}=P_m\times\{0,1\}^{s-2}.$$

Then

$$Q(P_{m,s})\simeq Q(P_m)\times T^{s-2}$$

and its Betti polynomial is

$$\boxed{\big(1+2t+(m-1)t^2\big)(1+t)^{s-2}}.\tag{95.14}$$

Equivalently,

$$\boxed{\beta_j = \binom{s-2}{j} + 2\binom{s-2}{j-1} + (m-1)\binom{s-2}{j-2}}, \tag{95.15}$$

with out-of-range binomial coefficients interpreted as zero. In particular

$$\beta_1 = s, \qquad \beta_s = m - 1.$$

Proof. The active ranks add under products, so $P_{m,s}$ has rank $2 + (s-2) = s$. By theorems 95.6 and 95.8,

$$Q(P_{m,s}) \simeq Q(P_m) \times T^{s-2}.$$

Its fundamental group is

$$\mathbb{Z}^2 \times \mathbb{Z}^{s-2} = \mathbb{Z}^s.$$

The two factors have torsion-free homology, so the integral Künneth formula has no Tor terms and multiplies Betti polynomials. This gives (95.14) and (95.15). For $2 \le j \le s$, the coefficient

$$\binom{s-2}{j-2}$$

of $m-1$ is positive, so every such β_j is unbounded. Finally, theorem 95.2 realizes each product configuration as a completion-total optimal fiber. □

Corollary 95.24 (Unbounded third homology in rank three)**.** *At active rank three there are completion-total systems with*

$$\pi_1 \cong \mathbb{Z}^3, \qquad (H_0, H_1, H_2, H_3) \cong (\mathbb{Z}, \mathbb{Z}^3, \mathbb{Z}^{m+1}, \mathbb{Z}^{m-1}).$$

Thus $\beta_3 = m - 1$ is unbounded at fixed active rank and fixed fundamental group. The case $m = 2$ is T^3.

95.11.1 The displacement map contains information beyond π_1

The minimal rank-three fiber from theorem 95.4 has

$$\pi_1 \cong F_3, \qquad H_1 \cong \mathbb{Z}^3 \xrightarrow{\cong} L \cong \mathbb{Z}^3,$$

so $K_{\mathrm{lin}} = 0$. For a rank-two comparison, take

$$P_F = \{(0,0), (0,1), (1,1), (3,2)\} \subset \mathbb{Z}^2.$$

Its convex hull is the triangle with vertices $(0,0), (0,1), (3,2)$ and normalized area 3; Pick's theorem shows that $(1,1)$ is its unique interior lattice point, so P_F is lattice-convex. The six unordered nonzero pair differences

$$(0,1), (1,1), (3,2), (1,0), (3,1), (2,1)$$

are distinct up to sign. Hence the presentation (95.1) has no nontrivial repeated-difference relation and is free on the three nonzero vertices:

$$\pi_1(Q(P_F)) \cong F_3.$$

But the displacement map on abelianization is a surjection

$$\mathbb{Z}^3 \longrightarrow \mathbb{Z}^2,$$

so its kernel is primitive of rank one. By theorem 95.2, P_F also occurs as a genuine completion-total optimal fiber. We therefore obtain the following intrinsic conclusion.

Corollary 95.25 (Same fundamental group, different displacement visibility)**.** *The abstract group $\pi_1(Q(P))$ does not determine the active rank or the linear zero-displacement subgroup $K_{\mathrm{lin}}(P)$. In particular F_3 occurs with both $K_{\mathrm{lin}} = 0$ and $K_{\mathrm{lin}} \cong \mathbb{Z}$.*

This statement is purely about the intrinsic completion quotient. It does not assert a physical stack-carrier lower bound; such claims require a separately specified transport model. Part VIII takes up exactly that representation-independent physical question.

95.12 The displacement-marked group in arbitrary rank

The preceding sections show that the abstract group $\pi_1(Q(P))$ and the active lattice L both matter. The correct intrinsic package retains the map between them.

Definition 95.26 (Displacement-marked group)**.** For a finite spanning lattice-convex configuration $P \subset L$, put

$$G_P = \pi_1(Q(P)), \qquad \mathfrak{M}(P) = (G_P, L, \delta_P),$$

where $\delta_P : G_P \twoheadrightarrow L$ is the displacement homomorphism of (95.2). Define

$$K_0(P) = \ker \delta_P, \qquad K_{\mathrm{lin}}(P) = \ker\bigl(H_1(Q(P);\mathbb{Z}) \xrightarrow{\delta_P^{\mathrm{ab}}} L\bigr).$$

The first is the full pure zero-displacement sector; the second is its abelian shadow.

Theorem 95.27 (Affine-lattice invariance of the marked group)**.** *Let $f : P \to P'$ be an affine lattice isomorphism $f(z) = Az + b$ from a spanning lattice-convex $P \subset L$ to $P' \subset L'$. Then the induced quotient equivalence satisfies*

$$\delta_{P'} \circ f_* = A \circ \delta_P.$$

Hence $\mathfrak{M}(P)$ is an affine-lattice invariant up to marked isomorphism.

Proof. Translate both configurations so that the chosen base vertex is zero. A generator X_p of the difference presentation maps to the generator associated with Ap. Since $\delta_P(X_p) = p$ and $\delta_{P'}(X_{Ap}) = Ap$, the displayed identity holds on generators and therefore on all of G_P. □

Theorem 95.28 (Universal visibility exact sequences)**.** *For every finite spanning lattice-convex $P \subset L$ there are exact sequences*

$$\boxed{1 \longrightarrow [G_P, G_P] \longrightarrow K_0(P) \longrightarrow K_{\mathrm{lin}}(P) \longrightarrow 0}$$

and

$$\boxed{0 \longrightarrow K_{\mathrm{lin}}(P) \longrightarrow H_1(Q(P);\mathbb{Z}) \xrightarrow{\delta_P^{\mathrm{ab}}} L \longrightarrow 0.}$$

Equivalently,

$$K_0(P)/[G_P, G_P] \cong K_{\mathrm{lin}}(P).$$

Proof. Because the target lattice is abelian, $[G_P, G_P] \subseteq K_0(P)$. Abelianization identifies $H_1(Q(P);\mathbb{Z})$ with $G_P/[G_P, G_P]$, and the image of $K_0(P)$ in that quotient is exactly the kernel of δ_P^{ab}. This gives the first sequence. The second is the defining kernel sequence, and surjectivity follows from the fact that P spans L. □

Corollary 95.29 (Linear visibility excess)**.** *Since L is free abelian, the second sequence splits noncanonically. Thus*

$$H_1(Q(P);\mathbb{Z}) \cong K_{\mathrm{lin}}(P) \oplus L, \qquad \operatorname{rank} K_{\mathrm{lin}}(P) = \beta_1(Q(P)) - \operatorname{rank} L.$$

In particular, if $K_{\mathrm{lin}}(P) = 0$ then $K_0(P) = [G_P, G_P]$; and $K_0(P) = 1$ exactly when G_P is abelian and $K_{\mathrm{lin}}(P) = 0$.

Positioning. The use of a fundamental-group representation as monodromy is classical in covering-space theory [11]; voltage and gain graphs provide related group-valued cycle bookkeeping [117, 118], and periodic-net theory likewise attaches translation data to quotient cycles [119, 120]. The completion-specific point here is the exact marked package $\mathfrak{M}(P)$ together with the product, prism, apex, and physical-transfer laws below.

95.13 Marked product, prism, and apex transfer

Define

$$(G, L, \delta) \boxtimes (H, M, \varepsilon) = (G \times H, L \oplus M, \delta \oplus \varepsilon).$$

Theorem 95.30 (Marked product transfer)**.** *For finite spanning lattice-convex $P \subset L$ and $R \subset M$,*

$$\boxed{\mathfrak{M}(P \times R) \cong \mathfrak{M}(P) \boxtimes \mathfrak{M}(R).}$$

Consequently

$$K_0(P \times R) \cong K_0(P) \times K_0(R), \qquad K_{\mathrm{lin}}(P \times R) \cong K_{\mathrm{lin}}(P) \oplus K_{\mathrm{lin}}(R).$$

Proof. The topological product theorem already gives $Q(P \times R) \simeq Q(P) \times Q(R)$. At the presentation level, with $A_p = X_{(p,0)}$ and $B_r = X_{(0,r)}$, repeated-difference relations give

$$X_{(p,r)} = A_p B_r, \qquad [A_p, B_r] = 1.$$

Thus the fundamental group splits as the direct product, and the displacement of $X_{(p,r)}$ is (p, r). The kernel formulas follow immediately. □

Corollary 95.31 (Prism preserves the pure sector)**.** *Let $I = \{0, 1\}$. Then*

$$\mathfrak{M}(P \times I) \cong \mathfrak{M}(P) \boxtimes (\mathbb{Z}, \mathbb{Z}, \mathrm{id}),$$

so

$$\boxed{K_0(P \times I) \cong K_0(P), \qquad K_{\mathrm{lin}}(P \times I) \cong K_{\mathrm{lin}}(P).}$$

A prism adds one displacement-visible direction but no new pure-loop sector.

Theorem 95.32 (Apex pure-holonomy creation)**.** *For the translation-isolated apex*

$$P^+ = (P \times \{0\}) \cup \{e\} \subset L \oplus \mathbb{Z}e,$$

write $G = \pi_1(Q(P))$ and let t denote the new apex circle. Then

$$\pi_1(Q(P^+)) \cong G * \langle t \rangle,$$

with marked displacement

$$\delta_+(g) = (\delta_P(g), 0), \qquad \delta_+(t) = (0, 1),$$

and

$$\boxed{K_0(P^+) = \langle\langle K_0(P), [G, t] \rangle\rangle .}$$

Moreover

$$\boxed{K_{\mathrm{lin}}(P^+) \cong K_{\mathrm{lin}}(P).}$$

Thus apex rank-raising can create new nonabelian pure holonomy while leaving the linear pure sector unchanged.

Proof. The topological apex theorem gives the free product. Every element of $K_0(P)$ and every cross-commutator $[G, t]$ has zero displacement, so their normal closure lies in $\ker \delta_+$. Conversely, quotient first by $K_0(P)$, obtaining $L * \langle t \rangle$, and then by all cross-commutators, obtaining $L \oplus \mathbb{Z}$. The resulting quotient is exactly the displacement target, proving the kernel formula. Abelianization of $G * \mathbb{Z}$ is $H_1(Q(P)) \oplus \mathbb{Z}$, and the new summand maps isomorphically to the new lattice direction, so the linear kernel is unchanged. □

Corollary 95.33 (Apex–prism separation)**.** *Prism and apex both raise active rank by one and preserve K_{lin}, but they have different full pure-loop behavior: prism preserves K_0, while apex adjoins the normal closure of cross-commutators. In particular, the minimal rank-s fiber has*

$$\pi_1 \cong F_s, \qquad K_{\mathrm{lin}} = 0, \qquad K_0 = [F_s, F_s] \quad (s \geq 2),$$

whereas the Boolean cube I^s has

$$\pi_1 \cong \mathbb{Z}^s, \qquad K_0 = 1, \qquad K_{\mathrm{lin}} = 0.$$

Corollary 95.34 (Fixed marked group with unbounded higher homology)**.** *For the family $P_{m,s} = P_m \times I^{s-2}$ of theorem 95.23,*

$$\boxed{\mathfrak{M}(P_{m,s}) \cong (\mathbb{Z}^s, \mathbb{Z}^s, \mathrm{id})}$$

for every m, hence $K_0 = 1$ and $K_{\mathrm{lin}} = 0$, while every β_j, $2 \leq j \leq s$, is unbounded with m.

This strengthens the earlier fixed-π_1 statement: even the full displacement-marked group can be held fixed while all higher Betti numbers vary without bound. Together with theorem 95.25, it gives the exact intrinsic hierarchy

$$\pi_1 \not\Rightarrow (L, K_{\mathrm{lin}}), \qquad \mathfrak{M}(P) \Rightarrow (K_0, K_{\mathrm{lin}}), \qquad \mathfrak{M}(P) \not\Rightarrow H_{\geq 2}(Q(P)).$$

95.14 Finite physical holonomy and literal carrier costs

The marked group identifies which pure loops exist. A physical implementation must still choose how those loops act.

Definition 95.35 (Marked finite-permutation transport)**.** Choose a homomorphism

$$\rho : G_P \longrightarrow S_n$$

and put

$$G_{\mathrm{phys}} = L \times S_n, \qquad \mathrm{Hol}_\rho(g) = (\delta_P(g), \rho(g)).$$

The physically realized pure image is

$$\Pi_\rho(P) = \rho(K_0(P)).$$

Proposition 95.36 (Flat local realization of a finite representation)**.** *Let X be a connected finite CW complex and $\rho : \pi_1(X, x_0) \to H$ a homomorphism to a finite group. After a finite subdivision, the oriented edges of X admit an H-labeling with trivial ordered product around every 2-cell and based loop holonomy exactly ρ.*

Proof. Choose a maximal tree, label its edges by the identity, and label each non-tree edge by the image under ρ of the associated based generator loop. Every 2-cell attaching word is null in π_1, hence has trivial label product. Tree collapse shows that loop holonomy recovers ρ. □

Theorem 95.37 (Physical pure-image theorem)**.** *For every marked finite-permutation transport,*

$$\boxed{\mathrm{Hol}_\rho(K_0(P)) = \{0\} \times \Pi_\rho(P).}$$

Thus $K_0(P) \neq 1$ means that pure loops are intrinsically available, not that every representation must realize them nontrivially.

Theorem 95.38 (Physical prism and apex transfer)**.** *For a prism extension, any homomorphic extension $\rho^{\square} : G_P \times \langle u \rangle \to S_n$ has $h = \rho^{\square}(u)$ centralizing $\rho(G_P)$ and*

$$\boxed{\Pi_{\rho^{\square}}(P \times I) = \Pi_{\rho}(P).}$$

For an apex extension, choose $h \in S_n$ and extend ρ to $\rho_h : G_P * \langle t \rangle \to S_n$ by $\rho_h(t) = h$. Then

$$\boxed{\Pi_{\rho_h}(P^+) = \langle\!\langle \Pi_{\rho}(P), [\rho(G_P), h] \rangle\!\rangle_{\langle \rho(G_P), h \rangle}.}$$

In particular, an apex creates no new realized pure holonomy when h centralizes the old image, and can create it exactly through failure of centralization.

The same separation appears at the carrier level. Let A be the integer counting matrix of a realization and let $z \in \mathbb{N}^d$ be a nonzero zero-semantic count vector, $Az = 0$. The number of distinct literal orderings with this Parikh vector is

$$M(z) = \frac{|z|!}{\prod_i z_i!}.$$

For $k \geq 2$ define the order-carrier girth

$$\boxed{g_k^{\mathrm{ord}}(A) = \min\{|z| : z \neq 0,\ Az = 0,\ M(z) \geq k\},}$$

with value $+\infty$ if no such z exists.

Theorem 95.39 (Persistent literal carrier bound)**.** *In the buffered literal-order category, if k persistent sheets share one zero-semantic Parikh vector z, then*

$$H = |z| \geq g_k^{\mathrm{ord}}(A).$$

In particular a regular-sheet realization of a finite image $H_\rho = \operatorname{im} \rho$ satisfies

$$H \geq g_{|H_\rho|}^{\mathrm{ord}}(A).$$

Proof. Persistent sheet identity is stored only in literal order, so one Parikh vector z can support at most $M(z)$ sheets. Hence $k \leq M(z)$ and the definition of $g_k^{\mathrm{ord}}(A)$ gives the bound. □

Theorem 95.40 (Exact buffered ancestry law)**.** *Let $H \leq S_n$ be a finite permutation image and let*

$$a(H) = \left| \bigcup_{\sigma \in H} \operatorname{supp}(\sigma) \right|.$$

In the ancestry-bijective one-top buffered macro category,

$$\boxed{b_{\min}(H) = \max\{a(H) - 1, 0\}.}$$

For nontrivial H this is exactly $a(H) - 1$.

Proof. Place the active support contiguously at the top of the distinguished block. Some group element moves the deepest active cell. Before that cell can be inspected, all $a(H) - 1$ cells above it have been removed from the live stack and their ancestries must be stored externally, giving the lower bound. Conversely, pop those ancestries into handles, rewrite the deepest active cell, and rebuild the block in the target order. The trivial image requires no handle. □

Relation to structured pushdown models. Visibly pushdown and height-deterministic pushdown automata constrain stack-height evolution in different ways [122, 123]. The present carrier law concerns a different frozen category: persistent sheet identity lives in literal order, while temporary ancestry handles are cleared at macro endpoints. The numerical bounds should therefore be read relative to this category, not as unrestricted PDA lower bounds.

95.15 Connection memory and repair

Carrier height and ancestry support do not determine all transient control. The local connection may merge histories that must later be distinguished.

Definition 95.41 (Terminal collision multiplicity)**.** For a deferred-carrier connection certificate $\mathcal{C}$, let $\chi(\mathcal{C})$ be the largest number of histories that reach one preterminal semantic vertex with the same exposed persistent sheet and the same next action, but carry pairwise distinct accumulated physical multipliers that require distinct terminal updates.

Theorem 95.42 (Deferred path-memory lower bound)**.** *Every deterministic deferred-carrier realization of $\mathcal{C}$ requires*

$$\boxed{q_{\mathrm{tr}} \geq \chi(\mathcal{C})}$$

transient path states. If at every preterminal vertex there are at most $\chi(\mathcal{C})$ accumulated-product classes, the bound is attained by recording that class in finite control.

Proof. Histories counted by $\chi(\mathcal{C})$ expose identical persistent data and the same next action. If two shared one transient state, determinism would force the same terminal update, contradicting the definition. The upper construction stores exactly the accumulated-product class. □

For repair, fix a finite cell model X, a finite physical group H, and a raw oriented-edge label $a : E^{\mathrm{or}}(X) \to H$ with $a(\bar{e}) = a(e)^{-1}$. Let $Z^1(X;H)$ be the set of flat labelings, equivalently nonabelian 1-cocycles. For a conjugation-invariant generating metric d_H define

$$C_0(a) = \min_{b \in Z^1(X;H)} |\{e : a_e \neq b_e\}|,$$

$$C_1(a) = \min_{b \in Z^1(X;H)} \sum_e d_H(a_e, b_e).$$

Proposition 95.43 (Repair is nearest-flat nonabelian cocycle optimization)**.** *Flatness is determined by the 2-skeleton of X: higher cells impose no additional relations on group-valued edge transport. The numerical values of C_0 and C_1 depend on the raw labeling, chosen trivialization, cell presentation, physical group, and metric, and therefore are not invariants of $\mathfrak{M}(P)$ alone.*

Proof. The fundamental group of a CW complex is presented by the 1-skeleton modulo the attaching relations of the 2-cells; higher cells do not change π_1. Thus group-valued edge flatness is exactly the satisfaction of the 2-cell relations. The optimization objectives explicitly contain the raw labels and the chosen metric, so changing those data can change the minimum without changing the intrinsic marked group. This is the same cycle-consistency distinction that appears in group synchronization, though the present objective is tied to the completion transport category [121]. □

The higher-rank transfer is therefore layered:

$$\boxed{\begin{array}{c}\text{marked topology} \to \text{chosen holonomy representation} \\ \to \text{carrier/support constraints} \to \text{connection memory} \\ \to \text{raw-policy repair cost}\end{array}}$$

95.16 An all-rank exact finite-holonomy family

Let $P_s = \{0, e_1, \dots, e_s\} \subset \mathbb{Z}^s$ be the minimal active rank-s configuration. By theorem 95.4,

$$G_s = \pi_1(Q(P_s)) \cong F_s, \qquad \delta_s : F_s \twoheadrightarrow \mathbb{Z}^s$$

is the abelianization map, so $K_0(P_s) = [F_s, F_s]$ and $K_{\mathrm{lin}}(P_s) = 0$.

Theorem 95.44 (Constant finite physical image in every rank)**.** *For every $s \geq 2$ there is a permutation representation $\rho_s : F_s \to S_3$ with*

$$\boxed{\operatorname{im} \rho_s = S_3, \qquad \Pi_{\rho_s}(P_s) = A_3.}$$

Hence genuine pure physical holonomy persists in arbitrary active rank with constant finite permutation image.

Proof. Let $r = (123)$ and $q = (13)$, and set

$$\rho_s(x_1) = r, \qquad \rho_s(x_2) = q, \qquad \rho_s(x_i) = 1 \quad (i \geq 3).$$

The image is S_3. Since $K_0(P_s) = [F_s, F_s]$, its image is the derived subgroup $[S_3, S_3] = A_3$; already $[x_1, x_2]$ maps to a nontrivial 3-cycle. □

Use the scalar realization

$$\omega_s = (\underbrace{1, \dots, 1}_{s+1 \text{ entries}}, -(s+1)).$$

A nonzero zero-weight vector $z \in \mathbb{N}^{s+2}$ satisfies

$$\sum_{i=1}^{s+1} z_i = (s+1) z_{s+2},$$

so

$$|z| = (s+2) z_{s+2} \geq s + 2.$$

The all-ones vector attains equality and has order capacity $(s+2)! \geq 6$.

Theorem 95.45 (All-rank exact buffered carrier family)**.** *For every $s \geq 2$, the minimal active rank-s fiber admits a marked finite-permutation transport and buffered literal-order realization with*

$$\boxed{\operatorname{im} \rho_s = S_3, \qquad \Pi_{\rho_s}(P_s) = A_3,}$$

$$\boxed{H_{\min} = s + 2, \qquad b_{\min} = 2.}$$

Thus active rank is unbounded while full image size 6*, pure image size* 3*, and ancestry cost* 2 *remain constant; the linear carrier-height growth comes from the zero-weight arithmetic of the chosen scalar realization rather than from increasing permutation complexity.*

Proof. The preceding theorem supplies the S_3 transport, and theorem 95.36 realizes it by a flat local connection. Every nonempty zero-weight literal carrier for ω_s has height at least $s + 2$, while the all-ones vector supplies at least six orderings at exactly that height. Choose three literal types as an active top block, use its six orderings as the regular S_3 sheets, and keep the remaining $s - 1$ types as a common suffix. This realizes the persistent transport at height $s + 2$. The image acts on exactly three active cells, so theorem 95.40 gives $b_{\min} = 3 - 1 = 2$. □

Remark 95.46 (Real-time scope). The preceding exact bounds hold in the buffered literal-order macro category. They do not by themselves assert the same all-rank bounds in the stricter no-ε real-time Stable-LIFO subclass. A separate scheduling construction would be required there. This is a category boundary, not an unresolved point in the marked-group calculus.

The chapter now separates five levels that were previously easy to conflate: intrinsic displacement-marked topology, realized finite holonomy, persistent literal carrier capacity, transient connection memory, and raw-policy repair. Higher homology lies outside this one-type transport chain: theorem 95.34 keeps the entire marked group fixed while higher Betti numbers diverge.

Part VIII

Representation-Independent Physical Complexity and Resource Synthesis

Part guide. **Question.** What resource lower bounds remain when the opponent is allowed to choose an arbitrary deterministic pushdown representation? **Main result.** The Part proves exact unrestricted state–stack endpoints, an exact one-work-symbol bounded-replacement staircase, filtered-collapse/source-stack exposure bounds, and protected-contract Pareto frontiers. **Why here.** It turns the semantic/physical distinction into representation-independent complexity theorems and extends the language-visible resource viewpoint introduced by the hierarchy theorem of Part I. **Dependency.** It uses the semantic invariants and collapse-capacity viewpoint developed earlier and leads directly to the decomposition question of Part IX.

Chapter 96

Protected Phase Pumping and Deferred Exposure

Fix coprime positive integers p, q and

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}.$$

The question is now unrestricted: an arbitrary deterministic pushdown recognizer may distribute the residual among control, visible stack, deep stack, long rewrites, and deterministic ε-normalization. The proof below reconstructs no canonical representation. Instead it uses quotient rigidity to label reachable physical configurations semantically and then forces synchronized phases above protected stack cuts.

96.1 Model and semantic quotient

96.1.1 Pushdown model

A deterministic pushdown automaton has finite control Q, finite stack alphabet Γ, and transitions that inspect the current state, the visible top stack symbol, and either one input symbol or ε. The stack top is written on the right. Stack words, including regular languages of stacks used later, are therefore written and read from bottom to top. Thus a transition

$$\delta(q, X, a) = (q', \beta)$$

acts as

$$(q, \eta X, av) \vdash (q', \eta\beta, v).$$

A single transition may replace X by an arbitrary fixed finite word $\beta \in \Gamma^*$. Determinism means that for each state/top pair there is at most one move on each input letter, and if an ε-move is enabled then no input-consuming move is enabled at that state/top pair.

For the lower bound we use a *regular-target* acceptance convention. Fix a regular set $T \subseteq Q \times \Gamma^*$ of accepting control–stack configurations. After the finite binary input has been consumed, the word is accepted exactly when the deterministic post-input computation reaches T without consuming any further binary input. Reaching T in zero post-input steps is allowed. Ordinary final-state, empty-stack, and final-state-plus-empty-stack acceptance are obtained by the corresponding regular choices of T. A mandatory right-endmarker convention is handled by treating the endmarker as the final symbol of the continuation. The fixed-continuation regularity consequence needed later is proved separately in Theorem 96.12; it is not part of the model definition.

The matching upper bound in Section 97.2 is stated explicitly for a mandatory right endmarker.

Remark 96.1 (Meaning of "unrestricted"). The lower-bound opponent may use an arbitrary finite stack alphabet, arbitrary finite replacement words, deterministic ε-moves, noncanonical intermediate configurations, and any physical encoding compatible with correct recognition. No raw-conjugacy, homogeneous-tail, or canonical normal-form assumption is imposed.

96.1.2 Residuals

For $w \in \{0,1\}^*$ define

$$\rho(w) = qN_1(w) - pN_0(w).$$

Then

$$w \in L_{p,q} \iff \rho(w) = 0,$$

and

$$\rho(wa) = \begin{cases} \rho(w) - p, & a = 0, \\ \rho(w) + q, & a = 1. \end{cases}$$

Lemma 96.2 (Integer residual realizability)**.** *If* $\gcd(p,q) = 1$*, then for every* $t \in \mathbb{Z}$ *there exists* $z \in \{0,1\}^*$ *with*

$$\rho(z) = t.$$

Proof. Choose integers x_0, y_0 with $qx_0 - py_0 = t$. All integer solutions are

$$x = x_0 + pk, \qquad y = y_0 + qk.$$

For sufficiently large k, both x, y are nonnegative. Any binary word with $N_1 = x$ and $N_0 = y$ then has residual t. □

Lemma 96.3 (Residual classification of right quotients)**.** *For all* $u, v \in \{0,1\}^*$*,*

$$u^{-1}L_{p,q} = v^{-1}L_{p,q} \quad \iff \quad \rho(u) = \rho(v).$$

Proof. If $\rho(u) = \rho(v)$, then for every continuation z,

$$uz \in L_{p,q} \iff \rho(z) = -\rho(u) \iff \rho(z) = -\rho(v) \iff vz \in L_{p,q}.$$

Conversely, if $\rho(u) \neq \rho(v)$, choose by Theorem 96.2 a word z with $\rho(z) = -\rho(u)$. Then $uz \in L_{p,q}$ while

$$\rho(vz) = \rho(v) - \rho(u) \neq 0,$$

so $vz \notin L_{p,q}$. □

Lemma 96.4 (Physical configurations determine residuals)**.** *If two consumed prefixes* u, v *lead to the same reachable physical configuration of a deterministic recognizer of* $L_{p,q}$*, then*

$$\rho(u) = \rho(v).$$

Consequently every reachable physical configuration C *has a well-defined semantic label* $\Theta(C) \in \mathbb{Z}$*.*

Proof. Starting from one and the same deterministic physical configuration, the set of accepting continuations is identical. Thus

$$u^{-1}L_{p,q} = v^{-1}L_{p,q},$$

and Theorem 96.3 gives $\rho(u) = \rho(v)$. □

For every reachable input-consuming transition,

$$\Theta \xrightarrow{1} \Theta + q, \qquad \Theta \xrightarrow{0} \Theta - p,$$

while ε-moves preserve Θ.

96.2 Unary nonrecurrence and height escape

Consider the computation induced by the infinite unary input 1^ω. For every $n \geq 0$, both 1^n and 1^{n+1} have accepting finite extensions: their residuals nq and $(n+1)q$ admit cancelling continuations by Theorem 96.2. Consequently, after consuming 1^n the deterministic machine cannot halt, reject permanently, or become trapped in an infinite ε-computation before consuming the next 1; otherwise no extension beginning with that next 1 could ever be accepted. Hence the unary computation consumes arbitrarily many input symbols, and every finite consumed unary prefix is live.

Lemma 96.5 (Raw nonrecurrence on the positive unary ray)**.** *No full raw configuration occurs twice on the live computation over* 1^ω*.*

Proof. Suppose the same configuration C occurs twice. If the two occurrences have consumed different numbers $m < n$ of input symbols, then Theorem 96.4 assigns C both residuals mq and nq, impossible.

If the two occurrences have consumed the same number of input symbols, the intervening segment consists entirely of ε-moves. Determinism then repeats the same ε-cycle forever, so the next input symbol can never be read. This contradicts liveness of the current unary prefix. □

Corollary 96.6 (Unary height escape)**.** *Along the computation over* 1^ω*, stack height eventually leaves every fixed finite bound.*

Proof. For each fixed height bound H, there are only finitely many state/stack configurations with height at most H. If infinitely many raw positions remained in that bounded region, some full configuration would repeat, contradicting Theorem 96.5. □

Definition 96.7 (Suffix minimum)**.** A raw position C_i on an infinite computation is a suffix minimum if

$$h(C_i) \leq h(C_t) \qquad \text{for every } t \geq i.$$

The following elementary consequence will be used repeatedly.

Lemma 96.8 (Unbounded suffix minima)**.** *If an integer-valued height sequence tends to infinity, then it has suffix-minimum positions at arbitrarily large heights.*

Proof. For every raw position u, the tail $\{h(t) : t \geq u\}$ has a minimum because $h(t) \to \infty$. Choose a position at which this tail minimum is attained. As u tends to infinity, the selected positions cannot remain in a finite set, and their heights must become arbitrarily large. □

96.3 Protected stack pumping and synchronized phases

A nonempty stack configuration $(q, \eta X)$ has *mode* (q, X).

Lemma 96.9 (Protected unary pump)**.** *There exist a state $r \in Q$, a symbol $X \in \Gamma$, an integer $c \geq 1$, a nonempty word $U \in \Gamma^+$, and a lower context η such that*

$$(r, \eta X) \overset{1^c}{\Longrightarrow} (r, \eta U X),$$

where the displayed macro includes all forced ε-moves and no raw configuration of the macro has stack height below $|\eta X|$.

Moreover the macro is context-independent: replacing the untouched lower context η by any other lower context leaves the raw transition sequence unchanged as long as that context is not exposed.

Proof. By Theorem 96.8, suffix-minimum positions occur at arbitrarily large heights. Since only $|Q||\Gamma|$ state/top modes exist, some mode (r, X) occurs at suffix minima of arbitrarily large height. Choose two such positions $t_1 < t_2$ with the same mode and with the later height strictly larger. Write the earlier stack as ηX.

Because t_1 is a suffix minimum, the computation from t_1 to t_2 never drops below height $|\eta X|$. Exposing any symbol of the lower prefix η would first require popping the visible X and reaching height $|\eta| < |\eta X|$. Hence η is never exposed or modified. At t_2 the state and visible top are again r, X, so the larger stack has the form

$$\eta U X$$

for some $U \in \Gamma^+$.

Let c be the number of input symbols 1 consumed between t_1 and t_2. Every transition of the segment sees only the active suffix above the untouched lower context, so replacing η by any other lower context reproduces the same deterministic raw transition sequence. This proves context independence.

To prove $c \geq 1$, suppose instead that $c = 0$. Then the segment is a context-independent, height-increasing ε-macro from mode (r, X) back to itself:

$$(r, \eta X) \overset{\varepsilon}{\Rightarrow} (r, \eta U X).$$

Context independence would then repeat the same forced macro from $(r, \eta U^m X)$ for every $m \geq 0$, producing an infinite deterministic ε-computation and preventing the next symbol of the live unary input from ever being consumed. This is impossible. Therefore $c \geq 1$. □

96.3.1 Iteration and protected cuts

By context independence, for every $k \geq 0$,

$$(r, \eta U^k X) \overset{1^c}{\Longrightarrow} (r, \eta U^{k+1} X).$$

Each copied raw segment is identical to the original segment above its lower context, and hence never goes below the height of its own starting configuration. Therefore the cuts

$$H_k = \eta U^k$$

are protected throughout the entire future computation on 1^ω starting at

$$C_k = (r, H_k X).$$

After discarding finitely many initial k and rebasing the index, we may assume every H_k is nonempty.

This explicit repetition argument is important: we do not infer protection of the synthetic endpoints merely from the fact that the original two selected positions were suffix minima.

Lemma 96.10 (Synchronized p-phase family)**.** *For each $j \in \{0, \ldots, p-1\}$ there exist a state q_j and a nonempty upper stack word W_j, independent of k, with the following property. Starting from C_k, consume exactly j additional symbols 1 and then take every forced ε-move until the computation is again ready to inspect the next unread input symbol. The resulting raw configuration is*

$$D_{k,j} = (q_j, H_k W_j).$$

There is a constant $A \in \mathbb{Z}$ such that

$$\Theta(D_{k,j}) = A + kcq + jq.$$

Proof. The future 1^ω-computation from every C_k remains above H_k. Hence the relative computation above the cut depends only on the common state/top data and not on k. Liveness excludes an infinite ε-divergence after the jth consumed symbol, so the stated next-input boundary is well defined. For fixed j its state and upper stack are therefore independent of k; call them q_j, W_j. The upper word is nonempty because exposing H_k would contradict protection of the cut along the 1^ω future. The residual formula follows because each pump block contributes cq and each of the additional j symbols contributes q. □

96.4 Exposure, deferral, and the diagonal obstruction

Fix a phase j and follow the unique computation on 0^ω from each $D_{k,j}$. Until the common buried cut H_k is exposed, the machine sees only the state and the stack portion above H_k. Thus the relative raw computation is independent of k.

We say that H_k is *exposed* when the full stack is exactly H_k at a raw position. This event cannot be skipped downward. If the current stack is H_kZ with a nonempty upper suffix, one PDA transition rewrites only the visible top symbol and therefore lowers total stack height by at most one. Consequently any computation that ever reaches below the height of H_k must pass through a raw configuration whose stack is exactly H_k first.

Lemma 96.11 (Uniform exposure dichotomy)**.** *For each phase j, exactly one of the following holds.*

(i) There exist $z_j \geq 0$ and $s_j \in Q$ such that for every k the first exposure of H_k occurs after exactly z_j consumed zeros, at configuration

$$(s_j, H_k).$$

(ii) For every k, the computation on 0^ω from $D_{k,j}$ never exposes H_k.

Proof. Before exposure, every transition acts strictly above H_k. The initial relative configuration (q_j, W_j) is independent of k, so the entire relative raw computation is the same for every k. Either its upper stack becomes empty after one fixed raw history and a fixed number z_j of consumed zeros, or it never becomes empty. In the former case the state at that moment is also fixed. □

96.4.1 Fixed-continuation regularity

Lemma 96.12 (Fixed-continuation stack regularity)**.** *Fix a control state s and a finite continuation v (including the mandatory right endmarker, when that convention is used). Let*

$$R_{s,v} = \{S \in \Gamma^* : \textit{starting from } (s, S),\ \textit{the fixed continuation } v \textit{ leads to acceptance}\}.$$

Then $R_{s,v}$ is regular.

Proof. Write $v = v_1 \cdots v_m$. Augment the finite control by an input-position coordinate $i \in \{0, \ldots, m\}$. Original ε-moves leave i fixed; at position $i < m$, only a transition consuming the required letter v_{i+1} advances the coordinate to $i + 1$. Acceptance of the fixed word v is therefore reduced to predecessor reachability of a regular target set in an ordinary pushdown system. If one uses the standard bounded-push presentation of pushdown systems, each transition that writes a longer fixed word is compiled into finitely many auxiliary control states; this normalization does not change the predecessor set on the original configurations. Regular sets of pushdown configurations are effectively closed under predecessor reachability by the standard saturation theorem [4]. Restricting the resulting regular configuration set to the fixed control state s yields a regular language of stacks. □

Lemma 96.13 (Deferred exposure is impossible)**.** *No phase $j \in \{0, \dots, p-1\}$ can satisfy alternative (ii) of Theorem 96.11.*

Proof. Assume phase j is deferred. Fix one k. The 0^ω-computation remains strictly above H_k. Every finite consumed zero-prefix is live, because its current residual has a finite cancelling continuation by Theorem 96.2. Hence the computation consumes arbitrarily many zeros and cannot become trapped in an ε-cycle.

No full raw configuration can repeat: a repetition after different numbers of consumed zeros would assign one physical configuration two residuals differing by a nonzero multiple of p, while a repetition at the same consumed input position would create a deterministic ε-cycle. Thus the relative upper-stack height above H_k escapes every finite bound.

Apply Theorem 96.9, with 0 in place of 1, to this relative zero computation. There exist a state s, a visible symbol Y, an integer $d \geq 1$, a nonempty word $V \in \Gamma^+$, and a fixed upper word β such that

$$(s, H_k \beta Y) \overset{0^d}{\Longrightarrow} (s, H_k \beta V Y).$$

Because the complete relative zero computation before exposure is identical for every k, the same second pump appears above every H_k. Iterating it gives reachable configurations

$$E_{k,\ell} = (s, \eta U^k \beta V^\ell Y), \qquad k, \ell \geq 0.$$

There is a constant $B \in \mathbb{Z}$ such that

$$\Theta(E_{k,\ell}) = B + cqk - pd\ell.$$

Choose, using Theorem 96.2, a fixed binary word v_0 with $\rho(v_0) = -B$. If the model uses a mandatory right endmarker, append it and write $v = v_0 \dashv$; otherwise put $v = v_0$. Since every $E_{k,\ell}$ is reachable and the automaton recognizes $L_{p,q}$,

$$E_{k,\ell} \overset{v}{\Rightarrow} \text{accept} \quad \Longleftrightarrow \quad cqk = pd\ell.$$

By Theorem 96.12, the stack language $R_{s,v}$ is regular. Hence so is

$$K = \{a^k \# b^\ell : \eta U^k \beta V^\ell Y \in R_{s,v}\}.$$

Indeed, if a DFA recognizes $R_{s,v}$, reading each fixed block η, U, β, V, Y induces a fixed transformation of the DFA state set. A finite automaton on the parameter alphabet $\{a, \#, b\}$ can therefore simulate the transformations corresponding to $\eta U^k \beta V^\ell Y$, proving that K is regular.

On the other hand correctness gives

$$K = \{a^k \# b^\ell : cqk = pd\ell\}.$$

With $g = \gcd(cq, pd)$, set

$$\alpha = \frac{pd}{g}, \qquad \beta = \frac{cq}{g}.$$

Then

$$K = \left\{a^{\alpha n} \# b^{\beta n} : n \geq 0\right\}.$$

This language is nonregular. Indeed, if N were a pumping length, the word $a^{\alpha N} \# b^{\beta N}$ would have every admissible pumping segment lying wholly in the initial a-block; pumping that segment changes the number of a's while leaving the number of b's fixed, destroying the equation $\beta k = \alpha \ell$. This contradiction eliminates the deferred alternative. □

Chapter 97

Exact Unrestricted State Complexity of $L_{p,q}$

Permanent deferral has now been ruled out, so every synchronized phase must expose the common protected cut. This immediately converts the arithmetic phases into distinct finite-control states. A symmetric argument gives the opposite side, and a three-symbol right-endmarker construction supplies the matching upper bound.

97.1 The unrestricted lower bound

Theorem 97.1 (Positive-side lower bound)**.** *Every deterministic ε-pushdown recognizer of $L_{p,q}$ in the stated model satisfies*

$$|Q| \geq p.$$

Proof. By Theorem 96.13, every phase $j \in \{0, \ldots, p-1\}$ exposes the common cut. Fix one k. For each phase,

$$D_{k,j} \xRightarrow{0^{z_j}} (s_j, H_k).$$

If $s_j = s_t$ for distinct j, t, the two exposure configurations are physically identical, because their stacks are both exactly H_k. By Theorem 96.4, their residuals are equal:

$$A + kcq + jq - pz_j = A + kcq + tq - pz_t.$$

Thus

$$q(j - t) = p(z_j - z_t).$$

Since $\gcd(p, q) = 1$, this implies $p \mid (j - t)$, impossible for $0 < |j - t| < p$. Therefore $s_0, \ldots, s_{p-1}$ are pairwise distinct and $|Q| \geq p$. □

Theorem 97.2 (Negative-side state lower bound)**.** *Every deterministic ε-pushdown recognizer of $L_{p,q}$ in the stated model satisfies*

$$|Q| \geq q.$$

Proof. Apply the proof of Theorem 97.1 after interchanging the two input letters, interchanging p and q, and replacing the residual by its negative. □

Theorem 97.3 (Unrestricted state lower bound)**.** *For coprime $p, q \geq 1$, every deterministic ε-pushdown recognizer of $L_{p,q}$ in the stated regular-target model satisfies*

$$\boxed{|Q| \geq \max(p, q)}.$$

Proof. Combine Theorems 97.1 and 97.2. □

Remark 97.4. The theorem is stronger than the product lower bound

$$|Q||\Gamma| \geq \max(p, q)$$

that one might first seek. The argument charges the ratio parameter directly to finite control, without any restriction on $|\Gamma|$.

97.2 A matching three-symbol construction

We now give a self-contained upper bound. This section is independent of the lower-bound representation and is included so that the exact state-complexity statement is proved entirely within the present Part.

Assume first that $p \geq q$. Let

$$Q = \{R_0, \ldots, R_{p-1}\}, \qquad \Gamma = \{A, B, Z_0\}.$$

We encode each residual $r \in \mathbb{Z}$ as follows. For $r \geq 0$, write uniquely

$$r = pk + s, \qquad k \geq 0, \quad 0 \leq s < p,$$

and define

$$\zeta(r) = (R_s, Z_0 A^k).$$

For $r < 0$, write uniquely

$$r = u - qh, \qquad h \geq 1, \quad 0 \leq u < q,$$

and define

$$\zeta(r) = (R_u, Z_0 B^h).$$

For $0 \leq s < p$, put

$$s^+ = (s + q) \bmod p, \qquad c_s = \frac{s + q - s^+}{p} \in \{0, 1\},$$

and

$$u_s = (s - p) \bmod q, \qquad h_s = \frac{u_s + p - s}{q} \geq 1.$$

For $0 \leq u < q$, put

$$u^- = (u - p) \bmod q, \qquad d_u = \frac{u^- - u + p}{q} \geq 1.$$

With the stack top on the right, define the following binary rules:

$$\delta(R_s, A, 0) = (R_s, \varepsilon), \qquad \delta(R_s, A, 1) = (R_{s^+}, A^{c_s+1}),$$

$$\delta(R_s, Z_0, 1) = (R_{s^+}, Z_0 A^{c_s}), \qquad \delta(R_s, Z_0, 0) = (R_{u_s}, Z_0 B^{h_s}),$$

and for $0 \leq u < q$,

$$\delta(R_u, B, 1) = (R_u, \varepsilon), \qquad \delta(R_u, B, 0) = (R_{u^-}, B^{d_u+1}).$$

No ε-transition is used.

A mandatory right endmarker $\dashv$ is accepted only from

$$\zeta(0) = (R_0, Z_0).$$

For example, declare R_0 final and define only

$$\delta(R_0, Z_0, \dashv) = (R_0, Z_0)$$

among endmarker moves.

Theorem 97.5 (Three-symbol residual realization)**.** *For every coprime $p, q \geq 1$, there is an ε-free right-endmarker DPDA recognizing $L_{p,q}$ with*

$$|Q| = \max(p, q), \qquad |\Gamma| = 3.$$

Proof. Assume $p \geq q$. We verify

$$\delta(\zeta(r), 0) = \zeta(r - p), \qquad \delta(\zeta(r), 1) = \zeta(r + q)$$

for every residual r.

Let $r = pk + s \geq 0$. If $k \geq 1$, input 0 pops one A, giving

$$pk + s - p = p(k - 1) + s.$$

For input 1,

$$pk + s + q = p(k + c_s) + s^+,$$

so the rule replacing A by A^{c_s+1} gives exactly $\zeta(r + q)$. When $k = 0$, the 1-rule uses the same carry identity, while

$$s - p = u_s - qh_s,$$

so the boundary 0-rule enters exactly the negative encoding.

Now let $r = u - qh < 0$. Input 1 pops one B. If $h > 1$ this gives $u - q(h - 1)$; if $h = 1$ it exposes Z_0 and leaves $(R_u, Z_0) = \zeta(u)$. For input 0,

$$u - p = u^- - qd_u,$$

and hence

$$r - p = u^- - q(h + d_u),$$

which is implemented by replacing one visible B by B^{d_u+1}.

Therefore every binary input symbol realizes the correct residual translation in one ordinary transition. Starting from $\zeta(0)$, after reading w the machine is in $\zeta(\rho(w))$. The right endmarker is accepted exactly at $\zeta(0)$, so the recognized language is $L_{p,q}$.

If $q > p$, exchange 0 and 1, exchange p and q, exchange A and B, and negate the residual. The resulting machine has $q = \max(p, q)$ states. □

97.3 Exact state complexity

Theorem 97.6 (Exact unrestricted right-endmarker state complexity)**.** *For coprime $p, q \geq 1$, among all deterministic pushdown automata in the lower-bound model whose inputs are terminated by the mandatory symbol $\dashv$,*

$$\boxed{Q_{\min}(L_{p,q}) = \max(p, q)}$$

under deterministic pushdown recognition with a mandatory right endmarker.

Proof. Theorem 97.3 gives the lower bound and Theorem 97.5 gives the matching upper bound. □

Corollary 97.7 (Arbitrary positive parameters)**.** *For arbitrary $p, q \geq 1$, let $d = \gcd(p, q)$. Since*

$$L_{p,q} = L_{p/d,q/d},$$

the right-endmarker state complexity is

$$\boxed{Q_{\min}(L_{p,q}) = \max\left(\frac{p}{d}, \frac{q}{d}\right).}$$

Chapter 98

Exact Unrestricted Bounded-Replacement State Complexity

Relation to the earlier binary-witness results. The binary-hierarchy $B_{m,k}$ resource geometry of Section 11.10 and the epsilon-free resource-existence decision of Section 11.11 have different target languages, acceptance conventions, and resource coordinates from the rational-balance $L_{p,q}$ staircase developed here. In particular, this chapter has a permanent bottom marker, a right endmarker, one work letter, and a bounded per-transition replacement parameter B, while the binary-hierarchy Pareto result uses a physically empty final stack and no fixed push-length bound. No coordinate value or lower bound passes between these settings merely by changing notation.

The preceding unrestricted-state branch removed the assumption that a recognizer must expose the residual in a prescribed physical coordinate system. The remaining bounded-replacement question is sharper: if the work-stack alphabet has only one non-bottom symbol and one transition may replace the visible work symbol by a word of length at most B, what is the exact state cost when *all* deterministic encodings and normalization histories are allowed?

Within the raw residual-conjugacy model, the bounded question has an exact staircase. The theorem developed here shows that the same law is representation-independent. Its proof retains only the language-level residual quotient and generic top-access locality; the decisive ingredients are protected one-symbol phases, finite-depth descent, outward first-hit bands, and a synchronization argument that converts bounded local stack growth into an exact global state lower bound.

Position relative to pushdown resource complexity. The resource parameter used here should not be presented as if bounded stack growth or bounded replacement were new ideas. Pushdown descriptional complexity already studies tradeoffs involving finite control and pushdown-alphabet size, including hierarchies obtained by limiting the number of pushdown symbols [85, 16]. Pushdown height and broader memory-use measures have also been studied quantitatively [87, 88], and push complexity counts the number of push operations used by accepting computations [86]. In standard PDA presentations that allow one transition to replace the top symbol by an arbitrary word, the maximum one-step replacement length is likewise part of the machine description; it can be normalized to unit pushes at a descriptional cost [87]. These neighboring parameters are not the theorem below. The claim of this chapter is the exact *representation-independent* state law, for the fixed rational-balance family $L_{p,q}$ and one work symbol, as a function of the per-transition replacement bound B, even against arbitrary deterministic ε-behavior and history-dependent encodings.

Put

$$w_{\max} = \max(p, q), \qquad w_{\min} = \min(p, q).$$

The chapter culminates in the exact law

$$\boxed{Q^{(B)}_{\text{unrestricted,1work}}(p, q) = w_{\max} + \max\left(w_{\min}, \left\lceil \frac{w_{\max}}{B-1} \right\rceil\right)} \qquad (B \geq 2),$$

and therefore upgrades the former typed raw staircase to an unrestricted theorem.

98.1 Model and semantic rigidity

98.1.1 Bounded-replacement one-work-symbol pushdown recognizers

A deterministic right-endmarker ε-pushdown automaton has a finite control set Q, binary input alphabet $\{0, 1\}$ followed by one mandatory symbol $\dashv$, and stack alphabet

$$\Gamma = \{Z_0, A\},$$

where Z_0 is a permanent bottom marker and A is the unique work symbol; the initial stack is Z_0. Permanence means that Z_0 is never removed and never occurs above the bottom: a rule reading Z_0 replaces it only by a word of the form Z_0A^t. Acceptance is by final state after the mandatory endmarker has been consumed. The stack top is written on the right. The *work height* of Z_0A^h is h. A raw transition has the form

$$(q, \eta X, av) \vdash (q', \eta\beta, v)$$

or its ε-analogue, where X is the visible top symbol. Determinism means that for each state/top pair there is at most one move on each symbol in $\{0, 1, \dashv\}$ and, whenever an ε-move is enabled, no input-consuming move is enabled at that state/top pair.

Fix $B \geq 2$. The machine is *B-bounded* if every replacement word β has length at most B. Thus one raw transition can decrease work height by at most one and can increase it by at most $B - 1$. The opponent may otherwise use arbitrary deterministic ε-moves, noncanonical intermediate configurations, and arbitrary history-dependent representations. A reachable configuration is called *settled* when no forced ε-move is enabled, so the machine is ready either to consume the next input symbol or the right endmarker. When comparing runs on different inputs, we call the pair consisting of the current control state and the complete stack Z_0A^h the *physical configuration*; thus "physically identical" suppresses the unread input suffix but requires equality of both state and complete stack.

The lower bound uses only the binary-prefix behavior of this model and the following liveness consequence: every finite prefix occurring below has a finite binary continuation followed by $\dashv$ that is accepted. Hence deterministic ε-divergence cannot permanently prevent the next required input symbol from being consumed.

Definition 98.1 (Unrestricted bounded one-work-symbol state complexity)**.** Let

$$Q^{(B)}_{\text{unrestricted,1work}}(p, q)$$

denote the least $|Q|$ over deterministic right-endmarker ε-pushdown recognizers of $L_{p,q}$ that are B-bounded, have a permanent bottom marker, and use exactly one non-bottom work symbol.

98.1.2 Residual quotient

For $w \in \{0, 1\}^*$ define

$$\rho(w) = qN_1(w) - pN_0(w).$$

Then $w \in L_{p,q}$ iff $\rho(w) = 0$.

Lemma 98.2 (Integer residual realizability)**.** *For every $t \in \mathbb{Z}$ there is a word $z \in \{0,1\}^*$ with $\rho(z) = t$.*

Proof. By Bezout, choose integers x_0, y_0 with $qx_0 - py_0 = t$. Replacing (x_0, y_0) by $(x_0 + pn, y_0 + qn)$ for sufficiently large n makes both entries nonnegative. Then $z = 1^{x_0+pn}0^{y_0+qn}$ has residual t. □

Lemma 98.3 (Residuals classify right quotients)**.** *For binary prefixes u, v,*

$$u^{-1}L_{p,q} = v^{-1}L_{p,q} \quad \Longleftrightarrow \quad \rho(u) = \rho(v).$$

Proof. If $\rho(u) = \rho(v)$, then for every continuation z,

$$uz \in L_{p,q} \iff \rho(z) = -\rho(u) \iff \rho(z) = -\rho(v) \iff vz \in L_{p,q}.$$

Conversely, if $\rho(u) \neq \rho(v)$, choose by Theorem 98.2 a word z with $\rho(z) = -\rho(u)$. Then $uz \in L_{p,q}$ but $vz \notin L_{p,q}$. □

Lemma 98.4 (Physical residual rigidity)**.** *If two reachable raw configurations of a deterministic recognizer are physically identical, then the consumed binary prefixes have the same residual.*

Proof. From one physical configuration, determinism gives the same future behavior on every finite continuation. Hence the two consumed prefixes have equal right quotients. Apply Theorem 98.3. □

It is therefore legitimate to write $\mathrm{Res}(C)$ for the unique residual carried by a reachable raw configuration C. Input-consuming transitions satisfy

$$\mathrm{Res}(C \xrightarrow{1} C') = \mathrm{Res}(C) + q, \qquad \mathrm{Res}(C \xrightarrow{0} C') = \mathrm{Res}(C) - p,$$

while ε-moves preserve Res.

98.2 A matching bounded one-work-symbol construction

By symmetry it is enough to assume $p \geq q$. Put

$$b = b_B := \max\left(q, \left\lceil \frac{p}{B-1} \right\rceil\right).$$

Use positive states $P_0, \ldots, P_{p-1}$ and negative states $N_0, \ldots, N_{b-1}$. Encode residuals as follows. For $r \geq 0$, write

$$r = pk + s, \qquad k \geq 0, \quad 0 \leq s < p,$$

and set

$$\eta(r) = (P_s, Z_0A^k).$$

For $r < 0$, write uniquely

$$r = u - bh, \qquad 0 \leq u < b, \quad h \geq 1,$$

and set

$$\eta(r) = (N_u, Z_0A^{h-1}).$$

Proposition 98.5 (Bounded residual construction)**.** *For $p \geq q$ and $B \geq 2$, the encoding above is realized by an ε-free deterministic B-bounded one-work-symbol pushdown automaton with exactly*

$$p + b_B$$

states. Hence, for arbitrary coprime p, q,

$$Q^{(B)}_{\mathrm{unrestricted,1work}}(p,q) \leq w_{\max} + \max\left(w_{\min}, \left\lceil \frac{w_{\max}}{B-1} \right\rceil\right).$$

Proof. Take (P_0, Z_0) as the initial configuration. We verify that the two residual translations can be implemented by one top rewrite.

On the positive side, input 0 sends $pk + s$ to $p(k-1) + s$ when $k \geq 1$, so one A is popped. At the boundary $k = 0$, write

$$s - p = u - bh, \qquad 0 \leq u < b, \quad h = \left\lceil \frac{p-s}{b} \right\rceil.$$

Since $p \leq b(B-1)$, one has $h \leq B-1$; replacing Z_0 by $Z_0 A^{h-1}$ has length at most B.

For input 1, put

$$s' = (s+q) \bmod p, \qquad c = \frac{s+q-s'}{p} \in \{0,1\}.$$

Then $pk+s+q = p(k+c)+s'$. Thus the stack height changes by at most one, implemented either by $A \mapsto A^{1+c}$ or by the corresponding bottom-marker rule.

On the negative side, write $r = u - bh$. For input 0, let

$$u' = (u-p) \bmod b, \qquad d = \frac{u'-u+p}{b} = \left\lceil \frac{p-u}{b} \right\rceil.$$

Then

$$r - p = u' - b(h+d).$$

Again $d \leq B-1$, so an existing top A is rewritten as A^{d+1}, or at height zero the bottom marker is rewritten as $Z_0 A^d$; both replacement words have length at most B.

Finally, because $b \geq q$, input 1 changes the negative quotient height by at most one. If $u+q < b$, the height is unchanged and the remainder becomes $u+q$. If $u+q \geq b$ and $h \geq 2$, one work symbol is popped and the remainder becomes $u+q-b$. If $u+q \geq b$ and $h = 1$, then

$$0 \leq u+q-b < q \leq p,$$

so the transition crosses directly to the positive state P_{u+q-b} while leaving the stack Z_0 unchanged. All binary rules are therefore deterministic, ε-free, and B-bounded, and they realize $r \mapsto r-p$ and $r \mapsto r+q$ exactly. Finally designate P_0 as the sole final state and, among right-endmarker moves, define only

$$\delta(P_0, Z_0, \dashv) = (P_0, Z_0).$$

Since $\eta(0) = (P_0, Z_0)$ and no nonzero residual representative has both state P_0 and top Z_0, the mandatory endmarker accepts exactly residual zero without introducing another state. □

98.3 Protected unary-stack phases and finite-depth descent

The first-hit block constructed later must be compared with a representation-independent family of opposite-sign phases living over one common unary stack. We therefore isolate the protected unary interface needed for the synchronization argument. All statements in this section concern an arbitrary deterministic recognizer in the model of Section 98.1; no bounded replacement property is used until Section 98.4.

Consider the live raw computation on 1^ω. Every finite prefix 1^n has a finite accepting extension by Theorem 98.2. Hence the run consumes arbitrarily many input symbols and cannot become trapped in a deterministic ε-cycle before the next required 1 is read.

Lemma 98.6 (Unary-stack height escape)**.** *Along the raw computation on 1^ω, work height eventually leaves every finite bound.*

Proof. No full raw state/stack configuration can recur. A recurrence after different numbers of consumed 1's would assign the same physical configuration two different residuals by Theorem 98.4; a recurrence at the same input position would create a deterministic ε-cycle and prevent the next live input symbol from being read. If work height failed to leave some finite bound, finitely many control states and unary stacks would force such a recurrence. □

Because the integer-valued height tends to infinity in the preceding sense, there are suffix minima at arbitrarily large heights. At a suffix minimum, the remainder of the 1^ω run never goes below the current complete unary stack.

Lemma 98.7 (Protected unary-stack pump)**.** *There exist a state r, integers $a, c, u \geq 1$, and an infinite family of reachable configurations*

$$(r, Z_0 A^{h_k}), \qquad h_k = a + uk,$$

such that one fixed raw macro consuming exactly c symbols 1 maps

$$(r, Z_0 A^{h_k}) \xRightarrow{1^c} (r, Z_0 A^{h_k+u})$$

for every $k \geq 0$. The macro never exposes a source cell below height h_k and is therefore independent of the buried unary context.

Proof. Among suffix minima of arbitrarily large height, some state/top mode occurs twice, the later occurrence at a strictly larger height. Between the two occurrences the computation never drops below the earlier suffix minimum, so the buried stack below that cut is not inspected. Since the work alphabet is unary, the later stack differs from the earlier one by $u \geq 1$ additional copies of A. The raw transition sequence is therefore valid over every copy of the same buried context and can be iterated. It must consume at least one input symbol: otherwise a deterministic height-increasing ε-macro from one mode back to itself would iterate forever, contradicting liveness. Writing the first chosen height as a gives $h_k = a + uk$. □

The next lemma is the unary form of finite-depth exposure. It is the point at which arbitrary history-dependent representations are neutralized: a fixed finite layer above a protected cut cannot remain hidden forever. We say that a family of configurations is obtained from the protected family by one *fixed relative history above height h_k* if one finite raw transition pattern, with the same sequence of input-consuming and ε-moves and the same rewrite rules, translates over every cut $Z_0 A^{h_k}$, never visits height below h_k, and ends in the stated common state and fixed height offset. Because all symbols above a positive cut are A, such a history is independent of the buried unary context.

Lemma 98.8 (Finite-depth unary-stack descent)**.** *Fix $d, D \geq 0$ and a reachable family*

$$(q, Z_0 A^{h_k+d}), \qquad h_k = a + uk \longrightarrow \infty,$$

obtained from the protected family by one fixed relative history above height h_k. Then, after discarding finitely many k so that $h_k \geq D$, there are an integer $e_{d,D} \geq 0$ and a state $q_{d,D}$, independent of k, such that following 0^ω until the first exposure of the shifted cut at height $h_k - D$ gives

$$(q, Z_0 A^{h_k+d}) \xRightarrow{0^{e_{d,D}}} (q_{d,D}, Z_0 A^{h_k-D}).$$

The displayed macro includes all forced ε-moves. In particular, $D = 0$ gives descent to the original protected cut.

Proof. Put $t_k = h_k - D$. Before the shifted cut at height t_k is exposed, the relative configuration above it starts from the fixed pair (q, A^{d+D}) and follows a deterministic raw history independent of k. Hence either this relative stack is emptied after one fixed finite history for every sufficiently large k, or it is never emptied for any sufficiently large k. Because one raw transition decreases work height by at most one, first exposure of the cut occurs at height exactly t_k.

Assume the second alternative. The relative 0^ω computation is live. Raw recurrence is impossible by Theorem 98.4, so its relative height escapes. Applying the suffix-minimum argument of Theorem 98.7 with 0 in place of 1 to this relative computation produces a second context-independent pump, now consuming $e \geq 1$ zeros and adding $v \geq 1$ work cells. After a fixed prefix, variation of the original protected-family parameter k and independent iteration of the new 0-pump yield reachable configurations in one fixed control state with residual

$$C + kcq - \ell ep$$

and complete unary height

$$H + uk + v\ell$$

for all sufficiently large k, ℓ, where the constants C, H absorb the fixed relative history and the shift D.

Compare the parameter pairs $(k+v, \ell)$ and $(k, \ell+u)$. Their unary heights, hence their complete stacks, are identical, and their control states are identical; but their residuals differ by

$$vcq + uep > 0.$$

This contradicts physical residual rigidity. Permanent nonexposure is therefore impossible, so the shifted cut is reached after one fixed relative history. The number $e_{d,D}$ of consumed zeros and the exposure state $q_{d,D}$ are consequently independent of k. □

We now extract the p positive residual phases used later. Starting at $(r, Z_0 A^{h_k})$, consume $j \in \{0, \ldots, p-1\}$ further symbols 1 and all forced ε-moves. Since the future on 1^ω is obtained by iterating the same protected macro, the computation above the cut is a fixed relative history for each j; thus for some state q_j and integer $d_j \geq 0$, independent of k, the resulting configuration is

$$(q_j, Z_0 A^{h_k + d_j}).$$

Apply Theorem 98.8 with $(d, D) = (d_j, 0)$.

Proposition 98.9 (Protected positive phases and simultaneous descent)**.** *There are affine protected heights $h_k = a + uk$ and p reachable configurations*

$$(s_j, Z_0 A^{h_k}), \qquad j = 0, \ldots, p-1,$$

whose states $s_0, \ldots, s_{p-1}$ are pairwise distinct. Their residuals tend to $+\infty$ as $k \to \infty$.

Moreover, for every fixed $D \geq 0$, after discarding finitely many k there are states $s_{j,D}$ and integers $z_{j,D} \geq 0$, independent of k, such that

$$(s_j, Z_0 A^{h_k}) \xrightarrow{0^{z_{j,D}}} (s_{j,D}, Z_0 A^{h_k - D}),$$

and the p states $s_{j,D}$ are pairwise distinct. For each fixed D, the descended residuals also tend to $+\infty$ with k. The conclusion holds simultaneously for all D in any prescribed finite set.

Proof. The first family is obtained from the preceding construction by descending the fixed upper depth d_j for each phase. If phases $j \neq t$ exposed the common stack $Z_0 A^{h_k}$ in the same state, and if z_j, z_t are the corresponding numbers of consumed zeros, then residual rigidity would give

$$q(j - t) = p(z_j - z_t).$$

Coprimality forces $p \mid (j-t)$, impossible for $0 < |j-t| < p$. Hence the exposure states are pairwise distinct. The protected pump contributes kcq to every phase residual, so all phase residuals tend to $+\infty$.

For a further fixed depth D, each exposed phase family is still obtained from the original protected family by one fixed relative history ending at height h_k. Apply Theorem 98.8 with $(d, D) = (0, D)$ to each of the p exposed phase families. Equality of two descended states at the common stack would again make their residuals equal. The residual difference between phases j and t always has the form

$$q(j-t) - pn$$

for some integer n, namely the difference of the total numbers of consumed zeros along the two phase histories. Equality would therefore force $q(j-t) = pn$, impossible by coprimality and $0 < |j-t| < p$. Thus phase separation survives descent. Since the additional zero history for a fixed D is independent of k, the common kcq term still drives every descended residual to $+\infty$. For a finite set of depths, take the maximum of the finitely many eventuality thresholds; all required descents then hold simultaneously. □

By exchanging 0 with 1 and p with q, one obtains a symmetric protected negative family with q phases and affine protected heights. We use that symmetric family only in the baseline $p+q$ theorem; the new bounded-replacement argument will construct its own negative block by first-hit geometry.

98.4 Outward first-hit bands

Assume $p \geq q$. Such settled configurations exist at arbitrarily negative residuals: start from a sufficiently long prefix 0^n and follow its forced ε-moves to the next settled configuration, which must be reached by liveness. Choose one whose residual R is sufficiently negative that

$$R + (p-1)q < 0.$$

For $j = 0, \ldots, p-1$, consume 1^j and all forced ε-moves until the machine is again ready to inspect the next binary input symbol. Denote the resulting reachable configuration by C_j. Its residual is

$$\mathrm{Res}(C_j) = R + jq < 0.$$

From every C_j follow the unique live computation on 0^ω, including all forced ε-moves. After n consumed zeros the residual is

$$R + jq - np.$$

The settled heights on each such ray are unbounded: otherwise finitely many states and bounded unary heights would force a repeated physical configuration carrying two different residuals.

Lemma 98.10 (Outward first-hit band)**.** *Let K exceed the heights of $C_0, \ldots, C_{p-1}$. For each j, let $E_j(K)$ be the first raw configuration on the outward 0^ω run from C_j whose work height is at least K. Then*

$$\mathrm{ht}(E_j(K)) \in \{K, K+1, \ldots, K+B-2\}.$$

Consequently there is a subset $J_K \subseteq \{0, \ldots, p-1\}$ with

$$|J_K| \geq \left\lceil \frac{p}{B-1} \right\rceil$$

such that all configurations $E_j(K)$, $j \in J_K$, have one common complete unary stack height. Their control states are pairwise distinct.

Proof. The first-hit configuration exists because the settled heights on the ray are unbounded. The raw configuration immediately preceding it has work height at most $K-1$. One B-bounded top rewrite can increase work height by at most $B-1$, so the first-hit height is at most $K+B-2$. This proves the band statement.

There are p phases and only $B-1$ possible first-hit heights, so pigeonhole gives the stated subset. Suppose two distinct indices $j, t \in J_K$ have the same control state. Since their stacks are unary and have the same height, the two full physical configurations are identical. Let z_j, z_t be the numbers of consumed zeros at the respective first-hit positions. Residual rigidity gives

$$R + jq - pz_j = R + tq - pz_t,$$

so

$$q(j-t) = p(z_j - z_t).$$

Because $\gcd(p,q) = 1$, one has $p \mid (j-t)$, impossible for $0 < |j-t| < p$. Hence the states are pairwise distinct. □

Remark 98.11. The lemma uses the replacement bound only through the finite overshoot band of width $B-1$. It makes no assumption about a residual section, a homogeneous tail, the timing of ε-moves, or the history by which the current unary height was created.

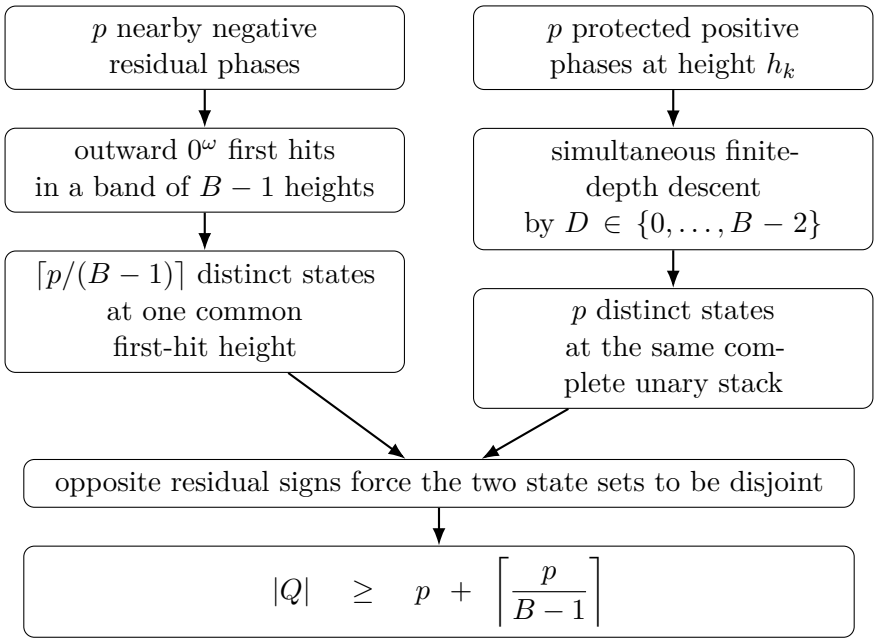


Figure 98.1: Architecture of the new unrestricted bounded-replacement lower bound for $p \geq q$. The bounded rewrite length is used only to create the finite first-hit band; all representation freedom before that crossing is absorbed by residual rigidity and protected finite-depth descent.

98.5 Synchronization and the exact theorem

We now synchronize the negative first-hit block with the positive protected family at one complete physical stack.

Theorem 98.12 (First-hit/protected-phase synchronization)**.** *Assume $p \geq q$. Every unrestricted deterministic B-bounded one-work-symbol recognizer of $L_{p,q}$ satisfies*

$$\boxed{|Q| \geq p + \left\lceil \frac{p}{B-1} \right\rceil.}$$

Proof. Take arbitrarily high positive protected heights h_k from Theorem 98.9. Put

$$K_k = h_k - (B-2).$$

For k large, K_k lies above all initial negative phase heights C_j, so Theorem 98.10 applies. Its common first-hit height has the form

$$K_k + d_k = h_k - D_k, \qquad 0 \le d_k \le B-2, \qquad D_k = (B-2) - d_k \in \{0, \dots, B-2\}.$$

By the simultaneous finite-depth form of Theorem 98.9, once k is large enough the positive protected family can be descended by whichever D_k occurs. Since only the finitely many depths $0, \dots, B-2$ are involved, enlarge k once more so that all descended phase residuals at all these depths are positive. At the common complete stack

$$Z_0 A^{h_k - D_k}$$

there are therefore

(i) p reachable positive-residual configurations in pairwise distinct states, and

(ii) at least $\lceil p/(B-1) \rceil$ reachable negative-residual first-hit configurations in pairwise distinct states.

The two state sets are disjoint. Indeed, equality of a positive state and a negative state at the same complete unary stack would identify two reachable physical configurations carrying residuals of opposite sign, contradicting Theorem 98.4. The displayed lower bound follows. □

We also use the representation-independent one-work-symbol bound.

Theorem 98.13 (Unrestricted one-work-symbol baseline)**.** *Every deterministic right-endmarker ε-pushdown recognizer of $L_{p,q}$ with a permanent bottom marker and one work symbol satisfies*

$$\boxed{|Q| \ge p + q.}$$

Proof. Let the positive and negative protected cut heights be

$$h_k^+ = a + uk, \qquad h_\ell^- = b + v\ell, \qquad u, v \ge 1.$$

Put $g = \gcd(u, v)$. Choose sufficiently large indices k_0, ℓ_0 and define

$$k_n = k_0 + \frac{v}{g} n, \qquad \ell_n = \ell_0 + \frac{u}{g} n.$$

Then

$$h_{k_n}^+ - h_{\ell_n}^- = D$$

is independent of n. If $D \ge 0$, apply the finite-depth descent theorem to all p positive phases with depth D and leave the q negative phases at depth zero. For all sufficiently large n, the resulting $p + q$ configurations have the identical complete unary stack

$$Z_0 A^{h_{\ell_n}^-}.$$

The p positive states are pairwise distinct by phase separation, and the symmetric argument separates the q negative states. Their residuals have opposite signs for sufficiently large n, so a positive and a negative configuration cannot share a state at the common stack by Theorem 98.4. Hence $|Q| \ge p + q$. If $D < 0$, descend the negative family by $-D$ instead. □

Theorem 98.14 (Exact unrestricted bounded-replacement one-work-symbol complexity)**.** *For every coprime $p, q \geq 1$ and every $B \geq 2$, with*

$$w_{\max} = \max(p, q), \qquad w_{\min} = \min(p, q),$$

one has

$$\boxed{Q^{(B)}_{\text{unrestricted,1work}}(p, q) = w_{\max} + \max\left(w_{\min}, \left\lceil \frac{w_{\max}}{B-1} \right\rceil\right).}$$

Proof. Assume first $p \geq q$. By Theorem 98.12,

$$|Q| \geq p + \left\lceil \frac{p}{B-1} \right\rceil,$$

while Theorem 98.13 gives $|Q| \geq p + q$. Therefore

$$|Q| \geq p + \max\left(q, \left\lceil \frac{p}{B-1} \right\rceil\right).$$

The case $q > p$ is obtained by symmetry. The matching upper bound is Theorem 98.5. □

Theorem 98.15 (Complete unrestricted state–stack frontier)**.** *For coprime $p, q \geq 1$, under deterministic right-endmarker recognition with a permanent bottom marker, the Pareto-minimal pairs*

$$(|Q|, |\Gamma|)$$

are exactly

$$\boxed{\text{Pareto}_{\text{unrestricted}}(L_{p,q}) = \{(p + q, 2), (w_{\max}, 3)\}.}$$

Here $|\Gamma|$ counts the permanent bottom marker.

Proof. With only one total stack symbol the store cannot carry unbounded information, so the resulting device has only finitely many physical configurations and cannot recognize the nonregular language $L_{p,q}$. With two total stack symbols there is exactly one work symbol; Theorem 98.13 gives $|Q| \geq p + q$, while Theorem 98.14 attains $p + q$ once $B \geq B_{\text{crit}}$. With three or more total stack symbols, the representation-independent state lower bound Theorem 97.3 gives $|Q| \geq w_{\max}$, and the three-symbol construction Theorem 97.5 attains $(w_{\max}, 3)$. Any point with more than three stack symbols and at least $w_{\max}$ states is dominated by $(w_{\max}, 3)$. Finally $w_{\max} < p + q$, so the two displayed endpoints are incomparable and hence both Pareto-minimal. □

98.6 The unrestricted state–push staircase

Put

$$Q_B := w_{\max} + \max\left(w_{\min}, \left\lceil \frac{w_{\max}}{B-1} \right\rceil\right).$$

The exact theorem turns the bounded replacement parameter into a completely explicit staircase. The restriction $B \geq 2$ is the first nontrivial regime: with $B = 1$, the initial work height zero can never increase, so only finitely many physical configurations are reachable. By Theorem 98.3, $L_{p,q}$ has infinitely many right quotients and is therefore nonregular; thus $B = 1$ cannot recognize it in this model.

Corollary 98.16 (The first two bounded levels)**.** *For every coprime p, q,*

$$\boxed{Q_2 = 2w_{\max},}$$

and

$$\boxed{Q_3 = w_{\max} + \max\left(w_{\min}, \left\lceil \frac{w_{\max}}{2} \right\rceil\right).}$$

Corollary 98.17 (Critical replacement threshold)**.** *The unbounded one-work-symbol optimum $w_{\max} + w_{\min} = p + q$ is attained if and only if*

$$\boxed{B \geq B_{\text{crit}} := 1 + \left\lceil \frac{w_{\max}}{w_{\min}} \right\rceil .}$$

Proof. By Theorem 98.14, $Q_B = w_{\max} + w_{\min}$ exactly when

$$\left\lceil \frac{w_{\max}}{B-1} \right\rceil \leq w_{\min},$$

which is equivalent to $B - 1 \geq w_{\max}/w_{\min}$. □

Corollary 98.18 (Inverse state-to-replacement law)**.** *Fix a state budget $S \geq w_{\max} + w_{\min}$. The least replacement bound permitting an unrestricted one-work-symbol recognizer with at most S states is*

$$\boxed{B_{\min}(S) = \begin{cases} 2, & S \geq 2w_{\max}, \\ 1 + \left\lceil \dfrac{w_{\max}}{S - w_{\max}} \right\rceil, & w_{\max} + w_{\min} \leq S < 2w_{\max}. \end{cases}}$$

Proof. If $S \geq 2w_{\max}$, Theorem 98.16 shows that $B = 2$ already suffices, and no smaller nontrivial replacement bound exists. Suppose $w_{\max}+w_{\min} \leq S < 2w_{\max}$. Since $S-w_{\max} \geq w_{\min}$, Theorem 98.14 gives $Q_B \leq S$ exactly when

$$\left\lceil \frac{w_{\max}}{B-1} \right\rceil \leq S - w_{\max},$$

which is equivalent to

$$B - 1 \geq \frac{w_{\max}}{S - w_{\max}}.$$

The least integer B satisfying this is the displayed value. □

Theorem 98.19 (Exact unrestricted one-work-symbol state–push Pareto staircase)**.** *At fixed total stack alphabet size $|\Gamma| = 2$, the Pareto-minimal pairs $(|Q|, B)$ are exactly*

$$\boxed{\{(Q_B, B) : 2 \leq B \leq B_{\text{crit}},\ B = 2 \text{ or } Q_B < Q_{B-1}\}.}$$

Proof. A plateau value is dominated by the same state count with the smaller replacement bound. Every point after B_{crit} is dominated by the critical-threshold point. At every strict drop, reducing B forces a larger state count by Theorem 98.14; hence every displayed breakpoint is Pareto-minimal. □

For the motivating 2:1 case, $w_{\max} = 2$ and $w_{\min} = 1$, so

$$Q_B = 2 + \max\left(1, \left\lceil \frac{2}{B-1} \right\rceil\right).$$

Thus $Q_2 = 4$, while $Q_B = 3$ for every $B \geq 3$. The bounded-push transition from four states to the unrestricted one-work-symbol optimum of three states therefore occurs exactly at $B = 3$.

98.7 Why the lower bound is genuinely unrestricted

The proof of Theorem 98.14 never assumes that the competitor stores the residual in a chosen state/height coordinate. In particular it uses none of the following:

- raw residual conjugacy;
- a prescribed translation-invariant high-stack regime;
- realtime computation;
- a history-independent representative of a residual;
- a prescribed normalization boundary;
- a canonical relationship between residual magnitude and stack height.

The bounded replacement parameter enters only at the first crossing of a physical height threshold. That crossing has at most $B-1$ possible overshoots. The semantic phase information is then separated by residual rigidity and coprimality, while the protected family supplies the opposite-sign block at the same complete unary stack. This is why a representation-relative state–push staircase becomes a language-intrinsic unrestricted theorem.

98.8 A separate three-symbol bounded-push optimum

The optimal three-symbol point can also be made replacement-at-most-two. Its interface is different from the one-work-symbol theorem above: a normalized macro-step consists of one input-consuming move is followed, when necessary, by a uniquely forced terminating ε-burst, after which the stable residual representative has been restored.

Assume $p \geq q$ and use the $w_{\max} = p$-state, three-symbol stable encoding from the preceding unrestricted-state construction. For $0 \leq u < q$, let

$$u^- = (u-p) \bmod q, \qquad d_u = \frac{u^- - u + p}{q},$$

and for a target remainder $v \in \mathbb{Z}_q$ put

$$\mathrm{pre}(v) = (v+p) \bmod q, \qquad L_v = d_{\mathrm{pre}(v)} - 1.$$

Since addition by p permutes $\mathbb{Z}_q$,

$$\sum_{v=0}^{q-1} L_v = \sum_{u=0}^{q-1} (d_u - 1) = p - q.$$

Introduce abstract suffix roles $H_{v,j}$, $1 \leq j \leq L_v$, meaning "push exactly j further copies of the negative work symbol B, then enter R_v." On a visible B their forced rules are

$$\delta(H_{v,1}, \varepsilon, B) = (R_v, BB), \qquad \delta(H_{v,j}, \varepsilon, B) = (H_{v,j-1}, BB) \quad (j > 1).$$

These are roles rather than new state names. Stable negative configurations use top B only in $R_0, \ldots, R_{q-1}$, so the $p-q$ state/top cells

$$(R_i, B), \qquad q \leq i < p,$$

are unused by stable semantics. Fix a bijection from the suffix roles onto those cells. A long negative input-0 write is replaced by one consuming $B \mapsto BB$ move into the appropriate suffix role; positive-to-negative boundary writes enter a suffix of the same chain. If $u_s = (s-p) \bmod q$ and $h_s = (u_s + p - s)/q$, then

$$d_{\mathrm{pre}(u_s)} = h_s + \left\lfloor \frac{s}{q} \right\rfloor,$$

so $h_s - 1 \leq L_{u_s}$ and every required boundary suffix exists.

Theorem 98.20 (State-preserving push-at-most-two macro compiler)**.** *For every coprime p, q, there is a deterministic right-endmarker DPDA whose stable configurations realize the residual graph and satisfy*

$$\boxed{|Q| = w_{\max}, \qquad |\Gamma| = 3, \qquad \max|\alpha| \le 2.}$$

Every nontrivial normalization burst is a uniquely forced finite ε-suffix chain. In particular, for every replacement bound $B \ge 2$, the minimum state count among deterministic recognizers with at most three total stack symbols is exactly $w_{\max}$.

Proof. The role overlay above proves the case $p \ge q$. Every long negative write is split into one consuming push and a strictly descending suffix chain. The number of roles is exactly $p-q$, all fit into state/top cells unused by stable negative semantics, and those cells carry no competing input transition, so determinism is preserved. Boundary crossings use suffixes of the same chains, every chain terminates, and its macro-result equals the original stable residual transition. The case $q > p$ is symmetric. The baseline lower bound $|Q| \ge w_{\max}$ from theorem 97.3 proves state optimality. For $B = 1$ stack height cannot increase, leaving only finitely many physical configurations, so no nonregular $L_{p,q}$ can be recognized. □

Corollary 98.21 (The bounded three-symbol point is Pareto-minimal)**.** *In the broad deterministic permanent-bottom model, the resource triple*

$$\boxed{(w_{\max}, 3, 2)}$$

is Pareto-minimal for $(|Q|, |\Gamma|, B)$.

Proof. It is attained by theorem 98.20. A dominating triple cannot use fewer than $w_{\max}$ states by theorem 97.3; with only two total stack symbols and replacement bound $B = 2$, theorem 98.14 requires $2w_{\max} > w_{\max}$ states; and $B = 1$ cannot recognize the nonregular language. Hence no strict coordinate improvement dominates $(w_{\max}, 3, 2)$. □

98.9 Resource-theoretic synthesis

The one-work-symbol staircase above and the three-symbol push-at-most-two compiler answer complementary resource questions. The former is now representation-independent at fixed $|\Gamma| = 2$; the latter shows that one additional work symbol collapses the state requirement back to $w_{\max}$ while keeping $B = 2$. The protected-contract synthesis later in the monograph uses these endpoint theorems without reintroducing a raw residual-section hypothesis.

Chapter 99

Filtered Collapse Capacity for Unrestricted Pushdown Stores

Research branch — Retained Information, Physical Exposure, and Protected Resource Synthesis. *The General Theory now contains the abstract filtered collapse-capacity theorem. This branch supplies the representation-independent pushdown specialization, then uses universal higher-order geodesic fibers as exponentially large semantic collapse witnesses before passing to protected resource synthesis.*

Chapter 65 established the abstract collapse interface: a finite source family $\mathcal{S} \subseteq U_a^{-1}(m^*)$, collapse load $N = |\mathcal{S}|$, debt count D, canonical count $K = N - D$, nested source observations ω_h of capacity $V(h)$, and the number X_h of canonical sources that cannot remain h-confined. Its Filtered Collapse-Capacity Theorem (57.1) states

$$\boxed{D + X_h + V(h) \geq N}$$

under hidden-tail noncoalescence.

The present chapter does not re-prove that counting theorem. Its task is the model-specific step that the abstract theory deliberately leaves open: prove hidden-tail noncoalescence for an unrestricted deterministic top-access ε-pushdown realization and compute the resulting observation-growth function from source-local state, alphabet, and exposure depth.

Position relative to stack-space and store-complexity literature. The exposure depth introduced below is deliberately narrower than classical stack-space measures. Pushdown height and its constant/nonconstant regimes are established resource measures [87, 88], and related stack-automaton models have been studied through weak, accepting, and strong space complexity [90]. Pushdown store languages and the descriptional complexity of representing reachable store contents are also classical objects [17]. None of those measures records which cells of the *initial source representation* must become visible during one exact semantic macrostep. The contribution here is therefore not a new claim about stack height in general; it is a representation-independent collapse-to-exposure theorem for this provenance-sensitive source-access resource, followed by sharp universal-k-fiber witnesses.

For this specialization we write the physical projection as

$$\pi : P \to M$$

and the chosen reachable section as

$$\tau : M \to P, \qquad \pi\tau = \mathrm{id},$$

to keep the pushdown notation of the original exposure argument distinct from the book-wide symbols Θ, σ. The one-symbol physical action is the settled macrostep of the General

Theory: it may contain uniquely forced finite internal moves, consumes exactly the selected semantic action, consumes no later input, and ends at the next settled boundary. All quantities $\mathcal{S}, N, D, K, X_h, V(h)$ therefore have exactly the meanings fixed in Chapter 65.

99.1 Unrestricted deterministic pushdown exposure

99.1.1 Raw model

We use the standard deterministic top-access ε-pushdown interface. A physical configuration is

$$(q, \alpha), \qquad q \in Q, \quad \alpha \in \Gamma^*,$$

with the leftmost symbol of α regarded as the top. A raw transition inspects the finite-control state, the visible top symbol when one exists, and either one input symbol or ε, and replaces the visible top symbol by an arbitrary fixed finite word. Determinism includes the usual exclusion that an enabled ε-move prevents a competing input-consuming move at the same state/top mode. No bound is imposed on replacement length. A permanent bottom marker may be used and is not counted among the work symbols; the same argument also covers the empty-stack convention. The lift may contain arbitrarily many history-dependent physical representatives of the same semantic state. The section τ merely selects one reachable reference representative for the collapse interface; uniqueness of representation is not assumed.

Only the initial source stack is charged to the exposure resource. Conceptually tag the work cells present at the source boundary, from top to bottom, by immutable provenance labels $1, 2, \ldots$. These labels are bookkeeping devices only and are not available to the machine. When a transition replaces a visible cell by a finite word, the removed source cell keeps no descendant provenance: every newly written cell is administrative, even if it carries the same stack symbol. Thus arbitrary replacement words cannot create new source cells for purposes of the resource measure.

Definition 99.1 (Source-stack exposure depth)**.** For a canonical source x, define $e(x)$ to be the largest provenance index of an initial source-stack cell that becomes physically visible as the top cell during the one-symbol macrostep, up to and including the settled canonical hit. If no initial work cell becomes visible, put $e(x) = 0$.

Thus $e(x)$ is not maximum stack height and not the number of pushes. It is a depth of *read access to the initial semantic representation.*

For the chosen canonical sources, let

$$q_{\mathrm{src}} = \big|\{\text{source control state of } \tau(x)\}\big|$$

and let g_{src} be the number of work-stack symbols that actually occur in their initial source stacks, with the convention $g_{\mathrm{src}} = 1$ if every canonical source stack is work-empty. If $K = 0$, set $q_{\mathrm{src}} = 0$ and $g_{\mathrm{src}} = 1$; then every exposure inequality below reduces immediately to $D = N$. Administrative states and symbols used only after leaving the source interface are not charged to these source-local parameters.

For $h \geq 0$ define the depth-h source signature as follows: record the source control state and, if the initial stack has fewer than h work cells, record it completely together with its end; otherwise record its top h source work symbols. The number of possible signatures is at most

$$C_h^{\mathrm{src}} = q_{\mathrm{src}} \sum_{r=0}^{h} g_{\mathrm{src}}^{r}.$$

Lemma 99.2 (Hidden suffix cannot coalesce)**.** *Two distinct canonical sources with the same depth-h source signature cannot both reach the same fixed canonical target while exposing at most h initial source cells.*

Proof. Suppose two source representatives share the same depth-h signature and both computations remain within the first h source cells. Until a deeper source cell becomes visible, every raw step sees the same control state, the same currently visible top symbol, and the same unread input status on both computations. Determinism therefore forces the same raw rule whenever the visible prefixes agree.

A raw rule replaces only the visible top symbol by a fixed finite word. Consequently it cannot alter the untouched source suffix below the exposed cut. Pushed administrative material may be created and later removed, but the hidden suffix remains literally unchanged until a deeper source cell is exposed. Hence two distinct hidden suffixes cannot be transformed into one identical complete target stack by computations that never expose those suffixes. If the hidden suffixes were already identical, equality of the complete source configurations together with section injectivity would identify the semantic sources. Both alternatives contradict distinctness. Therefore at most one shallow canonical source occupies each depth-h signature. □

Theorem 99.3 (Source-Stack Exposure Profile)**.** *For every finite semantic collapse family realized by an unrestricted deterministic ε-pushdown system and every $h \geq 0$,*

$$\boxed{D + X_h + q_{\mathrm{src}} \sum_{r=0}^{h} g_{\mathrm{src}}^{r} \geq N.}$$

No bounded-replacement, realtime, homogeneous-tail, canonical residual, or unique-representation hypothesis is required.

Proof. Use the depth-h source signatures as the filtration in Theorem 65.1. Lemma 99.2 verifies hidden-tail noncoalescence, and the displayed signature count bounds $V(h)$. □

The global machine-resource form follows immediately from $q_{\mathrm{src}} \leq |Q|$ and $g_{\mathrm{src}} \leq |\Gamma|$.

99.2 Optimal exposure spectrum and pointwise sharpness

Let $K = N - D$ be the number of canonical sources and sort their exposure depths:

$$e_{(1)} \leq e_{(2)} \leq \cdots \leq e_{(K)}.$$

Define

$$d_r = \min \left\{ h \geq 0 : q_{\mathrm{src}} \sum_{j=0}^{h} g_{\mathrm{src}}^{j} \geq r \right\}.$$

Theorem 99.4 (Exact exposure-rank lower bound)**.** *For every $1 \leq r \leq K$,*

$$\boxed{e_{(r)} \geq d_r.}$$

For $g_{\mathrm{src}} = g \geq 2$,

$$d_r = \left\lceil \log_g \left(1 + \frac{(g-1)r}{q_{\mathrm{src}}} \right) \right\rceil - 1,$$

whereas for $g_{\mathrm{src}} = 1$,

$$d_r = \left\lceil \frac{r}{q_{\mathrm{src}}} \right\rceil - 1.$$

Proof. At most C_h^{src} canonical sources can have exposure at most h by Theorem 99.3. If $e_{(r)} < d_r$, then with $h = e_{(r)}$ there are at least $r > C_h^{\mathrm{src}}$ shallow canonical sources, a contradiction. The closed forms invert the geometric sum. □

Corollary 99.5 (Aggregate exposure)**.**

$$\boxed{\sum_{x \text{ canonical}} e(x) \geq \sum_{r=1}^{K} d_r = \sum_{h \geq 0} \left(K - q_{\mathrm{src}} \sum_{j=0}^{h} g_{\mathrm{src}}^{j} \right)_{+}.}$$

For fixed $g \geq 2$ and $K/q_{\mathrm{src}} \to \infty$,

$$\bar{e} \geq \log_g(K/q_{\mathrm{src}}) - O_g(1),$$

while for $g = 1$ one has $\bar{e} = \Omega(K/q_{\mathrm{src}})$.

Theorem 99.6 (Pointwise sharpness)**.** *The rank lower bound is pointwise sharp. For every $q \geq 1$, $g \geq 1$, and $H \geq 1$, there exists a zero-debt deterministic ε-pushdown collapse family with*

$$K = q \sum_{r=0}^{H} g^r, \qquad q_{\mathrm{src}} = q, \qquad g_{\mathrm{src}} = g,$$

for which

$$\boxed{e_{(r)} = d_r \qquad (1 \leq r \leq K).}$$

Initial-segment truncations give every intermediate family size and remain rank-sharp with their actual source-local alphabet. The canonicalization uses no push operation.

Proof. Take source control states $q_1, \ldots, q_q$, one cleanup state c, a permanent bottom marker Z_0, and a work alphabet Γ of size g. Use every pair

$$(q_i, wZ_0), \qquad w \in \Gamma^*, \quad |w| \leq H,$$

as a distinct semantic source. One common semantic action a sends all sources to one target.

If $w \neq \varepsilon$, the physical a-transition removes the first source work symbol and enters c; deterministic ε-rules in c remove the remaining source word. If $w = \varepsilon$, the a-transition on Z_0 goes directly to the common canonical target. No rule pushes material. A source represented by a word of length r has exposure exactly r. There are exactly qg^r source representatives at depth r, hence exactly

$$q \sum_{j=0}^{h} g^j$$

with exposure at most h for $0 \leq h \leq H$. The ordered spectrum therefore equals the generalized inverse of this count at every rank. Truncating the list in nondecreasing word length preserves the same rank saturation; if the truncation has not yet used every work symbol, its actual g_{src} is smaller and the same counting argument applies to that actual source alphabet. □

Remark 99.7 (Why exposure is a distinct resource). The witness in Theorem 99.6 has zero push complexity during canonicalization but unbounded source exposure. Conversely, a machine may push arbitrarily much administrative material above the source without exposing any deeper source cell. Source exposure is therefore neither push count nor maximum current stack height.

99.3 Universal k-fibers as the semantic witness family

The universal witness need not be reconstructed here. Chapter 19 already proved the Universal k-Fiber Theorem theorem 19.30: for every $s \geq k \geq 2$, one fixed completion-total periodic action system contains $2^{\binom{s}{k}}$ labeled states with common exact distance and common complete compatibility through order $k-1$, while the order-k faces on a distinguished structural s-simplex range over every possible pattern.

For the present physical question, retain only the following interface. Write the distinguished actions as $a_1, \ldots, a_s$, and let v_A denote the witness corresponding to the k-uniform pattern $A \subseteq \binom{[s]}{k}$. For a structural prefix J, its action map has the periodic-translation form

$$T_J(z, \phi) = \big(z + G_J(\phi), \tau_J(\phi)\big),$$

so it is injective in the lattice coordinate. Adjoin a dead semantic state $\perp$ and define the geodesic-prefix transition by sending a distinguished action to its translated state when it remains on a shortest path and to $\perp$ otherwise. This converts the already-established universal fiber into a deterministic many-to-one semantic collapse family without repeating its construction.

99.3.1 From universal freedom to a common collapse

Turn geodesic compatibility into an ordinary deterministic semantic system by adjoining a dead state $\perp$. For a distinguished action a_i define

$$U_{a_i}(v) = \begin{cases} T_i v, & c_i + d(T_i v) = d(v), \\ \perp, & \text{otherwise}, \end{cases} \qquad U_{a_i}(\perp) = \perp.$$

A structural action word remains live exactly while it is a shortest prefix.

Lemma 99.8 (Affine prefix retention)**.** *For every executable action word w from a fixed phase ϕ,*

$$T_w(z, \phi) = \big(z + G_w(\phi), \tau_w(\phi)\big).$$

Hence T_w is injective in z. In particular, every lower-order structural prefix T_J preserves all distinctions among the universal witness states v_A.

Proof. Composition of translations is translation by the sum of the phase-dependent displacement vectors encountered along the fixed phase word. Subtracting the common displacement recovers the initial lattice coordinate. □

Theorem 99.9 (Universal face-collapse retention)**.** *Let a full universal periodic k-fiber on an s-simplex be witnessed by $\{v_A : A \subseteq \binom{[s]}{k}\}$. Fix $|J| = k-1$, $j \notin J$, and $E = J \cup \{j\}$. Then the sources*

$$x_A = T_J v_A, \qquad E \notin A,$$

are pairwise distinct and all satisfy

$$U_{a_j}(x_A) = \perp.$$

Therefore every full universal periodic k-fiber has a one-action collapse family of exact size

$$\boxed{N_{s,k}^{\text{face}} = 2^{\binom{s}{k}-1}},$$

and retained information

$$\boxed{I_{\text{ret}} = \binom{s}{k} - 1 \ \textit{bits.}}$$

Proof. Exactly half of the 2^m patterns omit E. By Theorem 99.8, applying T_J preserves their pairwise distinctness. Full universality makes every $(k-1)$-prefix J geodesic and makes the completed k-face E nongeodesic exactly on the branch $E \notin A$. Since J itself is geodesic,

$$c_J + d(T_J v_A) = d(v_A).$$

Thus geodesicity of the next action a_j at $T_J v_A$ is equivalent to geodesicity of the whole face E at v_A. On the branch $E \notin A$ the geodesic-prefix semantic transition therefore goes to $\bot$. □

Corollary 99.10 (Fixed-action star amplification)**.** *In the explicit universal construction, fix $j \in [s]$ and take all $(k-1)$-sets $J \subseteq [s] \setminus \{j\}$. Their collapse-source families are disjoint, and the common final action a_j has a collapse family of size*

$$\boxed{N^{\star}_{s,k} = \binom{s-1}{k-1} 2^{\binom{s}{k}-1}.}$$

Proof. The post-prefix private-coordinate pattern records J: coordinate p_i equals 0 exactly when $i \in J$ and remains -1 otherwise. Thus source families for distinct J are disjoint. Each family has the face-collapse size from Theorem 99.9. □

Chapter 100

Universal Retained Information Forces Physical Exposure

Combining Theorem 99.3 with Theorem 99.9 gives the main representation-independent bridge.

Theorem 100.1 (Universal Information-to-Physical Exposure Theorem)**.** *For every $s \geq k \geq 2$, every full universal periodic k-fiber, every chosen face-collapse interface, and every unrestricted deterministic exact ε-pushdown realization,*

$$\boxed{D + X_h + q_{\mathrm{src}} \sum_{r=0}^{h} g_{\mathrm{src}}^{r} \geq 2^{\binom{s}{k}-1}}$$

for every $h \geq 0$.

In particular, the global machine-resource bound

$$D + X_h + |Q| \sum_{r=0}^{h} |\Gamma|^{r} \geq 2^{\binom{s}{k}-1}$$

holds independently of the chosen stack encoding and normalization history.

Proof. Apply Theorem 99.3 to the face-collapse family supplied by Theorem 99.9, whose cardinality is $N = 2^{\binom{s}{k}-1}$. This gives the first displayed inequality. The global machine-resource form follows from $q_{\mathrm{src}} \leq |Q|$ and $g_{\mathrm{src}} \leq |\Gamma|$. □

Corollary 100.2 (Zero-debt bit budget)**.** *Assume zero debt and $g_{\mathrm{src}} = g \geq 2$. If every source has exposure at most h, then*

$$q_{\mathrm{src}} \frac{g^{h+1} - 1}{g - 1} \geq 2^{\binom{s}{k}-1}.$$

Equivalently, the exact depth bound is

$$h \geq \left\lceil \log_g \left(1 + \frac{(g-1)2^{\binom{s}{k}-1}}{q_{\mathrm{src}}} \right) \right\rceil - 1.$$

In particular,

$$\log_2 q_{\mathrm{src}} + (h+1) \log_2 g > \binom{s}{k} - 1 + \log_2(g-1).$$

Corollary 100.3 (Universal exposure spectrum)**.** *In the zero-debt case and $g \geq 2$,*

$$e_{(r)} \geq \left\lceil \log_g \left(1 + \frac{(g-1)r}{q_{\mathrm{src}}} \right) \right\rceil - 1, \qquad 1 \leq r \leq 2^{\binom{s}{k}-1}.$$

For fixed q_{src} and fixed $g \geq 2$, every fixed positive quantile has exposure $\Omega\big(\binom{s}{k}\big)$. In particular, for fixed k this is $\Omega(s^k)$. For $g = 1$, fixed positive quantiles are exponential in $\binom{s}{k}$.

Corollary 100.4 (Star-amplified pushdown bound)**.** *For the explicit universal witness and fixed final action a_j,*

$$D + X_h + q_{\mathrm{src}} \sum_{r=0}^{h} g_{\mathrm{src}}^{r} \geq \binom{s-1}{k-1} 2^{\binom{s}{k}-1}.$$

100.0.1 The graph case

For $k = 2$, the universal fiber realizes every graph on s distinguished vertices while exact distance and all order-1 compatibility data remain fixed. A single edge query retains

$$2^{\binom{s}{2}-1}$$

sources before collapse. Hence fixed finite control and fixed nonunary source alphabet force exposure $\Omega(s^2)$ on every fixed positive quantile. In the star-amplified construction, one final vertex action simultaneously supports

$$(s-1)2^{\binom{s}{2}-1}$$

collapse sources across its incident edge contexts.

Example 100.5 (The smallest nontrivial graph fiber)**.** Take $s = 3$ and $k = 2$. Then $m = 3$, so one common order-1 signature contains all eight graphs on the three distinguished vertices. Fix the edge $E = \{1, 2\}$, use prefix $J = \{1\}$, and test the final action a_2. Exactly four graphs omit E; after the prefix their four semantic states remain distinct, and all four collapse to $\perp$ on a_2. Thus $I_{\mathrm{ret}} = 2$ bits.

If a zero-debt realization uses one source control state and a binary source alphabet, the rank theorem gives

$$(d_1, d_2, d_3, d_4) = (0, 1, 1, 2).$$

Hence at least one of the four sources must expose two initial work cells. The example is intentionally small: the general theorem replaces these four sources by $2^{\binom{s}{k}-1}$.

Chapter 101

Strong Converse and Simultaneous-Query Hardness

101.0.1 Strong converse

For a universal face-collapse family put

$$m = \binom{s}{k}, \qquad N = 2^{m-1}, \qquad C_h^{\mathrm{src}} = q_{\mathrm{src}} \sum_{r=0}^{h} g_{\mathrm{src}}^{r}.$$

Then Corollary 65.3 becomes

$$\boxed{\frac{D + X_h}{2^{m-1}} \geq \left(1 - \frac{C_h^{\mathrm{src}}}{2^{m-1}}\right)_+.}$$

If

$$\log_2 C_h^{\mathrm{src}} \leq m - 1 - \lambda,$$

then

$$\boxed{\frac{D + X_h}{2^{m-1}} \geq 1 - 2^{-\lambda}.}$$

Thus a deficit of λ physical observation bits forces all but a $2^{-\lambda}$ fraction of the retained semantic sources to pay canonical debt or deeper exposure.

101.0.2 A direct-sum theorem across all k-face queries

For every $E \in \binom{[s]}{k}$ choose once and for all a decomposition

$$E = J_E \cup \{j_E\}, \qquad |J_E| = k - 1,$$

and use the face-collapse query obtained by applying J_E and then testing a_{j_E}. For a fiber pattern $A \subseteq \binom{[s]}{k}$, the query E is an *absent-face query* when $E \notin A$. Call an absent-face incidence (A, E) *h-hard* when its realization either has canonical debt or requires exposure $> h$.

Assume one common upper bound $V(h)$ on the depth-h source observation count for every face query. Put

$$\eta_h = \frac{V(h)}{2^{m-1}}.$$

Theorem 101.1 (Simultaneous-query direct sum)**.** *There exists a universal witness pattern A with at least*

$$\boxed{\frac{m}{2}(1 - \eta_h)_+}$$

h-hard absent-face queries. In particular, if $V(h) \le 2^{m-1-\lambda}$*, then one witness is hard on at least*

$$\boxed{\frac{m}{2}(1-2^{-\lambda})}$$

absent-face queries.

Proof. For each face E, exactly 2^{m-1} fiber patterns omit E. By the filtered capacity theorem, at most $V(h)$ of those absent-branch incidences can be both canonical and h-confined. Hence the number of hard incidences for this face is at least $\big(2^{m-1} - V(h)\big)_+$. Summing over the m faces gives at least

$$m\big(2^{m-1} - V(h)\big)_+$$

hard incidences. Averaging over the 2^m witness patterns yields one A with at least

$$\frac{m}{2}(1-\eta_h)_+$$

hard queries. □

Theorem 101.2 (Typical-case direct sum)**.** *Let A be uniformly random among the 2^m universal fiber patterns. For every $0 < \rho < 1/2$ and $0 < \zeta < 1/2 - \rho$, at least a fraction*

$$\boxed{\left(1 - e^{-2\rho^2 m} - \frac{\eta_h}{2\zeta}\right)_+}$$

of all patterns have at least

$$\boxed{m\left(\frac{1}{2} - \rho - \zeta\right)}$$

h-hard absent-face queries.

Proof. The number $Z(A)$ of absent faces is distributed as $\mathrm{Bin}(m, 1/2)$. Hoeffding's inequality [89] gives

$$\Pr\left[Z(A) < m\left(\frac{1}{2} - \rho\right)\right] \le e^{-2\rho^2 m}.$$

Across all patterns and faces, the number of easy absent incidences is at most $mV(h)$. Therefore the expected number $Y(A)$ of easy absent queries of a random pattern is at most

$$\mathbb{E}Y(A) \le \frac{mV(h)}{2^m} = \frac{m\eta_h}{2}.$$

Markov's inequality yields

$$\Pr[Y(A) > \zeta m] \le \frac{\eta_h}{2\zeta}.$$

Outside the union of these exceptional events, the number of hard absent queries is at least

$$Z(A) - Y(A) \ge m\left(\frac{1}{2} - \rho - \zeta\right).$$

□

Corollary 101.3 (Asymptotically almost-everywhere hardness)**.** *If $s \to \infty$ and the depth-h physical capacity deficit $\lambda = \lambda(s)$ tends to infinity so that $\eta_h \le 2^{-\lambda}$, choose for example $\rho = m^{-1/4}$ and, for all sufficiently large s, $\zeta = 2^{-\lambda/2}$. Then $\rho, \zeta \to 0$ and*

$$e^{-2\rho^2 m} + \frac{2^{-\lambda}}{2\zeta} \to 0.$$

Consequently, asymptotically almost every universal witness pattern is h-hard on

$$\left(\frac{1}{2} - o(1)\right)\binom{s}{k}$$

of its absent-face queries.

101.1 Adaptive protected-source readout and exact resource scaling

The protected-exposure principles of this chapter can also be posed as a sharp readout resource problem. The source record must be genuine and immutable, while its owner identity can be chosen adaptively during a single scan. This is a deliberately restricted one-pass model: all finite-control phases count and the bound does not transfer automatically to unrestricted pushdown machines.

101.1.1 Scope and prior work

One-pass reordering is traditionally measured by buffer occupancy, scheduling cost, or transducer expressiveness. The present statistic is different: the number of *finite-control states* when a single protected stage retains an actual named input record. The owner's immutable name can itself transport a bounded amount of information, even when its payload is just one symbol. We ask whether that additional information capacity changes the exponential state cost of all cyclicly ordered readouts from the same frozen stack.

Praveen [211] studies streaming string/register transducers and their origin semantics.

Bojańczyk and Stefański [212] study single-use register transducers, where reading a register empties it. In contrast, our protected source record remains physically readable until its required output; its owner name can be inspected repeatedly. These models are not identified without a formal translation.

Reordering-buffer management, studied for example by Adamaszek *et al.* [213], optimizes competitive scheduling cost rather than the finite-control state count considered here. These works supply essential context, not a confirmed historical priority result for the present special cost theorem.

101.1.2 Contributions and novelty

The paper-specific contribution is an exact asymptotic state-complexity characterization for a source-typed, one-pass cyclic readout model with a single repeatedly inspectable *actual* protected source record. The lower bound accounts for the information retained in a physical owner–payload pair, while the constructive upper bound uses the identity of a *future genuine owner* to encode a short block of previously read symbols. A direct literal prefix removes the logarithmic overhead in counting all decoding states. Under payload-independent fixed-owner staging the same task has a larger state cost, yielding a linear-in-n separation between adaptive and fixed owner identity at unchanged stage capacity. Streaming/register transducer frameworks and reordering-buffer problems provide context, but their results and their models are not claimed as new or formally equivalent to the protected-source model here. A definitive historical priority claim would require a broader specialist literature comparison.

101.1.3 Machine model and main theorem

Fix an integer $n \geq 2$, a finite payload alphabet Σ with $m = |\Sigma| \geq 2$, and a set of named owners $[n] = \{1, \ldots, n\}$. The physical source is a read-once stack with top-to-bottom records

$$(1, b_1), (2, b_2), \ldots, (n, b_n), \qquad b_i \in \Sigma.$$

All m^n payload assignments are possible. At the same frozen initial stack checkpoint the environment chooses a query label $j \in [n]$ *before the scan.* The prescribed readout is the payload sequence with owners

$$\sigma_j = (j, j+1, \ldots, n, 1, \ldots, j-1). \tag{101.1}$$

A deterministic finite controller may read and pop each original record exactly once and cannot rewrite or reinsert it. It has q internal states and one optional protected stage.

This stage is either empty or retains exactly one original, immutable record (i, b_i) already popped from the source; the controller may inspect both its genuine owner label and payload while it is present. It may stage a record in a payload-dependent manner, but cannot relabel a staged owner, change its payload, or stage two records. Each output move prints at most one payload symbol (a zero-output move is allowed). Every phase of output or scan, including query-specific modes, is included in q; there are no uncounted clocks, data registers, writable auxiliary stacks, or temporary staging positions. Owner labels on the current source record and the protected stage are observable symbols of the given typed model; the controller's transition table may depend on the fixed finite n, m.

Let $q_{\mathrm{cyc}}(n, 1; m)$ be the smallest q for one shared controller that accepts every $j \in [n]$ and produces the readout (101.1) for every payload assignment, with the same frozen source order.

Theorem 101.4 (Tight finite-alphabet cyclic readout frontier)**.** *For every fixed $m \geq 2$ there are positive constants c_m, C_m, n_m such that, for all $n \geq n_m$,*

$$c_m \frac{m^n}{n} \leq q_{\mathrm{cyc}}(n, 1; m) \leq C_m \frac{m^n}{n}. \tag{101.2}$$

The protected stage stores only one immutable genuine source record. In particular $q_{\mathrm{cyc}}(n, 1; 2) = \Theta(2^n/n)$.

101.1.4 The indistinguishability lower bound

Lemma 101.5 (Bottom-owner cut)**.** *For every n, m in the model above,*

$$q_{\mathrm{cyc}}(n, 1; m) \geq \left\lceil \frac{m^{n-1}}{1 + m(n-1)} \right\rceil. \tag{101.3}$$

Proof. Select query $j = n$ and fix the last owner's payload b_n to one given value. Immediately after the first $n - 1$ records have been popped, but before the bottom owner can be read, the required output has not started: owner n must be output first, and its payload is still unread. In particular, an earlier first output could not be valid simultaneously for all $m \geq 2$ possible bottom payloads. We may therefore fix b_n for the counting argument without allowing any early output. All m^{n-1} assignments to the first $n - 1$ bits/symbols have identical unread stack suffix, fixed output history, and identical externally controlled input position. If two assignments had the same controller state and the same stage configuration at this cut, determinism would force the same continuation and output, which is impossible. The stage has at most $1 + m(n - 1)$ physical configurations: empty, or one of the $n - 1$ already popped owners with one of m payload values. Hence $q[1 + m(n - 1)] \geq m^{n-1}$. This proves (101.3). The environment has fixed the query before scanning, so it is not an uncounted post-hoc channel. □

101.1.5 A genuine-owner online encoder

For a query $j = K + 1$, the K owners preceding j in the stack must be deferred. Once j and owners $j + 1, \ldots, n$ are emitted during the scan, the original deferred owners must be flushed *in increasing order* $1, \ldots, K$. We next build a one-stage encoder for such a prefix over any fixed m-symbol alphabet.

For sufficiently large K, let $d \geq 1$ be maximal with

$$m^d + 2d + \lceil \log_m(d + 1) \rceil \leq K, \tag{101.4}$$

and put

$$r = \lceil \log_m(d + 1) \rceil, \quad M = K - d - 1, \quad L = r + d + m^d - 1. \tag{101.5}$$

Equation (101.4) gives

$$L \leq M, \qquad m^r \geq d + 1. \tag{101.6}$$

For the finitely many K having no valid d, use an ordinary finite-control buffer storing all K symbols.

Lemma 101.6 (Online future-owner selection)**.** *For every payload word $b_1 \dots b_K \in \Sigma^K$, one can make a single left-to-right physical pass which stores one* unaltered actually occurring source record *and at most M other payload symbols in finite control at any time. From these resources, all K deferred symbols can later be output in their original increasing-owner order. No information from an unread source record is consulted by the stage-selection rule.*

Proof. Identify Σ with $\{0, 1, \dots, m-1\}$. Read the initial r payload symbols literally as a block A, and the following d symbols as a base-m integer $v \in \{0, \dots, m^d - 1\}$. At this time all required selector data are already read; choose the target source index

$$i = r + d + 1 + v \le K. \tag{101.7}$$

Continue scanning and accumulating the block X of the v genuine intermediate owners. When original owner i physically arrives, move *that exact record* (i, b_i) into the protected stage. Now remove the d selector symbols from the controller's buffer: the actual staged label i reversibly represents them by $v = i - r - d - 1$. Let Y denote the later $K - i$ symbols read after staging. At the end of the deferred scan the controller holds the word $A\,X\,Y$ of length

$$|A| + |X| + |Y| = r + v + (K - i) = K - d - 1 = M,$$

while the protected stage holds (i, b_i). Just before the latest possible staging event, the controller contains at most $r + d + (m^d - 1) = L \le M$ literal symbols; at all later times it contains at most M. Thus both online and post-stage memory bounds are valid.

When the blocking owner has been passed and deferred output may start, emit the r symbols of A. Then inspect the genuine stage label i and decode the d base-m digits of $v = i - r - d - 1$ in their natural order, emitting one per step. Finally emit the $|X| = v$ buffered symbols of X, then the actual stage payload b_i (releasing the stage), then the buffered symbols of Y. The output is exactly $A\,B\,X\,b_i\,Y = b_1 \dots b_K$. The lengths of the buffered suffix and the visible stage label determine when X is exhausted; no separate unbounded counter is hidden in the schedule. □

Lemma 101.7 (All phases have $O_m(m^M)$ finite-control states)**.** *For each deferred-prefix length K and the parameters* (101.5)*, a deterministic implementation of Lemma 101.6 exists with at most $C'_m m^M$ controller states, where C'_m depends only on m and not on K, counting the stage-selection, source-scan, all output symbols, and decode phases.*

Proof. A finite-control buffer storing an arbitrary word over Σ of length at most H takes $\sum_{h=0}^{H} m^h \le m^{H+1}/(m-1)$ distinct payload-word states. This count already distinguishes buffer lengths, so it must not be multiplied by another factor for a separate length counter. The present implementation uses a bounded number of phase types, with source-owner and protected-owner names read directly from their physical locations. Separate mode banks for the pre-stage scan ($H = L$), the post-stage scan ($H = M$), and the direct source-output segment $j, \dots, n$ ($H = M$) therefore cost $O_m(m^M)$ states. The output of the first r literal symbols reduces buffer length by one at every step, and all those states form another $O_m(m^M)$ bank.

The delicate part is the decoded base-m owner-index block. It requires exactly d one-symbol output steps while the untouched buffer has length $M - r$. Explicitly including a distinct mode for each decode position uses at most

$$d\,m^{M-r} \le m^M$$

controller states by (101.6). Once decoding ends, output of X, the single stage symbol, and Y uses a constant number of modes and words of length at most $M - r$, again $O_m(m^M)$ states. All decisions about the position $|X| = v$ use the visible original staged owner label i, and so require no additional family of controller states. These counts enumerate every scan, output, and decoding phase, including the deliberate r-symbol memory reduction. Their sum is at most $C'_m m^M$. □

101.1.6 Assembly for all cyclic queries

Proof of Theorem 101.4. Lemma 101.5 proves the lower bound of order $\Omega_m(m^n/n)$. For the upper bound, fix query $j = K+1$ before scanning, where $0 \le K \le n-1$. Use Lemmas 101.6 and 101.7 on the prefix $1, \dots, K$, output the rest of the source $j, \dots, n$ directly during the scan, and finally emit the deferred original prefix. For fixed m, maximality in (101.4) implies $m^d = \Theta_m(K)$ for all sufficiently large K. Indeed, $m^d \le K$, while the failure of the inequality at $d+1$ gives

$$K < m^{d+1} + 2(d+1) + \lceil \log_m(d+2) \rceil.$$

Since $d = O_m(\log K)$, the additive term is $o(K)$; hence $K = O_m(m^{d+1}) = O_m(m^d)$. Thus $m^d = \Theta_m(K)$. Since $M = K - d - 1$,

$$q_K \le C'_m m^{K-d-1} = O_m\left(\frac{m^K}{K+1}\right). \tag{101.8}$$

For bounded K ordinary finite-control buffering has a constant cost absorbed into this asymptotic inequality. Use disjoint, query-labelled finite-control banks for the n alternatives; these banks all operate on the *same frozen source order*, and the query's initial choice is itself counted. The total finite-control cost is bounded by

$$q_{\mathrm{cyc}}(n, 1; m) \le \sum_{K=0}^{n-1} q_K = O_m\left(\sum_{K=0}^{n-1} \frac{m^K}{K+1}\right) = O_m\left(\frac{m^n}{n}\right).$$

The final estimate follows by splitting the sum at $K = \lfloor n/2 \rfloor$ and using geometric decay in its upper half. Combined with the lower bound, this proves (101.2). □

Remark 101.8 (Scope and necessity of the short direct prefix). Without the literal prefix A, the d successive decode phases would each retain a controller buffer with M arbitrary symbols, apparently requiring dm^M states. Taking $r = \lceil \log_m(d+1) \rceil$ and emitting A first reduces the unchanged buffer to $M - r$ symbols, so all decode phases together cost at most m^M. The identity-bearing stage remains a genuine source record; this is neither a writable data register nor an extra stack.

101.1.7 A quantitative separation from fixed-owner staging

Suppose, in the same source-typed model, that the decision of *which owner* may be staged and whether the stage is occupied is fixed by the chosen query and source position, independently of all payload values. The protected stage can still hold the genuine symbol of the predetermined owner. Write $q^{\mathrm{fixed}}_{\mathrm{cyc}}(n, 1; m)$ for the minimum controller size under this restriction.

Proposition 101.9 (Linear advantage of identity-adaptive staging)**.** *For every fixed alphabet size $m \ge 2$,*

$$q^{\mathrm{fixed}}_{\mathrm{cyc}}(n, 1; m) = \Theta_m(m^{n-2}) \quad \textit{whereas} \quad q_{\mathrm{cyc}}(n, 1; m) = \Theta_m(m^n/n).$$

Consequently the adaptive choice of an actual protected source owner's identity saves a factor $\Theta_m(n)$ in control states, even though the stage's symbol alphabet and physical capacity are unchanged.

Proof. At the bottom-first query's indistinguishability cut, all m^{n-1} assignments of the first $n-1$ payloads must remain distinguishable. In the fixed-owner regime the stage configuration is either always empty or has a fixed owner with at most m possible payloads, giving $q \ge m^{n-2}$. Conversely, for a query with $K = j - 1 \ge 1$ deferred owners, preselect original owner K for the stage and store the first $K-1$ symbols literally in finite control. The stage record is read only when it actually appears. Once owner j and the later source

owners have been printed in order, emit the $K-1$ cached symbols and then the staged Kth symbol. A constant number of mode banks with variable-length controller words uses $O_m(m^{K-1})$ states for that query, including the scan and output phases. The query $j=1$ needs only constant control. Summing the query-specific banks gives $O_m(m^{n-2})$. The final ratio with Theorem 101.4 is $\Theta_m(n)$. □

101.1.8 Finite verification and scope

The mathematical lower and upper bounds are derived symbolically; finite checks are independent consistency tests, not substitutes for universal proofs or a certificate of exact finite-state minimality. The reported tests cover binary, ternary, and quaternary alphabets on short deferred prefixes, all cyclic source/query pairs for binary instances through $n = 12$ and ternary instances through $n = 8$ (a total of 163,921 pairs), and representative memory-inequality checks for fixed alphabet sizes and prefix lengths up to 4096. Independent bounded reconstructions also checked 125,096 short-prefix payload instances; these corroborate the stated encoder identities within the tested ranges. No claim about untested parameter ranges is inferred from these counts.

The theorem is relative to a model with a genuine, repeatedly inspectable physical owner label and a query available before scanning. These features are not automatically available in a general DPDA or in a single-use atom-register transducer. A fuller specialist literature comparison would be needed to settle historical priority; the statements here are mathematical claims about the specified model.

Open Problem

Exact finite-size cyclic readout state complexity. For fixed integers $m \geq 2$ and $n \geq 2$, determine the exact minimum $q_{\mathrm{cyc}}(n, 1; m)$ for the deterministic one-pass model of this paper with one adaptive, immutable, repeatedly inspectable genuine source record, all controller phases counted, and the query supplied before scanning. Theorem 101.4 determines its asymptotic order for fixed m, but not the exact value for every finite (n, m).

101.2 Transition to protected-contract synthesis

The conclusion of this branch is not that every semantic collapse must be realized in one canonical stack representation. It is the opposite: arbitrary exact pushdown encodings are allowed, and the invariant is a three-way alternative between source-local capacity, deeper exposure, and debt. The next chapter gives an independent exact example: the minimum future-stable readout-port state count for write-once sources, with its source-memory layer charged separately. The later protected-contract synthesis uses the broader distinction to compare resource points without confusing harmless recodings with genuine reductions.

Chapter 102

Future-Stable Readout Monitors and Exact Protected-Port Complexity

Research branch — Independent protected-realization study. *This chapter integrates the independent write-once-source readout study. Its exact index measures the autonomous observation port, not the uncharged exact source layer. The projection thresholds, cube-face counterexample, and partition-matroid structure are retained with their proofs. The protection and memory claims are not converted into unrestricted pushdown lower bounds.*

Connection to the General Theory. The source and its observation interface instantiate the protected-information and canonical-factor distinctions of Chapters 112 and 113. Unlike the same-model assembly profile of Chapter 113, the port cost counts only the independently charged finite-state monitor; it does not measure general one-stack compilation or δ_{asm}. The result is a separately publishable research manuscript [247].

102.1 Introduction

Monitoring concurrent sources requires more than recording their currently observable outputs: future assignments can distinguish source histories that presently look identical. The state complexity of deterministic interfaces is governed by future equivalence [246], while the combinatorics of Boolean projections and cube faces has an independent extremal theory [238, 239, 240]. We bring these ingredients together for a readout port that preserves the enabledness of every source assignment and recognizes designated complete labels.

The problem is formulated in a two-component architecture: an exact source layer retains its semantic configuration, and an autonomous, separately budgeted readout monitor provides protected observations. The resulting minimum port-state count is a concrete invariant with an exact projection formula. Work on semantic normalization and preservation of stack information provides broader representation-theoretic context [227, 228].

102.1.1 Main contributions

(i) An exact minimum-state index for future-stable readout ports that preserve both outputs and source-action enabledness.

(ii) Sharp minimum port budgets for two, three, and four protected complete labels in every source dimension, including the rigidity of four-label optimizers.

(iii) An all-dimensional pair of protected families with the same cube-face dimension vector but different readout-state complexity.

(iv) A truncated partition-matroid rank decomposition, submodularity, and an exact incremental cost formula for protected labels.

102.2 The physical readout-port contract

Fix $d \geq 2$, write $[d] = \{1, \dots, d\}$, and let $M = \{\bot, 0, 1\}^d$. The root is $\bot^d$. For each $i \in [d]$ and $a \in \{0, 1\}$, $X_{i,a}$ is defined precisely when coordinate i is unassigned and sets it to a. These actions are independent whenever both are enabled on distinct coordinates.

An uncharged exact source layer retains the full state $s \in M$. A separately charged readout port is a deterministic partial-action Moore machine $(H, h_0, (V_{i,a}), o)$ with a history-independent factor map $h : M \to H$ that intertwines the actions, reflects their enabledness, and outputs a distinct protected label k at the complete state $k \in K \subseteq \{0, 1\}^d$, with a generic output elsewhere. The port alone, not the exact source layer, answers protected observations. Its cost is $|H|$.

For $U \subseteq [d]$ let $\pi_U K$ be the set of restrictions of elements of K to U, and let $A_U = \{0, 1\}^U$. A protected set is *marginally surjective* if each coordinate assumes both values among its elements.

Theorem 102.1 (Exact autonomous-port index)**.** *For any nonempty $K \subseteq \{0, 1\}^d$, the least possible port-state count in the preceding category is*

$$I_d(K) = \sum_{U \subseteq [d]} \left(|\pi_U K| + \mathbf{1}_{\{\pi_U K \neq A_U\}} \right). \tag{102.1}$$

Proof. A partially assigned source state has a uniquely determined assigned-coordinate support U. Enabledness reflection prevents identification of states with distinct supports: an action on a coordinate in their symmetric difference is enabled at exactly one of them. Fix U. Every partial assignment in $\pi_U K$ has a protected completion; two distinct such assignments are distinguished by completing to one of their protected final labels, which is impossible from the other. A partial assignment outside $\pi_U K$ has no protected completion, and all such assignments have identical future protected outputs and enabledness, so they may be merged into one dead residual class for that support. These classes are stable under the action maps: extensions of a protected prefix are protected prefixes or become dead, and extensions of a dead prefix remain dead. Taking precisely these classes gives a well-defined deterministic port attaining the stated count. □

Remark 102.2 (Interpretation of the port cost). The port may have fewer states than M because it does not replace the exact source layer. In particular, $I_d(K)$ is not the minimum total physical state count of the composed implementation. This distinction is essential for interpreting any positive assembly defect.

102.3 Extremal protected budgets

For $2 \leq k \leq 2^d$ let

$$f_d(k) = \min\{I_d(K) : K \subseteq \{0, 1\}^d,\ |K| = k,\ K \text{ marginally surjective}\}.$$

For budgets admitting such K, define $\mathrm{Sat}_d(b) = \max\{k : f_d(k) \leq b\}$. The function Sat_d measures the jointly protectable label population in the stated readout-port architecture.

Theorem 102.3 (First two thresholds)**.** *For every $d \geq 2$,*

$$f_d(2) = 3 \cdot 2^d - d - 2, \qquad f_d(3) = 7 \cdot 2^{d-1} - d - 3.$$

Proof. A marginally surjective two-element set must consist of complementary words. A singleton projection has size two and costs two, and every support of size at least two projects to two words but omits an ambient assignment, costing three. Summing gives the first equality.

For three distinct rows, every nonconstant coordinate partitions the rows into a singleton versus a pair. Let $r \in \{2, 3\}$ be the number of occurring partition types, with multiplicities $n_1, \dots, n_r \geq 1$ summing to d. For a support of size at least two, the port contribution is three exactly when all its coordinates have one type and four otherwise; the empty-support contribution is one and singleton contributions are two. Thus

$$I_d(K) = 4 \cdot 2^d - 3 - d + r - \sum_{i=1}^{r} 2^{n_i}.$$

By strict convexity, the maximum of $\sum_i (2^{n_i} - 1)$ subject to $r \geq 2$ occurs for $r = 2$ with multiplicities $(d-1, 1)$. This gives the lower bound, attained by $\{0^d, 1^{d-1}0, 1^{d-1}1\}$. □

Theorem 102.4 (Four-label threshold and rigidity)**.** *For every $d \geq 2$,*

$$f_d(4) = 4 \cdot 2^d - 2d - 3.$$

Equality holds precisely for the protected sets equivalent, up to row relabeling and independent coordinate permutations and bit-complements, to $\{0^{d-1}, 1^{d-1}\} \times \{0, 1\}$.

Proof. Label the four protected rows and identify each nonconstant coordinate with its unordered nontrivial bipartition (cut type) of these rows. Let $\mathcal{M}$ contain supports of size at least two on which all coordinates have the same cut type. Let $\mathcal{F}$ contain supports of size at least three that distinguish all four rows. Directly comparing the contribution of each support in (102.1) with a baseline value four yields

$$I_d(K) = 4 \cdot 2^d - 2d - 3 + |\mathcal{F}| - |\mathcal{M}|.$$

For each occurring cut type q choose one representative coordinate r_q. Map a monochromatic support $U \in \mathcal{M}$ of type p to

$$\Phi(U) = U \cup \{r_q : q \neq p\}.$$

Since the full family of cut types distinguishes all four rows, $\Phi(U) \in \mathcal{F}$. Type p is the unique type repeated in $\Phi(U)$; all added types occur once. This recovers p and then U, so Φ is injective, proving $|\mathcal{F}| \geq |\mathcal{M}|$.

If at least three types occur, the set of the chosen representatives belongs to $\mathcal{F}$ but not to the image of Φ, giving strict inequality. With exactly two types, distinction of all four rows forces both cuts to be balanced and transversal. If each type occurs at least twice, a support with two coordinates of each type lies in $\mathcal{F}$ but outside the image. Thus equality forces multiplicities $(d-1, 1)$. Conversely, in this case every support in $\mathcal{F}$ is obtained by adjoining the unique singleton-type coordinate to a monochromatic support of the repeated type, and the injection is onto. □

102.4 Cube-face statistics and the missing-class correction

For a protected set K, let $c_j(K)$ count the j-dimensional faces of the d-cube meeting K and let

$$F_d(K) = \sum_{U \subseteq [d]} |\pi_U K| = \sum_{j=0}^{d} c_j(K), \qquad Q_d(K) = \#\{U \subseteq [d] : \pi_U K \neq A_U\}.$$

Then $I_d = F_d + Q_d$. The complete vector $(c_0, \dots, c_d)$ therefore does not necessarily determine the port complexity.

Proposition 102.5 (Face-vector non-determination in every dimension)**.** *Take* $K_A = \{000, 001, 010, 111\}$ *and* $K_B = \{000, 001, 110, 111\}$*. For each* $d \geq 3$ *define*

$$K_A^{(d)} = K_A \times \{0,1\}^{d-3}, \qquad K_B^{(d)} = K_B \times \{0,1\}^{d-3}.$$

Both sets are marginally surjective, and their entire face-dimension polynomials coincide:

$$\sum_j c_j(K_A^{(d)})t^j = \sum_j c_j(K_B^{(d)})t^j = (4 + 10t + 6t^2 + t^3)(2 + t)^{d-3}.$$

Nevertheless

$$I_d(K_A^{(d)}) - I_d(K_B^{(d)}) = 2^{d-3} > 0.$$

Proof. A face of a product cube is the product of faces, so its face-intersection polynomial multiplies by $2+t$ per added free coordinate. In dimension three both sets meet respectively 4, 10, 6, 1 faces of dimensions 0,1,2,3, yielding the claimed common polynomial. The extra coordinate projections of the Cartesian product contribute a factor 3^{d-3} to F and a factor 2^{d-3} to Q. In dimension three, $F_3(K_A) = F_3(K_B) = 21$, $Q_3(K_A) = 3$, and $Q_3(K_B) = 2$, which proves the assertion. □

102.5 Exact reduction to contained faces and VC shattering

The principal combinatorial quantities admit an exact classical interpretation. Let $C(K^c)$ be the number of all Boolean subcube faces entirely contained in the complement K^c, and let $\mathrm{Shat}(K)$ denote the family of coordinate supports U for which $\pi_U K = \{0,1\}^U$ (the usual shattered sets).

Proposition 102.6 (Classical-invariant decomposition)**.** *For every nonempty protected set,*

$$I_d(K) = 3^d + 2^d - C(K^c) - |\mathrm{Shat}(K)|.$$

Proof. Each Boolean coordinate of a face is fixed to zero, fixed to one, or free, so there are 3^d faces overall. A face meets K if and only if it is not entirely contained in K^c. Counting faces meeting K according to their fixed coordinate set gives $\sum_{U \subseteq [d]} |\pi_U K|$. The missing-class correction counts the coordinate subsets not shattered by K, which is $2^d - |\mathrm{Shat}(K)|$. Adding gives the identity. □

Remark 102.7 (Relation to cube-face inequalities). Bollobás–Radcliffe's face-isoperimetric theorem [238] characterizes extremal contained-face counts for fixed-size Boolean sets. The decomposition above separates that face statistic from the number of shattered coordinate supports. Our protected-family optimization requires both terms to be evaluated under the marginal-surjectivity condition.

102.6 The protected-set objective as a polymatroid rank

The exact index has a further structural expression. Although the matroid tools used below are classical, the decomposition records an additional property of this particular readout cost without relying on any fixed-cardinality case. For the empty protected set we extend the index by its all-generic Moore quotient, which has one state for every support, namely $I_d(\varnothing) = 2^d$.

Theorem 102.8 (Truncated-partition-matroid decomposition)**.** *For every* $K \subseteq \{0,1\}^d$*, including the empty set, set*

$$R_d(K) = \sum_{U \subseteq [d]} \min\big\{|\pi_U K|,\, 2^{|U|} - 1\big\}.$$

Then $I_d(K) = 2^d + R_d(K)$. Moreover R_d is a normalized, nondecreasing, integer-valued submodular function on the ground set $\{0,1\}^d$:

$$R_d(A) + R_d(B) \geq R_d(A \cup B) + R_d(A \cap B) \qquad (A, B \subseteq \{0,1\}^d).$$

More precisely, R_d is a sum of ranks of truncated partition matroids.

Proof. For integers $0 \leq m \leq n$ we have $m + \mathbf{1}_{\{m<n\}} = 1 + \min\{m, n-1\}$. Apply this to $m = |\pi_U K|$ and $n = 2^{|U|}$, and sum over all 2^d supports. For fixed U, partition the ground set $\{0,1\}^d$ into the fibers of π_U. The partition matroid which permits at most one representative from each fiber has rank $|\pi_U K|$ on K. Truncating this matroid to rank $2^{|U|} - 1$ gives the summand of R_d. Matroid ranks are normalized, nondecreasing and submodular, as are their sums. □

Proposition 102.9 (Exact diminishing returns and the final-label plateau)**.** *For $x \notin K$, the additional port cost is exactly*

$$I_d(K \cup \{x\}) - I_d(K) = \#\big\{U \subseteq [d] : \pi_U(x) \notin \pi_U K, \quad |\pi_U K| \leq 2^{|U|} - 2\big\}.$$

Consequently, if K is nonempty the increment is at most $2^d - d - 1$, and this upper bound is sharp for $K = \{0^d\}$ and $x = 1^d$. Every addition with $|K| \leq 2^d - 2$ has strictly positive cost, but the very last full word has cost zero: $I_d(\{0,1\}^d \setminus \{x\}) = I_d(\{0,1\}^d) = 3^d$.

Proof. In each rank summand of Theorem 102.8, adding x increases the number of projection classes precisely when $\pi_U(x) \notin \pi_U K$, and this increases the truncated rank by one precisely when the old rank is below $2^{|U|} - 1$. For nonempty K, supports of size at most one contribute no increment; the remaining number is $2^d - d - 1$, and the specified complementary singleton construction gains one on every such support. If $|K| \leq 2^d - 2$, the full support $U = [d]$ satisfies both requirements, giving an increment of at least one. If K consists of all but one complete word, every proper-support projection is already full, and the full-support rank is already at its cap. Thus the final addition costs zero. The complete source has precisely 3^d port classes, proving the final equality. □

Remark 102.10 (Submodularity and constrained extremals). The matroid truncation and rank submodularity used here are standard [242, 243]. They supply a structural description of the port objective. The fixed-cardinality problem with marginal surjectivity is a constrained extremal problem, and it is not solved by an unconstrained submodular-minimization routine.

Chapter 103

Contract Refinement and Protected-Information Capacity

Research branch — Protected Contracts and Pareto Synthesis. *Earlier chapters established exact semantic lifts, protected quotient/recoding theory, representation-independent state complexity, and several endpoint resource theorems. This branch reads those results together. Protected contracts order the information a realization must preserve; contract refinement then induces exact monotonicity and strictness principles for the attainable state–stack region.*

The preceding future-stable readout chapter (Chapter 102) gives a model-specific minimum for a separately budgeted observation port; it is not a state–stack Pareto bound for all exact lifts. The minimal-core theory becomes quantitatively useful once the protected interface is promoted to part of a representation contract. Refining the contract can only shrink the admissible realization class, and finite protected-role capacity gives explicit certificates that the Pareto frontier must move.

103.1 Protected representation contracts and refinement

Observable refinement alone cannot compare genuinely different physical re-coordinations. The representation class must also be part of the data.

Definition 103.1 (Observable refinement)**.** On one representation class, write

$$\lambda' \succeq_{\mathrm{obs}} \lambda$$

if there is a map h with

$$\lambda = h \circ \lambda'.$$

Then λ' remembers at least as much protected information as λ.

Proposition 103.2 (Quotient antitonicity)**.** *If $\lambda' \succeq_{\mathrm{obs}} \lambda$, every λ'-safe surjective quotient is λ-safe.*

Proof. If $\lambda' = \bar{\lambda}' F$, then $\lambda = h\bar{\lambda}' F$. □

Definition 103.3 (Protected representation contract)**.** A protected contract is

$$\mathfrak{K} = \big(\mathfrak{C}, \{\lambda_{\mathcal{L}}\}_{\mathcal{L}\in\mathfrak{C}}\big),$$

where $\mathfrak{C}$ is a BBR-closed class of finite-coordinate exact sectioned pushdown lifts of one semantic system and each $\lambda_{\mathcal{L}} : C_{\mathcal{L}}^{\mathrm{reach}} \to \Lambda$ is compatible with BBR conjugacy. We suppress the object subscript when harmless. Write $\mathfrak{K}' \succeq \mathfrak{K}$ if $\mathfrak{C}' \subseteq \mathfrak{C}$ and the protected interfaces refine objectwise on every lift of $\mathfrak{C}'$. Section 62.2 records the typing explicitly.

Define

$$\mathcal{R}(\mathfrak{K}) = \{(|Q_{\text{use}}(\mathcal{L})|, |\Gamma_{\text{use}}(\mathcal{L})|) : \mathcal{L} \in \mathfrak{C}\}$$

and

$$\text{Pareto}(\mathfrak{K}) = \text{Min}\, \mathcal{R}(\mathfrak{K}).$$

For finite antichains $A, B \subset \mathbb{N}^2$, write $A \preceq_P B$ if every $b \in B$ is componentwise dominated by some $a \in A$.

Theorem 103.4 (Contract-refinement Pareto monotonicity)**.** *If $\mathfrak{K}' \succeq \mathfrak{K}$, then*

$$\mathcal{R}(\mathfrak{K}') \subseteq \mathcal{R}(\mathfrak{K})$$

and

$$\boxed{\text{Pareto}(\mathfrak{K}) \preceq_P \text{Pareto}(\mathfrak{K}').}$$

Thus a stronger physical contract cannot improve the Pareto frontier.

Proof. Class inclusion gives resource-set inclusion. For any finer-contract Pareto point b, the same point belongs to the coarser resource set. Descending componentwise inside $\mathbb{N}^2$ yields a coarser minimal point $a \leq b$. □

103.2 Strictness from protected-information capacity

Weak monotonicity is automatic from class inclusion. Strictness requires an actual obstruction.

Definition 103.5 (Resource down-set)**.** For $r \in \mathbb{N}^2$, let

$$\downarrow r = \{s \in \mathbb{N}^2 : s \leq r \text{ componentwise}\}.$$

Theorem 103.6 (Strictness by down-set exclusion)**.** *Let $\mathfrak{K}' \succeq \mathfrak{K}$. If there exists an attainable coarse point $r_0 \in \mathcal{R}(\mathfrak{K})$ such that*

$$\mathcal{R}(\mathfrak{K}') \cap \downarrow r_0 = \varnothing,$$

then

$$\boxed{\text{Pareto}(\mathfrak{K}) \prec_P \text{Pareto}(\mathfrak{K}').}$$

Proof. Theorem 103.4 gives weak dominance. Since r_0 is coarse-attainable, some coarse Pareto point a satisfies $a \leq r_0$. Reverse dominance would give a fine Pareto point $b \leq a \leq r_0$, contradicting the down-set exclusion. □

Corollary 103.7 (Protected-role capacity certificate)**.** *Suppose a fine contract forces N protected roles to occupy distinct local-observation cells, while every realization with resource vector at most r has at most* $\text{cap}(r)$ *such cells, with* cap *monotone. If a coarse attainable point r_0 satisfies*

$$N > \text{cap}(r_0),$$

then r_0 is a down-set exclusion witness and Pareto refinement is strict.

This is the main conceptual bridge:

$$\boxed{\begin{array}{l}\text{protected-information separation} \\ \quad \Longrightarrow \text{ resource down-set exclusion} \\ \qquad \Longrightarrow \text{ strict Pareto penalty.}\end{array}}$$

Chapter 104

Exact Pareto Frontiers for Rational Balance Lifts

We now apply the contract order to the rational-balance endpoint theorems already established in the monograph. The point is synthesis rather than re-proving those endpoint results: broad, deferred, raw residual-faithful, and phase-local architectures become comparable only after their protected information and recoding conventions have been typed.

104.1 Rational balance application I: an exact contract state premium

The abstract SCQ/BBR/protected-contract results above are proved entirely within this development. The rational-balance applications now combine that new comparison framework with four exact endpoint resource theorems from the preceding normalization theory. Section 104.3 states those inputs and records their proof interfaces and scope; in particular, the unrestricted state lower bound and the raw/phase-local optima are not being re-labelled as new results of the present chapter sequence.

Fix coprime $p, q \geq 1$ and a mandatory right-endmarker convention. Let $\mathfrak{K}_{\mathrm{broad}}$ be the broad exact-pushdown contract for the residual semantics of $L_{p,q}$. The previously established broad-class state theorem gives

$$Q_{\min}(\mathfrak{K}_{\mathrm{broad}}) = \max(p, q).$$

A three-symbol exact realization attains this value.

Let $\mathfrak{K}_{\mathrm{def}}$ be the canonical deferred-SEARCH literal-demand contract. Its previously established exact state minimum is

$$D_{p,q} = 2\left\lfloor \frac{p+q-1}{2} \right\rfloor + 1,$$

also attained with three total stack symbols.

Theorem 104.1 (Exact broad/deferred state premium)**.** *The exact state premium of the deferred literal-demand contract over the broad exact-pushdown contract is*

$$\boxed{\Delta_Q(p, q) = D_{p,q} - \max(p, q) = \min(p, q) - \mathbf{1}_{\{p+q \text{ even}\}}.}$$

It is strict except when $\min(p, q) = 1$ *and* $p + q$ *is even.*

Proof. If $p+q$ is odd, $D_{p,q} = p+q$, giving $\Delta_Q = \min(p, q)$. If $p+q$ is even, $D_{p,q} = p+q-1$, giving $\Delta_Q = \min(p, q) - 1$. The equality classification follows immediately. □

The two attaining constructions use the same total stack alphabet size 3, so the displayed witness points isolate a control-state premium rather than a stack-alphabet artifact.

104.2 Rational balance application II: exact two-resource frontiers

Let

$$w_{\max} = \max(p, q).$$

In raw residual-faithful Model R, let C be the number of work symbols; the total stack alphabet has size $C + 1$ including Z_0. The exact per-C state minimum is

$$S_{\mathrm{raw}}(C) = \begin{cases} p + q, & C = 1, \\ w_{\max}, & C \geq 2. \end{cases}$$

Theorem 104.2 (Exact raw two-resource frontier)**.** *Across all finite work-alphabet sizes,*

$$\boxed{\mathrm{Pareto}(\mathfrak{K}_{\mathrm{raw}}) = \{(p + q, 2), (w_{\max}, 3)\}.}$$

Proof. The point $C = 1$ gives $(p + q, 2)$. The point $C = 2$ gives $(w_{\max}, 3)$ and dominates every $C > 2$ point $(w_{\max}, C + 1)$. Since $w_{\max} < p + q$, the two displayed points are incomparable. □

The same two endpoint pairs are now known to be the complete *unrestricted* state–stack frontier for $L_{p,q}$ by Theorem 98.15. The raw theorem above is retained because this chapter compares protected representation contracts and therefore needs the stronger statement that the endpoint is attained inside the raw residual-faithful category, not merely somewhere in the unrestricted category.

Now fix $B \geq w_{\max}$ and write every residual as $r = s + Bz$, $0 \leq s < B$. Define

$$h_+(B) = \frac{p}{\gcd(B, p)}, \qquad h_-(B) = \frac{q}{\gcd(B, q)}, \qquad H(B) = h_+(B) + h_-(B).$$

The normalized phase-local contract requires the current signed quotient phase to be visible through control state plus one work symbol. Its exact per-C state minimum is

$$Q^{\min}_{\mathrm{phase}}(B, C) = B \left\lceil \frac{H(B)}{C} \right\rceil.$$

Theorem 104.3 (Exact phase-local Pareto staircase)**.** *Let $H = H(B)$ and*

$$\mathcal{B}_H = \{1\} \cup \left\{ 2 \leq C \leq H : \left\lceil \frac{H}{C} \right\rceil < \left\lceil \frac{H}{C - 1} \right\rceil \right\}.$$

Then

$$\boxed{\mathrm{Pareto}(\mathfrak{K}_{\mathrm{phase}}(B)) = \left\{ \left(B \left\lceil \frac{H}{C} \right\rceil, C + 1 \right) : C \in \mathcal{B}_H \right\}.}$$

Proof. The state coordinate is nonincreasing in C, while the alphabet coordinate is strictly increasing. A point inside a plateau of $\lceil H/C \rceil$ is dominated by the smallest C in that plateau. At every strict drop, no earlier point has as small a state coordinate and no later point has as small an alphabet. For $C > H$ the state coordinate is already B, so $C = H$ dominates all later points. Exact attainability holds for every C by the normalized phase-local constructions. □

At fixed C, the phase state premium is

$$\Delta^{\mathrm{phase}}_Q(B, C) = B \left\lceil \frac{H(B)}{C} \right\rceil - S_{\mathrm{raw}}(C) \geq 0.$$

For $C \geq 2$, equality holds exactly when

$$B = w_{\max}, \qquad C \geq H(B).$$

Since p, q are coprime, $B = w_{\max}$ gives

$$H(B) = \min(p, q) + 1.$$

Thus additional stack symbols can absorb the stronger protected phase information while retaining raw-optimal state count.

104.2.1 The audited raw-safe nesting

There is one important scope distinction. Model R forbids epsilon-normalization on encoded residual configurations. The phase-local optimum is raw and epsilon-free for $C \geq 2$, but the $C = 1$ optimum uses one forced bottom-cell normalization. Hence full $C \geq 1$ phase-local and raw classes are not literally nested.

Restrict to at least two work symbols. Let $\mathfrak{K}_{\text{raw}}^{\geq 2}$ be Model R with $C \geq 2$, and let $\mathfrak{K}_{\text{phase}}^{\text{raw}}(B)$ be the phase-local contract restricted to $C \geq 2$.

Theorem 104.4 (Exact nested raw-safe comparison)**.** *The restricted phase-local contract refines the restricted raw contract, and*

$$\text{Pareto}(\mathfrak{K}_{\text{raw}}^{\geq 2}) = \{(w_{\max}, 3)\}.$$

The two restricted Pareto frontiers are equal exactly when

$$\boxed{B = w_{\max} \quad \textit{and} \quad \min(p, q) = 1.}$$

In every other case the refinement is strict.

Proof. The $C \geq 2$ phase constructions are raw, so the contract inclusion is genuine. Every phase point has state at least $B \geq w_{\max}$ and alphabet at least 3, hence $(w_{\max}, 3)$ dominates. Reverse dominance requires a phase point at most $(w_{\max}, 3)$, so $C = 2$, $B = w_{\max}$, and $\lceil H/2 \rceil = 1$, equivalently $H = 2$. At $B = w_{\max}$, coprimality gives $H = \min(p, q) + 1$, hence $\min(p, q) = 1$. □

Theorem 104.5 (Direct full-frontier comparison)**.** *Over all $C \geq 1$,*

$$\text{Pareto}(\mathfrak{K}_{\text{raw}}) \preceq_P \text{Pareto}(\mathfrak{K}_{\text{phase}}(B)),$$

and the dominance is strict except exactly at

$$(p, q, B) = (1, 1, 1),$$

where both frontiers equal

$$\{(2, 2), (1, 3)\}.$$

Proof. For the $C = 1$ phase point,

$$BH(B) \geq 2B \geq 2M \geq p + q,$$

so the raw point $(p + q, 2)$ dominates it. For every phase point with $C \geq 2$, the raw point $(w_{\max}, 3)$ dominates because its state coordinate is at least $B \geq w_{\max}$. Reverse dominance forces a phase point below $(p + q, 2)$, necessarily the $C = 1$ point with $BH(B) \leq p + q$. Equality throughout the displayed chain occurs only at $(1, 1, 1)$. □

104.3 Endpoint resource theorems used in the rational-balance application

The rational-balance application combines previously established endpoint optima with the new protected-contract formalism. The logical inputs are:

1. in the broad deterministic pushdown/right-endmarker exact-lift class, the exact state minimum is $\max(p, q)$ and is attained with two work symbols plus the bottom marker;

2. in the canonical deferred-SEARCH literal-demand class, the exact state minimum is

$$D_{p,q} = 2\left\lfloor \frac{p+q-1}{2} \right\rfloor + 1,$$

   again attainable with three total stack symbols;

3. in raw residual-faithful Model R, with C work symbols,

$$S_{\mathrm{raw}}(C) = \begin{cases} p+q, & C = 1, \\ \max(p, q), & C \geq 2; \end{cases}$$

4. in the normalized phase-local class,

$$Q^{\min}_{\mathrm{phase}}(B, C) = B\left\lceil \frac{H(B)}{C} \right\rceil, \qquad H(B) = \frac{p}{\gcd(B, p)} + \frac{q}{\gcd(B, q)}.$$

For reviewability, the proof interfaces are as follows. The deferred lower bound exposes the $2h + 1$ residual debts $-hm, \ldots, 0, \ldots, hm$ at the same exhausted work stack; a common distinguishing continuation then forces distinct control states, while the centered rail attains the bound. The raw frontier uses positive- and negative-side synchronization to give $|Q| \geq \max(p, q)$ for every nonempty work alphabet, strengthens this to $p+q$ with one work symbol by cross-sign unary separation, and is attained by the one- and two-work-symbol residual encodings. The normalized phase-local formula counts, in each residue class modulo B, the $H(B)$ signed phase roles that must be exposed through C work symbols and then sums the sharp per-class bound; the matching construction is raw for $C \geq 2$. The broad unrestricted state theorem combines the positive- and negative-side phase-exposure bounds $|Q| \geq p$ and $|Q| \geq q$ with the explicit three-symbol residual realization having $\max(p, q)$ states.

These endpoint results are inputs, not re-labelled contributions of the present chapter sequence. They are included here at the level needed to audit every numerical comparison, while the new results are the SCQ/BBR reduction theory, protected-contract order, Pareto monotonicity/strictness criteria, and the synthesis of the endpoint optima into exact contract frontiers. The $C = 1$ phase optimum uses one forced bottom-cell normalization and is therefore compared directly with the full raw frontier rather than being treated as a literal subclass of raw Model R. For $C \geq 2$ the phase construction is raw, which is precisely the nested scope of Theorem 104.4.

104.4 Synthesis and scope

The resulting frontiers are exact inside their declared contracts. In particular, the raw Model-R frontier and the normalized phase-local staircase are not promoted to an unrestricted two-resource theorem for all recognizers. The $C = 1$ phase optimum uses one forced bottom-cell normalization, so the full phase-local class is compared directly with raw Model R rather than asserted to be nested inside it; genuine raw/phase nesting is used only for the audited $C \geq 2$ subfamily. This closes the representation-Pareto synthesis. The next Part changes the question from resource minimization to existential decomposition and locates the Prime-DOCA algorithmic boundary; only after that boundary is established does the monograph return to the original 2:1 machine.

Part IX

State Decomposition and the Algorithmic Boundary

Part guide. **Question.** Beyond optimizing a fixed presentation, what is the algorithmic boundary between existential decomposition and comparison of fixed one-counter presentations? Can one decide whether a deterministic one-counter language presentation is an intersection of strictly smaller superlanguage presentations, how many factors are needed for hardness, what happens when a fixed deterministic one-counter net is compared directly with a fixed nondeterministic one-counter net, and what changes when an explicit zero test is restored on the deterministic side? **Main result.** Totzke's Prime-DOCA problem is undecidable and the hardness already appears at the smallest nontrivial fixed budget $k = 2$. In contrast, both fixed-comparison problems treated here are decidable: Problem 24.04(B) for deterministic versus nondeterministic one-counter nets, and Problem 24.04(A) for a deterministic zero-testing one-counter automaton versus a nondeterministic one-counter net; the latter remains decidable under the stated epsilon-transition extension. **Why here.** This is an algorithmic boundary, not another resource formula: fixed comparison, unrestricted existential decomposition, and fixed-budget decomposition separate sharply even within closely related one-counter models. **Dependency.** It follows the state/resource complexity program. The B branch introduces exact envelope/requirement and WQO machinery under global net monotonicity; the A branch then replaces that monotonicity by zero-cut/first-return compactness and, for silent moves, finite weak macros and stable/reset normalization.

Chapter 105

From State Complexity to State-Primality

Research branch — From exact resource cost to decomposition. *Exact state complexity asks how small one recognizer may be. State-primality asks whether a given presentation can instead be decomposed into finitely many strictly smaller superlanguage recognizers. The threshold is the number of control states of the presented automaton, so this is a presentation-relative decomposition problem.*

Totzke's Prime-DOCA formulation asks whether this property is decidable for deterministic one-counter automata [99]. The theorem below gives the unrestricted boundary result. Chapters 109–110 sharpen it by showing that unbounded factor supply is unnecessary: undecidability already holds for exactly two smaller factors.

Theorem 105.1 (Main theorem)**.** *The Prime-DOCA decision problem is undecidable. More precisely, from a deterministic two-counter Minsky machine M one can effectively construct a DOCA T_M such that*

$$\boxed{M \text{ halts} \iff T_M \text{ is prime.}}$$

The target T_M can be chosen complete, real-time, epsilon-free, with one work symbol, a permanent bottom marker, all control states final, and only unit counter updates $-1, 0, +1$.

The proof does not encode two source counters simultaneously in the target counter. Instead, it separates a proposed source computation into three predicates and lets the *intersection structure itself* enforce simultaneous correctness. This is the main conceptual shift.

The proof is by an effective reduction from deterministic two-counter halting. Its two structural ingredients are developed in the next chapter: a transcript decomposition into one regular and two one-counter predicates, and an elementary-abelian permutation core that is prime with one rejected point but admits exact half-size quotient factors as soon as an additional point fails.

105.1 Preliminaries

105.1.1 Deterministic one-counter automata

We use a counter presentation equivalent to a deterministic pushdown automaton with a permanent bottom marker Z and one work symbol A. A configuration is a pair

$$(q, h) \in Q \times \mathbb{N},$$

where h is the number of A symbols above Z. A transition can inspect whether $h = 0$ or $h > 0$, consume either one input symbol or ε, change the control state, and update the

counter by an allowed finite amount. In all automata constructed in this branch, transitions are real-time and use only updates $-1, 0, +1$.

For the ambient model we allow epsilon transitions. We assume the standard deterministic pushdown convention: from a fixed control/top situation there is at most one enabled epsilon transition, and if an epsilon transition is enabled then no input-consuming transition is enabled from the same situation. Thus, once the unread input is fixed, the continuation from a physical configuration is unique.

Acceptance is by final control and zero counter after the entire input has been consumed. Formally, after the last input symbol has been consumed, the unique deterministic post-input epsilon path (finite or infinite) is followed; a word is accepted exactly when some configuration on that path has the form $(q, 0)$ with q final. This is the reachability formulation of final-state-plus-zero acceptance and makes post-prefix epsilon cuts explicit. The constructed target and factors are epsilon-free and have every control state final, so for them acceptance is exactly the condition that the real-time run ends at height zero.

Definition 105.2 (Prime and composite DOCA instances)**.** Let A be a DOCA with $n = |Q_A|$ control states. We call A *composite* if there exist an integer $k \geq 1$ and DOCAs $B_1, \ldots, B_k$ such that

$$L(A) \subseteq L(B_i), \qquad |Q_{B_i}| < n \quad (1 \leq i \leq k),$$

and

$$L(A) = \bigcap_{i=1}^{k} L(B_i).$$

Otherwise A is *prime*. The decision problem asking whether a given DOCA is prime is denoted $\mathrm{PRIME}_{\mathrm{DOCA}}$.

Definition 105.3 (Fixed-budget DOCA compositionality)**.** Let A be a DOCA with $n = |Q_A|$. For a fixed integer $k \geq 1$, call A *k-factor composite* if there are DOCAs $B_1, \ldots, B_k$ such that

$$|Q_{B_i}| < n \quad (1 \leq i \leq k), \qquad L(A) = \bigcap_{i=1}^{k} L(B_i).$$

We write $\mathsf{Factor\text{-}DOCA}_k$ for the corresponding decision problem. The equality automatically implies $L(A) \subseteq L(B_i)$ for every factor. No distinctness condition is imposed on the factors.

Thus unrestricted primality quantifies over every finite factor budget, whereas $\mathsf{Factor\text{-}DOCA}_k$ fixes the budget in advance. The case $k = 2$ is the smallest budget capable of expressing a genuine intersection decomposition.

The superlanguage condition is stated separately because it is part of the original formulation; once the intersection equality holds it is of course automatic. The size measure is the number of control states of the presented automaton, exactly as in the original problem formulation [99]. We do not assume that an arbitrary input DOCA is minimal. Thus primality here is a property of the presentation $(A, |Q_A|)$ rather than an invariant defined using the minimal DOCA size of $L(A)$. In the halting branch of our reduction the constructed presentation will in fact be state-minimal (Corollary 108.3).

105.1.2 Deterministic two-counter machines

A deterministic two-counter Minsky machine has a finite control set and two nonnegative counters c_1, c_2. We use instructions of the forms

increment c_j and go to q',
if $c_j = 0$ go to q_0, else decrement c_j and go to q_+,
halt.

The machine starts in a designated state p_0 with both counters zero. Halting of deterministic two-counter machines from zero is r.e.-complete, hence undecidable; this is a standard consequence of the universality of two-counter Minsky machines [100]. The fixed-input r.e.-hardness form follows by compiling the finite input into the control of the simulated machine.

We use a normalization that makes the accepting counter values explicit. After the original halt instruction, enter fresh deterministic drain control, decrement c_1 to zero, then decrement c_2 to zero, and only then enter a fresh state p_f with no outgoing machine transition. Therefore

$$M \text{ halts} \iff (p_f, 0, 0) \text{ is reached by the normalized machine.} \tag{105.1}$$

We write Q_M for the control states of this normalized machine.

Chapter 106

Transcript Encoding and the Elementary-Abelian Prime Core

A single one-counter target does not attempt to simulate both source counters simultaneously. Instead the proposed source computation is separated into three independently checkable predicates, while the intersection structure of the eventual factors enforces conjunction. The finite prime/composite switch is supplied by an elementary-abelian translation core.

106.1 A transcript lemma for deterministic two-counter halting

The reduction will not attempt to store both source counters in one target run. Instead, a proposed source computation is split into one regular predicate and two one-counter predicates.

106.1.1 Branch-labelled transition alphabet

Let Δ_M contain one symbol for every *branch-labelled* transition of the normalized machine. An increment instruction on counter j from p to q contributes

$$\tau = (p, \mathrm{inc}_j, q).$$

A conditional instruction

$$\text{if } c_j = 0 \text{ goto } q_0 \text{ else } c_j := c_j - 1;\ \text{goto } q_+$$

contributes two distinct symbols

$$\tau_0 = (p, \mathrm{zero}_j, q_0), \qquad \tau_+ = (p, \mathrm{dec}_j, q_+).$$

For $\tau \in \Delta_M$, let $\mathrm{src}(\tau)$ and $\mathrm{tgt}(\tau)$ denote its source and target control states. For $j \in \{1, 2\}$, let $\mathrm{op}_j(\tau) \in \{\mathrm{inc}, \mathrm{zero}, \mathrm{dec}, \mathsf{skip}\}$ be the operation induced on counter c_j: a transition acting on the other counter has operation skip on c_j.

106.1.2 The three predicates

Definition 106.1 (Regular control predicate)**.** A word $z = \tau_1 \cdots \tau_k \in \Delta_M^*$ with $k \geq 1$ belongs to R_M if

$$\mathrm{src}(\tau_1) = p_0, \qquad \mathrm{tgt}(\tau_i) = \mathrm{src}(\tau_{i+1})\ (1 \leq i < k), \qquad \mathrm{tgt}(\tau_k) = p_f.$$

We normalize so that $p_0 \neq p_f$; in particular the empty word is not in R_M. Clearly R_M is regular.

Definition 106.2 (Counter predicates)**.** For $j \in \{1, 2\}$, define a deterministic height trajectory on $z = \tau_1 \cdots \tau_k$. Start with $h_0 = 0$. At step i:

- if $\mathrm{op}_j(\tau_i) = \mathrm{inc}$, set $h_i = h_{i-1} + 1$;
- if $\mathrm{op}_j(\tau_i) = \mathrm{zero}$, require $h_{i-1} = 0$ and set $h_i = 0$;
- if $\mathrm{op}_j(\tau_i) = \mathrm{dec}$, require $h_{i-1} > 0$ and set $h_i = h_{i-1} - 1$;
- if $\mathrm{op}_j(\tau_i) = \mathsf{skip}$, set $h_i = h_{i-1}$.

We put $z \in C_j$ if every guard is satisfied and $h_k = 0$.

Each C_j is a deterministic one-counter language. In partial form it needs only one logical checker state; a complete presentation routes illegal behavior to a permanent-positive poison role.

Lemma 106.3 (Two-Counter Transcript Lemma)**.** *For the normalized deterministic two-counter machine M,*

$$\boxed{M \text{ halts} \iff R_M \cap C_1 \cap C_2 \neq \varnothing.}$$

Proof. If M halts, take the branch-labelled transition sequence of the unique normalized computation from $(p_0, 0, 0)$ to $(p_f, 0, 0)$. The finite-control path lies in R_M, and its projections onto the two counters satisfy exactly the definitions of C_1 and C_2.

Conversely, let $z = \tau_1 \cdots \tau_k \in R_M \cap C_1 \cap C_2$. We prove by induction on i that after $\tau_1 \cdots \tau_i$, the heights maintained by C_1 and C_2 equal the two counter values of the genuine machine computation following the control path forced by R_M. The claim is immediate for increments and for operations on the other counter. A symbol τ with $\mathrm{op}_j(\tau) = \mathrm{zero}$ is admitted by C_j exactly when the corresponding height is zero, hence exactly when the zero branch is enabled in M. A symbol with $\mathrm{op}_j(\tau) = \mathrm{dec}$ is admitted exactly at positive height and then both semantics decrement once. Thus every branch-labelled transition appearing in z is genuinely enabled. The path ends at p_f, and both projected counters end at zero. By (105.1), the original machine halts. □

106.2 The elementary-abelian prime core and failure lattice

Choose a power of two

$$N = 2^d$$

large enough that

$$N - 3 \geq |Q_M|, \qquad N/2 \geq |Q_M| + 1. \tag{106.1}$$

Let

$$V = (\mathbb{Z}_2)^d, \qquad |V| = N,$$

and let $E = \{e_1, \ldots, e_d\}$ be an alphabet indexed by the standard basis. Each core letter e_i acts on V by translation $v \mapsto v + e_i$.

Define

$$P_V = \{x \in E^* : \delta(0, x) \neq 0\}. \tag{106.2}$$

Lemma 106.4 (Elementary-Abelian Prime-Core Lemma)**.** *The translation DFA for P_V is minimal of index N, is a permutation DFA, has exactly $N - 1$ accepting states, and P_V is DFA-prime.*

Proof. Every vector of V is reachable because the basis translations generate V. Let $u \neq v$ be two states. Choose a core word whose total translation is u. Since every element of V is self-inverse, appending this word sends u to 0 and sends v to $v + u \neq 0$. Hence the states are distinguishable and the DFA has index N.

Every input letter acts as a permutation, and the accepting set is $V \setminus \{0\}$, so there are exactly $N - 1$ accepting states. Theorem 7.13 of Kupferman and Mosheiff states that a permutation DFA of index n with $n - 1$ accepting states is prime [98]. Applying it with $n = N$ proves the claim. □

The next observation is elementary but is the structural heart of the reduction.

Lemma 106.5 (Quotient Failure-Factor Lemma)**.** *Let $R \subseteq V$ be a rejecting set with $0 \in R$. If $R = \{0\}$, the corresponding core section is P_V and is prime. If $|R| \geq 2$, then the section accepting $V \setminus R$ is an intersection of superlanguages recognized by quotient DFAs with exactly $N/2$ states.*

Proof. For every nonzero $r \in R$, put

$$H_r = \langle r \rangle = \{0, r\}.$$

The quotient translation action on V/H_r has $N/2$ states. Let the quotient factor reject exactly the zero coset H_r and accept all other cosets. Because $H_r \subseteq R$, every state accepted by the original section is accepted by this quotient factor, so the factor is a superlanguage.

Moreover,

$$\bigcup_{r \in R \setminus \{0\}} H_r = \bigcup_{r \in R \setminus \{0\}} \{0, r\} = R.$$

Hence the intersection of these quotient factors rejects exactly R. □

Remark 106.6 (Why exponent two is ideal). The identity $H_r = \{0, r\}$ is what makes every extra rejected point generate a two-point failure coset while keeping the quotient exactly half-size. No subgroup-intersection calculation is hidden in the argument: the rejected set is recovered simply as a union of these two-element subgroups.

Chapter 107

The Target DOCA: Primality, Failure Factors, and Exact Decomposition

We now couple the transcript predicates to the finite group core. In the halting case a fixed-suffix zero cut exposes the prime core section even against epsilon-enabled smaller factors. In the nonhalting case every nonzero failure point supplies an explicit quotient factor with exactly half the target states, and these factors intersect back to the target language.

107.1 Construction of the target DOCA

107.1.1 Coloring and alphabet

Partition the nonzero core points as

$$V \setminus \{0\} = G_R \sqcup G_1 \sqcup G_2, \tag{107.1}$$

with $G_1, G_2 \neq \varnothing$ and $|G_R| \geq |Q_M|$. This is possible by (106.1). Fix an injection

$$\phi : Q_M \hookrightarrow G_R.$$

The target state set is exactly V; there are no auxiliary control states. Every target state is final.

The full alphabet is

$$\Sigma = E \sqcup \{@\} \sqcup \Delta_M \sqcup \{\$\}. \tag{107.2}$$

The markers @ and \$ package a source transcript. The counter-color modes do not need a separate finite-control bit recording whether @ has appeared. This is intentional: the reduction needs exact behavior on valid transcript suffixes and exact factor simulation on arbitrary words, not a separate global syntax recognizer in each color.

We now define T_M. Its counter height is the number of work symbols above the permanent bottom marker. A positive configuration in control state 0 is the permanent poison mode: on every input symbol it remains in state 0 and preserves positive height.

107.1.2 Core and poison transitions

For every $v \in V$ and $e_i \in E$,

$$(v, 0) \xrightarrow{e_i} (v + e_i, 0). \tag{107.3}$$

At positive height, every core letter enters permanent poison while keeping the counter positive.

At the zero anchor $(0,0)$, every non-core symbol enters permanent-positive poison. In particular,

$$(0,0) \xrightarrow{@} (0,1).$$

A transition symbol or \$ at $(0,0)$ is totalized in the same way.

107.1.3 Counter-color transitions

Fix $v \in G_j$, where $j \in \{1,2\}$. At zero height the control state remains v except for poison or successful reset:

input	zero-height action at $v \in G_j$	purpose
@	stay $(v,0)$	begin/ignore marker
τ: $\mathrm{op}_j(\tau) = \mathrm{inc}$	$(v,0) \to (v,1)$	increment checked counter
τ: $\mathrm{op}_j(\tau) = \mathrm{zero}$	stay $(v,0)$	legal zero branch
τ: $\mathrm{op}_j(\tau) = \mathrm{dec}$	enter positive poison	illegal decrement at zero
τ: $\mathrm{op}_j(\tau) = \mathsf{skip}$	stay $(v,0)$	projection ignores other counter
\$	reset to $(0,0)$	successful zero terminal

At positive height while the control is v:

input	positive-height action	purpose
τ: $\mathrm{op}_j(\tau) = \mathrm{inc}$	increment	checked increment
τ: $\mathrm{op}_j(\tau) = \mathrm{dec}$	decrement	checked positive branch
τ: $\mathrm{op}_j(\tau) = \mathrm{zero}$	poison, preserve positivity	illegal zero branch
τ: $\mathrm{op}_j(\tau) = \mathsf{skip}$	preserve height	projection ignores other counter
@, \$, or $e_i \in E$	poison, preserve positivity	forbidden in positive checker mode

If a legal decrement returns to height zero, the next input symbol is governed again by the zero-height rules at control state v.

107.1.4 Regular-color transitions

Fix an anchor $v \in G_R$. At zero height, only @ starts regular mode:

$$(v,0) \xrightarrow{@} (\phi(p_0),1). \tag{107.4}$$

Any $\tau \in \Delta_M$ or \$ at zero height enters poison.

At positive height one, if the control is $\phi(p)$, then on a transition symbol τ with $\mathrm{src}(\tau) = p$ the target moves to $\phi(\mathrm{tgt}(\tau))$ and keeps height one. A wrong transition symbol, @ or a core letter enters poison. On \$, the machine pops the unique mode symbol and resets to $(0,0)$ exactly when $p = p_f$; otherwise it enters poison. Thus regular mode uses counter height one only as a mode bit.

Every transition not explicitly specified above is routed to state 0 while making or keeping the counter positive.

Lemma 107.1 (Target Construction Lemma)**.** *T_M is a complete real-time deterministic one-counter automaton with exactly N control states, one work symbol above a permanent bottom marker, all control states final, and counter updates in $\{-1,0,+1\}$.*

Proof. The state set is exactly V, hence has size N. The cases above are disjoint by control state, zero/positive status, and input symbol. Every unspecified case is totalized into the existing positive poison role. No epsilon transition or extra control state is introduced, and every update is a unit increment, unit decrement, or no-op. □

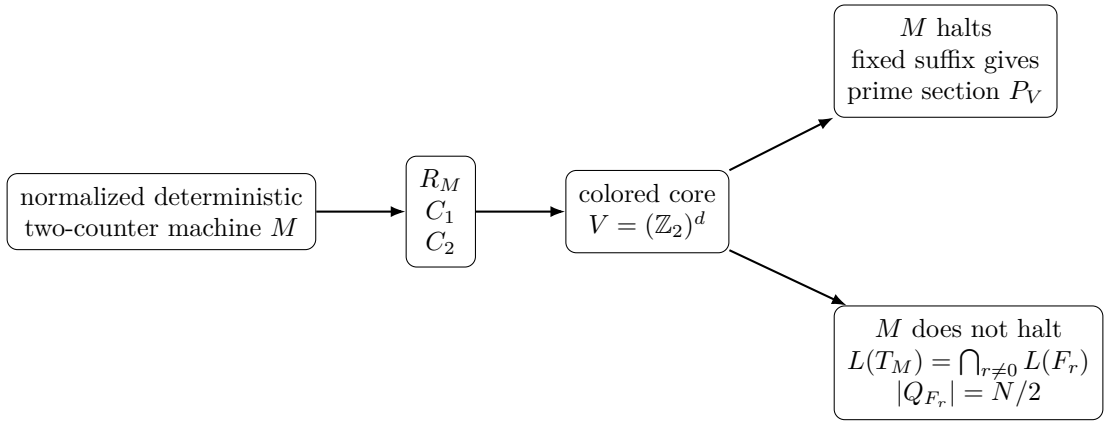


Figure 107.1: Proof architecture. The target never stores both source counters simultaneously. The halting direction extracts a prime finite section; the nonhalting direction is witnessed by explicit quotient factors.

107.2 Halting implies primality

Assume M halts. By Lemma 106.3, choose

$$z = \tau_1 \cdots \tau_k \in R_M \cap C_1 \cap C_2$$

and define the fixed suffix

$$u = @z\$. \tag{107.5}$$

For a core word $x \in E^*$, let $v = \delta(0, x)$ in the translation core.

If $v = 0$, the first symbol @ enters target poison, so xu is rejected. If $v \in G_1$, the target follows the valid counter-1 projection and returns to zero on \$. The case $v \in G_2$ is symmetric. If $v \in G_R$, regular mode follows the valid control path from p_0 to p_f and returns to zero on \$. Hence

$$\boxed{xu \in L(T_M) \iff \delta(0, x) \neq 0.} \tag{107.6}$$

Thus the fixed-suffix section is exactly P_V.

The remaining issue is that a hypothetical smaller factor in a Prime-DOCA decomposition may use epsilon transitions even though T_M does not. We therefore isolate the section argument carefully.

Lemma 107.2 (Epsilon-Robust Zero-Cut Section Lemma)**.** *Let B be a deterministic one-counter automaton in the ambient model, possibly with epsilon transitions, and suppose $E^* \subseteq L(B)$. Fix a suffix $u \in \Sigma^*$. Then*

$$S_u(B) = \{x \in E^* : xu \in L(B)\}$$

is recognized by a DFA with at most $|Q_B|$ states.

Proof. If $u = \varepsilon$, then $S_u(B) = E^*$ and the claim is trivial, so assume $u \neq \varepsilon$.

For each $x \in E^*$, consider the unique post-input epsilon trajectory that starts immediately after the last symbol of x has been consumed. Since $x \in L(B)$, this trajectory visits a final zero configuration. Let

$$c_x = (q_x, 0)$$

be the first such configuration.

We first isolate the point on which epsilon-robustness depends.

Forced-cut claim. For every continuation $y \in \Sigma^*$, the run of B on xy reaches the same physical configuration c_x after consuming exactly x and before consuming any symbol of y.

Indeed, every transition taken between the configuration immediately after x and the occurrence of c_x is an epsilon transition on the post-input run of x. Each such transition is forced by determinism; and whenever it is enabled, epsilon/input exclusivity forbids an input-consuming transition from the same control/top situation. Pending input therefore cannot pre-empt any step of this finite prefix of the epsilon trajectory. Thus the run on every extension xy follows exactly the same epsilon prefix and reaches c_x before reading y.

Now put

$$S = \{q_x : x \in E^*\} \subseteq Q_B.$$

We define a DFA D over E with state set S and initial state q_ε. For $q_x \in S$ and $e \in E$, set

$$\delta_D(q_x, e) = q_{xe}.$$

This is well-defined. If $q_x = q_y$, then $c_x = c_y = (q, 0)$ is literally the same physical configuration. By the forced-cut claim, the runs on xe and ye reach this same cut before reading e; from that point, determinism forces the same continuation on e and the same subsequent epsilon behavior. Hence their canonical first final zero cuts coincide, so $q_{xe} = q_{ye}$. The transition is defined for every $e \in E$ because $xe \in E^* \subseteq L(B)$.

Declare $q \in S$ accepting in D exactly when B, started from the physical configuration $(q, 0)$, accepts the fixed continuation u. An induction on $x \in E^*$ gives that the state reached by D after x is q_x. Applying the forced-cut claim once more, now with continuation u, shows that the run on xu reaches c_x before processing u. Therefore

$$x \in L(D) \iff B \text{ accepts } u \text{ from } (q_x, 0) \iff xu \in L(B).$$

Thus D recognizes $S_u(B)$ and has $|S| \leq |Q_B|$ states. □

Lemma 107.3 (Halting Implies Prime)**.** *If M halts, then T_M is prime.*

Proof. Suppose toward contradiction that

$$L(T_M) = \bigcap_{i=1}^{k} L(B_i), \qquad |Q_{B_i}| < N,$$

where every B_i is a DOCA superlanguage of $L(T_M)$. Every core-only word belongs to $L(T_M)$, so $E^* \subseteq L(B_i)$ for every i. Apply Lemma 107.2 to each B_i with the fixed suffix u from (107.5); this yields DFAs D_i with

$$|Q_{D_i}| \leq |Q_{B_i}| < N.$$

By (107.6),

$$P_V = \bigcap_{i=1}^{k} L(D_i).$$

This contradicts the DFA-primality of P_V from Lemma 106.4. Hence T_M is prime. □

107.3 Explicit half-size factors in the nonhalting case

Assume from now on that M does not halt. For every nonzero $r \in V$, let

$$H_r = \{0, r\}$$

and define a factor F_r with control-state set

$$Q_r = V/H_r.$$

Thus

$$|Q_r| = N/2 < N. \tag{107.7}$$

Every factor state is final. Write

$$S_r = H_r$$

for the special zero coset.

107.3.1 Core behavior and the special coset

At counter height zero, core letters act by quotient translation:

$$(v + H_r, 0) \xrightarrow{e_i} (v + e_i + H_r, 0). \tag{107.8}$$

Consequently, when the exact target zero anchor is v, the factor is in its special quotient state exactly when

$$v + H_r = H_r \iff v \in \{0, r\}.$$

At a nonzero anchor v, exactly one factor—F_v—is special.

107.3.2 Non-special zero cosets

At a zero-height coset $C \neq S_r$, the factor deliberately overaccepts query symbols:

- on @ or any $\tau \in \Delta_M$, remain at $(C, 0)$;
- on \$, reset to $(S_r, 0)$;
- on a core letter, use quotient translation (107.8).

No source correctness is checked in a non-special coset.

107.3.3 Special counter factors

Suppose $r \in G_j$. At $(S_r, 0)$, the factor uses the same counter-j transition rules as the target uses at a state of G_j, except that a successful \$ resets to $(S_r, 0)$. Legal positive counter behavior remains in control state S_r.

Choose one existing quotient state $\pi_r \in Q_r$ as a positive poison role. If illegal behavior occurs, enter (π_r, h) with $h > 0$ and thereafter preserve positive height forever. Height distinguishes this poison role from the ordinary zero-height quotient meaning of π_r, so no additional control state is required. Since $N/2 \geq 2$, such a distinct quotient state is available.

107.3.4 Special regular factors

Suppose $r \in G_R$. By (106.1),

$$N/2 \geq |Q_M| + 1.$$

Therefore choose an injection

$$\psi_r : Q_M \cup \{\bot\} \hookrightarrow Q_r,$$

where $\bot$ is a positive poison role. At $(S_r, 0)$, only @ starts regular mode:

$$(S_r, 0) \xrightarrow{@} (\psi_r(p_0), 1).$$

At height one, $\psi_r(p)$ follows exactly those branch-labelled transition symbols whose source is p. A successful \$ from $\psi_r(p_f)$ pops the mode symbol and resets to $(S_r, 0)$. Wrong transitions, malformed markers, core letters while positive, or a premature \$ enter $\psi_r(\bot)$ at positive height; this poison role preserves positivity forever.

Every otherwise undefined transition of F_r is routed to an existing positive poison role while making or preserving positive height.

Lemma 107.4 (Explicit Factor Construction)**.** *For every $r \neq 0$, F_r is a complete real-time DOCA with exactly $N/2$ control states, one work symbol, all states final, and unit counter updates.*

Proof. The quotient state set has size $N/2$. Counter factors use one quotient state as the legal special state and another, distinguished only at positive height, as poison. Regular factors fit an encoding of all Q_M states plus one poison role into Q_r by the size choice. Every remaining transition is totalized into an existing positive poison role. No new state is added. □

107.4 Exactness of the factorization

We first show that every F_r is a superlanguage, independently of the nonhalting assumption.

Lemma 107.5 (Zero-height quotient and owner invariant)**.** *Fix $r \neq 0$. For every consumed prefix $y \in \Sigma^*$ for which the target run is at a zero-height configuration $(v, 0)$ after reading y, the run of F_r on the same prefix is at*

$$(v + H_r, 0).$$

Consequently, at every nonzero zero anchor v the factor F_v is the unique special factor, while at anchor 0 every factor is special. If the target leaves a nonzero anchor v to execute a positive checker episode, the owner factor F_v follows the same checker semantics until the next successful zero return/reset or the corresponding permanent-positive failure; ownership remains v even when a regular checker changes its finite-control state through $\phi(Q_M)$.

Proof. We prove the zero-height statement by induction over successive target configurations of height zero. It holds initially at $(0, 0)$ because every factor starts at $(H_r, 0)$.

Assume the target is at $(v, 0)$ and F_r is at $(v + H_r, 0)$. A core letter sends the target to $(v + e_i, 0)$ and the factor to $(v + e_i + H_r, 0)$, so the invariant is preserved immediately. If $v = 0$, every non-core symbol sends the target to permanent-positive poison; hence no later target zero configuration can arise from that move, and there is no further zero-height obligation to verify.

Suppose next that $v \neq 0$. If $v = r$, then $v + H_r = H_r = S_r$, so F_r is special. For $v \in G_1 \cup G_2$, the special factor uses exactly the target's counter-checker operations. A zero-preserving step leaves both at their corresponding zero anchors; a legal positive segment is mirrored operation for operation and, if it returns to zero, returns the target to v and the factor to $S_r = v + H_r$; a successful \$ resets them to 0 and H_r, respectively. Any illegal positive behavior that sends the target to permanent-positive poison sends the owner factor to permanent-positive poison as well. For $v \in G_R$, the same statement holds for the regular checker: @ starts the two corresponding positive regular modes, the branch-labelled control path is mirrored under ϕ and ψ_r, and a successful \$ resets both sides to their zero anchors. The owner is still the original anchor $v = r$ throughout this positive episode, regardless of the target's current state in $\phi(Q_M)$.

If instead $v \notin \{0, r\}$, then $v + H_r \neq H_r$, so F_r is non-special. On @ and transition symbols it deliberately stays at the same zero coset, and on \$ it resets to H_r. Thus, while the target performs a legal counter or regular episode launched from v, the non-owner factor overaccepts at height zero; when the target next returns to zero at v or resets to 0, the factor is respectively at $v + H_r$ or H_r. If the target enters permanent-positive poison, again no later target zero configuration exists. This proves renewal of the invariant at every possible next zero configuration.

Finally,

$$v + H_r = H_r \iff v \in H_r = \{0, r\}.$$

Hence at a nonzero anchor v exactly F_v is special, whereas at anchor 0 all factors are special. The owner statement was established in the $v = r$ case above. □

Lemma 107.6 (Forward inclusion)**.** *For every nonzero $r \in V$,*

$$L(T_M) \subseteq L(F_r).$$

Proof. Fix $r \neq 0$ and let $w \in L(T_M)$. Because every target state is final, acceptance means that the real-time run of T_M ends at some zero-height configuration $(v, 0)$. Lemma 107.5, applied to the full prefix w, places F_r at $(v + H_r, 0)$ after the same input. Every state of F_r is final, so F_r accepts w. Since r was arbitrary, $L(T_M) \subseteq L(F_r)$ for every $r \neq 0$. □

The reverse inclusion uses the nonhalting assumption. It is useful to formulate it in terms of the *final positive episode*: because all target states are final, a target word is rejected exactly when its real-time run ends at positive height. Let a be the exact target control state at the last zero-height configuration before this final positive episode.

Lemma 107.7 (Nonhalting Implies Composite)**.** *If M does not halt, then*

$$\boxed{L(T_M) = \bigcap_{r \in V \setminus \{0\}} L(F_r)}.$$

Consequently T_M is composite.

Proof. The inclusion from left to right is Lemma 107.6. For the converse, take a word $w \notin L(T_M)$. The target ends at positive height. Since the run starts at height zero, there is a last zero-height configuration before the final positive episode; write its exact target control state as a. Lemma 107.5 fixes the simultaneous factor configurations at this anchor.

Case 1: $a \neq 0$. By Lemma 107.5, F_a is the unique special factor at that anchor and is the owner of the ensuing checker episode.

If $a \in G_1 \cup G_2$, the owner factor F_a performs exactly the same one-counter operations as the target during the final positive episode. Illegal branches, a forbidden marker or core letter while positive, or an unfinished positive trajectory leave both target and owner factor positive. Hence F_a rejects w.

If $a \in G_R$, the owner factor is still F_a even though the target finite control may later change through states $\phi(Q_M)$. There are two subcases. If the first symbol of the final positive episode is not @ (necessarily a transition symbol or \$), both the target and F_a enter positive poison immediately. If the episode starts with @ then both enter regular mode, and from that point they follow the same branch-labelled control path. A wrong transition, a second @ while positive, a core letter while positive, a premature \$, or an unfinished regular payload leaves the owner factor positive. In either subcase F_a rejects w.

Case 2: $a = 0$. By Lemma 107.5, every factor is simultaneously at its special zero coset S_r. Because a is the last zero anchor, the symbol that begins the final positive episode is non-core; the target enters poison on that symbol.

If that symbol is not @ then every regular-color factor immediately enters positive poison, so at least one factor rejects permanently.

Suppose the first non-core symbol is @. Every G_R-indexed factor starts the regular checker, every G_1-indexed factor starts the counter-1 checker, and every G_2-indexed factor starts the counter-2 checker. Now inspect the suffix after this @ until the first \$, if any.

If the input ends first, every regular-color factor is still positive and rejects. If an @ or a core letter occurs before the first \$, every regular-color factor enters permanent positive poison. If a transition symbol violates the control path, the regular-color factors enter poison as well. Hence the only remaining possibility is a block

$$@z\$, \qquad z \in \Delta_M^*,$$

whose transition sequence is control-consistent up to the terminal marker. At this first \$, a regular-color factor returns to zero exactly when $z \in R_M$, and a counter-j factor returns to zero exactly when $z \in C_j$. Thus all factors could return to zero simultaneously only if

$$z \in R_M \cap C_1 \cap C_2.$$

By Lemma 106.3, this would imply that M halts, contradicting the assumption. Therefore before or at the first terminal marker at least one factor becomes permanently positive. It remains positive on all later input, so it rejects w.

Thus every target rejection is witnessed by at least one F_r, proving

$$\bigcap_{r \neq 0} L(F_r) \subseteq L(T_M).$$

By (107.7), every factor has strictly fewer than N states, so the target is composite. □

Remark 107.8 (Owner factor versus current regular state). The owner of a query is determined by the zero anchor where the query began. In regular mode the target control subsequently moves through $\phi(Q_M)$, but this does not change the owner. Confusing the current regular control state with the owner index is an easy proof error; the construction and Lemma 107.7 avoid it explicitly.

Chapter 108

Undecidability of State-Primality and the Verification–Decomposition Boundary

The construction is now complete. The first section states the undecidability theorem and its exact restricted-class consequences; the second records the proof features most likely to be confused when the result is compared with the earlier exact-lift verification hierarchy.

108.1 Undecidability and consequences

Theorem 108.1 (Prime-DOCA Undecidability)**.** *The mapping $M \mapsto T_M$ is computable and satisfies*

$$\boxed{M \text{ halts} \iff T_M \text{ is prime.}}$$

Hence $\mathrm{PRIME}_{\mathrm{DOCA}}$ *is undecidable.*

Proof. Lemma 107.3 gives the forward implication. Lemma 107.7 gives

$$M \text{ does not halt} \Rightarrow T_M \text{ is composite,}$$

which is the reverse implication. The construction of N, the coloring, transition tables, and factors is effective. Undecidability follows from deterministic two-counter halting. □

Corollary 108.2 (Restricted-class undecidability)**.** *Prime-DOCA remains undecidable when the input presentation is restricted to complete real-time deterministic one-counter automata that use one work symbol and a permanent bottom marker, have all control states final, and use only updates $-1, 0, +1$. The same reduction also proves undecidability for the stricter variant in which admissible smaller factors are required to lie in this subclass as well.*

Proof. Every target T_M lies in the stated subclass. In the halting case Lemma 107.3 excludes all smaller ambient DOCAs, hence in particular all smaller factors from the subclass. In the nonhalting case the explicit factors F_r themselves lie in the subclass. Therefore both interpretations of the restriction are covered by the same mapping. □

Corollary 108.3 (State-minimality in the prime case)**.** *If M halts, then no DOCA with fewer than N control states recognizes $L(T_M)$.*

Proof. Otherwise let B be such a smaller recognizer. Since $E^* \subseteq L(T_M) = L(B)$, Lemma 107.2 applied to the valid halting suffix u would yield a DFA with fewer than N states recognizing the fixed section P_V. This contradicts the fact that P_V has minimal DFA index N. □

Corollary 108.4 (Recursion-theoretic consequences)**.** *There is a many-one reduction*

$$\mathrm{HALT}_2 \leq_m \mathrm{PRIME}_{\mathrm{DOCA}}.$$

Consequently:

(a) $\mathrm{PRIME}_{\mathrm{DOCA}}$ *is r.e.-hard and undecidable;*

(b) $\mathrm{PRIME}_{\mathrm{DOCA}}$ *is not co-r.e.;*

(c) $\mathrm{COMPOSITE}_{\mathrm{DOCA}}$ *is not r.e.*

No upper semidecidability classification is claimed.

Proof. The reduction is Theorem 108.1. If $\mathrm{PRIME}_{\mathrm{DOCA}}$ were co-r.e., then its computable preimage HALT_2 would be co-r.e., contradiction. Likewise,

$$M \notin \mathrm{HALT}_2 \iff T_M \in \mathrm{COMPOSITE}_{\mathrm{DOCA}},$$

so r.e.-ness of $\mathrm{COMPOSITE}_{\mathrm{DOCA}}$ would imply r.e.-ness of deterministic two-counter non-halting. □

108.2 Proof anatomy, edge cases, and verification points

The reduction is compact once the elementary-abelian core is in place, but several details are structurally important. This section records them explicitly both for readability and for independent verification.

108.2.1 Why the target does not need two counters

The same-witness coupling problem is delegated to the intersection. A single target run beginning at a nonzero anchor checks only one of R_M, C_1, C_2. This keeps the target within one-counter power. The common zero anchor is where all quotient factors become special simultaneously; only there does the intersection enforce all three predicates at once. Thus conjunction exists semantically across factors, not physically inside one automaton.

108.2.2 Why the state budget is exact

The target uses exactly the N core points. Counter colors need no separate checker state: the anchor state itself is the checker control, and positive height stores the unbounded counter value. Regular colors encode Q_M inside G_R while positive height serves as the mode bit. State 0 at positive height is reused as target poison.

Each factor has exactly $N/2$ quotient states. Counter factors reuse one quotient state as positive poison; regular factors reuse the positive-height meanings of quotient states to encode Q_M plus one poison role. The inequality $N/2 \geq |Q_M|+1$ is precisely the factor-side packing condition.

108.2.3 Malformed and unfinished inputs

The target language is not defined by a global "well-formed transcript" grammar. In counter colors, when the counter happens to be zero the machine may process symbols according to its zero-height transition table even if a human reader would regard the current substring as malformed. This causes no problem: the reduction only requires exact target/factor agreement on arbitrary inputs and exact source semantics on the fixed valid suffix used in the prime direction.

In the nonhalting reverse inclusion, malformed behavior launched from a nonzero anchor is caught by the owner factor because it mirrors the target transition system, not an abstract syntax predicate. At anchor zero, regular-color factors provide the strict syntax/control witness needed to prevent all factors from accepting a malformed payload simultaneously.

108.2.4 Underflow and zero tests

A decrement branch for counter j is legal only at positive height; a zero branch is legal only at zero. Illegal cases go to permanent-positive poison. Therefore the counter predicates implement exact Minsky guards rather than merely net counter effects.

108.2.5 Epsilon-enabled adversarial factors

The target and explicit factors are epsilon-free, but prime decompositions quantify over all smaller DOCAs in the ambient problem model. This is why Lemma 107.2 is stated for epsilon-enabled factors. The key object is a physical accepting zero configuration reached after a consumed core prefix. Determinism and epsilon/input exclusivity make that configuration a continuation-independent cut. No assumption is made about the factor's internal representation of the core translation.

108.2.6 One reject point versus many

The proof is not simply "embed a hard regular language." That alone would prove a lower bound in the halting case but would not produce explicit decompositions in the nonhalting case. The elementary-abelian core simultaneously supplies both sides of the reduction: the single-reject section is prime, while each additional reject point carries its own canonical half-size quotient factor. This dual role is the feature that the earlier residual-rigidity route was missing.

108.3 Related work and positioning

Kupferman and Mosheiff introduced intersection primality for regular languages and developed structural and algorithmic tools for DFA decomposition [98]. Their size threshold is the language index, i.e. the number of states of the minimal DFA. This distinction matters for the present problem: Totzke's Prime-DOCA formulation is presentation-relative, comparing every factor with the number of control states in the given DOCA presentation. Spenner later called the analogous presentation-relative DFA notion S-primality [103]. Our halting construction happens to remove this distinction internally, because the prime target is state-minimal by Corollary 108.3.

The regular-language theory has continued to develop after the original prime-language paper. Unary primality admits a structural LogSpace algorithm [101]; permutation automata admit specialized decomposition algorithms, including low-complexity results for commutative permutation DFAs [102]; primality of acyclic DFAs and finite languages is characterized in [103]; and the general Prime-DFA problem was recently shown NP-hard [104]. Our finite obstruction lies inside the especially structured commutative permutation class, but it is used only as a fixed-suffix section: the undecidable part comes from asking whether an infinite-state DOCA presentation has any finite intersection decomposition by smaller DOCAs.

The algorithmic theory of deterministic one-counter automata is classical. Valiant and Paterson proved decidability of language equivalence for DOCAs [106], and Böhm, Göller, and Jančar later proved the problem NL-complete [107]. These results make an instructive contrast with state-primality: equivalence of two given DOCAs is decidable, while deciding whether one DOCA language admits an exact finite intersection decomposition into arbitrary smaller DOCA superlanguages is undecidable.

Almagor, Avni, Sinclair-Banks, and Yeshurun study primality for counter nets and prove undecidability when "smaller" means lower counter dimension [105]. Counter nets differ from DOCAs in both the resource measure and operational model: they may have several counters, have no explicit zero tests, and compare dimensions rather than finite-control state counts. Their theorem and ours therefore establish undecidability for distinct decomposition axes.

108.4 The verification–decomposition boundary

The contrast with the earlier verification hierarchy is structural. Equivalence of two given deterministic one-counter automata is decidable [106, 107]; by contrast, the theorem above shows that deciding whether one presented DOCA language admits an exact finite intersection decomposition into arbitrary strictly smaller DOCA superlanguages is undecidable. Thus a decision procedure for comparing two fixed presentations does not extend to a decision procedure for existential decomposition over an unbounded family of smaller presentations.

This distinction is the reason the present branch belongs after the General Theory and after the exact state/resource frontiers. It is not an ingredient used to build those theories; it identifies the algorithmic decomposition boundary of the representation program. With optimization and decomposition now separated, the final Part can return to the original 2023 machine and synthesize which structures were forced, chosen, or subsequently generalized.

Chapter 109

From Unrestricted Decomposition to Two Factors

Research branch — The fixed-budget sharpening. *The unrestricted Prime-DOCA theorem allows an arbitrary finite family of smaller factors. The new question is whether that factor supply is essential. It is not: alternating sentinels distribute the natural regular-control witness across two one-counter factors, so the smallest genuine intersection budget already carries the undecidability.*

The transcript, elementary-abelian prime core, and epsilon-robust zero-cut arguments have already been established in Chapters 106–107. They are reused here rather than restated. In particular, Lemma 106.3 gives M halts iff $R_M \cap C_1 \cap C_2 \neq \varnothing$, Lemma 106.4 supplies the prime section P_V, and Lemma 107.2 applies against arbitrary smaller epsilon-enabled DOCAs. The genuinely new task is to enforce the three transcript predicates with only two smaller one-counter factors.

109.1 Phase coding and alternating sentinels

The natural three-checker split assigns R_M, C_1, C_2 to three separate factors. To use only two factors we must let the factors share the syntax/control burden without ever requiring one DOCA to store both unbounded counters.

For every $\tau \in \Delta_M$, introduce six phase-labelled symbols

$$O^1_\tau,\ A^1_\tau,\ C^1_\tau,\ O^2_\tau,\ A^2_\tau,\ C^2_\tau.$$

For $z = \tau_1 \cdots \tau_m$, let

$$\widehat{z} = \prod_{\ell=1}^{m} O^1_{\tau_\ell} A^1_{\tau_\ell} C^1_{\tau_\ell} O^2_{\tau_\ell} A^2_{\tau_\ell} C^2_{\tau_\ell}. \tag{109.1}$$

Add a query-start marker @ and terminal symbols $T_1, Z_1, T_2, Z_2, \$$. The canonical complete query is

$$u(z) = @\widehat{z} T_1 Z_1 T_2 Z_2 \$. \tag{109.2}$$

Both checkers represent a source counter value c_j at stable points by physical height

$$h_j = c_j + 1.$$

The extra unit is a sentinel. To test a source zero/decrement guard, checker j temporarily removes its sentinel, making the physical height exactly c_j. The construction staggers these exposures: checker 2 remains positive while checker 1 is exposed, and checker 1 remains positive while checker 2 is exposed.

109.1.1 Checker 1

Checker 1 carries the source-control state and verifies counter 1. Its logical roles are

$$P_p\ (p \in Q_M), \qquad K^A_\tau, K^C_\tau\ (\tau \in \Delta_M), \qquad U_1, U_2, U_3, \qquad \bot_1,$$

together with one raw/special role S_1.

At $(S_1, 0)$, @ starts the checker in $(P_{p_0}, 1)$. From (P_p, h) with $h > 0$, the symbol O^1_τ is legal only when $\mathrm{src}(\tau) = p$; it removes the sentinel and moves to raw role S_1. At raw S_1, A^1_σ performs $\mathrm{op}_1(\sigma)$ on the exposed height, with illegal zero/decrement branches entering permanent-positive poison. A symbol C^1_ρ restores the sentinel and moves to $P_{\mathrm{tgt}(\rho)}$.

During the second half, checker 1 becomes the lexical witness: O^2_τ moves from P_p to K^A_τ, only A^2_τ moves to K^C_τ, and only C^2_τ returns to $P_{\mathrm{tgt}(\tau)}$. The height stays positive throughout this half.

109.1.2 Checker 2

Checker 2 verifies counter 2 and carries the complementary lexical memory. Its logical roles are

$$B, \qquad A_\tau, C_\tau\ (\tau \in \Delta_M), \qquad W_1, V_2, \qquad \bot_2,$$

together with raw/special role S_2.

At $(S_2, 0)$, @ starts $(B, 1)$. From B, O^1_τ stores τ in A_τ; only A^1_τ reaches C_τ. The symbol C^1_τ preserves that lexical role, and only O^2_τ then removes checker 2's sentinel and enters raw S_2. At raw S_2, A^2_σ performs $\mathrm{op}_2(\sigma)$, and C^2_ρ restores the sentinel and returns to B. Checker 1 forces equality of the second-half labels.

The apparent reuse of C_τ before and after C^1_τ causes no ambiguity: if C^1_τ is omitted, checker 1 is still raw and rejects a premature O^2_τ.

109.1.3 Terminal block

From P_p, checker 1 permits T_1 only for $p = p_f$, removes its sentinel, and enters raw S_1. The next Z_1 is legal only at height zero; it restores the sentinel and enters U_1. Checker 1 then reads T_2, Z_2 through U_1, U_2, U_3 and finally removes the restored sentinel on \$.

Checker 2 reads T_1, Z_1 through B, W_1, B, then T_2 removes its sentinel and enters raw S_2. The next Z_2 is legal only at height zero; it restores the sentinel in V_2, and \$ removes it. For the checker pair used as the two quotient factors, a successful final \$ returns checker i to its special zero role S_i. When the same checker routine is instantiated inside the target, the successful final \$ instead returns the target to the exact core zero state 0. This is the only terminal-destination difference between the target checker and its quotient-factor copy. In either implementation, simultaneous terminal success is equivalent to source control p_f and both source counters zero.

109.2 Synchronization lemmas

Define a synchronized stable pair

$$\mathcal{G}(p, a, b) = ((P_p, a + 1), (B, b + 1)).$$

Lemma 109.1 (Block synchronization)**.** *Suppose the two checkers start in $\mathcal{G}(p, a, b)$, consume a non-poisoning source-transition macro, and next return to a synchronized stable pair. Then the consumed macro is uniquely*

$$O^1_\tau A^1_\tau C^1_\tau O^2_\tau A^2_\tau C^2_\tau$$

for a single $\tau \in \Delta_M$ with $\mathrm{src}(\tau) = p$. Both source-counter guards are legal at (a, b), and the new pair is

$$\mathcal{G}(\mathrm{tgt}(\tau), a', b'),$$

where (a', b') are exactly the source-counter values after τ.

Proof. Checker 1 source-checks the opening O^1_τ. Checker 2 stores that same label and forces A^1_τ, C^1_τ, so checker 1 performs exactly the counter-1 operation of τ. Checker 1 rejects O^2 while still raw, hence the closing C^1_τ cannot be omitted. Checker 2 then permits only O^2_τ and exposes counter 2. Checker 1 stores the same label and forces A^2_τ, C^2_τ, so checker 2 performs exactly the counter-2 operation. Both sentinels are restored and checker 1's control role is $P_{\mathrm{tgt}(\tau)}$. □

Lemma 109.2 (Common-legal-symbol property)**.** *Fix a non-poison paired phase after the opening @ and before the first simultaneous zero return. Apart from the choice of the next source transition τ at a stable source boundary, there is at most one next phase label that both checkers can process without poison. For a chosen τ, the common sequence is forced to be*

$$O^1_\tau \to A^1_\tau \to C^1_\tau \to O^2_\tau \to A^2_\tau \to C^2_\tau.$$

At a terminal boundary it is forced to be

$$T_1 \to Z_1 \to T_2 \to Z_2 \to \$.$$

Proof. This follows by inspecting the paired roles. After O^1_τ, checker 2's state A_τ forces A^1_τ. Next C_τ permits C^1_τ, while checker 1 rejects a premature O^2. Once C^1_τ restores checker 1's sentinel, checker 2 still remembers τ and forces O^2_τ. Checker 1 then forces A^2_τ and C^2_τ. The terminal roles force the terminal sequence analogously. A core letter during an exposed raw-zero phase may be non-poisoning for the exposed checker, but the complementary checker is positive and therefore poisons on that same core letter; hence it is never a common legal continuation. □

Lemma 109.3 (Alternating sentinel)**.** *After the opening @ and before the final common reset \$, every proper non-poison prefix leaves at least one of the two checker counters strictly positive.*

Proof. Checker 1 removes its sentinel only during the O^1A^1 interval, while checker 2 keeps its sentinel. The C^1 symbol restores checker 1's sentinel before checker 2 removes its sentinel on O^2. During O^2A^2, checker 1 therefore remains positive, and C^2 restores checker 2's sentinel. The terminal block is staggered in the same order: T_1Z_1 opens and recloses checker 1 before T_2Z_2 opens checker 2. Only the final \$ removes both restored sentinels on the same input symbol. □

Lemma 109.4 (First simultaneous zero return)**.** *Start both checkers in their special zero configurations and consume an opening @. A later prefix is the first prefix after that opening at which both checkers are again at height zero without poison if and only if that prefix is*

$$u(z) = @\widehat{z}T_1Z_1T_2Z_2\$$$

for some $z \in R_M \cap C_1 \cap C_2$.

Proof. By Lemma 109.2, every common non-poison continuation is forced block by block. By Lemma 109.1, each complete block realizes one genuinely enabled source transition and updates the two represented counters correctly. By Lemma 109.3, no proper prefix before the final common reset can have both counters zero. The terminal block succeeds for both checkers exactly at source state p_f with both source counters zero. Conversely, any normalized halting transcript follows all these transitions. □

Combining Lemmas 106.3 and 109.4 gives

$$\boxed{M \text{ halts} \iff \text{a first simultaneous zero-return query exists.}} \tag{109.3}$$

Chapter 110

Two Half-Size Factors and the Sharp Factor-Budget Boundary

The alternating-sentinel mechanism now replaces the unrestricted family of failure factors by two explicit quotient DOCAs. The halting branch still exposes the same prime finite section, while the nonhalting branch becomes an exact two-factor decomposition.

110.1 Target and two half-size factors

Let

$$q = |Q_M|, \qquad m = |\Delta_M|.$$

Choose a power of two $N = 2^d$ satisfying

$$N/2 \geq q + 2m + 5. \tag{110.1}$$

Choose distinct nonzero vectors $r_1, r_2 \in V$ and put

$$H_i = \{0, r_i\}, \qquad Q_i = V/H_i, \qquad S_i = H_i.$$

Thus $|Q_i| = N/2$.

Checker 1 needs at most $q + 2m + 5$ control names: q source-control roles, $2m$ lexical roles, three terminal roles, one poison role, and one raw/special role. Checker 2 needs at most $2m + 5$ names. Hence (110.1) packs both checkers into their quotient state sets. The target can use disjoint positive-role families of total size at most $q + 4m + 10 \leq N$. The same underlying state names may carry unrelated zero-height core meanings and positive-height checker meanings; physical height disambiguates them, so no extra control states are needed.

110.1.1 The target T_M

The target state set is exactly V, every state is final, and the input alphabet is the disjoint union of the core alphabet E and the phase/query alphabet.

At height zero, every core letter acts by translation. At exact anchor 0, every non-core symbol enters permanent-positive poison. At r_i, the special zero table of checker i applies. Every other nonzero zero-height state is generic: all non-reset query symbols leave the target at the same zero anchor, while the final reset \$ sends it to 0. A successful owner-checker terminal block also sends the target to exact state 0: after checker 1 reaches $(U_3, 1)$, its final \$ transition is $(U_3, 1) \xrightarrow{\$} (0, 0)$, and after checker 2 reaches $(V_2, 1)$, its final \$ transition is $(V_2, 1) \xrightarrow{\$} (0, 0)$. At positive height, the packed logical role determines the appropriate checker transition; every unspecified positive transition enters permanent-positive poison.

A core symbol read at height zero always performs core translation, even if the target has reached r_i as a temporary raw-zero checker configuration. A core symbol read at positive height poisons. Thus no hidden query-mode bit is assumed at height zero.

110.1.2 The factors F_1, F_2

Factor F_i has control set $Q_i = V/H_i$ and all states final. At height zero, core letters act by quotient translation:

$$(v + H_i, 0) \xrightarrow{e} (v + e + H_i, 0).$$

At the special coset S_i, checker i's special/raw transition table applies. In particular, a successful checker-i terminal block removes the restored sentinel on its final \$ and returns F_i to $(S_i, 0)$. Every non-special zero coset deliberately overaccepts query symbols: non-reset query symbols preserve that zero coset, while \$ resets to S_i. Thus every successful factor reset returns to the quotient zero state $H_i = S_i$, whereas every successful target reset returns to the exact zero state 0. Positive logical roles and a permanent-positive poison role are packed into existing quotient state names. No additional state is introduced.

Lemma 110.1 (Size and syntactic restrictions)**.** *T_M is a complete real-time ε-free DOCA with exactly N control states. Each F_i is a complete real-time ε-free DOCA with exactly $N/2$ control states. All three automata use one work symbol, have all states final, and use only updates in $\{-1, 0, +1\}$.*

Proof. The state counts follow from the construction and (110.1). Each transition case is determined by the current control role, zero/positive status, and input class. Undefined cases are totalized into an existing permanent-positive poison role, using $+1$ at zero and 0 at positive height. Hence no sink state or ε-transition is needed. □

110.2 The quotient invariant and forward inclusion

Lemma 110.2 (Quotient invariant)**.** *Fix $i \in \{1, 2\}$. Whenever, after a common consumed prefix, the target is at zero height in exact core state v, factor F_i is at*

$$(v + H_i, 0),$$

unless an earlier target rejection has already made the implication irrelevant.

Proof. Core letters preserve the relation by translation. At a generic nonzero anchor, the target and every non-special factor remain at their zero states throughout non-reset query symbols; a successful \$ sends the target to exact state 0 and every factor F_i to $H_i = 0 + H_i$.

At owner anchor r_i, factor F_i is special and mirrors the target checker throughout the body of the query. If a raw counter action returns to height zero before the query is complete, the target is again at exact owner state r_i and the owner factor is at $H_i = r_i + H_i$, so the quotient relation is preserved. On a successful terminal \$, the target resets to 0 while the owner factor resets to H_i, again giving the required relation $H_i = 0 + H_i$.

At the other active anchor r_j, $j \neq i$, factor F_i is non-special and remains at zero in coset $r_j + H_i$ while the target runs checker j; whenever checker j temporarily returns to raw zero, its exact state is again r_j, so the relation still holds. If checker j completes successfully, the target's final \$ resets to 0, while the non-owner factor's non-special reset sends it to $H_i = 0 + H_i$.

If a core letter interrupts an exposed raw-zero phase, the target translates from the exact owner anchor and every zero-height factor performs the corresponding quotient translation. Finally, a non-core symbol from exact anchor 0 makes the target permanently positive and creates no later zero-height obligation. Thus every successful query reset, whether from a generic anchor or either active anchor, renews the exact zero/quotient-zero synchronization, and these cases exhaust all ways to pass between zero-height target configurations. □

Remark 110.3 (Reset synchronization). The distinction between exact zero and quotient zero is intentional. A successful target query always ends at $(0, 0)$, whereas a successful run of factor F_i ends at $(H_i, 0)$. Since $H_i = 0 + H_i$, this is precisely the destination required by the quotient invariant. In particular, after any successful query episode, subsequent core letters or later query episodes start from a freshly synchronized exact/quotient zero configuration.

Lemma 110.4 (Forward inclusion)**.** *For $i = 1, 2$,*

$$L(T_M) \subseteq L(F_i).$$

Proof. An accepted target word ends at zero height in some exact state v. By Lemma 110.2, factor F_i ends at zero height in quotient state $v + H_i$. Since every factor state is final, F_i accepts. □

110.3 Halting exposes a prime section

Assume M halts and choose $z \in R_M \cap C_1 \cap C_2$. Fix the suffix $u = u(z)$ from (109.2). For a core word $x \in E^*$ let $v = \delta(0, x)$.

If $v = 0$, the first @ enters permanent-positive poison. If $v = r_1$, checker 1 successfully processes u and the final \$ resets the target to exact state 0; if $v = r_2$, checker 2 does the same. At every other nonzero anchor the query is deliberately overaccepted and the final \$ also resets the target to 0. Hence

$$\boxed{xu \in L(T_M) \iff \delta(0, x) \neq 0.} \tag{110.2}$$

The fixed-suffix section is exactly P_V.

Lemma 110.5 (Halting implies primality)**.** *If M halts, then T_M is prime: there is no finite family of DOCAs $B_1, \ldots, B_t$ with $|Q_{B_j}| < N$ and*

$$L(T_M) = \bigcap_{j=1}^{t} L(B_j).$$

Proof. Suppose such a family exists. Every core word belongs to $L(T_M)$, hence to every $L(B_j)$. Apply Lemma 107.2 to each B_j with the fixed suffix u. This yields DFAs D_j with

$$|Q_{D_j}| \leq |Q_{B_j}| < N$$

and, by (110.2),

$$P_V = \bigcap_{j=1}^{t} L(D_j).$$

This contradicts the DFA-primality of P_V from Lemma 106.4. □

110.4 Nonhalting gives an exact two-factor decomposition

Assume from now on that M does not halt. By (109.3), no query launched with both factors special can have a first simultaneous zero return.

Lemma 110.6 (Reverse inclusion)**.** *If M does not halt, then*

$$L(F_1) \cap L(F_2) \subseteq L(T_M).$$

Proof. Take $w \notin L(T_M)$. Since the target is complete, real-time, and all states are final, its run on w ends at positive height. Let a be the last zero-height exact target state before the final positive episode.

If $a = r_i$, then by Lemma 110.2 factor F_i is special at that point. The target and owner factor use identical checker transitions throughout the query body; their only successful terminal difference is that the target's final \$ goes to exact zero 0 while the factor's goes to quotient zero H_i. Since a was chosen as the last zero-height target state before the final positive episode, no successful terminal reset occurs afterward. Hence target and owner factor follow the same positive behavior throughout this final episode, and the owner factor

also remains positive and rejects. This includes the case where $a = r_i$ is a temporary raw-zero configuration reached inside a malformed or unfinished checker episode.

A generic nonzero a is impossible: generic non-core behavior never creates positive height.

It remains to consider $a = 0$. Both factors are then simultaneously special. A non-core symbol different from @ sends at least one special factor to permanent-positive poison; in particular, \$ is not an immediate reset from a special zero state. Suppose the final positive episode begins with @. By Lemma 109.2, a wrong phase, label mismatch, repeated marker, or premature terminal symbol poisons at least one factor. An illegal source zero/decrement branch poisons the exposed checker. A core symbol during an exposed raw-zero interval translates the exposed checker at height zero, but the complementary checker is still positive and therefore enters permanent-positive poison. If the input ends before a common reset, Lemma 109.3 leaves at least one factor positive. Finally, if both factors were to return to zero, their first simultaneous zero return would contradict nonhalting by Lemma 109.4. Thus at least one factor rejects w. □

Combining Lemmas 110.4 and 110.6 gives the exact decomposition.

Lemma 110.7 (Two factors suffice in the nonhalting case)**.** *If M does not halt, then*

$$\boxed{L(T_M) = L(F_1) \cap L(F_2),} \qquad |Q_{F_1}| = |Q_{F_2}| = N/2 < N.$$

110.5 Undecidability and consequences

Theorem 110.8 (Two-factor undecidability)**.** *The mapping $M \mapsto T_M$ is computable and satisfies*

$$M \text{ halts} \Longrightarrow T_M \text{ is prime},$$

while

$$M \text{ does not halt} \Longrightarrow L(T_M) = L(F_1) \cap L(F_2)$$

for two explicit factors with $N/2$ states each. Consequently Factor-DOCA$_2$ *is undecidable, even under the restrictions of Lemma 110.1.*

Proof. The halting implication is Lemma 110.5; the nonhalting implication is Lemma 110.7. A decider for Factor-DOCA$_2$ would therefore decide halting of deterministic two-counter machines. □

Corollary 110.9 (Unrestricted Prime-DOCA recovered from the two-factor theorem)**.** *The unrestricted Prime-DOCA problem is undecidable under the same restricted target class. This recovers Theorem 108.1 from the sharper two-factor construction.*

Proof. The reduction of Theorem 110.8 maps halting machines to prime targets and nonhalting machines to targets that already decompose into two smaller factors. Hence a decider for unrestricted primality would decide deterministic two-counter halting. □

Corollary 110.10 (Every fixed budget $k \geq 2$)**.** *For every fixed integer $k \geq 2$,* Factor-DOCA$_k$ *is undecidable under the same restrictions.*

Proof. The case $k = 2$ is Theorem 110.8. For $k > 2$, repeat either of the two factors. Repetition preserves the intersection language and Definition 105.3 does not require distinct factors. □

Corollary 110.11 (Bounded-factor input)**.** *The problem that receives a DOCA A together with an integer k and asks whether A is k-factor composite is undecidable, even when every reduction instance has $k = 2$.*

Corollary 110.12 (Non-semidecidability)**.** *The set of 2-factor-composite DOCA presentations is not recursively enumerable.*

Proof. The construction gives

$$M \notin \mathrm{HALT}_2 \iff T_M \in \mathsf{Factor\text{-}DOCA}_2.$$

Deterministic two-counter nonhalting is not recursively enumerable, so recursive enumerability of $\mathsf{Factor\text{-}DOCA}_2$ would give a recursive enumeration of nonhalting instances by computable preimage. □

110.6 Interpretation

The result identifies the exact nontrivial factor-budget boundary for this decomposition problem. Unrestricted primality permits an arbitrary finite number of smaller factors, but the fixed-budget theorem shows that unbounded factor supply is not the source of the hardness. Two factors, the smallest budget capable of expressing a genuine intersection decomposition, already support undecidability.

The finite prime obstruction and the infinite-state synchronization play different roles. The elementary-abelian core handles the halting branch by exposing a fixed regular section that no intersection of smaller finite-state sections can realize. The alternating-sentinel code handles the nonhalting branch by making two one-counter factors jointly certify a two-counter transcript. Neither factor stores two unbounded counters. Instead, each factor stores lexical information exactly while the other temporarily gives up its sentinel to expose an exact zero test. In this sense, the natural third regular checker has not been simulated wholesale inside either counter factor; its responsibility has been distributed across the pair.

This gives a sharp contrast with the finite-state landscape. Bounded-factor decomposition remains within classical complexity classes for DFAs, including NP-completeness for commutative permutation DFAs [102]. For deterministic one-counter automata, by contrast, the same bounded formulation is already undecidable at $k = 2$.

110.7 Construction reference tables

The tables below retain the non-poisoning phase transitions so the construction can be reconstructed entirely from the monograph. Every omitted non-core transition from a special zero state enters permanent-positive poison with update $+1$; every omitted transition at positive height enters permanent-positive poison with update 0. Core letters always translate at height zero and poison at positive height. At non-special factor cosets, query symbols overaccept at height zero and \$ resets to the special coset. The checker body is shared by target and factor implementations, but the successful final terminal reset is deliberately different: the target goes to exact state 0, whereas factor F_i goes to special coset $S_i = H_i$.

110.7.1 Checker 1

Mode	Input	Action
S_1, zero	@	$(P_{p_0}, +1)$
P_p, positive	O^1_τ, $\mathrm{src}(\tau) = p$	$(S_1, -1)$
S_1, raw	A^1_τ, $\mathrm{op}_1(\tau) = \mathrm{inc}$	raw, $+1$
S_1, raw	A^1_τ, $\mathrm{op}_1(\tau) = \mathsf{skip}$	raw, 0
S_1, zero raw	A^1_τ, $\mathrm{op}_1(\tau) = \mathrm{zero}$	raw, 0
S_1, positive raw	A^1_τ, $\mathrm{op}_1(\tau) = \mathrm{dec}$	raw, -1
S_1, raw	C^1_τ	$(P_{\mathrm{tgt}(\tau)}, +1)$
P_p, positive	O^2_τ	$(K^A_\tau, 0)$
K^A_τ, positive	A^2_τ	$(K^C_\tau, 0)$
K^C_τ, positive	C^2_τ	$(P_{\mathrm{tgt}(\tau)}, 0)$
P_{p_f}, positive	T_1	$(S_1, -1)$
S_1, zero raw	Z_1	$(U_1, +1)$
U_1, positive	T_2	$(U_2, 0)$
U_2, positive	Z_2	$(U_3, 0)$
U_3, height 1	\$	target: $(0, -1)$; factor F_1: $(S_1, -1)$

110.7.2 Checker 2

Mode	Input	Action
S_2, zero	@	$(B, +1)$
B, positive	O^1_τ	$(A_\tau, 0)$
A_τ, positive	A^1_τ	$(C_\tau, 0)$
C_τ, positive	C^1_τ	$(C_\tau, 0)$
C_τ, positive	O^2_τ	$(S_2, -1)$
S_2, raw	A^2_τ, $\mathrm{op}_2(\tau) = \mathrm{inc}$	raw, $+1$
S_2, raw	A^2_τ, $\mathrm{op}_2(\tau) = \mathsf{skip}$	raw, 0
S_2, zero raw	A^2_τ, $\mathrm{op}_2(\tau) = \mathrm{zero}$	raw, 0
S_2, positive raw	A^2_τ, $\mathrm{op}_2(\tau) = \mathrm{dec}$	raw, -1
S_2, raw	C^2_τ	$(B, +1)$
B, positive	T_1	$(W_1, 0)$
W_1, positive	Z_1	$(B, 0)$
B, positive	T_2	$(S_2, -1)$
S_2, zero raw	Z_2	$(V_2, +1)$
V_2, height 1	\$	target: $(0, -1)$; factor F_2: $(S_2, -1)$

110.8 Edge conditions for repeated queries

Several edge conditions are part of the construction rather than optional proof commentary. They are collected here to make malformed inputs, raw-zero interruptions, and repeated query episodes explicit.

1. The target never stores both source counters; conjunction is enforced by the two factors at the zero anchor.

2. The factor budget is exactly two, and both factors have exactly $N/2$ states.

3. No hidden query-mode bit is assumed at zero height: raw zero and ordinary owner zero are the same physical configuration.

4. During every raw-zero interval, the complementary factor is positive and carries the lexical/sentinel witness.

5. Core interruption at raw zero cannot create a false common success: the zero factor translates, while the positive factor poisons.

6. A premature O^2 before C^1 is rejected by raw checker 1; hence checker 2 may safely reuse C_τ before and after C_τ^1.

7. The terminal transition $B \xrightarrow{T_2,-1} S_2$ is explicit.

8. Immediate \$ is not a legal reset from a special zero state.

9. Successful reset renews the quotient invariant for repeated query episodes: the target returns to exact state 0, every owner factor returns to its special coset H_i, and every non-owner factor resets from its non-special coset to the same $H_i = 0 + H_i$.

10. The halting/primality argument is self-contained for arbitrary smaller deterministic one-counter factors, including ε-enabled factors, via the physical zero-cut Lemma 107.2.

Chapter 111

Fixed One-Counter Language Equivalence: Problems 24.04(B) and 24.04(A)

Research branch — The comparison boundary after decomposition. *The preceding chapters show that existential decomposition of a deterministic one-counter presentation is algorithmically hard even at the smallest nontrivial factor budget. This chapter keeps the two fixed-presentation comparison problems distinct. It first establishes the solution of Automata Exchange Problem 24.04(B), comparing a deterministic one-counter net with a nondeterministic one-counter net. It then adds the stronger zero-testing branch, Problem 24.04(A), where the deterministic machine is a one-counter automaton with an explicit zero test. Both fixed-comparison problems are decidable, but by different compactness mechanisms; the second branch also survives the stated epsilon-transition extension.*

The distinction is essential. Prime-DOCA and fixed-budget factorization quantify over unknown smaller recognizers. Language equivalence fixes both recognizers in advance. Nondeterminism enlarges the forward semantic state from one counter value to a finite envelope, but the monotonicity of one-counter nets still leaves enough order structure for a computable small-witness theorem.

The first fixed-presentation comparison theorem answers Automata Exchange Problem 24.04(B) [109]. The model is the one-counter *net*: the counter is nonnegative, transitions may be disabled by underflow, and there is no explicit zero test. This theorem itself does not cover an explicit zero test. The zero-testing asymmetric problem is taken up later in this same chapter as Problem 24.04(A); its proof requires a different compactness architecture rather than a cosmetic modification of the net argument.

111.1 Model, acceptance, and integer-weight normalization

A one-counter net (OCN) is a tuple

$$A = (Q, \Sigma, \delta, q_0, F),$$

where

$$\delta \subseteq Q \times \Sigma \times \mathbb{Z} \times Q.$$

A transition (q, a, z, r) is enabled from counter value $c \in \mathbb{N}$ exactly when $c + z \geq 0$, and then moves to $(r, c + z)$. A word is accepted when some run from $(q_0, 0)$ ends in a state of F; the final counter value is unrestricted. Determinism means that for every control state and input letter there is at most one outgoing transition.

For the structural proof it is convenient to work in unit-step form.

Lemma 111.1 (Unit-step normalization)**.** *Every finite integer-weight OCN can be effectively transformed, over the same input alphabet, into a unit-step OCN whose updates lie in* $\{-1, 0, +1\}$ *and which recognizes the same final-state language. Determinism and nondeterminism are preserved.*

Proof. Let W be the largest absolute transition weight and choose $M > W$; when every weight is zero take $M = 1$. Write each original counter value uniquely as

$$n = Mc + r, \qquad c \in \mathbb{N}, \quad 0 \le r < M.$$

Store r in finite control and c in the physical counter. For an original update z and stored remainder r, write

$$r + z = kM + r', \qquad 0 \le r' < M.$$

Since $|z| < M$, one has $k \in \{-1, 0, +1\}$. Replace (q, a, z, q') by

$$((q, r), a, k, (q', r')).$$

If $k = -1$ and the quotient counter is zero, the unit-step transition is disabled exactly when the original update would make $n + z < 0$. Otherwise the encoding invariant is preserved. Finality depends only on the original state component, so accepted words are unchanged. Refining the source control state by r cannot introduce nondeterministic choices into a deterministic presentation. □

From now until the final assembly, D denotes a deterministic unit-step OCN and N a nondeterministic unit-step OCN over the same alphabet, both started at counter value zero.

111.2 Exact residual semantics for the nondeterministic side

For a prefix u and $q \in Q_N$, define the exact forward envelope

$$H_u(q) = \max\{h \in \mathbb{N} : (q_{0,N}, 0) \xrightarrow{u} (q, h)\},$$

with value $\perp$ when q is unreachable. For a suffix v, define the backward completion requirement

$$R_v(q) = \min\{h \in \mathbb{N} : (q, h) \xrightarrow{v} (f, c) \text{ for some } f \in F_N, c \in \mathbb{N}\},$$

with value ∞ when no such counter exists.

The pair (H, R) is the finite-dimensional replacement for the scalar residuals used earlier in the monograph.

Proposition 111.2 (Exact cut semantics)**.** *For every factorization* $w = uv$*,*

$$uv \in L(N) \iff \exists q \in Q_N : \ H_u(q) \ge R_v(q).$$

Hence

$$uv \notin L(N) \iff H_u(q) < R_v(q) \quad (\forall q \in Q_N).$$

Proof. An accepting run cut after u has some state q and counter h with $h \le H_u(q)$ and $h \ge R_v(q)$. Conversely, if $H_u(q) \ge R_v(q)$, take a prefix run attaining the envelope value and execute an accepting v-run that needs only $R_v(q)$ units of credit. One-counter monotonicity permits the same transition sequence from any larger initial credit. □

For a fixed factor x, let T_x be the exact forward-envelope transformer and S_x the exact backward-requirement transformer. If a path $\pi : q \xrightarrow{x} r$ has total effect $\mathrm{eff}(\pi)$ and minimum entry requirement $\mathrm{req}(\pi)$, then

$$(T_x H)(r) = \max_{\substack{q,\pi: q \xrightarrow{x} r \\ H(q) \geq \mathrm{req}(\pi)}} \big(H(q) + \mathrm{eff}(\pi)\big),$$

and

$$(S_x R)(q) = \min_{r,\pi: q \xrightarrow{x} r} \max\{\mathrm{req}(\pi), R(r) - \mathrm{eff}(\pi)\}.$$

Theorem 111.3 (Exact residuation)**.** *For every finite envelope H, requirement vector R, and word x,*

$$T_x(H) < R \iff H < S_x(R)$$

componentwise, with the usual conventions for $\bot$ and ∞.

Proof. Fix a source coordinate q. The inequality $H(q) < S_x(R)(q)$ says that for every x-path $\pi : q \to r$, either $H(q)$ is below its entry requirement or the resulting credit remains below $R(r)$. This is exactly the assertion that no contribution from q can make any coordinate of $T_x(H)$ reach the corresponding completion requirement. □

111.3 Shortest counterexamples and bounded-height compactness

Assume the reverse-inclusion promise

$$L(N) \subseteq L(D)$$

and choose a shortest word

$$w \in L(D) \setminus L(N).$$

Let the unique deterministic run visit (p_t, d_t) and let H_t be the exact N-envelope at the same cut.

Lemma 111.4 (Mixed-domination deletion)**.** *Suppose $i < j$ satisfy*

$$p_i = p_j, \qquad d_i \geq d_j, \qquad H_i \leq H_j.$$

Then deleting the factor between the two cuts produces a shorter word in $L(D) \setminus L(N)$.

Proof. The unchanged deterministic suffix starts from at least the original height d_j, so it remains feasible and accepting. On the nondeterministic side, rejection at cut j gives $H_j < R_j$ for the exact requirement of the unchanged suffix. Since $H_i \leq H_j$, the shortened word is still rejected. □

This immediately gives the first effective compactness theorem.

Theorem 111.5 (Effective bounded-height compression)**.** *For every fixed B there is a total computable $F(B, D, N)$ such that every shortest counterexample with*

$$d_t \leq B \qquad (0 \leq t \leq |w|)$$

has length at most $F(B, D, N)$.

Proof. Encode $\perp$ by zero and a finite envelope value h by $h+1$. At cut t record

$$X_t = (p_t, B - d_t, \widehat{H}_t).$$

For each fixed deterministic control state, order the last $1+|Q_N|$ coordinates componentwise. Dickson's lemma makes the finite disjoint union of these orders a well-quasi-order. A comparable pair $X_i \leq X_j$ would satisfy the hypotheses of Lemma 111.4, so a shortest counterexample yields a bad sequence.

For effectivity, at depth t every finite forward-envelope coordinate is at most t in the unit-step model. Hence the tree of all bad sequences satisfying the depth-dependent coordinate bounds and $d_t \in \{0, \ldots, B\}$ is effectively finitely branching. Dickson excludes an infinite branch and König therefore makes the whole tree finite. Breadth-first generation computes its maximum depth. □

Thus all remaining difficulty is concentrated in high deterministic excursions.

111.4 Promise geometry: sources, sinks, and slope budgets

Long high regions expose bounded synchronized event cores and finite connector regimes. The following proofs make the event extraction and connector-length bound explicit, including the backward projective phase and the exact deletion-or-pumping alternative.

111.4.1 Protected exposure and bounded event cores

The hard case is therefore a shortest counterexample with large deterministic height.

Lemma 111.6 (Positive-credit exposure)**.** *Let x be a factor between two cuts with the same deterministic control state. Suppose*

$$\delta_D(x) = c > 0, \qquad H_{\text{before}} \leq H_{\text{after}}.$$

Then, in a shortest counterexample, the deterministic suffix after x must later visit a height at most $c-1$.

Proof. If the suffix stayed at height at least c, deleting x would lower the entire suffix by c but keep it nonnegative. The deterministic word would still be accepted. The nondeterministic envelope before x is no larger than after x, so the deleted word would remain rejected, contradicting shortestness. □

The proof repeatedly extracts short synchronized cycles. Put

$$M_2 = |Q_D||Q_N|, \qquad M_3 = |Q_D||Q_N|^2.$$

A simple cycle in the synchronized pair graph has length at most M_2, and a simple cycle in the synchronized triple graph has length at most M_3.

Definition 111.7 (Primitive event cores)**.** The primitive event universe consists of:

(i) simple deterministic control cycles with nonzero effect;

(ii) synchronized D/N pair cycles of length at most M_2 with nonzero relative effect

$$\Delta_{\text{rel}} = \Delta_N - \Delta_D;$$

(iii) synchronized $D/N/N$ triple cycles of length at most M_3 with nonzero strand-differential effect;

(iv) bounded boundary/support events.

All primitive core words have effectively bounded length; write B_{core} for a common bound.

Backward spread and finite connector phase

For a finite requirement vector R on support A, write

$$\operatorname{spread}(R) = \max_{q,r\in A} |R(q) - R(r)|.$$

Lemma 111.8 (Backward differential escape)**.** *Let*

$$C_R = 2(M_3 - 1).$$

Along a connector segment with stable finite backward support, if

$$\operatorname{spread}(R) > C_R,$$

then two Bellman-optimal completion strands expose a synchronized $D/N/N$ triple cycle with nonzero N-strand differential effect.

Proof. Choose two finite coordinates realizing a gap larger than C_R and follow Bellman-optimal strands through the connector. Record the synchronized triple consisting of the deterministic control state and the two N control states. If every repeated triple had zero differential effect between the two N strands, erase the corresponding closed subwalks. The remaining triple path is simple, has length at most M_3-1, and each step changes the strand gap by at most 2. Hence the total gap change is at most $2(M_3 - 1)$, a contradiction. □

Corollary 111.9 (Finite backward projective phase)**.** *If a connector exposes neither a boundary event nor a triple differential core, then its normalized backward requirement profile ranges over an effectively finite set, independently of the absolute requirement level.*

Proof. The support is finite, and Lemma 111.8 bounds every pairwise finite-coordinate difference by C_R. Normalize by subtracting the minimum finite coordinate. □

Theorem 111.10 (Connector compression)**.** *There is a total computable constant B_{conn} such that any segment of a shortest counterexample that exposes none of*

(a) an N-boundary event,

(b) a bounded differential core,

(c) a bounded nonzero-D sign core,

(d) an already available pump/deletion certificate,

has length $< B_{\mathrm{conn}}$.

Proof. By Corollary 111.9, throughout the connector the normalized finite part of the backward requirement belongs to an effectively finite set. Refine it to the connector color

$$\kappa_t = \big(p_t,\ A_t,\ R_t - \min_{A_t} R_t\,\mathbf{1}_{A_t},\ \text{threshold/support regime}\big),$$

where $A_t = \{q : R_t(q) < \infty\}$. The color set is finite.

Suppose $\kappa_i = \kappa_j$ for $i < j$, and let W be the intervening word. The deterministic control state is the same at both ends. If $\Delta_D(W) \neq 0$, a closed deterministic walk of nonzero total effect contains a simple nonzero-effect deterministic cycle, contrary to the connector hypotheses. Hence $\Delta_D(W) = 0$.

Equal projective backward profiles give

$$R_j = R_i + \beta\mathbf{1}_A$$

on the common finite support A, so R_i and R_j are componentwise comparable. Exact backward semantics gives $R_i = S_W(R_j)$. If $R_i \leq R_j$, deleting W preserves deterministic feasibility and the strict separator, contradicting shortestness. If $R_j \leq R_i = S_W(R_j)$, then W is a zero-effect backward-demand self-cover; monotonicity of S_W yields a pump certificate. Thus no connector color can repeat, and the number of cuts is bounded by the finite color count. This count is an effective choice of B_{conn}. □

Corollary 111.11 (Event-skeleton normal form)**.** *Every shortest counterexample decomposes into bounded event words separated by connectors of length* $< B_{\mathrm{conn}}$*, except when a local recurrence outcome has already been extracted.*

Local recurrence outcomes and certificate-free corridors. A local recurrence outcome is a finite structural certificate obtained from a bounded macro configuration: an endpoint-preserving deletion, a contradiction to the reverse-inclusion promise, a bounded decisive continuation, or a repeatable power/arithmetic family relative to the current completion support. A corridor is called *certificate-free* only while none of these outcomes has been extracted. The following reductions concern such a corridor; once an outcome is found, that branch is passed to its already established finite case instead of being counted again.

111.4.2 Interior full phases and sign locking

The event analysis repeatedly uses intervals on which support and threshold regimes are stable. We isolate the exact recurrence used later.

Definition 111.12 (Compatible full phase)**.** Two interior cuts $i < j$ are in the same compatible full phase if:

(i) the deterministic control states agree;

(ii) the forward supports E_i, E_j agree;

(iii) the finite backward supports A_i, A_j agree;

(iv) the normalized forward profiles agree on E_i;

(v) the normalized backward profiles agree on A_i;

(vi) the finite threshold/regime labels agree.

Then there are integers α, β such that

$$H_j = H_i + \alpha \mathbf{1}_E, \qquad R_j = R_i + \beta \mathbf{1}_A$$

on their finite supports.

Let W be the intervening factor and

$$e = \Delta_D(W).$$

Theorem 111.13 (Full-phase sign lock)**.** *In a shortest certificate-free counterexample, a nontrivial compatible full-phase return satisfies*

$$\operatorname{sgn}(e) = \operatorname{sgn}(\alpha) = \operatorname{sgn}(\beta) \neq 0.$$

Thus:

$$e > 0 \Rightarrow \alpha > 0,\ \beta > 0, \qquad e < 0 \Rightarrow \alpha < 0,\ \beta < 0,$$

and $e = 0$ *cannot survive.*

Proof. We first prove the forward sign.

If $e > 0$ and $\alpha \leq 0$, then

$$T_W(H_i) = H_j \leq H_i.$$

One executable copy of W raises the deterministic height, hence all further copies are executable and the fixed suffix remains deterministically feasible. Monotonicity gives

$$T_W^k(H_i) \leq T_W(H_i) = H_j \qquad (k \geq 1).$$

Since the original word is rejected at cut j,

$$H_j < R_j,$$

so every word uW^kv is in $L(D) \setminus L(N)$, contradicting certificate-freeness.

If $e < 0$ and $\alpha \geq 0$, deleting W starts the deterministic suffix from a higher counter value. Moreover

$$H_i \leq H_j < R_j,$$

so the deleted word remains N-rejected, contradicting shortestness.

If $e = 0$ and $\alpha \geq 0$, the same deletion argument applies. If $e = 0$ and $\alpha < 0$, then W is a neutral forward-contraction return; iterating it preserves deterministic feasibility and yields a neutral pump family. Hence no zero-effect same-phase return survives.

Now prove the backward sign. Exact backward semantics gives

$$R_i = S_W(R_j), \qquad R_j = R_i + \beta \mathbf{1}_A.$$

If $e > 0$ and $\beta \leq 0$, then

$$R_j \leq R_i = S_W(R_j).$$

Thus W is a backward-demand self-cover at phase R_j. The deterministic side can repeat W indefinitely, and monotonicity of S_W yields

$$R_j \leq S_W(R_j) \leq S_W^2(R_j) \leq \cdots .$$

Since the original separator satisfies

$$H_i < S_W(R_j),$$

every further copy preserves rejection, giving an infinite $D \setminus N$ family.

If $e < 0$ and $\beta \geq 0$, then $R_i \leq R_j$. Deleting W starts the deterministic suffix higher, and

$$H_i < R_i \leq R_j$$

shows that the shortened word remains rejected by N.

The case $e = 0$ has already been excluded by the forward argument. Therefore

$$\operatorname{sgn}(e) = \operatorname{sgn}(\alpha) = \operatorname{sgn}(\beta) \neq 0.$$

□

Remark 111.14. The theorem is used only on genuine interior full phases. Equality of an *active-only* backward projection is not sufficient and is never used as a substitute for this hypothesis.

Corollary 111.15 (Large clean excursion)**.** *Inside a clean interior corridor with finitely many full-phase colors, a sufficiently large deterministic rise contains a positive full-phase return, and a sufficiently large deterministic fall contains a negative full-phase return.*

Proof. Take first-hit cuts of successive deterministic heights. More height levels than full-phase colors force equal colors at distinct heights. Theorem 111.13 determines the corresponding scalar signs. □

111.4.3 Boundary and low-active branches

Boundary recurrence is used only as a local reduction; we do not require a uniform D, N-only count at unbounded entry height.

Proposition 111.16 (Boundary recurrence reduction)**.** *Refine a boundary occurrence by its deterministic control state, support data, and bounded forward boundary profile. For two consecutive occurrences of the same refined signature, exactly one of the following occurs:*

(i) the intervening deterministic return has positive effect, exposing a positive full-phase/core event;

(ii) it has zero effect, giving mixed-domination deletion or a neutral return handled by exact-forward recurrence; or

(iii) it has negative effect.

Consequently, on any segment whose deterministic height is already bounded by K, the number of unresolved boundary recurrences is effectively bounded in terms of D, N, K. No uniform multiplicity conclusion is asserted when the entry/peak height is unbounded.

Proof. There are finitely many refined signatures. Equality of two signatures fixes the finite semantic data used by the boundary analysis. The sign of the intervening deterministic return gives the three cases above. In the negative case each recurrence of one fixed signature lowers deterministic height by at least one; under a height bound K this can happen at most $K+1$ times for that signature. Taking the finite maximum over signatures gives the local bound. □

The opposite difficulty is a large backward requirement with small active forward credit.

Theorem 111.17 (Low-active exact recurrence)**.** *There is an effective constant B_{PB} such that every pure-backward/low-active branch either*

(i) yields a deletion,

(ii) yields a positive-return power certificate, or

(iii) contains at most B_{PB} relevant rejected cuts.

No bound on the absolute magnitude of the backward requirement is needed.

Proof. When all active forward coordinates are bounded, record the exact color

$$\chi(t) = \big(p_D(t), A_t, H_t|_{A_t}\big),$$

where A_t is the support of finite backward requirements. This color set is finite. Equality of two colors with nonpositive deterministic effect permits deletion. Equality with positive effect yields a repeatable deterministic return while the active forward envelope does not improve enough to cross the fixed suffix requirement; hence one obtains a power family of counterexamples. □

111.4.4 Anchored relative envelopes and the no-core case

Define the deterministic-height anchored envelope

$$F_t(q) = H_t(q) - d_t$$

on reachable coordinates.

Lemma 111.18 (Relative escape)**.** *Let*

$$C_{\mathrm{rel}} = 2(M_2 - 1).$$

If for some cut and state q,

$$|F_t(q)| > C_{\mathrm{rel}},$$

then a maximizing synchronized prefix path contains a simple pair cycle C with

$$\Delta_N(C) - \Delta_D(C) \neq 0.$$

Proof. If every repeated synchronized pair state had zero relative drift, erase those closed subwalks. The remaining simple pair path has length at most $M_2 - 1$, so the relative displacement is bounded by $2(M_2 - 1)$. □

If no relative-differential core occurs, the anchored phase space becomes finite.

Theorem 111.19 (No-core height bound)**.** *Assume a shortest certificate-free counterexample contains no relative-differential core. Then there is an effective finite anchored color set of size B_F, and*

$$d_t < H_* := B_F^2 + 2B_F + 1$$

at every cut.

Proof. Equal anchored colors at ascent cuts shift the full forward envelope exactly by the deterministic height difference. A positive anchored return either deletes or, by Lemma 111.6, forces a later low deterministic level. Pair each first-hit ascent level with the first later descent through the same level. More than B_F^2 such paired levels force two equal paired anchored colors, and deleting the equal-height positive and negative factors preserves deterministic feasibility and preserves rejection by monotonicity of the fixed middle transformer. This contradicts shortestness. □

Corollary 111.20. *In the no-relative-core branch, shortest counterexamples have an effective length bound*

$$F(H_* - 1, D, N).$$

Thus every remaining unbounded branch repeatedly exposes bounded relative/differential event cores.

The standing promise now imposes a numerical restriction on recurrent sources and sinks. Consider a compatible ordered route with an earlier synchronized source

$$X : (a, +A)$$

and a later synchronized sink

$$Z : (-b, -B),$$

where $A, B, b > 0$ and compatibility means that fixed bounded connectors lead from the prefix to X, from X to Z, and from Z to one accepting N-suffix.

Lemma 111.21 (Promise source sign)**.** *If a synchronized loop has positive N-effect and can be pumped to cross a fixed accepting-suffix requirement, then its deterministic effect is nonnegative.*

Proof. A negative deterministic effect together with positive N-effect would, after enough repetitions, supply the nondeterministic suffix while forcing deterministic underflow. That would produce a word in $L(N) \setminus L(D)$, contradicting the promise. □

Theorem 111.22 (Promise efficiency inequality)**.** *Every compatible ordered source–sink route satisfies*

$$aB \geq bA.$$

When $a > 0$ this is

$$\frac{A}{a} \leq \frac{B}{b}.$$

Proof. Let δ be the fixed N-credit needed by the connector and accepting suffix. For n sink repetitions take

$$m_n = \left\lceil \frac{nB + \delta}{A} \right\rceil$$

source repetitions. The selected nondeterministic route is then feasible. If $aB < bA$, the deterministic net effect $m_n a - nb$ tends to $-\infty$, so sufficiently large n again yields a word accepted by N but blocked in D. □

111.4.5 Slope potentials and SCC budgets

Fix one realizable source-to-sink route class. Let

$$\rho_+$$

be the maximum positive N-per-D ratio over recurrent source cycles, and let

$$\rho_-$$

be the minimum N-loss-per-D-loss ratio over recurrent sink cycles. By Theorem 111.22,

$$\rho_+ \le \rho_-.$$

Choose a rational

$$\lambda \in [\rho_+, \rho_-].$$

Write $\lambda = p/q$ in lowest terms and define

$$\Psi = q\,\Delta_N - p\,\Delta_D.$$

Theorem 111.23 (Promise slope potential)**.** *Every recurrent block cycle in the compatible source and sink phases has*

$$\Psi \le 0.$$

Proof. For a source cycle with $\Delta_D > 0$, the inequality is exactly its ratio being at most λ. Source cycles of nonpositive N effect also have $\Psi < 0$. For a sink cycle $(\Delta_D, \Delta_N) = (-d, -n)$ with $d, n > 0$, the inequality is equivalent to $n/d \ge \lambda$. □

The critical zero-slope case is handled by a max-plus potential.

Lemma 111.24 (Critical graph normalization)**.** *Let S be a finite weighted strongly connected graph whose cycle weights are all nonpositive integers. Then there is an integer vertex potential h such that every reduced edge weight*

$$\bar{\psi}(e) = \psi(e) + h(\mathrm{src}(e)) - h(\mathrm{tgt}(e))$$

satisfies

$$\bar{\psi}(e) \le 0.$$

Hence every reduced edge is either critical, with weight 0*, or decreases by at least* 1*.*

Proof. Fix a root and let $h(v)$ be the maximum weight of a root-to-v path. This maximum is finite because there is no positive cycle. The defining inequality for h gives $\bar{\psi}(e) \le 0$. □

Define the adjusted budget at a mode state r by

$$B_\lambda = qN - pD - h(r).$$

It is preserved on critical edges and decreases by at least 1 on every noncritical edge.

The remaining issue is to bound the budget entering a critical SCC.

Theorem 111.25 (SCC cone separation)**.** *Let S be a synchronized SCC before a later critical sink. Its cycle cone in coordinates (Δ_D, Ψ) contains no vector with*

$$\Delta_D \geq 0, \qquad \Psi > 0.$$

Consequently there exists a rational $\mu \geq 0$ such that every cycle C in S satisfies

$$\Psi(C) + \mu\, \Delta_D(C) \leq 0.$$

Proof. Let K_S be the rational cone generated by the finitely many simple-cycle vectors (Δ_D, Ψ) of S. Suppose that

$$K_S \cap \{(d, s) : d \geq 0,\ s > 0\} \neq \varnothing.$$

Choose a rational vector in the intersection and clear denominators.

If its deterministic coordinate is strictly positive, multiply the selected cycle multiplicities by a common factor and join their bases by a fixed closed tour inside the SCC. The connector contribution is $O(1)$ while the selected vector scales linearly; for a sufficiently large factor we obtain a closed synchronized macro W with

$$\Delta_D(W) > 0, \qquad \Psi(W) > 0.$$

Thus the N-per-D source ratio of W is strictly larger than λ, in contradiction with Theorem 111.22 and the later λ-critical sink.

It remains to justify the boundary case in which the cone vector has deterministic coordinate 0. If S contains any positive-D cycle, add an arbitrarily small positive rational multiple of that cycle; strict positivity of Ψ is preserved, so we reduce to the preceding case. Otherwise every simple cycle of S has nonpositive deterministic effect. A nonnegative combination with total deterministic effect 0 can then use only zero-D cycles. Since its total Ψ is positive, at least one used zero-D simple cycle has positive Ψ; that cycle itself is a D-neutral/positive-N pumpable source before a genuine negative-D sink, contradicting the promise efficiency inequality (equivalently, its $a = 0$ case).

Hence the two rational cones are disjoint. The rational separating-hyperplane/Farkas alternative gives an effective $\mu \geq 0$ with

$$\Psi(C) + \mu\Delta_D(C) \leq 0$$

for every cycle C in S. □

Corollary 111.26 (Entry-credit bound)**.** *For every path π inside S,*

$$\Psi(\pi) \leq C_S + \mu D_{\text{in}},$$

for an effectively computable constant C_S. Thus bounded deterministic entry credit implies bounded adjusted budget.

Proof. Erase cycle subwalks. They have nonpositive $\Psi + \mu\Delta_D$ drift. The residual simple path has bounded length and hence bounded drift. □

Proposition 111.27 (Finite-depth funding provenance)**.** *Consider the synchronized product graph and its condensation DAG. If an SCC needs an unbounded amount of deterministic (respectively N) credit that it cannot generate by a cycle-cone direction with nonnegative drift in the other resource, then the required unbounded credit is already present at the entry of that SCC. Iterating this alternative moves strictly backward in the condensation DAG. Hence every unresolved funding chain has depth at most $|Q_D||Q_N|$.*

Proof. For one SCC S, let

$$K_S = \mathrm{cone}\{(\Delta_D(C), \Delta_N(C)) : C \text{ a simple cycle of } S\}.$$

For the D resource, either K_S contains a vector with $\Delta_D > 0$ and $\Delta_N \geq 0$, or rational cone separation supplies a nonnegative linear ranking

$$\eta_D \Delta_D + \eta_N \Delta_N \leq 0, \qquad \eta_D > 0,\ \eta_N \geq 0,$$

on every cycle. In the ranking branch, cycle erasure bounds D growth by a linear function of the N credit and the bounded simple-path transient; if the necessary D credit is unbounded, the corresponding required resource must therefore already be unbounded at SCC entry. The symmetric alternative holds for N.

If both resources are internally required without an unbounded entry seed, the two productivity directions can be combined, after positive rational rescaling, into a cone vector with both coordinates strictly positive. Such a bi-productive SCC contains a bounded synchronized joint-funder macro and is recorded as a source event rather than as unexplained entry credit. Otherwise the unresolved resource is pushed to a predecessor SCC. A condensation-DAG edge is crossed at each push, so the process terminates after at most the number of synchronized product SCCs, bounded by $|Q_D||Q_N|$. □

Consequently, in the cone-separated branch of Corollary 111.26, choose the earliest SCC whose adjusted budget would be unbounded. Proposition 111.27 makes its deterministic entry credit bounded; hence the adjusted budget is bounded as well.

Theorem 111.28 (Joint source–sink reduction)**.** *Consider an ordered compatible source/sink route with*

$$X : (a, +A), \qquad Z : (-b, -B), \qquad a, A, b, B > 0.$$

Assume the physical D/N endpoint credits of the route lie in a fixed effective bounded box. Then the following dichotomy holds under the promise $L(N) \subseteq L(D)$.

(i) If the realized efficiency gap is strict, all recurrent source/sink multiplicities in the localized route are effectively bounded.

(ii) If the realized slope is exact, the noncritical multiplicities are effectively bounded and the critical source/sink relation powers have effective preperiods and periods. Above an effective balanced-period threshold, one source period and one later sink period package can be deleted while rejoining the same separator state*: the identical deterministic physical configuration together with the identical critical N relation state seen by the unchanged suffix. Hence a shortest nondecisive exact-slope route also has an effective local multiplicity and literal-length bound.*

Thus at bounded endpoints the route is either an effective bounded local route, an endpoint-preserving shortening, or a decisive local outcome. The later bounded-entry localization theorems are used separately when an endpoint credit escapes; they are not part of this theorem's hypothesis.

Proof. Let ρ_+ be the maximal recurrent source ratio and ρ_- the minimal compatible sink cost. Theorem 111.22 gives $\rho_+ \leq \rho_-$. In the strict case choose $\lambda = \rho_-$ and clear denominators in $\Psi = q\Delta_N - p\Delta_D$. Chronological matching pairs positive deterministic funding only with later negative-D consumption. The finite set of realized type pairs has a positive minimum gap $\varepsilon_{\mathrm{route}}$, so the bounded endpoint/entry budget bounds the total matched funding; unmatched funding is absorbed by the bounded endpoints and simple skeleton. Thus all recurrent multiplicities are bounded.

Suppose now $\rho_+ = \rho_- = p/q$. Normalize by Lemma 111.24. Noncritical edges decrease the adjusted coordinate

$$B_\lambda = qN - pD - h(\mathrm{state})$$

by at least one, so bounded localized endpoints bound their total number. The critical source and sink relations are finite and therefore ultimately periodic. Choose positive relation periods r, s whose deterministic effects balance, say $ra = sb = K$. Because the exact-slope N effects balance as well, deleting these r source periods and the corresponding s later sink periods preserves the net D and critical-N effects. All connectors and internal requirements are bounded, so above an effective protection threshold the middle route stays feasible; after the sink batch the same separator state is rejoined: both the deterministic physical configuration and the critical N relation state are identical to those before deletion. Hence the unchanged suffix is evaluated against exactly the same separator semantics. Such a deletion contradicts shortestness. Below the threshold, the preperiods, periods, noncritical edges, and unmatched endpoint funding range over a finite effective box. This proves the stated local bound. □

The quantitative form needed later is worth recording explicitly.

Corollary 111.29 (Quantitative relative-source localization)**.** *For every bounded deterministic entry range $0 \le d \le B$ there is a computable threshold*

$$R_{\mathrm{rel}}(D, N, B)$$

such that, whenever a reachable envelope coordinate satisfies

$$H(q) - d > R_{\mathrm{rel}}(D, N, B),$$

its provenance contains an earlier bounded-entry synchronized source P with

$$\Delta D(P) \ge 0, \qquad \Delta N(P) > 0, \qquad \Delta N(P) - \Delta D(P) > 0.$$

Both source-entry counters are bounded effectively from D, N, B.

Proof. For every synchronized SCC whose cycle cone misses the target cone

$$\{(d, r) : d \ge 0,\ r > 0\}, \qquad r = \Delta N - \Delta D,$$

rational separation bounds relative gain by a computable affine function of deterministic entry credit plus a bounded simple-path/connector transient. The condensation DAG has finite depth, and there are only finitely many SCC, connector, and regime types. Taking the maximum of these effective bounds over all provenance chains with deterministic entry at most B produces R_{rel}. Above it, some SCC must realize a positive-relative cone direction, which can be converted, after bounded connectors, into the asserted closed source macro. □

111.5 Full-envelope control inside same-level high returns

Selected-run source/sink arguments are insufficient by themselves: a different nondeterministic run may become the maximizing envelope owner later. The proof therefore works in the synchronized triple graph

$$G_3 = Q_D \times Q_N \times Q_N$$

with differential weight

$$g = \Delta N(\pi) - \Delta N(\rho).$$

111.5.1 Differential potentials and canonical owners

Lemma 111.30 (One-sign differential potential)**.** *Let S be an SCC of G_3. If S has no negative g-cycle, then there is a computable potential ϕ and a constant C_S such that every path inside S changes the strand gap by at least $-C_S$. Dually, if S has no positive g-cycle, every path changes the gap by at most C_S. If every cycle has $g = 0$, then g is an exact endpoint-potential difference.*

Proof. In the no-negative case, fix a root and let $\phi(v)$ be the minimum g-weight of a root-to-v path. The minimum is finite because there is no negative cycle, and a minimizing path may be taken simple. Hence for every edge $u \to v$,

$$g(e) \geq \phi(v) - \phi(u).$$

Summing along a path gives the claim. Apply the same argument to $-g$ for the dual statement. If both signs are excluded, the inequalities force equality on every cycle and hence exact potential behavior. □

Lemma 111.31 (Mixed-sign zero-differential package)**.** *If a triple SCC contains a positive and a negative differential cycle, then it contains an effectively bounded same-base closed macro Z on which the two strands have equal total N effect.*

Proof. Join a common base to bounded simple positive and negative differential cycles and back by simple connectors. Repeat the positive and negative cycles only enough to dominate the bounded connector error, obtaining same-base macros of differential effects $A > 0$ and $-B < 0$. Taking $B/\gcd(A, B)$ copies of the first and $A/\gcd(A, B)$ copies of the second gives total differential effect zero. □

Fix a total order on N transitions. At every suffix-relevant envelope update, if several predecessor runs attain the same maximum, choose the least one in this order as the *canonical owner*. When the canonical owner changes from one maximizing strand to another, follow the two competing strands in G_3. Inside a fixed triple SCC S, define the *switch signature* to consist of

$$(\text{triple vertex},\ \text{ordered old/new owner states},\ \gamma,\ \text{event/regime label}),$$

where γ is the integer height gap between the two competing strands at the switch cut. Boundary and relative events are recorded separately rather than included as ordinary signatures.

Theorem 111.32 (Owner stabilization outside mixed-sign SCCs)**.** *Arbitrarily many semantically relevant envelope-owner changes in a bounded-return excursion can occur only inside a mixed-sign triple SCC. In a one-sign or all-zero SCC, either the number of owner switches is effectively bounded, or a repeated switch signature exposes an effectively bounded same-base zero-differential package. The only other exits are an already recorded boundary event or relative event.*

Proof. Let S first have no negative differential cycle; the dual case is symmetric. By Lemma 111.30, along every path in S the strand gap can decrease by at most the computable potential oscillation

$$C_S = \max_S \phi - \min_S \phi.$$

Hence if one candidate leads another by more than C_S (enlarged by the finite one-edge/event correction), the trailing candidate cannot become the canonical maximum before the path exits S. Every owner switch inside S therefore occurs with γ in one fixed finite integer interval. The triple vertex, ordered owner pair, and event/regime label also range over finite sets, so there are only finitely many switch signatures.

If more switches occur than signatures, choose two occurrences with the same signature. The intervening synchronized triple walk begins and ends at the same triple vertex and the same gap value, hence has total differential weight 0. In a one-sign SCC every directed simple cycle has differential weight of the same weak sign. Simple-cycle erasure of this closed walk therefore contains a simple cycle of differential weight 0; its length is at most $|Q_D||Q_N|^2$. Together with the bounded connectors to its base this is the required bounded same-base zero-differential package. If every cycle of S already has weight 0, the same conclusion is immediate from the exact endpoint-potential case of Lemma 111.30.

The condensation graph of G_3 is acyclic, so an excursion can change triple SCC only finitely many times. A switch that leaves the stable support/requirement regime is, by definition, a boundary or relative event. Thus unbounded switching without one of the stated bounded packages can persist only in a mixed-sign SCC, where Lemma 111.31 supplies the separate mixed-sign reduction. □

111.5.2 Positive relative escape and lag/catch-up reduction

The positive-relative branch admits a full-envelope localization, not merely a selected-strand one.

Theorem 111.33 (Canonical positive-relative source cut)**.** *For every bounded deterministic entry range there are computable constants B_{src} and B_{full} with the following property. If at a bounded-D cut some reachable coordinate satisfies $H(q) - D$ above the corresponding positive-relative threshold, then at an earlier cut s there is a bounded positive-relative source and*

$$D(s) \le B_{\mathrm{src}}, \qquad \max_q H_s(q) \le B_{\mathrm{full}}.$$

Proof. Fix an a priori deterministic entry bound B for the cut in the theorem. Corollary 111.29 supplies the effective threshold $R_{\mathrm{rel}}(D, N, B)$ and, whenever that threshold is crossed, a computable bound on the deterministic and selected-N entry counters of the resulting localized source. Let B_{src} dominate those source-entry bounds over the finite source/regime types. Let C_* be the maximum bounded connector/simple-path correction occurring in the finite source-provenance construction, and choose one computable threshold R^* satisfying

$$R^* \ge R_{\mathrm{rel}}(D, N, B), \qquad R^* \ge R_{\mathrm{rel}}(D, N, B_{\mathrm{src}}) + C_*.$$

For every envelope coordinate whose anchored value $H(q) - D$ exceeds R^* at the given cut, apply Corollary 111.29 to a realizing predecessor strand. This produces a bounded-entry positive-relative source certificate in the prefix. Among all such certificates feeding all large coordinates, choose one whose *entry cut* s is leftmost; ties at the same cut are broken by the fixed finite order on source types. By construction,

$$D(s) \le B_{\mathrm{src}},$$

and the selected N entry coordinate of that certificate is bounded as well.

We claim that every other reachable coordinate at the same entry cut is bounded. Suppose not. Then for some r,

$$H_s(r) > B_{\mathrm{src}} + R^*.$$

Since $D(s) \le B_{\mathrm{src}}$, we have

$$H_s(r) - D(s) > R^* \ge R_{\mathrm{rel}}(D, N, B_{\mathrm{src}}) + C_*.$$

After absorbing the bounded connector/simple-path correction, the anchored seed still exceeds the quantitative threshold for deterministic entry range B_{src}. Apply Corollary 111.29 to a realizing strand for $H_s(r)$, now using the prefix ending *at* s. The large seed is already present before the chosen source macro begins, so its provenance certificate has an entry cut strictly to the left of s; a certificate tied at s cannot generate credit that is already present at its own entry. This contradicts the leftmost choice. The same argument covers ties among several maximum coordinates at the original cut because the leftmost minimization is taken simultaneously over all coordinates exceeding R^*.

Thus every finite $H_s(r)$ lies below the computable common bound

$$B_{\mathrm{full}} := B_{\mathrm{src}} + R^*.$$

Taking the maximum over the finite source/regime types gives one pair of constants $B_{\mathrm{src}}, B_{\mathrm{full}}$ for the stated bounded deterministic entry range. □

Call such a bounded full-envelope cut *POS-canonical*. Between POS-canonical cuts, any remaining unbounded spread is negative relative lag. Follow a predecessor chain that later becomes semantically relevant. A large catch-up contains a bounded positive-relative core. Outside canonical regeneration its deterministic effect must be negative; if its N effect were positive, pumping it would eventually accept in N while underflowing D, contradicting the promise. Thus surviving catch-up is N-nonpositive. Dually, a genuinely new lag generator has positive deterministic effect and negative relative effect.

Proposition 111.34 (Lag/catch-up sign reduction)**.** *Outside already classified drain/demand branches and POS-canonical regeneration, every recurrent lag/catch-up pair has the form*

$$C_{\mathrm{lag}} : (+a, -A), \qquad C_{\mathrm{up}} : (-b, +B),$$

with $a, A, b, B > 0$ and $\Delta_N(C_{\mathrm{up}}) = -b + B \leq 0$. Any positive-N recurrent mode on the lag side reduces to the strict/exact source–sink program or contradicts the standing promise. Hence the only unresolved recurrent pair has nonpositive useful N maximum mean on both sides, with zero-mean residues recorded by finite critical relation states.

Proof. Write a recurrent macro in coordinates (d, r), where $d = \Delta_D$ and $r = \Delta_N - \Delta_D$ is the anchored relative effect.

A genuine new lag generator has $r < 0$. If $d < 0$, it is already a negative-D drain/demand event; if $d = 0$, it is a zero-D losing recurrence handled by the neutral/mean classification. Thus an unresolved lag generator must have $d = a > 0$ and $r = -A < 0$, giving $C_{\mathrm{lag}} : (+a, -A)$.

A catch-up macro has $r > 0$. If $d \geq 0$, repetition creates an unbounded positive relative seed. Theorem 111.29 first localizes that seed to bounded deterministic entry, and Theorem 111.33 then regenerates a POS-canonical cut. Outside that branch we must therefore have $d = -b < 0$. Write $r = B > 0$. Its N effect is $-b + B$. If this were positive, the recurrent macro could be pumped: N would gain credit while D loses credit, eventually producing an N-accepting/D-underflowing word and contradicting $L(N) \subseteq L(D)$. Hence $-b + B \leq 0$.

Finally consider any other recurrent useful mode on the lag side. A mode with positive N effect and negative D effect contradicts the promise as above; with zero D effect its repetition creates a positive-relative seed and triggers bounded-entry localization followed by canonical regeneration; with positive D effect it is a genuine source whose later negative-D consumption is governed by the strict/exact source–sink alternatives. After removing these cases, no surviving useful block graph can have positive maximum N mean. Zero-mean critical behavior is retained only through its finite critical relation state; no multiplicity bound is inferred from relation periodicity itself. □

These finitely many surviving event modes are assembled into a raw graph M_{raw}. Each edge carries its deterministic effect $d(e)$, tracked nondeterministic effect $n(e)$, finite control/regime data, internal requirements, and a literal-length bound.

Zero-D closed-mode recurrence is handled from the *concrete exact entry envelope*: the local search state is a finite mode vertex together with the exact envelope. Dickson plus König gives an effective finite node-dependent search tree unless mixed-domination deletion occurs.

Positive-N cycles are then removed by sign.

Lemma 111.35 (Chronological compatible matching)**.** *In one same-level bounded-return excursion, after cycle erasure every recurrent positive deterministic unit can be matched in chronological order to later negative deterministic consumption, up to a computably bounded correction absorbed by the simple residual skeleton and bounded endpoint fragments. Every matched source/sink pair belongs to one of finitely many compatible ordered route classes.*

Proof. Give each positive-D recurrent occurrence one token per gained unit and each negative-D occurrence one consumption slot per lost unit. Scan from left to right and assign each slot to the earliest available earlier token. Since the excursion returns to its

starting deterministic level, only the bounded residual skeleton can account for unmatched tokens or initial deficits. Erasing recurrent subwalks between one matched source and sink leaves a simple synchronized connector. Accessibility comes from the original prefix and coaccessibility from the original suffix, while support, requirement regime, critical residue, and internal requirements range over finite event data. Hence only finitely many compatible route classes arise. □

A negative-D/positive-N useful cycle contradicts the promise. A zero-D/positive-N cycle belongs to the exact-entry WQO system. A positive-D/positive-N cycle is matched by Lemma 111.35 to later negative-D consumption and is handled by the strict or exact source–sink slope alternatives. Exact slopes use finite critical relation periods together with balanced deletion: above an effective protection threshold, one source period and one later sink period can be deleted while rejoining the same deterministic physical configuration and the same critical N relation state seen by the unchanged suffix.

After replacing these discharged episodes by finitely many exact bounded representatives, and retaining bounded-entry deferral transfers with exact scalar entry/exit labels, one obtains a finite reduced graph M_{red}.

Theorem 111.36 (No positive cycle in the reduced graph)**.** *Every directed cycle* Γ *of* M_{red} *satisfies*

$$n(\Gamma) \le 0.$$

Proof. Expand each reduced macro edge to its exact bounded representative. If the resulting closed useful walk had positive tracked N-effect, simple-cycle decomposition would expose a positive useful recurrent cycle of the raw graph. The preceding trichotomy sends such a recurrence to promise contradiction, zero-D exact-entry compactness, bounded strict/exact source–sink reduction, or bounded-entry deferral. The first three are removed rather than recurrent in M_{red}; retained deferral edges are split by exact scalar transfer, so any closed cycle of transfers telescopes and cannot create positive net scalar gain. □

Theorem 111.37 (Arbitrary-interleaving envelope bound)**.** *Fix a same-level bounded-return excursion entered with deterministic control* p *and exact envelope* H_0*. There is a computable constant*

$$C_{\mathrm{BR}}(D, N, p, H_0)$$

such that every suffix-relevant forward-envelope coordinate throughout any retained unresolved excursion is at most

$$\max_{q : H_0(q) \neq \bot} H_0(q) + C_{\mathrm{BR}}(D, N, p, H_0).$$

Proof. Inside each SCC of M_{red}, Theorem 111.36 gives a max-plus potential bounding every path's positive tracked gain by an endpoint constant. The SCC condensation graph is acyclic, so inter-SCC gain contributes only a finite DAG correction. Every collapsed zero-D, source–sink, or retained deferral edge has a computable internal peak above its endpoints. Taking the maximum internal correction over the finite reduced graph, rather than multiplying by the number of traversals, gives the asserted bound under arbitrary interleaving. □

111.6 Context-safe recurrence and exact-entry bounded-return compactness

The full-envelope bound makes the exact active color set finite at canonical cuts. An active color records the deterministic control, finite completion support A, and the exact restricted envelope $H|_A$.

Lemma 111.38 (Exact active-color deletion)**.** *If two canonical cuts of a rejected word have the same exact active color and the intervening deterministic effect is nonpositive, deleting that factor preserves deterministic acceptance and nondeterministic rejection of the unchanged suffix.*

Proof. The deterministic control state agrees, and nonpositive effect means the earlier cut has at least as much deterministic credit as the later cut. On the N side the active-envelope coordinates are identical, while states outside the completion support cannot accept the unchanged suffix. Thus the same strict cut separator survives deletion. □

A retained exact-color cycle therefore has strictly positive integral deterministic effect. This yields a global event bound inside a same-level return without any cumulative N-loss assumption.

Theorem 111.39 (Positive-return total multiplicity bound)**.** *For every concrete exact same-level entry (p, H_0), the total number of canonical event edges in a shortest retained bounded-return excursion is effectively bounded, uniformly over arbitrary interleaving of the remaining reduced modes.*

Proof. Let G_{col} be the finite exact-color graph induced after Theorem 111.37. Apply ordinary cycle erasure to the traced vertex sequence. It decomposes into a simple residual path σ and directed simple cycles $\Gamma_1, \dots, \Gamma_m$. Let C_{col} be the number of color vertices and D_{edge} the largest absolute deterministic effect of one reduced edge. Then

$$d(\sigma) \geq -(C_{\mathrm{col}} - 1)D_{\mathrm{edge}}.$$

The complete excursion has deterministic net effect zero, up to a fixed bounded endpoint correction C_{end}, so

$$\sum_{j=1}^{m} d(\Gamma_j) \leq (C_{\mathrm{col}} - 1)D_{\mathrm{edge}} + C_{\mathrm{end}}.$$

Lemma 111.38 makes every retained cycle effect a positive integer. Hence m is bounded, and each simple cycle has at most C_{col} edges. This bounds the whole canonical event trace. □

Neutral zero-D powers admit exact-envelope self-cover: for the computable sequence $H, T_x H, T_x^2 H, \dots$, Dickson yields $i < j$ with $T_x^i H \leq T_x^j H$, so mixed-domination supplies deletion on the retained shortest branch. The other recurrent cases retain their exact source distinctions below.

111.6.1 Negative and zero mean

Let C be a bounded recurrent core with $d = \Delta_D(C) \geq 0$ and concrete exact entry envelope H. Fix only the finite completion support A of the inherited suffix.

If the useful C-block graph has maximum mean $\mu_C < 0$, cycle erasure computes $\varepsilon_C > 0$ and K_C such that every useful k-block path has N effect at most $K_C - \varepsilon_C k$. Therefore there is a support-parametric kill threshold

$$K_{\mathrm{kill}} = 1 + \left\lceil \frac{\mathrm{maxfin}(H) + K_C}{\varepsilon_C} \right\rceil$$

(up to the finite internal-requirement correction) after which no useful route can reach A with nonnegative credit. If $d = 0$, excess copies delete. If $d > 0$, an excess copy either deletes or is indispensable only because its deterministic funding is consumed later, forcing a later bounded-D exposure and transfer to the LOW/source–sink machinery.

If $\mu_C = 0$, a critical potential bounds all gain relative to the exact entry. Suffix interaction reduces to a finite semantic quotient: either a large-deficit class, where no power can reach the completion support, or one of finitely many bounded-deficit critical relation states. Periodic recurrence at $d = 0$ deletes directly. At $d > 0$, failure of deletion again exposes later deterministic consumption.

Theorem 111.40 (Mean-recurrence context elimination)**.** *Negative-mean and zero-mean local recurrences from a concrete exact entry contribute only a bounded local transient, endpoint-preserving deletion, deterministic-funding transfer to a later bounded-D exposure, or ordinary continuation. No complete witness bound is obtained by appending an inherited suffix.*

111.6.2 Strict and exact source–sink routes

For a compatible positive-D source C_+ and negative-D sink C_- define

$$\rho(C_+) = \frac{\Delta_N(C_+)}{\Delta_D(C_+)}, \qquad \kappa(C_-) = \frac{-\Delta_N(C_-)}{-\Delta_D(C_-)}.$$

The promise gives $\rho \le \kappa$. If endpoints are bounded and the inequality is strict, chronological matching charges each unit of deterministic funding an N-budget loss of at least one positive computable gap $\varepsilon_{\mathrm{route}}$. Hence total matched funding, and thus route length, is bounded. If an endpoint N credit escapes while D is bounded, anchored relative escape localizes it to a bounded-entry source/drain episode, which returns to the already classified mean/source–sink cases rather than carrying an unbounded suffix scalar.

At exact slope $\rho = \kappa = p/q$, use the adjusted coordinate

$$B_\lambda = qN - pD - h(\text{state}).$$

Noncritical edges decrease it and critical relation powers have finite preperiods and periods. Above an effective protection threshold, balanced source and sink critical periods can be deleted together so that the same separator state — the same D physical configuration and the same critical N relation state — is rejoined before the unchanged suffix.

Theorem 111.41 (Source–sink context elimination)**.** *Every source–sink route is reduced, without bounding the inherited suffix, to one of:*

(i) a bounded strict local route;

(ii) a bounded exact-critical local route;

(iii) endpoint-preserving balanced-period shortening;

(iv) bounded-entry localization to an already classified local mechanism; or

(v) a contradiction to $L(N) \subseteq L(D)$.

Combining these mechanisms gives the local theorem on which the global construction rests.

Theorem 111.42 (Exact-entry bounded-return compactness)**.** *For every concrete same-level high entry (p, H) there is a total computable integer*

$$L^{\dagger}_{\mathrm{BR}}(D, N, p, H)$$

such that every same-level bounded-return factor in a shortest hard-direction counterexample has length at most this number, unless the local analysis already produces endpoint-preserving shortening, a contradiction to $L(N) \subseteq L(D)$, or a bounded decisive continuation from the current exact node.

Proof. First compute the exact-entry zero-D search trees and the positive-cycle reduction, then the reduced-graph potential constant C_{BR}, then the full-envelope bound, then the finite exact-color graph and its maximum local literal edge length, and finally the event-edge bound of Theorem 111.39. None of these quantities uses the terminal ladder, the later LOW-state tree, or the final witness bound. Multiplying the finite event-edge bound by the finite local edge-length bound and adding the bounded endpoint fragments gives $L^{\dagger}_{\mathrm{BR}}$. □

111.7 Terminal growth and the exact LOW continuation tree

Choose a computable low threshold B_0 above every primitive deterministic effect and every bounded internal requirement used by the local event decomposition. A cut is LOW when the deterministic counter is at most B_0.

After the last LOW cut, if the run never returns to the low band, let c_ℓ be the last occurrence of deterministic level ℓ. Between c_ℓ and $c_{\ell+1}$ there is one $+1$ deterministic step followed by a same-level bounded-return excursion. Thus Theorem 111.42 makes the successor set from one exact ladder state (p_ℓ, H_ℓ) effectively finite.

Lemma 111.43 (Terminal ladder deletion)**.** *If $\ell < \ell'$ satisfy*

$$p_\ell = p_{\ell'}, \qquad H_\ell \le H_{\ell'},$$

then deleting the whole factor from c_ℓ to $c_{\ell'}$ produces a shorter word in $L(D) \setminus L(N)$.

Proof. The deleted factor raises the deterministic level by $e = \ell' - \ell$. By last-occurrence minimality, the original suffix after $c_{\ell'}$ never falls below ℓ', so after deletion it remains nonnegative when shifted down by e. Rejection follows from $H_\ell \le H_{\ell'} < R_{\ell'}$. □

Hence the terminal ladder is a bad sequence in finitely many copies of $\mathbb{N}^{|Q_N|}$. Effective finite branching plus Dickson and König produce a total computable terminal-tail bound

$$L^{\dagger}_{\mathrm{term}}(D, N, p, H).$$

For the nonterminal part, record at every LOW cut the exact state

$$X = (p, d, H) \in Q_D \times \{0, \dots, B_0\} \times (\mathbb{N} \cup \{\bot\})^{Q_N}.$$

Order LOW states with the same p, d by componentwise envelope domination:

$$(p, d, H) \preceq (p', d', H') \iff p = p',\ d = d',\ H \le H'.$$

Lemma 111.44 (Exact LOW deletion)**.** *A shortest $D \setminus N$ witness contains no two LOW cuts $i < j$ with $X_i \preceq X_j$.*

Proof. Equal p, d means that the deterministic physical configurations coincide, so deleting the interval rejoins the identical accepting suffix. If R_j is the exact backward requirement of that suffix, rejection gives $H_j < R_j$; since $H_i \le H_j$, the shortened word remains rejected. □

111.7.1 Node-dependent successor enumeration

For one exact LOW state $X = (p, d, H)$ define $\mathrm{Succ}^{\dagger}(X)$ as follows. If $d < B_0$, or $d = B_0$ and the next deterministic effect is nonpositive, enumerate all feasible one-letter LOW successors. If $d = B_0$ and a feasible letter a has effect $+1$, compute the post-exit control and exact envelope

$$(p^+, H^+) = (p^+, T_a(H)).$$

If the actual run later returns to LOW for the first time, the high interior before the final -1 return step is a same-level bounded-return factor. Enumerate every D-feasible candidate interior of length at most

$$L^{\dagger}_{\mathrm{BR}}(D, N, p^+, H^+)$$

and every feasible final -1 letter, and compute the resulting exact LOW envelope. If no LOW return occurs, the branch is terminal and belongs to Section 111.7. If a bounded decisive continuation is produced, record its remaining length from the current node.

Theorem 111.45 (Effective LOW successors)**.** *For every exact LOW state X,* $\mathrm{Succ}^\dagger(X)$ *is finite and effectively enumerable, and contains the next LOW state of every shortest hard-direction counterexample whenever such a next LOW cut exists.*

Proof. The one-letter LOW cases are finite. In the $+1$ exit case the concrete exact post-exit envelope $H^+ = T_a(H)$ is computable, and Theorem 111.42 supplies a finite length bound for every retained same-level interior before the first return. Exhaustive enumeration up to that bound, followed by one feasible -1 return letter, therefore lists every possible next LOW state. Shortening and promise-contradiction outcomes cannot occur on the retained shortest promised branch; bounded decisive completions are recorded separately. No quantity defined from the later LOW tree is used in this enumeration. □

111.7.2 Finite bad-sequence tree

Build $T^\dagger_{\mathrm{low}}$ from the exact initial LOW state. A node is a finite sequence

$$X_0, \dots, X_k$$

with $X_{i+1} \in \mathrm{Succ}^\dagger(X_i)$ and no comparable pair $X_i \preceq X_j$ for $i < j$. Attach to every edge its literal realization length and let $w(X)$ be the accumulated path weight to node X.

Theorem 111.46 (Finite exact-LOW continuation tree)**.** *The tree $T^\dagger_{\mathrm{low}}(D, N)$ is finite and effectively constructible.*

Proof. Every node is effectively finitely branching by Theorem 111.45. Every branch is a bad sequence in a finite disjoint union of copies of $\mathbb{N}^{|Q_N|}$, so Dickson's lemma excludes infinite branches. König's lemma therefore makes the finitely branching tree finite. Breadth-first exhaustive generation terminates because every successor set is explicitly computable. □

111.7.3 Prefix-safe decisive accounting

For a constructed LOW-tree node X, let $w(X)$ be its accumulated literal edge length. Let $C_{\mathrm{dec}}(X)$ be the maximum remaining length of a genuinely bounded decisive continuation certified from that concrete node, and 0 if no such continuation occurs. Define

$$B^\dagger_{\mathrm{dec}}(D, N) = \max_{X \in T^\dagger_{\mathrm{low}}} (w(X) + C_{\mathrm{dec}}(X)).$$

A decisive continuation starting immediately after an exit letter receives that additional single-letter correction. This prefix term is essential: a short continuation from a node is not by itself a short complete word. Set

$$B^\dagger_{\mathrm{low}} = \max_{X \in T^\dagger_{\mathrm{low}}} w(X).$$

For all terminal $+1$ exits a from exact LOW nodes $X = (p, B_0, H)$, let $(p^+, H^+) = (p^+, T_a(H))$ and define

$$B^\dagger_{\mathrm{term}} = \max\{w(X) + 1 + L^\dagger_{\mathrm{term}}(D, N, p^+, H^+)\}.$$

These maxima are taken over an explicitly generated finite tree; the terminal bound itself depends on the exact post-exit state and is computed before this maximum. Write $B^\dagger_{\mathrm{tree}} = \max\{B^\dagger_{\mathrm{low}}, B^\dagger_{\mathrm{term}}, B^\dagger_{\mathrm{dec}}\}$. The independent bounded-height case is accounted for by Theorem 111.5, not by retroactively changing the local high-return bound.

Theorem 111.47 (Hard-direction small-witness theorem)**.** *Under the promise*

$$L(N) \subseteq L(D),$$

there is a total computable function $B^\dagger_{\mathrm{hard}}(D, N)$ such that

$$L(D) \not\subseteq L(N) \Longrightarrow \exists w \in L(D) \setminus L(N) : |w| \le B^\dagger_{\mathrm{hard}}(D, N).$$

Proof. Every branch of $T^{\dagger}_{\mathrm{low}}$ is a bad sequence in a finite disjoint union of copies of $\mathbb{N}^{|Q_N|}$, so Dickson excludes infinite branches. Theorem 111.45 supplies the finite successor sets, while Theorem 111.46 and König give an effectively finite whole tree. Attach each edge its literal realization length. Take the maximum accumulated LOW-tree weight, the maximum terminal-tail weight obtained after a high exit, and the maximum already-traversed prefix plus any genuinely bounded decisive continuation produced by the local analysis. These are maxima over a finished finite tree. Finally take the maximum with the independent bounded-height bound of Theorem 111.5. This yields $B^{\dagger}_{\mathrm{hard}}$. □

Corollary 111.48 (Hard inclusion under the reverse-inclusion promise)**.** *Under $L(N) \subseteq L(D)$, the inclusion $L(D) \subseteq L(N)$ is decidable.*

Proof. Compute $B^{\dagger}_{\mathrm{hard}}$ and exhaustively test all words up to that length. If none belongs to $L(D) \setminus L(N)$, Theorem 111.47 excludes any counterexample. □

111.8 Equality and the resolved comparison problem

The opposite inclusion orientation is classical. Hofman and Totzke proved decidability of deterministic-right OCN trace inclusion and their compact-witness method applies to monotone final-state acceptance relations [108]. Thus for nondeterministic N and deterministic D one can decide

$$L(N) \subseteq L(D).$$

Only decidability is used here; no complexity bound for the full equivalence algorithm is claimed.

Theorem 111.49 (Deterministic–nondeterministic OCN language equivalence)**.** *Given a deterministic one-counter net D and a nondeterministic one-counter net N over the same finite alphabet, with arbitrary integer transition weights, both started at counter value zero and accepting by final control state, it is decidable whether*

$$\boxed{L(D) = L(N).}$$

Proof. Normalize both nets by Lemma 111.1. First decide the deterministic-right inclusion

$$L(N) \subseteq L(D)$$

using the Hofman–Totzke theorem. If it fails, the languages are unequal. If it holds, the standing promise required by Corollary 111.48 is satisfied, so decide the reverse inclusion

$$L(D) \subseteq L(N).$$

The two normalized languages are equal exactly when both inclusions hold, and normalization preserves the original final-state languages. □

This resolves the fixed-presentation mixed equivalence question recorded as Automata Exchange Problem 24.04(B) [109]. The result should be read together with the decomposition chapters immediately preceding it:

fixed deterministic/nondeterministic OCN comparison is decidable

existential decomposition of a deterministic one-counter presentation is undecidable

These are different quantifier structures, not competing statements.

111.9 Scope of the net theorem and the zero-test frontier

The proof uses monotonicity at every scale: forward envelopes are upward-credit monotone, mixed-domination deletion lowers or preserves the relevant resources, and the terminal/LOW WQO arguments depend on the fact that added counter credit cannot invalidate a fixed transition sequence. An explicit zero test breaks exactly this global monotonicity. Therefore the theorem in this first branch does not by itself decide the corresponding problem in which the deterministic machine is a zero-testing OCA and the other machine is a nondeterministic counter net.

Conceptually, this is the one-counter analogue of a recurring distinction throughout the monograph. A physical feature may look syntactically small while changing the semantic order structure on which a normalization or compactness theorem depends. Here the absent zero test is not a cosmetic restriction: it is what makes the exact envelope order compatible with deletion and makes the two global bad-sequence trees possible. The next branch closes this zero-test frontier by introducing deterministic zero cuts, positive first-return compactness, and—for silent transitions—a finite weak-macro calculus.

Problem 24.04(A): deterministic zero-testing OCA versus nondeterministic OCN

Research branch — Closing the zero-test comparison frontier. *The preceding branch proves Problem 24.04(B) for deterministic versus nondeterministic one-counter nets, where global credit monotonicity drives the compactness argument. We now restore an explicit zero test on the deterministic side. This changes the proof architecture: zero cuts and positive first returns replace the global net-only deletion scheme, and the epsilon extension requires finite threshold-affine weak macros together with stable/reset deterministic normalization. The branch is logically independent of the proof of Problem 24.04(B), but it occupies the adjacent comparison frontier and completes the two Automata Exchange questions treated in this monograph.*

The earlier state-primality and two-factor chapters concern existential decomposition of one fixed deterministic presentation. Here both machines are supplied as input and the question is language equality. The left-hand device may test zero and is deterministic; the right-hand device has no zero test and may branch nondeterministically. The proof therefore requires a different invariant: exact forward envelopes and backward requirements replace the decomposition quotients, and shortest-counterexample compactness replaces factor construction.

111.10 Main theorems and proof architecture

A one-counter automaton (OCA) is a pushdown automaton with a singleton stack alphabet. Unlike a one-counter net (OCN), it may distinguish counter value 0 from positive values. We study the asymmetric language-equivalence problem in which the zero-testing machine is deterministic and the net is nondeterministic.

Problem 24.04(A) in the Automata Exchange collection asks whether language equivalence between a deterministic one-counter automaton and a nondeterministic one-counter net is decidable [109]. The letter-labelled formalization used in the surrounding one-counter-net literature is real time; we first solve that formulation and then prove that the same decidability survives a standard deterministic-epsilon extension on the OCA side together with arbitrary epsilon moves on the OCN side.

Theorem 111.50 (Real-time main theorem)**.** *Let A be a deterministic real-time unit-step one-counter automaton with zero tests and let N be a nondeterministic real-time unit-step one-counter net over the same finite alphabet. Both start with counter value* 0 *and accept finite words by final control state. It is decidable whether $L(A) = L(N)$.*

Theorem 111.51 (Real-time global small witness)**.** *There is a total computable function B such that for every pair (A, N) as in theorem 111.50,*

$$L(A) \not\subseteq L(N) \Longrightarrow \exists w \in L(A) \setminus L(N) \text{ with } |w| \leq B(A, N).$$

Theorem 111.52 (Epsilon-move extension)**.** *Given a deterministic unit-update zero-testing one-counter automaton A with standard deterministic epsilon moves and a nondeterministic unit-update one-counter net N with arbitrary epsilon moves and no zero tests, both initialized at counter 0 and using finite-word final-state acceptance, it is decidable whether $L(A) = L(N)$.*

Theorem 111.53 (Epsilon hard inclusion)**.** *For the machines of theorem 111.52, it is decidable whether $L(A) \subseteq L(N)$. Moreover a finite bound $B^{\varepsilon}_{\mathrm{global}}(A, N)$ is effectively computable such that failure of inclusion has a witness of visible length at most this bound.*

Theorem 111.54 (Epsilon reverse inclusion)**.** *For the machines of theorem 111.52, it is decidable whether $L(N) \subseteq L(A)$.*

Proof architecture. The real-time proof has four layers: exact residual semantics, positive first-return compactness, an effectively finite zero-state tree, and a terminal drain reduction. The epsilon extension does not eliminate silent moves syntactically. Instead it compiles each weak visible OCN step into a finite threshold-affine macro system, proves an internal guarded-macro robustness theorem for the compactness argument, and combines this with the standard stable/reset normalization of deterministic epsilon-OCA. The reverse inclusion in both settings is handled independently by one-zero-test VASS reachability [115, 116].

Relation to known results and novelty scope. Valiant and Paterson proved decidability of equivalence for deterministic one-counter automata [106], and Böhm, Göller, and Jančar later obtained the NL-completeness bound in the real-time setting [107]. This should be contrasted with deterministic-OCA language inclusion, which is already undecidable [110]; hence the present equality problem does not follow from a two-way inclusion test on deterministic OCAs.

On the net side, trace inclusion from a nondeterministic OCN into a deterministic OCN is decidable (indeed NL-complete in the standard trace setting), whereas general OCN trace inclusion is undecidable [108]; strong and weak simulation over OCNs remain decidable and PSPACE-complete [112]. General OCN determinizability is undecidable under most natural notions [113]. History-determinism gives a different positive frontier: history-deterministic OCNs can be converted to language-equivalent deterministic OCAs and their language inclusion/equivalence problems are decidable in PSPACE [114]. The same work notes that its HD-OCN results extend, with weak-simulation machinery, to epsilon-transitions; thus silent moves themselves are not the novelty claimed here.

The comparison class in this chapter is different from all of the preceding settings: the right-hand machine is an arbitrary nondeterministic OCN, not assumed deterministic or history-deterministic, while the left-hand machine may test zero and is deterministic. To the best of our knowledge, we are not aware of a prior published decidability theorem for finite-word final-state language equality for this exact asymmetric pair, either in the real-time formulation or under the epsilon conventions of theorem 111.52. We cite the Automata Exchange entry only as a record of Problem 24.04(A), not as evidence of priority or novelty [109]. Problem 24.04(B), proved in the preceding branch, is logically separate and is not used in any argument of the present zero-testing proof.

Organization. Sections 111.12–111.19 of this branch prove the real-time theorem. The subsequent epsilon-extension sections develop weak envelopes, guarded macros, reset-ending returns, the stable-zero tree, terminal tails, and the asynchronous one-zero-test product. The final ledger sections record both effectivity ledgers and the module-by-module transfer certificate.

111.11 Machines and acceptance

111.11.1 Deterministic one-counter automata

A *real-time unit-step one-counter automaton* is a tuple

$$A = (Q_A, \Sigma, \delta_A, q_A^0, F_A),$$

where the counter ranges over $\mathbb{N}$. Transitions are separated into zero and positive rules. A zero rule is enabled exactly when the current counter is 0; a positive rule is enabled exactly when the current counter is positive. Each rule reads one input letter and changes the counter by -1, 0, or $+1$, subject to nonnegativity. We assume that A is deterministic: for each control state, input letter, and zero/positive regime, at most one transition is enabled.

A configuration is $(p, c) \in Q_A \times \mathbb{N}$. The initial configuration is $(q_A^0, 0)$, and a word is accepted when the unique run is defined and ends in a control state in F_A, with arbitrary final counter value.

111.11.2 Nondeterministic one-counter nets

A *one-counter net* is a tuple

$$N = (Q_N, \Sigma, \delta_N, q_N^0, F_N)$$

with transitions (q, a, d, r), $d \in \{-1, 0, +1\}$. A transition is enabled at counter value c iff $c + d \geq 0$. There is no zero test. Acceptance is again by final control state with arbitrary final counter value.

For a word w, we write $L(A)$ and $L(N)$ for the corresponding final-state languages from initial counter 0.

111.11.3 Model scope and normalization

The theorem is stated for the standard *real-time* labelled one-counter model: every transition consumes exactly one input symbol. This is the convention used throughout the proof because literal factors of a shortest counterexample are repeatedly deleted and replaced. The Automata Exchange record for Problem 24.04(A) describes the left machine as "a pushdown automaton with one stack alphabet" and the right machine as a one-counter automaton with no zero tests, but does not itself print a transition syntax. The formal OCA/OCN definition used by Prakash and Thejaswini [114] matches the present convention: OCA transitions are given by

$$Q \times \{\mathsf{zero}, \mathsf{nonzero}\} \times \Sigma \times \{-1, 0, 1\} \times Q,$$

and OCN transitions by $Q \times \Sigma \times \{-1, 0, 1\} \times Q$, with determinism a partial function on the letter-labelled OCA rules. Their surrounding generic LTS preliminaries allow ε-labels, and one displayed OCA sentence inherits the notation Σ_ε; however the OCA transition relation itself is letter-labelled over Σ, and their paper explicitly distinguishes realtime LTSs. We therefore treat Theorem 111.50 as the standard real-time formulation of Problem 24.04(A). The stronger convention with silent transitions is treated separately later in the chapter: the deterministic side is first put into stable/reset normal form, while arbitrary silent behavior of the net is compiled into a weak threshold-affine macro calculus.

The unit-step assumption is inessential for finite integer weights.

Proposition 111.55 (Integer-weight normalization)**.** *Let M be a deterministic real-time one-counter automaton, or a nondeterministic real-time one-counter net, whose counter updates belong to a finite subset of $\mathbb{Z}$. One can effectively construct a language-equivalent unit-step machine of the same deterministic/nondeterministic and zero-testing/net type.*

Proof. Choose an integer K larger than every absolute transition weight and represent a counter value uniquely as $n = Kc + r$ with $0 \le r < K$. Store r in finite control and use c as the physical counter. For a transition of weight z, write $r + z = Kk + r'$ with $k \in \{-1, 0, 1\}$; update the physical counter by k and the finite remainder to r'. Underflow of the physical counter when $k = -1$ is exactly original underflow. For an OCA, original zero is characterized by the conjunction $c = 0$ and $r = 0$, which is expressible because r is finite control and $c = 0$ is the unique zero test. If $r > 0$, the original counter is positive irrespective of c, so the corresponding original positive rule is copied into both physical zero and physical positive regimes; if $r = 0$, it is copied only to the physical positive regime. Determinism is preserved because the remainder refines the source state. For an OCN no zero test is introduced. Acceptance controls are copied for every remainder. □

Remark 111.56 (Scope of Theorem 111.50). Consequently the real-time theorem covers arbitrary finite integer transition weights. Silent transitions are not folded into this normalization; they are handled by the separate epsilon-extension machinery developed later in the chapter.

111.12 Exact residual semantics of the net

The proof uses exact finite-dimensional summaries of the nondeterministic net, rather than a selected run.

Definition 111.57 (Forward envelope)**.** For a prefix u and $q \in Q_N$, define

$$H_u(q) = \max\{c \in \mathbb{N} : (q_N^0, 0) \xrightarrow{u} (q, c)\},$$

with $H_u(q) = \bot$ if q is unreachable. We order $\bot < h$ for every $h \in \mathbb{N}$.

Definition 111.58 (Backward requirement)**.** For a suffix v and $q \in Q_N$, define

$$R_v(q) = \min\{c \in \mathbb{N} : \exists f \in F_N,\ (q, c) \xrightarrow{v} (f, c')\},$$

with $R_v(q) = \infty$ if no such counter value exists.

Proposition 111.59 (Exact cut semantics)**.** *For every factorization $w = uv$,*

$$w \in L(N) \iff \exists q \in Q_N :\ H_u(q) \ge R_v(q).$$

Hence

$$w \notin L(N) \iff H_u(q) < R_v(q) \quad \textit{for all } q \in Q_N.$$

Proof. If w is accepted, cut an accepting run after u. If the cut configuration is (q, c) then $c \le H_u(q)$ and $c \ge R_v(q)$. Conversely, if $H_u(q) \ge R_v(q)$, choose a prefix run realizing $H_u(q)$ and use monotonicity of the net to execute an accepting v-run that needs only $R_v(q)$ units of credit. □

For a fixed word x, let T_x denote the exact forward-envelope transformer and S_x the exact backward-requirement transformer.

Theorem 111.60 (Exact residuation)**.** *For every forward envelope H, backward requirement R, and word x,*

$$T_x(H) < R \iff H < S_x(R)$$

componentwise.

Proof. Fix a source state q. The inequality $H(q) < S_x(R)(q)$ says that every x-path starting at q either requires more initial counter than $H(q)$ or exits below the corresponding requirement in R. This is exactly the statement that no contribution from q can make $T_x(H)$ meet R. Taking all source and target states gives the equivalence. □

111.13 Shortest counterexamples and zero cuts

Assume $L(A) \not\subseteq L(N)$ and choose a shortest word

$$w \in L(A) \setminus L(N).$$

Let (p_t, d_t) be the unique A-configuration after the prefix of length t, and let H_t and R_t be the exact net residuals at that cut. Proposition 111.59 gives

$$H_t < R_t$$

at every cut.

Lemma 111.61 (Zero-cut deletion)**.** *Suppose $i < j$, $d_i = d_j = 0$, $p_i = p_j$, and $H_i \le H_j$ componentwise. Then deleting the factor between cuts i and j produces a shorter word in $L(A) \setminus L(N)$.*

Proof. The deterministic run rejoins the identical physical configuration $(p_i, 0) = (p_j, 0)$, so the unchanged suffix remains accepting. At cut j, rejection gives $H_j < R_j$. Since $H_i \le H_j$, the unchanged suffix also rejects from H_i. □

Thus the zero-cut states (p_t, H_t) of a shortest counterexample form a bad sequence under equality of p and componentwise order on H.

Between consecutive zero cuts, a nontrivial factor has the form

$$0 \xrightarrow{+1} \text{strictly positive interior} \xrightarrow{-1} 0.$$

The interior contains no zero test. We call such a factor a *positive first return*.

111.14 Positive first-return compactness

This section contains the main technical theorem. We state it in a suffix-parametric form because the proof is local to one exact zero entry and uses the future suffix only through its exact backward requirement. This stronger formulation will also be needed for reset-ending returns in the epsilon extension.

Definition 111.62 (Residual-minimal first return)**.** Fix a concrete exact zero entry (C_0, H_0), a deterministic zero exit configuration C_1, and a backward requirement vector

$$R \in (\mathbb{N} \cup \{\infty\})^{Q_N}.$$

A positive first-return word P from C_0 to C_1 is *R-minimal* if its exact exit envelope H_P satisfies $H_P < R$ and there is no strictly shorter word Q that takes A from C_0 to the same physical exit configuration C_1 and whose exact exit envelope satisfies $H_Q < R$. The competitor Q need not itself be a first return.

Theorem 111.63 (Suffix-parametric positive first-return compactness)**.** *Fix a concrete exact zero-cut entry consisting of an A zero configuration C_0 and an exact net envelope H_0. There is a total computable integer*

$$B_{\mathrm{FR}}(A, N, C_0, H_0)$$

such that, for every deterministic zero exit configuration C_1 and every backward requirement vector R, every R-minimal positive first-return word from (C_0, H_0) to C_1 has literal length at most B_{FR}. In particular, every positive first-return corridor occurring from this entry inside a globally shortest word of $L(A) \setminus L(N)$ has length at most the same bound.

Indeed, if a corridor P occurs in a globally shortest counterexample and R is the exact backward requirement of the unchanged suffix at its exit, then any shorter competitor Q from the same entry to the same physical zero exit with $H_Q < R$ would splice with that suffix and produce a shorter global counterexample. Thus P is R-minimal. Conversely, the proof below uses global shortestness only through this residual-minimality implication. Throughout this section, "shortest witness" may therefore be read as "R-minimal witness" for one fixed but arbitrary exit requirement R.

111.14.1 Replacement principles and anchored shells

Two different operations occur in the proof and must not be conflated.

Definition 111.64 (Trace-preserving deletion)**.** A trace-preserving deletion removes one or more literal contiguous blocks from the *observed* R-minimal word. Every removed block is identified at its actual location in that word. The deterministic run on the remaining letters must be checked directly.

Definition 111.65 (Endpoint replacement)**.** An endpoint replacement substitutes an observed factor P by a possibly different word Q. It is valid when Q starts at the same cut, reaches the identical deterministic A exit configuration, and its exact N exit envelope H_Q satisfies $H_Q \le H_P$.

Lemma 111.66 (Extensional splice principle)**.** *Let P and Q start at the same exact entry and reach the same deterministic exit configuration, with $H_Q \le H_P$. For every backward requirement vector R,*

$$H_P < R \quad \Longrightarrow \quad H_Q < R.$$

Hence an R-minimal witness admits no strictly shorter endpoint-dominating replacement. Applied with R equal to the exact requirement of an actual suffix, this is precisely the usual splice argument in a globally shortest counterexample.

Proof. Componentwise $H_Q \le H_P < R$ immediately gives $H_Q < R$. The deterministic endpoint is identical, so the same future deterministic computation is available when R comes from a concrete suffix. □

Choose a peak cut z of a positive first return. For each level h below the peak, let u_h be the last height-h cut before z and v_h the first height-h cut after z. Then

$$u_h < u_k < z < v_k < v_h \qquad (h < k).$$

The inner factor from u_k to v_k stays at height at least k and can therefore be translated downward by $k - h$ without activating a zero rule.

For every finite envelope coordinate put

$$F_t(q) = H_t(q) - d_t.$$

For every fixed OCN word transformer T_x and $c \ge 0$,

$$T_x(H + c\mathbf{1}) \ge T_x(H) + c\mathbf{1}. \tag{111.1}$$

Lemma 111.67 (Anchored shell deletion)**.** *Let $1 \le h < k$. Assume the corresponding ascent and descent boundary controls agree and*

$$F_{u_h} \le F_{u_k}, \qquad F_{v_k} \le F_{v_h}.$$

Then translating the inner shell to the outer level gives a strictly shorter word reaching the same deterministic exit and still rejected against the fixed requirement R, contradicting R-minimality.

Proof. The translated factor follows the same positive-regime deterministic control path and has enough vertical reserve, hence rejoins the exact deterministic outer exit. The first anchored inequality gives no larger effective input envelope after lowering the inner shell; applying the same literal inner word and (111.1), then using the second anchored inequality, yields an outer exit envelope no larger than the original one. Lemma 111.66 finishes the shortening. □

Thus a deep deletion-free tower forces noncomparability of anchored endpoint data. All subsequent machinery exists to turn that noncomparability into either lower-rank escape or a bounded context.

111.14.2 Finite owner/provenance refinement and macro systems

The later guard and wall constructions use a finite owner-resolved macro system. We make that object explicit here.

Definition 111.68 (Bounded macro type)**.** Fix a positive-mode proof state. A *bounded macro type* records a fixed finite literal transition skeleton between two refined cuts: its source and target deterministic controls, its literal input word, its deterministic counter effect and internal entry requirement, and the finite family of N-paths over that word between states of the current finite support. Low counter values at which one of those paths changes feasibility are unfolded into finite control. At every proof state only finitely many bounded macro types are used.

For a macro e from support S to support S', exact max-envelope propagation has the form

$$H'(r) = \max_{q \in S} \{H(q) + w_e(q, r) : H(q) \geq \lambda_e(q, r)\}, \qquad r \in S',$$

where $w_e(q, r)$ and the finite entry requirement $\lambda_e(q, r)$ are effectively computable from the fixed literal macro. Fix once and for all an order on Q_N and break ties by the least maximizing predecessor.

Definition 111.69 (Owner frame and provenance color)**.** At an owner-resolved macro cut, the *owner map*

$$\sigma_e : S' \to S$$

sends each target coordinate to its canonical maximizing predecessor. The *owner frame* records the current support, the owner map on live coordinates, and—after rank stabilization—the induced permutation of live provenance classes. Starting from a concrete entry support S_0, give every $q \in S_0$ a distinct provenance atom $[q]$. The composed map

$$\mathrm{Prov}_t : S_t \to S_0$$

assigns to every live coordinate at cut t the entry atom obtained by composing the intervening owner maps. A *provenance color* records the current support, the partition of current coordinates by equality of Prov_t, and the finite permutation/frame data.

All these colors are finite: Q_N is finite, the macro alphabet is finite, there are finitely many supports, maps between finite supports, set partitions, threshold regimes, and permutations. On one fixed owner-resolved edge the chosen maximizing predecessor is exact, so the live owner values satisfy an affine update

$$x' = P_e x + c_e, \tag{111.2}$$

with P_e the 0–1 matrix induced by the owner map. The validity of that owner choice is exactly a finite conjunction of affine difference inequalities comparing the selected predecessor with all competitors. Thus the refinement is an exact encoding of the envelope, not a selected-run approximation.

Definition 111.70 (Provenance rank)**.** The provenance rank at a refined cut is

$$\nu(t) := |\operatorname{im} \mathrm{Prov}_t|.$$

A *provenance-rank drop* means that one entry provenance atom represented before a macro step is no longer represented afterwards. A raw support change that merely creates or removes coordinates carrying already represented atoms is a finite-mode change, not a rank drop.

Lemma 111.71 (Finite-depth provenance descent)**.** *Along owner-resolved macro cuts, ν is monotone nonincreasing. If the induced owner action on the live provenance classes is nonsurjective, then ν decreases strictly. Hence there are at most $|Q_N| - 1$ strict provenance-rank drops before the rank stabilizes; after stabilization every owner update induces a permutation of the live provenance classes.*

Proof. For consecutive macro cuts $t \to t'$, the provenance map factors as

$$S_{t'} \xrightarrow{\sigma_e} S_t \xrightarrow{\mathrm{Prov}_t} S_0,$$

so $\operatorname{im}\mathrm{Prov}_{t'} \subseteq \operatorname{im}\mathrm{Prov}_t$. Strict loss of a represented class gives strict inequality of cardinalities. If the cardinality is stable, the induced map on the finite set of live provenance classes is surjective and therefore bijective. Recording that bijection is the finite permutation-frame refinement. □

Proposition 111.72 (Fixed-route terminal module)**.** *Fix one owner-resolved bounded-expression macro schema*

$$\alpha_0\beta_1^{n_1}\alpha_1 \cdots \beta_m^{n_m}\alpha_m$$

from a concrete exact entry to one exact deterministic target configuration. Then feasibility of the schema is an effective Presburger condition on $(n_1, \ldots, n_m)$, the endpoint owner vector is affine in the exponents, and the componentwise-minimal reachable endpoint set together with shortest representatives is effectively computable. Consequently this fixed-route family is shortest-witness compact.

Proof. After the finite owner/mode refinement, each powered block β_i begins and ends in the same refined mode and acts on its live owner vector by an affine map

$$x \longmapsto P_i x + c_i,$$

where P_i is a 0–1 functional matrix induced by the owner map. The finite monoid generated by P_i is effectively eventually periodic: compute μ_i, λ_i with $P_i^{\mu_i+\lambda_i} = P_i^{\mu_i}$. Refining the exponent by the finitely many cases

$$n_i < \mu_i \quad \text{or} \quad n_i = \mu_i + s_i + \lambda_i k_i, \qquad 0 \le s_i < \lambda_i,$$

makes the endpoint vector affine in the quotient variable k_i. Indeed

$$P_i^{n_i} x + \sum_{j<n_i} P_i^j c_i$$

has a fixed residue-dependent linear part plus k_i times one fixed period-sum vector.

Every threshold guard, owner comparison, and deterministic-positivity condition at an intermediate repetition is an affine inequality in the current owner vector. After the same residue refinement its truth at repetition j is therefore a Presburger formula in the exponent quotients and the bounded repetition index. Requiring it at every repetition is still Presburger, since Presburger arithmetic is effectively closed under bounded and unbounded first-order quantification. The exact deterministic target is handled in the same way. Taking the finite disjunction over all residue branches yields an effective Presburger description of the feasible exponent family, and on each branch the endpoint owner vector is affine in the quotient variables (equivalently Presburger-piecewise affine in the original exponents).

Presburger arithmetic therefore decides exact endpoint slices and computes the finite semilinear feasible exponent family. Recursively minimize endpoint coordinates using exact-slice queries, exactly as in Proposition 111.78 below, and enumerate shortest realizing words. Endpoint domination then invokes Lemma 111.66. □

111.14.3 Exact projective modes and finite-projective compression

Let $E = \operatorname{supp}(H)$ be the finite support of an exact envelope. If $E \neq \varnothing$, define its common scalar and normalized profile by

$$z(H) = \min_{q\in E} H(q), \qquad \theta(H) = H - z(H)\mathbf{1}_E.$$

Thus $\min \theta(H) = 0$. A *projective mode* records the finite data that determine how the exact envelope is read from (z, θ): the deterministic positive-mode control, the support E, the maximizing-owner frame used by the current finite macro system, and the normalized profile θ when that profile belongs to a fixed finite palette.

Definition 111.73 (Finite exact projective mode system)**.** A positive factor lies in a finite exact projective mode system $\mathcal{P}$ if there is a finite set V of modes and fixed vectors θ_v such that at every cut in mode v,

$$H = z\mathbf{1} + \theta_v$$

on the recorded support, and every macro transition of the factor updates the deterministic A height and the common scalar z additively while changing only the finite mode.

The phrase *exact* means that support, threshold regime, and the affine readout of every finite coordinate are part of the mode; no selected-run approximation is used.

Proposition 111.74 (Finite-projective compression)**.** *Fix a concrete exact entry, a finite exact projective system $\mathcal{P}$, and one exact deterministic target configuration. There is an effective bound on every shortest factor staying in $\mathcal{P}$ and reaching that target. Moreover one can compute an endpoint-dominating shortest representative.*

Proof. Use deterministic height and the common scalar z as two ordinary VASS counters; the projective mode is finite control. Reachability sets of two-dimensional VASS with states are effectively semilinear [111]; hence, after fixing the target mode and exact deterministic target height, the one-dimensional set of reachable target scalars z is effectively semilinear and its emptiness and least element are decidable. Let $z_{\min}$ be that least element whenever the target slice is nonempty. Exhaustive word enumeration then computes a shortest word reaching $(\text{target mode}, \text{target A configuration}, z_{\min})$. Any observed factor exiting with scalar $z \geq z_{\min}$ is endpoint-dominated by this representative. Taking the maximum over the finitely many nonempty target modes gives the bound; empty target slices contribute nothing. □

Remark 111.75 (Safe use of alternative paths). Proposition 111.74 is exactly the place where an alternative VASS path is legitimate: it is used to *construct a replacement word* with the same deterministic endpoint and a smaller exact envelope. No claim is made that the observed literal word itself belongs to a particular VASS path scheme.

111.14.4 Owner guards, recurrent guard rank, and the rank-one base

After finite support/provenance and maximizing-owner refinement, the remaining owner changes are governed by finitely many affine comparisons

$$g_j(x) = \ell_j(x) - k_j,$$

where x denotes the projective owner/resource vector. Fix one recurrent finite owner/chamber strongly connected component S. Let W_S be the rational span of projective displacements of directed cycles of S, after common translation is factored out.

Definition 111.76 (Recurrent guard rank)**.** The recurrent guard rank of S is

$$\rho(S) = \dim_{\mathbb{Q}} \operatorname{span}\{\ell_j|_{W_S} : j \text{ switches recurrently in } S\}.$$

Only restrictions to the recurrent cycle space count. Many syntactic guards may therefore have rank one.

The base of the induction is stronger than mere compactness.

Lemma 111.77 (Common scalar for rank-one guards)**.** *If $\rho(S) = 1$, there is an effectively computable integer linear functional μ on the cycle lattice and integers a_j such that*

$$\ell_j(w) = a_j \mu(w)$$

for every recurrent cycle displacement $w \in W_S$.

Proof. The restricted functionals span a one-dimensional rational subspace of W_S^*. Choose a rational generator and clear denominators simultaneously. □

For an edge $e : u \to v$ of the finite owner SCC with translation c_e, the residual

$$r_j(e) = \ell_j(c_e) - a_j \mu(c_e)$$

has zero sum on every directed cycle. Hence it is a finite-control coboundary: $r_j(e) = \phi_j(v) - \phi_j(u)$. Along a path from a fixed entry, with

$$T = \mu(x - x_0),$$

every recurrent guard therefore has the form

$$g_j(x) = a_j T + b_{j,v} \tag{111.3}$$

for a finite table of constants $b_{j,v}$.

Proposition 111.78 (Rank-one compilation)**.** *Every recurrent owner SCC of guard rank at most one can be compiled effectively into a VASS with one zero-tested counter. Consequently, from a concrete entry and exact deterministic target configuration, exact target-slice nonemptiness and finite Pareto-minimal endpoint bases are computable; in particular the SCC is shortest-witness compact.*

Proof. For rank zero all recurrent guard truths are fixed after finite refinement and their slacks are ordinary counters. For rank one, the finitely many affine thresholds in (111.3) partition $\mathbb{Z}$ into two unbounded rays and finitely many bounded cells. On an unbounded ray store the distance of T from its boundary in one distinguished counter s; bounded cells store the exact value of T in finite control. A macro edge changes T by a fixed integer. A cell crossing can occur only when s belongs to a computably bounded source band. Drain s through a bounded finite-control chain, use the unique zero test to verify the crossing value, remember the consumed bounded amount in finite control, and reconstruct the target slack. All other resources are ordinary VASS counters. Reachability for VASS with one zero test is decidable. An endpoint slice fixing some counters exactly and leaving the others free reduces to one exact-configuration query by entering a private cleanup phase that decrements only the free counters and leaves the fixed counters unchanged; the queried cleanup target fixes the prescribed counters and sets every formerly free counter to zero. Hence exact endpoint-slice nonemptiness is decidable from the same oracle.

Effective Pareto algorithm. The basis computation uses exact-slice reachability directly and does not rely on detecting when a growing product box is large enough. More generally, let R_F be the reachable endpoint set after fixing some owner coordinates F exactly. First decide whether R_F is empty. If it is nonempty, enumerate one reachable endpoint $a \in R_F$. Every componentwise-minimal $x \in R_F$ satisfies either $x = a$ or $x_i < a_i$ for at least one still-free coordinate i. For every such i and every integer $0 \le \zeta < a_i$, recursively solve the lower-dimensional exact slice $R_{F \cup \{x_i = \zeta\}}$. Each recursive call fixes one additional coordinate and each branching range is finite, so the recursion terminates. Taking the union of the returned candidates with a and removing dominated vectors gives exactly $\mathrm{Min}(R_F)$. At the root this computes the full finite Pareto basis. For each minimum, breadth-first word enumeration yields a shortest representative. Thus effectivity follows from the one-zero-test exact-slice oracle, not from Dickson finiteness alone.

Since the exact envelope is an affine readout of the owner resources in the fixed refined mode, a Pareto-minimal representative endpoint-dominates any longer competitor and Lemma 111.66 applies. □

111.14.5 Directional walls and exact finite wall signatures

Assume now $\rho \geq 2$. Choose a basis $\mu_1, \ldots, \mu_\rho$ for the recurrent guard directions. A guard is μ_1-sensitive if its restriction has a nonzero μ_1 coefficient. A μ_1 *wall event* is a transition at which the truth value of at least one μ_1-sensitive guard changes.

Every such switch occurs in a finite source band: the guard changes by a fixed integer on a fixed macro edge, so its source value must lie between two consecutive bounded thresholds.

Definition 111.79 (Exact wall signature)**.** An exact μ_1-wall signature records

(a) the source and target refined finite modes and deterministic controls;

(b) the set of μ_1-sensitive guards that change truth and their orientations;

(c) the exact bounded source guard/slack values at the crossing;

(d) the current support/provenance color and owner frame;

(e) the crossing macro edge.

The set $\mathcal{W}$ of exact wall signatures is finite and effectively enumerable.

Finiteness follows because all control/frame/support components are finite and every switching guard source value belongs to an edge-dependent finite threshold band.

Lemma 111.80 (Wall-free rank descent)**.** *Every maximal μ_1-wall-free strip has recurrent guard rank at most $\rho - 1$.*

Proof. Every μ_1-sensitive guard has fixed truth throughout the strip and is carried as an ordinary nonnegative slack. Only guards whose μ_1 coefficient is zero can switch, and their restrictions lie in $\operatorname{span}(\mu_2, \ldots, \mu_\rho)$. □

A bounded number of walls can also be handled without creating a second zero test.

Proposition 111.81 (Bounded-wall compilation)**.** *Fix K. A rank-ρ factor containing at most K μ_1-wall events yields either a responsible rank-$\leq \rho - 1$ escape or an effective endpoint-dominating length bound, assuming the lower-rank compactness invariant.*

Proof. The factor is a concatenation of at most $K+1$ wall-free strips and K crossing macros. If normalized/projective/anchored/common-shift data escape in a strip, Lemma 111.80 localizes a genuine lower-rank escape. Otherwise every strip belongs to a finite lower-rank bounded-context palette and is compact by induction; the crossing macros are finite. Summing the finitely many bounds gives the factor bound. If one wants an exact compilation rather than merely an extensional replacement, an exact bounded wall value $s = c$ can be checked without a fresh zero test by subtracting c, retiring the old ordinary slack counter, and requiring it to be zero only at the final exact-reachability target. This is needed only finitely many times because K is fixed. □

Hence a genuinely rank-ρ obstruction must have unbounded wall multiplicity.

111.14.6 Dense same-signature atoms and carrier colors

Let $B = |\mathcal{W}|$. Write the wall-signature word of a factor as $w_1 \cdots w_N$.

Lemma 111.82 (Dense same-signature returns)**.** *If $N \geq 4B$, some signature $\sigma \in \mathcal{W}$ has at least*

$$\left\lfloor \frac{N}{4B} \right\rfloor - 1$$

pairwise interior-disjoint consecutive σ-to-σ return intervals, each containing at most

$$K_{\mathrm{dense}} = 4B + 1$$

wall events.

Proof. Some σ occurs at least $\lceil N/B \rceil$ times. The sum of the consecutive occurrence gaps is $< N$. If more than half of these gaps exceeded $4B$, their sum would exceed N. The stated coarse linear lower bound follows. □

Returning to the same exact signature fixes one independent μ_1-sensitive wall equation. Thus the endpoint displacement of every dense atom lies in $W_S \cap \ker \ell_\sigma$, and its endpoint recurrent guard rank is at most $\rho - 1$.

Definition 111.83 (Carrier color)**.** For a dense atom, a left or right *carrier* is the bounded neighborhood from the exact wall endpoint to the canonical shell/interface cut used for comparison. Under a fixed bounded-context cutoff, its carrier color records the exact wall signature, side (left/right), deterministic control, support/provenance color, owner frame, bounded vertical offset from the shell level, and all normalized envelope/guard data in the prescribed finite cutoff box.

Carrier colors are therefore finite whenever the context is bounded. If the boundary-to-canonical correction leaves this finite palette, the first escaping quantity occurs in one of finitely many wall-free strip positions inside a bounded-wall carrier, and Lemma 111.80 localizes the escape below rank ρ.

111.14.7 High walls and translated arches

If the heights of all recurrent walls are bounded, the exact physical wall contexts belong to a finite set and Proposition 111.81 closes the branch. Thus the residual case has arbitrarily high wall centers. Around a wall center t of height a, for each $1 \le \ell < a$ define the last/first height-ℓ cuts

$$u_\ell^t < t < v_\ell^t.$$

Their intervening factors are nested positive arches. Combining the dense atom skeleton, finite carrier colors, and ordinary finite Ramsey refinement yields an arbitrarily deep family

$$J_\ell = e_\ell^L \longrightarrow e_\ell^R$$

with fixed left and right carrier colors and fixed offsets

$$d(e_\ell^L) = \ell + r_L, \qquad d(e_\ell^R) = \ell + r_R.$$

The carriers have recurrent rank at most $\rho - 1$.

The next lemma makes the finite-color argument explicit.

Lemma 111.84 (Translated-arch localization)**.** *Let $J_{h_0}, \dots, J_{h_m}$ be a sufficiently long fixed-carrier translated-arch family with strictly increasing levels. Then either*

(i) two levels satisfy the anchored deletion inequalities of Lemma 111.67; or

(ii) after passing to a homogeneous subfamily, one fixed side $X \in \{L, R\}$ and one fixed live coordinate $q \in Q_N$ satisfy a strict chronological chain

$$F_{x_0}(q) > F_{x_1}(q) > \cdots > F_{x_n}(q),$$

where x_i are the selected canonical endpoints/carrier cuts on side X.

In case (ii), the escape is either carried inside one lower-rank carrier strip or is visible on the same-signature lower-rank endpoint system.

Proof. For each pair of levels $h < k$, first ask whether both anchored comparisons required by Lemma 111.67 hold. If yes, case (i) occurs. Otherwise color the pair by the first failed side (left or right) and the least envelope coordinate witnessing that failure. Support and side are already fixed, so this is a finite coloring by at most $2|Q_N|$ colors. Ramsey's theorem gives a long homogeneous set. If the homogeneous color is, for example, "left, q", then for

every earlier/later pair the left comparison fails in the same orientation, hence integrality and total order on the selected coordinate give a strict monotone chain after reversing orientation if necessary. The right-side case is symmetric.

Now compare each canonical endpoint with the corresponding exact wall endpoint through its fixed carrier. There are finitely many strip positions. If the correction in coordinate q or any other normalized coordinate is unbounded, one fixed wall-free strip position carries unbounded variation and has rank at most $\rho - 1$. If all corrections are bounded, pass to a finite refinement fixing them; the strict drift then transfers to the exact same-signature wall endpoints. Their endpoint displacement lies in the wall kernel, so their recurrent endpoint rank is at most $\rho - 1$. □

Absolute shell height is not itself a new resource on a fixed carrier: once the vertical range about level ℓ is bounded and all zero tests are inactive, translating the whole carrier upward preserves its transition structure. Hence only the normalized/projective, anchored-relative, common-shift, and relative A-transfer quantities can escape.

111.14.8 Finite scalar/projective reserve split and the vector shell defect

Lemma 111.84 leaves a deletion-free translated-arch family in which one endpoint side carries an arbitrarily long strict anchored drift. The final proof uses a finite reserve split; no infinitary Dickson extraction on normalized profiles is used.

Assume for definiteness that the responsible side is the left side. At the selected left endpoints write

$$F_\ell^L = H_{e_\ell^L} - d(e_\ell^L)\mathbf{1}.$$

After stable-support refinement, finite pigeonhole fixes one live coordinate r attaining the minimum of F_ℓ^L on an arbitrarily long subfamily. Put

$$a_\ell := F_\ell^L(r), \qquad \theta_\ell := F_\ell^L - a_\ell \mathbf{1}. \tag{111.4}$$

Then $\theta_\ell \in \mathbb{N}^S$, $\theta_\ell(r) = 0$, and $F_\ell^L = a_\ell \mathbf{1} + \theta_\ell$. By Lemma 111.84, on the selected chain one fixed coordinate q satisfies

$$F_{h_0}^L(q) > F_{h_1}^L(q) > \cdots > F_{h_n}^L(q).$$

Lemma 111.85 (Effective projective cutoff)**.** *For the concrete first-return entry and fixed finite support/regime palette there is a computable constant C_{proj} with the following property. If a normalized endpoint profile has*

$$\max_{p \in S} \theta(p) > C_{\mathrm{proj}},$$

then a bounded nonzero N/N differential core is exposed on a fixed refined route, unless a provenance-rank drop occurs first. The branch therefore transfers to the projective/lower-rank localization machinery. In the complementary branch every normalized profile lies in the finite box $[0, C_{\mathrm{proj}}]^S$.

Proof. Track a fixed difference $H(p) - H(r)$ in the finite synchronized owner/product graph. After cycle erasure, a simple residual has effectively bounded length and hence bounded differential effect. If the difference exceeds the concrete entry spread plus twice the maximal simple-residual contribution (enlarged over the finite support/regime event palette), some primitive recurrent cycle has nonzero differential effect. Finite refinement fixes its sign and route type. If the responsible owner/provenance data cannot be stabilized before that extraction, the provenance branch has already descended. Taking the maximum of these finitely many simple-path/event bounds gives C_{proj}. □

Lemma 111.86 (Finite scalar/projective reserve split)**.** *For every n-step strict endpoint chain as above,*

$$(a_{h_0} - a_{h_n}) + (\theta_{h_0}(q) - \theta_{h_n}(q)) \geq n. \tag{111.5}$$

Consequently either the chain crosses the projective cutoff of Lemma 111.85, or finite pigeonhole extracts an arbitrarily long subchain on which one exact profile

$$\theta_\ell = \bar{\theta}$$

is fixed and the scalar sequence a_ℓ decreases strictly.

Proof. Integrality gives $F^L_{h_0}(q) - F^L_{h_n}(q) \geq n$. Substitute $F^L(q) = a + \theta(q)$ to obtain (111.5). If some profile leaves the finite projective box, Lemma 111.85 transfers the branch. Otherwise only finitely many exact profiles occur, so finite pigeonhole—not Dickson's lemma—gives an arbitrarily long constant-profile subchain. Strict decrease of $F^L(q)$ then forces strict decrease of a. □

From now on we work in this canonical no-projective-escape branch and write $\theta_\ell = \bar{\theta}$. Take $h < k$ in the fixed-profile subchain and put

$$c := k - h, \qquad \delta := a_h - a_k > 0.$$

The fixed endpoint offsets give the exact left-envelope identity

$$H^L_k = H^L_h + (c - \delta)\mathbf{1}. \tag{111.6}$$

Lemma 111.87 (Translated input domination)**.** *If $\delta \leq c$ and H' is the exact net envelope obtained by reading the literal inner word J_k from the outer left envelope H^L_h, then*

$$H' \leq H^R_k - (c - \delta)\mathbf{1}. \tag{111.7}$$

Proof. Equation (111.6) gives

$$H^L_h + (c - \delta)\mathbf{1} = H^L_k.$$

For a one-counter net, common input credit is monotone under a fixed word: for every $t \geq 0$,

$$T_x(H + t\mathbf{1}) \geq T_x(H) + t\mathbf{1}.$$

Applying this to $x = J_k$ gives

$$H' + (c - \delta)\mathbf{1} \leq T_{J_k}(H^L_k) = H^R_k.$$

□

Define the *vector shell defect*

$$Y_\ell := F^R_\ell - a_\ell \mathbf{1}, \qquad F^R_\ell := H^R_\ell - d(e^R_\ell)\mathbf{1}. \tag{111.8}$$

Because $d(e^R_k) - d(e^R_h) = c$,

$$H^R_k - (c - \delta)\mathbf{1} = H^R_h + (Y_k - Y_h). \tag{111.9}$$

Lemma 111.88 (Vector shell-defect pinning)**.** *Let $h < k$ belong to the fixed-profile subchain and assume $\delta \leq c$. Exactly one of the following sufficient outcomes occurs.*

(i) $Y_k \leq Y_h$ componentwise, in which case replacing the outer arch by the translated inner arch gives an endpoint-dominating strict shortening;

(ii) there is a live coordinate s such that

$$1 \leq R^R_h(s) - H^R_h(s) \leq Y_k(s) - Y_h(s). \tag{111.10}$$

Proof. Combine Lemma 111.87 with (111.9). If $Y_k \le Y_h$, the translated exit envelope is at most H_h^R, so Lemma 111.66 shortens the shortest witness. Otherwise the translated word is still accepted by A and is strictly shorter, hence R-minimality forces its exit envelope to satisfy the requirement. At the outer right cut the original word has $H_h^R < R_h^R$, while the translated exit reaches the suffix requirement in some coordinate s, giving (111.10). □

A pair is *nonsuperlevel* when $\delta \le c$ and *superlevel* when $\delta > c$. Finite Ramsey refinement yields a long homogeneous subfamily of one kind. In the nonsuperlevel case, Lemma 111.88 and a finite coloring by the least positive coordinate of $Y_k - Y_h$ give a fixed coordinate with strict Y-growth. The superlevel case is threshold-sensitive and is developed next.

111.14.9 Superlevel activation and the vector-defect/activation split

Fix a superlevel pair $h < k$ and put

$$d := \delta - c > 0. \tag{111.11}$$

Then (111.6) becomes

$$H_k^L = H_h^L - d\mathbf{1}, \tag{111.12}$$

and the corresponding right identity is

$$H_k^R = H_h^R - d\mathbf{1} + (Y_k - Y_h). \tag{111.13}$$

Suppose first that $Y_k \le Y_h$. Then the inner right envelope is strictly dominated by the outer right envelope. Let

$$Q_{h,k} := S_{J_k}(R_h^R) \tag{111.14}$$

be the exact entry requirement for reading the literal inner word J_k and then the inherited outer suffix.

Lemma 111.89 (Exact superlevel activation certificate)**.** *If $h < k$ is superlevel and $Y_k \le Y_h$, then*

$$H_k^L < Q_{h,k}$$

componentwise, while shortestness implies $H_h^L \not< Q_{h,k}$. Hence there is a live coordinate p such that

$$H_k^L(p) < Q_{h,k}(p) \le H_h^L(p). \tag{111.15}$$

Writing

$$\eta := Q_{h,k}(p) - H_k^L(p),$$

one has the exact bound

$$1 \le \eta \le d. \tag{111.16}$$

Proof. Since $Y_k \le Y_h$, equation (111.13) gives $H_k^R < H_h^R < R_h^R$. Exact residuation, Theorem 111.60, yields $H_k^L < S_{J_k}(R_h^R) = Q_{h,k}$. Replacing J_h by the downward translation of J_k is a legal strict deterministic shortening. Hence R-minimality forces the shortened replacement to meet the inherited requirement, equivalently $H_h^L \not< Q_{h,k}$. Choose p satisfying (111.15). Since $H_h^L(p) - H_k^L(p) = d$ by (111.12), the gap satisfies (111.16). □

This is an exact threshold statement: the actual inner input is short of the virtual continuation threshold by η, while the outer input crosses that threshold. It is not yet a statement about a forward maximizing strand of the original rejected word.

For every sufficiently separated superlevel pair color the pair by

- A if $Y_k \le Y_h$ componentwise;
- otherwise V_s, where s is the least coordinate with $Y_k(s) > Y_h(s)$.

There are finitely many colors.

Lemma 111.90 (Activation-or-vector-defect dichotomy)**.** *Every arbitrarily deep superlevel tower contains arbitrarily long subfamilies of exactly one of the following forms.*

(i) Activation-critical*: for every earlier/later pair Y is componentwise nonincreasing and Lemma 111.89 supplies an exact threshold gap η;*

(ii) Vector-defect*: one fixed coordinate s satisfies*

$$Y_{h_0}(s) < Y_{h_1}(s) < \cdots .$$

Proof. Apply finite Ramsey to the coloring above. Homogeneous color A gives the first alternative and homogeneous color V_s gives the second. □

The vector-defect alternative is not left abstract. Fix a live right coordinate s and, for each ℓ, choose one maximizing N-path $\pi_\ell : p_\ell \to s$ over the literal arch J_ℓ realizing $H^R_\ell(s)$. Pass to a subsequence with fixed input state p. Put

$$\kappa_A := r_R - r_L = \Delta_A(J_\ell).$$

Because the left projective profile is fixed to $\bar\theta$, the exact owner/path identity is

$$Y_\ell(s) = \bar\theta(p) + \omega_\ell, \qquad \omega_\ell := \Delta_N(\pi_\ell) - \kappa_A. \tag{111.17}$$

Lemma 111.91 (Actual maximizing-strand source of vector defect)**.** *If $Y_\ell(s)$ is unbounded above in the canonical no-projective-escape branch, then $\omega_\ell \to \infty$ along a subsequence. Cycle erasure on the actual maximizing synchronized paths exposes a fixed primitive synchronized cycle with positive relative effect $\Delta_N - \Delta_A > 0$, unless a provenance-rank drop occurs first.*

Proof. The term $\bar\theta(p)$ in (111.17) is fixed, so unbounded Y is exactly unbounded positive relative gain on the chosen literal maximizing strands. Cycle erasure in the finite synchronized positive-mode product leaves a bounded simple residual. Hence recurrent primitive cycles carry unbounded positive relative contribution, and finite pigeonhole fixes one positive type unless the owner/provenance refinement changes first. □

Thus only the activation-critical branch is new.

111.14.10 Literal activation annuli and fixed affine cells

Let $h < k$ be an activation-critical pair. The outer arch factors literally as

$$J_h = L_{h,k}\, J_k\, R_{k,h},$$

where $L_{h,k}$ runs from e^L_h to e^L_k and $R_{k,h}$ from e^R_k to e^R_h. The deterministic effects are $+c$ and $-c$, respectively.

Lemma 111.92 (Matched anchored drift in an activation annulus)**.** *For every activation-critical pair,*

$$F^L_k(r) - F^L_h(r) = -\delta,$$

and, in chronological order on the right flank,

$$F^R_h - F^R_k = (Y_h - Y_k) + \delta\mathbf{1} \geq \delta\mathbf{1}.$$

Equivalently, in absolute envelopes,

$$H^L_h(r) - H^L_k(r) = d, \qquad H^R_h - H^R_k \geq d\mathbf{1}. \tag{111.18}$$

Proof. The left identity follows from $F^L_\ell(r) = a_\ell$. The right identity follows from $F^R_\ell = Y_\ell + a_\ell\mathbf{1}$ and $Y_h \geq Y_k$. Restoring the deterministic height difference c converts δ into $d = \delta - c$ in the absolute envelopes. □

The two literal flanks of one annulus therefore carry a matched credit valley: the left loses anchored mass and the right restores at least the corresponding common amount. If adjacent cells have unbounded values of δ or unbounded endpoint increments, the responsible mass is already localized to one of finitely many bounded-wall flank positions and hence to lower rank. We may therefore restrict to bounded cells.

After finite refinement fix integers

$$c, \delta, d = \delta - c > 0, \qquad p, \qquad 1 \le \eta \le d$$

and define the right adjacent-cell increment

$$v_i := Y_i - Y_{i+1} \ge 0.$$

The left profile is already fixed by Lemma 111.86, so the endpoint recurrences are

$$H^L_{i+1} = H^L_i - d\mathbf{1}, \qquad H^R_i = H^R_{i+1} + d\mathbf{1} + v_i. \tag{111.19}$$

Lemma 111.93 (Finite affine-cell refinement)**.** *If some coordinate of v_i is unbounded, a bounded-height annulus carries unbounded normalized right-endpoint change and hence localizes to a lower-rank projective/relative escape or a provenance-rank drop. Otherwise, after finite refinement, $v_i = v$ is one fixed vector for every recurrent cell.*

Proof. A cell has fixed shell width c and fixed wall/carrier types, hence only finitely many wall-free strip positions. Unbounded normalized endpoint change forces one fixed position to carry unbounded change; Lemma 111.80 lowers the recurrent rank there. In the no-escape case v_i ranges over a finite box and may be fixed by finite refinement. □

Fixed endpoint affine data do not imply that the cell interiors form one lower-rank machine. We use only the endpoint recurrences and the extensional replacement principle.

111.14.11 Pure-scalar recurrence and the virtual threshold shadow

The terminal fixed-cell survivor simplifies further.

Lemma 111.94 (Scalarization of stable cells)**.** *In a recurrent fixed affine cell family with no endpoint-visible projective escape or provenance-rank drop, one has*

$$v = \lambda\mathbf{1}$$

for some fixed $\lambda \ge 0$. Hence

$$H^L_{i+1} = H^L_i - d\mathbf{1}, \qquad H^R_{i+1} = H^R_i - (d+\lambda)\mathbf{1}. \tag{111.20}$$

Proof. The left normalized profile is already fixed. If two coordinates of v differ, repeated cells produce linearly unbounded right projective spread, contradicting the no-projective-escape hypothesis. Hence v is a common scalar vector. □

Let

$$Q_i := S_{J_{i+1}}(R^R_i)$$

be the replacement-entry requirement of cell i, and retain the fixed activation coordinate p with

$$Q_i(p) - H^L_{i+1}(p) = \eta.$$

Equation (111.20) gives the exact threshold ladder

$$Q_{i+1}(p) = Q_i(p) - d, \qquad H^L_i(p) - Q_i(p) = d - \eta. \tag{111.21}$$

Put $K := d - \eta \ge 0$ and $q_i := Q_i(p)$. Then

$$H^L_{i+1}(p) = q_i - \eta, \qquad H^L_i(p) = q_i + K. \tag{111.22}$$

For each i choose canonically an N-path $\pi_i : p \to s_i$ over J_{i+1} attaining the minimum in the backward transformer at coordinate p:

$$q_i = \max\{\mathrm{req}(\pi_i), R_i^R(s_i) - \mathrm{eff}(\pi_i)\}. \tag{111.23}$$

After finite refinement the exit state $s_i = s$ is fixed.

There are two formal ways for the maximum in (111.23) to equal q_i. If $\mathrm{req}(\pi_i) = q_i$, the threshold run reaches counter 0 at an internal prefix. If instead the terminal term equals q_i and $\mathrm{req}(\pi_i) \le q_i - 1$, minimality forces exact tightness along the path.

Lemma 111.95 (Virtual tight-shadow principle)**.** *In the terminal-tight case, let $C_i(x)$ be the counter value of the threshold run on π_i at a cut x, and let $B_i(x)$ be the exact backward requirement at that path state for the remaining virtual continuation J_{i+1} followed by the inherited outer suffix. Then*

$$B_i(x) = C_i(x) \tag{111.24}$$

for every cut x of π_i.

Proof. The chosen suffix is feasible from $C_i(x)$, so $B_i(x) \le C_i(x)$. If $B_i(x) \le C_i(x) - 1$, then because the whole selected path has internal requirement at most $q_i - 1$, starting the prefix one unit lower remains feasible up to x, and the exact continuation is feasible from the resulting $C_i(x) - 1$. This contradicts the minimality of $q_i = Q_i(p)$. □

Thus the accepting translated replacement, which starts K units above threshold, stays exactly K above the virtual backward requirement along a terminal-tight path.

111.14.12 Collapse of the tight shadow to a bounded high contact

The tight-shadow alternative is not a separate terminal branch.

Lemma 111.96 (Terminal tightness forces a contact)**.** *In the terminal-tight case,*

$$q_i - \eta < \mathrm{req}(\pi_i) \le q_i - 1.$$

Consequently some prefix of π_i attains threshold counter

$$j_i := q_i - \mathrm{req}(\pi_i) \in \{1, \dots, \eta - 1\}.$$

At that cut the exact virtual backward requirement equals j_i, while the actual translated replacement path has counter $K + j_i$.

Proof. Assume instead that $\mathrm{req}(\pi_i) \le q_i - \eta = H_{i+1}^L(p)$. Then the selected path is internally feasible from the actual inner envelope coordinate. Its endpoint counter is $R_i^R(s) - \eta$. But the pure-scalar right recurrence raises the outer right envelope over the inner one by $d + \lambda$, and $\eta \le d$, giving $H_i^R(s) \ge R_i^R(s)$, contradicting rejection at the outer right cut. The rest follows from Lemma 111.95. □

In the internal-contact case the threshold run itself reaches counter 0, so the virtual requirement at that cut is 0 and the actual replacement counter is exactly K. Combining both cases yields the following uniform normal form.

Theorem 111.97 (Bounded virtual contact normal form)**.** *After finite refinement, every recurrent terminal activation cell contains a cut on a concrete accepting translated-replacement path with*

- *one fixed N contact state t;*
- *one fixed virtual backward requirement $b \in \{0, \dots, \eta - 1\}$ at t;*
- *one fixed actual replacement counter $b + K$.*

Moreover the deterministic A height at these contacts tends to infinity along every arbitrarily deep translated-arch tower.

Proof. The first three items follow from finite refinement of the two contact cases above. High-shell geometry gives a constant M_* such that every translated inner arch in the fixed carrier family stays at height at least its shell level minus M_*. The shell levels tend to infinity, so the contact height does as well. □

Let B_i be the full exact backward-requirement vector at the virtual contact. Since $B_i(t) = b$ is fixed, either some normalized difference $B_i(q) - B_i(t)$ escapes, in which case the backward differential argument localizes a lower-rank/projective escape, or the complete vectors B_i lie in a finite set and may be fixed to one exact vector B_*. In the latter branch the contact has fixed finite N context but arbitrarily high deterministic height.

111.14.13 From bounded contacts to an actual-trace relative valley

The threshold ladder (111.21) implies that in a tower of $n + 1$ recurrent cells the outer threshold is at least nd. Thus along a selected outer sequence

$$q_i \longrightarrow \infty. \tag{111.25}$$

Fix the bounded contact normal form of Theorem 111.97. Let P_i be the literal prefix of the translated inner arch from its translated start to the contact. On the chosen accepting replacement path, the N counter starts at $q_i + K$ and reaches $b + K$, so

$$\Delta_N(P_i) = b - q_i. \tag{111.26}$$

Let $\alpha_i := \Delta_A(P_i)$. High-shell geometry bounds α_i from below by a constant independent of i. Therefore

$$\omega(P_i) := \Delta_N(P_i) - \Delta_A(P_i) \longrightarrow -\infty. \tag{111.27}$$

Lemma 111.98 (Contact-prefix negative relative mass)**.** *Every arbitrarily deep terminal activation tower contains literal synchronized prefixes P_i of the translated/original inner-arch words with $\omega(P_i) \to -\infty$. Cycle erasure on these actual synchronized paths exposes a fixed primitive cycle type with negative relative effect, unless the responsible mass has already localized to a lower-rank wall-free strip or support/provenance drop.*

Proof. Equation (111.27) is exact. The synchronized positive-mode product is finite. Removing all cycle occurrences leaves a simple path of uniformly bounded relative effect, so unbounded negative relative mass must be carried by recurrent primitive cycles. Finite pigeonhole fixes one responsible type. If its occurrences have unbounded internal wall-free variation, Lemma 111.80 already gives the lower-rank alternative; otherwise bounded-wall packet refinement applies. □

Let T_i be the literal suffix of the same concrete accepting translated-replacement run from the bounded contact to the zero-return cut z terminating the positive first-return corridor. At the contact the deterministic counter is $D_i \to \infty$ and the selected N counter is the fixed value $b + K$. At z the deterministic counter is 0, while the selected accepting N run has some counter $c_i \geq 0$. Hence

$$\Delta_A(T_i) = -D_i, \qquad \Delta_N(T_i) = c_i - (b + K) \geq -(b + K),$$

and therefore

$$\omega(T_i) = \Delta_N(T_i) - \Delta_A(T_i) \geq D_i - (b + K) \longrightarrow +\infty. \tag{111.28}$$

This is the canonical post-contact argument: no split according to the A-transfer to the contact is needed.

Theorem 111.99 (Relative valley)**.** *After excluding strict lower-rank/projective/support escapes, every terminal activation tower contains, on one concrete accepting translated-replacement run and in chronological order,*

$$P_i \longrightarrow (t, b+K;\ A = D_i) \longrightarrow T_i$$

with a fixed bounded N contact, $D_i \to \infty$, and

$$\omega(P_i) \to -\infty, \qquad \omega(T_i) \to +\infty.$$

The suffix T_i ends at the actual zero-return cut of the positive first-return corridor. All factors are literal factors of the same accepting translated-replacement word; no alternative VASS path is used.

Proof. The negative side is Lemma 111.98. The positive side is the exact zero-return calculation above. The fixed contact state/counter and diverging deterministic height are supplied by Theorem 111.97. No path-scheme or fixed-transfer assumption enters the argument. □

Theorem 111.99 is the only input needed for the actual-trace cycle analysis below. Importantly, it does not say that the observed trace belongs to a fixed path scheme.

111.14.14 Trace fidelity: actual cycles versus alternative paths

The activation branch needs a second distinction in addition to the projective-mode and recurrent-guard-rank definitions. A generic 2-VASS flattening theorem says that if one synchronized configuration reaches another, some alternative path belonging to a finite family of path schemes reaches it. It does *not* imply that the observed literal input factor itself belongs to one of those schemes. Therefore path-scheme periods may not be deleted from the observed word unless their literal occurrences have been independently located.

We use the following two rules throughout.

(T1) **Observed-word operation.** Any literal deletion package is extracted by cycle erasure on the actual synchronized trace. Its blocks occur at identified positions of the original word.

(T2) **Replacement-word operation.** VASS reachability or flattening may construct a different word only when Lemma 111.66 is invoked at the common deterministic endpoint, as in Proposition 111.74.

Consider now an actual synchronized positive-mode trace before a bounded contact. For a simple synchronized cycle C put

$$v(C) = (\Delta_N(C), \omega(C)), \qquad \omega(C) = \Delta_N(C) - \Delta_A(C).$$

Cycle-erasing the actual trace leaves a bounded simple residual and a multiset of primitive simple cycle occurrences. Let $C_1, \dots, C_M$ be the finite simple-cycle types and $x_{i,j}$ their multiplicities in the ith actual pre-contact path. Pass to a subsequence so that, for each j, either $x_{i,j}$ is bounded or $x_{i,j} \to \infty$. Let I_∞ be the latter types and put

$$K_\infty = \mathrm{cone}_{\mathbb{Q}_{\geq 0}}\{v(C_j) : j \in I_\infty\}.$$

Lemma 111.100 (Actual-trace unbounded-multiplicity cone)**.** *If $\Delta_N(P_i) \to -\infty$ and $\omega(P_i) \to -\infty$ on the actual pre-contact paths, then K_∞ meets the open southwest quadrant*

$$\mathrm{SW} = \{(x, y) : x < 0,\ y < 0\}.$$

Every generator used in K_∞ occurs with unbounded actual multiplicity in the observed literal traces.

Proof. The bounded-multiplicity cycle types and the simple residual contribute only $O(1)$. If K_∞ were disjoint from the open southwest quadrant, the rational polyhedral separation theorem would give a nonzero nonnegative linear functional that is nonnegative on K_∞. Applied to the cycle decomposition, that functional would bound from below the corresponding nonnegative combination of $(\Delta_N(P_i), \omega(P_i))$, contradicting that both coordinates tend to $-\infty$. The last statement is the definition of I_∞. □

Lemma 111.101 (Support-two southwest reduction)**.** *If a finitely generated rational cone in* $\mathbb{R}^2$ *meets* SW*, it contains a southwest vector supported on at most two generators. If the support is two and neither generator is southwest, then after relabeling the two actual cycle types* S, D *satisfy*

	Δ_A	Δ_N	ω
S	> 0	≥ 0	< 0
D	< 0	< 0	≥ 0.

Proof. Apply the conic form of Carathéodory's theorem in dimension two to the cone K_∞ from Lemma 111.100. The selected one or two generators therefore belong to I_∞ and occur with unbounded actual multiplicity. For a support-minimal two-generator southwest combination, neither generator can have both coordinates negative. One must supply the negative N coordinate and the other the negative relative coordinate; substituting $\Delta_A = \Delta_N - \omega$ gives the stated opposite deterministic signs. □

Write $a = \Delta_A(S) > 0$, $d = -\Delta_A(D) > 0$, $x = \Delta_N(S) \geq 0$, and $y = -\Delta_N(D) > 0$. Define

$$\rho = x/a, \qquad \kappa = y/d.$$

A positive combination is southwest exactly when $\rho < \kappa$. With $g = \gcd(a, d)$ set

$$m = d/g, \qquad n = a/g.$$

Then the primitive deterministic balance

$$\Pi = mS + nD$$

satisfies

$$\Delta_A(\Pi) = 0, \qquad \Delta_N(\Pi) = \frac{ad}{g}(\rho - \kappa) < 0. \tag{111.29}$$

After clearing rational coefficients, m, n are fixed positive integers. By Lemma 111.100 and the proof of Lemma 111.101, both selected types lie in I_∞; sufficiently deep actual traces therefore contain the required finite number of literal occurrences of each type.

Proposition 111.102 (Deep trace-faithful neutral package deletion)**.** *In a sufficiently high translated arch, delete from the* original word *the selected actual occurrences of one package* Π*, leaving the intervening connectors untouched. The modified deterministic run stays positive, rejoins the exact original* A *configuration after the last removed block, and strictly shortens the word.*

Proof. Each removed block is an actual A-control cycle in positive mode. The package contains a fixed number of fixed cycle types, so the maximum intermediate height discrepancy caused by deleting them in chronological order is bounded by a package constant. High-shell reserve eventually exceeds that constant. Equation (111.29) gives zero total deterministic effect, hence exact rejoining after the final deletion. □

No statement about preserving an N run is made. Instead shortestness converts the A-safe deletion into a full-envelope certificate.

Lemma 111.103 (Critical deletion certificate)**.** *Let* H *be the original exact envelope and* H' *the deleted-word envelope at the common splice after the last removed block; let* R *be the backward requirement of the unchanged suffix. Then for some live coordinate* q*,*

$$H(q) < R(q) \leq H'(q).$$

Proof. The original factor is rejected against the fixed residual, so $H < R$. The deletion word is shorter and still accepted by A; R-minimality therefore gives $H' \not< R$. The same coordinate witnesses both inequalities. □

The singleton southwest case with $\Delta_A = 0$ is identical. Suppose instead that one actual primitive southwest cycle C occurs with unbounded multiplicity and $\Delta_A(C) \neq 0$. The literal translated endpoint arch containing these occurrences has one fixed total deterministic effect κ_A, independent of shell depth. Cycle-erasing that same actual synchronized arch leaves a simple residual of uniformly bounded deterministic effect and finitely many primitive cycle types. In the exact identity

$$\kappa_A = \mathrm{res}_A(i) + \sum_j N_j(i)\Delta_A(C_j),$$

the contribution of C tends to one infinity sign. Therefore the sum of contributions of the opposite deterministic sign tends to the other; by finiteness one fixed actual primitive cycle type D of opposite A-effect also has unbounded multiplicity. With

$$g = \gcd(|\Delta_A(C)|, |\Delta_A(D)|), \quad m = |\Delta_A(D)|/g, \quad n = |\Delta_A(C)|/g,$$

the sparse literal package $mC + nD$ has total deterministic effect zero. Its cardinality is fixed, so high-shell reserve makes its chronological deletion A-safe and exact-rejoining. No sign condition on its net N effect is needed: shortestness alone yields Lemma 111.103. Thus the nonzero-A singleton folds into the same critical-deletion branch without any trace-changing path scheme.

111.14.15 Activation gain, threshold contacts, and noncircular return

At a critical deletion splice fix the decisive coordinate and write

$$\kappa_{\mathrm{act}} = H'(q) - H(q) \geq 1.$$

There are two cases.

Bounded activation gain. If $\kappa_{\mathrm{act}} \leq M$, then

$$1 \leq R(q) - H(q) \leq M.$$

The chronological rejection margin before the splice is therefore confined to a finite band. After support and normalized-profile refinement, every no-escape segment lies in finitely many exact diagonal/projective phases. Absorbing their finite interfaces produces one finite exact projective mode system, which closes by Proposition 111.74.

Unbounded activation gain. The deleted package contains only a fixed number of fixed literal blocks. Telescoping between successive deletion variants selects one fixed block whose removal causes unbounded later envelope sensitivity. Immediately after a fixed word C, two envelopes with stable support and bounded normalized entry data differ by at most an effective constant L_C: above the finite entry thresholds, every C-path is affine in the common scalar. Moreover, as long as two one-letter transformers enable the same transition family, max-plus evaluation is sup-norm nonexpansive. Therefore unbounded amplification must have a first step at which enabled-edge families differ, or support changes.

Lemma 111.104 (First-threshold contact)**.** *Excluding an earlier support/projective escape, unbounded activation gain forces an aligned cut of the actual original shortest word, a fixed N state r, and an effective constant B_* such that*

$$H(r) \leq B_*.$$

The corresponding deterministic height tends to infinity along the high-shell family.

Proof. Choose the first suffix step where the previous uniform sup-distance bound fails. If support has not changed, some transition is enabled from the larger envelope and disabled from the smaller. Its fixed negative requirement pins the source coordinate of both envelopes into a bounded threshold band. There are only finitely many earlier deleted slots; bounded pairwise differences before the first responsible slot transfer the bounded coordinate back to the actual original witness by the triangle inequality. High-shell geometry supplies the diverging A height. □

Thus at the contact

$$F(r) = H(r) - d \longrightarrow -\infty.$$

If the normalized N profile relative to r stays bounded, the contact belongs to a finite exact projective context. If it is unbounded, a genuine N/N projective difference escapes and localizes through a bounded-wall carrier to rank at most $\rho - 1$ or to a support/provenance drop.

Finally transfer the bounded contact to the canonical left endpoint of the translated arches. If that bounded correction itself escapes, it is already a lower-rank carrier escape. Otherwise after finite refinement one fixed reference coordinate has

$$H_\ell^L(r) = b, \qquad d(e_\ell^L) = \ell + r_L.$$

Hence

$$a_\ell := F_\ell^L(r) = b - (\ell + r_L).$$

For $h < k$, with $c = k - h$,

$$a_h - a_k = c. \tag{111.30}$$

Lemma 111.105 (Contact-anchor noncircularity)**.** *The superlevel activation condition $a_h - a_k > k - h$ cannot recur after a threshold contact is returned to the canonical endpoint system.*

Proof. Equation (111.30) gives equality for every pair. □

This is the strict well-foundedness invariant missing from a naive recurring-wall argument.

111.14.16 Bounded-context domination and the rank lift

The induction does not need a global lower-rank relation or an arbitrary exact-slice oracle. It needs only the endpoint replacement guaranteed by Lemma 111.66.

Definition 111.106 (Bounded-context domination compactness, WCI(ρ))**.** Fix an actual positive-mode factor P in an R-minimal witness, one concrete exact entry, and one exact deterministic A exit. Assume its recurrent owner-guard rank is at most ρ, and assume that all normalized/projective, anchored-relative, common-shift, support/provenance, and deterministic transfer/range data relevant to the current carrier belong to fixed finite palettes. Then WCI(ρ) asserts that there is an effective uniform bound on $|P|$; equivalently, every longer P has a strictly shorter endpoint replacement Q with the same exact deterministic exit and $H_Q \le H_P$.

If one of the listed quantities leaves its finite palette, that first exit is not hidden inside WCI: it is declared a responsible escape and localized separately.

Proposition 111.78 proves WCI(ρ) for $\rho \le 1$.

Theorem 111.107 (Structural rank lift)**.** *For every finite $\rho \ge 1$,*

$$\mathrm{WCI}(\rho - 1) \Longrightarrow \mathrm{WCI}(\rho).$$

Proof. Fix a rank-ρ bounded-context factor.

Wall-free branch. Lemma 111.80 lowers the rank. If a tracked normalized quantity escapes, this is a responsible lower-rank escape; otherwise $\mathrm{WCI}(\rho - 1)$ bounds the strip.

Bounded-wall branch. Proposition 111.81 gives the same dichotomy for every fixed wall budget.

Unbounded-wall branch. Lemma 111.82 supplies linearly many bounded-wall same-signature atoms, with lower-rank endpoint systems. Bounded carrier corrections belong to the finite palette of Definition 111.83; unbounded corrections localize to a lower-rank wall-free strip.

Translated-arch branch. Lemma 111.84 gives either immediate anchored deletion or one fixed-coordinate endpoint/carrier escape. The latter is lower rank unless the only remaining mass is the activation/debt configuration.

Activation/contact branch. Sections 111.14.8–111.14.13 derive the activation mechanism from exact translated-arch endpoint identities, reduce the terminal survivor to a bounded high contact, and obtain the actual contact-centered relative valley of Theorem 111.99. Section 111.14.14 then extracts literal cycle packages from the observed synchronized trace, without trace-changing flattening. Bounded activation enters a finite projective system. Unbounded activation produces an actual threshold contact, and Lemma 111.105 prevents return to the same-rank superlevel branch. The return therefore ends in finite-projective compactness, support/provenance decrease, or a genuine lower-rank projective/relative escape.

All branches either terminate in Proposition 111.74, strictly lower recurrent guard rank, strictly lower provenance rank, or advance to a later same-(ν, ρ) stage in the finite stage order made explicit in Table 111.1. A responsible escape is therefore not a same-state recursive call: at its first occurrence it is reclassified as one of these successor types. Lemma 111.105 rules out the only possible return from the contact stage to the earlier activation stage at unchanged (ν, ρ). Hence the rank-ρ bounded-context factor is structurally reduced to terminal compactness or a strictly smaller proof state. The computability of the resulting numerical bound is established separately in Proposition 111.108. □

111.14.17 Quantitative effectivity and the proof-state order

The preceding theorem is structural. We now explain why its bound is a total computable integer rather than only an abstract finiteness statement.

Every tracked integer observable changes by at most a computable constant on the finite primitive macro alphabet. Therefore, for any cutoff M, the *first* exit from $[-M, M]$ occurs in a finite band

$$M < |Z| \le M + D$$

for a computable drift bound D. Before this first exit the factor lies in a finite normalized context box, so the relevant lower-rank WCI bound applies. Wall-density constants are explicit, exact wall and carrier colors are finite, and every Ramsey or pigeonhole refinement is invoked only after choosing a finite required chain length.

A potentially dangerous infinitary Dickson extraction is avoided. Along a strict endpoint chain, after choosing a fixed reference coordinate, one obtains

$$(a_0 - a_n) + (\theta_0(q) - \theta_n(q)) \ge n. \tag{111.31}$$

Thus either common anchored scalar debt is large or projective spread is large. Projective spread above the explicit synchronized-product simple-path bound exposes a bounded nonzero differential core; below that bound normalized profiles lie in a finite box.

For complete algorithmic bookkeeping define the lexicographic measure

$$\mathcal{M} = (\nu, \rho, s),$$

where ν is the provenance rank from Definition 111.70, ρ is recurrent guard rank, and s belongs to the following finite stage set, ordered from top to bottom. Support changes

that preserve the represented provenance atoms are finite-mode refinements; a recursive "support/provenance drop" means a strict decrease of ν as in Lemma 111.71.

s	Stage	Allowed same-(ν, ρ) successor
7	bounded-context entry/refinement	wall decomposition or terminal finite projective
6	directional wall decomposition	fixed-K terminal or unbounded-wall skeleton
5	dense same-signature atoms	translated high-wall carrier stage
4	translated arches / vector shell defect	anchored deletion, lower rank, or activation cells
3	activation cells / bounded contact / actual trace	relative valley or lower-rank terminal
2	cycle package / activation gain	critical deletion, finite-projective phase, or threshold contact
1	contact return	finite-projective terminal or strict lower-rank/support escape
0	terminal compactness	no recursive same-rank call

Table 111.1: Finite same-rank stage component of the proof-state measure.

A genuine provenance-rank drop strictly decreases ν by Lemma 111.71; wall-free responsible escapes strictly decrease ρ; every other nonterminal transition in Table 111.1 decreases s. Lemma 111.105 is precisely what prevents a contact return from jumping back from stage 1 to stage 4 at the same (ν, ρ).

Proposition 111.108 (Effective proof recursion)**.** *For a concrete exact first-return entry, all thresholds and terminal bounds can be computed by well-founded recursion on $\mathcal{M}$. In particular, the bound $B_{\mathrm{FR}}(A, N, C_0, H_0)$ is total computable.*

Proof. The terminal modules are effective: rank at most one uses Proposition 111.78; finite-projective branches use Proposition 111.74; fixed owner-resolved bounded-expression route schemas use Proposition 111.72; fixed wall and carrier alphabets are explicit finite sets. At a nonterminal proof state, compute the finitely many smaller-measure successor bounds first, choose the finite first-exit/Ramsey/wall thresholds required to force one of those successors, and take the finite maximum/sum dictated by the decomposition. Lexicographic well-foundedness guarantees termination. □

Theorem 111.63 follows from Theorem 111.107 and Proposition 111.108.

111.15 Effective next-zero successors

Fix a concrete zero state

$$Z = (p, H),$$

meaning deterministic configuration $(p, 0)$ and exact net envelope H. The first-return theorem is node dependent: the bound used at Z is computed from this exact H, not from a global uniform envelope bound.

Definition 111.109 (Next-zero factor)**.** A nonempty word u is a next-zero factor from Z if the unique A-run starts at $(p, 0)$, ends at some $(p', 0)$, and has no zero cut at a strict internal word boundary.

Thus a next-zero factor is either a one-letter zero-to-zero transition or a nontrivial positive first return. Put

$$B_0(Z) = \max\{1, B_{\mathrm{FR}}(A, N, Z) + 2\},$$

where the harmless $+2$ absorbs whether the boundary exit/return letters are counted inside the local first-return convention.

For a fixed finite word $u = a_1 \cdots a_k$, the exact transformer $T_u(H)$ is computable without any reachability oracle. For each source coordinate q and each length-k N-path π labelled by u, let $\mathrm{eff}(\pi)$ be its total counter effect and let

$$\mathrm{req}(\pi) = \max\{0, -\min \text{ prefix effect of } \pi\}$$

be the least initial credit required to execute it. Then

$$(T_u H)(r) = \max_{\substack{q,\pi: q \xrightarrow{u} r \\ H(q) \geq \mathrm{req}(\pi)}} \big(H(q) + \mathrm{eff}(\pi)\big).$$

The maximum is over a finite set of paths.

Definition 111.110 (Raw zero-successor set)**.** $\mathrm{RawSucc}_0(Z)$ is the set of records

$$e = (u, |u|, p_u, H_u)$$

obtained by enumerating all words $u \in \Sigma^{1 \dots B_0(Z)}$, retaining exactly the next-zero factors, and putting $H_u = T_u(H)$ and $(p_u, 0)$ for the exact deterministic exit.

Theorem 111.111 (Effective zero-successor enumeration)**.** $\mathrm{RawSucc}_0(Z)$ *is finite, effectively computable, and literally complete for consecutive zero cuts of every shortest counterexample passing through* Z*.*

Proof. The enumeration is finite. Exact deterministic simulation decides the next-zero property and the formula above computes H_u. If u is the actual factor to the next zero cut of a shortest counterexample, then either $|u| = 1$ or Theorem 111.63 bounds its first-return length by B_{FR}; hence u occurs in the enumeration. □

There is also a safe finite pruning order, useful for implementations though not needed in the global proof.

Definition 111.112 (Cost-envelope order)**.** For two raw edges with the same target zero control,

$$e_i = (u_i, \ell_i, p', H_i),$$

write $e_1 \preceq_0 e_2$ when $\ell_1 \leq \ell_2$ and $H_1 \leq H_2$ componentwise.

Lemma 111.113 (Zero-edge domination)**.** *If* $e_1 \preceq_0 e_2$ *and* $\ell_1 < \ell_2$*, then* e_2 *cannot be used between consecutive zero cuts of a shortest counterexample. If* $\ell_1 = \ell_2$*, replacing* e_2 *by* e_1 *preserves shortestness and rejection.*

Proof. Both factors start from the same zero configuration and end in the same deterministic zero control. The smaller envelope preserves rejection by Lemma 111.66. A strict cost decrease contradicts shortestness; equal cost gives another shortest witness. □

The finite set of $\preceq_0$-minimal edges may therefore be used as a canonical successor antichain when only existence of a shortest witness matters. For the global tree below we keep the raw set, so the actual shortest-witness zero prefix embeds literally rather than only up to replacement.

111.16 The finite zero-state tree

The local bound of Section 111.15 is converted into a global bound by a second well-quasi-order argument.

Definition 111.114 (Zero-state order)**.** For zero states (p, H) and (p', H') put

$$(p, H) \leq_Z (p', H') \quad \Longleftrightarrow \quad p = p' \text{ and } H \leq H'.$$

Encode $\perp$ by 0 and a finite value h by $h + 1$. For each fixed control p, this is isomorphic to componentwise order on $\mathbb{N}^{|Q_N|}$.

Hence the finite disjoint union of all zero-control components is a WQO by Dickson's lemma.

Construct a rooted edge-labelled tree T_0 as follows. The root is the initial exact zero state. At a node Z, compute $\mathrm{RawSucc}_0(Z)$. For a candidate child $Z' = (p', H')$, inspect only the ancestors on the current root-to-node path. If some ancestor (p', H_a) satisfies $H_a \leq H'$, prune Z'; otherwise insert it and retain the actual edge word and its cost.

Lemma 111.115 (No infinite branch)**.** *The pruned zero tree has no infinite branch.*

Proof. Along a branch no later zero state is above an earlier zero state with the same deterministic control. If the branch were infinite, some control would occur infinitely often. Dickson's lemma applied to the corresponding infinite sequence of encoded envelopes would produce an increasing pair, contradicting the pruning rule. □

Theorem 111.116 (Finite zero tree)**.** *The pruned tree T_0 is finite and effectively constructible. The complete zero-cut sequence of every shortest counterexample embeds literally as one root-to-node path of T_0.*

Proof. Theorem 111.111 gives finite branching at every generated node. Lemma 111.115 rules out infinite branches. König's lemma therefore implies finite depth; finite branching at each finite depth gives a finite tree. Exhaustive BFS or DFS terminates because it is enumerating this finite object.

For literal completeness, let $Z_0, \ldots, Z_m$ be the actual zero states of a shortest counterexample. Theorem 111.111 inserts the actual factor $Z_i \to Z_{i+1}$ whenever Z_i is reached. Lemma 111.61 says that no Z_j can be pruned against an earlier Z_i with the same control, because such a pair would itself shorten the witness. Induction on i embeds the whole zero prefix. □

Two computable quantities are extracted from the explicit finite tree:

$$D_0 = \max\{\text{root-to-node edge depth}\},$$

and, retaining literal edge costs,

$$L_0 = \max\{\text{root-to-node total input length}\}.$$

Thus every shortest counterexample has at most $D_0 + 1$ zero cuts, and its prefix through the last zero cut has length at most L_0.

Remark 111.117 (Why König is effective here)*.* König's lemma is used only to prove that the explicitly generated finitely branching tree is finite. The algorithm itself is ordinary exhaustive generation; it does not require a closed-form Dickson length function. Once mathematical finiteness is established, the generation necessarily reaches an empty frontier.

111.17 The terminal positive tail

A shortest counterexample may end strictly after its last zero cut. This case has no final zero return, so it is not directly covered by Theorem 111.63. A fresh-symbol construction turns it into one.

Fix a zero node $Z = (p, H)$ of T_0.

Definition 111.118 (Positive terminal tail)**.** A word u is a positive terminal tail from Z if the A-run starts in $(p, 0)$, is positive at every noninitial cut, and ends in an accepting control at height $h \geq 1$. It is rejecting if N has no accepting continuation over u from exact input envelope H.

Introduce three fresh letters $\#, d, \$$. Form A_Z^{tail} with a start-zero copy of p and positive-phase copies q^+ of the original controls. The start copy permits exactly the original zero rules that leave zero; the positive copies permit exactly the original positive rules. From each accepting f^+ add

$$f^+ \xrightarrow{\#,0} r,$$

where r is a drain control. At r add the positive rule $r \xrightarrow{d,-1} r$ and the zero rule $r \xrightarrow{\$,0} \mathsf{acc}$. The only accepting control is acc.

Extend N to N^{tail} by adding zero-effect self-loops on $\#, d, \$$ at every state, leaving the final states unchanged.

Lemma 111.119 (Identity of the fresh suffix)**.** *For every original-alphabet word u and every $k \geq 0$,*

$$T_{u\#d^k\$}^{N^{\mathrm{tail}}}(H) = T_u^N(H).$$

In particular the extended net accepts $u\#d^k\$$ from H iff the original net accepts u from H.

Proof. Each fresh letter has exactly one zero-effect self-loop at every net control and no other fresh-letter transition, so its exact envelope transformer is the identity. □

Lemma 111.120 (Exact tail/corridor correspondence)**.** *A rejecting positive terminal tail u from Z ending at deterministic height h exists iff*

$$u\#d^h\$ \in L(A_Z^{\mathrm{tail}}) \setminus L_H(N^{\mathrm{tail}}),$$

where the net side starts from envelope H. Every accepted word of the extended deterministic machine has this unique form.

Proof. The extended control graph forces the original positive phase, then the marker from an accepting original control, then exactly h decrements to reach zero, then the zero-only final marker. Lemma 111.119 gives the net equivalence. □

Theorem 111.121 (Terminal-tail reduction)**.** *For every zero node $Z = (p, H)$ there is a total computable integer $B_{\mathrm{tail}}(Z)$ such that, if any rejecting positive terminal tail exists from Z, then one exists with length at most $B_{\mathrm{tail}}(Z)$.*

Proof. If a rejecting tail exists, choose a shortest extended counterexample $u\#d^h\$$ given by Lemma 111.120. The factor $u\#d^h$ is a strictly positive first return from zero to zero: the original part stays positive, the marker has zero effect, and only the last drain letter reaches zero. Therefore Theorem 111.63 applied to the extended instance gives

$$|u\#d^h| \leq B_{\mathrm{FR}}(A_Z^{\mathrm{tail}}, N^{\mathrm{tail}}, (s_p, 0), H).$$

Take this first-return bound as $B_{\mathrm{tail}}(Z)$; then $|u| \leq B_{\mathrm{tail}}(Z)$. □

The theorem is existential: it need not bound the literal terminal suffix of every shortest original witness. For global decidability it is enough that any obstruction at the same exact last-zero state has a bounded replacement tail.

Because T_0 is finite, compute

$$B_{\mathrm{tail}}^{\max} = \max_{Z \in T_0} B_{\mathrm{tail}}(Z).$$

111.18 Global small witnesses and decidability

Define

$$B_{\mathrm{global}} = L_0 + B_{\mathrm{tail}}^{\max}.$$

Theorem 111.122 (Global small witness, detailed form)**.** *If $L(A) \not\subseteq L(N)$, then there is a word $w \in L(A) \setminus L(N)$ with*

$$|w| \le B_{\mathrm{global}}.$$

Proof. Choose a shortest counterexample. If it ends at a zero cut, its entire word is a root-to-node path in T_0, hence has length at most L_0. Otherwise write it xu, where x ends at the last zero state $Z = (p, H)$ and u is a rejecting positive terminal tail. The literal zero prefix x embeds in T_0, so $|x| \le L_0$. Theorem 111.121 supplies another rejecting positive tail v from the same exact deterministic configuration $(p, 0)$ and the same exact net envelope H, with $|v| \le B_{\mathrm{tail}}(Z)$. Consequently xv is still accepted by A and rejected by N, and

$$|xv| \le L_0 + B_{\mathrm{tail}}^{\max}.$$

□

Corollary 111.123 (Hard inclusion)**.** *It is decidable whether $L(A) \subseteq L(N)$.*

Proof. Compute the finite integer B_{global} and enumerate $\Sigma^{\le B_{\mathrm{global}}}$. Deterministic membership in A is direct. Membership in N is a finite dynamic program over the word. If one enumerated word lies in $L(A) \setminus L(N)$, report noninclusion; if none does, Theorem 111.122 proves inclusion. □

111.19 The reverse inclusion

The hard part of the chapter is $L(A) \subseteq L(N)$. Equality also requires the opposite orientation; we spell out the reduction because the deterministic machine has zero tests.

Proposition 111.124 (Deterministic-right inclusion)**.** *It is decidable whether $L(N) \subseteq L(A)$.*

Proof. **1. Total deterministic complement.** The deterministic OCA A may be partial. Add one dead control † and, for every missing letter transition in either the zero or positive regime, add a zero-effect transition to †; add zero-effect self-loops on all letters in both regimes at †. The resulting deterministic OCA is total on every word. Complement the set of final controls to obtain $\overline{A}$, so

$$L(\overline{A}) = \Sigma^* \setminus L(A).$$

2. Synchronous product. Now

$$L(N) \not\subseteq L(A) \iff L(N) \cap L(\overline{A}) \neq \varnothing.$$

Synchronize one transition of N with the unique $\overline{A}$ transition on each letter. The product has two nonnegative counters: the N counter is never zero-tested and the A counter is the only tested counter.

3. Compile the positive regime. A positive-regime OCA rule must be disabled at A-counter value zero. Simulate it by first decrementing the A counter by one, which is possible exactly when it is positive, while performing the synchronized N update. Then use a bounded finite-control chain to add back the amount needed to realize the original update $-1, 0$, or $+1$ (equivalently use the corresponding fixed VASS vector if arbitrary finite vectors are allowed). Thus positivity is enforced without any zero test.

4. Serialize all zero-regime rules through one zero-test gate. The product may contain many zero-regime rules, whereas the standard VAS_0 presentation uses one

distinguished zero-test transition. This difference is harmless and we spell out the reduction. For every product zero-rule e, introduce one fresh ordinary tag counter t_e. From the source control of e, an ordinary transition moves to a common pre-gate control and places one token in t_e, without changing the tested A counter. A single distinguished transition

$$g_0 : \; q_{?0} \longrightarrow q_{=0}$$

is enabled exactly when the A counter is zero. After g_0, the unique post-gate transition corresponding to e consumes the token in t_e, performs the synchronized N update and the legal zero-regime A update (necessarily 0 or $+1$ in unit-step form), and enters the target product control. All other post-gate branches are disabled because their tag counter is zero. If the source A counter was positive, the run can enter the pre-gate gadget but cannot pass g_0, so it produces no spurious reachable target. Hence every zero-regime product edge is simulated exactly while all tests share one single distinguished zero-test transition.

The resulting counter system is therefore effectively a VASS/VAS with one zero-test in the sense of Reinhardt and Bonnet, for which exact configuration reachability is decidable [115, 116].

5. Final counters need not be zero. The product accepts upon reaching a control pair in $F_N \times F_A$ with arbitrary nonnegative counter values. From every such product control enter a drain phase with ordinary decrement loops for each counter. Then some accepting product control is reachable with arbitrary counters iff one fixed drain target with both counters equal to zero is exactly reachable. No additional zero test is required because exact reachability asks for the target vector itself.

Therefore noninclusion is decidable, and so is $L(N) \subseteq L(A)$. □

Combining Proposition 111.124 and Corollary 111.123 proves Theorem 111.50.

111.20 The epsilon-move extension: models and deterministic reset normalization

111.20.1 Deterministic epsilon one-counter automata

A deterministic epsilon one-counter automaton (epsilon-OCA) has a finite control set Q_A, an initial control q_A^0, a set F_A of accepting controls, a nonnegative counter initialized to 0, a visible alphabet Σ, and transitions labelled by $\Sigma \cup \{\varepsilon\}$. Transitions may distinguish counter zero from counter positive. We use the standard unit-update OCA model, with counter effects in $\{-1, 0, +1\}$. This unit-update convention is part of the theorem statement; no unproved expansion of larger deterministic updates is used below.

The determinism convention is the standard one: in a fixed control/sign regime, if an epsilon transition is enabled then no visible transition competes with it. A visible word is accepted when the unique complete deterministic computation is defined, includes all forced silent behavior, and ends in a control of F_A; the final counter may be any nonnegative integer. If the computation becomes undefined, the word is rejected.

We use the standard stable/reset normalization of deterministic epsilon-OCA from the deterministic one-counter equivalence literature [107]. In this presentation, stable controls consume visible letters and reset controls perform forced silent behavior. A reset control s carries a finite period per_s and a map

$$\mathrm{goto}_s : \{0, \ldots, \mathrm{per}_s - 1\} \to Q_A^{\mathrm{st}},$$

with semantics

$$(s, n) \xRightarrow{\varepsilon^*} \big(\mathrm{goto}_s(n \bmod \mathrm{per}_s), 0\big).$$

Initial and accepting configurations are arranged stable. We take this normalization as an external standard input and perform the remainder of the hard-direction proof in the stable/reset presentation.

Remark 111.125. The deterministic side of theorem 111.52 is not allowed arbitrary nondeterministic silent behavior. The theorem covers the standard deterministic-epsilon convention described above.

111.20.2 Nondeterministic epsilon one-counter nets

A nondeterministic epsilon one-counter net (epsilon-OCN) has a finite control set Q_N, an initial control q_N^0, a set F_N of accepting controls, one nonnegative counter initialized to 0, visible and epsilon transitions, and no zero test. Primitive transitions have unit effects $d \in \{-1, 0, +1\}$ and are enabled exactly when the resulting counter remains nonnegative. Weak macro effects introduced below are arbitrary integers because they sum finitely many primitive effects. A word is accepted when some weak run ends in a control of F_N, with arbitrary nonnegative final counter value.

For a visible word w, a weak run is any run whose visible projection is exactly w; arbitrary epsilon moves may occur before, between, and after visible letters. The empty visible word is included.

111.20.3 Path requirement and effect

For a finite OCN path π with counter effects $d_1, \ldots, d_m$, define

$$\mathrm{eff}(\pi) = \sum_{i=1}^{m} d_i$$

and

$$\mathrm{req}(\pi) = \max\left(0, -\min_{0 \le j \le m} \sum_{i=1}^{j} d_i\right).$$

Lemma 111.126. *A path π is feasible from counter c if and only if $c \ge \mathrm{req}(\pi)$. In that case its exit counter is $c + \mathrm{eff}(\pi)$.*

Proof. Let $s_j = \sum_{i=1}^{j} d_i$, with $s_0 = 0$. Starting from counter c, the counter after the first j transitions is $c + s_j$. Thus the whole path is feasible exactly when $c + s_j \ge 0$ for every j, equivalently when $c \ge -\min_j s_j$ and $c \ge 0$. This is precisely $c \ge \mathrm{req}(\pi)$. The final counter is $c + s_m = c + \mathrm{eff}(\pi)$. □

This elementary formula is the basis of the weak-envelope construction.

111.20.4 Stable cuts and visible length

A stable cut is taken after one visible input letter has been consumed and every forced deterministic reset closure triggered by that letter has completed. A stable zero configuration is a stable deterministic configuration $(p, 0)$.

All length bounds in this chapter count visible input letters only. Silent microsteps have zero word cost.

111.21 Weak epsilon-OCN envelope calculus

For controls $p, q \in Q_N$ and $a \in \Sigma$, write

$$(p, c) \Rightarrow_a (q, d)$$

when there is an OCN path of shape

$$\varepsilon^* \, a \, \varepsilon^*$$

from (p, c) to (q, d). Define

$$U_{p,a,q}(c) = \sup\{d : (p, c) \Rightarrow_a (q, d)\},$$

with value $\perp$ when no such weak path exists and ω when reachable exit counters are unbounded.

111.21.1 Cycle deletion and pumping

Lemma 111.127 (Nonpositive epsilon-cycle deletion)**.** *If a weak-a path contains an epsilon cycle C with $\mathrm{eff}(C) \le 0$, deleting that literal occurrence of C does not increase the path's entry requirement and does not decrease its exit counter.*

Proof. Write the path as $\alpha C \beta$ and let $e = \mathrm{eff}(C) \le 0$. Prefix sums before C are unchanged after deletion. Prefix sums after the deleted block increase by $-e \ge 0$, while the internal prefixes of C disappear. Hence the minimum prefix sum cannot decrease, so the entry requirement cannot increase. The total effect increases by $-e$, and therefore the exit counter does not decrease. □

Lemma 111.128 (Positive epsilon-cycle pumping)**.** *If a feasible weak-a path contains an epsilon cycle C with $\mathrm{eff}(C) > 0$ in its epsilon prefix or suffix, then above an effectively computable entry threshold the corresponding target upper envelope is unbounded.*

Proof. Write one weak path containing the positive epsilon cycle as $\rho C \tau$, where C is silent and $g = \mathrm{eff}(C) > 0$. Put

$$r_C = \mathrm{req}(\rho C).$$

For every source credit $c \ge r_C$, the prefix ρ and one traversal of C are feasible. After that traversal the counter at the cycle base has increased by g, so the same cycle remains feasible; after k repetitions it has gained kg. The fixed continuation τ has a finite entry requirement, hence sufficiently many repetitions make τ feasible and the target counter can be made arbitrarily large. Thus r_C is the exact entry threshold for this fixed pump witness.

Conversely, if pumping is possible from a source credit c, some feasible weak run contains a positive epsilon closed walk. Decomposing that walk into simple cycles yields a simple positive epsilon cycle. Choose the first such cycle on the run and delete all preceding nonpositive silent cycles; this produces one of the finitely enumerable pump witnesses with $r_C \le c$. Taking the minimum r_C over all such witnesses therefore gives the exact onset of the pumping branch. □

Theorem 111.129 (Finite weak-envelope normal form)**.** *For every p, a, q one can effectively compute:*

- *an optional pumping threshold r_∞, and*
- *a finite set $S_{p,a,q} \subseteq \mathbb{N} \times \mathbb{Z}$ of threshold/effect pairs,*

such that

$$U_{p,a,q}(c) = \omega$$

whenever the pump threshold applies, and otherwise

$$U_{p,a,q}(c) = \max\{c + e : (r, e) \in S_{p,a,q},\ r \le c\},$$

with $\bot$ when the set of enabled clauses is empty.

Proof. Positive epsilon cycles give the pumping branch by theorem 111.128. Below all pump thresholds, take any feasible weak a-path and repeatedly delete every nonpositive epsilon cycle using theorem 111.127. If an epsilon prefix or suffix still contained a closed walk, it would contain a simple cycle; because positive cycles are excluded below the pump threshold, such a cycle would be nonpositive and could be deleted. Thus both silent parts become simple. There are only finitely many resulting simple $\varepsilon^* a \varepsilon^*$ skeletons. By theorem 111.126, each skeleton contributes one pair $(\mathrm{req}(\pi), \mathrm{eff}(\pi))$, and every deleted path is dominated by its reduced skeleton. Taking the maximum over enabled skeletons gives the displayed finite branch. Enumerating simple connectors and simple positive epsilon cycles makes both the finite clause set and the pumping thresholds effective; Section 111.32 gives the explicit construction. □

For all later proof bookkeeping we use a *skeleton-refined palette*: if several simple weak paths induce the same threshold/effect pair, they are retained as distinct macro types whenever their concrete skeleton identity is needed. Likewise a pump type retains one concrete lasso certificate (prefix, positive silent cycle, and suffix) witnessing its threshold. This finite refinement does not change the envelope transformer.

111.21.2 Forward envelopes and backward requirements

Let

$$D = \{\bot\} \cup \mathbb{N} \cup \{\omega\},$$

ordered by

$$\bot < 0 < 1 < 2 < \cdots < \omega.$$

A forward envelope is a vector $H \in D^{Q_N}$. The value $H(q) = \omega$ means that arbitrarily large finite counters are reachable at q; it does not mean that every sufficiently large integer is reachable.

For $a \in \Sigma$, define

$$T_a^{\varepsilon}(H)(q) = \max_{p \in Q_N} U_{p,a,q}(H(p)),$$

with the natural interpretation of a source value ω through the finite threshold/pump description.

For a visible suffix v and state q, define

$$R_v(q) = \min \left\{ c \in \mathbb{N} : \begin{array}{l} \text{some weak run from } (q,c) \text{ reading } v \\ \text{reaches a final control} \end{array} \right\},$$

with $R_v(q) = \infty$ when the set is empty.

Definition 111.130 (Acceptance predicate)**.** For a forward envelope H and requirement vector R, write $\mathrm{Acc}(H, R)$ if there exists q such that $R(q) < \infty$ and either $H(q) = \omega$ or $H(q) \in \mathbb{N}$ with $H(q) \geq R(q)$. Write $\mathrm{Rej}(H, R)$ for the negation.

We deliberately do not encode rejection by a global expression $H < R$ once ω and ∞ coexist.

Theorem 111.131 (Exact weak residuation)**.** *For every visible letter a, there is an effective backward transformer S_a^{ε} such that*

$$\mathrm{Acc}(T_a^{\varepsilon}(H), R) \iff \mathrm{Acc}(H, S_a^{\varepsilon}(R)).$$

Equivalently,

$$\mathrm{Rej}(T_a^{\varepsilon}(H), R) \iff \mathrm{Rej}(H, S_a^{\varepsilon}(R)).$$

Proof. For each source control p, define $S_a^{\varepsilon}(R)(p)$ semantically as the least finite counter from which some weak a-run reaches a target q with enough exit credit to satisfy the finite requirement $R(q)$; use ∞ if no such counter exists. The feasible source set is upward closed because N has no zero tests. Therefore a forward envelope H can realize an R-accepting weak a-continuation exactly when one of its coordinates meets this least source requirement; an ω coordinate meets every finite requirement because it represents unbounded reachable finite credit. This is precisely the displayed equivalence. Effectivity follows from the finite simple-skeleton and pump description of theorem 111.129; the concrete formula is recorded below and in Section 111.32. □

For a finite simple weak skeleton $\pi : p \Rightarrow_a q$ and finite target requirement $R(q)$, the corresponding source requirement is

$$\max\big(\mathrm{req}(\pi),\ R(q) - \mathrm{eff}(\pi),\ 0\big).$$

A pump lasso contributes one finite entry threshold independent of the magnitude of the finite target requirement.

111.21.3 Initial and terminal epsilon closure

Let H_ε be the exact epsilon saturation of the initial N configuration. Let

$$R_\varepsilon(q)$$

be the least counter from which an epsilon-only run reaches an accepting control, or ∞ if impossible. If $w = a_1 \cdots a_n$, then

$$H_w = T^\varepsilon_{a_n} \cdots T^\varepsilon_{a_1}(H_\varepsilon),$$

and

$$w \in L(N) \iff \mathrm{Acc}(H_w, R_\varepsilon).$$

The same formula handles the empty visible word.

Lemma 111.132 (Active-omega exclusion)**.** *If uv is rejected and $R_v(q) < \infty$, then $H_u(q) \neq \omega$.*

Proof. If $H_u(q) = \omega$, arbitrarily large finite counters are reachable at q after u. Since $R_v(q) < \infty$, choose one reachable counter $c \geq R_v(q)$. By definition of $R_v(q)$, the suffix v then has an accepting weak continuation, contradicting rejection of uv. □

111.22 From unit edges to threshold-affine weak macros

This section records the epsilon-specific transfer lemmas needed to lift the real-time positive first-return compactness theorem.

111.22.1 Epsilon separator margin

For $c \in \mathbb{N}$, define $H + c$ by leaving $\bot$ and ω unchanged and adding c to finite coordinates. At a rejected cut set

$$m_\varepsilon(H, R) = \min\{c \in \mathbb{N} : \mathrm{Acc}(H + c, R)\},$$

with value ∞ if no such c exists.

Lemma 111.133 (Weak shift inequality)**.** *For every visible factor x and $c \geq 0$,*

$$T^\varepsilon_x(H + c) \geq T^\varepsilon_x(H) + c.$$

Proof. Every concrete weak run feasible from a finite source counter h remains feasible from $h+c$ and exits exactly c higher. Taking coordinatewise suprema preserves the inequality. If a coordinate on the right is ω, then arbitrarily large finite outputs from H remain arbitrarily large after the source shift, so the left coordinate is also ω. □

Lemma 111.134 (Margin monotonicity)**.** *If*

$$H' = T^\varepsilon_x(H), \qquad R = S^\varepsilon_x(R'),$$

and the factor lies in a rejected word, then

$$m_\varepsilon(H, R) \leq m_\varepsilon(H', R').$$

Proof. Let $c = m_\varepsilon(H', R') < \infty$. Then $\mathrm{Acc}(H' + c, R')$. By theorem 111.133, $T^\varepsilon_x(H + c) \geq H' + c$, and acceptance is monotone in the forward envelope. Hence $\mathrm{Acc}(T^\varepsilon_x(H + c), R')$. Applying exact residuation theorem 111.131 gives $\mathrm{Acc}(H + c, S^\varepsilon_x(R')) = \mathrm{Acc}(H + c, R)$, so $m_\varepsilon(H, R) \leq c$. The case $c = \infty$ is immediate. □

111.22.2 Threshold contacts and protected propagation

A finite weak macro is a tuple

$$e = (p, a, q, r, \delta, \sigma),$$

where r is its entry threshold, δ its integer effect, and σ identifies one concrete simple weak skeleton $\varepsilon^* a \varepsilon^*$ realizing the clause. Pump clauses are recorded separately.

Lemma 111.135 (Pump-or-threshold contact)**.** *Suppose a state q is critical at one step of a constant finite separator margin $\eta \geq 1$. If a backward-minimizing witness is a pump clause with threshold r_{pump}, then*

$$H(q) = r_{\mathrm{pump}} - \eta < r_{\mathrm{pump}}.$$

Hence the source lies in a computably finite threshold band.

Proof. A pump witness can satisfy any finite target requirement once its source counter reaches its fixed threshold, so its backward candidate is exactly r_{pump}. If it attains the backward minimum at the critical source, then $R(q) = r_{\mathrm{pump}}$. Criticality gives $R(q) - H(q) = \eta$, hence $H(q) = r_{\mathrm{pump}} - \eta < r_{\mathrm{pump}}$. □

Lemma 111.136 (Protected finite-macro propagation)**.** *Suppose a backward-minimizing witness is a finite macro with threshold r and the critical source satisfies $H(q) \geq r$. Then the macro is forward-enabled, forward-maximizing, backward-tight, and transports the critical gap exactly to the next cut.*

Proof. Let the macro go from q to q' with effect δ, and let R' be the next-cut requirement. At the critical source $R(q) = H(q) + \eta > H(q) \geq r$. Since the macro attains the backward minimum, its candidate is $\max(r, R'(q') - \delta) = R(q)$. The maximum therefore cannot be attained by r, and so $R(q) = R'(q') - \delta$. The source threshold is satisfied, hence the macro is forward-enabled and

$$H'(q') \geq H(q) + \delta = R'(q') - \eta.$$

Constant separator margin η at the next cut gives the opposite inequality $H'(q') \leq R'(q') - \eta$. Thus equality holds: the macro is forward-maximizing, the backward equality is tight, and q' is critical with the same gap η. □

Lemma 111.137 (Threshold-contact recurrence is margin-independent)**.** *Work inside one constant-margin phase. If a backward-minimizing finite macro with threshold r is not forward-enabled at a critical source, then its source credit is one of the finitely many values*

$$c = H(q) \in \{0, \ldots, r - 1\}.$$

For recurrence/pigeonhole bookkeeping it is sufficient to color such a contact by the refined finite mode, the critical source state, the chosen macro type, the bounded value c, and the finite support/omega data. The numerical separator margin η need not be included in the contact color. Consequently, if deterministic height and projective spread relative to the pinned value c are bounded, only finitely many exact contact states occur; a repeated exact state admits the usual shortest-witness deletion. All contact multiplicity bounds are therefore independent of the future backward requirement.

Proof. The inequality $H(q) < r$ gives the finite source-credit set. Inside the chosen phase the separator margin is already constant, so coloring by its numerical value contributes nothing to identifying the forward exact state. After fixing the finite mode and omega mask, bounded deterministic height and bounded projective spread around the pinned finite coordinate c bound every remaining finite envelope coordinate. Hence the exact state space is finite. If the same deterministic configuration and exact envelope recur, deleting the intervening visible factor rejoins both machines at the same exact forward state and preserves rejection against the inherited suffix, contradicting shortestness. No numerical bound on η is used. □

Thus every constant-margin factor either exposes a bounded threshold contact or carries a coherent tight critical strand through finite weak macros.

111.22.3 High-zone pump obstruction

Lemma 111.138 (High-zone pump obstruction)**.** *Inside a bounded projective/relative mode, a pump macro into a backward-live target is either disabled below a computable finite deterministic-height threshold or contradicts rejection. Consequently, above a computable height, all backward-live active coordinates evolve only by finite threshold-affine macros.*

Proof. In a fixed bounded projective/relative mode, every finite present source has form $H(p) = d + \kappa_p$, where d is the deterministic height parameter and κ_p ranges over a finite computable set. Consider a pump macro from p to a target q with threshold r_p. If q is backward-live and the pump is enabled, then the next forward value at q is ω, contrary to theorem 111.132 for a rejected continuation. Hence every such pump must satisfy $d + \kappa_p < r_p$, which bounds d by $r_p - 1 - \kappa_p$. Taking the maximum over the finite pump palette gives the claimed computable high-zone cutoff. Above it, only finite affine macros can contribute to backward-live targets. □

111.22.4 First amplification and trace fidelity

Lemma 111.139 (Regime nonexpansiveness and first split)**.** *Fix a visible letter a. Let H, K have the same finite support/omega mask and enable the same finite and pump macro clauses. On every target coordinate that is finite in both outputs,*

$$\left|T_a^{\varepsilon}(H)(q) - T_a^{\varepsilon}(K)(q)\right| \le \|H - K\|_{\infty}$$

where the norm ranges over their common finite source coordinates. Moreover, if two previously identical finite semantic regimes first diverge on the next visible step, then the divergence is caused by a support/omega event or by crossing one fixed finite or pump threshold; if the pre-split finite sup-distance is at most L, the responsible source values lie in a computably bounded L-neighborhood of that threshold.

Proof. For a finite target in a fixed regime, the weak transformer is the maximum of the same finite family of affine candidates $H(p) + \delta_e$; taking a maximum of functions with slope one in one source coordinate is 1-Lipschitz in the sup norm. Enabled pump clauses produce ω on both sides and therefore do not enter the finite-coordinate estimate. If the next regimes differ for the first time, the two inputs had the same support/omega mask and the same clause status immediately before that step. Hence the first disagreement is either a support/omega event or the enabling status of some clause with fixed threshold r. In the latter case one source value is below r and the other is at least r; a pre-split distance at most L places both values in the finite band $[r - L, r + L]$ after intersecting with $\mathbb{N}$. □

Lemma 111.140 (Exact finite-composition closure)**.** *Fix a refined route $e_1 \cdots e_k$ of weak macro choices with matching controls. If every e_j is finite, write r_j, δ_j for its threshold and effect and $E_{j-1} = \sum_{i<j} \delta_i$. Then the composition is exactly one threshold-affine route*

$$c \longmapsto c + \sum_{j=1}^{k} \delta_j \quad \textit{for} \quad c \ge r_{\mathrm{comp}} := \max\Big(0, \max_{1 \le j \le k} (r_j - E_{j-1})\Big).$$

If the first pump on the route occurs at position j, then the whole route is a pump route with exact source threshold

$$r_{\mathrm{comp}} = \max\Big(0, \max_{i<j}(r_i - E_{i-1}), r_j^{\mathrm{pump}} - E_{j-1}\Big).$$

The construction is effective. A finite composition retains the concatenated concrete weak skeleton; a pump composition retains the finite prefix/suffix skeletons together with one concrete positive-cycle lasso certificate.

Proof. Before the first pump, the entry credit at macro e_j is $c + E_{j-1}$, so simultaneous feasibility of all finite prefixes is exactly the displayed maximum of shifted thresholds. With no pump, the endpoint formula follows by summing effects. If the first pump is enabled, it produces arbitrarily large reachable finite credit at its target. Every fixed remaining suffix has only finitely many finite thresholds (or another pump), so sufficiently large pumped credit executes that suffix and leaves unbounded final credit. Conversely, before the first pump no smaller source credit can violate any of the listed prefix/first-pump requirements. The skeleton and lasso certificates are obtained by concatenation. □

Lemma 111.141 (Residue-refined powered-schema realization)**.** *Fix an owner-resolved bounded-expression schema over the skeleton-refined finite weak-macro palette. After an effective finite refinement of every powered exponent into residue classes, its threshold feasibility, owner comparisons, support/mode constraints, and deterministic positivity are Presburger conditions on the exponent variables; on each residue branch the endpoint finite owner vector is affine in the quotient exponent variables. The refinement and all formulas are independent of the future backward requirement.*

Proof. First apply theorem 111.140 to each fixed connector and powered block. On a fixed owner-resolved block β, the finite owner vector has an affine update $x \mapsto Px + c$ for one effectively known 0–1 functional matrix P. Since the powers of a finite matrix chosen from this finite transformation monoid are eventually periodic, compute μ, λ with $P^{\mu+\lambda} = P^{\mu}$. For $n = \mu + s + \lambda k$ with $0 \le s < \lambda$,

$$P^n x + \sum_{j<n} P^j c$$

is a fixed residue-dependent affine function of x and k: the tail of the matrix-power sequence is periodic and each full period contributes one fixed integer vector. The finitely many cases $n < \mu$ are handled separately.

Every macro threshold and every owner comparison is an affine inequality in the current finite owner vector. After splitting the repetition index by the same finite residues, its validity at an intermediate repetition is a Presburger formula in the quotient exponents and the repetition index. Requiring validity for all repetitions is therefore Presburger by effective closure under first-order quantification. Repeating this construction for the finitely many powered blocks and taking the finite product of residue branches proves the claim. No backward requirement appears in the construction. □

For trace arguments, macro types retain the concrete simple skeleton identifier σ. Concatenating selected actual macro occurrences thus gives one actual N micro-run over the visible factor. In particular, an actual synchronized macro cycle consumes a nonempty contiguous visible word. No generic alternative VASS path is used to infer literal structure of the observed word.

111.23 Guarded-macro robustness of positive first-return compactness

The real-time compactness theorem has already been proved in theorem 111.63. We now isolate the exact features of that proof that are used by the epsilon compilation. This turns the epsilon extension into a self-contained internal robustness theorem.

Definition 111.142 (Admissible guarded macro system)**.** Fix the two machines and one concrete exact zero entry. An *admissible guarded macro system* is a finite, effectively computed family of visible macro types (for all visible letters, not for one preselected corridor) such that:

(G1) every finite macro has a source threshold r, an integer endpoint effect δ, and a concrete weak skeleton, and acts as $c \mapsto c + \delta$ whenever $c \ge r$;

(G2) every pump macro has one finite enabling threshold and one concrete lasso certificate and, when enabled, produces unbounded exit credit, represented by ω;
(G3) the induced forward and backward transformers are exact and monotone and satisfy exact residuation; moreover, on a constant-margin rejected phase, every enabled finite backward-minimizing critical macro is simultaneously forward-maximizing and transports the same tight separator equality to the next cut;
(G4) common-credit shifts are monotone: for every visible factor x and $c \geq 0$, $T_x(H+c) \geq T_x(H) + c$ on finite coordinates, with $\bot$ and ω fixed under the shift;
(G5) on each fixed finite support/omega/threshold regime the transformer is nonexpansive in finite coordinates, and a first regime split is localized to a support/omega event or to a bounded neighborhood of one of finitely many thresholds; threshold-contact recurrence uses only the bounded source credit and finite mode data, not the numerical separator margin;
(G6) every tracked integer observable used by the compactness proof has a computable one-macro drift bound; moreover every fixed bounded-expression macro schema $\alpha_0\beta_1^{n_1}\alpha_1\cdots\beta_m^{n_m}\alpha_m$ has effectively Presburger threshold/guard/owner feasibility, and after an effective finite exponent-residue refinement its endpoint data are affine in the quotient exponent variables;
(G7) support, owner/provenance, guard, threshold, and omega data range over finite effectively enumerable mode alphabets, while bounded-height exact-state boxes are well-quasi-ordered after fixing the finite omega mask;
(G8) every macro used in a literal deletion argument retains a concrete skeleton from the observed weak run, so a selected synchronized macro cycle represents a nonempty contiguous visible factor of the observed word;
(G9) the palette is effectively closed under every fixed finite composition used as a proof macro: after finitely refining clause and pump choices, each composed route is again either threshold-affine with one computable source threshold and total integer effect, or a pump route; finite routes retain the concatenated concrete skeleton, while pump routes retain a concrete lasso certificate and finite prefix/suffix skeletons.

All palettes, mode alphabets, drift bounds, and thresholds in (G1)–(G9) are computed from the machines and the concrete exact zero entry and are independent of the future suffix and its backward requirement.

Theorem 111.143 (Weak-palette realization)**.** *For the skeleton-refined weak semantics of the epsilon-OCN, the macro palette computed from the machines satisfies (G1)–(G9) of theorem 111.142. The construction is uniform over all ordinary positive first-return corridors from the fixed exact zero entry and is independent of the future suffix, exit requirement, and numerical separator margin.*

Proof. The skeleton-refined finite weak-envelope normal form theorem 111.129 gives (G1) and (G2), including concrete finite skeletons and pump lasso certificates. The exact forward transformer is a finite maximum of those clauses, and theorem 111.131 gives the exact monotone backward transformer. On a constant-margin rejected phase, theorem 111.136 shows that every forward-enabled finite backward minimizer is also forward-maximizing and transports the same tight gap; this is precisely (G3). The shift property (G4) is theorem 111.133.

For (G5), fixed enabled-clause/support/omega regimes are nonexpansive by theorem 111.139; the first loss of that regime is a support/omega event or a crossing of one of finitely many finite/pump thresholds. Theorem 111.137 shows that recurrent threshold contacts are colored by bounded source credit and finite mode data only—the numerical separator margin is not a color and therefore cannot enter the bound. Pump minimizers are threshold contacts by theorem 111.135, while theorem 111.138 removes enabled pumps from backward-live protected high zones.

The finite macro palette gives a computable drift bound for every fixed integer observable. The powered part of (G6) is exactly theorem 111.141: eventual periodicity of the finite owner-map monoid yields a finite exponent-residue refinement on which endpoint data

are affine and all threshold/owner/positivity constraints are Presburger. For (G7), support masks, threshold regimes, owner/provenance frames, guard data, and omega masks range over finite effectively enumerable sets. After fixing the bottom/omega masks, the remaining finite coordinates lie in a finite power of $\mathbb{N}$ and are well-quasi-ordered by Dickson; bounded context boxes are effectively finite when the relevant integer ranges are bounded.

For (G8), every finite target value is a maximum over a finite family of finite skeleton clauses and is therefore attained. Choosing the canonical maximizing predecessor and one attaining skeleton at each observed macro occurrence makes consecutive owner edges concatenate to an actual weak run; no omega coordinate is used on a backward-live rejected strand by theorem 111.132. Finally, theorem 111.140 proves (G9), with exact shifted thresholds for finite routes and for the first pump on a pump route. Every construction in this paragraph uses only the machines, the fixed exact entry for its provenance naming, and finite syntactic refinements; no backward requirement or future suffix is an input. □

Theorem 111.144 (Guarded-macro transfer theorem)**.** *Fix a concrete exact zero entry for the deterministic machine and an admissible guarded macro system in the sense of theorem 111.142. Then the suffix-parametric real-time positive first-return compactness proof of section 111.14 remains effective after replacing the unit net edges by these guarded macros. More precisely, one can compute from the machines and the exact entry a single integer B_{GM} such that, for every deterministic zero exit configuration C_1 and every exact backward requirement R at that exit, every strictly positive first-return corridor that is R-minimal in the sense of theorem 111.62 has visible length at most B_{GM}. In particular, B_{GM} is independent of R and hence of the future suffix.*

Proof. The real-time proof uses only the interfaces listed in (G1)–(G9); the quantitative replacements are recorded in Section 111.33. Briefly, (G1), (G6), (G7), and (G9) provide the finite owner/provenance data, additive guarded effects, composition closure, Presburger powered-route checks, and bounded drift; (G4) supplies the common-credit shift inequalities; (G3) and (G5) supply exact critical propagation together with bounded threshold/contact localization; and (G8) preserves literal observed-trace deletion. Hence the wall, carrier, translated-arch, activation, contact, bounded-context, and WQO modules of Sections 111.14.2–111.14.16 carry over after finite rescaling.

The terminal modules remain in the same decidability classes: a finite route with threshold r and effect δ is compiled by subtracting r and then adding $r + \delta$, so threshold guards add no zero test to the finite-projective 2-VASS module or to the rank-one one-zero-test module. Pumps create no new recursive branch: a critical pump is a threshold contact, a protected high-zone pump into a backward-live target contradicts rejection, and an ω coordinate in a rejected factor is backward-dead. Therefore the same lexicographic proof measure (ν, ρ, s) is well founded and the effective recursion of theorem 111.108 computes one bound from the machines and exact entry alone. Since the real-time theorem is already suffix-parametric, the backward requirement R is never a recursive bound parameter. This yields the stated uniform B_{GM}. □

Theorem 111.145 (Robust positive first-return compactness)**.** *Fix a concrete stable zero entry*

$$Z = (p, H)$$

of the deterministic stable/reset automaton together with the exact ε-saturated net envelope H. There is a total computable integer

$$B^{\varepsilon}_{\mathrm{FR}}(Z)$$

such that, for every stable zero exit configuration C_1 and every exact backward requirement vector R, every ordinary positive first-return factor P from Z to C_1 that is visible-length-minimal among all words from the same exact entry to C_1 whose exact exit envelope is rejected against R has visible length at most $B^{\varepsilon}_{\mathrm{FR}}(Z)$. In particular, the same bound holds for every ordinary positive first return occurring inside a globally shortest rejected continuation from Z.

Proof. By theorem 111.143, the skeleton-refined weak palette is an admissible guarded macro system satisfying (G1)–(G9), uniformly in the future requirement. Apply theorem 111.144 at the exact entry Z. Its bound B_{GM} is already uniform over all stable zero exits and all exact backward requirements, so we may take $B^{\varepsilon}_{\mathrm{FR}}(Z) = B_{\mathrm{GM}}$. □

111.24 Reset-ending first returns and effective zero successors

We now return to the deterministic stable/reset structure. The local compactness theorem theorem 111.145 applies directly to ordinary positive first returns. The only new deterministic phenomenon is a visible step that enters a reset control and thereby returns silently to stable counter zero. We show that this case can be reduced to the same compactness theorem by a genuine auxiliary construction that stays inside the epsilon-OCA/epsilon-OCN model.

111.24.1 Consecutive stable zero cuts

Lemma 111.146 (Stable-zero trichotomy)**.** *Let $u \neq \varepsilon$ be the visible factor between two consecutive stable zero cuts of the deterministic stable/reset automaton. Exactly one of the following occurs.*

1. Direct zero step: $|u| = 1$*, and after the unique visible step and its forced reset closure, if any, the stable counter is zero.*
2. Ordinary positive first return: *after the first visible letter the stable counter is positive, all strict internal stable cuts are positive, and the final visible step reaches stable zero without entering a reset control.*
3. Reset-ending positive first return: *after the first visible letter the stable counter is positive, all strict internal stable cuts are positive, and the final visible letter enters a reset control whose forced silent behavior ends at stable zero.*

Proof. If a reset occurs at a strict internal stable boundary, then its forced silent closure produces an earlier stable zero cut, contrary to consecutive- ness. Hence all strict internal stable cuts are positive and reset-free. The final visible step either reaches stable zero directly or enters a reset control. The remaining case is the one-letter direct zero step. □

111.24.2 Reset signatures

Fix a reset-ending factor of the form ua. Immediately before the final letter a, suppose the deterministic machine is in stable control p at height $h > 0$. Let the a-rule have deterministic counter effect j, enter reset control s, and let

$$m = \mathrm{per}_s, \qquad r \equiv h + j \pmod{m}, \qquad q = \mathrm{goto}_s(r).$$

The finite tuple

$$\sigma = (p, a, j, s, m, r, q)$$

is called the *reset signature*. There are only finitely many such signatures. While reading the reset-free prefix u, the residue $h \bmod m$ can be carried in finite control.

111.24.3 A genuine reset-drain auxiliary instance

Fix a stable zero entry $Z = (p_0, H)$ and a reset signature σ as above. Construct a deterministic auxiliary automaton $A^{\mathrm{reset}}_{Z,\sigma}$ as follows.

From the fresh stable-zero entry representing Z, the prefix phase first copies the deterministic zero-regime visible step of A whenever its forced closure reaches a stable positive configuration without entering a reset. Thereafter it copies only the positive-regime, reset-free stable dynamics of A, while carrying the current counter residue modulo m in finite

control. Every visible transition entering any reset control before the designated final marker is omitted, and no zero-regime transition is provided from a positive-copy control. Consequently, if the copied run reaches counter zero before the marker, it is stuck and cannot later reach the private drain exit. From the copy of the designated pre-reset control p, and only in the positive regime and in residue classes satisfying

$$h + j \equiv r \pmod{m},$$

a fresh visible marker $\#$ with counter effect 0 enters a private drain control. The marker deliberately does not reproduce the original update j: the reset discards the post-a value $h + j$, while the only information about that value needed to select the correct reset target is its residue, which is already enforced by the guard above. Keeping effect 0 preserves the pre-reset height h so that the drain exposes it as an ordinary visible first return. At the drain control the fresh letter d is available only in the positive regime, decrements the counter by one, and remains in the drain control. Thus every run reaching the drain-zero exit is automatically a strictly positive first return, and its first return to counter zero occurs after a word

$$u \# d^h,$$

where the corresponding original factor ua has reset signature σ. The zero drain control is declared stable and is regarded as the fixed auxiliary exit representing the stable reset target q.

The net side realizes the final visible a semantically. Construct $N_{a\mapsto\#}$ by retaining all original epsilon transitions and, for every original visible transition

$$x \xrightarrow{a,w} y,$$

adding the copied transition

$$x \xrightarrow{\#,w} y.$$

All original visible transitions on the whole alphabet Σ are retained, independently of the eventual witness u, and d is a zero-effect self-loop at every N-control.

Lemma 111.147 (Copied-marker/drain identity)**.** *For every epsilon-saturated envelope K,*

$$T^{\varepsilon}_{\#}(K) = T^{\varepsilon}_{a}(K), \qquad T^{\varepsilon}_{d}(K) = K.$$

Consequently, whenever $u\#d^h$ is an auxiliary reset-drain first return,

$$T^{\varepsilon}_{u\#d^h}(H) = T^{\varepsilon}_{ua}(H).$$

Proof. A weak $\#$ path in the auxiliary net is exactly a path $\varepsilon^* \# \varepsilon^*$ whose unique visible edge is a copied a-edge with the same source, target, and counter effect as in N; replacing it by the original a-edge gives a bijection preserving requirements and exit credit. The fresh letter d is a zero-effect self-loop at every N-control, so on an already epsilon-saturated envelope its weak transformer is the identity. □

Lemma 111.148 (Reset-auxiliary uniformity certificate)**.** *For fixed machines A, N, a concrete stable zero entry $Z = (p_0, H)$, and a reset signature σ, the auxiliary pair*

$$(A^{\mathrm{reset}}_{Z,\sigma}, N_{a\mapsto\#})$$

and its exact entry envelope are effectively determined before any reset-ending witness, pre-reset height, or future backward requirement is chosen. In particular, the height h is not compiled into the machines: it is the runtime counter value, while its residue modulo m is tracked in finite control. Thus the guarded-macro palette and the compactness bound for the auxiliary pair are uniform over all factors of signature σ and all inherited suffix requirements.

Proof. The deterministic auxiliary control graph is obtained from the finite stable/reset presentation of A by product with the residue set $\mathbb{Z}/m\mathbb{Z}$, deletion of reset-entering prefix edges, and addition of the fixed marker/drain gadget. The net auxiliary is obtained from N by copying the finitely many a-edges to $\#$ and adding the fixed d-loops. All of these operations depend only on A, N, Z, σ. The exact entry is Z with envelope H. Neither the literal prefix u, nor its runtime height h, nor the later backward vector R enters the construction. The suffix-parametric guarded-macro theorem therefore computes one bound for this fixed auxiliary instance, uniformly in R. □

For an auxiliary reset-drain first return put

$$\mathrm{rcost}(u\#d^h) = |u| + 1.$$

Thus d-letters have zero reset cost, and the reset cost is exactly the visible length of the corresponding original factor ua.

Lemma 111.149 (Residual-minimal reset comparison)**.** *Let a reset-ending factor ua of signature σ occur between consecutive stable zero cuts of a globally shortest word in $L(A) \setminus L(N)$, and let R be the exact weak backward requirement of the unchanged suffix after the reset. If the pre-reset height is h, then $P = u\#d^h$ is reset-cost-minimal among all auxiliary positive first returns Q from the same exact entry to the drain-zero exit satisfying*

$$\mathrm{Rej}(T^{\varepsilon}_{Q}(H), R).$$

Proof. By theorem 111.147, the exit envelope of P is exactly the original envelope after ua, so it rejects the requirement R. Every auxiliary positive first return to the drain-zero exit has the form $Q = u'\#d^{h'}$. The residue guard on $\#$ guarantees that the original word $u'a$ has the same reset signature σ and therefore reaches the same stable zero control q after the forced reset. Again by theorem 111.147, its net envelope is exactly the exit envelope of Q.

If such a Q rejected R and had $\mathrm{rcost}(Q) < \mathrm{rcost}(P)$, replacing ua by $u'a$ and keeping the original suffix would give a strictly shorter word accepted by A and rejected by N, contradicting global shortestness. Hence P is reset-cost-minimal. □

Theorem 111.150 (Effective reset-ending first-return bound)**.** *For every concrete stable zero entry Z and reset signature σ there is a total computable bound $B^{\varepsilon}_{\mathrm{reset}}(Z, \sigma)$ such that every reset-ending factor of signature σ occurring between consecutive stable zero cuts of a globally shortest counterexample has visible length at most this bound.*

Proof. Fix such a factor and its suffix requirement R. Among all auxiliary positive first returns from the concrete auxiliary entry to the drain-zero exit that reject R, choose one, say Q_*, of minimum ordinary visible length. It exists because $P = u\#d^h$ is one such return. The suffix-parametric form of robust compactness, theorem 111.145, applied to the fixed auxiliary pair $(A^{\mathrm{reset}}_{Z,\sigma}, N_{a\mapsto\#})$, gives a computable number $\widehat{B}^{\varepsilon}_{\mathrm{FR}}(Z, \sigma)$ such that

$$|Q_*| \le \widehat{B}^{\varepsilon}_{\mathrm{FR}}(Z, \sigma).$$

By theorem 111.148, this number is computed uniformly from A, N, Z, σ and does not depend on the chosen factor, its pre-reset height, or R.

By theorem 111.149, P is reset-cost-minimal. Therefore

$$|ua| = \mathrm{rcost}(P) \le \mathrm{rcost}(Q_*) \le |Q_*| \le \widehat{B}^{\varepsilon}_{\mathrm{FR}}(Z, \sigma).$$

Taking the last quantity as $B^{\varepsilon}_{\mathrm{reset}}(Z, \sigma)$ proves the claim. Notice that this argument does not assert that the particular drain word $u\#d^h$ is ordinary-length-minimal; this is exactly why the separate reset cost is introduced. □

111.24.4 Effective next-zero successors

For a concrete stable zero node $Z = (p, H)$, let $B^{\varepsilon}_{\mathrm{ord}}(Z)$ be the suffix-independent ordinary first-return bound from theorem 111.145. For each syntactically feasible reset signature σ of A, let $B^{\varepsilon}_{\mathrm{reset}}(Z, \sigma)$ be the bound from theorem 111.150; if there is no such signature, the corresponding maximum below is defined as 0. Define

$$B^{\varepsilon}_{0}(Z) = \max\Big(1, B^{\varepsilon}_{\mathrm{ord}}(Z), \max_{\sigma} B^{\varepsilon}_{\mathrm{reset}}(Z, \sigma)\Big),$$

up to the fixed boundary-letter convention already absorbed into the auxiliary constructions.

Theorem 111.151 (Effective next-zero successors)**.** *From every concrete stable zero node $Z = (p, H)$ one can effectively compute a finite set*

$$\mathrm{Succ}^{\varepsilon}_{0}(Z)$$

of exact next-stable-zero successors, complete for every shortest word in $L(A) \setminus L(N)$ passing through Z.

Proof. Enumerate every visible word u with $1 \le |u| \le B^{\varepsilon}_{0}(Z)$. Simulate the deterministic stable/reset machine exactly, retaining u iff it is a literal factor from the current stable zero configuration to the next stable zero configuration and contains no earlier stable zero cut. By theorems 111.146 and 111.150, every consecutive-zero factor of a shortest counterexample is covered by the direct, ordinary, or reset-ending bound and therefore appears in this finite enumeration.

For each retained word compute the exact target envelope

$$H' = T^{\varepsilon}_{u}(H)$$

using theorems 111.129 and 111.131, and record the target stable control, exact envelope, literal word, and visible cost. All steps are effective and the enumeration is finite. □

111.25 The finite stable-zero tree

We next organize the finitely branching zero-successor relation by an extended Dickson order.

Definition 111.152 (Stable zero state)**.** A stable zero state is a pair

$$Z = (p, H),$$

where p is a stable deterministic zero control and $H \in D^{Q_N}$. For equal deterministic control define

$$(p, H) \preceq_0 (p, K) \quad \Longleftrightarrow \quad H \le K$$

componentwise. States with different deterministic controls are incomparable.

Lemma 111.153 (Extended Dickson property)**.** *For every finite set of N-controls, D^{Q_N} is a well quasi-order under componentwise order.*

Proof. Take an infinite sequence of vectors in D^{Q_N}. There are only finitely many masks describing the coordinates equal to ω, so pass to an infinite subsequence with one fixed omega mask. On the remaining coordinates, $D \setminus \{\omega\} = \{\bot\} \cup \mathbb{N}$ is order-isomorphic to $\mathbb{N}$ via $\bot \mapsto 0$ and $n \mapsto n + 1$. Dickson's lemma gives two comparable vectors on those coordinates, while the omega coordinates agree. □

The key deletion statement uses only monotonicity of weak N semantics, not arithmetic involving ω and ∞.

Lemma 111.154 (Stable zero-cut domination)**.** *Let $i < j$ be two stable zero cuts on a rejected deterministic accepting run. Suppose the deterministic stable zero control is the same at both cuts and that the exact envelopes satisfy*

$$H_i \leq H_j.$$

Then deleting the visible factor between the two cuts preserves deterministic acceptance and N-rejection.

Proof. The deterministic run rejoins at the identical physical configuration $(p, 0)$, so its entire later behavior, including every forced reset closure, is unchanged. Let v be the inherited visible suffix after cut j. If v were accepted from H_i, monotonicity and $H_i \leq H_j$ would imply acceptance from H_j, contradicting rejection of the original word. Thus the shortened word remains rejected. □

Starting from the initial stable zero context, generate the raw successor tree using theorem 111.151. Prune a candidate node (p, H') if an ancestor on the same branch has the same deterministic control p and an envelope $H \leq H'$.

Theorem 111.155 (Finite stable-zero tree)**.** *The pruned zero-successor tree $\mathcal{T}_0^{\varepsilon}$ is finite and effectively constructible. Moreover, the complete stable-zero sequence of every shortest counterexample embeds literally as one root-to-node path in $\mathcal{T}_0^{\varepsilon}$.*

Proof. Every node has finitely many children by theorem 111.151. An infinite unpruned branch would be a bad sequence for the wqo of theorem 111.153, which is impossible. Hence the finitely branching tree has no infinite branch and is finite by König's lemma. Exhaustive generation therefore terminates.

For completeness, consider the literal stable-zero sequence of a shortest counterexample. Each consecutive factor appears among the raw successors by theorem 111.151. If a later literal node were pruned by an earlier dominated node on that same path, theorem 111.154 would give a strictly shorter counterexample, contradicting shortestness. □

Let

$$D_0^{\varepsilon}$$

be the maximum root-to-node edge depth of $\mathcal{T}_0^{\varepsilon}$, and let

$$L_0^{\varepsilon}$$

be the maximum sum of visible edge costs along a root-to-node path. Both are total computable integers. Every shortest counterexample has at most $D_0^{\varepsilon} + 1$ stable zero cuts, and its visible prefix through the last stable zero cut has length at most L_0^{ε}.

111.26 Terminal tails and the global hard-inclusion bound

It remains to control the suffix after the last stable zero cut. Such a suffix cannot contain a reset: every reset ends at stable counter zero and would create a later stable zero cut.

111.26.1 A syntactic terminal auxiliary automaton

Fix a stable zero node $Z = (p, H)$ from $\mathcal{T}_0^{\varepsilon}$. Construct a deterministic auxiliary automaton A_Z^{term} syntactically. From a fresh zero start representing p, copy exactly those zero-regime visible transitions whose target is a stable positive configuration. Omit transitions entering reset controls or returning immediately to stable zero. For every stable control q, create a positive copy q^+ and copy every positive-regime visible transition whose target is stable; transitions entering reset controls are omitted. If a copied decrement reaches zero, the run is stuck because no zero-regime original transitions are available inside the positive copy.

From every copied accepting stable control f^+ add a fresh marker $\#$ *only in the positive regime*, with counter effect 0, into a fresh stable drain control. At the drain control, the fresh letter d is available only in the positive regime and decrements the counter by one; the fresh letter \$ is available only in the zero regime and enters the unique accepting control. No zero-regime $\#$ rule is present at any positive-copy state. Hence a copied decrement that reaches zero before the marker is truly stuck, while an accepted run reaches zero for the first time during the private drain. Thus the accepted words are exactly

$$u \# d^h \$,$$

where u is a reset-free positive terminal tail ending at an original accepting stable control at height $h > 0$.

On the N side retain all original transitions and add zero-effect self-loops on $\#, d, \$$ at every control. Both auxiliary machines are constructed uniformly from (A, N, Z) alone; neither the particular terminal tail, its ending height, nor any future backward requirement is used in their control graphs.

Lemma 111.156 (Fresh identity letters)**.** *If K is epsilon-saturated and $x \in \{\#, d, \$\}$ is one of the fresh identity letters, then*

$$T_x^\varepsilon(K) = K.$$

Consequently,

$$T_{u\#d^h\$}^\varepsilon(H) = T_u^\varepsilon(H).$$

Proof. A weak x path has the form $\varepsilon^* x \varepsilon^*$ and the visible x-edge is a zero-effect self-loop at every control. Hence its transformer is $C_\varepsilon \circ \mathrm{Id} \circ C_\varepsilon$, which equals the identity on an already epsilon-saturated envelope. Iterating over the fresh suffix gives the second equality. □

Theorem 111.157 (Effective terminal-tail bound)**.** *For every concrete zero node Z, one can compute an integer $B_{\mathrm{tail}}^\varepsilon(Z)$ such that, if there exists any accepting terminal tail from Z rejected by N, then there exists one of visible length at most $B_{\mathrm{tail}}^\varepsilon(Z)$.*

Proof. Choose a shortest counterexample of the auxiliary terminal instance. It has shape $u\#d^h\$$. The prefix $u\#d^h$ starts at stable zero, is positive after its first visible letter, contains no internal reset or stable zero cut, and reaches zero for the first time on the final d. It is therefore an ordinary positive first-return corridor. By theorem 111.145, its visible length is bounded by a total computable integer depending only on the concrete auxiliary entry. By theorem 111.156, rejection of the auxiliary word is equivalent to rejection of the original tail u from the same exact envelope H. Taking the first-return bound as $B_{\mathrm{tail}}^\varepsilon(Z)$ proves the claim. □

Define

$$B_{\mathrm{tail,max}}^\varepsilon = \max_{Z \in \mathcal{T}_0^\varepsilon} B_{\mathrm{tail}}^\varepsilon(Z)$$

and

$$B_{\mathrm{global}}^\varepsilon = L_0^\varepsilon + B_{\mathrm{tail,max}}^\varepsilon.$$

Both are computable because the zero tree is finite.

Theorem 111.158 (Global small counterexample)**.** *If*

$$L(A) \not\subseteq L(N),$$

then there exists

$$w \in L(A) \setminus L(N)$$

with

$$|w| \le B_{\mathrm{global}}^\varepsilon.$$

Proof. Take a shortest counterexample w. Let x be the prefix ending at its last stable zero cut, with zero node $Z = (p, H)$, and write $w = xu$. By theorem 111.155, $|x| \le L_0^\varepsilon$.

If $u = \varepsilon$, the claim follows immediately. Otherwise u is a reset-free positive terminal tail from Z. By theorem 111.157, there exists an accepting tail v from the same deterministic physical configuration and the same exact N envelope H, rejected by N, with $|v| \le B^\varepsilon_{\mathrm{tail}}(Z)$. Thus xv is again in $L(A) \setminus L(N)$ and

$$|xv| \le L_0^\varepsilon + B^\varepsilon_{\mathrm{tail,max}} = B^\varepsilon_{\mathrm{global}}.$$

□

Proof of theorem 111.53. Compute $B^\varepsilon_{\mathrm{global}}$ and enumerate all visible words of lengths

$$0, 1, \ldots, B^\varepsilon_{\mathrm{global}}.$$

The length-0 case is included explicitly. For each word, simulate the unique deterministic stable/reset run of A and compute the exact weak N envelope from theorem 111.129; final N-acceptance is tested after trailing epsilon saturation. By theorem 111.158, the inclusion fails if and only if one of these finitely many words belongs to $L(A) \setminus L(N)$. □

111.27 Reverse inclusion

We now decide $L(N) \subseteq L(A)$. This direction does not use the small- witness theorem.

111.27.1 A marked noninclusion reduction

Introduce a fresh visible endmarker $\dashv \notin \Sigma$. First totalize the visible stable semantics of A by sending every missing stable visible rule to one fresh stable dead control, with dead self-loops on all original visible letters in both sign regimes.

Construct a marked deterministic rejection automaton A_{bad} by adding $\dashv$-rules from stable controls: an original accepting stable control reads $\dashv$ into a nonaccepting sink GOOD, whereas every nonaccepting stable control, including the dead control, reads $\dashv$ into an accepting sink BAD. The marked sinks have no outgoing transitions.

Construct N_{end} by retaining all original transitions and, from every original accepting N-control q, adding

$$q \xrightarrow{\dashv,0} f_{\mathrm{end}},$$

where f_{end} is the unique marked accepting control.

Lemma 111.159 (Marked noninclusion)**.**

$$L(N) \not\subseteq L(A)$$

if and only if there exists $w \in \Sigma^*$ such that both marked systems have accepting runs on $w \dashv$.

Proof. The totalized deterministic stable/reset machine has one unique stable endpoint after every visible word. Hence A_{bad} accepts $w \dashv$ exactly when the original A rejects w. On the net side, arbitrary trailing epsilon moves may occur before the endmarker, so N_{end} accepts $w \dashv$ exactly when N accepts w. The case $w = \varepsilon$ is represented by the one-letter marked word $\dashv$. □

111.27.2 Reset decompression

For a reset control s of period p, replace the semantic reset

$$(s, n) \xRightarrow{\varepsilon^*} (\mathrm{goto}_s(n \bmod p), 0)$$

by finite micro-controls

$$s[0], \ldots, s[p-1].$$

At positive counter add

$$s[i] \xrightarrow{\varepsilon,-1} s[(i+1) \bmod p],$$

and at zero add

$$s[i] \xrightarrow{\varepsilon,0} \mathrm{goto}_s(i).$$

Starting from height n, exactly n silent decrements occur and the zero exit is taken in phase $n \bmod p$.

Lemma 111.160 (Commutation with N-epsilon moves)**.** *Inside the asynchronous product, N-epsilon moves commute with the decompressed A-reset microsteps without changing component endpoints or visible projection.*

Proof. An N-epsilon move changes only the N-control and N-counter, while an A-reset microstep changes only the A-control and A-counter. Neither consumes visible input. Their feasibility conditions therefore concern disjoint counters, so adjacent independent moves may be swapped. Repeating the swap moves all N-epsilon steps to either side of the finite A-reset microsequence while preserving both component endpoints. □

111.27.3 The asynchronous marked product

A genuine product control stores one control of N_{end} and one stable or reset-micro control of A_{bad}. The two main counters are denoted by x_N and x_A.

N-epsilon moves change only (q_N, x_N). Decompressed A-epsilon moves change only (q_A, x_A). An original visible letter synchronizes one chosen N-edge with the unique deterministic A-edge in the current A sign regime. The endmarker $\dashv$ is synchronized in the same way. By theorem 111.160, allowing arbitrary interleavings of the two silent components introduces no new visible behavior.

Lemma 111.161 (Product/run correspondence)**.** *A marked accepting product control is reachable if and only if there exists $w \in L(N) \setminus L(A)$.*

Proof. Projecting a product run gives weak component runs with the same visible word $w \dashv$. Conversely, interleave the silent moves of any two such component runs and synchronize their visible letters; theorem 111.160 handles reset microsteps. Apply theorem 111.159. □

111.27.4 Compilation to one zero-test VASS

The low-level construction is given in Section 111.34; we record here only the correctness interfaces used by the theorem. A positive-regime synchronized edge with effects (d_N, d_A) is compiled through a private intermediate control as

$$(d_N, -1) \quad \text{then} \quad (0, d_A + 1),$$

so ordinary nonnegativity enforces $x_A > 0$ and the private control makes the gadget atomic. Positive reset microsteps already have effect $(0, -1)$.

All zero-regime semantic edge types are serialized through one common gate. Each type e has an ordinary tag counter t_e: its source transition records e, one distinguished transition g_0 tests only $x_A = 0$, and the tagged post-gate branch consumes t_e, performs the fixed ordinary vector update of e, and returns to a genuine product control.

Lemma 111.162 (Single-gate exactness)**.** *A complete gate path for a semantic zero edge e reaches its genuine target if and only if the edge is taken at its proper source with $x_A = 0$ and its ordinary counter update is feasible. At every genuine product control all tag counters are zero. Thus g_0 is the unique zero-test transition.*

Proof. The unique tag identifies the chosen edge; all other completion branches are disabled by ordinary nonnegativity. If the vector update is infeasible the run is trapped inside the private gadget. A successful completion consumes the tag and restores the all-zero tag invariant. See Section 111.34 for the explicit transition schema. □

From the marked accepting product control, a private cleanup phase supplies ordinary decrement loops for every counter and one final zero-effect edge to a fixed target.

Lemma 111.163 (Cleanup equivalence)**.** *Some marked accepting product configuration with arbitrary nonnegative counters is reachable if and only if the fixed all-zero cleanup target is exactly reachable.*

Proof. Drain all counters before taking the final edge. Conversely cleanup is reachable only from marked acceptance, and taking the final edge early cannot reach the queried all-zero configuration. □

Exact reachability for VASS/Petri nets with one zero test (equivalently, one inhibitor arc in the standard formulation) is decidable [115, 116]. Our compiled system matches that model exactly: g_0 is the single zero-test transition, all remaining counters are ordinary VASS counters, and every other transition is an ordinary vector addition.

Proof of theorem 111.54. By theorems 111.161 to 111.163, $L(N) \not\subseteq L(A)$ if and only if the fixed all-zero cleanup target is reachable in the compiled VASS with one zero test. The latter reachability problem is decidable, so $L(N) \subseteq L(A)$ is decidable. □

Proof of theorem 111.52. By theorem 111.53, the inclusion $L(A) \subseteq L(N)$ is decidable. By theorem 111.54, the reverse inclusion $L(N) \subseteq L(A)$ is decidable. Therefore one decides $L(A) = L(N)$ by deciding both inclusions and accepting exactly when both hold. □

111.28 Conclusion

Language equality between deterministic zero-testing one-counter automata and nondeterministic one-counter nets is decidable in the standard real-time setting and remains decidable under the epsilon-transition conventions of theorem 111.52. The common mechanism is a shortest-counterexample compactness theory organized around exact residual semantics and deterministic zero returns. Silent net behavior is absorbed by a finite threshold-affine weak-macro calculus, while deterministic silent behavior is handled by stable/reset normalization.

The resulting bounds are effective but intentionally complexity-agnostic. They combine well-founded rank recursions, finite Ramsey and Dickson arguments, Presburger checks, and one-zero-test reachability subroutines.

111.29 Real-time effectivity ledger

The real-time global small-witness theorem needs a *computable* witness bound. The following list records the external or finite-search ingredients used to compute that real-time bound. The epsilon extension does *not* use the finite-path statement in item 1; its corresponding effectivity is supplied instead by the exact weak-envelope construction and guarded-macro ledgers in Appendices 111.32 and 111.33.

1. Exact net transformers T_x and S_x for a fixed word are finite dynamic programs over the finitely many paths of length $|x|$.

2. Support, owner frames, primitive synchronized simple cycles, exact wall signatures, and bounded-context carrier colors are finite and effectively enumerable.

3. Rank-one recurrent owner switching is compiled to a VASS with one zero-tested scalar (Proposition 111.78); exact reachability for that class is decidable [115, 116].

4. Finite-projective systems reduce to exact 2-VASS reachability and shortest representative search (Proposition 111.74).

5. The dense-wall constants are explicit: $K_{\mathrm{dense}} = 4|\mathcal{W}| + 1$, and a requested number of dense atoms is forced by a computable linear wall-count threshold.

6. Every finite Ramsey use is on an explicitly finite coloring (for example side $\times$ envelope coordinate in Lemma 111.84); only finite Ramsey numbers are required.

7. Every tracked integer observable has a computable one-macro drift bound; first exit from a cutoff occurs in a finite band.

8. The effectivity proof avoids an uncontrolled infinite Dickson subsequence by the reserve split (111.31); below the explicit projective cutoff the normalized profile space is finite.

9. The translated-arch activation bridge is quantitative: bounded-cell refinement fixes the finite tuple $(c, \delta, d, p, \eta, u, v)$; the pure-scalar survivor fixes (d, λ, η, K); the bounded-contact state, requirement, and counter lie in finite computable sets. Hence the contact/relative-valley transition introduces no infinitary choice.

10. The proof-state recursion uses the explicit finite stage set in Table 111.1 and the lexicographic measure (ν, ρ, s).

11. Next-zero successors are obtained by literal word enumeration up to the computed first-return bound; exact envelopes are computed by dynamic programming.

12. The pruned zero-state tree is effectively generated. It terminates because it is finitely branching and every branch is a Dickson bad sequence.

13. Terminal tails reduce to another concrete first-return instance by the fresh-marker drain construction, after which a finite maximum is taken over the finite zero tree.

111.30 Real-time canonical dependency graph and excluded proof routes

The real-time compactness proof uses only the corrected canonical dependency chain below. The epsilon theorem invokes this core only through the guarded-macro transfer theorem of Section 111.23. Exploratory routes that were later superseded are not premises of any

theorem in this chapter.

exact residuals and extensional splice
$\Downarrow$
shell deletion + finite-projective terminal module + rank-one base
$\Downarrow$
wall-free descent + bounded-wall compactness
$\Downarrow$
dense same-signature atoms + translated arches
$\Downarrow$
monotone endpoint debt + vector shell defect
$\Downarrow$
activation cells + pure-scalar recurrence + bounded high contact
$\Downarrow$
actual contact-centered relative valley + literal cycle packages
$\Downarrow$
critical deletion + activation/contact noncircularity
$\Downarrow$
WCI rank lift + effective B_{FR}
$\Downarrow$
finite next-zero successors + finite zero tree
$\Downarrow$
terminal drain + global small witness.

For referee clarity, several historically tempting routes are *not* dependencies of the theorem.

Excluded shortcut	**Why it is insufficient**	**Canonical replacement**
Uniform separator "gap spending"	the rejection margin is monotone but need not decrease as a consumable budget	endpoint domination, anchored shell deletion, and rank descent
Global compilation of recurrent wall equality into one lower-rank machine	unboundedly many exact wall equalities may require independent equality checks	fixed-K compilation plus dense same-signature returns and translated arches
Treating 2-VASS flattening as trace preserving	reachability endpoints may be represented by a different input word	actual-trace cycle erasure for literal deletions; VASS alternatives only for endpoint replacements
Strong arbitrary lower-rank exact-slice/Pareto oracle	stronger than the shortest-witness induction can justify at every rank	bounded-context domination WCI(ρ) plus responsible-escape alternative
Infinitary Dickson extraction for normalized endpoint profiles	abstract WQO alone does not yield the desired quantitative threshold without growth control	explicit scalar/projective reserve split and finite profile boxes
Nested-zero-test shortcut at wall boundaries	simultaneous-zero compatibility of high-rank wall tests with lower-rank tests is not established	extensional fixed-K wall handling and literal high-shell deletion

111.31 Real-time trace-preservation checklist

Every use of cycle or VASS machinery in the real-time compactness proof falls into one of the following two audited categories. The epsilon transfer preserves the same distinction by retaining concrete weak skeletons, as recorded in Section 111.33.

1. **Literal operation on the observed witness.** The factor to be removed is an actual contiguous cycle occurrence (or a finite sparse set of such occurrences) found by cycle erasure on the observed synchronized trace. Deterministic safety is verified on that literal word. No alternative path scheme is substituted silently.

2. **Constructive endpoint replacement.** A reachability theorem may return a different word. The proof records its exact deterministic endpoint and exact N exit envelope and invokes Lemma 111.66. Proposition 111.74 and the bounded-projective terminal branches are of this form.

The support-two southwest reduction and the A-neutral package in Section 111.14.14 belong to category 1. Finite-projective compression belongs to category 2. No theorem crosses from category 2 to a literal deletion claim without an independent trace-faithful extraction.

111.32 Effective weak-envelope construction

This ledger section supplies the algorithmic details behind theorems 111.129 and 111.131. The purpose is not to eliminate silent moves syntactically, but to compile their exact effect on upper envelopes and backward requirements into a finite collection of clauses.

111.32.1 Effective finite and pump enumeration

Fix p, q and a visible letter a. Enumerate every simple epsilon-prefix, one a-edge, and every simple epsilon-suffix. Each resulting skeleton π contributes the exact finite clause

$$c \mapsto c + \mathrm{eff}(\pi) \qquad (c \geq \mathrm{req}(\pi)).$$

For pumping, enumerate a simple positive epsilon cycle together with simple connectors from the source, through the visible edge, and to the target, with the cycle on either silent side. For a lasso $\rho C \tau$, the pump threshold is $\mathrm{req}(\rho C)$. The proofs of theorems 111.127 to 111.129 show that these finite families are complete: every finite maximizing run is dominated by a simple skeleton, while every pumpable run contains one of the enumerated positive-cycle witnesses. Hence both the finite clause palette and the exact onset of the pumping branch are effectively computable.

111.32.2 Exact propagation from an omega source

If $H(p) = \omega$, the set of reachable finite counters at p is unbounded. For any finite clause with threshold r, some reachable finite source counter is at least r, and applying the fixed clause to arbitrarily large reachable source counters produces arbitrarily large target counters. Thus the target value is ω. No congruence or eventual-density assumption is used.

111.32.3 Backward transformer

Let R be a target requirement vector. For a finite simple skeleton $\pi : p \Rightarrow_a q$ with $R(q) < \infty$, the least source credit that both executes π and leaves enough target credit is

$$b(\pi, q, R) = \max\{\mathrm{req}(\pi),\, R(q) - \mathrm{eff}(\pi),\, 0\}.$$

A pump witness leading to a target with finite $R(q)$ contributes its pump entry threshold itself: after the positive cycle is executable, enough repetitions can manufacture any prescribed finite target credit. Targets with $R(q) = \infty$ contribute no candidate. Therefore

$$S_a^{\varepsilon}(R)(p)$$

is the minimum over a finite effectively computable candidate set, or ∞ when that set is empty. Upward closure of OCN feasibility then gives

$$\mathrm{Acc}(T_a^{\varepsilon}(H), R) \iff \mathrm{Acc}(H, S_a^{\varepsilon}(R)),$$

including the ω case. This also proves that the initial saturation H_ε and terminal requirement R_ε are effectively computable by the same finite epsilon-path/cycle analysis with the visible edge omitted.

111.33 Macro-transfer and constant-rescaling ledger

We collect here the quantitative interface between the guarded-macro transfer theorem theorem 111.144 and the weak-macro system of section 111.22. The real-time argument is contained earlier in this chapter: owner/provenance and finite-projective setup in Sections 111.14.2–111.14.4, wall and translated-arch localization in Sections 111.14.5–111.14.9, threshold/contact and literal-trace analysis in Sections 111.14.10–111.14.15, the structural rank lift in theorem 111.107, and the quantitative recursion in theorem 111.108. The point of this ledger section is to isolate every place where those arguments used unit real-time semantics and to give its weak-macro replacement.

Let $\mathcal{M}$ be the finite palette of finite weak macro types after refining by source, target, visible letter, threshold, effect, and concrete simple skeleton identity. Let $\mathcal{P}$ be the finite palette of pump witness types. Put

$$R_* = \max\big(\{r_e : e \in \mathcal{M}\} \cup \{r_p : p \in \mathcal{P}\} \cup \{0\}\big)$$

and

$$E_* = \max\big(\{|\delta_e| : e \in \mathcal{M}\} \cup \{0\}\big).$$

Both are effectively computable.

111.33.1 Per-step observable bounds

On a fixed finite regime:

- one N/N projective difference changes by at most $2E_*$;
- one anchored A/N relative coordinate changes by at most $E_* + 1$, since the deterministic stable step has unit effect;
- a common anchored N-credit observable changes by at most E_*;
- support, provenance, owner, and omega masks range over finite sets.

Thus one conservative universal first-exit constant is

$$D_* = \max\{1, 2E_* + 2\}.$$

Whenever a tracked integer observable first exits $[-M, M]$, it lies in the finite band

$$M < |z| \le M + D_*.$$

This is the only change required in first-exit localization.

111.33.2 Threshold events

Every finite macro enabling change occurs at one of the finitely many thresholds $r_e \le R_*$. Every pump enabling change occurs at one of the finitely many pump thresholds $r_p \le R_*$. Hence a first semantic-regime split is localized either by a finite support/omega event or by a source counter in a bounded band around one of these thresholds. For a recurrent threshold contact the color records this bounded source credit and finite mode data, but not the numerical separator margin; by theorem 111.137 the resulting contact alphabet is finite and independent of the future backward requirement.

111.33.3 Projective cutoffs

Any real-time estimate that bounded the accumulated effect along a simple synchronized path by the number of vertices is replaced by the same path length multiplied by the appropriate macro differential bound. In particular, if a simple residual path contains at most $|V| - 1$ macro edges, its N/N differential contribution has absolute value at most

$$(|V| - 1)\, 2E_*.$$

Thus every explicit projective-spread cutoff remains computable after one finite rescaling.

111.33.4 Transfer certificate

The structural use of these constants is summarized in the proof of theorem 111.144. No additional semantic assumption is hidden in this ledger section: all owner/provenance, wall/contact, WQO, terminal-oracle, and powered-route requirements are exactly (G1)–(G9). The purpose of the present ledger is only to make the finite rescalings R_*, E_*, and D_* explicit.

111.33.5 Literal-trace interface

The transferred proof may delete only cycles/packages extracted from the actual observed trace. Accordingly, each macro edge used by an owner/provenance strand stores one concrete simple weak skeleton. Consecutive selected macro edges concatenate to an actual N micro-run. A nonempty synchronized macro cycle therefore consumes a nonempty contiguous visible factor. Alternative VASS paths may be used for decidability or fixed-route checks, but never as a substitute for the literal observed trace in a shortest-word deletion.

111.34 Formal one-zero-test product gadget

This ledger section records the low-level compilation used in section 111.27. It is included to make the exact match with the external one-zero-test reachability theorem transparent.

111.34.1 Semantic product edges

Before compilation, a semantic product edge records:

- one source product control and one target product control;
- the chosen N transition, when the edge is visible;
- the unique deterministic A rule in the current sign regime;
- the ordinary vector effect on (x_N, x_A);
- whether the A source guard is positive or zero.

N-epsilon and positive reset-decrement edges are already ordinary VASS transitions and need no special treatment.

111.34.2 Positive-source gadget

For a positive-source semantic edge with effect (d_N, d_A) and $d_A \in \{-1, 0, +1\}$, introduce one private intermediate control and replace the edge by

$$(d_N, -1) \quad \text{followed by} \quad (0, d_A + 1).$$

The first transition is enabled exactly when the original N update is feasible and $x_A \geq 1$. The second transition only adds a nonnegative amount to x_A. No other transition is incident from the private intermediate control, so the gadget is atomic with respect to all asynchronous behavior.

111.34.3 Unique zero-test gate

Let E_0 be the finite set of zero-source semantic product-edge types. For each $e \in E_0$ introduce an ordinary tag counter t_e, initially zero. Every genuine product configuration has all tags zero.

For an edge e from genuine source s_e to genuine target t_e^{ctl}:

1. an ordinary source transition from s_e enters the common pre-gate control and increments only the tag counter t_e;
2. the common pre-gate has one outgoing distinguished transition g_0, enabled exactly when $x_A = 0$, with zero vector effect;
3. the common post-gate has one completion branch per semantic edge type; branch e decrements the tag t_e, performs the original ordinary vector update of e, and enters t_e^{ctl}.

All unchosen tags are zero, so their completion branches are disabled by ordinary VASS nonnegativity. The pre/post-gate controls expose no product, epsilon, visible, endmarker, or cleanup transitions other than the gadget moves just listed.

If the ordinary N component of the semantic update is infeasible, a run may cross g_0 but cannot complete the chosen branch and is trapped away from all genuine accepting controls. Hence incomplete gate attempts create no spurious accepting reachability.

It follows that a genuine source-to-target zero-edge simulation exists if and only if the chosen semantic edge is at its proper source, $x_A = 0$, and its ordinary vector update is feasible. The compiled system therefore has exactly one zero-test transition, namely g_0.

111.34.4 Reset decompression and asynchronous commutation

For a reset state of period p, the phase controls

$$s[0], \ldots, s[p-1]$$

perform positive silent decrements

$$s[i] \xrightarrow{\varepsilon,-1} s[i+1 \bmod p]$$

until $x_A = 0$, followed by the appropriate zero exit. N-epsilon steps modify only x_N and the N control; reset microsteps modify only x_A and the A control. Since neither consumes visible input, any interleaving of these two families commutes to an interleaving with the same component endpoints and counter values. Thus decompression does not change the synchronized visible language of the product.

111.34.5 Exact cleanup target

After the unique marked accepting genuine product control, enter a private cleanup phase. Each ordinary counter, including x_N, x_A, and all tags, has a decrement loop. A zero-effect edge leads to one fixed target control. The reachability query is for that control with the exact all-zero vector. Taking the final edge before complete drainage does not satisfy the query, and no edge returns from cleanup to the product. Hence arbitrary nonnegative accepting counter values are equivalent to exact reachability of one fixed all-zero configuration.

Part X

Unified Realization Theory and External Scalar-Field Generativity

Part guide. **Question.** Can the geometric, chronological, normalization, protected-information, and holonomy branches be placed in one typed exact-realization architecture without identifying objects that carry different mathematical meanings, and does that architecture generate useful structure outside pushdown automata? **Main result.** USNRT supplies exact path-outcome factorization, fixed-template section debt, protected-obligation capacity, product and transfer calculus, same-model saturation, typed holonomy/resource bridges, and four-axis source recovery; the independent guarded two-layer graph-action result then gives an exact stack-alphabet cost in its fixed physical category, and the linear-modular and signed-winding two-source guard results classify canonical one-stack realization in their respective models; EF–FG subsequently provides an independent scalar-field theory with observer-record normal form, exact common-realization semantics, a polynomial/NP-complete local–global boundary, and all-orders obstruction families; its second stage adds shallow spatial-frame NP-completeness, universality of finite common-realization complexes, fixed-parameter tractable ambiguity regimes, correlated rational-polyhedral semantics, and finite layered transfer; the independent physical third stage now realizes those intrinsic relations on controlled heat networks, with a sharp shared-power obstruction, nonsemialgebraic optimum face, exactly correlated terminal caps, and Krylov-rigidity transition. **Why here.** Every source axis required by the recovery theorem has now been developed, so the synthesis can be stated without forward dependence. **Dependency.** USNRT depends only on previously proved source structures; The two-layer alphabet-cost and guarded-Dyck results use the exact-realization interface but retain distinct physical models, hypotheses, and proofs; EF–FG is logically independent as a scalar-field theory and is used afterward as an external generativity test; the thermal chapter is an additional model-specific construction, not a theorem derivable from the abstract EF–FG complexity results alone.

Chapter 112

Exact Realizations, Path Outcomes, and Fixed-Template Debt

Research branch — Unified Semantic Normalization and Realization Theory. *The preceding General Theory and post-theory branches developed exact lifts, completion geometry, chronology, protected information, topology, and physical exposure separately. This chapter begins a typed synthesis. Its purpose is not to replace those source theories, but to state which objects may be identified, which must remain distinct, and which exact factorization statements survive after the source-specific coordinates are forgotten.*

The exact-realization object below is the same semantic/physical architecture already developed in the General Theory: a reachable deterministic semantic action system, a deterministic physical realization, a surjective semantic projection, domain reflection, and exact action intertwining. We restate the data because the present Part uses them as a typed host for several earlier branches. A canonical section remains optional extra structure rather than semantic identity. The independent future-stable monitor result of Chapter 102 charges only its readout port; the subsequent guarded two-layer action study (Chapter 114) instead charges physical stack symbols in a specified one-control-state category. Neither resource measure is intrinsic to the untyped exact-realization axioms.

112.1 Typed architecture and theorem scope

An enriched USNRT object may carry, in addition to the exact realization, independent geometry, chronology, observation, protected-contract, category, and resource data, which we denote schematically by

$$\mathfrak{U} = (R; \mathcal{G}_\bullet, \mathcal{J}_\bullet, \mathcal{O}_\bullet, \mathcal{B}, \mathsf{Cat}, \varrho).$$

The notation is packaging only: the enrichments are not required simultaneously by every theorem. Geometry may be absent from a purely chronological specialization; chronology may be absent from a purely geometric one; a resource evaluation is meaningful only after a physical category has been declared.

Proposition 112.1 (Typing and non-collapse principle)**.** *The USNRT axioms do not identify any of the following pairs by definition: semantic state and physical representative; representation fiber and compatibility fiber; section debt and behavioral failure; exact chronology and its shadow; intrinsic marked topology and physical transport; protected obligation pressure and resource cost. Any theorem transferring information between two such layers must state the required bridge data explicitly.*

Proof. The primitive data are typed independently. The exact realization contains only the semantic/physical projection and action laws. A section is optional extra structure. Geometry, chronology, observation, protected contracts, physical category, and resource evaluation are independent enrichments. The derived objects used later—path-outcome fibers, obligation complexes, holonomy orbits, and Pareto/resource bounds—are formed only after the corresponding enrichments are supplied. Therefore none of the listed identifications follows from the base definitions. □

This proposition is the main scope discipline of the present Part. It is deliberately negative, but it is mathematically useful: it tells us exactly which hypotheses must appear when a geometric, chronological, or topological distinction is converted into protected information or physical cost.

112.2 Semantic action systems and exact realizations

Definition 112.2 (Semantic action system)**.** A rooted deterministic semantic action system is a tuple

$$S = (M, m_0, \Sigma, U),$$

where M is a set, $m_0 \in M$, Σ is an action alphabet, and each $U_a : M \rightharpoonup M$ is a partial function. We restrict all statements to the reachable part of M.

A semantic morphism $f : S \to S'$ preserves roots, definedness on reachable states, and actions: $f(U_a(m)) = U'_a(f(m))$ whenever $U_a(m)$ is defined.

Definition 112.3 (Exact realization)**.** An exact realization of S is

$$R = (S; C, c_0, \delta; \Theta),$$

where C is a physical state set with root c_0, each $\delta_a : C \rightharpoonup C$ is deterministic and partial, and

$$\Theta : C_{\mathrm{reach}} \to M_{\mathrm{reach}}$$

is surjective, root preserving, domain reflecting, and action intertwining:

$$c \in \mathrm{Dom}(\delta_a) \iff \Theta(c) \in \mathrm{Dom}(U_a), \qquad \Theta(\delta_a(c)) = U_a(\Theta(c)).$$

An exact-realization morphism $(f, g) : R \to R'$ consists of a semantic morphism f and a rooted physical map g that intertwines physical actions and satisfies $\Theta' g = f\Theta$.

Proposition 112.4. *Exact realizations and exact-realization morphisms form a category* Real_Σ.

Proof. Identity pairs satisfy the defining conditions. If $(f, g) : R \to R'$ and $(f', g') : R' \to R''$, then roots and physical intertwining are preserved by composition and

$$\Theta'' g' g = f' \Theta' g = f' f \Theta.$$

Associativity is inherited from function composition. □

Theorem 112.5 (Identity realization embedding)**.** *The assignment*

$$I(S) = (S; M, m_0, U; \mathrm{id}_M), \qquad I(f) = (f, f)$$

defines a fully faithful functor $I : \mathsf{SemAct}_\Sigma \to \mathsf{Real}_\Sigma$.

Proof. Exactness is immediate for the identity projection. A morphism between identity realizations is determined by its semantic component, so the hom-set map is bijective. □

This embedding is useful conceptually: semantics is not a quotient imposed after the fact. It sits fully faithfully inside the realization category, while nontrivial realizations add physical multiplicity.

112.2.1 Sections are optional structure

Definition 112.6 (Sectioned realization)**.** A sectioned realization is an exact realization together with a map

$$\sigma : M_{\mathrm{reach}} \to C_{\mathrm{reach}}, \qquad \Theta\sigma = \mathrm{id}.$$

When root-zero debt is required we also impose $\sigma(m_0) = c_0$.

The representation fiber is

$$\mathrm{Fib}_{\mathrm{rep}}(m) = \Theta^{-1}(m).$$

A section chooses one representative from each fiber, but no theorem identifies this choice with semantic identity. This distinction will matter below.

112.3 Path outcomes, canonicalization, and fixed-template debt

Fix a rooted exact realization. For an admissible word w, let

$$e_M(w) = U_w(m_0), \qquad e_C(w) = \delta_w(c_0).$$

Define two equivalence relations on rooted admissible words by equality of semantic and physical endpoints, respectively. Exactness gives

$$e_C(u) = e_C(v) \Longrightarrow e_M(u) = e_M(v),$$

but the converse may fail.

For a reachable semantic state m, define its path-outcome fiber

$$H_{c_0}(m) = \{\delta_w(c_0) : U_w(m_0) = m\}.$$

This is distinct from the full representation fiber $\Theta^{-1}(m)$: it records only physical representatives actually produced by rooted semantic paths.

Theorem 112.7 (Rooted endpoint factorization)**.** *The following are equivalent.*

(a) There is a rooted transition-closed section σ through c_0.

(b) The physical endpoint map factors through the semantic endpoint map: $e_C = \sigma e_M$.

(c) Semantic endpoint equivalence equals physical endpoint equivalence.

(d) Every path-outcome fiber $H_{c_0}(m)$ is a singleton.

When these conditions hold, the section is uniquely determined by c_0.

Proof. Transition closure implies $\delta_w(c_0) = \sigma(U_w(m_0))$, giving the factorization. The factorization identifies physical endpoints whenever semantic endpoints agree; the reverse implication already follows from exactness. Equality of endpoint equivalences is precisely singleton path-outcome fibers. Finally, if each fiber is a singleton, define $\sigma(m)$ to be its unique rooted path outcome; exact intertwining makes the section transition closed. □

Assume the relevant fibers are finite and define $\mu_{c_0}(m) = |H_{c_0}(m)|$. For a finite semantic test set T, set

$$\Delta_T(c_0) = \sum_{m \in T} (\mu_{c_0}(m) - 1).$$

For any rooted section σ, let $\mathrm{Debt}_T(\sigma)$ count path outcomes over T that do not lie on the selected section.

Theorem 112.8 (Exact section-defect formula)**.** *For every finite test region T,*

$$\min_{\sigma} \mathrm{Debt}_T(\sigma) = \Delta_T(c_0).$$

Proof. At a state m, a single section value can cover at most one element of $H_{c_0}(m)$, so at least $\mu_{c_0}(m) - 1$ outcomes remain off-section. Summing gives the lower bound. Choose one path outcome at each $m \in T$ as the section value and extend arbitrarily on other reachable states; then exactly one path outcome at each m lies on the section. □

Corollary 112.9 (Fixed-template debt)**.** *If $\mu_{c_0}(m) \geq 2$ for some reachable m, then no transition-closed section through c_0 exists. Refining only the observation map cannot remove this defect while the physical transition template and root representative remain fixed.*

Thus canonicalization failure is a path-factorization obstruction, not an observation-capacity statement. Observation becomes relevant only after additional protected future distinctions are specified.

Chapter 113

Protected Obligations, Products, and Same-Model Saturation

Research branch — Protected obligation and transfer calculus. *The abstract collapse-capacity theorem and the later protected-contract/resource chapters already show that semantic distinctions can force physical pressure. The present chapter packages that mechanism as an obligation complex, develops its product and asymptotic calculus, and then separates abstract Cartesian product realization from simultaneous compilation inside one restricted physical category.*

113.1 Protected obligations and compatibility complexes

A protected obligation problem adds an observation level, confinement semantics, and a protected future contract to an exact realization. We suppress inessential data and write P_h for a finite protected problem with finite semantic source family S.

Definition 113.1 (Obligation complex)**.** A subset $A \subseteq S$ is *one-cell compatible at level h* if the sources in A can be realized within one common level-h observation cell, under the declared confinement condition and while satisfying the protected contract. Empty sets and singletons are compatible by convention. The downward-closed family of compatible subsets is the obligation complex

$$\mathrm{Ob}_h(P_h).$$

Its face-cover number $\kappa_h(P_h)$ is the minimum number of faces whose union covers S.

Definition 113.2 (Protected distinguishability certificate)**.** A map $d : S \to Z$ is a protected distinguishability certificate if every face of $\mathrm{Ob}_h(P_h)$ lies in one fiber of d.

Let D be the number of source vertices allowed to leave the declared canonical interface, X_h the number of remaining canonical sources whose obligation-realizing computation is not level-h confined, and

$$V_S(h) = |\omega_h(\sigma(S))|$$

the number of observation cells occupied by canonical confined sources.

Theorem 113.3 (Obligation-cover capacity)**.** *If every actual level-h observation class of canonical confined sources is a face of* $\mathrm{Ob}_h(P_h)$*, then*

$$D + X_h + V_S(h) \geq \kappa_h(P_h).$$

Proof. Remove the D sources permitted to leave the canonical interface and the X_h canonical sources not satisfying confinement. The remaining sources are partitioned by their actual observation cells, and each such cell is a face by hypothesis. Together with the exceptional singletons, these faces cover all source vertices, so their number is at least the minimum face-cover number. □

Corollary 113.4 (Distinguishability-capacity transfer)**.** *If $d : S \to Z$ is a protected distinguishability certificate, then*

$$\kappa_h(P_h) \geq |d(S)|, \qquad D + X_h + V_S(h) \geq |d(S)|.$$

Proof. A face lies inside one fiber of d, so a face cover must contain at least one face for each represented value of d. Apply the preceding theorem. □

This inequality is intentionally not a physical resource lower bound by itself. A resource conclusion requires a category-specific capacity envelope relating $D + X_h + V_S(h)$ to the resource evaluation.

113.1.1 Refinement and resource exclusion

Strengthening a protected contract can delete compatible faces but cannot create new ones. Hence obligation complexes are contravariantly monotone under contract refinement, while their minimum face-cover number is nondecreasing. If a physical category has a resource bound r forcing

$$D + X_h + V_S(h) \leq B_h(r),$$

then any problem with $\kappa_h(P_h) > B_h(r)$ is excluded below resource r. This is the generic route from protected incompatibility to a category-specific Pareto exclusion.

113.2 Products, amplification, and asymptotic obligation rate

Given finite simplicial complexes K on S and L on T, define their rectangular compatibility product $K \boxtimes L$ on $S \times T$ by declaring $A \subseteq S \times T$ a face when both coordinate projections are faces.

Theorem 113.5 (Rectangular product cover bounds)**.** *For nonempty finite complexes,*

$$\max\{\kappa(K), \kappa(L)\} \leq \kappa(K \boxtimes L) \leq \kappa(K)\kappa(L).$$

Both bounds are sharp in general.

The upper bound comes from products of faces in optimal covers. Projecting any cover of $K \boxtimes L$ yields covers of each factor, giving the lower bound. No universal exact product law exists without additional structure.

For repeated rectangular powers $K^{\boxtimes n}$, submultiplicativity implies existence of the asymptotic rate

$$h_{\mathrm{obl}}(K) = \lim_{n\to\infty} \frac{1}{n} \log \kappa(K^{\boxtimes n}).$$

Let $\kappa_f(K)$ be the fractional face-cover number. The classical linear-programming fractional cover formalism [144, 147] yields exact multiplicativity under rectangular powers. A standard probabilistic rounding argument then gives the exact asymptotic identity

$$h_{\mathrm{obl}}(K) = \log \kappa_f(K). \tag{113.1}$$

The combinatorial shell of (113.1) is classical; its role here is to quantify repeated protected-realization pressure.

The equality (113.1) is a statement about the derived compatibility complex. A physical interpretation requires an additional product-realization theorem.

Definition 113.6 (Iterated rectangular realization)**.** Let P_h be a protected problem with $K = \mathrm{Ob}_h(P_h)$. An iterated rectangular realization family is a sequence $P_h^{(n)}$ such that

$$\mathrm{Ob}_h(P_h^{(n)}) \cong K^{\boxtimes n}$$

for every $n \geq 1$, with the declared protection and confinement semantics preserved coordinatewise.

Theorem 113.7 (Asymptotic capacity-rate exclusion)**.** *Suppose P_h admits an iterated rectangular realization family and suppose resource bounds r_n imply certified capacity bounds*

$$D_n + X_n + V_n \leq B_n.$$

If

$$\limsup_{n\to\infty} \frac{1}{n} \log B_n < h_{\mathrm{obl}}(K) = \log \kappa_f(K),$$

then, for all sufficiently large n, no admissible n-fold realization can satisfy resource bound r_n.

Proof. Let $\kappa_n = \kappa(K^{\boxtimes n})$. By (113.1), $(1/n)\log \kappa_n$ converges to $h_{\mathrm{obl}}(K)$. Choose a number λ strictly between the two exponential rates. Eventually $\kappa_n > e^{\lambda n} > B_n$. This contradicts the one-shot obligation-capacity inequality $\kappa_n \leq D_n + X_n + V_n \leq B_n$ for any admissible realization. □

Corollary 113.8 (Mixed protected rate)**.** *If one protected problem carries two jointly independent finite certificates of ranks m and n, and their iterated physical product realization exists, then*

$$h_{\mathrm{obl}} \geq \log m + \log n.$$

This is a protected-realization rate statement, not an identification with Shannon entropy.

113.3 Transfer calculus and product realization

A finite distinction transfer $\tau : X \to Y$ has logarithmic loss

$$L(\tau; X) = \log |X| - \log |\tau(X)|.$$

This measures cardinality loss only; reversibility is a stronger structured property.

Theorem 113.9 (Exact transfer-loss composition)**.** *For a chain of effective-image-normalized transfers*

$$X_0 \xrightarrow{\tau_1} X_1 \xrightarrow{\tau_2} \cdots \xrightarrow{\tau_k} X_k,$$

$$\log |X_k| = \log |X_0| - \sum_{i=1}^{k} L(\tau_i; X_{i-1}).$$

Equivalently, retention ratios multiply.

Proof. The logarithmic differences telescope. □

When the surviving terminal distinctions are protected, the obligation-capacity theorem converts the surviving rank into physical pressure. This gives a precise way to account for lossy bridges rather than asserting that information is "preserved" informally.

113.3.1 Exact product realization

Two exact realizations admit an asynchronous stutter-completed product: semantic and physical states are Cartesian products, and actions update one or both coordinates with the other coordinate stuttering. The projection and, when present, the section are coordinatewise products.

Theorem 113.10 (Exact product realization and protected product)**.** *The asynchronous product is an exact realization. For coordinate-separable protected contracts,*

$$\mathrm{Ob}_h(P_1 \otimes P_2) = \mathrm{Ob}_h(P_1) \boxtimes \mathrm{Ob}_h(P_2).$$

Proof. Surjectivity, domain reflection, and action intertwining hold coordinatewise. For protected compatibility, any product witness projects to witnesses in the factors. Conversely, factor witnesses can be run asynchronously with stutters, preserving coordinatewise confinement, observation, and protected obligations. □

Suppose the first problem has protected certificate $g : S_G \to G$ with $|g(S_G)| = m$, and the second has $j : S_J \to J$ with $|j(S_J)| = n$. Then the product certificate (g, j) has full image $g(S_G) \times j(S_J)$ and hence rank mn. Therefore

$$D + X_h + V_h \geq mn.$$

This is an abstract mixed-axis theorem. It does not assert that a restrictive one-stack pushdown category is closed under such products.

113.4 Same-model transfer saturation

The product theorem separates abstract coordinate products from compilation inside one restrictive physical model. Fix finite protected label sets A, B, with $|A| = m$ and $|B| = n$, and fix a declared realization category Cat together with its confinement and repair contract.

Definition 113.11 (Same-model compiler class)**.** An object of $\mathrm{Comp}_{\mathsf{Cat}}(A, B)$ is a tuple

$$P = (R_P, S_P, \lambda_P, \Pi_A, \Pi_B)$$

where R_P is one realization object in Cat, S_P is a finite protected source family inside that same realization, and

$$\lambda_P : S_P \longrightarrow A \times B$$

is a joint protected-label map. The contracts Π_A and Π_B require the coordinate projections $\pi_A\lambda_P$ and $\pi_B\lambda_P$ to realize the prescribed marginal protected families under the declared confinement/repair semantics. No product closure of Cat is assumed. The joint image is

$$J_P := \lambda_P(S_P) \subseteq A \times B.$$

Define

$$\mathrm{Sat}_{\mathsf{Cat}}(A, B) = \max_{P \in \mathrm{Comp}_{\mathsf{Cat}}(A,B)} |J_P|,$$

when the class is nonempty and the finite maximum is attained. The normalized saturation ratio is $\mathrm{Sat}_{\mathsf{Cat}}/(mn)$, and the logarithmic loss of one compiler is $\log(mn/|J_P|)$. Thus the optimal same-model loss is

$$L^*_{\mathsf{Cat}} = \log(mn) - \log \mathrm{Sat}_{\mathsf{Cat}}(A, B).$$

Proposition 113.12 (Full-saturation criterion)**.** *The following are equivalent:*

(a) $\mathrm{Sat}_{\mathsf{Cat}}(A, B) = mn$*;*

(b) the optimal loss is zero;

(c) the category contains a marginal-preserving compiler whose joint image is all of $A \times B$*.*

Product closure is sufficient but not necessary: a category may contain a bespoke full-rank compiler for one pair without admitting arbitrary Cartesian products.

113.4.1 Pairwise defect versus assembly defect

Let

$$E_{\mathsf{Cat}}(A,B) = \bigcup_{P \in \mathrm{Comp}_{\mathsf{Cat}}(A,B)} J_P$$

be the pairwise envelope and $p_{\mathsf{Cat}} = |E_{\mathsf{Cat}}(A,B)|$. Define

$$\delta_{\mathrm{pair}} = mn - p_{\mathsf{Cat}}, \qquad \delta_{\mathrm{asm}} = p_{\mathsf{Cat}} - \mathrm{Sat}_{\mathsf{Cat}}(A,B).$$

Proposition 113.13 (Saturation-defect identity)**.**

$$mn - \mathrm{Sat}_{\mathsf{Cat}}(A,B) = \delta_{\mathrm{pair}} + \delta_{\mathrm{asm}}.$$

Both defects are nonnegative, and full saturation occurs exactly when both vanish.

Proof. By definition,

$$mn - \mathrm{Sat}_{\mathsf{Cat}} = (mn - p_{\mathsf{Cat}}) + (p_{\mathsf{Cat}} - \mathrm{Sat}_{\mathsf{Cat}}).$$

The envelope contains every joint image, so $p_{\mathsf{Cat}} \geq \mathrm{Sat}_{\mathsf{Cat}}$; also $p_{\mathsf{Cat}} \leq mn$. Hence both defects are nonnegative. Their simultaneous vanishing is equivalent to $p_{\mathsf{Cat}} = \mathrm{Sat}_{\mathsf{Cat}} = mn$. □

The interpretation is important. A positive pairwise defect means that some label pairs are never jointly realizable. A positive assembly defect means that every pair may occur somewhere, while no single physical realization assembles all available pairs simultaneously. Thus pairwise compatibility and simultaneous compilation are different questions even when they use the same marginal labels.

Proposition 113.14 (Elementary saturation bounds)**.** *Whenever the compiler class is nonempty and every compiler is marginal-surjective,*

$$\max\{m,n\} \leq \mathrm{Sat}_{\mathsf{Cat}}(A,B) \leq mn.$$

Moreover, if $\mathsf{Cat}_1 \subseteq \mathsf{Cat}_2$ *with the same protected witness interpretation, then*

$$\mathrm{Sat}_{\mathsf{Cat}_1}(A,B) \leq \mathrm{Sat}_{\mathsf{Cat}_2}(A,B).$$

Proof. For any marginal-surjective joint image $J \subseteq A \times B$, both coordinate projections are onto, so $|J| \geq m$ and $|J| \geq n$. The upper bound is $|J| \leq |A \times B| = mn$. Enlarging the admissible compiler class can only enlarge the set over which the maximum is taken. □

113.4.2 Shared-interface bounds

If every compiler factors its joint certificate through an interface of capacity at most C, then

$$\mathrm{Sat}_{\mathsf{Cat}}(A,B) \leq C.$$

Hence $C < mn$ forces positive same-model information loss, with

$$L^*_{\mathsf{Cat}} \geq \log \frac{mn}{C}.$$

More generally, if a resource bound r gives a capacity envelope $B_h(r)$, then the resource-bounded saturation profile obeys

$$\mathrm{Sat}_{\mathsf{Cat}}(r; A,B) \leq B_h(r).$$

Proposition 113.15 (Optimal same-model transfer loss)**.** *The minimum logarithmic loss among marginal-preserving compilers is exactly*

$$L^*_{\mathsf{Cat}} = \log \frac{mn}{\mathrm{Sat}_{\mathsf{Cat}}(A, B)}.$$

An independent capacity bound $\mathrm{Sat}_{\mathsf{Cat}} \le C$ *gives a lower bound* $L^*_{\mathsf{Cat}} \ge \log(mn/C)$*; an explicit compiler of joint rank* R *gives* $L^*_{\mathsf{Cat}} \le \log(mn/R)$*. Matching bounds certify exact loss optimality.*

Proof. Every admissible compiler has joint rank $R \le \mathrm{Sat}_{\mathsf{Cat}}$ and hence loss at least

$$\log \frac{mn}{\mathrm{Sat}_{\mathsf{Cat}}}.$$

By definition of saturation, some admissible compiler attains rank $\mathrm{Sat}_{\mathsf{Cat}}$ whenever the finite maximum is attained, and its loss is exactly the displayed value. The capacity and construction estimates follow monotonically from the same formula. □

113.4.3 A source-stack capacity specialization

The abstract theory does not prescribe a stack or memory measure. The required capacity law has already been proved inside this monograph. The filtered collapse-capacity theorem, Theorem 65.1, gives

$$D + X_h + V(h) \ge N$$

for a collapse family of size N under hidden-tail noncoalescence. The representation-independent pushdown specialization, Theorem 99.3 in Chapter 99, then gives

$$\boxed{D + X_h + q_{\mathrm{src}} \sum_{j=0}^{h} g^j_{\mathrm{src}} \ge N.}$$

Its proof verifies hidden-tail noncoalescence for deterministic top-access ε-pushdown sources and counts depth-h source signatures.

Choose one protected source representative for each distinct joint label in a same-stack compiler of joint rank R; the resulting collapse family has size $N = R$. Applying Theorem 99.3 gives the necessary inequality

$$\boxed{R \le D + X_h + q_{\mathrm{src}} \sum_{j=0}^{h} g^j_{\mathrm{src}}.}$$

Consequently, a zero-debt, h-confined compiler with full joint rank mn must satisfy

$$q_{\mathrm{src}} \sum_{j=0}^{h} g^j_{\mathrm{src}} \ge mn.$$

For $g_{\mathrm{src}} \ge 2$, this yields

$$h \ge \left\lceil \log_{g_{\mathrm{src}}} \left(1 + \frac{(g_{\mathrm{src}} - 1)mn}{q_{\mathrm{src}}} \right) \right\rceil - 1.$$

Remark 113.16 (Necessary, not sufficient). The source-stack exposure theorem is a lower-bound/capacity statement. It neither constructs a one-stack compiler meeting the bound nor asserts full rectangular independence for any particular pair of geometric and chronological witnesses. Those are category-specific construction questions.

113.5 Source-native shortest fibers and protected paired-session comparison

The existence of optimal fibers and their marked comparison groups was established in the geometric chapters. A single unit-cost source can realize both a prescribed support complex and its literal shortest Parikh fiber, with a separate operational question of reading marked comparisons from genuine source words. A paired-session service has a designated root reset and fixed block length; this is not an unframed single-start machine.

113.5.1 Native source and comparison semantics

Let $A \in \mathbb{Z}^{d\times n}$ be an integer counting source on letters $a_1, \dots, a_n$, with each letter costing one. For $b \in \mathbb{Z}^d$ with nonnegative solutions set

$$\ell_A(b) = \min\{\mathbf{1}^{\mathsf{T}} z : z \in \mathbb{N}^n,\ Az = b\},$$
$$P_A(b) = \{z \in \mathbb{N}^n : Az = b,\ \mathbf{1}^{\mathsf{T}} z = \ell_A(b)\}.$$

The source is *completion-total* when $Ah = 0$ for some strictly positive integer vector h: any integral solution may then be shifted by a positive multiple of h to become nonnegative. For an optimal fiber P, its active lattice is $L(P) = \langle P - P\rangle_{\mathbb{Z}}$. Choose a vertex $p_0 \in P$ and write $\widehat{P} = P - p_0 \subset L(P)$. Its periodic nerve and quotient are

$$\mathcal{N}(\widehat{P}) = \{F \subset L(P) : F \text{ finite},\ \exists c \in L(P),\ c - F \subset \widehat{P}\},$$
$$Q(P) = |\mathcal{N}(\widehat{P})|/L(P).$$

The quotient is independent of the chosen translation vertex, up to its natural identification. The geometric compression of shortest fibers and the marked difference presentation for $Q(P)$ are established in [186, 187, 188]. In particular, a primitive fiber with m affinely independent vertices has a comparison group F_{m-1}, with displacement onto the rank-$(m-1)$ active lattice given by abelianization; this background is not reproved as a new topological theorem here. Prior work on semigroup realizations and isolated factorizations supplies related constructions and structural restrictions [71, 193]; it does not by itself assert the exact simultaneous source-and-service contract defined below.

Three types of operational statements must be distinguished. A *whole-language recognizer* answers whether $A\,\mathrm{Parikh}(w) = 0$ for an arbitrary word. A *paired-session comparator* receives fixed-length successive rooted shortest sessions, with externally specified session resets. A *single uninterrupted source trace* has neither those resets nor a separate mode selector. A pushdown realization in one category is not silently promoted to either of the others.

113.6 An exact facet-and-fiber compiler

Let K have labeled vertices $1, \dots, s$ and facets $F_0, \dots, F_{m-1}$, with $m \geq 1$. Write $v_{ji} = \mathbf{1}_{\{i\in F_j\}}$ and $d_i = \sum_j v_{ji}$. On $(x_i^+, x_i^-)_{i=1}^s, (y_j)_{j=0}^{m-1}, t$, impose

$$x_i^+ - \sum_j v_{ji} y_j + (d_i - 1)t = 0, \tag{113.2}$$

$$x_i^- - \sum_j (1 - v_{ji}) y_j + (m - d_i - 1)t = 0, \tag{113.3}$$

$$\sum_j y_j - mt = 1. \tag{113.4}$$

These are the row equations for a single $(2s+1) \times (2s+m+1)$ integer matrix A_K and one target b.

Theorem 113.17 (Simultaneous source-native realization)**.** *The source A_K is completion-total with the all-ones kernel. Its unit-cost shortest completion length is $s+1$, and the entire optimal Parikh fiber consists of exactly m vectors $p_0, \dots, p_{m-1}$, with*

$$t = 0, \quad y_j = 1,\ y_k = 0\ (k \neq j), \quad x_i^+ = v_{ji}, \quad x_i^- = 1 - v_{ji}.$$

On the designated letters x_i^+, the optimal support complex is precisely K. The fiber is a primitive integral $(m-1)$-simplex in its active lattice. Consequently its native quotient has $\pi_1 Q(P) \cong F_{m-1}$ with natural displacement the abelianization to $\mathbb{Z}^{m-1}$.

Proof. Every row vanishes at the all-ones vector. Adding (113.2) and (113.3) gives $x_i^+ + x_i^- = 1 + 2t$, while (113.4) gives $\sum_j y_j = 1 + mt$. Thus every nonnegative solution has length

$$s(1+2t) + (1+mt) + t = (s+1) + (2s+m+1)t.$$

The minimum forces $t = 0$, where the selector y is one-hot, giving exactly the stated p_j. Its positive-letter supports are exactly the facets, and their downward closure is K. The differences $p_j - p_0$ are an integral basis of their span because their selector coordinates $y_1, \dots, y_{m-1}$ project to the standard basis. The quotient and its marked group follow from the minimal-active-fiber theorem theorem 95.4, proved earlier, with the basis $p_j - p_0$ fixing the displacement coordinates. □

113.7 Whole-language recognition after lossless scalar folding

For a matrix source $A \in \mathbb{Z}^{d \times n}$, target b, and known shortest length L, let

$$M = \max\{1, \max_{j,i} |A_{ji}|, \max_j |b_j|\}, \qquad R > (L+1)M$$

with integer R. Define scalar letter effects and scalar target

$$\lambda_i = \sum_{j=0}^{d-1} A_{ji} R^j, \qquad \beta = \sum_{j=0}^{d-1} b_j R^j. \tag{113.5}$$

Theorem 113.18 (Literal preservation of every shortest completion)**.** *For every nonnegative z of length at most L, $\sum_i \lambda_i z_i = \beta$ if and only if $Az = b$. Hence the scalar source has the same shortest length and* exactly the same *optimal Parikh fiber, supports, active lattice, and marked periodic quotient. If $Ah = 0$ with $h > 0$, then also $\lambda h = 0$.*

Proof. Write $e_j = \sum_i A_{ji} z_i - b_j$. Then $|e_j| \leq (L+1)M < R$. The scalar equality is $\sum_j e_j R^j = 0$. Taking residues modulo R forces $e_0 = 0$. Divide by R and repeat; every $e_j = 0$. All original length-L solutions persist and no shorter scalar solutions are introduced. The remaining claims are literal invariance of the fiber and linearity of (113.5). □

Proposition 113.19 (Real-time one-stack zero-return recognition)**.** *For any integer scalar weights $(\lambda_1, \dots, \lambda_n)$, the language $\{w : \sum_i \lambda_i \#_{a_i}(w) = 0\}$ has a deterministic real-time one-stack recognizer, using at most one push or pop per input symbol and no empty-input transitions.*

Proof. Set $D = \max(1, \max_i |\lambda_i|)$. Represent the running total uniquely as $Dq + r$ with $0 \leq r < D$. Keep r in finite control and q as a signed-unary stack above a bottom marker. On input letter a_i, write $r + \lambda_i = D\delta + r'$ with $0 \leq r' < D$. Since $0 \leq r < D$ and $|\lambda_i| \leq D$, necessarily $\delta \in \{-1, 0, 1\}$. Thus the quotient changes by at most one; make the corresponding push, pop, or no-op and update the remainder to r'. The sign is represented by the stack alphabet $\{\bot, +, -\}$. Only the zero residue and empty signed stack accept. □

Remark 113.20. For general completion-total vector sources of rational rank at least two, whole-language context-freeness already fails by a standard two-independent-counters reduction, consistent with the group-language literature [194]. The folding theorem changes the matrix and retains only the specified shortest fiber: it does not preserve the entire vector zero-return language. Standard base encoding and one-counter recognition themselves are not claimed as new methods.

113.8 A source-faithful protected comparison service

Take the primitive optimal fiber $P = \{p_0, \dots, p_{m-1}\}$ from Theorem 113.17 or its folded scalar source, with minimum length L. Fix the source-native base vertex p_0. One *paired block* consists of two successive genuine shortest source words, each exactly L letters, and an interface reset between them to the same root. Restrict to pairs (p_0, p_j) and (p_j, p_0), $j \geq 1$, assigning symbols X_j and X_j^{-1} respectively. These are comparison labels extracted from source words, not new source input letters. Let $B = \bigcup_{p \in P} \{z \in \mathbb{N}^n : z \leq p\}$ be the finite set of possible shortest-prefix Parikh vectors.

Theorem 113.21 (Protected paired-session comparison)**.** *With this explicitly selected service, there is a deterministic real-time single-stack machine on the* original source alphabet *that checks all session words, updates the freely reduced word in F_{m-1}, and accepts precisely the full paired streams representing the identity. It uses at most $(m+1)|B|+1$ verifier states, stack alphabet of size $1+2(m-1)$, and at most one push or pop on the final letter of a pair, with no empty-input moves. For $m \geq 3$ the service records nonidentity comparisons with zero natural displacement.*

Proof. The first-half verifier stores a prefix Parikh vector $z \in B$ in finite control, rejecting an impossible extension. The completed first half uses $|B|$ states; the second-half states are pairs (i, z) with $i \in \{0, \dots, m-1\}$ and $z \in B$, plus one rejecting sink. Thus the advertised $(m+1)|B|+1$ bound includes both phases and the sink. After L letters, z must equal one unique p_i; remember i and restart the verifier component at zero by the service clock. Check the second half similarly, deriving p_j. If the pair is allowed, read its single $X_k^{\pm 1}$ comparison label at the final source-letter transition. Cancel against the inverse top stack letter or push it; this maintains free reduction inductively. Accept at a completed block boundary if and only if only the bottom marker remains. There is no input alphabet extension and one stack suffices. The chosen group generators correspond to the source's primitive optimal-fiber comparisons. For $m \geq 3$, $[X_1, X_2]$ is nontrivial in the free group but has zero displacement since its exponents sum to zero on every generator. □

Proposition 113.22 (Unbounded comparison requires memory)**.** *For $m \geq 2$, the identity language of complete admissible paired blocks cannot be recognized by a finite-state-only device, although the single-stack construction above recognizes it.*

Proof. Select two fixed genuine paired blocks U, V that produce X_1 and X_1^{-1}. The concatenation $U^a V^b$ represents identity precisely when $a = b$. For distinct a, a', the prefixes U^a and $U^{a'}$ are distinguished by suffix V^a. Thus infinitely many finite-state residuals are needed by the Myhill–Nerode criterion. The proof concerns faithful unbounded pair streams and does not give a minimum number of pushdown states or stack symbols. □

Proposition 113.23 (Sharp semantic restrictions)**.** *Suppose a shortest rooted session has nonzero target displacement b. A concatenation of $k \geq 2$ such sessions has displacement kb, and cannot be one original rooted shortest execution to b. Further, for any completion-total source with $h > 0$, every finite source word is the prefix of some zero-return source word. Thus no finite prefix can be reserved exclusively for irreversible comparison-mode routing while still recognizing every zero-return continuation.*

Proof. Displacements and lengths add, so the concatenation ends at kb and costs kL, not b in length L. For the prefix claim, if a word has Parikh vector c, choose N with $Nh \geq c$ and

append any suffix of Parikh vector $Nh - c$. The completed word has net effect $ANh = 0$. The routing consequence follows immediately. □

Remark 113.24 (Precisely what the construction certifies). The whole-language counter recognizer and the paired comparison machine are *host-selected alternative services* using one physical stack per run. The theorem does not prove one fixed-start device that concurrently emits both outputs for every unframed raw input, or a physical directed path literally executing formal inverses. Its push and state counts are constructive upper bounds, not general resource minima. These type distinctions are part of the result, not merely implementation commentary.

113.9 A seven-letter exact certificate

The folded unit-cost scalar source

$$\lambda = (1, 13, 169, 2041, 2029, 2197, -6450), \qquad \beta = 2211$$

has $\lambda \cdot (1, \dots, 1) = 0$ and precisely three shortest Parikh vectors of length three:

$$p_0 = (1, 1, 0, 0, 0, 1, 0), \quad p_1 = (1, 0, 1, 1, 0, 0, 0), \quad p_2 = (0, 1, 1, 0, 1, 0).$$

The corresponding source supports are $\{1, 2, 6\}$, $\{1, 3, 4\}$, $\{2, 3, 5\}$; the first three action types induce the boundary of a triangle. The active fiber is a primitive rank-two triangle, and its comparison group is the free group $F(X, Y)$ with the natural rank-two displacement marking. The global zero-return language is recognized by Proposition 113.19 using $D = 6450$ remainder-control states and signed-unary stack. The paired comparison service has $|B| = 19$, at most 77 verifier states and five stack symbols. Four pairs built from the genuine words $a_1a_2a_6$, $a_1a_3a_4$, and $a_2a_3a_5$ yield the reduced word $XYX^{-1}Y^{-1}$, which is nontrivial but has zero displacement. Indeed, the seven-coordinate nonnegative vectors of length at most three number $\sum_{j=0}^{3} \binom{6+j}{j} = 120$. Direct substitution into $\lambda \cdot z = \beta$ gives no solutions of length below three and exactly the three displayed solutions at length three. Each has six distinct orderings; their downsets have union of size 19, and the four allowed ordered types of pairs therefore yield $4 \cdot 6^2 = 144$ six-letter words. The general assertions do not rely on this finite example.

Interface to guarded-family saturation. The next chapter studies the separate family profile Sat_g and Δ_g for a budget-dependent set of guarded two-layer graph sources. These quantities range over source masks that change with g; they are not the fixed-marginal $\mathrm{Sat}_{\mathsf{Cat}}(A, B)$ and δ_{asm} defined here. The connection is a comparison of physical resource constraints, not an identification of the two deficits.

Chapter 114

Guarded Two-Layer Graph Actions: Stack-Alphabet Cost and Saturation Gaps

Research branch — Independent protected-realization study. *This chapter integrates the independent one-control-state, two-layer graph-action study. It keeps the exact stack-alphabet and depth-two constructions, all-budget saturation formulas, nonmonotone gap witnesses, and vertex-cover complexity proof in their stated guarded model.*

Connection to the General Theory. The exact rooted source/action representation is a restrictive physical category within the semantic-realization interface of Chapter 112. The attainable population, saturation Sat_g, and gap Δ_g are a concrete budgeted family profile related to, but not literally identical with, the fixed-family δ_{asm} in Chapter 113: here the source mask varies with the budget. The result is a separately publishable research manuscript [255].

114.1 Introduction

A physical realization of a labelled transition system must do more than recognize a language: it must encode every named source state and preserve the domain and result of each input action. In a pushdown implementation, such requirements impose structural constraints on the configurations available at a given stack-alphabet budget. We study those constraints for a two-layer family built from rooted, two-vertex-connected graphs.

Descriptional complexity of deterministic pushdown machines and labelled transition-system synthesis provide the general background [8, 251, 252]. Previous work on semantic normalization and exact LIFO lifting [227, 228], as well as a state–stack product lower bound for deterministic pushdown automata [253], motivate the resource-sensitive viewpoint. The present model fixes one physical control state and minimizes the stack alphabet while retaining the full labelled partial-action structure.

114.1.1 Main contributions

(i) An exact formula for the least stack-alphabet size of every guarded upper mask, together with a canonical realization of height two.

(ii) A complete all-budget formula for attainable population, joint saturation, and their potentially nonmonotone gap.

(iii) A vertex-cover identity and NP-completeness theorem at the first feasible alphabet budget, with a corresponding next-budget formula.

(iv) A sharp height-one versus height-two cost comparison and a complexity transition for the protected-family profile.

114.2 Guarded partial actions and physical devices

Let $H = (V, E)$ be a finite simple undirected two-vertex-connected graph, let $M = |V| \geq 3$, and choose a root $r \in V$. Every edge $e = \{u, v\}$ has a private input letter h_e. For a mask $S \subseteq V$ containing r, define

$$J_S = (V \times \{0\}) \cup (S \times \{1\}).$$

For $e = \{u, v\}$, the action h_e interchanges (u, b) and (v, b) when both endpoints belong to J_S and is undefined elsewhere. The vertical action z interchanges $(v, 0)$ and $(v, 1)$ exactly for $v \in S$. The root is $(r, 0)$. These actions are partial involutions, and all source states are root reachable.

A physical device consists of exactly one control state and an alphabet Γ of g stack symbols. A physical step reads one input and the current top symbol (or emptiness), then either pushes one symbol, pops the top, rewrites the top, keeps the stack unchanged, or rejects. There are no epsilon steps. A valid realization is an injective root-respecting map $\sigma : J_S \to \Gamma^*$, with $\sigma(r, 0) = \varepsilon$, which is history-independent and transports *and reflects* the domain and result of every input action. Rules depend only on the pair (input, visible top/empty); they need not be specified on unreachable configurations. Set

$$g_{\min}(H, S) = \min\{|\Gamma| : J_S \text{ admits such a realization}\}.$$

Let

$$A(S) = \{v \in S : \deg_{H[S]}(v) > 0\}, \qquad I(S) = S \setminus A(S).$$

114.3 An exact alphabet-size theorem

Lemma 114.1 (Stack-clique obstruction)**.** *The undirected one-step configuration graph of a one-control-state top-rewrite/push/pop stack is a block graph whose maximal cliques are $C_w = \{w\} \cup \{w\gamma : \gamma \in \Gamma\}$, for $w \in \Gamma^*$. Every finite two-vertex-connected source subgraph represented injectively by single physical steps lies inside one such clique. If it contains the empty root, it lies in C_ε.*

Proof. For any two adjacent distinct words, either one extends the other by one symbol, or both differ only in their top symbol and share the same shorter prefix. Hence every edge belongs to a prefix clique C_w. Distinct maximal prefix cliques intersect in at most one word. The resulting clique-intersection structure has no cycles of blocks, so an injective two-connected subgraph lies in a single block. The empty root appears only in C_ε. □

Theorem 114.2 (Exact alphabet cost)**.** *For every allowed rooted graph and every $r \in S \subseteq V$,*

$$\boxed{g_{\min}(H, S) = M - 1 + |A(S) \cup \{r\}| + \mathbf{1}_{\{r \in A(S),\ I(S) \neq \varnothing\}}.}$$

In particular, deciding realizability for a specified mask S and bound g takes $O(M + |E|)$ time.

Proof. The lower layer is two-vertex-connected, so Lemma 114.1 places its M distinct codes in C_ε. The root code is empty, forcing $M - 1$ stack symbols. The upper root is a distinct neighbor of the empty root, so its code also lies in C_ε. If $uv \in E(H[S])$, the square on $(u, 0), (v, 0), (u, 1), (v, 1)$ attaches along its lower edge to the two-connected lower layer. The union is again two-connected. Thus every upper state indexed by $A(S)$ lies in C_ε, accounting for $|A(S) \cup \{r\}|$ additional symbols. When $r \in A(S)$ and $I(S)$ is nonempty, all the symbols already counted are used at lower states or upper states with at least one

enabled horizontal action. Each upper state indexed by $I(S)$ rejects all horizontal actions; its nonempty code therefore needs a different visible top symbol, adding one symbol.

For attainment, assign the lower root to ε and every other lower state $(v, 0)$ to a distinct one-symbol code c_v. Assign a fresh one-symbol code d_v to each upper state indexed by $A(S)$. If $r \notin A(S)$, assign a fresh symbol b to upper root $(r, 1)$; then every other upper isolated state $(v, 1)$ gets code $c_v b$. If $r \in A(S)$ but $I(S) \neq \varnothing$, introduce one further symbol b and likewise code isolated upper states by $c_v b$. The upper root is now coded by d_r. With no isolated upper states no b is necessary. All source transitions have physically adjacent codes: the lower layer uses pushes, pops, and top rewrites between ε, c_v; upper active horizontal edges use rewrites between d_u, d_v; z acts by a rewrite or a pop on the upper side; and every upper isolated state ends in b, whose horizontal instructions always reject and whose z instruction always pops. Rules on different visible tops cannot conflict, and both enabledness and targets match the source. This attains the lower bound. □

Corollary 114.3 (Depth-two optimal normal form and exact depth-one penalty)**.** *Let $g^{(h)}_{\min}(H, S)$ denote the least stack-alphabet size of a valid realization whose reachable stack height is at most h, under precisely the same single-step physical instructions and rooted source-faithfulness contract. For every $r \in S$,*

$$g^{(1)}_{\min}(H, S) = M + |S| - 1, \qquad g^{(h)}_{\min}(H, S) = g_{\min}(H, S) \quad (h \geq 2).$$

Moreover the exact penalty for insisting on stack height at most one is

$$\boxed{g^{(1)}_{\min}(H, S) - g_{\min}(H, S) = \max\{|I(S)| - 1, 0\}.}$$

Proof. The source has $M + |S|$ distinct named states, and an injective rooted realization of height at most one has only $1 + g$ possible codes, forcing $g \geq M + |S| - 1$. This is attained by coding the root by the empty word and each other source state by a different stack symbol: every legal source transition can then be implemented by one push, pop, or top rewrite, and each illegal action is rejected using its unique visible code. The constructive coding in Theorem 114.2 uses words of length at most two, so it attains $g_{\min}(H, S)$ for every $h \geq 2$. Finally, if $r \notin A(S)$ then $r \in I(S)$ and $|A(S) \cup \{r\}| = |A(S)| + 1$; if $r \in A(S)$ then the isolated states are disjoint from the root. Substitution into Theorem 114.2 proves the stated penalty in both cases. □

Remark 114.4 (Depth, rather than total stack height, can save alphabet)*.* The precise saving comes from reusing one top marker for all but the first horizontally isolated upper states, each with a different lower prefix. Thus unbounded physical stack depth is unnecessary for the optimal alphabet count in this family: height two suffices.

114.4 Protected saturation profiles

For fixed H, r, g, let $\mathcal{S}_g = \{S \subseteq V : r \in S,\ g_{\min}(H, S) \leq g\}$. If this family is nonempty, define the accessible population envelope, saturation, and family gap by

$$E_g = \bigcup_{S \in \mathcal{S}_g} J_S, \quad p_g = |E_g|, \quad \mathrm{Sat}_g = \max_{S \in \mathcal{S}_g} |J_S|, \quad \Delta_g = p_g - \mathrm{Sat}_g\,.$$

No saturation/gap is assigned when $\mathcal{S}_g = \varnothing$. These are family-profile invariants: J_S is *not* constant as S changes.

Remark 114.5 (Only the gap can be nonmonotone)*.* Because $\mathcal{S}_g \subseteq \mathcal{S}_{g+1}$, both p_g and Sat_g are nondecreasing whenever defined. The difference $\Delta_g = p_g - \mathrm{Sat}_g$, however, need not be monotone: at the first feasible budget its population envelope can expand faster than the largest jointly realizable mask. Accordingly, the nonmonotonicity claimed in the title is a nonmonotonicity of the *gap*, never of Sat_g itself.

Theorem 114.6 (All-budget characterization)**.** *For $g < M$, $\mathcal{S}_g = \varnothing$. For $g \geq M$,*

$$\mathrm{Sat}_g = M + \max_{S \in \mathcal{S}_g} |S|.$$

Writing $d_r = \deg_H(r)$ and $U = V \setminus N_H[r]$, the envelopes are

$$p_M = 2M - d_r, \qquad p_g = 2M \quad (g \geq M+1).$$

In particular,

$$\boxed{\Delta_M = \tau(H[U])}, \qquad \boxed{\Delta_g = M - \max_{S \in \mathcal{S}_g} |S| \quad (g \geq M+1)}.$$

For all $g \geq 2M-1$, $\Delta_g = 0$.

Proof. The empty-core mask $S = \{r\}$ costs exactly M, so the family is nonempty from that budget onward. At $g = M$, Theorem 114.2 permits precisely the independent masks $S \ni r$. These have form $\{r\} \cup T$ with T an independent subset of $H[U]$. Their union reaches upper vertices $\{r\} \cup U$ and their maximum size is $1 + \alpha(H[U])$. Thus $p_M = M + 1 + |U| = 2M - d_r$, $\mathrm{Sat}_M = M + 1 + \alpha(H[U])$, and $\Delta_M = \tau(H[U])$ by the vertex cover/independent-set identity.

At $g = M+1$, every upper vertex u becomes individually available: if u is not adjacent to r, take the independent mask $\{r, u\}$; otherwise take the adjacent two-vertex mask $\{r, u\}$, for which Theorem 114.2 gives cost $M+1$. Hence E_g contains both layers in full for every $g \geq M+1$. The saturation is the maximum source-state population by definition and the exact cost theorem. For $S = V$, $A(S) = V$, $I(S) = \varnothing$, so $g_{\min}(H, V) = 2M - 1$, yielding the last claim. □

Corollary 114.7 (A strict nonmonotone gap)**.** *For every such graph, $\Delta_{M+1} > \Delta_M$. More precisely, with $d_r = \deg_H(r)$,*

$$\Delta_{M+1} - \Delta_M = \begin{cases} d_r, & U \neq \varnothing, \\ d_r - 1, & U = \varnothing. \end{cases}$$

Thereafter Δ_g is nonincreasing for $g \geq M+1$, and first vanishes at $g = 2M-1$.

Proof. At $g = M+1$, every eligible upper mask is either independent or has exactly two adjacent vertices, one of which is r: an activated upper edge forces two upper vertices into the root clique, and any further upper vertex would require an additional root symbol or an unused horizontally silent symbol. Thus $\max |S| = \max\{1 + \alpha(H[U]), 2\}$. If U is nonempty, then $\alpha(H[U]) \geq 1$ and the maximum is $1 + \alpha(H[U])$; subtracting the expression for Δ_M gives d_r. If U is empty, the maximum is 2 and the difference is $M - 2 = d_r - 1 > 0$. For $g \geq M+1$ the envelope is constant, while the maximum feasible $|S|$ is nondecreasing, so Δ_g is nonincreasing. The full upper mask requires precisely $2M-1$ symbols by Theorem 114.2. □

Corollary 114.8 (Hardness persists at the next alphabet budget)**.** *Let $U = V \setminus N_H[r]$ and $d_r = \deg_H(r)$. At $g = M+1$, for every stack-height bound $h \geq 2$,*

$$\Delta^{(h)}_{M+1} = \begin{cases} \tau(H[U]) + d_r, & U \neq \varnothing, \\ d_r - 1, & U = \varnothing. \end{cases}$$

The decision problem whether $\Delta^{(h)}_{M+1} \leq t$ is NP-complete, even when $h = 2$ is fixed. This claim concerns the variable-mask family profile, not a fixed-source assembly defect.

Proof. By Corollary 114.7, the difference between the gaps at $M+1$ and M is d_r if $U \neq \varnothing$ and $d_r - 1$ otherwise. Combine with $\Delta_M = \tau(H[U])$ and $\tau(H[\varnothing]) = 0$. For NP-hardness use the rooted graph $H(X)$ of Theorem 114.10 with a nonempty input graph X. The root has degree two and $U \cong V(X)$, so $\Delta^{(2)}_{M+1} = \tau(X) + 2$. Restriction of VERTEX COVER to nonempty graphs remains NP-complete, and the construction is polynomial. Membership in NP follows from an independent-set certificate for the corresponding bound on $\tau(X)$, or directly by the exact characterization and a vertex-cover certificate for $H[U]$. The fixed-depth-two normal form extends the result to every $h \geq 2$. □

Corollary 114.9 (Connectivity tail)**.** *Let $\kappa(H)$ be vertex connectivity. For $1 \leq j \leq \min\{\kappa(H), M-2\}$,*

$$\Delta_{2M-1-j} = j.$$

Proof. Delete j vertices outside the root, retaining a connected induced subgraph on $M - j \geq 2$ vertices. One may choose the deleted vertices so that the remainder is connected and contains the root: iteratively delete non-root leaves of a spanning tree, recomputing a spanning tree when necessary. The retained connected induced graph has no isolated upper vertex and hence costs $M - 1 + (M - j) = 2M - 1 - j$. This gives $\Delta \leq j$. Conversely, any mask with more than $M - j$ vertices excludes fewer than $j \leq \kappa(H)$ vertices. Its induced graph remains connected on at least two vertices and has cost $M - 1 + |S| > 2M - 1 - j$, a contradiction. □

114.5 Complexity at the first feasible budget

Theorem 114.10 (Restricted vertex-cover transfer)**.** *Given a rooted two-vertex-connected graph H and integer t, deciding whether the guarded-family gap at budget M is at most t is NP-complete.*

Proof. Theorem 114.6 gives $\Delta_M = \tau(H[V \setminus N_H[r]])$, so membership in NP follows by a standard vertex-cover certificate. For hardness, let X be an arbitrary finite simple graph. Adjoin vertices r, p, q; connect r, p, q as a triangle; connect both p and q to every vertex of X; and preserve all edges of X. The resulting graph $H(X)$ is two-vertex-connected, $N_{H(X)}[r] = \{r, p, q\}$, and $H(X)[V \setminus N[r]] \cong X$. Thus $\Delta_{|V(H(X))|} = \tau(X)$, providing a polynomial-time reduction from vertex cover. □

114.6 A height-induced complexity transition

For a physical stack-depth bound $h \geq 1$ and a stack-alphabet budget g, write $\mathcal{S}^{(h)}_g = \{S \ni r : g^{(h)}_{\min}(H, S) \leq g\}$. Define $p^{(h)}_g$, $\mathrm{Sat}^{(h)}_g$ and $\Delta^{(h)}_g$ from the same family-envelope definitions as in Section 114.4. The two height regimes admit an exact comparison at *every* alphabet budget.

Theorem 114.11 (Depth-one profile and depth-two activation of vertex cover)**.** *For any rooted two-vertex-connected graph H on $M \geq 3$ vertices, $\mathcal{S}^{(1)}_g = \varnothing$ for $g < M$. At $g = M$,*

$$(p^{(1)}_M, \mathrm{Sat}^{(1)}_M, \Delta^{(1)}_M) = (M+1, M+1, 0).$$

For $g \geq M+1$,

$$\boxed{p^{(1)}_g = 2M, \qquad \mathrm{Sat}^{(1)}_g = M + \min\{M, g - M + 1\}, \qquad \Delta^{(1)}_g = \max\{0, 2M - g - 1\}.}$$

For every $h \geq 2$, these family invariants agree with the unrestricted-height profile of Theorem 114.6. In particular, at the same minimum feasible alphabet budget *$g = M$,*

$$\boxed{\Delta^{(1)}_M = 0, \qquad \Delta^{(h)}_M = \tau\big(H[V \setminus N_H[r]]\big) \quad (h \geq 2).}$$

Consequently, computing $\Delta_g^{(1)}$ *is constant-time once* M, g *are known, whereas deciding whether* $\Delta_M^{(h)} \le t$ *for variable* H, r, t *and any fixed* $h \ge 2$ *is NP-complete.*

Proof. By Corollary 114.3, height one realizes precisely the masks $S \ni r$ with $M + |S| - 1 \le g$. Since $|S| \ge 1$, no mask is realizable below M. At $g = M$ the only realizable mask is $S = \{r\}$, with $M + 1$ named states; the population envelope and saturation therefore coincide and the gap is zero. For $g \ge M + 1$, each vertex $v \ne r$ occurs in the mask $\{r, v\}$, so the population envelope contains both layers and has size $2M$. The largest feasible mask has cardinality $\min\{M, g - M + 1\}$, yielding the stated saturation and gap. The optimum depth-two construction from Corollary 114.3 proves $\mathcal{S}_g^{(h)} = \mathcal{S}_g$ for every $h \ge 2$. The first-budget identity is Theorem 114.6. The constant-time evaluation is immediate from the closed profile; NP-completeness for the depth-two (hence every $h \ge 2$) case follows from the explicit polynomial reduction in Theorem 114.10. All decision claims concern the *variable-mask family profile*, not minimization of a fixed semantic source. □

Remark 114.12 (Algorithmic interpretation). At depth one the physical alphabet has to name every semantic state; no graph optimization remains in the family profile. A second cell allows upper states with all horizontal actions disabled to share the same top marker over distinct lower prefixes; choosing which upper states are simultaneously feasible introduces maximum-independent-set optimization. Thus a second cell activates graph-dependent *family-profile* complexity, even though each fixed-mask alphabet minimum is computable in linear time. The statement is not an NP-completeness result for arbitrary one-state pushdown automata or a single fixed-source assembly defect.

114.7 Related work

The physical machine reads the current input and the visible stack top; push, pop and rewrite are all available according to that pair. Accordingly, the same input letter may induce different stack operations at different source states. This is distinct from the input-partitioned transition types of visibly pushdown automata [8]. Work on accessible-stack learning [250] and bounded-height pushdown language recognition [87] provides additional automata-theoretic context.

Exact synthesis of labelled transition systems has also been studied for Petri nets and Boolean nets [251, 252]. Our target is a single-control-state stack and our measured resource is its alphabet size. Classical block-graph structure [248] supplies the geometric obstruction used in the lower bound, and classical vertex cover [254] supplies the complexity problem recovered from saturation. The author's earlier work concerns semantic normalization, LIFO lifting, and state–stack tradeoffs [227, 228, 253].

114.8 An illustrative four-cycle and concluding remarks

For a rooted four-cycle, the successive gaps at alphabet budgets four through seven are

$$(\Delta_4, \Delta_5, \Delta_6, \Delta_7) = (0, 2, 1, 0).$$

The attainable population grows from $p_4 = 6$ to $p_5 = 8$; the largest jointly representable population grows at a different rate. Hence although both population quantities are non-decreasing, their difference can first increase and subsequently fall to zero. The exact alphabet cost, depth-two normal form, and vertex-cover characterization provide a unified description of this resource effect for guarded two-layer actions.

Chapter 115

Marked Transport, Four-Axis Recovery, and Non-Transfer

Research branch — Typed synthesis and recovery. *A geometric fiber, a chronological shadow, a path-outcome fiber, a protected compatibility class, and a physical resource coordinate are not the same object. This chapter records the exact bridges that permit information to pass between them, proves source-faithful recovery for four theorem-bearing branches already developed in the monograph, and records the corresponding non-transfer principles.*

115.1 Marked topology, transport, and holonomy

Intrinsic topology does not itself specify a physical action. Let a marked displacement system be

$$\mathfrak{M} = (G, L, \partial),$$

where G is a loop group, L is a displacement group, and $\partial : G \to L$ is a homomorphism. The pure sector is $K_0 = \ker \partial$. A chosen physical transport representation is a homomorphism $\rho : G \to H$.

Theorem 115.1 (Pure-holonomy factorization)**.** *The following are equivalent:*

(a) $\rho(K_0) = \{e\}$*;*

(b) $K_0 \subseteq \ker \rho$*;*

(c) there is a unique homomorphism $\bar{\rho} : \mathrm{im}(\partial) \to H$ *such that* $\rho = \bar{\rho}\,\partial$*.*

Now suppose loops act on a physical carrier set Ω, and fix $c \in \Omega$. The orbit $A = G \cdot c$ is a set of physical path outcomes with a common semantic endpoint.

Theorem 115.2 (Orbit-to-section defect)**.** *If the orbit A is finite, then the path-outcome multiplicity at the common semantic state is at least $|A|$, and every section of the fixed realization incurs at least $|A| - 1$ off-section outcomes there. If the action is transitive on the orbit,*

$$|A| = [G : \mathrm{Stab}_G(c)].$$

A resource conclusion requires another bridge.

Theorem 115.3 (Protected holonomy-orbit transfer)**.** *Assume the orbit alternatives carry a protected readout $\beta : A \to Y$ whose distinct values cannot share one level-h observation cell under the declared confined repair contract. Then*

$$D + X_h + V_A(h) \geq |\beta(A)|.$$

If β is injective, the right side is $|A|$.

Thus the valid chain is

$$\begin{aligned}\text{marked topology} &\longrightarrow \text{chosen transport} \longrightarrow \text{orbit/holonomy}\\ &\longrightarrow \text{protected readout} \longrightarrow \text{obligation pressure}\\ &\longrightarrow \text{resource, if a capacity law is supplied.}\end{aligned}$$

Deleting any of the middle hypotheses is not justified in general.

115.1.1 A typed transfer theorem

The previous branches can be summarized without collapsing them into one implication.

Theorem 115.4 (Typed transfer chain)**.** *Let X be a finite structural witness family. Suppose a declared bridge produces a finite protected label image Y, and suppose every level-h observation class of confined canonical sources lies in one protected-compatible face. Then*

$$D + X_h + V_h \geq |Y|.$$

If, in addition, a category-specific resource envelope gives

$$D + X_h + V_h \leq B_h(r)$$

for all realizations of resource at most r, then $|Y| > B_h(r)$ excludes resource r. The conclusion depends on the witness-to-label bridge and on the resource envelope; neither is a consequence of the bare structural witness alone.

Proof. The bridge produces a protected distinguishability certificate on Y, so the obligation-cover theorem gives the first inequality. The second statement is immediate by contradiction from the category-specific upper envelope. □

Three common instances are now transparent. A geometric compatibility witness requires a protected realization/readout. An exact chronological distinction requires a protected decoder or future obligation; a common shadow alone is insufficient. A holonomy class requires a physical transport action, then an orbit, then a protected readout. In all three cases the last step to a physical resource is category-sensitive.

115.2 Recovery of four source theories

USNRT was extracted from four previously developed theorem packages. The recovery claim in this section is therefore not that USNRT re-proves those source theorems from weaker axioms. It is a source-faithfulness statement: the source data, hypotheses, and theorem-bearing distinctions embed into typed USNRT components and are recovered by an explicit forgetful map. The source results used in this audit occur in earlier Parts of this monograph; their external provenance is recorded at the corresponding source locations and in the bibliography where appropriate.

115.2.1 Recovery datum

For a source class Src_i, let USNRT_i be the typed specialization carrying exactly the source structures listed below. A recovery datum consists of

$$I_i : \mathrm{Src}_i \longrightarrow \mathrm{USNRT}_i, \qquad R_i : \mathrm{im}(I_i) \longrightarrow \mathrm{Src}_i$$

with $R_i I_i \simeq \mathrm{id}$ up to the native source isomorphism. In addition, for each audited source theorem we record the source statement, the USNRT component that carries each hypothesis, and the reconstruction that returns the original theorem.

115.2.2 Axis I: completion geometry

Source. The source package is the periodic shortest-path compatibility theory developed in Chapters 20–23. The audited source statements are the exact universal row-complexity and zero-shell-defect theorem, the universal k-fiber theorem, and the extension-fiber completeness consequences.

USNRT data and bridge. The fixed Bellman/completion system, exact distance, lower-order compatibility signature, higher-order extension fibers, and optional marked data are copied into the geometric enrichment $\mathcal{G}_\bullet$. Purely geometric instances use the identity realization; no physical resource is inferred until a separate protected-readout/capacity bridge is supplied.

Reconstruction. The map R_I forgets unused realization/protection layers and returns the original periodic system, exact distance, compatibility hierarchy, and extension fibers. In particular a universal k-fiber is not identified with a representation fiber. The Bellman coefficient construction and common-shell hypotheses remain source-specific.

115.2.3 Axis II: exact pushdown normalization

Source. The source package is the exact-normalization and General-Theory development of the preceding Parts, together with the confluence specialization in Chapter 63. The audited theorem family comprises finite-observation/common-target collapse capacity, the bounded-local three-stage obstruction decomposition, and rooted path-outcome coherence.

USNRT data and bridge. Semantic dynamics, physical configurations, the exact projection, a chosen section when present, bounded observations, protected future obligations, and local schedule/action data are carried literally by the exact-realization and protection layers. Under the source tail-injectivity/noncoalescence and transition-closure hypotheses, the same-target predecessor family becomes a complete-conflict protected-obligation instance, so the USNRT capacity inequality returns the source finite-observation bound. The later allocation and local-alignment stages remain source-specific rather than consequences of the cover bound.

Reconstruction. The forgetful map returns the original semantic/physical policy, section, observation map, and local action data. It preserves the distinctions "representation debt versus behavioral correctness", "observation collision versus hidden-tail noncoalescence", and "capacity screen versus exact local alignment".

115.2.4 Axis III: chronological arithmetic

Source. The source is the CCR-k General Theory in Chapter 34. The audited package consists of terminal-block separation, higher-band lower-Hessenberg realization, the subdiagonal light-cone theorem, normalized forward transport, inverse source reconstruction, the stationary parameter lattice, and the universal raw Stirling shadow.

USNRT data and bridge. The deterministic chronology is placed in $\mathcal{J}_\bullet$: exact connected source sequences, Hessenberg bands, normalized transport, inverse reconstruction, stationary parameters, and raw shadow maps remain separate typed coordinates. A common shadow is not itself a protected-information conclusion; a protected decoder/readout is an additional bridge.

Reconstruction. Forgetting unrelated realization layers returns the exact source sequences and shadow maps separately. Thus exact chronological data are not identified with their mod-p shadow, and the CCR-k theorem retains the source hypotheses under which its parameter lattice and shadow statements are valid.

115.2.5 Axis IV: information and resource complexity

Source. The source is the retained-information and physical-exposure branch beginning at Chapter 99. Its filtered collapse-capacity theorem gives the inequality

$$D + X_h + V(h) \geq N$$

under hidden-tail noncoalescence, and its source-stack exposure profile gives the deterministic top-access pushdown specialization

$$D + X_h + q_{\mathrm{src}} \sum_{j=0}^{h} g_{\mathrm{src}}^{j} \geq N.$$

USNRT data and bridge. The source family, protected contract, observation filtration, confinement predicate, and debt/escape counts are represented by the protected-obligation branch. In the same-stack saturation specialization, choose one protected source representative for each distinct joint label. Then the collapse-family size is $N = R$, where R is the compiler's joint rank, and the source-stack exposure theorem gives exactly the bound used in Chapter 113.

Reconstruction. The map returns the source family, observation filtration, exposure depths, debt counts, control/alphabet parameters, and protected recoding contract. Source exposure is not identified with maximum stack height, and a capacity inequality is not identified with a matching compiler.

Theorem 115.5 (Source-faithful four-axis recovery)**.** *For the four cited source packages above, the displayed interpretations define recovery data (I_i, R_i) satisfying*

$$R_i I_i \simeq \mathrm{id}_{\mathrm{Src}_i}.$$

For each audited theorem family, every source hypothesis is assigned to an explicit USNRT component or to an explicitly retained source-specific enrichment, and reconstruction returns the source theorem without adding a converse.

Proof. The four interpretations are componentwise injections of the source data listed above. In each case R_i is the projection onto those components, so $R_i I_i$ is the identity up to the source's native renaming/isomorphism. The theorem-level claim is checked axis by axis: Bellman/common-shell data are retained on Axis I; tail noncoalescence, transition closure, and local alignment are retained on Axis II; exact Hessenberg transport and shadow maps remain distinct on Axis III; and the observation filtration plus top-access capacity law are retained on Axis IV. Since no source-specific hypothesis is deleted or promoted to a generic USNRT axiom, reconstruction yields the original implication and no additional converse. □

115.2.6 Recovery/transfer consistency

The recovery maps also preserve the direction of the cross-axis bridges used in this synthesis. Four checks are especially important.

(R1) *Axis I to Axis IV.* A universal compatibility fiber becomes a physical lower bound only after a protected collapse/readout bridge and the Axis-IV capacity theorem are inserted.

(R2) *Axis II to obligation capacity.* Common-target finite-observation collapse is the complete-conflict specialization of protected-obligation capacity; the subsequent allocation and local-alignment stages remain source-specific.

(R3) *Axis III to protected information.* Exact chronological alternatives with one common raw shadow create protected pressure only when a future decoder/readout distinguishes those exact alternatives.

(R4) *Axis I topology to transport/resource.* Marked topology is distinct from a chosen physical action; resource consequences additionally require protected readout and a category-specific capacity envelope.

Corollary 115.6 (Non-collapse of the unification)**.** *On the cited source packages, USNRT is a typed host rather than a quotient that identifies theorem-bearing source data. The recovery statement does not assert that arbitrary objects satisfying the bare USNRT axioms must satisfy the source-specific Bellman, pushdown, CCR-k, or exposure theorems.*

115.2.7 Dependency boundary

The proved USNRT core uses only the source packages and hypotheses explicitly cited above. Later arithmetic refinements, unresolved resource-frontier calculations, or category-specific compiler constructions are specialization questions and are not premises of the exact-realization, protected-obligation, saturation, holonomy, or recovery results in this Part.

115.3 Non-transfer principles and scope

Several tempting implications are invalid without additional hypotheses. They are recorded here because the boundary of a transfer theorem is part of the theorem.

(N1) **Topology alone does not imply physical cost.** A physical transport representation must first be chosen; a protected readout and capacity law are additionally required for resource consequences.

(N2) **Chronological shadow loss alone does not imply physical cost.** Lost exact chronology becomes a resource issue only if the lost distinctions remain protected by a future decoder/readout contract.

(N3) **Geometry alone does not imply physical cost.** A geometric witness needs a protected-realization bridge and then a category-specific capacity envelope.

(N4) **Pairwise availability does not imply global assembly.** This is measured exactly by the assembly defect δ_{asm}.

(N5) **A capacity lower bound does not imply a matching construction.** Lower bounds are necessary conditions; exact optimality additionally needs an upper-bound construction.

(N6) **Abstract product realization does not imply one-stack pushdown product closure.** A Cartesian product may carry two independent physical memories.

These non-transfer principles are not limitations of the framework; they are the mechanism by which it prevents category errors between semantic, informational, topological, and physical statements.

115.4 Relation to neighboring frameworks

USNRT has substantial classical neighbors, and the distinctions matter for the scope of the contribution. Refinement mappings explicitly relate lower-level state machines to higher-level specifications by state-space maps preserving steps and behaviors [139]; refinement calculi formalize correctness-preserving program and data refinement [140]; abstract interpretation constructs composable hierarchies of semantics through abstraction maps [141]; and coalgebra provides a broad categorical theory of transition and dynamical systems [142]. Accordingly, the existence of an abstract/concrete state map, a semantics-preserving refinement relation, or a categorical transition-system shell is not claimed as new here.

Rewriting theory studies confluence, normal forms, and canonicalization [143, 61]. Fractional covers and zero-error/confusability methods provide classical combinatorial and information-theoretic lower-bound shells [144, 145, 146, 147]. Sheaf-theoretic contextuality identifies local/global incompatibility with global-section obstructions [148]; valuation algebras generalize disagreement across information sources [149]; bundle approaches relate contextuality and holonomy [150]; and resource theories formalize monotones and free operations in operational settings [151]. Thus simplicial compatibility, local/global obstruction, holonomy, fractional covering, and resource monotonicity are likewise not claimed as inventions of USNRT.

The contribution of this synthesis Part is the combined typed architecture and its theorem package: (i) exact realization with an optional rather than intrinsic section; (ii) an exact path-outcome criterion and debt formula for fixed-template canonicalization; (iii) protected-obligation compatibility as a separate information layer; (iv) explicit conditional bridges from geometry, chronology, or transported holonomy into that layer; and (v) category-relative transfer saturation that separates missing pairwise compatibility from failure of simultaneous assembly in one physical model. The novelty claim is therefore a claim about this specialized synthesis and the resulting transfer/non-transfer theorems, not about the classical ingredients individually.

To the best of our knowledge after the targeted literature comparison summarized above, no single neighboring framework contains this complete typed package with the same theorem roles. We do not treat that comparison as a historical-priority certificate: later prior art may revise the novelty classification without changing the mathematical statements proved here.

115.5 Theorem-scope and bridge ledger

The purpose of this ledger is to make the logical status of the main theorem families explicit. A statement is *intrinsic/formal* when its conclusion follows from the typed mathematical data stated in the theorem and does not need a model-specific physical capacity theorem. A statement is *conditional* when an explicit bridge must be supplied. A statement is *category-sensitive* when its numerical physical consequence depends on the admissible realization category or resource evaluation.

115.5.1 Intrinsic/formal results

The following results belong to the formal USNRT core once their displayed inputs are given:

(I1) the exact-realization category and the fully faithful identity-realization embedding;

(I2) the rooted endpoint-factorization theorem and the fixed-template section-defect formula;

(I3) formation of the protected obligation complex and its finite face-cover invariants;

(I4) rectangular compatibility-product bounds and the fractional-cover rate identity at the derived-complex level;

(I5) exact finite transfer-loss telescoping;

(I6) abstract asynchronous product realization for coordinate products;

(I7) transfer-saturation arithmetic, pair/assembly defect decomposition, and optimal same-model loss once a compiler category has been fixed.

None of these results, by itself, asserts a stack-height, state-count, alphabet-size, exposure, or Pareto lower bound.

115.5.2 Conditional bridges

Four bridge types recur throughout this Part.

(C1) A geometric witness produces protected information only when a witness constructor and protected realization/readout are provided.

(C2) Exact chronology produces protected information only when a future decoder or equivalent protected obligation distinguishes the exact alternatives. A common shadow is insufficient.

(C3) Path-outcome multiplicity produces obligation pressure only when the alternatives receive protected repair or continuation labels.

(C4) Marked loops produce physical orbit information only after a transport action on a carrier is specified.

These hypotheses are not technical decoration. Removing them changes the type of the statement and can make the intended conclusion false.

115.5.3 Category-sensitive conclusions

A protected-information lower bound becomes a physical resource statement only after a category-specific capacity envelope is supplied. Thus observation capacity, debt/escape envelopes, source-stack exposure, state/stack/alphabet lower bounds, and Pareto exclusions are category-sensitive. The source-stack formula in Section 113.4.3 is one such specialization; it is not promoted to an invariant of bare semantics.

Likewise, a physical repeated-product interpretation of h_{obl} requires an iterated rectangular realization family. Without that hypothesis, $h_{\mathrm{obl}}(K) = \log \kappa_f(K)$ remains a theorem about the derived protected compatibility complex.

115.6 Four-axis recovery ledger

This section gives the recovery audit in the same five-field format for each axis. It is intentionally a verification ledger rather than a second proof of the source theorems.

115.6.1 Axis I: completion geometry

Source. Chapters 20–23.
Audited theorem family. Exact universal row complexity, zero shell defect, the universal k-fiber theorem, and extension-fiber completeness.
Embedding. Fixed periodic/Bellman system, exact distance, compatibility hierarchy, lower-order signature, higher-order extension fibers, and marked data are placed in the geometric enrichment $\mathcal{G}_\bullet$; pure geometric examples use the identity realization.
Bridge. No physical conclusion is drawn without an additional protected-readout/capacity bridge.
Reconstruction. Forgetting unrelated USNRT layers returns the source system and its theorem-bearing compatibility data; in particular, compatibility fibers are not identified with representation fibers.

115.6.2 Axis II: exact pushdown normalization

Source. The preceding exact-normalization/General-Theory Parts and Chapter 63.
Audited theorem family. Finite-observation/common-target collapse capacity, the bounded-local three-stage obstruction decomposition, and rooted path-outcome coherence.
Embedding. Semantic dynamics, physical configurations, exact projection, chosen section, bounded observations, protected obligations, and local action/schedule data are carried by the realization and protection layers.
Bridge. Tail-injectivity/noncoalescence and transition closure turn the same-target predecessor family into the complete-conflict protected-obligation specialization; later allocation and local-alignment stages remain source-specific.
Reconstruction. Projection returns the original policy, section, observations, and local action data while preserving the distinction between a capacity screen and exact local realizability.

115.6.3 Axis III: chronological arithmetic

Source. Chapter 34.
Audited theorem family. Terminal-block separation, higher-band lower-Hessenberg realization, subdiagonal light cone, normalized forward transport, inverse source reconstruction, stationary parameter lattice, and raw Stirling shadow.
Embedding. Exact connected sources, Hessenberg bands, transport/inverse data, stationary parameters, and shadow maps are stored as separate chronological coordinates in $\mathcal{J}_\bullet$.
Bridge. A common shadow creates protected-information pressure only when a future decoder/readout distinguishes the exact alternatives.
Reconstruction. Forgetting unrelated layers returns exact and shadow data separately, with all source hypotheses for stationarity, divisibility, and transport retained.

115.6.4 Axis IV: information and resource complexity

Source. The retained-information/exposure branch beginning at Chapter 99.
Audited theorem family. Filtered collapse-capacity and source-stack exposure theorems.
Embedding. The collapse source family, protected contract, observation filtration, confinement predicate, debt/escape counts, and resource parameters are represented in the protected-obligation/resource layers.
Bridge. The abstract collapse theorem gives $D + X_h + V(h) \geq N$ under hidden-tail noncoalescence, while the pushdown specialization gives $D + X_h + q_{\mathrm{src}} \sum_{j=0}^{h} g_{\mathrm{src}}^{j} \geq N$ for deterministic top-access pushdown sources. Selecting one source representative per distinct joint protected label sets $N = R$ in the saturation specialization.
Reconstruction. Projection returns the source filtration, exposure depths, debt data, and source-local state/alphabet parameters; exposure is not identified with stack height and a capacity inequality is not identified with a matching compiler.

115.6.5 Ledger conclusion

For every axis, the reconstruction map returns the audited source package after interpretation, while source-specific hypotheses remain explicit. The ledger therefore certifies source faithfulness; it does not assert that bare USNRT axioms imply any of the four source theorem families.

115.7 Right-turn spectra and deterministic recognition limits

For tree-walking automata, determinism, unambiguity, and visible-pebble access must be distinguished as recognition contracts. The three-marker non-determinization witness of Bojanczyk and Colcombet [156] provides a fixed starting obstruction, not a new separation claim. For this family the relations $\mathsf{DTWA} \subseteq \mathsf{UTWA} \subsetneq \mathsf{NTWA} \subsetneq \mathsf{REG}$ are known, while strictness of $\mathsf{DTWA} \subseteq \mathsf{UTWA}$ is not established here. The one-pebble class is not an intermediate member of that chain.

115.7.1 Spectrum model and exact regularity boundary

Let $\mathcal{W}_3$ be the set of finite ordered full binary trees whose internal nodes are blank and whose leaves are labelled a or b, with exactly three a-labelled leaves. Denote those three leaves, in left-to-right order, by a_1, a_2, a_3. A basic deterministic tree-walking automaton recognizes $\mathcal{W}_3$: perform a standard complete depth-first tour and saturate the number of marked leaves at four.

For $t \in \mathcal{W}_3$, define

$$\begin{aligned} &r(t) = \mathrm{lca}(a_1, a_2, a_3), && q(t) = \mathrm{lca}(a_2, a_3), \\ &\kappa(t) = \#\{\text{right-child edges on } r(t) \rightsquigarrow q(t)\}. \end{aligned} \tag{115.1}$$

The downward path can be empty. In the left-pair branching case $((a_1, a_2), a_3)$, $q = r$ and $\kappa = 0$; in the right-pair case $(a_1, (a_2, a_3))$, q lies in the right subtree of r, so $\kappa \geq 1$. This remains true with arbitrary blank subtrees inserted. For $E \subseteq \mathbb{N} = \{0, 1, 2, \ldots\}$, define

$$\mathcal{L}_E = \{t \in \mathcal{W}_3 : \kappa(t) \in E\}, \quad \mathcal{Z} = \mathcal{L}_{\{0\}}, \quad \mathcal{B} = \mathcal{L}_{\{1,2,3,\ldots\}}.$$

The three-marker separation from [156] is $\mathcal{B} \notin \mathsf{DTWA}$. Because $\mathcal{W}_3 \in \mathsf{DTWA}$ and deterministic tree-walking languages are Boolean-closed [159], also $\mathcal{Z} \notin \mathsf{DTWA}$.

A set $E \subseteq \mathbb{N}$ is *ultimately periodic* if there are $N \geq 0$ and $p \geq 1$ with $\mathbf{1}_E(n+p) = \mathbf{1}_E(n)$ for all $n \geq N$.

Theorem 115.7 (Exact regularity characterization)**.** *For every $E \subseteq \mathbb{N}$,*

$$\boxed{\mathcal{L}_E \in \mathsf{REG} \iff E \text{ is ultimately periodic.}}$$

If N, p witness ultimate periodicity, a deterministic bottom-up automaton with at most $3 + 2(N + p)$ states can be constructed.

Proof. For necessity let $D = \mathrm{Node}(a, a)$, $C[z] = \mathrm{Node}(b, z)$, and $R[z] = \mathrm{Node}(a, z)$, where each displayed b is an unmarked leaf. For $n \geq 0$ let $T_n = R[C^n[D]]$. The three marked leaves are ordered as required and $\kappa(T_n) = n + 1$. In any deterministic bottom-up finite automaton, the state reached on $C^n[D]$ evolves by iterating a single self-map of a finite state set. Its state sequence is eventually periodic. Acceptance of the fixed outer context $R[-]$ is then eventually periodic, so $\mathbf{1}_E(n+1)$, and thus $\mathbf{1}_E(n)$, is ultimately periodic.

Conversely, choose an ultimately periodic representation with threshold N and period p. Let $\rho(v)$ retain v exactly for $v < N$ and retain the residue v mod p in a disjoint periodic part for $v \geq N$. The domain $\mathcal{D}_{N,p}$ contains $N+p$ values, supports a deterministic increment, and determines whether $v \in E$.

At each subtree, store its number c of marked leaves capped at four. When $c = 2$, additionally store $\rho(d)$, where d is the number of right-child edges from the subtree root to the LCA of its two marked leaves. When $c = 3$, instead store $\rho(\kappa)$ for its three marked leaves. The other three count classes $c \in \{0, 1, \geq 4\}$ need no auxiliary value. The complete transition rules at an internal node with left and right child counts c_L, c_R are

count split for $c = 2$	$(1,1)$	$(2,0)$	$(0,2)$
d_{new}	0	d_L	$1 + d_R$

count split for $c = 3$	$(3,0)$	$(0,3)$	$(2,1)$	$(1,2)$
κ_{new}	κ_L	κ_R	0	$1 + d_R$

The additions are performed in $\mathcal{D}_{N,p}$. At the root accept precisely if the marked-leaf count is three and the represented statistic belongs to E. There are at most $3+2(N+p)$ states. □

Theorem 115.8 (Linear-order bottom-up state complexity)**.** *Suppose E is ultimately periodic. Let $p \geq 1$ be its* least eventual period*, and let $N \geq 0$ be the least threshold from which membership is periodic with period p. If s_E denotes the minimum number of states of a deterministic bottom-up tree automaton recognizing $\mathcal{L}_E$, then*

$$\boxed{p + \max\{N-1, 0\} \;\leq\; s_E \;\leq\; 3 + 2(N+p).}$$

Consequently the deterministic bottom-up state complexity of this family has tight linear order *in $N+p$; this is not a lower bound on the state count of a deterministic tree-*walking *automaton.*

Proof. The upper bound is the explicit construction in Theorem 115.7. For the lower bound retain the fixed binary tree $D = \mathrm{Node}(a, a)$ and unary context $C[z] = \mathrm{Node}(b, z)$ from that proof. If a deterministic bottom-up recognizer has state set Q, the state u_n at $C^n[D]$ is obtained from u_0 by n iterations of a *single* map $f : Q \to Q$. Its acceptance by the fixed outer context $R[z] = \mathrm{Node}(a, z)$ is precisely

$$\mathbf{1}_E(n+1) \qquad (n \geq 0).$$

The state orbit of a finite map has an initial tail of length μ followed by a cycle of length λ, using exactly $\mu + \lambda$ distinct states. The displayed Boolean output sequence then has eventual period dividing λ and becomes periodic no later than index μ. Its *least* eventual period is p, while its least preperiod is $\max\{N-1, 0\}$: shifting a unary sequence by one index removes at most its initial exceptional value. Indeed, any periodicity of the shifted sequence from an earlier index, possibly with a nonminimal period, would extend the original eventually-p periodic tail back to an index earlier than N. Thus $p \mid \lambda$ and $\mu \geq \max\{N-1, 0\}$. Therefore

$$|Q| \geq \mu + \lambda \geq p + \max\{N-1, 0\}.$$

This holds for every deterministic bottom-up recognizer and hence for the minimum s_E. □

Corollary 115.9 (Growing numerical thresholds need growing bottom-up control)**.** *For the threshold spectrum $E = \{0, \ldots, B\}$ with $B \geq 1$, a deterministic bottom-up recognizer of $\mathcal{L}_E$ needs at least $B+1$ states and the construction of Theorem 115.7 uses at most $2B+7$ states. For the pure modulus-m spectrum $E = m\mathbb{N}$, $m \geq 2$, every such recognizer needs at least m states.*

Proof. For the threshold spectrum the least preperiod is $N = B + 1$ and eventual period is $p = 1$; for $m\mathbb{N}$ they are $N = 0$ and $p = m$. Substitute in Theorem 115.8. □

Corollary 115.10. *If E is not ultimately periodic, $\mathcal{L}_E$ is not recognized by any ordinary or finite-pebble tree-walking automaton. In particular it is not in* DTWA*.*

Proof. Ordinary tree-walking languages are regular [156, 158]; languages of the standard nested finite-pebble model are regular [160]. Apply Theorem 115.7. □

115.8 Right-edge dilation and a nonrecognizability criterion

For $d \geq 1$, define $R_0(z) = z$, $R_{j+1}(z) = \mathrm{Node}(b, R_j(z))$ and the tree transformation

$$H_d(a) = a, \quad H_d(b) = b, \qquad H_d(\mathrm{Node}(u, v)) = \mathrm{Node}\big(H_d(u), R_{d-1}(H_d(v))\big). \tag{115.2}$$

Every original right-child edge becomes a chain of d right-child edges with blank left sibling leaves; original left-child edges are unchanged.

Lemma 115.11 (Exact dilation)**.** *H_d preserves the number of marked leaves and, for every $t \in \mathcal{W}_3$, $\kappa(H_d(t)) = d\,\kappa(t)$.*

Proof. The marked-leaf order and the corresponding original lowest common ancestors are unchanged. The path joining the two relevant ancestor nodes is obtained from the old path by replacing each right-child edge with exactly d such edges; a left-child edge remains left-child. □

Lemma 115.12 (Finite-control inverse simulation)**.** *For fixed $d \geq 1$, $K \in \mathsf{DTWA}$ implies $H_d^{-1}(K) \in \mathsf{DTWA}$.*

Proof. A machine reading t can simulate a deterministic walker on the virtual tree $H_d(t)$ without constructing that tree. Encode a virtual head position by a physical input node plus one of finitely many tags: an original node, one of the $d-1$ wrapper nodes replacing a right edge out of the current original node, or one of the corresponding blank left leaves. For a wrapper node or its blank left leaf, the physical position is the original parent of the expanded edge. Local parent/left/right/stay transitions of the virtual machine translate into a bounded number of physical movements and tag changes, determined by local input node type and the fixed d. Labels and node types in the virtual tree can also be recovered from this representation. Consequently, the product of the original finite control with the tag set supplies a deterministic simulation. This is a direct special-case inverse transformation argument, consonant with [161]. □

Theorem 115.13 (Spectrum dilation obstruction)**.** *Let $E \subseteq \mathbb{N}$. If some $d \geq 1$ satisfies*

$$\mathbf{1}_E(dn) = 1 - \mathbf{1}_E(0) \qquad (n \geq 1), \tag{115.3}$$

then $\mathcal{L}_E \notin \mathsf{DTWA}$.

Proof. Inside $\mathcal{W}_3$, Lemma 115.11 identifies $H_d^{-1}(\mathcal{L}_E)$ with $\mathcal{Z}$ if $0 \in E$ and with $\mathcal{B}$ otherwise. Either case is outside DTWA by the classical separation and Boolean closure. Were $\mathcal{L}_E$ deterministically recognizable, Lemma 115.12 would recognize this preimage, a contradiction. □

Corollary 115.14 (Uniform threshold nonrecognizability)**.** *For every fixed $B \geq 0$,*

$$\boxed{\mathcal{L}_{\{0,\ldots,B\}}, \quad \mathcal{L}_{\{B+1,B+2,\ldots\}} \in \mathsf{REG} \setminus \mathsf{DTWA}.}$$

Proof. For each spectrum use $d = B + 1$ in Theorem 115.13, and apply Theorem 115.7 for regularity. □

Proposition 115.15 (Sharp factor within uniform dilation)**.** *For every $d \geq 1$ and $B \geq 0$,*

$$H_d^{-1}(\mathcal{L}_{\{0,\ldots,B\}}) = \mathcal{L}_{\{0,\ldots,\lfloor B/d \rfloor\}}.$$

Consequently, the least dilation factor whose inverse image is exactly $\mathcal{Z}$ is $B + 1$.

Proof. The inequality $d\,\kappa(t) \leq B$ is equivalent to $\kappa(t) \leq \lfloor B/d \rfloor$. This bound is zero precisely when $d > B$. Every nonnegative value of κ occurs in $\mathcal{W}_3$, so smaller d cannot give exactly $\mathcal{Z}$. □

Corollary 115.16 (Periodic spectra, including infinite examples)**.** *For every $m \geq 2$ and $R \subseteq \{0, \dots, m-1\}$ not containing 0, the spectrum*

$$E = \{0\} \cup \{n \geq 1 : n \bmod m \in R\}$$

satisfies $\mathcal{L}_E \in \mathsf{REG} \setminus \mathsf{DTWA}$. The same holds for $E = \{n \geq 1 : n \bmod m \in R\}$ whenever $0 \in R$.

Proof. Set $d = m$ in Theorem 115.13 and use ultimate periodicity for regularity. In the first displayed family the choice $R = \varnothing$ gives the finite spectrum $\{0\}$; to obtain an infinite spectrum there take $R \neq \varnothing$. □

115.9 Exactly what the dilation method proves

For ultimately periodic E with threshold N and period p, put

$$b_E = \mathbf{1}_E(0), \qquad e_E = \mathbf{1}_E(np) \quad \text{for sufficiently large } n.$$

Let $\mathcal{A}_E$ be the Boolean algebra of subsets of $\mathcal{W}_3$ generated by $H_d^{-1}(\mathcal{L}_E)$ for all positive integers d, with complements taken relative to $\mathcal{W}_3$. For clarity, this construction allows *only* these inverse images and finite Boolean operations; it does not encompass arbitrary DTWA proof techniques.

Theorem 115.17 (Exact reach of the dilation–Boolean method)**.** *For every ultimately periodic E, the following are equivalent:*

(i) $b_E \neq e_E$;
(ii) *for some $d \geq 1$, $H_d^{-1}(\mathcal{L}_E)$ is $\mathcal{Z}$ or $\mathcal{B}$;*
(iii) $\mathcal{Z} \in \mathcal{A}_E$ *(equivalently, $\mathcal{B} \in \mathcal{A}_E$).*

Proof. If $b_E \neq e_E$, choose $d \geq N$ divisible by p. At $\kappa = 0$, the inverse image has membership b_E; at every positive κ, $d\kappa$ is beyond the threshold and congruent to 0 modulo p, so the membership is $e_E = 1 - b_E$. This proves (i)⇒(ii), and (ii)⇒(iii) follows by Boolean closure.

For the converse suppose $b_E = e_E$. Choose a positive multiple q of p with $q \geq N$. For every $d \geq 1$,

$$\mathbf{1}_E(d \cdot 0) = b_E = e_E = \mathbf{1}_E(dq).$$

Thus each generator, and every Boolean combination of them, agrees on the nonempty fibers $\kappa = 0$ and $\kappa = q$. Neither $\mathcal{Z}$ nor $\mathcal{B}$ does. Therefore (iii) fails. □

Remark 115.18 (A method frontier, not a class characterization). Theorem 115.17 says exactly when this particular construction can recover the known branching separator. If $b_E = e_E$, it does *not* follow that $\mathcal{L}_E$ belongs to DTWA. In particular the pure even spectrum survives this test. The saturated-pattern arguments of [156] alone do not supply a parity-changing equivalent pair by the reductions used here.

115.10 One visible pebble recognizes every regular spectrum

In the standard deterministic one-pebble tree-walking model, the machine has a single movable head and finite control, and can place one visible pebble at its current node, detect it when revisiting that node, and lift it there. The pebble must be lifted before acceptance. Let $\mathsf{1pDTWA}$ denote the resulting language class. The nested-pebble regularity inclusion and early anchored-walk techniques were established by Engelfriet and Hoogeboom [160]; a later account of deterministic one-pebble branching tests is given in [157]. The following theorem is a classification of our *particular* numerical family, not the first known zero-/one-pebble class gap.

Lemma 115.19 (Bounded subtree probe)**.** *Given its head at an internal node v, a deterministic one-pebble walker can count, up to any fixed bound, the marked leaves in either chosen child subtree and return to v with its pebble removed.*

Proof. Place the pebble at v and enter the chosen child. Execute a complete local depth-first tour with a finite saturated count. The first upward return to the pebble identifies the boundary v; lift the pebble and resume. The local DFS requires no unbounded stack since the parent/left/right node type is visible at each step. □

Theorem 115.20 (Exact deterministic one-pebble characterization)**.** *For every $E \subseteq \mathbb{N}$,*

$$\boxed{\mathcal{L}_E \in \mathsf{1pDTWA} \iff \mathcal{L}_E \in \mathsf{REG} \iff E \text{ is ultimately periodic.}}$$

Proof. Assume E is ultimately periodic. Begin with a deterministic full-tree DFS verifying membership in $\mathcal{W}_3$. On failure reject. On success, start a fresh preorder DFS, with a finite counter of the marked leaves already visited. At each internal node v, probe both child subtrees via Lemma 115.19; the global preorder counter is untouched during the probes.

If the global counter is one and each child subtree contains exactly one marked leaf, then $v = q(t)$: the visited leaf is a_1 and the two unvisited marked leaves are separated at v. Conversely, in the right-pair case this situation occurs at q, and in the left-pair case no node satisfies it. In the latter case $r = q$, hence $\kappa = 0$, and the machine decides according to $0 \in E$.

In the right-pair case, suspend preorder at q and move upwards. On an ascent from a *left* child, do not probe that parent and do not increment the statistic. On an ascent from a *right* child, increment the finite counter and probe the left subtree of the parent v. This distinction is essential: after a left-child ascent below r, the parent's left subtree can contain a_2, a_3, whereas for a right-child ascent strictly below r the left sibling subtree has no marked leaf. At r the path from r to q begins with a right-child edge and the left subtree contains a_1; therefore the first positive probe performed after a right-child ascent identifies r. The increments are exactly the right-child edges on $q \rightsquigarrow r$, read downward, so their number is $\kappa(t)$.

Implement the counter by exact values below the threshold N and residues modulo p after that threshold, as in Theorem 115.7; increments are monotone on the ascent, and the finite-state summary determines membership in E. The pebble is always lifted after each subtree probe, so only one pebble is used. If the acceptance convention requires a final visit to the root, move there while retaining the decision in one state bit.

Conversely, every language recognized by the standard one-pebble model is regular [160], and Theorem 115.7 then forces E to be ultimately periodic. □

Corollary 115.21 (Explicit deterministic pebble gap)**.** *For every fixed $B \geq 0$,*

$$\boxed{\mathcal{L}_{\{0,\dots,B\}} \in \mathsf{1pDTWA} \setminus \mathsf{DTWA}.}$$

Proof. Combine Theorem 115.20 and Corollary 115.14. □

115.11 The pure even spectrum: resolved and unresolved aspects

The spectrum $E_{\text{even}} = 2\mathbb{N}$ is purely periodic with period two, and contains zero. Hence

$$\mathcal{L}_{\text{even}} = \{t \in \mathcal{W}_3 : \kappa(t) \equiv 0 \pmod 2\} \in \mathsf{REG} \cap \mathsf{1pDTWA} \tag{115.4}$$

by Theorems 115.7 and 115.20. Its membership is also unproblematic for a different *zero-pebble* model that accepts according to the parity of the number of accepting nondeterministic computations; this is not ordinary deterministic acceptance.

Lemma 115.22 (Right-turn witness bijection)**.** *On every $t \in \mathcal{W}_3$, the number of left-to-right DFS turns $v : v_L \to v_R$ made after exactly one marked leaf has been visited and with a_3 in the right subtree of v is precisely $\kappa(t)$.*

Proof. If $\kappa(t) = 0$, the leftmost two marked leaves lie on the same side of the top branching node and no such right-subtree turn occurs after seeing only a_1. Otherwise the a_2, a_3 pair lies in the right child subtree of r, and its common ancestor is q. A successful turn's right subtree contains a_3 and occurs after exactly a_1 was visited; it must therefore also contain a_2 (which precedes a_3) and cannot contain a_1. Thus the turning node is an ancestor of q at which the path to q follows the right child, and is at or below r. Conversely, every right-child edge on the path $r \rightsquigarrow q$ produces exactly one such turn: at that time a_1 has been visited, whereas a_2, a_3 remain in the corresponding right subtree. This gives the bijection. □

Proposition 115.23 (Exact ambiguity and an auxiliary parity-of-runs model)**.** *There is an ordinary nondeterministic tree-walking automaton A_{turn} with no pebbles such that*

$$\#\,\mathrm{Acc}_{A_{\mathrm{turn}}}(t) = \begin{cases} \kappa(t), & t \in \mathcal{W}_3, \\ 0, & t \notin \mathcal{W}_3. \end{cases}$$

Consequently an automaton with odd-number-of-accepting-runs semantics and no pebbles recognizes $\mathcal{L}_{\mathrm{even}}$. This conclusion is not *deterministic recognizability.*

Proof. For completeness, a zero-pebble nondeterministic walker can be arranged to have exactly $\kappa(t)$ accepting runs on $t \in \mathcal{W}_3$. After checking the three-leaf promise, it performs a deterministic left-to-right DFS. At each transition from a left subtree to the corresponding right subtree, provided exactly one marked leaf has been seen, it may guess that turn, then nondeterministically descend to a marked leaf in the right subtree. Once a marked leaf z is reached, the walker continues the ordinary left-to-right depth-first tour from z to the end of the tree. This continuation starts in the *leave-leaf* phase and accepts the guess precisely when no later marked leaf is encountered. Thus z must be the rightmost marked leaf a_3. Both the successor tour and its one-bit flag for a later marked leaf use finite control, require no pebble, and have no further branching. For a fixed candidate turn, the unique downward path to a_3 is therefore the only successful guessed path, if a_3 lies in its right subtree. By Lemma 115.22, the successful turns are in bijection with the right edges on $r \rightsquigarrow q$, giving $\kappa(t)$ accepting runs. Adding one independent accepting run yields $1 + \kappa(t)$ runs, odd precisely when $\kappa(t)$ is even. No pebble is involved, but the acceptance semantics is *parity-of-runs*, not DTWA. □

The exact remaining question within the family is therefore:

Question 115.24 (Unannotated, zero-pebble deterministic parity)**.** *Is $\mathcal{L}_{\mathrm{even}}$ recognized by an ordinary deterministic tree-walking automaton with one head, no pebbles, and standard single-run acceptance?*

$$\boxed{\mathcal{L}_{\mathrm{even}} \overset{?}{\in} \mathsf{DTWA}.}$$

Theorem 115.17 explains why uniform dilation and Boolean combinations do not decide Question 115.24: here $b_E = e_E = 1$. The one-pebble construction does not decide it either, since its reusable subtree probe relies on an explicitly marked return point. The preceding parity-of-runs construction proves membership only in a different computational model. These are precise limitations of the present arguments, *not* an assertion of general literature-wide openness.

115.12 Discussion, limitations and future work

The family $\{\mathcal{L}_E : E \subseteq \mathbb{N}\}$ illustrates how a simple numerical statistic can separate three questions: regularity, zero-pebble deterministic recognition, and one-pebble deterministic recognition. Exact regularity and one-pebble membership both reduce to ultimate periodicity. The dilation obstruction transfers the classical three-leaf deterministic lower bound to fixed thresholds and many infinite spectra; its exact Boolean closure boundary is expressible by the value of the spectrum at zero and at sufficiently large multiples of an eventual period.

The proof dependencies matter. The fundamental three-leaf nonrecognizability is due to [156]; Boolean closure and inverse-transformation principles are established machinery [159, 161]; finite-pebble regularity and the general utility of pebbles were known by [160, 157]. The explicit 2025 one-pebble example [163, Theorem 20] concerns a different language; it is not a previous proof of our full spectrum classification. The explicit right-turn family sharpens the parameterized consequences of those tools. Its deterministic lower bounds transfer an established separation rather than constituting a new baseline separation of machine classes.

The results leave the separate DTWA–UTWA equality question untouched: a language outside DTWA is not a counterexample to that equality unless it is also shown to belong to UTWA.

Future work is deliberately limited to Question 115.24: either a concrete ordinary zero-pebble DTWA for $\mathcal{L}_{\text{even}}$ or a proof that every such device fails on some parity-distinguishing tree pair. No positive or negative answer to that question is claimed here, and its resolution is not a prerequisite for the theorems proved here.

A small comparison of the established boundaries. For the fixed threshold spectrum $E = \{0, \dots, B\}$, $\mathcal{L}_E$ is regular and one-pebble recognizable but not in DTWA. For the pure even spectrum $E = 2\mathbb{N}$, regularity and one-pebble recognition are established, while zero-pebble deterministic membership is undecided by the present results. Finally, if E is not ultimately periodic, $\mathcal{L}_E$ lies outside all standard finite-pebble tree-walking language classes because it is nonregular. The distinction between these three cases prevents the general regularity theorem from being mistaken for a DTWA membership decision.

115.13 Finite-control realization details

The following two virtual-head and pebble implementation lemmas supply the finite-control details used in the preceding proofs.

115.13.1 Local virtual-head transitions for dilation

Fix $d \geq 2$. An original input node x is represented by (x, O). For every original internal node x, the $d-1$ wrapper nodes inserted on the edge from x to its right child are represented by (x, W_j) for $1 \leq j < d$ (numbered from top to bottom). The blank left leaf attached to wrapper j is represented by (x, D_j). This uses precisely $1 + 2(d-1)$ virtual-position tags. The virtual downward neighbors are

virtual position	left child	right child
(x, O), x internal	(x_L, O)	(x, W_1)
(x, W_j), $j < d-1$	(x, D_j)	(x, W_{j+1})
(x, W_{d-1})	(x, D_{d-1})	(x_R, O)

Their inverse parent operations are

virtual position	parent
(x, D_j)	(x, W_j)
(x, W_1)	(x, O)
$(x, \mathsf{W}_j),\ j > 1$	(x, W_{j-1})
$(x, \mathsf{O}),\ x \neq \text{root},\ x = x_{\text{par},L}$	$(x_{\text{par}}, \mathsf{O})$
$(x, \mathsf{O}),\ x \neq \text{root},\ x = x_{\text{par},R}$	$(x_{\text{par}}, \mathsf{W}_{d-1})$

A virtual movement between original nodes induces the corresponding physical parent/child move; a movement inside a fixed original right-edge gadget only changes the finite tag. The virtual node label, root/left/right-child type, and leaf/internal type are all functions of the physical local type and tag. All virtual transitions can therefore be implemented by a bounded microprogram. For $d = 1$ only the tag O is needed, and the simulator is the original input walker. This supplies the promised deterministic implementation of Lemma 115.12 with no pebble or stack.

115.13.2 Anchored one-pebble probes and the monotone ascent

The one-pebble recognizer has a fixed sequence of phases: (i) check $\mathcal{W}_3$ by root-to-root DFS; (ii) root-to-root preorder with a saturated count of a leaves visited, interrupting at a candidate node for two bounded subtree probes; (iii) if q is found, abandon preorder and climb from q to r, probing left siblings only after an ascent along a right-child edge; and (iv) lift the pebble, if present, return to the root if required, and halt according to the final finite counter. Every subtree probe begins by placing the sole pebble at the candidate parent v and ends on the first return to v from the chosen child, when that pebble is lifted. Thus the ordinary preorder count and the probe count belong to disjoint phases and are not conflated.

The control can be organized into finitely many traversal phases: preorder entry, anchored left-child probe, anchored right-child probe, and upward ascent. A probe saves the saturated preorder count and the ascent summary in finite control; its own temporary marked-leaf count is reset when the pebble is placed and discarded after the return to that pebble. The ascent phase inspects the type of the child just left, which the standard tree-walking model exposes: leaving a left child continues upward without either testing the parent or incrementing the right-edge count; leaving a right child increments that count and starts a fresh anchored probe of the parent's left child. No probe remains active across successive ascent edges, and acceptance is tested only after an eligible right-child ascent reveals the leftmost marker. Thus there is no unbounded recursion or second pebble.

The test for q uses the preorder count 1 and child counts $(1, 1)$. For three marks, it can first succeed only at $q = \operatorname{lca}(a_2, a_3)$ in the right-pair case. During the climb, a positive left-sibling probe on a right-child ascent occurs first exactly at r; all lower such left siblings contain no marked leaf. Each qualifying ascent increments the finite E-counter once, and no backtracking probe changes that counter. Its final value is therefore the right-edge count on $r \rightsquigarrow q$. This justifies the one-pebble implementation independently of any bottom-up automaton construction.

115.13.3 Reproducibility scope

The finite consistency range contains $C_{n-1}\binom{n}{3}$ marked ordered full binary trees with n leaves, where $C_j = \frac{1}{j+1}\binom{2j}{j}$. Hence there are exactly $\sum_{n=3}^{8} C_{n-1}\binom{n}{3} = 29{,}646$ inputs for $3 \leq n \leq 8$. Direct lowest-common-ancestor counting, the bottom-up recurrence, the one-pebble search-and-ascent invariant, and the right-turn witness bijection describe the same statistic. For $d = 1, 2, 3, 4$ there are 118,584 corresponding finite dilation identities. These finite counts illustrate the constructions; the unrestricted statements are proved by the symbolic arguments above.

115.14 Bridge to the external scalar-field realization

The independent guarded two-layer stack-alphabet chapter precedes this synthesis and illustrates a resource-specific physical category; its budget-dependent family gap is not the assembly defect of Chapter 113. After the intervening independent height-guard realization chapters, the EF–FG chapters provide a deliberately non-pushdown test of this architecture. Extrema–Frame Foreground–Background theory is developed as an independent scalar-field theory, not as an axiom of USNRT. After its native theorems are proved, the all-orders common-realization obstruction families will be read back through the protected-obligation language of this chapter sequence. No EF–FG theorem is used as a premise of the USNRT core above.

Chapter 116

Linear Modular Height Guards: Exact One-Stack Realizability

Research branch — Independent two-source realization programme. *This chapter gives the full scoped arithmetic classification for two independently tagged Dyck sources under a linear modular height guard. Its original standalone article is preserved separately; the book presentation retains the definitions, constructive proofs, negative certificates, and exact logical scope rather than treating the article as a preface to the next one.*

Relation to the General Theory. The decoder and unique rooted physical representation are instances of the exact semantic projection and canonical section developed in Chapters 112 and 115. The claim is specific to disjoint single-source input actions, free axes, tagged stacks, and strict real-time one-stack implementation. It is not a general assembly-deficit computation from Chapter 113. The independent manuscript is recorded bibliographically as a 2026 research manuscript [236]. The literature background includes classical Presburger-semilinear languages [69], real-time strict deterministic recognition [230], periodic counter walks [231], bounded and scope-sensitive multistack restrictions [232, 233], and scheduler-dependent shuffles and geometric-intersection studies [234, 235]. These are comparison points, not claims of discovery of buffering or crossed-equality techniques. Earlier pushdown-normalization and resource studies [227, 228, 229] motivate but do not replace the self-contained scoped proof below.

Height region. With $F_0 = \{(x, y) \in \mathbb{N}^2 : xy = 0\}$ and $p \geq 2$, define

$$S_{p,a,b} = F_0 \cup \{(x, y) \in \mathbb{N}^2_{>0} : ax + by \not\equiv 0 \pmod{p}\}. \tag{116.1}$$

The all-prefix source condition is imposed on the live pair of stack heights, not solely on the input scheduler word.

116.1 Model and main theorem

Let T_X, T_Y be arbitrary nonempty finite tag sets. For $t \in T_X$ write x_t for a push on X and $\bar{x}_t$ for its matching pop. Let $y_s, \bar{y}_s$ be the corresponding Y letters. The alphabets are disjoint. Each source stack is initially empty and a pop is enabled only when its tag agrees with the current top. Let $\mathrm{ht}_X(w), \mathrm{ht}_Y(w)$ be the source heights after a legal prefix w. A word is *guarded legal* when every prefix, including the empty prefix, has both source operations defined and satisfies (116.1). Let $L_{p,a,b}$ consist of the guarded legal words whose two final source stacks are empty.

Definition 116.1 (Exact rooted source-canonical real-time realization)**.** A one-stack implementation is *source faithful* if its reachable configurations at consumed-input boundaries admit a decoder to the complete ordered pair of tagged source stacks, and the decoder

both reflects transition definedness and intertwines each input action with the corresponding source action. It is *rooted canonical* when, on reachable source configurations, there is a unique encoding $\sigma(U, V)$, independent of the input history, with the empty source pair encoded at the physical root. It is *strict real-time* when no ε-transition is used and each input symbol is processed in one pushdown transition with at most one physical top-stack edit. The state and physical alphabet may depend on the fixed parameters p, a, b, T_X, T_Y.

Theorem 116.2 (Sharp modular-guard dichotomy)**.** *For every $p \geq 2$ and $a, b \in \mathbb{Z}/p\mathbb{Z}$ the following are equivalent:*

(i) *$a = 0$, $b = 0$, or $a + b = 0$ in $\mathbb{Z}/p\mathbb{Z}$;*
(ii) *$L_{p,a,b}$ admits an exact rooted source-canonical strict-real-time one-stack implementation;*
(iii) *$L_{p,a,b}$ is context-free.*

If the equivalent conditions fail, $L_{p,a,b}$ is not context-free; the obstruction excludes even nondeterministic PDAs. All congruences in (i) are tested separately, including when p is composite.

We prove the constructive direction in Sections 116.2–116.3 and the negative direction in Section 116.4. In the nonzero diagonal case the machinery works with the period $q = p/\gcd(p, a)$; consequently nonunits pose no exceptional case.

116.2 Canonical positive realizations

We first state the two normal forms underlying the positive half. A stack frame may store, in addition to its tag(s), any fixed finite summary such as the previous height modulo q and whether a frame is the first of its kind. These annotations are uniquely determined by the source contents and the chosen normal form.

116.2.1 Axis and vertical-stripe cases

If $a = b = 0$, the positive quadrant is forbidden. At most one source stack is nonempty; the physical stack stores that stack verbatim, with a finite control specifying the active source. Every attempted transition to two nonempty sources is rejected. This is immediately real-time and canonical.

Suppose $b = 0$, $a \neq 0$, and put $q = p/\gcd(p, a) \geq 2$. In the positive quadrant,

$$ax \not\equiv 0 \pmod{p} \quad \Longleftrightarrow \quad x \not\equiv 0 \pmod{q}. \tag{116.2}$$

Define $H_q = \{0\} \cup \{x \geq 1 : x \not\equiv 0 \pmod{q}\}$. Its nearest-neighbor components are

$$[0, q-1],\ [q+1, 2q-1],\ [2q+1, 3q-1], \ldots. \tag{116.3}$$

For $x \in H_q$ let $\lambda(x)$ denote the left endpoint of the component containing x, and let $d(x) = x - \lambda(x)$. Always $0 \leq d(x) \leq q - 1$.

Lemma 116.3 (Canonical stripe normal form)**.** *For the guard $F_0 \cup \{(x, y) \in \mathbb{N}^2_{>0} : x \not\equiv 0 \pmod{q}\}$, the tagged source pair admits a history-independent one-stack normal form. Every input step, including normalization, is realizable by a deterministic sequence of at most $q + O(1)$ physical top operations. All decisions depend only on finite control and the physical top frame.*

Proof. When $y = 0$, the physical stack is the complete tagged X stack; Y is empty. When $y > 0$, the physical stack, bottom to top, consists of the first $\lambda(x)$ tagged X frames followed by all Y frames except its bottommost one. The finite controller retains the last $d(x)$ tags of X and the bottommost tag of Y. If $x = 0$ set $\lambda(0) = d(0) = 0$. These data specify a unique, injective encoding of (X, Y).

A transition $y = 0 \to 1$ is allowed only if $x \in H_q$. It removes the last $d(x) \le q - 1$ physical X frames into the finite tail and caches the new Y tag. The reverse transition $y = 1 \to 0$ reattaches exactly the bounded tail after verifying the last Y tag. For $y > 0$, every legal change of x remains within the same connected component of H_q: any move to another component would cross a forbidden height divisible by q. Hence $\lambda(x)$ is fixed and only the bounded finite X tail changes. A purported X pop below the left endpoint is rejected from the known height/residue boundary, before examining any buried frame. In this mode Y pushes and pops use physical frames above the frozen X base, retaining the bottom Y tag as a boundary marker in the controller. When $y > 1 \to 1$, a distinguished first excess frame records the crossover. The controller knows whether the physical top is a Y frame and its tag, or the exposed base.

Modulo-q counters and finite predecessor summaries stored on the frames suffice to check all height conditions without inspecting the buried stack. Each case produces exactly the specified normal form for the updated sources; rejected actions leave it unchanged. The only multi-operation conversions move at most $q - 1$ X frames. Induction from the empty source pair proves exact domain reflection, decoding, and path-independent canonicality. □

Exact boundary transitions. For clarity, let t_X denote the finite X-tail and let Y_0 denote the cached bottom Y tag in the $y > 0$ mode. Every consumed symbol falls into one of the following mutually exclusive cases (the reverse, disabled operations have their usual typed-top checks):

Source condition	Quiescent update before real-time packing
$y = 0$, X operation	Update the physical X stack and the height summary; reject an illegal typed pop.
$y = 0$, Y push	If the new interior height is allowed, transfer the last $d(x)$ X tags to t_X and store the new Y_0 in control; otherwise reject without mutation.
$y = 1$, Y pop	Verify Y_0, restore t_X to the X stack, clear the Y mode.
$y \ge 1$, X operation	Check the new guard and source-top tag using the finite tail and its component boundary. Update only t_X; no crossing to another H_q component is legal.
$y \ge 1$, Y operation, not at $1 \to 0$	Push/pop the top excess-Y frame, or change between $y = 1$ and $y = 2$ using its distinguished boundary frame.

The components in (116.3) are separated by a forbidden positive X height; hence no legal interior update of X requires access to the buried X base. This is the point at which a global height restriction becomes a bounded top-of-stack operation.

If $a = 0$, $b \neq 0$, the same lemma applies after exchanging the X and Y roles.

116.2.2 Diagonal-stripe case

Suppose $a + b = 0$ and $a \neq 0$. Write $b = -a$ and $q = p/\gcd(p, a) \ge 2$. Then

$$ax + by \not\equiv 0 \pmod{p} \quad \Longleftrightarrow \quad x - y \not\equiv 0 \pmod{q}. \tag{116.4}$$

When $x, y > 0$, this implies $x \neq y$. Put $d = |x - y| = kq + r$ with $k \ge 0$ and $1 \le r \le q - 1$.

Lemma 116.4 (Canonical paired-frame normal form)**.** *Under the guard $F_0 \cup \{x, y > 0 : x - y \not\equiv 0 \pmod{q}\}$, there exists a unique, history-independent one-stack normal form preserving all tags. Each source action updates it using at most $q + O(1)$ deterministic top operations.*

Proof. At the origin the physical stack is empty. On either axis it stores exactly the nonempty source stack as single-source frames. In the positive quadrant, assume $x > y$; the other case is symmetric. The physical stack consists, bottom to top, of the first kq

tagged X symbols as single frames, followed by y paired frames. The ith pair stores the $(kq+i)$th X tag and the ith Y tag, both indexed from the bottom. The remaining r top X tags form a finite-control tail. A marker on the lowest paired frame permits the controller to recognize when the last pair is removed. This decomposition is unique for the source pair.

In the interior, a majority-side push/pop changes r by ± 1. A minority push takes the lowest tag from the finite majority tail and puts it together with the new minority tag into a top paired frame, reducing r by one. A minority pop removes the top pair and places its majority tag at the beginning of the finite tail, increasing r by one. Whenever this would give a forbidden difference divisible by q, the action is rejected *before* altering the representation. The block of kq majority tags cannot change while the run stays in this interior band.

For a minority pop $1 \to 0$, the entire pair/tail interface is converted into ordinary majority frames. The number of frames restored is at most $r + 1 \le q$. Conversely, moving from an axis into the interior consumes at most q top majority tags to form the first pair and its finite tail. These are deterministic top-only conversions; the chosen normal form is restored at every input boundary. The residue class, mode, and first-pair marker are finite data. All actions, source-top tag tests, and guard checks are therefore exact and implementable with a constant number of microsteps depending on q. □

Reversible pairing at the band boundary. For $x > y > 0$, the paired-frame count is precisely y and the control tail has length r. The four interior updates can be checked algebraically: a majority push/pop changes d by ± 1 and extends/shortens the tail; a minority push/pop changes d by ∓ 1 and transfers one majority tag between the beginning of the tail and the top pair. At $r = 1$ a minority push is rejected unless it is an axis operation, since it would create a positive-positive difference divisible by q; at $r = q - 1$ a minority pop is rejected unless it empties the minority source. These exceptions are exactly the axis crossover cases already treated in the lemma. In particular no allowed interior operation moves the kq-frame base or changes the identity of its boundary. All stored finite summaries may be reconstructed deterministically from the final tagged source pair, so the encoding is not a trace-dependent normal form.

116.3 From bounded repairs to a strict real-time PDA

The preceding two lemmas give deterministic quiescent one-stack encodings with bounded intermediate normalization. For clarity we spell out the (classical-style) stack-blocking step required to obtain one consumed-input transition per source letter; we do not claim this acceleration technique as an original independent result; compare the classical real-time strict-deterministic equivalence in [230], whose hypotheses and conclusions are not being silently substituted for the explicit construction below.

Lemma 116.5 (Block-packing acceleration)**.** *Suppose a deterministic one-stack implementation over a fixed finite alphabet has a unique quiescent configuration at each input boundary, and each input letter induces a deterministic sequence of at most C stack-top microsteps from one quiescent configuration to another (or detects rejection), where C does not depend on input length or stack depth. Then it admits a deterministic ε-free real-time simulation, preserving its quiescent source decoder and history-independent encoding if the original encoding had these properties.*

Proof. Choose $B \ge C + 2$. Partition the simulated stack into a physical stack of full blocks of B old symbols and a top word cached in finite control. Normalize the packing as follows: if the old stack has fewer than B symbols, cache all of it; otherwise cache the final $B + ((n - B) \bmod B)$ symbols of a stack of length n and partition the lower prefix into blocks of exactly B. Thus, whenever a hidden block exists, the cache has length between B and $2B - 1$.

On reading one input symbol, execute the complete bounded microprogram in the finite controller using the cached top word, and inspect the (single) visible top block if needed. The microprogram can remove at most $C < B$ old symbols, so it never needs a symbol below the starting cache. Afterward the cache has changed in length by at most C. Restore normalized packing by at most one physical block operation: push the bottom B cached symbols if the cache is too long, pop the top hidden block and prepend its symbols if the cache is too short, or do nothing. Because $B \geq C + 2$, one correction is enough. Before committing a physical push/pop, the control determines the entire bounded microprogram on a temporary copy of its finite cache. If that program would reject, the compiled transition is undefined and no physical state is changed. If it succeeds, at most one packed-stack block is pushed or popped. The block is a *single symbol* of the new finite stack alphabet, so this is one permitted PDA edit, not B edits. The initial packing is canonical because its block count and cache length are uniquely determined by n; the post-step packing is uniquely normalized by the same rule. All possible cached words, microprogram branches, and full-block symbols range over finite sets, yielding a finite deterministic real-time transition table. The packing uniquely encodes the prior quiescent stack and preserves the full decoder and canonical section by composition. □

Proposition 116.6 (Positive half of Theorem 116.2)**.** *If $a = 0$ or $b = 0$ or $a + b = 0$ modulo p, then $L_{p,a,b}$ has an exact rooted source-canonical strict-real-time one-stack implementation.*

Proof. The zero-zero case uses the axis construction. In the one-zero cases use (116.2), Lemma 116.3, and symmetry. In the remaining case use (116.4) and Lemma 116.4. Lemma 116.5 converts each bounded-repair implementation to strict real time while maintaining unique source encoding and exact definedness. Both tagged sources are empty exactly in the root normal form, which is the acceptance condition. □

116.4 Two independent impossibility mechanisms

We write $x, \bar{x}, y, \bar{y}$ for a fixed single tag of each source, so all negative witnesses use a regular subalphabet of any richer tag set. For a signed source word w, let w^{-1} denote reversal with all pushes and matching pops exchanged, so its signed-height path is traversed backward.

116.4.1 Anti-diagonal transfer: three equal blocks

Lemma 116.7 (Three-block obstruction)**.** *If $a, b, a + b$ are nonzero modulo p and $a = b$ modulo p, then $L_{p,a,b}$ is not context-free.*

Proof. Let $q = p/\gcd(p, a)$. Since $a \neq 0$ and $2a = a + b \neq 0 \pmod{p}$, one has $q \geq 3$. The forbidden positive-quadrant points are exactly those satisfying $x + y \equiv 0 \pmod{q}$. For each $n \geq 0$, set $N = qn + 2$ and consider

$$x^N (\bar{x}y)^N \bar{y}^N. \tag{116.5}$$

The first and last blocks lie on axes; during the middle block the sum of heights alternates between $N - 1$ and N, which are 1 and 2 modulo q. Thus every word in (116.5) is guarded legal and ends with both stacks empty.

Intersect $L_{p,a,b}$ with the regular pattern $x^2 (x^q)^* (\bar{x}y)^* \bar{y}^*$. After treating each $\bar{x}y$ as a single letter by inverse homomorphism, the resulting language is

$$\{x^N z^N \bar{y}^N : N \equiv 2 \pmod{q},\ N \geq 2\}.$$

Indeed, the final emptiness of X and Y forces all three exponents to coincide, while the preceding prefix check verifies all equal-count words. This language is not context-free by the standard three-equal-block pumping argument, which is unchanged by restricting N to one infinite arithmetic progression. Closure under regular intersection and inverse homomorphism gives the claim; applying the inverse map to the three-block macro alphabet is legitimate because each transfer factor $\bar{x}y$ is a fixed word. □

116.4.2 Isolated holes: periodic detours

Lemma 116.8 (Detour certificates)**.** *Suppose $a, b, a+b, a-b$ are all nonzero modulo p. There are two explicitly constructible source words P, Q, each of length at most $2p$, with signed height displacements $(p, 0)$ and $(0, p)$, such that their paths are guard legal from $(1, 1)$ and all translations of these paths by (ip, jp) for $i, j \geq 0$ are guard legal. Consequently $L_{p,a,b}$ is not context-free.*

Proof. A forbidden positive-quadrant point has $ax + by \equiv 0 \pmod{p}$. Two such points cannot be eight-neighbors, since their displacement would give one of $\pm a$, $\pm b$, $\pm(a+b)$, or $\pm(a-b)$ equal to zero modulo p. Thus deleted cells are isolated even under diagonal adjacency.

The point $(1, 1)$ is permitted because $a + b \neq 0$. Starting there, traverse the row $y = 1$ from $x = 1$ to $x = p + 1$. If the next cell $(h, 1)$ is forbidden, replace the two horizontal moves through it by the four-step bypass $yxx\bar{y}$, traveling from $(h-1, 1)$ to $(h+1, 1)$. Isolation guarantees that the bypass's intermediate cells, including those on the row $y = 2$, are permitted. All deleted cells are nonadjacent, so bypasses do not conflict. The resulting word P has displacement $(p, 0)$. Construct Q symmetrically along $x = 1$, replacing two steps through a forbidden $(1, h)$ by $xyy\bar{x}$; its displacement is $(0, p)$. The endpoints $(p+1, 1)$ and $(1, p+1)$ are permitted by periodicity.

If h bypasses occur on a route of p elementary axial moves, the length is $p + 2h$. There are at most $\lfloor p/2 \rfloor$ bypasses, so $|P|, |Q| \leq 2p$. The membership congruence and the positive-quadrant condition are invariant under translations by (ip, jp), $i, j \geq 0$, establishing the claimed path property.

Let $u = xy$ be the legal entry to $(1, 1)$. For all $m, n \geq 0$, the word

$$uP^m Q^n (P^{-1})^m (Q^{-1})^n u^{-1} \tag{116.6}$$

is guarded legal and empties both sources, by periodicity and reversal of the appropriate translated paths. For arbitrary nonnegative exponents m, n, k, ℓ, replace the two inverse-block exponents in (116.6) by k, ℓ. Final source emptiness forces $p(m-k) = p(n-\ell) = 0$, hence $m = k$ and $n = \ell$. Conversely balanced exponents give (116.6) and are accepted.

Using four fresh macro letters and the homomorphism mapping them to P, Q, P^{-1}, Q^{-1}, followed by fixed left/right quotients with u, u^{-1} and intersection with the regular four-block pattern, one obtains exactly

$$\{\alpha^m \beta^n \gamma^m \delta^n : m, n \geq 0\},$$

which is not context-free. Hence $L_{p,a,b}$ cannot be context-free. □

Proof of Theorem 116.2. Proposition 116.6 establishes (i)⇒(ii); (ii)⇒(iii) is immediate. If (i) fails, then $a, b, a+b$ are all nonzero. When $a - b = 0$, Lemma 116.7 excludes context-freeness; when $a - b \neq 0$, Lemma 116.8 excludes it. This exhaustive split proves (iii)⇒(i). □

116.5 Consequences and action-sensitive scope

Corollary 116.9 (Coefficient census)**.** *For any $p \geq 2$, among the p^2 ordered coefficient pairs (a, b) there are exactly $3p-2$ realizable pairs and $(p-1)(p-2)$ non-context-free pairs. In particular the realizable proportion is $3/p - 2/p^2 \to 0$.*

Proof. The three length-p lines $a = 0$, $b = 0$, and $a + b = 0$ intersect pairwise and jointly only at $(0, 0)$. Inclusion–exclusion gives $3p - 3 + 1 = 3p - 2$. Subtracting from p^2 yields the second count. □

Proposition 116.10 (A schedule word does not determine the guard)**.** *There are two correctly tagged, independently balanced source words with the same X/Y scheduling word*

such that one belongs to a modularly guarded language of the above family and the other does not. Thus, even for a fixed pair of Dyck source languages, the guard cannot in general be represented solely by a scheduling-language restriction, whether regular or otherwise.

Proof. Take $p = 3$ and $a = b = 1$ and use one tag per source. The two words

$$w_1 = x\bar{x}yx\bar{x}\bar{y}, \qquad w_2 = xxy\bar{x}\bar{x}\bar{y}$$

have the common scheduling sequence $XXYXXY$. Both projections are Dyck words. The first never meets a forbidden positive-positive sum divisible by three; the second passes through $(2, 1)$ and fails the guard. Any membership restriction depending only on the scheduling word would treat both alike. □

It is essential not to silently allow one letter to alter both source stacks. Such simultaneous actions change the accessible height graph and can change context-freeness even under the same geometric guard. Likewise the theorem is not a classifier for all Presburger height guards or all two-source control systems. For a source-preserving implementation, the decoder and canonical section satisfy the usual rooted intertwining and domain-reflection equations; such an instance does not by itself establish a new universal realization axiom or a quantitative assembly-deficit formula.

A finite arithmetic decision and a constructive negative witness. The input p, a, b is first reduced modulo p. Three modular equality tests decide the positive branch, including composite moduli; no multiplication-based zero-divisor test is sound here. In the negative branch the test $a = b$ chooses the three-block witness; otherwise the two detour words are constructed by inspecting at most p consecutive row/column positions, each with constant-length replacements. The algebraic decision is polynomial in the binary size of the coefficients, whereas explicitly writing these periodic witnesses can require length proportional to p. The positive device is effective for each fixed parameter triple but its alphabet and control size need not be bounded polynomially in $\log p$.

Finite checks. Independent scripts checked parameter instances (including composite moduli), detour paths, and representative tagged-source transitions. These checks are supplementary consistency tests; the results above rest on their symbolic proofs. The explicit detour words have length $O(p)$, which is not polynomial in the binary input length $\log p$.

Chapter 117

Periodic Height Guards: Signed-Winding Realizability

Research branch — Independent periodic-mask realization programme. *This chapter treats explicit periodic masks by a finite signed residue multigraph and full-source strict-real-time encoding. Its standalone article is a separate publication candidate with distinct geometric proof mechanisms. Common source semantics are restated deliberately; its general signed-winding theorem is not presented as an edition or sequel of the preceding arithmetic manuscript.*

Relation to the General Theory. The source decoder, section, domain reflection, and same-input top-access obligations are the exact-realization interface of Chapter 112. This chapter constructs category-specific realizations and certificates; the source-information lower bound is about finite physical information capacity, not an independently computed assembly deficit. Its distinct standalone source is cited as a research manuscript [237]. The periodic-graph and counter-machine background [231, 69], bounded/scope multistack work [232, 233], real-time automata [230], and recent trajectory/intersection studies [234, 235] are prior technical contexts. The article's claim concerns their combination under this exact full-tag source-preserving contract, not the invention of voltage graphs or generic context-free obstructions.

117.1 The guarded two-source model and its finite quotient

Fix integers $p, q \geq 1$, two nonempty *finite* tag alphabets T_X, T_Y, and a residue mask

$$R \subseteq \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}.$$

Let X_t^+, X_t^- and Y_s^+, Y_s^- denote tagged push and matching-pop input letters ($t \in T_X$, $s \in T_Y$). A source configuration is a pair $(U, V) \in T_X^* \times T_Y^*$, written bottom-to-top. A push appends its tag; a pop is defined exactly when the source top has the prescribed tag. Each input letter changes *one* of U, V, and never the other.

The height safety set is

$$S_R = \{(x, y) \in \mathbb{N}^2 : xy = 0\} \ \cup \ \{(x, y) \in \mathbb{N}^2 : x, y \geq 1, \ (x \bmod p, y \bmod q) \in R\}.$$

A word belongs to L_R if every prefix has defined source-stack action and a height in S_R, and the terminal configuration is $(\varepsilon, \varepsilon)$. Write $\mathcal{C}_R$ for the full *root-reachable* source transition system; preserving only heights would not preserve its tags.

Define the signed residue *multigraph* G_R: its vertex set is R, and for each $r \in R$ and each $e \in \{\pm(1,0), \pm(0,1)\}$ it has an arc $r \xrightarrow{e} r + e$ when $r + e \in R$ modulo (p, q). Parallel arcs, including $+1$ and -1 arcs with the same endpoints when a period is 1 or 2, are distinct. Every arc has its integer displacement e. For a connected component K, its

voltage subgroup $H_K \le p\mathbb{Z} \times q\mathbb{Z}$ consists of all displacements of closed quotient walks based at a fixed vertex. It is independent of the basepoint and has rank $0, 1$ or 2.

Choose a finite spanning tree $\mathcal{T}_K$ and integer potentials $t(r) = (t_x(r), t_y(r))$ obtained by summing the edge labels along tree paths from a root with potential zero. The vector

$$\omega(r \xrightarrow{e} s) = t(r) + e - t(s)$$

is a cycle voltage; the nonzero such vectors generate H_K. If $N = \max(1, |R|)$, tree potentials have coordinate absolute values at most N, and chord voltages at most $2N + 1$ in absolute value.

Theorem 117.1 (Signed-winding realization dichotomy)**.** *For the model above, the following statements are equivalent.*

(i) *For every connected component K of G_R, either* $\operatorname{rank} H_K = 0$*, or* $\operatorname{rank} H_K = 1$ *with a nonzero generator (A_K, B_K) satisfying $A_K B_K \ge 0$.*

(ii) *The reachable guarded source system $\mathcal{C}_R$ has an effective, rooted, history-independent, full-source-injective, transition-equivariant, domain-reflecting, same-input, epsilon-free real-time one-physical-stack representation. Every input letter requires at most one physical top-stack replacement.*

(iii) $L_R \in$ DCFL*.*

(iv) $L_R \in$ CFL*.*

If these conditions hold, a finite graph certificate and the resulting physical machine can be constructed effectively. If they fail, an explicit regular-macro non-context-free witness can be extracted from an offending component.

The ordinary logical implications (ii)⇒(iii)⇒(iv) are immediate. Sections 117.2 and 117.3 establish the substantive implications (iv)⇒(i) and (i)⇒(ii).

117.2 Two unconditional negative obstructions

Here the argument can be restricted to a *single tag* in each source, since a regular restriction of a context-free language must remain context-free. Replace each legal unit-step word W by its reverse *inverse* $\overline{W}$, obtained by reversing its letter sequence and exchanging the corresponding pushes and pops. When lifted from the endpoint of W, $\overline{W}$ retraces the path.

Lemma 117.2 (Automatic accessibility of a positive axis loop)**.** *Let K contain a based closed quotient walk Q with net displacement $(0, D)$, $D > 0$. Its repeated lift can be placed sufficiently far to the right so that a positive suffix begins at a vertex $(x_e, 1)$ and stays in $\mathbb{N}^2_{>0}$, reaching a lift of the chosen basepoint at arbitrarily large height. The suffix is reachable from the empty source by a free-X-axis prefix followed by one Y push.*

Proof. The bi-infinite lift $Q^{\mathbb{Z}}$ has bounded horizontal deviation and tends to opposite infinities vertically in its two time directions. Translate it horizontally by a sufficiently large multiple of p so every horizontal coordinate is positive. There are only finitely many visits to height 1 at nonnegative time; choose the *last* such visit after fixing an arbitrary origin of time. Every later vertex has $y \ge 1$, since any later $y = 0$ excursion would recross $y = 1$. A full turn reaches the basepoint; additional turns raise the height by D. Travel on the free X axis to $(x_e, 0)$ and push a Y symbol to enter at $(x_e, 1)$. The remaining fixed suffix is legal. □

Theorem 117.3 (Rank two excludes every PDA)**.** *If some H_K has rank two then $L_R \notin$* CFL*.*

Proof. Let U, V be based closed quotient words with independent net displacements $g = (g_x, g_y)$, $h = (h_x, h_y)$. By concatenation and inversion, integer linear combinations of these based loops yield based quotient words P, Q with displacements $(D, 0)$ and $(0, D)$, respectively, for $D = |\det(g, h)| > 0$. Since $g, h \in p\mathbb{Z} \times q\mathbb{Z}$, D is divisible by both periods. Thus every translation by $i(D, 0) + j(0, D)$ preserves allowed interior residues.

Lemma 117.2 applied to Q supplies a legal entry word u from the root to a lift z of the base vertex. Choose z sufficiently deep and far right that the finite paths P, Q and their reversals, based there, remain in the strictly positive quadrant. Then for every $m, n \geq 0$ the word

$$uP^m Q^n \overline{P}^m \overline{Q}^n \overline{u}$$

(where exponentiation applies to the full inverse words) is accepted: its pieces are legal nonnegative period translates of the anchored paths, and it returns to the root. Conversely a word of the regular macro shape

$$uP^m Q^n \overline{P}^k \overline{Q}^\ell \overline{u}$$

can be accepted only if its terminal height is zero, which implies $(m-k)D = (n-\ell)D = 0$. Hence the macro exponent language is exactly

$$\{\alpha^m \beta^n \gamma^m \delta^n : m, n \geq 0\}.$$

The inverse homomorphic preimage of L_R under the four fixed macros, after taking fixed left/right contexts and intersecting with the regular block expression, is this standard non-CFL. Closure of CFLs under these operations yields the contradiction. □

Theorem 117.4 (Opposite rank-one winding excludes every PDA)**.** *If some component has a rank-one voltage direction with negative coordinate product, then* $L_R \notin$ CFL.

Proof. Reverse an appropriate based closed quotient walk if needed to get net displacement $(-A, B)$ with $A, B > 0$ and $p \mid A$, $q \mid B$. Lift its base z deep into the positive quadrant. Repeating the inverse walk backward decreases y and increases x without bound; the first hit of $y = 1$ on this backward trajectory provides a fixed positive path U from $(x_e, 0)$, after one Y push, to z. Repeating the original walk forward decreases x and increases y without bound; the first hit of $x = 1$ gives a positive exit path V to $(0, y_f)$, after one X pop. The finite entry and exit paths may be translated by horizontal multiples of A and vertical multiples of B.

Choose fixed source words u for the entry after a^{x_e} and v for the exit to the Y axis. Consider the regular macro family

$$a^{x_e}(a^A)^m \, u \, T^n \, v \, (C^B)^\ell C^{y_f}, \qquad m, n, \ell \geq 0.$$

Here a pushes X, C pops Y, and the fixed u, v contain all necessary boundary steps. Terminal X and Y heights are respectively $A(m-n)$ and $B(n-\ell)$; therefore acceptance forces $m = n = \ell$. If they are equal, the T loops traverse translates indexed by $(A(m-1-j), Bj)$, and the translated entry/exit paths are allowed; the final Y pops occur on the free axis. Thus all and only matched triple exponents are accepted. Fixed-context rational transduction would make $\{\alpha^m \beta^m \gamma^m : m \geq 0\}$ context-free if L_R were context-free, a contradiction. □

117.3 The canonical positive construction

Assume now that every voltage component passes the sign/rank test. All source tags and their original order must be recoverable, not merely their heights. We first define a *logical* source-labelled word $W(U, V)$ and finite auxiliary tails; then prove uniform top locality and compile it into a physical word using one stack edit per input.

Set $N = \max(1, |R|)$, $D = 2N + 1$, $M = 4N^2 + 20$, and

$$E = M + 2N^2 + 2N + 1, \qquad C = 6N^2 + 7N + 70. \tag{117.1}$$

These are intentionally generous explicit constants, independent of source heights or tag contents.

117.3.1 Rank-zero lifted islands and short-source modes

Lemma 117.5 (Sharp bounded-minimum lemma)**.** *In any root-reachable rank-zero interior lifted component K, $\min(x, y) \le |K| \le N$.*

Proof. A zero-voltage lift contains exactly one vertex for each quotient residue in K, hence exactly $|K|$ vertices and graph diameter at most $|K| - 1$. The last positive-quadrant segment of a rooted path enters at $(x_e, 1)$ or $(1, y_e)$. Every vertex of that segment is at graph distance at most $|K| - 1$ from its entry, so respectively $y \le |K|$ or $x \le |K|$. □

Define the rank-zero normal form Z_F when $|U|, |V| \le N$ by storing both complete strings in finite control and setting $W = \varepsilon$. In Z_X when $|U| > N \ge |V|$, let $W = U$ and store V in control; in Z_Y, exchange roles. Mode and bounded stack tails are uniquely determined by the full source configuration. Any threshold crossing at N changes a top suffix of length at most $N + 1$, and no one-source-letter step can jump directly from Z_X to Z_Y.

117.3.2 Axis-parallel rank-one components

For a horizontal-winding component, choose tree potential t, and set

$$L_H(r) = N + 4 + t_y(r) - \min_{s \in K} t_y(s), \qquad b_Y = \max(0, |V| - L_H(r)).$$

Define $W = V[:\, b_Y]\, U$ and cache $V[b_Y :]$ in finite control. The latter tail has length at most $2N + 3$. Every interior edge $r \xrightarrow{(d_x, d_y)} s$ satisfies $t_y(s) = t_y(r) + d_y$, since all cycle voltages have zero vertical coordinate. Therefore b_Y is an exact interior invariant: changing X touches the exposed U top, and changing Y touches the cached Y tail only.

For a vertical-winding component symmetrically put

$$L_V(r) = N + 4 + t_x(r) - \min_{s \in K} t_x(s), \quad b_X = \max(0, |U| - L_V(r)),$$

and take $W = U[:\, b_X]\, V$, caching the bounded remaining X suffix. On an axis crossing, the old axis word and new interior word agree below all but at most $2N + 3$ source cells, since the frozen base is zero at the incident short-coordinate axis, or all but a bounded transverse tail at the opposite axis.

117.3.3 Same-sign rank-one components: aligned tagged frames

Let the primitive positive winding direction of K be (A, B), $A, B \ge 1$. All fundamental cycle voltages are integer multiples of this vector. It follows that $A, B \le D$. For $r = (x \bmod p, y \bmod q)$ define the *transverse invariant*

$$\chi = B(x - t_x(r)) - A(y - t_y(r)). \tag{117.2}$$

Every allowed interior edge preserves χ: the failure of $t(s) - t(r)$ to equal the edge displacement is a multiple of (A, B).

If $\chi \ge 0$, set

$$b_X = \max\{0, \lfloor \chi / B \rfloor - M\}, \quad b_Y = 0;$$

if $\chi < 0$, put

$$b_X = 0, \quad b_Y = \max\{0, \lfloor -\chi / A \rfloor - M\}.$$

Both base lengths are nonnegative, do not exceed their source heights, and remain constant on every interior step. Write

$$n_f = \min\left\{\left\lfloor \frac{x - b_X}{A} \right\rfloor, \left\lfloor \frac{y - b_Y}{B} \right\rfloor\right\}.$$

The logical word starts with the first b_X X-tags or b_Y Y-tags, labelled by their source. Above this *frozen base*, it has n_f ordered *paired frames*, frame i containing the next A X-tags and B Y-tags. Store the remaining top tails

$$U[b_X + n_f A :], \qquad V[b_Y + n_f B :]$$

in finite control. The first frame has a special marker; the remaining frames have an ordinary marker. The first original source cell of each nonempty stack likewise carries a finite depth-one marker wherever it resides. These finite marks make exact source emptiness and singleton-stack detection possible without scanning the physical stack.

Lemma 117.6 (Bounded tagged tails)**.** *The Northeast normal form above is defined for every positive source pair with residue in K; each finite cached tail has length at most $E + D + 1 < C$. The frozen base is invariant under interior edges, and n_f changes by at most one under any such edge.*

Proof. It suffices to analyze $\chi \geq 0$; the other case exchanges X and Y. Put $e = (x - b_X) - (A/B)y$. Since $|t_x|, |t_y| \leq N$ and $A, B \leq D$, if $b_X > 0$, then

$$e = M + t_x - (A/B)t_y + \{\chi/B\} > M - N - DN > 0,$$

and $e \leq E$. Hence $n_f = \lfloor y/B \rfloor$; the remaining Y suffix has length less than B, and the remaining X suffix has length less than $E + A$. When $b_X = 0$, we have $\lfloor \chi/B \rfloor \leq M$, so $-(N + DN) \leq e \leq E$. If X becomes the limiting source, its remainder is $< A$ and the remaining Y suffix is at most $B + (B/A)(N + (A/B)N) \leq D + DN + N$. Otherwise the previous bounds apply. All are smaller than $E + D + 1$. Invariance of b_X follows from invariance of χ; one source height changes by one on an interior step, so a floor quotient in the definition of n_f changes by at most one. Finally $M > N + DN$ prevents a nonempty frozen base from being the source top: above such a base there is a positive, mask-bounded-margin portion. □

117.3.4 Free-axis interfaces, top locality and one-stack packing

For an axis configuration (U, ε) or (ε, V) take W to be its sole nonempty tagged source word and store the usual finite mode/residue flags. A transition to an interior rank-zero component is top-local by Lemma 117.5. The horizontal/vertical case follows from the frozen-base formula. The remaining axis interface is explicit:

Lemma 117.7 (Northeast entry from either free axis)**.** *At a legal X-axis entry $y : 0 \to 1$ into a northeast component, with $r = (x \bmod p, 1 \bmod q)$ and $t(r) = (t_x, t_y)$, the new frozen X-prefix when $\chi \geq 0$ is*

$$b_X = \max\{0, x - k_X(r)\}, \qquad k_X(r) = M + t_x + \left\lceil \frac{A(1 - t_y)}{B} \right\rceil.$$

If $\chi < 0$, then the old X height is bounded by a mask-dependent constant. The reverse exit is top-local. The statements with X and Y interchanged hold on the Y axis.

Proof. The displayed formula follows directly from $\lfloor \chi/B \rfloor = x - t_x - \lceil A(1 - t_y)/B \rceil$. If $\chi < 0$, then $x < t_x + (A/B)(1 - t_y)$, which is bounded solely in N, D. In the active positive branch all but $k_X(r)$ cells of the axis X word form an unchanged physical prefix; if inactive, the entire old word has bounded length. The new logical word contains at most one paired

frame since $y = 1$. All content to be rearranged is therefore in a fixed-length top suffix. A reverse crossing has $y = 1$ and removes its lone Y source tag. The first-paired-frame marker, source-bottom tag markers and bounded caches distinguish this event from a deeper pop, even when a lower physical block is hidden. No complete source word is consulted. The Y-axis proof is symmetric. □

Lemma 117.8 (Finite top-local transduction)**.** *For all root-reachable source states satisfying the signed-winding condition, every defined or undefined one-letter source action can be decided from finite control and a bounded logical top suffix. On a defined action, the logical normal forms before and after have a common prefix and each changed suffix has length at most C in* (117.1)*. The necessary replacement suffix and new finite control are computable from those top data alone.*

Proof. Interior-to-interior edges remain in the same quotient component by construction of G_R. In rank-zero modes, bounded minimum height and threshold flags suffice. In horizontal/vertical modes, the transverse frozen base is unchanged and the other source is fully exposed above it, while the transverse source's top is in the bounded cache. In northeast modes the frozen base and its direction are invariant, and only the top paired frame and the two bounded tails may change; Lemma 117.6 bounds their sizes. Source top-tag matching is therefore visible. On an axis boundary, Lemma 117.7 or its rank-zero/axis-parallel analog gives a uniformly bounded common-prefix rewrite and a finite threshold test depending only on the residue and visible source-bottom/first-frame marks. The constants N, D, M, E, C were selected to dominate all individual cache and axis rewrite bounds (the largest is $O(N^2)$). Rejection is detected before changing the physical configuration. Thus a finite transition table over fixed finite cached strings and logical suffixes exists. □

Lemma 117.9 (One-operation block packing)**.** *Suppose a logical word W has a top-local transition that replaces a suffix of length at most C by one of length at most C. There is a unique fixed-block representation with a finite top buffer and a physical stack of C-cell blocks in which every logical input transition uses at most one physical stack push or pop, or a no-op, and no epsilon move.*

Proof. Store all of W in finite control if $|W| < 2C$. Otherwise place as many full C-cell blocks as possible on the physical stack while keeping exactly $C, \dots, 2C-1$ remaining top cells in finite control. The canonical choice is $k = \lfloor (|W| - C)/C \rfloor$ blocks; then $|W| - kC \in [C, 2C-1]$. The last C old cells lie inside the top buffer, so the source operation is evaluated without a hidden scan. After the bounded suffix replacement, the temporary buffer has length strictly less than $3C$. If it has length $\geq 2C$, push one full C-block; if it has length $< C$ and at least one physical block exists, pop one C-block and prepend it to the buffer; otherwise do nothing. This restores the unique canonical packing, including the case of zero blocks. A single stored block is one finite physical stack symbol; its push or pop is one real-time operation. □

Proof of the positive direction of Theorem 117.1. Define $\mathrm{NF}(U, V)$ by the axis, rank-zero, horizontal, vertical and northeast clauses. The original source tags, their bottom-to-top order, current residues, component and mode are explicitly recoverable from the combined control and labelled logical word. This proves full-source injectivity and history independence. Lemma 117.8 provides an exact finite top-local transition on every reachable source state. Lemma 117.9 turns it into a physical deterministic real-time transition reading the same input letter. Illegal pops, wrong tags, underflows and post-step mask violations are rejected without modifying the physical state. At the root the normal form is the initial physical state. Induction on guarded input length gives

$$\Delta_a\big(\mathrm{NF}(U, V)\big) = \mathrm{NF}\big(\delta_a(U, V)\big)$$

for every defined input action a, together with exact reflection of undefined actions and terminal empty-stack acceptance. □

117.4 An effective signed-winding test and examples

Proposition 117.10 (Explicit-mask decision complexity)**.** *For an explicitly supplied $p \times q$ mask, the signed-winding condition is computable using $O(pq)$ word-RAM operations and $O(pq)$ words of memory. The coordinates manipulated have $O(\log(pq+2))$ bits; a conservative schoolbook bit-operation upper bound is $O(pq\log^2(pq+2))$.*

Proof. Process each vertex and each of its at most four signed arcs using a spanning-forest search. Every non-tree arc contributes a chord voltage. In each component, save the first nonzero voltage v. Another voltage with nonzero determinant with v certifies rank two; otherwise all nonzero voltages are collinear and their common sign type decides the rank-one condition. Each edge is processed constantly many times and each determinant costs a constant number of word operations. The potential magnitude is $O(pq)$ by the tree length bound. □

The explicit-mask graph criterion does *not* imply a small compiled machine. With a fixed logical alphabet of g cells and the bound C, physical stack symbols may be C-tuples, so a coarse physical alphabet bound is $1+g^C$; the controller has similarly large, but finite, bounded-tag caches.

Example 117.11 (A non-affine periodic example)**.** For periods $(p,q)=(4,4)$, define the mask R by integer bitset 6599, where bit $i+4j$ indicates membership of residue (i,j). Its row patterns at $j=0,1,2,3$ are respectively 1110, 0011, 1001, and 1000 in increasing i order. An exhaustive check of all $4^3=64$ coefficient triples shows that it is not representable as an affine-complement condition $ax+by+c\not\equiv 0 \pmod 4$, yet its graph satisfies the positive signed-winding condition and has a northeast rank-one component. Thus the example is genuinely outside the affine-complement modular family. The mask is explicitly specified here, and its signed residue graph can be verified directly from the definition.

For the important special case $R=\{(x,y) : ax+by\not\equiv 0 \pmod m\}$, the coefficient condition is $a=0$ or $b=0$ or $a+b=0 \pmod m$. The linear-modular coefficient condition is proved separately by the explicit arithmetic and path arguments in Chapter 116. The signed-winding criterion concerns a larger guard family; identifying the two tests for every composite modulus requires a separate symbolic specialization argument, not merely finite computations. The present paper requires no primality assumption on any period.

117.4.1 A sharp island bound and a physical-information lower bound

Proposition 117.12 (Uniform sharpness for rank-zero masks)**.** *For every integer $m\ge 1$ there exists a rank-zero mask with precisely m allowed residues and a root-reachable point satisfying $\min(x,y)=m$. Thus $|R|$ is the optimal universal bound in Lemma 117.5, viewed as a function of $|R|$ alone.*

Proof. Take $p=2$, $q=m+1$ and $R_m=\{(1,j) : 1\le j\le m\}$. Every connected interior lift is a vertical chain of m points, so every voltage vanishes. Choose an odd $x_m\ge m$, move along the free X axis from the origin to $(x_m,0)$, and push Y successively to height m. All intervening positive points have permitted residues and the endpoint has $\min(x_m,m)=m=|R_m|$. □

Proposition 117.13 (Source-information capacity; elementary counting bound)**.** *Suppose that, at a root-reachable height (x,y), all s^xt^y assignments of independent source tags are reachable, where $s=|T_X|$, $t=|T_Y|$. Any full-source-injective encoding with at most Q finite-control states and a physical stack alphabet of $g\ge 2$ symbols must attain stack depth at least*

$$H(x,y) \;\ge\; \log_g\left(1+\frac{(g-1)s^xt^y}{Q}\right)-1$$

for some configuration at that height. In particular, when $s,t\ge 2$ and a legal family has $x,y=\Theta(n)$, every such representation has worst-case depth $\Omega(n)$ for fixed Q,g.

Proof. The $s^x t^y$ distinct source pairs require distinct physical configurations. There are at most $Q \sum_{k=0}^{H} g^k = Q(g^{H+1} - 1)/(g - 1)$ physical configurations with stack height at most H. Rearranging gives the claimed inequality. This standard information-counting argument is included only to quantify the physical meaning of full source preservation; it is not claimed as a novel combinatorial method. □

Finite checks. Independent implementations have tested signed-voltage classification, tagged-source transitions, and physical block thresholds for representative periodic masks. These finite checks are supplementary consistency evidence; the all-period assertions depend on the symbolic proofs in the preceding sections.

Chapter 118

Extrema–Frame Semantics, Observer Records, and Exact Realizations

Research branch — External generativity: finite scalar fields. *This chapter begins a theory that is independent of pushdown automata. Its role in the monograph is twofold: it develops a self-dual extrema-based foreground/background semantics in its own native language, and it tests whether the exact-realization discipline of the preceding US-NRT chapters transfers to a genuinely different physical setting. The resulting scalar-field theorems do not depend on the original p:q automata.*

Relation to neighboring scalar-field theories. Merge trees, hierarchical region trees, self-dual morphology, extinction/dynamics, and uncertainty in scalar-field topology are established subjects [125, 126, 127, 128, 129, 130, 131, 136, 135]. The contribution here is narrower: observer-relative foreground ownership under exact same-realization semantics, the observer-record normal form, the local/global complexity separation on shallow graph classes, and explicit all-orders common-realization obstruction families.

118.1 EF-FG model

118.1.1 Filtration conventions and birth atoms

For $t \in \mathbb{R}$, let $G^{\geq t}$ be the induced subgraph on $\{v : F(v) \geq t\}$. A **birth plateau** at level B is a connected local-maximum plateau $P \subseteq F^{-1}(B)$ whose superlevel component appears when the threshold reaches B. The corresponding component identity is a **primitive birth atom** λ, with birth level $B_\lambda = B$. A **birth-carrier representative** is any vertex of P; all such representatives have the same maximin attachment to any observer because they are connected inside $G^{\geq B}$.

As the threshold decreases, birth components merge. The elder rule keeps the atom of larger birth level. If several atoms have equal birth level, an **admissible elder/tie resolution** is a total order on those equal-birth atoms, used consistently at later merges. Exactly one atom survives to the bottom of the connected filtration; it is the **essential atom**. Every other atom is **finite** (or **nonessential**), and its **elder death level** $d(\lambda)$ is the merge level at which it is killed by an older atom. The dark filtration is the same construction applied to $-F$.

118.1.2 Scalar fields, births, and maximin attachment

Fix a finite connected graph $G = (V, E)$ and a scalar field $F : V \to \mathbb{R}$. For the bright polarity, let λ range over primitive nonessential birth atoms of the superlevel filtration.

Write

$$B_\lambda = \text{birth level of } \lambda, \qquad d(\lambda) = \text{elder death level.}$$

For an observer $x \in V$, define

$$a_\lambda(x) = \kappa_F(\lambda, x),$$

where κ_F is maximin connectivity. If $O(\lambda)$ denotes the elder atoms of λ, define the observer elder pressure

$$D_x(\lambda) = \max_{\mu \in O(\lambda)} a_\mu(x).$$

The dark polarity is the bright construction applied to $-F$.

118.1.3 Observer records

A nonessential birth λ is an **observer record at** x if

$$a_\lambda(x) > D_x(\lambda).$$

Thus records are precisely the births that strictly raise the running elder attachment available at the observer.

A basic maximin identity drives the theory.

Lemma 118.1 (maximin equality)**.** *If*

$$\kappa_F(p, x) > \kappa_F(q, x),$$

then

$$\boxed{\kappa_F(p, q) = \kappa_F(q, x).}$$

Proof sketch. The strong-triangle inequality gives the lower bound $\kappa_F(p, q) \geq \kappa_F(q, x)$. If the left side were larger, concatenating a maximizing p–q path with a q–x path would force q to attach to x above $\kappa_F(q, x)$, a contradiction. A complete proof is given in the observer-record proof package later in this chapter. □

118.1.4 Observer-core saliency

Fix $0 < \theta < 1$. A finite birth λ contributes to the observer core when

$$a_\lambda(x) > \theta B_\lambda + (1 - \theta) D_x(\lambda).$$

The one-polarity observer saliency is

$$\Sigma_\theta(x; F) = \max\Big(\{B_\lambda - D_x(\lambda) : a_\lambda(x) > \theta B_\lambda + (1 - \theta) D_x(\lambda)\} \cup \{0\}\Big).$$

Set

$$\Sigma_\theta^+(x; F) = \Sigma_\theta(x; F), \qquad \Sigma_\theta^-(x; F) = \Sigma_\theta(x; -F).$$

For a significance threshold $\Delta \geq 0$, the polarity-core ownership label is

$$\mathrm{Own}_{\Delta,\theta}(x; F) \in \{\text{bright}, \text{dark}, \text{background}, \text{ambiguous}\},$$

with background defined by

$$\max\{\Sigma_\theta^+(x; F), \Sigma_\theta^-(x; F)\} \leq \Delta.$$

The remaining labels are determined by which polarity exceeds Δ, with an explicit ambiguous state when both do.

118.1.5 Spatial-frame projection

Fix a spatial seed threshold $\Delta_{\text{seed}} \geq 0$. For one polarity, the selected seed family consists of the finite birth atoms with persistence $B_\lambda - d(\lambda) > \Delta_{\text{seed}}$; the essential atom is excluded. Choose one birth-carrier representative for each selected atom p and put

$$\kappa_p(v) = \kappa_F(p, v), \qquad \beta(v) = \max_r \kappa_r(v), \qquad W(v) = \{p : \kappa_p(v) = \beta(v)\}.$$

The **rooted co-winner basin** $\mathcal{B}_p$ is the connected component containing p in $\{v : p \in W(v)\}$. Two selected seeds p, q are **carrier-adjacent** when $\mathcal{B}_p \cap \mathcal{B}_q \neq \varnothing$. Their gate scalar is

$$s(p, q) = \max_{v \in \mathcal{B}_p \cap \mathcal{B}_q} \beta(v),$$

and the connected components attaining this maximum are the top gate components. Route incidence through an observer, midpoint margins, designated boundary terminals, and correlated residual terminal regions then form the spatial-frame state; their exact definitions are given in the spatial-frame construction later in this chapter.

Scope principle. Polarity-core ownership is an observer/persistence projection, whereas spatial frame is a carrier/sector projection. The two are evaluated from the same scalar realization, but neither determines the other in general. The full fixed-realization EF-FG state stores both rather than identifying them.

118.1.6 Interval uncertainty and exact semantics

An interval-valued scalar field is

$$\mathcal{P} = \prod_{v \in V} [\ell_v, u_v].$$

A **realization** consists of one $F \in \mathcal{P}$ together with an admissible resolution of ties/elder order. A property is **possible** if one realization satisfies it and **robust** if every realization satisfies it.

This quantifier discipline is essential. If $A(\omega)$ and $B(\omega)$ are observables on the realization space, then generally

$$\{(A(\omega), B(\omega)) : \omega \in \Omega\} \subsetneq \{A(\omega) : \omega \in \Omega\} \times \{B(\omega) : \omega \in \Omega\}.$$

Thus one must **join before projecting**: a possible ownership label cannot in general be inferred by combining independently attainable extrema of births, attachments, or saliencies.

118.2 Hierarchy-free observer normal form

The full merge hierarchy can be eliminated from pointwise observer-core evaluation.

118.2.1 Exact observer-record death

Theorem 118.2 (exact observer-record death)**.** *Let λ be a nonessential primitive birth atom. If*

$$a_\lambda(x) > D_x(\lambda),$$

then λ is an observer record and

$$\boxed{d(\lambda) = D_x(\lambda).}$$

If instead $a_\lambda(x) \leq D_x(\lambda)$, then λ cannot make a positive observer-core contribution.

Proof sketch. For a record, choose the elder atom attaining $D_x(\lambda)$. Lemma 118.1 identifies the merge level between the younger record and that elder atom with the elder's attachment at x, so the ordinary elder death is exactly $D_x(\lambda)$. If λ is not a record, its attachment does not exceed elder pressure and therefore cannot satisfy the strict core inequality. □

A complete proof is given in the observer-record proof package.

An immediate consequence is the formula already used in the observer-record definition above: the saliency maximum can be taken over all births because nonrecords eliminate themselves automatically.

118.2.2 Record chains

Order the finite observer records by elder-to-younger birth order,

$$r_1, r_2, \dots, r_m,$$

and write $a_i = a_{r_i}(x)$. Include the essential elder supply as a_0. Then

$$\boxed{d(r_i) = a_{i-1}.}$$

Thus the complete finite record history is a chain of adjacent transitions. If B_i is the birth of r_i, then the i-th potential contribution is determined by the triple

$$(B_i, a_{i-1}, a_i).$$

For an H-safety query, a record is dangerous precisely when

$$B_i - a_{i-1} > H$$

and

$$a_i > \theta B_i + (1-\theta)a_{i-1}.$$

Equivalently, define the required predecessor supply

$$L_i(H) = \min\left\{B_i - H, \frac{a_i - \theta B_i}{1-\theta}\right\}$$

with the obvious convention that only the active branch matters. Then r_i is H-safe iff a_{i-1} exceeds the corresponding threshold. This is the local origin of the later scheduling formulation.

118.2.3 A single elder-supply profile

The record chain may be represented by one rooted supply profile $E_x(b)$: the greatest observer attachment supplied by births older than threshold b. Its jumps are exactly the observer records. At a finite jump (B, D, A),

$$D = \text{exact elder pressure}, \qquad A = \text{record attachment},$$
$$B - D = \text{exact persistence}.$$

Hence

$$\boxed{\begin{array}{c}\Sigma_\theta(x;F) \le H \iff \text{every finite jump } (B,D,A) \text{ satisfies} \\ B - D \le H \quad \text{or} \quad A \le \theta B + (1-\theta)D.\end{array}}$$

On rooted trees the profile obeys a max/min recurrence over child subtrees, giving a direct deterministic evaluation algorithm. A proof of the record-chain reduction and the rooted supply recurrence is given in the observer-record proof package.

118.2.4 Self-duality

All statements above are polarity-normalized. Applying them to $-F$ gives the dark theory with no change of form. In particular, grayscale inversion transports the entire common-realization relation bijectively: bright-possible becomes dark-possible, bright-robust becomes dark-robust, and the full ownership relation is conjugated realization by realization.

118.3 Exact common-realization perturbation theory

118.3.1 Why marginal envelopes are insufficient

Suppose two admissible realizations ω_0, ω_1 give

$$(A(\omega_0), B(\omega_0)) = (0, 1), \qquad (A(\omega_1), B(\omega_1)) = (1, 0).$$

Then each coordinate individually attains both 0 and 1, but $(0, 0)$ and $(1, 1)$ are unrealizable. This elementary observation becomes structural in EF-FG because births, attachments, elder pressure, record identity, and ownership are coupled by the same field values.

We therefore define the **exact EF-FG realization relation** by evaluating all requested coordinates on one realization first and projecting only afterward. Possible and robust states are existential and universal statements over that relation.

118.3.2 Ownership relation

For a fixed observer x, a realization determines

$$\big(\Sigma^+_\theta(x), \Sigma^-_\theta(x), \mathrm{Own}_{\Delta,\theta}(x), \mathsf{SF}_x, \text{record data}\big).$$

The exact perturbation object is the image of the realization space under this joint map. In particular, a claim that x is possibly background means

$$\exists F, \rho : \ \Sigma^+_\theta(x; F, \rho) \le \Delta \quad \text{and} \quad \Sigma^-_\theta(x; F, \rho) \le \Delta$$

for the **same** F and resolution ρ.

118.3.3 Realization-wise self-dual conjugacy

Theorem 118.3 (realization-wise self-duality)**.** *Grayscale inversion $F \mapsto -F$ induces a bijection between the bright exact relation over $\mathcal{P}$ and the dark exact relation over $-\mathcal{P}$. Consequently possibility, robustness, background status, and deterministic bright/dark ambiguity are preserved under polarity exchange in the expected self-dual sense.*

The theorem follows because inversion is a bijection on realizations and every dark construction is defined as the bright construction on $-F$.

Proof details are included in the observer-record proof package.

118.3.4 Local witness versus global synchronization

The exact relation already suggests two algorithmic regimes. A local exceedance needs one feature witness and a bounded set of separator constraints. A global safety condition quantifies over the entire realized record history. Later sections make this distinction exact: the former is polynomial on interval-valued cacti, while the latter is NP-complete already on radius-two trees.

118.4 Proof package: observer records and exact common-realization conjugacy

This proof package supplies the details used by the observer-record and exact-common-realization results above.

118.4.1 Maximin equality

For vertices p, q, x, if $\kappa_F(p, x) > \kappa_F(q, x)$, then

$$\boxed{\kappa_F(p, q) = \kappa_F(q, x).}$$

Indeed, the maximin strong-triangle inequality gives

$$\kappa_F(p, q) \geq \min\{\kappa_F(p, x), \kappa_F(q, x)\} = \kappa_F(q, x).$$

If the inequality were strict, applying the same inequality to $q \to p \to x$ would give

$$\kappa_F(q, x) \geq \min\{\kappa_F(q, p), \kappa_F(p, x)\} > \kappa_F(q, x),$$

a contradiction.

118.4.2 Proof of Theorem 118.2

Assume first that λ is an observer record, so $a_\lambda(x) > a_\mu(x)$ for every elder atom μ. Choose birth-carrier representatives e_λ, e_μ. By A.1,

$$\kappa_F(e_\lambda, e_\mu) = a_\mu(x).$$

The ordinary elder death of λ is the largest merge level at which it meets an elder atom, hence

$$d(\lambda) = \max_{\mu \in O(\lambda)} \kappa_F(e_\lambda, e_\mu) = \max_{\mu \in O(\lambda)} a_\mu(x) = D_x(\lambda).$$

If λ is not a record, choose an elder μ with $a_\mu(x) = D_x(\lambda) \geq a_\lambda(x)$. Then

$$\kappa_F(e_\lambda, e_\mu) \geq \min\{a_\lambda(x), a_\mu(x)\} = a_\lambda(x),$$

so $d(\lambda) \geq a_\lambda(x)$. Its observer-relative depth is therefore nonpositive and it cannot satisfy the strict core inequality for any $\theta > 0$. This proves both record death and nonrecord elimination.

118.4.3 Adjacent-record chain and supply profile

Order the records $r_1, \ldots, r_m$ from elder to younger. Because r_i strictly raises the running attachment, its elder pressure is the previous record value: $D_x(r_i) = a_{i-1}$. By A.2,

$$\boxed{d(r_i) = a_{i-1}.}$$

Thus each finite contribution is determined by (B_i, a_{i-1}, a_i). The danger conditions are exactly

$$B_i - a_{i-1} > H, \qquad a_i > \theta B_i + (1 - \theta) a_{i-1}.$$

Define $E_x(b)$ to be the greatest observer attachment supplied by births older than level b. The function is monotone stepwise; a jump occurs exactly when a younger birth strictly raises the running maximum, i.e. exactly at an observer record. Its left value is D, its right value is A, and A.2 gives persistence $B - D$. Hence the one-profile criterion in the supply-profile section above follows by checking the displayed two danger inequalities at

every finite jump. On a rooted tree, maximin connectivity to a parent is the minimum of the edge/vertex bottleneck and the child-subtree value, while independent child branches combine by maximum; induction on the rooted tree gives the stated max/min recurrence.

118.4.4 Self-dual conjugacy

The map $F \mapsto -F$ is a bijection between realizations of $\mathcal{P}$ and $-\mathcal{P}$. Superlevel births of $-F$ are precisely sublevel births of F; maximin connectivity, elder order, record status, and all strict/equality cases are transported under the polarity relabeling. Therefore the joint tuple

$$(\Sigma_\theta^+, \Sigma_\theta^-, \mathrm{Own}_{\Delta,\theta}, \mathsf{SF}, \text{record data})$$

is conjugated realization by realization. Existential and universal projections of this exact relation are consequently exchanged in the claimed bright/dark fashion. This proves Theorem 118.3.

118.5 Exact spatial-frame construction

The ownership projection does not encode route geometry. This proof package fixes one polarity-normalized realization and gives the second EF-FG projection used in the spatial-frame synthesis chapter.

Fix a spatial seed threshold $\Delta_{\mathrm{seed}} \geq 0$. Let S be the finite set of nonessential birth atoms satisfying

$$B_p - d(p) > \Delta_{\mathrm{seed}}.$$

Choose one representative vertex for each $p \in S$. For $v \in V$ define

$$\kappa_p(v) = \kappa_F(p, v), \qquad \beta(v) = \max_{r \in S} \kappa_r(v), \qquad W(v) = \{p \in S : \kappa_p(v) = \beta(v)\}.$$

The *rooted co-winner basin* $\mathcal{B}_p$ is the connected component containing p in the induced set $\{v : p \in W(v)\}$. The rooted unique-winner basin is defined analogously from $W(v) = \{p\}$. Selected seeds p, q are *carrier-adjacent* iff

$$\mathcal{B}_p \cap \mathcal{B}_q \neq \varnothing.$$

For an adjacent pair put

$$s(p, q) = \max_{v \in \mathcal{B}_p \cap \mathcal{B}_q} \beta(v).$$

The *top gate carrier* is the induced set

$$\{v \in \mathcal{B}_p \cap \mathcal{B}_q : \beta(v) = s(p, q)\},$$

retained componentwise rather than collapsed to one arbitrary point.

Fix an ordered pair (p, q) and $x \in \mathcal{B}_p$. For a top gate component C, a simple path in $\mathcal{B}_p$ from p to a vertex of C is an admissible top route. The component is *weakly incident through* x when at least one such route contains x, and *strongly incident through* x when every such route contains x. Thus strong incidence implies weak incidence; on a cactus the two conditions reduce respectively to route existence and a no-bypass (vertex-dominator) test in the rooted carrier subgraph.

For a bright sector define the midpoint margin

$$\mu^+_{p,q,x} = 2F(x) - F(p) - s(p, q).$$

The three signs > 0, $= 0$, and < 0 are retained, and are interpreted together with weak/strong incidence. The dark sector state is the same construction applied to $-F$.

To make boundary information explicit, let $T \subseteq V$ be a designated terminal set supplied with the spatial query. An exposed terminal for seed p is a vertex $t \in T \cap \mathcal{B}_p$. Its gate is

$$s_\partial(p,t) = \kappa_{F|\mathcal{B}_p}(p,t).$$

Put $s = s_\partial(p,t)$ and, for $v \in \mathcal{B}_p$,

$$\alpha(v) = \kappa_{F|\mathcal{B}_p}(p,v), \qquad \gamma(v) = \kappa_{F|\mathcal{B}_p}(t,v).$$

Vertices with $\alpha > s$ are forced to the peak side; vertices with $\gamma > s$ are forced to the terminal side; only the tie zone $\alpha = \gamma = s$ can be assigned by residual resolution. A *residual terminal resolution* is a connected terminal-side region R containing t, containing every forced terminal-side vertex in its connected terminal component, excluding forced peak-side vertices, and using only tie-zone vertices for the remaining choice. The intrinsic family of all such regions is denoted $\mathfrak{R}^{\mathrm{res}}_{p,t}$.

For $R \in \mathfrak{R}^{\mathrm{res}}_{p,t}$ define its floor and cap

$$L(R) = \min_{z \in R} F(z), \qquad U(R) = \max_{z \in R} F(z).$$

For the canonical terminal side $L(R) = s_\partial(p,t)$. At threshold Δ, define

$$T^+_\Delta(L,U) = \max\left\{L + \Delta, \frac{L+U}{2}\right\}, \qquad T^-_\Delta(L,U) = \min\left\{U - \Delta, \frac{L+U}{2}\right\}.$$

A query value y is terminal-dark when $y < T^-_\Delta$, neutral when $T^-_\Delta \le y \le T^+_\Delta$, and terminal-bright when $y > T^+_\Delta$. Grayscale inversion exchanges the bright and dark inequalities and fixes the neutral band.

Under interval uncertainty, these quantities must remain correlated. The exact residual object is therefore

$$\mathrm{ResRel}_{p,t,x} = \left\{(s,U,y,\chi) : \begin{array}{l}\text{one admissible realization and one } R \in \mathfrak{R}^{\mathrm{res}}_{p,t} \\ \text{realize the tuple}\end{array}\right\},$$

where $y = F(x)$ and $\chi = \mathbf{1}[x \in R]$. Independent extrema of s, U, y, χ may not be recombined. On cactus graphs the block-cut decomposition gives constant-width route interfaces, so pairwise gates, weak/strong incidence, terminal exposure, residual connectedness, and the correlated floor/cap/query relation admit polynomial exact dynamic programs. This is a spatial-frame statement and is distinct from the polarity-core complexity theorems below.

Chapter 119

Threshold-Band Scheduling and Ownership Complexity

Research branch — Local witness versus global synchronization. *Exact common-realization semantics creates an algorithmic boundary. A local exceedance witness can be certified through bounded separator data on cactus graphs, whereas one globally synchronized realization that suppresses every dangerous record is NP-complete already on radius-two trees. The proof passes through a two-coordinate scheduling kernel and an exact physical lift.*

Book-wide notation. We use τ for the local critical birth floor and call the corresponding scheduling kernel the *τ-band*, or *threshold-band*. This distinguishes the local floor from the global balance-ratio parameter q.

119.1 The τ-band owner-scheduling kernel

We now isolate the arithmetic core of one-polarity common-realization safety.

119.1.1 Hard-band state

Write the running state as (M, D), where M is the current birth cap and D is the current elder supply. For one event i, let its birth interval be $[\ell_i, u_i]$ and its attachment interval be $[\alpha_i, \beta_i]$. In the hard band, safety is

$$a \le \theta B + (1-\theta)D.$$

Fix a critical birth floor τ. Events with $\ell_i = \tau$ must be resolved before the cap drops below τ. When $\tau \le D + H$, the band is easy: an H-safe suffix is immediately reachable. The nontrivial regime is therefore

$$\tau > D + H.$$

For a hard-band event define the maximal safe choice

$$U_i = \min\{M, u_i\}, \qquad a_i^* = \min\{\beta_i, \theta U_i + (1-\theta)D\}.$$

Every other τ-preserving safe realization of the event is dominated by this choice.

119.1.2 τ-band coordinates

Set

$$h = M - \tau, \qquad \xi = \tau - D,$$

and for event i define

$$h_i = u_i - \tau, \qquad A_i = \tau - \alpha_i, \qquad B_i = \tau - \beta_i.$$

For general θ let $c = 1 - \theta$ and $s_i = \min\{h, h_i\}$. Then an owner use is legal exactly when

$$\boxed{B_i < \xi}$$

and

$$\boxed{c\xi - \theta s_i \le A_i},$$

with deterministic transition

$$\boxed{(h, \xi) \longmapsto (s_i, \max\{B_i, c\xi - \theta s_i\})\,.}$$

For the hardness specialization $\theta = 1/2$ this becomes

$$(h, \xi) \longmapsto \left(s_i, \max\left\{B_i, \frac{\xi - s_i}{2}\right\}\right),$$

legal when

$$B_i < \xi, \qquad \frac{\xi - s_i}{2} \le A_i.$$

119.1.3 Fixed-order reachability

The scalar transition is monotone for a fixed effective headroom, but a used event can lower the future headroom. Consequently one scalar minimum does not suffice for an arbitrary fixed order. The correct state retains the current headroom as well.

Theorem 119.1 (fixed-order absorbing-tail reachability)**.** *For any fixed event order, whether some admissible subsequence reaches the absorbing safe tail $\xi \le H$ is decidable by a headroom-indexed dynamic program using $O(n^2)$ event-state transitions. If additionally*

$$h_0 \ge h_1 \ge \cdots \ge h_n,$$

then every used event has effective headroom h_i, independent of previous skips, and the one-pass prefix-minimum greedy is exact in $O(n)$ transitions.

A proposed owner word is still verified in $O(n)$ exact transitions, so NP membership does not rely on the stronger greedy special case.

Remark 119.2 (why the headroom hypothesis matters). At $\theta = 1/2$, $H = 0$, start from $(h, \xi) = (10, 10)$ and fix the order

$$e_1 = (0, 10, 0), \qquad e_2 = (10, 0, 0).$$

Using e_1 first gives $(0, 5)$, after which e_2 is infeasible because $(5 - 0)/2 > 0$. Skipping e_1 and using e_2 gives $(10, 0)$ and reaches the safe tail. Hence a single prefix minimum in ξ is not valid without retaining headroom information.

The dynamic program and the nonincreasing-headroom greedy proof are given in the threshold-band derivation later in this chapter.

119.2 NP-completeness of τ-band owner scheduling

119.2.1 Decision problem

The τ-band owner-scheduling completion problem asks whether a finite event family admits an order and legal owner-use pattern that reaches the absorbing safe tail. We use the fixed specialization

$$\theta = \frac{1}{2}, \qquad H = 0.$$

Membership in NP follows from a rational schedule certificate: one checks the event order and the deterministic τ-band transitions in polynomial time.

119.2.2 Reduction spine

NP-hardness is by reduction from Monotone NAE-3-Sat-E4 [134]. The main logical spine is as follows; the complete reduction, including the binding scale and phase-leak repairs, is given in the complete threshold-band reduction later in this chapter.

Each variable module has exactly two successful traversal modes. The choice designates one of two payload banks for deletion and leaves the other available for clause auditing. A two-gate **headroom certificate** prevents a payload token from being replaced by a token belonging to another variable or another truth bank. Clause modules use a three-token not-all-high audit, together with directed self-policing, so that a clause succeeds exactly when its three incoming truth states are not all equal. A homogeneous scale hierarchy isolates variables from one another and global audit floors prevent cross-module substitutions. A final finite control wrapper closes premature phase entry and early-pivot leakage.

Completeness maps a satisfying NAE assignment to the corresponding branch of each variable module and then executes all clause/audit modules. Soundness reconstructs a Boolean choice from every successful schedule and uses the headroom certificates plus the self-policing clause gadgets to show that every clause is NAE-satisfied.

Theorem 119.3 (τ-band NP-completeness)**.** *τ-band owner-scheduling completion is NP-complete even for*

$$\boxed{\theta = \frac{1}{2}, \qquad H = 0,}$$

with rational dyadic event parameters and a linear number of events.

The result isolates the source of hardness: **global order selection under one-use owner resources**, not evaluation of a fixed order.

The complete iff reduction and NP-membership proof are in the complete threshold-band reduction later in this chapter.

119.3 Exact lift to interval-valued scalar trees

The scheduling kernel is not merely an abstract surrogate. It is exactly realized by radius-two scalar graphs.

119.3.1 Affine rectangle conjugacy

Choose τ larger than all τ-band coordinates and apply

$$M = \tau + h, \qquad D = \tau - \xi,$$

and, for every event,

$$\ell_i = \tau, \qquad u_i = \tau + h_i, \qquad \alpha_i = \tau - A_i, \qquad \beta_i = \tau - B_i.$$

Under this change of variables, the maximal-safe rectangle transition is conjugate to the τ-band transition. Before the physical tree is built we apply the positive-floor normalization proved in the physical-lift proof package later in this chapter: every convenience floor $B_i = 0$ is replaced by one common rational $\varepsilon > 0$ chosen below every intended positive transition scale. the physical-lift proof package later in this chapter proves by a dominance coupling that this normalization preserves YES/NO equivalence; in particular it neither destroys the intended completeness schedules nor creates a false YES instance. After normalization,

$$0 < \varepsilon \le B_i \le A_i$$

for every event.

119.3.2 Radius-two physical realization

For each event create a branch

$$x - s_i - p_i,$$

where p_i is an interval-valued peak and s_i an interval-valued saddle. After positive-floor normalization,

$$F(s_i) \le \tau - \varepsilon = F(x) < \tau \le F(p_i).$$

Since the branch path is unique and the observer is no lower than the saddle,

$$\kappa_F(p_i, x) = F(s_i).$$

An initial essential branch sets the starting elder pressure, and a fixed observer guard prevents unintended critical changes. Peak heights encode the schedule order; saddle intervals encode the admissible attachments.

Theorem 119.4 (exact physical lift)**.** *For the constructed radius-two interval tree,*

$$\boxed{\begin{array}{c}\textit{threshold-band instance is YES}\\ \Longleftrightarrow\\ \textit{interval-tree one-polarity observer-safety instance is YES.}\end{array}}$$

The construction is polynomial and uses rational endpoints.

The exact conjugacy and tree-realization proof are in the physical-lift proof package later in this chapter.

119.3.3 Physical complexity theorem

A YES instance has a polynomial-bit rational realization and polynomial-size elder/tie certificate. On a tree, all maximin attachments are unique-path minima, so verification is polynomial.

Theorem 119.5 (physical observer-safety complexity)**.** *The problem of deciding whether an interval-valued rooted tree and observer x admit one realization and one admissible elder/tie resolution for which the one-polarity EF-FG saliency is H-safe at x is NP-complete even when*

$$\boxed{\theta = \frac{1}{2}, \qquad H = 0,}$$

the tree has radius two about the observer, and all interval endpoints are rational.

Thus the hardness does not require cycles, deep tree topology, or variable model parameters.

Polynomial witness size and verification are proved in the physical-lift proof package later in this chapter.

119.4 Self-dual ownership complexity

119.4.1 Dark-anchor neutralization

To lift one-polarity safety to the full self-dual ownership predicate, add one constant-size low-valued leaf d, the **dark anchor**. It is chosen so that it is not a bright primitive maximum and hence leaves every bright observer quantity unchanged.

In the inverted field $-F$, the anchor becomes the essential dark atom. Every other primitive dark birth reaches the observer through a path whose bottleneck equals the observer value. Consequently every finite dark birth has the same observer attachment as the anchor's running elder pressure.

Theorem 119.6 (dark-anchor neutralization)**.** *For every realization and every admissible resolution,*

$$\boxed{\Sigma_\theta^-(x;F) = 0.}$$

The statement is realization-wise; no independent choice of a dark realization is made.

A realization-wise proof is given in the dark-anchor proof package later in this chapter.

119.4.2 Possible background ownership

At $\Delta = 0$, after adding the dark anchor,

$$\mathrm{Own}_{0,\theta}(x;F) = \text{background} \iff \Sigma_\theta^+(x;F) = 0.$$

Thus the one-polarity safe-realization problem is exactly the self-dual possible-background decision problem.

Membership in NP follows from a polynomial-bit scalar realization and polynomial-size resolved elder/tie data for both polarities.

Theorem 119.7 (possible-background ownership complexity)**.** *Possible Background Polarity-Core Ownership is NP-complete even on interval-valued radius-two trees with fixed*

$$\boxed{\theta = \frac{1}{2}, \qquad \Delta = 0.}$$

The only additional construction beyond Theorem 119.5 is the constant-size dark anchor.

The reduction equivalence and NP-membership argument are in the dark-anchor proof package later in this chapter.

119.5 The local/global algorithmic boundary

The complexity theorem is best understood by contrasting it with local witness problems.

119.5.1 Local possible exceedance

For fixed θ and H, a positive saliency witness can be represented by one candidate birth together with a bounded set of separator/connectivity conditions. On interval-valued cactus graphs those constraints are decidable by polynomial exact routines.

Theorem 119.8 (possible one-polarity exceedance)**.** *On interval-valued cactus graphs,*

$$\exists F, \rho : \Sigma_\theta^\sigma(x;F,\rho) > H$$

is polynomially decidable for either polarity $\sigma \in \{+,-\}$.

A proof by a finite separator witness and cactus block-cut dynamic program is given in the cactus-witness proof package later in this chapter.

119.5.2 Complexity separation

Combining the local witness theorem, the fixed-order collapse, and the hardness results gives:

Theorem 119.9 (EF-FG local/global separation)**.** *For polarity-core saliency:*

1. *possible one-polarity exceedance is polynomially decidable on interval-valued cacti;*
2. *arbitrary fixed-order absorbing-tail reachability is polynomial (quadratic by Theorem 119.1), and the nonincreasing-headroom subclass is linear;*
3. *global possible one-polarity safety is NP-complete on radius-two trees;*
4. *self-dual possible-background ownership is NP-complete on radius-two trees.*

The transition is therefore neither "tree versus cactus" nor merely "deterministic versus uncertain." It occurs when one realization must coordinate all potential observer records under a global no-danger requirement.

The conclusion follows from Theorem 119.8, Theorem 119.1, and the hardness proofs completed in Appendices D–F.

119.6 Proof package: threshold-band derivation and fixed-order theorem

Let $c = 1 - \theta$. In a critical band with floor τ, no schedule may lower the running birth cap below τ while an event with minimum birth τ remains unresolved. If $\tau \le D + H$, every remaining birth $B \le \tau$ is persistence-safe, so only the hard regime $\tau > D + H$ matters.

For one hard-band event, put $U_i = \min\{M, u_i\}$. It has a τ-preserving safe realization iff

$$\alpha_i \le \theta U_i + cD.$$

The coordinatewise maximal safe attachment is

$$a_i^* = \min\{\beta_i, \theta U_i + cD\},$$

and every other τ-preserving realization is dominated by $(U_i, \max\{D, a_i^*\})$.

Translate by

$$h = M - \tau, \qquad \xi = \tau - D,$$

and $h_i = u_i - \tau$, $A_i = \tau - \alpha_i$, $B_i = \tau - \beta_i$. With $s_i = \min\{h, h_i\}$, owner legality becomes

$$B_i < \xi, \qquad c\xi - \theta s_i \le A_i,$$

and the transition is

$$(h, \xi) \mapsto (s_i, \max\{B_i, c\xi - \theta s_i\})\,.$$

The absorbing safe tail is $\xi \le H$. For a fixed legal owner word $i_1, \dots, i_k$ with realized headrooms s_t, repeated substitution gives

$$\xi_k = \max\left\{B_{i_k}, cB_{i_{k-1}} - \theta s_k, \dots, c^k \xi_0 - \theta \sum_{t=1}^{k} c^{k-t} s_t\right\}.$$

Thus evaluation of a fixed word is linear-size max-affine arithmetic.

For an arbitrary fixed order, let

$$\mathcal{H} = \{h_0\} \cup \{\min(h_0, h_i) : 1 \le i \le n\}.$$

After every prefix, the current headroom belongs to $\mathcal{H}$. For $\eta \in \mathcal{H}$, let $g_j(\eta)$ be the minimum hard deficit $\xi > H$ reachable after processing the first j events with current headroom exactly η; use $+\infty$ if no such state exists, and retain one absorbing flag when some subsequence has reached $\xi \le H$. Initially $g_0(h_0) = \xi_0$.

Process event i from every finite state (η, g) in two ways. The skip transition retains (η, g). For the use transition set

$$s = \min\{\eta, h_i\}.$$

It is legal exactly when

$$B_i < g, \qquad cg - \theta s \le A_i,$$

and then outputs

$$y = \max\{B_i, cg - \theta s\}.$$

If $y \le H$, set the absorbing flag; otherwise update $g_i(s) \leftarrow \min\{g_i(s), y\}$.

For a fixed headroom η, a smaller hard deficit dominates every larger one for future absorbing-tail reachability. If an event is feasible at the smaller value, monotonicity gives no larger output there. If the smaller value is at or below an event floor B_i, any use from a larger live predecessor outputs at least B_i and cannot beat the smaller state. If the smaller value violates the upper feasibility inequality, every larger value violates it as well. Hence keeping only $g_j(\eta)$ is exact. There are at most $n + 1$ headroom values and n prefixes, so the dynamic program uses $O(n^2)$ event-state transitions. This proves the first part of Theorem 119.1.

Now assume

$$h_0 \ge h_1 \ge \cdots \ge h_n.$$

Whenever event i is used, its effective headroom is exactly h_i, independent of earlier skips. Put

$$U_i = \frac{A_i + \theta h_i}{c}, \qquad T_i(\xi) = \max\{B_i, c\xi - \theta h_i\}.$$

At a hard state event i is usable exactly when $B_i < \xi \le U_i$, and then $T_i(\xi) < \xi$. Scan left to right and use the event whenever it is usable. If the current greedy value is g, every alternative predecessor is at least g. If i is usable at g, monotonicity of T_i makes $T_i(g)$ the smallest successor. If $g \le B_i$, every live use from a larger state outputs at least $B_i \ge g$. If $g > U_i$, the event is infeasible from every larger state. Induction proves the prefix-minimum invariant and the $O(n)$ greedy specialization.

119.7 Complete threshold-band NP-completeness reduction

We now give the reduction underlying Theorem 119.3. Fix $\theta = c = 1/2$ and $H = 0$, so

$$(h, \xi) \mapsto \left(s_i, \max\left\{B_i, \frac{\xi - s_i}{2}\right\}\right), \qquad s_i = \min\{h, h_i\},$$

with legality $B_i < \xi$ and $(\xi - s_i)/2 \le A_i$. We reduce from Monotone NAE-3-SAT-E4.

119.7.1 The normalized Boolean interface

We first give the exact-one control used for a Boolean variable. Work at three consecutive caps

$$h_0, \qquad h_1 = h_0 - 1, \qquad h_2 = h_0 - 2,$$

and let both marker banks have size $m = 8$. Fix $Z = 1$, take

$$h_2 = 2^{80}, \qquad h_0 = h_2 + 2,$$

and set

$$y_0 = 2^{28}(Z + h_0 + 10), \qquad X = y_0 - h_0.$$

The module enters at (h_0, X). Each A-marker has headroom h_0 and each B-marker headroom h_1; all markers have the common large attachment threshold

$$A_{\text{big}} = X.$$

Choose their floors in two disjoint ordered intervals far below every assignment-time core value, with every B-marker floor larger than every A-marker floor. Distinct dyadic values in, say, $(-2, -3/2)$ for the A bank and $(-1, -1/2)$ for the B bank suffice for the local construction. Since every intended marker step is then core-limited, the numerical values of these floors do not enter the displayed recurrences. In the final composed reduction they may be translated to a small positive ordered interval, still below all assignment-time core outputs; this preserves every comparison below and permits the exact cleanup used in Section D.4.

At cap h_0, put $y = \xi + h_0$. An A-marker acts by $y \mapsto y/2$, so after t such markers

$$\xi_A(t) = \frac{y_0}{2^t} - h_0.$$

The **early A pivot** is

$$E_A : \qquad h = h_1, \quad B = X - 1, \quad A = A_{\text{big}}.$$

At prefix zero it sends (h_0, X) to $(h_1, X - 1)$. After even one A-marker we have $\xi_A(1) < X - 1$, so E_A is permanently owner-dead. Thus E_A is exactly the retain-A exit.

For the **late A pivot**, define

$$C_A(t) = \frac{\xi_A(t) - h_1}{2}$$

and choose

$$B_{L_A} = 0, \qquad A_{L_A} = \frac{C_A(7) + C_A(8)}{2}.$$

Hence L_A is infeasible before all eight A-markers have been consumed and becomes feasible at $t = 8$. The two first-fork outputs are therefore

$$\xi_H = X - 1$$

on the retain-A branch and

$$\boxed{\xi_L = C_A(8) = \frac{y_0}{2^9} - h_0 + \frac{1}{2}}$$

on the consume-A branch.

Now work at cap h_1. After t B-markers write

$$\xi_H(t) = \frac{\xi_H + h_1}{2^t} - h_1, \qquad \xi_L(t) = \frac{\xi_L + h_1}{2^t} - h_1.$$

The chosen scale gives $\xi_H(8) > \xi_L$. We use the sharpened second fork needed later in the global reduction. Set

$$B_{E_B} = \xi_L - \frac{1}{4}, \qquad U_{E_B} = \xi_H(8) - \frac{1}{4}, \qquad A_{E_B} = \frac{U_{E_B} - h_2}{2}.$$

At the low input ξ_L, the early pivot E_B is live and feasible and is floor-limited, so it sends the branch exactly to

$$\boxed{\xi_- = \xi_L - \frac{1}{4}}.$$

After one low-branch B-marker it is owner-dead. On every high-branch prefix $0 \le t \le 8$, the scalar is at least $\xi_H(8) > U_{E_B}$, so the corresponding pivot candidate exceeds A_{E_B} and E_B is infeasible there. Thus the consume-A branch can complete only by retaining the whole B bank through immediate E_B.

For the **late B pivot**, put

$$B_{L_B} = \xi_L + 1, \qquad C_H(t) = \frac{\xi_H(t) - h_2}{2}, \qquad A_{L_B} = \frac{C_H(7) + C_H(8)}{2}.$$

The floor B_{L_B} is above every scalar on the low branch, so L_B is owner-dead there. On the high branch it is infeasible through the first seven B-markers and becomes feasible after the eighth. At that point

$$C_H(8) = \xi_L + 1 - 2^{-8} < B_{L_B},$$

so it is floor-limited and sends the branch exactly to

$$\boxed{\xi_+ = \xi_L + 1}.$$

Consequently the only two local completions to cap h_2 are

$$E_A,\ B_1, \dots, B_8,\ L_B$$

and

$$A_1, \dots, A_8,\ L_A,\ E_B.$$

The first retains the entire A bank and consumes the entire B bank; the second consumes the entire A bank and retains the entire B bank. There is no completion retaining both banks, consuming both banks, or retaining a proper nonempty subbank. Thus the module is an exact-one Boolean selector.

The two exits share cap h_2 and satisfy

$$\xi_+ - \xi_- = \frac{5}{4}.$$

Consume eighteen copies of the explicit plateau normalizer

$$N: \qquad h_N = h_2, \qquad B_N = 0, \qquad A_N = \frac{\xi_+ - h_2}{2}.$$

At every intended normalizer input $\xi \le \xi_+$, its core candidate is $(\xi - h_2)/2 \le A_N$; it is positive throughout these eighteen uses, so the zero floor is inactive. Hence, in shifted coordinate $y = \xi + h_2$, each copy acts exactly by $y \mapsto y/2$. The normalized pair is

$$S_- = (h_2, \ell), \qquad S_+ = (h_2, \ell + \delta),$$

where

$$\delta = \frac{5}{2^{20}}, \qquad \ell = h_2 + 26 - \frac{7}{2^{20}}.$$

Finally choose the low-branch adapter cap $C = h_2 - 1$. A low adapter with threshold strictly between the two branch candidates is feasible only on S_-, while a high adapter with floor $\ell + \delta/2$ is owner-dead on S_- and live on S_+. The resulting transfer-mode entry scalars are

$$a = \frac{27}{2} - \frac{7}{2^{21}}, \qquad b = h_2 + 26 - \frac{9}{2^{21}},$$

with $b/a > 2^{75}$. Hence a completed variable selects exactly one Boolean transfer mode and designates exactly one of two eight-token payload banks for deletion, leaving the other for the final clause audits.

119.7.2 Exact headroom certification

At a transfer state (C, X) let the intended payload have effective headroom $0 < h < C$, every wrong candidate have headroom $t \neq h$, and assume $X > 2h$. A candidate gives $y(t) = (X - t)/2$. After a high-headroom first gate the core candidate is $z(t) = (X - 3t)/4$. Choose

$$\max\{\max_{t>h} y(t), \max_{t\leq h} z(t)\} < Q < y(h).$$

A first gate with floor Q kills every $t > h$ substitute and floor-normalizes all $t \leq h$ survivors to common scalar Q. The target then has next core candidate $c_h = (Q - h)/2$, whereas every $t < h$ gives $c_t = (Q - t)/2 > c_h$. Choose

$$c_h < R < A_2 < \min_{t<h} c_t.$$

A second gate with floor R and threshold A_2 accepts and floor-normalizes only the target. Thus payload identity is certified entirely by headroom; its own floor B remains free for clause logic.

For the eight-step deletion word use, at stage j,

$$X_j = \frac{S}{16^{j-1}}, \qquad h_j = \frac{X_j}{4},$$

$$Q_j = \frac{17}{64}X_j, \qquad R_j = \frac{1}{16}X_j, \qquad A_j^{(2)} = \frac{3}{32}X_j.$$

For the first gate choose the explicit attachment threshold

$$A_j^{(1)} = X_j.$$

Every intended or competing Gate-1 core candidate considered here is at most $X_j/2 < A_j^{(1)}$, so this threshold never removes an intended Gate-1 use.

The intended branch satisfies

$$\frac{X_j}{128} < \frac{X_j}{16} < \frac{3X_j}{32} < \frac{X_j}{8},$$

so the second gate accepts it and outputs $X_{j+1} = X_j/16$, while higher substitutes are rejected by Gate 1 and lower substitutes by Gate 2. This deletes exactly the designated eight-token bank.

119.7.3 Clause audit

At cap $h = 0$, an audit token acts by $\xi \mapsto \max\{B, \xi/2\}$. For a clause scale H let high tokens have floor H and low tokens floor $L \leq H/2$, and enter at $\xi_0 = 4H$. Three high tokens fail because

$$4H \to 2H \to H$$

and the third high token is owner-dead. If at least one token is low, all three designated uses are possible and the state reaches $H/2$; a progress event at floor/threshold $H/4$

certifies completion. Hence one orientation checks *not all True*. Reversing the high/low truth orientation checks *not all False*. The conjunction is exactly NAE.

Audit stages are chained by

$$H_{s+1} = H_s/16, \qquad L_s = H_s/2.$$

After the two audit copies define the final positive audit output

$$\boxed{\varepsilon := H_{2m}/4.}$$

The explicit scale inequalities of the control construction place every intended input and every intended core output of a convenience zero-floor event at or above this final scale, with strict input liveness and attachment allowance $A_i \geq \varepsilon$. The successful word terminates with τ-deficit exactly ε.

Earlier-stage tokens are already owner-dead at later stages. Therefore any foreign substitution points strictly forward. Following the missing-token deficit reaches a maximal stage with no later substitute, so no globally successful schedule can have net foreign-token displacement.

119.7.4 Global scale separation and phase-leak repair

The τ-band kernel is positively homogeneous. Put

$$\lambda_i = \sigma^{i-1}, \qquad \sigma = 2^{-512},$$

and let $X_0 = X$ denote the normalized variable-input scalar from Section D.1.

Lemma 119.10 (scale separation and exact inter-variable composition)**.** *For every variable $i < n$, the two truth branches can be reconverged at one explicit positive raw port and transferred to the input port of variable $i+1$ by a constant number of τ-band events. Moreover, at every certified deletion stage of variable i, payloads belonging to earlier or later variables cannot substitute for the intended payload. All event coordinates are dyadic rationals of polynomial bit length.*

Proof. After the eight certified deletions, the high and low transfer modes have scalars

$$S_H = \frac{b}{16^8}, \qquad S_L = \frac{a}{16^8}.$$

Define the normalized raw port

$$C_{\mathrm{raw}} = \frac{S_L}{2}, \qquad P_{\mathrm{raw}} = \frac{3S_L}{4}.$$

Use eighty copies of the reconvergence normalizer

$$N_{\mathrm{rec}}: \qquad h = C_{\mathrm{raw}}, \qquad B = P_{\mathrm{raw}}, \qquad A = S_H.$$

On the low branch the first copy is floor-limited and reaches $(C_{\mathrm{raw}}, P_{\mathrm{raw}})$. On the high branch, with $y = \xi + C_{\mathrm{raw}}$, the recurrence is

$$y \longmapsto \max\left\{P_{\mathrm{raw}} + C_{\mathrm{raw}}, \frac{y}{2}\right\}.$$

Since $b < 2^{81}$, $a > 2^3$, and hence $S_H/S_L = b/a < 2^{78}$, eighty copies are more than enough to reach the same floor. Once the floor is reached, the unused copies are owner-dead. Thus both truth branches reach exactly the same raw port.

Scale variable i by λ_i. From the normalized constants,

$$2^{-31} < C_{\mathrm{raw}} < P_{\mathrm{raw}} < 2^{-28}, \qquad h_0 < 2^{81},$$
$$2^{107} < X_0 < 2^{108}, \qquad X_0 + h_0 = y_0 > 2^{108}.$$

Hence

$$\sigma h_0 < 2^{-431} < C_{\mathrm{raw}}, \qquad \sigma X_0 < 2^{-404} < P_{\mathrm{raw}},$$

so the first tail event is both cap-lowering and owner-live. Append 384 copies of

$$T_i : \qquad h_T = \lambda_{i+1} h_0, \quad B_T = \lambda_{i+1} X_0, \quad A_T = \lambda_i P_{\mathrm{raw}}.$$

The first and every later intended use is attachment-feasible because its core candidate is below $\lambda_i P_{\mathrm{raw}}/2 < A_T$. After the first copy the cap is $\lambda_{i+1} h_0$; in the shifted coordinate $y = \xi + \lambda_{i+1} h_0$ the recurrence is exactly

$$y \longmapsto \max\left\{\lambda_{i+1}(X_0 + h_0), \frac{y}{2}\right\}.$$

At the start of the tail,

$$y < \lambda_i 2^{-27},$$

whereas the target floor satisfies

$$\lambda_{i+1}(X_0 + h_0) > \lambda_i 2^{-404}.$$

Consequently

$$\frac{y}{2^{384}} < \lambda_i 2^{-411} < \lambda_{i+1}(X_0 + h_0),$$

so the target floor is reached by or before the 384th copy. At that moment the state is exactly

$$\lambda_{i+1}(h_0, X_0),$$

and every unused tail copy is owner-dead.

For separation, every transfer headroom of variable i lies in

$$\lambda_i(2^{-30}, 2^{80}).$$

Hence every future-variable transfer headroom is at most

$$2^{80}\lambda_{i+1} = 2^{-432}\lambda_i,$$

whereas every current intended transfer headroom exceeds $2^{-30}\lambda_i$. The ratio is therefore greater than 2^{402}. Future payloads are lower/over-drop candidates and fail Gate 2. A retained payload from the immediately preceding variable already has headroom greater than $2^{-30}\lambda_{i-1} = 2^{482}\lambda_i$, while the current cap is below $h_0\lambda_i < 2^{81}\lambda_i$. Hence every earlier retained payload appears at the current cap as a higher/no-drop candidate and fails Gate 1. Thus no foreign payload can replace an intended deletion token. Finally, the exponent of λ_i is $512(i-1)$, and all other constants have fixed dyadic encoding, so the complete construction has polynomial bit length. □

The same scale bounds keep future transfer and audit gates attachment-infeasible throughout the preceding handoff: before the next canonical input one has $\xi > \lambda_{i+1} X_0$, with $X_0 > 2^{107}$, while the normalized transfer/audit headrooms and attachment thresholds are below 2^{81}. Thus their core candidate is greater than $2^{105}\lambda_{i+1}$, far above the corresponding attachment threshold.

It remains to exclude premature use of the high-threshold assignment controls. Choose the residual RN136 marker floors positive and in their required order, but below

$$\varepsilon_{\mathrm{ctl}} = \frac{P_{\mathrm{raw}}}{2^{13}}.$$

At the common raw port use one drop event

$$D : \qquad h_D = 0, \quad B_D = 0, \quad A_D = P_{\mathrm{raw}}/2,$$

which sends the canonical scalar to $P_{\mathrm{raw}}/2$. Eight residual controls are then core-limited and yield

$$\xi_8 = \frac{P_{\mathrm{raw}}}{2^9}.$$

The cleanup gate

$$G_{\mathrm{ctl}}: \qquad h = 0, \quad B = 0, \quad A = \frac{P_{\mathrm{raw}}}{2^{10}}$$

is feasible after eight controls and infeasible after only seven. Therefore premature consumption of an assignment marker destroys both future truth branches. Likewise, the branch adapter requires exactly eighteen common post-choice normalizers; using one during the preceding handoff leaves only seventeen after the scale floor erases its scalar effect, so the certified adapter input cannot be reached.

The only remaining high-threshold leak is a premature early pivot E_A. Write its output as

$$\xi_E = X - 1 + \xi, \qquad 0 \le \xi \le 1.$$

After it fires, all remaining assignment markers act as $y \mapsto y/2$ in shifted coordinate. No truth exit is possible before eight contractions, and more than eight destroys the cleanup count. After exactly eight contractions, either the low route $L_A \to E_B$ reaches exactly ξ_-, or the direct mixed E_B value lies in

$$\xi_L + 1 - \frac{1}{256} \le \xi_{\mathrm{mid}} \le \xi_L + 1 - \frac{1}{512}.$$

Since the truth midpoint is $\xi_L + 3/8$, every mixed state lies strictly on the high side and the branch adapter routes it to the canonical high mode. Thus premature phase entry cannot create a third truth value or retain both payload banks.

119.7.5 Correctness and NP membership

Given a NAE assignment, choose the corresponding truth branch in each variable, delete the opposite bank by the certified eight-step words, traverse the scale ladder, and run both audit copies. Every clause contains both truth values, so both not-all audits pass.

Conversely, every successful completion induces exactly one retained truth bank per variable. Headroom certification prevents foreign substitution during deletion; phase-leak repair preserves the two-valued interpretation; directed audit self-policing removes net foreign displacement. The first audit copy implies no clause is all True and the second no clause is all False. Hence the induced assignment is NAE-satisfying.

Each variable and clause contributes only constantly many events, so the construction has $O(n + m)$ events. The dyadic scale exponents and all rational coordinates have polynomial encoding length. A YES certificate is the owner-event order. Verification simulates at most N exact rational transitions; at $\theta = 1/2$ denominator exponent grows by at most one per non-floor step. Therefore the problem is in NP. This proves Theorem 119.3.

119.8 Proof package: exact physical lift and physical NP membership

The radius-two realization requires event saddles not to exceed the observer. The hardness construction initially uses a few zero floors only for notational convenience, so we first remove them.

Lemma 119.11 (positive-floor normalization)**.** *For the τ-band hardness family, there is a polynomial-bit rational $\varepsilon > 0$ such that replacing every convenience floor $B_i = 0$ by $B_i = \varepsilon$ preserves YES/NO equivalence. After normalization,*

$$0 < \varepsilon \le B_i \le A_i$$

for every event.

Proof. Use the same final audit scale fixed in the complete threshold-band reduction later in this chapter,

$$\varepsilon = H_{2m}/4.$$

It is a positive dyadic rational of polynomial encoding length. The explicit scale inequalities in the complete threshold-band reduction later in this chapter give, for every intended use of a convenience zero-floor event, input $\xi > \varepsilon$, attachment allowance $A_i \geq \varepsilon$, and intended core output at least ε. Thus the intended completeness schedules are literally unchanged after replacing $B_i = 0$ by $B_i = \varepsilon$; in particular the successful word still ends at τ-deficit exactly ε.

For soundness, denote the original and normalized floors by $B_i^{(0)}$ and $B_i^{(\varepsilon)}$, respectively. Thus

$$B_i^{(0)} \leq B_i^{(\varepsilon)},$$

with equality except for convenience zero-floor events, where $B_i^{(0)} = 0$ and $B_i^{(\varepsilon)} = \varepsilon$. Compare any normalized trajectory with the original instance and couple states in the product order

$$h^{(0)} \geq h^{(\varepsilon)}, \qquad \xi^{(0)} \leq \xi^{(\varepsilon)}.$$

This holds initially. Suppose the normalized schedule next uses event i. If $B_i^{(0)} < \xi^{(0)}$, then the event is owner-live in the original state; moreover

$$\frac{\xi^{(0)} - \min\{h^{(0)}, h_i\}}{2} \leq \frac{\xi^{(\varepsilon)} - \min\{h^{(\varepsilon)}, h_i\}}{2} \leq A_i,$$

so the original use is feasible. Its new deficit is no larger because both its floor and its core candidate are no larger, while its new headroom is no smaller. If instead $\xi^{(0)} \leq B_i^{(0)}$, then $B_i^{(0)} \leq A_i$ implies $\xi^{(0)} \leq A_i$, so event i is already passively resolved in the original instance and may be skipped. The product order is preserved in either case. Keeping the larger old headroom only makes every later core-feasibility inequality easier, while the smaller old deficit never worsens terminal safety. Induction therefore turns every normalized successful completion into a successful completion of the original instance. Hence normalization cannot create a false YES. Together with the unchanged intended completeness schedules, YES/NO equivalence is preserved. □

Choose

$$\tau > \max\{\xi_0, \max_i A_i\} + 1$$

and truncate h_i to $\bar{h}_i = \min\{h_i, h_0\}$, which changes no reachable effective headroom. Map

$$M = \tau + h, \qquad D = \tau - \xi,$$

and

$$\ell_i = \tau, \qquad u_i = \tau + \bar{h}_i, \qquad \alpha_i = \tau - A_i, \qquad \beta_i = \tau - B_i.$$

By Lemma 119.11,

$$0 \leq \alpha_i \leq \beta_i \leq \tau - \varepsilon < \tau = \ell_i \leq u_i.$$

Thus the physical birth/attachment constraint is automatic and every event lies in the same critical τ-band.

At $(M, D) = (\tau + h, \tau - \xi)$, with $s_i = \min\{h, \bar{h}_i\}$,

$$U_i = \min\{M, u_i\} = \tau + s_i.$$

The physical core-feasibility inequality

$$\alpha_i \leq \frac{1}{2}U_i + \frac{1}{2}D$$

is exactly $(\xi - s_i)/2 \le A_i$, while the event raises the owner exactly when

$$\beta_i > D \iff B_i < \xi.$$

The owner-changing update translates to

$$\xi' = \max\left\{B_i, \frac{\xi - s_i}{2}\right\}, \qquad h' = s_i.$$

Hence rectangle-event and τ-band owner dynamics are exactly conjugate; owner-dead events correspond to $\beta_i \le D$ and may be processed passively.

For the tree realization create an observer x with

$$F(x) = \tau - \varepsilon$$

and, for every event, a branch $x - s_i - p_i$ with

$$F(p_i) \in [\tau, \tau + \bar{h}_i], \qquad F(s_i) \in [\tau - A_i, \tau - B_i].$$

Then

$$F(s_i) \le \tau - \varepsilon = F(x) < \tau \le F(p_i).$$

Thus p_i is a primitive local maximum, s_i is not, and the unique p_i–x path has minimum exactly $F(s_i)$:

$$\boxed{\kappa_F(p_i, x) = F(s_i).}$$

Different event rectangles are independent because their branches are vertex-disjoint away from x.

Put $M_0 = \tau + h_0$ and $D_0 = \tau - \xi_0$. Add an essential branch $x - s_0 - e$ with $F(e) = M_0$ and $F(s_0) = D_0$, resolving e before equal-height event peaks. Add one guard leaf g adjacent to x with $F(g) = \tau$ and resolve it after event peaks born at τ. Event peak heights realize the schedule order and saddle values realize the attachments. After a prefix $i_1, \dots, i_t$ the observer elder pressure is

$$D_t = \max\{D_0, F(s_{i_1}), \dots, F(s_{i_t})\},$$

exactly the rectangle running supply. By the observer-record proof package, a peak that raises this maximum is an observer record whose death is the previous elder pressure, so its safety inequality is exactly the rectangle/τ-band inequality.

In the successful normalized construction the final τ-deficit is ε, hence final elder pressure is $\tau - \varepsilon = F(x)$. The guard has observer attachment $F(x)$ and is therefore a nonrecord. It creates no positive saliency. Conversely, the guard only adds a constraint, so any observer-safe tree realization already induces a feasible rectangle schedule. Therefore

$$\boxed{\tau\text{-band YES} \iff \text{radius-two interval-tree observer-safety YES.}}$$

The construction uses $2N + O(1)$ vertices and polynomial rational endpoints.

For NP membership, use as certificate a rational scalar realization, total preorder/tie resolution, bottleneck identities, and record/nonrecord pattern. Once these discrete choices are fixed on a tree, all remaining conditions are rational linear equalities and inequalities: interval bounds, order/tie relations, path-minimum identities, local-maximum conditions, record inequalities, and either $B - D \le H$ or $a \le \theta B + (1-\theta)D$. Replace strict inequalities by one common positive rational slack δ. A nonempty rational polyhedron has a basic feasible rational point of polynomial bit length. The verifier checks all path minima and inequalities exactly in polynomial time. This proves Theorems 7.1–7.2.

119.9 Proof package: dark-anchor lift and ownership NP-completeness

Start from the bright hardness tree and attach one fixed sufficiently low leaf d at the observer-side construction so that it is never a bright primitive maximum. Therefore all bright births, attachments, records, and saliencies are unchanged.

In the inverted field $-F$, the leaf d is the essential dark atom. Every finite dark birth reaches the observer through a path whose bottleneck is the observer value, so all finite dark observer attachments equal the running essential dark pressure. Consequently no finite dark atom strictly raises the running elder supply. By the nonrecord elimination theorem in the observer-record proof package,

$$\boxed{\Sigma_\theta^-(x;F) = 0}$$

for every realization and every admissible tie/elder resolution.

At $\Delta = 0$ we therefore have, realization by realization,

$$\mathrm{Own}_{0,\theta}(x;F) = \text{background} \iff \Sigma_\theta^+(x;F) = 0.$$

The construction adds only constantly many vertices and leaves the radius-two bound unchanged. Thus the bright one-polarity safe-realization problem reduces exactly to Possible Background Polarity-Core Ownership. NP membership uses the same polynomial-bit realization certificate as the physical-lift proof package later in this chapter, evaluated for both polarities. This proves Theorems 8.1–8.2.

119.10 Proof package: local possible-exceedance witness on cactus graphs

Fix one polarity and one finite birth atom λ. Write

$$B = B_\lambda, \qquad a = a_\lambda(x), \qquad D = D_x(\lambda).$$

By the observer-record proof package, λ contributes more than H iff

$$D < B - H, \qquad D < \frac{a - \theta B}{1 - \theta}.$$

Equivalently, there exists a separator scalar ℓ such that

$$D < \ell, \qquad B - \ell > H, \qquad a > \theta B + (1-\theta)\ell. \tag{G.1}$$

The last inequality means that a marked representative p and the observer x lie in the same component of

$$\{v : F(v) > t_\theta(B,\ell)\}, \qquad t_\theta(B,\ell) = \theta B + (1-\theta)\ell.$$

The inequality $D < \ell$ means that the x-component of $\{F \geq \ell\}$ contains no elder capable of suppressing the marked birth. Ties require only two modes: either a strict-higher elder exists, or one marked disconnected equal-B plateau is resolved elder. Thus a witness uses a constant number of marked vertices and the two retained parameters (B, ℓ).

Root the cactus block-cut tree at the block containing x. For each threshold graph in the certificate, a block interface stores only:

(i) whether each articulation port is active;

(ii) the connectivity partition of active ports;

(iii) whether the active component is connected to the marked root;

(iv) constant-many flags for the required presence/absence of an elder/high vertex and for the equal-height tie mode.

The number of thresholds and marks is constant, so every interface has constant state width. A bridge has a constant transition table. A simple cycle is scanned as a two-port finite automaton; child-block messages combine without increasing the state width.

For a vertex interval $[l_v, u_v]$, local modes are determined by comparisons with the affine forms

$$B, \qquad \ell, \qquad \theta B + (1-\theta)\ell,$$

together with the fixed offset $B - H$. Their truth pattern changes only when one of these forms meets l_v or u_v, or on the relevant equality faces. Hence $O(n)$ rational lines in the two-dimensional (B, ℓ) plane induce an arrangement with $O(n^2)$ cells and faces. On each cell every vertex has a fixed finite local-mode set, so one bottom-up block-cut DP decides feasibility in $O(n)$ state transitions.

Enumerate the marked birth representative p and, in equal-elder mode, one elder mark q. There are $O(n^2)$ marked choices. For each, inspect the $O(n^2)$ arrangement cells/faces and run the $O(n)$ cactus DP. A straightforward implementation therefore uses

$$\boxed{O(n^5)}$$

rational arithmetic/state-transition operations; no optimality of this bound is claimed. Coefficients have polynomial bit length, so rational cell representatives and comparisons have polynomial bit complexity.

Soundness. An accepted state supplies an actual birth plateau, an admissible strict/equal elder mode, and a separator ℓ satisfying (G.1), so the marked atom contributes saliency $> H$.

Completeness. If one realization has $\Sigma_\theta(x) > H$, choose an atom attaining the contribution, a representative of its birth plateau, and a separator ℓ strictly between D and the two upper bounds in (G.1). That realization lies in one enumerated arrangement cell/face and induces an accepted block-cut state.

This proves Theorem 119.8. Applying the same construction to $-F$ gives the opposite polarity. The local/global separation theorem then follows from Theorem 119.1 and Appendices D–F.

Chapter 120

All-Orders Common-Realization Obstructions and EF–FG Synthesis

Research branch — Unbounded obstruction order and external generativity. *The complexity theorem is not the only global obstruction. This chapter constructs protected strict-record families whose feasible subfamilies form simplex boundaries of arbitrary dimension, first under the canonical threshold-band section and then in a section-free radius-two scalar realization. The chapter closes by translating those native EF–FG families into the USNRT protected-obligation language.*

120.1 Unbounded common-realization obstruction order

NP-completeness is an algorithmic statement. EF-FG also exhibits an all-orders **structural** local-to-global obstruction.

120.1.1 Protected record complexes

Let P be a finite family of named strict observer-record obligations. Define

$$K_{\mathrm{raw}}(P) = \left\{ A \subseteq P : \begin{array}{l} \text{one admissible EF–FG realization} \\ \text{satisfies every obligation in } A \end{array} \right\}.$$

When the native maximal-safe hard-band rule is imposed, define

$$K_{\sigma}(P) = \left\{ A \subseteq P : \begin{array}{l} \text{one admissible canonical realization} \\ \text{satisfies every obligation in } A \end{array} \right\}.$$

Both are downward-closed simplicial complexes. Their minimal nonfaces measure the order at which local joint feasibility ceases to assemble into one realization.

120.1.2 Canonical boundary theorem

Fix $r \geq 1$. Start from

$$h_0 = 1, \qquad x_0 = 2^r,$$

and take $r+1$ parameter-identical events

$$(h_i, A_i, B_i) = (0, 2^{r-1}, 1).$$

For a canonical owner use,

$$x' = \max\{1, x/2\}.$$

After k protected uses,

$$x_k = 2^{r-k}, \qquad 0 \le k \le r.$$

Thus any proper protected subfamily, having size at most r, may be scheduled first and every selected event is a strict canonical record. After r uses, however, $x = 1 = B_i$, so one further strict owner record would require $1 < 1$.

Theorem 120.1 (canonical record boundary)**.** *For every $r \ge 1$ there exists an interval-valued radius-two EF-FG tree with $|P_r| = r + 1$ such that*

$$\boxed{K_\sigma(P_r) = \partial\Delta^r.}$$

This theorem is genuinely section-dependent for the symmetric family: with arbitrary safe attachments, the analogous raw claim can fail already at $r = 2$.

The displayed construction is the full proof: every proper face uses at most r protected events, while a $(r + 1)$st strict owner use is blocked by $B_i = x = 1$.

120.1.3 Raw boundary theorem

The section-free construction uses shared maximin geometry. Fix $r \ge 1$ and form a radius-two graph with observer x, gates $g_1, \dots, g_r$, individual peaks $p_1, \dots, p_r$, and one shared peak p_*. Each p_i is connected to x through g_i, while p_* is connected through every gate. Add one fixed essential branch to initialize observer pressure.

Choose

$$F(x) = 5, \qquad F(s_0) = 1, \qquad F(e) = 8, \qquad F(g_i) = u_i \in [2, 3],$$

and allow event peak values in $[6, 7]$. Then

$$\kappa_F(p_i, x) = u_i, \qquad \kappa_F(p_*, x) = \max_i u_i.$$

Let

$$P_r = \{p_1^{\mathrm{ob}}, \dots, p_r^{\mathrm{ob}}, p_*^{\mathrm{ob}}\}.$$

Every proper subfamily is realizable. If the shared obligation is absent, choose distinct selected gates and process the selected individual peaks in increasing attachment order. If the shared obligation is present, at least one individual obligation is absent; let an omitted gate supply the maximum value 3, choose all selected individual gate values distinct below 3, process them first, and then process p_*.

The full family is impossible. In every realization, some index j attains the maximum gate value, so

$$\kappa_F(p_*, x) = u_j = \kappa_F(p_j, x).$$

All $r + 1$ events cannot then be strict records in one elder order, because strict records require strictly increasing and hence pairwise distinct attachments.

Theorem 120.2 (raw protected-record boundary)**.** *For every $r \ge 1$,*

$$\boxed{K_{\mathrm{raw}}(P_r) = \partial\Delta^r.}$$

Hence every proper protected observer-record family admits a common scalar realization while the full family does not.

The preceding proper-face construction and duplicate-maximum argument form the full proof.

120.1.4 Consequence

For every fixed locality order k, choose $r > k$. Every protected subfamily of size at most k is realizable, yet the whole family is not. Therefore

Local-to-global consequence. No bounded order of protected local-feasibility tests certifies global common realizability in general.

The statement is specific to the EF-FG observer-record semantics; it does not claim novelty for simplex boundaries, Helly-type language, or local/global inconsistency in the abstract.

120.2 Spatial-frame layer and complete EF-FG output

The complexity results above concern the polarity-core ownership projection. EF-FG retains a second, spatial projection because scalar foreground/background structure contains information that observer saliency alone does not encode.

For a fixed realization, competitor extrema determine maximin gates toward the observer. Weak/strong sector incidence, terminal directions, and residual terminal regions produce a spatial label recording which side of a local extremal frame the observer occupies. On trees, much of this structure is rigid; on cactus graphs, multiple route channels require a correlated residual state. The spatial layer is self-dual under $F \mapsto -F$.

The **complete deterministic EF-FG state** is therefore a pair

$$\boxed{\big(\mathsf{PO}_x(F), \mathsf{SF}_x(F)\big),}$$

accompanied when needed by witness/certificate data. Under interval uncertainty the exact state is not the Cartesian product of marginal possibilities; both projections must be computed from one common realization before projection.

The exact spatial-frame construction in Chapter 118 supplies pairwise gates, route incidence, terminal thresholds, correlated residual states, and the cactus closure statement. None of those details is used to strengthen the ownership hardness claim; they are retained because they are part of the EF-FG theory rather than an implementation artifact.

120.3 Finite evaluation and structural consistency checks

The fixed-realization evaluation and the interval-valued decision questions are mathematically distinct. For a specified grayscale field, the deterministic evaluation consists of locating bright and dark extrema, their persistence and record chains, observer attachments, saliency values, polarity-core ownership, route incidence, and the resulting fixed spatial frame. Every comparison is made for the same specified realization, so the associated construction is polynomial; this does not settle the existential ownership questions for interval inputs.

120.3.1 Fixed-realization evaluation

The evaluation data are:

- bright and dark extrema and persistence values;
- observer attachments, record chains, and saliency;
- polarity-core ownership and fixed-realization spatial-frame output;
- finite witnesses for the stated route and terminal conditions.

These objects are determined by the definitions and the indicated finite comparisons. Their existence is separate from the global NP-completeness statement about choosing a common realization.

120.3.2 Interval-valued instances

For interval-valued inputs, local witness conditions and globally coupled common-realization conditions must be treated separately. A local possible-exceedance predicate admits the stated finite witness criterion; correlated safety and ownership conditions demand one common choice of all uncertain quantities, not independent coordinatewise extrema. The interval-cactus theorem is proved by its combinatorial decomposition; no implementation hypothesis is part of its statement.

120.3.3 Finite consistency checks

The finite checks address observer-record death identities, grayscale-inversion duality, ownership partitions, record examples, deterministic ambiguity, weak/strong route incidence, terminal sectors, interval path and cactus cases, common-realization/order instances, and the smallest raw boundary seed at $r = 2$. These ranges provide consistency evidence for the definitions and boundary conventions. They do not replace the symbolic proofs or extend any theorem beyond its stated hypothesis.

120.4 Discussion

120.4.1 What the theory adds

EF-FG is not a replacement for merge trees, persistence, mathematical morphology, or uncertain scalar-field topology. It uses those neighboring structures to define a specific foreground/background semantics. Three aspects are central.

First, observer-relative saliency admits a hierarchy-free record normal form. Second, uncertainty is treated by exact common realization rather than by independent coordinate ranges. Third, the same semantics supports both computational and structural local/global separation theorems.

The NP-completeness result and the simplex-boundary theorem are complementary. The former says that choosing one globally safe realization can encode hard combinatorial synchronization. The latter says that even arbitrarily high-order local feasibility can fail to certify global feasibility. Neither theorem is implied by the other.

120.4.2 Why radius two matters

Hardness on radius-two trees rules out an explanation based on graph-topological complexity. The reduction uses shallow geometry; the combinatorial burden lies in coordinating the scalar realization and event order. Conversely, the raw boundary theorem uses radius-two fan graphs because shared gates are the mechanism that forces the duplicate maximum. The canonical boundary theorem already works on radius-two trees.

120.4.3 Scope and limitations

Several boundaries should remain explicit.

- The NP-complete ownership predicate is the **polarity-core ownership projection**, not the full spatial-frame mask.
- Deterministic evaluation of one fixed scalar field is not NP-complete.
- The tractable local-exceedance result is proved for the stated structured graph class; it is not a universal polynomial theorem for arbitrary graphs and arbitrary EF-FG predicates.
- The raw boundary theorem is category-specific; it does not assert that every simplicial complex is realizable as an EF-FG obligation complex.

- The present image-analysis experiments are validation and demonstration, not a benchmark claim against learned segmentation systems.

120.4.4 Second-stage program

The foundational theorem package leaves five explicit directions: characterize which finite complexes occur as EF–FG common-realization complexes; determine parameterized complexity in the number of incomparable records or uncertain gates; extend exact perturbation semantics beyond boxes; develop multichannel and spatiotemporal versions; and determine the complexity of the full spatial-frame projection under uncertainty.

These questions are retained here as the historical handoff from EF–FG I, but they are not left open in the monograph. The three chapters immediately following this one form EF–FG II and close all five within explicit finite scopes: shallow global spatial-frame NP-completeness, raw and canonical finite-complex universality, two fixed-parameter tractable ambiguity regimes, correlated rational-polyhedral common-realization semantics, and finite multichannel/spatiotemporal transfer.

A broader exact-realization viewpoint also motivates comparing EF-FG obligation complexes with protected-realization invariants developed in other domains in earlier chapters. The EF–FG theorems proved here, however, are self-contained and require no external realization-theory axiom.

120.5 Conclusion

EF-FG provides a self-dual mathematical theory in which foreground and background structure is organized by extrema, persistence, maximin observer geometry, and exact common-realization uncertainty. Its two projections are deliberately separate: polarity-core ownership captures persistent observer-relative explanation, while the spatial-frame layer captures carrier and terminal geometry.

For the ownership layer, observer records eliminate the need to carry the full merge hierarchy. This normal form leads to an exact threshold-band scheduling kernel. Fixed schedules are easy to evaluate, but choosing a globally safe order is NP-complete. The kernel lifts exactly to interval-valued radius-two scalar trees, and a constant-size dark anchor yields NP-completeness of Possible Background Polarity-Core Ownership at fixed $\theta = 1/2$ and $\Delta = 0$.

The same common-realization semantics also supports an all-orders structural obstruction: for every r, explicit EF-FG families realize the simplex boundary $\partial\Delta^r$ as a protected record-feasibility complex, both canonically and, with shared maximin geometry, in a section-free raw form. Hence no bounded local consistency order can certify global common realizability in general.

Together these results identify a precise boundary: **individual danger may be witnessed locally, but global absence of danger and global assembly of protected records can require genuinely global coordination inside one scalar realization.**

120.6 External generativity: interval-valued scalar fields

The EF–FG scalar-field theory proved in the preceding chapters supplies a downstream test of the USNRT architecture. The application is intentionally asymmetric: EF-FG provides native scalar-field theorems, while USNRT supplies a typed interpretation of their common-realization obstruction; the USNRT layer does not claim those scalar-field boundary constructions as a second primary home.

For a finite protected family P, let $K_\sigma(P)$ denote EF-FG feasibility under its native maximal-safe canonical section and $K_{\mathrm{raw}}(P)$ raw common-realization feasibility without imposing that section. The preceding EF–FG development proves two all-orders boundary

results:

$$K_\sigma(P_r) = \partial\Delta^r, \qquad K_{\mathrm{raw}}(P_r) = \partial\Delta^r \quad (r \geq 1),$$

using respectively the canonical threshold-band family and the raw radius-two fan construction.

Proposition 120.3 (USNRT interpretation of the EF-FG boundary families)**.** *Under the protected-obligation interpretation, both EF-FG families are instances of unbounded common-realization obstruction order: every proper protected subfamily is jointly realizable, while the full family is not. The canonical family additionally depends on a chosen section, whereas the raw family does not. No new USNRT axiom is required to state this distinction.*

Proof. The preceding EF–FG development supplies the realizability statements and their proofs. In USNRT, place the protected observer-record obligations as vertices of $\mathrm{Ob}_h(P_r)$. A face is present exactly when one common EF-FG realization satisfies the corresponding protected obligations. Therefore the two cited EF-FG identities translate directly to $\mathrm{Ob}_h(P_r) \cong \partial\Delta^r$. The only difference between the two cases is whether the EF-FG section is part of the realization contract. □

This application is evidence of expressiveness, not a logical premise of the core theory. It also avoids theorem duplication: the scalar constructions, physical lift, and NP-completeness results remain in the preceding EF–FG chapters, while the present USNRT interpretation records only the typed consequence relevant to its obligation-complex language.

Chapter 121

EF–FG II: Spatial-Frame Complexity and Finite-Complex Universality

The preceding EF–FG chapters established the foundational semantics, the observer-record normal form, exact common-realization uncertainty, the threshold-band ownership kernel, shallow-tree NP-completeness, and all-orders obstruction families. They also isolated five theorem-sized directions. The present second-stage development closes those directions within explicit finite scopes.

The first two closures concern the spatial projection and the topology of common realizability. Complete finite categorical spatial-frame realization is NP-complete even on shallow bipartite graphs, while protected-obligation complexes are universal: every finite simplicial complex occurs under both raw and canonical EF–FG common-realization semantics. These results sharpen the earlier local/global boundary without attributing hardness to deep graph topology.

Relation to prior work. Merge trees, component trees, and trees of shapes provide established scalar-field and morphological summaries [125, 126, 130, 131]. Uncertainty in scalar-field topology has been studied through mandatory critical points and statistical/ensemble summaries [128, 129, 136, 135]. Multivariate topology uses genuinely multivariate objects such as Reeb spaces and related simplifications [152], while temporal scalar-field analysis includes merge-tree alignment, tracking, and time-varying contour topology [153, 154]. The claims here concern the specific EF–FG common-realization decision problems, finite-complex universality inside EF–FG semantics, and the exact parameterized/polyhedral/layered transfer theorems developed in the following chapters.

121.1 Preliminaries from EF–FG I

We use the scalar-field and observer semantics of the preceding foundational EF–FG chapters. Only the ingredients required for the new theorems are recalled here.

Let $G = (V, E)$ be a finite graph and $F : V \to \mathbb{R}$ a scalar field. For vertices u, v, the maximin connectivity is

$$\kappa_F(u, v) = \max_{\pi : u \rightsquigarrow v} \min_{w \in \pi} F(w).$$

A resolved bright birth atom has a birth value, an elder order, and an observer attachment. Dark quantities are defined by applying the same construction to $-F$. The observer-record theorem from EF–FG I reduces pointwise polarity-core saliency to a record chain: a finite birth is relevant exactly when its observer attachment strictly raises the running elder pressure.

For spatial-frame structure, fix the selected finite birth atoms. For a seed p define

$$\kappa_p(v) = \kappa_F(p, v), \qquad \beta(v) = \max_r \kappa_r(v), \qquad W(v) = \{p : \kappa_p(v) = \beta(v)\}.$$

The rooted co-winner basin B_p is the component containing p in $\{v : p \in W(v)\}$. If p, q are carrier-adjacent, their gate floor is

$$s(p, q) = \max_{v \in B_p \cap B_q} \beta(v).$$

For a boundary terminal $t \in B_p$, the terminal gate is

$$s_\partial(p, t) = \kappa_{F|B_p}(p, t).$$

For an observer x, the bright midpoint margin against a gate floor s is

$$\mu^+_{p,q,x} = 2F(x) - F(p) - s,$$

with the dark analogue obtained from $-F$. Weak incidence means that at least one admissible top route passes through x, while strong incidence means that every admissible top route does.

Under uncertainty, a claimed state must arise from one common realization. For a realization family $\mathcal{P}$ and a finite query Q,

$$\text{Possible}_{\mathcal{P}}(Q) \iff \exists F \in \mathcal{P}, \exists \rho : \ Q(F, \rho),$$

where ρ denotes the finite elder/tie resolution. Robust semantics use universal quantification over the same common-realization space.

For polarity-core complexity we also use the threshold-band kernel. At $\theta = 1/2$, a hard-band state is (h, x) and event i has parameters (h_i, A_i, B_i). With

$$s_i = \min\{h, h_i\},$$

owner use is legal exactly when

$$B_i < x, \qquad \frac{x - s_i}{2} \le A_i,$$

and the transition is

$$(h, x) \mapsto \left(s_i, \max\left\{ B_i, \frac{x - s_i}{2} \right\} \right).$$

For a fixed admissible event order, the prefix-minimum theorem of EF–FG I evaluates existential completion in linear time. The NP-completeness therefore lies in choosing a successful order among incomparable one-use owner events, not in evaluating a proposed order.

121.2 Global spatial-frame realization complexity

The ownership hardness of EF–FG I does not determine the complexity of the spatial-frame projection. We now classify a natural finite categorical realization problem for that projection.

121.2.1 Terminal mixedness

For one polarity and observer x, let

$$M_T(x) = (W^+, W^-, S^+, S^-)$$

be the terminal peak-side mask. The bit W^+ is one iff some weakly incident exposed terminal sector has positive midpoint margin; W^- is one iff some such sector has nonpositive margin; S^+ and S^- use strong incidence.

Definition 121.1 (Possible Terminal-Mixed Mask (PTMM))**.** The input consists of a finite graph, rational interval scalar values, a boundary-terminal set T, an observer set X, and a rational seed threshold. The question is whether one common scalar realization and one elder/tie resolution satisfy

$$M_T(x) = (1, 1, 1, 1) \qquad \text{for every } x \in X.$$

Theorem 121.2 (Terminal-mask hardness)**.** *PTMM is NP-complete. NP-hardness holds simultaneously when the graph is bipartite, has radius two about a specified hub, the seed threshold is fixed at* 2*, uncertainty occurs only on boundary terminals, every uncertain terminal has interval* $[5, 7]$*, and every other scalar value belongs to* $\{0, 8, 10, 20\}$*.*

Proof. Reduce from Monotone NAE-3-SAT-E4 [134]. Let h be a hub with $F(h) = 0$ and let e be an essential peak with $F(e) = 20$. For each clause c, create an observer x_c with value 8 and a clause peak p_c with value 10, with edges

$$h - x_c - p_c.$$

For each Boolean variable i, create one shared boundary terminal z_i with

$$F(z_i) \in [5, 7],$$

and join x_c to z_i iff variable i occurs in clause c.

The graph is bipartite with hub radius two. The selected finite seeds are exactly the clause peaks. At x_c the unique rooted seed is p_c: its score is 8, while any other clause peak reaches x_c through a bottleneck at most 7. A terminal z_i belongs to B_{p_c} exactly when $i \in c$. In that case the path $p_c - x_c - z_i$ lies in the basin, the terminal gate is

$$s_\partial(p_c, z_i) = F(z_i),$$

and x_c is strongly incident because no basin route can bypass it.

Therefore the terminal midpoint margin is

$$\mu_{c,i} = 2F(x_c) - F(p_c) - s_\partial(p_c, z_i) = 16 - 10 - F(z_i) = 6 - F(z_i).$$

Hence a clause observer has mask 1111 exactly when its three terminals contain at least one value below 6 and at least one value at least 6. A NAE assignment gives such a realization by mapping the two truth values to 5 and 7. Conversely, any successful realization induces a Boolean assignment by the threshold $F(z_i) < 6$ versus $F(z_i) \geq 6$, and every clause is NAE-satisfied.

NP membership follows from a polynomial-bit rational realization certificate together with a polynomial-size deterministic comparison/maximin trace. Once the trace is fixed, all scalar requirements are rational linear equalities and strict/non-strict inequalities; a nonempty trace cell has a polynomial-bit rational point. Verification of the deterministic spatial mask is polynomial. □

121.2.2 Complete categorical spatial-frame realization

PTMM constrains a native coordinate block but is still a cylinder query. We now lift it to a fully prescribed finite categorical state.

For one polarity define

$$\mathsf{CSF}^{\sigma}(x) = (\rho^{\sigma}(x), M^{\sigma}_{\mathrm{aug}}(x), N^{\sigma}(x), R^{\sigma}(x)),$$

where ρ records unique-rooted/co-winner/no-finite-root status, M_{aug} is the complete competitor+terminal four-bit mask, N is the exact-neutral flag, and R is the categorical residual-terminal state. The complete self-dual state is

$$\mathsf{CSF}(x) = (\mathsf{CSF}^{+}(x), \mathsf{CSF}^{-}(x)).$$

Exact real-valued gate/cap coordinates are deliberately excluded from this finite quotient.

Definition 121.3 (Complete Categorical Spatial-Frame Realization (CCSFR))**.** Given an interval-valued graph, boundary terminals, observer set X, fixed seed threshold, and one prescribed finite categorical target τ_x for each observer, decide whether one common realization and one elder/tie resolution satisfy

$$\mathsf{CSF}(x) = \tau_x \qquad (x \in X).$$

Theorem 121.4 (Complete spatial-state hardness)**.** *CCSFR is NP-complete. Hardness holds on bipartite graphs of radius at most three with fixed seed threshold* 2*, uncertainty only on identical terminal intervals* $[5, 7]$*, and all fixed scalar values in the constant rational set*

$$\{-8, 0, 5/2, 4, 8, 19/2, 12, 20\}.$$

Proof. Use the same Boolean coordinates $z_i \in [5, 7]$ in two mirror clause gadgets. In the bright gadget choose

$$F(x_c^{+}) = 8, \qquad F(p_c^{+}) = 19/2.$$

The active terminal margin is

$$\mu^{+}_{c,i} = \frac{13}{2} - z_i.$$

Thus the required bright nonpositive bit exists exactly when some clause variable has $z_i > 6.5$ once exact neutrality is separately excluded.

For the dark mirror use inversion constant 12 with

$$F(x_c^{-}) = 4, \qquad F(p_c^{-}) = 5/2.$$

In dark-normalized coordinates the corresponding margin is

$$\mu^{-}_{c,i} = z_i - \frac{11}{2},$$

so the required dark nonpositive bit exists exactly when some clause variable has $z_i < 5.5$.

The remaining categorical coordinates are frozen throughout $z_i \in [5, 7]$: the selected seed family is unchanged; same-polarity query roots remain unique; opposite-polarity query roots remain co-winners/nonunique; the observer stays on the forced peak side of each active terminal gate, so the residual coordinate is empty; and no unintended neutral sector appears under the prescribed target. Hence each bright observer can be prescribed the full target

$$((\text{unique}, 1111, 0, \text{empty}), (\text{co-winner}, 0000, 0, \text{empty}))$$

and each dark observer the mirror target.

If the NAE instance is satisfiable, map the two truth values to 7 and 5. Every clause then has one value above 6.5 and one below 5.5, so all full targets are realized. Conversely, a realization of every full target gives in each clause one $z_i > 6.5$ and one $z_j < 5.5$. Assign these two ranges opposite Boolean values and assign middle values arbitrarily. Every clause already has both truth values and is NAE-satisfied.

Membership in NP uses the same polynomial trace-cell argument as Theorem 121.2. □

Corollary 121.5. *Exact arbitrary querying of the richer numerical full spatial-frame realization relation is NP-hard.*

The mechanism of Theorems 121.2–121.4 is independent of polarity-core owner scheduling. It uses terminal exposure, gate floors, route incidence, midpoint signs, and synchronization through one common scalar realization.

121.3 Finite-complex realization universality

Let P be a finite family of named protected strict observer-record obligations. As in EF–FG I, define

$$K_{\mathrm{raw}}(P) = \left\{A \subseteq P : \begin{smallmatrix}\text{one admissible raw realization}\\ \text{satisfies every obligation in } A\end{smallmatrix}\right\}$$

and let $K_\sigma(P)$ be the corresponding complex under the canonical maximal-safe rule. Both are finite simplicial complexes.

Earlier constructions produced all simplex boundaries. We now classify the realizability classes completely.

121.3.1 Facet-selector construction

Let K be a finite simplicial complex on vertices $1, \dots, n$ with facets

$$F_1, \dots, F_m.$$

Create one observer x with $F(x) = 4$. For every facet create a selector gate g_j with uncertain value

$$y_j := F(g_j) \in [2, 3].$$

Choose rational caps

$$2 < c_1 < \cdots < c_n < 3.$$

For every complex vertex i, create a protected peak p_i and a private elder blocker q_i. Their maximin attachments to x are arranged so that, for every selector realization y,

$$a_i(y) = \kappa(p_i, x) = \min\left(c_i, \max_{j:i\in F_j} y_j\right),$$

and

$$b_i(y) = \kappa(q_i, x) = \min\left(c_i, \max\left(2, \max_{j:i\notin F_j} y_j\right)\right).$$

Fix an essential elder with observer attachment 1 and choose finite births in $(6, 7)$ with elder order

$$q_1 > p_1 > q_2 > p_2 > \cdots > q_n > p_n.$$

The graph is bipartite and has radius at most three about the observer.

Lemma 121.6 (Maximizer-facet soundness)**.** *For every selector realization, every protected peak that is a strict record belongs to every facet attaining the global maximum selector value. In particular the protected record set is a face of K.*

Proof. If p_i is a strict record against its private blocker, then $a_i > b_i$. Since the same cap c_i is applied to both sides, this forces

$$\max_{j:i\in F_j} y_j > \max\left(2, \max_{j:i\notin F_j} y_j\right).$$

Let F_{j^*} be any facet with globally maximal y_{j^*}. If $i \notin F_{j^*}$, the right-hand maximum is at least y_{j^*}, while the left-hand maximum cannot exceed it, contradicting the strict inequality. Thus every strict protected record lies in every maximizing facet. □

Lemma 121.7 (Facet completeness)**.** *For every facet F_t there is a realization whose protected strict-record set is exactly F_t.*

Proof. Set

$$y_t = 3, \qquad y_j = 2 \quad (j \neq t).$$

Then for $i \in F_t$ we obtain $a_i = c_i$ and $b_i = 2$, while for $i \notin F_t$ the blocker is not below the protected attachment. With increasing caps and the fixed elder birth order, exactly the p_i with $i \in F_t$ are strict observer records. □

Theorem 121.8 (Raw universality)**.** *For every finite simplicial complex K there exists a finite interval-valued EF–FG graph and protected obligation family P such that*

$$\boxed{K_{\mathrm{raw}}(P) \cong K.}$$

The construction may be chosen with one observer, a bipartite graph of radius at most three, uncertainty only on facet selectors, and identical selector interval $[2, 3]$.

Proof. Lemma 121.6 gives $K_{\mathrm{raw}}(P) \subseteq K$. Lemma 121.7 realizes every facet, hence every subface, so $K \subseteq K_{\mathrm{raw}}(P)$. □

121.3.2 Canonical freezing

The raw construction becomes canonical because its geometry already forces every relevant attachment once the selector vector is fixed.

Lemma 121.9 (Forced-safe canonical fiber)**.** *Fix one selector realization y. For every protected peak and private blocker, the feasible attachment fiber is the singleton $\{a_i(y)\}$ or $\{b_i(y)\}$. Moreover every such attachment lies strictly below the canonical hard-band safety ceiling at $\theta = 1/2$ and $H = 0$.*

Proof. The maximin construction itself fixes the attachment values. Every attachment is below 3, whereas all finite births exceed 6 and the essential elder pressure is 1. Hence the canonical safety ceiling satisfies

$$\frac{B + D}{2} > \frac{6 + 1}{2} = 3.5 > 3.$$

Thus every geometrically forced attachment is strictly safe. The maximal-safe canonical section has no alternative value to choose and therefore equals the forced raw attachment. □

Theorem 121.10 (Canonical universality)**.** *For every finite simplicial complex K there exists a finite interval-valued EF–FG instance with*

$$\boxed{K_{\sigma}(P) \cong K.}$$

Consequently

$$\boxed{\{K_{\mathrm{raw}}(P)\} = \{K_{\sigma}(P)\} = \{\textit{all finite simplicial complexes}\}.}$$

Proof. By Lemma 121.9, realization by realization the canonical and raw elder scans coincide on the protected family. The proof of Theorem 121.8 therefore transfers verbatim. □

Remark 121.11. The equality in Theorem 121.10 is an equality of realizability classes. It does not assert $K_{\mathrm{raw}}(P) = K_{\sigma}(P)$ for every fixed EF–FG instance. The canonical/raw distinction remains genuine when an instance has nontrivial intermediate attachment freedom.

Chapter 122

EF–FG II: Parameterized Ownership and Correlated Uncertainty

The shallow-tree hardness results show that global common-realization synchronization can remain difficult even when the ambient graph is simple. This chapter separates that hardness from two finite ambiguity parameters and then enlarges the uncertainty model from independent interval boxes to correlated rational-polyhedral realization spaces.

122.1 Parameterized complexity of the ownership kernel

The threshold-band kernel is NP-complete, but fixed-order evaluation is linear. We now identify two natural parameters that measure the remaining global choice.

122.1.1 Elder-order ambiguity

For event birth intervals

$$I_i = [\ell_i, u_i],$$

define forced elder precedence

$$i \prec_B j \iff \ell_i > u_j.$$

Let C_B be the set of events incomparable with at least one other event under this relation, and put

$$k = |C_B|.$$

Lemma 122.1 (Linear extensions are exactly realizable elder orders)**.** *A total event order is realizable by scalar choices from the birth intervals iff it is a linear extension of $\prec_B$.*

Proof. Necessity is immediate. For sufficiency, write the proposed elder-to-younger order as $i_1, \dots, i_n$ and construct birth values backwards. Set the last value to any point in its interval and recursively choose

$$b_{i_t} = \max\{\ell_{i_t}, b_{i_{t+1}}\}.$$

If this exceeded u_{i_t}, then $b_{i_{t+1}} > u_{i_t}$, and some later event would be forced elder to i_t, contradicting the linear-extension property. Ties are resolved in the proposed order. □

Lemma 122.2 (Ambiguity-core enumeration)**.** *The events outside C_B form one total chain and are comparable with every event. Fixing a linear order of the k ambiguity-core events determines at most one full linear extension. Hence the number of admissible elder orders is at most $k!$.*

Proof. If two events outside C_B were incomparable, each would belong to C_B. Thus the outside set is a chain. For each core event, its comparisons with the outside chain determine a unique insertion slot. A chosen core permutation is either inconsistent with the internal precedence constraints or yields one full extension. □

Theorem 122.3 (Order-ambiguity FPT)**.** *Threshold-band completion is fixed-parameter tractable in k:*

$$\boxed{T(n,k) = O(k!\,\mathrm{poly}(n)).}$$

For one hard band the bound is $O(k!\,n)$ after the admissible orders have been generated.

Proof. Enumerate the at most $k!$ core permutations, reconstruct the corresponding full linear extension when one exists, and evaluate that fixed order by the EF–FG I prefix-minimum scan. □

122.1.2 Uncertain attachment gates

For event $e_i = (h_i, A_i, B_i)$, the corresponding physical attachment interval is

$$[q - A_i, q - B_i].$$

Call it uncertain iff $A_i > B_i$, and let

$$g = \#\{i : A_i > B_i\}.$$

Lemma 122.4 (Deterministic-gate reset)**.** *If $A_i = B_i = b_i$ and event i is owner-legal, then the post-event deficit is exactly*

$$x' = b_i.$$

Proof. Owner legality gives $(x - s_i)/2 \le b_i$, so

$$x' = \max\{b_i, (x - s_i)/2\} = b_i.$$

□

Lemma 122.5 (Deterministic-gate chain)**.** *If two deterministic-gate events satisfy $B_i > B_j$, then any legal owner word using both must place i before j. Equal-B deterministic events cannot both be owner uses.*

Proof. If j were used first, Lemma 122.4 would leave $x = B_j$, after which event i would require $B_i < x = B_j$, impossible. For equal B, the first reset makes the second fail the strict owner-live inequality. □

Define state dominance by

$$(h_1, x_1) \succeq (h_2, x_2) \iff h_1 \ge h_2, \quad x_1 \le x_2.$$

A dominating state can simulate every successful future continuation of the dominated state, possibly skipping an event that has already become owner-dead.

Theorem 122.6 (Uncertain-gate FPT)**.** *Threshold-band completion is fixed-parameter tractable in g:*

$$\boxed{T(n,g) = 2^g\,\mathrm{poly}(n).}$$

A straightforward exact dynamic program runs in $O(2^g g n^3)$ arithmetic operations.

Proof. Sort the deterministic-gate events into the forced chain $d_1, \dots, d_m$ from Lemma 122.5, and let U be the g uncertain-gate events. Use DP cells

$$DP[j, S], \qquad 0 \le j \le m, \quad S \subseteq U,$$

storing the Pareto frontier of reachable hard states after deciding the first j deterministic events and using exactly the uncertain owner events in S. From each state one may skip d_{j+1}, use it if legal, or use any unused uncertain event if legal. Every legal owner word is represented because its deterministic subsequence must respect the fixed chain.

Reachable headroom values belong to the finite set $\{h_0, h_1, \dots, h_n\}$. For a fixed headroom, only the smallest deficit survives dominance pruning, so each frontier has at most $n+1$ states. There are $(m+1)2^g$ cells and at most $g+1$ owner choices per state, yielding the displayed FPT bound. □

Corollary 122.7 (Parameterized boundary)**.** *On the threshold-band/radius-two family carrying the ownership hardness theorem:*

$$\boxed{\textit{small } k \textit{ or small } g \Longrightarrow \textit{FPT.}}$$

By contrast, band count, graph treewidth, and graph radius alone are para-NP-hard parameters: NP-hardness already holds at one hard band, treewidth one, and radius two.

Proof. The positive statements are Theorems 122.3–122.6 and transfer through the exact radius-two physical lift and dark-anchor lift of EF–FG I. The negative statements follow immediately from the restrictions of the existing NP-completeness construction. □

122.2 Correlated rational-polyhedral uncertainty

Independent interval boxes are useful but cannot express shared calibration, conservation, monotonicity, or other cross-vertex correlations. We now replace the box by an exact correlated class.

Definition 122.8 (Rational-polyhedral realization model)**.** Let $F \in \mathbb{R}^V$ be the scalar field and $z \in \mathbb{R}^r$ auxiliary latent variables. A realization model is given in extended form by

$$\widehat{\mathcal{P}} = \{(F, z) : AF + Bz \le c,\ CF + Dz = e\},$$

with rational data. The admissible scalar fields are

$$\mathcal{P} = \operatorname{proj}_F \widehat{\mathcal{P}}.$$

The class contains interval boxes, shared offsets, linear calibration models, Lipschitz constraints, monotonicity constraints, affine conservation laws, and finite intersections of rational halfspaces and hyperplanes.

Theorem 122.9 (Polyhedral trace theorem)**.** *Fix a finite categorical or rational-threshold EF–FG query Q. There exists a finite family of discrete deterministic traces $\mathfrak{T}_Q$ such that the realization set of Q is a finite union of projections of relatively open rational polyhedra:*

$$\{F \in \mathcal{P} : \exists \rho\ Q(F, \rho)\} = \bigcup_{\tau \in \mathfrak{T}_Q} \operatorname{proj}_F \widehat{\mathcal{P}}_\tau.$$

For a fixed trace τ, feasibility of $\widehat{\mathcal{P}}_\tau$ is decidable in polynomial time by rational linear programming.

Proof. The deterministic EF–FG evaluator is a polynomial-time finite comparison program built from scalar coordinates, rational affine combinations, min/max operations, comparisons, graph searches whose active subgraphs are comparison-determined, and finite elder/tie choices. A trace records every comparison branch, chosen min/max attainer, elder/tie decision, and resulting query branch. Once the trace is fixed, every numerical intermediate value is a rational affine function of F. Trace compatibility is therefore a rational system

$$A_\tau F \le b_\tau, \qquad C_\tau F = d_\tau, \qquad S_\tau F < t_\tau.$$

Intersecting with the extended uncertainty model gives a relatively open rational polyhedron.

For strict inequalities introduce one common slack $0 \le \delta \le 1$ and replace $r < s$ by $r + \delta \le s$. Maximizing δ yields strict feasibility iff the optimum is positive. This is a rational LP of polynomial size for a fixed trace. □

Corollary 122.10 (Polynomial rational witnesses)**.** *Every feasible existential finite EF–FG query under rational-polyhedral uncertainty has a polynomial-bit rational common realization and polynomial-size finite trace/tie certificate. Hence such possible queries lie in NP whenever deterministic verification is polynomial. Their universal robust complements lie in coNP.*

Proof. A feasible strict rational trace LP with positive slack has a polynomial-bit basic feasible solution by standard determinant bounds for rational linear systems. Verification replays the deterministic trace and checks the joint polyhedral constraints exactly. □

Proposition 122.11 (Self-duality)**.** *If $\mathcal{P}$ is rational-polyhedral, then*

$$-\mathcal{P} = \{-F : F \in \mathcal{P}\}$$

is rational-polyhedral, and the realization-wise EF–FG duality

$$\mathcal{E}(-F) = \mathcal{D}(\mathcal{E}(F))$$

extends exactly to possible and robust polyhedral semantics.

Proof. Substitute $F = -F'$ in the rational extended formulation and use the deterministic inversion theorem layerwise. □

Corollary 122.12 (Complexity transfer)**.** *All interval-box NP-hardness and NP-completeness results proved in EF–FG I and Section 121.2 remain valid for rational-polyhedral uncertainty.*

Proof. An interval box is a rational polytope. NP membership follows from Corollary 122.10. □

Remark 122.13 (Scope). The old interval-cactus dynamic programs exploit coordinatewise independence and local separator factorization. Arbitrary dense linear correlations can couple distant blocks, so we do not claim that those specific DPs remain polynomial. The generic exact replacement is finite trace enumeration with LP feasibility.

Chapter 123

EF–FG II: Layered Semantics and Closure of the Open Program

The final second-stage chapter transfers exact common-realization semantics to finitely many coupled scalar layers and then records the closure of the five directions posed by the foundational EF–FG stage. The layered construction is deliberately finite and scalar-by-layer: it does not identify EF–FG with a multivariate Reeb-space theory, but couples ordinary scalar EF–FG layers through one joint realization and explicit cross-layer predicates.

123.1 Finite multichannel and spatiotemporal transfer

We now extend the scalar theory to a finite family of coupled scalar layers without imposing a vector-valued total order.

Definition 123.1 (Finite layered EF–FG)**.** Let $\Lambda = \{1, \dots, L\}$ be a finite layer set. Layer λ has graph $G_\lambda = (V_\lambda, E_\lambda)$, scalar field F_λ, and ordinary deterministic EF–FG output $\mathcal{E}_\lambda(F_\lambda, \rho_\lambda)$. The uncoupled product output is

$$\mathcal{E}_\Lambda(F, \rho) = (\mathcal{E}_1, \dots, \mathcal{E}_L).$$

All layer scalar coordinates and optional latent variables are jointly constrained by one rational extended polyhedron.

A finite cross-layer specification Γ may contain categorical relation atoms

$$(s_{\lambda,a}, s_{\mu,b}) \in R$$

for an explicitly supplied finite relation R, and rational affine atoms in named numerical observables,

$$\sum_j q_j y_j \ \bowtie \ b, \qquad \bowtie \in \{<, \leq, =, \geq, >\}.$$

For time layers, finite correspondence relations between named observers/features may be supplied as part of the input. No automatic feature-identity theorem is assumed.

Theorem 123.2 (Deterministic layered transfer)**.** *For fixed realizations and elder/tie resolutions, finite layered EF–FG is evaluable in*

$$\boxed{O\left(\sum_{\lambda=1}^{L} T_\lambda + |\Gamma|\right),}$$

where T_λ is the deterministic evaluation time of layer λ.

Proof. Evaluate every scalar layer independently and then evaluate the finite cross-layer predicates. □

Theorem 123.3 (Joint layered trace theorem)**.** *Fix one deterministic trace for each layer and one Boolean branch of every predicate in* Γ*. The joint realization cell is a relatively open rational polyhedron in the concatenated layer and latent variables. Fixed joint-trace feasibility is rational LP feasibility, and every feasible possible query has a polynomial-bit rational joint witness.*

Proof. Apply Theorem 122.9 to each layer. On a fixed layer trace every named numerical observable is rational affine in that layer's scalar coordinates. The cross-layer affine atoms remain affine in the concatenated vector; categorical relation atoms become fixed truth values. One common slack handles all strict inequalities. Corollary 122.10 then gives a polynomial-bit rational joint solution. □

Theorem 123.4 (Layered self-duality)**.** *Under simultaneous inversion*

$$\mathfrak{I}(F_1, \dots, F_L) = (-F_1, \dots, -F_L),$$

finite layered EF–FG satisfies

$$\boxed{\mathcal{E}_\Lambda(\mathfrak{I}F) = \mathcal{D}_\Lambda(\mathcal{E}_\Lambda(F)),}$$

with $\mathcal{D}_\Lambda = \mathcal{D}_1 \times \cdots \times \mathcal{D}_L$*. If* Γ *is transformed by the corresponding duality action, possible and robust statements are preserved; if* Γ *is duality-invariant, the entire model is self-dual.*

Proof. Apply the scalar inversion theorem to every layer and the linear substitution $F \mapsto -F$ to the joint polyhedral model. □

Theorem 123.5 (Independent-product factorization)**.** *Suppose*

$$\mathcal{P} = \prod_{\lambda=1}^{L} \mathcal{P}_\lambda$$

and the query is a conjunction $Q = \bigwedge_\lambda Q_\lambda$ *of layer-local predicates with no cross-layer atom. Then*

$$\mathrm{Possible}_{\mathcal{P}}(Q) \iff \bigwedge_\lambda \mathrm{Possible}_{\mathcal{P}_\lambda}(Q_\lambda),$$

and

$$\mathrm{Robust}_{\mathcal{P}}(Q) \iff \bigwedge_\lambda \mathrm{Robust}_{\mathcal{P}_\lambda}(Q_\lambda).$$

Proof. For possible semantics, combine independent layer witnesses into their product realization; the converse is projection. The robust statement follows from universal quantification over a Cartesian product. □

Theorem 123.6 (Two-layer replication transfer)**.** *Let* Q *be any single-layer EF–FG decision query. Construct two layer copies and impose exact equality*

$$F_2(v) = F_1(v)$$

for corresponding scalar coordinates. Require the same target query in both layers. Then the single-layer and two-layer instances are equivalent. Hence every NP-hard single-layer EF–FG query remains NP-hard already with exactly two channels or exactly two time slices; when the single-layer query is in NP, the replicated construction is NP-complete.

Proof. A single-layer witness can be copied to both layers with the same resolution. Conversely, a successful two-layer realization projects to a successful realization of either layer. The equality coupling is rational-polyhedral and the transformation is polynomial. □

A temporal state-persistence requirement is a special categorical relation. For example, if S is a finite queried state space, the relation

$$R_{\mathrm{id}} = \{(s, s) : s \in S\}$$

enforces exact state persistence across adjacent frames. Theorem 123.6 remains valid with this explicit temporal consistency predicate.

123.2 Synthesis: the closed theorem package

The new results complete the theorem-sized directions left by EF–FG I.

Theorem 123.7 (EF–FG II completion theorem)**.** *Within the finite scopes defined in this second-stage development, the following statements hold.*

(i) Full finite categorical spatial-frame realization is NP-complete on severely restricted shallow bipartite graphs (Theorem 121.4).

(ii) Every finite simplicial complex occurs as a raw EF–FG common-realization complex and as a canonical EF–FG common-realization complex (Theorems 121.8–121.10).

(iii) Threshold-band ownership completion is FPT both in elder-order ambiguity k and in uncertain-gate count g, while one hard band, treewidth one, and radius two alone do not remove NP-hardness (Corollary 122.7).

(iv) Exact common-realization semantics extends from independent interval boxes to correlated rational-polyhedral uncertainty with polyhedral trace cells, LP fixed-trace feasibility, polynomial-bit rational witnesses, and preserved self-duality (Theorem 122.9 and Proposition 122.11).

(v) Finite multichannel and spatiotemporal EF–FG admits exact joint-realization semantics, deterministic product evaluation, polyhedral trace certification, self-duality, factorization in the uncoupled case, and two-layer complexity transfer (Theorems 123.2–123.6).

These statements sharpen the original local/global picture. Global difficulty does not come from deep graph topology: ownership hardness survives on radius-two trees, while spatial-frame hardness survives on shallow bipartite graphs. At the same time, difficulty is not uniform. Order ambiguity and gate uncertainty expose finite-parameter routes back to exact tractability. The realization-complex theorem shows that structural compatibility is maximally expressive at finite scale, while the polyhedral and layered theorems show that exact common-realization semantics is stable under a substantial enlargement of the uncertainty and data model.

The scope boundaries are part of the theorem package. The finite layered theory is not a multivariate Reeb-space theory, and the polyhedral theorem does not assert preservation of interval-cactus factorization under dense correlations. Likewise, the equality of raw and canonical realizability classes does not identify the two complexes for each individual instance. Stating these distinctions explicitly keeps the sequel within the semantics actually proved.

123.3 Conclusion

This second-stage development turns the open theorem program of EF–FG I into a closed second-stage theory. The spatial-frame projection now has a direct NP-completeness classification at the complete finite categorical level. Protected common-realization complexes are universal among finite simplicial complexes under both raw and canonical semantics. The ownership hardness kernel admits two exact FPT escape parameters, distinguishing

genuine order/gate ambiguity from shallow graph structure. Common-realization semantics extends exactly to correlated rational-polyhedral uncertainty, and finite scalar layers can be coupled into multichannel or spatiotemporal models without abandoning self-duality or exact certification.

The combined picture is therefore not simply that global EF–FG questions are hard. Rather, the theory identifies three complementary regimes: local or fixed-order calculations with direct polynomial algorithms; globally synchronized realization problems that are NP-complete even on shallow graph families; and intermediate instances whose global uncertainty is concentrated in a small ambiguity core and is fixed-parameter tractable. At the structural level, finite compatibility patterns are fully universal. At the modeling level, exact semantics survives correlated and layered coupling by remaining faithful to one common realization before projection.

Chapter 124

Joint Physical Realization of Relational Extrema States: Sharp Resource Obstructions, Correlated Terminal Fibers, and Spectral Rigidity

Research branch — EF–FG III: physical common realization on thermal graphs. *The preceding EF–FG chapters establish observer and spatial-frame semantics, common-realization complexity, universality, and layered transfer. This chapter introduces a different, explicit physical constraint: one positive heat equation on a rational weighted network, a fixed common initial condition, and a single shared actuation budget. It derives sharp global feasibility thresholds, exposed-face geometry, correlated terminal caps, and a spectral rigidity transition while preserving the intrinsic elder rule and connected-region semantics.*

Placement and proof boundary. The foundational EF–FG I and II semantics are treated in the earlier chapters of the EF–FG Part, culminating in Chapter 123, with their intermediate complexity, uncertainty and universality results. The independent mathematical study [268] supplies the precise physical realization theorems proved here. Their assumptions concern rational finite weighted thermal networks, measurable nonnegative shared controls, one fixed observation horizon and source-native connected-region witnesses; the conclusions are not new DPDA lower bounds, generic graph-algorithm improvements, or physical laboratory measurements. This is a concrete *external* physical realization test related to, but not formally derived without additional hypotheses from, the general exact-lift principles. The supporting control and algebraic tools are cited in their own right.

Principal results. Theorems 124.6, 124.9, 124.13, 124.14 and 124.16 respectively establish a sharp minimal joint-power obstruction, a finite active-tie Krylov criterion, a nonsemialgebraic optimal physical face, an exact correlated terminal-cap range, and a one-percent perturbation-induced rank jump and optimum-face collapse. The strict-versus-weak distinction, positive-budget condition and independent terminal-region choices are part of the statements and cannot be dropped.

124.1 Introduction

A controlled temperature field can satisfy an individually specified target without satisfying several targets together. The difficulty becomes particularly visible when the targets are not point temperatures but *relational states*: an observer is dominated by a persistent local maximum, a designated terminal owns a separator, or a connected terminal region has a

prescribed floor, cap, and membership flag. Each observable may be possible in isolation. Their juxtaposition is a valid physical claim only if all observables are evaluated on the *same* terminal temperature field.

We examine this issue in a finite graph realization of extrema–frame foreground/background (EF–FG) semantics. EF–FG specifies a superlevel-component elder rule, observer-relative saliency and ownership, spatial maximin gates, and intrinsic connected residual terminal regions. The foundational EF–FG I and II articles define the observer and spatial-frame outputs and their common-realization consistency requirements [269, 270]. The physical plant is conventional positive linear heat diffusion. Its novelty is *not* presumed: support-function duality, the bang–bang principle, finite-dimensional Krylov subspaces, and spectral graph perturbation are established ideas [271, 272, 274, 275]. What is investigated here is the exact coupling of these tools to a *joint, source-faithful relational realization* problem, including the terminal caps and their correlations. Related continuous-time reachability and spectral decision problems have already been studied in substantially more general control-theoretic settings [273].

Contributions. The chapter supplies four linked mathematical outcomes. First, it expresses physical common-witness feasibility in a dual form while keeping strict ownership inequalities distinct from weak terminal conditions. Second, on a nonsymmetric degree-three thermal tree, it computes a sharp resource threshold and its unique two-switch realizing control, together with witnesses for every proper subfamily of the three requested predicates. Third, it gives an active-tie Krylov formula for the dimension of an optimal physical face and, subject to a necessary relative-interior qualification, an exact cap-fiber dimension formula. Fourth, on a controlled pair of eight-vertex networks, it identifies a nonsemialgebraic optimum face with a *sharp correlated* cap segment and demonstrates how a one-percent conductance perturbation collapses that face to one physical field without removing the intrinsically multiple terminal-region choices.

The goal is therefore both a physical application and a mathematical test of the utility of relational semantics: the answers identify *what can be realized jointly, why a joint demand can fail, how much physical resource removes the failure, and which terminal outputs remain coupled*. All actuation examples are exact mathematical solutions of ideal network heat equations; no apparatus measurements are claimed.

The mathematical contribution and the model boundary

The word *realization* has two meanings in nearby literatures. In abstract EF–FG complexity theory, a combinatorial structure can be encoded by a small-radius scalar-field gadget. Here, by contrast, a *physical realization* is an actual solution of a constrained network heat equation, with nonnegative heater powers, one initial field, and one observation time. Neither type of theorem automatically implies the other. We use the EF–FG formalism as a prescribed output semantics, not as a substitute for the physical dynamics. The underlying definitions are developed in the foundational EF–FG I article [269] and extended in the EF–FG II article [270], especially for exact common-realization semantics, observer records, spatial frames, and correlated relational outputs.

More precisely, the full deterministic EF–FG state contains an observer component and a spatial-frame component, schematically $\big(PO_x(F), SF_x(F)\big)$. We investigate *selected projections* of that state—bright ownership, existence of an intrinsic connected terminal resolution with a given membership flag, and the floor/cap data of chosen resolutions. We do not claim a universal physical realization theorem for every component of every categorical EF–FG output. This distinction is especially relevant when the joint query involves two terminals: the terminal regions are part of the common witness, but their identities need not be unique even after the physical field is fixed.

The paper is organized around three progressively finer questions. First, is a joint semantic demand possible under a given physical budget? Second, if it is possible, can we determine an exact least budget and a realizing control? Third, when many controls realize

the optimum, what restrictions do they impose on the *joint* terminal tuples, rather than on their unrelated coordinate-wise ranges?

124.2 Physical model and source-faithful EF–FG queries

Let $G = (V, E)$ be finite, connected and undirected, with strictly positive edge conductances $c_{ij} = c_{ji}$. Its Laplacian is L, and $P_s = e^{-sL}$. For the permitted heater vertices $J \subseteq V$ and a pointwise shared power bound $Q > 0$, consider

$$\dot{T}(t) = -LT(t) + \sum_{j \in J} e_j u_j(t), \quad T(0) = Z > 0, \quad u_j(t) \geq 0, \quad \sum_{j \in J} u_j(t) \leq Q \text{ a.e.} \tag{124.1}$$

For observation time $\tau > 0$ and backward kernel time s, the final field is

$$F(u) = P_\tau Z + \sum_{j \in J} \int_0^\tau P_s e_j u_j(\tau - s)\, ds, \qquad \mathcal{K}_Q = \{F(u) : u \text{ obeys (124.1)}\}. \tag{124.2}$$

The set $\mathcal{K}_Q$ is compact and convex. Indeed, the control simplex is weak-star compact in the bounded L^∞ ball, the endpoint map is weak-star continuous, and the pointwise constraints are convex. Moreover P_s is a nonnegative stochastic matrix.

To state the required piece of EF–FG semantics self-containedly, define for vertices x, y

$$\kappa_F(x, y) = \max_{\pi : x \rightsquigarrow y} \min_{z \in \pi} F(z). \tag{124.3}$$

Local maxima of the superlevel filtration of F generate primitive birth components. On merging components, the elder rule retains the older (higher-birth) component, with a consistent resolution of equal-birth ties. A finite seed p has birth B_p and elder death $d(p)$. The spatial selected seeds have $B_p - d(p) > \Delta_{\text{seed}}$, and the rooted co-winner basin $\mathcal{B}_p$ is the connected component through p of the set on which p maximizes seed attachment. The two observer polarities are obtained using F and $-F$. Ownership is *bright* only if bright saliency exceeds its threshold and dark saliency does not; it is not a synonym for the inequality at an arbitrary vertex pair.

For a designated terminal $t \in \mathcal{B}_p$, put

$$s_t = \kappa_{F|\mathcal{B}_p}(p, t), \quad \alpha(v) = \kappa_{F|\mathcal{B}_p}(p, v), \quad \gamma_t(v) = \kappa_{F|\mathcal{B}_p}(t, v). \tag{124.4}$$

Vertices with $\alpha > s_t$ belong to the peak side and must be excluded from a terminal region; vertices with $\gamma_t > s_t$ belong to the terminal side; the remaining eligible vertices satisfy $\alpha = \gamma_t = s_t$. An *intrinsic residual terminal region* $R \in \mathrm{Res}_{p,t}(F)$ is connected, contains t and the forced terminal component, excludes the forced peak vertices, and selects any other vertices only from the tie zone. Its real-valued data are

$$\mathrm{ResRel}_{p,t,x}(F, R) = \left(s_t,\ U(R) = \max_{z \in R} F(z),\ y = F(x),\ \chi_x = \mathbf{1}_{\{x \in R\}} \right). \tag{124.5}$$

We also retain $L(R) = \min_{z \in R} F(z)$, as the floor need not equal s_t for every admissible R. When the terminal classification margin is zero, the terminal-neutral band is determined by $L(R), U(R)$ and y; in particular, $y = L(R) = U(R)$ is neutral, whereas $y = L(R) < U(R)$ is dark.

Definition 124.1 (Joint source-native witness)**.** For two terminals t_1, t_2 and a separator b, set $E_i(F)$ to mean $\exists R_i \in \mathrm{Res}_{p,t_i}(F)$ with $b \in R_i$. A *joint witness* for ownership O_Δ and E_1, E_2 consists of one admissible control u, its single field $F(u)$, one consistent elder choice, and an admissible region R_i for each terminal. No additional condition $R_1 \cap R_2 = \varnothing$ is imposed: it is not part of the intrinsic per-terminal definition.

For the trees below we certify an invariant EF–FG semantic chamber in which p is the sole selected finite bright seed, $\mathcal{B}_p = V$, the essential elder component is fixed, and the two intrinsic queries have the *proved* simplifications

$$O_\Delta(F) \iff g^\top F > \Delta, \qquad E_i(F) \iff h_i^\top F \geq 0, \qquad g = e_p - e_w, \quad h_i = e_{t_i} - e_b. \tag{124.6}$$

The reductions in (124.6) are *conclusions of model-specific extrema and connectedness checks*; they must not be taken as the general definition of EF–FG.

124.2.1 What is retained from the intrinsic spatial semantics

The gate, floor, cap, query value, and membership flag carry different information. The gate s_t depends on the path geometry in the rooted basin, whereas $U(R)$ and $L(R)$ depend on the *selected connected region*. For terminal margin Δ_{term}, the source-native thresholds are

$$\begin{aligned} T^+_{\Delta_{\mathrm{term}}}(L,U) &= \max\{L + \Delta_{\mathrm{term}},\ (L+U)/2\}, \\ T^-_{\Delta_{\mathrm{term}}}(L,U) &= \min\{U - \Delta_{\mathrm{term}},\ (L+U)/2\}. \end{aligned} \tag{124.7}$$

A query y is dark below T^-, neutral in the closed band $[T^-, T^+]$, and bright above T^+. In particular, at zero terminal margin, if $y = L = s_t < U$, then $T^- = (L+U)/2 > L$, so the same terminal membership flag can be paired with a *dark* label; if $U = L = y$, its label is neutral. Both outcomes can arise from different admissible regions on one temperature field.

This distinction explains why the marginal statements "terminal i can be neutral" and "terminal j can have a large cap" cannot simply be multiplied: there must exist a *single* F, a consistent elder resolution, and regions R_i, R_j realizing the full tuples together. For example, imposing disjointness of R_1 and R_2 when both must contain the separator b would create an immediate contradiction; but that disjointness is an *additional global gluing contract*, not part of the intrinsic two-terminal relation analyzed here.

Remark 124.2 (Scope of the semantic chamber). The equivalent temperature inequalities in (124.6) are used only after verifying the selected birth atoms, essential component, co-winner basin, maximin gates, peak-side exclusions, terminal-side requirements, and connectedness of the displayed regions. At an unverified birth or gate tie, the simple affine conditions must not replace the original EF–FG construction. The semantic checks below enumerate the intrinsic connected candidates using the native gate, forced-side, and tie-zone rules rather than assuming the guard equivalence everywhere.

124.3 Shared-control duality and exact joint infeasibility

For any $v \in \mathbb{R}^V$, write $\sigma_Q(v) = \max_{F \in \mathcal{K}_Q} v^\top F$. The control simplex yields the exact support formula

$$\boxed{\sigma_Q(v) = v^\top P_\tau Z + Q \int_0^\tau \max\left(0, \max_{j \in J} v^\top P_s e_j\right) ds.} \tag{124.8}$$

This is classical support optimization, but it is particularly useful when the h_i in (124.6) are *actual* terminal-membership guards rather than ad hoc temperature targets.

Proposition 124.3 (Common-witness dual certificate)**.** *Fix a chamber in which* (124.6) *holds. For $\lambda_i \geq 0$, let $v = g + \lambda_1 h_1 + \lambda_2 h_2$. If $\sigma_Q(v) \leq \Delta$, then the joint EF–FG query $O_\Delta \wedge E_1 \wedge E_2$ is impossible for every measurable control in* (124.1)*. Conversely, if one admissible F satisfies $g^\top F > \Delta$ and both $h_i^\top F \geq 0$, it is an exact common witness, irrespective of whether the R_i choices are unique.*

Proof. If E_i hold then $h_i^\top F \geq 0$, hence $g^\top F \leq v^\top F \leq \sigma_Q(v) \leq \Delta$, contradicting strict ownership. The converse follows from the verified equivalences (124.6) and the intrinsic connected-region witnesses. □

Remark 124.4. For a general finite family of weak affine guards $AF \geq a$ and strict guards $CF > c$, the exact same-field feasibility test has two margins: $d_w = \max_{F \in \mathcal{K}_Q} \min_i((AF)_i - a_i)$, and, when $d_w \geq 0$, $d_s = \max_{F \in \mathcal{K}_Q, AF \geq a} \min_j((CF)_j - c_j)$. A witness exists precisely when $d_w \geq 0$ and $d_s > 0$. Minimax duality expresses these margins through (124.8); zero-margin equality faces are not replaced by approximate tolerances. This is a convex-analytic characterization, not a claim of a terminating bit algorithm for arbitrary real spectral input.

124.3.1 Exact support equality and complementary slackness

The support integral can be interpreted pointwise: in backward time s, the controller selects an action in $\mathcal{A} = \{0\} \cup \{e_j : j \in J\}$ with the largest value of $v^\top P_s a$; a nonzero winner is assigned the full available power. Tied winners permit convex sharing, but ties at isolated times do not alter the final field. Consequently the maximum in (124.8) is an *attained* global bound over all measurable controls, not a numerical discretization of time.

Lemma 124.5 (Equality in a two-terminal dual bound)**.** *Assume* (124.6)*, and let* $\lambda_1, \lambda_2 > 0$*. If a feasible field* F *satisfies* $g^\top F = \sigma_Q(g + \lambda_1 h_1 + \lambda_2 h_2)$ *and* $h_i^\top F \geq 0$*, then both* $h_i^\top F = 0$ *and the corresponding control attains the pointwise support maximizing action almost everywhere. If this action is unique except at finitely many switching instants, the realizing control is unique almost everywhere and the exposed face is a singleton.*

Proof. The nonnegative sum $\lambda_1 h_1^\top F + \lambda_2 h_2^\top F$ must vanish since $g^\top F \leq (g + \lambda_1 h_1 + \lambda_2 h_2)^\top F \leq \sigma_Q$. Positive multipliers force the two individual equalities. The difference between σ_Q and the actual dual value is the integral of a nonnegative pointwise control-allocation loss. An integral of a nonnegative measurable function is zero only when the loss vanishes almost everywhere. Uniqueness of the pointwise maximizing action therefore transfers to the control and its endpoint. □

A finite-dimensional certificate workflow. For a proposed active-port sequence, one solves the exact integral constraints for its switching times, actuation scale, and positive dual multipliers. The candidate is *not* accepted merely because a local solver converges. One must also prove (i) all required EF–FG guard identities on the same field; (ii) positivity of all dual multipliers; (iii) dominance of the proposed heater over *every* alternate heater and idle on the entire relevant time intervals; and (iv) a certified root enclosure. Steps (ii)–(iii) convert the local candidate into a global proof against arbitrary measurable controls. We implement this workflow in the three networks below.

124.4 A sharp minimal resource obstruction on a nonsymmetric tree

Consider the seven-vertex tree in Figure 124.1, with vertex order $(r, w, p, b, t_1, c, t_2)$ and rational conductances

$$(c_{rw}, c_{wp}, c_{pb}, c_{bt_1}, c_{bc}, c_{ct_2}) = (1.01, 1.02, 0.97, 1.03, 0.96, 1.04). \tag{124.9}$$

Take $Z = (25, 5, 15, 11, 10.8, 11.5, 10.8)$, $\tau = 0.01$, heaters p, t_1, t_2, observer parameter $\theta = 0.4$, spatial seed threshold 5, terminal margin 0, and strict ownership margin $\Delta = 10.16$.

A uniform positive-semigroup/Taylor interval argument verifies, for all admissible $Q \leq 109$, that r remains essential and p the only selected finite bright seed with $\mathcal{B}_p = V$. The ownership and terminal-region conditions are exactly (124.6); for terminal t_2 the intervening node c is forced into every R_2 containing b. See Section 124.10 for the model-specific certification.

Define $K(\alpha, \beta) = \int_\alpha^\beta P_s \, ds$ and, for $0 < a < b < \tau$,

$$F(Q, a, b) = P_\tau Z + Q\left[K(0, a)e_p + K(a, b)e_{t_1} + K(b, \tau)e_{t_2}\right]. \tag{124.10}$$

Here the source order is p, t_1, t_2 in *backward* kernel time, equivalently t_2, t_1, p in real physical time.

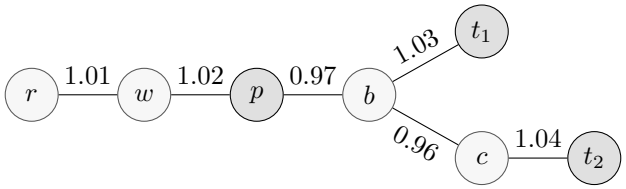


Figure 124.1: Seven-vertex nonsymmetric rational thermal tree. Shaded nodes are controllable heater ports. Both terminal queries concern the same separator b and final field.

Theorem 124.6 (Sharp resource threshold and unique realizing schedule)**.** *There exist exact real numbers Q_*, a_*, b_* with*

$$\begin{aligned} 108.32865113354069 < Q_* < 108.32865113374070,\\ a_* = 0.00555636232738007475\ldots,\\ b_* = 0.00779845184741640996\ldots, \end{aligned}$$

uniquely specified in the certified local root box by

$$g^\top F(Q_*, a_*, b_*) = 10.16, \qquad h_1^\top F(Q_*, a_*, b_*) = h_2^\top F(Q_*, a_*, b_*) = 0. \tag{124.11}$$

For the actual source-native EF–FG queries of this graph and every *available pointwise budget $Q \geq 0$,*

$$\boxed{O_{10.16} \wedge E_1 \wedge E_2 \text{ is physically realizable} \iff Q > Q_*.} \tag{124.12}$$

At the critical budget, the solution of the associated weak *Bright problem is the two-switch control in* (124.10)*, unique up to a set of times of measure zero; the strict Bright condition fails by equality.*

Proof. The three equations in (124.11) are analytic in (Q, a, b). A rational Taylor expansion of P_s through degree 42, a rational approximate inverse Jacobian, and the contraction bounds listed in Section 124.10 give existence and uniqueness of the exact root within an explicit rational cube. Set $v = g + \lambda_1 h_1 + \lambda_2 h_2$, where λ_1, λ_2 make the source gains equal at a_* (ports p, t_1) and at b_* (ports t_1, t_2). Exact interval bounds give $0.989 < \lambda_1 < 0.991$ and $0.973 < \lambda_2 < 0.976$.

Let $D_1(s) = v^\top P_s(e_p - e_{t_1})$ and $D_2(s) = v^\top P_s(e_{t_1} - e_{t_2})$. The full-interval certified derivative estimates give $D_1'(s) < 0$ and $D_2'(s) < 0$, with $D_1(a_*) = D_2(b_*) = 0$ and $a_* < b_*$. It follows that the maximizing heater is p before a_*, t_1 between a_* and b_*, and t_2 afterward; both heater gains remain strictly positive, excluding the idle action. Thus (124.10) at Q_* attains the *global* support $\sigma_{Q_*}(v)$. The two terminal contrasts in (124.11) are exactly zero, so

$$\max_{F \in \mathcal{K}_{Q_*},\ h_i^\top F \geq 0} g^\top F = 10.16.$$

Proposition 124.3 proves impossibility for $Q \leq Q_*$ (reachable sets are nested) and shows that equality is not sufficient for strict ownership. Since the optimal action is strictly unique except at the two switching instants, any control attaining the support agrees with the stated two-switch schedule almost everywhere.

For any $Q > Q_*$, choose $Q' \in (Q_*, \min\{Q, 109\})$. Apply the same switching times at power Q'. By linearity, from $h_i^\top F(Q_*) = 0$ and $h_i^\top P_\tau Z < 0$ we get $h_i^\top F(Q') > 0$; similarly $g^\top F(Q') > 10.16$. The bound $Q' < 109$ preserves the verified EF–FG chamber. This produces a genuine connected-region common witness for every larger available budget. □

At $Q = 109$, the same schedule gives $g^\top F \approx 10.163699289185$, $h_1^\top F \approx 0.001474729213$, and $h_2^\top F \approx 0.001444220901$, all on the required strict sides. Thus the theorem distinguishes an approximate numerical target from an exact resource threshold.

Proposition 124.7 (Minimal three-query obstruction)**.** *At $Q = 100$, the family $\{O_{10.16}, E_1, E_2\}$ is minimally unrealizable: every proper subfamily has a common physical witness, whereas the three-query family has none.*

Proof. Impossibility is Theorem 124.6. Every proper pair is witnessed by a constant rational power allocation $u = (u_p, u_{t_1}, u_{t_2})$: $O \wedge E_1$ by $(70, 30, 0)$, $O \wedge E_2$ by $(70, 0, 30)$, and $E_1 \wedge E_2$ by $(0, 50, 50)$. Single predicates are inherited from these pair witnesses. The exact endpoint inequalities follow from rational-Taylor interval evaluation of (124.2); their certified margins are summarized in Table 124.1. No three independent final fields are recombined. □

Family	u_p	u_{t_1}	u_{t_2}	$g^\top F$	$h_1^\top F$	$h_2^\top F$
O	100	0	0	10.54820	−0.24272	−0.23781
E_1, E_2	0	50	50	9.56309	0.25693	0.26183
O, E_1	70	30	0	10.25267	0.05565	−0.23790
O, E_2	70	0	30	10.25266	−0.24129	0.06207

Table 124.1: Mathematical physical witnesses for every proper subfamily at $Q = 100$. Values shown are rounded, while signs are separately certified.

Even at the unique critical physical field there are different admissible terminal resolutions. Specifically, the regions containing b for the first terminal are $\{b, t_1\}$, $\{b, c, t_1\}$, and $\{b, c, t_1, t_2\}$; for the second terminal they are $\{b, c, t_2\}$ and $\{b, c, t_1, t_2\}$. There are six valid per-terminal pairs evaluated on the *same* final field and elder resolution, with two neutral–dark and four dark–dark label combinations. Physical control uniqueness therefore does not force semantic uniqueness.

124.4.1 From controls to one physical state: the seven-vertex case

The resource threshold separates three logically distinct statements: there are individual physical witnesses, there are witnesses for every *proper* subfamily, and there is no witness for the full family at $Q = 100$. The first two follow from the explicitly displayed constant controls; the third requires a bound valid for *every* possible time-varying allocation. Theorem 124.6 proves the third without restricting the control to a chosen number of switches.

In forward physical time the critical controller activates t_2 on $[0, \tau - b_*)$, then t_1 on $[\tau - b_*, \tau - a_*)$, and then p on $[\tau - a_*, \tau]$, with power Q_* at each active port. The approximate switches are 0.00220154815258 and 0.00444363767262. The exact values are defined by (124.11); these decimals should not be used as a substitute for the exact construction when verifying terminal equalities. Both switching instants and all three heater allocations are physically admissible in the ideal graph model. No instantaneous negative heating, off-budget pulse, or different temperature field for the two terminals is used.

The two terminals probe the *same* separator b but different routes. For t_1, the connected terminal region can include b without passing through c. For t_2, the path $t_2 - c - b$ makes c mandatory whenever b is included. In the critical field $F(c) > F(b)$. Thus the source-native $\chi_b = 1$ region for t_2 has a cap strictly above its floor, whereas t_1 admits the neutral choice $\{b, t_1\}$. The six possible pairs of b-containing resolutions split into two neutral–dark and four dark–dark cases. This difference is a direct geometric statement about connected regions, not an artifact of the heating optimization.

Remark 124.8 (What the power number does and does not mean). The value Q_* is a sharp *pointwise aggregate heater-power* threshold for the specified ideal graph, initial data, observation time, and EF–FG predicates. It is not a universal material energy scale, not an

experimentally measured wattage, and not a bound on the time integral of power. Converting it to physical watts requires a choice of thermal capacitances, conductance units, and a dimensional normalization of (124.1).

124.5 A finite spectral test for the dimension of an optimal physical fiber

Let $\mathcal{A} = \{0\} \cup \{e_j : j \in J\}$ be the allowed instantaneous actions (the zero vector denotes idle), and let Π_λ be the orthogonal spectral projectors of L. For a dual direction v set $k_j(s) = v^\top P_s a_j$. A pair (i, j) is called an *active tie* if both gains are equal to the *maximum* over all allowed actions on a nonempty open time interval. The qualification "maximum" is essential: a tie between two suboptimal ports cannot create an optimal-face direction.

Theorem 124.9 (Active-tie Krylov criterion)**.** *Suppose L has d distinct eigenvalues. The following conditions are equivalent for any two actions a_i, a_j:*

(i) $v^\top P_s(a_i - a_j) = 0$ *on a set of positive measure;*

(ii) this gain difference vanishes for every real s;

(iii) $v^\top L^k(a_i - a_j) = 0$ *for $k = 0, \dots, d-1$;*

(iv) v is orthogonal to every $\Pi_\lambda(a_i - a_j)$.

For the face-dimension statement below, assume explicitly that $Q > 0$. Let $\mathcal{T}_v$ collect the active *tied pairs, and define*

$$S_v = \operatorname{span}\{\Pi_\lambda(a_i - a_j) : (i, j) \in \mathcal{T}_v,\ \lambda \in \operatorname{spec}(L)\}. \tag{124.13}$$

For the exposed face $\mathcal{K}_{Q,v} = \{F \in \mathcal{K}_Q : v^\top F = \sigma_Q(v)\}$,

$$\operatorname{aff}(\mathcal{K}_{Q,v}) = F_0 + S_v, \qquad \dim \mathcal{K}_{Q,v} = \dim S_v, \tag{124.14}$$

for any $F_0 \in \mathcal{K}_{Q,v}$. In particular, the face is a singleton precisely when there is no nontrivial active tie.

Proof. The gain difference is the real-analytic exponential sum $\sum_\lambda e^{-\lambda s} v^\top \Pi_\lambda(a_i - a_j)$. An analytic function vanishing on a positive-measure set is identically zero. Linear independence of exponentials and the Vandermonde system from its derivatives at zero give equivalence of the four conditions.

For the face formula, the assumption $Q > 0$ allows nonzero bidirectional transfers of power between pointwise tied actions. Equality in the support integral (124.8) forces any optimal control to put its instantaneous mass only on actions tied for the pointwise maximum (including idle if it ties). Differences between any two such controls therefore lie in S_v. Conversely, on an open active-tie interval choose a baseline that assigns strictly positive mass to each tied action, and make small positive and negative measurable perturbations that transfer mass between those actions. The linear span of all resulting endpoint differences contains $\operatorname{span}\{P_s(a_i - a_j) : s \in I\}$. Analyticity and linear independence of $e^{-\lambda s}$ imply this span equals $\operatorname{span}\{\Pi_\lambda(a_i - a_j)\}_\lambda$. Summing over the finitely many active-tie classes gives S_v, proving both inclusions. Nonidentical analytic gains can coincide only at isolated instants on $[0, \tau]$ and hence contribute no face dimension. □

Corollary 124.10 (Rank of the cap image after terminal equalities)**.** *Assume $Q > 0$. Let $HF = \eta$ encode terminal equalities and suppose* $\operatorname{relint}(\mathcal{K}_{Q,v}) \cap \{F : HF = \eta\} \neq \varnothing$. *Then*

$$\operatorname{aff}(\mathcal{K}_{Q,v} \cap \{HF = \eta\}) = F_0 + (S_v \cap \ker H). \tag{124.15}$$

Let $\mathcal{U}$ be a nonempty relatively open subset of this sliced face on which a prescribed connected-region selection stays admissible and the maximizer of every cap is a fixed vertex. On $\mathcal{U}$ the cap vector is a linear map CF, and its attainable image has affine dimension $\operatorname{rank}(C|_{S_v \cap \ker H})$.

Proof. The relative-interior intersection contains a relative neighborhood of F_0 within the affine slice $F_0 + S_v \cap \ker H$, so that slice is precisely the affine hull. Since $\mathcal{U}$ is relatively open and nonempty in the affine slice, $\mathrm{aff}(\mathcal{U})$ is that whole slice; a linear image has dimension equal to the rank of its restricted direction space. □

Remark 124.11. The positivity hypothesis $Q > 0$ is necessary for the face-dimension formula: at $Q = 0$ the reachable set is a singleton even if pointwise actions tie and S_v is nonzero. Without the relative-interior hypothesis, the rank can be only an upper bound; a boundary slice may be smaller or empty. A fixed region R alone does not fix $U(R)$ as a linear function when its maximizing vertex changes; the active-cap chamber qualification is essential. Caps of *different* connected-region branches must never be multiplied as independent coordinate ranges.

124.5.1 Why active ties, not graph automorphisms alone, govern optimal fibers

Theorem 124.9 distinguishes a symmetry of the physical network from a symmetry that is actually *visible in the dual objective*. A graph automorphism may exchange two sources, but if the dual direction is not invariant under the exchange, their gains need not tie. Conversely, equal dual gains can be certified by vanishing Krylov moments even in the absence of an obvious vertex automorphism.

For a candidate pair of heater actions, form the $n \times n$ Krylov matrix $[d, Ld, \dots, L^{n-1}d]$, where $d = a_i - a_j$. Its column rank determines how large the subspace of endpoint perturbations *could* be under sustained source sharing, whereas the test $v^\top L^k d = 0$ determines whether a specified dual direction actually observes them as tied. Only pairs that also attain the *maximum* gain on a nonempty open interval contribute to the exposed face $\mathcal{K}_{Q,v}$. If terminal equality constraints are then imposed, the permissible perturbations lie in $S_v \cap \ker H$.

The cap image needs one further check: the selected connected regions must remain admissible on an open portion of that affine slice, and their cap-maximizing vertices must not change. With these conditions the matrix rank in Corollary 124.10 is exact; without them a number obtained by rank alone can overstate the actual cap freedom. This qualification is central to interpreting EF–FG data as *joint relations* rather than independent coordinate intervals.

124.6 A nonsemialgebraic optimal face with a sharp correlated cap segment

Consider the symmetric eight-vertex tree in Figure 124.2. Its vertex order is $(r, w, p, b, x_1, t_1, x_2, t_2)$; the backbone conductances $(r, w), (w, p), (p, b)$ are $(1.01, 1.02, 0.97)$, and all four arm conductances equal 1. Let

$$Z = (25, 5, 15, 11, 11.5, 10.8, 11.5, 10.8), \quad \tau = 0.01, \quad Q = 100, \tag{124.16}$$

with heater ports p, t_1, t_2. The EF–FG parameters are $\theta = 0.4$, spatial seed threshold 5 and terminal margin 0. Exact uniform bounds again certify (124.6) with both arms satisfying $F(x_i) > F(b)$ for all permitted controls.

The arm-swap symmetry preserves both the objective $g^\top F = F_p - F_w$ and the control simplex and exchanges the two terminal inequalities. Averaging a feasible control with its swapped counterpart preserves $g^\top F$ and turns $h_1^\top F, h_2^\top F$ into their common average. This exact group-averaging argument reduces the maximum of g subject to both real terminal guards to a symmetric two-port problem with $e_T = (e_{t_1} + e_{t_2})/2$ and $h = (h_1 + h_2)/2$. It does *not* average the connected regions themselves.

Lemma 124.12 (Finite-group reduction for affine source-native guards)**.** *Let a finite group Γ act by vertex permutations preserving L, Z, the permitted control set, and the aggregate*

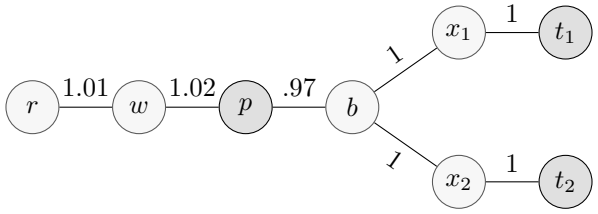


Figure 124.2: An eight-node thermal tree with exchangeable terminal arms. Replacing only the x_2t_2 conductance 1 by 1.01 yields the nonsymmetric comparison in Section 124.7.

power bound. Suppose the objective $g^\top F$ is invariant and the affine guards $h_i^\top F \geq 0$ form a permutation orbit under Γ. Then an optimizer of $g^\top F$ subject to all guards can be chosen Γ-invariant. On this invariant subspace all orbit guards have the same value, so the family reduces to their average.

Proof. Given an admissible control u, average its orbit of permuted controls: $\bar{u} = |\Gamma|^{-1} \sum_{\gamma \in \Gamma} \gamma u$. Nonnegativity and the pointwise aggregate power budget are preserved by convexity. The endpoint map is equivariant because P_s commutes with Γ and Z is invariant, hence $F(\bar{u}) = |\Gamma|^{-1} \sum_\gamma \gamma F(u)$. The invariant objective is unchanged. Since every orbit of a satisfied guard stays satisfied under the corresponding permutation, the average obeys each guard. Finally each guard has the same value on a symmetric field, so it is equivalent there to the orbit average. □

The lemma concerns *affine temperature guards after a verified semantic reduction.* The connected sets R_i themselves are not averaged into a fictitious connected set: once the common symmetric field is obtained, each genuine V18 terminal region is selected and checked on that field. In the eight-node tree, the group has two elements, exchanging the two terminal arms and leaving the observer p fixed.

Theorem 124.13 (Sharp two-terminal obstruction and nonsemialgebraic optimal face)**.** *For the symmetric model* (124.16)*, the exact maximum of $g^\top F$ subject to E_1 and E_2 is*

$$G_* = 10.072177752864946627341898\ldots, \qquad 10.07217775286 < G_* < 10.07217775287. \tag{124.17}$$

In particular $O_{10.07} \wedge E_1 \wedge E_2$ is realizable, while $O_{10.08} \wedge E_1 \wedge E_2$ is not; every proper subfamily of the latter three-query family is realizable at $Q = 100$. The exposed physical support face *of the corresponding optimal dual direction is not a semialgebraic set.*

Proof. Define the symmetric one-switch field $F_s = P_\tau Z + 100[K(0,s)e_p + K(s,\tau)e_T]$. The scalar $H(s) = h^\top F_s$ decreases strictly from positive to negative and has one exact root

$$s_* = 0.0051299779322837138844\ldots.$$

Choose the positive multiplier $\lambda_* = -g^\top P_{s_*}(e_p - e_T)/[h^\top P_{s_*}(e_p - e_T)] \approx 1.96017386923015$. Certified whole-interval bounds for $v = g + \lambda_* h$ yield a strictly decreasing source-gain difference, unique zero s_*, and positive gains for both ports. Proposition 124.3 then gives the global bound $g^\top F \leq g^\top F_{s_*} = G_*$ whenever $h_1^\top F, h_2^\top F \geq 0$. The symmetric source allocation $u_{t_1} = u_{t_2} = 50$ in backward time $s \in [s_*, \tau]$, followed by $p = 100$ on $[0, s_*]$, realizes both terminal equalities and this value. The proper subfamily witnesses are the constant source triples $(100, 0, 0)$, $(0, 50, 50)$, $(70, 30, 0)$, and $(70, 0, 30)$.

To prove the semialgebraic claim, note that on the support face the source p uniquely wins for $s < s_*$, while t_1, t_2 are tied as maximum for $s > s_*$. Thus the total terminal heating there is 100, but its signed allocation $q(s) = u_{t_1}(\tau - s) - u_{t_2}(\tau - s)$ can vary through all measurable values $[-100, 100]$ on $[s_*, \tau]$. The antisymmetric arm Laplacian is $B = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 1 \end{smallmatrix}\right)$, with eigenvalues $a = (3 - \sqrt{5})/2$ and $b = (3 + \sqrt{5})/2$, whose ratio $(7 + 3\sqrt{5})/2$

is irrational. The linear projection onto the two antisymmetric spectral modes is, up to a fixed translation, exactly

$$\left\{ \left(\int_{s_*}^{\tau} e^{-as} q(s) ds,\ \int_{s_*}^{\tau} e^{-bs} q(s) ds \right) : |q(s)| \le 100 \right\}. \tag{124.18}$$

The boundary of this two-moment body contains an analytic irrational-power arc $Y = \alpha + \beta(\gamma + \delta X)^{b/a}$ on an open interval (after converting q to a nonnegative bounded input). Such an arc cannot satisfy any nonzero polynomial relation: the functions $x^{i+j(b/a)}$ have pairwise distinct exponents, and are linearly independent on a positive interval. An analytic semialgebraic arc would satisfy a polynomial relation [277]. Hence the projection (124.18) is not semialgebraic. If the exposed face were semialgebraic, its linear image would be semialgebraic by the Tarski–Seidenberg theorem, a contradiction. □

The nonsemialgebraic face of Theorem 124.13 contains optimal states that do *not* satisfy both terminal constraints. The genuine *jointly feasible* optimum fiber is obtained by intersecting it with the two equalities $h_1^\top F = h_2^\top F = 0$. This is the distinction required when reporting physically correlated relational outputs.

124.6.1 Why the optimal physical face can be nonsemialgebraic

For clarity, the source of the irrational-power boundary in (124.18) can be made explicit, rather than left as a general algebraic-geometric assertion. Fix $A < B$, rates $0 < a < b$, and a scalar input $0 \le u(s) \le Q$ on $[A, B]$, and put

$$X = \int_A^B e^{-as} u(s) ds, \qquad Y = \int_A^B e^{-bs} u(s) ds.$$

For a prescribed X, the largest Y is obtained by allocating full power at the *earliest* values of s until the required X is reached. Indeed, the ratio of the Y-kernel to the X-kernel, $e^{-(b-a)s}$, strictly decreases with s; shifting an infinitesimal amount of weighted input from a later to an earlier time increases Y without changing X. The rearrangement is exact over measurable inputs, including arbitrary fractional controls.

Thus the upper boundary is achieved by $u = Q$ on $[A, A + \ell]$ and zero afterward. Eliminating ℓ yields

$$Y_{\max}(X) = \frac{Q}{b} \left[e^{-bA} - \left(e^{-aA} - \frac{aX}{Q} \right)^{b/a} \right], \quad 0 < X < \frac{Q}{a}(e^{-aA} - e^{-aB}). \tag{124.19}$$

If b/a is irrational, this real-analytic arc is not the zero set of any nonzero polynomial in (X, Y): a polynomial relation would produce a finite linear combination of distinct real powers $z^{i+j(b/a)}$, which are linearly independent on any positive interval. Since an analytic semialgebraic arc is Nash, this boundary and hence the associated two-dimensional moment body are not semialgebraic. Conversely a rational exponent gives an algebraic power relation and a semialgebraic moment body. This concrete boundary is sufficient for the present proof; it is not a statement that every irrational-spectrum reachability query is undecidable.

The antisymmetric source allocation of our physical tree satisfies $-Q \le q \le Q$. Replacing q by $2u - Q$ is an affine translation and scaling of its two modal moments. Therefore the preceding argument applies unchanged to the exact linear image of the physical optimal support face. The use of a *linear* image matters: if the whole face were semialgebraic, this image would be semialgebraic by projection, contradicting (124.19).

Theorem 124.14 (Exactly attainable correlated terminal caps)**.** *Fix on the optimal jointly feasible fiber the two intrinsic admissible connected regions*

$$R_1 = \{b, x_1, t_1\}, \qquad R_2 = \{b, x_2, t_2\}, \tag{124.20}$$

whose caps are $U_1 = F(x_1)$ and $U_2 = F(x_2)$. Then the complete attainable cap set is

$$\boxed{(U_1, U_2) = (C + z/2, C - z/2), \quad -\delta \le z \le \delta,} \tag{124.21}$$

where $C = 11.490213858830921509\ldots$ and $\delta = 0.000584009198972739483\ldots$. The endpoints are realized by full-power two-switch schedules, and every intermediate pair by admissible convex mixing. In particular, the two individual cap intervals may not *be combined independently.*

Proof. Every optimizer obeying both terminal guards has $h_1 = h_2 = 0$ because the two dual multipliers are strictly positive and g attains the support bound. The symmetric component of the final field is uniquely fixed by the winning action p for $s < s_*$ and total terminal source 100 for $s > s_*$. Therefore the cap sum is $2C$.

Let q be the signed difference of terminal heating on $[s_*, \tau]$. For the two antisymmetric arm eigenvalues $a < b$ in the preceding proof, set

$$k_y(s) = \frac{(b-1)e^{-as} + (1-a)e^{-bs}}{b-a} > 0, \tag{124.22}$$
$$k_x(s) = \frac{e^{-as} - e^{-bs}}{b-a} > 0.$$

The terminal equality is $\int_{s_*}^{\tau} k_y(s)q(s)ds = 0$ and the cap difference is $U_1 - U_2 = \int_{s_*}^{\tau} k_x(s)q(s)ds$. The ratio k_x/k_y strictly increases with s: setting $z = e^{-(b-a)s}$ turns it into $(1-z)/[(b-1) + (1-a)z]$.

Let $c \in (s_*, \tau)$ be the unique weighted median satisfying

$$\int_{s_*}^{c} k_y(s)ds = \int_{c}^{\tau} k_y(s)ds.$$

Among all $|q| \le 100$ with zero k_y-moment, the rearrangement inequality (or the pointwise sign of $k_x - \mu k_y$) gives the sharp extremes $q = \pm 100$ on $[s_*, c]$ and the opposite sign on $[c, \tau]$. Thus

$$\delta = 100\left(\int_{c}^{\tau} k_x(s)ds - \int_{s_*}^{c} k_x(s)ds\right) > 0, \quad c = 0.00756204666625562686\ldots.$$

These controls satisfy the terminal equality exactly and yield the two endpoints of (124.21). Convex interpolation gives every intermediate z without changing the optimal objective or terminal equalities. Since $F(x_i) > F(b) = F(t_i)$ in the certified EF–FG chamber, (124.20) are valid intrinsic connected terminal regions, their floors are $F(b)$, and their caps are $F(x_i)$, as claimed. □

For the symmetric optimum one obtains $U_1 = U_2 = C$. At a two-switch endpoint the actual field has $U_1 \approx 11.4899218542$ and $U_2 \approx 11.4905058634$; both have the same $g^\top F = G_*$ and terminal equalities. The result is a physical explanation of why a pair of individually possible terminal cap values need not be jointly possible.

124.6.2 An exact relational obstruction inside an optimal face

The line segment in (124.21) gives more information than an inequality bounding each terminal separately. For example, the marginal values $U_1 = C + \delta/2$ and $U_2 = C + \delta/2$ are both physically realizable *individually* at the optimal Bright score and with both terminal membership guards. Nevertheless the pair $(C + \delta/2, C + \delta/2)$ is impossible: it violates the exact joint invariant $U_1 + U_2 = 2C$.

Similarly, the endpoint $(C + \delta/2, C - \delta/2)$ is witnessed by one full-power terminal switching order, and the reversed endpoint by interchanging the two arms. Their midpoint

has a symmetric power-sharing realization. Every intermediate pair is obtained by convexly mixing the two endpoint controls, which is admissible because the power constraint is pointwise convex. This constructive argument proves both *necessity* and *sufficiency* of the correlation law, with no relaxation to a rectangular product of marginal cap intervals.

Corollary 124.15 (A minimal correlated-cap obstruction)**.** *On the jointly feasible optimal fiber of Theorem 124.14, let A assert $U_1 = C + \delta/2$ and B assert $U_2 = C + \delta/2$, for the fixed true regions* (124.20)*. Each assertion separately has a physical joint-terminal witness attaining G_*, but $A \wedge B$ has none on the same final field.*

Proof. The two endpoints of (124.21) witness each individual assertion. Their simultaneous conjunction would require $U_1 + U_2 = 2C + \delta > 2C$, contradicting the sharp cap invariant. □

This is a second kind of obstruction. Theorem 124.6 concerns the *resources required to obtain joint predicates*; Corollary 124.15 concerns the *coupling of two numerical relational outputs after physical optimality and terminal membership have already been achieved.* Neither is obtained by performing independent optimizations of the two terminal caps.

124.7 Conductance perturbation and relational rigidity

Keep the data of Section 124.6, changing only the conductance $c_{x_2t_2} = 1$ to $101/100$. The tree topology, allowable heaters, initial field, and two intrinsic terminal requests are unchanged. Put $d = e_{t_1} - e_{t_2}$ and $\mathcal{C}_d = [d, Ld, \dots, L^7 d]$.

Theorem 124.16 (Exact Krylov rank jump and collapse of the optimum face)**.** *The symmetric graph has* $\operatorname{rank} \mathcal{C}_d = 2$*, while the one-percent-perturbed graph has* $\operatorname{rank} \mathcal{C}_d = 7$*. For the latter graph a 7×7 subdeterminant, using rows $2, \dots, 8$ (one-based vertex indexing) and columns $d, \dots, L^6 d$, equals*

$$\frac{12710725656263913231}{125000000000000000000000000000} \neq 0.$$

For the perturbed network at $Q = 100$, the sharp joint-terminal Bright maximum is

$$G_{\mathrm{pert}} = 10.0722304026414387016940\ldots. \tag{124.23}$$

Its global dual-optimal physical field is unique; a pure full-power two-switch schedule, in forward physical time $t_1 \to t_2 \to p$, realizes it with both terminal equality conditions. The pair of caps corresponding to (124.20) *is uniquely*

$$(U_1, U_2) = (11.4905058628528755\ldots,\ 11.4898687382819038\ldots). \tag{124.24}$$

Nevertheless, each terminal still admits multiple distinct connected R_i choices, so the source-native complete ResRel *relation is not a singleton.*

Proof. The stated ranks and minor follow by exact rational multiplication of the two graph Laplacians; the vector d has coordinate sum zero, so rank seven is maximal here. The subdeterminant as a function of the edge conductance is a nonzero polynomial, giving rank seven for all but finitely many conductance values, although no claim is made for every perturbed weight.

To obtain the actual optimum, in backward time choose p on $[0, a]$, t_2 on $[a, b]$, and t_1 on $[b, \tau]$, where $(a, b, \lambda_1, \lambda_2)$ are the unique certified local solution to

$$\begin{aligned} h_1^\top F(a,b) = h_2^\top F(a,b) &= 0, \\ (g + \lambda_1 h_1 + \lambda_2 h_2)^\top P_a (e_p - e_{t_2}) &= 0, \\ (g + \lambda_1 h_1 + \lambda_2 h_2)^\top P_b (e_{t_2} - e_{t_1}) &= 0. \end{aligned} \tag{124.25}$$

Here

$$F(a,b) = P_\tau Z + 100\big[K(0,a)e_p + K(a,b)e_{t_2} + K(b,\tau)e_{t_1}\big].$$

The approximate root is

$$a = 0.00513051262657\ldots, \qquad b = 0.00756204696120\ldots,$$
$$\lambda_1 = 0.98006094114\ldots, \qquad \lambda_2 = 0.98013504755\ldots.$$

A four-dimensional rational Taylor/Banach certificate establishes the exact root. On the entire time interval, both relevant dual source-gain differences decrease strictly, and every winning port gain is positive. Thus the maximizing sequence is unique almost everywhere and its exact physical support equals (124.23). The full support bound with $\lambda_i > 0$ and the terminal equalities proves the asserted optimum, as in Theorem 124.6.

The strict uniqueness of the pointwise winning port gives a singleton exposed face. Equivalently Theorem 124.9 yields $S_v = 0$, because there is no active positive-measure tie. The two fixed-region caps must consequently take the single values (124.24). Intrinsic connected-region alternatives remain: there are five valid regions for each terminal, three containing b, hence nine source-native pairs with both $\chi_b = 1$ on this unique field. The two forms of uniqueness must not be conflated. □

124.7.1 A controlled comparison of the two physical regimes

Table 124.2 summarizes the distinction between the two nearby eight-vertex thermal plants. The comparison changes a single rational conductance; all semantic definitions, heater locations, initial temperatures, and power constraints are held fixed.

Property	Symmetric tree	One-percent perturbed tree
x_2t_2 conductance	1	101/100
Terminal-difference Krylov rank	2	7
Optimal physical endpoint fields	continuum	singleton
Optimal Bright score at $Q = 100$	$10.0721777528649\ldots$	$10.0722304026414\ldots$
Caps for fixed true R_1, R_2	exact diagonal segment	one ordered pair
Both terminal membership flags	attainable	attainable
Distinct intrinsic region choices	retained	retained

Table 124.2: Controlled physical comparison: the network perturbation changes physical optimal-face geometry, but not the definition of the EF–FG terminal records.

The distinction between the physical fiber and the semantic fiber can also be expressed as a two-stage map. Let $\mathcal{U}$ be the set of admissible controls and $\mathcal{R}(F)$ the set of compatible intrinsic connected-region tuples over the shared F. Then the realized records form the relational image

$$\mathcal{J} = \big\{\big(F, \mathrm{ResRel}(F,R_1), \mathrm{ResRel}(F,R_2)\big) : F = F(u),\ u \in \mathcal{U},\ (R_1,R_2) \in \mathcal{R}(F)\big\}. \tag{124.26}$$

Its fiber over a physical field can contain several R-choices, even when F itself is unique among optimal controls. Its image under a fixed-region cap map can instead shrink from a segment to a point when spectral ties are removed. Both effects are present in Table 124.2; neither can be read from a collection of independently attained marginal records.

124.8 Physical meaning and comparison with existing methods

The application in Sections 124.4–124.7 has three concrete levels. At the *state* level, heat diffusion maps a constrained power schedule to a final temperature field. At the *semantic* level, EF–FG extracts ownership and connected terminal records from that same field. At the *joint feasibility* level, the support dual determines whether all requested records can coexist and, in the resource-threshold model, the minimum power required. This is more informative than independently displaying separate temperature fields for each request, which would not establish one common physical witness.

The sharp threshold is actionable inside the mathematical plant model: below Q_* no allowable switching or sharing schedule succeeds; above it a physically specified two-switch schedule does. The fiber statements add a distinct operational message: even when a chosen ownership score is optimal, separately reported terminal caps may be incompatible because their pair is constrained to a diagonal segment. Symmetry breaking can remove the continuum of physical optima without eliminating the plurality of valid semantic region choices.

These results should not be confused with a benchmark demonstrating that EF–FG numerically outperforms standard optimal control. The support dual and switching principles are rooted in established optimal-control theory [271, 272]; the rank test is based on familiar graph control and spectral theory [274, 275]; the nonsemialgebraic proof uses classical real algebraic geometry [277]; and the use of symmetry in optimal control has an independent literature [276]. What EF–FG contributes *within this study* is the specified family of structurally meaningful output predicates and the constraint that all tuples and connected-region choices be evaluated on one physical realization. The resulting sharp combined statements, rather than the component tools, are the object of this chapter.

The model has important boundaries. Actuators are ideal and share a pointwise power limit; edge conductances and initial conditions are exact; measurements, finite switching delays, and experimental device imperfections are not incorporated. We prove fixed-time exact realization, not equality of complete trajectories. General finite input EF–FG atlas compilation, complexity bounds for arbitrary irrational spectra, and experimental improvements over classical controllers are not claimed.

124.9 Conclusion

A network heat equation supports exact mathematical realization of selected EF–FG relational predicates. On one rational nonsymmetric tree, three individually and pairwise feasible semantic queries have a sharp common-power obstruction, resolved by a certified optimal two-switch schedule. On another tree, the actual jointly feasible optimal terminal caps obey a necessary-and-sufficient line-segment law despite a nonsemialgebraic physical support face. A one-percent perturbation of a single conductance yields a finite Krylov-rank jump and a unique physical optimum while leaving multiple admissible terminal resolutions. The common mathematical mechanism is the combination of one shared heat field, exact semantic connectedness and cap constraints, and a global physical support certificate.

124.10 Certificate construction and independent reproduction

This chapter's quoted decimals are display approximations of mathematically defined exact real roots or kernel integrals. Verification consists of certified rational interval arithmetic and exact symbolic calculations; numerical validation consists of independent high-precision heat-flow evaluations using a different representation of the same finite thermal systems.

The calculations, not separate executable files, are specified by the following reproducible steps.

A. Semantics and full-control chamber certification. For a Laplacian with $\|L\|_\infty < 6$ and $\tau = 0.01$, evaluate P_s by $\sum_{k=0}^{N}(-sL)^k/k!$ with rational interval arithmetic. Because P_s is nonnegative and stochastic, $\|P_s\|_\infty = 1$ and the truncation errors can be bounded uniformly. Upper and lower interval bounds for all vertex differences, births, maximin attachment levels, and terminal-side component constraints certify the semantic chambers. A separate exact finite enumeration of all connected candidate R_i checks the forced peak, forced terminal, and tie-zone conditions; no disjoint global partition is inserted. For Section 124.4, the chamber is valid for all pointwise budgets through 109. For Sections 124.6–124.7, it is verified for $Q = 100$.

B. Seven-vertex resource threshold. With $q = Q/100$, $A = a/\tau$ and $B = b/\tau$, use the rational center

$$\begin{aligned} q &= 1.083286511336406946, \\ A &= 0.555636232738007475, \\ B &= 0.779845184741640996, \end{aligned}$$

and a cube of ℓ_∞ radius 10^{-12}. The kernel Taylor truncation through degree 42 has uniform remainder less than $4.79 \cdot 10^{-106}$. A rational approximate inverse B_0 of the Jacobian gives

$$\|\Phi(x_0)\|_\infty < 2.73 \cdot 10^{-19}, \quad \|I - B_0 D\Phi(x_0)\|_\infty < 1.27 \cdot 10^{-12}, \quad \|B_0\|_\infty < 4.$$

With an explicit Hessian bound the Newton-type map $x \mapsto x - B_0\Phi(x)$ has Lipschitz constant below $9.22 \cdot 10^{-11}$ on that cube and maps it into itself. Banach's theorem proves existence and local uniqueness of the exact root. The positive dual multiplier intervals are $(0.989, 0.991)$ and $(0.973, 0.976)$; the two source-gain differences have full-interval strictly negative derivatives, and all winning source gains exceed 0.893. These inequalities were checked using rational enclosures of the matrix-exponential series; the displayed floating-point values are not used to certify their signs. As an independent numerical validation, the original seven-vertex heat equation was also evaluated at 105-digit precision with a separate matrix-Taylor construction.

Constant-control witnesses in Table 124.1. For a fixed allocation $u = (u_p, u_{t_1}, u_{t_2})$ at every time, set $\widetilde{u} = u_p e_p + u_{t_1} e_{t_1} + u_{t_2} e_{t_2}$. Then the endpoint has an entirely rational Taylor approximation

$$F^{(12)} = \sum_{k=0}^{12} \frac{(-\tau L)^k}{k!} Z + \sum_{k=0}^{12} \frac{(-L)^k \tau^{k+1}}{(k+1)!} \widetilde{u}.$$

For the seven-vertex graph $\|L\|_\infty < 6$, $\tau = 0.01$, $\|Z\|_\infty = 25$ and $\|\widetilde{u}\|_\infty \le 100$. The matrix-exponential remainder and its time integral give

$$\|F - F^{(12)}\|_\infty \le (25 + 100\tau) e^{6\tau} \frac{(6\tau)^{13}}{13!},$$

so each displayed two-vertex contrast differs from the corresponding rational Taylor value by less than $1.2 \cdot 10^{-24}$. The four rows in Table 124.1 are the correctly rounded five-decimal evaluations of these rational expressions. Their strict sign margins are therefore certifiable without relying on floating-point signs.

C. Symmetric eight-vertex optimum and cap endpoints. Degree-33 exact rational Taylor bounds have remainder below $9.73 \cdot 10^{-81}$. They bracket $0.00512997793228 < s_* < 0.00512997793230$, certify $1.959 < \lambda_* < 1.962$, and establish strict source-gain monotonicity throughout $[0, 0.01]$. Exact symbolic calculation on the arm-swap antisymmetric space produces eigenvalues $(3 \pm \sqrt{5})/2$. An independent 116-digit spectral-integration evaluation also reproduced the root, the switching median c, and the endpoint cap difference δ; this numerical check is distinct from the analytic proof of nonsemialgebraicity.

D. Perturbed eight-vertex optimum. A direct rational Laplacian Krylov calculation verifies the displayed 7×7 minor (rows 2–8 counted from one, not rows 1–7). The four unknowns $(a/\tau, b/\tau, \lambda_1, \lambda_2)$ have the rational center

$$(0.513051262657132927,\ 0.756204696120191786,\\ 0.980060941142877874,\ 0.980135047554554104)$$

and a 10^{-12} local cube. The degree-34 matrix-Taylor error is below $1.67 \cdot 10^{-83}$, the rational contraction constant below $2.08 \cdot 10^{-10}$, and all source-gain ordering inequalities hold over the *entire* time interval. An independent 115-digit direct spectral evaluation reconstructed the root and cross-checked the EF–FG connected-region classification and the physical field.

E. Reproduction of the verified quantities without external code. An independent calculation begins with the rational edge list and the initial vector specified for each graph in Sections 124.4 and 124.6–124.7. Form the weighted Laplacian $L_{ii} = \sum_{j \neq i} c_{ij}$, $L_{ij} = -c_{ij}$, and approximate $P_s = e^{-sL}$ using

$$P_s^{(N)} = \sum_{k=0}^{N} \frac{(-sL)^k}{k!}, \qquad \|P_s - P_s^{(N)}\|_\infty \le e^{s\|L\|_\infty} \frac{(s\|L\|_\infty)^{N+1}}{(N+1)!}.$$

Integrate this polynomial term by term over each stated switching interval, retaining rational lower and upper enclosures for all entries. For the seven-vertex system, solve its three equations for the two switches and Q, check the contraction estimates in part B, and then verify the pointwise dual-port ordering on $[0, \tau]$. For the symmetric eight-vertex model, solve the terminal equality and the scalar switching median equation, evaluate both cap endpoints, and check the interval bounds of part C. In the perturbed model, solve the four equations for two switches and two dual multipliers, verify the rational Krylov minor, and check the uniform source-order inequalities in part D.

The remaining EF–FG verification is finite: enumerate connected subsets containing each terminal, reject subsets violating the peak-side exclusion or forced-terminal inclusion, and calculate their floors, caps, query values, and membership flags from the *same* terminal field. The numerical validation reported above uses independent high-precision evaluation of the original heat equation to cross-check the certified quantities, without replacing the exact inequalities by numerical tests. These specifications are sufficient to write independent implementations; no software artifact or external dataset is required to interpret or rederive the claims. Hardware or laboratory validation is not claimed.

Chapter 125

Actuation-Resolution Language Collapse and Exact Causal Power in a Relational Heat Network

Research branch — EF–FG IV: sampled quantitative records and temporal physical control. *The preceding single-observation conservative heat model establishes joint source-native witness and resource obstructions. This chapter studies a different leaky sampled network with restricted one- or two-hold inputs: the declared numerical terminal-record projection, not the full source-native categorical color output. It establishes sharp all-word language/power obstructions, causal policies, exact modular lifts and a certified crossed-network critical phase. CCR-k arithmetic is not used as a control-speed assumption.*

Placement and proof boundary. The chapter integrates the independent full-article proof and certificate tables of [279]. All claims are conditional on the displayed positive leaky heat equation, local source-native projection, sampled-time guards and control-port budget. Its matrices, finite rational witnesses and interval verification rules are retained below. No external field experiment is asserted as direct validation of its six- or twelve-node plant. The conservative one-shot EF–FG III chapter and the CCR-k information chapters have different hypotheses and resource currencies.

125.1 Introduction

A continuous physical network may admit individual relational observations without admitting arbitrary sequences of observations under shared actuation constraints. The difference between changing the power budget and changing the temporal resolution of an actuator is particularly important: an additional within-period switch changes the reachable input-to-state map even when no new heater is added.

We provide a fully specified test case in which a one-piece-per-period schedule loses every word of length two, whereas a two-piece schedule realizes the entire free monoid of finite words. The second assertion holds at an *exact*, analytically characterized common-power threshold and without knowing future target symbols.

The basic tools are established: positive-system reachability and linear-programming certificates [284, 285]; positive-state reachability with sign-unrestricted inputs [286]; finite-switch positive reachability on time scales [297]; controllers meeting regular-language specifications [290, 287]; dynamically reconfigurable finite-state control under regular-language specifications [289, 288]; finite-memory temporal control [300]; reachability-based STL synthesis [301]; and abstraction robustness margins [299]. A recent AP-observation automata treatment addresses continuous-time logic and control with interval-wise atomic-proposition information [304]; our guarantees, by contrast, concern only sample times. We therefore make no claim to have introduced these general methodologies. Our contribution

is the exact *joint special-model phenomenon* of language collapse, causal power equality, and explicit safe perturbations.

Switching-count and discretization predecessors. The dependence of reachability on switching and sampling deserves explicit priority attribution. Valcher and Santesso [296] study reachability and bounds on switchings in *single-input positive switched systems*; their plant switches among subsystem dynamics, whereas the present comparison fixes one thermal drift matrix $-K$ and refines the number of *held heater inputs within one observation interval*. Bartosiewicz [298] studies when positivity, stability, reachability, and observability survive *time-scale discretization*, including Euler-type discretization; here the sampled maps are the *exact* matrix-exponential images of a continuous-time fixed plant under one or two held inputs. Neither switching-count sensitivity nor sampling sensitivity is therefore claimed as a new general phenomenon. The separate, checkable statement here is a complete language obstruction, a matched exact all-word power optimum and a causal witness for this particular source-defined observation chamber.

The relational predicates follow a protected chamber of the foundational EF–FG ownership and spatial-frame formalism [269, 270] as specialized in the distinct physical joint-realization study [268]. Lemma 125.1 verifies intrinsic witnesses and observer ownership, Proposition 125.2 proves region-independent readout, and Proposition 125.3 classifies every admissible intrinsic region and its possible native color. The three numeric bands are *not* native EF–FG colors (Propositions 125.3 and 125.5), and the projected labels do not require choosing a canonical intrinsic region. The separate CCR computational study [261] concerns a different arithmetic/archival problem. Our controller is executable as an ordinary finite-state lookup; neither a CCR-specific speedup nor a memory optimality claim follows from it.

Comparison with dynamic regular-language controllers. Masciulli and Pola [289, 288] already study finite-state control of continuous-time nonlinear plants when regular-language specifications are only partially known and later revised. Thus the nonanticipative use of the current target letter, finite memory, and regular languages cannot alone be claimed as new. Their central task is synthesis and reconfiguration under changing specifications, with approximation and computational complexity questions; our narrowly specified task is a *sharp input-hold resolution obstruction* on one positive graph, together with the exact physical common-power optimum for realizing every requested word. Our result is a certified special-model phenomenon, not a competing general synthesis method or a faster reconfiguration algorithm. Similarly, the positive-state reachable-space characterization of Guiver, Rebarber and Townley [286] permits sign-indefinite inputs in its central positive-state formulation, whereas every heater input here is constrained to be nonnegative; the reachability questions therefore should not be identified.

Source-native semantics versus a categorical language. The exact source-native objects (seed selection, co-winner basin, observer-record ownership, and intrinsic terminal regions) precede this work in EF–FG I and II [269, 270]. The one-time physical support-function problem also precedes it [268]. The two terminal colors can vary with the selected intrinsic regions even when the field, full-record query-minus-gate numbers, and all numerical target labels remain unchanged. We do not claim a new EF–FG observer definition, new categorical three-color reachability theorem, or realization of an arbitrary full native tuple. The temporal language and sharp all-word heating threshold concern the declared fixed-query numerical projection only.

Measured-building context and boundary of validation. The broader CCR–k/EF–FG research program has also examined logged building-automation measurements from a 35-zone building and from an operational CSIRO office building with controlled AHU fault interventions [280, 281, 282, 283]. Those separate, exploratory analyses addressed

event-sequence reconstruction, the size of finite prediction contexts, and fault-related observability; they did not yield a reliable CCR-specific fault-diagnosis advantage over the strong classical comparators considered there. Crucially, neither archive implements or directly measures the present six- or twelve-node protected sample-guard contract. Thus those field-data investigations support the broader *application motivation*, not the physical experimental validation of the H1/H2 language theorems established below. The results of this chapter are mathematical theorems supported by exact-arithmetic certificates and finite in-model numerical diagnostics, not field-deployment claims.

Contributions. We specify the physical and observation contract; derive a region-choice-invariant native-record projection and an exact classification of admissible terminal regions; prove one-piece language collapse via 17 nonnegative rational Farkas certificates; determine the sharp two-piece all-word power by an analytic primal–dual basis and a staged invariant controller; establish a parameter-uniform robustness theorem with an explicit power–tolerance region; and prove exact language and online/offline power scaling for arbitrarily many coupled copies under a synchronous, locally evaluated source-native observation contract. For mismatched modules we prove all-size, coupling-strength-independent quantitative H1/H2 persistence and a two-sided power estimate, without claiming an unchanged heterogeneous sharp optimum. The modular lift concerns a structured unbounded family, not arbitrary graphs or global EF–FG semantics on the enlarged graph. Subsequent sections treat independent local words under weak cross-type links, certify a locally unique analytic cross-link threshold, and close the associated H1/H2 vector-word phase for every horizon using an 81-action memoryless controller and 81 two-sample Farkas duals.

125.2 Plant, semantics, and languages

Let $V = (r, w, p, b, t_1, t_2)$ be ordered as written. Give the undirected edges rw, wp, pb, bt_1, bt_2 conductances $2, 3, 3, \kappa_1, \kappa_2$, respectively, with no other edges. Let $L(\kappa)$ be the weighted graph Laplacian and put

$$K(\kappa) = I_6 + L(\kappa), \qquad \dot{x}(t) = -K(\kappa)x(t) + u(t), \qquad x(0) = 0. \tag{125.1}$$

The heater u_b is disabled; the five available ports are $P = (r, w, p, t_1, t_2)$. Every held input has nonnegative components and satisfies $\sum_{j\in P} u_j \le Q$. A sample interval of duration τ is subdivided into H equally long constant-input phases. We compare $H = 1$ and $H = 2$. The budget Q bounds each held phase's *power rate*, not its time-integrated energy.

Define the common guard family, evaluated only at sample times $k\tau$, by

$$x_r - x_p \ge 1, \quad x_p - x_w \ge \frac{5751}{1000}, \quad x_b - x_w \ge 1, \quad x_p - x_{t_i} \ge 1 \quad (i = 1, 2). \tag{125.2}$$

For $i \in \{1, 2\}$ and $s_i \in \{0, 1, 2\}$ impose $x_{t_i} - x_b \in I_{s_i}$, where

$$I_0 = [1/2, 1], \qquad I_1 = [2, 3], \qquad I_2 = [4, 9/2]. \tag{125.3}$$

The target alphabet is $\Sigma = \{0, 1, 2\}^2$, ordered lexicographically. For $s = (s_1, s_2)$, let $H_s x \ge b_s$ comprise the five inequalities in (125.2) and two lower and two upper terminal-band inequalities. We use nine rows per H_s. All constraints are affine *state-difference* guards. The ownership condition $x_p - x_w > 23/4$ follows from the stronger $5751/1000$ inequality. The letter is a *declared quantitative binning* of the region-choice-independent full-record coordinate $y_i - s_i^{\mathrm{gate}} = x_{t_i} - x_b$, not one of the three native polarity/terminal-classification colors. We assert neither these guards at unsampled times nor a characterization of all possible EF–FG source-native tuples (s, U, y, χ).

Lemma 125.1 (Source-native witness and projected observable on the guarded tree)**.** *Fix a sampled field x satisfying* (125.2)–(125.3)*, the root observer p, selected-seed threshold*

$\Delta_{\text{seed}} = 5$, *observer interpolation parameter* $\theta = 2/5$, *and bright-ownership threshold* $\Delta_{\text{own}} = 23/4$. *Compute primitive extrema, superlevel elders, maximin attachments, and the two observer polarities using the source-native rules of [268]. Then* r *is the essential bright maximum,* p *is the sole selected finite bright seed, and the bright and dark observer saliencies at* p *are* $x_p - x_w$ *and* 0*, respectively. Hence* p *has strict bright ownership. For each terminal* t_i*, the intrinsic residual-region family admits the connected witness*

$$R_i^{\text{path}} = \{b, t_i\}, \qquad b \in R_i^{\text{path}}, \quad s_i^{\text{gate}} = x_b, \quad L(R_i^{\text{path}}) = x_b, \quad U(R_i^{\text{path}}) = x_{t_i}.$$

At the native query vertex t_i*, the complete selected-region record is*

$$(s_i^{\text{gate}}, U, y, \chi_{t_i}) = (x_b, x_{t_i}, x_{t_i}, 1), \qquad U - L = x_{t_i} - x_b.$$

Consequently the two numerical band labels are well-defined readings of one source-native joint witness. Proposition 125.2 strengthens this to all *admissible region choices at the same terminal. The labels are not the native dark/neutral/bright classifications.*

Proof. The guards imply $x_r > x_p > x_{t_i} > x_b > x_w$. Hence the only local maxima are r, p, t_1, t_2 and the only local minima are b, w. On the tree the maxima t_i merge into the older component of p at level x_b, while p merges into the older component of r at level x_w. Thus p has finite persistence $x_p - x_w > 5$ and the terminal maxima have persistence $x_{t_i} - x_b \le 9/2 < 5$; r is essential. For the bright observer at p, the older r attaches through w at level x_w, whereas the seed p attaches at its own birth x_p. Later terminal maxima attach through b below x_p, so no extra bright observer record occurs. Under the negative field, the minima w and b both attach to the observer through a path whose largest original-field value is x_p; the younger dark atom therefore fails the strict record-attachment test. The saliencies are as stated. For the path $p - b - t_i$, the maximin terminal gate is $s_i^{\text{gate}} = \min\{x_p, x_b, x_{t_i}\} = x_b$. The vertex p is forced onto the peak side, t_i onto the terminal side, and b lies in the equal-attachment tie zone, so $\{b, t_i\}$ is a connected intrinsic admissible terminal region. The construction uses the *same* field and elder choice for $i = 1, 2$; the source-native definition imposes no disjointness on the two regions, which intersect at b. Since $x_{t_i} > x_b$, its floor, cap, gate and native query tuple have the displayed values. Finally $x_p - x_w \ge 5751/1000 > 23/4$ yields strict bright ownership. □

Proposition 125.2 (Region-choice invariance of the native numerical projection)**.** *Under the conditions of Lemma 125.1, let* R_i *be* any *intrinsic residual terminal region for* t_i *that contains the common separator* b*. Its full source-native record evaluated at the terminal is*

$$\text{ResRel}_{p,t_i,t_i}(x, R_i) = \big(x_b, U(R_i), x_{t_i}, 1\big).$$

Thus the numerical quantity

$$\pi_i\big(\text{ResRel}_{p,t_i,t_i}(x, R_i)\big) := y_i - s_i^{\text{gate}} = x_{t_i} - x_b$$

is independent of R_i*. Consequently the nine-symbol labeling from* (125.3) *does not require the prior choice of a particular admissible region. In contrast, the cap width* $U(R_i) - L(R_i)$ *and the native terminal color may depend on the selected region.*

Proof. The gate $s_i^{\text{gate}} = \kappa_{x|\mathcal{B}_p}(p, t_i) = x_b$ depends on the field and terminal but not on a residual region. At the fixed query point t_i, the value $y_i = x_{t_i}$ is also independent of that region, and every admissible region contains the forced terminal, so $\chi_{t_i} = 1$. Subtracting these two native record coordinates gives $\pi_i = x_{t_i} - x_b$ for every allowed R_i. For noninvariance of the cap and color, take the exact admissible field in vertex order

$$x = (38, 30, 37, 31, 63/2, 35)$$

with labels 02. For terminal t_1, both $R_1^{\text{path}} = \{b, t_1\}$ and $R_1^{\max} = \{b, t_1, t_2\}$ are admissible intrinsic regions. The first has $(L, U, y) = (31, 63/2, 63/2)$, cap width $1/2$, and a bright terminal query at zero native margin. The second has $(L, U, y) = (31, 35, 63/2)$, cap width 4, and a dark terminal query since $63/2 < (31 + 35)/2 = 33$. Both records nevertheless yield $y - s = 1/2$. This also demonstrates why the nine numerical labels must not be equated with native color labels. □

Proposition 125.3 (Complete intrinsic-region and native-color classification)**.** *For the protected tree of Lemma 125.1, the rooted co-winner basin of the selected seed p is $\mathcal{B}_p = \{w, p, b, t_1, t_2\}$. For $i \in \{1, 2\}$ and $j \neq i$, the* complete *intrinsic terminal-region family is*

$$\mathcal{R}_{p,t_i}(x) = \big\{\{t_i\},\ \{b, t_i\},\ \{b, t_i, t_j\}\big\}.$$

Exactly two of these regions satisfy the common-separator query $b \in R_i$. Put $g_i = x_{t_i} - x_b > 0$ and take the original terminal classification margin $\Delta_{\mathrm{term}} = 0$. For the smaller region $R_i^- = \{b, t_i\}$ the source-native terminal color at the fixed query vertex t_i is always bright*. For the larger region $R_i^+ = \{b, t_i, t_j\}$ it is* bright*,* neutral*, or* dark *according as $g_j < 2g_i$, $g_j = 2g_i$, or $g_j > 2g_i$, respectively. The region without b, namely $\{t_i\}$, has native color* neutral *but does not satisfy the common-separator query. In all cases the full-record coordinate $y_i - s_i^{\mathrm{gate}} = g_i$ is unchanged.*

Proof. The only selected finite bright seed is p and the essential older seed is r. On the tree, the seed attachments satisfy $\kappa_x(p, v) > \kappa_x(r, v)$ for $v = p, b, t_1, t_2$, equality for $v = w$, and the reverse inequality for $v = r$. Thus the connected co-winner basin containing p is precisely $\mathcal{B}_p = V \setminus \{r\}$. In this basin the gate $s_i = \kappa_{x|\mathcal{B}_p}(p, t_i) = x_b$. The two maximin functions $\alpha(v) = \kappa_{x|\mathcal{B}_p}(p, v)$ and $\gamma_i(v) = \kappa_{x|\mathcal{B}_p}(t_i, v)$ have the following classes: $\alpha(p) > s_i$ makes p forced peak-side, $\gamma_i(t_i) > s_i$ makes t_i forced terminal-side, and $\alpha = \gamma_i = s_i$ holds exactly at the two eligible tie vertices b, t_j. The other basin vertex w has both values strictly below s_i, hence is in neither the forced terminal component nor the tie zone. Every intrinsic admissible terminal region must therefore contain t_i, exclude p, w, and may include b, t_j only subject to connectedness. The three listed subsets are exactly the connected possibilities.

At terminal query $x = t_i$ each admissible region has $y = x_{t_i}$, $\chi = 1$, and the same gate $s_i = x_b$. For a separator-containing region its floor is $L = x_b$, whereas its cap is either $U = x_b + g_i$ or $U = x_b + \max(g_i, g_j)$. At zero terminal margin the original EF–FG midpoint rule compares y with $(L + U)/2$: values above, at, and below the midpoint are bright, neutral and dark, respectively. The smaller region has $y = U > L$ and is always bright. The larger region is bright exactly when $2g_i > \max(g_i, g_j)$, equivalent to $g_j < 2g_i$; equality and reverse inequality give the other two colors. For the singleton $L = U = y$, hence neutral. Finally $y - s_i = g_i$ does not depend on any region choice. □

Corollary 125.4 (Nine-band possibilities for the maximal region)**.** *For $s_i \in \{0, 1, 2\}$ the band interval of g_i is $I_0 = [1/2, 1]$, $I_1 = [2, 3]$, $I_2 = [4, 9/2]$. Writing B, N and D for the three native colors possible under R_i^+, the* complete *possibilities are*

	$s_j = 0$	$s_j = 1$	$s_j = 2$
$s_i = 0$	B, N	N, D	D
$s_i = 1$	B	B	B, N, D
$s_i = 2$	B	B	B

At the smaller region R_i^-, the native color remains B in every cell. Thus the nine numerical labels are not equivalent to the intrinsic categorical color records.

Proof. The affine difference $g_j - 2g_i$ ranges over the exact interval $[\inf I_j - 2\sup I_i,\ \sup I_j - 2\inf I_i]$. Applying Proposition 125.3 to its negative, zero and positive values gives precisely the table; the interval endpoints are attainable. □

Proposition 125.5 (The three numerical levels do not encode three native colors)**.** *Assume zero terminal-classification margin in the native EF–FG threshold rules. For each of the nine projected letters, the source-native color of the query at $x = t_i$* when the minimal region R_i^{path} is chosen *is* bright *for both terminals, whereas the color of the query at $x = b$ is* dark*. The complete native query record evaluated at $x = t_i$, however, retains the numerical band information through $y_i - s_i^{\mathrm{gate}} = x_{t_i} - x_b$.*

Proof. The native zero-margin thresholds are $T^+ = T^- = (L+U)/2$. Since $U = x_{t_i} > x_b = L$, at the terminal $y_i = U > T^+$, so its native color is bright regardless of the value of $U - L$. At b the query value is $y = x_b = L < T^-$, so it is dark. In the full native record at t_i both y_i and s_i^{gate} are retained, and their difference determines the declared numerical band. □

Remark 125.6 (Distinction between the native output and the tested language). The existential predicates $E_i(F; b)$ assert existence of intrinsic regions containing b. The nine-letter observable adds only the three closed, disjoint numerical intervals for the region-independent native-record projection $y_i - s_i^{\mathrm{gate}}$ at query vertex t_i. Unlike $U - L$, this projection needs no prescribed region-selection rule. Its quantization is an *extra output contract*, not a native three-class color. The existence or exact control power of this projected language is not a theorem about every full source-native output tuple.

For fixed H, Q, let $\mathcal{L}_n(H, Q)$ consist of words $s_1 \cdots s_n$ for which some admissible input schedule makes $H_{s_k} x(k\tau) \geq b_{s_k}$ for each $1 \leq k \leq n$. Write $\mathcal{L}_n(H, \infty) = \bigcup_{Q<\infty} \mathcal{L}_n(H, Q)$. For a required *universal* all-word capability, define $Q_{\mathrm{on},\infty}$ as the infimum of budgets admitting one nonanticipative rule for every word at every finite horizon. A rule may use the current target symbol at the start of its interval, but not future symbols. Define $Q_{\mathrm{off},\infty}$ similarly when the entire requested word is supplied in advance. These are worst-case all-word budgets, not a total sum of realized heater powers over time. Here *online* means that the controller receives the requested letter s_k before selecting the held controls on interval $[(k-1)\tau, k\tau)$; no observation of $s_{k+1}, s_{k+2}, \ldots$ is available then. It does not mean that an unknown future physical disturbance is being rejected by output feedback. Each word has its own trajectory driven by the same causal lookup rule; distinct words are not claimed to occur simultaneously on one trajectory.

125.3 Sampled input operators and the one-piece obstruction

At the nominal parameters $\kappa_1 = \kappa_2 = 211/100$, $\tau = 1$, set

$$\begin{aligned} E &= e^{-K/2}, \qquad C = \int_0^{1/2} e^{-Ks}\, ds = K^{-1}(I - E), \\ A &= E^2 = e^{-K}, \quad D = \begin{bmatrix} EC|_P & C|_P \end{bmatrix}. \end{aligned} \tag{125.4}$$

Here $D \in \mathbb{R}^{6\times 10}$ maps two held, five-port input phases to one sampled state increment: $x_k = Ax_{k-1} + Du_k$. In the one-piece case, $x_k = Ax_{k-1} + C_1 u_k$ with $C_1 = K^{-1}(I - A)|_P \in \mathbb{R}^{6\times 5}$.

Theorem 125.7 (Exact one-piece language collapse)**.** *At nominal parameters, for every finite $Q \geq 0$, $\mathcal{L}_1(1, Q) \subseteq \{22\}$ and $\mathcal{L}_n(1, Q) = \varnothing$ for every $n \geq 2$. Moreover, $\mathcal{L}_1(1, \infty) = \{22\}$; the target 22 is realized by a positive one-piece control of total power* 233.258410.

Proof. For each of the eight non-22 one-letter requests and each of the nine length-two requests $22t$, a nonnegative rational Farkas vector y certifies infeasibility. For a length-$m \leq 2$ request, stack the guard rows and write the sampled outputs as linear functions of the m nonnegative five-port controls U. If $\mathcal{G}U \geq b$ denotes the resulting guard system, the corresponding vector in the certificate tables of Section 125.18 satisfies $y \geq 0$, $y^T b > 0$ and $y^T \mathcal{G} < 0$ componentwise; hence $y^T \mathcal{G} U \leq 0 < y^T b$ for all $U \geq 0$. Each strict sign is checked using outward rational enclosures for e^{-K}. The 17 vectors and their input-column sign summaries are printed in Section 125.18, as exact decimal rationals. No power constraint is used in the contradiction.

The one-letter candidate 22 is achieved by a single control in vertex order,

$$u_{22}^{(1)} = (57.280683,\ 0,\ 76.951713,\ 0,\ 49.513007,\ 49.513007)^T,$$

which uses total available-port power 233.258410. Its nine guard margins are strictly positive under certified outward interval evaluation. Any longer admissible word would have a feasible length-two prefix; only 22 can be the first symbol, and each $22t$ is excluded. □

Remark 125.8. The language assertion for $n = 1$ is a statement about the *union of budgets*. It does not claim that 22 is feasible for every small budget. The impossibility for $n \geq 2$, however, is independent of how large the finite phase-power budget is.

125.4 The sharp two-piece budget

For the symmetric terminal symbol 22, exchange t_1 and t_2 and average any feasible two-piece input with its exchange. This preserves the constraints and cannot increase the maximum phase power. The reduced eight inputs are ordered

$$(u_r^E, u_w^E, u_p^E, u_t^E, u_r^L, u_w^L, u_p^L, u_t^L),$$

where each terminal receives u_t and therefore costs $2u_t$ in the corresponding phase. The active nonzero controls occupy reduced indices $\mathcal{I} = \{1, 4, 5, 7, 8\}$ (one-based). The active weak guards have normals

$$H_* = \begin{bmatrix} 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 \end{bmatrix}, \quad b_* = (1, 1, 1, 4)^T. \tag{125.5}$$

Define $D_{\mathrm{sym}} \in \mathbb{R}^{6\times 8}$ by summing the two terminal columns of D within each phase (with other columns unaltered) and $G_* = H_* D_{\mathrm{sym}}$. Let $w = (1, 1, 1, 2, 1, 1, 1, 2)$ and w_E, w_L be its phase-separated restrictions to $\mathcal{I}$. The following 6×6 linear system fixes the active controls $v \in \mathbb{R}^5$ and common phase power q:

$$T \begin{pmatrix} v \\ q \end{pmatrix} = \begin{pmatrix} b_* \\ 0 \\ 0 \end{pmatrix}, \qquad T = \begin{bmatrix} (G_*)_{:,\mathcal{I}} & 0_{4\times 1} \\ w_E^T & -1 \\ w_L^T & -1 \end{bmatrix}. \tag{125.6}$$

In (125.6), $w_E, w_L \in \mathbb{R}^5$ have zero entries in inactive phases. Let $T_{6\leftarrow b}$ replace its last column by $b = (b_*^T, 0, 0)^T$.

Theorem 125.9 (One-sample analytic optimum)**.** *The smallest two-piece budget for the single initial symbol* 22 *equals*

$$Q_0 = \frac{\det(T_{6\leftarrow b})}{\det T}, \tag{125.7}$$

with the strictly certified bounds

$$\texttt{201.131345777023810433521487478546245111831804926982562}7 \\ < Q_0 < \\ \texttt{201.131345777023810433564841709247362174559932503567920}9.$$

Proof. The unique solution of (125.6) has five strictly positive active controls and both phase budgets tight. Its four active guard equalities imply $r - p = 1$, $b - w = 1$, $p - t_i = 1$, and $t_i - b = 4$, hence $p - w = 6 > 5751/1000$. It is physically feasible. For the reduced dual, find $y \in \mathbb{R}^4_{\geq 0}$ and $(\lambda_E, \lambda_L) \in \mathbb{R}^2_{\geq 0}$ satisfying $\lambda_E + \lambda_L = 1$ and $(G_*)_{:,j}^T y \leq \lambda_{\mathrm{phase}(j)} w_j$, with equality for $j \in \mathcal{I}$. The associated six-by-six dual active matrix is nonsingular, and outward rational interval arithmetic proves all six dual variables positive and all three inactive reduced costs strictly positive. Weak duality makes the primal optimum global. Terminal-exchange averaging lifts the reduced result to all nonsymmetric feasible actions. Cramer's rule gives (125.7); the displayed enclosure is certified by a rational Taylor/uniformization bound and perturbation bounds for both six-by-six inverse systems. The active primal/dual construction and certified strict bounds are detailed in Section 125.18. □

125.5 An exact causal all-word policy

Let $a = e^{-1}$, $\|A\|_\infty = a < 1/2$, and $A\mathbf{1} = a\mathbf{1}$. The controller specified by Section 125.18 contains initial actions u^0_s ($s \in \Sigma$), bootstrap actions $u^B_{s,t}$ ($s, t \in \Sigma$), and persistent actions $u^T_{s,t}$ ($s, t \in \Sigma$). There are $9 + 81 + 81 = 171$ actions. The initial 22 action is the analytic optimizer of Theorem 125.9; the other 170 actions consist of nonnegative rationals on the 10^{-10} lattice. Their complete ten-port values are printed in Section 125.18.

Let $M \in \mathbb{Q}$ be a common mean parameter, let $z^0_s \in \mathbb{Q}^6$ be nine initial reference states, and let $d_s \in \mathbb{Q}^6$ be nine persistent shape modes. All are explicitly given in Section 125.18. The intended states are $z_{k,s} = (1 - a^k)M\mathbf{1} + d_s$ for $k \geq 2$. The policies satisfy the approximately affine identities

$$Az^0_s + Du^B_{s,t} = (1 - a^2)M\mathbf{1} + d_t + r^B_{s,t}, \tag{125.8}$$

$$Ad_s + Du^T_{s,t} = (1 - a)M\mathbf{1} + d_t + r^T_{s,t}, \tag{125.9}$$

with explicit uniformly small residuals. The exact plant runs with these finite lists; the ideal identities are only used for error certification.

Theorem 125.10 (Exact all-horizon causal power and zero preview gap)**.** *At the nominal parameters, for every $n \geq 1$, every word in Σ^n is realized by the above single nonanticipative two-piece policy with phase power at most Q_0, and*

$$\boxed{Q_{\mathrm{on},\infty} = Q_{\mathrm{off},\infty} = Q_0.}$$

Proof. Certified initial reference errors are at most δ_0; the bootstrap and persistent residuals in (125.8)–(125.9) are at most δ_B and δ_T in $\|\cdot\|_\infty$. Taking

$$R = 2(\delta_0 + \delta_B + \delta_T) < 4.425 \cdot 10^{-10},$$

we have $R \geq \delta_0$, $R \geq a\delta_0 + \delta_B$, and $R \geq aR + \delta_T$. Induction yields $\|x_k - z_{k,s_k}\|_\infty \leq R$ for every $k \geq 2$, with the correspondingly defined initial ideal state. Every guard is a difference of two coordinates, so a reference margin exceeding $2R$ remains positive in the physical state. The interval proof yields a uniform guard margin $> 7.84967 \cdot 10^{-5}$ for all first symbols other than 22 and all $k \geq 2$. The special initial 22 satisfies its four weak guards *exactly*, by the analytic active-basis equations; the stronger ownership guard remains strict. Every rational action has two phase sums strictly less than the *certified lower bound* on Q_0, while the analytic 22 action meets Q_0 exactly. Thus the policy realizes every word without preview at power Q_0.

Conversely any all-word policy, online or offline, must realize the one-letter word 22 and therefore needs at least Q_0 by Theorem 125.9. This proves the claimed equality. □

125.6 Strict and parameter-robust language separation

The exact optimal action for the first symbol 22 lies on a weak-guard boundary and is not itself a strictly robust witness. Let u^{210} be a second, strictly feasible initial 22 action with each held-phase power below 210. Its ten exact decimal-rational entries are printed in Section 125.18. Let u^* denote the analytic optimum and $\Delta = D(u^{210} - u^*)$. Rational intervals show all six entries of Δ positive, $\mathrm{osc}(\Delta) < 401/1000$, and the u^{210} initial-22 guard margin exceeds $m_0 = 999/10000$. The exact causal policy of Theorem 125.10 has all other sampled-guard margins exceeding $G_0 = 7849/10^8$.

Lemma 125.11 (Oscillation contraction)**.** *If $A \geq 0$ and $A\mathbf{1} = a\mathbf{1}$ for $0 < a < 1$, then $\mathrm{osc}(A^j d) \leq a^j \,\mathrm{osc}(d)$ for all vectors d and $j \geq 1$. In particular $|(e_i - e_l)^T A^j d| \leq a^j \,\mathrm{osc}(d)$.*

Proof. Multiply $\min(d)\mathbf{1} \leq d \leq \max(d)\mathbf{1}$ by the nonnegative matrix A^j and use $A^j\mathbf{1} = a^j\mathbf{1}$. □

Replace only the special initial action by $u^{\alpha} = (1-\alpha)u^{*} + \alpha u^{210}$. The first 22 guard margin is greater than αm_0. Every later state differs from the original exact causal policy by $\alpha A^{k-1}\Delta$. By Lemma 125.11, $a < 3/8$ and $\operatorname{osc}(\Delta) < 401/1000$, each subsequent guard loses less than αH_0, where $H_0 = 151/1000$. The resulting nominal uniform margin therefore exceeds

$$\mu(\alpha) = \min\{\alpha m_0,\ G_0 - \alpha H_0\}, \quad m_0 = \frac{999}{10000}, \quad G_0 = \frac{7849}{10^{8}}, \quad H_0 = \frac{151}{1000}. \tag{125.10}$$

The mixed first action costs less than $Q_0 + 10\alpha$ in each phase.

Theorem 125.12 (Explicit parameter-robust phase and power tradeoff)**.** *For $0 < \varepsilon < 1/100$, choose α, η so that*

$$0 < \alpha < \min\{\varepsilon/10, G_0/H_0\}, \qquad 0 \le \eta < \min\left\{10^{-8}, \frac{\alpha m_0}{24240}, \frac{G_0 - \alpha H_0}{24240}\right\}. \tag{125.11}$$

For every plant with $|\kappa_i - 211/100| \le \eta$ and $|\tau - 1| \le \eta$, the same finite-action, nonanticipative $H = 2$ policy realizes every word in Σ^n, for every n, within phase power $Q_0 + \varepsilon$ and with strictly satisfied sampled guards. In the same parameter box, $\mathcal{L}_n(1, Q) = \varnothing$ for all finite Q when $n \ge 2$, while $\mathcal{L}_1(1, \infty) = \{22\}$.

Proof. All phases have power less than $Q_0 + \varepsilon < 202$. Positive leakage and nonnegative controls give $\sup_t \|x(t)\|_\infty < 202$. Uniform Duhamel estimates for $\eta < 10^{-8}$ yield $\|A_\theta - A_0\|_\infty \le 20\eta$ and $\|D_\theta - D_0\|_\infty \le 10\eta$. Since $\|A_\theta\|_\infty < 1/2$, the sample-to-sample state error for identical actions is bounded by 12120η at every horizon, and any difference guard changes by less than 24240η. The inequalities in (125.11), together with (125.10), guarantee positive residual guard margin.

For $H = 1$, all 17 strict Farkas certificates are stable under the same parameter variation: their nominal maximum reduced cost is less than $-1.6286883 \cdot 10^{-6}$ and its drift is bounded by 49η, which cannot reach zero for $\eta < 10^{-8}$. A strictly feasible one-letter 22 witness has margin > 0.003425 and loses less than 1872η in the same box. Thus the one-piece language assertion persists. The common power cap is a rate cap, unchanged by variation in the sampling interval. □

Corollary 125.13 (Numerical rational robustness cube)**.** *Set $\alpha = 1/3200$, $\varepsilon = 1/250$ and $\eta = 1/800000000$. Then throughout*

$$|\kappa_1 - 2.11|, \quad |\kappa_2 - 2.11|, \quad |\tau - 1| \le 1.25 \cdot 10^{-9},$$

$H = 2$ realizes every word online at power $Q_0 + 0.004$, with sampled guard margin $> 9.1875 \cdot 10^{-7}$; the one-piece collapse persists. This is a sufficient *cube, not a computation of the maximum physical tolerance or of the sharp perturbed power threshold.*

Audit of the perturbation constants

Write $\theta = (\kappa_1, \kappa_2, \tau)$ and $\eta = \|\theta - \theta_0\|_\infty$, interpreting the norm coordinatewise. The Laplacian has row-sum zero; hence $K(\kappa)\mathbf{1} = \mathbf{1}$, $-K(\kappa)$ is Metzler, and every $e^{-K(\kappa)s}$ is nonnegative with infinity norm exactly e^{-s}. Moreover $\|K(\kappa) - K(\kappa_0)\|_\infty \le 4\eta$ for changes of the two pendant conductances. Duhamel's formula yields

$$e^{-K's} - e^{-Ks} = \int_0^s e^{-K'(s-v)}(K - K')e^{-Kv}\,dv, \quad \|e^{-K's} - e^{-Ks}\|_\infty \le 4s\eta.$$

For $|\tau - 1| \le \eta < 10^{-8}$, $\|K'\|_\infty < 16$ and $h = \tau/2$, this estimate and the time-variation estimate imply the conservative bounds

$$\|e^{-K'h} - e^{-K/2}\|_\infty < 10\eta, \quad \|\textstyle\int_0^h e^{-K's}ds - \int_0^{1/2} e^{-Ks}ds\|_\infty < 2\eta.$$

Consequently the sampled two-half maps obey $\|D' - D\|_\infty < 10\eta$, $\|A' - A\|_\infty < 20\eta$; the corresponding one-piece maps $C_1' = \int_0^\tau e^{-K's} ds$ and $A'C_1'$ obey $\|C_1' - C_1\|_\infty < 4\eta$ and $\|A'C_1' - AC_1\|_\infty < 24\eta$.

Because $\|A'\|_\infty = e^{-\tau} < 1/2$ and the shared heater sum is below 202, the state error $e_k = x_k' - x_k$ obeys $\|e_k\|_\infty \le \frac{1}{2}\|e_{k-1}\|_\infty + (20+10)202\eta$, giving the horizon-uniform estimate $\|e_k\|_\infty < 12120\eta$. For one-piece infeasibility, each guard normal has ℓ_1 norm 2; each Farkas vector printed in Section 125.18 is nonnegative and has total multiplier mass no more than $1 + 10^{-12}$. Thus the perturbation of each reduced input coefficient is bounded by

$$2 \max\{4\eta, 24\eta\}(1 + 10^{-12}) < 49\eta.$$

The certified strict negative coefficients and positive right sides retain their signs over the stated robustness region. This argument checks *all* perturbed parameter points, not only the eight cube vertices.

125.7 An unbounded family of physically coupled thermal modules

The six-node example is not confined to one isolated plant. This section gives an exact, all-budget transfer to connected networks of unbounded size, while stating precisely which observation language is preserved. The projection relies on classical diffusion averaging and synchronous invariant subspaces, which are extensively studied in network dynamics [293, 294, 295]. The new claim here is the application of this exact intertwining to the particular certified $H = 1/H = 2$ language phase and its sharp all-word power.

Fix an integer $m \ge 1$ and m identical six-node modules indexed by $a = 1, \dots, m$. Each local module has the weighted edges and leakage of (125.1), including its individually disabled b-heater. For each node type $v \in V$, choose *arbitrary symmetric nonnegative intermodule conductances* $\gamma_{ab}^v = \gamma_{ba}^v \ge 0$ for $a \ne b$, independently for the six node types. These edges connect (a, v) to (b, v); no cross-type intermodule edge is permitted. Let L_v be the resulting $m \times m$ weighted Laplacian. With e_v the six standard coordinate vectors, the full $6m$-node thermal plant is

$$K_m = I_m \otimes K_0 + \sum_{v \in V} L_v \otimes e_v e_v^{\mathsf{T}}, \quad \dot{X} = -K_m X + U, \quad X(0) = 0, \tag{125.12}$$

where $K_0 = I_6 + L(211/100, 211/100)$ at nominal parameters. At every held phase, $U^a \ge 0$, $U_b^a = 0$, and the *global*, not per-module, power satisfies $\sum_{a=1}^m \sum_{v \ne b} U_v^a \le Q$. When at least one of the intermodule graphs is connected, the resulting entire thermal graph is connected; the couplings may otherwise be arbitrary, including arbitrarily large.

Synchronous local observation contract. A letter $s_k \in \Sigma$ is required in *every* module a at sample time $k\tau$, with exactly the same affine guards $H_{s_k} X_k^a \ge b_{s_k}$. Its region-independent native-record projection is calculated on the *induced local six-node tree* G_a, at the query vertices (a, t_i). This local-graph choice is part of the mathematical task. It does not assert that the full $6m$-node graph's single global EF–FG record has the same terminals, ownership, or elder order. Different target words for different modules are not part of this theorem. Write $\mathcal{L}_n^{(m)}(H, Q)$ for the common-word language in this network, using the same H-piece hold class as before.

Define the averaging and synchronous-embedding maps

$$\mathsf{R}_m = \frac{1}{m}(\mathbf{1}_m^{\mathsf{T}} \otimes I_6), \qquad \mathsf{S}_m = \mathbf{1}_m \otimes I_6. \tag{125.13}$$

Lemma 125.14 (Exact thermal mean and synchronized-copy intertwining)**.** *For all m and all the allowed intermodule conductances,*

$$\mathsf{R}_m \mathsf{S}_m = I_6, \qquad \mathsf{R}_m K_m = K_0 \mathsf{R}_m, \qquad K_m \mathsf{S}_m = \mathsf{S}_m K_0. \tag{125.14}$$

Thus the mean $\bar{x} = \mathsf{R}_m X$ obeys $\dot{\bar{x}} = -K_0 \bar{x} + \bar{u}$, $\bar{u} = \mathsf{R}_m U$, while for common copies $X = \mathsf{S}_m x$, $U = \mathsf{S}_m u$ the intermodule heat currents vanish identically.

Proof. For each symmetric graph Laplacian L_v, both $L_v\mathbf{1}_m = 0$ and $\mathbf{1}_m^{\mathsf{T}} L_v = 0$. Left- and right-multiplication of (125.12) by the Kronecker maps in (125.13) annihilates every coupling term. The $I_m \otimes K_0$ term gives the displayed identities. Multiplying the full differential equation and using uniqueness proves the dynamic assertions. The equal-modules subspace is invariant even when the six intermodule graphs are mutually different. □

Theorem 125.15 (Exact language and sharp power scaling on coupled $6m$-node graphs)**.** *For every $m \geq 1$, every allowed symmetric intermodule coupling, every $Q \geq 0$, every $n \geq 1$, and $H \in \{1, 2\}$,*

$$\boxed{\mathcal{L}_n^{(m)}(H, Q) = \mathcal{L}_n^{(1)}(H, Q/m).} \tag{125.15}$$

The equality also transfers existence of a single nonanticipative controller for all words; both the causal and full-preview all-word power thresholds scale by m. In particular, at the nominal plant,

$$\mathcal{L}_1^{(m)}(1, \infty) = \{22\}, \qquad \mathcal{L}_n^{(m)}(1, Q) = \varnothing \quad (n \geq 2,\ Q < \infty), \tag{125.16}$$

$$\mathcal{L}_n^{(m)}(2, mQ_0) = \Sigma^n \quad (n \geq 1), \qquad \boxed{Q_{\mathrm{on},\infty}^{(m)} = Q_{\mathrm{off},\infty}^{(m)} = mQ_0}.$$

The broadcast implementation uses the same 171 control templates and no future-letter preview, with the prescribed ten-port action copied simultaneously to all m modules. The 171 templates are not a controller-size lower bound.

Proof. Suppose an admissible m-module input realizes a common word $s_1 \cdots s_n$; the module controls may be completely unequal and may depend on their individual measured states. Lemma 125.14 gives a physically valid base-plant control $\bar{u}(t) = m^{-1} \sum_a U^a(t)$ with the same H-piece hold resolution, nonnegative available ports, and phase sum at most Q/m. Since each guard $H_{s_k} X_k^a \geq b_{s_k}$ is affine with an identical right-hand side, averaging gives $H_{s_k} \bar{x}_k \geq b_{s_k}$. Hence the full-system word is also a base-plant word at Q/m; this proves the first inclusion in (125.15) for *arbitrary nonuniform* module inputs.

Conversely, given any base-plant realization with phase budget Q/m, copy that control vector to all modules. By the second intertwining identity and the common zero initial field, the unique full-network solution remains $X(t) = \mathsf{S}_m x(t)$, regardless of coupling strength. Every module then satisfies all local sampled guards. Its total phase input is m times the base input, hence at most Q, proving the other inclusion.

For nonanticipative controllers, averaging a deterministic common-word policy's chosen inputs for every observed target prefix defines a base policy on the same prefixes, since there are no exogenous unknown disturbances or hidden adversarial inputs in the model. Replication carries any base policy back to the module network. The same argument works with full preview. Apply Theorems 125.7 and 125.10 to obtain (125.16); the first-letter 22 lower bound is inherited by averaging even if the full-network designer allocates heater power nonuniformly across modules. □

Corollary 125.16 (Network-size-independent robustness for homogeneous cell perturbations)**.** *Let all modules share the same perturbed κ_1, κ_2, τ from Corollary 125.13, while the symmetric intermodule conductances remain arbitrary and nonnegative. Then for every m, the $H = 1$ language remains empty after one sample at every finite global power, whereas the strict broadcast $H = 2$ controller realizes all common words at global power $m(Q_0 + 0.004)$. The certified sampled local-guard margin $> 9.1875 \cdot 10^{-7}$ is independent of m and of intermodule coupling strength. This is a sufficient global power budget; the sharp* perturbed *base threshold is not supplied.*

Proof. The two intertwining identities hold verbatim with the common perturbed six-node matrix $K(\kappa)$ in place of K_0. Replicate the strict base policy of Theorem 125.10, so that each local state is identically the base trajectory. Averaging the altered plant establishes the robust one-piece no-go. The per-cell control and margin are exactly those of the base theorem. □

Corollary 125.17 (Time-varying undirected intermodule diffusion)**.** *The exact language identity* (125.15)*, the nominal mQ_0 threshold, and the homogeneous-cell robustness conclusion continue to hold if each γ_{ab}^{v} varies arbitrarily with time, provided the intermodule conductances remain nonnegative, symmetric for each time, and locally bounded and piecewise continuous. The broadcast controller need not know the coupling schedule.*

Proof. At each time t, every symmetric intermodule Laplacian $L_v(t)$ has both left and right constant-vector kernels. Hence (125.14) holds pointwise with $K_m(t)$, and uniqueness for the corresponding nonautonomous linear system gives the same averaging and synchronous-embedding trajectories. All phase budget and affine-guard arguments are unchanged. □

Remark 125.18 (Scope and non-priority of the general synchrony principle)*.* The operator identities (125.14) are classical properties of balanced diffusive couplings and equitable/quotient dynamics [293, 294, 295]. We do not claim a new general synchronization result or an arbitrary-graph EF–FG theorem. The exact mQ_0 conclusion is for the structured, identical-module, synchronous-*local*-observation family. It need not hold for different target words per module, cross-type intermodule edges, heterogeneous within-module plants, or EF–FG computed intrinsically on the entire joined graph. Those cases require separate analysis.

125.8 Nonidentical modules: a uniform robust language phase

The synchrony equalities of Section 125.7 cease to hold when the modules have different internal dynamics. Nevertheless, the strict one-piece Farkas signs and a positive-semigroup perturbation argument allow a uniform theorem without any upper bound on the number or strength of intermodule links.

Nonidentical/time-dependent local plant. Keep the locally evaluated, simultaneous, same-letter observation contract and unit sampling period of Section 125.7. For $a = 1, \dots, m$, put

$$\mathcal{K}_\eta(t) = I_m \otimes K_0 + C(t) + \Delta(t), \quad \Delta(t) = \bigoplus_{a=1}^{m} \sum_{i=1}^{2} d_i^{(a)}(t)(e_b - e_{t_i})(e_b - e_{t_i})^T, \quad |d_i^{(a)}(t)| \le \eta. \tag{125.17}$$

Here $C(t)$ is any matching-node-type, nonnegative symmetric diffusive intermodule Laplacian, locally bounded and piecewise continuous. The $d_i^{(a)}(t)$ may be different across modules and may vary with time; $0 \le \eta \le 3/200000000$ ensures every terminal conductance stays strictly positive. The common leak remains exactly one, the starting field is zero and each b-heater is still closed. The total instantaneous power limit Q applies across *all* modules. In particular the local and intermodule coupling schedules are not assumed known to the controller. All statements below concern local sampled EF–FG projected bands, not global EF–FG states of the joined graph.

Lemma 125.19 (Uniform positive-semigroup row perturbation)**.** *Write $P_\eta(T,t)$ and $P_0(T,t)$ for the state-transition operators generated by $-\mathcal{K}_\eta$ and by the same system with $\Delta = 0$. Both are entrywise nonnegative, with every row sum and every column sum equal to $e^{-(T-t)}$. Furthermore, $\|\Delta(t)\|_1 = \|\Delta(t)\|_\infty \le 4\eta$. For any $q \in \mathbb{R}^6$ and $\widehat{q} = (\mathbf{1}_m^T \otimes q^T)/m$,*

$$\|\widehat{q}(P_\eta(T,t) - P_0(T,t))\|_{\max} \le \frac{4\eta}{m}\|q\|_\infty (T-t)e^{-(T-t)}. \tag{125.18}$$

Here $\|\cdot\|_{\max}$ is the largest absolute row entry.

Proof. Each edge perturbation is a symmetric signed Laplacian; at the common separator b the absolute column and row sums are at most 4η, and elsewhere at most 2η. Both full drift matrices are symmetric M-matrices with all row and column sums equal to one. Their nonautonomous transition operators are therefore positive and doubly substochastic, with the asserted exact mass sums. The unperturbed intermodule network has the exact averaging intertwiner $R_m P_0(T,s) = e^{-K_0(T-s)}R_m$, even for changing $C(s)$. Apply the Duhamel formula $P_\eta - P_0 = -\int_t^T P_0(T,s)\Delta(s)P_\eta(s,t)\,ds$. Using the column-sum bound for the positive base 6×6 heat semigroup, the column 1-norm bound on Δ, and the column sums of $P_\eta(s,t)$ gives (125.18) after integration. No graph-size or coupling-norm factor occurs. □

Theorem 125.20 (Heterogeneous all-size H1/H2 persistence)**.** *For every $m \ge 1$, all intermodule couplings $C(t)$ above, and every family of local terminal conductances satisfying*

$$|\kappa_i^{(a)}(t) - 211/100| \le \frac{3}{200000000} = 1.5\cdot 10^{-8}, \qquad i = 1,2, \tag{125.19}$$

we have, independently of the total heater budget,

$$\mathcal{L}_1^{(m)}(H1,Q) \subseteq \{22\}, \qquad \mathcal{L}_n^{(m)}(H1,Q) = \varnothing \quad (n \ge 2).$$

The unbounded-budget first-letter union is exactly $\{22\}$; the broadcast initial H1 witness realizes 22 at global budget $m\cdot 233.258410$. Conversely, the same 171-action, nonanticipative H2 lookup policy as in the nominal plant, broadcasting the input to every module and using the strict initial-22 mixture of $\alpha = 1/3200$, realizes $\mathcal{L}_n^{(m)}(H2,Q) = \Sigma^n$ for all $n \ge 1$ at common global budget

$$Q = m(Q_0 + 1/250). \tag{125.20}$$

Every sampled local scalar guard has a certified strict margin

$$> \frac{3121875}{10^{11}} - 8\cdot 202\cdot \frac{3}{200000000} = 6.97875\cdot 10^{-6}. \tag{125.21}$$

The policy requires neither future letters nor knowledge of the local parameter deviations and intermodule coupling schedules.

Proof. For the H1 obstruction, apply each of the 17 tabulated one-step or two-step nonnegative Farkas vectors to the *mean* guard arrays, which are valid whenever all modules must realize the same word. Each dual has coefficient sum $s \le 1 + 10^{-12}$, positive right-hand side, and nominal input-column coefficient at most $-\gamma/m$, with $\gamma > 1.6286883\cdot 10^{-6}$. By Lemma 125.19, an H1 held-input coefficient changes, for a one-step constraint, by at most $2\eta s/m$; for a two-step constraint, the first and second input blocks change by at most $6\eta s/m$ and $2\eta s/m$, respectively. These inequalities integrate $4\eta(T-t)/m$ on the relevant unit periods. Consequently all input coefficients remain strictly negative because $6\eta(1+10^{-12}) < 9.0001\cdot 10^{-8} < \gamma$. The positive Farkas right-hand sides are unchanged. This excludes the eight non-22 first letters and all nine possible successors after 22, *without using any power bound*. The nominal strict one-step H1 22 witness has minimum guard margin > 0.00342526, heater sum 233.258410 per module, and its broadcast error is bounded by $4\eta\cdot 234$ per coordinate. Its relational guard loss is at most $8\eta\cdot 234$, so the first letter is still attainable.

For H2, let $z(t) = \mathbf{1}_m \otimes x^0(t)$ be the sampled-word trajectory of the strictly feasible nominal 171-action policy and let $X_\eta(t)$ be the heterogeneous trajectory under its broadcast controls. The difference satisfies a positive-system variation-of-constants formula with forcing $-\Delta(t)z(t)$. Since every local heater component is nonnegative and the phase sum is at most $Q_0 + 0.004 < 202$, positivity and unit leakage give $\sup_t \|z(t)\|_\infty < 202$. Thus $\sup_t \|X_\eta(t) - z(t)\|_\infty \le 4\eta\cdot 202$, regardless of m and $C(t)$. Each guard compares two coordinates, so its loss is at most $8\eta\cdot 202$. The previous exact nominal strictification certifies every local sampled guard > 0.00003121875. Subtracting the uniform loss gives (125.21) for every horizon. □

Theorem 125.21 (Two-sided heterogeneous power bounds)**.** *Let* $Q^{(m)}_{\mathrm{on},\infty}(\eta)$ *and* $Q^{(m)}_{\mathrm{off},\infty}(\eta)$ *denote the smallest (infimal) budgets for all local synchronous words of all lengths for any fixed admissible nonidentical plant. Then at* $0 \le \eta \le 1.5 \cdot 10^{-8}$*,*

$$\frac{mQ_0}{1+168\eta} \le Q^{(m)}_{\mathrm{off},\infty}(\eta) \le Q^{(m)}_{\mathrm{on},\infty}(\eta) \le m(Q_0 + 0.004). \tag{125.22}$$

Moreover, for every $0 < \eta \le 10^{-8}$ *the upper bound improves to*

$$Q^{(m)}_{\mathrm{on},\infty}(\eta) \le m(Q_0 + 210000\eta), \tag{125.23}$$

with a guard-strict broadcast policy obtained by mixing the first 22 *action with* $\alpha = 20000\eta$*. At* $\eta = 0$ *the sharp identity* $Q^{(m)}_{\mathrm{on},\infty} = Q^{(m)}_{\mathrm{off},\infty} = mQ_0$ *is recovered. Therefore the normalized online and offline optimal powers converge to* Q_0 *uniformly in* m *and the allowable coupling functions as* $\eta \downarrow 0$*.*

Proof. Any all-word controller must realize the first letter 22 in every module. Apply the exact six-node first-22 analytical dual from Section 4 to the terminal-symmetrized *mean* output at time one. Its four nonnegative guard multipliers sum to $S < 84$ and the two phase multipliers sum to one; weak duality bounds its nominal value by Q/m even when the module heater allocations are unequal. The averaged physical field differs from the nominal averaged field, for each dual guard row, by at most $2\eta Q/m$: this is the time integral over $0 \le t \le 1$ of the row estimate (125.18) against $\|U(t)\|_1 \le Q$. Since the physical averaged first-22 guards hold, the nominal dual yields $Q_0 \le (Q/m)(1 + 2\eta S) < (Q/m)(1 + 168\eta)$. This proves the first inequality in (125.22); the remaining ones follow from Theorem 125.20 and the trivial comparison of preview with causal control.

For the near-nominal upper bound, choose $\alpha = 20000\eta$ and phase budget excess $\varepsilon = 210000\eta$. The original strictification argument gives initial-22 margin at least 0.0999α and every other margin at least $0.00007849 - 0.151\alpha$, while power remains strictly below $Q_0 + \varepsilon$ (since $10\alpha < \varepsilon$). The heterogeneous plant loses at most 1616η in any guard, because $Q_0 + \varepsilon < 202$. Thus the first guard retains at least 382η and the others at least $0.00007849 - 4636\eta > 0$ whenever $0 < \eta \le 10^{-8}$. The same causal policy then realizes all words at the budget of (125.23). At $\eta = 0$, use the exact nominal boundary policy, not the mixed strictly interior one. □

Relation to established results. Perturbation bounds for nonidentical diffusively coupled networks are well established [291, 292]; our Duhamel and norm inequalities are not a claimed general invention. The special-model contribution is that *both* a budget-free Farkas language obstruction and a common all-word causal control survive independently of module count and intermodule coupling strength. The exact equality mQ_0 need not persist under heterogeneity; (125.22) is a certified interval, not an assertion of an exact nonidentical-network optimum. This model still uses local synchronous words, a fixed sampling clock, fixed leakage, matching-type symmetric couplings and sampled-only constraints.

125.9 Cross-type coupling and independently prescribed local words

The exact modular equality in Theorem 125.15 assumes the same word in every module and links only matching node types. We now remove both restrictions, at the price of a quantitative bound on intermodule coupling strength. This is a *different language contract*: at each sampling period k, every module a independently receives a letter $s_k^{(a)} \in \Sigma$. The combined alphabet is Σ^m, with $(9^m)^n = 9^{mn}$ length-n word sequences. Local EF–FG projections are evaluated on the induced six-node trees, not on the full coupled graph's global intrinsic basin.

Write $K_{\mathrm{iso}}(t) = \bigoplus_{a=1}^{m} K_a(t)$ with terminal edge perturbations as in (125.17), $|\kappa_i^{(a)}(t) - 211/100| \le \eta$. Let $C_\times(t)$ be an arbitrary *symmetric nonnegative intermodule graph Laplacian*, now allowed to link *different* local vertex types. Assume it is locally bounded and piecewise continuous and define its maximum instantaneous weighted degree

$$\Gamma = \sup_{t\ge 0} \max_{(a,v)} \sum_{(b,w):\ b\neq a} c_{(a,v),(b,w)}(t). \tag{125.24}$$

The plant is $\dot X = -(K_{\mathrm{iso}}(t) + C_\times(t))X + u$, $X(0) = 0$, with $u \ge 0$, every b-port closed, and a *single global* peak-power constraint $\mathbf{1}^T u(t) \le Q = mq$. No bound on the number of modules, the number of cross edges, or their change frequency beyond local regularity is required. The maximum weighted degree Γ, however, must be small for the next theorem.

Lemma 125.22 (Cross-type positive-semigroup perturbation)**.** *Let $Z(t)$ solve the uncoupled, possibly nonidentical block-diagonal plant $K_{\mathrm{iso}}(t)$ under the* same *nonnegative control u as the coupled trajectory $X(t)$. Then for any input with total peak power at most Q,*

$$\sup_{t\ge 0} \|X(t) - Z(t)\|_1 \le 2\Gamma Q. \tag{125.25}$$

If, in addition, every module's own peak input sum is at most q, and Y is the reference trajectory of m independent nominal *copies driven by these same per-module controls, then*

$$\sup_{t\ge 0} \|X(t) - Y(t)\|_\infty \le (\Gamma + 4\eta)q. \tag{125.26}$$

Both bounds hold uniformly in m, over arbitrary target-word histories, and for time-varying link weights.

Proof. All drift matrices are symmetric M-matrices with row and column sums one. Therefore the nonautonomous evolutions are nonnegative and have every row and column sum $e^{-(t-s)}$. Nonnegative control and zero initial temperature imply $\|Z(t)\|_1 \le Q$ and $0 \le Y_i(t) \le q$. The cross Laplacian has induced column norm $\|C_\times(t)\|_1 \le 2\Gamma$. Duhamel's formula for $X - Z$, using the positive ℓ_1 contraction, yields $\|X(t) - Z(t)\|_1 \le \int_0^t e^{-(t-s)} 2\Gamma Q\, ds \le 2\Gamma Q$. For the second statement, variation of constants relative to the nominal reference gives forcing $-[C_\times(t) + \Delta(t)]Y(t)$. At each vertex, the cross term is a weighted sum of coordinate differences of Y, so its magnitude is at most Γq (all nominal coordinates lie in $[0, q]$); the signed local terminal perturbation has infinity norm at most 4η and hence contributes at most $4\eta q$. The positive full-network evolution has infinity norm $e^{-(t-s)}$, which integrates to (125.26). □

Theorem 125.23 (Independent-word phase separation under weak cross-type links)**.** *Let $m \ge 1$, let every local terminal conductance satisfy $\eta \le 3/200000000 = 1.5 \cdot 10^{-8}$, and allow all symmetric cross-type intermodule connections satisfying*

$$0 \le \Gamma \le 10^{-9}, \qquad Q = m(Q_0 + 1/250), \quad Q_0 + 1/250 < 202. \tag{125.27}$$

For the complete vector-word *language with independently prescribed module letters:*

1. *Under H2, one copy of the original 171-action nonanticipative policy in each cell realizes* every *vector word in $(\Sigma^m)^n$ at all finite horizons $n \ge 1$, using no future letters or knowledge of the changing links. Every sampled guard remains strictly greater than*

$$0.00003121875 - 2 \cdot 202(4\eta + \Gamma) \ge 0.00000657475. \tag{125.28}$$

2. *Under H1 and at the* same stated global budget *Q, no vector word of length two is realizable, hence no longer vector word is realizable. This holds even when the heater budget is allocated arbitrarily and unevenly among modules and the controller knows the future word.*

The H1 assertion is finite-budget*; unlike Theorems 125.7 and 125.20, it is not an all-budgets impossibility claim for arbitrary cross-type coupling.*

Proof. H2: the nominal strict initial-22 mixture with $\alpha = 1/3200$ and the other 170 tabulated actions implement each requested word using its own local phase inputs; all have per-module peak sum below $q = Q_0 + 1/250$. This is the same causal policy with independent, not broadcast, inputs. The six-node rational certificate guarantees all nine guards of each independent reference trajectory a margin $> g_* := 0.00003121875$. By Lemma 125.22, every local coordinate difference changes by at most $2(\Gamma + 4\eta)q$, which is strictly smaller than $2 \cdot 202(\Gamma + 4\eta)$. The right-hand side of (125.28) is positive. Thus the same controls work for every array of requested words, all m, and all n.

H1: for each module a, fix its requested two-letter prefix. If its first letter is not 22, take the tabulated one-sample Farkas certificate for that first letter; otherwise take the two-sample certificate for its $22 \to s_2^{(a)}$ prefix. Thus *every* module has one of the 17 certificates. Its multipliers are nonnegative, their sum is at most $s_* := 1 + 10^{-12}$, and its rational right-hand side is at least

$$\delta_* := \frac{26059}{16000000000} = 1.6286875 \cdot 10^{-6} > 0. \tag{125.29}$$

The nominal signed input coefficients are all strictly negative, and Theorem 125.20 certifies their persistence for an *isolated* nonidentical local plant with $\eta \le 1.5 \cdot 10^{-8}$, even for time-varying conductances. Hence each certificate functional applied to its uncoupled local temperature observations $Z^{(a)}$ is nonpositive, for arbitrary nonnegative H1 controls.

If the coupled plant were to realize the two-letter vector word, the weighted observed guard inequalities would yield at least $m\delta_*$ after summing the m local Farkas functionals. Each weighted guard row compares two coordinates; because the nonnegative multipliers in a cell sum to at most s_*, the resulting coefficient at any coordinate/time has absolute value at most s_*. Consequently Lemma 125.22, at times one and two, bounds the discrepancy between the summed functionals on X and Z by

$$s_*\big(\|X(1) - Z(1)\|_1 + \|X(2) - Z(2)\|_1\big) \le 4\Gamma Q s_* = 4\Gamma m q s_*.$$

But $4 \cdot 10^{-9} \cdot 202(1 + 10^{-12}) < 8.081 \cdot 10^{-7} < \delta_*$, a contradiction after dividing by m. Thus every length-two vector word is excluded at the specified total budget. □

Theorem 125.24 (Strong-coupling obstructions are topology- and word-dependent)**.** *Consider two nominal local cells under zero initial temperature, unit leak, and arbitrary time-varying nonnegative heaters with* total *instantaneous budget Q. The H1/H2 distinction is irrelevant to both lower bounds below.*

1. *Add only the two* cross-type *links (r_1, w_2) and (r_2, w_1) of conductance g. If both cells satisfy the protected sampling guards at even one positive sample time, then*

$$Q \ge \frac{13.502(5 + 2g)}{4}. \tag{125.30}$$

   At $g = 60$ this is $Q \ge 421.9375 > 2(Q_0 + 0.004)$, so even the first common symbol $(22, 22)$ is impossible at the previous power budget.
2. *Instead connect* every corresponding node type *between the two cells by an edge of conductance g. If the first requested terminal-*1 *band in cell one is* 0 *and in cell two is* 2*, then*

$$Q \ge \frac{3}{2}(1 + 2g). \tag{125.31}$$

   At $g = 200$ this gives $Q \ge 601.5 > 2(Q_0 + 0.004)$. Hence even matching-type links can prevent independently different requested words at a fixed budget, although they preserve exactly the common-word *language of Theorem 125.15.*

These are analytical necessary conditions for any admissible time-varying control, not merely infeasibility reports from a finite H2 linear program.

Proof. For the crossed-link case set $Y(t) = (x_{r_1} - x_{w_1}) + (x_{r_2} - x_{w_2})$. Directly subtracting the local r, w heat equations and adding the two crossed-edge flows yields

$$\dot{Y} = -(5 + 2g)Y + 3\sum_{a=1}^{2}(x_{w_a} - x_{p_a}) + \sum_{a=1}^{2}(u_{r_a} - u_{w_a}).$$

The total thermal mass $S(t) = \mathbf{1}^T X(t)$ obeys $\dot{S} = -S + \mathbf{1}^T u \le -S + Q$, so $S(t) \le Q$. Positivity gives $\sum(x_{w_a} - x_{p_a}) \le S(t) \le Q$, and nonnegative inputs imply $\sum(u_{r_a} - u_{w_a}) \le Q$. Starting from $Y(0) = 0$ we get $Y(t) \le 4Q/(5 + 2g)$. But the two protected observations require $x_{r_a} - x_{w_a} \ge 1 + 5751/1000 = 6.751$, hence $Y(t) \ge 13.502$, proving (125.30).

For the matching-type case let $Z = X^{(2)} - X^{(1)}$. All six cross-links contribute $-2gZ$, so $\dot{Z} = -(K_0 + 2gI_6)Z + (u^{(2)} - u^{(1)})$. The local positive semigroup has infinity norm $e^{-(1+2g)t}$. Since $\|u^{(2)} - u^{(1)}\|_\infty \le Q$, the variation-of-constants formula gives $\|Z(t)\|_\infty \le Q/(1 + 2g)$. The two different first terminal bands require $g_1^{(2)} - g_1^{(1)} \ge 4 - 1 = 3$, whereas $|g_1^{(2)} - g_1^{(1)}| = |Z_{t_1} - Z_b| \le 2\|Z\|_\infty$. This implies (125.31). □

Interpretation against prior work. Positive semigroup and input-to-state sensitivity bounds, small-gain arguments for compositional temporal contracts [302], and synchronization constraints in diffusive networks [303] are existing methods, not new general principles claimed here. The additional certified fact is a concrete *topology-dependent phase separation* for region-independent quantitative EF–FG local observations: weak arbitrary-type links permit independent words and preserve the two-step H1 obstruction at a finite power, whereas two explicit strong-coupling networks force a larger power already at the first observation. Neither a global EF–FG color theorem for the joined graph nor a CCR-specific advantage follows.

125.10 A certified first-sample cross-link resolution phase

The preceding weak-link theorem guarantees all horizons at a very small degree cap; its strong-link examples supply coarse necessary power bounds. We now fix two *nominal, identical* six-node cells and replace the degree-bound assumption by two precise symmetric cross-type edges:

$$(r_1, w_2), \quad (r_2, w_1), \qquad \text{each of conductance } g > 0.$$

The local observations are computed on the induced six-node trees as in Section 125.9. Let K_g be the resulting 12×12 matrix $I + L_g$, and let P_{10} select the ten open local heater ports. Put

$$C_g = K_g^{-1}(I - e^{-K_g})P_{10}, \tag{125.32}$$

$$E_{g,h} = e^{-K_g/2}, \quad C_{g,h} = K_g^{-1}(I - E_{g,h})P_{10}, \tag{125.33}$$

$$D_g = [\, E_{g,h}C_{g,h} \mid C_{g,h} \,]. \tag{125.34}$$

For a first vector letter $(s, t) \in \Sigma^2$, the 18×12 guard matrix $H_{s,t}$ is the direct sum of the two local nine-row systems, with right side $b_{s,t}$. Hence the exact first-sample constraints are $H_{s,t}C_g v \ge b_{s,t}$ for H1 and $H_{s,t}D_g(v^-, v^+) \ge b_{s,t}$ for H2, with nonnegative controls and one global power cap applied to each held phase.

Theorem 125.25 (The 81-letter first-sample resolution phase)**.** *For either of the* two exact *cross-conductances*

$$g = \frac{7}{1000} \qquad \textit{or} \qquad g = \frac{1}{100},$$

H1 has empty *first-sample vector language at every finite peak-power budget, while H2 realizes* all *81 first vector letters at global per-phase power* $Q = 421$*:*

$$\mathcal{L}_1^{(2)}(H1, Q) = \varnothing \quad (Q < \infty), \qquad \mathcal{L}_1^{(2)}(H2, 421) = \Sigma^2.$$

At $g = 7/1000$ the 81 exact nonnegative Farkas vectors have a positive right-side pairing $> 2.03\cdot10^{-6}$ and every reduced input coefficient $< -2.03\cdot10^{-6}$. The 81 rational H2 actions have every sampled guard margin > 0.1047. At $g = 1/100$ the corresponding bounds are $> 6.03 \cdot 10^{-6}$, $< -6.03 \cdot 10^{-6}$, and > 0.1039, respectively.

Proof. For each (s,t), the certificate table in Section 125.27 supplies $y_{s,t} \in \mathbb{Q}^{18}_{\geq 0}$ such that

$$y_{s,t}^{\mathsf{T}} b_{s,t} > 0, \qquad y_{s,t}^{\mathsf{T}} H_{s,t} C_g < 0 \quad \text{coordinatewise on all ten heater columns.}$$

If $v \geq 0$ satisfied the H1 constraints, multiplying by $y_{s,t}$ would give $0 < y_{s,t}^T b_{s,t} \leq y_{s,t}^T H_{s,t} C_g v \leq 0$, a contradiction. This proves the H1 statement *without* a power bound. For H2 the certificate table in Section 125.27 supplies a nonnegative rational pair $(v_{s,t}^-, v_{s,t}^+) \in \mathbb{Q}_+^{20}$ with each ten-port phase sum < 421 and all 18 sampled guards strictly satisfied. Each of the 81 dual/primal pairs is checked using rational upper/lower envelopes for the matrix exponentials by the outward-rational verification procedure in Section 125.22; no floating-point solver output is used to certify a sign. □

Proposition 125.26 (A certified change between two coupling strengths)**.** *At $g = 1/200 = 0.005$, H1 realizes the common first vector letter $(22, 22)$ by applying the same full-interval five-port held heater to both modules:*

$$v^{(a)} = \big(57.31316857, 0, 76.95531716, 49.53599898, 49.53599898\big), \quad a = 1, 2.$$

The total power is exactly 466.68096738, and the least sampled guard exceeds 0.0002973. In contrast, Theorem 125.25 excludes every *first vector letter at $g = 0.007$, for all power budgets. Therefore feasible first-step H1 behavior changes between these two exhibited coupling values. This early two-point comparison alone does* not *establish a unique threshold; a subsequent local theorem sharpens it analytically.*

Proof. Under the symmetric broadcast input, both cell temperatures are identical, and the cross-links act exactly as an added local r–w conductance g. With the quotient drift $K_0 + g(e_r - e_w)(e_r - e_w)^T$, the displayed rational held input yields nine strict local sampled guards in both cells; the least is bounded as stated by the exact positive-series exponential envelope. The other endpoint is Theorem 125.25. □

Theorem 125.27 (Strong-link structural cone obstruction)**.** *For the same two crossed links but $g = 5$, H1 cannot satisfy even the five common structural guards in both modules at the* first *sample, under any finite nonnegative heater power. Nevertheless, H2 realizes the common first vector letter $(22, 22)$ with total per-phase power at most 702 and minimum strict guard > 0.096.*

Proof. For arbitrary, not necessarily identical, module controls, the averaged state $\bar{x} = (x^{(1)} + x^{(2)})/2$ obeys the *exact* quotient identity

$$\dot{\bar{x}} = -(K_0 + g(e_r - e_w)(e_r - e_w)^T)\bar{x} + \bar{u}, \qquad \bar{u} = (u^{(1)} + u^{(2)})/2 \geq 0. \tag{125.35}$$

If both module states obey the five common affine difference guards at the first sample, their average does too. At $g = 5$, the quotient's r–w conductance is 7. For these five guards take the nonnegative rational multiplier

$$\lambda = (0.287949535,\ 0,\ 0.473411111,\ 0.119319677,\ 0.119319677).$$

It has $\lambda^T b > 0$ and all five coefficients of $\lambda^T H_{\text{struct}}(K_0 + 5L_{rw})^{-1}(I - e^{-(K_0+5L_{rw})})P_5$ are strictly less than -0.0025, as certified with rational matrix-exponential enclosures. Thus no constant nonnegative input can meet the averaged structural guards. For H2 use the same two five-port phase controls in both modules,

$$v^- = (0, 0, 0, 34.77637637, 315.22362363),$$
$$v^+ = (132.78157732, 0, 96.03257593, 59.22748101, 0).$$

The early/late sums are 350 and 288.04163426, respectively, per module. The quotient under this symmetric broadcast has all nine requested 22 guards strictly above 0.096, using the same rigorous exponential enclosure. The global phase-power cap 702 follows. □

Remark 125.28 (What has and has not changed). The two-cell language here is the vector of *locally evaluated* EF–FG-compatible numerical terminal records. Joining the graphs can change the global elder/merge geometry: for example, two equal-height local elder maxima become competing equal-height maxima in one connected graph, whose global elder/tie resolution is not two separate local elder resolutions. No full-global-EF–FG equivalence, all-horizon H2 claim at $g = 0.007$ or 0.01, or *global* monotone sharp threshold on all $g > 0$ follows. The exact quotient map (125.35) is a standard balanced-diffusion argument; the special-model certified language consequences are the claim here [294, 297, 286].

125.11 An exact local cross-link critical threshold

The preceding theorem certifies two separated cross-link strengths. We now obtain a sharp *local* phase classification, including a uniquely defined analytic critical value. Set

$$g_- = \frac{54757560}{10^{10}}, \qquad g_+ = \frac{54757561}{10^{10}}, \qquad I_g = [g_-, g_+].$$

These are the decimal values 0.0054757560 and 0.0054757561. We continue to require two identical, six-node nominal modules with exactly the two crossed r–w links defined above, positive input ports other than b, and the nine *local* EF–FG-compatible sample guards per module. No assertion here concerns the full joined graph's global extrema records.

Three-variable exact quotient. Write $v = e_r - e_w$, $\widetilde{K}(g) = K_0 + gvv^{\mathsf{T}}$, $T(g) = \widetilde{K}(g)^{-1}(I - e^{-\widetilde{K}(g)})$, and $S = [\, e_r \;\; e_p \;\; e_{t_1} + e_{t_2} \,]$. Define the three active-guard rows and the terminal-gap row by

$$A(g) = \begin{bmatrix} (e_r - e_p)^{\mathsf{T}} \\ (e_b - e_w)^{\mathsf{T}} \\ (e_p - e_{t_1})^{\mathsf{T}} \end{bmatrix} T(g)S, \qquad c(g) = (e_{t_1} - e_b)^{\mathsf{T}} T(g) S, \tag{125.36}$$

$$G(g) = c(g)A(g)^{-1}\mathbf{1}_3, \qquad \mathcal{D}(g) = \det \begin{bmatrix} A(g) & \mathbf{1}_3 \\ c(g) & 9/2 \end{bmatrix}. \tag{125.37}$$

Here $\mathbf{1}_3 = (1,1,1)^{\mathsf{T}}$. The exact equation $\mathcal{D}(g) = 0$ is equivalent to $G(g) = 9/2$ wherever $A(g)$ is invertible. It is an analytic matrix-exponential equation, not a claim that the root is rational.

Theorem 125.29 (Complete local first-sample phase and unique critical root)**.** *There exists a* unique $g_* \in (g_-, g_+)$ *with* $\mathcal{D}(g_*) = 0$. *For every* $g \in I_g$ *the following first-sample vector-language assertions hold:*

$$\mathcal{L}_1^{(2)}(H1, 467; g) = \begin{cases} \{(22,22)\}, & g_- \le g \le g_*, \\ \varnothing, & g_* < g \le g_+; \end{cases} \tag{125.38}$$

$$\mathcal{L}_1^{(2)}(H1, Q; g) = \varnothing \qquad (g_* < g \le g_+, \ \textit{all finite } Q), \tag{125.39}$$

$$\mathcal{L}_1^{(2)}(H2, 421; g) = \Sigma^2 \qquad (g_- \le g \le g_+). \tag{125.40}$$

A high-precision numerical estimate *is* $g_* \simeq 0.00547575602911761758$*; only the stated rational bracket, uniqueness in that bracket, and the language statements are claimed as rigorously certified. No uniqueness or monotonicity assertion is made for* all $g > 0$*, and the H2 claim here is for* one observation.

Proof. Symmetric reduction. The plant, the desired first vector letter $(22, 22)$, the control constraints, and the guard polyhedra are invariant under interchange of the two modules and interchange of their two terminals. Any feasible nonnegative H1 control for this common letter can therefore be averaged over these permutations. The averaged input uses the same $(u_r, u_w, u_p, u_t, u_t)$ in both modules. The module trajectory then obeys the exact six-node drift $\widetilde{K}(g)$.

Primal face. Exact rational-exponential enclosures give $\|A(g)^{-1}\|_\infty < 80$ on I_g. The candidate three-input control $u(g) = A(g)^{-1}\mathbf{1}_3$ has strictly positive coordinates in the $r, p, t_1 + t_2$ grouping. Set $u_w = u_b = 0$ and apply u_t at each terminal in both modules. By construction,

$$x_r - x_p = x_b - x_w = x_p - x_{t_i} = 1, \quad x_{t_i} - x_b = G(g), \quad x_p - x_w = G(g) + 2.$$

The finite interval certificate below gives $4 < G(g) < 5$ and $2(u_r + u_p + 2u_t) < 467$ throughout I_g. Consequently all guards for $(22, 22)$ hold whenever $G(g) \le 9/2$. At the critical point an upper terminal band is tight, which is permitted by our closed sampled-band convention.

Matching dual. For a symmetric held input, use four of the nine local guard rows: $r - p \ge 1$, $b - w \ge 1$, $p - t_1 \ge 1$, and $b - t_1 \ge -9/2$. With the same three controlled columns, set $z(g) = c(g)A(g)^{-1}$ and take the nonnegative multiplier $y = (z_1, z_2, z_3, 1)$. The exact interval check yields $z_i > 1/2$ throughout I_g; its reduced coefficients vanish on the r, p, t inputs, and the reduced coefficient on the remaining w input is $< -1/5$. Its right-side pairing is exactly

$$y^{\mathsf{T}} b = z(g)\mathbf{1}_3 - \frac{9}{2} = G(g) - \frac{9}{2}.$$

If $G(g) > 9/2$, the Farkas contradiction excludes $(22, 22)$ even at unbounded heater power. Symmetrization transfers the contradiction to arbitrary two-module controls for this common letter.

Unique root. For $C(g) = \int_0^1 e^{-t\widetilde{K}(g)}\, dt$, positivity, the row sum e^{-t}, and Duhamel differentiation show $\|C'(g)\|_\infty \le 1$ and $\|C''(g)\|_\infty < 4/3$. Thus $\|A'\|_\infty \le 2$, $\|A''\|_\infty < 3$, $\|c\|_1, \|c'\|_1 \le 2$, and $\|c''\|_1 < 3$. Twice differentiating $G = cA^{-1}\mathbf{1}_3$ with $\|A^{-1}\|_\infty < 80$ yields

$$|G''(g)| \le 3(80) + 14(80)^2 + 16(80)^3 < 9 \cdot 10^6.$$

The pure-rational endpoint evaluation, including an absolute G-error less than $2.1 \cdot 10^{-20}$, establishes

$$G(g_-) - 9/2 < -1.0009 \cdot 10^{-10}, \qquad G(g_+) - 9/2 > 2.4365 \cdot 10^{-10},$$
$$\frac{G(g_+) - G(g_-)}{g_+ - g_-} > 3.43744549.$$

Since $g_+ - g_- = 10^{-10}$, the curvature bound implies $G'(g) > 3.4365 > 0$ for the entire interval. Therefore $G = 9/2$ has exactly one root there, and $\mathcal{D} = \det A\,(9/2 - G)$ has the same unique root.

All other 80 first vector letters. At g_+ there are 80 nonnegative rational Farkas multipliers, one for each vector letter other than $(22, 22)$, whose right-side pairings are > 0.00051730 and whose every held-port reduced cost is < -0.00047918. Their total multiplier mass is < 2. For the twelve-node cross-link generator, the g-derivative of its H1 input map has infinity norm at most one; hence a reduced coefficient can drift by at most $4|g - g_+| < 4 \cdot 10^{-10}$. The same 80 duals thus exclude these letters *uniformly* on I_g. Together with the primal–dual equivalence for $(22, 22)$, this proves (125.38) and (125.39).

H2 throughout the bracket. At g_+ the 81 rational two-phase witnesses each respect the global power cap 421 and have a strict minimum first-sample guard margin > 0.09644. The continuous-time Duhamel formula bounds the first-sample state change for these fixed inputs by $421|g - g_+|$ in infinity norm; each two-coordinate guard changes by at most $842|g - g_+|$. This is below $8.42 \cdot 10^{-8}$ throughout I_g, so all 81 witnesses remain valid on

the whole interval. All numerical signs used here are verified with outward exact-rational matrix-exponential bounds and the rational certificate tables of Section 125.27. □

Remark 125.30 (Scope of the critical value). The number g_* is a *unique local critical coupling* for the *first-sample* H1 language in the precise two-cross-link network. It is not a globally unique H2-power threshold, a globally unique switching value for arbitrary g, a sharp power budget for the connected network, or a result for a global EF–FG observation on the combined graph.

125.12 Complete all-horizon language phase for the crossed network

The first-step classification above leaves open whether the crossed two-cell plant can sustain independently chosen vector letters indefinitely. Here the answer is exact on the critical interval $I_g = [g_-, g_+]$ of Theorem 125.29, and yields a smaller explicit controller than the original six-node staged policy. The following result uses the *same* two cross links, held half-period heater model, initially zero thermal field, and *local* source-native numeric EF–FG readouts.

For each $s \in \Sigma^2$, let H be the common list of 18 two-coordinate guard normals and let $b_s \in \mathbb{Q}^{18}$ be the corresponding band-dependent right-hand side, so $Hx \geq b_s$ is precisely the sampled observation predicate. The normals do not depend on s; only b_s does. At fixed g put

$$E_g = e^{-K_g}, \quad C_{g,h} = K_g^{-1}(I - e^{-K_g/2})P_{10}, \qquad D_g = \begin{bmatrix} e^{-K_g/2}C_{g,h} & C_{g,h} \end{bmatrix}.$$

The input in the first ten columns of D_g acts on the first half-period, and the second ten columns on the second half-period.

Theorem 125.31 (A memoryless complete vector-word controller)**.** *For every $g \in I_g$, a fixed library of* 81 *nonnegative rational two-phase input vectors $u_s \in \mathbb{Q}^{20}_{\geq 0}$, indexed by the* current *vector target letter $s \in \Sigma^2$, has the following property. Both phase-power sums are strictly below* 421 *for each u_s, and for every finite $n \geq 1$ and every independently specified vector word $s_1 \cdots s_n \in (\Sigma^2)^n$ the zero-initial-state trajectory under $u_{s_1}, \ldots, u_{s_n}$ satisfies*

$$Hx_k \geq b_{s_k} \quad (1 \leq k \leq n), \qquad \min_{k,h}\big((Hx_k - b_{s_k})_h\big) > 0.0293355.$$

In particular,

$$\boxed{\mathcal{L}_n^{(2)}(H2, 421; g) = (\Sigma^2)^n \qquad (n \geq 1).} \tag{125.41}$$

The control at period k is a function only of s_k, not of any earlier or later target, and requires neither a state observer nor model-dependent online optimization.

Proof. It suffices first to work at g_+. With the rational actions in the action table in Section 125.27, define for each signed guard normal h and integer lag $j \geq 1$ the worst independent predecessor contribution

$$m_{j,h} = \min_{a \in \Sigma^2} hE_{g_+}^j D_{g_+} u_a, \qquad a_{s,h} = hD_{g_+} u_s.$$

For any vector word, positivity and linearity give the exact sample formula

$$x_k = \sum_{j=0}^{k-1} E_{g_+}^j D_{g_+} u_{s_{k-j}}, \tag{125.42}$$

so, for $k \leq 12$, each current guard is bounded below by $a_{s_k,h} + \sum_{j=1}^{k-1} m_{j,h} - b_{s_k,h}$. The rational inequalities prove, for *every* s, h and all $\ell = 0, \ldots, 11$,

$$a_{s,h} + \sum_{j=1}^{\ell} m_{j,h} - b_{s,h} > 0.0397542, \tag{125.43}$$

including verified outward matrix-exponential and rational-action errors.

For $k > 12$, the omitted lags $j \geq 12$ satisfy $|hE_{g_+}^j D_{g_+} u_a| \leq 2 \cdot 421 e^{-j}$: the heat semigroup is positive with infinity norm e^{-j} and $0 \leq D_{g_+} u_a \leq 421\mathbf{1}$. Since $e^{-1} < 3/8$, their entire potential negative contribution is bounded by the *rational* tail

$$T = 2(421)\frac{(3/8)^{12}}{1-3/8} = \frac{223736661}{21474836480} < 0.010419. \tag{125.44}$$

Using all eleven explicit lag minima and this tail, the rational bounds prove the more conservative uniform guard margin > 0.02933572 at g_+.

Finally, for any $g \in I_g$, or even measurable time-varying $g(t)$ with values in I_g, the two crossed edges yield $\|K_{g(t)} - K_{g_+}\|_\infty \leq 2|g(t) - g_+|$. All inputs have pointwise total power at most 421, hence all physical coordinates are in $[0, 421]$. Positivity and Duhamel give $\|x_{g(\cdot)}(t) - x_{g_+}(t)\|_\infty \leq 842 \sup |g(t) - g_+|$. Each guard compares two coordinates, so its variation is at most $1684(g_+ - g_-) = 1.684 \cdot 10^{-7}$. This leaves more than 0.0293355 guard margin at every sample, uniformly in horizon. □

Theorem 125.32 (Power-independent two-sample H1 extinction)**.** *For every $g \in I_g$ and every finite peak-power budget Q,*

$$\boxed{\mathcal{L}_n^{(2)}(H1, Q; g) = \varnothing \qquad (n \geq 2).} \tag{125.45}$$

Proof. By Theorem 125.29, the only possibly feasible first vector letter is $(22, 22)$. At g_-, fix this first letter. For each of the 81 second vector letters t, let H_{22} and H_t be the corresponding 18-row local guard systems. The two-sample map with nonnegative constant heater vectors $v_1, v_2 \in \mathbb{R}_{\geq 0}^{10}$ is

$$x_1 = C_g v_1, \quad x_2 = E_g C_g v_1 + C_g v_2, \qquad C_g = K_g^{-1}(I - E_g)P_{10}.$$

Section 125.27 supplies an explicit $y_t \in \mathbb{Q}_{\geq 0}^{36}$ with

$$y_t^T(b_{22}, b_t) > 4.4101 \cdot 10^{-6}, \qquad y_t^T \begin{bmatrix} H_{22}C_{g_-} & 0 \\ H_t E_{g_-} C_{g_-} & H_t C_{g_-} \end{bmatrix} < -4.4101 \cdot 10^{-6} \mathbf{1}_{20}^T.$$

The inequalities are strict for all twenty input columns and are certified with exact-rational matrix-exponential enclosures. Farkas' lemma forbids any nonnegative two-step control, without assuming a finite upper bound on its magnitude. Across I_g, Duhamel estimates change any reduced coefficient by less than $7 \cdot 10^{-10}$; the multipliers remain valid, with every reduced coefficient still $< -4.4094 \cdot 10^{-6}$. Thus no $(22, 22) \to t$ can occur. Other first letters were already impossible, so every length-two word is excluded; every longer word has an excluded length-two prefix. □

Corollary 125.33 (Complete local all-horizon H1/H2 phase)**.** *Let g_* denote the unique root of Theorem 125.29. For every $g \in I_g$ and every $n \geq 1$, H2 realizes all 81^n independently requested vector words at global peak power 421. For H1, no length $n \geq 2$ word is possible at* any *finite power. At the first observation, the only H1 word is $(22, 22)$ whenever $g \leq g_*$ and $Q \geq 467$; for $g > g_*$, H1 has no first word at any finite power. The sufficient value 467 is not asserted to be the sharp first-word H1 budget, and 421 is not asserted to be the sharp all-word H2 budget.*

Corollary 125.34 (Time-varying cross-link robustness)**.** *The same 81-action H2 policy realizes* every *vector word of every length if the symmetric conductance of each of the two crossed edges changes together in time, measurably, within $|g(t) - g_+| \leq 10^{-5}$, at total phase power 421. Its sampled-guard margin exceeds 0.01249. No corresponding time-varying H1 extinction is claimed.*

Proof. Repeat the Duhamel comparison in Theorem 125.31: $1684 \cdot 10^{-5} = 0.01684 < 0.02933572 - 0.01249$. □

Significance and limits. The nonanticipative controller of Theorem 125.31 uses only the current vector letter, rather than a previous-letter mode. Its existence is certified by a finite polyhedral min-plus inequality and the positive semigroup tail, not by a finite simulation. Positive-system bounds, input-hold reachability, and affine certificates are standard tools; the special-model statement here is the full and exact first-vs-two-hold language classification in a nontrivially cross-coupled thermal graph. The theorem does not extend to arbitrary cross-link strengths or to the single global EF–FG color/ownership output of the joined graph.

125.13 Reproducibility and comparisons

No executable program is distributed with this monograph chapter. Instead, the governing matrices, exact rational witnesses, finite input tables and strictly bounded verification method are included in this chapter itself. Section 125.18 prints the 17 one-cell H1 Farkas certificates, the 170 rational two-hold controls (with the final initial action defined analytically), the 18 reference states, and a strict initial action for robustness. Section 125.27 prints all 81 crossed-network memoryless actions, all 81 two-step H1 duals and the first-step primal–dual families underlying the crossing-threshold and illustrative stronger-coupling results. Section 125.22 states a language-independent rational-arithmetic verification procedure and the exact matrix-exponential enclosure needed to check these finite certificates. The arithmetic signs and the contraction/tail arguments, rather than a finite prefix enumeration, establish the all-horizon assertions.

A reader may implement the deterministic checks in any programming language supporting exact integers and rational arithmetic: (i) form the graph Laplacians and port-incidence matrices as specified in Section 125.2 (or directly from the displayed edge lists); (ii) enclose each exponential with the positive Taylor/uniformization expansion and rational remainder; (iii) reconstruct the held-input maps; (iv) test every printed primal control and Farkas multiplier using rational interval comparisons; (v) verify the 171-action reference recurrence and the 81-action finite-lag minimum; and (vi) apply the proved positive-semigroup contraction for all remaining horizons. This is an independently reconstructible *algorithmic specification*; it does not assert a unique software implementation or release any unpublished source code. No external experimental dataset is used in deriving or testing the results of this chapter. The public measured-building datasets mentioned in the Introduction are cited to establish application context only; the preceding separate exploratory studies are not used as evidence for the exact H1/H2 claims. Finite full-network simulations may provide additional checks but are not needed for the stated theorems.

Interpretation of the resource certificate. The determinant ratio Q_0 is an exact expression in matrix-exponential entries, rather than an assertion that the number is rational or has a short closed form in elementary rational constants. The first 22 action is optimal with some weak guards tight, so nominal optimality alone does not imply parameter robustness at exactly Q_0. The robustness cube requires extra power and retains only the stated word-language properties, not the perturbed exact optimum. The 171-action table is a certificate for existence, not a minimal controller-size theorem. It does not improve on classical finite-state implementation in a machine-independent resource metric. The feasibility claims are quantified over every prescribed finite word with an interval-initially-revealed target letter; no unknown disturbances, intermediate-time guards, or arbitrary temporal formula are included.

Theorem-level relation to previous synthesis results. The 2020/2021 regular-language papers [289, 288, 290, 287] already establish general finite-state and data-driven synthesis frameworks for regular-language specifications. Robust finite abstractions [299], finite-memory LTL synthesis [300], positive-network LP control [284], and continuous-time AP-observation synthesis [304] likewise provide methodological precedent. Our claimed

added result is not a faster synthesis algorithm, a new language formalism, or continuous-time-between-samples satisfaction: it is the matching of 17 fixed-plant no-go certificates, a closed-form first-letter power optimum, one explicit all-word causal invariant policy at exactly that optimum, and a quantitative robustness family. This comparison identifies a distinct theorem statement, but does not establish universal bibliographic priority.

The six-node causal policy uses a finite phase marker and the previous/current target labels after its initial phase. The new crossed-network H2 policy instead uses only the current two-cell target letter. It is directly executable by a conventional finite-state controller. The 171 listed actions are a constructive upper bound on a representation of the policy, not a proven lower bound on the smallest action library, memory, runtime, or device energy. Nor do our results imply any CCR-specific computational advantage. The exact nominal online/offline equality does not extend automatically to perturbed parameters. No guarantee is given between sample times, in the conservative $\dot{x} = -Lx + u$ model, or for arbitrary graph topologies outside the explicitly lifted modular family.

125.14 Conclusion

For a region-choice-independent numeric projection of source-native terminal records in one explicitly specified positive heat network, increasing held-input temporal resolution from one to two pieces changes the complete achievable language from a single feasible first letter and no extensions to all finite nine-symbol words. The two-piece language has a sharp, analytically certified all-horizon power threshold and needs no future target preview. Rational witness data and a positive-contraction argument also provide a concrete open perturbation family at slightly larger power. Moreover, a mean-mode projection and synchronous embedding lift the exact word languages, causal and preview thresholds, and robustness guarantee to an arbitrarily large family of physically interconnected thermal modules. This is a structured, broadcast-observation construction; no claim is made for the entire native EF–FG state of the full joined graph. For weak cross-type links, a new aggregated-dual argument still forbids all two-sample independent-vector words at a fixed total power while a componentwise causal two-hold policy realizes every vector word. Strong cross-type links can destroy even the common first symbol at that power; strong matching-type links may instead destroy incompatible independent first symbols. A further exactly certified cross-link phase identifies a locally unique determinant-defined critical coupling. On its rigorously certified narrow interval, 81 exact two-sample Farkas multipliers prove H1 extinction for every word of length at least two and every finite budget; a new 81-action memoryless H2 policy realizes every independently targeted vector word at every horizon with power 421, with a quantitative time-varying-coupling tolerance. The critical point is the unique root of a three-control determinant equation only on that narrow interval. These are special-model consequences of classical positivity, Farkas, Duhamel and synchronization methods, and do not amount to a universal language theorem for arbitrary thermally coupled graphs.

125.15 Integrated exact certificate proofs and construction data

Every item needed for the stated finite primal–dual checks appears within this monograph chapter. The following certificate sections form an integral part of this monograph chapter, not a separate supplement. Intermediate rational values written in decimal notation are exact decimal fractions unless explicitly declared as approximate. The mathematical theorems are proved by inequalities with outward rational error estimates, rather than by floating-point solver tolerances.

Explicit strict initial (22) action for robustness

In the port order (r, w, p, t_1, t_2), the early and late halves of u^{210} are

$$u^{210,-} = \big(81.216809,\ 0.010000,\ 0.010000\big) \oplus \big(0.010000,\ 128.267448\big),$$
$$u^{210,+} = \big(49.399026,\ 0.029877,\ 76.369600\big) \oplus \big(55.401182,\ 28.314571\big).$$

Here $\oplus$ means concatenation of coordinate tuples. The entries are exact six-place decimal rationals, both phase sums are below 210, and the strict initial guard and oscillation bounds required by the robustness theorem follow from the rational enclosure method in Section 125.22.

125.16 Exact modular lift certificate

For each $m \geq 1$, take m identical local six-node thermal trees G_a and connect only matching node types (a, v) across modules by independent, arbitrarily weighted undirected thermal edges. The $m \times m$ coupling Laplacians L_v have zero row and column sums. These identities also hold pointwise if the symmetric conductances vary with time; local boundedness and piecewise continuity give the needed ODE uniqueness, so the language and power equalities are unchanged. In module-major order, write $K_m = I_m \otimes K_0 + \sum_v L_v \otimes e_v e_v^T$, and define $R_m = \frac{1}{m}(\mathbf{1}_m^T \otimes I_6)$ and $S_m = \mathbf{1}_m \otimes I_6$. Then exactly (no floating-point approximation)

$$R_m S_m = I_6, \qquad R_m K_m = K_0 R_m, \qquad K_m S_m = S_m K_0.$$

The first identity transports the nonuniform-input mean to the same six-node heat equation; the second transports a six-node control to a synchronous $6m$-node trajectory with all coupling heat flows identically zero. The aggregate power transforms as $Q \mapsto Q/m$ under averaging and $q \mapsto mq$ under replication. For each common sample letter s_k, the affine guards $H_{s_k} x_k^a \geq b_{s_k}$ are preserved by averaging and by replication. Consequently, for $H = 1, 2$ and all n, Q, the synchronous local-observation language satisfies $\mathcal{L}_n^{(m)}(H, Q) = \mathcal{L}_n^{(1)}(H, Q/m)$. The same proof transfers online/all-preview policies since there are no unknown disturbances. Thus $H = 1$ has no word of length at least two, while the $H = 2$ all-word nominal threshold is exactly mQ_0 and has zero preview gap.

The lifted theorem concerns the *same target word in every induced six-node local tree*; it does not assert the corresponding claim for the full joined graph's global intrinsic EF–FG labels or independently different requested words. The averaging/synchrony technique is not a new general control-theory tool. For illustration, the matrix intertwining may be checked over rational numbers on independently assembled 6-, 12-, 18-, 24-, and 42-node networks. Direct $6m$-dimensional matrix exponentials on coupled examples, including the eight corners of the stated parameter cube, provide independent finite diagnostics. They are secondary to the symbolic all-m proof.

125.17 Uniform heterogeneous-network certificate

Plant. For m six-node cells, symmetric nonnegative and locally bounded intermodule conductances may change with time. In every cell and at every time, $|\kappa_i^{(a)} - 2.11| \leq \eta$ for $i = 1, 2$, while the other three local edges, common leakage and sampling clock are unchanged. Writing Δ for the direct sum of two signed edge-Laplacian perturbations per cell, its row and column absolute sums are both at most 4η. The full positive evolution operator retains both row and column sums $e^{-(T-t)}$.

Mean-observation Duhamel inequality. For an arbitrary signed single-cell observation row q and $\widehat{q} = (\mathbf{1}_m^T \otimes q^T)/m$,

$$\|\widehat{q}[P_\eta(T, t) - P_0(T, t)]\|_{\max} \leq (4\eta/m)\|q\|_\infty (T - t) e^{-(T-t)}.$$

For H1 Farkas vectors the sum of multipliers is $\le 1 + 10^{-12}$, and the original worst reduced input coefficient is less than $-1.6286883 \times 10^{-6}$ (after factoring $1/m$). For two sampled times the perturbed input coefficient moves by at most $(6\eta/m)(1 + 10^{-12})$. At $\eta = 3/200000000$ it stays less than $-1.5386883 \times 10^{-6}/m$. The dual right sides stay strictly positive; the no-go remains independent of heater budget.

H2 broadcast action. The analytic initial-22 action is mixed with the tabulated strict Q210 action at coefficient 1/3200. The other 170 tabulated actions are unchanged. The nominal local guard margin is at least 0.00003121875 and the per-cell phase budget is less than $Q_0 + 0.004 < 202$. The actual nonidentical state stays within $4\eta \cdot 202$ in ℓ_∞ of the nominal broadcast state, independent of time, network size and coupling weights. Every two-coordinate guard thus retains a margin greater than

$$0.00003121875 - 1616\eta \ge 0.00000697875 \quad (\eta \le 1.5 \times 10^{-8}).$$

The original H1 initial-22 nominal strict witness has margin > 0.00342526, and its distributed perturbation loses at most 1872η.

Two-sided exact-dual power bounds. The certified one-letter first-22 active guard dual has four nonnegative multipliers of sum

$$S \approx 83.071254203673391146293502469 1659 < 84$$

and two nonnegative phase multipliers summing to one. For arbitrary uneven full-network H2 heat inputs with instantaneous total power Q, the nominal averaged first-22 dual value cannot exceed Q/m. Duhamel shifts each of its averaged guard outputs by at most $2\eta Q/m$; therefore

$$mQ_0/(1 + 168\eta) \le Q^{(m)}_{\mathrm{off},\infty} \le Q^{(m)}_{\mathrm{on},\infty} \le m(Q_0 + 0.004).$$

For $0 < \eta \le 10^{-8}$, the stricter mixed policy with $\alpha = 20000\eta$, $\varepsilon = 210000\eta$ has nominal first margin $\ge 1998\eta$, subsequent margin $\ge 0.00007849 - 3020\eta$, and heterogeneous guard loss $< 1616\eta$, proving the improved online upper bound $m(Q_0 + 210000\eta)$. For $\eta = 0$ use the exact nominal Q_0 policy.

Exact verification. The perturbation constants, original Farkas and dual margins, and strict inequalities can all be recalculated by rational arithmetic from the matrices and printed tables in this chapter. Direct numerical trajectories are illustrative; they are not required to prove the dimension-independent bound.

125.18 One-cell exact Farkas duals and causal input ledgers

The nine state guards, in the required row order, are

$$\begin{aligned}
&h_1 = e_r - e_p, && h_2 = e_p - e_w, && h_3 = e_b - e_w,\\
&h_4 = e_p - e_{t_1}, && h_5 = e_p - e_{t_2}, &&\\
&h_6 = e_{t_1} - e_b, && h_7 = e_b - e_{t_1}, &&\\
&h_8 = e_{t_2} - e_b, && h_9 = e_b - e_{t_2}. &&
\end{aligned}$$

The associated right sides are $(1, 5751/1000, 1, 1, 1, L_{s_1}, -U_{s_1}, L_{s_2}, -U_{s_2})$, where $(L_0, U_0) = (1/2, 1)$, $(L_1, U_1) = (2, 3)$, and $(L_2, U_2) = (4, 9/2)$. For a one-letter word the dual vector has 9 entries; for a two-letter word it has 18 entries, ordered first-sample then second-sample. The vector components not listed are zero. Each component below is given as an exact decimal rational in this chapter.

For each word w, the following table reports the exact y and a conservative bound on $\max_j (y^T \mathcal{G}_w)_j < 0$. This proves infeasibility even with unbounded nonnegative inputs.

word	$y^T b$	max reduced cost	nonzero dual entries ($j : y_j$)
00	1.013e-01	-6.294e-03	1:0.121099994839, 3:0.429530748355, 7:0.224684628403, 9:0.224684628403
01	4.223e-03	-4.223e-03	1:0.132558612598, 2:0.050714387924, 3:0.404486529776, 7:0.206120234851, 9:0.206120234851
02	2.886e-03	-2.886e-03	1:0.139953368585, 3:0.471767020387, 5:0.083442639411, 7:0.194139805514, 9:0.110697166103
10	4.223e-03	-4.223e-03	1:0.132558612598, 2:0.050714387924, 3:0.404486529776, 7:0.206120234851, 9:0.206120234851
11	2.042e-03	-2.042e-03	1:0.144623334131, 2:0.104111314225, 3:0.378117596847, 7:0.186573877399, 9:0.186573877399
12	1.128e-03	-1.128e-03	1:0.149681213084, 3:0.493559824797, 5:0.126496844546, 7:0.178379481059, 9:0.051882636513
20	2.886e-03	-2.886e-03	1:0.139953368585, 3:0.471767020387, 4:0.083442639411, 7:0.110697166103, 9:0.194139805514
21	1.128e-03	-1.128e-03	1:0.149681213084, 3:0.493559824797, 4:0.126496844546, 7:0.051882636513, 9:0.178379481059
2200	2.003e-03	-2.003e-03	1:0.037195497542, 3:0.100890039158, 7:0.066931878723, 9:0.066931878723, 10:0.102438528711, 11:0.063164254858, 12:0.281527844756, 16:0.140460038764, 18:0.140460038764
2201	1.470e-03	-1.470e-03	1:0.026753512969, 3:0.060191000891, 7:0.047396243996, 9:0.047396243996, 10:0.119448554593, 11:0.090097098897, 12:0.306976587710, 16:0.150870378474, 18:0.150870378474
2202	1.174e-03	-1.174e-03	1:0.020944204009, 3:0.037548438453, 7:0.036527760416, 9:0.036527760416, 10:0.128911937255, 12:0.426215735280, 14:0.105080956718, 16:0.156662082085, 18:0.051581125368
2210	1.470e-03	-1.470e-03	1:0.026753512969, 3:0.060191000891, 7:0.047396243996, 9:0.047396243996, 10:0.119448554593, 11:0.090097098897, 12:0.306976587710, 16:0.150870378474, 18:0.150870378474
2211	8.530e-04	-8.530e-04	1:0.014639788939, 3:0.012976133410, 7:0.024732995818, 9:0.024732995818, 10:0.139181849335, 11:0.121341833820, 12:0.336499620127, 16:0.162947391366, 18:0.162947391366
2212	5.329e-04	-5.329e-04	1:0.011606245494, 4:0.003166024156, 7:0.013410312219, 9:0.016576336375, 10:0.145992289250, 12:0.478213219423, 14:0.134159716821, 16:0.165517786541, 18:0.031358069720

word	$y^T b$	max reduced cost	nonzero dual entries $(j : y_j)$
2220	1.174e-03	-1.174e-03	`1:0.020944204009, 3:0.037548438453, 7:0.036527760416, 9:0.036527760416, 10:0.128911937255, 12:0.426215735280, 13:0.105080956718, 16:0.051581125368, 18:0.156662082085`
2221	5.329e-04	-5.329e-04	`1:0.011606245494, 4:0.003166024156, 7:0.013410312219, 9:0.016576336375, 10:0.145992289250, 12:0.478213219423, 13:0.134159716821, 16:0.031358069720, 18:0.165517786541`
2222	1.629e-06	-1.629e-06	`1:0.012650846879, 4:0.009764870472, 5:0.004585966142, 9:0.005178904330, 10:0.150892615262, 12:0.491181525832, 13:0.149106289719, 16:0.013766345822, 18:0.162872635541`

125.19 Analytic initial 22 action

Let $(v_1, v_2, v_3, v_4, v_5, Q_0)$ solve $T(v_1, v_2, v_3, v_4, v_5, Q_0)^T = (1, 1, 1, 4, 0, 0)^T$. The **exact** row I–22 is

$$(v_1, 0, 0, v_2, v_2 \mid v_3, 0, v_4, v_5, v_5),$$

where the vertical stroke separates early and late half-interval port vectors. Its two phase sums equal Q_0 exactly. There is no rounding of this row to a ten-decimal fixed grid; the policy uses these real analytic coefficients.

125.20 Exact common mean and reference shapes

The common mean is $M = \frac{164347980223}{5000000000}$. The nine vectors z_s^0 and d_s are shown as integers in units of 10^{-10}. Coordinate order: (r, w, p, b, t_1, t_2).

type	sym	r	w	p	b	t1	t2
z0	00	285185195436	167541458990	237719053301	178737308005	188736523029	188736523029
z0	01	289761207936	164615670333	227262641380	182052012922	192051227946	212051227946
z0	02	245554608451	173525632236	235553823475	183526417212	193525632236	225553038499
z0	10	250041410954	172282582576	240040625978	182283367552	212282582576	192282582576
z0	11	243315227169	155421997946	233314442193	193113997783	223113212807	223113212807
z0	12	243643722154	173640582250	233642937178	183641367226	203642152202	223642152202
z0	20	248496710078	167965302083	238495925102	183495925102	228495140126	193495140126
z0	21	243643722154	173640582250	233642937178	183641367226	223642152202	203642152202
z0	22	241898764484	171898764484	231898764484	181898764484	221898764484	221898764484
d	00	76961557465	-38932581080	28771198459	-28931796104	-18932581080	-18932581080
d	01	80205117876	-41542694250	15968090726	-31541909274	-21542694250	-1542694250
d	02	49828526159	-37966003887	22036351041	-27965218911	-17966003887	12035566065
d	10	80335412275	-41568753130	15942031846	-31567968154	-1568753130	-21568753130
d	11	37926624313	-37086995537	20423789439	-27086210561	2913004463	2913004463
d	12	33335439382	-36667700522	23334654406	-26666915546	-6666130570	13333869430
d	20	50607131176	-38121724891	21880630037	-28120939915	11879845061	-18121724891
d	21	32327479837	-37675660067	22326694861	-27674875091	12325909885	-1626332847
d	22	34124974753	-35878165151	24124189777	-25877380175	14123404801	14123404801

125.21 Exact held-input tables

E denotes first half-interval and L denotes second half-interval. Every row has ten port numbers in the order $(r, w, p, t_1, t_2 \mid r, w, p, t_1, t_2)$. Initial rows have label `I-ss`; second-sample bootstraps are `B-ss-tt`; recurrent rows are `T-ss-tt`. The special analytic `I-22` row is given in Section B and is omitted from the following integer table. Every other control is exactly the integer vector shown divided by 10^{10}; zeros are printed explicitly.

Early phase (values in units of 10^{-10})

ID	r	w	p	t_1	t_2
I-00	0	0	0	1685834602178	240884723729
I-01	0	0	0	0	1996235849540
I-02	0	0	1053140108347	0	898868703837
I-10	0	0	796180122390	0	1122198021806
I-11	0	0	0	2011312672794	0
I-12	126865321333	0	964378972332	0	880097171130
I-20	430209356296	0	0	0	1535229056332
I-21	126865321333	0	964378972332	0	880097171130
B-00-00	0	0	0	299526561492	1711786111302
B-00-01	0	0	131157482245	0	1801845652849
B-00-02	0	442716494935	0	1568596177859	0
B-00-10	3573357728	0	0	1904085068592	0
B-00-11	0	0	894475165670	0	1013183260649
B-00-12	424861334062	0	0	0	1586451338733
B-00-20	0	0	665192774425	0	1258182052170
B-00-21	0	646855856073	0	0	1260802570246
B-00-22	737302046995	0	0	0	1274010625799
B-01-00	0	0	0	291018425249	1687710258661
B-01-01	0	4631342199	0	0	1870443095236
B-01-02	154591214025	0	0	0	1720483223410
B-01-10	0	5533709403	0	1869540728032	0
B-01-11	239327029169	0	0	1635747408266	0
B-01-12	0	454624957685	0	0	1420449479750
B-01-20	0	0	665192774425	0	1225598063286
B-01-21	0	672581933492	0	0	1215608119057
B-01-22	737302046995	0	0	1241426636915	0
B-02-00	0	0	63851419511	244384672407	1666187318074
B-02-01	0	130352723701	0	1711868827422	79154581114
B-02-02	0	0	745458517972	0	1147243521193
B-02-10	0	128516619917	0	106476594164	1684986796591
B-02-11	282472456611	0	0	0	1608872198612

ID	r	w	p	t_1	t_2
`B-02-12`	0	521440058700	0	0	1369904596522
`B-02-20`	201218690640	0	0	0	1690125964583
`B-02-21`	0	713670957089	0	1177673698134	0
`B-02-22`	780447474436	0	0	1214551427261	0
`B-10-00`	0	0	55125985934	237303594044	1697177546479
`B-10-01`	0	124520251327	0	1698944364529	111964904082
`B-10-02`	0	297084957962	0	0	1606631728607
`B-10-10`	40822869868	0	0	1862893816700	0
`B-10-11`	5049549705	0	1014821232867	0	883845903997
`B-10-12`	79574815209	0	939021385153	0	885120486207
`B-10-20`	0	481493467734	0	1513488884921	0
`B-10-21`	0	709430622698	0	1196779120690	0
`B-10-22`	774551559136	0	0	1232819373907	0
`B-11-00`	0	0	91733479532	195641740646	1658109586886
`B-11-01`	0	148990347078	0	1679022878361	68036261282
`B-11-02`	12087870658	0	755451597719	0	1101988515744
`B-11-10`	7114895307	0	209614716978	1652798371838	0
`B-11-11`	298953374253	0	0	1570574609869	0
`B-11-12`	353197526249	0	0	0	1516330457873
`B-11-20`	0	337130077995	0	0	1532397906127
`B-11-21`	9461881546	724540651368	0	1135525451208	0
`B-11-22`	795381688272	0	5780743399	0	1172019798925
`B-12-00`	0	0	68063755859	1682793572333	215096612417
`B-12-01`	0	0	199221238104	0	1688423164805
`B-12-02`	0	0	749670854320	0	1134561715463
`B-12-10`	0	0	196474409559	18285796681	1671843910639
`B-12-11`	0	33937868730	984460518984	0	865834182069
`B-12-12`	88317073686	0	939021385153	856894110944	0
`B-12-20`	0	490141655243	0	0	1482863523807
`B-12-21`	0	718078810207	0	0	1166153759576
`B-12-22`	783293817613	0	0	1204592998645	0
`B-20-00`	0	0	63053791276	214620497070	1692065675589
`B-20-01`	0	0	194211273522	0	1697219152714
`B-20-02`	0	0	737018381522	0	1149385799794
`B-20-10`	0	0	191464444976	1156787330	1697768907899
`B-20-11`	0	436602798781	0	1449801382535	0
`B-20-12`	84931760329	0	939021385153	0	862451035834
`B-20-20`	0	310773043388	0	0	1575631137928
`B-20-21`	0	709748479708	7452288264	1169203413344	0
`B-20-22`	779908504256	0	0	0	1210149923535
`B-21-00`	0	45496876715	0	241844477818	1687429846611
`B-21-01`	0	0	199221238104	1671871386656	16551778150

ID	r	w	p	t_1	t_2
B-21-02	0	0	749670854320	1134561715463	0
B-21-10	0	0	196474409559	1632902677	1688496804643
B-21-11	0	33937868730	984460518984	0	865834182069
B-21-12	0	525847911819	0	0	1358384657965
B-21-20	13035532322	0	713965094500	0	1157231942961
B-21-21	0	718078810207	0	1166153759576	0
B-21-22	783293817613	0	0	0	1204592998645
B-22-00	0	0	74121327501	210136405910	1665038863786
B-22-01	0	82193031858	0	0	1787334187375
B-22-02	0	316961973623	0	0	1552565245611
B-22-10	0	83095399062	0	103817484894	1682614335277
B-22-11	17885006419	0	1014821232867	836820979948	0
B-22-12	0	532186647344	0	0	1337340571889
B-22-20	208158232053	0	0	0	1661368987180
B-22-21	467788032669	0	0	1401739186565	0
B-22-22	787387015850	0	0	0	1185794449858
T-00-00	0	0	0	299541395583	1711771277212
T-00-01	2990659833	0	0	0	1904667766487
T-00-02	0	0	681607098461	0	1247984203507
T-00-10	0	0	128410653700	1803552195364	0
T-00-11	0	0	967168134227	0	955846034273
T-00-12	0	454624957685	0	1453033468635	0
T-00-20	0	0	665192774425	0	1258182052170
T-00-21	245823996863	0	642391114032	1019443315425	0
T-00-22	737302046995	0	0	0	1274010625799
T-01-00	0	4105409156	6641151817	1714796838262	285769273559
T-01-01	0	0	143940364807	0	1788267145076
T-01-02	0	246257428859	0	1661400997461	0
T-01-10	0	0	29904595627	1877753830693	0
T-01-11	0	0	979951016788	0	942267526500
T-01-12	43993675824	0	949335223483	0	914329527012
T-01-20	0	0	677975656987	0	1244603544398
T-01-21	422131021611	0	0	0	1485527404708
T-01-22	738970419752	0	10313838330	1262028414713	0
T-02-00	0	0	57297531057	261555668241	1676743073220
T-02-01	0	59199831574	0	0	1848458594746
T-02-02	0	0	738904629519	1174970272175	0
T-02-10	0	0	185708184757	48091958501	1682446305530
T-02-11	0	1860988098	1021681609570	884115828651	0
T-02-12	328808436862	0	0	1578849989457	0
T-02-20	0	0	722490305482	0	1185168120838
T-02-21	278780235159	0	650916325014	0	977961866147

ID	r	w	p	t_1	t_2
`T-02-22`	766385580749	9529666935	0	1235397425111	0
`T-10-00`	0	0	12613364316	283522353694	1714441683320
`T-10-01`	0	0	143770846561	1703519497642	84977519427
`T-10-02`	0	0	694220462777	0	1234635567727
`T-10-10`	0	0	141024018016	70058643954	1720144915629
`T-10-11`	0	0	979781498542	0	942497398493
`T-10-12`	0	107944982606	852785747249	946927696465	0
`T-10-20`	0	0	606886562673	0	1300771863647
`T-10-21`	0	653524378789	0	0	1254134047531
`T-10-22`	741608203217	0	0	1269704469577	0
`T-11-00`	0	0	80078273341	1680350817743	228607908147
`T-11-01`	0	0	211235755586	0	1699491705945
`T-11-02`	0	509144771973	0	1496842730820	0
`T-11-10`	53213942894	0	0	157493961666	1696950521760
`T-11-11`	14510632440	0	1025771953975	867375839905	0
`T-11-12`	89035897944	0	949972106261	868650422114	0
`T-11-20`	339004001261	0	0	0	1672308671534
`T-11-21`	292534591739	0	653341835140	0	961781999440
`T-11-22`	786942632161	0	0	1224370040633	0
`T-12-00`	0	0	84901350365	231797353503	1670309546182
`T-12-01`	0	0	216058832610	1692639879741	0
`T-12-02`	0	12237132540	748201583350	1147219710430	0
`T-12-10`	0	0	213312004065	18333643763	1676012778492
`T-12-11`	0	656789859001	0	0	1354522813794
`T-12-12`	0	726305886511	0	0	1277621267254
`T-12-20`	18008507674	0	723443124987	1166206793659	0
`T-12-21`	667438304	728673559584	0	1178317428432	0
`T-12-22`	783961255917	10594749377	0	1216756667500	0
`T-20-00`	0	0	0	279516148827	1704551175130
`T-20-01`	0	0	187442016905	1682851442804	47353933206
`T-20-02`	0	292841616863	0	1614816809456	0
`T-20-10`	0	0	142332784300	55677197310	1709648444710
`T-20-11`	0	0	1023452668886	0	884205757434
`T-20-12`	328080582722	0	0	1579577843598	0
`T-20-20`	0	298233925644	0	1609424500675	0
`T-20-21`	0	710205048791	0	1202504581949	0
`T-20-22`	769502009431	0	8631436020	0	1233179227344
`T-21-00`	0	0	89928903348	1670767821698	225271238974
`T-21-01`	0	0	221086385593	1676498530024	10073510702
`T-21-02`	210754737433	0	0	1696903688887	0
`T-21-10`	59736881135	0	0	1847921545185	0
`T-21-11`	0	660913511320	0	0	1350399161474

ID	r	w	p	t_1	t_2
T-21-12	0	729666529122	0	1273871628625	0
T-21-20	20189917776	0	725242383806	1162226124738	0
T-21-21	2031508189	730684810368	0	1174942107763	0
T-21-22	790448203068	0	11277289305	0	1209587180421
T-22-00	0	0	74119533738	1665039046316	210137535984
T-22-01	0	0	205277015984	0	1665709562355
T-22-02	0	10069259041	740662925886	1118793975125	0
T-22-10	53657114508	0	0	1815869045543	0
T-22-11	0	448182160775	0	0	1421343999276
T-22-12	0	719098835186	0	0	1245719463433
T-22-20	17127518484	0	713965094500	0	1138433547067
T-22-21	295907753644	0	642391114032	931227292376	0
T-22-22	787385803776	0	0	0	1185794602750

Late phase (values in units of 10^{-10})

ID	r	w	p	t_1	t_2
I-00	886197076577	0	808399510823	0	305158072777
I-01	907150110441	0	727833742294	359529871363	16798948697
I-02	642294749975	0	680139192275	376218786809	312659943735
I-10	681621675133	0	741389272953	452075463069	136226261639
I-11	738168411852	0	765206094146	41585142566	466353024231
I-12	602315737156	0	669938058953	423036580458	316022296227
I-20	609962237989	0	792827541649	535369767141	73153126015
I-21	602315737156	0	669938058953	501889474462	237169402223
B-00-00	860880120834	0	789308851011	298254222453	0
B-00-01	867337508782	0	704837964488	362721316813	61043539778
B-00-02	679760502944	2121217079	705653510747	55675787172	505232176356
B-00-10	876562270809	13532753045	716643886506	40652053153	363921709281
B-00-11	637700803141	7989902658	655083594947	462255959683	248282412364
B-00-12	555190453729	40986764988	730944940796	438754909799	182566124986
B-00-20	706425694179	0	688369793884	495492463860	111492242243
B-00-21	578021131965	4230161833	700077283923	525130473690	203853621384
B-00-22	463065292907	0	731576240526	542864328451	273806810910
B-01-00	860662551887	0	789297241969	298483400442	0
B-01-01	876124770126	13654377353	716424756821	362395255721	42713512773
B-01-02	704743045564	49771440739	739263079790	383057140067	134477966635
B-01-10	876535624778	13204088128	716321702439	46396956841	358854300608
B-01-11	630312211692	67375054748	721313021966	125188406742	467123977647

ID	r	w	p	t_1	t_2
`B-01-12`	611132888672	35340974606	714191936629	437647370994	212999501893
`B-01-20`	706208125232	0	688358184843	493924813808	113289070284
`B-01-21`	573569809469	0	697511615893	523762701604	208513523140
`B-01-22`	462847723960	0	731564631484	279120553355	537779763995
`B-02-00`	868785985981	0	786235369579	305901505296	0
`B-02-01`	868313049418	0	706751686420	0	418032925391
`B-02-02`	709766646106	0	683867715266	373243509488	243611506956
`B-02-10`	869174575624	0	706920504672	417849371468	0
`B-02-11`	629889486274	67375054748	721238051853	469109344306	123700735613
`B-02-12`	613015404283	31372609440	710367022271	436322965566	220234671234
`B-02-20`	706589435014	47400897589	738874922854	499650744736	18796672601
`B-02-21`	579686078628	261796667	696240760524	273526236949	461597800026
`B-02-22`	462424998542	0	731489661372	283264764546	534133248335
`B-10-00`	867766898493	0	786657624788	304792964344	0
`B-10-01`	867604050780	0	706950795345	0	417523020924
`B-10-02`	714719048512	32126603264	722528541998	379719798372	162218680648
`B-10-10`	876070740003	13532753045	716571395951	43525184160	361612599635
`B-10-11`	638174088788	0	646098785210	455090877604	271948921193
`B-10-12`	602701699267	0	670106578114	421531318025	316973077388
`B-10-20`	687713421149	0	704140168481	179619745814	384483912690
`B-10-21`	578715061039	0	696281803598	266774235282	468029457478
`B-10-22`	462573762101	0	731503749971	276679941918	540555218805
`B-11-00`	871263634370	0	785121214917	308857681580	0
`B-11-01`	869799811123	0	706350594096	0	419076163108
`B-11-02`	709849953381	0	683590388294	366157537334	251714793786
`B-11-10`	877075626192	0	701381949706	81327261561	351527835335
`B-11-11`	629362651298	67758844415	721585870880	130458311976	462146994226
`B-11-12`	594497950436	62726412372	739980248634	427744030313	186364031039
`B-11-20`	717694193433	27761440454	720301585185	493733576123	51821877599
`B-11-21`	579534267634	0	695978526629	275498847690	460301030841
`B-11-22`	461944507883	0	731409383066	532738459812	285220322034
`B-12-00`	869270148284	0	786030907086	0	306445100465
`B-12-01`	875727536233	0	701560020563	356598400407	75357334195
`B-12-02`	710250808409	0	683663252772	371118691752	246279919860
`B-12-10`	876491444263	0	701799086852	431583725656	0
`B-12-11`	638873204688	0	646420932642	456195218124	269823317340
`B-12-12`	602473300078	0	670085075688	241225689575	497528607454
`B-12-20`	688756881961	0	703844324339	499917817269	64950339963
`B-12-21`	579758521850	0	695985959456	520186567152	215381624336
`B-12-22`	462345362912	0	731482247544	283302362512	534182699826
`B-20-00`	868633459180	0	786267371474	305866134930	0
`B-20-01`	875090847128	0	701796484951	352401969962	78974799105

ID	r	w	p	t_1	t_2
`B-20-02`	710183240280	840011048	684602344743	366992511611	248694565112
`B-20-10`	875854755158	0	702035551240	431004760121	0
`B-20-11`	644907298582	41443801013	696728757739	157947778162	470285037298
`B-20-12`	602507167036	0	670087179200	417925667081	320792659477
`B-20-20`	717265894689	28943080404	721427723386	494281736083	49394238232
`B-20-21`	579564722560	0	695903711269	268887513788	466956725177
`B-20-22`	462379229870	0	731484351057	533432653457	284016438410
`B-21-00`	866851264268	0	787771585532	301775367186	0
`B-21-01`	875727536233	0	701560020563	0	431955734602
`B-21-02`	710250808409	0	683663252772	127994389092	489404222521
`B-21-10`	876491444263	0	701799086852	431583725656	0
`B-21-11`	638873204688	0	646420932642	452678303720	273340231744
`B-21-12`	613087847504	31110812773	710112221204	430754200105	226247591208
`B-21-20`	712233669598	0	685994283044	486124218062	126960502090
`B-21-21`	579758521850	0	695985959456	270390360199	465177831289
`B-21-22`	462345362912	0	731482247544	534182699826	283302362512
`B-22-00`	869935675423	0	785734472048	307259951882	0
`B-22-01`	878153181385	9047737495	711976216717	355374323399	56761213798
`B-22-02`	715031196738	30946042951	721378389719	376778404174	167178639212
`B-22-10`	878564036037	8597448271	711873162334	412278026152	0
`B-22-11`	637800444261	0	646064214580	275360153909	452087860045
`B-22-12`	613161299931	30734334748	709743396524	430626438672	227047202918
`B-22-20`	706364554046	47400897589	738854440823	493920900924	24771879413
`B-22-21`	554838267424	42649025341	736274491656	218263595401	459287292972
`B-22-22`	462200117574	0	731469179340	534035286842	283608089038
`T-00-00`	860885199960	0	789310037490	298247956848	0
`T-00-01`	876187628572	13929447760	716697482729	363945414735	40552698999
`T-00-02`	701865860085	0	686942383177	377238475356	231962992640
`T-00-10`	868106495938	0	705078217256	60674200672	362712381367
`T-00-11`	632292594906	0	648401624101	461584629615	259720095737
`T-00-12`	611355536745	35340974606	714204732150	132346647046	518064782248
`T-00-20`	706430773305	0	688370980364	495489331058	111489109441
`T-00-21`	560858582288	0	688809978687	300678621630	460965490189
`T-00-22`	463070372033	0	731577427005	542861195649	273803678108
`T-01-00`	860863731075	0	789300928460	0	298278534762
`T-01-01`	867539386942	0	704672971876	363268798446	60941743681
`T-01-02`	713245662268	35830161406	726167992871	35214994456	500853861794
`T-01-10`	876590740480	12232101952	715143584814	46577190850	360769054698
`T-01-11`	632489393941	0	648235445010	462135244051	259621432442
`T-01-12`	603064640312	0	670129575350	427939582936	310178874196
`T-01-20`	706627572339	0	688204801272	496039945493	111390446146
`T-01-21`	555492166780	43333771974	737095858635	523822070585	151568804821

ID	r	w	p	t_1	t_2
`T-01-22`	462936703146	0	731526747207	276919567533	539929654908
`T-02-00`	866902332033	0	786925138838	304148202054	0
`T-02-01`	877651009021	10979375873	713902899571	362788485731	45990902598
`T-02-02`	707882992158	0	684557484525	126975291853	488126421349
`T-02-10`	874123628012	0	702693318604	429286827245	0
`T-02-11`	638210785807	0	646087925543	272786866542	454227094903
`T-02-12`	595091038802	62908620477	740248155406	103025628612	510039229497
`T-02-20`	712447905378	0	685986081712	493367393957	119511291747
`T-02-21`	560347808947	0	688706597894	514538906125	247719359828
`T-02-22`	462820042562	0	731570542215	280647458083	536274629934
`T-10-00`	861058325000	0	789152778556	298678055429	0
`T-10-01`	867515712948	0	704681892033	0	424188689566
`T-10-02`	702038985125	0	686785124243	374285522369	235346044207
`T-10-10`	868279620978	0	704920958322	423816680620	0
`T-10-11`	632465719946	0	648244365167	458631676628	263103147304
`T-10-12`	606016149173	0	671222769066	225861435263	508212319293
`T-10-20`	711885125021	7794983760	694733838763	493188274228	103710451022
`T-10-21`	578041183805	4520023483	700418799344	522342141434	205990524728
`T-10-22`	462855268732	685926804	732292021465	271907219161	543572236632
`T-11-00`	869426263522	0	785935120927	0	306592688831
`T-11-01`	875883651471	0	701464234404	356082516573	76020806395
`T-11-02`	683337765499	0	703046986850	65442748911	499845571912
`T-11-10`	875768280672	14259782999	717314834590	403969774533	0
`T-11-11`	637959421573	0	646031259901	272070711587	455251279734
`T-11-12`	602487032052	0	670039052805	238241973778	500544614159
`T-11-20`	665961113193	26772069825	730858382127	498155497924	26696131229
`T-11-21`	560147305141	0	688668776344	510311736935	252184854374
`T-11-22`	462271302770	727029954	732247188610	278746599228	537320552233
`T-12-00`	870230326781	0	785638828781	307315416297	0
`T-12-01`	876687714729	0	701167942258	11249799	432814800635
`T-12-02`	710560388306	0	683739358536	129981843594	487031082358
`T-12-10`	877451622759	0	701407008547	432454041488	0
`T-12-11`	611894810760	7639577392	676248860243	471118848562	181541097340
`T-12-12`	582312414970	0	691366871783	436863759177	242379692097
`T-12-20`	712208811136	0	685946477407	244334668319	368822715931
`T-12-21`	580023215747	0	696045434690	272460771180	462783251178
`T-12-22`	462610056808	0	731541722778	281855859089	535305034119
`T-20-00`	870952249432	6186402372	792153062128	295676619771	0
`T-20-01`	873218250426	0	702507556827	0	429528251042
`T-20-02`	714497615889	32951679194	723377148417	37946442891	502539786403
`T-20-10`	877136794606	4656179154	706641286061	422878412973	0
`T-20-11`	638168257424	0	646070029961	454266802887	272807582522

ID	r	w	p	t_1	t_2
`T-20-12`	595098519240	62915672870	740256970492	97661449445	515380060748
`T-20-20`	716952775098	30260869367	722761322565	157227111076	384110594687
`T-20-21`	577667851936	0	696632251157	264862873628	469085985725
`T-20-22`	462567930738	0	731474994722	536214229684	281055517651
`T-21-00`	870613778105	0	785474786039	0	307726973087
`T-21-01`	877071166054	0	701003899517	0	433237607224
`T-21-02`	704076777833	50520151901	739955011882	18826857890	497933873288
`T-21-10`	875678434131	14281464207	717324209557	45083303218	358945261682
`T-21-11`	611974021483	7514115026	676137642487	467776969180	185040446121
`T-21-12`	582517194586	0	691331404860	164487788801	514822287790
`T-21-20`	712160173134	0	685933450873	241800659806	371418388981
`T-21-21`	580029544822	0	696052773785	269825174751	465405179437
`T-21-22`	462271866448	0	731421415373	535308959343	282310431631
`T-22-00`	869935440044	0	785734556950	0	307259751771
`T-22-01`	876392827992	0	701263670427	352848833018	79921552890
`T-22-02`	710380758592	0	683752145393	131308414566	485871354244
`T-22-10`	875697100182	13532753045	716536826312	50453312287	355092680968
`T-22-11`	645127699786	40756065215	696063704559	464768982481	164596220754
`T-22-12`	582400697860	0	691186863246	432078669220	247848838798
`T-22-20`	712088428967	0	685981215831	485976818374	127266209622
`T-22-21`	559988332536	0	688701732014	310483013566	452139594678
`T-22-22`	462200122281	0	731469180331	534035300138	283608070044

125.22 Proof-verification protocol

For each tested plant generator $K = I + L$ and held duration $h \in \{1, 1/2\}$, form P by selecting only the five open heater columns in each six-node cell. The exact transition operators are $E_h = e^{-hK}$ and $C_h = K^{-1}(I - E_h)P$. Use a rational number $s \geq \max_i (hK)_{ii}$ and set $B = sI - hK \geq 0$ componentwise, so that

$$e^{-hK} = \frac{\sum_{k\geq 0} B^k/k!}{\sum_{k\geq 0} s^k/k!}.$$

For a fixed integer cutoff $N = 115$, replace the two series by their rational partial sums. For $z \in [0, N+2)$, an upper bound on the omitted positive tail is

$$R_N(z) = \frac{z^{N+1}}{(N+1)!\,[1 - z/(N+2)]}.$$

Because $B \geq 0$ and its row sums equal $\beta = s - h$, its tail has entrywise upper bound $R_N(\beta)$. Combining the numerator and denominator tails gives the fully rational entrywise enclosure

$$\left| e^{-hK} - \widehat{E}_h \right|_{ij} \leq \left(\frac{1}{S_N(s)} - \frac{1}{S_N(s) + R_N(s)} \right) S_N(\beta) + \frac{R_N(\beta)}{S_N(s)},$$

$$\widehat{E}_h = \frac{\sum_{k=0}^{N} B^k/k!}{S_N(s)}, \qquad S_N(z) = \sum_{k=0}^{N} \frac{z^k}{k!}.$$

For the 12-node crossed plant one may take $(s, h) = (9, 1)$ or $(5, 1/2)$; the resulting enclosure has error less than 10^{-45}. Matrix inversion is exact over $\mathbb{Q}$, and ordinary outward-rational interval propagation through all matrix products yields certified lower and upper entries of C_h, $E_h C_h$ and the two-hold input map. Other sample rates use any larger rational s with $s < N + 2$.

For every printed Farkas row y, form the relevant stacked guard matrix H, right-hand side b and held-input map $\mathcal{C}$ in the displayed row order. Test $y_i \geq 0$, the rational inequality $y^T b > 0$ and upper bounds on every component of $y^T H\mathcal{C}$ strictly below zero. For every printed H2 action, sum each five- or ten-port half-phase vector to check its budget, multiply the rational exponential enclosures by the action and test the specified guard rows with lower endpoints. The residual bounds and contraction estimates appearing in the main proofs then extend finite tests to all words and all times specified in those theorems. In the crossed-network 81-action certificate, compute $m_{h,j} = \min_s hE^j Du_s$ for $1 \leq j \leq 11$, then apply the displayed geometric bound for all $j \geq 12$. All such operations involve rational integers, comparisons, additions, multiplications and exact linear algebra; no released software or external table is needed.

125.23 Arbitrary-type intermodule couplings

Independent word alphabet. For m cells, each sample letter is the vector $(s^{(1)}, \dots, s^{(m)}) \in \Sigma^m$. The cross-module links may join any different node types but have symmetric nonnegative conductances with weighted degree at most Γ at each vertex, even when they vary with time. Each module has its own five open actuator ports, the same closed b port, and the system's peak heater sum is at most $Q = mq$. The local terminal conductance deviations obey $|\kappa_i^{(a)}(t) - 2.11| \leq \eta$.

Exact constant audit. The prior fixed strict initial-22 mixture with $\alpha = 1/3200$ has nominal all-guard margin $g_* = 0.00003121875$, and the exact nominal power satisfies $q = Q_0 + 0.004 < 202$. Choose

$$\eta = \frac{3}{200000000} = 1.5 \cdot 10^{-8}, \qquad \Gamma = \frac{1}{10^9}, \qquad q < 202.$$

By leaky positivity, independent nominal state coordinates under each per-module H2 control lie in $[0, q]$. The time-varying total cross-Laplacian has at most Γ total conductance incident to each vertex. Its action on the nominal state is bounded coordinatewise by Γq; signed terminal edge perturbations add at most $4\eta q$. The positive coupled semigroup has row sum e^{-t}, so the state infinity-norm deviation is at most $(\Gamma + 4\eta)q$. Each guard thus remains strictly larger than

$$g_* - 2(\Gamma + 4\eta)q > 0.00003121875 - 404(6.1 \cdot 10^{-8}) = \boxed{0.00000657475}.$$

The 171-action lookup is applied separately to each module's word; the resulting physical controls meet the global power cap and require no future labels.

Finite-power H1 obstruction for independent words. Fix a requested two-sample word in every module. Every module has a certified nonnegative Farkas multiplier vector: choose its first-symbol no-go if the first symbol is not 22, or the $22 \to s$ two-symbol no-go otherwise. All 17 witnesses have $\sum y_j \le 1 + 10^{-12}$ and $y^T c \ge \delta_* = 26059/16000000000$. The isolated nonidentical-cell H1 reduced input coefficients remain negative for $\eta \le 1.5 \cdot 10^{-8}$ by the earlier case. For the same input distribution, the coupled and isolated trajectories differ in ℓ_1 by at most $2\Gamma Q$ at every instant, since $\|C_\times\|_1 \le 2\Gamma$ and the isolated total heat is at most Q. Summing the cellwise duals over the first two observations, their entire cumulative perturbation is bounded by $4\Gamma Q(1 + 10^{-12})$. At $Q = mq$ and $q < 202$ this becomes per module

$$4\Gamma q(1 + 10^{-12}) < 8.081 \cdot 10^{-7} < 1.6286875 \cdot 10^{-6} \le \delta_*.$$

Hence no length-two vector word is feasible under H1 at this *fixed power budget*. The claim is not the prior all-budget no-go, which was proved only with matching-type module links.

Two strong-link lower certificates (without linear programming). For the two edges (r_1, w_2) and (r_2, w_1) of weight g, direct subtraction of the local r, w equations gives

$$Y' = -(5 + 2g)Y + 3\sum_a (w_a - p_a) + \sum_a (u_{r_a} - u_{w_a}), \quad Y = \sum_a (r_a - w_a).$$

Nonnegative heat and $\sum_i x_i(t) \le Q$ imply $Y' \le -(5 + 2g)Y + 4Q$. At any valid sample $Y \ge 13.502$, so $Q \ge 13.502(5 + 2g)/4$; at $g = 60$, this is exactly 421.9375. Even the first common $(22, 22)$ symbol fails at $Q = 2q < 404$.

For six corresponding-type edges of weight g between the two cells, the cell contrast $z = x^{(2)} - x^{(1)}$ obeys $z' = -(K_0 + 2gI)z + (u^{(2)} - u^{(1)})$. Thus $\|z\|_\infty \le Q/(1 + 2g)$, but the first symbols $(00, 22)$ require $g_1^{(2)} - g_1^{(1)} \ge 3$, giving $Q \ge 3(1 + 2g)/2$; at $g = 200$ this is 601.5.

Numerical cross-check. The adjacent Direct full-network matrix-exponential reconstruction independently recomputes complete matrix exponentials under different words, random cross-type edges and changing local parameters on 12, 18 and 30-node networks. It also solves a separate one-sample H2 linear feasibility problem for the two strong-link examples, both infeasible at the old two-module budget. The all-size/all-word guarantee rests on the exact inequalities and tabulated rational witnesses, *not* on these finite tests.

125.24 Cross-link first-sample phase and quotient obstruction

Physical and logical contract. Two nominal six-node cells are joined by exactly the symmetric cross edges (r_1, w_2) and (r_2, w_1), each with weight g. Both separator heater ports remain shut, all other heater phases are nonnegative, and local terminal numerics are sampled on the induced six-node trees. The two-vector alphabet contains $9^2 = 81$ possible first letters; each supplies 18 affine guards. The claim for H2 in this section is *one sample only*, not all word lengths.

Complete printed certificates. At $g = 7/1000$ Section 125.27 contains exactly 81 rows, indexed by $(s,t) \in \Sigma^2$. At $g = 1/100$ the same appendix contains the corresponding 81-row family. Each row has an H1 rational nonnegative 18-vector y, a 10-vector held-input reduced-cost target, and two H2 rational nonnegative ten-port phase actions. The rational-enclosure inequalities yield the following conservative strict bounds:

g	$\min y^T b$	$\max_j (y^T H C_g)_j$	H2 least guard
$7/1000$	$> 2.03 \cdot 10^{-6}$	$< -2.03 \cdot 10^{-6}$	> 0.1047
$1/100$	$> 6.03 \cdot 10^{-6}$	$< -6.03 \cdot 10^{-6}$	> 0.1039

In both cases the sum of all ten open heater inputs is strictly < 421 in each H2 half interval. H1 has no applicable power cap in the contradiction. The complete rational dual and action records for both conductances are printed in Section 125.27.

Rigorous exponential interval algorithm. For either $A = K_g$ or $A = K_g/2$, select $s = 9$ and $s = 5$, respectively, so $B = sI - A \geq 0$ entrywise and all row sums are $\beta = s - 1$ or $s - 1/2$. Let $N = 115$. Construct the exact rational matrix polynomial $P_N(B) = \sum_{k=0}^{N} B^k/k!$ using integer matrix products with a common denominator. For any $z \geq 0$, the scalar Taylor remainder obeys

$$R_N(z) = \sum_{k=N+1}^{\infty} \frac{z^k}{k!} \leq \frac{z^{N+1}}{(N+1)!} \frac{1}{1 - z/(N+2)}.$$

Set $S_N(s) = \sum_{k=0}^{N} s^k/k!$ and $\widehat{E} = P_N(B)/S_N(s)$. Positivity gives the *entrywise*, rigorous error bound

$$\left|e^{-A} - \widehat{E}\right|_{ij} \leq \varepsilon_A := \left(\frac{1}{S_N(s)} - \frac{1}{S_N(s) + R_N(s)}\right) S_N(\beta) + \frac{R_N(\beta)}{S_N(s)}.$$

The rational enclosure evaluation yields $\varepsilon_A < 10^{-45}$ in both cases (actual bounds are much smaller). The inverse K_g^{-1} is computed as an exact rational nonnegative matrix with row sums one. Therefore the coefficient matrix $C_g = K_g^{-1}(I - e^{-K_g})P_{10}$ is within ε_{K_g} per entry of its rational approximation. The H1 signed guard multipliers add at most $2\varepsilon_{K_g} \sum_j y_j$ to any reduced coefficient. For H2, the early response $E_{g,h} K_g^{-1}(I - E_{g,h})P_{10}$ has entrywise error at most $2\varepsilon_{K_g/2}$, and the late response at most $\varepsilon_{K_g/2}$. Thus the loss in any two-coordinate guard under one control pair is bounded by $2\varepsilon_{K_g/2}(2\sum v^- + \sum v^+)$. All inequalities are checked with exact `Fraction` against these upper bounds; solver tolerances are not part of the proof.

A certified nonmonotonicity-free bracket. At $g = 1/200$ both modules can take the same rational full-period action

$$(57.31316857, 0, 76.95531716, 49.53599898, 49.53599898).$$

The full two-cell first letter $(22, 22)$ then has 9×2 guard margins greater than 0.0002973 under H1, and the global power is 466.68096738. The fraction proof uses $N = 115$ with the local mean quotient. At $g = 7/1000$, all 81 first letters are ruled out. This is an explicit change of feasible behavior between the *two tested conductances*; no uniqueness of a critical g or global monotonicity is inferred.

Large-link quotient obstruction. For $g = 5$ and arbitrary controls, the two-cell average $\bar{x}$ obeys

$$\dot{\bar{x}} = -(K_0 + 5(e_r - e_w)(e_r - e_w)^T)\bar{x} + \bar{u}.$$

The effective single-cell r–w conductance is 7. The common structural five-guard multiplier is

$$\lambda = (0.287949535, 0, 0.473411111, 0.119319677, 0.119319677).$$

Its rational-exponential reduced costs on the five allowed input ports are all < -0.0025 and its guard right-side pairing is > 0. Hence even the common structural chamber is

H1-impossible at the first sample for unlimited nonnegative held inputs. For H2, replicate in both cells the rational phases

$$v^- = (0, 0, 0, 34.77637637, 315.22362363),$$
$$v^+ = (132.78157732, 0, 96.03257593, 59.22748101, 0).$$

The global per-phase power is at most 702, and each local 22 guard has a rigorous margin > 0.096.

Reproduction. All the displayed coefficient and guard inequalities follow from exact rational matrix enclosures and the certificate tables printed in this chapter. Floating-point candidate discovery, finite simulations and LP tolerances are not part of these sign proofs.

125.25 Certified unique local coupling threshold

Exact analytic contract. On the interval $[g_-, g_+] = [0.0054757560, 0.0054757561]$, the two-module plant has exactly two cross-edges $(r_1, w_2), (r_2, w_1)$ of common weight g. The quotient matrices $T(g), A(g), c(g)$ and $G(g) = c(g)A(g)^{-1}\mathbf{1}_3$ are defined in the main text. The critical equation is

$$\det \begin{bmatrix} A(g) & \mathbf{1}_3 \\ c(g) & 9/2 \end{bmatrix} = 0.$$

The symmetrized common-first-letter candidate uses $u_w = u_b = 0$ and three independent nonnegative heater controls u_r, u_p, u_t, with u_t applied at both terminal ports. The active rows are $r - p$, $b - w$, and $p - t_i$, all equal to one. The remaining upper terminal guard is $t_i - b \le 9/2$. Its value is exactly $G(g)$.

Endpoint and curvature certificate. The pure-rational The exact endpoint certificate recomputes $e^{-\widetilde{K}(g)}$ via a positive Taylor-uniformization series, with a rigorously bounded tail, at both exact rational endpoints. After rounding the resulting exact rational approximations to a 10^{-25} grid and allowing the rounding error, the full matrix T has an entrywise enclosure of radius less than $2 \cdot 10^{-25}$. The inverse of the rounded 3×3 matrix is computed exactly, and matrix-perturbation estimates give:

Certified quantity	Rigorous inequality
$G(g_-) - 9/2$	$< -1.0009 \cdot 10^{-10}$
$G(g_+) - 9/2$	$> 2.4365 \cdot 10^{-10}$
Endpoint error in G	$< 2.1 \cdot 10^{-20}$
Secant slope	> 3.43744549
$\|A(g)^{-1}\|_\infty$ on the interval	< 80
$\|G''(g)\|$ on the interval	$< 9 \cdot 10^6$
$G'(g)$ on the interval	> 3.4365

The derivative bounds follow from $T(g) = \int_0^1 e^{-t\widetilde{K}(g)}dt$ and $\|\partial_g T\|_\infty \le 1$, $\|\partial_g^2 T\|_\infty < 4/3$, using the positivity and unit-row-sum property of the heat semigroup after leakage. The unique root g_* therefore lies strictly between the two rational endpoints; no global monotonicity outside that interval is assumed.

Exact local Farkas completion. Let $a(g) = c(g)A(g)^{-1}$. The displayed inequalities establish on the entire interval that all three components $a_i(g) > 1/2$ and that the unheated-w reduced coefficient of the four-guard dual $(a_1, a_2, a_3, 1)$ is $< -1/5$; the other three controlled-column reduced coefficients vanish identically. The dual right-hand side is exactly $G(g) - 9/2$. This is a complete primal–dual characterization of the common first vector letter $(22, 22)$ in the interval. The controls have combined power < 467 wherever $G(g) \le 9/2$, and at the root the closed upper band remains allowed.

All other 80 vector letters. Section 125.27 prints the complete 81-entry nonnegative rational Farkas family at g_+: 80 previously certified multipliers and one new high-precision symmetrized critical-pair multiplier. For the other 80 pairs, all right-hand sides are $>$

0.00051730, all held-port reduced coefficients are < -0.00047918, and each multiplier's ℓ_1 norm is < 2. Since the derivative of the first-sample input map has infinity norm at most one, their reduced coefficients change by at most $4(g_+ - g_-) = 4 \cdot 10^{-10}$. Thus the same 80 Farkas witnesses exclude all these letters *throughout* the interval. The new $(22, 22)$ Farkas vector at g_+ has a positive right-side pairing $> 1.88 \cdot 10^{-13}$ and every reduced input coefficient $< -9.4 \cdot 10^{-14}$, using 115 exact positive-series terms and a tail error smaller than 10^{-45}.

All 81 H2 first letters. The right endpoint's 81 exact rational nonnegative twenty-input half-period actions have each global phase power < 421 and minimum first-sample guard margin > 0.09644. They remain valid for every g in the narrow interval, because Duhamel gives a coordinate error at the first sample of at most $421|g - g_+|$, hence a guard error of at most $842 \cdot 10^{-10}$. This proves a *one-sample* H2 statement; it does not assert an all-horizon controller at these couplings.

Additional explicit primal at the left endpoint. The printed left-endpoint witness certifies one rational H1 held control applied in each module,

$$\begin{aligned} u_r &= 57.316257207698142887177967 26, \quad u_w = 0, \\ u_p &= 76.955659700849080528348534 65, \\ u_{t_1} &= u_{t_2} = 49.538184786128927699334308 44. \end{aligned}$$

The two-cell peak power is less than 467 and the minimum sampled-guard margin is rigorously greater than 10^{-12}. The printed right-endpoint certificate family certifies 81 H1 duals and 81 H2 first-step controls at the opposite endpoint. Only the rational enclosures, algebraic derivative bounds, and Farkas/positive control witnesses certify the phase. The high-precision decimal estimate of g_* is separately labeled exploratory and is *not* asserted with more certified digits than the displayed rational bracket.

125.26 Exact all-horizon two-cross-link certificates

Let K_g be the 12-node leaky generator with the two cross edges $r_1 w_2$ and $r_2 w_1$ each of conductance g. Use the exact rational endpoints

$$g_- = 54757560/10^{10}, \quad g_+ = 54757561/10^{10}, \qquad Q = 421.$$

The main text proves the determinant-root first-sample phase on this interval. The following two independent certificate families complete the phase for all word lengths.

A. H2 symbol-only controller for all 81^n vector words

The complete action table in Section 125.27 lists 81 rational vectors $u_s \in \mathbb{Q}^{20}_{\geq 0}$ with eight digits of decimal precision: ten first-half and ten second-half heater rates. Each half-phase sum is strictly below 421. Index s by the current pair of three-band terminal words; no previous symbol, future target or measured state is used.

At g_+ let $E = e^{-K_{g_+}}$ and $D = [e^{-K_{g_+}/2} C_{1/2}, C_{1/2}]$ with $C_{1/2} = K_{g_+}^{-1}(I - e^{-K_{g_+}/2}) P_{10}$. The exact state after n samples is

$$x_n = \sum_{j=0}^{n-1} E^j D u_{s_{n-j}}.$$

For each of the 18 guard normals h and each lag $j \geq 1$ put $m_{h,j} = \min_{s \in \Sigma^2} hE^j D u_s$. Section 125.22 prescribes the computation of exact-rational outward matrix-exponential enclosures from 115 nonnegative Taylor terms and checks the finite family

$$\min_{s,h} \left[hDu_s + \sum_{j=1}^{k} m_{h,j} - b_{s,h} \right] > 0.0397542, \qquad 0 \leq k \leq 11.$$

The 12-lag cutoff is guaranteed by positivity:

$$\sum_{j=12}^{\infty} |hE^j Du_{s_{n-j}}| \le 2(421) \sum_{j=12}^{\infty} (3/8)^j = \frac{223736661}{21474836480} < 0.010419.$$

All proxy entries are rational multiples of 10^{-16}, each entry enclosing the exact matrix coefficient with error below 10^{-16}. For the first 12 lags the matrix product row error is below $3 \cdot 10^{-14}$, yielding a guard expression error at most $2(421)(3 \cdot 10^{-14})$ per lag. These errors are *included* in the reported guard margins.

Quantity	Verified bound
81 input vectors, each of 20 coordinates	Nonnegative rationals
Largest first-half power	< 420.999958
Largest second-half power	< 420.999958
Minimum finite-prefix guard margin ($k \le 11$)	> 0.0397542
Bound on all remaining lags	< 0.010419
Uniform all-horizon guard margin at g_+	> 0.02933572
Uniform all-horizon margin throughout $[g_-, g_+]$	> 0.0293355
Margin under $\lvert g(t) - g_+ \rvert \le 10^{-5}$	> 0.01249

The last two lines use the global leaky-semigroup Duhamel bound $\|x_{g(\cdot)}(t) - x_{g_+}(t)\|_\infty \le 2Q \sup |g(t) - g_+|$ and a factor two for each signed guard. All horizons are covered without enumerating target sequences.

B. H1, every two-sample vector word impossible

At g_- the exact input–output map for the first joint vector symbol $(22, 22)$ followed by any $t \in \Sigma^2$ is

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} C & 0 \\ EC & C \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}, \quad C = K_{g_-}^{-1}(I - e^{-K_{g_-}})P_{10}.$$

The full two-step dual table in Section 125.27 contains *exact decimal-rational* 36-component nonnegative Farkas vectors y_t for all 81 second letters. Their entries were initially discovered numerically but the final proof checks each integer decimal as an exact fraction. The rational-arithmetic two-step certificate verifies

$$y_t^T b_t > 4.4101 \cdot 10^{-6}, \qquad \max_{i \le 20} (y_t^T A_{g_-})_i < -4.4101 \cdot 10^{-6}.$$

After including the exact matrix-exponential error, the worst coefficient is below $-4.41010665 \cdot 10^{-6}$. The Duhamel variation on $[g_-, g_+]$ is below $7 \cdot 10^{-10}$ per reduced coefficient; hence all 81 certificates retain coefficients below $-4.4094 \cdot 10^{-6}$ uniformly over the bracket. The first letter cannot be anything but $(22, 22)$, by the critical-coupling theorem. Thus every two-sample word is impossible at every finite H1 budget; all longer words are excluded by their first two letters.

C. Diagnostic physical replay

An independent full-network matrix-exponential computation computes the actual 12-dimensional matrix exponential for 37,500 samples across 15 target/coupling protocols, including alternating extreme symbols and conductances switched every half-period. Its lowest observed guard margin is above 0.03972. These finite numerical experiments are diagnostics only; the arbitrary-word result follows from Sections A and B above.

125.27 Full finite certificate tables for the crossed network

All vectors below are printed as exact rationals; no auxiliary data or software is needed to recover their entries. Each listed coordinate is identified by its *one-based* position, and any unlisted coordinate equals exactly zero. In a row with scale 10^{-d}, `index:numerator` means that the entry at that coordinate is the displayed integer divided by 10^d. A parenthesized rational `index:(a/b)` denotes exactly a/b without further scaling. All signed guard rows follow the fixed order of the nine inequalities for each cell: $(r-p,\ p-w,\ b-w,\ p-t_1,\ p-t_2,\ t_1-b,\ b-t_1,\ t_2-b,\ b-t_2)$, with lower and upper band right sides as in Section 125.2; the 18 two-cell rows list cell 1 first, then cell 2. For a two-sample H1 dual, the 36 entries list the 18 first-sample guards and then the 18 second-sample guards. The twenty heater coordinates in H2 are first the ten cell-major open ports (r, w, p, t_1, t_2) at the early half, then the same ten ports at the late half. All actions are held powers, not energy integrals.

125.28 Eighty-one exact all-horizon H2 controls

This is the memoryless policy of the all-horizon theorem. Each $u_s \in \mathbb{Q}^{20}_{\geq 0}$ is the integer numerator printed in the table times 10^{-8}; use the current vector letter $s = (s_1, s_2)$ to index the row. The two 10-port phase sums are strictly less than 421.

Label	Nonzero coordinates written as `index:numerator`
00/00	4:13102581772 5:6685307180 9:13113964668 10:9198142168 11:5828848705 13:5629309273 15:1355260259 16:5872256544 18:5383928769 20:826980136
00/01	3:433603236 4:12510644908 5:12510644908 7:470445039 9:14305773675 10:1868884022 11:5946721885 13:5169169354 16:6131261282 18:6441825884 20:3520413858
00/02	4:13122192009 5:9465696269 10:19512107511 11:5878721760 13:5361540963 15:772213077 16:7879403195 18:8448038811 19:3941154556 20:1094984174
00/10	4:10799850150 5:13121815266 9:3753129602 10:13611752761 11:5900166445 13:5227695393 14:490374379 16:5776389346 18:5756138548 19:2568087235
00/11	4:12640787094 5:13358693733 6:1556202898 9:11181492558 11:6244514636 13:5169617826 14:151614259 16:5434394232 18:6367023155 19:1692844103 20:4054256614
00/12	4:12401377411 5:12401377411 7:194628920 10:4528079885 11:5920319433 13:5189087045 16:5936193743 18:7294967076 19:4205635029 20:3735403330
00/20	4:13122192009 5:9465696269 9:850399677 10:18661707833 11:5878721760 13:5361540963 15:772213077 16:7879403195 18:8448038811 19:5036138731
00/21	4:8135554684 5:12493904659 10:4320299084 11:5846119898 13:5570498287 14:1025776614 16:5929974628 18:7294232140 19:4672246066 20:3273794073
00/22	4:13154051719 5:12655260668 9:1712702553 11:6122757071 13:5154492514 16:6093927221 18:7731165946 19:4668502278 20:5030206873
01/00	4:13607423669 5:2595895104 9:13152326795 10:12653504610 11:5756659020 13:5869582302 15:2811575191 16:6120367011 18:5153793216
01/01	4:13381383558 9:13610799783 10:3750645172 11:5820921376 13:6255761306 14:164961186 15:3779496180 16:5776518490 18:5755898206 20:2568410661
01/02	1:2053038903 4:3686460353 5:17048334578 9:18605311909 10:706850044 11:5626095406 12:67380861 13:5870490395 14:2335828696 16:7833667542 18:8426949681 20:5054544669
01/10	4:13611427664 5:3751790981 7:83033590 9:13497197171 11:5776672373 13:5756008622 15:2568307855 16:5827962418 18:6259211395 19:938967483 20:3000903360
01/11	4:1447775020 5:15911996304 10:10421142318 11:5776564524 13:5755763310 14:2568632691 16:5639644914 18:6253938117 19:3834494725 20:1331181948
01/12	4:13611000541 5:3750756288 9:4318675286 11:5776964840 13:5755925154 15:2568436168 16:5929978106 18:7294115038 19:3274042664 20:4672151728
01/20	1:2053038903 4:3686460353 5:17048334578 9:706850044 10:18605311909 11:5626095406 12:67380861 13:5870490395 14:2335828696 16:7833667542 18:8426949681 19:5054544669
01/21	4:1449233050 5:15912523779 9:4318675286 11:5776964840 13:5755925154 14:2568436168 16:5929978106 18:7294115038 19:3760093360 20:4186101031
01/22	4:13611346362 5:3751701794 9:1706684368 11:5777179825 13:5756027576 15:2568309521 16:6093723059 18:7730728337 19:4669424694 20:5029858312
02/00	5:19524887432 9:9453412756 10:13121695600 11:7883133387 13:8449111065 14:3941931117 15:1093068327 16:5878368564 18:5362592945 19:774702389
02/01	4:18602471049 5:701085874 8:3826493504 9:1468731309 10:16862502048 11:7833190647 13:8426279820 15:5055168616 16:5996034108 18:5667904050 19:2764950242
02/02	1:935756442 4:18898070887 5:1157523413 6:975009318 10:20133635727 11:7620705192 13:8408906847 15:5021201401 16:7620391772 18:8413255765 19:3996482675 20:1019052133
02/10	4:18473518836 5:377636276 9:20662430603 10:2586410074 11:7734952738 13:8380390928 15:5096244399 16:7077185227 17:1017447400 18:6918841065 20:3028934755
02/11	5:19529467021 10:12053937314 11:7884322980 13:8449634292 14:3942204530 15:1092374579 16:5981316891 17:308361262 18:6507025132 19:4075255765 20:1529592185

Label	Nonzero coordinates written as `index:numerator` (cont.)
02/12	4:18549689237 5:565405543 6:3851905298 10:11037461083 11:7789888647 13:8406016452 15:5072675903 16:5744376041 18:7607651189 19:4662165725 20:2817221829
02/20	1:935756442 4:18898070887 5:1157523413 6:975009318 9:20133635727 11:7620705192 13:8408906847 15:5021201401 16:7620391772 18:8413255765 19:1019052133 20:3996482675
02/21	5:19027960476 6:5587490335 10:11721233679 11:7769182272 13:8396540457 14:3912355259 15:1168438166 16:5297823931 18:7488141036 19:5219889063 20:2258438262
02/22	1:885795227 5:20030552438 6:8156258448 9:13027389675 11:7629432510 13:8408987816 14:3989053666 15:1033399796 16:4959916138 18:7671615130 19:2828172174 20:5579418557
10/00	3:752403888 4:3585993457 5:13182304709 9:13012175270 10:11567118464 11:5808118069 13:5838106196 14:2618035694 16:5913587490 18:5187464078 20:199834705
10/01	4:3751683351 5:13611379405 9:13610919846 10:3750939586 11:5776555404 13:5755999699 14:2568320394 16:5776398924 18:5755922009 20:2568373840
10/02	3:3642484417 4:16817629266 5:1469387269 9:18603246839 10:703186132 11:5985721867 13:5672304704 15:2755328480 16:7832961113 18:8426818707 20:5054882327
10/10	4:15912676653 5:1449566932 9:15482758908 11:5776558664 13:5755941333 15:2568397941 16:5744001128 18:5938306457 19:88831422 20:2872575021
10/11	4:15801246325 5:2012000495 10:10421375502 11:5786665768 13:5735213739 15:2320745667 16:5639644517 18:6253954982 19:3532030099 20:1633624405
10/12	4:1436821441 5:14021622921 9:4321871101 11:6083659385 13:6416227187 14:3551644690 16:5930307809 18:7294360810 19:3273550161 20:4672334147
10/20	4:8481574362 5:14428794355 9:619754222 10:18569872849 11:6431355783 12:535795907 13:6072740155 14:1847386364 16:7808463695 18:8415150784 19:5065454028
10/21	4:4831393847 5:13009202559 6:2581309083 9:11266867411 11:5788019427 13:5736724968 14:2318463019 16:6013325534 18:7573238552 19:2614349059 20:4205262971
10/22	4:3751870883 5:13611410470 9:2160188170 11:5777179429 13:5756044442 14:2568287350 16:6103824933 18:7710177484 19:4550258826 20:4901121683
11/00	2:397326304 4:9544036846 9:12039411475 10:12512280750 11:5623497743 13:6314483473 14:1925811183 15:3941410433 16:5913181836 18:5189517608 19:205210919
11/01	1:1559994020 5:7807378089 9:13613226761 10:3755315374 11:5375624792 13:6694058679 14:4050869828 15:2402034469 16:5776816698 18:5756108392 20:2567936916
11/02	2:3316699046 5:13956531002 10:19328252409 11:5747848678 13:6266157472 14:4165452949 15:915510193 16:7836288795 18:8428642325 19:3930179882 20:1122837746
11/10	4:15161084006 10:13385530074 11:5728184765 13:5861390831 14:266563397 15:3468423368 16:5821786159 18:6256344360 19:3779736333 20:164325639
11/11	4:10419131296 9:10417850920 11:5639811843 13:6253771936 14:1633990542 15:3531928859 16:5639654034 18:6253701730 19:1331685649 20:3834303318
11/12	5:9119006784 10:4314714325 11:5611207436 13:6312939838 14:3901671675 15:1975834262 16:5929988364 18:7293830226 19:4185870751 20:3760699593
11/20	5:12059579187 9:853615383 10:18662974925 11:5982026218 12:308825193 13:6507379869 14:4075607882 15:1528752799 16:7880570564 18:8448552291 19:5035727204
11/21	4:10421116801 9:4315383888 11:5640212159 13:6253933780 14:1331211032 15:3834511845 16:5929987226 18:7293878652 19:3760597061 20:4185909624
11/22	4:15163004357 9:1705561499 11:5728715218 13:5861292592 14:266176990 15:3468442519 16:6093725075 18:7730647177 19:4669596631 20:5029793111
12/00	1:97969285 4:4452915297 9:12047411873 10:12512540059 11:5919057886 13:7297838425 14:3255583421 15:4682049126 16:5913762396 18:5188922514 19:203576083
12/01	5:4320204528 9:13610541166 10:3750013016 11:5930185177 13:7294229051 14:4186217954 15:3759893899 16:5776808403 18:5755847514 20:2568489549
12/02	1:5462368720 5:11557534380 10:19030464453 11:5312228663 13:7483950891 14:4721720582 15:2766949303 16:7768917353 18:8397071671 19:3912490894 20:1168038412
12/10	4:8118700406 9:3754273882 10:13612138250 11:6346696905 13:7513438850 14:2254191810 15:4757303998 16:5777130717 18:5756284188 19:2567926986
12/11	5:4322653690 6:1014195029 9:7095228249 11:5930462400 13:7294214266 14:4186424434 15:3759583141 16:5438003399 18:6674050695 19:2498898886 20:3997335760
12/12	5:4324705582 10:8091364033 11:5930913748 13:7294587365 14:4186472298 15:3759197668 16:6343092191 18:7510515233 19:3967074757 20:3046793444
12/20	1:3488870706 3:2910492180 4:9669859006 9:500507494 10:18523792528 11:5701748095 13:7347686107 14:2638107777 15:5166336603 16:7768291285 18:8396919922 19:5080905997
12/21	4:8053361904 10:4323094903 11:6337166744 13:7506228037 14:2264006941 15:4753320331 16:5930701602 18:7294467259 19:4672401388 20:3273358947
12/22	1:5869654410 4:12912677466 9:2270984696 11:5283334899 13:7457762575 14:1793509566 15:5202569318 16:5885289587 18:7391057654 19:4303432166 20:4783040022
20/00	4:19524887432 9:9453412756 10:13121695600 11:7883133387 13:8449111065 14:1093068327 15:3941931117 16:5878368564 18:5362592945 19:774702389
20/01	4:19303556924 8:3826493504 9:1468731309 10:16862502048 11:7833190647 13:8426279820 14:1126524240 15:3928644376 16:5996034108 18:5667904050 19:2764950242
20/02	1:935756442 4:20055594300 6:975009318 9:18923691265 10:1209944462 11:7620705192 13:8408906847 14:1030129477 15:3991071923 16:7620391772 18:8413255765 20:5015534809
20/10	4:701085874 5:18602471049 8:3826493504 9:16862502048 10:1468731309 11:7833190647 13:8426279820 14:5055168616 16:5996034108 18:5667904050 20:2764950242
20/11	4:868806309 5:18669784156 6:1907126899 9:11662949382 11:7884683881 13:8449907765 14:5033963655 16:5396224077 18:6380477007 19:1626511386 20:4089602466
20/12	4:359872683 5:18467733187 6:2774551856 8:6335545474 9:10206940956 11:7725396373 13:8376618198 14:5098774013 16:5991598324 18:7213627566 19:2452179904 20:5241801408

Label	Nonzero coordinates written as `index:numerator` (cont.)
20/20	1:935756442 4:1157523413 5:18898070887 6:975009318 9:20133635727 11:7620705192 13:8408906847 14:5021201401 16:7620391772 18:8413255765 19:1019052133 20:3996482675
20/21	4:363885736 5:18470016118 6:5370885628 10:15415087836 11:7726693917 13:8376780888 14:5098408630 16:5767112163 18:7720131294 19:5294794823 20:1250763079
20/22	1:885795227 4:20030552438 6:8156258448 9:13027389675 11:7629432510 13:8408987816 14:1033399796 15:3989053666 16:4959916138 18:7671615130 19:2828172174 20:5579418557
21/00	2:986500466 4:5371374889 9:13354707677 10:12631819408 11:5962082381 13:7297824201 14:3636311186 15:4284631260 16:6241092039 18:5168914709 20:152666327
21/01	4:4253310544 9:13602311072 10:1901058526 11:6150630032 13:7615395070 14:4000354584 15:4307211967 16:5744956376 18:5936554290 20:2957230998
21/02	1:3488870706 3:2910492180 4:9669859006 9:18523792528 10:500507494 11:5701748095 13:7347686107 14:3124165049 15:4680279331 16:7768291285 18:8396919922 20:5080905997
21/10	5:4320204528 9:15912033270 10:1448520912 11:5930185177 13:7294229051 14:4672275226 15:3273836627 16:5776808403 18:5755847514 20:2568489549
21/11	5:4319082463 9:15162350099 11:5930187190 13:7294147948 14:4672210072 15:3274008441 16:5728375656 18:5860935433 19:266138916 20:3468266273
21/12	5:8053361904 9:4323094903 11:6337166744 13:7506228037 14:4753320331 15:2264006941 16:5930701602 18:7294467259 19:3273358947 20:4672401388
21/20	1:5462368720 5:11557534380 9:19030464453 11:5312228663 13:7483950891 14:5207777854 15:2280892031 16:7768917353 18:8397071671 19:1168038412 20:3912490894
21/21	5:4330458435 6:1488621373 10:6283168812 11:5930977899 13:7294704117 14:4672961615 15:3272357500 16:5761467364 18:7345249600 19:4817633465 20:3004644226
21/22	1:6602305789 4:13099590559 6:7790046049 9:12497297258 11:5183623147 13:7524271395 14:2555390552 15:4835827723 16:5221964585 18:7980520715 19:3026877613 20:5771520055
22/00	4:2165033944 9:11556651259 10:13011569355 11:6104007302 13:7710512160 14:4549540647 15:4901433456 16:5914081143 18:5188277176 19:201917305
22/01	4:2161693230 9:13610951047 10:3751127482 11:6104017614 13:7710272729 14:4550051801 15:4901239086 16:5777022978 18:5755966789 20:2568340748
22/02	1:8101072166 5:12956428340 6:939725909 9:18913714788 10:1189054584 11:4966397842 13:7669851558 14:5574174729 15:2837914626 16:7622625176 18:8411192252 20:5017839603
22/10	4:2230990577 9:1441527042 10:14023382575 11:5883906831 13:7389296085 14:4309219786 15:4780381309 16:6084872698 18:6417062284 19:3551029112
22/11	4:2160575338 9:15173445258 11:6104019610 13:7710191923 14:4550222977 15:4901174175 16:5728755243 18:5860125771 19:263892011 20:3468362547
22/12	1:1736229340 5:4574332512 6:7093527805 10:13697133998 11:5688794490 13:7451320236 14:4953543386 15:3987492927 16:5125903581 18:7538623405 19:4880478168 20:2473839592
22/20	1:8101072166 4:12956428340 6:939725910 9:20102769373 11:4966397842 13:7669851558 14:2837914626 15:5574174729 16:7622625176 18:8411192252 19:1023463854 20:3994375749
22/21	1:7726138118 4:12414902379 6:6665965831 10:13181631804 11:5229425520 13:7978440959 14:3038179573 15:5765414516 16:5176175874 18:7526323750 19:5327953486 20:2058082107
22/22	1:6436703150 4:10800971889 6:9734090459 9:15128230291 11:5156093381 13:7613915155 14:3133415746 15:5414466331 16:4780869372 18:7726013676 19:2540150254 20:5735071870

125.29 Eighty-one exact two-sample H1 Farkas duals

The first vector letter is $(22, 22)$ and the row label is the second vector letter t. The 36-component nonnegative multiplier uses exact scale 10^{-14}. For each row, $y_t^T b_t > 4.4101 \cdot 10^{-6}$ and every reduced input coefficient is less than $-4.4101 \cdot 10^{-6}$ at g_- after rigorous exponential error; the uniform drift over $[g_-, g_+]$ is at most $7 \cdot 10^{-10}$.

Label	Nonzero coordinates written as `index:numerator`
00/00	1:1889697863377 3:6011462750779 7:3449429134682 9:3449429134682 10:1842196894516 12:4104731362981 16:3256720888236 18:3256720888236 19:1904935676228 21:6698809417144 25:3512504402318 27:3512504402318 28:8346219614973 29:6312131937065 30:21417091204354 34:10517707214053 36:10517707214054
00/01	1:1725650061209 3:3570979857976 7:3034237487489 9:3034237487489 10:1777590700325 12:5655923581863 16:3244957178547 18:3244957178547 19:8833554098180 20:6902094312158 21:22382545085954 25:10958575183271 27:10958575183271 28:1790236613710 30:6288621508133 34:3298632240938 36:3298632240938
00/02	1:1647147192266 3:3211458940453 7:2884378544478 9:2884378544478 10:1702078276016 12:5416441787384 16:3107230268950 18:3107230268950 19:9161809735682 20:7299477387257 21:23032849323193 25:11255532204119 27:11255532204119 28:1712978354374 30:6012329765589 34:3154573601346 36:3154573601346
00/10	1:1725650061209 3:3570979857977 7:3034237487489 9:3034237487489 10:1777590700325 12:5655923581863 16:3244957178548 18:3244957178548 19:8833554098181 20:6902094312159 21:22382545085954 25:10958575183271 27:10958575183271 28:1790236613710 30:6288621508133 34:3298632240938 36:3298632240938

Label	Nonzero coordinates written as `index:numerator` (cont.)
00/11	1:1622481437385 3:3098496765363 7:2837292571635 9:2837292571635 10:1678352124684 12:5341196139413 16:3063956204721 18:3063956204721 19:9264948298672 20:7424335926272 21:23237176185774 25:11348836680572 27:11348836680572 28:1688703658262 30:5925518359972 34:3109310095173 36:3109310095173
00/12	1:1552555911334 3:2778257650249 7:2703807444727 9:2703807444727 10:1611090302736 12:5127880490137 16:2941277546798 18:2941277546798 19:9557338225826 20:7778300322023 21:23816427192707 25:11613347725543 27:11613347725544 28:1619886753878 30:5679414673558 34:2980991521708 36:2980991521708
00/20	1:1647147192266 3:3211458940453 7:2884378544478 9:2884378544478 10:1702078276016 12:5416441787385 16:3107230268950 18:3107230268950 19:9161809735683 20:7299477387257 21:23032849323195 25:11255532204119 27:11255532204119 28:1712978354373 30:6012329765587 34:3154573601345 36:3154573601345
00/21	1:1552555911334 3:2778257650249 7:2703807444726 9:2703807444726 10:1611090302736 12:5127880490137 16:2941277546798 18:2941277546798 19:9557338225825 20:7778300322023 21:23816427192707 25:11613347725544 27:11613347725544 28:1619886753878 30:5679414673558 34:2980991521708 36:2980991521708
00/22	1:1488173172718 3:2483402932560 7:2580903284827 9:2580903284827 10:1549160124361 12:4931473734755 16:2828323259389 18:2828323259389 19:9826551277440 20:8104207018161 21:24349762703766 25:11856891770633 27:11856891770633 28:1556524759803 30:5452818890733 34:2862844378002 36:2862844378002
01/00	1:1777590700325 3:5655923581863 7:3244957178547 9:3244957178547 10:1725650061209 12:3570979857976 16:3034237487489 18:3034237487489 19:1790236613711 21:6288621508134 25:3298632240939 27:3298632240939 28:8833554098180 29:6902094312158 30:22382545085953 34:10958575183271 36:10958575183271
01/01	1:1377629701799 3:4387478309117 7:2515469378486 9:2515469378485 10:1309849798713 12:1666732358582 16:2240490855287 18:2240490855287 19:1381028581634 21:4825207742194 25:2535607641143 27:2535607641143 28:10572201132078 29:9006883463987 30:25826962802015 34:12531445180025 36:12531445180025
01/02	1:1241274613129 3:1352677420557 7:2109583476145 9:2109583476145 10:1311666781199 12:4178282024681 16:2395159783272 18:2395159783272 19:10858944681249 20:9354012409590 21:26395027821238 25:12790848219550 27:12790848219550 28:1313540608987 30:4583856594451 34:2409767043492 36:2409767043492
01/10	1:1377629701799 3:4387478309117 7:2515469378486 9:2515469378485 10:1309849798713 12:1666732358582 16:2240490855287 18:2240490855287 19:1381028581634 21:4825207742194 25:2535607641143 27:2535607641143 28:10572201132077 29:9006883463987 30:25826962802015 34:12531445180025 36:12531445180025
01/11	1:1219904198941 3:1254806973016 7:2068788181336 9:2068788181336 10:1291110439685 12:4113089182593 16:2357667126729 18:2357667126729 19:10948303949610 20:9462189868842 21:26572056649423 25:12871687233678 27:12871687233678 28:1292509008184 30:4508643177993 34:2370550734114 36:2370550734114
01/12	1:1159772050205 3:979418753736 7:1953998233248 9:1953998233248 10:1233268931583 12:3929649327728 16:2252170154141 18:2252170154141 19:11199743382127 20:9766579994949 21:27070181006463 25:13099152344018 27:13099152344018 28:1233330213776 30:4297007394724 34:2260203740947 36:2260203740947
01/20	1:1241274613129 3:1352677420557 7:2109583476145 9:2109583476145 10:1311666781199 12:4178282024681 16:2395159783272 18:2395159783272 19:10858944681249 20:9354012409590 21:26395027821238 25:12790848219550 27:12790848219550 28:1313540608987 30:4583856594451 34:2409767043492 36:2409767043492
01/21	1:1159772050205 3:979418753736 7:1953998233248 9:1953998233248 10:1233268931583 12:3929649327728 16:2252170154141 18:2252170154141 19:11199743382127 20:9766579994949 21:27070181006463 25:13099152344018 27:13099152344018 28:1233330213776 30:4297007394724 34:2260203740947 36:2260203740947
01/22	1:1104989095619 3:728528329442 7:1849419690071 9:1849419690071 10:1180572848404 12:3762527788761 16:2156057908367 18:2156057908367 19:11428815439290 20:10043892394477 21:27523993561030 25:13306382770195 27:13306382770195 28:1179415805683 30:4104198161276 34:2159672919376 36:2159672919376
02/00	1:1702078276016 3:5416441787385 7:3107230268950 9:3107230268950 10:1647147192266 12:3211458940453 16:2884378544478 18:2884378544478 19:1712978354374 21:6012329765588 25:3154573601345 27:3154573601345 28:9161809735682 29:7299477387257 30:23032849323194 34:11255532204119 36:11255532204119
02/01	1:1311666781199 3:4178282024681 7:2395159783272 9:2395159783272 10:1241274613129 12:1352677420557 16:2109583476145 18:2109583476145 19:1313540608987 21:4583856594451 25:2409767043492 27:2409767043492 28:10858944681249 29:9354012409590 30:26395027821238 34:12790848219550 36:12790848219550
02/02	1:1093003450683 3:3484808238641 7:1996340317073 9:1996340317073 10:1013951772748 12:311603161117 16:1675632958912 18:1675632958912 19:1089821817660 21:3783791265173 25:1992612618755 27:1992612618755 28:11809483239822 30:38782856177707 32:10504725123255 34:13650754543485 36:3146029420230
02/10	1:1311666781199 3:4178282024681 7:2395159783272 9:2395159783272 10:1241274613129 12:1352677420557 16:2109583476145 18:2109583476145 19:1313540608987 21:4583856594451 25:2409767043492 27:2409767043492 28:10858944681249 29:9354012409590 30:26395027821238 34:12790848219550 36:12790848219550

Label	Nonzero coordinates written as `index:numerator` (cont.)
02/11	1:995670331974 3:227879337401 7:1640734394975 9:1640734394975 10:1075418412947 12:3429038655935 16:1964267013579 18:1964267013579 19:11885926127534 21:39026837497217 23:10597266137994 25:13719908731887 27:3122642593893 28:1071830216699 30:3719449525875 34:1959064806769 36:1959064806769
02/12	1:946416520463 5:2172532791 7:1544921496807 9:1542748964016 10:1025891399065 12:3271946159180 16:1873928833956 18:1873928833956 19:12098850909823 21:39706272041157 23:10855564262279 25:13912114030045 27:3056549767766 28:1021171858465 30:3538300597848 34:1864610896192 36:1864610896192
02/20	1:1093003450683 3:3484808238641 7:1996340317073 9:1996340317073 10:1013951772748 12:311603161117 16:1675632958912 18:1675632958912 19:1089821817660 21:3783791265173 25:1992612618755 27:1992612618755 28:11809483239822 30:38782856177707 31:10504725123255 34:3146029420230 36:13650754543485
02/21	1:946416520463 5:2172532791 7:1544921496807 9:1542748964016 10:1025891399065 12:3271946159180 16:1873928833956 18:1873928833956 19:12098850909823 21:39706272041157 23:10855564262279 25:13912114030045 27:3056549767766 28:1021171858465 30:3538300597848 34:1864610896192 36:1864610896192
02/22	1:961334336781 5:66403358728 7:1520499190483 9:1454095831755 10:976692569594 12:3115282750652 16:1784018099228 18:1784018099228 19:12242615867970 21:40160586955394 23:11045376388565 25:14029828812963 27:2984452424399 28:971240960344 30:3360209918306 34:1771672217805 36:1771672217805
10/00	1:1777590700325 3:5655923581863 7:3244957178548 9:3244957178548 10:1725650061209 12:3570979857976 16:3034237487489 18:3034237487489 19:1790236613710 21:6288621508134 25:3298632240938 27:3298632240938 28:8833554098180 29:6902094312158 30:22382545085953 34:10958575183271 36:10958575183271
10/01	1:1377629701799 3:4387478309117 7:2515469378486 9:2515469378486 10:1309849798713 12:1666732358582 16:2240490855287 18:2240490855287 19:1381028581634 21:4825207742194 25:2535607641143 27:2535607641143 28:10572201132077 29:9006883463987 30:25826962802015 34:12531445180025 36:12531445180025
10/02	1:1241274613129 3:1352677420557 7:2109583476145 9:2109583476145 10:1311666781199 12:4178282024681 16:2395159783272 18:2395159783272 19:10858944681249 20:9354012409590 21:26395027821238 25:12790848219550 27:12790848219550 28:1313540608987 30:4583856594451 34:2409767043492 36:2409767043492
10/10	1:1377629701799 3:4387478309118 7:2515469378486 9:2515469378486 10:1309849798713 12:1666732358582 16:2240490855287 18:2240490855287 19:1381028581634 21:4825207742194 25:2535607641143 27:2535607641143 28:10572201132077 29:9006883463987 30:25826962802015 34:12531445180025 36:12531445180025
10/11	1:1219904198941 3:1254806973016 7:2068788181336 9:2068788181336 10:1291110439685 12:4113089182593 16:2357667126729 18:2357667126729 19:10948303949610 20:9462189868842 21:26572056649423 25:12871687233678 27:12871687233678 28:1292509008184 30:4508643177993 34:2370550734114 36:2370550734114
10/12	1:1159772050205 3:979418753736 7:1953998233248 9:1953998233248 10:1233268931583 12:3929649327728 16:2252170154141 18:2252170154141 19:11199743382127 20:9766579994949 21:27070181006463 25:13099152344018 27:13099152344018 28:1233330213776 30:4297007394724 34:2260203740947 36:2260203740947
10/20	1:1241274613129 3:1352677420557 7:2109583476145 9:2109583476145 10:1311666781199 12:4178282024681 16:2395159783272 18:2395159783272 19:10858944681249 20:9354012409590 21:26395027821238 25:12790848219550 27:12790848219550 28:1313540608987 30:4583856594451 34:2409767043492 36:2409767043492
10/21	1:1159772050205 3:979418753736 7:1953998233248 9:1953998233248 10:1233268931583 12:3929649327728 16:2252170154141 18:2252170154141 19:11199743382127 20:9766579994949 21:27070181006463 25:13099152344018 27:13099152344018 28:1233330213776 30:4297007394724 34:2260203740947 36:2260203740947
10/22	1:1104989095619 3:728528329442 7:1849419690071 9:1849419690071 10:1180572848404 12:3762527788761 16:2156057908367 18:2156057908367 19:11428815439290 20:10043892394477 21:27523993561030 25:13306382770195 27:13306382770195 28:1179415805683 30:4104198161276 34:2159672919376 36:2159672919376
11/00	1:1678352124684 3:5341196139413 7:3063956204721 9:3063956204721 10:1622481437385 12:3098496765364 16:2837292571635 18:2837292571636 19:1688703658263 21:5925518359975 25:3109310095175 27:3109310095175 28:9264948298671 29:7424335926273 30:23237176185770 34:11348836680571 36:11348836680549
11/01	1:1291110439685 3:4113089182593 7:2357667126729 9:2357667126729 10:1219904198941 12:1254806973016 16:2068788181336 18:2068788181336 19:1292509008184 21:4508643177993 25:2370550734114 27:2370550734114 28:10948303949610 29:9462189868842 30:26572056649422 34:12871687233678 36:12871687233678
11/02	1:1075418412947 3:3429038655935 7:1964267013579 9:1964267013579 10:995670331974 12:227879337401 16:1640734394975 18:1640734394975 19:1071830216699 21:3719449525875 25:1959064806769 27:1959064806769 28:11885926127534 30:39026837497217 32:10597266137994 34:13719908731887 36:3122642593893
11/10	1:1291110439685 3:4113089182593 7:2357667126729 9:2357667126729 10:1219904198941 12:1254806973016 16:2068788181336 18:2068788181336 19:1292509008184 21:4508643177993 25:2370550734114 27:2370550734114 28:10948303949610 29:9462189868842 30:26572056649422 34:12871687233678 36:12871687233678

Label	Nonzero coordinates written as `index:numerator` (cont.)
11/11	1:754930625509 3:1343953267049 7:1315179425741 9:1315179425741 10:721449725841 16:1179349660176 18:1179349660176 19:4693414945761 20:3841309911943 21:11609542821053 25:5665887633644 27:5665887633644 28:9233532472994 29:8290395081019 30:21983674709715 34:10603481499996 36:10603481499996
11/12	1:1038036673442 5:396656415204 7:1394927999521 9:998271584317 10:723728925259 12:2309772798859 16:1321727687719 18:1321727687719 19:12981806375786 20:12021325770304 21:30475193922062 25:14635078181688 27:14635078181688 28:714513268049 30:2444528228239 34:1293813150072 36:1293813150072
11/20	1:1075418412947 3:3429038655935 7:1964267013579 9:1964267013579 10:995670331974 12:227879337401 16:1640734394975 18:1640734394975 19:1071830216699 21:3719449525875 25:1959064806769 27:1959064806769 28:11885926127534 30:39026837497217 31:10597266137994 34:3122642593893 36:13719908731887
11/21	1:1038036673442 5:396656415204 7:1394927999521 9:998271584317 10:723728925259 12:2309772798859 16:1321727687719 18:1321727687719 19:12981806375786 20:12021325770304 21:30475193922062 25:14635078181688 27:14635078181688 28:714513268049 30:2444528228239 34:1293813150072 36:1293813150072
11/22	1:1049189420308 5:444676187495 7:1376669576539 9:931993389044 10:686947262347 12:2192649266754 16:1254509293044 18:1254509293044 19:13089286864365 20:12163231700194 21:30672939528822 25:14723083232044 27:14723083232044 28:677184301419 30:2311385399171 34:1224331026683 36:1224331026683
12/00	1:1611090302736 3:5127880490137 7:2941277546798 9:2941277546798 10:1552555911334 12:2778257650249 16:2703807444726 18:2703807444726 19:1619886753878 21:5679414673558 25:2980991521708 27:2980991521708 28:9557338225825 29:7778300322023 30:23816427192707 34:11613347725543 36:11613347725543
12/01	1:1233268931583 3:3929649327728 7:2252170154141 9:2252170154141 10:1159772050205 12:979418753736 16:1953998233248 18:1953998233248 19:1233330213776 21:4297007394724 25:2260203740947 27:2260203740947 28:11199743382127 29:9766579994949 30:27070181006463 34:13099152344018 36:13099152344018
12/02	1:1025891399065 3:3271946159180 7:1873928833956 9:1873928833956 10:946416520463 14:2172532791 16:1544921496807 18:1542748964016 19:1021171858465 21:3538300597848 25:1864610896192 27:1864610896192 28:12098850909823 30:39706272041157 32:10855564262279 34:13912114030045 36:3056549767766
12/10	1:1233268931583 3:3929649327728 7:2252170154141 9:2252170154141 10:1159772050205 12:979418753736 16:1953998233248 18:1953998233248 19:1233330213776 21:4297007394724 25:2260203740947 27:2260203740947 28:11199743382127 29:9766579994949 30:27070181006463 34:13099152344018 36:13099152344018
12/11	1:723728925259 3:2309772798859 7:1321727687719 9:1321727687719 10:1038036673442 14:396656415204 16:1394927999521 18:998271584317 19:714513268049 21:2444528228239 25:1293813150072 27:1293813150072 28:12981806375786 29:12021325770304 30:30475193922061 34:14635078181688 36:14635078181688
12/12	1:438454931468 7:716756453340 9:716756453340 10:720471262563 14:300830366573 16:949319158673 18:648488792100 19:5618101013407 21:18374822559519 23:5053747280618 25:6429411122742 27:1375663842124 28:9004018336698 30:29429643744572 32:8371774722950 34:10111757341132 36:1739982618183
12/20	1:1025891399065 3:3271946159180 7:1873928833956 9:1873928833956 10:946416520463 14:2172532791 16:1544921496807 18:1542748964016 19:1021171858465 21:3538300597848 25:1864610896192 27:1864610896192 28:12098850909823 30:39706272041156 31:10855564262279 34:3056549767766 36:13912114030045
12/21	1:438454931468 7:716756453340 9:716756453340 10:720471262563 14:300830366573 16:949319158673 18:648488792100 19:5618101013407 21:18374822559519 23:5053747280618 25:6429411122742 27:1375663842124 28:9004018336698 30:29429643744572 31:8371774722950 34:1739982618183 36:10111757341132
12/22	1:1124750686198 5:770016197956 7:1252966459248 9:482950261292 10:437746859563 12:1399122572591 16:799096195824 18:799096195824 19:13817480723073 21:45137353205017 23:13124662263327 25:15319328541787 27:2194666278460 28:424275846951 30:1409325948749 34:753580882070 36:753580882070
20/00	1:1702078276016 3:5416441787385 7:3107230268950 9:3107230268950 10:1647147192266 12:3211458940453 16:2884378544478 18:2884378544478 19:1712978354374 21:6012329765588 25:3154573601346 27:3154573601346 28:9161809735682 29:7299477387257 30:23032849323194 34:11255532204119 36:11255532204119
20/01	1:1311666781199 3:4178282024681 7:2395159783272 9:2395159783272 10:1241274613129 12:1352677420557 16:2109583476145 18:2109583476145 19:1313540608987 21:4583856594451 25:2409767043492 27:2409767043492 28:10858944681249 29:9354012409590 30:26395027821238 34:12790848219550 36:12790848219550
20/02	1:1093003450683 3:3484808238641 7:1996340317073 9:1996340317073 10:1013951772748 12:311603161117 16:1675632958912 18:1675632958912 19:1089821817660 21:3783791265173 25:1992612618755 27:1992612618755 28:11809483239822 30:38782856177707 32:10504725123255 34:13650754543485 36:3146029420230
20/10	1:1311666781199 3:4178282024681 7:2395159783272 9:2395159783272 10:1241274613129 12:1352677420557 16:2109583476145 18:2109583476145 19:1313540608987 21:4583856594451 25:2409767043492 27:2409767043492 28:10858944681249 29:9354012409590 30:26395027821238 34:12790848219550 36:12790848219550

Label	Nonzero coordinates written as `index:numerator` (cont.)
20/11	1:995670331974 3:227879337401 7:1640734394975 9:1640734394975 10:1075418412947 12:3429038655935 16:1964267013579 18:1964267013579 19:11885926127534 21:39026837497217 22:10597266137994 25:3122642593893 27:13719908731887 28:1071830216699 30:3719449525875 34:1959064806769 36:1959064806768
20/12	1:946416520463 5:2172532791 7:1544921496807 9:1542748964016 10:1025891399065 12:3271946159180 16:1873928833956 18:1873928833956 19:12098850909823 21:39706272041157 22:10855564262279 25:3056549767766 27:13912114030045 28:1021171858465 30:3538300597848 34:1864610896192 36:1864610896192
20/20	1:1093003450683 3:3484808238641 7:1996340317073 9:1996340317073 10:1013951772748 12:311603161117 16:1675632958912 18:1675632958912 19:1089821817660 21:3783791265173 25:1992612618755 27:1992612618755 28:11809483239822 30:38782856177707 31:10504725123255 34:3146029420230 36:13650754543485
20/21	1:946416520463 5:2172532791 7:1544921496807 9:1542748964016 10:1025891399065 12:3271946159180 16:1873928833956 18:1873928833956 19:12098850909823 21:39706272041157 22:10855564262279 25:3056549767766 27:13912114030045 28:1021171858465 30:3538300597848 34:1864610896192 36:1864610896192
20/22	1:961334336781 5:66403358728 7:1520499190483 9:1454095831755 10:976692569594 12:3115282750652 16:1784018099228 18:1784018099228 19:12242615867970 21:40160586955394 22:11045376388565 25:2984452424399 27:14029828812963 28:971240960344 30:3360209918306 34:1771672217805 36:1771672217805
21/00	1:1611090302736 3:5127880490137 7:2941277546798 9:2941277546798 10:1552555911334 12:2778257650249 16:2703807444726 18:2703807444726 19:1619886753878 21:5679414673558 25:2980991521708 27:2980991521708 28:9557338225825 29:7778300322023 30:23816427192707 34:11613347725543 36:11613347725543
21/01	1:1233268931583 3:3929649327728 7:2252170154141 9:2252170154141 10:1159772050205 12:979418753736 16:1953998233248 18:1953998233248 19:1233330213776 21:4297007394724 25:2260203740947 27:2260203740947 28:11199743382127 29:9766579994949 30:27070181006463 34:13099152344018 36:13099152344018
21/02	1:1025891399065 3:3271946159180 7:1873928833956 9:1873928833956 10:946416520463 14:2172532791 16:1544921496807 18:1542748964016 19:1021171858465 21:3538300597848 25:1864610896192 27:1864610896192 28:12098850909823 30:39706272041157 32:10855564262279 34:13912114030045 36:3056549767766
21/10	1:1233268931583 3:3929649327728 7:2252170154141 9:2252170154141 10:1159772050205 12:979418753736 16:1953998233248 18:1953998233248 19:1233330213776 21:4297007394725 25:2260203740947 27:2260203740947 28:11199743382127 29:9766579994949 30:27070181006463 34:13099152344018 36:13099152344018
21/11	1:723728925259 3:2309772798859 7:1321727687719 9:1321727687719 10:1038036673442 14:396656415204 16:1394927999521 18:998271584317 19:714513268049 21:2444528228239 25:1293813150072 27:1293813150072 28:12981806375786 29:12021325770304 30:30475193922061 34:14635078181688 36:14635078181688
21/12	1:720471262563 5:300830366573 7:949319158673 9:648488792100 10:438454931468 16:716756453340 18:716756453340 19:9004018336698 21:29429643744573 22:8371774722950 25:1739982618183 27:10111757341132 28:5618101013407 30:18374822559518 32:5053747280618 34:6429411122742 36:1375663842123
21/20	1:1025891399065 3:3271946159180 7:1873928833956 9:1873928833956 10:946416520463 14:2172532791 16:1544921496807 18:1542748964016 19:1021171858465 21:3538300597848 25:1864610896192 27:1864610896192 28:12098850909823 30:39706272041157 31:10855564262279 34:3056549767766 36:13912114030045
21/21	1:438454931468 7:716756453340 9:716756453340 10:720471262563 14:300830366573 16:949319158673 18:648488792100 19:5618101013407 21:18374822559519 22:5053747280618 25:1375663842124 27:6429411122742 28:9004018336698 30:29429643744572 31:8371774722950 34:1739982618183 36:10111757341132
21/22	1:1124750686198 5:770016197956 7:1252966459248 9:482950261292 10:437746859563 12:1399122572591 16:799096195824 18:799096195824 19:13817480723073 21:45137353205017 22:13124662263327 25:2194666278460 27:15319328541787 28:424275846951 30:1409325948749 34:753580882070 36:753580882070
22/00	1:1549160124361 3:4931473734755 7:2828323259389 9:2828323259389 10:1488173172718 12:2483402932560 16:2580903284827 18:2580903284827 19:1556524759803 21:5452818890733 25:2862844378002 27:2862844378002 28:9826551277440 29:8104207018161 30:24349762703766 34:11856891770633 36:11856891770633
22/01	1:1180572848404 3:3762527788761 7:2156057908367 9:2156057908367 10:1104989095619 12:728528329442 16:1849419690071 18:1849419690071 19:1179415805683 21:4104198161276 25:2159672919376 27:2159672919376 28:11428815439290 29:10043892394477 30:27523993561030 34:13306382770195 36:13306382770195
22/02	1:976692569594 3:3115282750652 7:1784018099228 9:1784018099228 10:961334336781 14:66403358728 16:1520499190483 18:1454095831755 19:971240960344 21:3360209918306 25:1771672217805 27:1771672217805 28:12242615867970 30:40160586955394 32:11045376388565 34:14029828812963 36:2984452424399
22/10	1:1180572848404 3:3762527788761 7:2156057908367 9:2156057908367 10:1104989095619 12:728528329442 16:1849419690071 18:1849419690071 19:1179415805683 21:4104198161276 25:2159672919376 27:2159672919376 28:11428815439290 29:10043892394477 30:27523993561030 34:13306382770195 36:13306382770195

Label	Nonzero coordinates written as `index:numerator` (cont.)
22/11	1:686947262347 3:2192649266754 7:1254509293044 9:1254509293044 10:1049189420308 14:444676187495 16:1376669576539 18:931993389044 19:677184301419 21:2311385399171 25:1224331026683 27:1224331026683 28:13089286864365 29:12163231700194 30:30672939528822 34:14723083232044 36:14723083232044
22/12	1:437746859563 3:1399122572591 7:799096195824 9:799096195824 10:1124750686198 14:770016197956 16:1252966459248 18:482950261292 19:424275846951 21:1409325948749 25:753580882070 27:753580882070 28:13817480723073 30:45137353205017 32:13124662263327 34:15319328541787 36:2194666278460
22/20	1:976692569594 3:3115282750652 7:1784018099228 9:1784018099228 10:961334336781 14:66403358728 16:1520499190483 18:1454095831755 19:971240960344 21:3360209918306 25:1771672217805 27:1771672217805 28:12242615867970 30:40160586955394 31:11045376388565 34:2984452424399 36:14029828812963
22/21	1:437746859563 3:1399122572591 7:799096195824 9:799096195824 10:1124750686198 14:770016197956 16:1252966459248 18:482950261292 19:424275846951 21:1409325948749 25:753580882070 27:753580882070 28:13817480723073 30:45137353205017 31:13124662263327 34:2194666278460 36:15319328541787
22/22	1:159371454105 4:102920701986 5:102920701986 10:1104045392072 13:331597814865 14:881939157472 16:550341342607 19:1885573977785 21:6035447075239 23:1903322478895 25:1970718217858 27:67395738963 28:13227427263386 30:43065931876793 32:13017764107359 34:14305523402995 36:1287759295636

125.30 First-sample H1 duals at the right critical endpoint

The 18 components for each first vector letter are exact fractions. Ordinary rows use scale 10^{-12}, except the specially constructed critical $(22, 22)$ row: entries not divisible by 10^{-12} are printed as their unreduced exact rational fractions. The first 80 rows remain strictly excluding throughout the critical bracket; the right critical row certifies the strict obstruction at g_+.

Label	Nonzero coordinates written as `index:numerator`
00/00	1:60678259969 3:214679144971 7:112321297530 9:112321297530 10:60678259969 12:214679144971 16:112321297530 18:112321297530
00/01	1:70906276745 2:16375506907 3:229814438420 7:118451770793 9:118451770793 10:54155492165 12:191453605516 16:100195569331 18:100195569331
00/02	1:81540285500 2:33401024918 3:245550513809 7:124825587710 9:124825587710 10:47373808925 12:167306149042 16:87588521193 18:87588521193
00/10	1:70906276745 2:16375506907 3:229814438420 7:118451770793 9:118451770793 10:54155492165 12:191453605516 16:100195569331 18:100195569331
00/11	1:84514218035 2:38162422347 3:249951302605 7:126608104721 9:126608104721 10:45477227035 12:160553013126 16:84062803705 18:84062803705
00/12	1:92154274819 2:50394490925 3:261256964682 7:131187405368 9:131187405368 10:40604892789 12:143204152233 16:75005206909 18:75005206909
00/20	1:81540285500 2:33401024918 3:245550513809 7:124825587710 9:124825587710 10:47373808925 12:167306149042 16:87588521193 18:87588521193
00/21	1:92154274819 2:50394490925 3:261256964682 7:131187405368 9:131187405368 10:40604892789 12:143204152233 16:75005206909 18:75005206909
00/22	1:98308071803 2:60246991878 3:270363277785 7:134875870953 9:134875870953 10:36680398947 12:129230255064 16:67709631308 18:67709631308
01/00	1:54155492165 3:191453605516 7:100195569331 9:100195569331 10:70906276745 11:16375506907 12:229814438420 16:118451770793 18:118451770793
01/01	1:92141419776 2:50373909434 3:261237941948 7:131179700307 9:131179700307 10:40613090903 12:143233343160 16:75020447082 18:75020447082
01/02	1:100841451557 2:64303044832 3:274112141992 7:136394329241 9:136394329241 10:35064773137 12:123477516023 16:64706206989 18:64706206989
01/10	1:92141419776 2:50373909434 3:261237941948 7:131179700307 9:131179700307 10:40613090903 12:143233343160 16:75020447082 18:75020447082
01/11	1:103233144246 2:68132250534 3:277651339423 7:137827863073 9:137827863073 10:33539506175 12:118046516627 16:61870758424 18:61870758424
01/12	1:109296688073 2:77840252192 3:286624097014 7:141462232680 9:141462232680 10:29672569962 12:104277564203 16:54682181598 18:54682181598
01/20	1:100841451557 2:64303044832 3:274112141992 7:136394329241 9:136394329241 10:35064773137 12:123477516023 16:64706206989 18:64706206989
01/21	1:109296688073 2:77840252192 3:286624097014 7:141462232680 9:141462232680 10:29672569962 12:104277564203 16:54682181598 18:54682181598
01/22	1:114097889872 2:85527188472 3:293728856395 7:144339979159 9:144339979159 10:26610673878 12:93375108305 16:48990162380 18:48990162380
02/00	1:47373808925 3:167306149042 7:87588521193 9:87588521193 10:81540285500 11:33401024918 12:245550513809 16:124825587710 18:124825587710
02/01	1:35064773137 3:123477516023 7:64706206989 9:64706206989 10:100841451557 11:64303044832 12:274112141992 16:136394329241 18:136394329241

Label	Nonzero coordinates written as `index:numerator` (cont.)
02/02	1:27526969899 3:96637752328 7:50693543036 9:50693543036 10:112661093200 12:374829515846 14:83226813498 16:143478791328 18:60251977830
02/10	1:35064773137 3:123477516023 7:64706206989 9:64706206989 10:100841451557 11:64303044832 12:274112141992 16:136394329241 18:136394329241
02/11	1:114494176933 3:380476938685 5:86161661571 7:144577505874 9:58415844303 10:26357947613 12:92475228987 16:48520348017 18:48520348017
02/12	1:119157889455 3:394845052997 5:93628471442 7:147372843968 9:53744372526 10:23383733300 12:81884980499 16:42991327906 18:42991327906
02/20	1:27526969899 3:96637752328 7:50693543036 9:50693543036 10:112661093200 12:374829515846 13:83226813498 16:60251977830 18:143478791328
02/21	1:119157889455 3:394845052997 5:93628471442 7:147372843968 9:53744372526 10:23383733300 12:81884980499 16:42991327906 18:42991327906
02/22	1:122867446130 3:406273573536 5:99567635630 7:149596279705 9:50028644074 10:21018017895 12:73461406624 16:38593498203 18:38593498203
10/00	1:54155492165 3:191453605516 7:100195569331 9:100195569331 10:70906276745 11:16375506907 12:229814438420 16:118451770793 18:118451770793
10/01	1:92141419776 2:50373909434 3:261237941948 7:131179700307 9:131179700307 10:40613090903 12:143233343160 16:75020447082 18:75020447082
10/02	1:100841451557 2:64303044832 3:274112141992 7:136394329241 9:136394329241 10:35064773137 12:123477516023 16:64706206989 18:64706206989
10/10	1:92141419776 2:50373909434 3:261237941948 7:131179700307 9:131179700307 10:40613090903 12:143233343160 16:75020447082 18:75020447082
10/11	1:103233144246 2:68132250534 3:277651339423 7:137827863073 9:137827863073 10:33539506175 12:118046516627 16:61870758424 18:61870758424
10/12	1:109296688073 2:77840252192 3:286624097014 7:141462232680 9:141462232680 10:29672569962 12:104277564203 16:54682181598 18:54682181598
10/20	1:100841451557 2:64303044832 3:274112141992 7:136394329241 9:136394329241 10:35064773137 12:123477516023 16:64706206989 18:64706206989
10/21	1:109296688073 2:77840252192 3:286624097014 7:141462232680 9:141462232680 10:29672569962 12:104277564203 16:54682181598 18:54682181598
10/22	1:114097889872 2:85527188472 3:293728856395 7:144339979159 9:144339979159 10:26610673878 12:93375108305 16:48990162380 18:48990162380
11/00	1:45477227035 3:160553013126 7:84062803705 9:84062803705 10:84514218035 11:38162422347 12:249951302605 16:126608104721 18:126608104721
11/01	1:33539506175 3:118046516627 7:61870758424 9:61870758424 10:103233144246 11:68132250534 12:277651339423 16:137827863073 18:137827863073
11/02	1:26357947613 3:92475228987 7:48520348017 9:48520348017 10:114494176933 12:380476938685 14:86161661571 16:144577505874 18:58415844303
11/10	1:33539506175 3:118046516627 7:61870758424 9:61870758424 10:103233144246 11:68132250534 12:277651339423 16:137827863073 18:137827863073
11/11	1:19280262431 3:67273802021 7:35363036677 9:35363036677 10:125592331113 11:103930296920 12:310738189195 16:151229522483 18:151229522483
11/12	1:129352534136 2:109950548150 3:316302491388 7:153483314650 9:153483314650 10:16882248060 12:58735221612 16:30905163677 18:30905163677
11/20	1:26357947613 3:92475228987 7:48520348017 9:48520348017 10:114494176933 12:380476938685 13:86161661571 16:58415844303 18:144577505874
11/21	1:129352534136 2:109950548150 3:316302491388 7:153483314650 9:153483314650 10:16882248060 12:58735221612 16:30905163677 18:30905163677
11/22	1:132270842402 2:114622888665 3:320620968074 7:155232491562 9:155232491562 10:15021139732 12:52108395976 16:27445391014 18:27445391014
12/00	1:40604892789 3:143204152233 7:75005206909 9:75005206909 10:92154274819 11:50394490925 12:261256964682 16:131187405368 18:131187405368
12/01	1:29672569962 3:104277564203 7:54682181598 9:54682181598 10:109296688073 11:77840252192 12:286624097014 16:141462232680 18:141462232680
12/02	1:23383733300 3:81884980499 7:42991327906 9:42991327906 10:119157889455 12:394845052997 14:93628471442 16:147372843968 18:53744372526
12/10	1:29672569962 3:104277564203 7:54682181598 9:54682181598 10:109296688073 11:77840252192 12:286624097014 16:141462232680 18:141462232680
12/11	1:16882248060 3:58735221612 7:30905163677 9:30905163677 10:129352534136 11:109950548150 12:316302491388 16:153483314650 18:153483314650
12/12	1:139616474211 3:457874513543 5:126383569848 7:159635319636 9:33251749788 10:10336570698 12:35428108375 16:18736846951 18:18736846951
12/20	1:23383733300 3:81884980499 7:42991327906 9:42991327906 10:119157889455 12:394845052997 13:93628471442 16:53744372526 18:147372843968
12/21	1:139616474211 3:457874513543 5:126383569848 7:159635319636 9:33251749788 10:10336570698 12:35428108375 16:18736846951 18:18736846951
12/22	1:141352947970 3:463224297291 5:129163741082 7:160676128053 9:31512386971 10:9229160007 12:31484964687 16:16678186969 18:16678186969
20/00	1:47373808925 3:167306149042 7:87588521193 9:87588521193 10:81540285500 11:33401024918 12:245550513809 16:124825587710 18:124825587710
20/01	1:35064773137 3:123477516023 7:64706206989 9:64706206989 10:100841451557 11:64303044832 12:274112141992 16:136394329241 18:136394329241
20/02	1:27526969899 3:96637752328 7:50693543036 9:50693543036 10:112661093200 12:374829515846 14:83226813498 16:143478791328 18:60251977830

Label	Nonzero coordinates written as `index:numerator` (cont.)
20/10	1:35064773137 3:123477516023 7:64706206989 9:64706206989 10:100841451557 11:64303044832 12:274112141992 16:136394329241 18:136394329241
20/11	1:114494176933 3:380476938685 4:86161661571 7:58415844303 9:144577505874 10:26357947613 12:92475228987 16:48520348017 18:48520348017
20/12	1:119157889455 3:394845052997 4:93628471442 7:53744372526 9:147372843968 10:23383733300 12:81884980499 16:42991327906 18:42991327906
20/20	1:27526969899 3:96637752328 7:50693543036 9:50693543036 10:112661093200 12:374829515846 13:83226813498 16:60251977830 18:143478791328
20/21	1:119157889455 3:394845052997 4:93628471442 7:53744372526 9:147372843968 10:23383733300 12:81884980499 16:42991327906 18:42991327906
20/22	1:122867446130 3:406273573536 4:99567635630 7:50028644074 9:149596279705 10:21018017895 12:73461406624 16:38593498203 18:38593498203
21/00	1:40604892789 3:143204152233 7:75005206909 9:75005206909 10:92154274819 11:50394490925 12:261256964682 16:131187405368 18:131187405368
21/01	1:29672569962 3:104277564203 7:54682181598 9:54682181598 10:109296688073 11:77840252192 12:286624097014 16:141462232680 18:141462232680
21/02	1:23383733300 3:81884980499 7:42991327906 9:42991327906 10:119157889455 12:394845052997 14:93628471442 16:147372843968 18:53744372526
21/10	1:29672569962 3:104277564203 7:54682181598 9:54682181598 10:109296688073 11:77840252192 12:286624097014 16:141462232680 18:141462232680
21/11	1:16882248060 3:58735221612 7:30905163677 9:30905163677 10:129352534136 11:109950548150 12:316302491388 16:153483314650 18:153483314650
21/12	1:139616474211 3:457874513543 4:126383569848 7:33251749788 9:159635319636 10:10336570698 12:35428108375 16:18736846951 18:18736846951
21/20	1:23383733300 3:81884980499 7:42991327906 9:42991327906 10:119157889455 12:394845052997 13:93628471442 16:53744372526 18:147372843968
21/21	1:139616474211 3:457874513543 4:126383569848 7:33251749788 9:159635319636 10:10336570698 12:35428108375 16:18736846951 18:18736846951
21/22	1:141352947970 3:463224297291 4:129163741082 7:31512386971 9:160676128053 10:9229160007 12:31484964687 16:16678186969 18:16678186969
22/00	1:36680398947 3:129230255064 7:67709631308 9:67709631308 10:98308071803 11:60246991878 12:270363277785 16:134875870953 18:134875870953
22/01	1:26610673878 3:93375108305 7:48990162380 9:48990162380 10:114097889872 11:85527188472 12:293728856395 16:144339979159 18:144339979159
22/02	1:21018017895 3:73461406624 7:38593498203 9:38593498203 10:122867446130 12:406273573536 14:99567635630 16:149596279705 18:50028644074
22/10	1:26610673878 3:93375108305 7:48990162380 9:48990162380 10:114097889872 11:85527188472 12:293728856395 16:144339979159 18:144339979159
22/11	1:15021139732 3:52108395976 7:27445391014 9:27445391014 10:132270842402 11:114622888665 12:320620968074 16:155232491562 18:155232491562
22/12	1:9229160007 3:31484964687 7:16678186969 9:16678186969 10:141352947970 12:463224297291 14:129163741082 16:160676128053 18:31512386971
22/20	1:21018017895 3:73461406624 7:38593498203 9:38593498203 10:122867446130 12:406273573536 13:99567635630 16:50028644074 18:149596279705
22/21	1:9229160007 3:31484964687 7:16678186969 9:16678186969 10:141352947970 12:463224297291 13:129163741082 16:31512386971 18:160676128053
22/22	1:(31246001677865809/400000000000000000) 3:(25376615812862827/1000000000000000000) 5:(154419493535266239/2000000000000000000) 7:(84059418838353439/1000000000000000000) 9:(13699344141440577/2000000000000000000) 10:(31246001677865809/400000000000000000) 12:(25376615812862827/1000000000000000000) 14:(154419493535266239/2000000000000000000) 16:(84059418838353439/1000000000000000000) 18:(13699344141440577/2000000000000000000)

125.31 First-sample H2 primal controls at the right critical endpoint

The first 10 coordinates are the early control phase and the last 10 the late phase. Every exact entry is a multiple of 10^{-9}, and each half-phase total is less than 421. These controls certify all 81 first-letter pairs at g_+ and throughout the narrow critical bracket by the stated Duhamel estimate.

Label	Nonzero coordinates written as `index:numerator`
00/00	4:129787204019 5:9483735295 9:9483735295 10:129787204019 11:59650516008 13:64538317461 15:25406815289 16:59650516008 18:64538317461 19:25406815289
00/01	4:129782608580 5:9471614609 10:113816256706 11:59650544930 13:64537108553 15:25408404542 16:59083615046 18:65692304686 19:28723055776 20:10600237838
00/02	4:129804047428 5:9529356485 10:90724436776 11:59658191024 13:64543158402 15:25400737727 16:64791215814 18:75658955486 19:33211064353 20:27850283011
00/10	4:129782608580 5:9471614609 10:113816256706 11:59650544930 13:64537108553 15:25408404542 16:59083615046 18:65692304686 19:34637022826 20:4686270787

Label	Nonzero coordinates written as `index:numerator` (cont.)
00/11	4:129778013141 5:9459493923 10:88361574098 11:59650573852 13:64535899646 15:25409993794 16:58516714084 18:66846291910 19:35950399110 20:17289372827
00/12	4:129799451989 5:9517235799 10:65269754168 11:59658219946 13:64541949494 15:25402326980 16:64224314852 18:76812942711 19:40438407688 20:34539418001
00/20	4:129804047428 5:9529356485 10:90724436776 11:59658191024 13:64543158402 15:25400737727 16:64791215814 18:75658955486 19:47010320804 20:14051026560
00/21	4:129799451989 5:9517235799 10:65269754168 11:59658219946 13:64541949494 15:25402326980 16:64224314852 18:76812942711 19:48323697088 20:26654128600
00/22	4:129793324737 5:9501074884 10:31330177357 11:59658258508 13:64540337618 15:25404445983 16:63468446902 18:78351592344 19:50074865467 20:43458264653
01/00	4:113816256706 9:9471614609 10:129782608580 11:59083615046 13:65692304686 14:4686270787 15:34637022826 16:59650544930 18:64537108553 19:25408404542
01/01	4:113799540581 10:113799540581 11:59083643968 13:65691095778 14:4688830548 15:34636052318 16:59083643968 18:65691095778 19:28722085268 20:10602797598
01/02	4:113878721305 10:90707720651 11:59091290062 13:65697145627 14:4676636076 15:34640579975 16:64791244736 18:75657746579 19:33210093846 20:27852842771
01/10	4:113799540581 10:113799540581 11:59083643968 13:65691095778 14:4688830548 15:34636052318 16:59083643968 18:65691095778 19:34636052318 20:4688830548
01/11	4:113782824455 10:88344857973 11:59083672890 13:65689886871 14:4691390308 15:34635081810 16:58516743006 18:66845083003 19:35949428602 20:17291932587
01/12	4:113862005180 10:65253038043 11:59091318984 13:65695936719 14:4679195837 15:34639609467 16:64224343774 18:76811733803 19:40437437180 20:34541977761
01/20	4:113878721305 10:90707720651 11:59091290062 13:65697145627 14:4676636076 15:34640579975 16:64791244736 18:75657746579 19:47009350296 20:14053586321
01/21	4:113862005180 10:65253038043 11:59091318984 13:65695936719 14:4679195837 15:34639609467 16:64224343774 18:76811733803 19:48322726580 20:26656688360
01/22	4:113839717013 10:31313461232 11:59091357546 13:65694324843 14:4682608850 15:34638315457 16:63468475824 18:78350383436 19:50073894959 20:43460824413
02/00	4:90724436776 9:9529356485 10:129804047428 11:64791215814 13:75658955486 14:14051026560 15:47010320804 16:59658191024 18:64543158402 19:25400737727
02/01	4:90707720651 10:113878721305 11:64791244736 13:75657746579 14:14053586321 15:47009350296 16:59091290062 18:65697145627 19:28726612925 20:10590603127
02/02	4:90786901375 10:90786901375 11:64798890830 13:75663796427 14:14041391849 15:47013877953 16:64798890830 18:75663796427 19:33214621503 20:27840648300
02/10	4:90707720651 10:113878721305 11:64791244736 13:75657746579 14:14053586321 15:47009350296 16:59091290062 18:65697145627 19:34640579975 20:4676636076
02/11	4:90691004526 10:88424038697 11:64791273658 13:75656537671 14:14056146081 15:47008379788 16:58524389099 18:66851132851 19:35953956259 20:17279738116
02/12	4:90770185250 10:65332218767 11:64798919752 13:75662587520 14:14043951610 15:47012907445 16:64231989867 18:76817783652 19:40441964837 20:34529783290
02/20	4:90786901375 10:90786901375 11:64798890830 13:75663796427 14:14041391849 15:47013877953 16:64798890830 18:75663796427 19:47013877953 20:14041391849
02/21	4:90770185250 10:65332218767 11:64798919752 13:75662587520 14:14043951610 15:47012907445 16:64231989867 18:76817783652 19:48327254238 20:26644493889
02/22	1:104024441422 4:205368647530 5:22551269251 10:32423706555 11:53000966026 13:79228912663 15:52408345705 16:63510309433 18:78405428469 19:50146790137 20:43299247567
10/00	4:105331362503 9:9467574380 10:129781076767 11:58894648059 13:66076967094 14:14801271851 15:29160847870 16:59650554571 18:64536705584 19:25408934292
10/01	4:105314646378 10:113793968539 11:58894676980 13:66075758186 14:14803831611 15:29159877362 16:59083653609 18:65690692809 19:28721761765 20:10603650851
10/02	4:105393827102 10:90702148609 11:58902323074 13:66081808035 14:14791637140 15:29164405019 16:64791254377 18:75657343609 19:33209770343 20:27853696025
10/10	4:105314646378 10:113793968539 11:58894676980 13:66075758186 14:14803831611 15:29159877362 16:59083653609 18:65690692809 19:34635728815 20:4689683801
10/11	4:105297930253 10:88339285931 11:58894705902 13:66074549279 14:14806391371 15:29158906854 16:58516752646 18:66844680034 19:35949105099 20:17292785841
10/12	4:105377110977 10:65247466001 11:58902351996 13:66080599127 14:14794196900 15:29163434512 16:64224353414 18:76811330834 19:40437113677 20:34542831014
10/20	4:105393827102 10:90702148609 11:58902323074 13:66081808035 14:14791637140 15:29164405019 16:64791254377 18:75657343609 19:47009026794 20:14054439574
10/21	4:105377110977 10:65247466001 11:58902351996 13:66080599127 14:14794196900 15:29163434512 16:64224353414 18:76811330834 19:48322403078 20:26657541614
10/22	4:105354822811 10:31307889190 11:58902390559 13:66078987251 14:14797609914 15:29162140501 16:63468485465 18:78349980467 19:50073571457 20:43461677667
11/00	4:79876679895 9:9455453694 10:129776481328 11:58327747096 13:67230954319 14:21490406840 15:36388191205 16:59650583493 18:64535496677 19:25410523545
11/01	4:79859963770 10:113777252414 11:58327776018 13:67229745411 14:21492966601 15:36387220697 16:59083682530 18:65689483902 19:28720791257 20:10606210612
11/02	4:79939144494 10:90685432484 11:58335422112 13:67235795260 14:21480772129 15:36391748354 16:64791283299 18:75656134702 19:33208799835 20:27856255785
11/10	4:79859963770 10:113777252414 11:58327776018 13:67229745411 14:21492966601 15:36387220697 16:59083682530 18:65689483902 19:34634758307 20:4692243561
11/11	4:98895683495 10:98895683495 11:62161924164 13:69203549934 14:17634518603 15:38520236946 16:62161924164 18:69203549934 19:38520236946 20:17634518603

Label	Nonzero coordinates written as `index:numerator` (cont.)
11/12	4:79922428369 10:65230749876 11:58335451034 13:67234586352 14:21483331890 15:36390777846 16:64224382336 18:76810121927 19:40436143169 20:34545390774
11/20	4:79939144494 10:90685432484 11:58335422112 13:67235795260 14:21480772129 15:36391748354 16:64791283299 18:75656134702 19:47008056286 20:14056999334
11/21	4:79922428369 10:65230749876 11:58335451034 13:67234586352 14:21483331890 15:36390777846 16:64224382336 18:76810121927 19:48321432570 20:26660101374
11/22	4:79900140203 10:31291173065 11:58335489596 13:67232974476 14:21486744903 15:36389483835 16:63468514386 18:78348771560 19:50072600949 20:43464237427
12/00	4:56784859965 9:9513195571 10:129797920176 11:64035347864 13:77197605119 14:30855162613 15:48761489183 16:59658229587 18:64541546525 19:25402856731
12/01	4:56768143840 10:113856433138 11:64035376786 13:77196396212 14:30857722374 15:48760518675 16:59091328624 18:65695533750 19:28725318914 20:10594016140
12/02	1:14634625344 4:76145300007 10:90909668004 11:62383228368 13:77704399455 14:28199259248 15:50194305198 16:64803739046 18:75669077413 19:33222945742 20:27823045531
12/10	4:56768143840 10:113856433138 11:64035376786 13:77196396212 14:30857722374 15:48760518675 16:59091328624 18:65695533750 19:34639285964 20:4680049090
12/11	4:56751427715 10:88401750530 11:64035405708 13:77195187304 14:30860282134 15:48759548167 16:58524427662 18:66849520975 19:35952662249 20:17283151130
12/12	1:104024441422 4:194002628137 10:66340995199 11:52245059514 13:80769174173 14:12038134409 15:58923396726 16:64266215946 18:76865166959 19:40509038347 20:34383813928
12/20	1:14634625344 4:76145300007 10:90909668004 11:62383228368 13:77704399455 14:28199259248 15:50194305198 16:64803739046 18:75669077413 19:47022202192 20:14023789080
12/21	1:104024441422 4:194002628137 10:66340995199 11:52245059514 13:80769174173 14:12038134409 15:58923396726 16:64266215946 18:76865166959 19:48394327747 20:26498524527
12/22	1:141911282127 4:248865494150 10:29223223723 11:46551271131 13:80091369703 14:864830665 15:61307870136 16:61936953878 18:76831052225 19:49294143420 20:43122508843
20/00	4:90724436776 9:9529356485 10:129804047428 11:64791215814 13:75658955486 14:27850283011 15:33211064353 16:59658191024 18:64543158402 19:25400737727
20/01	4:90707720651 10:113878721305 11:64791244736 13:75657746579 14:27852842771 15:33210093846 16:59091290062 18:65697145627 19:28726612925 20:10590603127
20/02	4:90786901375 10:90786901375 11:64798890830 13:75663796427 14:27840648300 15:33214621503 16:64798890830 18:75663796427 19:33214621503 20:27840648300
20/10	4:90707720651 10:113878721305 11:64791244736 13:75657746579 14:27852842771 15:33210093846 16:59091290062 18:65697145627 19:34640579975 20:4676636076
20/11	4:90691004526 10:88424038697 11:64791273658 13:75656537671 14:27855402532 15:33209123338 16:58524389099 18:66851132851 19:35953956259 20:17279738116
20/12	4:90770185250 10:65332218767 11:64798919752 13:75662587520 14:27843208060 15:33213650995 16:64231989867 18:76817783652 19:40441964837 20:34529783290
20/20	4:90786901375 10:90786901375 11:64798890830 13:75663796427 14:27840648300 15:33214621503 16:64798890830 18:75663796427 19:47013877953 20:14041391849
20/21	4:90770185250 10:65332218767 11:64798919752 13:75662587520 14:27843208060 15:33213650995 16:64231989867 18:76817783652 19:48327254238 20:26644493889
20/22	1:104024441422 4:227919916781 10:32423706555 11:53000966026 13:79228912663 14:9036667820 15:43371677886 16:63510309433 18:78405428469 19:50146790137 20:43299247567
21/00	4:65269754168 9:9517235799 10:129799451989 11:64224314852 13:76812942711 14:34539418001 15:40438407688 16:59658219946 18:64541949494 19:25402326980
21/01	4:65253038043 10:113862005180 11:64224343774 13:76811733803 14:34541977761 15:40437437180 16:59091318984 18:65695936719 19:28725642417 20:10593162887
21/02	4:65332218767 10:90770185250 11:64231989867 13:76817783652 14:34529783290 15:40441964837 16:64798919752 18:75662587520 19:33213650995 20:27843208060
21/10	4:65253038043 10:113862005180 11:64224343774 13:76811733803 14:34541977761 15:40437437180 16:59091318984 18:65695936719 19:34639609467 20:4679195837
21/11	4:65236321918 10:88407322572 11:64224372696 13:76810524896 14:34544537521 15:40436466672 16:58524418021 18:66849923944 19:35952985751 20:17282297876
21/12	1:74227836062 4:163196174254 10:66051230640 11:55813425760 13:79362517223 14:21110284662 15:47690294552 16:64256413684 18:76851535821 19:40489778759 20:34425749538
21/20	4:65332218767 10:90770185250 11:64231989867 13:76817783652 14:34529783290 15:40441964837 16:64798919752 18:75662587520 19:47012907445 20:14043951610
21/21	1:74227836062 4:163196174254 10:66051230640 11:55813425760 13:79362517223 14:21110284662 15:47690294552 16:64256413684 18:76851535821 19:48375068160 20:26540460138
21/22	1:141911282127 4:248865494150 10:29223223723 11:46551271131 13:80091369703 14:8750120066 15:53422580735 16:61936953878 18:76831052225 19:49294143420 20:43122508843
22/00	4:31330177357 9:9501074884 10:129793324737 11:63468446902 13:78351592344 14:43458264653 15:50074865467 16:59658258508 18:64540337618 19:25404445983
22/01	4:31313461232 10:113839717013 11:63468475824 13:78350383436 14:43460824413 15:50073894959 16:59091357546 18:65694324843 19:28724348406 20:10596575901
22/02	1:104024441422 4:168564661654 10:91778961682 11:51678129629 13:81924370305 14:24638676688 15:60237743518 16:64833145830 18:75709970827 19:33280724505 20:27697238698
22/10	4:31313461232 10:113839717013 11:63468475824 13:78350383436 14:43460824413 15:50073894959 16:59091357546 18:65694324843 19:34638315457 20:4682608850
22/11	4:31296745107 10:88385034405 11:63468504746 13:78349174529 14:43463384174 15:50072924451 16:58524456584 18:66848312067 19:35951691741 20:17285710890
22/12	1:141911282127 4:214948205506 10:63140512367 11:45795364618 13:81631631212 14:17665553705 15:63060332525 16:62692860390 18:75290790715 19:39656391630 20:34207075204

Label	Nonzero coordinates written as `index:numerator` (cont.)
22/20	1:104024441422 4:168564661654 10:91778961682 11:51678129629 13:81924370305 14:24638676688 15:60237743518 16:64833145830 18:75709970827 19:47079980955 20:13897982248
22/21	1:141911282127 4:214948205506 10:63140512367 11:45795364618 13:81631631212 14:17665553705 15:63060332525 16:62692860390 18:75290790715 19:47541681030 20:26321785804
22/22	1:162357141269 4:235063077854 10:22579780877 11:40436188556 13:79281033441 14:13732456344 15:63375282638 16:58903376283 18:73788811335 19:47625730694 20:42857120710

125.32 First-sample H1 duals at $g = 7/1000$

Exactly 81 nonnegative multipliers for $g = 7/1000$. Each 18-vector is printed in integer units of 10^{-12}; the corresponding right-side pairing is strictly positive and the reduced input coefficients strictly negative using the rational-exponential error enclosure.

Label	Nonzero coordinates written as `index:numerator`
00/00	1:60678259969 3:214679144971 7:112321297530 9:112321297530 10:60678259969 12:214679144971 16:112321297530 18:112321297530
00/01	1:70906276745 2:16375506907 3:229814438420 7:118451770793 9:118451770793 10:54155492165 12:191453605516 16:100195569331 18:100195569331
00/02	1:81540285500 2:33401024918 3:245550513809 7:124825587710 9:124825587710 10:47373808925 12:167306149042 16:87588521193 18:87588521193
00/10	1:70906276745 2:16375506907 3:229814438420 7:118451770793 9:118451770793 10:54155492165 12:191453605516 16:100195569331 18:100195569331
00/11	1:84514218035 2:38162422347 3:249951302605 7:126608104721 9:126608104721 10:45477227035 12:160553013126 16:84062803705 18:84062803705
00/12	1:92154274819 2:50394490925 3:261256964682 7:131187405368 9:131187405368 10:40604892789 12:143204152233 16:75005206909 18:75005206909
00/20	1:81540285500 2:33401024918 3:245550513809 7:124825587710 9:124825587710 10:47373808925 12:167306149042 16:87588521193 18:87588521193
00/21	1:92154274819 2:50394490925 3:261256964682 7:131187405368 9:131187405368 10:40604892789 12:143204152233 16:75005206909 18:75005206909
00/22	1:98308071803 2:60246991878 3:270363277785 7:134875870953 9:134875870953 10:36680398947 12:129230255064 16:67709631308 18:67709631308
01/00	1:54155492165 3:191453605516 7:100195569331 9:100195569331 10:70906276745 11:16375506907 12:229814438420 16:118451770793 18:118451770793
01/01	1:92141419776 2:50373909434 3:261237941948 7:131179700307 9:131179700307 10:40613090903 12:143233343160 16:75020447082 18:75020447082
01/02	1:100841451557 2:64303044832 3:274112141992 7:136394329241 9:136394329241 10:35064773137 12:123477516023 16:64706206989 18:64706206989
01/10	1:92141419776 2:50373909434 3:261237941948 7:131179700307 9:131179700307 10:40613090903 12:143233343160 16:75020447082 18:75020447082
01/11	1:103233144246 2:68132250534 3:277651339423 7:137827863073 9:137827863073 10:33539506175 12:118046516627 16:61870758424 18:61870758424
01/12	1:109296688073 2:77840252192 3:286624097014 7:141462232680 9:141462232680 10:29672569962 12:104277564203 16:54682181598 18:54682181598
01/20	1:100841451557 2:64303044832 3:274112141992 7:136394329241 9:136394329241 10:35064773137 12:123477516023 16:64706206989 18:64706206989
01/21	1:109296688073 2:77840252192 3:286624097014 7:141462232680 9:141462232680 10:29672569962 12:104277564203 16:54682181598 18:54682181598
01/22	1:114097889872 2:85527188472 3:293728856395 7:144339979159 9:144339979159 10:26610673878 12:93375108305 16:48990162380 18:48990162380
02/00	1:47373808925 3:167306149042 7:87588521193 9:87588521193 10:81540285500 11:33401024918 12:245550513809 16:124825587710 18:124825587710
02/01	1:35064773137 3:123477516023 7:64706206989 9:64706206989 10:100841451557 11:64303044832 12:274112141992 16:136394329241 18:136394329241
02/02	1:27526969899 3:96637752328 7:50693543036 9:50693543036 10:112661093200 12:374829515846 14:83226813498 16:143478791328 18:60251977830
02/10	1:35064773137 3:123477516023 7:64706206989 9:64706206989 10:100841451557 11:64303044832 12:274112141992 16:136394329241 18:136394329241
02/11	1:114494176933 3:380476938685 5:86161661571 7:144577505874 9:58415844303 10:26357947613 12:92475228987 16:48520348017 18:48520348017
02/12	1:119157889455 3:394845052997 5:93628471442 7:147372843968 9:53744372526 10:23383733300 12:81884980499 16:42991327906 18:42991327906
02/20	1:27526969899 3:96637752328 7:50693543036 9:50693543036 10:112661093200 12:374829515846 13:83226813498 16:60251977830 18:143478791328
02/21	1:119157889455 3:394845052997 5:93628471442 7:147372843968 9:53744372526 10:23383733300 12:81884980499 16:42991327906 18:42991327906
02/22	1:122867446130 3:406273573536 5:99567635630 7:149596279705 9:50028644074 10:21018017895 12:73461406624 16:38593498203 18:38593498203
10/00	1:54155492165 3:191453605516 7:100195569331 9:100195569331 10:70906276745 11:16375506907 12:229814438420 16:118451770793 18:118451770793

Label	Nonzero coordinates written as `index:numerator` (cont.)
10/01	1:92141419776 2:50373909434 3:261237941948 7:131179700307 9:131179700307 10:40613090903 12:143233343160 16:75020447082 18:75020447082
10/02	1:100841451557 2:64303044832 3:274112141992 7:136394329241 9:136394329241 10:35064773137 12:123477516023 16:64706206989 18:64706206989
10/10	1:92141419776 2:50373909434 3:261237941948 7:131179700307 9:131179700307 10:40613090903 12:143233343160 16:75020447082 18:75020447082
10/11	1:103233144246 2:68132250534 3:277651339423 7:137827863073 9:137827863073 10:33539506175 12:118046516627 16:61870758424 18:61870758424
10/12	1:109296688073 2:77840252192 3:286624097014 7:141462232680 9:141462232680 10:29672569962 12:104277564203 16:54682181598 18:54682181598
10/20	1:100841451557 2:64303044832 3:274112141992 7:136394329241 9:136394329241 10:35064773137 12:123477516023 16:64706206989 18:64706206989
10/21	1:109296688073 2:77840252192 3:286624097014 7:141462232680 9:141462232680 10:29672569962 12:104277564203 16:54682181598 18:54682181598
10/22	1:114097889872 2:85527188472 3:293728856395 7:144339979159 9:144339979159 10:26610673878 12:93375108305 16:48990162380 18:48990162380
11/00	1:45477227035 3:160553013126 7:84062803705 9:84062803705 10:84514218035 11:38162422347 12:249951302605 16:126608104721 18:126608104721
11/01	1:33539506175 3:118046516627 7:61870758424 9:61870758424 10:103233144246 11:68132250534 12:277651339423 16:137827863073 18:137827863073
11/02	1:26357947613 3:92475228987 7:48520348017 9:48520348017 10:114494176933 12:380476938685 14:86161661571 16:144577505874 18:58415844303
11/10	1:33539506175 3:118046516627 7:61870758424 9:61870758424 10:103233144246 11:68132250534 12:277651339423 16:137827863073 18:137827863073
11/11	1:19280262431 3:67273802021 7:35363036677 9:35363036677 10:125592331113 11:103930296920 12:310738189195 16:151229522483 18:151229522483
11/12	1:129352534136 2:109950548150 3:316302491388 7:153483314650 9:153483314650 10:16882248060 12:58735221612 16:30905163677 18:30905163677
11/20	1:26357947613 3:92475228987 7:48520348017 9:48520348017 10:114494176933 12:380476938685 13:86161661571 16:58415844303 18:144577505874
11/21	1:129352534136 2:109950548150 3:316302491388 7:153483314650 9:153483314650 10:16882248060 12:58735221612 16:30905163677 18:30905163677
11/22	1:132270842402 2:114622888665 3:320620968074 7:155232491562 9:155232491562 10:15021139732 12:52108395976 16:27445391014 18:27445391014
12/00	1:40604892789 3:143204152233 7:75005206909 9:75005206909 10:92154274819 11:50394490925 12:261256964682 16:131187405368 18:131187405368
12/01	1:29672569962 3:104277564203 7:54682181598 9:54682181598 10:109296688073 11:77840252192 12:286624097014 16:141462232680 18:141462232680
12/02	1:23383733300 3:81884980499 7:42991327906 9:42991327906 10:119157889455 12:394845052997 14:93628471442 16:147372843968 18:53744372526
12/10	1:29672569962 3:104277564203 7:54682181598 9:54682181598 10:109296688073 11:77840252192 12:286624097014 16:141462232680 18:141462232680
12/11	1:16882248060 3:58735221612 7:30905163677 9:30905163677 10:129352534136 11:109950548150 12:316302491388 16:153483314650 18:153483314650
12/12	1:139616474211 3:457874513543 5:126383569848 7:159635319636 9:33251749788 10:10336570698 12:35428108375 16:18736846951 18:18736846951
12/20	1:23383733300 3:81884980499 7:42991327906 9:42991327906 10:119157889455 12:394845052997 13:93628471442 16:53744372526 18:147372843968
12/21	1:139616474211 3:457874513543 5:126383569848 7:159635319636 9:33251749788 10:10336570698 12:35428108375 16:18736846951 18:18736846951
12/22	1:141352947970 3:463224297291 5:129163741082 7:160676128053 9:31512386971 10:9229160007 12:31484964687 16:16678186969 18:16678186969
20/00	1:47373808925 3:167306149042 7:87588521193 9:87588521193 10:81540285500 11:33401024918 12:245550513809 16:124825587710 18:124825587710
20/01	1:35064773137 3:123477516023 7:64706206989 9:64706206989 10:100841451557 11:64303044832 12:274112141992 16:136394329241 18:136394329241
20/02	1:27526969899 3:96637752328 7:50693543036 9:50693543036 10:112661093200 12:374829515846 14:83226813498 16:143478791328 18:60251977830
20/10	1:35064773137 3:123477516023 7:64706206989 9:64706206989 10:100841451557 11:64303044832 12:274112141992 16:136394329241 18:136394329241
20/11	1:114494176933 3:380476938685 4:86161661571 7:58415844303 9:144577505874 10:26357947613 12:92475228987 16:48520348017 18:48520348017
20/12	1:119157889455 3:394845052997 4:93628471442 7:53744372526 9:147372843968 10:23383733300 12:81884980499 16:42991327906 18:42991327906
20/20	1:27526969899 3:96637752328 7:50693543036 9:50693543036 10:112661093200 12:374829515846 13:83226813498 16:60251977830 18:143478791328
20/21	1:119157889455 3:394845052997 4:93628471442 7:53744372526 9:147372843968 10:23383733300 12:81884980499 16:42991327906 18:42991327906
20/22	1:122867446130 3:406273573536 4:99567635630 7:50028644074 9:149596279705 10:21018017895 12:73461406624 16:38593498203 18:38593498203
21/00	1:40604892789 3:143204152233 7:75005206909 9:75005206909 10:92154274819 11:50394490925 12:261256964682 16:131187405368 18:131187405368
21/01	1:29672569962 3:104277564203 7:54682181598 9:54682181598 10:109296688073 11:77840252192 12:286624097014 16:141462232680 18:141462232680

Label	Nonzero coordinates written as `index:numerator` (cont.)
21/02	1:23383733300 3:81884980499 7:42991327906 9:42991327906 10:119157889455 12:394845052997 14:93628471442 16:147372843968 18:53744372526
21/10	1:29672569962 3:104277564203 7:54682181598 9:54682181598 10:109296688073 11:77840252192 12:286624097014 16:141462232680 18:141462232680
21/11	1:16882248060 3:58735221612 7:30905163677 9:30905163677 10:129352534136 11:109950548150 12:316302491388 16:153483314650 18:153483314650
21/12	1:139616474211 3:457874513543 4:126383569848 7:33251749788 9:159635319636 10:10336570698 12:35428108375 16:18736846951 18:18736846951
21/20	1:23383733300 3:81884980499 7:42991327906 9:42991327906 10:119157889455 12:394845052997 13:93628471442 16:53744372526 18:147372843968
21/21	1:139616474211 3:457874513543 4:126383569848 7:33251749788 9:159635319636 10:10336570698 12:35428108375 16:18736846951 18:18736846951
21/22	1:141352947970 3:463224297291 4:129163741082 7:31512386971 9:160676128053 10:9229160007 12:31484964687 16:16678186969 18:16678186969
22/00	1:36680398947 3:129230255064 7:67709631308 9:67709631308 10:98308071803 11:60246991878 12:270363277785 16:134875870953 18:134875870953
22/01	1:26610673878 3:93375108305 7:48990162380 9:48990162380 10:114097889872 11:85527188472 12:293728856395 16:144339979159 18:144339979159
22/02	1:21018017895 3:73461406624 7:38593498203 9:38593498203 10:122867446130 12:406273573536 14:99567635630 16:149596279705 18:50028644074
22/10	1:26610673878 3:93375108305 7:48990162380 9:48990162380 10:114097889872 11:85527188472 12:293728856395 16:144339979159 18:144339979159
22/11	1:15021139732 3:52108395976 7:27445391014 9:27445391014 10:132270842402 11:114622888665 12:320620968074 16:155232491562 18:155232491562
22/12	1:9229160007 3:31484964687 7:16678186969 9:16678186969 10:141352947970 12:463224297291 14:129163741082 16:160676128053 18:31512386971
22/20	1:21018017895 3:73461406624 7:38593498203 9:38593498203 10:122867446130 12:406273573536 13:99567635630 16:50028644074 18:149596279705
22/21	1:9229160007 3:31484964687 7:16678186969 9:16678186969 10:141352947970 12:463224297291 13:129163741082 16:31512386971 18:160676128053
22/22	1:153769087237 3:500850243630 4:151131490102 7:15343160753 9:166474650855 10:2524774690 12:6630999948 13:1637796393 14:1637796393

125.33 First-sample H2 controls at $g = 7/1000$

These 81 independent one-sample witnesses for $g = 7/1000$ are exact decimal rationals at scale 10^{-9}. Each ten-port half-phase control has total power below 421; the guard inequalities are strict.

Label	Nonzero coordinates written as `index:numerator`
00/00	4:129787204019 5:9483735295 9:9483735295 10:129787204019 11:59650516008 13:64538317461 15:25406815289 16:59650516008 18:64538317461 19:25406815289
00/01	4:129782608580 5:9471614609 10:113816256706 11:59650544930 13:64537108553 15:25408404542 16:59083615046 18:65692304686 19:28723055776 20:10600237838
00/02	4:129804047428 5:9529356485 10:90724436776 11:59658191024 13:64543158402 15:25400737727 16:64791215814 18:75658955486 19:33211064353 20:27850283011
00/10	4:129782608580 5:9471614609 10:113816256706 11:59650544930 13:64537108553 15:25408404542 16:59083615046 18:65692304686 19:34637022826 20:4686270787
00/11	4:129778013141 5:9459493923 10:88361574098 11:59650573852 13:64535899646 15:25409993794 16:58516714084 18:66846291910 19:35950399110 20:17289372827
00/12	4:129799451989 5:9517235799 10:65269754168 11:59658219946 13:64541949494 15:25402326980 16:64224314852 18:76812942711 19:40438407688 20:34539418001
00/20	4:129804047428 5:9529356485 10:90724436776 11:59658191024 13:64543158402 15:25400737727 16:64791215814 18:75658955486 19:47010320804 20:14051026560
00/21	4:129799451989 5:9517235799 10:65269754168 11:59658219946 13:64541949494 15:25402326980 16:64224314852 18:76812942711 19:48323697088 20:26654128600
00/22	4:129793324737 5:9501074884 10:31330177357 11:59658258508 13:64540337618 15:25404445983 16:63468446902 18:78351592344 19:50074865467 20:43458264653
01/00	4:113816256706 9:9471614609 10:129782608580 11:59083615046 13:65692304686 14:4686270787 15:34637022826 16:59650544930 18:64537108553 19:25408404542
01/01	4:113799540581 10:113799540581 11:59083643968 13:65691095778 14:4688830548 15:34636052318 16:59083643968 18:65691095778 19:28722085268 20:10602797598
01/02	4:113878721305 10:90707720651 11:59091290062 13:65697145627 14:4676636076 15:34640579975 16:64791244736 18:75657746579 19:33210093846 20:27852842771
01/10	4:113799540581 10:113799540581 11:59083643968 13:65691095778 14:4688830548 15:34636052318 16:59083643968 18:65691095778 19:34636052318 20:4688830548
01/11	4:113782824455 10:88344857973 11:59083672890 13:65689886871 14:4691390308 15:34635081810 16:58516743006 18:66845083003 19:35949428602 20:17291932587
01/12	4:113862005180 10:65253038043 11:59091318984 13:65695936719 14:4679195837 15:34639609467 16:64224343774 18:76811733803 19:40437437180 20:34541977761

Label	Nonzero coordinates written as `index:numerator` (cont.)
01/20	4:113878721305 10:90707720651 11:59091290062 13:65697145627 14:4676636076 15:34640579975 16:64791244736 18:75657746579 19:47009350296 20:14053586321
01/21	4:113862005180 10:65253038043 11:59091318984 13:65695936719 14:4679195837 15:34639609467 16:64224343774 18:76811733803 19:48322726580 20:26656688360
01/22	4:113839717013 10:31313461232 11:59091357546 13:65694324843 14:4682608850 15:34638315457 16:63468475824 18:78350383436 19:50073894959 20:43460824413
02/00	4:90724436776 9:9529356485 10:129804047428 11:64791215814 13:75658955486 14:14051026560 15:47010320804 16:59658191024 18:64543158402 19:25400737727
02/01	4:90707720651 10:113878721305 11:64791244736 13:75657746579 14:14053586321 15:47009350296 16:59091290062 18:65697145627 19:28726612925 20:10590603127
02/02	4:90786901375 10:90786901375 11:64798890830 13:75663796427 14:14041391849 15:47013877953 16:64798890830 18:75663796427 19:33214621503 20:27840648300
02/10	4:90707720651 10:113878721305 11:64791244736 13:75657746579 14:14053586321 15:47009350296 16:59091290062 18:65697145627 19:34640579975 20:4676636076
02/11	4:90691004526 10:88424038697 11:64791273658 13:75656537671 14:14056146081 15:47008379788 16:58524389099 18:66851132851 19:35953956259 20:17279738116
02/12	4:90770185250 10:65332218767 11:64798919752 13:75662587520 14:14043951610 15:47012907445 16:64231989867 18:76817783652 19:40441964837 20:34529783290
02/20	4:90786901375 10:90786901375 11:64798890830 13:75663796427 14:14041391849 15:47013877953 16:64798890830 18:75663796427 19:47013877953 20:14041391849
02/21	4:90770185250 10:65332218767 11:64798919752 13:75662587520 14:14043951610 15:47012907445 16:64231989867 18:76817783652 19:48327254238 20:26644493889
02/22	1:104024441422 4:205368647530 5:22551269251 10:32423706555 11:53000966026 13:79228912663 15:52408345705 16:63510309433 18:78405428469 19:50146790137 20:43299247567
10/00	4:105331362503 9:9467574380 10:129781076767 11:58894648059 13:66076967094 14:14801271851 15:29160847870 16:59650554571 18:64536705584 19:25408934292
10/01	4:105314646378 10:113793968539 11:58894676980 13:66075758186 14:14803831611 15:29159877362 16:59083653609 18:65690692809 19:28721761765 20:10603650851
10/02	4:105393827102 10:90702148609 11:58902323074 13:66081808035 14:14791637140 15:29164405019 16:64791254377 18:75657343609 19:33209770343 20:27853696025
10/10	4:105314646378 10:113793968539 11:58894676980 13:66075758186 14:14803831611 15:29159877362 16:59083653609 18:65690692809 19:34635728815 20:4689683801
10/11	4:105297930253 10:88339285931 11:58894705902 13:66074549279 14:14806391371 15:29158906854 16:58516752646 18:66844680034 19:35949105099 20:17292785841
10/12	4:105377110977 10:65247466001 11:58902351996 13:66080599127 14:14794196900 15:29163434512 16:64224353414 18:76811330834 19:40437113677 20:34542831014
10/20	4:105393827102 10:90702148609 11:58902323074 13:66081808035 14:14791637140 15:29164405019 16:64791254377 18:75657343609 19:47009026794 20:14054439574
10/21	4:105377110977 10:65247466001 11:58902351996 13:66080599127 14:14794196900 15:29163434512 16:64224353414 18:76811330834 19:48322403078 20:26657541614
10/22	4:105354822811 10:31307889190 11:58902390559 13:66078987251 14:14797609914 15:29162140501 16:63468485465 18:78349980467 19:50073571457 20:43461677667
11/00	4:79876679895 9:9455453694 10:129776481328 11:58327747096 13:67230954319 14:21490406840 15:36388191205 16:59650583493 18:64535496677 19:25410523545
11/01	4:79859963770 10:113777252414 11:58327776018 13:67229745411 14:21492966601 15:36387220697 16:59083682530 18:65689483902 19:28720791257 20:10606210612
11/02	4:79939144494 10:90685432484 11:58335422112 13:67235795260 14:21480772129 15:36391748354 16:64791283299 18:75656134702 19:33208799835 20:27856255785
11/10	4:79859963770 10:113777252414 11:58327776018 13:67229745411 14:21492966601 15:36387220697 16:59083682530 18:65689483902 19:34634758307 20:4692243561
11/11	4:98895683495 10:98895683495 11:62161924164 13:69203549934 14:17634518603 15:38520236946 16:62161924164 18:69203549934 19:38520236946 20:17634518603
11/12	4:79922428369 10:65230749876 11:58335451034 13:67234586352 14:21483331890 15:36390777846 16:64224382336 18:76810121927 19:40436143169 20:34545390774
11/20	4:79939144494 10:90685432484 11:58335422112 13:67235795260 14:21480772129 15:36391748354 16:64791283299 18:75656134702 19:47008056286 20:14056999334
11/21	4:79922428369 10:65230749876 11:58335451034 13:67234586352 14:21483331890 15:36390777846 16:64224382336 18:76810121927 19:48321432570 20:26660101374
11/22	4:79900140203 10:31291173065 11:58335489596 13:67232974476 14:21486744903 15:36389483835 16:63468514386 18:78348771560 19:50072600949 20:43464237427
12/00	4:56784859965 9:9513195571 10:129797920176 11:64035347864 13:77197605119 14:30855162613 15:48761489183 16:59658229587 18:64541546525 19:25402856731
12/01	4:56768143840 10:113856433138 11:64035376786 13:77196396212 14:30857722374 15:48760518675 16:59091328624 18:65695533750 19:28725318914 20:10594016140
12/02	1:14634625344 4:76145300007 10:90909668004 11:62383228368 13:77704399455 14:28199259248 15:50194305198 16:64803739046 18:75669077413 19:33222945742 20:27823045531
12/10	4:56768143840 10:113856433138 11:64035376786 13:77196396212 14:30857722374 15:48760518675 16:59091328624 18:65695533750 19:34639285964 20:4680049090
12/11	4:56751427715 10:88401750530 11:64035405708 13:77195187304 14:30860282134 15:48759548167 16:58524427662 18:66849520975 19:35952662249 20:17283151130
12/12	1:104024441422 4:194002628137 10:66340995199 11:52245059514 13:80769174173 14:12038134409 15:58923396726 16:64266215946 18:76865166959 19:40509038347 20:34383813928
12/20	1:14634625344 4:76145300007 10:90909668004 11:62383228368 13:77704399455 14:28199259248 15:50194305198 16:64803739046 18:75669077413 19:47022202192 20:14023789080

Label	Nonzero coordinates written as `index:numerator` (cont.)
12/21	1:104024441422 4:194002628137 10:66340995199 11:52245059514 13:80769174173 14:12038134409 15:58923396726 16:64266215946 18:76865166959 19:48394327747 20:26498524527
12/22	1:141911282127 4:248865494150 10:29223223723 11:46551271131 13:80091369703 14:864830665 15:61307870136 16:61936953878 18:76831052225 19:49294143420 20:43122508843
20/00	4:90724436776 9:9529356485 10:129804047428 11:64791215814 13:75658955486 14:27850283011 15:33211064353 16:59658191024 18:64543158402 19:25400737727
20/01	4:90707720651 10:113878721305 11:64791244736 13:75657746579 14:27852842771 15:33210093846 16:59091290062 18:65697145627 19:28726612925 20:10590603127
20/02	4:90786901375 10:90786901375 11:64798890830 13:75663796427 14:27840648300 15:33214621503 16:64798890830 18:75663796427 19:33214621503 20:27840648300
20/10	4:90707720651 10:113878721305 11:64791244736 13:75657746579 14:27852842771 15:33210093846 16:59091290062 18:65697145627 19:34640579975 20:4676636076
20/11	4:90691004526 10:88424038697 11:64791273658 13:75656537671 14:27855402532 15:33209123338 16:58524389099 18:66851132851 19:35953956259 20:17279738116
20/12	4:90770185250 10:65332218767 11:64798919752 13:75662587520 14:27843208060 15:33213650995 16:64231989867 18:76817783652 19:40441964837 20:34529783290
20/20	4:90786901375 10:90786901375 11:64798890830 13:75663796427 14:27840648300 15:33214621503 16:64798890830 18:75663796427 19:47013877953 20:14041391849
20/21	4:90770185250 10:65332218767 11:64798919752 13:75662587520 14:27843208060 15:33213650995 16:64231989867 18:76817783652 19:48327254238 20:26644493889
20/22	1:104024441422 4:227919916781 10:32423706555 11:53000966026 13:79228912663 14:9036667820 15:43371677886 16:63510309433 18:78405428469 19:50146790137 20:43299247567
21/00	4:65269754168 9:9517235799 10:129799451989 11:64224314852 13:76812942711 14:34539418001 15:40438407688 16:59658219946 18:64541949494 19:25402326980
21/01	4:65253038043 10:113862005180 11:64224343774 13:76811733803 14:34541977761 15:40437437180 16:59091318984 18:65695936719 19:28725642417 20:10593162887
21/02	4:65332218767 10:90770185250 11:64231989867 13:76817783652 14:34529783290 15:40441964837 16:64798919752 18:75662587520 19:33213650995 20:27843208060
21/10	4:65253038043 10:113862005180 11:64224343774 13:76811733803 14:34541977761 15:40437437180 16:59091318984 18:65695936719 19:34639609467 20:4679195837
21/11	4:65236321918 10:88407322572 11:64224372696 13:76810524896 14:34544537521 15:40436466672 16:58524418021 18:66849923944 19:35952985751 20:17282297876
21/12	1:74227836062 4:163196174254 10:66051230640 11:55813425760 13:79362517223 14:21110284662 15:47690294552 16:64256413684 18:76851535821 19:40489778759 20:34425749538
21/20	4:65332218767 10:90770185250 11:64231989867 13:76817783652 14:34529783290 15:40441964837 16:64798919752 18:75662587520 19:47012907445 20:14043951610
21/21	1:74227836062 4:163196174254 10:66051230640 11:55813425760 13:79362517223 14:21110284662 15:47690294552 16:64256413684 18:76851535821 19:48375068160 20:26540460138
21/22	1:141911282127 4:248865494150 10:29223223723 11:46551271131 13:80091369703 14:8750120066 15:53422580735 16:61936953878 18:76831052225 19:49294143420 20:43122508843
22/00	4:31330177357 9:9501074884 10:129793324737 11:63468446902 13:78351592344 14:43458264653 15:50074865467 16:59658258508 18:64540337618 19:25404445983
22/01	4:31313461232 10:113839717013 11:63468475824 13:78350383436 14:43460824413 15:50073894959 16:59091357546 18:65694324843 19:28724348406 20:10596575901
22/02	1:104024441422 4:168564661654 10:91778961682 11:51678129629 13:81924370305 14:24638676688 15:60237743518 16:64833145830 18:75709970827 19:33280724505 20:27697238698
22/10	4:31313461232 10:113839717013 11:63468475824 13:78350383436 14:43460824413 15:50073894959 16:59091357546 18:65694324843 19:34638315457 20:4682608850
22/11	4:31296745107 10:88385034405 11:63468504746 13:78349174529 14:43463384174 15:50072924451 16:58524456584 18:66848312067 19:35951691741 20:17285710890
22/12	1:141911282127 4:214948205506 10:63140512367 11:45795364618 13:81631631212 14:17665553705 15:63060332525 16:62692860390 18:75290790715 19:39656391630 20:34207075204
22/20	1:104024441422 4:168564661654 10:91778961682 11:51678129629 13:81924370305 14:24638676688 15:60237743518 16:64833145830 18:75709970827 19:47079980955 20:13897982248
22/21	1:141911282127 4:214948205506 10:63140512367 11:45795364618 13:81631631212 14:17665553705 15:63060332525 16:62692860390 18:75290790715 19:47541681030 20:26321785804
22/22	1:162357141269 4:235063077854 10:22579780877 11:40436188556 13:79281033441 14:13732456344 15:63375282638 16:58903376283 18:73788811335 19:47625730694 20:42857120710

125.34 First-sample H1 duals at $g = 1/100$

Exactly 81 nonnegative multipliers for $g = 1/100$. Each 18-vector is printed in integer units of 10^{-12}; the corresponding right-side pairing is strictly positive and the reduced input coefficients strictly negative using the rational-exponential error enclosure.

Label	Nonzero coordinates written as `index:numerator`
00/00	1:60733111416 3:214642267055 7:112312310765 9:112312310765 10:60733111416 12:214642267055 16:112312310765 18:112312310765
00/01	1:70981696035 2:16342756338 3:229922410211 7:118512328380 9:118512328380 10:54184627355 12:191286542309 16:100128655497 18:100128655497

Label	Nonzero coordinates written as `index:numerator` (cont.)
00/02	1:81629976894 2:33322882470 3:245798481096 7:124954147553 9:124954147553 10:47380751494 12:167019941016 16:87469835962 18:87469835962
00/10	1:70981696035 2:16342756338 3:229922410211 7:118512328380 9:118512328380 10:54184627355 12:191286542309 16:100128655497 18:100128655497
00/11	1:84605280817 2:38067407579 3:250234515102 7:126754097245 9:126754097245 10:45479637239 12:160239455510 16:83932754631 18:83932754631
00/12	1:92243618025 2:50247770865 3:261622905827 7:131375010962 9:131375010962 10:40599009198 12:142832280960 16:74852196601 18:74852196601
00/20	1:81629976894 2:33322882470 3:245798481096 7:124954147553 9:124954147553 10:47380751494 12:167019941016 16:87469835962 18:87469835962
00/21	1:92243618025 2:50247770865 3:261622905827 7:131375010962 9:131375010962 10:40599009198 12:142832280960 16:74852196601 18:74852196601
00/22	1:98390547987 2:60049883285 3:270787680600 7:135093678022 9:135093678022 10:36671337781 12:128823907165 16:67544643569 18:67544643569
01/00	1:54184627355 3:191286542309 7:100128655497 9:100128655497 10:70981696035 11:16342756338 12:229922410211 16:118512328380 18:118512328380
01/01	1:92312674051 2:50357890052 3:261725865016 7:131416787324 9:131416787324 10:40554884834 12:142674907664 16:74770101868 18:74770101868
01/02	1:101010258092 2:64227365886 3:274693541451 7:136678506504 9:136678506504 10:34997435431 12:122853792226 16:64430296953 18:64430296953
01/10	1:92312674051 2:50357890052 3:261725865016 7:131416787324 9:131416787324 10:40554884834 12:142674907664 16:74770101868 18:74770101868
01/11	1:103398443298 2:68035650675 3:278254209893 7:138123270917 9:138123270917 10:33471469326 12:117411304470 16:61591190252 18:61591190252
01/12	1:109447626615 2:77681892832 3:287273249156 7:141782804845 9:141782804845 10:29606254566 12:103625687650 16:54399839746 18:54399839746
01/20	1:101010258092 2:64227365886 3:274693541451 7:136678506504 9:136678506504 10:34997435431 12:122853792226 16:64430296953 18:64430296953
01/21	1:109447626615 2:77681892832 3:287273249156 7:141782804845 9:141782804845 10:29606254566 12:103625687650 16:54399839746 18:54399839746
01/22	1:114231901802 2:85311067645 3:294406371677 7:144677115771 9:144677115771 10:26549271482 12:92722697771 16:48712229041 18:48712229041
02/00	1:47380751494 3:167019941016 7:87469835962 9:87469835962 10:81629976894 11:33322882470 12:245798481096 16:124954147553 18:124954147553
02/01	1:34997435431 3:122853792226 7:64430296953 9:64430296953 10:101010258092 11:64227365886 12:274693541451 16:136678506504 18:136678506504
02/02	1:112913799085 3:375650318711 5:83209174327 7:143879711996 9:60670537669 10:27391492623 12:95726551023 16:50279207283 18:50279207283
02/10	1:34997435431 3:122853792226 7:64430296953 9:64430296953 10:101010258092 11:64227365886 12:274693541451 16:136678506504 18:136678506504
02/11	1:114738640620 3:381281026032 5:86119131327 7:144983674160 9:58864542833 10:26225483252 12:91567879540 16:48109811117 18:48109811117
02/12	1:119376612498 3:395591891777 5:93515005697 7:147789476953 9:54274471256 10:23261982823 12:80998303488 16:42596127754 18:42596127754
02/20	1:112913799085 3:375650318711 5:83209174327 7:143879711996 9:60670537669 10:27391492623 12:95726551023 16:50279207283 18:50279207283
02/21	1:119376612498 3:395591891777 5:93515005697 7:147789476953 9:54274471256 10:23261982823 12:80998303488 16:42596127754 18:42596127754
02/22	1:123060811832 3:406959809356 5:99389960557 7:150018282355 9:50628321799 10:20907909405 12:72602300649 16:38216302023 18:38216302023
10/00	1:54184627355 3:191286542309 7:100128655497 9:100128655497 10:70981696035 11:16342756338 12:229922410211 16:118512328380 18:118512328380
10/01	1:92312674051 2:50357890052 3:261725865016 7:131416787324 9:131416787324 10:40554884834 12:142674907664 16:74770101868 18:74770101868
10/02	1:101010258092 2:64227365886 3:274693541451 7:136678506504 9:136678506504 10:34997435431 12:122853792226 16:64430296953 18:64430296953
10/10	1:92312674051 2:50357890052 3:261725865016 7:131416787324 9:131416787324 10:40554884834 12:142674907664 16:74770101868 18:74770101868
10/11	1:103398443298 2:68035650675 3:278254209893 7:138123270917 9:138123270917 10:33471469326 12:117411304470 16:61591190252 18:61591190252
10/12	1:109447626615 2:77681892832 3:287273249156 7:141782804845 9:141782804845 10:29606254566 12:103625687650 16:54399839746 18:54399839746
10/20	1:101010258092 2:64227365886 3:274693541451 7:136678506504 9:136678506504 10:34997435431 12:122853792226 16:64430296953 18:64430296953
10/21	1:109447626615 2:77681892832 3:287273249156 7:141782804845 9:141782804845 10:29606254566 12:103625687650 16:54399839746 18:54399839746
10/22	1:114231901802 2:85311067645 3:294406371677 7:144677115771 9:144677115771 10:26549271482 12:92722697771 16:48712229041 18:48712229041
11/00	1:45479637239 3:160239455510 7:83932754631 9:83932754631 10:84605280817 11:38067407579 12:250234515102 16:126754097245 18:126754097245
11/01	1:33471469326 3:117411304470 7:61591190252 9:61591190252 10:103398443298 11:68035650675 12:278254209893 16:138123270917 18:138123270917
11/02	1:26225483252 3:91567879540 7:48109811117 9:48109811117 10:114738640620 12:381281026032 14:86119131327 16:144983674160 18:58864542833

Label	Nonzero coordinates written as `index:numerator` (cont.)
11/10	1:33471469326 3:117411304470 7:61591190252 9:61591190252 10:103398443298 11:68035650675 12:278254209893 16:138123270917 18:138123270917
11/11	1:125895083174 2:103909590111 3:311795610058 7:151732912511 9:151732912511 10:19096909994 12:66143217393 16:34846882123 18:34846882123
11/12	1:129612036869 2:109836776237 3:317337407944 7:153981533101 9:153981533101 10:16721907695 12:57672569920 16:30428117567 18:30428117567
11/20	1:26225483252 3:91567879540 7:48109811117 9:48109811117 10:114738640620 12:381281026032 13:86119131327 16:58864542833 18:144983674160
11/21	1:129612036869 2:109836776237 3:317337407944 7:153981533101 9:153981533101 10:16721907695 12:57672569920 16:30428117567 18:30428117567
11/22	1:132492414848 2:114429929070 3:321631911965 7:155724056081 9:155724056081 10:14881447769 12:51108413235 16:27003885476 18:27003885476
12/00	1:40599009198 3:142832280960 7:74852196601 9:74852196601 10:92243618025 11:50247770865 12:261622905827 16:131375010962 18:131375010962
12/01	1:29606254566 3:103625687650 7:54399839746 9:54399839746 10:109447626615 11:77681892832 12:287273249156 16:141782804845 18:141782804845
12/02	1:23261982823 3:80998303488 7:42596127754 9:42596127754 10:119376612498 12:395591891777 14:93515005697 16:147789476953 18:54274471256
12/10	1:29606254566 3:103625687650 7:54399839746 9:54399839746 10:109447626615 11:77681892832 12:287273249156 16:141782804845 18:141782804845
12/11	1:16721907695 3:57672569920 7:30428117567 9:30428117567 10:129612036869 11:109836776237 12:317337407944 16:153981533101 18:153981533101
12/12	1:10100338798 3:34056181899 7:18108485723 9:18108485723 10:139975001566 12:459150013359 14:126361927104 16:160250746466 18:33888819362
12/20	1:23261982823 3:80998303488 7:42596127754 9:42596127754 10:119376612498 12:395591891777 13:93515005697 16:54274471256 18:147789476953
12/21	1:10100338798 3:34056181899 7:18108485723 9:18108485723 10:139975001566 12:459150013359 13:126361927104 16:33888819362 18:160250746466
12/22	1:141658693750 3:464345192077 5:129046802345 7:161269318453 9:32222516108 10:9024518883 12:30219178776 16:16106889804 18:16106889804
20/00	1:47380751494 3:167019941016 7:87469835962 9:87469835962 10:81629976894 11:33322882470 12:245798481096 16:124954147553 18:124954147553
20/01	1:34997435431 3:122853792226 7:64430296953 9:64430296953 10:101010258092 11:64227365886 12:274693541451 16:136678506504 18:136678506504
20/02	1:112913799085 3:375650318711 4:83209174327 7:60670537669 9:143879711996 10:27391492623 12:95726551023 16:50279207283 18:50279207283
20/10	1:34997435431 3:122853792226 7:64430296953 9:64430296953 10:101010258092 11:64227365886 12:274693541451 16:136678506504 18:136678506504
20/11	1:114738640620 3:381281026032 4:86119131327 7:58864542833 9:144983674160 10:26225483252 12:91567879540 16:48109811117 18:48109811117
20/12	1:119376612498 3:395591891777 4:93515005697 7:54274471256 9:147789476953 10:23261982823 12:80998303488 16:42596127754 18:42596127754
20/20	1:112913799085 3:375650318711 4:83209174327 7:60670537669 9:143879711996 10:27391492623 12:95726551023 16:50279207283 18:50279207283
20/21	1:119376612498 3:395591891777 4:93515005697 7:54274471256 9:147789476953 10:23261982823 12:80998303488 16:42596127754 18:42596127754
20/22	1:123060811832 3:406959809356 4:99389960557 7:50628321799 9:150018282355 10:20907909405 12:72602300649 16:38216302023 18:38216302023
21/00	1:40599009198 3:142832280960 7:74852196601 9:74852196601 10:92243618025 11:50247770865 12:261622905827 16:131375010962 18:131375010962
21/01	1:29606254566 3:103625687650 7:54399839746 9:54399839746 10:109447626615 11:77681892832 12:287273249156 16:141782804845 18:141782804845
21/02	1:23261982823 3:80998303488 7:42596127754 9:42596127754 10:119376612498 12:395591891777 14:93515005697 16:147789476953 18:54274471256
21/10	1:29606254566 3:103625687650 7:54399839746 9:54399839746 10:109447626615 11:77681892832 12:287273249156 16:141782804845 18:141782804845
21/11	1:16721907695 3:57672569920 7:30428117567 9:30428117567 10:129612036869 11:109836776237 12:317337407944 16:153981533101 18:153981533101
21/12	1:10100338798 3:34056181899 7:18108485723 9:18108485723 10:139975001566 12:459150013359 14:126361927104 16:160250746466 18:33888819362
21/20	1:23261982823 3:80998303488 7:42596127754 9:42596127754 10:119376612498 12:395591891777 13:93515005697 16:54274471256 18:147789476953
21/21	1:10100338798 3:34056181899 7:18108485723 9:18108485723 10:139975001566 12:459150013359 13:126361927104 16:33888819362 18:160250746466
21/22	1:141658693750 3:464345192077 4:129046802345 7:32222516108 9:161269318453 10:9024518883 12:30219178776 16:16106889804 18:16106889804
22/00	1:36671337781 3:128823907165 7:67544643569 9:67544643569 10:98390547987 11:60049883285 12:270787680600 16:135093678022 18:135093678022
22/01	1:26549271482 3:92722697771 7:48712229041 9:48712229041 10:114231901802 11:85311067645 12:294406371677 16:144677115771 18:144677115771
22/02	1:20907909405 3:72602300649 7:38216302023 9:38216302023 10:123060811832 12:406959809356 14:99389960557 16:150018282355 18:50628321799
22/10	1:26549271482 3:92722697771 7:48712229041 9:48712229041 10:114231901802 11:85311067645 12:294406371677 16:144677115771 18:144677115771

Label	Nonzero coordinates written as `index:numerator` (cont.)
22/11	1:14881447769 3:51108413235 7:27003885476 9:27003885476 10:132492414848 11:114429929070 12:321631911965 16:155724056081 18:155724056081
22/12	1:9024518883 3:30219178776 7:16106889804 9:16106889804 10:141658693750 12:464345192077 14:129046802345 16:161269318453 18:32222516108
22/20	1:20907909405 3:72602300649 7:38216302023 9:38216302023 10:123060811832 12:406959809356 13:99389960557 16:50628321799 18:150018282355
22/21	1:9024518883 3:30219178776 7:16106889804 9:16106889804 10:141658693750 12:464345192077 13:129046802345 16:32222516108 18:161269318453
22/22	1:3447026915 3:8984069343 4:2225388677 5:2225388677 10:152972384744 12:498396753118 14:149931904799 16:165874494263 18:15942589463

125.35 First-sample H2 controls at $g = 1/100$

These 81 independent one-sample witnesses for $g = 1/100$ are exact decimal rationals at scale 10^{-9}. Each ten-port half-phase control has total power below 421; the guard inequalities are strict.

Label	Nonzero coordinates written as `index:numerator`
00/00	4:129837598946 5:9617425470 9:9617425470 10:129837598946 11:59691271192 13:64551313959 15:25389224217 16:59691271192 18:64551313959 19:25389224217
00/01	4:129831048873 5:9600149759 10:114008888449 11:59691313818 13:64549591091 15:25391489358 16:59124769693 18:65705874903 19:28734204639 20:10570704922
00/02	4:129861638513 5:9682537720 10:90904471112 11:59702231672 13:64558222718 15:25380550103 16:64833938291 18:75671375971 19:33221521965 20:27822719282
00/10	4:129831048873 5:9600149759 10:114008888449 11:59691313818 13:64549591091 15:25391489358 16:59124769693 18:65705874903 19:34648171689 20:4656737872
00/11	4:129824498801 5:9582874047 10:88562752481 11:59691356443 13:64547868224 15:25393754499 16:58558268194 18:66860435846 19:35962053963 20:17258540942
00/12	4:129855088440 5:9665262009 10:65458335145 11:59702274297 13:64556499850 15:25382815244 16:64267436792 18:76825936914 19:40449371289 20:34510555302
00/20	4:129861638513 5:9682537720 10:90904471112 11:59702231672 13:64558222718 15:25380550103 16:64833938291 18:75671375971 19:47020778416 20:14023462832
00/21	4:129855088440 5:9665262009 10:65458335145 11:59702274297 13:64556499850 15:25382815244 16:64267436792 18:76825936914 19:48334660690 20:26625265901
00/22	4:129846355009 5:9642227727 10:31530153855 11:59702331131 13:64554202693 15:25385835432 16:63512101460 18:78365351505 19:50086503722 20:43427669994
01/00	4:114008888449 9:9600149759 10:129831048873 11:59124769693 13:65705874903 14:4656737872 15:34648171689 16:59691313818 18:64549591091 19:25391489358
01/01	4:113985062665 10:113985062665 11:59124812319 13:65704152035 14:4660386319 15:34646788383 16:59124812319 18:65704152035 19:28732821333 20:10574353369
01/02	4:114098040265 10:90880645328 11:59135730173 13:65712783661 14:4642986856 15:34653248590 16:64833980916 18:75669653103 19:33220138659 20:27826367729
01/10	4:113985062665 10:113985062665 11:59124812319 13:65704152035 14:4660386319 15:34646788383 16:59124812319 18:65704152035 19:34646788383 20:4660386319
01/11	4:113961236880 10:88538926697 11:59124854944 13:65702429167 14:4664034766 15:34645405078 16:58558310820 18:66858712978 19:35960670657 20:17262189389
01/12	4:114074214481 10:65434509361 11:59135772798 13:65711060793 14:4646635303 15:34651865285 16:64267479417 18:76824214046 19:40447987983 20:34514203749
01/20	4:114098040265 10:90880645328 11:59135730173 13:65712783661 14:4642986856 15:34653248590 16:64833980916 18:75669653103 19:47019395110 20:14027111278
01/21	4:114074214481 10:65434509361 11:59135772798 13:65711060793 14:4646635303 15:34651865285 16:64267479417 18:76824214046 19:48333277384 20:26628914348
01/22	4:114042446769 10:31506328071 11:59135829632 13:65708763636 14:4651499899 15:34650020877 16:63512144085 18:78363628637 19:50085120416 20:43431318441
02/00	4:90904471112 9:9682537720 10:129861638513 11:64833938291 13:75671375971 14:14023462832 15:47020778416 16:59702231672 18:64558222718 19:25380550103
02/01	4:90880645328 10:114098040265 11:64833980916 13:75669653103 14:14027111278 15:47019395110 16:59135730173 18:65712783661 19:28739281540 20:10556953906
02/02	4:90993622929 10:90993622929 11:64844898771 13:75678284729 14:14009711816 15:47025855317 16:64844898771 18:75678284729 19:33226598866 20:27808968266
02/10	4:90880645328 10:114098040265 11:64833980916 13:75669653103 14:14027111278 15:47019395110 16:59135730173 18:65712783661 19:34653248590 20:4642986856
02/11	4:90856819544 10:88651904298 11:64834023542 13:75667930235 14:14030759725 15:47018011804 16:58569228674 18:66867344605 19:35967130864 20:17244789926
02/12	4:90969797145 10:65547486961 11:64844941396 13:75676561861 14:14013360262 15:47024472011 16:64278397272 18:76832845673 19:40454448190 20:34496804286
02/20	4:90993622929 10:90993622929 11:64844898771 13:75678284729 14:14009711816 15:47025855317 16:64844898771 18:75678284729 19:47025855317 20:14009711816
02/21	4:90969797145 10:65547486961 11:64844941396 13:75676561861 14:14013360262 15:47024472011 16:64278397272 18:76832845673 19:48339737591 20:26611514885

Label	Nonzero coordinates written as `index:numerator` (cont.)
02/22	1:104687731875 4:205610915572 5:22984574214 10:33097152450 11:52983883952 13:79244206153 15:52368000736 16:63572065929 18:78442410959 19:50189599732 20:43199832539
10/00	4:105526843126 9:9594391188 10:129828865516 11:58935935860 13:66090728550 14:14771305945 15:29172165397 16:59691328026 18:64549016802 19:25392244405
10/01	4:105503017342 10:113977120736 11:58935978486 13:66089005682 14:14774954392 15:29170782091 16:59124826527 18:65703577745 19:28732360231 20:10575569518
10/02	4:105615994943 10:90872703400 11:58946896340 13:66097637309 14:14757554930 15:29177242298 16:64833995125 18:75669078813 19:33219677557 20:27827583878
10/10	4:105503017342 10:113977120736 11:58935978486 13:66089005682 14:14774954392 15:29170782091 16:59124826527 18:65703577745 19:34646327282 20:4661602468
10/11	4:105479191558 10:88530984769 11:58936021111 13:66087282815 14:14778602839 15:29169398785 16:58558325028 18:66858138689 19:35960209555 20:17263405537
10/12	4:105592169159 10:65426567433 11:58946938965 13:66095914441 14:14761203376 15:29175858992 16:64267493626 18:76823639757 19:40447526882 20:34515419898
10/20	4:105615994943 10:90872703400 11:58946896340 13:66097637309 14:14757554930 15:29177242298 16:64833995125 18:75669078813 19:47018934008 20:14028327427
10/21	4:105592169159 10:65426567433 11:58946938965 13:66095914441 14:14761203376 15:29175858992 16:64267493626 18:76823639757 19:48332816282 20:26630130497
10/22	4:105560401446 10:31498386143 11:58946995799 13:66093617284 14:14766067972 15:29174014584 16:63512158294 18:78363054348 19:50084659314 20:43432534590
11/00	4:80080707159 9:9577115477 10:129822315443 11:58369434361 13:67245289494 14:21459141965 15:36400014721 16:59691370651 18:64547293934 19:25394509546
11/01	4:80056881375 10:113953294952 11:58369476987 13:67243566626 14:21462790412 15:36398631415 16:59124869152 18:65701854877 19:28730976925 20:10579217965
11/02	4:80169858975 10:90848877616 11:58380394841 13:67252198252 14:21445390949 15:36405091622 16:64834037750 18:75667355946 19:33218294251 20:27831232325
11/10	4:80056881375 10:113953294952 11:58369476987 13:67243566626 14:21462790412 15:36398631415 16:59124869152 18:65701854877 19:34644943976 20:4665250915
11/11	4:99095187658 10:99095187658 11:62206271054 13:69217557568 14:17603938592 15:38531790097 16:62206271054 18:69217557568 19:38531790097 20:17603938592
11/12	4:80146033191 10:65402741648 11:58380437466 13:67250475384 14:21449039396 15:36403708317 16:64267536251 18:76821916889 19:40446143576 20:34519068344
11/20	4:80169858975 10:90848877616 11:58380394841 13:67252198252 14:21445390949 15:36405091622 16:64834037750 18:75667355946 19:47017550702 20:14031975874
11/21	4:80146033191 10:65402741648 11:58380437466 13:67250475384 14:21449039396 15:36403708317 16:64267536251 18:76821916889 19:48331432976 20:26633778944
11/22	4:80114265479 10:31474560358 11:58380494300 13:67248178227 14:21453903992 15:36401863909 16:63512200919 18:78361331480 19:50083276008 20:43436183037
12/00	4:56976289822 9:9659503438 10:129852905082 11:64078602959 13:77210790562 14:30825866925 15:48772621448 16:59702288506 18:64555925561 19:25383570291
12/01	4:56952464038 10:114066272553 11:64078645584 13:77209067694 14:30829515371 15:48771238142 16:59135787007 18:65710486504 19:28737437132 20:10561818502
12/02	1:15162880780 4:77003629452 10:91175905269 11:62371609878 13:77734766560 14:28078664940 15:50254980955 16:64852053300 18:75686148138 19:33238951462 20:27782824767
12/10	4:56952464038 10:114066272553 11:64078645584 13:77209067694 14:30829515371 15:48771238142 16:59135787007 18:65710486504 19:34651404183 20:4647851452
12/11	4:56928638254 10:88620136585 11:64078688210 13:77207344826 14:30833163818 15:48769854836 16:58569285508 18:66865047447 19:35965286457 20:17249654522
12/12	1:104687731875 4:194699076209 10:66993566028 11:52228491786 13:80785917901 14:11943441459 15:58975786214 16:64327458095 18:76900699211 19:40550622892 20:34287582442
12/20	1:15162880780 4:77003629452 10:91175905269 11:62371609878 13:77734766560 14:28078664940 15:50254980955 16:64852053300 18:75686148138 19:47038207912 20:13983568316
12/21	1:104687731875 4:194699076209 10:66993566028 11:52228491786 13:80785917901 14:11943441459 15:58975786214 16:64327458095 18:76900699211 19:48435912292 20:26402293041
12/22	1:141757584328 4:248269774254 10:29972641418 11:46603861345 13:80044885536 14:938836468 15:61256066222 16:61975951974 18:76847533079 19:49331384507 20:43001480869
20/00	4:90904471112 9:9682537720 10:129861638513 11:64833938291 13:75671375971 14:27822719282 15:33221521965 16:59702231672 18:64558222718 19:25380550103
20/01	4:90880645328 10:114098040265 11:64833980916 13:75669653103 14:27826367729 15:33220138659 16:59135730173 18:65712783661 19:28739281540 20:10556953906
20/02	4:90993622929 10:90993622929 11:64844898771 13:75678284729 14:27808968266 15:33226598866 16:64844898771 18:75678284729 19:33226598866 20:27808968266
20/10	4:90880645328 10:114098040265 11:64833980916 13:75669653103 14:27826367729 15:33220138659 16:59135730173 18:65712783661 19:34653248590 20:4642986856
20/11	4:90856819544 10:88651904298 11:64834023542 13:75667930235 14:27830016176 15:33218755353 16:58569228674 18:66867344605 19:35967130864 20:17244789926
20/12	4:90969797145 10:65547486961 11:64844941396 13:75676561861 14:27812616713 15:33225215560 16:64278397272 18:76832845673 19:40454448190 20:34496804286
20/20	4:90993622929 10:90993622929 11:64844898771 13:75678284729 14:27808968266 15:33226598866 16:64844898771 18:75678284729 19:47025855317 20:14009711816
20/21	4:90969797145 10:65547486961 11:64844941396 13:75676561861 14:27812616713 15:33225215560 16:64278397272 18:76832845673 19:48339737591 20:26611514885
20/22	1:104687731875 4:228595489787 10:33097152450 11:52983883952 13:79244206153 14:8945158412 15:43422842324 16:63572065929 18:78442410959 19:50189599732 20:43199832539

Label	Nonzero coordinates written as `index:numerator` (cont.)
21/00	4:65458335145 9:9665262009 10:129855088440 11:64267436792 13:76825936914 14:34510555302 15:40449371289 16:59702274297 18:64556499850 19:25382815244
21/01	4:65434509361 10:114074214481 11:64267479417 13:76824214046 14:34514203749 15:40447987983 16:59135772798 18:65711060793 19:28737898234 20:10560602353
21/02	4:65547486961 10:90969797145 11:64278397272 13:76832845673 14:34496804286 15:40454448190 16:64844941396 18:75676561861 19:33225215560 20:27812616713
21/10	4:65434509361 10:114074214481 11:64267479417 13:76824214046 14:34514203749 15:40447987983 16:59135772798 18:65711060793 19:34651865285 20:4646635303
21/11	4:65410683576 10:88628078514 11:64267522043 13:76822491178 14:34517852195 15:40446604678 16:58569271299 18:66865621737 19:35965747559 20:17248438373
21/12	1:74846114843 4:163941364018 10:66580242381 11:55798379192 13:79383439517 14:21007753812 15:47745139868 16:64313475122 18:76881276790 19:40523143292 20:34347392408
21/20	4:65547486961 10:90969797145 11:64278397272 13:76832845673 14:34496804286 15:40454448190 16:64844941396 18:75676561861 19:47024472011 20:14013360262
21/21	1:74846114843 4:163941364018 10:66580242381 11:55798379192 13:79383439517 14:21007753812 15:47745139868 16:64313475122 18:76881276790 19:48408432692 20:26462103007
21/22	1:141757584328 4:248269774254 10:29972641418 11:46603861345 13:80044885536 14:8824125869 15:53370776821 16:61975951974 18:76847533079 19:49331384507 20:43001480869
22/00	4:31530153855 9:9642227727 10:129846355009 11:63512101460 13:78365351505 14:43427669994 15:50086503722 16:59702331131 18:64554202693 19:25385835432
22/01	4:31506328071 10:114042446769 11:63512144085 13:78363628637 14:43431318441 15:50085120416 16:59135829632 18:65708763636 19:28736053826 20:10565466949
22/02	1:104687731875 4:169276766026 10:92415876211 11:51661947662 13:81942201712 14:24541596082 15:60291051794 16:64894002219 18:75744415400 19:33321390262 20:27603394869
22/10	4:31506328071 10:114042446769 11:63512144085 13:78363628637 14:43431318441 15:50085120416 16:59135829632 18:65708763636 19:34650020877 20:4651499899
22/11	4:31482502286 10:88596310801 11:63512186711 13:78361905769 14:43434966888 15:50083737110 16:58569328133 18:66863324579 19:35963903151 20:17253302968
22/12	1:141757584328 4:214373360676 10:63869054996 11:45848469179 13:81586597284 14:17736375965 15:63009753661 16:62731344140 18:75305821331 19:39692407667 20:34089230773
22/20	1:104687731875 4:169276766026 10:92415876211 11:51661947662 13:81942201712 14:24541596082 15:60291051794 16:64894002219 18:75744415400 19:47120646712 20:13804138418
22/21	1:141757584328 4:214373360676 10:63869054996 11:45848469179 13:81586597284 14:17736375965 15:63009753661 16:62731344140 18:75305821331 19:47577697067 20:26203941372
22/22	1:162207042597 4:234397342960 10:23395614444 11:40488929299 13:79230509630 14:13816797821 15:63319027974 16:58942906920 18:73807804432 19:47667464647 20:42726559276

Part XI

Synthesis and Return to the Original Machine

Part guide. **Question.** What does the original 2023 machine look like after the full semantic, geometric, structural, enumerative, resource, topological, and algorithmic theory has been developed? **Main result.** The final Part reconstructs the research chain, distinguishes forced structure from coordinate choice, and identifies which theorem families genuinely descend from the original completion invariant. **Why here.** The monograph ends by explaining the starting machine rather than merely stopping after the last branch. **Dependency.** This Part synthesizes all preceding Parts and introduces no new foundational assumptions.

Chapter 126

The 2023 DPDA Revisited

The monograph began with two diagrams.

They represented the same $C/L/R$ recognition core but different empty-word wrappers:

$$\boxed{L_{2,1} \setminus \{\varepsilon\}}$$

and

$$\boxed{L_{2,1}.}$$

At the beginning, the figures were concrete machines to be understood.

We now return to them after developing:

- shortest-completion geometry and exact residual-faithful resource frontiers;
- representation-independent hierarchy methods;
- higher-dimensional completion geometry;
- finite geodesic compilation, higher-order optimal-compatibility structure theory, and full simplicial universality with exact Boolean-lattice-width row complexity;
- cancellation/defect geometry and chronological arithmetic refinement;
- the general real-time p:q machine and WSC Structure Theory;
- exact semantic lifts and physical lift freedom;
- arbitrary-coprime causal-schedule compilation and exact fatal-support classification;
- strict coordinate quotients, protected interfaces, bounded bi-recoding, and minimal cores;
- General Theory;
- operational CANCEL/WRITE enumeration and exact one-counter compression;
- explicit rational-Dyck run–return and ratio-run–return enumeration, followed by contact-preserving barrier compression;
- Farey-recursive shortest-completion arithmetic and universal spectra;
- quantitative LIFO chronology;
- rank-two topology, physical holonomy, and sharp carrier costs;
- exact unrestricted state–stack and bounded-replacement complexity;
- filtered collapse capacity, retained-information bounds, and source-stack exposure;
- protected-contract Pareto synthesis;
- Unified Semantic Normalization and Realization Theory: rooted path outcomes, protected-obligation capacity, same-model transfer saturation, typed holonomy/resource bridges, and four-axis source recovery;
- independent height-guarded two-source realization classifications: the linear modular arithmetic dichotomy and the explicit periodic signed-winding criterion, including source-canonical compilers and non-context-freeness witnesses;
- Extrema–Frame Foreground–Background theory: observer-record normal form, exact common-realization scalar uncertainty, the cactus/radius-two local–global complexity boundary, shallow spatial-frame NP-completeness, finite-complex universality,

parameterized ambiguity bounds, correlated polyhedral uncertainty, finite layered semantics, all-orders obstruction families, and exact controlled-heat common realization with a sharp power threshold, correlated caps, nonsemialgebraic optimal faces, and spectral rigidity;

- DOCA state-primality and exact two-factor decomposition, followed by decidability of deterministic zero-testing OCA–nondeterministic OCN language equivalence, including the epsilon extension, and the resulting separation between fixed comparison and existential decomposition.

The purpose of this chapter is not to redraw the machine as an abstract object.

It is to show what every part of the original design means after the full theory has been developed.

126.1 The two origin figures remain distinct

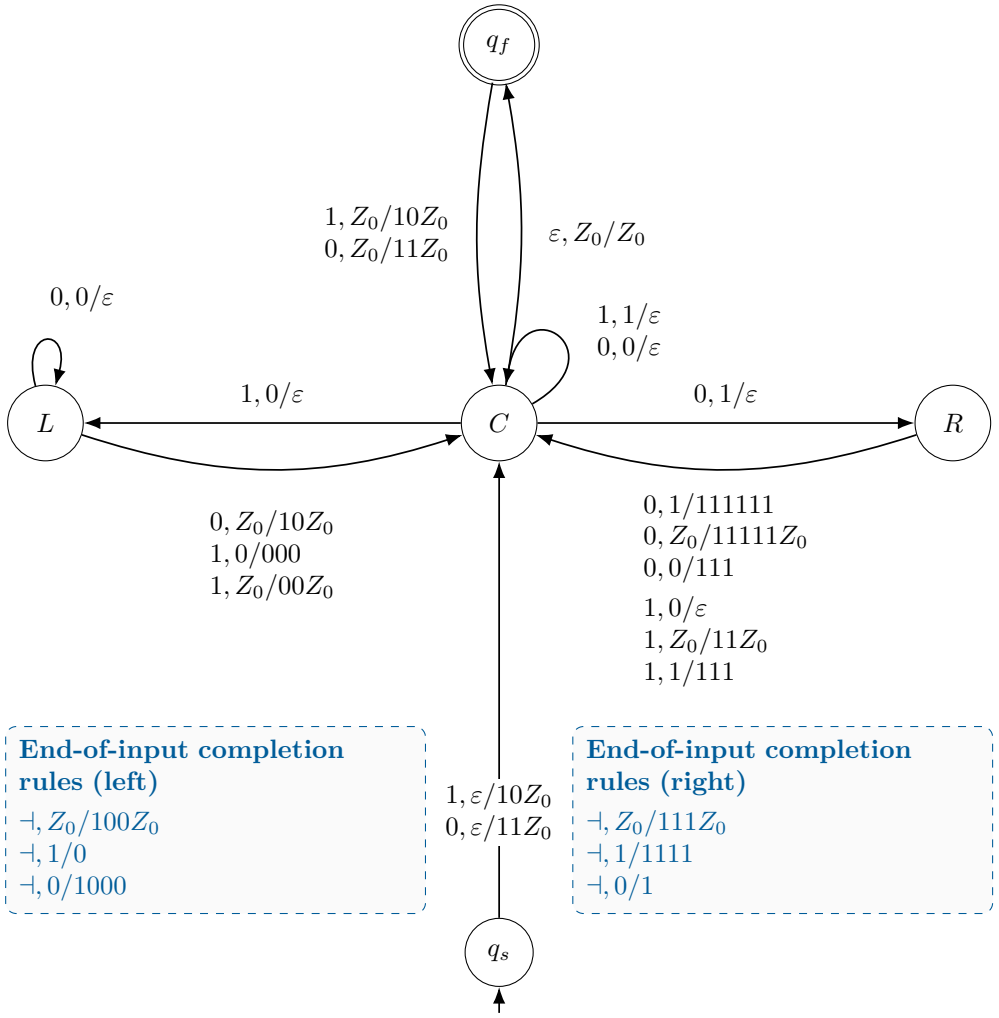


Figure 126.1: The empty-input-excluded original DPDA revisited after the general theory. This is the same formalized machine as Figure 1.1, deliberately repeated so that the final interpretation can be read against the original object.

This variant recognizes

$$L_{2,1} \setminus \{\varepsilon\}.$$

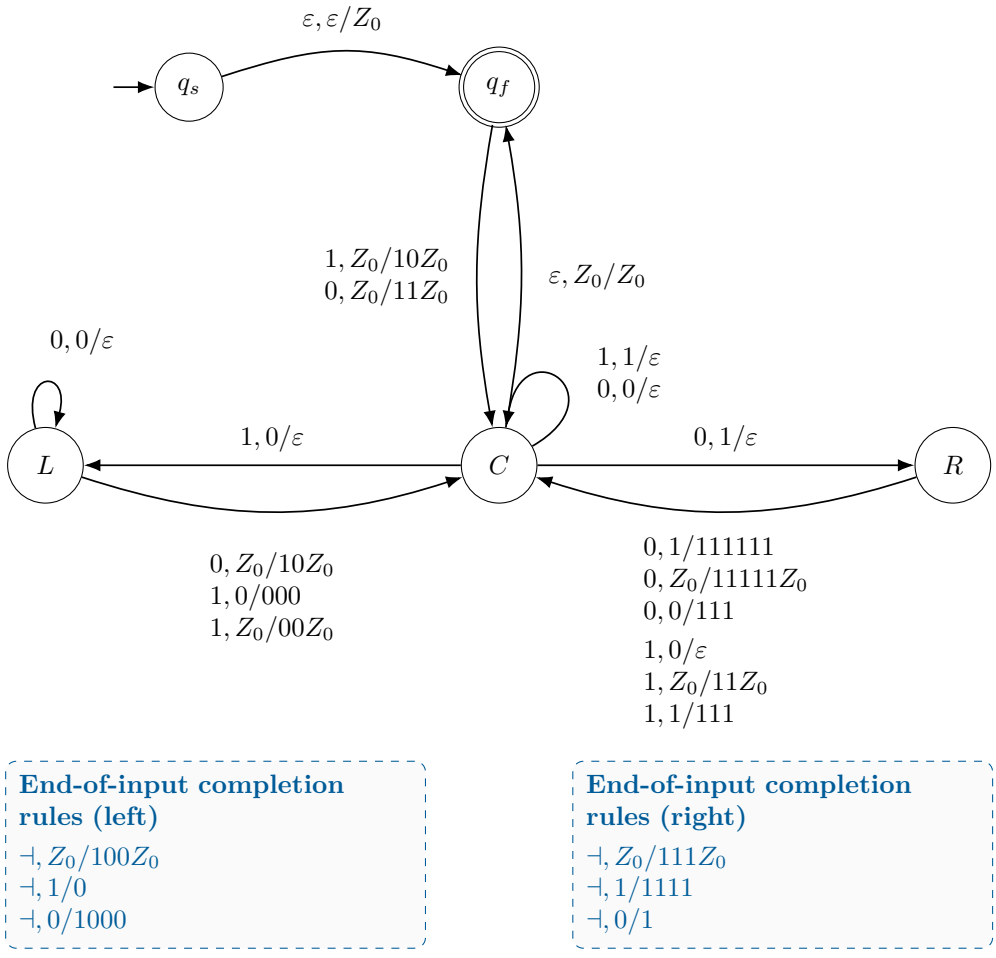


Figure 126.2: The empty-input-included original DPDA revisited. This is the same recognition/completion core as Figure 1.2; the wrapper accepts ε.

This variant recognizes

$$L_{2,1}.$$

The common $C/L/R$ operational core is the same.
The difference is the acceptance/initialization wrapper.
This is the first lesson of the completed theory:

$$\boxed{\text{acceptance convention} \neq \text{semantic core.}}$$

126.2 What the stack was storing

The original stack looked like an operational memory structure.
Chapter 1 proved that its reachable form is highly restricted.
The state-potential invariant then gave

$$\boxed{\rho(w) = D(S) + \Phi(q).}$$

For 2:1,

$$D(S) = 2N_0(S) - N_1(S)$$

under the original machine's stack-weight convention, while the state potentials are

$$\Phi(C) = 0, \qquad \Phi(L) = 3, \qquad \Phi(R) = -3.$$

Thus the machine stores the arithmetic imbalance **distributed between control and stack**.
That observation became the prototype for every later semantic/physical decomposition.

126.3 What the completion rules were doing

The right-endmarker completion rules do not create new semantic information.
They materialize information already present in the final state/stack configuration.
After normalization, the physical word decodes a shortest completion demand.
The machine therefore contains an implicit map

$$\boxed{\begin{array}{c}\text{physical configuration} \to \text{residual} \to \text{minimum demand} \\ \to \text{canonical shortest continuation.}\end{array}}$$

This was the discovery that opened the entire completion-geometry branch.

126.4 Why depth two appeared

The original normalized stores lie in only a few structural families.
Therefore the qualitative question

> Which demand symbols occur anywhere in the remaining stack?

is determined by the first two cells.
The general centered $p{:}q$ machine later recovered this two-cell qualitative visibility through a completely different canonical ordering.
Thus the original depth-two property was not merely a graphical accident.
It was an early manifestation of a broader principle:

$$\boxed{\text{unbounded quantitative demand can have bounded qualitative visibility.}}$$

126.5 The cancellation transition

The transition

$$\boxed{(L, 0S) \xrightarrow{0} (L, S)}$$

acquired three meanings.

Machine meaning

Delete one matching stack symbol.

Geodesic meaning

Consume one required zero along a shortest completion.

Combinatorial meaning

In the nonnegative 2-Dyck sector, mark a down-step landing at an exact odd-height family.

That third interpretation generated:

$$\begin{aligned}\text{correction signatures} &\to \text{defect} \to \text{Catalan boundary} \\ &\to \text{binary skeleton} \to \text{enumeration.}\end{aligned}$$

Thus one local transition produced an entire independent research branch.

126.6 Why the improved raw DPDA looked so different

The three-symbol raw residual-conjugate DPDA of Chapters 7–8 is not a cleaned-up drawing of the original machine.

It solves a different physical representation problem.

It asks:

> If the residual itself is the semantic state, what is the most efficient stable raw encoding within the stated model?

That representation moves finite phase information into control and leaves unary magnitude on the stack.

It achieves the exact raw residual-faithful frontier.

The original machine instead stores a richer literal completion-demand process with normalization chronology.

Therefore the two machines are not rivals for one undefined notion of optimality.

They are optimal or explanatory in different representation categories.

126.7 Why the general $p{:}q$ machine returned to the original architecture

The centered real-time $p{:}q$ construction deliberately preserved the operational idea of the original machine:

- literal demand on the stack;
- one-cell LIFO blocker displacement;
- signed debt in finite control;
- later WRITE materialization.

When

$$(p, q) = (2, 1),$$

the general debt rail becomes

$$D_{-1}, D_0, D_1$$

with

$$\boxed{D_{-1} = R, \qquad D_0 = C, \qquad D_1 = L.}$$

The general potentials become

$$-3, 0, +3.$$

The historical WRITE arithmetic is recovered from the kernel-carry compiler.
Thus the original architecture was not merely generalized by analogy.
It was reconstructed from the general arithmetic.

126.8 Why the historical table was not literally the centered specialization

Two reachable right-state matching families remained different.
The centered machine keeps debt lazily in R.
The historical machine immediately rematerializes the same semantic demand in C.
Therefore:

$$\boxed{\text{historical} = \text{eager normalization},}$$

$$\boxed{\text{centered specialization} = \text{lazy normalization}.}$$

This small transition-table discrepancy turned out to be one of the most important facts in the research program.
It proved concretely that:

$$\boxed{\text{same semantics} \not\Rightarrow \text{unique physical chronology}.}$$

That observation led directly to Structure Theory and Uniqueness Theory.

126.9 C/L/R after Structure Theory

For 2:1,

$$r = p + q = 3.$$

The WSC exchange-sheet coordinate gives

$$\boxed{s(C) = 0, \qquad s(L) = 1, \qquad s(R) = -1.}$$

Thus the original states become SEARCH sheets.
The characteristic stack-length changes

$$-1, +2, +5$$

become materialization levels

$$M = 0, 1, 2.$$

And the historical identity

$$b + 2c = n$$

becomes a specialization of the global materialization budget.

The original table has therefore been reinterpreted from local syntax into semantic/physical accounting.

126.10 C/L/R after Uniqueness Theory

The original eager machine is one reduced lift.

The centered lazy machine is another.

They are:

- semantically equivalent;
- section-preserving incomparable;
- coordinate-reduced;
- non-isomorphic.

Within the selected three-site family, there are six admissible reduced normalization schedules.

The historical machine is therefore not “the” unique physical realization of the semantics.

It is a distinguished point in a nontrivial lift space.

126.11 The original machine after General Theory

The General Theory gives a final vocabulary.

The original configuration space is a physical lift.

The residual is semantic state.

The canonical completion word is a section choice.

The side-state configurations occupy noncanonical semantic fibers.

The one-top LIFO rule creates locality constraints.

Leaving the canonical section creates representation debt.

Some schedule choices preserve the behavioral hull.

Some expose fatal frontiers.

Alternative coordinate systems can remove obstructions that are essential in the literal policy.

Thus the full research program can be read backward into the original figure.

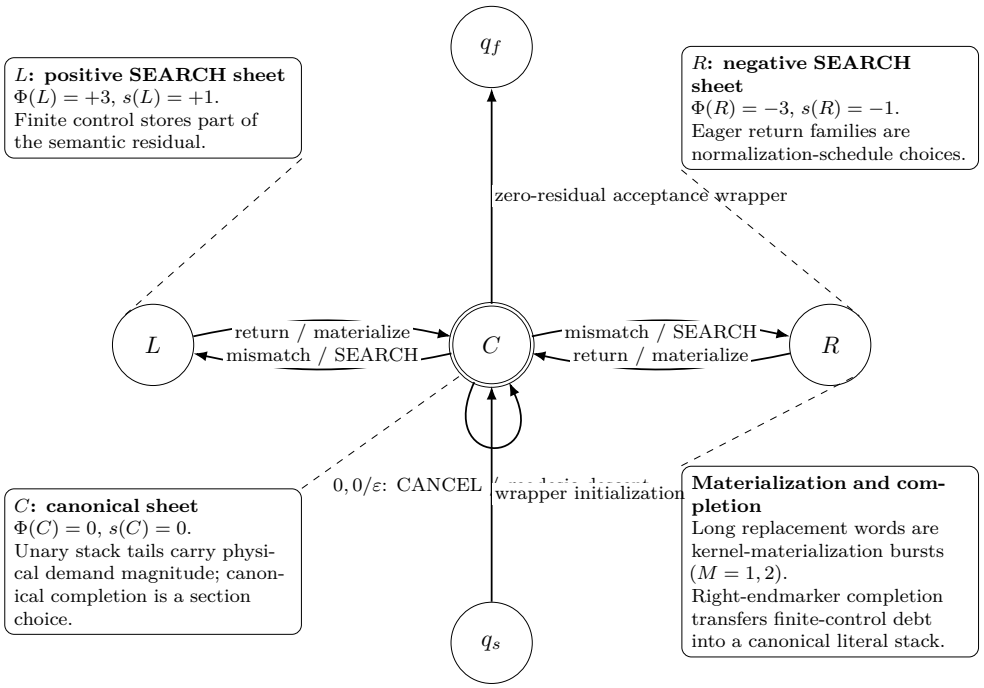


Figure 126.3: The original $C/L/R$ architecture annotated by the complete theory. The figure is interpretive rather than a replacement transition table: it overlays the residual/state-potential invariant, WSC SEARCH-sheet coordinates, literal-demand magnitude, kernel materialization, end-of-input completion, schedule freedom, and the semantic-lift viewpoint on the original machine architecture.

Suggested annotations:

- C: canonical sheet / zero SEARCH charge;
- L: positive SEARCH sheet;
- R: negative SEARCH sheet;
- unary stack tails: physical demand magnitude;
- $0, 0/\varepsilon$: CANCEL / geodesic descent / correction event;
- long replacement words: kernel-materialization bursts;
- right-endmarker completion: final canonical materialization;
- eager right-state families: schedule choice;
- wrapper: acceptance convention, not semantic core.

The final visual object of the monograph should be this reinterpreted original machine.

Chapter 127

What Was Forced, What Was Chosen, What Became General

The entire monograph can now be organized by one question:

$$\boxed{\text{Which features were mathematically forced, and at what level?}}$$

The answer requires several categories.

127.1 Language-semantic necessities

These are determined by $L_{p,q}$ itself.

Residual quotient

$$\boxed{u^{-1}L_{p,q} = v^{-1}L_{p,q} \iff \rho(u) = \rho(v).}$$

The right-residual system is canonical.

Minimum completion demand

Each residual determines one unique minimum nonnegative completion Parikh vector.

Primitive kernel

$$\boxed{K = (q, p).}$$

Every equivalent completion vector differs by a kernel packet.

Semantic carry chronology

For a fixed input word,

$$(\rho_t), \quad (\kappa_t), \quad (\mu_t), \quad (\chi_t)$$

are unique.
These facts do not depend on a pushdown representation.

127.2 Necessities of a chosen locality category

These are not language-intrinsic, but become forced once the physical category is fixed.
Examples include:

One-top suffix obstruction

A hidden suffix cannot be arbitrarily rearranged by one top rewrite.

Literal-policy fiber branching

For nontrivial $p{:}q$, the fixed canonical literal section cannot remain transition-closed.

Deferred-SEARCH control width

$$2\left\lfloor \frac{p+q-1}{2} \right\rfloor + 1$$

states are exact in the canonical deferred-SEARCH model.

Bounded-local collapse capacity

Too many semantic predecessors cannot be distinguished by too few local observations.

These are **representation-category necessities**.

They must always be stated with their category.

127.3 Coordinate artifacts

Some difficulties disappear when the physical coordinates change.

The clearest example is the four-symbol normalized lift.

Its

$$p+q-1$$

raw boundary defects were exact in those coordinates.

But a boundary-compatible coordinate shift removed them and produced a three-symbol raw DPDA.

Therefore:

$$\boxed{\text{locality defect} \not\Rightarrow \text{semantic necessity.}}$$

A defect may belong to the coordinate system.

127.4 Normalization-schedule choices

The semantic chronology can be unique while physical materialization chronology varies.

Examples:

- historical eager 2:1;
- centered lazy 2:1;
- the six admissible reduced 2:1 schedules.

The total kernel budget remains rigid even when its timing changes.

Thus:

$$\boxed{\text{semantic WRITE arrivals are forced;}}$$

$$\boxed{\text{physical WRITE timing may be chosen.}}$$

127.5 Special 2:1 features

Several concrete facts are specializations rather than universal principles.

For

$$p = 2, \qquad q = 1 :$$

$$K = (1, 2),$$

$$m = 3,$$

$$h = 1,$$

so the centered debt rail has exactly three states.

These become:

$$R, C, L.$$

The state potentials are

$$-3, 0, +3.$$

The historical materialization changes are

$$-1, +2, +5.$$

The correction transition in the nonnegative path sector lands at

$$3, 5, 7, \ldots$$

These numerical values are 2:1-specific.

The principles they instantiate are broader.

127.6 What generalized to all coprime p:q

The following extend uniformly:

- residual semantics;
- shortest completion demand;
- primitive kernel;
- geodesic rectangles;
- bounded canonical action visibility;
- raw residual-conjugate DPDA;
- exact raw residual-faithful frontier;
- centered real-time literal-demand DPDA;
- SEARCH debt;
- minimal kernel-carry WRITE;
- canonical-segment invariant;
- WSC semantic carry factorization;
- materialization/backlog accounting.

The original 2:1 machine was therefore a genuine first member of a general structural family.

127.7 What generalized beyond rational binary counting

Several branches moved beyond $L_{p,q}$.

Matrix completion geometry

The unique minimum pair became an optimal Parikh fiber.

The rectangle became a cubical quotient by the equal-length kernel.

Topological hierarchy

Kernel rank controlled local/global shortest-schedule topology.

Finite geodesic compilation

The shortest-completion viewpoint generalized further to positive-cost finite-phase periodic systems. Finite arithmetic data compile exact distance, optimal branching, arbitrary shortest-prefix membership, and the complete local optimal-compatibility complex. Across the class, that compatibility hierarchy is strictly non-collapsing at every prescribed order and admits Boolean amplification under products. The universal k-fiber theorem sharpens this: even after exact distance and the complete hierarchy through order $k-1$ are fixed, all $2^{\binom{s}{k}}$ next-order patterns on a distinguished structural s-simplex can occur, including arbitrarily overlapping k-faces. The all-rank theorem removes the distinguished rank altogether: one fixed coefficient bank of exact size $\binom{s}{\lfloor s/2 \rfloor}$ realizes every labeled simplicial complex, the same optimum survives common-shell synchronization, and one periodic Bellman system realizes the entire simplicial universe at one canonical distance.

General exact-lift theory

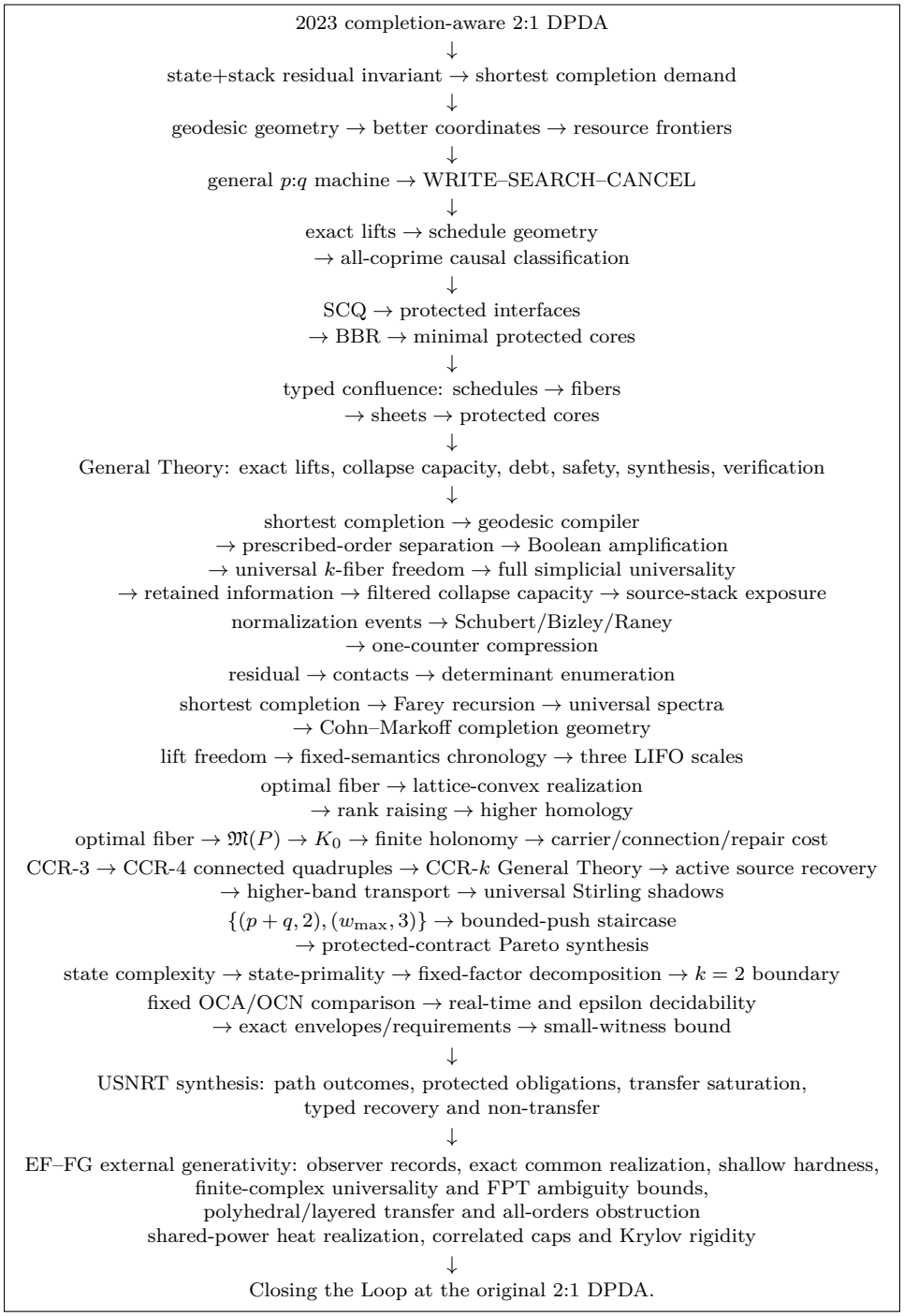
Semantic fibers, canonical sections, bounded-local obstruction, behavioral hulls, containment, causality, synthesis, and verification no longer depend specifically on weighted binary balance.

Thus the project eventually generalized not just the machine, but the distinction that the machine exposed:

> what must be remembered semantically
>
> *versus*
>
> how a LIFO machine can physically remember it.

127.8 The dependency structure in one chain

The completed monograph has one main mathematical spine together with consequence branches that reuse the same semantic objects without being prerequisites of the earlier construction:



The displayed branches are consequences and syntheses rather than a single linear causation claim. The large consequence array appears before USNRT because the recovery theorem uses theorem-bearing geometry, chronology, and exposure branches developed there. USNRT then acts as a typed synthesis of those already established branches; EF–FG is a logically independent scalar-field theory placed after that synthesis because it supplies a genuine external generativity test rather than another reformulation of the pushdown model. The geodesic-compiler branch and its full-simplicial completion are placed immediately after the higher-dimensional completion geometry to which they mathematically belong; the other consequence branches occur after the General Theory. State-primality and its two-factor sharpening follow the exact state/resource results because they mark an algorithmic decomposition boundary rather than a prerequisite of the General Theory; the one-counter comparison chapter then preserves the fixed OCN/OCN solution of Problem 24.04(B) and

adds the zero-testing OCA/OCN solution of Problem 24.04(A), completing the contrast with two fixed-presentation comparison problems that remain decidable. The A branch moreover survives the stated epsilon-transition extension. This order preserves mathematical prerequisites and narrative continuity: operational statistics precede intrinsic path statistics, topology precedes physical holonomy, and endpoint resource theorems precede Pareto synthesis.

Six points on this map are explicit resolutions of questions posed in the literature: the two Dai–Fu–Qiu joint-enumeration questions (run with return and ratio-run with return) [23], the fixed-state stack-alphabet hierarchy case left open after Masopust's realtime theorem [16], Totzke's Prime-DOCA decidability question [99], Automata Exchange Problem 24.04(B) on deterministic–nondeterministic one-counter-net language equivalence, and Automata Exchange Problem 24.04(A) on deterministic zero-testing one-counter automata versus nondeterministic one-counter nets [109]. The remaining branches are stated as new consequences, abstractions, or synthesis results rather than retroactively being labeled as external open-problem resolutions.

127.9 Final perspective

The research program began with a concrete deterministic pushdown automaton.

Its most productive feature was not simply that it recognized

$$L_{2,1}.$$

It distributed one arithmetic quantity across:

- finite control;
- a LIFO store;
- completion operations.

Trying to understand that distribution revealed:

$$\boxed{\text{semantic information} \neq \text{physical representation.}}$$

Everything that followed can be traced to that separation. The same semantic extraction later reached beyond the first pushdown representation itself: shortest-completion geometry admitted a finite cubical geodesic compiler whose local optimal-compatibility hierarchy is strict at every prescribed order; normalization history became an operational enumerative object; the residual became rational-path rank and boundary contact; the shortest-completion map became a Farey-recursive arithmetic object with universal scalar spectra and then an exact Cohn–Markoff completion geometry; optimal-fiber topology acquired a rank-raising realization calculus, a displacement-marked pure-loop calculus, and a distinct finite physical holonomy/carrier/connection/repair layer, while unequal simplex-strand optimal fibers obey marked product rigidity beyond literal Cartesian decompositions; chronological carry progressed from its third-order refinement through an exact fourth-order deformation parameter to an all-order CCR-k source–band–transport–shadow theory with an explicit integral Stirling obstruction lattice, exceptional local folds, Galois specializations, adelic resonance laws and a sharp third-source near-leading coefficient refinement, followed by restricted-input active observability with exact nonresetting adaptive intervention bounds and by a sharp five-step forecast archive, finite-ring response profiles and lossless source-mask coding; a separate Nörlund–Stirling quotient analysis contributes single-site common-root rigidity and one exactly classified sporadic exception; and the state lower-bound branch gained binary-hierarchy Pareto corner and width theorems and an epsilon-free fixed-resource decision theorem, then closed to an unrestricted state–stack frontier and an exact unrestricted bounded-replacement staircase, while universal higher-order semantic

freedom acquired a representation-independent source-stack exposure cost. The geodesic branch reaches full simplicial universality with exact Boolean-lattice-width row complexity and zero common-shell defect. A source-native facet construction realizes the entire shortest Parikh fiber, preserves it under bounded scalar folding, and admits both a real-time zero-return recognizer and an explicitly clocked, separately selected paired-session comparison service; neither contract implies a universal unframed dual-service machine. The final protected-contract synthesis then makes explicit which resource reductions are genuine and which are only bounded changes of coordinates; the subsequent DOCA state-primality theorem and its two-factor sharpening show that existential decomposition is undecidable already at the smallest genuine intersection budget, while the final one-counter comparison chapter first proves deterministic–nondeterministic OCN language equivalence decidable for two fixed net presentations and then closes the adjacent zero-testing frontier by proving deterministic OCA–nondeterministic OCN language equivalence decidable in real time and under the stated epsilon-transition extension. The contrast isolates the quantifier boundary rather than attributing hardness to one-counter storage alone.

A separate tree-walking boundary establishes an exact regularity criterion for right-turn spectra, family-specific deterministic nonrecognizability and a one-visible-pebble characterization, while leaving the UTWA–DTWA question unresolved. After those domain-specific and algorithmic branches are in place, the late USNRT synthesis makes the bookkeeping principle itself mathematical: representation fibers, path-outcome fibers, protected-obligation complexes, transported holonomy, and resource coordinates remain typed separately, with transfer theorems stating exactly which bridges are needed. EF–FG then supplies the deliberately external test: the same common-realization discipline organizes observer records in finite scalar fields, separates polynomial local witnessing from NP-complete global synchronization on shallow graphs, classifies complete categorical spatial-frame realization as NP-complete on shallow bipartite graphs, realizes every finite simplicial complex as a raw and canonical common-realization complex, exposes two fixed-parameter tractable ambiguity regimes, extends exact semantics to correlated rational-polyhedral uncertainty and finite coupled layers, and supports obstruction families of unbounded order. The subsequent controlled-heat realization chapter further proves a sharp three-query power threshold on a rational tree, a nonsemialgebraic optimum face with exact correlated terminal-cap segment, and a rank-2 to rank-7 spectral rigidity transition under a one-percent conductance change. These results are model-specific physical statements, not automatic consequences of the combinatorial EF–FG theorems. The following sampled leaky-heat study is different again: it proves a budget-independent H1 language collapse, an exactly powered causal H2 all-word controller and a local critical cross-link threshold with a complete two-module all-horizon phase. Its numerical bands are source-native record projections rather than intrinsic terminal colors; the one-time power Q_* and sampled-power Q_0 are not directly comparable across the two models. This last step is not a new encoding of the original pushdown language; it is evidence that the realization viewpoint extracted from that language has become reusable mathematics.

The final lesson is therefore not one specific DPDA construction or one specific theorem. It is a method:

extract the semantic object from the machine,

identify which physical constraints are truly forced,

change coordinates when the obstruction is representational,

retain debt when locality forces it,

and distinguish semantic correctness from normalization history.

One additional lesson of the geodesic-compiler synthesis is that

shortest-path geometry can be finite in description

while

its optimal compatibility is unbounded in informational order across the class.

The original 2023 DPDA remains the first concrete object in that story—and, after the full theory, the final object through which the story can be read.

Appendix A

Complete Transition Audit of the Original 2023 DPDA

A.1 Purpose and scope

This appendix closes the finite verification step that is deliberately compressed in Chapter 1.

The object audited here is the **original 2023 completion-aware** 2:1 **DPDA**, in the formalized presentation used throughout this monograph. The language convention is

$$L_{2,1} = \{\, w \in \{0,1\}^* : N_1(w) = 2N_0(w) \,\},$$

with signed prefix residual

$$\boxed{\rho(w) = N_1(w) - 2N_0(w).}$$

For a logical stack word $S \in \{0,1\}^*$, define

$$\boxed{D(S) = 2N_0(S) - N_1(S)}$$

and assign the core-state potentials

$$\boxed{\Phi(C) = 0, \qquad \Phi(L) = 3, \qquad \Phi(R) = -3.}$$

The Chapter 1 invariant is

$$\boxed{\rho(w) = D(S) + \Phi(q)} \tag{A.1}$$

for every processed prefix w, where (q, S) is the corresponding core-projected configuration.

Since reading a binary symbol changes the residual by

$$\Delta\rho = \begin{cases} -2, & a = 0, \\ +1, & a = 1, \end{cases} \tag{A.2}$$

it is enough to verify, for every reachable recognition transition,

$$\boxed{\Delta D + \Delta\Phi = \Delta\rho.} \tag{A.3}$$

The audit also checks that every transition preserves the state-dependent reachable-store families

$$\boxed{\mathcal{S}_C = 0^* \cup 1^* \cup \{10\}, \qquad \mathcal{S}_L = 0^*, \qquad \mathcal{S}_R = 1^* \cup \{0\}.} \tag{A.4}$$

The bottom marker Z_0 is omitted from logical-stack words below. The leftmost logical symbol is the top.

A.2 Wrapper projection

The two formal variants differ only in how initialization and empty-input acceptance are wrapped around the semantic core.

The projection used throughout the monograph is

$$\pi(C, SZ_0) = (C, S), \qquad \pi(L, SZ_0) = (L, S), \qquad \pi(R, SZ_0) = (R, S),$$

together with

$$\boxed{\pi(q_s, \varepsilon) = (C, \varepsilon), \qquad \pi(q_f, Z_0) = (C, \varepsilon).} \tag{A.5}$$

Thus the wrappers carry no additional residual semantics.

Empty-input-excluded wrapper

The first binary symbol is consumed directly at the source wrapper.

input	projected source	projected target	ΔD	$\Delta\Phi$	$\Delta(D+\Phi)$	$\Delta\rho$
1	(C, ε)	$(C, 10)$	$+1$	0	$+1$	$+1$
0	(C, ε)	$(C, 11)$	-2	0	-2	-2

Indeed,

$$D(10) = 2 - 1 = 1, \qquad D(11) = -2.$$

Empty-input-included wrapper

The initialization move

$$\varepsilon, \varepsilon / Z_0$$

takes the physical source state to q_f. Under (A.5), both endpoints represent (C, ε), so this move has semantic change 0.

When the first binary symbol is later consumed from q_f,

$$1, Z_0 / 10 Z_0 \quad \text{or} \quad 0, Z_0 / 11 Z_0,$$

the same two projected checks above apply.

Finally, the core-to-final move

$$\varepsilon, Z_0 / Z_0$$

is available only at the balanced empty logical stack and projects

$$(C, \varepsilon) \longrightarrow (C, \varepsilon).$$

It therefore also preserves (A.1).

A.3 Center-state recognition transitions

A.3.1 Matching 1-cancellation

$$(C, 1T) \xrightarrow{1} (C, T).$$

One 1 is popped, hence

$$\Delta D = +1, \qquad \Delta\Phi = 0.$$

Therefore

$$\Delta(D + \Phi) = +1 = \Delta\rho.$$

The target remains in $0^* \cup 1^* \cup \{10\}$: from a homogeneous 1-stack it remains homogeneous, while the exceptional source 10 becomes 0.

A.3.2 Matching 0-cancellation

$$(C, 0T) \xrightarrow{0} (C, T).$$

One 0 is popped, so

$$\Delta D = -2, \qquad \Delta\Phi = 0,$$

and therefore

$$\Delta(D + \Phi) = -2 = \Delta\rho.$$

Reachability forces the source to lie in 0^+, so the target lies in 0^*.

A.3.3 Center-to-left mismatch

$$(C, 0T) \xrightarrow{1} (L, T).$$

Popping the top 0 gives

$$\Delta D = -2.$$

The state potential changes by

$$\Delta\Phi = \Phi(L) - \Phi(C) = 3.$$

Thus

$$\boxed{\Delta(D + \Phi) = -2 + 3 = +1 = \Delta\rho.}$$

Because the source is a reachable center configuration with top 0, it belongs to 0^+; hence $T \in 0^*$, exactly the required left-state family.

A.3.4 Center-to-right mismatch

$$(C, 1T) \xrightarrow{0} (R, T).$$

Popping one 1 yields

$$\Delta D = +1,$$

while

$$\Delta\Phi = \Phi(R) - \Phi(C) = -3.$$

Therefore

$$\boxed{\Delta(D+\Phi) = 1 - 3 = -2 = \Delta\rho.}$$

There are two reachable source shapes:

$$1^{m+1} \longmapsto 1^m,$$

which enters the 1^* part of $\mathcal{S}_R$, and

$$10 \longmapsto 0,$$

which produces the exceptional right-state store 0.
These are exactly the two right-state families in (A.4).

A.4 Left-state recognition transitions

Every reachable left-state logical store lies in 0^*.

A.4.1 Left-state 0-cancellation

$$(L, 0T) \xrightarrow{0} (L, T).$$

The stack change is one popped 0:

$$\Delta D = -2, \qquad \Delta\Phi = 0.$$

Hence

$$\Delta(D+\Phi) = -2 = \Delta\rho.$$

The target remains in 0^*.

A.4.2 Empty-left state on input 0

The bottom-marker rule is

$$0, Z_0/10Z_0,$$

so logically

$$(L, \varepsilon) \xrightarrow{0} (C, 10).$$

The source value is

$$D(\varepsilon) + \Phi(L) = 3.$$

The target value is

$$D(10) + \Phi(C) = 1.$$

Thus

$$\Delta(D+\Phi) = 1 - 3 = -2 = \Delta\rho.$$

The target 10 is the exceptional center word permitted by $\mathcal{S}_C$.

A.4.3 Nonempty-left state on input 1

For $T \in 0^*$,

$$(L, 0T) \xrightarrow{1} (C, 000T).$$

Replacing one 0 by three 0's adds two zeros, so

$$\Delta D = +4.$$

The state potential drops by 3:

$$\Delta\Phi = 0 - 3 = -3.$$

Hence

$$\boxed{\Delta(D + \Phi) = 4 - 3 = +1 = \Delta\rho.}$$

The target is again homogeneous 0^*.

A.4.4 Empty-left state on input 1

The bottom-marker rule

$$1, Z_0/00Z_0$$

gives

$$(L, \varepsilon) \xrightarrow{1} (C, 00).$$

Now

$$\Delta D = +4, \qquad \Delta\Phi = -3,$$

so again

$$\Delta(D + \Phi) = +1 = \Delta\rho.$$

The target lies in 0^*.

A.5 Right-state recognition transitions

Every reachable right-state logical store lies in

$$1^* \cup \{0\}.$$

A.5.1 Nonempty homogeneous-right state on input 0

For $T \in 1^*$,

$$(R, 1T) \xrightarrow{0} (C, 111111T).$$

Replacing one 1 by six 1's adds five ones, therefore

$$\Delta D = -5.$$

The state potential rises from -3 to 0:

$$\Delta\Phi = +3.$$

Thus

$$\boxed{\Delta(D+\Phi) = -5+3 = -2 = \Delta\rho.}$$

The target is in 1^*.

A.5.2 Empty-right state on input 0

The bottom-marker rule

$$0, Z_0/11111Z_0$$

gives

$$(R,\varepsilon) \xrightarrow{0} (C, 11111).$$

The stack contribution changes by

$$\Delta D = -5,$$

and the potential changes by $+3$. Hence

$$\Delta(D+\Phi) = -2 = \Delta\rho.$$

A.5.3 Exceptional right store 0 on input 0

The rule

$$0, 0/111$$

gives

$$(R,0) \xrightarrow{0} (C, 111).$$

The stack weight changes from

$$D(0) = 2$$

to

$$D(111) = -3,$$

so

$$\Delta D = -5.$$

Again $\Delta\Phi = +3$, and

$$\Delta(D+\Phi) = -2 = \Delta\rho.$$

A.5.4 Exceptional right store 0 on input 1

The rule

$$1, 0/\varepsilon$$

gives

$$(R, 0) \xrightarrow{1} (C, \varepsilon).$$

The stack contribution changes by

$$\Delta D = -2,$$

while the state-potential change is $+3$. Therefore

$$\boxed{\Delta(D + \Phi) = -2 + 3 = +1 = \Delta\rho.}$$

A.5.5 Empty-right state on input 1

The bottom-marker rule

$$1, Z_0/11Z_0$$

gives

$$(R, \varepsilon) \xrightarrow{1} (C, 11).$$

Here

$$\Delta D = -2, \qquad \Delta\Phi = +3,$$

and therefore

$$\Delta(D + \Phi) = +1 = \Delta\rho.$$

A.5.6 Nonempty homogeneous-right state on input 1

For $T \in 1^*$,

$$(R, 1T) \xrightarrow{1} (C, 111T).$$

Replacing one 1 by three 1's adds two ones:

$$\Delta D = -2.$$

Together with $\Delta\Phi = +3$,

$$\boxed{\Delta(D + \Phi) = +1 = \Delta\rho.}$$

The target remains homogeneous 1^*.

A.6 Complete recognition-phase audit table

The preceding cases may be compressed into one table.

source	input / top	logical action	target	ΔD	$\Delta\Phi$	total	required
C	$1/1$	pop 1	C	$+1$	0	$+1$	$+1$
C	$0/0$	pop 0	C	-2	0	-2	-2
C	$1/0$	pop 0	L	-2	$+3$	$+1$	$+1$
C	$0/1$	pop 1	R	$+1$	-3	-2	-2
L	$0/0$	pop 0	L	-2	0	-2	-2
L	$0/Z_0$	write 10	C	$+1$	-3	-2	-2
L	$1/0$	$0 \mapsto 000$	C	$+4$	-3	$+1$	$+1$
L	$1/Z_0$	write 00	C	$+4$	-3	$+1$	$+1$
R	$0/1$	$1 \mapsto 111111$	C	-5	$+3$	-2	-2
R	$0/Z_0$	write 11111	C	-5	$+3$	-2	-2
R	$0/0$	$0 \mapsto 111$	C	-5	$+3$	-2	-2
R	$1/0$	pop 0	C	-2	$+3$	$+1$	$+1$
R	$1/Z_0$	write 11	C	-2	$+3$	$+1$	$+1$
R	$1/1$	$1 \mapsto 111$	C	-2	$+3$	$+1$	$+1$

Thus every reachable recognition transition satisfies (A.3).
This proves the finite transition step omitted from the short proof in Chapter 1.

A.7 Reachable-store closure

The same table proves the state-dependent store normal form.

Center

The center can contain only

$$0^*, \qquad 1^*, \qquad 10.$$

Its two matching loops shorten an existing block. A mismatch can leave the center only by popping the visible blocking symbol. Every side-to-center transition returns to a homogeneous block except the left-empty/input-0 case, which creates exactly 10.

Left

The only transition entering L is

$$(C, 0T) \xrightarrow{1} (L, T).$$

Reachability forces the source center stack to belong to 0^+, so

$$T \in 0^*.$$

Therefore no reachable left configuration has top 1.
This proves, in particular, that the listed completion rule

$$\dashv, 1/0$$

has an unreachable source condition.

Right

The only transition entering R is

$$(C, 1T) \xrightarrow{0} (R, T).$$

If the source is 1^{m+1}, then $T = 1^m$.
If the source is the exceptional center word 10, then $T = 0$.
Hence

$$\mathcal{S}_R = 1^* \cup \{0\}.$$

No other right-state store can be produced.

A.8 Audit of the end-of-input completion rules

The endmarker $\dashv$ is not a binary input symbol, so completion normalization must preserve the residual already accumulated by the prefix.

The side-state potential is transferred into literal stack content.

A.8.1 Left-state completion

Every reachable left store is 0^m.

Empty left store.

$$(L, \varepsilon) \xrightarrow{\dashv} 100.$$

Before completion,

$$D(\varepsilon) + \Phi(L) = 0 + 3 = 3.$$

After completion,

$$D(100) = 4 - 1 = 3.$$

Nonempty left store. For $m \geq 1$,

$$(L, 0^m) \xrightarrow{\dashv} 1\,0^{m+2}.$$

Indeed the physical rule replaces the top 0 by 1000, leaving 0^{m-1} untouched.
Before completion,

$$D(0^m) + \Phi(L) = 2m + 3.$$

After completion,

$$D(1\,0^{m+2}) = 2(m+2) - 1 = 2m + 3.$$

Thus left completion preserves the represented residual exactly.
The listed left rule with top 1 is unreachable and plays no role in any run.

A.8.2 Right-state completion

There are three reachable right forms.

Empty right store.

$$(R, \varepsilon) \xrightarrow{\dashv} 111.$$

Before completion,

$$D(\varepsilon) + \Phi(R) = -3,$$

and after completion,

$$D(111) = -3.$$

Nonempty homogeneous right store. For $m \geq 1$,

$$(R, 1^m) \xrightarrow{\dashv} 1^{m+3}.$$

Before completion,

$$D(1^m) + \Phi(R) = -m - 3.$$

After completion,

$$D(1^{m+3}) = -m - 3.$$

Exceptional right store.

$$(R, 0) \xrightarrow{\dashv} 1.$$

Before completion,

$$D(0) + \Phi(R) = 2 - 3 = -1.$$

After completion,

$$D(1) = -1.$$

Therefore every reachable endmarker normalization rule preserves the prefix residual.

A.9 From residual preservation to the canonical shortest completion

After the side-state correction, every normalized logical stack belongs to

$$0^* \cup 1^* \cup 10^+.$$

The three residual cases are therefore:

$$\widehat{S}(w) = \begin{cases} 0^m, & \rho(w) = 2m, \quad m \geq 0, \\ 1\,0^{m+1}, & \rho(w) = 2m+1, \quad m \geq 0, \\ 1^m, & \rho(w) = -m, \quad m \geq 1. \end{cases} \tag{A.6}$$

For any continuation containing x zeros and y ones,

$$wz \in L_{2,1} \quad \Longleftrightarrow \quad 2x - y = \rho(w). \tag{A.7}$$

The words in (A.6) realize the unique minimum nonnegative Parikh solution of (A.7):

- $2m$: $(x, y) = (m, 0)$;

- $2m+1$: $(x, y) = (m+1, 1)$;
- $-m$: $(x, y) = (0, m)$.

Thus the transition audit not only proves recognition-phase arithmetic consistency; together with the endmarker audit it recovers the shortest-completion semantics of the original design.

A.10 Two representative runs

A.10.1 Accepted word 110

Using the core projection and omitting Z_0,

$$(C, \varepsilon) \xrightarrow{1} (C, 10) \xrightarrow{1} (C, 0) \xrightarrow{0} (C, \varepsilon).$$

The residual sequence is

$$0,\ 1,\ 2,\ 0,$$

while $D(S) + \Phi(q)$ gives the same sequence:

$$0,\ 1,\ 2,\ 0.$$

The final empty center stack therefore represents zero residual and the word is accepted.

A.10.2 Prefix 111

The run is

$$(C, \varepsilon) \xrightarrow{1} (C, 10) \xrightarrow{1} (C, 0) \xrightarrow{1} (L, \varepsilon).$$

The residual is

$$\rho(111) = 3.$$

The final logical stack is empty, but

$$D(\varepsilon) + \Phi(L) = 3.$$

The endmarker rule transfers this finite-control contribution into the literal completion word:

$$(L, \varepsilon) \xrightarrow{\dashv} 100.$$

Since

$$2N_0(100) - N_1(100) = 4 - 1 = 3,$$

the normalized stack represents exactly the outstanding residual. Appending 100 gives

$$111100 \in L_{2,1}.$$

This is the canonical shortest completion selected by the machine.

A.11 Determinism of the completion extension

The completion callouts should not be interpreted as ordinary ε-moves.

For a fixed state/top pair:

- binary recognition transitions are selected by next input 0 or 1;
- completion is selected only by the distinct right endmarker $\dashv$.

Direct all completion rules to a dedicated completion sink with no outgoing transitions.

Then no configuration has two enabled successors.

Hence the completion-extended formalization is deterministic in the standard DPDA sense.

A.12 Finite transition consistency checks

The symbolic audit above is complete: it covers every reachable transition family.

Independent finite-range checks complement the symbolic transition audit. Their scope comprises the following cases and invariants:

- 524,287 original-machine binary inputs;
- 8,912,898 prefix transition checks;
- the residual invariant;
- the reachable-store normal form;
- completion normalization;
- independent shortest-completion length;
- depth-two support visibility.

All listed comparisons agree with the symbolic formulas.

The computational audit is supporting evidence only; the proof of (A.1) is the finite symbolic transition audit in Sections A.2–A.8.

A.13 Conclusion

The original 2023 design is not merely a correct recognizer.

Its state and stack form an exact split representation of the prefix residual:

$$\boxed{\rho = D + \Phi.}$$

The center state stores the residual entirely in the stack.

The side states store part of the same residual in finite control:

$$\boxed{\Phi(L) = +3, \qquad \Phi(R) = -3.}$$

Every binary transition preserves that semantic quantity with the correct residual increment, and every reachable endmarker rule transfers the finite-control offset back into literal stack form.

Thus the three operational ideas visible in the original machine—

$$\boxed{\text{cancellation,} \qquad \text{side-state mismatch,} \qquad \text{completion,}}$$

—are already an exact semantic normalization mechanism.

This is the local machine-level fact from which the later completion geometry, residual quotients, WRITE–SEARCH–CANCEL theory, lift theory, and General Theory are developed.

Appendix B

General Construction Transition Schemas

B.1 Purpose and model separation

The main text deliberately develops the general constructions conceptually before presenting every transition rule. This appendix collects the exact machine syntax in one place so that the book remains reproducible without forcing long transition tables into the narrative chapters.

Fix coprime integers $p, q \geq 1$ and write

$$\rho(w) = qN_1(w) - pN_0(w), \qquad T_0(r) = r - p, \qquad T_1(r) = r + q.$$

The primitive zero-weight kernel is

$$K = (q, p).$$

Three constructions must be kept separate.

1. The *paired-sign four-symbol lift* realizes the residual graph exactly after a deterministic normalization macro-step. It is a normalized residual–configuration conjugacy, not a raw one-step realization.

2. The *boundary-compatible three-symbol lift* changes coordinates so that every residual edge is one ordinary binary-input transition. It is ε-free and gives raw residual–configuration conjugacy with $\max(p, q)$ states.

3. The *centered literal-demand construction* stores the unique shortest completion demand itself, uses a finite-control debt rail to repair the LIFO ordering obstruction, consumes exactly one binary symbol per transition, and is real-time during binary input.

The exact resource-frontier theorems later compare these constructions only inside their stated representation models. In particular, no table below should be read as an unrestricted minimum-state theorem for every DPDA recognizing $L_{p,q}$.

B.2 Four-symbol normalized paired-sign lift

This is the first compact general physical lift used for the residual-geometry and configuration-rectangle results.

Put

$$m_{\mathrm{st}} = \max(p, q), \qquad Q_4 = \{R_0, \ldots, R_{m_{\mathrm{st}}-1}\}, \qquad \Gamma_4 = \{0, 1, X, Z_0\}.$$

For $0 \le s < p$ and $0 \le t < q$, define the finite residual bases

$$a_s = \begin{cases} 0, & s = 0, \\ s - p, & 1 \le s < p, \end{cases} \qquad b_t = \begin{cases} 0, & t = 0, \\ q - t, & 1 \le t < q. \end{cases}$$

The stable encoding $\eta_4 : \mathbb{Z} \to Q_4 \times \Gamma_4^*$ is

$$\eta_4(0) = (R_0, Z_0).$$

If $r > 0$, let $s = r \bmod p$ and write uniquely

$$r = pk + a_s, \qquad k \ge 1;$$

then

$$\eta_4(r) = (R_s, 0^k Z_0).$$

If $r < 0$, let $t = (-r) \bmod q$ and write uniquely

$$r = -qk + b_t, \qquad k \ge 1;$$

then

$$\eta_4(r) = (R_t, 1^k X).$$

The two sign roles may share a control state; their work symbols and sentinels keep the stable encodings disjoint.

Positive stable rules

For $0 \le s < p$, put

$$s' = (s + q) \bmod p, \qquad c_s = \frac{a_s + q - a_{s'}}{p}.$$

Then

$$\boxed{\delta(R_s, 0, 0) = (R_s, \varepsilon)}$$

and

$$\boxed{\delta(R_s, 1, 0) = (R_{s'}, 0^{c_s+1})}.$$

When the last positive work symbol is removed, a unique forced boundary normalization is used:

$$\boxed{\delta(R_s, \varepsilon, Z_0) = \eta_4(a_s)} \qquad (1 \le s < p).$$

For $s = 0$, popping the final positive 0 already exposes the zero encoding (R_0, Z_0), so no boundary normalization is required.

Negative stable rules

For $0 \le t < q$, put

$$t' = (t + p) \bmod q, \qquad d_t = \frac{p - b_t + b_{t'}}{q}.$$

Then

$$\boxed{\delta(R_t, 1, 1) = (R_t, \varepsilon)}$$

and

$$\boxed{\delta(R_t, 0, 1) = (R_{t'}, 1^{d_t+1})}.$$

When the negative sentinel is exposed,

$$\boxed{\delta(R_t, \varepsilon, X) = \eta_4(b_t)}.$$

Zero rules and acceptance

At residual zero,

$$\boxed{\delta(R_0, 1, Z_0) = \eta_4(q)}, \qquad \boxed{\delta(R_0, 0, Z_0) = \eta_4(-p)}.$$

A mandatory right endmarker is accepted only at the stable zero configuration $\eta_4(0)$.

An expression of the form $\delta(\cdot) = \eta_4(r')$ is semantic shorthand: the exposed boundary symbol is replaced by the stack word of the stable encoding while control enters the corresponding state. Long fixed writes may be compiled into deterministic forced ε-chains if a bounded-write PDA syntax is required. Such a compilation preserves the stable target but is no longer a raw one-binary-transition realization.

Define a normalized input macro-step $c \overset{a}{\Rightarrow} c'$ to mean one consumed binary symbol followed by the unique forced normalization chain. The transition schema satisfies

$$\boxed{\eta_4(r) \overset{0}{\Rightarrow} \eta_4(r-p), \qquad \eta_4(r) \overset{1}{\Rightarrow} \eta_4(r+q).}$$

This is the exact normalized residual–configuration conjugacy used in the configuration-rectangle chapters.

B.3 One-work-symbol raw conjugacy

The one-work-symbol construction is included because it is the matching upper bound for the exact $p+q$-state result in the permanent-bottom raw-conjugacy model.

Let

$$\Gamma_2 = \{A, Z_0\},$$

with disjoint state families

$$P_0, \dots, P_{p-1}, \qquad N_0, \dots, N_{q-1}.$$

For $r \geq 0$, write

$$r = pk + s, \qquad k \geq 0, \quad 0 \leq s < p,$$

and define

$$\xi(r) = (P_s, A^k Z_0).$$

For $r < 0$, write uniquely

$$r = u - qh, \qquad 0 \leq u < q, \quad h \geq 1,$$

and define

$$\xi(r) = (N_u, A^{h-1} Z_0).$$

For $0 \leq s < p$, set

$$s^+ = (s+q) \bmod p, \qquad c_s = \frac{s+q-s^+}{p} \geq 0,$$

and write

$$s - p = u_s - qh_s, \qquad 0 \leq u_s < q, \quad h_s \geq 1.$$

The positive and boundary rules are

$$\boxed{\delta(P_s, 0, A) = (P_s, \varepsilon)}, \qquad \boxed{\delta(P_s, 1, A) = (P_{s^+}, A^{c_s+1})},$$

$$\boxed{\delta(P_s, 1, Z_0) = (P_{s^+}, A^{c_s} Z_0)}, \qquad \boxed{\delta(P_s, 0, Z_0) = (N_{u_s}, A^{h_s-1} Z_0)}.$$

For $0 \le u < q$, put

$$u^- = (u - p) \bmod q, \qquad d_u = \frac{u^- - u + p}{q} \ge 0.$$

The negative rules are

$$\boxed{\delta(N_u, 1, A) = (N_u, \varepsilon)}, \qquad \boxed{\delta(N_u, 0, A) = (N_{u^-}, A^{d_u+1})},$$

$$\boxed{\delta(N_u, 1, Z_0) = \xi(u)}, \qquad \boxed{\delta(N_u, 0, Z_0) = (N_{u^-}, A^{d_u} Z_0)}.$$

The mandatory right endmarker is accepted only at

$$\xi(0) = (P_0, Z_0).$$

There are no ε-transitions. Direct substitution gives

$$\boxed{\delta(\xi(r), 0) = \xi(r - p), \qquad \delta(\xi(r), 1) = \xi(r + q).}$$

The construction uses exactly $p + q$ states and two total stack symbols, one of which is the permanent bottom marker.

B.4 Boundary-compatible three-symbol raw DPDA

The four-symbol normalized lift shows that the residual graph has a compact physical realization, but it has a finite set of boundary edges that require normalization. The following coordinate change removes those defects completely.

Assume first that $p \ge q$. Let

$$Q_3 = \{R_0, \dots, R_{p-1}\}, \qquad \Gamma_3 = \{0, 1, Z_0\}.$$

For $r \ge 0$, write uniquely

$$r = pk + s, \qquad k \ge 0, \quad 0 \le s < p,$$

and set

$$\boxed{\zeta(r) = (R_s, 0^k Z_0)}.$$

For $r < 0$, write uniquely

$$r = u - qh, \qquad h \ge 1, \quad 0 \le u < q,$$

and set

$$\boxed{\zeta(r) = (R_u, 1^h Z_0)}.$$

The state sets overlap deliberately. The sign is visible from the top of stack: negative encodings have top 1, whereas nonnegative encodings have top 0 or Z_0.

For $0 \le s < p$, define

$$s^+ = (s + q) \bmod p, \qquad c_s = \frac{s + q - s^+}{p} \in \{0, 1\},$$

$$u_s = (s - p) \bmod q, \qquad h_s = \frac{u_s + p - s}{q} \ge 1.$$

For $0 \le u < q$, define

$$u^- = (u - p) \bmod q, \qquad d_u = \frac{u^- - u + p}{q} \ge 1.$$

Complete transition table for $p \geq q$

region	input/top	target	replacement
nonnegative interior	$0/0$	R_s	ε
nonnegative interior	$1/0$	R_{s^+}	0^{c_s+1}
nonnegative boundary	$1/Z_0$	R_{s^+}	$0^{c_s}Z_0$
nonnegative boundary	$0/Z_0$	R_{u_s}	$1^{h_s}Z_0$
negative	$1/1$	R_u	ε
negative	$0/1$	R_{u^-}	1^{d_u+1}

Equivalently,

$$\boxed{\begin{aligned}\delta(R_s,0,0) &= (R_s,\varepsilon),\\ \delta(R_s,1,0) &= (R_{s^+},0^{c_s+1}),\\ \delta(R_s,1,Z_0) &= (R_{s^+},0^{c_s}Z_0),\\ \delta(R_s,0,Z_0) &= (R_{u_s},1^{h_s}Z_0),\\ \delta(R_u,1,1) &= (R_u,\varepsilon),\\ \delta(R_u,0,1) &= (R_{u^-},1^{d_u+1}).\end{aligned}}$$

The right endmarker is accepted only at

$$\zeta(0) = (R_0, Z_0).$$

No ε-transition is used. Every binary transition consumes exactly one input symbol, and

$$\boxed{\delta(\zeta(r),0) = \zeta(r-p), \qquad \delta(\zeta(r),1) = \zeta(r+q).}$$

Hence the raw binary-input configuration graph is label-isomorphic to the residual graph.

If $q > p$, use the symmetric construction obtained by simultaneously exchanging

$$0 \leftrightarrow 1, \qquad p \leftrightarrow q, \qquad r \leftrightarrow -r.$$

The resulting state count is always

$$\boxed{|Q_3| = \max(p,q)},$$

with three total stack symbols including the permanent bottom marker.

B.5 Centered real-time literal-demand construction

The raw residual encodings above store a signed residual coordinate. The centered construction solves a different problem: it keeps a physical representation whose semantic projection is the unique shortest completion demand and repairs one-cell LIFO ordering conflicts by moving a blocker into finite control.

Let

$$m = p + q, \qquad h_{\mathrm{ctr}} = \left\lfloor \frac{m-1}{2} \right\rfloor.$$

For a residual r, let

$$\mu(r) = (x_r, y_r)$$

be the unique minimum nonnegative solution of

$$px - qy = r.$$

For a demand vector (x, y) use the canonical two-visible ordering

$$\eta(x,y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & y > x. \end{cases}$$

For debt $d \in \mathbb{Z}$, define the virtual demand

$$V_d = \mu(dm), \qquad \Phi(D_d) = dm.$$

The centered state set is

$$\boxed{Q_{\mathrm{ctr}} = \{D_{-h_{\mathrm{ctr}}}, \ldots, D_{-1}, D_0, D_1, \ldots, D_{h_{\mathrm{ctr}}}\},}$$

with $D_0 = C$, and

$$\Gamma_{\mathrm{ctr}} = \{0, 1, Z_0\}.$$

For a logical stack S, the semantic projection is

$$\Theta(D_d, S) = \mathrm{red}_{p,q}\big(P(S) + V_d\big).$$

The residual invariant is

$$\boxed{ho = W(P(S)) + dm.}$$

MATCH

For $z \in \{0, 1\}$ and every $|d| \leq h_{\mathrm{ctr}}$,

$$\boxed{(D_d, zT) \xrightarrow{z} (D_d, T).}$$

Interior SEARCH

For $d < h_{\mathrm{ctr}}$,

$$\boxed{(D_d, 0T) \xrightarrow{1} (D_{d+1}, T),}$$

and for $d > -h_{\mathrm{ctr}}$,

$$\boxed{(D_d, 1T) \xrightarrow{0} (D_{d-1}, T).}$$

The displaced blocker is not restored immediately. Its signed effect is represented by the debt coordinate d.

Outer-boundary WRITE

Put

$$U_+ = \eta(\mu((h_{\mathrm{ctr}} + 1)m)), \qquad U_- = \eta(\mu(-(h_{\mathrm{ctr}} + 1)m)).$$

Then

$$\boxed{(D_{h_{\mathrm{ctr}}}, 0T) \xrightarrow{1} (C, U_+ T),}$$

and

$$\boxed{(D_{-h_{\mathrm{ctr}}}, 1T) \xrightarrow{0} (C, U_- T).}$$

Bottom WRITE

For every $|d| \leq h_{\mathrm{ctr}}$,

$$\boxed{(D_d, Z_0) \xrightarrow{1} (C, \eta(\mu(dm + q))Z_0),}$$

$$\boxed{(D_d, Z_0) \xrightarrow{0} (C, \eta(\mu(dm - p))Z_0).}$$

Acceptance

A mandatory right endmarker is permitted only at the balanced empty configuration:

$$\boxed{(C, Z_0) \xrightarrow{\dashv} (C, \varepsilon).}$$

There is no right-endmarker rule from any other reachable configuration. Every binary transition consumes exactly one input symbol; the centered construction has no ε-move during binary processing.

B.6 Minimal kernel-carry compiler identity

The WRITE strings in the centered construction are not ad hoc. Suppose a transition consumes input symbol a, removes visible top symbol z, changes state from D_s to D_t, and writes a replacement word u. Let

$$e_0 = (1, 0), \qquad e_1 = (0, 1).$$

Ignoring the untouched suffix, invariant preservation forces

$$P(u) = e_z - e_a + V_s - V_t + \kappa K.$$

At the bottom marker, omit the term e_z. If

$$B = e_z - e_a + V_s - V_t,$$

then the smallest kernel carry making the replacement nonnegative is

$$\boxed{\kappa_* = \max\left(\left\lceil -\frac{B_0}{q} \right\rceil, \left\lceil -\frac{B_1}{p} \right\rceil\right).}$$

The unique shortest compatible replacement vector is

$$\boxed{R_* = B + \kappa_* K.}$$

Every other nonnegative invariant-preserving replacement is $R_* + \ell K$ for some $\ell \geq 1$ and is longer by exactly $\ell(p + q)$ symbols.

This identity is the common arithmetic compiler behind the large literal WRITE words of the original 2:1 machine and the general centered family.

B.7 The 2:1 specialization and the eager–lazy distinction

For $(p, q) = (2, 1)$,

$$m = 3, \qquad h_{\mathrm{ctr}} = 1,$$

so

$$D_{-1} = R, \qquad D_0 = C, \qquad D_1 = L,$$

with potentials

$$-3, \quad 0, \quad +3.$$

The centered schema reproduces the historical center/left/right control skeleton, the center mismatch pops, the left cancellation family, and the arithmetic of the characteristic WRITE words.

Two reachable right-state matching families differ in schedule, not semantics. The centered machine keeps the resulting debt in finite control (lazy materialization), whereas the historical machine immediately materializes the same canonical demand on the stack (eager materialization). If $\mathcal{N}$ denotes canonical materialization, then on exactly those families

$$\delta_{\mathrm{hist}}(c, a) = \mathcal{N}(\delta_{\mathrm{cent}}(c, a)).$$

Accordingly, direct specialization is not claimed to be literal transition-table identity. The remaining difference is precisely the eager–lazy normalization-schedule choice.

B.8 Determinism and implementation conventions

The constructions in this appendix use three different transition disciplines and should not be conflated.

- In the four-symbol paired-sign lift, forced ε-normalization is enabled only on transient boundary pairs where binary input is disabled. The normalization macro-step is therefore deterministic.
- In the one-work-symbol and boundary-compatible three-symbol raw lifts, there are no ε-transitions at all; the right endmarker is the only nonbinary input symbol.
- In the centered literal-demand construction, every binary step is real-time and the right endmarker is enabled only at the balanced empty configuration.

All replacement words are fixed finite constants once (p, q) is fixed. If a separate PDA convention imposes a bound on replacement length, long fixed writes can be compiled using auxiliary deterministic microstates. Such compilation changes the physical transition granularity and must not be silently substituted into a theorem whose statement concerns raw one-input transitions or exact state counts in a fixed model.

B.9 Summary table

construction	states	stack alphabet	exact role
paired-sign normalized lift	$\max(p, q)$	$\{0, 1, X, Z_0\}$	exact residual conjugacy after a unique normalization macro-step
one-work-symbol raw lift	$p + q$	$\{A, Z_0\}$	exact raw residual conjugacy; matching upper bound for the one-work-symbol state theorem
boundary-compatible raw lift	$\max(p, q)$	$\{0, 1, Z_0\}$	ε-free raw residual conjugacy by ordinary binary transitions
centered literal-demand lift	$2\lfloor(p + q - 1)/2\rfloor + 1$	$\{0, 1, Z_0\}$	real-time shortest-demand normalization with finite-control SEARCH debt; exact width only in the canonical deferred-SEARCH model

The purpose of the table is organizational. The constructions optimize different interfaces, and their state counts are not interchangeable unrestricted complexity claims.

Appendix C

Scoped Lower Bounds and Finite Certificates

C.1 Why this appendix is necessary

Several chapters use the word *optimal*, but they do so in different representation categories. The purpose of this appendix is to make the scope of every principal lower bound explicit and to place the one genuinely computer-assisted finite elimination in an explicitly auditable finite form.

The central distinction is:

$$\boxed{\text{representation-relative lower bound} \neq \text{unrestricted recognition lower bound.}}$$

A machine may recognize the same language while abandoning the residual coordinates, stack-tail form, literal-demand interface, or normalization schedule used by another construction. A lower bound obtained by counting coordinates in one architecture therefore cannot automatically be promoted to all DPDAs.

C.2 Lower-bound scope table

result	scope, exact bound, and excluded overclaim
one-work-symbol raw residual conjugacy	**Opponent:** permanent bottom, one work symbol, ε-free raw residual–configuration conjugacy. **Bound:** $\lvert Q\rvert \geq p + q$. **Not:** an unrestricted recognition bound for $L_{p,q}$.
homogeneous-tail raw frontier	**Opponent:** permanent bottom, sign-separated homogeneous tails, at least two work symbols. **Bound:** $\lvert Q\rvert \geq \max(p, q)$. **Not:** a theorem about arbitrary stack encodings.
arbitrary three-symbol raw residual encoding	**Opponent:** permanent bottom, three total stack symbols, arbitrary injective raw residual encoding. **Bound:** $\lvert Q\rvert \geq \max(p, q)$ for every coprime (p, q). **Not:** an unrestricted recognition bound; Theorem 8.7 is representation-shape-free within Model R.
canonical deferred-SEARCH width	**Opponent:** literal shortest-demand representation with deferred SEARCH interface. **Bound:** $2\lfloor (p + q - 1)/2 \rfloor + 1$ states. **Not:** an unrestricted state minimum for every DPDA recognizing $L_{p,q}$.

result	scope, exact bound, and excluded overclaim
phase-local capacity	**Opponent:** specified local observation depth and remainder architecture. **Bound:** $\lvert Q\rvert g^{\lambda} \geq BH(B)$. **Not:** a universal Pareto frontier.
binary stack-alphabet hierarchy witness	**Opponent:** arbitrary general DPDA in the stated single-initial-symbol model; deterministic ε-moves allowed. **Bound:** $\lvert Q\rvert\lvert\Gamma\rvert \geq mk+1$ for $B_{m,k}$. **Not:** automatically a bound for $L_{p,q}$; it applies to the engineered witness language.
unrestricted rational-balance state complexity	**Opponent:** arbitrary deterministic right-endmarker ε-DPDA in the stated regular-target model. **Bound:** $\lvert Q\rvert \geq w_{\max}$, attained with three total stack symbols. **Scope:** representation-independent for $L_{p,q}$; no residual coordinate is assumed.
unrestricted rational-balance state–stack frontier	**Opponent:** the same broad class, with the permanent-bottom convention. **Frontier:** $\{(p+q,2),(w_{\max},3)\}$. **Scope:** the explicit global endpoint, superseding any raw-only reading of the same pair.
unrestricted bounded replacement, one work symbol	**Opponent:** arbitrary history-dependent deterministic encodings, one work symbol, replacement bound $B \geq 2$. **Bound:** $w_{\max} + \max(w_{\min}, \lceil w_{\max}/(B-1)\rceil)$. **Scope:** exact only for this one-work-symbol bounded-replacement resource question.
filtered collapse / source exposure	**Opponent:** exact deterministic lifts with a finite collapse family and hidden-tail noncoalescence; the pushdown specialization is arbitrary. **Bounds:** $D + X_h + V(h) \geq N$ and $V(h) \leq q_{\mathrm{src}} \sum_{r=0}^{h} g_{\mathrm{src}}^{r}$. **Not:** a state minimum for every language; this is an information/exposure tradeoff.

C.3 Raw residual-conjugacy frontier

For the permanent-bottom raw residual-conjugacy models developed in the resource chapters, the exact structured frontier is:

$$\boxed{S_{\mathrm{raw}}(g) = \begin{cases} \infty, & g = 0, \\ p+q, & g = 1, \\ \max(p,q), & g \geq 2, \end{cases}}$$

where g denotes the number of work symbols above the permanent bottom marker in the homogeneous-tail model.

The upper bounds are realized by the constructions collected in Appendix B. The lower bounds rely on the raw-conjugacy interface: every residual must have one injective physical configuration, and each binary residual edge must be realized by one ordinary input transition.

The $g = 1$ lower bound is exact because a one-work-symbol store can encode unbounded magnitude only through height, while the p positive-end and q negative-end arithmetic phases require disjoint recurrent state resources. The resulting state requirement is

$$\boxed{\lvert Q\rvert \geq p+q.}$$

With two distinct work symbols, sign can be placed in the visible top symbol. The boundary-compatible construction then attains

$$\boxed{\lvert Q\rvert = \max(p,q), \qquad \lvert\Gamma\rvert = 3}$$

including the permanent bottom marker.

The homogeneous-tail hypothesis is removed completely in Chapter 8. The representation-shape-free Model-R argument proves the same three-symbol state optimum $|Q| = \max(p, q)$ for every coprime pair. The earlier small-ratio eliminations remain useful finite certificates, but they are no longer the endpoint of the theory.

C.4 The 3:2 finite reduction

The first genuinely two-sided arbitrary-encoding case is $(p, q) = (3, 2)$. The theorem concerns the full permanent-bottom three-symbol model

$$\Gamma_3 = \{A, B, Z_0\}$$

with no ε-moves and an injective raw residual–configuration conjugacy. No homogeneous-tail or sign-separated assumption is imposed.

Assume, for contradiction, that two control states suffice. Choose an encoded configuration of minimum work-stack height and write the two parity chains relative to its residual r_* as

$$A_k = \eta(r_* + 2k), \qquad B_k = \eta(r_* + 1 + 2k).$$

Let G denote the input-1 action and F the input-0 action. Since input 1 adds 2 to the residual and input 0 subtracts 3,

$$G(A_k) = A_{k+1}, \qquad G(B_k) = B_{k+1},$$

$$F(A_{k+2}) = B_k, \qquad F(B_{k+1}) = A_k.$$

Therefore, on the encoded residual graph,

$$\boxed{F^2G^3 = G^3F^2 = \mathrm{id}\,.} \tag{C.1}$$

Equation (C.1) is the key finiteness mechanism. One consuming PDA transition deletes at most one work symbol. If a work-top rule writes m work symbols in place of one, its first step increases work height by $m - 1$. The remaining four moves of either five-step inverse identity can delete at most four symbols before the original configuration must be recovered. Hence

$$\boxed{m \le 5}$$

for every work-top replacement encountered in the search. From the bare bottom, the first step creates rather than replaces its work word, so every bottom write has length at most four.

Thus an a priori infinite transition search reduces to a finite deterministic rule-synthesis problem.

C.5 Normalization of the minimum configuration

There are exactly two minimum-height cases.

Positive minimum height. The suffix strictly below the visible top of a minimum encoded configuration can never be exposed by another encoded configuration; it is inert and may be stripped. Rename states and work symbols so that

$$\boxed{A_0 = (0, A\#),}$$

where # denotes the fixed inert suffix.

Zero minimum height. Normalize instead to

$$\boxed{A_0 = (0, \#),}$$

where # is the bare bottom marker.

These two cases exhaust the possible minimum work heights.

C.6 Legal branching and pruning rules

The finite certificate assigns a transition rule only when a state/top pair is encountered for the first time. At that first occurrence it branches over every legal target state and every replacement word satisfying the proven length bounds. A repeated state/top pair must reuse the rule already fixed by determinism.

Branches are rejected only by consequences already forced by the raw conjugacy and the inverse identities:

1. a prescribed F-relation or G-relation has the wrong target;
2. suffix preservation makes the required replacement impossible;
3. two distinct residuals obtain the same encoded configuration, violating injectivity;
4. a configuration is too tall to return to a prescribed target in the remaining number of one-pop transitions.

Representative deletion-capacity bounds are

$$F^2(A_3) = A_0, \qquad F(B_1) = A_0,$$

which imply

$$|A_3| \le |A_0| + 2, \qquad |B_1| \le |A_0| + 1.$$

More generally,

$$F(B_j) = A_{j-1}, \qquad F(A_{j+2}) = B_j,$$

and the corresponding two-deletion relations

$$F^2(B_j) = B_{j-3}, \qquad F^2(A_{j+2}) = A_{j-1}$$

provide the remaining legal height-pruning bounds.

No heuristic pruning is permitted: every discarded branch violates one of these proved necessary conditions.

C.7 Exact survivor vectors

For positive minimum height, the exact numbers of surviving partial rule systems at successive synthesis stages are

$$\boxed{1, 123, 11643, 1691, 2120, 1591, 348, 43, 13, 0.} \tag{C.2}$$

The search examines exactly

$$\boxed{1{,}130{,}430}$$

branch attempts in this case.

For minimum height zero, the survivor vector is

$$\boxed{1, 61, 7556, 1134, 2146, 2966, 6846, 7032, 2376, 650, 190, 26, 0.} \tag{C.3}$$

This case examines exactly

$$\boxed{1{,}696{,}120}$$

branch attempts.

Both exhaustive searches terminate with zero survivors. Since the symbolic reduction proves that every possible two-state raw encoding must occur in one of the two finite searches, two states are impossible. The explicit three-state boundary-compatible construction supplies the matching upper bound. Hence

$$\boxed{|Q|_{\min} = 3}$$

for the arbitrary permanent-bottom three-symbol raw-conjugacy model at $(3, 2)$, and symmetrically at $(2, 3)$.

C.8 Computational certificate validation

An independent finite evaluation of the structure and resource conditions yields

$$\boxed{80{,}717{,}789 \text{ checks} - \text{PASS.}}$$

Among the independently checked components are

component	count
brute minimum-demand cases	11,011
broad arithmetic cases	1,548,547
compact normalized macro cases	667,110
normalized rectangle vertices	1,859,269
normalized rectangle diamonds	1,286,746
raw four-symbol immediate cases	1,111,110
one-work-symbol raw cases	1,111,110
three-symbol raw cases	1,111,110
raw rectangle classifications	61,455
three-symbol raw rectangle vertices	731,091
three-symbol raw rectangle diamonds	474,684

The same validation also confirms the exact branch counts and survivor vectors in (C.2)–(C.3).

The later resource-frontier computational validation was likewise rerun and terminates with

$$\boxed{79{,}606{,}679 \text{ checks} - \text{PASS.}}$$

These computational audits support the symbolic results and certify the bounded search. Except for the explicitly computer-assisted $(3, 2)$ elimination, they are regression evidence rather than substitutes for proof.

C.9 Architecture-relative deferred-SEARCH width

For the centered literal-demand machine, put

$$h_{\mathrm{ctr}} = \left\lfloor \frac{p + q - 1}{2} \right\rfloor .$$

In the canonical deferred-SEARCH model, after a witness demand block has been consumed, the remaining work stack is the same bare bottom marker for every debt value

$$-h_{\mathrm{ctr}}, \ldots, h_{\mathrm{ctr}}.$$

Distinct residuals are continuation-distinguishable. Determinism therefore forces distinct control states for all these debt values. The exact width is

$$\boxed{|Q|_{\min}^{\text{deferred}} = 2h_{\text{ctr}} + 1 = 2\left\lfloor \frac{p+q-1}{2} \right\rfloor + 1.}$$

The centered construction attains the bound.

The deferred condition is essential: the final consumed witness cell is a pure pop, and materialization is postponed to a later transition. Eager encodings, alternate stack coordinates, or machines that do not store literal minimum demand are outside this lower-bound category.

C.10 Representation-independent hierarchy product bound

A different chapter deliberately leaves the counting-language representation class. For every $m \geq 2$ and $k \geq 1$, the binary witness language $B_{m,k}$ is recognized by a realtime DPDA with exactly m states and $k+1$ pushdown symbols. Every competing DPDA in the stated *general* deterministic model—including opponents with arbitrary deterministic ε-moves and unrelated internal stack encodings—satisfies

$$\boxed{|Q|\,|\Gamma| \geq mk + 1.} \tag{C.4}$$

Consequently the fixed-state stack-alphabet classes obey the strict hierarchy

$$\boxed{\mathcal{D}_{m,k} \subsetneq \mathcal{D}_{m,k+1} \qquad (m \geq 2,\ k \geq 1).}$$

The proof does not count residual phases in a prescribed code. It isolates $mk+1$ externally distinguishable affine completion slopes. Repetition, height-escape, and step-pump arguments show that one recurrent state/top-symbol mode cannot realize two distinct slopes. Since a DPDA has at most $|Q||\Gamma|$ nonempty modes, (C.4) follows.

This is representation-independent for the witness language $B_{m,k}$, but it must not be retroactively read as an unrestricted product lower bound for the rational counting languages $L_{p,q}$.

C.11 Final scope rule

The lower-bound hierarchy in this monograph is intentionally layered, and the final endpoint contains two genuinely unrestricted branches:

coordinate-relative exact frontiers and protected contracts

$\Downarrow$

architecture-relative literal-demand and local-observation bounds

$\Downarrow$

representation-independent witness hierarchy

$\Downarrow$

unrestricted $L_{p,q}$ state/state–stack/bounded-replacement theorems

$\Downarrow$

representation-independent collapse/exposure tradeoffs.

The arrows record the research route rather than logical implication. In particular, raw residual-faithful frontiers remain useful typed results, but they are no longer the strongest

statements available for the natural rational-balance family. The unrestricted endpoint is stated explicitly in Chapter 97 and Theorem 98.15; the abstract information-capacity endpoint is Theorem 65.1 and its pushdown specialization. Every lower bound remains attached to the opponent model printed in its theorem.

Appendix D

Computational Validation

Additional validation provenance. The newly integrated binary-hierarchy Pareto and epsilon-free resource chapters come from the RN28 papers. Their general mathematical theorems are justified in the chapter proofs; finite numerical checks play a separate role. In the $B_{3,1}$ case study the reported finite elimination remains expressly conditional on all necessary case-reduction and coverage claims. The original finite-search programs are not part of the publication package; no independent rerun from the PDF alone is claimed.

D.1 Role of computation

The mathematical arguments proved in the chapters form the basis of the monograph. Computational audits serve three narrower purposes:

1. detect transcription errors in long transition tables and closed forms;
2. test boundary cases across large finite parameter ranges;
3. certify a finite bounded elimination when a theorem explicitly has a computer-assisted final step.

The discussion below records the scope and numerical outcome of each audit. Only the mathematical method, finite ranges, and resulting certificates are reported; no implementation-specific identifiers are needed.

D.2 Validation-status terminology

Three statuses are used throughout this appendix.

Independently rerun.
The reported validation was repeated during the monograph audit and the stated outcome was reproduced.

Independently recomputed.
The stated finite mathematical conditions were evaluated afresh from the explicit definitions, and the recorded counts were reproduced.

Reported finite count.
The stated finite count is recorded as supporting evidence but has not been independently reproduced in the present review.

This distinction concerns the provenance of the computational evidence only; it does not alter the symbolic theorem statements.

D.3 Independently rerun audits

Completion/frontier/original-machine audit

The independent audit passes and includes:

check family	count
original-machine binary inputs	524,287
prefix transition checks	8,912,898
independent quotient-distance checks	25,281,628
compact-DPDA runs	1,282,610
parameter formula audits	3,931

The audit checks the original-machine residual invariant, reachable-store structure, completion normalization, shortest-completion formulas, compact realization, and related finite-frontier claims.

Structure/resource audit

The independent audit terminates with

80,717,789 checks – PASS.

Principal reported subcounts are:

check family	count
brute minimum-demand cases	11,011
broad arithmetic cases	1,548,547
compact macro cases	667,110
configuration-rectangle vertices	1,859,269
configuration-rectangle diamonds	1,286,746
phase samples	44,047
raw four-symbol cases	1,111,110
raw four-symbol defects encountered	16,247
one-work-symbol cases	1,111,110
raw three-symbol cases	1,111,110
raw rectangle cases	61,455
raw three-symbol rectangle vertices	731,091
raw three-symbol rectangle diamonds	474,684
ladder paths	11,778

The same audit contains the bounded $(3, 2)$ rule-synthesis certificate recorded in Appendix C.

Exact resource-frontier audit

The independent audit terminates with

79,606,679 checks – PASS.

It independently rechecks the minimum-demand arithmetic, normalized and raw configuration rectangles, raw-conjugacy boundaries, and the state–alphabet resource formulas used in the exact frontier analysis.

Centered real-time family audit

The independent audit covers every ordered coprime pair

$$1 \le p, q \le 20$$

and reports

$$\boxed{255 \text{ ordered coprime ratios}},$$

$$\boxed{1{,}909{,}191 \text{ reached configurations}},$$

and

$$\boxed{3{,}738{,}864 \text{ binary edges}}.$$

The audit checks transition totality on reached binary configurations, the residual invariant, residual-collision consistency, two-cell visibility, and zero-residual correctness. The rerun passes.

D.4 Independent finite-range comparisons

WRITE–SEARCH–CANCEL audit

The WRITE–SEARCH–CANCEL validation study reports

$$\boxed{477{,}033 \text{ assertion-level checks.}}$$

These checks evaluate the stated finite mathematical conditions directly over the ranges below:

- all coprime $1 \le p, q \le 8$ and all binary prefixes of length at most nine for the closed-form minimum-demand formula, semantic carry criterion, and demand clock;
- the bounded canonical one-top direct-realizability classification;
- WSC defect-lattice coordinate formulas;
- count-shadow fiber inequalities;
- physical-factorization reconstruction formulas;
- bounded carry-order debt instances;
- supplementary kernel-fiber identities.

The combined number of checked conditions is:

$$\boxed{477{,}033 \text{ checks} - \text{PASS}.}$$

These are finite consistency comparisons; the symbolic arguments establish the general claims.

Semantic-uniqueness/lift audit

The semantic-uniqueness/lift validation study reports

$$\boxed{458{,}776 \text{ assertion-level checks.}}$$

The same finite method applies to the schedule and causality conditions, including:

- the six admissible and two inadmissible 2:1 schedules;

- pairwise distinct reduced schedule fingerprints;
- the two-maxima/non-lattice schedule-poset structure;
- the directed $A \rightsquigarrow B$ causal certificate;
- the one-kernel excess at the eager B successor;
- the first forbidden recognition witness.

The number of checked conditions is:

458,776 checks – PASS.

The finite checks support the stated symbolic conclusions but do not replace their proofs.

D.5 General-Theory computational support

The General Theory is formulated and proved symbolically. Its verification chapter imports classical decision mechanisms only after reducing correctness to an exact structural target. No external computation is required to establish the main General-Theory proofs.

Finite catalogues used as stress tests, including the 3:2 and odd-p:2 schedule families, are recorded separately in the General-Theory catalogue appendix rather than being treated as substitutes for the abstract theorems.

D.6 Enumerative and arithmetic computational audits

The enumerative and arithmetic results have symbolic proofs as their primary evidence. Their numerical audits are recorded below as source-reported unless an independent rerun is explicitly stated.

Explicit run–return determinants

The source audit reports direct coarea enumeration against the weighted determinant formula and an independent coefficient-free check. The final source record lists 33 sampled-run determinant cases and 86 coefficient-free cases with no mismatches; an earlier full audit record additionally reports an extended 220-case determinant/direct-enumeration pass. These computations audit indexing and signs and are not proof substitutes.

Contact-preserving barrier compression

The source audit reports 88,032 determinant/enumeration comparisons for the general exact-contact theorem. Its rational-path specializations report 755 ordinary fixed-composition instances with 9,983 statistic queries, 628 ratio-run instances with 54,664 exact-return queries, and 501 simultaneous instances with 246,392 queries, all without discrepancy. The fixed-signature audit covers 2,357 classes over 12,081 paths. A larger $(23, 11)$ stress test enumerates all 63,920 fixed-composition paths; 616 paths of signature 11111110 split into four classes of 154 each, matching the determinant.

Farey recursion and spectra

The reviewer-oriented source audit exhaustively checks all 24,064 ordered coprime pairs with $2 \le p, q \le 200$ and adds 30,000 randomized coprime pairs with $p, q \le 5000$ for the unit-lift laws, closed profiles, all four ordered-recursion identities, and the complementary index partitions, with no counterexample. The scalar audit covers all 13,715 coprime ratios with $1 \le p, q \le 150$ through residual shells $k = 0, \dots, 7$, again with no counterexample.

The novelty stress test separately confirms that the local walk is a unimodular image of classical mechanical/Christoffel geometry and therefore keeps the novelty claim on the cross-parameter minimizer recursion.

D.7 Active chronological-source recovery certificates

The active CCR-k observability theorem in Chapter 35 uses both exact symbolic arguments and finite exact-arithmetic audits, which have different logical roles. The difference-set criterion for a fixed pair of pulses and the posterior-set criterion for two adaptive pulses are identities over finite fields; the general assertion for all primes $q \geq 41$ relies on the rational circle-difference lemma and its explicitly tabulated finite interval decomposition (Section 35.7). It is not extrapolated from a finite list of prime moduli.

For the separately specified CCR-10 source contracts, exact checks of 625 source–nuisance trajectories and 256 nonzero slope pairs in $\mathbb{F}_{17}$, and 16,440 full-row two-slot comparisons over $q = 41, 43, 47$, serve as finite consistency certificates. The lower bounds for full-row observation also require the book's expressly stated freedom to choose later high-degree nuisance sources: the triangular completion lemma (Section 35.8) supplies that mathematical bridge. These checks do not certify an unrestricted CCR source model or observability when hidden, unmasked earlier rows are revealed. The standalone paper is cited in the bibliography and its mathematical statement is integrated into the main text, without requiring separate code-file references.

D.8 EF–FG controlled-heat realization: exact-certificate ledger

The independent physical common-realization chapter (Chapter 124) uses global heat-control support identities, positive-semigroup bounds, exact rational series enclosures, and finite connected-region enumeration. Its source states explicit rational Taylor degrees, remainder estimates, contraction cubes, dual multiplier intervals, and exact Krylov minor computations. These certificates and reconstruction steps are retained, rather than replaced by plots or unspecified numerical optimization, in Section 124.10.

The strongest general inference is an exact *within-model* statement: on the seven-vertex rational network the strict joint query holds if and only if $Q > Q_*$, while every proper pair is feasible already at $Q = 100$. On the eight-vertex pair the optimal connected terminal caps are correlated, and a one-percent conductance change induces a Krylov-rank jump and physical-face collapse. The physical plant is an ideal finite weighted heat equation with measurable controls; no laboratory validation is claimed. Independent checks of selected decimal values, exact rational Krylov ranks, and witness signs must not be confused with a new full rerun of every published interval and Banach certificate. The cited chapter explains its exact certificate construction sufficiently for targeted independent reproduction.

D.9 Interpretation rule

Throughout the book, the phrase "validated computationally" must be read according to the hierarchy

symbolic theorem first,	computational evidence second.

The sole exception is an explicitly labeled computer-assisted theorem whose symbolic argument reduces the remaining obligation to a finite exhaustive search. In that case, the finite certificate is part of the proof and its pruning assumptions, survivor vectors, and terminal zero-survivor result must be preserved together.

D.10 Post-theory validation

D.10.1 Fixed-Parikh quantitative LIFO chronology

The fixed-semantics chronology results are symbolic. Computation was used only as an independent transcription and boundary-case audit of the finite physical-lift kernels and small-parameter constants. Direct probability propagation and kernel calculations were compared on overlapping horizons and parameter pairs; the numerical checks do not enter any proof premise.

D.10.2 Optimal-fiber rank-two topology

The rank-two optimal-fiber classification is proved symbolically. Finite-grid computations were used as consistency checks for active-fiber saturation, nerve compression, cellular homology, and the explicit scalar realization families. These calculations serve only as audits of examples and boundary cases.

D.10.3 Unrestricted state complexity

The unrestricted state lower bound is entirely symbolic. During preparation, the matching three-symbol construction was independently sanity-checked on finite ranges of coprime parameters and residuals by verifying the two translation identities for every tested configuration. No computational assertion is used in the proof of the lower bound or exactness theorem.

D.11 Geodesic-compiler validation

Prescribed-order separation and Boolean amplification

The geodesic-compiler and non-factorization theorems are proved symbolically. Independent finite checks use the explicit translation actions in the theorem statements. Exact shortest distances are obtained by reverse breadth-first search and compared with, rather than inferred from, the stated closed distance formulas.

For every $1 \le k \le 10$, the audit verifies the Binary-Deficit Distance Lemma on every pair

$$0 \le D < 2^k, \qquad k \le \ell \le 2k,$$

for a total of

$$\boxed{\text{20,480 binary-deficit instances.}}$$

For the same ten blocks it reconstructs the complete structural/optimal face data and confirms

$$\mathcal{H}(r) = 2^{A_-}, \qquad \mathcal{H}(r') = 2^{A_-} \setminus \{J_\star, A_-\},$$

including exact agreement through order $k-1$ and the unique order-k discrepancy.

A separate product-graph reverse-BFS audit checks

$$(k, m) = (1, 3), (2, 2), (3, 2), (4, 2).$$

In every case all 2^m selected product states have the same exact distance and the same order-$(k-1)$ signature, while the order-k signatures are pairwise distinct and each coordinate toggles exactly its distinguished face $J_\star^{(t)}$. The finite comparison gives

$$\boxed{\text{OVERALL PASS.}}$$

These computations are independent regression evidence only and are not premises of the compiler, non-factorization, or amplification theorems.

D.12 Universal k-fiber validation

The Universal k-Fiber Theorem and exact capacity theorem in Chapter 19 are proved symbolically. As an independent regression check, the explicit two-phase construction was exhaustively evaluated for several small parameter pairs. For every witness state in each tested family, the audit checked the common closed-form distance, reconstructed the complete structural/optimal truncation through order $k-1$, and tested every distinguished k-face against the Bellman equality.

s	k	$\binom{s}{k}$	witnesses	common distance	patterns
2	2	1	2	6	2
3	2	3	8	11	8
4	2	6	64	18	64
5	2	10	1024	27	1024
3	3	1	2	8	2
4	3	4	16	18	16
5	3	10	1024	37	1024
4	4	1	2	10	2
5	4	5	32	27	32

In every tested case, all witness states had one common exact distance and one common complete lower-order compatibility truncation, while the distinguished order-k layer ranged over all $2^{\binom{s}{k}}$ prescribed patterns. The $s = 3, k = 2$ row independently reproduces all eight simple graphs in the worked universal fiber of Chapter 19. These computations are regression evidence only and are not premises of any theorem.

D.13 Finite-horizon and epsilon-free resource-certificate provenance

The canonical calibration was separately checked by a transition-level interpreter on representative (m, k, H) tuples. These finite tests corroborate selected source identities but do not establish the general normalization and compactness theorems, whose mathematical proofs are presented in the main text. The resource decision in Theorem 11.28 is a terminating *conceptual* algorithm; its finite-table search and deterministic-PDA equivalence component may have prohibitive complexity. No explicit uniform small cutoff for the number of ones is claimed.

The finite-refutation format has been independently tested at $(m, k, q, g) = (2, 1, 1, 1)$: 18 tables at $H = 0$, 50 at $H = 1$ and 128 at $H = 2$ were enumerated, and the three-word $H = 2$ packet refutes the final level. The archived G88/G89 two-state/three-letter obstruction for $B_{3,1}$ was replayed successfully together with an independently constructed epsilon-free $(2, 4)$ automaton. The G85–G87 original programs have been re-executed; an independent full G86 constraint scan, a full G88 candidate-generation check, and an independent full G89 zero-table census additionally matched their archival counts and terminal physical witnesses. The exact Pareto consequence nevertheless remains conditional on independent proof-to-code validation of the full G71–G89 case coverage, especially G75/G78, in Section 11.11.8; the G72/G74 mathematical arguments have been reproduced. The outstanding questions include obtaining useful complexity or witness-size bounds, classifying further languages with effective zero-drain calibrations, exhibiting practical finite-horizon certificates for nontrivial (q, g) pairs, and comparing this criterion with related equivalence and descriptional-complexity literature. In particular the epsilon-enabled resource-existence problem and the complete numerical Pareto frontier of $B_{m,k}$ remain open. No statement here conflates an epsilon-free NO certificate with an unrestricted epsilon-DPDA impossibility.

D.14 Temporal relational heat control: finite rational certificates

The EF–FG temporal heat chapter (Chapter 125) contains its own finite rational certificate arrays, exact matrix-exponential enclosure prescription, Farkas multipliers, nominal and crossed-network constructive heater tables, and the geometric tail bound that turns finite history checks into a statement for all word lengths. The chapter does not release executable code or separate proof files. Programs for repeating the finite arithmetic checks can be written directly from its public mathematical specification. Its independently studied measured-building context is not a physical field validation of the exactly prescribed H1/H2 model.

Appendix E

Lagrange–Bürmann Derivation of the Exact Correction Triangle

E.1 The marked first-return grammar

This appendix supplies the coefficient derivation deferred from Chapter 27. Fix an integer $d \geq 1$. Let $F(z,t)$ denote the correction-marked excursion generating function at baseline zero, and let $A(z,t)$ and $B(z,t)$ be the two periodic tail classes introduced by the first-return decomposition. They satisfy

$$\boxed{F = 1 + zA^d F,} \tag{E.1}$$

$$\boxed{A = 1 + zA^d B,} \tag{E.2}$$

$$\boxed{B = 1 + ztA^{d-1}B^2.} \tag{E.3}$$

The variable z marks down-steps and t marks the distinguished correction events.

Define

$$\boxed{Q_d(n,k) = [z^n t^k]F(z,t).}$$

The goal is to derive the closed form for $Q_d(n,k)$ directly from (E.1)–(E.3).

E.2 The auxiliary kernel variable

Set

$$Y = zA^{d-1}B.$$

Equation (E.2) gives

$$A = 1 + AY,$$

so

$$\boxed{A = \frac{1}{1-Y}.} \tag{E.4}$$

Similarly, divide (E.3) by B and use $Y = zA^{d-1}B$:

$$1 = B^{-1} + tY,$$

whence

$$\boxed{B = \frac{1}{1-tY}.} \tag{E.5}$$

Substituting (E.4)–(E.5) into the definition of Y yields

$$Y = \frac{z}{(1-Y)^{d-1}(1-tY)}.$$

Equivalently,

$$\boxed{Y(1-Y)^{d-1}(1-tY) = z.} \tag{E.6}$$

Thus Y is the unique formal power series with zero constant term satisfying

$$\boxed{Y = z\,\Phi(Y), \qquad \Phi(y) = (1-y)^{-(d-1)}(1-ty)^{-1}.} \tag{E.7}$$

This is exactly the form required for Lagrange inversion.

E.3 Coefficient formula for the auxiliary kernel

Define

$$K_d(z,t) = \frac{Y}{z}.$$

Since

$$[z^n t^k]K_d = [z^{n+1}t^k]Y,$$

the basic Lagrange formula applied to (E.7) gives

$$[z^{n+1}t^k]Y = \frac{1}{n+1}[y^n t^k]\Phi(y)^{n+1}.$$

Now

$$\Phi(y)^{n+1} = (1-y)^{-(d-1)(n+1)}(1-ty)^{-(n+1)}.$$

Extracting the t^k coefficient gives

$$[t^k](1-ty)^{-(n+1)} = \binom{n+k}{k} y^k.$$

Therefore

$$[z^n t^k]K_d = \frac{1}{n+1}\binom{n+k}{k}[y^{n-k}](1-y)^{-(d-1)(n+1)}.$$

Using

$$[y^r](1-y)^{-a} = \binom{a+r-1}{r},$$

we obtain

$$\boxed{[z^n t^k]K_d = \frac{1}{n+1}\binom{n+k}{k}\binom{dn+d-k-2}{n-k}.} \tag{E.8}$$

For $d = 2$ this reduces to

$$\boxed{[z^n t^k]K_2 = \frac{1}{n+1}\binom{n+k}{k}\binom{2n-k}{n-k}.} \tag{E.9}$$

This is the auxiliary kernel triangle identified in the main text. It is not the actual correction distribution.

E.4 Recovering the correction generating function

From (E.1),

$$F = \frac{1}{1 - zA^d}.$$

Using (E.4),

$$\boxed{F = \sum_{m \geq 0} z^m A^{dm} = \sum_{m \geq 0} z^m (1 - Y)^{-dm}.} \tag{E.10}$$

This expansion is the convenient starting point for the exact correction triangle.

For fixed $n \geq 1$,

$$Q_d(n, k) = \sum_{m=0}^{n} [z^{n-m} t^k](1 - Y)^{-dm}. \tag{E.11}$$

The term $m = n$ contains no positive power of Y, so it contributes

$$\boxed{\mathbf{1}_{\{k=0\}}.} \tag{E.12}$$

The term $m = 0$ contributes nothing when $n \geq 1$. It remains to treat $1 \leq m \leq n - 1$.

E.5 Lagrange–Bürmann extraction for a fixed summand

Fix $1 \leq m \leq n - 1$ and set

$$N = n - m > 0.$$

Apply Lagrange–Bürmann to

$$f(y) = (1 - y)^{-dm}.$$

Since

$$f'(y) = dm(1 - y)^{-dm-1},$$

we obtain

$$[z^N](1 - Y)^{-dm} = \frac{dm}{N} [y^{N-1}](1 - y)^{-dm-1} \Phi(y)^N.$$

Substitute (E.7):

$$[z^N](1 - Y)^{-dm} = \frac{dm}{N} [y^{N-1}](1 - y)^{-dm-1-(d-1)N} (1 - ty)^{-N}. \tag{E.13}$$

Because $N = n - m$,

$$dm + 1 + (d - 1)N = (d - 1)n + m + 1.$$

Thus

$$[z^N t^k](1 - Y)^{-dm} = \frac{dm}{N} [y^{N-1} t^k](1 - y)^{-((d-1)n+m+1)} (1 - ty)^{-N}.$$

Now

$$[t^k](1 - ty)^{-N} = \binom{N + k - 1}{k} y^k,$$

so

$$[z^N t^k](1 - Y)^{-dm} = \frac{dm}{N} \binom{N + k - 1}{k} [y^{N-k-1}](1 - y)^{-((d-1)n+m+1)}.$$

A final binomial extraction gives

$$[y^{N-k-1}](1 - y)^{-((d-1)n+m+1)} = \binom{dn - k - 1}{N - k - 1}.$$

Hence

$$\boxed{[z^{n-m}t^k](1-Y)^{-dm} = \frac{dm}{n-m}\binom{n-m+k-1}{k}\binom{dn-k-1}{n-m-k-1}.} \tag{E.14}$$

The coefficient is automatically zero unless

$$n - m \geq k + 1.$$

E.6 Exact correction triangle

Insert (E.12) and (E.14) into (E.11). Make the change of variable

$$j = n - m.$$

Then

$$m = n - j,$$

and the nonzero range is

$$k + 1 \leq j \leq n - 1.$$

Therefore

$$\boxed{Q_d(n,k) = \mathbf{1}_{\{k=0\}} + \sum_{j=k+1}^{n-1} \frac{d(n-j)}{j}\binom{j+k-1}{k}\binom{dn-k-1}{j-k-1}.} \tag{E.15}$$

This is the formula stated in Chapter 27.

The upper support bound follows immediately. If $k > n - 2$, then there is no integer j with

$$k + 1 \leq j \leq n - 1,$$

and the indicator term vanishes because $k > 0$. Thus

$$\boxed{Q_d(n,k) = 0 \qquad (k > n - 2).} \tag{E.16}$$

E.7 Equivalent elimination through $U = A - 1$

For completeness, the scalar equation quoted in the main chapter follows directly from the same grammar. Put

$$U = A - 1.$$

From (E.2),

$$U = zA^dB = AY.$$

Since $A = 1 + U$, this gives

$$Y = \frac{U}{1+U}.$$

Equation (E.3) can be written as

$$B = 1 + tYB,$$

so

$$B = \frac{1}{1-tY} = \frac{1+U}{1+(1-t)U}.$$

Substituting into $U = zA^dB$ with $A = 1 + U$ gives the convenient form

$$\boxed{U\big(1 + (1-t)U\big) = z(1+U)^{d+1}.} \tag{E.17}$$

Equation (E.17) and the kernel equation (E.6) are related by

$$\boxed{Y = \frac{U}{1+U}.}$$

Thus the two inversion routes are exactly the same formal system in different coordinates.

E.8 Row sum

Setting $t = 1$ forgets the correction marking. Equation (E.17) becomes

$$U = z(1 + U)^{d+1}.$$

At $t = 1$, the correction generating function simplifies to

$$F(z, 1) = 1 + U.$$

Lagrange inversion gives, for $n \geq 1$,

$$[z^n]U = \frac{1}{n}[u^{n-1}](1 + u)^{(d+1)n} = \frac{1}{n}\binom{(d+1)n}{n-1}.$$

Using

$$\frac{1}{n}\binom{(d+1)n}{n-1} = \frac{1}{dn+1}\binom{(d+1)n}{n},$$

we recover the Fuss–Catalan row sum

$$\boxed{\sum_{k\geq 0} Q_d(n, k) = \frac{1}{dn+1}\binom{(d+1)n}{n}.} \tag{E.18}$$

E.9 Why the kernel and correction arrays differ

The kernel coefficients (E.8) arise from the single implicit series Y and record the algebraic kernel introduced to solve the periodic grammar. The correction coefficients (E.15) arise only after the full zero-baseline excursion series

$$F = \sum_{m\geq 0} z^m (1 - Y)^{-dm}$$

is reconstructed.

Therefore

$$\boxed{K_d \text{ is an algebraic waypoint, whereas } Q_d \text{ is the path statistic.}}$$

The distinction is not terminological: the two coefficient arrays have different combinatorial meanings and diverge as soon as the correction marking becomes nontrivial.

Appendix F

General-Theory Stress-Test Catalogues

F.1 Purpose and scope

The abstract General Theory developed in the General Theory macro-Part is not defined by the weighted-balance family. Its semantic system may be non-additive and need not have a residual space isomorphic to $\mathbb{Z}$. The centered $p{:}q$ constructions are instead a deliberately rigid stress-test family in which safe-reset domains, source-fiber expansion, directed exposure, and robust fatal supports can all be calculated exactly.

This appendix records the full finite 3:2 certificate and the infinite odd-p:2 arithmetic/source catalogues used in the stress-test theorems. The purpose is reproducibility: the main chapters can state the structural consequences without interrupting the narrative with long case catalogues.

Throughout this appendix, for a canonical literal-demand pair (x, y) we use

$$\eta(x,y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & y > x. \end{cases}$$

For the weighted-balance residual, the primitive nonnegative kernel vector is $K = (q, p)$.

F.2 The exact centered 3:2 certificate

Set

$$p = 3, \qquad q = 2, \qquad m = p + q = 5,$$

and use the centered states

$$D_{-2}, D_{-1}, \mathsf{C}, D_1, D_2.$$

The six natural eager sites are indexed by the current centered state, the next input symbol, and the visible top symbol. Their eager replacement words are

site	(d, a, X)	γ
A	$(-2, 0, 0)$	11111
B	$(-2, 1, 0)$	10111
C	$(-2, 1, 1)$	11111
D	$(1, 0, 0)$	10100
E	$(1, 1, 0)$	10000
F	$(2, 0, 0)$	10000

(F.1)

F.2.1 Finite reset repertoires

Define

$$\boxed{\begin{aligned}\mathcal{R}_0^{(3,2)} = {} & \{(2,2)\} \cup \{(0,y) : y \geq 4\} \cup \{(1,y) : y \geq 3\} \\ & \cup \{(x,1) : x \geq 3\} \cup \{(x,0) : x \geq 4\}\end{aligned}} \tag{F.2}$$

and

$$\boxed{\mathcal{R}_D^{(3,2)} = \mathcal{R}_0^{(3,2)} \cup \{(x,2) : x \geq 3\}.} \tag{F.3}$$

Before the first reset event of an episode begun from one of these canonical pairs, the relevant eager-source shapes are

local context	$\mathcal{R}_0^{(3,2)}$	$\mathcal{R}_D^{(3,2)}$
D_{-2}, top 0	0	0^k
D_{-2}, top 1	1^k	1^k
D_1, top 0	0^k	0^k
D_2, top 0	0^k	0^k

$(k \geq 1).$ (F.4)

The bottom-WRITE targets are

d	0	1
-2	$(1,8)$	$(0,4)$
-1	$(0,4)$	$(1,3)$
0	$(1,3)$	$(2,2)$
1	$(2,2)$	$(3,1)$
2	$(3,1)$	$(4,0)$

(F.5)

and every one lies in $\mathcal{R}_0^{(3,2)}$.

Writing the unary source tail as $t \geq 0$, the remaining reset targets are

reset action	canonical target pair
positive boundary WRITE	$(t+5, 0)$
negative boundary WRITE	$(1, t+9)$
A	$(0,5)$
B	$(1,4)$
C	$(0, t+5)$
D	$(t+3, 2)$
E	$(t+4, 1)$
F	$(t+4, 1)$

(F.6)

These tables give two closed reset regimes:

(a) with D disabled, every baseline reset and every enabled eager action among A, B, C, E, F returns to $\mathcal{R}_0^{(3,2)}$;

(b) with D enabled and A, B disabled, every baseline reset and every enabled eager action among C, D, E, F returns to $\mathcal{R}_D^{(3,2)}$.

The only new reset ray in the second regime is

$$\{(x,2) : x \geq 3\},$$

and this is exactly the ray that enlarges the D_{-2}-top-0 source from the singleton stack 0 to the unary family 0^k.

F.2.2 Direct verification of reset closure

The canonical words in $\mathcal{R}_0^{(3,2)}$ are respectively

$$1010, \qquad 1^y, \qquad 10\,1^{y-1}, \qquad 1\,0^x, \qquad 0^x,$$

while the added D-ray has

$$\eta(x,2) = 101\,0^{x-1}.$$

Before the first reset, MATCH and interior SEARCH only remove the visible top cell, so the current stack is always a suffix of the starting canonical word. The source catalogue in (F.4) follows immediately. In particular, without the D-ray only the word 1010 can reach D_{-2} while a zero remains visible, and exactly its final zero remains.

For the non-bottom reset events, the centered boundary blocks and the six eager replacement words yield the canonical words

$$0^{t+5}, \quad 10\,1^{t+8}, \quad 1^5, \quad 10\,1^3, \quad 1^{t+5}, \quad 101\,0^{t+2}, \quad 1\,0^{t+4}.$$

These are precisely the encodings listed in (F.6). Every row except D lies in $\mathcal{R}_0^{(3,2)}$; the D row is precisely the additional ray of $\mathcal{R}_D^{(3,2)}$.

F.2.3 Exact schedule classification and fatal witnesses

The admissible schedules are exactly

$$\boxed{\mathrm{Adm}_{3,2} = \{H : \{A,D\} \not\subseteq H,\ \{B,D\} \not\subseteq H\}.} \tag{F.7}$$

Therefore

$$\boxed{\text{40 schedules are admissible and 24 are inadmissible.}} \tag{F.8}$$

The minimal directed fatal fork is

$$\boxed{D \to A, \qquad D \to B.} \tag{F.9}$$

The maximal admissible faces are

$$\{A,B,C,E,F\}, \qquad \{C,D,E,F\},$$

so the admissible complex is non-pure and the schedule poset is not a lattice. Its independence polynomial is

$$\boxed{(1+x)^3(1+3x+x^2).} \tag{F.10}$$

Sufficiency follows from the two reset-closed regimes above: the $D = 0$ regime contributes $2^5 = 32$ schedules, while $D = 1$ with $A = B = 0$ contributes $2^3 = 8$ more.

Necessity is certified by two complete balanced words,

$$\boxed{w_B = 1111000011}, \qquad \boxed{w_A = 011111110000011}. \tag{F.11}$$

Their critical portions are

branch	reaching prefix	victim step	diagnostic
$D \to B$	$11110000 : (D_{-2}, 00)$	$1 : (\mathsf{C}, 101110)$	$1 : (\mathsf{C}, 01110)$
$D \to A$	$011111110000 : (D_{-2}, 000)$	$0 : (\mathsf{C}, 1111100)$	$11 : (\mathsf{C}, 11100)$

(F.12)

In the first row, the earlier eager D step occurs on prefix 11110; in the second it occurs on prefix 011111110. Before the victim step the only eager optional site encountered is therefore D. After the victim step the listed diagnostic uses only baseline center transitions. Both terminal configurations have semantic residual zero but a nonempty center stack, so they are nonaccepting under the fixed centered acceptance convention. The witnesses are therefore extension-stable under arbitrary choices at C, E, F and at the other victim site.

F.3 The infinite odd-p:2 family

Fix

$$\boxed{p = 2h - 1, \qquad q = 2, \qquad h \geq 2,} \tag{F.13}$$

so that

$$m = p + q = 2h + 1.$$

For $t \geq 1$ set

$$\epsilon_t = t \bmod 2, \qquad k_t = \left\lceil \frac{t}{2} \right\rceil, \qquad \beta_t = t + p\, k_t,$$

and define

$$\boxed{\Gamma_t = \eta(\epsilon_t, \beta_t).} \tag{F.14}$$

The six eager sites are

$$\begin{aligned}
&A : (D_{-h}, 0T) \xrightarrow{0} (\mathsf{C}, \Gamma_h T),\\
&B : (D_{-h}, 0T) \xrightarrow{1} (\mathsf{C}, \Gamma_{h-1} T),\\
&C : (D_{-h}, 1T) \xrightarrow{1} (\mathsf{C}, \Gamma_h T),\\
&D : (D_{h-1}, 0T) \xrightarrow{0} (\mathsf{C}, (10)^h 0T),\\
&E : (D_{h-1}, 0T) \xrightarrow{1} (\mathsf{C}, (10)^{h-1} 000T),\\
&F : (D_h, 0T) \xrightarrow{0} (\mathsf{C}, (10)^{h-1} 000T).
\end{aligned} \tag{F.15}$$

F.3.1 Arithmetic catalogue

For the minimum-demand map μ the exact identities are:

(i) for every $t \geq 1$,

$$\boxed{\mu(-tm) = (\epsilon_t, \beta_t);} \tag{F.16}$$

(ii) for $0 \leq j \leq h$,

$$\boxed{\mu(jm + 2) = (j + 2, p - j - 1);} \tag{F.17}$$

(iii) for $1 \leq t \leq h + 1$,

$$\boxed{\mu(-tm + 2) = (\epsilon_t, \beta_t - 1);} \tag{F.18}$$

(iv) at the positive outer boundary,

$$\boxed{\mu((h + 1)m) = (h + 3, h - 2).} \tag{F.19}$$

For completeness, if $t = 2r$ then $(0, \beta_t)$ has weight

$$-2(2r + pr) = -2r(p + 2) = -tm,$$

while if $t = 2r + 1$ then $(1, \beta_t)$ has weight

$$p - 2(2r + 1 + p(r + 1)) = -(2r + 1)(p + 2) = -tm.$$

In both cases the first coordinate is below the kernel step $q = 2$, so the pair is kernel-reduced. Equation (F.17) follows from

$$p(j + 2) - 2(p - j - 1) = j(p + 2) + 2 = jm + 2,$$

with second coordinate below p. Equation (F.18) is obtained by lowering the second coordinate in (F.16) by one, and (F.19) follows from

$$p(h + 3) - 2(h - 2) = (h + 1)(2h + 1).$$

The complete family of canonical baseline reset pairs is therefore

$$\boxed{\begin{aligned}\mathcal{R}_h^{\mathrm{base}} = {} & \{\mu(jm+2) : -h-1 \le j \le h\} \\ & \cup \{(x, h-2) : x \ge h+3\} \\ & \cup \{(\epsilon_{h+1}, y) : y \ge \beta_{h+1}\}.\end{aligned}} \tag{F.20}$$

The two schedule-regime repertoires are

$$\boxed{\begin{aligned}\mathcal{R}_{\neg D,h} = \mathcal{R}_h^{\mathrm{base}} & \cup \{(\epsilon_h, y) : y \ge \beta_h\} \\ & \cup \{(\epsilon_{h-1}, \beta_{h-1})\} \cup \{(x, h-1) : x \ge h+2\},\end{aligned}} \tag{F.21}$$

and

$$\boxed{\begin{aligned}\mathcal{R}_{D,h} = \mathcal{R}_h^{\mathrm{base}} & \cup \{(\epsilon_h, y) : y \ge \beta_h\} \\ & \cup \{(x, h-1) : x \ge h+2\} \cup \{(x, h) : x \ge h+1\}.\end{aligned}} \tag{F.22}$$

Every pair in either repertoire has minority size at most h.

F.3.2 Source-geometry catalogue

Consider an episode beginning at a canonical center pair in one of the two repertoires and stopping at its first reset event. The exact source geometry is:

(i) in $\mathcal{R}_{\neg D,h}$,

$$D_{-h} \text{ with top } 0 \Longrightarrow \boxed{\text{complete remaining stack } 0}; \tag{F.23}$$

(ii) in $\mathcal{R}_{D,h}$,

$$D_{-h} \text{ with top } 0 \Longrightarrow \boxed{\text{complete remaining stack } 0^n,\ n \ge 1}, \tag{F.24}$$

and lengths $n \ge 2$ arise only from the additional D-ray $\{(x, h) : x \ge h+1\}$;

(iii) in either repertoire,

$$\begin{aligned} D_{-h} \text{ with top } 1 &\Longrightarrow 1^n, \\ D_{h-1} \text{ with top } 0 &\Longrightarrow 0^n, \\ D_h \text{ with top } 0 &\Longrightarrow 0^n \end{aligned} \tag{F.25}$$

for some $n \ge 1$.

The key observation is that, before the first reset event, MATCH and interior SEARCH only delete the visible top cell, so the current stack is a suffix of the canonical reset word from which the episode began. Reaching D_{-h} requires at least h more negative than positive SEARCHes. In $\mathcal{R}_{\neg D,h}$ the only canonical pair with enough of both symbols to leave a zero visible at that boundary is (h, h), whose word is $(10)^h$; all h one-cells must have been negative SEARCHes, leaving exactly its final zero. The only new family in $\mathcal{R}_{D,h}$ that can leave a longer zero suffix is (x, h), whose word is $(10)^h 0^{x-h}$.

The remaining unary-source conclusions follow from the same minority-size bound. A suffix containing the opposite symbol beyond the visible boundary cell would force minority size at least $h+1$, contradicting the defining families in (F.21)–(F.22).

F.3.3 Reset-target catalogue and closure

Writing the unary source stack length as $n \geq 1$, all eager reset targets are

site	source stack	canonical target pair
A	0	(ϵ_h, β_h)
B	0	$(\epsilon_{h-1}, \beta_{h-1})$
C	1^n	$(\epsilon_h, \beta_h + n - 1)$
D	0^n	$(n + h, h)$
E	0^n	$(n + h + 1, h - 1)$
F	0^n	$(n + h + 1, h - 1)$

(F.26)

The identities are literal: for instance

$$(10)^h 0\, 0^{n-1} = \eta(n + h, h)$$

and

$$(10)^{h-1}000\, 0^{n-1} = \eta(n + h + 1, h - 1).$$

Consequently:

(a) if D is disabled, baseline resets and enabled actions among A, B, C, E, F preserve $\mathcal{R}_{\neg D,h}$;

(b) if D is enabled and A, B are disabled, baseline resets and enabled actions among C, D, E, F preserve $\mathcal{R}_{D,h}$.

F.3.4 Uniform D-induced source-fiber expansion

For every $x \geq h + 1$,

$$\boxed{(\mathsf{C}, \eta(x, h)) \xrightarrow{0^{2h-1}} (D_{-h}, 0^{x-h+1}).} \tag{F.27}$$

Indeed,

$$\eta(x, h) = (10)^h 0^{x-h}.$$

The first $2h-2$ zeros remove $h-1$ complete 10 pairs, and the final zero SEARCHes through the last 1, leaving the last alternating zero together with the zero tail. Thus enabling D expands the common A/B source fiber from

$$\boxed{\{0\}}$$

to

$$\boxed{\{0, 00, 000, \ldots\}}. \tag{F.28}$$

F.3.5 Uniform reachability of the length-three frontier

There is, for every $h \geq 2$, a word u_h that reaches

$$\boxed{(D_{-h}, 000)} \tag{F.29}$$

from the initial centered configuration and encounters no eager optional site other than D.

Define

$$v_h = \begin{cases} 1^{(h-1)m/2}, & h \text{ odd}, \\ 0\, 1^{\,2h+(h-2)m/2}, & h \text{ even}. \end{cases} \tag{F.30}$$

Then

$$\boxed{u_h = v_h\, 1^{\,2h-1}\, 0^{\,2h}.} \tag{F.31}$$

The useful increment block is

$$(D_d, \varepsilon) \xrightarrow{1^m} (D_{d+2}, \varepsilon), \qquad 0 \le d \le h-2. \tag{F.32}$$

The first symbol bottom-WRITEs

$$\mu(dm+2) = (d+2, p-d-1),$$

whose canonical word has length $p+1 = 2h$; the remaining $2h$ ones erase the word, and its $d+2$ zeros contribute the required positive SEARCHes. This builds v_h so that the machine reaches (D_{h-1}, ε). The next 1^{2h-1} segment leaves $(D_{h-1}, 00)$, the following zero invokes D and produces $(\mathsf{C}, (10)^h 00)$, and the remaining 0^{2h-1} reaches $(D_{-h}, 000)$. No site among A, B, C, E, F is encountered before that endpoint.

F.3.6 Uniform kernel-suffix fatality

For every $t \ge 1$, Γ_t has suffix 1^p. Hence there is a unique word z_t such that

$$\boxed{\Gamma_t = z_t 1^p.} \tag{F.33}$$

If $t = 2r$ then

$$\Gamma_t = 1^{\beta_t}, \qquad \beta_t = rm \ge m > p,$$

while if $t = 2r+1$ then

$$\Gamma_t = 10\, 1^{\beta_t - 1}, \qquad \beta_t - 1 = p + rm \ge p.$$

Thus the suffix claim is uniform in t.

At the common source $(D_{-h}, 000)$, eager A and B yield

$$A : (D_{-h}, 000) \xrightarrow{0} (\mathsf{C}, \Gamma_h 00) = (\mathsf{C}, z_h 1^p 00),$$

$$B : (D_{-h}, 000) \xrightarrow{1} (\mathsf{C}, \Gamma_{h-1} 00) = (\mathsf{C}, z_{h-1} 1^p 00).$$

Reading z_h or z_{h-1} from the center uses only MATCHes and reaches

$$\boxed{(\mathsf{C}, 1^p 00).} \tag{F.34}$$

The stack has two zeros and p ones, hence weight

$$p \cdot 2 - 2 \cdot p = 0,$$

but is nonempty. Under centered acceptance it is therefore a site-free fatal diagnostic.

F.3.7 Exact odd-p:2 classification

For every odd $p \ge 3$,

$$\boxed{\mathrm{Adm}_{p,2} = \mathrm{Ind}\big\{\{A, D\}, \{B, D\}\big\}.} \tag{F.35}$$

Equivalently,

$$\boxed{H \in \mathrm{Adm}_{p,2} \iff \neg(A \wedge D) \wedge \neg(B \wedge D).} \tag{F.36}$$

Every member of the family therefore has exactly 40 admissible and 24 inadmissible schedules, with facets

$$\{A, B, C, E, F\}, \qquad \{C, D, E, F\},$$

and the same fatal fork

$$\boxed{D \to A, \qquad D \to B.} \tag{F.37}$$

Sufficiency is the reset-closure induction: $D = 0$ gives $2^5 = 32$ schedules, and $D = 1$ with $A = B = 0$ gives another $2^3 = 8$. Necessity uses the common reaching prefix u_h from (F.31) together with the complete counterexample words

$$\boxed{w_A^{(h)} = u_h\, 0\, z_h,} \qquad \boxed{w_B^{(h)} = u_h\, 1\, z_{h-1}.} \tag{F.38}$$

The prefix u_h encounters no optional eager site except D, and the post-victim suffixes are site-free. The two forbidden supports are therefore robust under arbitrary additions of C, E, F and of the other victim site.

F.4 Interpretive boundary

The finite 3:2 calculation and the infinite odd-p:2 family have a single role in the monograph: they certify that the abstract mechanisms of the General Theory macro-Part are not artifacts of one numerical instance. They exhibit, in exact families,

$$\begin{array}{c}\text{safe reset} \longrightarrow \text{fiber expansion} \\ \longrightarrow \text{directed exposure} \longrightarrow \text{robust fatal support.}\end{array}$$

They do *not* define the General Theory. The General Theory applies to a rooted exact physical lift over a deterministic partial semantic system. Its settled macro-updates reflect defined domains exactly and satisfy

$$\Theta\delta_a = U_a\Theta$$

on that common domain, with no requirement that the semantic system be a weighted-balance residual space or even an additive group.

Bibliography

[1] A. Adelberg and T. Lengyel, New results on the p-adic valuation of Stirling numbers, *Integers* **22** (2022), Paper A82, `https://math.colgate.edu/~integers/w82/w82.pdf`.

[2] R. Alur, V. Kumar, P. Madhusudan, and M. Viswanathan, Congruences for visibly pushdown languages, in *ICALP 2005*, Lecture Notes in Computer Science, vol. 3580, Springer, 2005, pp. 1102–1114. doi:10.1007/11523468_89.

[3] J. Autry, P. Graves, J. Loucks, C. O'Neill, V. Ponomarenko, and S. Yih. Squarefree divisor complexes of certain numerical semigroup elements. *Involve*, 14(1):1–9, 2021.

[4] A. Bouajjani, J. Esparza, and O. Maler, Reachability analysis of pushdown automata: Application to model-checking, in *CONCUR 1997: Concurrency Theory*, Lecture Notes in Computer Science, vol. 1243, Springer, 1997, pp. 135–150. doi:10.1007/3-540-63141-0_10.

[5] A. Carayol and O. Serre. Marking shortest paths on pushdown graphs does not preserve MSO decidability. *Information Processing Letters*, 116(10):638–643, 2016. doi:10.1016/j.ipl.2016.04.015.

[6] D. Chistikov, W. Czerwiński, P. Hofman, M. Pilipczuk, and M. Wehar. Shortest paths in one-counter systems. *Logical Methods in Computer Science*, 15(1):19:1–19:28, 2019. doi:10.23638/LMCS-15(1:19)2019.

[7] P. Diaconis and J. Fulman, Carries, shuffling, and symmetric functions, arXiv:0902.0179 [math.PR], 2009.

[8] O. Gauwin, A. Muscholl, and M. Raskin. Minimization of visibly pushdown automata is NP-complete. *Logical Methods in Computer Science*, 16(1):14, 2020. doi:10.23638/LMCS-16(1:14)2020.

[9] J. Goldstine, M. Kappes, C. M. Kintala, H. Leung, A. Malcher, and D. Wotschke. Descriptional complexity of machines with limited resources. *Journal of Universal Computer Science*, 8(2):193–234, 2002. DOI: 10.3217/jucs-008-02-0193.

[10] J. Goldstine, J. K. Price, and D. Wotschke. On reducing the number of stack symbols in a PDA. *Mathematical Systems Theory*, 26:313–326, 1993. DOI: 10.1007/BF01189852.

[11] A. Hatcher. *Algebraic Topology*. Cambridge University Press, 2002.

[12] S. Hong and M. Qiu, On the p-adic properties of Stirling numbers of the first kind, *Acta Mathematica Hungarica* 161 (2020), 366–395. doi:10.1007/s10474-020-01037-2.

[13] P. Jančar. A short decidability proof for DPDA language equivalence via first-order grammars. CoRR abs/1010.4760, 2011.

[14] D. Kozlov. *Combinatorial Algebraic Topology*. Algorithms and Computation in Mathematics, vol. 21, Springer, 2008.

[15] O. Martynova. Exact descriptional complexity of determinization of input-driven pushdown automata. In *Implementation and Application of Automata (CIAA 2024)*, pp. 249–260, 2024. arXiv:2404.10516.

[16] T. Masopust. A note on limited pushdown alphabets in stateless deterministic pushdown automata. *International Journal of Foundations of Computer Science*, 24(3):319–328, 2013. DOI: 10.1142/S0129054113500068; arXiv:1208.5002.

[17] A. Malcher, K. Meckel, C. Mereghetti, and B. Palano. Descriptional complexity of pushdown store languages. *Journal of Automata, Languages and Combinatorics*, 17(2–4):225–244, 2012. DOI: 10.25596/jalc-2012-225.

[18] D. A. Moj, Biorthogonal eigenvectors of the Holte carry matrix and cascade-free enumeration, arXiv:2604.04591 [math.CO], 2026.

[19] The OEIS Foundation, The On-Line Encyclopedia of Integer Sequences, A371395, triangle $T(n,k) = \binom{n+k}{k}\binom{2n-k}{n-k}/(n+1)$, accessed August 2026.

[20] G. Pighizzini. Deterministic pushdown automata and unary languages. *International Journal of Foundations of Computer Science*, 20(4):629–645, 2009. doi:10.1142/S0129054109006784.

[21] G. Sénizergues. $L(A) = L(B)$? Decidability results from complete formal systems. *Theoretical Computer Science*, 251(1–2):1–166, 2001. doi:10.1016/S0304-3975(00)00285-1.

[22] D. Armstrong, N. A. Loehr, and G. S. Warrington, Rational parking functions and Catalan numbers, *Annals of Combinatorics* 20 (2016), 21–58. doi:10.1007/s00026-015-0293-6.

[23] L. Dai, S. Fu, and D. Qiu, Symmetric statistics on rational Dyck paths, *Advances in Applied Mathematics* 177 (2026), 103066. doi:10.1016/j.aam.2026.103066; arXiv:2503.20155.

[24] I. Gessel and G. Viennot, Binomial determinants, paths, and hook length formulae, *Advances in Mathematics* 58 (1985), no. 3, 300–321. doi:10.1016/0001-8708(85)90121-5.

[25] G. Kreweras, Sur une classe de problèmes de dénombrement liés au treillis des partitions des entiers, *Cahiers du Bureau universitaire de recherche opérationnelle, Série Recherche* 6 (1965), 9–107.

[26] R. Pemantle and H. S. Wilf, Counting nondecreasing integer sequences that lie below a barrier, *Electronic Journal of Combinatorics* 16 (2009), no. 1, Research Paper R60. doi:10.37236/149.

[27] J. Berstel, A. Lauve, C. Reutenauer, and F. V. Saliola, *Combinatorics on Words: Christoffel Words and Repetitions in Words*, CRM Monograph Series, vol. 27, American Mathematical Society, 2009.

[28] V. Berthé, A. de Luca, and C. Reutenauer, On an involution of Christoffel words and Sturmian morphisms, *European Journal of Combinatorics* 29 (2008), no. 2, 535–553. doi:10.1016/j.ejc.2007.03.001.

[29] F. de Vieilleville and J.-O. Lachaud, Revisiting digital straight segment recognition, in *Discrete Geometry for Computer Imagery (DGCI 2006)*, LNCS 4245, Springer, 2006, pp. 355–366. doi:10.1007/11907350_30.

[30] R. Graham and K. O'Bryant, A discrete Fourier kernel and Fraenkel's tiling conjecture, *Acta Arithmetica* 118 (2005), no. 3, 283–304. doi:10.4064/aa118-3-4.

[31] H. Rademacher and E. Grosswald, *Dedekind Sums*, Carus Mathematical Monographs, vol. 16, Mathematical Association of America, 1972.

[32] R. Klette and A. Rosenfeld, Digital straightness—a review, *Discrete Applied Mathematics* 139 (2004), no. 1–3, 197–230. doi:10.1016/j.dam.2002.12.001.

[33] J.-O. Lachaud and M. Said, Two efficient algorithms for computing the characteristics of a subsegment of a digital straight line, *Discrete Applied Mathematics* 161 (2013), no. 15, 2293–2315. doi:10.1016/j.dam.2012.08.038.

[34] L. Takács, On generalized Dedekind sums, *Journal of Number Theory* 11 (1979), no. 2, 264–272. doi:10.1016/0022-314X(79)90044-1.

[35] L. Takács. On the fluctuations of election returns. *Journal of Applied Probability*, 7(1):114–123, 1970. doi:10.2307/3212153.

[36] F. B. Knight. The uniform law for exchangeable and Lévy process bridges. In *Hommage à P. A. Meyer et J. Neveu*, *Astérisque* 236, pages 171–188, 1996. doi:10.24033/ast.337.

[37] P. J. Fitzsimmons and R. K. Getoor. Occupation time distributions for Lévy bridges and excursions. *Stochastic Processes and their Applications*, 58(1):73–89, 1995. doi:10.1016/0304-4149(95)00013-W.

[38] C. Banderier and M. Wallner. Local time for lattice paths and the associated limit laws. In *Proceedings of GASCom 2018*, pages 69–78, 2018. arXiv:1805.09065.

[39] G. Latouche and V. Ramaswami. *Introduction to Matrix Analytic Methods in Stochastic Modeling*. ASA-SIAM Series on Statistics and Applied Probability, SIAM, Philadelphia, 1999.

[40] P. Flajolet and R. Sedgewick. *Analytic Combinatorics*. Cambridge University Press, Cambridge, 2009.

[41] N. Alon, G. Kalai, J. Matoušek, and R. Meshulam, Transversal numbers for hypergraphs arising in geometry, *Advances in Applied Mathematics* **29** (2002), no. 1, 79–101. DOI: 10.1016/S0196-8858(02)00003-9.

[42] I. Peeva and B. Sturmfels, Syzygies of codimension 2 lattice ideals, *Mathematische Zeitschrift* **229** (1998), no. 1, 163–194. DOI: 10.1007/PL00004645.

[43] R. G. Swan, Projective modules over Laurent polynomial rings, *Transactions of the American Mathematical Society* **237** (1978), 111–120. DOI: 10.1090/S0002-9947-1978-0469906-4.

[44] D. Bayer and B. Sturmfels, Cellular resolutions of monomial modules, *Journal für die reine und angewandte Mathematik* **502** (1998), 123–140.

[45] T. Fernós, D. Futer, and M. Hagen, Homotopy equivalent boundaries of cube complexes, *Geometriae Dedicata* **218** (2024), article 33. DOI: 10.1007/s10711-023-00877-w.

[46] J. R. Norris. *Markov Chains*. Cambridge University Press, Cambridge, 1997.

[47] T. Tao and V. Vu, *Additive Combinatorics*, Cambridge Studies in Advanced Mathematics, vol. 105, Cambridge University Press, Cambridge, 2006.

[48] R. P. Stanley, *Enumerative Combinatorics, Vol. 1*, 2nd ed., Cambridge University Press, 2012.

[49] L. E. Dickson, "Finiteness of the odd perfect and primitive abundant numbers with n distinct prime factors," *American Journal of Mathematics* 35 (1913), 413–422.

[50] A. Asinowski, A. Bacher, C. Banderier, and B. Gittenberger, *Analytic Combinatorics of Lattice Paths with Forbidden Patterns, the Vectorial Kernel Method, and Generating Functions for Pushdown Automata*, Algorithmica 82 (2020), 386–428. DOI: 10.1007/s00453-019-00623-3.

[51] D. Armstrong, B. Rhoades, and N. Williams, *Rational Associahedra and Noncrossing Partitions*, Electronic Journal of Combinatorics 20(3) (2013), P54. DOI: 10.37236/3432.

[52] C. Banderier and P. Flajolet, *Basic analytic combinatorics of directed lattice paths*, Theoretical Computer Science 281 (2002), 37–80. DOI: 10.1016/S0304-3975(02)00007-5.

[53] D. Birmajer, J. B. Gil, and M. D. Weiner, *On rational Dyck paths and the enumeration of factor-free Dyck words*, Discrete Applied Mathematics 244 (2018), 36–43. DOI: 10.1016/j.dam.2018.02.020.

[54] M. T. L. Bizley, *Derivation of a new formula for the number of minimal lattice paths from $(0,0)$ to (km,kn) having just t contacts with the line $my = nx$ and having no points above this line; and a proof of Grossman's formula for the number of paths which may touch but do not rise above this line*, Journal of the Institute of Actuaries 80(1) (1954), 55–62. DOI: 10.1017/S002026810005424X.

[55] J. E. Bonin, A. de Mier, and M. Noy, *Lattice path matroids: enumerative aspects and Tutte polynomials*, Journal of Combinatorial Theory, Series A 104(1) (2003), 63–94. DOI: 10.1016/S0097-3165(03)00122-5.

[56] P. Brändén and J. Huh, *Lorentzian polynomials*, Annals of Mathematics 192(3) (2020), 821–891. DOI: 10.4007/annals.2020.192.3.4.

[57] P. Duchon, *On the enumeration and generation of generalized Dyck words*, Discrete Mathematics 225 (2000), 121–135. DOI: 10.1016/S0012-365X(00)00150-3.

[58] M. Latteux, *Langages à un compteur*, Journal of Computer and System Sciences 26(1) (1983), 14–33. DOI: 10.1016/0022-0000(83)90018-1.

[59] G. N. Raney, *Functional composition patterns and power series reversion*, Transactions of the American Mathematical Society 94 (1960), 441–451.

[60] B. Lindström, *On the Vector Representations of Induced Matroids*, Bulletin of the London Mathematical Society 5(1) (1973), 85–90. DOI: 10.1112/blms/5.1.85.

[61] F. Baader and T. Nipkow, *Term Rewriting and All That*, Cambridge University Press, 1998.

[62] M. H. A. Newman, "On theories with a combinatorial definition of equivalence," *Annals of Mathematics* 43 (1942), 223–243.

[63] D. Quillen, "Homotopy properties of the poset of nontrivial p-subgroups of a group," *Advances in Mathematics* 28 (1978), 101–128.

[64] G.-C. Rota, "On the foundations of combinatorial theory I: Theory of Möbius functions," *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 2 (1964), 340–368.

[65] T. P. Speed, "Cumulants and partition lattices," *Australian Journal of Statistics* 25 (1983), 378–388.

[66] J. Coussement and W. Van Assche, "Gaussian quadrature for multiple orthogonal polynomials," *Journal of Computational and Applied Mathematics* 178 (2005), 131–145.

[67] R. L. Graham, D. E. Knuth, and O. Patashnik, *Concrete Mathematics: A Foundation for Computer Science*, 2nd ed., Addison–Wesley, 1994.

[68] R. J. Parikh, "On context-free languages," *Journal of the ACM* 13(4) (1966), 570–581. DOI: 10.1145/321356.321364.

[69] S. Ginsburg and E. H. Spanier, "Semigroups, Presburger formulas, and languages," *Pacific Journal of Mathematics* 16(2) (1966), 285–296. DOI: 10.2140/pjm.1966.16.285.

[70] F. Höfting and E. Wanke, "Minimum cost paths in periodic graphs," *SIAM Journal on Computing* 24(5) (1995), 1051–1067. DOI: 10.1137/S0097539792234378.

[71] W. Bruns and J. Herzog, "Semigroup rings and simplicial complexes," *Journal of Pure and Applied Algebra* 122(3) (1997), 185–208. DOI: 10.1016/S0022-4049(97)00051-0.

[72] R. Ghrist and V. Peterson, "The geometry and topology of reconfiguration," *Advances in Applied Mathematics* 38(3) (2007), 302–323. DOI: 10.1016/j.aam.2005.08.009.

[73] A. Botea, "Fast, optimal pathfinding with compressed path databases," in *Proceedings of the Fifth Annual Symposium on Combinatorial Search*, vol. 3, no. 1, 2012, pp. 204–205. DOI: 10.1609/socs.v3i1.18251.

[74] S. Bhattacharya, "Towards optimal path computation in a simplicial complex," *The International Journal of Robotics Research* 38(8) (2019), 981–1009. DOI: 10.1177/0278364919855422.

[75] T. Bogart, J. Goodrick, and K. Woods, "Periodic behavior in families of numerical and affine semigroups via parametric Presburger arithmetic," *Semigroup Forum* 102(2) (2021), 340–356. DOI: 10.1007/s00233-021-10164-3.

[76] B. Shen, M. A. Cheema, D. D. Harabor, and P. J. Stuckey, "Contracting and compressing shortest path databases," in *Proceedings of the Thirty-First International Conference on Automated Planning and Scheduling*, 2021, pp. 322–330.

[77] A. Stjerna and P. Rümmer, "A constraint solving approach to Parikh images of regular languages," *Proceedings of the ACM on Programming Languages* 8(OOPSLA1) (2024), 1235–1263. DOI: 10.1145/3649855.

[78] C. Haase, S. N. Krishna, K. Madnani, O. S. Mishra, and G. Zetzsche, "An efficient quantifier elimination procedure for Presburger arithmetic," in *51st International Colloquium on Automata, Languages, and Programming (ICALP 2024)*, LIPIcs 297, Article 142, 2024, pp. 142:1–142:17. DOI: 10.4230/LIPIcs.ICALP.2024.142.

[79] M. Aigner, *Markov's Theorem and 100 Years of the Uniqueness Conjecture: A Mathematical Journey from Irrational Numbers to Perfect Matchings*, Springer, Cham, 2013. doi:10.1007/978-3-319-00888-2.

[80] H. Cohn, "Approach to Markoff's minimal forms through modular functions," *Annals of Mathematics* 61 (1955), 1–12. doi:10.2307/1969618.

[81] H. Cohn, "Markoff forms and primitive words," *Mathematische Annalen* 196 (1972), 8–22. `https://eudml.org/doc/162238`.

[82] C. Lagisquet, E. Pelantová, S. Tavenas and L. Vuillon, "On the Markov numbers: fixed numerator, denominator, and sum conjectures," *Advances in Applied Mathematics* 130 (2021), 102227. doi:10.1016/j.aam.2021.102227.

[83] C. Reutenauer, *From Christoffel Words to Markoff Numbers*, 2nd ed., Oxford University Press, Oxford, 2026. doi:10.1093/9780197907535.001.0001.

[84] A. P. Veselov, "Markov fractions and Cohn matrices," arXiv:2604.17401, 2026. doi:10.48550/arXiv.2604.17401.

[85] A. Meduna, L. Vrábel, and P. Zemek, "An Infinite Hierarchy of Language Families Resulting from Stateless Pushdown Automata with Limited Pushdown Alphabets," in *Descriptional Complexity of Formal Systems (DCFS 2012)*, Lecture Notes in Computer Science 7386, Springer, 2012, pp. 236–243. doi:10.1007/978-3-642-31623-4_18.

[86] G. Pighizzini, *Push complexity: Optimal bounds and decidability*, Theoretical Computer Science 1071 (2026), 115839. `https://doi.org/10.1016/j.tcs.2026.115839`

[87] G. Pighizzini and L. Prigioniero, "Pushdown Automata and Constant Height: Decidability and Bounds," *Acta Informatica* 60 (2023), 123–144. doi:10.1007/s00236-022-00434-0.

[88] G. Pighizzini and L. Prigioniero, *Pushdown and one-counter automata: Constant and non-constant memory usage*, Information and Computation 306 (2025), 105329. `https://doi.org/10.1016/j.ic.2025.105329`

[89] W. Hoeffding, *Probability inequalities for sums of bounded random variables*, Journal of the American Statistical Association 58 (1963), 13–30.

[90] O. H. Ibarra, J. Jirásek, I. McQuillan, and L. Prigioniero, *Space complexity of stack automata models*, International Journal of Foundations of Computer Science 32(6) (2021), 801–823. `https://doi.org/10.1142/S0129054121420090`

[91] E. Sperner, Ein Satz über Untermengen einer endlichen Menge, *Mathematische Zeitschrift* 27 (1928), 544–548.

[92] N. G. de Bruijn, C. van Ebbenhorst Tengbergen, and D. Kruyswijk, On the set of divisors of a number, *Nieuw Archief voor Wiskunde*, Ser. 2, 23 (1951), 191–193.

[93] D. Lubell, A short proof of Sperner's lemma, *Journal of Combinatorial Theory* 1(2) (1966), 299.

[94] D. J. Kleitman, On Dedekind's problem: The number of monotone Boolean functions, *Proceedings of the American Mathematical Society* 21(3) (1969), 677–682.

[95] R. Geoghegan, *Topological Methods in Group Theory*, Graduate Texts in Mathematics 243, Springer, New York, 2008.

[96] M. O. Rabin, Recursive unsolvability of group theoretic problems, *Annals of Mathematics* 67(1) (1958), 172–194.

[97] M. Filakovský, P. Franek, U. Wagner, and S. Zhechev, Computing simplicial representatives of homotopy group elements, *Journal of Applied and Computational Topology* 2 (2018), 177–231.

[98] O. Kupferman and J. Mosheiff, Prime languages, *Information and Computation* 240 (2015), 90–107. doi:10.1016/j.ic.2014.09.010.

[99] P. Totzke, Prime DOCAs, Automata Exchange, Problem 20.03, July 1, 2020.

[100] M. L. Minsky, *Computation: Finite and Infinite Machines*, Prentice-Hall, Englewood Cliffs, NJ, 1967.

[101] I. Jecker, O. Kupferman, and N. Mazzocchi, Unary prime languages, in *MFCS 2020*, LIPIcs 170, Article 51, 2020. doi:10.4230/LIPIcs.MFCS.2020.51.

[102] I. Jecker, N. Mazzocchi, and P. Wolf, Decomposing permutation automata, *Journal of Computer and System Sciences* 156 (2026), Article 103721. doi:10.1016/j.jcss.2025.103721.

[103] D. A. Spenner, Decomposing finite languages, in *MFCS 2023*, LIPIcs 272, Article 83, 2023. doi:10.4230/LIPIcs.MFCS.2023.83.

[104] D. A. Spenner, Deciding DFA-primality is NP-hard, in *ICALP 2026*, LIPIcs 374, Article 192, 2026. doi:10.4230/LIPIcs.ICALP.2026.192.

[105] S. Almagor, G. Avni, H. Sinclair-Banks, and A. Yeshurun, Dimension-minimality and primality of counter nets, in *FoSSaCS 2024*, LNCS 14575, Springer, 2024, pp. 229–249. doi:10.1007/978-3-031-57231-9_11.

[106] L. G. Valiant and M. S. Paterson, Deterministic one-counter automata, *Journal of Computer and System Sciences* 10(3) (1975), 340–350. doi:10.1016/S0022-0000(75)80005-5.

[107] S. Böhm, S. Göller, and P. Jančar, Equivalence of deterministic one-counter automata is NL-complete, in *Proceedings of STOC 2013*, ACM, 2013, pp. 131–140. doi:10.1145/2488608.2488626.

[108] P. Hofman and P. Totzke, Trace inclusion for one-counter nets revisited, *Theoretical Computer Science* 735 (2018), 50–63. doi:10.1016/j.tcs.2017.05.009.

[109] A. Prakash, *Problem 24.04: Language Equivalence between Nondeterministic and Deterministic One-Counter Machines*, Automata Exchange, 2024. Problem 24.04 webpage.

[110] L. G. Valiant, *Decision Procedures for Families of Deterministic Pushdown Automata*, Ph.D. thesis, University of Warwick, 1973.

[111] R. R. Howell, L. E. Rosier, D. T. Huynh, and H.-C. Yen, Some complexity bounds for problems concerning finite and 2-dimensional vector addition systems with states, *Theoretical Computer Science* 46 (1986), 107–140. doi:10.1016/0304-3975(86)90026-5.

[112] P. Hofman, S. Lasota, R. Mayr, and P. Totzke, Simulation problems over one-counter nets, *Logical Methods in Computer Science* 12(1) (2016), Article 6. doi:10.2168/LMCS-12(1:6)2016.

[113] S. Almagor and A. Yeshurun, Determinization of one-counter nets, in *CONCUR 2022*, LIPIcs 243, Article 18, 2022. doi:10.4230/LIPIcs.CONCUR.2022.18.

[114] A. Prakash and K. S. Thejaswini, On history-deterministic one-counter nets, in *FoSSaCS 2023*, LNCS 13992, Springer, 2023, pp. 218–239. doi:10.1007/978-3-031-30829-1_11.

[115] K. Reinhardt, Reachability in Petri nets with inhibitor arcs, *Electronic Notes in Theoretical Computer Science* 223 (2008), 239–264. doi:10.1016/j.entcs.2008.12.042.

[116] R. Bonnet, The reachability problem for vector addition system with one zero-test, in *MFCS 2011*, LNCS 6907, Springer, 2011, pp. 145–157. doi:10.1007/978-3-642-22993-0_16.

[117] J. L. Gross and T. W. Tucker, Generating all graph coverings by permutation voltage assignments, *Discrete Mathematics* 18(3) (1977), 273–283. doi:10.1016/0012-365X(77)90131-5.

[118] T. Zaslavsky, Biased graphs. I. Bias, balance, and gains, *Journal of Combinatorial Theory, Series B* 47(1) (1989), 32–52. doi:10.1016/0095-8956(89)90063-4.

[119] J.-G. Eon, Graph-theoretical characterization of periodicity in crystallographic nets and other infinite graphs, *Acta Crystallographica Section A* 61(5) (2005), 501–511. doi:10.1107/S0108767305019963.

[120] O. Delgado-Friedrichs and M. O'Keeffe, Crystal nets as graphs: Terminology and definitions, *Journal of Solid State Chemistry* 178(8) (2005), 2480–2485. doi:10.1016/j.jssc.2005.06.011.

[121] G. Lerman and Y. Shi, Robust Group Synchronization via Cycle-Edge Message Passing, *Foundations of Computational Mathematics* 22 (2022), 1665–1741. doi:10.1007/s10208-021-09532-w.

[122] R. Alur and P. Madhusudan, Visibly pushdown languages, in *Proceedings of STOC 2004*, ACM, 2004, pp. 202–211. doi:10.1145/1007352.1007390.

[123] D. Nowotka and J. Srba, Height-Deterministic Pushdown Automata, in *MFCS 2007*, LNCS 4708, Springer, 2007, pp. 125–134. doi:10.1007/978-3-540-74456-6_13.

[124] A. Björner, Topological methods, in R. L. Graham, M. Grötschel, and L. Lovász (eds.), *Handbook of Combinatorics*, Vols. 1–2, Elsevier, Amsterdam, 1995, pp. 1819–1872.

[125] P. Monasse and F. Guichard, Fast Computation of a Contrast-Invariant Image Representation, *IEEE Transactions on Image Processing* 9(5) (2000), 860–872. doi:10.1109/83.841532.

[126] T. Liu, M. Seyedhosseini, and T. Tasdizen, Image Segmentation Using Hierarchical Merge Tree, *IEEE Transactions on Image Processing* 25(10) (2016), 4596–4607. doi:10.1109/TIP.2016.2592704.

[127] A. G. Silva and R. de Alencar Lotufo, Efficient Computation of New Extinction Values from Extended Component Tree, *Pattern Recognition Letters* 32(1) (2011), 79–90. doi:10.1016/j.patrec.2010.07.019.

[128] D. Günther, J. Salmon, and J. Tierny, Mandatory Critical Points of 2D Uncertain Scalar Fields, *Computer Graphics Forum* 33(3) (2014), 31–40. doi:10.1111/cgf.12359.

[129] L. Yan, Y. Wang, E. Munch, E. Gasparovic, and B. Wang, A Structural Average of Labeled Merge Trees for Uncertainty Visualization, *IEEE Transactions on Visualization and Computer Graphics* 26(1) (2020), 832–842. doi:10.1109/TVCG.2019.2934242.

[130] T. Géraud, N. Boutry, S. Crozet, E. Carlinet, and L. Najman, A Proof of the Tree of Shapes in n-D, arXiv:2206.05109, 2022. doi:10.48550/arXiv.2206.05109.

[131] J. Mendes Forte, N. Passat, A. Shimizu, and Y. Kenmochi, Consistent Connected Operators Based on Trees of Shapes, *SIAM Journal on Imaging Sciences* 18(4) (2025), 2547–2579. doi:10.1137/25M174704X.

[132] E. Carlinet, B. Esteban, N. Boutry, and G. Romero-García, Leveraging the Geodesic Distance Transform on Interval-Valued Maps to Compute the Tree of Shapes, Research Square preprint, 2026. doi:10.21203/rs.3.rs-8888553/v1.

[133] D. Attali, U. Bauer, O. Devillers, M. Glisse, and A. Lieutier, Homological Reconstruction and Simplification in $\mathbb{R}^3$, *Computational Geometry* 48(8) (2015), 606–621. doi:10.1016/j.comgeo.2014.08.010.

[134] A. Darmann and J. Döcker, On a simple hard variant of Not-All-Equal 3-Sat, *Theoretical Computer Science* 815 (2020), 147–152. doi:10.1016/j.tcs.2020.02.010.

[135] T. M. Athawale, Z. Wang, D. Pugmire, K. Moreland, Q. Gong, S. Klasky, C. R. Johnson, and P. Rosen, Uncertainty Visualization of Critical Points of 2D Scalar Fields for Parametric and Nonparametric Probabilistic Models, *IEEE Transactions on Visualization and Computer Graphics* 31(1) (2025), 108–118. doi:10.1109/TVCG.2024.3456393.

[136] T. M. Athawale, D. Maljovec, L. Yan, C. R. Johnson, V. Pascucci, and B. Wang, Uncertainty Visualization of 2D Morse Complex Ensembles Using Statistical Summary Maps, *IEEE Transactions on Visualization and Computer Graphics* 28(4) (2022), 1955–1966. doi:10.1109/TVCG.2020.3022359.

[137] A. Nigmetov and D. Morozov, Local–Global Merge Tree Computation with Local Exchanges, in *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC '19)*, Art. 60, 13 pp., 2019. doi:10.1145/3295500.3356188.

[138] J. Brüggemann, On Merge Trees and Discrete Morse Functions on Paths and Trees, *Journal of Applied and Computational Topology* 7 (2023), 103–138. doi:10.1007/s41468-022-00101-w.

[139] M. Abadi and L. Lamport, The existence of refinement mappings, *Theoretical Computer Science* 82 (1991), 253–284. doi:10.1016/0304-3975(91)90224-P.

[140] R.-J. Back and J. von Wright, *Refinement Calculus: A Systematic Introduction*, Springer, New York, 1998. doi:10.1007/978-1-4612-1674-2.

[141] P. Cousot, Constructive design of a hierarchy of semantics of a transition system by abstract interpretation, *Electronic Notes in Theoretical Computer Science* 6 (1997), 77–102. doi:10.1016/S1571-0661(05)80168-9.

[142] J. J. M. M. Rutten, Universal coalgebra: a theory of systems, *Theoretical Computer Science* 249 (2000), 3–80. doi:10.1016/S0304-3975(00)00056-6.

[143] G. Huet, Confluent reductions: abstract properties and applications to term rewriting systems, *Journal of the ACM* 27 (1980), 797–821. doi:10.1145/322217.322230.

[144] L. Lovász, On the ratio of optimal integral and fractional covers, *Discrete Mathematics* 13 (1975), 383–390. doi:10.1016/0012-365X(75)90058-8.

[145] M. Karchmer, E. Kushilevitz, and N. Nisan, Fractional covers and communication complexity, *SIAM Journal on Discrete Mathematics* 8 (1995), 76–92. doi:10.1137/S0895480192238482.

[146] H. S. Witsenhausen, The zero-error side information problem and chromatic numbers, *IEEE Transactions on Information Theory* 22 (1976), 592–593. doi:10.1109/TIT.1976.1055607.

[147] E. R. Scheinerman and D. H. Ullman, *Fractional Graph Theory*, Wiley, 1997.

[148] S. Abramsky and A. Brandenburger, The sheaf-theoretic structure of non-locality and contextuality, *New Journal of Physics* 13 (2011), 113036. doi:10.1088/1367-2630/13/11/113036.

[149] S. Abramsky and G. Carù, Non-locality, contextuality and valuation algebras: a general theory of disagreement, *Philosophical Transactions of the Royal Society A* 377 (2019), 20190036. doi:10.1098/rsta.2019.0036.

[150] S. B. Montanhano, Contextuality in the bundle approach, n-contextuality, and the role of holonomy, arXiv:2105.14132, 2021.

[151] B. Amaral, Resource theory of contextuality, *Philosophical Transactions of the Royal Society A* 377 (2019), 20190010. doi:10.1098/rsta.2019.0010.

[152] A. Chattopadhyay, H. Carr, D. Duke, Z. Geng, and O. Saeki, Multivariate Topology Simplification, *Computational Geometry* 58 (2016), 1–24. doi:10.1016/j.comgeo.2016.05.006.

[153] W. Köpp and T. Weinkauf, Temporal Merge Tree Maps: A Topology-Based Static Visualization for Temporal Scalar Data, *IEEE Transactions on Visualization and Computer Graphics* 29(1) (2023), 1157–1167. doi:10.1109/TVCG.2022.3209387.

[154] M. Li, X. Yan, L. Yan, T. Needham, and B. Wang, Flexible and Probabilistic Topology Tracking With Partial Optimal Transport, *IEEE Transactions on Visualization and Computer Graphics* 31(10) (2025), 7951–7969. doi:10.1109/TVCG.2025.3561300.

[155] A. E. Bütün, *Semantic Normalization and Realization: A Unified General Theory—Geometry, Chronology, Information, and Resources*, research monograph, 2026, Chapters 108–109. doi:10.5281/zenodo.23221519.

[156] M. Bojańczyk and T. Colcombet, "Tree-walking automata cannot be determinized," *Theoretical Computer Science* **350** (2006), 164–173. doi:10.1016/j.tcs.2005.10.031.

[157] M. Bojańczyk, "Tree-walking automata," survey in *Language and Automata Theory and Applications (LATA 2008)*. author-hosted preprint.

[158] M. Bojańczyk and T. Colcombet, "Tree-walking automata do not recognize all regular languages," *SIAM Journal on Computing* **38** (2008), 658–701. doi:10.1137/050645427.

[159] A. Muscholl, M. Samuelides, and L. Segoufin, "Complementing deterministic tree-walking automata," *Information Processing Letters* **99** (2006), 33–39. doi:10.1016/j.ipl.2005.09.017.

[160] J. Engelfriet and H. J. Hoogeboom, "Tree-walking pebble automata," in *Jewels Are Forever*, Springer (1999), 72–83. doi:10.1007/978-3-642-60207-8_7.

[161] O. Martynova and A. Okhotin, "Homomorphisms and inverse homomorphisms on graph-walking automata," *Theoretical Computer Science* **979** (2023), 114197. doi:10.1016/j.tcs.2023.114197.

[162] M. Bojańczyk, "It is undecidable if two regular tree languages can be separated by a deterministic tree-walking automaton," *Fundamenta Informaticae* **154** (2017), 37–46. doi:10.3233/FI-2017-1551.

[163] O. Martynova and A. Okhotin, "Nondeterministic tree-walking automata are not closed under complementation," *ICALP 2025*, LIPIcs **334**, article 168. doi:10.4230/LIPIcs.ICALP.2025.168.

[164] A. E. Bütün, *Chronological Carry Refinement and p-Adic Root Geometry: A Third-Order Arithmetic Lift of DPDA-Derived Correction Geometry*, Zenodo preprint (2026), doi:10.5281/zenodo.22709258.

[165] A. E. Bütün, *Fourth-Order Chronological Carry: Connected Quadruples, Extension Parameters, and the Stirling Shadow*, Zenodo preprint (2026), doi:10.5281/zenodo.22709298.

[166] A. E. Bütün, *CCR-k General Theory: Higher-Order Chronological Carry, Hessenberg Hierarchies, and Universal Stirling Shadows*, Zenodo preprint (2026), doi:10.5281/zenodo.22734188.

[167] F. Bencherif and T. Garici, *On a property of Stirling polynomials*, Publ. Inst. Math. (Beograd) 102(116) (2017), 149–153. doi:10.2298/PIM1716149B.

[168] A. T. Lundell, *A divisibility property for Stirling numbers*, J. Number Theory 10 (1978), 35–54. doi:10.1016/0022-314X(78)90005-7.

[169] A. Adelberg, *A finite difference approach to degenerate Bernoulli and Stirling polynomials*, Discrete Math. 140 (1995), 1–21. doi:10.1016/0012-365X(93)E0188-A.

[170] A. Adelberg, *Arithmetic properties of the Nörlund polynomial* $B_n^{(x)}$, Discrete Math. 204 (1999), 5–13. doi:10.1016/S0012-365X(98)00363-X.

[171] NIST Digital Library of Mathematical Functions, §26.8, Eq. 26.8.36 (signed surjection/Stirling-number identity), `https://dlmf.nist.gov/26.8.E36`.

[172] NIST Digital Library of Mathematical Functions, §24.16, *Nörlund polynomials*, `https://dlmf.nist.gov/24.16`.

[173] A. D. Sokal, *Multiple orthogonal polynomials, d-orthogonal polynomials, production matrices, and branched continued fractions*, Trans. Amer. Math. Soc. Ser. B 11 (2024), 762–797. doi:10.1090/btran/133.

[174] A. Krelifa and E. Zerouki, *Riordan arrays and d-orthogonality*, Linear Algebra Appl. 515 (2017), 331–353. doi:10.1016/j.laa.2016.11.039.

[175] R. Cluckers and A. Herremans, *The fundamental theorem of prehomogeneous vector spaces modulo* p^m *(with an appendix by F. Sato)*, Bull. Soc. Math. France 135 (2007), 475–495, §3 (a p-adic analogue of Morse's lemma). `https://www.numdam.org/article/BSMF_2007__135_4_475_0.pdf`.

[176] A. Dwivedi and N. Saxena, *Computing Igusaś local zeta function of univariates in deterministic polynomial-time*, preprint (2020), arXiv:2006.08926.

[177] Mertens' theorem on reciprocal primes and the corresponding Euler product; see e.g. Z. Teitler, *Analytic Number Theory Notes*, Section 6.3, `https://zteitler.github.io/analytic-number-theory-notes/sec-mertens.html`.

[178] *Chebotarev density theorem*, classical theorem; see, e.g., H. W. Lenstra Jr., "Chebotarev and his density theorem".

[179] V. P. Kostov, *A property of discriminants*, Vietnam J. Math. (2019), doi:10.1007/s10013-018-0296-9.

[180] S. D. Cohen, *The Galois group of a polynomial with two indeterminate coefficients*, Pacific J. Math. 90 (1980), 63–76, doi:10.2140/pjm.1980.90.63.

[181] S. D. Cohen, *Corrections to: The Galois group of a polynomial with two indeterminate coefficients*, Pacific J. Math. 97 (1981), 483–486, doi:10.2140/pjm.1981.97.483.

[182] F. Naccarato, *The arithmetic of critical values I: equicritical quartic polynomials*, International Mathematics Research Notices 2026 (2026), rnag141, doi:10.1093/imrn/rnag141.

[183] J.-P. Serre, *Topics in Galois Theory*, 2nd ed., A K Peters, 2008, §4.4.

[184] J. C. Rosales and P. A. García-Sánchez, *Numerical Semigroups*, Springer, 2009.

[185] A. E. Bütün, "Higher-Dimensional Completion Geometry for Linear Counting Languages: Shortest-Prefix Quotients, Rank-One Topology, and Eventual Geodesic Structure," 2026. `https://doi.org/10.5281/zenodo.22709372`.

[186] A. E. Bütün, "Beyond Kernel Rank: Optimal-Fiber Geometry and Rank-Two Topology of Minimum-Completion Quotients," 2026. `https://doi.org/10.5281/zenodo.22709322`.

[187] A. E. Bütün, “Lattice-Convex Realization and Rank-Raising Topology of Optimal Completion Fibers,” 2026. https://doi.org/10.5281/zenodo.22709306.

[188] A. E. Bütün, “Higher-Rank Completion Holonomy: Displacement-Marked Optimal Fibers and Physical LIFO Transport,” 2026. https://doi.org/10.5281/zenodo.22863525.

[189] U. Bauer, M. Kerber, F. Roll, and A. Rolle, “A unified view on the functorial nerve theorem and its variations,” *Expositiones Mathematicae* **41** (2023), art. 125503, https://doi.org/10.1016/j.exmath.2023.04.005.

[190] E. J. González and J. González, “Equivariant Nerve Lemma, simplicial difference, and models for configuration spaces on simplicial complexes,” *Topology and its Applications* **341** (2024), art. 108749, https://doi.org/10.1016/j.topol.2023.108749.

[191] B. Sturmfels, “On vector partition functions,” *Journal of Combinatorial Theory, Series A* **72** (1995), 302–309, https://doi.org/10.1016/0097-3165(95)90067-5.

[192] H. Charalambous, A. Katsabekis, and A. Thoma, “Minimal systems of binomial generators and the indispensable complex of a toric ideal,” *Proceedings of the American Mathematical Society* **135** (2007), 3443–3452, https://doi.org/10.1090/S0002-9939-07-09037-5.

[193] P. A. García-Sánchez and A. Herrera-Poyatos, “Isolated factorizations and their applications in simplicial affine semigroups,” *Journal of Algebra and Its Applications* **19** (2020), https://doi.org/10.1142/S0219498820500826.

[194] S. A. M. Jones and R. M. Thomas, “Word problems of groups: Formal languages, characterizations and decidability,” *Theoretical Computer Science* (2018), https://doi.org/10.1016/j.tcs.2018.05.007.

[195] A. E. Bütün, *CCR-k Arithmetic Refinement: p-Adic Resonance, Ramification, and Arithmetic Density Laws*, Zenodo preprint, 2026. doi:10.5281/zenodo.23441311.

[196] I. Gessel and R. P. Stanley, Stirling polynomials, *J. Combin. Theory Ser. A* **24** (1978), 24–33. doi:10.1016/0097-3165(78)90042-0.

[197] T. Komatsu and P. T. Young, Exact p-adic valuations of Stirling numbers of the first kind, *J. Number Theory* **177** (2017), 20–27. doi:10.1016/j.jnt.2017.01.023.

[198] P. T. Young, Congruences for Bernoulli, Euler, and Stirling numbers, *J. Number Theory* **78** (1999), 204–227. doi:10.1006/jnth.1999.2401.

[199] Y. H. H. Kwong, Minimum periods of $S(n,k)$ modulo M, *Fibonacci Quarterly* **27** (1989), 217–221.

[200] Z.-W. Sun, Combinatorial congruences and Stirling numbers, *Acta Arithmetica* **126** (2007), 387–398. doi:10.4064/aa126-4-7.

[201] The Stacks Project, Kummer extensions, Tag 09I6, https://stacks.math.columbia.edu/tag/09I6.

[202] J. Goldstine, J. K. Price, and D. Wotschke, On reducing the number of states in a PDA, *Mathematical Systems Theory* **15** (1982), 315–321. doi:10.1007/BF01786988.

[203] L. Kiss and B. Rovan, Descriptional complexity of push down automata, *ITAT 2020*, CEUR Workshop Proceedings **2718** (2020), 59–66.

[204] P. Jančar, Equivalence of pushdown automata via first-order grammars, *J. Comput. System Sci.* **115** (2021), 86–112. doi:10.1016/j.jcss.2020.07.004.

[205] G. Paquin and C. Reutenauer, *On the superimposition of Christoffel words.* Theoretical Computer Science 412 (2011), 402–418. DOI: 10.1016/j.tcs.2010.10.007.

[206] X. Provençal and L. Vuillon, *Discrete segments of $\mathbb{Z}^3$ constructed by synchronization of words.* Discrete Applied Mathematics 183 (2015), 102–117. DOI: 10.1016/j.dam.2014.03.010.

[207] L. Tarsissi, A. A. Menaa, L. Vuillon, L. Najman, *A geometric point of view on the synchronization of three Christoffel words*. Discrete Applied Mathematics, online 2026. DOI: 10.1016/j.dam.2026.06.028.

[208] F. Aguiló-Gost and P. A. García-Sánchez, *Factoring in embedding dimension three numerical semigroups*, Electronic Journal of Combinatorics **17** (2010), R138. doi:10.37236/410.

[209] D. Gómez, J. Gutierrez and À. Ibeas, *Cayley digraphs of finite abelian groups and monomial ideals*, SIAM Journal on Discrete Mathematics **21** (2007), 763–784. doi:10.1137/050646056.

[210] J. Autry, T. Gomes, C. O'Neill and V. Ponomarenko, *Elasticity in Apéry sets*, The American Mathematical Monthly **127** (2020), 744–749. doi:10.1080/00029890.2020.1790926.

[211] M. Praveen, *What You Must Remember When Transforming Datawords*, FSTTCS 2020, LIPIcs **182**, 55:1–55:14, doi:10.4230/LIPIcs.FSTTCS.2020.55.

[212] M. Bojańczyk and R. Stefański, *Single-Use Automata and Transducers for Infinite Alphabets*, ICALP 2020, LIPIcs **168**, 113:1–113:14, doi:10.4230/LIPIcs.ICALP.2020.113.

[213] A. Adamaszek, A. Czumaj, M. Englert and H. Räcke, *Almost Tight Bounds for Reordering Buffer Management*, SIAM J. Comput. **51** (2022), 701–722, doi:10.1137/20M1326167.

[214] C. Reutenauer, *On quadratic numbers and forms, and Markoff theory*, J. Number Theory **227** (2021), 265–305. doi:10.1016/j.jnt.2021.03.005.

[215] G. McShane and H. Parlier, *Multiplicities of simple closed geodesics and hypersurfaces in Teichmüller space*, Geom. Topol. **12** (2008), 1883–1919. doi:10.2140/gt.2008.12.1883.

[216] Z. Greenberg, D. Kaufman and A. Wienhard, *SL_2-like properties of matrices over noncommutative rings and generalizations of Markov numbers*, Adv. Math. **482** (2025), 110556. doi:10.1016/j.aim.2025.110556.

[217] P. L. Clark, P. Pollack, J. Rouse and K. Thompson, *Densities of integer sets represented by quadratic forms*, J. Number Theory **256** (2024), 290–328. doi:10.1016/j.jnt.2023.09.010.

[218] A. E. Bütün, *Sharp p-Adic Rigidity and Source-Tunable Resonance in Stationary CCR Polynomial Transports*, research manuscript (2026).

[219] A. E. Bütün, *Single-Site Rigidity and a Sporadic Exception for Prime-Indexed Nörlund–Stirling Quotient Polynomials*, research manuscript (2026).

[220] A. Adelberg, Universal higher order Bernoulli numbers and Kummer and related congruences, *J. Number Theory* 84 (2000), 119–135. DOI: 10.1006/jnth.2000.2526.

[221] D. Gómez-Pérez, J. Gutiérrez, À. Ibeas, D. Sevilla, Common factors of resultants modulo p, *Bull. Aust. Math. Soc.* 79 (2009), 299–302. DOI: 10.1017/S0004972708001275.

[222] F. Clarke and I. Sh. Slavutskii, "The integrality of the values of Bernoulli polynomials and of generalised Bernoulli numbers," *Bulletin of the London Mathematical Society* 29 (1997), 22–24.

[223] W. Feit, "On large Zsigmondy primes," *Proc. Amer. Math. Soc.* **102** (1988), 29–36. doi:10.1090/S0002-9939-1988-0915710-1.

[224] M. Roitman, "On Zsigmondy primes," *Proc. Amer. Math. Soc.* **125** (1997), 1913–1919. doi:10.1090/S0002-9939-97-03981-6.

[225] P. J. McCarthy, "Irreducibility of Bernoulli polynomials of higher order," *Canad. J. Math.* **14** (1962), 565–567. doi:10.4153/CJM-1962-047-2.

[226] J.-P. Serre, *Topics in Galois Theory*, 2nd ed., A K Peters, 2008; see the Frobenius/Chebotarev splitting principle.

[227] A. E. Bütün, "A General Theory of Real-Time Pushdown Normalization: Semantic Quotients, Locality, Representation Debt, and Causal Scheduling," Zenodo preprint (2026). doi:10.5281/zenodo.22709302.

[228] A. E. Bütün, "Semantic Uniqueness and LIFO Lift Theory for Real-Time Pushdown Normalization," Zenodo preprint (2026). doi:10.5281/zenodo.22709334.

[229] A. E. Bütün, "Exact Structure and Resource Bounds in Deterministic Pushdown Automata for Binary Counting Languages," Zenodo preprint (2026). doi:10.5281/zenodo.22709367.

[230] M. A. Harrison and I. M. Havel, "Real-time strict deterministic languages," *SIAM Journal on Computing* **1** (1972), 333–349. doi:10.1137/0201024.

[231] N. Jonoska, M. Krajčevski, and G. McColm, "Counter machines and crystallographic structures," *Natural Computing* **15** (2016), 97–113. doi:10.1007/s11047-015-9527-0.

[232] H. J. S. Basten, "Context-free ambiguity detection using multi-stack pushdown automata," in *Developments in Language Theory, DLT 2016*, LNCS 9840, Springer, 2016, pp. 1–12. doi:10.1007/978-3-662-53132-7_1.

[233] S. La Torre, M. Napoli, and G. Parlato, "Scope-bounded pushdown languages," *International Journal of Foundations of Computer Science* **27** (2016), 215–233. doi:10.1142/S0129054116400074.

[234] C. Barloy, M. Cadilhac, and K. Ockerlund, "Shuffles of context-free languages along regular trajectories," *ICALP 2026*, LIPIcs **374** (2026), 163:1–163:17. doi:10.4230/LIPIcs.ICALP.2026.163.

[235] J. M. Silva, "Geometric characterization of context-free intersections via the inner segment dichotomy," preprint (2026), arXiv:2602.14722.

[236] A. E. Bütün, *A Sharp Real-Time Pushdown Realizability Dichotomy for Linear Modular Height Guards*, standalone research manuscript (October 2026).

[237] A. E. Bütün, *A Signed-Winding Dichotomy for Canonical Real-Time Pushdown Realization under Periodic Height Guards*, standalone research manuscript (October 2026).

[238] B. Bollobás and A. J. Radcliffe, Isoperimetric inequalities for faces of the cube and the grid, *European J. Combin.* 11 (1990), 323–333.

[239] B. Bollobás and I. Leader, Exact face-isoperimetric inequalities, *European J. Combin.* 11 (1990), 335–340.

[240] V. F. Lev and M. Rudnev, Minimising the sum of projections of a finite set, *Discrete Comput. Geom.* 60 (2018), 493–511.

[241] Z. Füredi and A. Sali, Optimal multivalued shattering, *SIAM J. Discrete Math.*, DOI 10.1137/110850104.

[242] J. Oxley, *Matroid Theory*, 2nd ed., Oxford University Press, 2011.

[243] S. Fujishige, *Submodular Functions and Optimization*, 2nd ed., Elsevier, 2005.

[244] N. Alon, Traces of hypergraphs, *J. London Math. Soc.* (2019).

[245] R. P. Anstee and N. A. Nikov, Shattering and more: extending the complete object, *Electron. J. Combin.* **29**(2) (2022), P2.42.

[246] A. Nerode, Linear automaton transformations, *Proc. Amer. Math. Soc.* 9 (1958), 541–544.

[247] A. E. Bütün, *Future-Stable Readout Monitors for Independent Write-Once Sources: Exact State Bounds, Matroid-Rank Structure, and Constrained Projection Extremals*, independent research manuscript (October 2026).

[248] F. Harary, A characterization of block-graphs, *Canadian Mathematical Bulletin* **6** (1963), 1–6. DOI: 10.4153/CMB-1963-001-x.

[249] D. E. Muller and P. E. Schupp, Pushdown automata, graphs, ends, second-order logic, and reachability problems, in *STOC* (1981), 46–54.

[250] J. Michaliszyn and J. Otop, Learning deterministic visibly pushdown automata under accessible stack, in *MFCS 2022*, LIPIcs vol. 241, paper 74. `https://doi.org/10.4230/LIPIcs.MFCS.2022.74`.

[251] R. Devillers and R. Tredup, Some basic techniques allowing Petri net synthesis: complexity and algorithmic issues, *Fundamenta Informaticae* **187** (2022), 167–196. `https://doi.org/10.46298/fi.8813`.

[252] R. Devillers and R. Tredup, On the complexity of techniques that make transition systems implementable by Boolean nets, *Fundamenta Informaticae* **189** (2023), 261–296. `https://doi.org/10.46298/fi.9231`.

[253] A. E. Bütün, A binary stack-alphabet hierarchy for deterministic pushdown automata with ε-moves: completion slopes, up-modes, and a product lower bound, author manuscript (2026).

[254] R. M. Karp, Reducibility among combinatorial problems, in *Complexity of Computer Computations* (1972), 85–103.

[255] A. E. Bütün, *Exact Stack-Alphabet Costs and Nonmonotone Saturation Gaps in Guarded Two-Layer Graph Actions*, independent research manuscript (October 2026).

[256] A. E. Bütün, *Active Observability of Chronological Carry Recurrences: Masking, Probing, and Sharp Intervention Costs*, independent research manuscript (November 2026).

[257] R. M. Hierons, G.-V. Jourdan, H. Ural, and H. Yenigün, "Using adaptive distinguishing sequences in checking sequence constructions," *Proc. ACM Symposium on Applied Computing* (2008), 682–687. doi:10.1145/1363686.1363850.

[258] S. Jia, F. Navidi, V. Nagarajan, and R. Ravi, "Optimal Decision Tree and Adaptive Submodular Ranking with Noisy Outcomes," *Journal of Machine Learning Research* **25** (2024), no. 382, 1–42. `https://www.jmlr.org/papers/v25/23-1484.html`.

[259] R. M. Hierons and U. C. Türker, "Incomplete Distinguishing Sequences for Finite State Machines," *The Computer Journal* **58** (2015), 3089–3113. doi:10.1093/comjnl/bxv041.

[260] M. N. Chatzis, K. Maes, and G. Lombaert, "Observability of nonlinear discrete-time systems with unmeasured inputs," *Mechanical Systems and Signal Processing* **238** (2025), 113142. doi:10.1016/j.ymssp.2025.113142.

[261] A. E. Bütün, *The Computational Power of Typed Chronological Carry Refinement: Sharp Forecast-Information Bounds, Fast Exact Algorithms, Finite-Ring Profiles, and Lossless Source Coding*, independent research manuscript (November 2026).

[262] W. Steeman, *BPI Challenge 2013, Incidents*, 4TU.ResearchData. DOI: 10.4121/500573e6-accc-4b0c-9576-aa5468b10cee.

[263] The FLINT development team, "Hermite and Smith Normal Forms," software and algorithm documentation. `https://fredrikj.net/fdoc/html/algorithms/linear-algebra/hermite-smith.html`.

[264] I. Verenich, *Helpdesk*, Mendeley Data, version 1 (2016). DOI: 10.17632/39bp3vv62t.1.

[265] S. Kiefer, "Notes on Equivalence and Minimization of Weighted Automata," arXiv:2009.01217 (2020). `https://arxiv.org/abs/2009.01217`.

[266] M. Mohri, “Weighted Automata Algorithms,” in *Handbook of Weighted Automata*, M. Droste, W. Kuich and H. Vogler (eds.), Springer, 2009, pp. 213–254. DOI: 10.1007/978-3-642-01492-5_6.

[267] F. Simon, P. Tittmann and M. Trinks, “Counting Connected Set Partitions of Graphs,” *Electronic Journal of Combinatorics* 18(1), P14 (2011). DOI: 10.37236/501.

[268] A. E. Bütün, *Joint Physical Realization of Relational Extrema States: Sharp Resource Obstructions, Correlated Terminal Fibers, and Spectral Rigidity*, corrected independent research manuscript (November 2026).

[269] A. E. Bütün, *Extrema–Frame Foreground–Background Theory: Self-Dual Structure, Exact Common-Realization Semantics, and Ownership Complexity*, Zenodo preprint (24 September 2026). doi:10.5281/zenodo.22944661.

[270] A. E. Bütün, *Extrema–Frame Foreground–Background Theory II: Realization Universality, Global Complexity, and Correlated Layered Semantics*, Zenodo preprint (7 October 2026). doi:10.5281/zenodo.23220689.

[271] V. J. Mizel and T. I. Seidman, An abstract bang–bang principle and time-optimal boundary control of the heat equation, *SIAM Journal on Control and Optimization* **35** (1997), 1204–1216. doi:10.1137/S0363012996265470.

[272] K. Kunisch and L. Wang, Time optimal control of the heat equation with pointwise control constraints, *ESAIM: Control, Optimisation and Calculus of Variations* **19** (2013), 460–485. doi:10.1051/cocv/2012017.

[273] M. Dantam and A. Pouly, On the decidability of reachability in continuous time linear time-invariant systems, *Proceedings of HSCC 2021*, article 15 (2021), 1–12. doi:10.1145/3447928.3456705.

[274] A. Farrugia and I. Sciriha, Controllability of undirected graphs, *Linear Algebra and its Applications* **454** (2014), 138–157. doi:10.1016/j.laa.2014.04.022.

[275] C. Poignard, T. Pereira, and J. P. Pade, Spectra of Laplacian matrices of weighted graphs: structural genericity properties, *SIAM Journal on Applied Mathematics* **78** (2018), 372–394. doi:10.1137/17M1124474.

[276] K. Flaßkamp, S. Maslovskaya, S. Ober-Blöbaum, and B. Wembe, Trim turnpikes for optimal control problems with symmetries, *Mathematics of Control, Signals, and Systems* **37** (2025), 247–277. doi:10.1007/s00498-025-00408-w.

[277] S. Basu, R. Pollack, and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, 2nd ed., Springer, 2006. doi:10.1007/3-540-33099-2.

[278] R. J. Aumann, Integrals of set-valued functions, *Journal of Mathematical Analysis and Applications* **12** (1965), 1–12. doi:10.1016/0022-247X(65)90049-1.

[279] A. E. Bütün, *Actuation-Resolution Language Collapse and Exact Causal Power in a Relational Heat Network*, full self-contained independent research manuscript, 6 November 2026.

[280] H. B. Gunay, Z. Shi, G. Newsham and R. Moromisato, Detection of zone sensor and actuator faults through inverse greybox modelling, *Building and Environment* **171** (2020), Article 106659, `https://doi.org/10.1016/j.buildenv.2020.106659`.

[281] H. B. Gunay, *Data for: Detection of zone sensor and actuator faults through inverse greybox modelling*, Mendeley Data, version 1 (2021), `https://doi.org/10.17632/fbp7d2mxj3.1`.

[282] Y. Guo, J. Wall, J. Li and S. West, Intelligent model based fault detection and diagnosis for HVAC system using statistical machine learning methods, *ASHRAE 2013 Winter Conference* (2013), pp. 119–124, `http://hdl.handle.net/102.100.100/98252`.

[283] CSIRO Energy Systems, *AHU fault-detection dataset: fault-generation experiments in an operational Newcastle office building*, public data repository (2013 measurements), https://github.com/csiro-energy-systems/ahu-fault-detection-dataset.

[284] A. Rantzer and M. E. Valcher, Scalable control of positive systems, *Annual Review of Control, Robotics, and Autonomous Systems* **4** (2021), 319–341, https://doi.org/10.1146/annurev-control-061520-010621.

[285] Y. Zeinaly, J. H. van Schuppen and B. De Schutter, Linear positive systems may have a reachable subset from the origin that is either polyhedral or nonpolyhedral, *SIAM Journal on Matrix Analysis and Applications* **41** (2020), 279–307, https://doi.org/10.1137/19M1268161.

[286] C. Guiver, R. Rebarber and S. Townley, Positive state reachability for positive linear control systems in continuous time, *Journal of the Franklin Institute* **363**(1) (2026), Article 108244, https://doi.org/10.1016/j.jfranklin.2025.108244.

[287] G. Pola, T. Masciulli, E. De Santis and M. D. Di Benedetto, Data-driven controller synthesis for abstract systems with regular language specifications, *Automatica* **134** (2021), Article 109903, https://doi.org/10.1016/j.automatica.2021.109903.

[288] T. Masciulli and G. Pola, Symbolic control design of incrementally stable nonlinear systems with dynamic regular language specifications, *Automatica* **130** (2021), Article 109712, https://doi.org/10.1016/j.automatica.2021.109712.

[289] T. Masciulli and G. Pola, On symbolic control design of nonlinear systems with dynamic regular language specifications, *IFAC-PapersOnLine* **53**(2) (2020), 1844–1849, https://doi.org/10.1016/j.ifacol.2020.12.2352.

[290] G. Pola, T. Masciulli, E. De Santis and M. D. Di Benedetto, On data-driven controller synthesis with regular language specifications, *IFAC-PapersOnLine* **53**(2) (2020), 3928–3933, https://doi.org/10.1016/j.ifacol.2020.12.2245.

[291] J. Zhao, D. J. Hill and T. Liu, Global bounded synchronization of general dynamical networks with nonidentical nodes, *IEEE Transactions on Automatic Control* **57**(10) (2012), 2656–2662, https://doi.org/10.1109/TAC.2012.2190206.

[292] T. Trummel, Z. Liu and O. Stursberg, Computing bounds for the synchronization errors of nonidentical nonlinear oscillators with time-varying diffusive coupling, in *ICINCO 2024*, vol. 1, 612–622, https://doi.org/10.5220/0013071100003822.

[293] L. Scardovi and R. Sepulchre, Synchronization in networks of identical linear systems, *Automatica* **45** (2009), 2557–2562, https://doi.org/10.1016/j.automatica.2009.07.006.

[294] N. O'Clery, Y. Yuan, G.-B. Stan and M. Barahona, Observability and coarse graining of consensus dynamics through the external equitable partition, *Physical Review E* **88** (2013), 042805, https://doi.org/10.1103/PhysRevE.88.042805.

[295] M. Aguiar and A. Dias, Synchrony and antisynchrony in weighted networks, *SIAM Journal on Applied Dynamical Systems* (2021), https://doi.org/10.1137/20M1348248.

[296] M. E. Valcher and P. Santesso, Reachability properties of single-input continuous-time positive switched systems, *IEEE Transactions on Automatic Control* (2010), https://doi.org/10.1109/TAC.2010.2045442.

[297] Z. Bartosiewicz, Linear positive control systems on time scales; controllability, *Mathematics of Control, Signals, and Systems* **25** (2013), 327–343, https://doi.org/10.1007/s00498-013-0106-6.

[298] Z. Bartosiewicz, Invariance under discretization for positive systems, *Mathematics of Control, Signals, and Systems* **33** (2021), 315–329, https://doi.org/10.1007/s00498-021-00283-1.

[299] J. Liu and N. Ozay, Finite abstractions with robustness margins for temporal logic-based control synthesis, *Nonlinear Analysis: Hybrid Systems* **22** (2016), 1–15, https://doi.org/10.1016/j.nahs.2016.02.002.

[300] Y. Li, Z. Sun and J. Liu, A specification-guided framework for temporal logic control of nonlinear systems, *IEEE Transactions on Automatic Control* **68**(4) (2023), 2002–2017, https://doi.org/10.1109/TAC.2022.3168489.

[301] W. Ren and R. Jungers, Reachability-based control synthesis under signal temporal logic specifications, *American Control Conference* (2022), https://doi.org/10.23919/ACC53348.2022.9867813.

[302] E. S. Kim, M. Arcak and S. A. Seshia, A small gain theorem for parametric assume–guarantee contracts, in *Hybrid Systems: Computation and Control* (2017), https://doi.org/10.1145/3049797.3049805.

[303] A. V. Proskurnikov, Diffusive coupling as a quadratic constraint: Synchronization and stability in heterogeneous networks, *Systems & Control Letters* (2026), https://doi.org/10.1016/j.sysconle.2026.106502.

[304] S. Pruekprasert and C. Eberhart, Symbolic verification and control of continuous-time systems via AP-observation automata, *Theoretical Computer Science* (2026), Article 116203, https://doi.org/10.1016/j.tcs.2026.116203.

[305] A. E. Bütün, *Exact Future-Archive Complexity and Exceptional-Characteristic Recovery in Typed Chronological Carry Refinement*, independent research manuscript (November 2026).

[306] M. Petreczky, R. Tóth, and G. Mercère, "Realization theory for LPV state-space representations with affine dependence," *IEEE Transactions on Automatic Control* **62**(9), 4667–4674 (2017). doi:10.1109/TAC.2016.2629989.

[307] E. D. Sontag, *Discrete-time polynomial systems with inputs and outputs: structure, reachability, observability, and minimal realizations*, arXiv:2609.36423 (2026). https://arxiv.org/abs/2609.36423.

Result Index

k-fiber 213
$p = 5$ denominator resonance 291
2:1 causal certificate. 505
2:1 specialization and recovery. 454

Absorbing-fiber harmlessness. 557
Activation tightness. 561
Active collision lemma 697
Active trade lattice 697
Active-fiber saturation 698
Adaptive advantage for CCR-10 382
Adelic CCR prime-by-column product law 330
Admissibility and fatal supports 510
Affine completion ray. 19
Affine prefix retention 776
Affine-lattice invariance of the marked group 742
Aggregate exposure 775
Algebraicity and regular kernels 695
All-odd-order fixed-source nonresonance 321
All-odd-order rare-resonance statistics 311
All-order non-identifiability 294
All-order terminal-block separation . . 293
All-prime root and factor stratification 304
All-rank exact buffered carrier family 747
Alternating sentinel. 818
Ambient lattice versus word orbit 661
Apex branch with unbounded second homology 740
Apex pure-holonomy creation 743
Apex–prism separation 744
Arbitrary positive parameters. 758
Arbitrary-boundary presence and exact contacts. 617
Arbitrary-coprime causal-hypergraph classification 517
Arbitrary-interleaving envelope bound 842
Arbitrary-order fixed-column polynomiality 298
Arbitrary-order raw Stirling shadow . 300
Arbitrary-scale operational enumeration 590
Arbitrary-state classification 231
Arithmetic carrier girth 720
Arithmetic distance predicate 197
Asymptotic Defect Hierarchy. 262
Asymptotically almost-everywhere hardness 781
Attainability of complete local splitting 314
Automatic compatibility detection. . . 613
Auxiliary structural isolation 215

Backlog law. 477
Balanced-cycle reduction. 45
Bare semantic modulo-confluence is vacuous 531
Baseline-portable site 511
BBR is an equivalence relation 529
Behavioral hull characterization. 554
Binary adjacent stack-alphabet hierarchy. 124
Binary Cancellation Skeleton. 250
Binary-deficit distance lemma 207
Bit strong converse. 550
Block Purity. 244
Block synchronization 817
Block-order test. 468
Boolean fiber amplification 212
Bottom-move bound 725
Bottom-WRITE Rayleigh law 682
Boundary and transport equations . . . 296
Boundary families. 622
Boundary fibers 518
Boundary guard 676
Boundary guard. 441
Boundary obstruction for the fixed encoding. 74
Bounded bi-recoding 528
Bounded residual construction 761
Bounded subtree probe 935
Bounded-d-top capacity. 551
Bounded-factor input 822

Cancellation-free primitive orbit 580
Cancellation–diamond identity. 39
Canonical absorption. 483, 547
Canonical and complete generation . . 198
Canonical behavioral core. 489
Canonical carry dichotomy. 478
Canonical critical-bridge representation 677
Canonical deferred-SEARCH realization. 446
Canonical direct-realizability classification. 464
Canonical Farey Unit-Lift Theorem. . 623
Canonical fiber and retraction. 546
Canonical interior. 571
Canonical minimum realization. 246
Canonical retraction and absorption; baseline 530
Canonical two-block shortest completion. 21
Canonical unit lifting. 624
Canonical visibility depth. 49
Canonical witness. 447
Canonical-demand agreement theorem 580
Canonical-hit capacity. 571
Canonical-normal-form criterion 531
Canonical-section closure dichotomy. . 494
Canonical-segment invariant. 442
Canonical-segment property 676
Canonical-strip characterization. 622
Carry-onset hierarchy 583
Category-relative minimality. 493
Causal hypergraph invariance. 506
CCR-3 arithmetic geometry. 282
CCR-3 baseline 284
CCR-3. 268
Centered phase-normal form 588
Centered-fiber reconstruction 663
Central diagnostic exclusion 518
Character-twisted Taylor slice 702
Characteristic divisibility 638
Characteristic root determines the oriented Farey parents 662
Classical tree, new decoration. 629
Closed form of the odd-order source-lattice index 320
Closed vector profiles. 624
Coarea characterization. 595
Coarea values and component columns. 612
Coefficient-free explicit formula. 601
Coefficient-free scope. 601
Coherence notions 531
Common cardinality on the fixed-signature skew diagonal. 615
Common-legal-symbol property 818
Complete joint law for the fifth column 316
Complete lower-order invariance 216
Complete minimal-onset algebraicity . 586
Complete rank-two homotopy classification 711
Complete schedule classification for $\min(p, q) \geq 3$ 515
Complete third-column valuation law. 278
Complete unrestricted state–stack frontier 768
Completion equation. 17
Completion frontier. 52
Completion totality 696
Completion totality. 155
Completion–Beatty identification. 631
Completion-phase metric 215
Completion-total system 728
Completion-totality 206, 215
Completion–cancellation boundary duality. 38
Composition-refined labeled run–return expansion. 618
Composition-refined ratio-run with exact returns. 612
Composition-refined ratio-run–return enumeration. 612
Compressed row 226
Computational verification 211
Computer-assisted exact 3:2 result. . . . 87
Conditional all-odd orbit-count mean 326
Conditional discrimination bound 213
Configuration-level separation of the three defect channels. 468
Confluence does not imply section-complete normalization 536
Connected chronological support 285
Connected interval support 292
Connected-quadruple translation law 290
Constant finite physical image in every rank 747
Constant lower-order signature 212
Contact-preserving value compression. 604
Contract-refinement Pareto monotonicity 793
Contractible rank-one saturation. 176
Contractible translate intersections . . 698
Coordinate identity 229
Coordinate reducedness. 497
Coordinate-shift improvement. 88

Coordinate/semantic separation principle. 506
Coordinatewise geodesic criterion 216
Core downward closure 514
Correction. 239
Count-exact carry-order repair. 471
Count-shadow fiber region. 475
Counter predicates 803
Counting synthesized sections. 568
Critical replacement threshold 769
Cross-parameter Ordered Farey Recursion. 627
Cross-sign unary separation. 94
Cubical Bellman equivalence 200
Cubical geodesic compiler 201

Debt birth dichotomy. 555
Debt distinguishability witness. 448
Defect coordinates. 473
Defect quotient. 474
Defect–Signature Characterization. . . 244
Defect–Signature Master Equation. . . 258
Deferral capacity bound. 481
Deferred exposure is impossible 755
Deferred path-memory lower bound . . 746
Demand-length clock. 462
Determinant-two parity completion . . 705
Deterministic–nondeterministic OCN language equivalence 847
Dimension of the continuous optimum polytope. 159
Diophantine completion interpretation 641
Direct full-frontier comparison 796
Direct-realization factorization 678
Directed cause recovers the conflict. . 505
Directed completion metric. 43
Discriminant and ramified field. 280
Displacement-compatible physical transport 717
Displacement-marked group 742
Distance and first-action queries 203
Distance does not determine topology 232
Distinguished k-fiber capacity 218
Distinguished separating face 208
Distinguished structural simplex 215
Dominating kernel level. 459
DPDA-equivalence behavioral fallback. 574
Dyck forced-debt frontier. 572

Eager-rule portability criterion. 501
Eager–lazy section incomparability. . . 496
Eager–lazy semantic equivalence. 455
EF–FG Actuation-resolution language collapse and causal power . . 1017
Effective bounded-height compression 828
Effective transfer principle 233
Effective universality 233
Effective-barrier interpretation. 609
Effective-barrier regularization. 608
Eight-Site Theorem 512
Elementary-Abelian Prime-Core Lemma 803
Empty stack only at block boundaries 680
Endmarker determinism. 9
Endpoint-bridge exclusion 707
Epsilon global small counterexample . 892
Epsilon-Robust Zero-Cut Section Lemma 807
Equal distance 207
Equal witness distance 216
Essential branching. 495
Every fixed DOCA factor budget $k \geq 2$ 822
Every integer residual is reachable. 41
Every lattice residual is completable. 155
Exact $\min(p, q) = 2$ classification 516
Exact $A \to B$ category bifurcation . . . 523
Exact k-face decoder 217
Exact $q = 1$ schedules 516
Exact 73-adic global count and covariance law 314
Exact 3:1 three-symbol state complexity. 86
Exact 3:2 finite reset certificate. 563
Exact 3:2 schedule classification. 563
Exact ε-transition count. 101
Exact m-step overlap. 175
Exact adjacent-swap factorization . . . 658
Exact almost-cycle frontier. 52
Exact ambiguity and an auxiliary parity-of-runs model 936
Exact arbitrary-coprime schedule-confluence dichotomy 533
Exact Bellman encoding 229
Exact bounded visibility. 49
Exact broad/deferred state premium . 794
Exact buffered ancestry law 745
Exact bulk LIFO-obstruction rates . . 685
Exact category bifurcation for the classical $A \to B$ edge. 508
Exact CCR source-obstruction lattice and coherent integral witness 319
Exact chain separator 223
Exact common lower-order truncation 208
Exact commutation potential. 59

Exact completion lattice 641
Exact completion metric 228
Exact contact-set determinant. 606
Exact correction triangle. 256
Exact criterion for downward-closed schedule families 532
Exact cut semantics for one-counter nets 827
Exact deferred-SEARCH control width. 449
Exact depth-two geodesic locality of the original machine. 48
Exact deterministic one-pebble characterization 935
Exact dilation 933
Exact distinguished-layer capacity . . . 218
Exact exposure-rank lower bound 774
Exact fiber characterization 729
Exact finite schedule classification. . . . 573
Exact finite-precision residue counts . 308
Exact first LIFO-obstruction survival law 678
Exact first-order source obstruction . . 319
Exact fixed-policy root verification. . . 574
Exact four-symbol raw-defect set. 73
Exact geodesic domain. 46
Exact geodesic ladder. 34
Exact geodesic-fork classification. 61
Exact geodesic-prefix criterion 697
Exact geodesic-prefix criterion. 160
Exact geodesic-word growth. 63
Exact global toral correction Pareto law 722
Exact Hessenberg realization 294
Exact homogeneous-tail frontier. 84
Exact information versus shadow information 301
Exact length and counting profile 735
Exact minimal eventual distance period. 190
Exact minimal local-geometry period. 192
Exact nested raw-safe comparison . . . 796
Exact no-endmarker cost in the early raw models. 87
Exact no-endmarker raw cost. 96
Exact normalized configuration rectangle. 71
Exact normalized phase-local frontier. 99
Exact north-step return positions. . . . 607
Exact north-step return set. 613
Exact one-work-symbol raw-conjugacy complexity. 78
Exact order filtration 295
Exact order-k variation 212
Exact parameter density 307
Exact parameter moments and rare-resonance limit 309
Exact phase-local Pareto staircase . . . 795
Exact prefix metric 228
Exact primary dependency classification. 504
Exact product reduction 535
Exact pure-zero slice. 114
Exact quadratic normal form 305
Exact quotient spheres. 46
Exact rank-one cubical f-vector. 183
Exact rank-one quotient vertex count. 182
Exact raw residual-faithful frontier. . . . 95
Exact raw two-resource frontier 795
Exact reach of the dilation–Boolean method 934
Exact real-time literal-demand lift. . . 463
Exact regularity characterization 931
Exact residuation for one-counter nets 828
Exact root classification. 573
Exact root counts at every prime-power precision 308
Exact sectioned lift 530
Exact sectioned lift. 492
Exact semantic-cell count. 100
Exact shortest-completion count. 58
Exact small-order Galois groups 325
Exact source reconstruction 296
Exact terminal decoder 637
Exact three-site schedule classification. 498
Exact three-symbol state complexity for 2:1. 85
Exact total transition count. 101
Exact two-stage source identification . 384
Exact Universal Row Complexity and Zero Shell Defect 221, 227
Exact unrestricted bounded-replacement one-work-symbol complexity 768
Exact unrestricted one-work-symbol state–push Pareto staircase . 769
Exact unrestricted right-endmarker state complexity 758
Exact WRITE normalization. 443
Exact-contact specialization. 606
Exact-entry bounded-return compactness 844
Example. 612
Excursion reward mean and exponential moments 681
Explicit $q = 1$ fatal certificates 520
Explicit composition-refined run–return enumeration. 611
Explicit deterministic pebble gap 935

Explicit Factor Construction 809
Explicit fixed-ratio-signature cardinality. 615
Explicit fixed-signature polynomial. . . 616
Explicit normalized fourth column . . . 290
Explicit scalar realization 729
Exponential signature fibers 212
Exposure-rank form. 550
Extension-Ideal Classification 230
Extremal compression preserves the complex. 226

Face-query complexity 204
Farey completion tree. 628
Farey residual 640
Farey sign of the swap 659
Fatal causality implies conflict. 561
Fatal Frontier. 560
Fatal-support criterion 532
Fiber congruence and semantic quotient. 482
Filled shortest-geodesic quotient. 164
Filtered collapse-capacity theorem. . . 550
Finite k-fiber compilation 214
Finite containment-core computation. 574
Finite containment-failure certificate. 574
Finite cubical geodesic compilation . . 201
Finite diagnostic basis. 562
Finite fundamental-group presentation 699
Finite geodesic compiler 197
Finite minimal-core existence 526
Finite physical-lift kernel 693
Finite protected Pareto envelope 528
Finite rooted synthesis. 568
Finite symbolic skeleton and global path-outcome coherence. 571
Finite two-digit certificate 317
Finite weak-envelope normal form . . . 879
Finite-control inverse simulation 933
Finite-depth unary-stack descent 763
Finite-prime simultaneous tuning versus fixed-source rigidity 322
Finite-to-macroscopic operational transfer 688
First actions. 161
First homology as the Freiman shadow 700
First simultaneous zero return 818
First third homology 731
First three fixed columns. 273
First three normalized columns 285
First-hit/protected-phase synchronization 766
Fixed marked group with unbounded higher homology 744
Fixed rank and fixed fundamental group with unbounded higher homology 740
Fixed-action star amplification 777
Fixed-budget DOCA compositionality 800
Fixed-composition component encoding. 607
Fixed-composition forbidden-column determinant. 608
Fixed-continuation stack regularity . . 754
Fixed-defect algebraicity. 261
Fixed-rank unbounded complexity . . . 714
Fixed-semantics operational limit 688
Fixed-signature skew diagonal. 614
Fixed-source prime support and exact two-digit stability 317
Fixed-source seventh/ninth prime-axis laws 326
Fixed-system rank-two bifurcation . . . 712
Flat local realization of a finite representation 744
Forbidden-value compression. 596
Forced kernel-excess return. 502
Forced separator and free suffix. 615
Forced unit SEARCH for buried cancellation. 469
Forward inclusion 810, 821
Forward invariance of behavioral safety. 555
Four-band Hessenberg realization 288
Fourth-band Hessenberg matrix 288
Fourth-column transport 289
Fourth-jet Stirling shadow 291
Fourth-order non-identifiability from CCR-3 286
Fourth-source reconstruction 286
Full completion-fiber confluence 534
Full completion-fiber rewrite 534
Full free-episode realization 719
Full toral realization and exact repair/debt tradeoff 723
Full transverse-face preimage 733
Full-complex description 209
Full-Shell Universality 231
Fundamental directed cycle. 44
Fundamental quadratic returns 661
Fundamental-group universality 233

Gauge persistence of pure holonomy . 717
General differential-operator realization 298
General Hessenberg matrix 294
General higher-order chronological-carry theorem 301

General physical phase clock. 480
Geodesic Parikh synchronization. 37
Geodesic-phase metric 215
Geometry does not determine arithmetic 293
Global fifth-column congruence moments 312
Global materialization budget. 480
Good-cover intersections. 176
Graphs inside one 2-fiber 217
Growing numerical thresholds need growing bottom-up control . . 932
Guarded-macro transfer theorem 886

Halting implies primality 821
Halting Implies Prime 808
Hard-direction OCN small-witness theorem. 846
Hessenberg realization. 269
Hidden suffix cannot coalesce 773
Higher even moments. 635
Higher-torus realization. 730
Historical eager refinement. 456
Historical stack actions as materialization bursts. 487
Historical states as SEARCH sheets. . 486
Homogenized semigroup correspondence 202
Horizontal interval complexes 732

Infinite-prime adaptive pulse gap 387
Infinitely many steps. 117
Integer discriminant certificate and effective resonance bound . . . 306
Integer residual realizability 751, 761
Integral homology profiles 711
Integral invariants 735
Integral-source finiteness and the height-limit dichotomy 331
Inverse state-to-replacement law 769
Iterated products 730

Kernel economy. 474
Kernel lift equals quadratic-return translation 660
Kernel reduction. 622
Kernel–correction separation. 256

Labeling scope. 619
Labelings of a fixed composition. 618
Last-height crossing. 92
Lattice-convex realization 728
Lattice-convexity and alignment 732
Lex-normalized protected-core rewrite 537
Line-locality of repeated differences . . 706
Linear visibility excess 742
Linear-order bottom-up state complexity 932
Literal preservation of every shortest completion 914
Literal-box criterion. 166
Local carry visibility criterion. 501
Local commuting diamonds. 37
Local Gap Cost. 242
Local, not global, Markoff rigidity . . . 659
Local–global Farey correspondence. . . 628
Locality–debt–recoordination trilemma. 552
Lossless Signature Theorem. 252
Lower-order block separation 286
Lower-orthant shattering bound 226

Main enumeration theorem. 600
Main theorem 799
Mandatory-WRITE reset envelope . . . 511
Marked finite-permutation transport . 744
Marked product transfer 743
Materialization amount from stack length. 477
Matrix residual quotient classification. 156
Maximal fiberwise semantic containment core. 558
Maximal interface-safe quotient criterion 527
Maximum antichains are middle levels 225
Mean and variance of rematerialization 683
Meaning of “unrestricted” 751
Minimal active fiber 729
Minimal kernel-carry rewrite. 434
Minimal parallelogram lemma 704
Minimal-conflict participation. 562
Minimal-return prefix collapse. 614
Minimal-return suffix barrier. 610
Minimum carry-order debt. 471
Minimum completion demand. 20
Minimum number of globally altered endpoint comparisons 722
Minimum-demand geodesic descent. . . . 31
Mixed-domination deletion 828
Model R. 91
Modular quotient/remainder synthesis. 570
Motivating $d = 2$ correspondence. 238
Multivariate contact determinant. . . . 605

Near-diagonal central fibers 690
Near-diagonal predecessor bound 691

Near-diagonal universal Rayleigh scale 692
Negative synchronization. 94
Negative unit-lift theorem. 624
Negative-side state lower bound 756
No fixed lower-order reconstruction . . 209
No later WRITE in the primitive sector 579
No negative outer carry. 589
No positive cycle in the reduced OCN mode graph 842
No raw configuration repetition. 116
Non-semidecidability 822
Nondegeneracy of the negative family 206
Nonhalting Implies Composite 811
Nonnegative integer sources and exact reachability 324
Normalized correction signature. 239
Normalized input macro-step. 70
Normalized residual–configuration conjugacy. 70
Normalized shadow and denominator resonances 300
Normalized shadow away from $p = 5$. 291

Observable refinement 792
Observer chain 704
Obstruction monotonicity. 507
Odd-p:2 fatal-fork family. 566
Odd-p:2 reset closure. 564
Odd-column Haar product law 328
One root representative determines the section. 568
One-kernel trapping 515
One-or-two normal-form law 533
One-row identifiability 290, 299
One-sided reflection. 632
One-top specialization. 552
Online bit complexity after normalization 205
Operator form 289
Operator form. 270
Optimal Parikh fiber. 157
Optimal simpliciality 200
Optimal-compatibility complex 200
Optimal-edge criterion 197
Optimal-Fiber Nerve Compression . . . 699
Optimal-fiber product theorem 730
Optimal-Fiber Taylor Slice. 701
Ordinary run as a first-missing component column. 613
Ordinary run, $(m, n) = (12, 4)$. 601
Ordinary run, $(m, n) = (13, 5)$. 601
Ordinary run–return. 600
Ordinary-run scope. 612
Original-machine shortest-completion extraction. 12
Original-machine state-potential invariant. 11
Output-sensitive geodesic enumeration 204
Outward first-hit band. 765
Overlapping-face independence 217
Owner factor versus current regular state 811

Palindromic sandwich identity 641
Parallelogram–abelian fundamental group 705
Parallelogram–commutator principle . 700
Parallelogram-branch homotopy theorem 709
Parallelogram-free homotopy theorem 708
Parent recovery. 626
Parity propagation 704
Periodic signed-winding realization dichotomy 948
Periodic spectra, including infinite examples. 934
Persistent literal carrier bound. 745
Phase capacity. 98
Phase coherence. 472
Physical configurations determine residuals 751
Physical length formula. 476
Physical materialization level. 476
Physical prism and apex transfer 745
Physical pure-image theorem 744
Physical rematerialization constant . . 681
Physical residual rigidity. 761
Physical-lift duality. 690
Physical-lift invariant measure 680
Pointwise sharpness 775
Policy-essential but broad-class removable branching. 508
Positive complete-homogeneous expansion. 611
Positive odd residuals. 237
Positive synchronization. 93
Positive-return total multiplicity bound 843
Positive-side lower bound 756
Prefix components determined by the signature. 614
Prefix criterion 197
Prefix quotient classification. 42
Prefix-query complexity 203
Prescribed-order adjacent-state non-factorization 209
Presence–absence contact calculus. . . . 606

Prime and composite DOCA instances 800
Prime-DOCA Undecidability 812
Primitive balanced CANCEL distribution 579
Primitive same-sign CCR-4 288
Primitive same-sign hierarchy 297
Primitive SEARCH law 679
Principal truncation lemma 702
Prism law 738
Prism preserves the pure sector 743
Problem 24.04(A), epsilon hard inclusion 849
Problem 24.04(A), epsilon reverse inclusion 849
Problem 24.04(A), epsilon-move extension 849
Problem 24.04(A), real-time global small witness 849
Problem 24.04(A), real-time language equivalence 848
Product distance 211
Product lower bound. 123
Product-join lemma 211
Projectivity of the π_2 module 709
Promise efficiency inequality 834
Protected exact-lift class. 536
Protected modulo-equivalence 537
Protected paired-session comparison . 915
Protected positive phases and simultaneous descent 764
Protected representation contract 792
Protected signature certificate 538
Protected unary pump. 752
Protected unary-stack pump 763
Protected-core confluence criterion . . . 537
Protected-interface lift 526
Protected-role capacity certificate 793
Pure nonabelian Stable-LIFO holonomy 718
Pure-loop sector 716
Pushdown residual interpretation. . . . 596

Quantitative forced-debt frontier. 571
Quantitative relative-source localization 838
Quantity defect. 467
Quotient antitonicity 792
Quotient Failure-Factor Lemma. 804
Quotient invariant 820
Quotient-protected branching. 495

Ramified square class. 281
Raney signature law 584
Rank-one global homotopy dichotomy. 176
Rank-one local action topology. 172
Rank-one macroscopic fluctuations . . . 689
Rank-one rigidity 699
Rank-three branches 738
Rank-two displacement-kernel visibility trichotomy 716
Rank-two first-Betti bound 700
Rank-Two Homological-Dimension Theorem 703
Rank-two seed family 738
Ratio-run, $(m, n) = (10, 4)$. 602
Ratio-run, $(m, n) = (11, 5)$. 601
Ratio-run–return. 600
Rational Dyck path. 595
Raw nonrecurrence on the positive unary ray 752
Raw rectangle criterion in the four-symbol orientation. 75
Raw residual–configuration conjugacy. 73
Raw-height escape. 117
Reachable semantic containment. 558
Reachable-containment root shortcut. 574
Reachable-store normal form. 10
Reading the minimal return from the signature. 615
Real-time centered rich DPDA. 444
Real-time one-stack zero-return recognition 914
Real-time scope of all-rank carrier family 747
Realtime upper construction. 113
Reconstruction. 479
Rectangular shortest-completion geometry. 56
Recursion-theoretic consequences 813
Recursive coherence. 628
Reduced lift nonuniqueness. 497
Reduced non-isomorphism with common normalized behavior. 500
Regular control predicate 802
Regular immediate-disagreement saturation. 573
Regular structural safe-reset certification. 573
Relation to flip methods 659
Relation with cumulant language 286
Relation with the $a = 0$ Cohn normalization 639
Relative completion coordinates 640
Repair is nearest-flat nonabelian cocycle optimization 746
Representation-obstruction separation. 508
Reset synchronization 820
Reset-ray fork template. 562

Residual classification of right quotients 751
Residual distinguishability. 447
Residual invariant. 438
Residuals classify right quotients 761
Resource down-set 793
Restricted-class undecidability 812
Return components are barrier contacts. 607
Return polynomial. 599
Return–contact correspondence. 595
Reversal transpose 638
Reverse inclusion 821
Rich-line incidence 707
Rich-to-demand projection. 433
Right-residual classification. 489
Right-turn witness bijection 936
Robust fatal supports give an independence complex. 562
Robust positive first-return compactness 886
Root correctness via reachable containment. 558
Root correctness. 554
Rooted path-outcome coherence criterion. 567
Rooted zero-debt/raw-conjugacy equivalence. 547
Rooted-sheet canonicalization 536

Safe-reset certificate. 560
Same fundamental group, different displacement visibility 741
Sampled run. 595
Scalar Farey profiles. 630
Scalar jump law and affine periodicity. 630
Scaled integrality defect. 187
Schedule activation 532
Schedule poset. 499
Schedule presentation 510
Schedule-induced direct-canonicality obstruction. 502
Schedule-induced fiber expansion. 502
SCQ reduction theorem 526
Semantic chronology uniqueness. 462
Semantic non-determination of causal structure. 506
Semantic preservation and exact kernel debt 534
Semantic trap containment. 557
Semantic WRITE count. 461
Semilinear path relation 196
Shadow blindness versus exact identifiability 291
Sharp factor within uniform dilation . 933
Sharp free literal carrier 721
Sharp linear-modular two-source realization dichotomy 941
Sharp mixed-sign lower-band threshold 324
Sharp semantic restrictions 915
Sheet-separable normalization 535
Shell generating polynomial. 632
Short parallel-path obstruction certificate. 568
Shortest completion. 622
Shortest-completion stratum 535
Shortest-demand quotient. 445
Shortest-prefix down-set. 160
Shortest-prefix quotient theorem. 165
Signature Persistence. 246
Signed integer multisnapshot criterion 323
Simple Hensel tower. 276
Simplex-strand product rigidity 733
Simultaneous All-Rank Saturation . . . 222, 231
Simultaneous first-missing statistics. . 606
Simultaneous run and ratio-run. 613
Simultaneous run–ratio-run–return-set enumeration. 613
Simultaneous source-native realization 914
Simultaneous-query direct sum 780
Single-face difference through order k 208
Singular extinction. 277
Site-free diagnostic frontier. 561
Six coordinate-reduced schedule lifts. 499
Six-schedule terminal coherence. 500
Size and syntactic restrictions 820
Slope separation. 121
Source-independent prime-axis densities 318
Source-stack exposure depth 773
Source-Stack Exposure Profile 774
Spectrum dilation obstruction 933
Squarefree divisor identification. 171
Stabilized defect profile. 189
Star-amplified pushdown bound 779
State-minimality in the prime case . . . 812
State-preserving bounded-push compiler. 97
State-preserving push-at-most-two macro compiler 771
Static two-visible obstruction. 431
Stationarity and one-parameter classification 287
Stationary p-divisible fourth polarization 287

Stationary p-divisible class 297
Stationary extension classification . . . 297
Stationary translation law 299
Step-pump lemma. 118
Stirling shadow. 274
Strict coordinate quotient 525
Strictly canonicalizing physical rewrite 531
Strictness by down-set exclusion 793
Strong normalization of canonical protected-core descent 537
Structural cube 199
Structural derivation of the historical identity. 487
Structural simpliciality 200
Subdiagonal light cone 295
Suffix minimum 752
Suffix-parametric positive first-return compactness 852
Surviving return contacts. 608
Symbol-level affine tail periodicity . . . 686
Synchronized p-phase family 753

Tail escape. 91
Target Construction Lemma 806
Target-indexed topological transition 736
Terminal block form through order four 285
Terminal collision multiplicity 746
Terminal Cost. 243
Terminal factorization. 554
Terminal protected fork 537
Terminal sign 641
Terminal transport formula 660
Termination 532
The F_3 branch 714
The base $p = 0$ 659
The bounded three-symbol point is Pareto-minimal 771
The canonical 2:1 fork 533
The exceptional prime 5 over parameters 310
The first two bounded levels 768
The fixed signature 11111110. 616
The pair 7:4. 624
The signature is not generally a prefix. 614
The smallest nontrivial graph fiber . . . 779
Third-order connected support. 268
Three-channel canonicality. 468
Three-scale LIFO realization 689
Three-stage bounded-local obstruction decomposition. 551
Three-symbol raw configuration rectangles. 84
Three-symbol raw residual–configuration conjugacy. 81
Three-symbol residual realization 758
Three-way diagnosis. 502
Tight depth-two visibility. 14
Top-λ annotation compiler. 101
Top-rewrite suffix criterion. 464
Torus-point injectivity 703
Total bridge-kernel perturbation 687
Translation-isolated apex 737
Transport system. 272
Transverse determinant lemma 706
Tree-coordinate lemma. 251
Two factors suffice in the nonhalting case 822
Two-cell visibility. 443
Two-channel physical debt 535
Two-Counter Transcript Lemma 803
Two-factor undecidability 822
Two-fold normal form and critical-value collision 313
Two-site failure of universal downward closure 522
Two-source critical-value isometry . . . 315
Typed CCR-k
  Event-equality source-mask coding 401
  Finite-ring response profiles 400
  Future-source support budget . . . 399
  Selective monitors and exact algorithms 396
  Sharp five-step forecast archive . . 398
Typed terminal-block CCR-k 293
Typical-case direct sum 781
Typing of named causal edges 523

Ultra-log-concavity 580
Unary completion uniqueness. 120
Unary height escape 752
Unary-stack height escape 762
Unbounded comparison requires memory 915
Unbounded rank-two sphere family . . 713
Unbounded suffix minima. 752
Unbounded third homology in rank three 741
Uniform D-induced fiber expansion. . 564
Uniform centered eager-return compiler 511
Uniform exposure dichotomy 754
Uniform kernel-suffix fatality. 565
Uniform LIFO-obstruction law 689
Uniform one-top rewrite criterion. . . . 494
Uniform prime-density law 307
Uniform reachability of the length-three frontier. 565

Uniform source-height threshold 322
Uniform threshold nonrecognizability 933
Uniform-hypergraph realization 217
Unique connected quadruple 285
Unique physical WSC count-shadow factorization. 478
Unique semantic CANCEL–carry factorization. 460
Unit semantic-shell mass 680
Unit-step normalization for one-counter nets 827
Universal k-Fiber Theorem 217
Universal Bellman Compatibility 222
Universal exposure spectrum 779
Universal Extension-Fiber Completeness 222, 230
Universal face-collapse retention 776
Universal fixed-source finite prime-support theorem 321
Universal Fundamental Spectrum. . . . 631
Universal Information-to-Physical Exposure Theorem 778
Universal moment hierarchy. 632
Universal odd-column CCR p-adic fold 311
Universal Optimal-Compatibility Theorem 229
Universal realization 223
Universal representation conservation. 467
Universal semantic quotient. 490
Universal Shell Spectrum. 632
Universal visibility exact sequences . . 742
Universal-cover higher homology 708
Unrestricted bounded one-work-symbol state complexity 760
Unrestricted one-work-symbol baseline 767
Unrestricted physical presentation nonuniqueness. 484
Unrestricted Prime-DOCA 822
Unrestricted state lower bound 756

Vector jump law. 626
Vector-valued Beatty shuffle. 628
Virtual-demand projection. 433

Weighted forbidden-value determinant polynomial. 598
Weighted forbidden-value determinant. 598
What $\lambda = 1$ means 288
Why exponent two is ideal 804
Why exposure is a distinct resource . . 775
Word-level swap factorization 658
Word-to-Farey coordinate conjugacy . 639
Wrapped trace defect equals reversal defect 663
WRITE frontier bound 675
WRITE-frontier minority bound. 439
WRITE-weight conservation 583
WSC defect lattice. 472
WSC semantic core equals the residual core. 490

Zero-curvature return rigidity 659
Zero-debt bit budget 778
Zero-debt finite-observation obstruction. 551
Zero-defect semigroup. 188
Zero-defect signatures. 247
Zero-height quotient and owner invariant 810
Zero-residual correctness 676
Zero-residual correctness. 443
Zero-weight loop coherence. 571

Notation Index

$L_A(r)$, liii
$\mathrm{Adm}(\mathcal{E})$, li

$\mathcal{B}_H$, li
b_i, lii

C, L, R, l
$\mathcal{U}(P)$, lii
$\alpha \vDash n$, lii

$D(S)$, l
D_d, l
δ_a, li
$\mathcal{D}_H$, li
$\mathcal{D}_{m,n}$, lii

$\mathsf{SF}_x, \mathsf{PO}_x$, li
$\Sigma_\theta^{\pm}(x; F)$, li
$\eta(x, y)$, xlix

$\mathsf{Factor\text{-}DOCA}_k$, fixed-budget decomposition, 800
$((c, u), (v, d))$, lii
$\mathcal{F}_{gq,gp}$, liii

h_{ctr}, l
$\sigma_Q(v)$, lii
$\mathbf{h}$, l

K, xlix
$\kappa_{p,q}$, liii
$\mathcal{C}_d$, lii

$L_{p,q}$, xlix
$L_{p,q}(r)$, lii

M, l
m, xlix
$\mathfrak{M}(P)$, liii
$\kappa_F(p, x)$, li

M_t, l
$\mu(r)$, xlix
$\mu_{p,q}(r)$, lii

N, li
$N_a(w)$, xlix

$\mathrm{Ob}_h(P_h)$, li
Ω, l
$\mathcal{O}_A(r)$, liii
$g_k^{\mathrm{ord}}(A)$, liii

P, l
$H_{c_0}(m)$, li
$\Pi_\rho(P)$, liii
K_0, K_{lin}, liii

Q, l
$Q_{\min}(L_{p,q})$, liii

$\rho_{p,q}(w)$, xlix
$\mathrm{run}, \widehat{\mathrm{run}}, \mathrm{ret}$, lii

$\mathrm{Sat}_{\mathsf{Cat}}(A, B)$, li
$\delta_{\mathrm{pair}}, \delta_{\mathrm{asm}}$, li
Q_*, lii
$I_A(r)$, liii
Σ, l
$\sigma : M \to C$, li

$\chi(\mathcal{C})$, liii
$\mathcal{K}_Q$, lii
Θ, l
$\Theta : C \to M$, li
ξ, li
τ, li

U_a, l

$w_{\max}, w_{\min}$, xlix